

LOCAL STUDY OF SCHEMES AND THEIR MORPHISMS (EGA IV)

CUMULATIVE ENGLISH TRANSLATION — §§1–21

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SUMMARY

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- §19. Regular immersions and normal flatness.
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- §21. Divisors.

The subjects discussed in the chapter call for the following remarks.

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- (a) The common property of all the subjects discussed is that they all related to *local* properties of preschemes or morphisms, i.e. considered at a point, or the points of a fibre, or on a (non-specified) neighbourhood of a point or of a fibre. These properties are generally of a *topological*, *differential*, or *dimensional* nature (i.e. bringing the ideas of *dimension* and *depth* into play), and are linked to the properties of the *local rings* at the points considered. One type of problem is the relating, for a given morphisms $f : X \rightarrow Y$ and point $x \in X$, of the properties of X at x with those of Y at $y = f(x)$ and those of the fibre $X_y = f^{-1}(y)$ at x . Another is the determining of the topological nature (for example, the constructibility, or the fact of being open or closed) of the set of points $x \in X$ at which X has a certain property, or for which the fibre $X_{f(x)}$ passing through x has a certain property at x . Similarly, we are interested in the topological nature of the set of points $y \in Y$ such that X has a certain property at all the points of the fibre X_y , or those such that this fibre itself has a certain property.
- (b) The most important idea for the following chapters is that of *flat morphisms of finite presentation*, as well as the particular cases of *smooth morphisms* and *étale morphisms*. Their detailed study (as well as that of connected questions) really starts in §11.
- (c) Sections §§1–10 can be considered as being preliminary in nature, and as developing three types of techniques, used, not only in the other sections of the chapter, but also, of course, in the follow chapters:
 - (c1) Sections §§1–4 are envisaged as treating the diverse aspects of the idea of *change of base*, above all in relation with the conditions of *finiteness* or *flatness*; we there initiate the technique of *descent*, with its most elementary aspects (the questions of “effectiveness” linked to this technique will be studied in Chapter V).
 - (c2) Sections §§5–7 are focused on what we may call *Noetherian* techniques, since the preschemes considered are always locally Noetherian, whereas, on the contrary, there is generally no finiteness condition imposed on the *morphisms*; this is essentially due to the fact that the ideas of dimension and depth are hardly manageable except in the case of Noetherian local rings. Recall that §7 constitutes a “delicate” theory of Noetherian local rings, not much used in what follows in the chapter.
 - (c3) Sections §§8–10 describe, amongst other things, the means of *eliminating the Noetherian hypotheses* on the preschemes considered, by substituting such hypotheses for suitable ones of *finiteness* (“finite presentation”) on the *morphisms* considered: the advantage of this substitution is that the latter such hypotheses (those of finiteness on the morphisms) are *stable under base change*, which is not the case for the Noetherian hypotheses on the preschemes. The technique permitting this substitution relies, in some part, on the use of the idea of the *projective limit* of preschemes, thanks to which we can reduce a question to the same question with *Noetherian* hypotheses; on the other hand, it relies on the systematic use of *constructible sets*, which have the double interest of being preserved under taking inverse images (of arbitrary morphisms) and by direct images (of morphisms of finite presentation), and having manageable topological properties in locally Noetherian preschemes. The same techniques often even allow to restrict to the case of more specific Noetherian rings, for example the **Z**-algebras of *finite type*, and it is here that the properties of “excellent” rings (studied in §7) intervene in a decisive

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manner. Independently of the question of elimination of Noetherian hypotheses, the techniques of §§8–10, elementary in nature, find constant use in nearly all applications.

§1. RELATIVE FINITENESS CONDITIONS. CONSTRUCTIBLE SETS IN PRESCHEMES

In this section, we will resume the exposé of “finiteness conditions” for a morphism of preschemes $f : X \rightarrow Y$ given in (I, 6.3 and 6.6). There are essentially two notions of “finiteness” of a *global* nature on X , that of *quasi-compact* morphism (defined in (I, 6.6.1)) and that of a *quasi-separated* morphism; on the other hand, there are two notions of “finiteness” of a *local* nature on X , that of a morphism *locally of finite type* (defined in (I, 6.6.2)) and that of a morphism *locally of finite presentation*. By combining these local notions with the preceding global notions, we obtain the notion of a morphism *of finite type* (defined in (I, 6.3.1)) and of a morphism *of finite presentation*. For the convenience of the reader, we will give again in this section the properties stated in (I, 6.3 and 6.6), referring to their labels in Chapter I for their proofs.

In n^{os} 1.8 and 1.9, we complete, in the context of preschemes, and making use of the previous notions of finiteness, the results on constructible sets given in (0_{III}, §9).

1.1. Quasi-compact morphisms

Definition (1.1.1). — We say that a morphism of preschemes $f : X \rightarrow Y$ is *quasi-compact* if the continuous map f from the topological space X to the topological space Y is quasi-compact (0, 9.1.1), in other words, if the inverse image $f^{-1}(U)$ of every quasi-compact open subset U of Y is quasi-compact (cf. (I, 6.6.1)).

If $\mathfrak{B}$ is a basis for the topology of Y consisting of affine open sets, then for f to be quasi-compact, it is necessary and sufficient that for all $V \in \mathfrak{B}$, $f^{-1}(V)$ is a *finite union of affine open sets*. For example, if Y is affine and X is quasi-compact, *every* morphism $f : X \rightarrow Y$ is quasi-compact (I, 6.6.1).

If $f : X \rightarrow Y$ is a quasi-compact morphism, then it is clear that for every open subset V of Y , the restriction of f to $f^{-1}(V)$ is a quasi-compact morphism $f^{-1}(V) \rightarrow V$. Conversely, if (U_α) is an open cover of Y and $f : X \rightarrow Y$ is a morphism such that the restrictions $f^{-1}(U_\alpha) \rightarrow U_\alpha$ are quasi-compact, then f is quasi-compact. As a result, if $f : X \rightarrow Y$ is an S -morphism of S -preschemes, and if there exists an open cover (S_λ) of S such that the restrictions $g^{-1}(S_\lambda) \rightarrow h^{-1}(S_\lambda)$ of f (where g and h are the structure morphisms) are quasi-compact, then f is quasi-compact.

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Proposition (1.1.2). — (i) *An immersion $X \rightarrow Y$ is quasi-compact if it is closed, or if the underlying space of Y is locally noetherian, or if the underlying space of X is noetherian.*

(i) *The composite of two quasi-compact morphisms is quasi-compact.*

(i) *If $f : X \rightarrow Y$ is an S -morphism quasi-compact, then the same is true of $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ for every base extension $g : S' \rightarrow S$ of the base prescheme.*

(i) *If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are two S -morphisms quasi-compact,*

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is quasi-compact.

(i) *If the composite $g \circ f$ of two morphisms $f : X \rightarrow Y$, $g : Y \rightarrow Z$ is quasi-compact, and if g is separated, or X locally noetherian, f is quasi-compact.*

(i) *For a morphism f to be quasi-compact, it is necessary and sufficient that f_{red} be so.*

For the proof, see (I, 6.6.4). We note that (vi) is also a consequence of the following more general proposition:

Proposition (1.1.3). — *Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms. If $g \circ f$ is quasi-compact and f surjective, g is quasi-compact.*

Indeed, if V is a quasi-compact open subset in Z , $f^{-1}(g^{-1}(V))$ is quasi-compact by hypothesis, and since $g^{-1}(V) = f(f^{-1}(g^{-1}(V)))$, since f is surjective, $g^{-1}(V)$ is quasi-compact.

Corollary (1.1.4). — *Let $f : X \rightarrow Y$, $g : Y' \rightarrow Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$; suppose g quasi-compact and surjective (resp. dominant). Then, for f to be quasi-compact (resp. dominant), it suffices that f' be so.*

If $g' : X' \rightarrow X$ is the canonical projection, g' is a surjective morphism (I, 3.5.2), and so $f \circ g' = g \circ f'$; if f' is quasi-compact (resp. dominant), the same is true of $g \circ f'$ since g is quasi-compact (1.1.2) (resp. surjective); as $f \circ g'$ is quasi-compact (resp. dominant) and g' is surjective, we deduce that f is quasi-compact (1.1.3) (resp. dominant).

To abbreviate, we will call a *maximal point* of a prescheme X any generic point of an irreducible component of X ; if $X = \text{Spec}(A)$ is affine, the maximal points of X are the *minimal prime ideals* of A .

Proposition (1.1.5). — *Let $f : X \rightarrow Y$ be a quasi-compact morphism. The following conditions are equivalent:*

- a) f is dominant;
- a) for every maximal point y of Y , $f^{-1}(y) \neq \emptyset$;
- a) for every maximal point y of Y , $f^{-1}(y)$ contains a maximal point of X .

The equivalence of a) and c) has been proved in (I, 6.6.5). It is clear that c) implies b); on the other hand, if X' is an irreducible component of X meeting $f^{-1}(y)$, $f(X')$ is contained in the irreducible component $Y' = \overline{\{y\}}$ of Y (0I, 2.1.6); if x is the generic point of X' , we necessarily have $f(x) = y$ (0I, 2.1.5); thus b) implies c).

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Proposition (1.1.6). — *Let $f' : X' \rightarrow Y$, $f'' : X'' \rightarrow Y$ be two morphisms, X the prescheme sum $X' \amalg X''$, $f : X \rightarrow Y$ the morphism which coincides with f' in X' and with f'' in X'' . For f to be quasi-compact, it is necessary and sufficient that f' and f'' be so.*

This follows immediately from the definition.

1.2. Quasi-separated morphisms

Definition (1.2.1). — Let X, Y be two preschemes. We say that a morphism $f : X \rightarrow Y$ is *quasi-separated* (or that X is *quasi-separated over Y*) if the diagonal morphism $\Delta_f = (1_X, 1_X)_Y : X \rightarrow X \times_Y X$ is quasi-compact. We say that a prescheme X is *quasi-separated* if it is quasi-separated over $\text{Spec}(\mathbf{Z})$.

By definition, every separated morphism is quasi-separated, Δ_f being a closed immersion (1.1.2), (i)); a *scheme* X is a prescheme quasi-separated, being separated over $\text{Spec}(\mathbf{Z})$ (I, 5.4.1).

Proposition (1.2.2). — (i) *Every monomorphism of preschemes $f : X \rightarrow Y$ (in particular every immersion) is quasi-separated.*

- (i) *If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are two quasi-separated morphisms, $g \circ f : X \rightarrow Z$ is quasi-separated.*
- (i) *Let X, Y be two S -preschemes, $f : X \rightarrow Y$ an S -morphism quasi-separated. Then, for every base change $g : S' \rightarrow S$, the morphism $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is quasi-separated.*
- (i) *If $f : X \rightarrow Y$, $f' : X' \rightarrow Y'$ are two S -morphisms quasi-separated, then*

$$f \times_S f' : X \times_S X' \longrightarrow Y \times_S Y'$$

is quasi-separated.

- (i) *If $f : X \rightarrow Y$, $g : Y \rightarrow Z$ are two morphisms such that $g \circ f$ is quasi-separated, then f is quasi-separated.*
- (i) *If $f : X \rightarrow Y$ is a quasi-separated morphism, $f_{\text{red}} : X_{\text{red}} \rightarrow Y_{\text{red}}$ is quasi-separated.*

By virtue of (I, 5.5.12), it suffices to prove (i), (ii), and (iii). The assertion (i) is trivial, every monomorphism being separated (I, 5.5.1). To prove (iii), we immediately reduce to the case where $Y = S$ (I, 3.3.11) and the assertion follows from the fact that $\Delta_{f_{(S')}} = (\Delta_f)_{(S')}$ (I, 5.3.4) and from (1.1.2), (iii). To prove (ii), consider the projections p and q of $X \times_Y X$ onto X ; if $i = (p, q)_Z$, we know that the diagram

$$\begin{array}{ccc} X & \xrightarrow{\Gamma_f} & X \times_Z Y \\ f \downarrow & & \downarrow 1 \times_Z f \\ Y & \xrightarrow{\Delta_g} & Y \times_Z Y \end{array}$$

is commutative and identifies $X \times_Y X$ as the product of the $(Y \times_Z Y)$ -preschemes Y and $X \times_Z X$ (I, 5.3.5). If g is quasi-separated, Δ_g is a quasi-compact morphism, and the same is true of Γ_f (1.1.2), (iii); the same is true of $i \circ \Delta_f$ (1.1.2), (ii)) which equals $\Delta_{g \circ f}$.

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Corollary (1.2.3). — (i) Let X be a quasi-separated prescheme. Then every morphism $f : X \rightarrow Y$ is quasi-separated.
 (ii) Let Y be a quasi-separated prescheme. For a morphism $f : X \rightarrow Y$ to be quasi-separated, it suffices that the prescheme X be quasi-separated.

This follows immediately from (1.2.2), (ii) and (v)) applied to X , Y and $\text{Spec}(\mathbf{Z})$.

Proposition (1.2.4). — Let $f : X \rightarrow Y$ be a morphism, $g : Y \rightarrow Z$ a morphism quasi-separated. If $g \circ f$ is quasi-compact, then the same is true of f .

We know that f is the composite morphism $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{\text{pr}_2} Y$ (I, 5.3.13); on the other hand, pr_2 is identified with $(g \circ f) \times_Z 1_Y$ (I, 3.3.4) and if $g \circ f$ is quasi-compact, the same is true of pr_2 (1.1.2), (iv)). Finally, we have the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\Gamma_f} & X \times_Z Y \\ f \downarrow & & \downarrow f \times 1_Y \\ Y & \xrightarrow{\Delta_g} & Y \times_Z Y \end{array}$$

(1.2.4.1)

which identifies X as the product of the $(Y \times_Z Y)$ -preschemes Y and $X \times_Z Y$ (I, 5.3.7). Since by hypothesis Δ_g is a quasi-compact morphism, the same is true of Γ_f (1.1.2), (iii)); the conclusion follows from (1.1.2), (ii)).

Proposition (1.2.5). — Let $f : X \rightarrow Y$, $g : Y' \rightarrow Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. If g is quasi-compact and surjective and f' quasi-separated, then f is quasi-separated.

Indeed $(X' \times_{Y'} X') = (X \times_Y X)_{(Y')}$ and $\Delta_{f'} = (\Delta_f)_{(Y')}$ (I, 5.3.4); since the projection $X' \times_{Y'} X' \rightarrow X \times_Y X$ is quasi-compact (1.1.2), (iii)) and surjective (I, 3.5.2), one can apply (1.1.4), which shows that since $\Delta_{f'}$ is a quasi-compact morphism, the same is true of Δ_f .

Proposition (1.2.6). — Let $f : X \rightarrow Y$ be a morphism, (U_α) an open cover of Y such that the preschemes induced U_α are quasi-separated. For f to be quasi-separated, it suffices that each of the preschemes induced by X on the open sets $f^{-1}(U_\alpha)$ be quasi-separated.

The inverse image $X_\alpha \times_Y X_\alpha$ of U_α is $X_\alpha \times_{U_\alpha} X_\alpha$, with $X_\alpha = f^{-1}(U_\alpha)$, and the restriction $X_\alpha \times_{U_\alpha} X_\alpha$ to X_α of Δ_f is none other than Δ_{f_α} , where f_α denotes the restriction $X_\alpha \rightarrow U_\alpha$ of f ; by virtue of the definition (1.2.1) and of the local character of the notion of quasi-compact morphism (1.1.1), for f to be quasi-separated, it suffices that each f_α be quasi-separated. But as by hypothesis the morphism $U_\alpha \rightarrow \text{Spec}(\mathbf{Z})$ is quasi-separated, it amounts to the same to say that f_α is quasi-separated or that the composite $X_\alpha \rightarrow U_\alpha \rightarrow \text{Spec}(\mathbf{Z})$ is (1.2.2), (ii) and (v)), whence the proposition.

To verify that a morphism is quasi-separated, one is thus led to give criteria for a prescheme to be quasi-separated:

Proposition (1.2.7). — Let X be a prescheme, (U_α) an open cover of X consisting of quasi-compact open sets. The following properties are equivalent:

- a) X is quasi-separated.
- a) For every quasi-compact open subset U of X , the canonical immersion $U \rightarrow X$ is quasi-compact (in other words, U is retrocompact in X).
- b') The intersection of two quasi-compact open subsets of X is quasi-compact.
- a) For every pair of indices (α, β) , the intersection $U_\alpha \cap U_\beta$ is quasi-compact.

The properties b) and b') are trivially equivalent by definition of a quasi-compact morphism. Since a quasi-compact open is a finite union of affine open sets, for two quasi-compact open sets U , V of X , $U \times_Z V$ is a quasi-compact open subset of $X \times_Z X$ (I, 3.2.7), whose inverse image by Δ_X is $U \cap V$, so a) implies b'). It is trivial that b') implies c); finally, if c) is satisfied, the $U_\alpha \times_Z U_\beta$ form a cover of $X \times_Z X$ by quasi-compact open sets, and we know that for the morphism $\Delta_X : X \rightarrow X \times_Z X$ to be quasi-compact, it suffices that the inverse images of $U_\alpha \times_Z U_\beta$ by Δ_X be quasi-compact (1.1.1); hence c) implies a).

Corollary (1.2.8). — *Every prescheme X whose underlying space is locally noetherian (for example a locally noetherian prescheme) is quasi-separated; every morphism $f : X \rightarrow Y$ is then quasi-separated.*

Proposition (1.2.9). — *Let X', X'' be two preschemes, $X = X' \amalg X''$ their sum, $f' : X' \rightarrow Y, f'' : X'' \rightarrow Y$ two morphisms, $f : X \rightarrow Y$ the morphism which coincides with f' in X' and with f'' in X'' . For f to be quasi-separated, it is necessary and sufficient that f' and f'' be so.*

Indeed, $X \times_Y X$ is the sum of the four preschemes $X' \times_Y X', X' \times_Y X'', X'' \times_Y X'$ and $X'' \times_Y X''$, and Δ_f is the morphism which coincides with $\Delta_{f'}$ in X' and with $\Delta_{f''}$ in X'' ; the proposition therefore follows immediately from the definitions.

1.3. Morphisms locally of finite type

(1.3.1). Recall that if A is a ring, a A -algebra (commutative) B is said to be of *finite type* if it is generated by a finite number of elements of B , or, what amounts to the same, if it is isomorphic to a quotient of a polynomial algebra $A[T_1, \dots, T_n]$ by an ideal of this algebra. It is clear that for every A -algebra (commutative) $A', B \otimes_A A'$ is then an A' -algebra of finite type. If B is an A -algebra of finite type and C a B -algebra of finite type, C is an A -algebra of finite type, for C is a quotient of $B[T_1, \dots, T_n]$ and B a quotient of $A[S_1, \dots, S_m]$, $B[T_1, \dots, T_n] = B \otimes_A A[T_1, \dots, T_n]$ is isomorphic to $A[S_1, \dots, S_m, T_1, \dots, T_n]$ and is a quotient of $A[S_1, \dots, S_m, T_1, \dots, T_n]$, so the same is true of C .

Definition (1.3.2). — Let $f : X \rightarrow Y$ be a morphism of preschemes, x a point of $X, y = f(x)$. We say that f is of *finite type in x* if there exists an affine open neighbourhood V of y and an affine open neighbourhood U of x such that $f(U) \subset V$ and such that the ring $A(U)$ is an $A(V)$ -algebra of finite type. We say that f is *locally of finite type* if it is of finite type at every point of X .

Proposition (1.3.3). — *If Y is a locally noetherian prescheme and $f : X \rightarrow Y$ a morphism locally of finite type, then X is locally noetherian.*

For the proof, see (I, 6.3.7).

Proposition (1.3.4). — (i) *Every local immersion is locally of finite type.*

(i) *If two morphisms $f : X \rightarrow Y, g : Y \rightarrow Z$ are locally of finite type, the same is true of $g \circ f$.*

(i) *If $f : X \rightarrow Y$ is an S -morphism locally of finite type, $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is locally of finite type for every base extension $S' \rightarrow S$ of the base prescheme.*

(i) *If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are two S -morphisms locally of finite type, $f \times_S g$ is locally of finite type.*

(i) *If the composite $g \circ f$ of two morphisms is locally of finite type, f is locally of finite type.*

(i) *If a morphism f is locally of finite type, the same is true of f_{red} .*

For the proof, see (I, 6.3.6).

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Corollary (1.3.5). — *Let $f : X \rightarrow Y$ be a morphism locally of finite type. For every morphism $Y' \rightarrow Y$ such that Y' is locally noetherian, $X \times_Y Y'$ is locally noetherian.*

Indeed, $f_{(Y')} : X \times_Y Y' \rightarrow Y'$ is locally of finite type by (1.3.4), (iii)), and it suffices to apply (1.3.3).

Proposition (1.3.6). — *Let $\varphi : A \rightarrow B$ be a homomorphism of rings. For the corresponding morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ to be locally of finite type, it is necessary and sufficient that B be an A -algebra of finite type.*

For the proof, see (I, 6.3.3), taking into account that f is quasi-compact and (I, 6.6.3).

Proposition (1.3.7). — *For a morphism locally of finite type $f : X \rightarrow Y$ to be surjective, it is necessary and sufficient that, for every algebraically closed field Ω , the map $X(\Omega) \rightarrow Y(\Omega)$ corresponding to f (I, 3.4.1) be surjective.*

For the proof, see (I, 6.3.10).

(1.3.8). Let A be a ring. We say that an A -algebra B is *essentially of finite type* if B is A -isomorphic to an A -algebra of the form $S^{-1}C$, where C is an A -algebra of finite type and S a multiplicative subset of C .

Proposition (1.3.9). — (i) *If B is an A -algebra essentially of finite type and C a B -algebra essentially of finite type, then C is an A -algebra essentially of finite type.*

(i) Let B be an A -algebra essentially of finite type, A' an A -algebra; then $B' = B \otimes_A A'$ is an A' -algebra essentially of finite type.

(i) Let $B = S'^{-1}B'$ and $C = T'^{-1}C'$, where B' (resp. C') is an A -algebra (resp. a B -algebra) of finite type and S' (resp. T') a multiplicative subset of B' (resp. C'); then C' is also the form $S'^{-1}C''$, where C'' is a B' -algebra of finite type, so C is also a ring of fractions of C'' ; as C'' is an A -algebra of finite type, this proves our assertion.

(ii) If B is a ring of fractions of an A -algebra of finite type C , B' is a ring of fractions of the A' -algebra of finite type $C \otimes_A A'$, whence (ii).

(1.3.10). If B is an A -algebra locally essentially of finite type, B is of the form $C_{\mathfrak{q}}$, where C is an A -algebra of finite type and $\mathfrak{q}$ a prime ideal of C (Bourbaki, *Alg. comm.*, chap. II, §5, prop. 11). Let $\mathfrak{p}$ be the inverse image in A of the ideal $\mathfrak{q}$; if we set $S = A - \mathfrak{p}$, $C_{\mathfrak{q}}$ is also a local ring at a prime ideal of $S^{-1}C$; as $S^{-1}C$ is an algebra of finite type over $A_{\mathfrak{p}} = S^{-1}A$, we see that B is also an $A_{\mathfrak{p}}$ -algebra essentially of finite type, and furthermore the homomorphism $A_{\mathfrak{p}} \rightarrow B$ is local.

Proposition (1.3.11). — If B is an A -algebra locally essentially of finite type, B is A -isomorphic to an A -algebra quotient of a localisation $C_{\mathfrak{q}}$, where C is a polynomial algebra $A[T_1, \dots, T_n]$ and $\mathfrak{q}$ a prime ideal of C .

Indeed, by definition, B is isomorphic to $C'_{\mathfrak{q}'}$, where C' is an A -algebra of finite type; but $C' = C/\mathfrak{b}$, where $C = A[T_1, \dots, T_n]$ and $\mathfrak{b}$ is an ideal of C ; so $\mathfrak{q}' = \mathfrak{q}/\mathfrak{b}$, where $\mathfrak{q}$ is a prime ideal of C , and $C'_{\mathfrak{q}'}$ is isomorphic to $C_{\mathfrak{q}}/\mathfrak{b}C_{\mathfrak{q}}$.

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1.4. Morphisms locally of finite presentation

(1.4.1). Given a ring A , we say that an A -algebra B (commutative) is of *finite presentation* if it is isomorphic to $A[T_1, \dots, T_n]/\mathfrak{a}$ for a *finitely generated* ideal $\mathfrak{a}$ of $A[T_1, \dots, T_n]$. For every A -algebra (commutative) A' , $B \otimes_A A'$ is then an A' -algebra of finite presentation, being isomorphic to the quotient of $A'[T_1, \dots, T_n]$ by the image of $\mathfrak{a}$ via the canonical map $A[T_1, \dots, T_n] \rightarrow A'[T_1, \dots, T_n]$, which is then an ideal of finite type. If B is an A -algebra of finite presentation and C a B -algebra of finite presentation, then C is an A -algebra of finite presentation. Indeed, let $B = A[S_1, \dots, S_m]/\mathfrak{a}$ and $C = B[T_1, \dots, T_n]/\mathfrak{b}$, where $\mathfrak{a}$ (resp. $\mathfrak{b}$) is a finitely generated ideal of $A[S_1, \dots, S_m]$ (resp. $B[T_1, \dots, T_n]$); the ring $B[T_1, \dots, T_n]$ is isomorphic to $A[S_1, \dots, S_m, T_1, \dots, T_n]/\mathfrak{a}'$, where $\mathfrak{a}'$ is the image of $\mathfrak{a}$ via the canonical map of $A[S_1, \dots, S_m]$ to $A[S_1, \dots, S_m, T_1, \dots, T_n]$, and thus an ideal of finite type; the ideal $\mathfrak{b}$ of $B[T_1, \dots, T_n]$ is of the form $\mathfrak{b}'/\mathfrak{a}'$, where $\mathfrak{b}'$ is an ideal of $A[S_1, \dots, S_m, T_1, \dots, T_n]$; as $\mathfrak{a}'$ and $\mathfrak{b}'/\mathfrak{a}'$ are finitely generated modules, the same is true of $\mathfrak{b}'$, and C , isomorphic to $A[S_1, \dots, S_m, T_1, \dots, T_n]/\mathfrak{b}'$, is indeed an A -algebra of finite presentation. Note finally that if A is *noetherian*, the same is true of $A[T_1, \dots, T_n]$, so every A -algebra of finite type is then of finite presentation.

Definition (1.4.2). — Let $f : X \rightarrow Y$ be a morphism of preschemes, x a point of X , $y = f(x)$. We say that f is of *finite presentation in x* if there exists an affine open neighbourhood V of y and an affine open neighbourhood U of x such that $f(U) \subset V$ and such that the ring $A(U)$ is an $A(V)$ -algebra of finite presentation. We say that f is *locally of finite presentation* if it is of finite presentation at every point of X .

If Y is locally noetherian, it amounts to the same to say that f is locally of finite type or locally of finite presentation.

Proposition (1.4.3). — (i) Every local isomorphism is locally of finite presentation.

(i) If two morphisms $f : X \rightarrow Y$, $g : Y \rightarrow Z$ are locally of finite presentation, the same is true of $g \circ f$.

(i) If $f : X \rightarrow Y$ is an S -morphism locally of finite presentation, $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is locally of finite presentation for every base extension $S' \rightarrow S$ of the base prescheme.

(i) If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are two S -morphisms locally of finite presentation, $f \times_S g$ is locally of finite presentation.

(i) If the composite $g \circ f$ of two morphisms is locally of finite presentation and if g is locally of finite type, then f is locally of finite presentation.

The assertion (i) is trivial. To prove (iii), one can restrict to the case where $S = Y$ (I, 3.3.11); let x' be a point of $X_{(S')}$, s' and x its projections onto S' and X respectively, s the common projection of s' and x onto S . By hypothesis, there exists an affine open neighbourhood V (resp. V') of s (resp. of x) such that the image of U be contained in V and such that $A(U)$ be an $A(V)$ -algebra of finite presentation; let V' be an affine open neighbourhood of s' whose image in S is contained in V , and $U' = \text{Spec}(A \otimes_{A(V)} A(V'))$ an affine open neighbourhood of x' whose image in S' is contained in V' and whose ring $A(U) \otimes_{A(V)} A(V')$ is an $A(V')$ -algebra of finite presentation.

One knows that (iv) follows from (i), (ii), and (iii) (I, 3.5.1). To prove (v), let us consider a point $x \in X$, and let $y = f(x)$, $z = g(f(x))$. There exists an affine open neighbourhood $W = \text{Spec}(C)$ (resp. $V = \text{Spec}(B)$) of z (resp. of y) such that $g(V) \subset W$ and such that B is a C -algebra of finite type. On the other hand, there exists an affine open neighbourhood $V' = \text{Spec}(B')$ (resp. $U = \text{Spec}(A)$) of y (resp. of x) such that $f(U) \subset V'$ and such that A is a B' -algebra of finite presentation. Let $s \in B$ be such that the open set $D(B_s)$ in V is a neighbourhood of y contained in V' , and let $U' = f^{-1}(D(s)) \subset U$; we have $U' = \text{Spec}(A \otimes_{B'} B_s)$, and $A \otimes_{B'} B_s$ is a B_s -algebra of finite presentation (1.4.1); on the other hand $B_s = B[T]/(1 - sT)$ is a B -algebra of finite presentation by definition; so by (1.4.1), $A \otimes_{B'} B_s$ is a C -algebra of finite presentation. One sees therefore that on the other hand f is also locally of finite presentation.

On the other hand, consider the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\Gamma_f} & X \times_Z Y \\ f \downarrow & & \downarrow f \times 1 \\ Y & \xrightarrow{\Delta_g} & Y \times_Z Y \end{array}$$

which identifies X as the product of the $(Y \times_Z Y)$ -preschemes Y and $X \times_Z Y$ (I, 5.3.7). If Δ_g is locally of finite presentation, the same follows for Γ_f by (iii); on the other hand, we have the factorisation of f as $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$ (I, 5.3.13), where the projection p_2 equals $(g \circ f) \times_Z 1_Y$; as $g \circ f$ is assumed locally of finite presentation, the same is true of p_2 by (iv), and it will follow from (ii) that f is also locally of finite presentation. One sees therefore that we are led to proving the

Corollary (1.4.3.1). — *Let $g : Y \rightarrow Z$ be a morphism locally of finite type; then the diagonal morphism $\Delta_g : Y \rightarrow Y \times_Z Y$ is locally of finite presentation.*

One can restrict to the case where $Z = \text{Spec}(A)$, $Y = \text{Spec}(B)$, B being an A -algebra of finite type; the morphism Δ_g corresponds then to the *augmentation homomorphism* $\delta : B \otimes_A B \rightarrow B$ such that $\delta(x \otimes y) = xy$. In view of the definition (1.4.2), it suffices to show that the kernel $\mathfrak{J}$ of δ is an ideal of finite type, which follows from (0_{IV}, 20.4.4).

Proposition (1.4.4). — *Let A be a ring, B an A -algebra, B' a polynomial algebra $A[T_1, \dots, T_n]$, $u : B' \rightarrow B$ a surjective homomorphism of A -algebras. For B to be an A -algebra of finite presentation, it is necessary and sufficient that the kernel $\mathfrak{J}$ of u be a finitely generated ideal of B' .*

The condition is sufficient by definition (1.4.1). To see that it is necessary, note that the morphism $g : \text{Spec}(B') \rightarrow \text{Spec}(A)$ is locally of finite type; if B is an A -algebra of finite presentation, the morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(B')$ corresponding to u is locally of finite presentation, so $g \circ f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ is locally of finite presentation. But, to prove that the ideal $\mathfrak{J}$ is of finite type, it suffices to prove that $\tilde{\mathfrak{J}}$ is a B' -Module of finite type (II, 6.1.4.1). Returning to the definition (1.4.2), we are finally led to proving the

Lemma (1.4.4.1). — *Let $\mathfrak{a}$ be an ideal of a ring C , s an element of C , such that $C_s/\mathfrak{a}_s$ is a C -algebra of finite presentation; then $\mathfrak{a}_s$ is a finitely generated ideal of C_s .*

By hypothesis, there exists a C -algebra of polynomials $C' = C[T_1, \dots, T_n]$ and a C -homomorphism surjective $v : C' \rightarrow C_s/\mathfrak{a}_s$ whose kernel $\mathfrak{b}$ is of finite type. Let $\rho : C_s \rightarrow C_s/\mathfrak{a}_s$ be the canonical homomorphism; for each i , there exists $t_i \in C$ such that $\rho(t_i/s^k) = v(T_i)$ (one may suppose that the exponent k is the same for all indices i). Consider then the C_s -algebra $D = C_s[T_1, \dots, T_n]$, and the homomorphism of C_s -algebras $w : D \rightarrow C_s$ such that $w(T_i) = t_i/s^k$; w is evidently surjective,

and by the same construction $v' = \rho \circ w : D \rightarrow C_s/\mathfrak{a}_s$. On the other hand, every polynomial of D can be written P/s^m , where $P \in C' = C[T_1, \dots, T_n]$, and we have $v'(P/s^m) = (1/s^m)v(P)$; as $1/s$ is invertible in C_s , the relation $v'(P/s^m) = 0$ in $C_s/\mathfrak{a}_s$ is equivalent to $v(P) = 0$, and by the same the kernel $\mathfrak{b}'$ of v' is generated by $\mathfrak{b}$ in the ring D . But as $\mathfrak{b}'$ is the image of $\mathfrak{b}$ by the canonical map of $\mathfrak{b}$ to D , and the kernel of w is the ideal generated by the $F_i(T_i)$ in D ; this proves that $\mathfrak{a}_s$ is a finitely generated ideal of C_s .

Corollary (1.4.5). — *Let X, Y be two preschemes, $j : X \rightarrow Y$ an immersion, U an open subset of Y such that $j(X)$ is closed in U , $\mathcal{J}$ the quasi-coherent Ideal of $\mathcal{O}_U$ defining the closed sub-prescheme of Y associated to j . For j to be locally of finite presentation, it is necessary and sufficient that $\mathcal{J}$ be an $\mathcal{O}_U$ -Module of finite type.*

Proposition (1.4.6). — *Let $\varphi : A \rightarrow B$ be a homomorphism of rings. For the corresponding morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ to be locally of finite presentation, it is necessary and sufficient that B be an A -algebra of finite presentation.*

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Proof. The condition being trivially sufficient, it remains to prove that it is necessary. If f is locally of finite presentation, it already follows from (1.3.6) that B is an A -algebra of finite type, so there exists a surjective homomorphism $u : B' = A[T_1, \dots, T_n] \rightarrow B$ of A -algebras; it follows from ((1.4.3), (v)) that the immersion $j : \text{Spec}(B) \rightarrow \text{Spec}(B')$ is locally of finite presentation; hence (1.4.5), if $\mathfrak{J}$ is the kernel of u , the B' -Module $\tilde{\mathfrak{J}}$ is of finite type, and consequently (II, 6.1.4.1) $\mathfrak{J}$ is an ideal of finite type. $\square$

Proposition (1.4.7). — *Let $\rho : A \rightarrow B$ be a homomorphism of rings making B a finite A -algebra. For B to be an A -algebra of finite presentation, it is necessary and sufficient that B be an A -module of finite presentation.*

We will use the following

Lemma (1.4.7.1). — *If B is a finite A -algebra, there exists a surjective A -homomorphism of algebras $u : B' \rightarrow B$, where B' is a finite A -algebra of finite presentation that is a free A -module.*

Proof. Indeed, there is a finite system $(a_i)_{1 \leq i \leq m}$ of generators of the A -algebra B such that each a_i satisfies a relation $F_i(a_i) = 0$, where $F_i \in A[X]$ is a monic polynomial of degree > 0 ; the quotient A -algebra $B'_i = A[X]/(F_i)$ is therefore free of finite rank over A ; it suffices then to take for B' the tensor product $B'_1 \otimes_A B'_2 \otimes \dots \otimes_A B'_m$ and to consider the homomorphism $u_1 \otimes u_2 \otimes \dots \otimes u_m : B' \rightarrow B$, which is surjective by construction. Moreover, if $B'' = A[T_1, \dots, T_n]$ and if b'_i denotes the canonical image of c_i in B' , the A -homomorphism of algebras $v : B'' \rightarrow B'$ such that $v(T_i) = b'_i$ ($1 \leq i \leq m$) is surjective, and its kernel $\mathfrak{b}'$ is the ideal of B'' generated by the $F_i(T_i)$, hence of finite type; this proves that B' is an A -algebra of finite presentation. $\square$

Proof of (1.4.7). This lemma being proved, keep the same notation, and let $w = v \circ u : B'' \rightarrow B$. If $\mathfrak{b}$ (resp. $\mathfrak{a}$) is the kernel of w (resp. u), we have $\mathfrak{a} = v(\mathfrak{b})$ since v is surjective. But (1.4.4), if B is an A -algebra of finite presentation, $\mathfrak{b}$ is an ideal of finite type of B'' , so $\mathfrak{a}$ is a finitely generated A -module of finite type since B is an A -algebra of finite type; as B' is an A -module of finite presentation, $\mathfrak{a}$ is *a fortiori* a finitely generated ideal of B' ; by the same, B is a B' -algebra of finite presentation, and as B' is an A -algebra of finite presentation, B is an A -algebra of finite presentation. $\square$

1.5. Morphisms of finite type

Proposition (1.5.1). — *Let $f : X \rightarrow Y$ be a morphism, (U_α) an open cover of Y consisting of affine open sets. The following conditions are equivalent:*

- a) *f is locally of finite type and quasi-compact.*
- a) *For each α , $f^{-1}(U_\alpha)$ is a finite union of affine open sets $V_{\alpha i}$ such that each ring $A(V_{\alpha i})$ is an $A(U_\alpha)$ -algebra of finite type.*
- a) *For every affine open subset U of Y , $f^{-1}(U)$ is a finite union of affine open sets V_j such that the rings $A(V_j)$ are $A(U)$ -algebras of finite type.*

For the proof, see (I, 6.3.2) and (I, 6.6.3).

Definition (1.5.2). — We say that a morphism f is of *finite type* if it satisfies the equivalent conditions of the proposition (1.5.1).

Proposition (1.5.3). — *Let $f : X \rightarrow Y$ be a morphism of finite type; if Y is noetherian, the same is true of X .*

For the proof, see (I, 6.3.7).

Proposition (1.5.4). — (i) *Let $j : X \rightarrow Y$ be a morphism of immersion. If j is quasi-compact (which is the case if j is a closed immersion, or if the underlying space of X is noetherian, or if the underlying space of Y is locally noetherian), j is of finite type.*

- (i) *The composite of two morphisms of finite type is of finite type.*
- (i) *If $f : X \rightarrow Y$ is an S -morphism of finite type, $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is of finite type for every base extension $S' \rightarrow S$ of the base prescheme.*
- (i) *If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are two S -morphisms of finite type, $f \times_S g$ is of finite type.*
- (i) *If the composite $g \circ f$ of two morphisms is of finite type and if g is quasi-separated or if X is noetherian, f is of finite type.*
- (i) *If a morphism f is of finite type, the same is true of f_{red} .*

This follows immediately (except the case (v) where X is noetherian, proved in (I, 6.3.6)) from the definitions and from (1.1.2), (1.2.4) and (1.3.4).

Corollary (1.5.5). — *Let $f : X \rightarrow Y$ be a morphism of finite type. For every morphism $Y' \rightarrow Y$ such that Y' is noetherian, $X \times_Y Y'$ is noetherian.*

This follows from (1.5.4), (iii)) and from (1.5.3).

Corollary (1.5.6). — *Let X be a prescheme of finite type over a locally noetherian prescheme S . Then every S -morphism $f : X \rightarrow Y$ is of finite type.*

For the proof, see (I, 6.3.9).

Proposition (1.5.7). — *Let $\varphi : A \rightarrow B$ be a homomorphism of rings. For the corresponding morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ to be of finite type, it is necessary and sufficient that φ make B an A -algebra of finite type.*

For the proof, see (I, 6.3.3).

1.6. Morphisms of finite presentation

Definition (1.6.1). — *Let X, Y be two preschemes. We say that a morphism $f : X \rightarrow Y$ is of finite presentation if it satisfies the following three conditions:*

- (i) *f is locally of finite presentation;*
- (i) *f is quasi-compact (which, together with (i), is equivalent to saying that f is of finite type (1.5.2));*
- (i) *f is quasi-separated.*

We then also say that X is of finite presentation over Y , or a Y -prescheme of finite presentation.

The condition (iii) is automatically satisfied if f is separated, or if X is locally noetherian (1.2.8); when Y is locally noetherian, it amounts to the same to say that f is of finite presentation or that f is of finite type (the latter condition implying that X is locally noetherian (1.3.3)).

Proposition (1.6.2). — (i) *Every quasi-compact immersion (in particular every quasi-compact open immersion) is of finite presentation.*

- (i) *The composite of two morphisms of finite presentation is of finite presentation.*
- (i) *Let X, Y be two S -preschemes, $f : X \rightarrow Y$ an S -morphism of finite presentation. Then, for every base change $S' \rightarrow S$, the morphism $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is of finite presentation.*
- (i) *Let $f : X \rightarrow Y, f' : X' \rightarrow Y'$ be two S -morphisms of finite presentation; then $f \times_S f' : X \times_S X' \rightarrow Y \times_S Y'$ is of finite presentation.*
- (i) *If the composite $g \circ f$ of two morphisms is of finite presentation and if g is quasi-separated and locally of finite type, then f is of finite presentation.*

This follows immediately from (1.1.2), (1.2.2), (1.2.4) and (1.4.3).

It follows in particular from (1.6.2), (iii)) that if f is a morphism of finite presentation and U an open subset of Y , the restriction $f^{-1}(U) \rightarrow U$ of f is a morphism of finite presentation. Conversely,

let (U_α) be a cover of Y by affine open sets, and suppose that the restrictions $f^{-1}(U_\alpha) \rightarrow U_\alpha$ of f are morphisms of finite presentation; then f is evidently locally of finite presentation, quasi-compact by virtue of (1.1.1) and quasi-separated by virtue of (1.2.6). One can therefore say that the property for a morphism $f : X \rightarrow Y$ of being of finite presentation is *local on Y* .

If X is a quasi-separated prescheme, every morphism $f : X \rightarrow Y$ is quasi-separated (1.2.3). So, if f is quasi-compact and locally of finite presentation, f is of finite presentation.

In particular:

Corollary (1.6.3). — *Let $\varphi : A \rightarrow B$ be a homomorphism of rings. For the corresponding morphism $f : \operatorname{Spec}(B) \rightarrow \operatorname{Spec}(A)$ to be of finite presentation, it is necessary and sufficient that φ make B an A -algebra of finite presentation.*

The necessity of the condition follows indeed from (1.4.6).

Remark (1.6.4). — In the definition (1.6.1), the condition (iii) is not a consequence of the other two. Take for example Y an affine scheme whose underlying space is not noetherian, and let U be an open non quasi-compact subset in Y . Let X be the prescheme obtained by gluing two preschemes Y_1, Y_2 isomorphic to Y along the open sets U_1, U_2 corresponding to U (0I, 4.1.7), so that X is the union of two open subsets isomorphic to Y_1 and Y_2 (which we identify with the latter), with $Y_1 \cap Y_2 = U$. Furthermore, there is a morphism $f : X \rightarrow Y$ which coincides in Y_i with the isomorphism $Y_i \rightarrow Y$ given. It is clear that f satisfies condition (i) of (1.6.1), being a local isomorphism; it also satisfies (ii), as the inverse image of an open quasi-compact set is quasi-compact; but as $Y_1 \cap Y_2$ is not quasi-separated (1.2.7), f is not quasi-separated.

Proposition (1.6.5). — *Let X', X'' be two preschemes, $X = X' \amalg X''$ their sum, $f' : X' \rightarrow Y, f'' : X'' \rightarrow Y$ two morphisms, $f : X \rightarrow Y$ the morphism which coincides with f' in X' and with f'' in X'' . For f to be of finite presentation, it is necessary and sufficient that f' and f'' be so.*

It suffices to show that for f to possess one of the three properties of the definition (1.6.1), it is necessary and sufficient that f' and f'' possess it; this is evident for the property of being locally of finite presentation, which is local on X ; for the property of being quasi-compact, this was seen in (1.1.6) and for the property of being quasi-separated, in (1.2.9).

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1.7. Improvements of earlier results We will give in this number a list of propositions proved in earlier chapters, whose statement can be improved with the help of the new finiteness conditions introduced above.

(1.7.1). In the statements of (I, 6.4.2), (I, 6.4.3) and (I, 6.4.9), one can, instead of supposing that X and Y are of finite type over the field K , suppose only that they are *locally of finite type* over K ; for (I, 6.4.2), it suffices to observe that every point of a prescheme X that is closed in every closed subset of X is a closed subset in X , and an affine scheme locally of finite type over K is *ipso facto* of finite type (1.5.1). For (I, 6.4.9), one uses (1.3.7) and (1.3.4), (v)).

(1.7.2). In (I, 6.5.1), (ii)), one can, instead of supposing that S is locally noetherian and Y of finite type over S , suppose only that Y is *locally of finite presentation over S* , as one sees immediately from the proof (by applying the definition (1.4.2)). Similarly in (I, 6.5.4), (ii)) and (I, 6.5.5), (ii)) it suffices to suppose that f is an S -morphism, Y being *locally of finite presentation over S* .

(1.7.3). In (I, 7.1.11), for the second assertion, instead of supposing S locally noetherian and Y of finite type over S , one can suppose only Y locally of finite presentation over S (the proof depending on (I, 6.5.1), (ii)); similarly in (I, 7.1.12) to (I, 7.1.15).

(1.7.4). In (I, 9.2.2), one can replace the three hypotheses by the single hypothesis (implied by each of the others) that f is *quasi-compact and quasi-separated*, by virtue of (1.2.6) and (1.2.7). We note that in (I, 9.2.1), the hypothesis means exactly that f is quasi-compact and quasi-separated.

(1.7.5). In (I, 9.3.1), (I, 9.3.2) and (I, 9.3.3), one can replace the two hypotheses on X by the (weaker) hypothesis that X is a *quasi-compact and quasi-separated* prescheme, the proof using exactly these two properties (by virtue of (1.2.7)).

(1.7.6). In (I, 9.3.4) and (I, 9.3.5), it suffices to suppose X quasi-compact, $\mathcal{F}$ and $\mathcal{G}$ quasi-coherent and of finite type.

(1.7.7). In (I, 9.4.7), one can weaken the hypothesis b) by supposing only X *quasi-separated*, since it suffices that the canonical immersion $U \rightarrow X$ be quasi-compact (1.2.7). One can make the same modification in (I, 9.4.8), (I, 9.4.9) and (I, 9.4.10).

(1.7.8). In (I, 9.5.1), the conditions in parentheses in the statement can be replaced by the condition that f be *quasi-compact and quasi-separated*; the current proof is indeed insufficient (with the hypotheses made), because of the fact that for every quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ such that $R^1 f_*(\mathcal{F}) = 0$, it does not necessarily follow, for an affine open subset U of Y , that the same property holds for the $(\mathcal{O}_X|f^{-1}(U))$ -Modules quasi-coherent, if one does not know whether such a Module is the *restriction* of an $\mathcal{O}_X$ -Module quasi-coherent (the same remark applying to conditions b) and c'); when f is quasi-compact and X quasi-separated, this extension is possible by virtue of (I, 9.4.7), modified above (1.7.7).

(1.7.9). In (II, 5.3.2), it suffices to suppose Y *quasi-compact and quasi-separated* (this being what intervenes in (II, 4.4.6) and (II, 4.4.7)). Similarly in (II, 5.5.3), (ii).

(1.7.10). In (II, 1.7.15), one can suppose only that Y is quasi-compact and quasi-separated, the proof using only (I, 9.6.5).

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(1.7.11). In the second assertion of (II, 3.4.5), one can substitute for the two hypotheses on Y the weaker hypothesis that Y is quasi-compact and quasi-separated. Similarly in (II, 3.4.8).

(1.7.12). In (II, 3.8.5), one can similarly suppose only that Y is quasi-compact and quasi-separated, this hypothesis being that which intervenes in (I, 9.4.7).

(1.7.13). In (II, 4.4.1), (II, 4.4.6) and (II, 4.4.7), it suffices to suppose that Y is *quasi-compact and quasi-separated*, taking into account (1.2.3) in the proof of (II, 4.4.7). Similarly, in (II, 4.4.10.1), it suffices to suppose Z quasi-compact and quasi-separated.

(1.7.14). In (II, 4.5.2) and (II, 4.5.5), it suffices to suppose X *quasi-compact and quasi-separated*, this hypothesis being that which intervenes in (I, 9.3.1) and (I, 9.3.2).

(1.7.15). In (II, 4.6.8), it suffices again to suppose X quasi-compact and quasi-separated, this hypothesis being that which intervenes in (I, 9.4.7).

(1.7.16). In (II, 5.1.2), it suffices to suppose that X is quasi-compact and quasi-separated (the hypothesis intervening in (II, 4.5.2) and (II, 4.5.5)). Similarly in (II, 5.1.9), where one uses (I, 9.6.5).

(1.7.17). In (II, 5.2.1), it suffices always to suppose X quasi-compact and quasi-separated (the hypothesis intervening in (II, 4.5.2)).

(1.7.18). In (II, 5.2.2), one must suppose f *quasi-compact* and X *quasi-separated*; the current proof is indeed insufficient (with the hypotheses made), because of the fact that for every quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ such that $R^1 f_*(\mathcal{F}) = 0$, it does not necessarily follow, for an affine open subset U of Y , that the same property holds for the $(\mathcal{O}_X|f^{-1}(U))$ -Modules quasi-coherent, if one does not know whether such a Module is the restriction of an $\mathcal{O}_X$ -Module quasi-coherent (the same remark applying to conditions b) and c'); when f is quasi-compact and X quasi-separated, this extension is possible by virtue of (I, 9.4.7), modified above (1.7.7).

(1.7.19). In (II, 5.3.2), it suffices to suppose Y *quasi-compact and quasi-separated* (this being what intervenes in (II, 4.4.6) and (II, 4.4.7)). Similarly in (II, 5.5.3), (ii).

(1.7.20). In (III, 1.2.2, 1.2.3 and 1.2.4), one can replace the two hypotheses on X by the weaker hypothesis that X is a prescheme *quasi-compact and quasi-separated*, the proof using only (I, 9.3.3).

(1.7.21). In (III, 1.4.10 to 1.4.14), one can replace “separated” by “quasi-separated”, this hypothesis serving only to be able to apply (I, 9.2.2) (see above (1.7.4)). Similarly, in (III, 1.4.15), it suffices to suppose that the morphism u is *quasi-separated and quasi-compact*.

(1.7.22). In (III, 2.1.3), one can again replace the hypotheses by the hypothesis that X is a prescheme quasi-compact and quasi-separated, the reasoning being the same as in (III, 1.2.2).

1.8. Morphisms of finite presentation and constructible sets

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(1.8.1). Let X be a prescheme; one knows that every quasi-compact open subset in X is a finite union of affine open sets, and conversely. For an open subset U of X to be *retrocompact* in X (0III, 9.1.1), it is necessary and sufficient that for every *affine* open subset V of X , $U \cap V$ be quasi-compact. When X is *quasi-separated* (1.2.1), every quasi-compact open subset in X is retrocompact in X (1.2.7), so every locally constructible subset of X is retrocompact in X (0III, 9.1.13); if in addition X is *quasi-compact*, there is identity between constructible subsets and locally constructible subsets in X (0III, 9.1.12), and between constructible open subsets and quasi-compact open subsets (0III, 9.1.5).

Proposition (1.8.2). — *Let X, Y be two preschemes, $f : X \rightarrow Y$ a morphism. For every constructible (resp. locally constructible) subset Z of Y , $f^{-1}(Z)$ is a constructible (resp. locally constructible) subset of X .*

Suppose Z locally constructible; for every $x \in X$, there will be an affine open neighbourhood V of $f(x)$ such that $Z \cap V$ be constructible in V ; as the proposition is proved for constructible subsets, $f^{-1}(Z) \cap f^{-1}(V)$ will be constructible in $f^{-1}(V)$, so $f^{-1}(Z)$ will be locally constructible. It suffices therefore to consider the case where Z is constructible, and taking into account (0III, 9.1.3), we are reduced to the case where Z is open and retrocompact in Y , in other words such that the canonical injection $Z \rightarrow Y$ is a quasi-compact morphism; it suffices then to see that $f^{-1}(Z)$ is *retrocompact* in X , in other words that the injection $f^{-1}(Z) \rightarrow X$ is a quasi-compact morphism; but this follows from (1.1.2), (iii)), since $f^{-1}(Z) = Z \times_Y X$ (I, 4.4.1).

Lemma (1.8.3). — *Let X be a quasi-compact and quasi-separated prescheme, Z a constructible subset of X . There exist then an affine scheme X' and a morphism of finite presentation $f : X' \rightarrow X$ such that $f(X') = Z$.*

Since X is quasi-compact, it is a finite union of affine open sets X_i ($i \in I$), and since X is quasi-separated, it follows from (1.2.7) that each of the canonical immersions $g_i : X_i \rightarrow X$ is a morphism of finite presentation; one concludes that if $f_i : X'_i \rightarrow X_i$ is a morphism of finite presentation, the same is true of $g_i \circ f_i : X'_i \rightarrow X$ (1.6.2), and if X' is the prescheme sum of the X'_i , the morphism $h : X' \rightarrow X$ which coincides with $g_i \circ f_i$ in each X'_i is also of finite presentation (1.6.5). Since $Z \cap X_i$ is constructible in X_i (0III, 9.1.8, (i)), one sees that one is led to proving the lemma when X_i is *affine*, that is a ring A . As Z is then a finite union of subsets of the form $U \cap \mathbb{A}^n$, where U and V are quasi-compact open subsets (0III, 9.1.3), one sees, by considering a suitable sum of preschemes, that one can even reduce to the case where $Z = U \cap \mathbb{A}^n$; as U is a finite union of open sets of the form X_{f_i} , where $f_i \in A$; let $\mathfrak{J}$ be the ideal of A generated by the f_i , and let Z' be the closed affine sub-scheme of X which it defines (and which is by construction of finite presentation on X); we have by definition $V = \mathbb{A}^n$, so $U \cap \mathbb{A}^n = U \cap Z'$. Consider the affine scheme $X' = Z' \times_X U$ and let $f : X' \rightarrow X$ be the structural morphism; one sees that the canonical immersion $U \rightarrow X$, which is of finite presentation, and the same is true of the open immersion $U \rightarrow X$, which is quasi-compact since U and X are affine; we conclude therefore from (1.6.2), (iv)) that f is of finite presentation, and we have $f(X') = Z' \cap U$ (I, 3.4.8), which finishes the proof.

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Theorem (1.8.4). — (Chevalley). — *Let $f : X \rightarrow Y$ be a morphism of finite presentation. For every locally constructible subset Z of X , $f(Z)$ is locally constructible in Y .*

Let $y \in Y$ and V an affine open neighbourhood of y ; as f is quasi-compact and quasi-separated, the same is true of its restriction $f^{-1}(V) \rightarrow V$, so $f^{-1}(V)$ is a prescheme quasi-compact and quasi-separated (1.2.3); since $Z \cap f^{-1}(V)$ (1.8.1) is constructible in $f^{-1}(V)$, one sees that it suffices to consider the case where Y is affine and Z constructible; but then it will suffice to prove the

Lemma (1.8.4.1). — *Let Y be an affine scheme, $f : X \rightarrow Y$ a morphism quasi-compact and locally of finite presentation; then $f(X)$ is a constructible subset of Y .*

Since X is quasi-compact, it is a finite union of affine open sets; one can therefore reduce to the case where $X = \text{Spec}(A)$, $Y = \text{Spec}(B)$, B being an A -algebra of finite presentation. As Y is quasi-compact and separated, it is also quasi-separated, whence (1.8.3), there exists a morphism of finite presentation $g : X' \rightarrow X$ such that $Z = g(X')$; as $f \circ g$ is of finite presentation (1.6.2), and $f(Z) = f(g(X'))$, we have seen that the question is reduced to proving that $f(X)$ is constructible; since f is of finite presentation (1.6.2), (iv)), it suffices to apply (1.8.4.1).

Lemma (1.8.4.2). — *Let A_0 be a ring, $(A_\lambda)_{\lambda \in L}$ an inductive system of A_0 -algebras, $A = \varinjlim A_\lambda$; if B is an A -algebra of finite presentation, there exist a λ and an A_λ -algebra of finite presentation B_λ such that B is isomorphic to $B_\lambda \otimes_{A_\lambda} A$.*

By hypothesis, B is isomorphic to an algebra of the form $A[T_1, \dots, T_n]/\mathfrak{J}$, where the T_i are indeterminates and $\mathfrak{J}$ an ideal of finite type; let (F_j) ($1 \leq j \leq m$) be a system of generators of $\mathfrak{J}$. Let $\varphi_\lambda : A_\lambda \rightarrow A$ be the canonical homomorphism. Since L is filtering, there exists λ such that each of the coefficients of the polynomials F_j is the image by φ_λ of an element of A_λ . In other words, there exist a system of polynomials $(F_{j\lambda})_{1 \leq j \leq m}$ of $A_\lambda[T_1, \dots, T_n]$ such that $F_j = \varphi_\lambda(F_{j\lambda})$ for $1 \leq j \leq m$. If $\mathfrak{J}_\lambda$ is the ideal of $A_\lambda[T_1, \dots, T_n]$ generated by the $F_{j\lambda}$, $\mathfrak{J}$ is the image of $\mathfrak{J}_\lambda \otimes_{A_\lambda} A$ in $A[T_1, \dots, T_n] = A_\lambda[T_1, \dots, T_n] \otimes_{A_\lambda} A$; the ring $B_\lambda = A_\lambda[T_1, \dots, T_n]/\mathfrak{J}_\lambda$ answers the question.

We will apply this lemma by considering A as an inductive limit of its sub- $\mathbf{Z}$ -algebras of finite type. One sees therefore that there exist such a sub- $\mathbf{Z}$ -algebra A_0 of A , and an A_0 -algebra of finite presentation B_0 such that B is isomorphic to $B_0 \otimes_{A_0} A$; setting $X_0 = \text{Spec}(B_0)$, $Y_0 = \text{Spec}(A_0)$, so that $X = X_0 \times_{Y_0} Y$, f being the projection $X \rightarrow Y$; it follows from (I, 3.4.8) that $f(X) = g_0^{-1}(f_0(X_0))$; taking into account (1.8.2), it suffices therefore to show that $f_0(X_0)$ is constructible. In other words, we are finally reduced to proving the

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Corollary (1.8.5). — *Let Y be a noetherian prescheme, $f : X \rightarrow Y$ a morphism of finite type. For every constructible subset Z of X , $f(Z)$ is a constructible subset of Y .*

One reduces as above to proving that $f(X)$ is constructible. Applying the criterion (0III, 9.2.3): one must prove that for every irreducible closed subset T of Y , $T \cap f(f^{-1}(T))$ is rare in T or contains a non-empty open subset of T ; by taking a sub-prescheme of Y having T as underlying space (I, 5.2.1) and considering its inverse image by f , one is finally (taking into account (1.5.4), (iii)) reduced to proving that if Y is irreducible and f dominant, $f(X)$ contains a non-empty open subset of Y . One can in addition suppose Y affine; then X is a finite union of affine open sets V_i , and as $f(X)$ is dense in Y , the same is true of at least one of the $f(V_i)$. One can therefore also suppose X affine; if (X_j) is the family (finite) of the irreducible components of X , one sees as above that at least one of the $f(X_j)$ is dense in Y ; one can therefore suppose X irreducible. Finally, replacing f by f_{red} (1.5.4), (vi) one can suppose X and Y integral. Then $X = \text{Spec}(B)$, $Y = \text{Spec}(A)$, where A is a noetherian integral ring and B an integral A -algebra of finite type containing A (1.2.7). It suffices to show that there exists $g \in A$ such that, for every $y \in D(g)$ (that is to say $g \notin \mathfrak{j}_y$) the ideal $\mathfrak{j}_y$ the ideal of A such that y is the intersection of A and of a prime ideal of B , it will be shown that $D(g) \subset f(X)$. Finally it suffices to prove the

Lemma (1.8.5.1). — *Let A be an integral ring, B an integral A -algebra of finite type. There exists $g \in A$ such that every homomorphism of A into an algebraically closed field Ω , non-zero on g , extends to a homomorphism of B into Ω .*

But this is a classical result of commutative algebra (Bourbaki, *Alg. comm.*, chap. V, §3, n° 1, cor. 3 of th. 1).

Corollary (1.8.6). — *Let Y be an irreducible prescheme, η its generic point, $f : X \rightarrow Y$ a morphism locally of finite type. If $f^{-1}(\eta)$ is not empty, there exists an open neighbourhood U of η in Y such that $f^{-1}(y)$ is not empty for every $y \in U$. If $\mathcal{F}$ is an $\mathcal{O}_X$ -Module quasi-coherent of finite type, and if $\mathcal{F}_\eta = \mathcal{F} \otimes_{\mathcal{O}_Y} k(\eta)$ is not zero, then there exists an open neighbourhood U' of η in Y such that $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$ is not zero for every $y \in U'$.*

If p_y is the canonical projection of the fibre $f^{-1}(y) = X \times_Y \text{Spec}(k(y))$ into X , we have $\mathcal{F}_y = p_y^*(\mathcal{F})$, so $\text{Supp}(\mathcal{F}_y) = p_y^{-1}(\text{Supp}(\mathcal{F})) = \text{Supp}(\mathcal{F}) \cap f^{-1}(y)$ by virtue of (I, 9.13.1) and (I, 3.6.1), since $\mathcal{F}$ is of finite type; moreover $\text{Supp}(\mathcal{F})$ is closed in X (0I, 5.2.2), and if Z is a closed sub-prescheme of X having $\text{Supp}(\mathcal{F})$ as underlying space and j the canonical immersion $Z \rightarrow X$, $f \circ j$ is locally of finite type (1.3.4), (vi)); this proves that the first assertion implies the second. To prove the first assertion, remark first that one can suppose X and Y reduced (1.3.4), (vi)), in other words Y integral. Furthermore, if Z is the closed sub-prescheme reduced of X having $\{\xi\}$ as underlying space (for a point ξ of $f^{-1}(\eta)$); one can, as above, replace X by Z , in other words suppose B integral. Since the morphism f is then dominant, the corresponding homomorphism $\varphi : A \rightarrow B$ is injective (I, 1.2.7); the corollary therefore follows finally from the lemma (1.8.5.1).

Proposition (1.8.7). — *Let S be a prescheme, $g : X \rightarrow S$, $h : Y \rightarrow S$ two morphisms of finite presentation, $f : X \rightarrow Y$ an S -morphism. For every $s \in S$, let $X_s = g^{-1}(s) = X \times_S \operatorname{Spec}(k(s))$, $Y_s = h^{-1}(s) = Y \times_S \operatorname{Spec}(k(s))$, $f_s = f \times 1 : X_s \rightarrow Y_s$. Then the set of $s \in S$ such that f_s is surjective (resp. radiciel) is locally constructible.*

Let E be the set of s such that f_s is surjective; the set $S - E$ is equal to $h(Y - f(X))$; but f is of finite presentation (1.6.2), (v)) so $f(X)$ is locally constructible in Y (1.8.4); as h is of finite presentation, $h(Y - f(X))$ is also locally constructible in S , hence also E .

To show that the set E' of s such that f_s is radiciel is locally constructible, we will use the

Lemma (1.8.7.1). — *Let U, V be two preschemes; for a morphism $f : U \rightarrow V$ to be radiciel, it suffices that the diagonal morphism $\Delta_f : U \rightarrow U \times_V U$ be surjective. Moreover, every radiciel morphism is separated.*

Indeed, Δ_f being an immersion (I, 5.3.9), is a morphism locally of finite type, and $\Delta_{f_s} : X_s \rightarrow X_s \times_{Y_s} X_s$ is deduced from $\Delta_f : X \rightarrow X \times_Y X$ by the base change $\operatorname{Spec}(k(s)) \rightarrow S$. As f is of finite presentation, Δ_f is also of finite presentation (1.6.2), (v)); the same is true of Δ_{f_s} (1.6.2), (iv)), so it suffices to apply the first part of the proposition to Δ_f , using the lemma (1.8.7.1).

1.9. Pro-constructible and ind-constructible sets

Lemma (1.9.1). — *Let $(A_\alpha)_{\alpha \in I}$ be an inductive system of rings whose index set is right filtering, and let $A = \varinjlim A_\alpha$. For $A = 0$, it is necessary and sufficient that there exist an index γ such that $A_\gamma = 0$ (and then $A_\beta = 0$ for all $\beta \geq \gamma$).*

Indeed, for every $\alpha \in I$, the canonical homomorphism $\varphi_\alpha : A_\alpha \rightarrow A$ transforms the unit element into the unit element; to say that $A = 0$ means that $\varphi_\alpha(1) = \varphi_\alpha(0)$, or equivalently that $\varphi_\alpha(1) = 0$; as the set is filtering, this is equivalent to saying that there exists an index $\gamma \geq \alpha$ such that $\varphi_{\gamma\alpha} : A_\alpha \rightarrow A_\gamma$ satisfies $\varphi_{\gamma\alpha}(1) = \varphi_{\gamma\alpha}(0)$, and this last relation is equivalent to $A_\gamma = 0$, whence the lemma.

Proposition (1.9.2). — *Let B be a ring, $(A_\alpha)_{\alpha \in I}$ an inductive system of B -algebras whose index set I is right filtering, and let $A = \varinjlim A_\alpha$; set $Y = \operatorname{Spec}(B)$, $X_\alpha = \operatorname{Spec}(A_\alpha)$, $X = \operatorname{Spec}(A)$, and let $u : X \rightarrow Y$, $u_\alpha : X_\alpha \rightarrow Y$ be the morphisms corresponding to the structural homomorphisms $B \rightarrow A$, $B \rightarrow A_\alpha$. Then:*

- (i) *For $X = \emptyset$, it is necessary and sufficient that there exist $\gamma \in I$ such that $X_\gamma = \emptyset$, and then $X_\alpha = \emptyset$ for $\alpha \geq \gamma$.*
- (i) *We have*

$$(1.9.2.1) \quad u(X) = \bigcap_{\alpha \in I} u_\alpha(X_\alpha).$$

The assertion (i) is none other than the translation of (1.9.1). To prove (ii), note that, since u factorises as $X \xrightarrow{u_\alpha} X_\alpha \rightarrow Y$, the first member is contained in the second. Conversely, let $y \in Y - u(X)$; on a $u^{-1}(y) = \emptyset$, and $u^{-1}(y)$ is the $k(y)$ -prescheme $\operatorname{Spec}(A \otimes_B k(y))$; as in the category of B -modules the functor $\varinjlim$ commutes with the tensor product, $A \otimes_B k(y)$ is the inductive limit of the system $(A_\alpha \otimes_B k(y))$; the lemma (1.9.1) shows that there exists α such that $A_\alpha \otimes_B k(y) = 0$, so $u_\alpha^{-1}(y) = \emptyset$, and hence $y \notin u_\alpha(X_\alpha)$. Q.E.D.

Proposition (1.9.3). — *Let X be a quasi-compact and quasi-separated prescheme, $(X_\alpha)_{\alpha \in I}$ an open cover of X , and for every $\alpha \in I$, set $E_\alpha = E \cap X_\alpha$. The following conditions are equivalent:*

- a) *There exists a quasi-compact morphism $f : X' \rightarrow X$ such that $E = f(X')$.*
- a') *For every α , there exists a quasi-compact morphism $f_\alpha : X'_\alpha \rightarrow X_\alpha$ such that $E_\alpha = f_\alpha(X'_\alpha)$.*
- a'') *For every α , there exists an affine scheme X''_α and a morphism $f_\alpha : X'_\alpha \rightarrow X_\alpha$ such that $f_\alpha(X'_\alpha) = E_\alpha$.*
- a) *E is an intersection of constructible subsets of X .*
- b') *$F = X - E$ is a union of constructible subsets of X .*

It is clear that b) and b') are equivalent. The condition a) implies a') by taking for X'_α the prescheme induced on $f^{-1}(X_\alpha)$ by X' and for f_α the restriction of f . To see that a') implies a''), it suffices to remark that X''_α is a finite union of affine open sets; let $X''_{\alpha\lambda}$ be the prescheme sum of the $X''_{\alpha\lambda}$ ($\lambda \in L_\alpha$), which is an affine scheme; if $g_\alpha : X''_\alpha \rightarrow X'_\alpha$ is the morphism coinciding with the identity in each of the $X''_{\alpha\lambda}$, it is clear that $f'_\alpha = f_\alpha \circ g_\alpha$ on a $E_\alpha = f'_\alpha(X''_\alpha)$ since g_α is surjective. To

show that a'') implies a), note that since X_α covers X for $\alpha \in J$, and let $f : X' \rightarrow X$ be the morphism which coincides with $g_\alpha \circ f_\alpha$ in each X'_α , it is clear that $E = f(X')$ and the hypothesis that X is quasi-separated implies that j_α is quasi-compact (1.2.7), hence f is quasi-compact (1.1.6) and finality of f (1.1.2).

It remains to prove the equivalence of a'') and b). Suppose first that a'') holds; for every $\alpha \in J$, where $X_\alpha = \text{Spec}(A)$ is affine, and $E = f(X')$ where $X' = \text{Spec}(A')$ is affine, $f : X' \rightarrow X$ a morphism corresponding to a homomorphism of rings $A \rightarrow A'$. Note now that A' is the inductive limit of the family (A'_λ) of its sub- A -algebras of *finite type*, ordered by inclusion; if $X'_\lambda = \text{Spec}(A'_\lambda)$, f factorises as $X' \rightarrow X'_\lambda \xrightarrow{f_\lambda} X$ and it follows from (1.9.2), (ii) that $f(X') = \bigcap_\lambda f_\lambda(X'_\lambda)$; if one shows that $f_\lambda(X'_\lambda)$ is an intersection of constructible subsets of X , the same will be true of $f(X')$. One can therefore restrict to the case where A' is an A -algebra of *finite type*. But then, by the theorem of Chevalley (1.8.5), $f(X')$ is constructible in X , whence b).

Let us show finally that b) implies a''). If E is an intersection of a family $(G_\lambda)_{\lambda \in L}$ of constructible subsets of X , each E_α is an intersection of constructible subsets $G_\lambda \cap X_\alpha$, which are constructible in X_α (0III, 9.1.8, (i)), hence we are reduced to the case where $X = \text{Spec}(A)$ is affine. We know then (1.8.3) that for each $\lambda \in L$ there exists a morphism $f_\lambda : X'_\lambda \rightarrow X$, where $X'_\lambda = \text{Spec}(A'_\lambda)$ is affine, such that $f_\lambda(X'_\lambda) = G_\lambda$. It suffices then to take for B' the product tensor $A'_J = \bigotimes_{\lambda \in J} A'_\lambda$ (product tensor of A -algebras), $X'_J = \text{Spec}(A'_J)$, and let $f_J : X'_J \rightarrow X$ be the structural morphism. If $A' = \varinjlim A'_J$, J ranging over the filtering set of finite subsets of L , $X' = \text{Spec}(A')$, and if $f : X' \rightarrow X$ is the structural morphism, we have $f(X') = \bigcap_{\lambda \in L} f_\lambda(X'_\lambda)$.

Lemma (1.9.3.1). — *Let A be a ring, A' an A -algebra of finite type. Then A' is a limit $\varinjlim A''_\mu$ where the A''_μ are A -algebras of finite presentation (1.4.1).*

Indeed, one can write $A' = B/\mathfrak{J}$, with $B = A[T_1, \dots, T_n]$ and $\mathfrak{J}$ an ideal of B . But $\mathfrak{J}$ is the inductive limit of the *finitely generated* ideals $\mathfrak{J}'_\mu$ of B , contained in $\mathfrak{J}$, ordered by inclusion; as in the category of B -modules the functor $\varinjlim$ is exact, we conclude that $A' = \varinjlim B/\mathfrak{J}'_\mu$ up to A -isomorphism.

Lemma (1.9.3.2). — *Let $(A'_\lambda)_{\lambda \in L}$ be a family of A -algebras, $X = \text{Spec}(A)$, $X'_\lambda = \text{Spec}(A'_\lambda)$, and let $f_\lambda : X'_\lambda \rightarrow X$ be the structural morphism. For every finite subset J of L , set $A'_J = \bigotimes_{\lambda \in J} A'_\lambda$ (tensor product of A -algebras), $X'_J = \text{Spec}(A'_J)$, and let $f_J : X'_J \rightarrow X$ be the structural morphism. If $A' = \varinjlim A'_J$, J ranging over the filtering set of finite subsets of L , $X' = \text{Spec}(A')$, and if $f : X' \rightarrow X$ is the structural morphism, we have $f_J(X'_J) = \bigcap_{\lambda \in J} f_\lambda(X'_\lambda)$. If $A' = \varinjlim A'_J$, $X' = \text{Spec}(A')$, and $f : X' \rightarrow X$ the structural morphism, we have $f(X') = \bigcap_{\lambda \in L} f_\lambda(X'_\lambda)$.*

The first assertion follows from (I, 3.4.7); the second follows from (1.9.2), (ii)).

Definition (1.9.4). — Let X be a topological space. We say that a subset E of X is *pro-constructible* (resp. *ind-constructible*) in X if, for every $x \in X$, there exists an open neighbourhood U of x such that $E \cap U$ is an intersection (resp. union) of locally constructible subsets of U .

Taking into account (1.2.7) and (0III, 9.1.8 and 9.1.12), the equivalent conditions of the proposition (1.9.3) are expressed by saying that E is *pro-constructible* in X , the equivalent conditions on the locally constructible subsets in X being identical to the constructible subsets when X is quasi-compact and quasi-separated (0III, 9.1.5).

Proposition (1.9.5). — *Let X be a prescheme.*

- (i) *For a subset E of X to be pro-constructible, it is necessary and sufficient that $X - E$ be ind-constructible.*
- (i) *Every finite union and every finite intersection of pro-constructible (resp. ind-constructible) subsets of X are pro-constructible (resp. ind-constructible).*
- (i) *Every intersection (resp. union) of locally constructible subsets of X is pro-constructible (resp. ind-constructible). Conversely, if X is quasi-compact and quasi-separated, every pro-constructible (resp. ind-constructible) subset of X is an intersection (resp. union) of constructible subsets.*
- (i) *Let (U_α) be an open cover of X . For a subset E of X to be pro-constructible (resp. ind-constructible) in X , it is necessary and sufficient that for every α , $E \cap U_\alpha$ be pro-constructible (resp. ind-constructible) in U_α .*
- (i) *Every pro-constructible closed subset E of X is retrocompact in X .*

- (i) Let $f : X' \rightarrow X$ be a morphism of preschemes; for every pro-constructible (resp. ind-constructible) subset E of X , $f^{-1}(E)$ is pro-constructible (resp. ind-constructible) in X' .
- (ii) Let $f : X' \rightarrow X$ be a quasi-compact morphism; for every pro-constructible subset E' of X' , $f(E')$ is pro-constructible in X ; in particular $f(X')$ is pro-constructible in X .
- (iii) Let $f : X' \rightarrow X$ be a morphism locally of finite presentation; for every ind-constructible subset E' of X' , $f(E')$ is ind-constructible in X ; in particular $f(X')$ is ind-constructible in X .
- (iv) Suppose X quasi-compact and quasi-separated; then, for a subset E of X to be pro-constructible, it is necessary and sufficient that there exist an affine scheme X' and a (necessarily quasi-compact) morphism $f : X' \rightarrow X$ such that $E = f(X')$.

The properties (i), (ii), (iv), and (iii) (first assertion) follow from the definition (1.9.4) and are valid for any topological space, using the distributivity of intersection and union, and the fact that if T is locally constructible in X , $T \cap U$ is locally constructible in U for every open U of X . If X is quasi-compact and quasi-separated and E is pro-constructible in X , there is a finite open cover (U_i) of X by affine open sets such that $E \cap U_i$ is intersection of constructible subsets in X_i for every i ; E being a union of the $E \cap U_i$, is intersection of finite unions $\bigcup_i F_{i,\lambda(i)}$, for all choices of $\lambda(i) \in L_i$; these unions are constructible in X ; this proves the second assertion of (iii) and also (ix) by (1.9.3).

To prove (v), one can restrict to the case where X is affine, and prove that E is quasi-compact for the induced topology of X . The properties (vi) and (viii) translate respectively (1.9.5), (vi) and (vii).

Similarly, to prove (vii), one can restrict to the case where X is affine; then X' is a finite union of affine open sets X'_i and $f(E')$ is a union of the $f(E' \cap X'_i)$, so one can even reduce to X' affine, in which case X' is a union of affine open sets, and $f(E')$ is a union of the $f(E' \cap X'_i)$; hence $f(E')$ is pro-constructible by an application continued.

To prove (viii), one can also restrict to the case where X is affine; then X' is a finite union of affine open sets X'_i and $f(E')$ is a union of the $f(E' \cap X'_i)$, so from (ii) one can restrict to the case where X' is affine; this then follows from the theorem of Chevalley (1.8.4) and (iii).

Finally, to prove (viii), one can still suppose X affine and, by (iv), also restrict to the case where E' is ind-constructible in X' ; the conclusion then follows from (iii) and (1.8.4).

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Examples (1.9.6). — Every finite subset of a prescheme X is pro-constructible: indeed, it suffices to consider the singleton $\{x\}$ reduced to a single point (1.9.5), (ii) and $\{x\}$ is the image of $\text{Spec}(k(x))$ by the canonical morphism $\text{Spec}(k(x)) \rightarrow X$, which is quasi-compact; whence the conclusion by (1.9.5), (vii). On the other hand, a subset $\{x\}$ reduced to a point is not necessarily ind-constructible; for example, let A be a noetherian integral ring having infinitely many maximal ideals, and let x be the generic point of $X = \text{Spec}(A)$; if $\{x\}$ were ind-constructible in X , it would be constructible by virtue of (1.9.5), (iii) and would therefore contain a non-empty open subset of X (0_{III}, 9.2.2); but this contradicts the hypothesis that A has infinitely many maximal ideals, by virtue of the theorem of Artin–Tate (0_I, 16.3.3).

Every closed subset Y of a prescheme X is pro-constructible, by (1.9.5), (v)). Every open subset of X is therefore ind-constructible, but an open subset U of X is not pro-constructible in general unless it is *retrocompact*, by virtue of (1.9.5), (v)).

Finally, if X is a noetherian prescheme, hence quasi-separated (1.2.8), it follows from (1.9.5), (iii) and (0_{III}, 9.1.7) that the ind-constructible subsets of X are the unions of locally closed subsets of X .

Theorem (1.9.7). — Let X be a quasi-compact prescheme, E an ind-constructible subset, $(F_\lambda)_{\lambda \in L}$ a family of pro-constructible subsets of X such that $\bigcap_{\lambda \in L} F_\lambda \subset E$; then there exists a finite subset J of L such that $\bigcap_{\lambda \in J} F_\lambda \subset E$.

Note that the sets $F = X - E$ and $F \cap F_\lambda$ are pro-constructible, so we are reduced to the

Corollary (1.9.8). — Let X be a quasi-compact prescheme, $(F_\lambda)_{\lambda \in L}$ a family of pro-constructible subsets of X whose intersection is empty; then there exists a finite sub-family $(F_\lambda)_{\lambda \in J}$ whose intersection is empty.

Covering X by a finite number of affine open sets, and using (1.9.5), (iv)), we are reduced to the case where X is affine; by virtue of (1.9.5), (ix)), we have $F_\lambda = f_\lambda(X'_\lambda)$, where $X'_\lambda = \text{Spec}(A'_\lambda)$ is affine, and f_λ is a morphism $X'_\lambda \rightarrow X$; then, with the notation of (1.9.3.2), on a by hypothesis $f(X') = \emptyset$, so $X' = \emptyset$; but this implies by (1.9.2), (i)) that there exists a finite subset J of L such that $X'_J = \emptyset$, hence $f_J(X'_J) = \bigcap_{\lambda \in J} F_\lambda = \emptyset$.

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Corollary (1.9.9). — *Let X be a quasi-compact prescheme, F a pro-constructible subset, $(E_\lambda)_{\lambda \in L}$ a family of ind-constructible subsets of X such that $F \subset \bigcup_{\lambda \in L} E_\lambda$; then there exists a finite subset J of L such that $F \subset \bigcup_{\lambda \in J} E_\lambda$. In particular, from every cover of X by ind-constructible subsets, one can extract a finite cover of X .*

This follows from (1.9.7) by passing to complements.

Proposition (1.9.10). — *Let X be a prescheme. For a subset E of X to be ind-constructible, it is necessary that, for every $x \in X$, the intersection $E \cap \overline{\{x\}}$ be a neighbourhood of x in $\overline{\{x\}}$. If X is locally noetherian, this condition is sufficient.*

Suppose E ind-constructible, and let Y be the intersection of $\overline{\{x\}}$ and an affine open subset in X , containing x ; there is then a sub-prescheme of X having Y as underlying space (I, 5.2.1), and if $j : Y \rightarrow X$ is the canonical injection, $E \cap Y = j^{-1}(E)$ is ind-constructible in Y (1.9.5), (vi)). Since Y is quasi-compact and separated, $E \cap Y$ is therefore a union of constructible subsets of Y (1.9.5), (iii)); furthermore, there are two open subsets U, V of Y such that $x \in U \cap \overline{V} \subset E \cap Y$ (0_{III}, 9.1.3). But as x is the generic point of the irreducible space Y , V is necessarily empty and we have $U \subset E \cap Y$.

Now suppose X locally noetherian, and let E be a subset of X satisfying the condition of the statement. By virtue of the definition (1.9.4), one can restrict to the case where X is noetherian. Then, for every $x \in E$, $E \cap \overline{\{x\}}$ contains a non-empty open subset of the form $U \cap \overline{\{x\}}$, where U is open in X ; this shows that E is a union of locally closed subsets, hence is ind-constructible (1.9.6).

Proposition (1.9.11). — *Let X be a prescheme, E a subset of X . The following properties are equivalent:*

- a) E is locally constructible.
- a) E is ind-constructible and pro-constructible.
- a) E and $X - E$ are pro-constructible.
- a) E and $X - E$ are ind-constructible.

It suffices evidently to prove that d) implies a), and one can restrict to the case where X is affine. Then (1.9.5), (iii)), E (resp. $X - E$) is a union of a family (E_λ) (resp. (E'_μ)) of constructible subsets of X ; as the E_λ and the E'_μ form a cover of X , by (1.9.9) it follows that there are finitely many λ_i, μ_j such that the E_{λ_i} and the E'_{μ_j} form a cover of X ; this implies that E is the union of the E_{λ_i} , hence is constructible.

Corollary (1.9.12). — *Let $f : X \rightarrow Y$ be a surjective morphism, either quasi-compact, or locally of finite presentation. For a subset E of Y to be locally constructible (resp. pro-constructible, resp. ind-constructible), it is necessary and sufficient that $f^{-1}(E)$ be so in X .*

One knows that the condition is necessary (1.8.2) and (1.9.5), (vi)); to show that it is sufficient, one can restrict (by virtue of (1.9.5), (iv))) to the case where X is affine, or the case where $f^{-1}(E)$ is pro-constructible, or the case where $f^{-1}(E)$ is ind-constructible. If f is surjective and quasi-compact, and $f^{-1}(E)$ pro-constructible, it follows from (1.9.5), (vii)) that $E = f(f^{-1}(E))$ is pro-constructible. If f is surjective and locally of finite presentation, and $f^{-1}(E)$ ind-constructible, $E = f(f^{-1}(E))$ is ind-constructible by (1.9.5), (viii)), which finishes the proof.

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(1.9.13). Let X be a prescheme, $\mathcal{T}$ its topology; it follows from (1.9.5), (i) and (ii)) that the ind-constructible (resp. pro-constructible) subsets of X are the open subsets (resp. closed subsets) for a topology on X , called the *constructible topology* and which we will denote by $\mathcal{T}^{\text{cons}}$; we will denote by X^{cons} the set X equipped with the topology $\mathcal{T}^{\text{cons}}$.

Proposition (1.9.14). — *Let X be a prescheme, $\mathcal{T}$ its topology, $\mathcal{T}^{\text{cons}}$ the constructible topology on X .*

- (i) The topology $\mathcal{T}^{\text{cons}}$ is finer than $\mathcal{T}$.
- (i) The locally constructible subsets of X are identical to the subsets that are both open and closed in X^{cons} .
- (i) For every morphism $f : X \rightarrow Y$, the underlying map of X^{cons} to Y^{cons} is continuous; we denote it f^{cons} .
- (i) If the morphism $f : X \rightarrow Y$ is quasi-compact, f^{cons} is a closed map; in particular, if f is quasi-compact and bijective, f^{cons} is a homeomorphism.
- (i) If the morphism $f : X \rightarrow Y$ is locally of finite presentation, f^{cons} is an open map; in particular, if f is bijective and locally of finite presentation, f^{cons} is a homeomorphism.

(i) For every open subset U of X , the topology induced by $\mathcal{T}^{\text{cons}}$ on U is identical to the topology of U^{cons} .

Indeed, (i) follows from the fact that every open set for $\mathcal{T}$ is an open set for $\mathcal{T}^{\text{cons}}$ (1.9.6), and (ii) is a translation of (1.9.11); (iii), (iv) and (v) translate respectively (1.9.5), (vi), (vii) and (viii). Finally, to prove (vi), it suffices to remark that the canonical injection $j : U \rightarrow X$ is locally of finite presentation (1.4.3), so $j^{\text{cons}} : U^{\text{cons}} \rightarrow X^{\text{cons}}$ is a continuous open map.

Proposition (1.9.15). — *Let X be a prescheme.*

- (i) For X^{cons} to be quasi-compact, it is necessary and sufficient that X be quasi-compact.
- (i) For X^{cons} to be separated, it is necessary and sufficient that X be quasi-separated, and then X^{cons} is locally compact and totally disconnected.
- (i) For X^{cons} to be compact, it is necessary and sufficient that X be quasi-compact and quasi-separated.
- (i) Every point of X admits, for the topology $\mathcal{T}^{\text{cons}}$, an open and compact neighbourhood.
- (i) For a quasi-compact morphism $f : X \rightarrow Y$ to be quasi-compact, it is necessary and sufficient that the continuous map $f^{\text{cons}} : X^{\text{cons}} \rightarrow Y^{\text{cons}}$ be proper.

(1.9.16). We will show later how one can, for every prescheme X , equip the space X^{cons} with a sheaf of rings which in fact makes it a prescheme whose local rings are all fields, identical to the residue fields at the points of X . Such preschemes arise naturally for example in the translation, in the language of schemes, of Néron's constructions in his theory of the reduction of abelian varieties.

1.10. Application to open morphisms

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Theorem (1.10.1). — *Let X be a prescheme, E an ind-constructible subset of X (1.9.4), x a point of X . For x to be interior to E , it is necessary and sufficient that every generalisation $(0_{\mathbf{I}}, 2.1.2)$ x' of x belong to E , in other words that $\text{Spec}(\mathcal{O}_{X,x}) \subset E$.*

The condition is evidently necessary, every neighbourhood of x containing the generalisations of x . To see that it is sufficient, one can evidently restrict to the case where $X = \text{Spec}(A)$ is affine, x being a prime ideal $\mathfrak{p}$ of A . One knows then (1.9.5), (ix)) that there exist an affine scheme $X' = \text{Spec}(A')$ and a morphism $f : X' \rightarrow X$ such that $f(X') = X - E$. Set $Y = \text{Spec}(\mathcal{O}_{X,x})$ and $Y' = X' \times_X Y$; the hypothesis means (by virtue of (I, 3.6.5)) that $Y' = \emptyset$, in other words $A' \otimes_A A_{\mathfrak{p}} = 0$. As $A_{\mathfrak{p}} = \varinjlim A_t$, where t ranges over $A - \mathfrak{p}$ (0_I, 1.4.5), on a $\varinjlim A_t = 0$, the functor $\varinjlim$ commuting with the tensor product. By (1.9.1), there exists $t \in A - \mathfrak{p}$ such that $A'_t = 0$, and $D(t)$ is then an open neighbourhood of x contained in E .

Corollary (1.10.2). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, and let $y \in Y$. For y to be interior to $f(X)$ it is necessary and sufficient that every generalisation y' of y belong to $f(X)$.*

One knows indeed (1.9.5), (viii)) that $f(X)$ is ind-constructible in Y and it suffices to apply (1.10.1).

We will say that a continuous map $f : X \rightarrow Y$ is *open at a point* $x \in X$ if the image by f of every neighbourhood of x in X is a neighbourhood of $f(x)$ in Y .

Proposition (1.10.3). — *Let $f : X \rightarrow Y$ be a morphism, x a point of X , $y = f(x)$. Consider the following conditions:*

- a) f is open at the point x .
- a) For every generalisation y' of y , there exists a generalisation x' of x such that $f(x') = y'$.
- b') The image by f of $\text{Spec}(\mathcal{O}_{X,x})$ is $\text{Spec}(\mathcal{O}_{Y,y})$.
- a) For every irreducible closed subset Y' of Y containing y , there exists an irreducible component of $X' = f^{-1}(Y')$ containing x and dominating Y' .

Then we have the implications $a) \Rightarrow b) \Leftrightarrow b') \Rightarrow c)$. If in addition f is locally of finite presentation, the four conditions are equivalent.

By definition of a generalisation, the image by f of $\text{Spec}(\mathcal{O}_{X,x})$ is always contained in $\text{Spec}(\mathcal{O}_{Y,y})$, so $b)$ and $b')$ are equivalent. It is immediate that $c)$ implies $b)$, for if y' is a generalisation of y , $Y' = \{y'\}$ is an irreducible closed subset of Y containing y ; if x' is the generic point of an irreducible component of $f^{-1}(Y')$ containing x and dominating Y' , we have $f(x') = y'$ (0_I, 2.1.5) and x' is a generalisation of x . Conversely, $b)$ implies $c)$: indeed, let y' be the generic point of Y' and let x' be a generalisation of x such that $f(x') = y'$; let Z' be an irreducible component of $f^{-1}(Y')$ containing x' (hence also x); as $f(x')$ is already dense in Y' , it is thus *a fortiori* so for $f(Z')$.

To see that $a)$ implies $b')$, one can evidently restrict to the case where $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$ are affine, x being a prime ideal $\mathfrak{q}$ of B , so that $\mathcal{O}_{X,x} = B_{\mathfrak{q}} = \varinjlim B_t$, where t ranges over $B - \mathfrak{q}$. We have, by (1.9.2), $f(\operatorname{Spec}(\mathcal{O}_{X,x})) = \bigcap_t f(\operatorname{Spec}(B_t)) = \bigcap_t f(D(t))$, and by hypothesis, for every $t \in B - \mathfrak{q}$, y is interior to $f(D(t))$, so $f(D(t))$ contains $\operatorname{Spec}(\mathcal{O}_{Y,y})$; whence $\operatorname{Spec}(\mathcal{O}_{Y,y}) \subset f(\operatorname{Spec}(\mathcal{O}_{X,x}))$, which establishes our assertion. Finally, when f is locally of finite presentation, $b)$ implies $a)$ by virtue of (1.10.2) applied to the restriction $U \rightarrow Y$ of f to an arbitrary open neighbourhood U of x . IV-1 | 250

Corollary (1.10.4). — *Let $f : X \rightarrow Y$ be a morphism. Consider the following conditions:*

- a) f is open.
- a) For every $x \in X$ and every generalisation y' of $y = f(x)$, there exists a generalisation x' of x such that $f(x') = y'$.
- b') For every $x \in X$, the image by f of $\operatorname{Spec}(\mathcal{O}_{X,x})$ is $\operatorname{Spec}(\mathcal{O}_{Y,f(x)})$.
- a) For every irreducible closed subset Y' of Y , every irreducible component of $f^{-1}(Y')$ dominates Y' .

Then we have the implications $a) \Rightarrow b) \Leftrightarrow b') \Rightarrow c)$. If in addition f is locally of finite presentation, the four conditions are equivalent.

To say that f is open means that f is open at every point $x \in X$, hence the implications $a) \Rightarrow b) \Leftrightarrow b')$ follow from the analogous implications in (1.10.3), as well as the implication $b) \Rightarrow a)$ when f is locally of finite presentation. The implication $c) \Rightarrow b)$ also follows from the analogous implication in (1.10.3); let us prove finally that $b)$ implies $c)$. Indeed, let x' be the generic point of an irreducible component of $f^{-1}(Y')$, and let us show that $y' = f(x')$ is the generic point of Y' . Let y'' be a generalisation of y' ; it exists by hypothesis a generalisation x'' of x' such that $f(x'') = y''$, and as $x'' \in f^{-1}(Y')$, we necessarily have $x'' = x'$, hence $y'' = y'$, which finishes the proof.

§2. BASE CHANGE AND FLATNESS

This section (contrary to §6) only exceptionally uses Noetherian techniques. Nos. 1 and 2 are scarcely more than translations of elementary properties of flatness in commutative algebra (cf. Bourbaki, *Alg. comm.*, chap. I), and are included here for convenience of reference. The following numbers are primarily devoted to properties of “descent” by flat or faithfully flat morphisms: if $g : Y' \rightarrow Y$ is such a morphism, the goal is to be able to assert that a part of Y , or an $\mathcal{O}_Y$ -Module, or a morphism $X \rightarrow Y$, has a certain property, when one knows that its inverse image by g possesses this property. We restrict ourselves here to properties that do not make use of the general technique of “descent”, which will be developed in Chapter V. IV-2 | 5

2.1. Flat modules on preschemes

(2.1.1). Let $f : X \rightarrow Y$ be a morphism of preschemes, $\mathcal{F}$ an $\mathcal{O}_X$ -Module; recall (0_I, 6.7.1) that $\mathcal{F}$ is said to be f -flat (or Y -flat) at a point $x \in X$ if $\mathcal{F}_x$ is a flat $\mathcal{O}_{f(x)}$ -module, f -flat (or Y -flat) if it is f -flat at every $x \in X$, and finally that the morphism f is said to be flat at the point $x \in X$ (resp. flat) if $\mathcal{O}_X$ is f -flat at the point x (resp. f -flat). When $f = 1_X$, we will simply say that an $\mathcal{O}_X$ -Module $\mathcal{F}$ is flat at the point x (resp. flat) if it is X -flat at this point (resp. at every point $x \in X$), that is to say if $\mathcal{F}_x$ is a flat $\mathcal{O}_x$ -module (resp. if this holds for every $x \in X$). Recall that we have shown (III, 1.4.15.1) the following property:

Proposition (2.1.2). — *Let A, B be two rings, $\varphi : A \rightarrow B$ a ring homomorphism, $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(A)$, $f : X \rightarrow Y$ the morphism corresponding to φ , and M a B -module. For $\mathcal{F} = \tilde{M}$ to be f -flat, it is necessary and sufficient that M be a flat A -module.*

Proposition (2.1.3). — *Let $f : X \rightarrow Y$ be a morphism of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. The following conditions are equivalent:* IV-2 | 6

- a) For every base change $g : Y' \rightarrow Y$, if we set $X' = X \times_Y Y'$, the functor $\mathcal{G}' \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ from the category of quasi-coherent $\mathcal{O}_{Y'}$ -Modules to that of quasi-coherent $\mathcal{O}_{X'}$ -Modules, is exact.
- a') Condition a) is satisfied for all the canonical morphisms $g : \operatorname{Spec}(\mathcal{O}_y) \rightarrow Y$ (I, 2.4.1), where y ranges over Y .
- b) $\mathcal{F}$ is f -flat.

Proof. The questions being local on X and Y , we can reduce to the case where $Y = \operatorname{Spec}(A)$, $\mathcal{F} = \widetilde{M}$, where M is a B -module. It is clear that a) implies a'); condition a') implies that for every $x \in X$, the functor $N \mapsto M_n \otimes_{B_n} N$ is exact in the category of B_n -modules, n being the ideal $j_{f(x)}$ of B ; this means that M_n is a flat B_n -module, and it follows from (0_I, 6.3.3) and from (2.1.2) that $\mathcal{F}$ is f -flat. Finally, to see that b) implies a), we can also reduce to the case where $Y' = \operatorname{Spec}(A')$ and $\mathcal{G}' = \widetilde{N'}$, where N' is an A' -module; the conclusion then follows again from (2.1.2) and the definition of flatness, since $(M \otimes_A N')^\sim = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$. $\square$

Proposition (2.1.4). — Let $f : X \rightarrow Y$, $g : Y' \rightarrow Y$ be two morphisms of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$, $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$, and let g' be the canonical projection $X' \rightarrow X$. Let x' be a point of X' , $x = g'(x')$, $y' = f'(x')$, $y = g(y') = f(x)$. If $\mathcal{F}$ is f -flat at the point x , then $\mathcal{F}'$ is f' -flat at the point x' ; in particular if $\mathcal{F}$ is f -flat, $\mathcal{F}'$ is f' -flat; if f is flat, f' is flat.

Proof. It suffices to prove the first assertion; applying (I, 3.6.5) three times, as well as (I, 2.4.4), we can reduce to the case where $Y = \operatorname{Spec}(\mathcal{O}_y)$, $X = \operatorname{Spec}(\mathcal{O}_x)$, $Y' = \operatorname{Spec}(\mathcal{O}_{y'})$, $\mathcal{F} = \widetilde{M}$, where $M = \mathcal{F}_x$; the hypothesis and (2.1.2) then imply that $\mathcal{F}$ is f -flat; we are thus reduced to proving a particular case of the second assertion, which follows immediately from (2.1.3). $\square$

Proposition (2.1.5). — Consider a commutative diagram of morphisms of preschemes

$$\begin{array}{ccc} X & \xleftarrow{g'} & X' \\ f \downarrow & & \downarrow f' \\ Y & \xleftarrow{g} & Y' \\ & & \downarrow h \\ & & Z \end{array}$$

where $X' = X \times_Y Y'$ and $f' = f_{(Y')}$. Let x' be a point of X' , and set $x = g'(x')$, $y' = f'(x')$, $y = f(x) = g(y')$, $z = h(y')$. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -Module f -flat at the point x (resp. f -flat), and $\mathcal{G}$ a quasi-coherent $\mathcal{O}_{Y'}$ -Module h -flat at the point y' (resp. h -flat); then $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ is a quasi-coherent $\mathcal{O}_{X'}$ -Module $(h \circ f')$ -flat at the point x' (resp. $(h \circ f')$ -flat).

Proof. As in (2.1.4), we can reduce to the case where $Y' = \operatorname{Spec}(\mathcal{O}_{y'})$ and $Z = \operatorname{Spec}(\mathcal{O}_z)$, and it then suffices to prove that $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ is $(h \circ f')$ -flat. Taking into account (2.1.2), the proposition follows from Bourbaki, *Alg. comm.*, chap. I, §2, no. 7, prop. 8. $\square$

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Corollary (2.1.6). — Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms of preschemes, $\mathcal{F}$ an $\mathcal{O}_X$ -Module. If $\mathcal{F}$ is f -flat at the point $x \in X$ and if g is flat at the point $f(x)$, then $\mathcal{F}$ is $(g \circ f)$ -flat at the point x . In particular, if f and g are flat morphisms, the same is true of $g \circ f$.

Proof. This is the particular case of (2.1.5) with $Y' = Y$, $\mathcal{G} = \mathcal{O}_Y$. $\square$

Corollary (2.1.7). — If $f : X \rightarrow X'$, $g : Y \rightarrow Y'$ are two flat S -morphisms, then

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is a flat morphism.

Proof. This follows from (2.1.4) and (2.1.6) (cf. (I, 3.5.1)). $\square$

Proposition (2.1.8). — Let $f : X \rightarrow Y$ be a morphism of preschemes,

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

an exact sequence of quasi-coherent $\mathcal{O}_X$ -Modules, such that $\mathcal{F}''$ is Y -flat.

(i) For every $g : Y' \rightarrow Y$ and every quasi-coherent $\mathcal{O}_{Y'}$ -Module $\mathcal{G}'$, the sequence

$$0 \longrightarrow \mathcal{F}' \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow \mathcal{F}'' \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow 0$$

of $\mathcal{O}_{X'}$ -Modules (where $X' = X \times_Y Y'$) is exact.

(ii) For $\mathcal{F}$ to be Y -flat, it is necessary and sufficient that $\mathcal{F}'$ be so.

Proof. We can evidently suppose X, Y, Y' affine; the conclusion then follows from (2.1.2) and (0_I, 6.1.2). $\square$

Corollary (2.1.9). — *Let $\mathcal{L}^\bullet$ be a complex of quasi-coherent $\mathcal{O}_X$ -Modules, i an index such that, denoting by $d^i : \mathcal{L}^i \rightarrow \mathcal{L}^{i+1}$ the derivation operator, $\mathcal{B}^{i+1}(\mathcal{L}^\bullet) = \text{Im}(d^i)$ and $\mathcal{Z}^{i+1}(\mathcal{L}^\bullet) = \text{Coker}(d^i)$ are Y -flat. Then, with the notations of (2.1.8), the canonical homomorphism*

$$\mathcal{H}^i(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow \mathcal{H}^i(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}')$$

is bijective.

Proof. Since the tensor product is right exact, we have

$$\mathcal{Z}^{i+1}(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' = \text{Coker}(d^i \otimes 1) = \mathcal{Z}^{i+1}(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}')$$

and $\mathcal{Z}^i(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' = \mathcal{Z}^i(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}')$. Furthermore, in the exact sequence

$$0 \longrightarrow \mathcal{B}^{i+1}(\mathcal{L}^\bullet) \longrightarrow \mathcal{L}^{i+1} \longrightarrow \mathcal{Z}^{i+1}(\mathcal{L}^\bullet) \longrightarrow 0$$

$\mathcal{Z}^{i+1}(\mathcal{L}^\bullet)$ is Y -flat, so it follows from (2.1.8) (i) that we have the exact sequence

$$0 \longrightarrow \mathcal{B}^{i+1}(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow \mathcal{L}^{i+1} \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow \mathcal{Z}^{i+1}(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}') \longrightarrow 0$$

whence $\mathcal{B}^{i+1}(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' = \text{Im}(d^i \otimes 1) = \mathcal{B}^{i+1}(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}')$. Then, since in the exact sequence

$$0 \longrightarrow \mathcal{H}^i(\mathcal{L}^\bullet) \longrightarrow \mathcal{Z}^i(\mathcal{L}^\bullet) \longrightarrow \mathcal{B}^{i+1}(\mathcal{L}^\bullet) \longrightarrow 0$$

$\mathcal{B}^{i+1}(\mathcal{L}^\bullet)$ is Y -flat, it follows from (2.1.8) (i) and from what precedes that the sequence

$$0 \longrightarrow \mathcal{H}^i(\mathcal{L}^\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' \longrightarrow \mathcal{Z}^i(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}') \longrightarrow \mathcal{B}^{i+1}(\mathcal{L}^\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}') \longrightarrow 0$$

is exact, which proves the corollary. $\square$

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Corollary (2.1.10). — *Let $f : X \rightarrow Y$ be a morphism of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module that is Y -flat, $\mathcal{L}_\bullet = (\mathcal{L}_i)$ a left resolution of $\mathcal{F}$ by quasi-coherent $\mathcal{O}_X$ -Modules that are Y -flat. Then, for every morphism $g : Y' \rightarrow Y$ and every quasi-coherent $\mathcal{O}_{Y'}$ -Module $\mathcal{G}'$, the complex $\mathcal{L}_\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}' = (\mathcal{L}_i \otimes_{\mathcal{O}_Y} \mathcal{G}')_{i \geq 0}$ is a left resolution of $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$.*

Furthermore, if $\mathcal{Z}_i(\mathcal{L}_\bullet) = \text{Ker}(\mathcal{L}_i \rightarrow \mathcal{L}_{i-1})$, the $\mathcal{Z}_i(\mathcal{L}_\bullet)$ are Y -flat, and we have $\mathcal{Z}_i(\mathcal{L}_\bullet) \otimes_{\mathcal{O}_Y} \mathcal{G}' = \mathcal{Z}_i(\mathcal{L}_\bullet \otimes_{\mathcal{O}_Y} \mathcal{G}') = \text{Ker}(\mathcal{L}_i \otimes_{\mathcal{O}_Y} \mathcal{G}' \rightarrow \mathcal{L}_{i-1} \otimes_{\mathcal{O}_Y} \mathcal{G}')$.

Proof. Set $\mathcal{R}_i = \text{Im}(\mathcal{L}_{i+1} \rightarrow \mathcal{L}_i) = \mathcal{Z}_i(\mathcal{L}_\bullet)$; we then have the exact sequences

$$\begin{aligned} 0 \leftarrow \mathcal{F} \leftarrow \mathcal{L}_0 \leftarrow \mathcal{R}_0 \leftarrow 0 \\ 0 \leftarrow \mathcal{R}_i \leftarrow \mathcal{L}_{i+1} \leftarrow \mathcal{R}_{i+1} \leftarrow 0 \end{aligned}$$

and since $\mathcal{F}$ and the $\mathcal{L}_i$ are Y -flat, we deduce from (2.1.8) (ii) by induction that all the $\mathcal{R}_i$ are also Y -flat; using (2.1.8) (i), we therefore have the exact sequences

$$\begin{aligned} 0 \leftarrow \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow \mathcal{L}_0 \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow \mathcal{R}_0 \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow 0 \\ 0 \leftarrow \mathcal{R}_i \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow \mathcal{L}_{i+1} \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow \mathcal{R}_{i+1} \otimes_{\mathcal{O}_Y} \mathcal{G}' \leftarrow 0 \quad (i \geq 0) \end{aligned}$$

which prove the corollary. $\square$

Proposition (2.1.11). — *Let $f : X \rightarrow Y$ be a flat morphism, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_Y$ -Module of finite presentation. If $\mathcal{J}$ is the ideal of $\mathcal{O}_Y$ that annihilates $\mathcal{F}$, then $f^*(\mathcal{J})$ is the ideal of $\mathcal{O}_X$ that annihilates $f^*(\mathcal{F})$.*

Proof. Indeed, by definition we have an exact sequence (0_I, 5.3.7)

$$0 \longrightarrow \mathcal{J} \longrightarrow \mathcal{O}_Y \longrightarrow \mathcal{H}om_{\mathcal{O}_Y}(\mathcal{F}, \mathcal{F})$$

whence, since f is flat, an exact sequence

$$0 \longrightarrow f^*(\mathcal{J}) \longrightarrow \mathcal{O}_X \longrightarrow f^*(\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{F}, \mathcal{F}))$$

and since by hypothesis $\mathcal{F}$ is a finitely presented $\mathcal{O}_Y$ -Module, $f^*(\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{F}, \mathcal{F}))$ is canonically identified with $\mathcal{H}om_{\mathcal{O}_X}(f^*(\mathcal{F}), f^*(\mathcal{F}))$ (0_I, 6.7.6), whence the conclusion. $\square$

Proposition (2.1.12). — *Let X be a prescheme, $\mathcal{F}$ an $\mathcal{O}_X$ -Module of finite presentation, x a point of X . The following conditions are equivalent:*

- a) $\mathcal{F}_x$ is a flat $\mathcal{O}_x$ -module.
- a) There exists an open neighbourhood U of x such that $\mathcal{F}|_U$ is a locally free $(\mathcal{O}_X|_U)$ -Module.

Proof. Indeed, $\mathcal{F}_x$ is a finitely presented $\mathcal{O}_x$ -module and $\mathcal{O}_x$ a local ring; it therefore amounts to the same to say that $\mathcal{F}_x$ is a flat $\mathcal{O}_x$ -module or a free $\mathcal{O}_x$ -module (Bourbaki, *Alg. comm.*, chap. II, §3, no. 2, cor. 2 of prop. 5); whence the conclusion, taking into account (0_I, 5.2.7). We note that the proposition is valid for an arbitrary locally ringed space. $\square$

Proposition (2.1.13). — *Let $f : X \rightarrow Y$ be a morphism of preschemes. If f is flat at a point $x \in X$ and if the ring $\mathcal{O}_x$ is reduced (resp. integral and integrally closed), then the ring $\mathcal{O}_{f(x)}$ is reduced (resp. integral and integrally closed). If f is faithfully flat and if X is reduced (resp. normal), then Y is reduced (resp. normal).*

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Proof. Set $\mathcal{O}_{f(x)} = A$, $\mathcal{O}_x = B$. If B is a flat A -module, it is also a faithfully flat A -module (0_I, 6.6.2), so A is identified with a subring of B ; if B is reduced, the same is therefore true of A . Suppose now that B is integral and integrally closed, and let L be its field of fractions; denote by $K \subset L$ the field of fractions of A . The hypothesis implies that $B \cap K = A$ (Bourbaki, *Alg. comm.*, chap. I, §3, no. 5, prop. 10). If then $t \in K$ is integral over A , it is also integral over B , and so belongs to B by hypothesis, and therefore $t \in A$, which proves that A is integrally closed. $\square$

Proposition (2.1.14). — *Let $f : X \rightarrow Y$ be a faithfully flat morphism. If X is integral and if the underlying space of Y is locally noetherian, then Y is locally integral.*

Proof. Indeed, every $y \in Y$ is of the form $f(x)$ for some $x \in X$ and by hypothesis $\mathcal{O}_y$ is identified with a subring of $\mathcal{O}_x$ (0_I, 6.6.1); since $\mathcal{O}_x$ is integral, the same is true of $\mathcal{O}_y$, and this proves the proposition (I, 5.1.4). $\square$

2.2. Faithfully flat modules on preschemes

Proposition (2.2.1). — *Let $f : X \rightarrow Y$ be a morphism of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. The following conditions are equivalent:*

- a) For every base change $g : Y' \rightarrow Y$, if we set $X' = X \times_Y Y'$, the functor $\mathcal{G}' \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$, from the category of quasi-coherent $\mathcal{O}_{Y'}$ -Modules to that of quasi-coherent $\mathcal{O}_{X'}$ -Modules, is exact and faithful.
- a') Condition a) is satisfied for all the canonical morphisms $g : \text{Spec}(\mathcal{O}_y) \rightarrow Y$ (I, 2.4.1), where y ranges over Y .
- a'') Condition a) is satisfied for all the canonical immersions $Y' \rightarrow Y$, where Y' ranges over the set of affine open subschemes of Y .
- a) $\mathcal{F}$ is Y -flat and, for every $y \in Y$, denoting by X_y the fibre $f^{-1}(y)$, the $\mathcal{O}_{X_y}$ -Module $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$ is $\neq 0$.

Proof. It is clear that a) implies a') and a''); condition a') implies first that $\mathcal{F}$ is Y -flat (2.1.3); on the other hand, it implies that for every $y \in Y$, the functor $N \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \tilde{N}$ is faithful in the category of $\mathcal{O}_y$ -modules; taking in particular $N = k(y)$, we obtain the second assertion of b). For the second assertion of b) to imply a), on the other hand, it suffices to restrict to the case where Y is affine, whence the question is local on Y , and one can also assume X affine, and the condition then reduces to showing that the functor $\mathcal{G}' \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ is exact and faithful, which proves that b') implies a). For showing that b) implies a''), we can restrict to the case where Y is affine; then f is faithfully flat in the category of $\mathcal{O}_Y$ -modules, and so the functor $\mathcal{G}' \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ is exact and faithful, which shows that b') implies c) and concludes the proof. $\square$

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Proposition (2.2.2). — *Let $Y = \text{Spec}(A)$ be an affine scheme, $f : X \rightarrow Y$ a morphism of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. Condition a) of (2.2.1) is equivalent to each of the following:*

- b') $\mathcal{F}$ is Y -flat and, for every closed point y of Y , $\mathcal{F}_y \neq 0$.
- c) The functor $\mathcal{G} \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ from the category of quasi-coherent $\mathcal{O}_Y$ -Modules to that of quasi-coherent $\mathcal{O}_X$ -Modules, is exact and faithful.

Proof. If b') is satisfied, there is at least one $x \in f^{-1}(y)$ such that $(\mathcal{F}_y)_x \neq 0$; let $U = \operatorname{Spec}(B)$ be an affine neighbourhood of x , so that $\mathcal{F}|_U = \tilde{M}$, where M is a B -module. Then b') implies that $M/\mathfrak{m}_y M \neq 0$ (since M is an A -module that is flat (2.1.2)), which implies that $M \otimes_A A_y \neq 0$ for every closed point $y \in Y$, that $M \otimes_A A_y \tilde{N}$ is exact and faithful (0_I, 6.4.5). The relation $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G} = 0$ implies $\mathcal{G} = 0$, otherwise said the functor $\mathcal{G} \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ is faithful; since it is also exact (2.1.3), this shows that b') implies c), and it is clear that c) implies a). $\square$

Suppose now that c) holds and that $g : Y' \rightarrow Y$ is also an affine morphism; moreover, the functor $\mathcal{H}' \mapsto g'_*(\mathcal{H}')$ is then exact in the category of quasi-coherent $\mathcal{O}_{Y'}$ -Modules (I, 1.6.4), and if $g'_*(\mathcal{H}') = \mathcal{H}'$, on has $\mathcal{H}' = \tilde{\mathcal{H}}$; the preceding functor is also faithful; to see that $\mathcal{G}' \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ is exact and faithful, it suffices therefore to see that the functor $\mathcal{G}' \mapsto g'_*(\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}')$ is so. Or, if $f' = f_{(Y')} : X' \rightarrow Y'$, the fact that g is affine implies that we have a canonical isomorphism

$$(1) \quad \mathcal{F} \otimes_{\mathcal{O}_X} f^*(g'_*(\mathcal{G}')) \xrightarrow{\sim} g'_*(g'^*(\mathcal{F}) \otimes_{\mathcal{O}_{X'}} f'^*(\mathcal{G}')).$$

Indeed, we know (II, 1.5.2) that there is a canonical isomorphism $f^*(g'_*(\mathcal{G}')) \xrightarrow{\sim} g'_*(f'^*(\mathcal{G}'))$, and on the other hand a canonical homomorphism $\mathcal{F} \rightarrow g'_*(g'^*(\mathcal{F}))$ (0_I, 4.4.3.2); composing the homomorphism $\mathcal{F} \otimes_{\mathcal{O}_X} f^*(g'_*(\mathcal{G}')) \rightarrow g'_*(g'^*(\mathcal{F})) \otimes_{\mathcal{O}_X} g'_*(f'^*(\mathcal{G}'))$ with the canonical morphism (0_I, 4.2.2.1), we deduce the homomorphism (2.2.2.1); the verification that it is an isomorphism is immediate, reducing to the case where X is affine. This being the case, the functor $\mathcal{G}' \mapsto g'_*(\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}')$ is exact and faithful and by hypothesis the same is true of $\mathcal{G} \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$; their composite is therefore exact and faithful (2.2.1) and (2.2.2).

Corollary (2.2.3). — *Let $X = \operatorname{Spec}(B)$, $Y = \operatorname{Spec}(A)$ be two affine schemes, $f : X \rightarrow Y$ a morphism, $\mathcal{F} = \tilde{M}$ a quasi-coherent $\mathcal{O}_X$ -Module. For $\mathcal{F}$ to satisfy the equivalent conditions of (2.2.1) (or (2.2.2)), it is necessary and sufficient that the A -module M be faithfully flat.*

Proof. Indeed, condition c) of (2.2.2) means that the functor $N \mapsto M \otimes_A N$ from the category of A -modules to that of B -modules is exact and faithful, and the conclusion follows from (0_I, 6.4.1). $\square$

Definition (2.2.4). — When the equivalent conditions of (2.2.1) are satisfied, we say that the quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ is *faithfully flat relative to f* (or relative to Y).

We note that this notion is local on Y , but *not* on X ; in particular one can have $\mathcal{F}_x = 0$ for certain $x \in X$, in other words $\operatorname{Supp}(\mathcal{F})$ is not necessarily equal to X . However, it follows immediately from the criterion (2.2.1) b) that for every $y \in Y$, there exists at least one $x \in f^{-1}(y)$ such that $(\mathcal{F}_y)_x = \mathcal{F}_x \otimes_{\mathcal{O}_Y} k(y) \neq 0$, and a fortiori $\mathcal{F}_x \neq 0$; in other words:

Corollary (2.2.5). — *If $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -Module, faithfully flat relative to f , then $f(\operatorname{Supp}(\mathcal{F})) = Y$, and a fortiori f is a surjective morphism.*

This result admits a partial converse:

Corollary (2.2.6). — *Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -Module of finite type. For $\mathcal{F}$ to be faithfully flat relative to f , it is necessary and sufficient that $\mathcal{F}$ be f -flat and that $f(\operatorname{Supp}(\mathcal{F})) = Y$.*

Proof. Indeed (I, 9.1.13) and (I, 3.6.1), we have $\operatorname{Supp}(\mathcal{F}_y) = f^{-1}(y) \cap \operatorname{Supp}(\mathcal{F})$, so in this case the criterion (2.2.1) b) is none other than the condition of the corollary. $\square$

In particular, the $\mathcal{O}_X$ -Module $\mathcal{O}_X$ is faithfully flat relative to f if and only if it is f -flat and f is surjective, in other words if and only if the morphism f is *faithfully flat* (0_I, 6.7.8).

Let us make explicit the properties intervening in Definition (2.2.4):

Proposition (2.2.7). — *Let $f : X \rightarrow Y$ be a morphism of preschemes, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module faithfully flat relative to f . Then, for a sequence $\mathcal{G}' \rightarrow \mathcal{G} \rightarrow \mathcal{G}''$ of quasi-coherent $\mathcal{O}_Y$ -Modules to be exact, it is necessary and sufficient that the corresponding sequence $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}' \rightarrow \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G} \rightarrow \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}''$ be so. In particular, for a homomorphism $u : \mathcal{G} \rightarrow \mathcal{G}'$ of quasi-coherent $\mathcal{O}_Y$ -Modules to be injective (resp. surjective, bijective, zero), it is necessary and sufficient that $1_{\mathcal{F}} \otimes f^*(u) : \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G} \rightarrow \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ be so. For a quasi-coherent $\mathcal{O}_Y$ -Module $\mathcal{G}$ to be zero, it is necessary and sufficient that $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G} = 0$. For every quasi-coherent $\mathcal{O}_Y$ -Module $\mathcal{G}$, the map $\mathcal{G}' \mapsto \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ from the set of quasi-coherent sub- $\mathcal{O}_Y$ -Modules of $\mathcal{G}$ to the set of quasi-coherent sub- $\mathcal{O}_X$ -Modules of $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ is injective.*

Proof. To prove the last assertion, that is to say that for two quasi-coherent sub- $\mathcal{O}_Y$ -Modules $\mathcal{G}', \mathcal{G}''$ of $\mathcal{G}$, the relation $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}''$ implies $\mathcal{G}' = \mathcal{G}''$, we can reduce to the case where $\mathcal{G}' \subset \mathcal{G}''$, and it then suffices to apply the second assertion of the statement to the injection $u : \mathcal{G}' \rightarrow \mathcal{G}''$. $\square$

Corollary (2.2.8). — *Let $f : X \rightarrow Y$ be a faithfully flat morphism. For every quasi-coherent $\mathcal{O}_Y$ -Module $\mathcal{G}$, the canonical map*

$$(2) \quad \Gamma(Y, \mathcal{G}) \longrightarrow \Gamma(X, f^*(\mathcal{G}))$$

is injective.

Proof. Indeed, $\Gamma(Y, \mathcal{G})$ is canonically identified with $\text{Hom}_{\mathcal{O}_Y}(\mathcal{O}_Y, \mathcal{G})$ (0I, 5.1.1) and similarly $\Gamma(X, f^*(\mathcal{G}))$ with $\text{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, f^*(\mathcal{G}))$. By virtue of (2.2.1) and (2.2.4), the hypothesis that the functor $\mathcal{G} \mapsto f^*(\mathcal{G})$ is exact and faithful in the category of $\mathcal{O}_Y$ -Modules implies that a homomorphism $u : \mathcal{O}_Y \rightarrow \mathcal{G}$ is zero if and only if $f^*(u) : f^*(\mathcal{O}_Y) = \mathcal{O}_X \rightarrow f^*(\mathcal{G})$ is zero. $\square$

Remark (2.2.9). — The results of (2.2.7) and (2.2.8) are still true when the $\mathcal{O}_Y$ -Modules $\mathcal{G}', \mathcal{G}, \mathcal{G}''$ that appear there are arbitrary (not necessarily quasi-coherent). Indeed, for every $y \in Y$, there exists $x \in f^{-1}(y)$ such that $\mathcal{F}_x$ is a faithfully flat $\mathcal{O}_y$ -module and consequently the functor $\mathcal{G}_y \mapsto \mathcal{G}_y \otimes_{\mathcal{O}_y} \mathcal{F}_x$ is faithful; since moreover for every $x \in f^{-1}(y)$ the functor $\mathcal{G}_y \mapsto \mathcal{G}_y \otimes_{\mathcal{O}_y} \mathcal{F}_x$ is exact, we immediately deduce the conclusion. IV-2 | 12

Proposition (2.2.10). — *Let $f : X \rightarrow Y, g : Y' \rightarrow Y, h : Y' \rightarrow Z$ be three morphisms of preschemes, $X' = X \times_Y Y'$, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module faithfully flat relative to Y , $\mathcal{G}$ a quasi-coherent $\mathcal{O}_{Y'}$ -Module faithfully flat relative to Z . For $\mathcal{G}$ to be flat (resp. faithfully flat) relative to Z , it is necessary and sufficient that $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ be a flat $\mathcal{O}_{X'}$ -Module (resp. faithfully flat) relative to Z .*

Proof. We already know that if $\mathcal{G}$ is Z -flat, the same is true of $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}'$ (2.1.5). Consider an arbitrary base change $Z'' \rightarrow Z$ and set $X'' = X' \times_Z Z''$; if $\mathcal{G}$ is faithfully flat relative to Z , the functor

$$(3) \quad \mathcal{H}'' \mapsto \mathcal{H}'' \otimes_{\mathcal{O}_Z} \mathcal{G}' \longrightarrow (\mathcal{H}'' \otimes_{\mathcal{O}_Z} \mathcal{G}') \otimes_{\mathcal{O}_{Y'}} \mathcal{F} = \mathcal{H}'' \otimes_{\mathcal{O}_Z} (\mathcal{G}' \otimes_{\mathcal{O}_{Y'}} \mathcal{F})$$

from the category of $\mathcal{O}_{X''}$ -Modules quasi-coherent to that of $\mathcal{O}_{X''}$ -Modules quasi-coherent is composed of two exact and faithful functors, so is exact and faithful. Conversely, if this composed functor is exact (resp. exact and faithful), the same is true of $\mathcal{H}'' \mapsto \mathcal{H}'' \otimes_{\mathcal{O}_Z} \mathcal{G}'$, since the functor $\mathcal{H}' \mapsto \mathcal{H}' \otimes_{\mathcal{O}_{Y'}} \mathcal{F}$ (from the category of quasi-coherent $\mathcal{O}_{Y'}$ -Modules to that of quasi-coherent $\mathcal{O}_{X'}$ -Modules) is exact and faithful by hypothesis. $\square$

Corollary (2.2.11). — (i) *Let $f : X \rightarrow Y, g : Y' \rightarrow Y$ be two morphisms, $X' = X \times_Y Y'$, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. If $\mathcal{F}$ is faithfully flat relative to Y , $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$ is faithfully flat relative to Y' .*

(ii) *Let $f : X \rightarrow Y, g : Y \rightarrow Z$ be two morphisms, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module faithfully flat relative to Y . For g to be a faithfully flat morphism, it is necessary and sufficient that $\mathcal{F}$ be faithfully flat relative to Z .*

(iii) *Let $f : X \rightarrow Y, g : Y \rightarrow Z$ be two morphisms, $\mathcal{G}$ a quasi-coherent $\mathcal{O}_Y$ -Module. Suppose the morphism f is faithfully flat. For $\mathcal{G}$ to be flat (resp. faithfully flat) relative to Z , it is necessary and sufficient that $f^*(\mathcal{G})$ be flat (resp. faithfully flat) relative to Z .*

(iv) *Let $f : X \rightarrow Y, g : Y \rightarrow Z$ be two morphisms, x a point of X . Suppose f is flat at the point x . For $g \circ f$ to be flat at the point x , it is necessary and sufficient that g be flat at the point $f(x)$.*

Proof. To show (i), apply (2.2.10) replacing Z by Y' and $\mathcal{G}'$ by $\mathcal{O}_{Y'}$. To show (ii), apply (2.2.10) taking $Y' \rightarrow Y$ the identity and replacing $\mathcal{G}'$ by $\mathcal{O}_Y$. To show (iii), apply (2.2.10) by replacing $Y' \rightarrow Y$ by the identity, $\mathcal{F}$ by $\mathcal{O}_X$ and $\mathcal{G}'$ by $\mathcal{G}$. Finally, (iv) follows from (ii) by replacing $\mathcal{F}$ by $\mathcal{O}_X$ and $\mathcal{G}$ by $\mathcal{O}_Y$, replacing X by $\text{Spec}(\mathcal{O}_x)$, $\mathcal{F}$ by $\mathcal{O}_X$, Y by $\text{Spec}(\mathcal{O}_{f(x)})$ and Z by $\text{Spec}(\mathcal{O}_{g(f(x))})$, and taking into account (0I, 6.6.2). $\square$

Corollary (2.2.12). — *Let Y be an affine scheme, $f : X \rightarrow Y$ a quasi-compact morphism, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. If $\mathcal{F}$ is faithfully flat relative to f , there exists an affine scheme X' and a surjective local isomorphism $g : X' \rightarrow X$ such that $g^*(\mathcal{F})$ is faithfully flat relative to $f \circ g$.*

Proof. Since X is quasi-compact, and therefore a finite union of affine opens X_i ; it suffices to take for X' the affine scheme *sum* of the preschemes X_i , f being the canonical morphism (I, 3.1), the hypothesis implies that $g^*(\mathcal{F})$ is faithfully flat relative to $f \circ g$, by virtue of (2.2.11) (iii). $\square$

Corollary (2.2.13). — (i) Let $f : X \rightarrow Y$, $g : Y' \rightarrow Y$ be two morphisms, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. If f is a faithfully flat morphism, the same is true of f' .

(i) If $f : X \rightarrow X'$, $g : Y \rightarrow Y'$ are two faithfully flat S -morphisms,

$$f \times_S g : X \times_S Y \longrightarrow X' \times_S Y'$$

is faithfully flat.

(i) Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms such that f is faithfully flat (resp. faithfully flat). For g to be a flat (resp. faithfully flat) morphism, it is necessary and sufficient that $g \circ f$ be a flat (resp. faithfully flat) morphism.

Proposition (2.2.14). — Let $f : X \rightarrow Y$ be a faithfully flat and quasi-compact morphism. If X is locally noetherian, the same is true of Y .

Proof. The question being local on Y , we can suppose $Y = \operatorname{Spec}(A)$ affine; since f is quasi-compact, we can also restrict to the case where $X = \operatorname{Spec}(B)$ is affine. Then B is a noetherian ring by hypothesis, and a faithfully flat A -module (2.2.3); therefore A is noetherian (0_I, 6.5.2). $\square$

Proposition (2.2.15). — Let $f : X \rightarrow Y$ be a faithfully flat morphism. If $\mathfrak{S}(X)$ and $\mathfrak{S}(Y)$ denote respectively the sets of sub-preschemes of X and of Y , the map $Z \mapsto f^{-1}(Z)$ from $\mathfrak{S}(Y)$ to $\mathfrak{S}(X)$ is injective.

Proof. Since f is surjective, we have $f(f^{-1}(Z)) = Z$. On the other hand, if U is an open of Y containing Z and in which Z is closed, $f^{-1}(Z)$ is closed in $f^{-1}(U)$, and the restriction $f^{-1}(U) \rightarrow U$ of f is faithfully flat. We can therefore restrict to considering only closed sub-preschemes of Y . Now, if Z is a sub-prescheme of Y corresponding to a quasi-coherent ideal $\mathcal{J}$ of $\mathcal{O}_Y$, we know that $f^{-1}(Z)$ corresponds to the quasi-coherent ideal $f^*(\mathcal{J})\mathcal{O}_X$ of $\mathcal{O}_X$ (I, 4.4.5), and since f is flat, $f^*(\mathcal{J})\mathcal{O}_X$ is identified with $f^*(\mathcal{J})$. But the map $\mathcal{J} \mapsto f^*(\mathcal{J})$ from the set of quasi-coherent ideals of $\mathcal{O}_Y$ to the set of quasi-coherent ideals of $\mathcal{O}_X$ is injective (2.2.7), whence the conclusion. $\square$

Corollary (2.2.16). — Let X, Y be two S -preschemes; if $S' \rightarrow S$ is a faithfully flat morphism, the map $f \mapsto f_{(S')}$ from $\operatorname{Hom}_S(X, Y)$ to $\operatorname{Hom}_{S'}(X_{(S')}, Y_{(S')})$ is injective.

Proof. Since $X_{(S')} \times_{S'} Y_{(S')} = (X \times_S Y)_{(S')}$ (I, 3.3.10), the morphism projection $X_{(S')} \times_{S'} Y_{(S')} \rightarrow X \times_S Y$ is faithfully flat (2.2.13). The elements of $\operatorname{Hom}_S(X, Y)$ correspond bijectively to the sub-preschemes of $X \times_S Y$ which are the graphs of these S -morphisms (I, 5.3.11), and if Z_f is the graph of $f \in \operatorname{Hom}_S(X, Y)$, then $Z_{f_{(S')}} = g^{-1}(Z_f)$ (I, 5.3.12). It therefore suffices to apply g to Proposition (2.2.15). $\square$

Proposition (2.2.17). — Let A be a ring, B an A -algebra such that B is a faithfully flat A -module of finite presentation. Then the structural homomorphism $\varphi : A \rightarrow B$ is an isomorphism of the A -module A onto a direct summand of the A -module B . If A is a local ring, B is a free A -module of finite type, and there exists a basis of this module containing the unit element.

Proof. By virtue of Bourbaki, *Alg. comm.*, chap. II, §3, no. 3, prop. 12, it suffices to prove the proposition when A is a local ring; one then has (*loc. cit.*, no. 2, cor. 2 of prop. 5) that B is a free A -module of finite type, and the conclusion follows from *loc. cit.*, prop. 5. $\square$

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2.3. Topological properties of flat morphisms

Lemma (2.3.1). — Let $f : X \rightarrow Y$ be a quasi-compact and quasi-separated morphism, $g : Y' \rightarrow Y$ a flat morphism; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. Then, for every quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, the canonical homomorphism

$$(4) \quad g^*(f_*(\mathcal{F})) \longrightarrow f'_*(\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'})$$

is bijective.

Proof. This is a particular case of (III, 1.4.15) (improved in (1.7.21)). $\square$

Proposition (2.3.2). — Let S be a prescheme, $f : X \rightarrow Y$ an S -morphism quasi-compact and quasi-separated, $j : Z \rightarrow Y$ the closed image of X by f (I, 9.5.3) and (1.7.8), and let $g : X \rightarrow Z$ be a morphism such that $f = j \circ g$, where $g : X \rightarrow Z$ is a morphism (*loc. cit.*). Let $h : S' \rightarrow S$ be a flat morphism, and set $f' = f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$; then $j' = j_{(S')} : Z_{(S')} \rightarrow Y_{(S')}$, the closed image of $X_{(S')}$ by f' .

Proof. As the morphism $Y_{(S')} \rightarrow Y$ is flat (2.1.4), we can restrict to the case where $S = Y$ (I, 3.3.11); we can then set $f' = (\psi, \theta)$, on sees easily (for example by reducing to the case where Y and Y' are affine and using (I, 2.2.4)) that $\theta' : \mathcal{O}_{S'} \rightarrow f'_*(\mathcal{O}_{X'})$ is composed of the canonical homomorphism (2.3.1) $h^*(f_*(\mathcal{O}_X)) \rightarrow f'_*(\mathcal{O}_{X'})$ and of $h^*(\theta) : h^*(\mathcal{O}_Y) \rightarrow h^*(f_*(\mathcal{O}_X))$; the conclusion thus follows from (2.3.1) and from the fact that $\text{Ker}(\theta)$, being an $\mathcal{O}_Y$ -Module quasi-coherent, commutes with flat base change (0_I, 6.7.8). $\square$

(2.3.3). We will say that a morphism $f : X \rightarrow Y$ is *quasi-flat* if there exists a quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ of finite type which is f -flat and whose support is equal to X . We will say that f is *quasi-faithfully flat* if it is quasi-flat and surjective. Every flat (resp. faithfully flat) morphism is quasi-flat (resp. quasi-faithfully flat), since then $\mathcal{F} = \mathcal{O}_X$ satisfies the preceding conditions.

It follows immediately from (2.1.4) and from (I, 9.1.13), that if f is quasi-flat, then for every morphism $Y' \rightarrow Y$, the morphism $f_{(Y')} : X \times_Y Y' \rightarrow Y'$ is quasi-flat. Similarly (taking into account (I, 3.5.2)), if f is quasi-faithfully flat, the same is true of $f_{(Y')}$.

Proposition (2.3.4). — *Let $f : X \rightarrow Y$ be a quasi-flat morphism (2.3.3). Then f has the following properties (which are equivalent by virtue of (1.10.4)):*

- (i) *For every $x \in X$ and every generisation y' of $y = f(x)$, there exists a generisation x' of x such that $f(x') = y'$.*
- (ii) *For every $x \in X$, the image by f of $\text{Spec}(\mathcal{O}_{X,x})$ is $\text{Spec}(\mathcal{O}_{Y,y})$.*
- (iii) *For every closed irreducible part Y' of Y , every irreducible component of $f^{-1}(Y')$ dominates Y' .*

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Proof. It suffices, for example, to prove (ii). By hypothesis, there is a quasi-coherent $\mathcal{O}_X$ -Module of finite type $\mathcal{F}$, which is f -flat and such that $\text{Supp}(\mathcal{F}) = X$. For every $x \in X$, $\mathcal{F}_x$ is then a finitely generated $\mathcal{O}_x$ -module, non-reduced to 0, and moreover $\mathcal{F}_x$ is a flat $\mathcal{O}_y$ -module flat, and consequently the homomorphism $\rho : \mathcal{O}_y \rightarrow \mathcal{O}_x$. Since the latter is local and $\mathcal{F}_x \neq 0$, the Nakayama lemma shows that $\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y) \neq 0$, so $\mathcal{F}_x$ is a faithfully flat $\mathcal{O}_y$ -module (0_I, 6.4.1). It follows that for every prime ideal $\mathfrak{q}$ of $\mathcal{O}_y$, there exists a prime ideal $\mathfrak{p}$ of $\mathcal{O}_x$ such that $\mathfrak{q} = \rho^{-1}(\mathfrak{p})$ (0_I, 6.5.1), which proves (ii). $\square$

Corollary (2.3.5). — *Let $f : X \rightarrow Y$ be a morphism satisfying the equivalent conditions (i), (ii), (iii) of (2.3.4) (in particular a quasi-flat or open morphism (1.10.4)).*

- (i) *Let Z, Z' be two closed irreducible parts of Y such that $Z \subset Z'$, and let T be an irreducible component of $f^{-1}(Z)$; then there exists an irreducible component T' of $f^{-1}(Z')$ containing T (and dominating Z').*
- (ii) *For every irreducible component Y' of Y , $\overline{f(T)}$ is an irreducible component of Y .*
- (iii) *Suppose Y irreducible, and, if y is its generic point, suppose that $f^{-1}(y)$ is irreducible. Then X is irreducible.*

Proof. (i) It suffices to apply (2.3.4) (i) by taking for x the generic point of T (the generic point of Z) and for y' the generic point of Z' .

(ii) It is clear that $\overline{f(T)} = Z$ is irreducible, and by virtue of (i), Z cannot be strictly contained in a closed irreducible component Z' of Y .

(iii) By virtue of (ii), every irreducible component of X dominates Y , so every such component meets $f^{-1}(y)$; the conclusion then follows from (0_I, 2.1.8). $\square$

Proposition (2.3.6). — *Let Y be a prescheme whose set of irreducible components is locally finite (which is the case if the underlying space of Y is locally noetherian (cf. (I, 6.1.9))).*

- (i) *Every stable closed part W of Y by generisation (0_I, 2.1.2) is open. In particular, every connected component of Y is open.*
- (ii) *Let $f : X \rightarrow Y$ be a closed morphism satisfying moreover the equivalent conditions (i), (ii), (iii) of (2.3.4) (which will be the case if f is quasi-flat or open). Then the image by f of every connected component C of X is a connected component of Y .*
- (iii) *Let $f : X \rightarrow Y$ be a morphism satisfying the equivalent conditions (i), (ii), (iii) of (2.3.4) and such that moreover for every $y \in Y$, the set $f^{-1}(y)$ is finite (which will be the case if f is quasi-finite or radiciel). Then the set of irreducible components of X is locally finite.*

Proof. (i) If $y \in W$, the generic points η_i ($1 \leq i \leq m$) of the irreducible components Y_i of Y containing y belong by hypothesis to W , so, since W is closed, these components are themselves contained in W ;

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since there is by hypothesis a neighbourhood U of y such that U is a union of the $U \cap Y_i$ (0_I, 2.1.6), then $U \subset W$, so W is open. Since for every $y \in Y$, $\overline{\{y\}}$ is connected, a connected component of Y is stable by generisation, so the second assertion follows immediately from the first.

(ii) Since C is closed in X , $f(C)$ is closed in Y by hypothesis. Moreover, as C is stable by generisation, the hypothesis on f implies that $f(C)$ is stable by generisation; therefore, by virtue of (i), $f(C)$ is open and closed, and since it is connected, it is a connected component of Y .

(iii) Let $x \in X$; by hypothesis, there is a neighbourhood U of $y = f(x)$ which intersects only a finite number of irreducible components Y_i of Y ($1 \leq i \leq n$); let y_i be the generic point of Y_i (2.3.4). Since each of these sets is finite by hypothesis, this proves our assertion. $\square$

Proposition (2.3.7). — *Let $f : X \rightarrow Y$ be a morphism, $g : Y' \rightarrow Y$ a quasi-flat morphism, $f' = f_{(Y')} : X' \rightarrow Y'$.*

- (i) *If f is quasi-compact and dominant, the same is true of f' .*
- (i) *If every irreducible component of X dominates an irreducible component of Y , then every irreducible component of X' dominates an irreducible component of Y' .*

Proof. Denote by $g' : X' \rightarrow X$ the canonical projection, which is a quasi-flat morphism (2.3.3).

(i) We already know (1.1.2) that f' is quasi-compact; moreover, if y' is a maximal point (1.1.4) of Y' , $y = g(y')$ is a maximal point of Y , as it follows from (2.3.4) (iii). By hypothesis (1.1.5), there exists $x \in X$ such that $f(x) = y$; therefore (I, 3.4.7), there exists $x' \in X'$ such that $f'(x') = y'$.

(ii) Let x' be a maximal point of X' , and let $x = g'(x')$; it follows from (2.3.4) (ii) that x is a maximal point of X , and by hypothesis $y = f(x)$ is a maximal point of Y . Set $y' = f'(x')$, so that $g(y') = y$; it suffices to show that y' is a maximal point of Y' . By virtue of (I, 5.1.7) and (2.3.3), we can restrict to the case where X and Y are reduced, so $\mathcal{O}_x$ and $\mathcal{O}_y$ are fields, and by virtue of (I, 2.4.4) applied twice and of (I, 3.6.5), we can restrict to the case where $Y = \text{Spec}(\mathcal{O}_y)$ and $Y' = \text{Spec}(\mathcal{O}_{y'})$. Then f is a flat morphism since $\mathcal{O}_y$ is a field (2.1.2), and it follows likewise from (2.1.4) that $y' = f'(x')$ is a maximal point of Y' ; therefore it follows from (2.3.4) (ii) that $y' = f'(x')$ is a maximal point. $\square$

Corollary (2.3.8). — (Zariski). — *Let A, B be two noetherian local rings, $\mathfrak{m}, \mathfrak{n}$ their maximal ideals respectively, $\varphi : A \rightarrow B$ a local homomorphism. Suppose the following hypotheses are satisfied:*

- 1° *B is an A -algebra essentially of finite type (1.3.8).*
- 2° *The completion $\hat{A}$ of A for the $\mathfrak{m}$ -adic topology is integral.*
- 3° *φ is injective.*

Then the $\mathfrak{m}$ -adic topology on A is induced by the $\mathfrak{n}$ -adic topology of B .

Proof. Set $B' = B \otimes_A \hat{A}$, where C is an A -algebra of finite type and S is a multiplicative part of C , so B' is an A -algebra of finite type. Since A is identified with a subring of $\hat{A}$ (0_I, 7.3.5), A is integral by 2°. Hypothesis 3° implies that there exists a prime ideal $\mathfrak{a}$ of B inducing the ideal $\mathfrak{o}$ of A (0_I, 1.5.8), and therefore the local homomorphism $A \rightarrow B/\mathfrak{a}$ is injective. We can therefore restrict to proving (2.3.8) by adding the hypothesis that B is a local integral ring. Let us apply (2.3.7) (ii) to $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$, $Y' = \text{Spec}(\hat{A})$ and $X' = \text{Spec}(B')$; since the morphism $Y' \rightarrow Y$ is flat and X dominates Y , every irreducible component of X' dominates Y' ; since by hypothesis Y' is integral, X' is also integral. Identifying A and $\hat{A}$ as subrings of $\mathcal{O}_{x'}$, and denoting by $\mathfrak{r}$ the maximal ideal of $\mathcal{O}_{x'}$, the intersection of the ideals $\mathfrak{r}^k \cap \hat{A}$ are open in $\hat{A}$ (0_I, 7.3.5). Moreover, the $\mathfrak{n}$ -preadic topology of B induces on A a topology that is finer than the $\mathfrak{m}$ -preadic topology of A (0_I, 7.3.5). On the other hand, $\mathfrak{n}^k \cap \mathfrak{r}^k \cap A$, so the $\mathfrak{n}$ -preadic topology of B induces on A the $\mathfrak{m}$ -preadic topology; but $\mathfrak{m}^k \subset \mathfrak{n}^k \cap A$, so these two topologies are identical. $\square$

Remark (2.3.9). — We will see further on (IV, 7.8.3) (vii), that for the most common noetherian local rings A in algebraic geometry, the hypothesis that A is integral and integrally closed implies that $\hat{A}$ is integral. This is why, in algebraic geometry over a base field, one generally states (2.3.8) under the hypothesis that A is *integral and integrally closed*.

Theorem (2.3.10). — *Let $f : X \rightarrow Y$ be a quasi-flat morphism (2.3.3). Then, for every pro-constructible part (1.9.4) Z of Y , we have $f^{-1}(\overline{Z}) = \overline{f^{-1}(Z)}$.*

Proof. Since f is continuous, we have $f^{-1}(\overline{Z}) \supset \overline{f^{-1}(Z)}$ and it suffices to prove that for every $x \in X$ such that $f(x) \in \overline{Z}$, x is adherent to $f^{-1}(Z)$; it is clear that the question is local on Y , so we can suppose Y affine. By virtue of the hypothesis, there exists an affine scheme Y' and a morphism $g : Y' \rightarrow Y$ such that $g(Y') = Z$ (1.9.5) (ix). Let Y_1 be the image by f of Y' (I, 9.5.3), and let X_1 be the closed sub-prescheme $f^{-1}(Y_1)$ of X ; if $f_1 : X_1 \rightarrow Y_1$ is the restriction of f to X_1 , we know that f_1 is quasi-flat (2.3.3); we can therefore replace X, Y by X_1, Y_1 respectively, in other words suppose that g is *dominant*. Set then $X' = X \times_Y Y'$, and let f' and g' be the projections of X' to Y' and X respectively, so that we have the commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{g'} & X' \\ f \uparrow & & \uparrow f' \\ Y & \xleftarrow{g} & Y' \end{array}$$

Since f is quasi-flat, g quasi-compact and dominant, it follows from (2.3.7) (where the roles of f and g are interchanged) that g' is a dominant morphism, which proves the theorem. $\square$

Corollary (2.3.11). — *Let f be a quasi-flat and quasi-compact morphism, F a closed part of X such that $F = f^{-1}(f(F))$; then $F = f^{-1}(\overline{f(F)})$.*

Proof. Let Y' be the reduced sub-prescheme of X having F as underlying space (I, 5.2.1) and let $j : Y' \rightarrow X$ be the canonical injection; then $f \circ j$ is quasi-compact (1.1.2), so $Z = f(F)$ is pro-constructible in Y (1.9.5) (vii), and the corollary follows from the fact that F is closed. $\square$

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We can also write the result of (2.3.11) in the form $F = f^{-1}(f(X) \cap \overline{f(F)})$, in other words, the closed subsets of the subspace $f(X)$ of Y are the parts $Z \subset f(X)$ such that $f^{-1}(Z)$ is closed in X ; this also means that *the topology induced by Y on $f(X)$ is the quotient of that of X by the equivalence relation defined by f* . In particular:

Corollary (2.3.12). — *Let $f : X \rightarrow Y$ be a quasi-faithfully flat morphism (2.3.3) and quasi-compact. Then the topology of Y is the quotient of that of X by the equivalence relation defined by f (in other words, for $Z \subset Y$ to be open (resp. closed) in Y , it is necessary and sufficient that $f^{-1}(Z)$ be open (resp. closed) in X).*

Proof. Indeed, we then have $f(X) = Y$. $\square$

Corollary (2.3.13). — *Let X, Y be two S -preschemes, $f : X \rightarrow Y$ an S -morphism faithfully flat and quasi-compact. For Y to be separated over S , it is necessary and sufficient that the canonical immersion $j : X \times_Y X \rightarrow X \times_S X$ (I, 5.3.10) be a closed immersion.*

Proof. Note for this that we have the commutative diagram (I, 5.3.5)

$$\begin{array}{ccc} X \times_Y X & \xrightarrow{j} & X \times_S X \\ \pi \downarrow & & \downarrow f \times_S f \\ Y & \xrightarrow{\Delta_Y} & Y \times_S Y \end{array}$$

identifying $X \times_Y X$ with the product of the $(Y \times_S Y)$ -preschemes Y and $X \times_S X$. Since f is surjective, the same is true of π and of $f \times_S f$, so the diagonal $\Delta_Y(Y)$ has as inverse image by $f \times_S f$ the image $j(X \times_Y X)$ (I, 3.4.8). Since $f \times_S f$ is faithfully flat and quasi-compact (1.1.2) and (2.2.13), it suffices to apply (2.3.12) to this morphism. $\square$

Corollary (2.3.14). — *Let $f : X \rightarrow Y$ be a quasi-faithfully flat and quasi-compact morphism, and let Z be a part of Y . For Z to be a locally closed pro-constructible part of Y , it is necessary and sufficient that $f^{-1}(Z)$ be a locally closed pro-constructible part of X .*

Proof. We already know (1.9.12) that, for Z to be a pro-constructible part of Y , it is necessary and sufficient that $f^{-1}(Z)$ be a pro-constructible part of X . The condition is evidently necessary. To prove that it is sufficient, consider the closed reduced sub-prescheme Y_1 of Y having $\overline{Z}$ as underlying space and, let X_1 be its inverse image $f^{-1}(Y_1)$, which has $\overline{f^{-1}(Z)}$ as underlying space by virtue

of (2.3.10). Since $f^{-1}(Z)$ is locally closed in X , it is open in $\overline{f^{-1}(Z)}$, and therefore in X_1 ; or, the morphism $f_1 : X_1 \rightarrow Y_1$ induced by f by restriction is quasi-faithfully flat and quasi-compact (1.1.2) and (2.3.3); we deduce from (2.3.12) that Z is open in $\overline{Z}$, which shows that Z is locally closed in Y . $\square$

Remark (2.3.15). — It does not suffice, in (2.3.12), to suppose only that f is *faithfully flat*. For example, take for Y a reduced irreducible algebraic curve (II, 7.4.2), for X the prescheme *sum* of the schemes $\text{Spec}(\mathcal{O}_y)$, where y ranges over Y (I, 3.1), and for f the canonical morphism, which is faithfully flat (I, 2.4.2); if η denotes the generic point of Y , $Z = \{\eta\}$ is not open in Y (II, 7.4.3), but $f^{-1}(Z) = \text{Spec}(\mathcal{O}_\eta)$ since $\mathcal{O}_\eta$ is a field, and $f^{-1}(Z)$ is open in X .

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2.4. Universally open morphisms and flat morphisms

(2.4.1). We have already defined (II, 5.4.9) the notion of a *universally closed* morphism; in the same manner, we introduce the following definitions:

Definition (2.4.2). — We say that a morphism of preschemes $f : X \rightarrow Y$ is *universally open* (resp. *universally bicontinuous*, resp. a *universal homeomorphism*) if, for every morphism $g : Y' \rightarrow Y$, the morphism $f_{(Y')} : X \times_Y Y' \rightarrow Y'$ is open (resp. a homeomorphism onto its image, resp. a homeomorphism onto Y').

We will see further on (IV, 14.3.2) that when Y is locally noetherian, the definition of universally open morphism given here is equivalent to Definition (III, 4.3.9) for morphisms of finite type; the reader will be able to verify that we do not use this last definition before §14.

Proposition (2.4.3). — (i) An immersion (resp. an open immersion, resp. a closed immersion) is universally bicontinuous (resp. universally open, resp. universally closed).

(i) The composite of two universally open (resp. universally closed, resp. universally bicontinuous, resp. universal homeomorphisms) is so as well.

(i) If $f : X \rightarrow Y$ is an S -morphism universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), the same is true of $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ for every base change $S' \rightarrow S$.

(i) If $f : X \rightarrow X'$, $g : Y \rightarrow Y'$ are two S -morphisms universally open (resp. universally closed, resp. universally bicontinuous, resp. universal homeomorphisms), the same is true of $f \times_S g : X \times_S Y \rightarrow X' \times_S Y'$.

(i) Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms such that f is surjective; if $g \circ f$ is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), the same is true of g .

(i) For a morphism f to be universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), it is necessary and sufficient that f_{red} be so.

(i) Let (U_α) be an open cover of Y . For $f : X \rightarrow Y$ to be universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), it is necessary and sufficient that for every α , its restriction $f^{-1}(U_\alpha) \rightarrow U_\alpha$ be so.

Proof. Assertion (i) follows from (I, 4.3.2). Assertion (ii) also follows from the definitions, and it is the same for (iii) by reducing to the case where $Y = S$, $Y' = S'$, which one can do using (I, 3.3.11); one knows that (iv) follows from (ii) and (iii) (I, 3.5.1). To prove (v), note that for every morphism $Z' \rightarrow Z$, $f_{(Z')} : X_{(Z')} \rightarrow Y_{(Z')}$ is surjective (I, 3.5.2), and it is then a purely topological question. For the case where $g \circ f$ is open (resp. closed), the fact that g is then open (resp. closed) follows from Bourbaki, *Top. gén.*, chap. I, 3^e éd., §5, no. 1, prop. 1; for the two other cases, we can reduce to supposing that $g(f(X)) = g(Y) = Z$, in other words that $g \circ f$ is a homeomorphism from X to Z ; since f is surjective, g is necessarily bijective, and since g is a continuous open map by what precedes, g is indeed a homeomorphism from Y to Z .

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To prove (vi), note that to say that a morphism g is open (resp. closed, resp. a homeomorphism onto its image, resp. a homeomorphism bijective) is equivalent to saying that g_{red} has the same property. On the other hand (I, 5.1.8), for every morphism $Y' \rightarrow Y$, one has $(X_{\text{red}} \times_{Y_{\text{red}}} Y'_{\text{red}})_{\text{red}} = (X \times_Y Y')_{\text{red}}$, so the previous remark shows that if f_{red} is universally open (resp. universally closed, resp. universally bicontinuous, resp. a universal homeomorphism), the same is true of f . The

converse is proved in the same way, noting here that for every morphism $Y'' \rightarrow Y_{\text{red}}$, one has $(X_{\text{red}} \times_{Y_{\text{red}}} Y'')_{\text{red}} = (X \times_Y Y'')_{\text{red}}$ (I, 5.1.8).

Finally, the necessity of (vii) follows immediately from (iii). Conversely, suppose (vii) is fulfilled, and let $g : Y' \rightarrow Y$ be a morphism; then the $g^{-1}(U_\alpha) = U'_\alpha$ form an open cover of Y' , and if we denote f_α the restriction $f^{-1}(U_\alpha) \rightarrow U_\alpha$ of f , and f' the morphism $f_{(Y')}$, the restriction $f'^{-1}(U'_\alpha) \rightarrow U'_\alpha$ is none other than $(f_\alpha)_{(U'_\alpha)}$. We can therefore restrict to proving that f is open (resp. closed, resp. a homeomorphism onto its image, resp. a homeomorphism onto Y), which is immediate. $\square$

Proposition (2.4.4). — *A universally bicontinuous morphism $f : X \rightarrow Y$ is radiciel (hence separated (1.7.7.1)).*

Proof. Indeed, f is universally injective by hypothesis (I, 3.5.11). $\square$

Proposition (2.4.5). — (i) *A morphism $f : X \rightarrow Y$ which is integral, surjective and radiciel is a universal homeomorphism.*

(i) *Inversely, suppose Y locally noetherian. Then, if a morphism of finite type $f : X \rightarrow Y$ is a universal homeomorphism, it is finite, surjective and radiciel.*

Proof. (i) It suffices to observe that the three properties of f are conserved by base change (I, 3.5.2), (I, 3.5.7) and (II, 6.1.5), and as an integral morphism is closed (II, 6.1.10), it is clear that f is a universal homeomorphism from X to Y .

(ii) Since f is of finite type and closed by hypothesis and separated by (2.4.4), it is proper (II, 5.4.4), and for every $y \in Y$, $f^{-1}(y)$ is reduced to a single element; therefore (III, 4.4.2) f is finite; it is clear that f is surjective, and it is radiciel since it is universally injective (I, 3.5.11). $\square$

Theorem (2.4.6). — *Let $f : X \rightarrow Y$ be a quasi-flat morphism (2.3.3) and locally of finite presentation. Then f is universally open. In particular, a flat morphism locally of finite presentation is universally open.*

Proof. We know that for every base change $Y' \rightarrow Y$, $f_{(Y')}$ is quasi-flat (2.3.3) and locally of finite presentation (1.4.3) (iii). It therefore suffices to prove that f is an open morphism. But this follows from the criterion (1.10.4) for morphisms locally of finite presentation, the conditions b), b'), and c) of (1.10.4) being none other than the conditions (i), (ii), (iii) of (2.3.4). $\square$

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Corollary (2.4.7). — *For every prescheme Y , the structural morphism $\mathbf{V}_Y^n \rightarrow Y$ (where $\mathbf{V}_Y^n = Y \otimes_{\mathbf{Z}} \text{Spec}(\mathbf{Z}[T_1, \dots, T_n])$), also written $Y[T_1, \dots, T_n]$ is universally open.*

Proof. Indeed, for $Y = \text{Spec}(A)$, $Y[T_1, \dots, T_n] = \text{Spec}(A[T_1, \dots, T_n])$, and $A[T_1, \dots, T_n]$ is a free A -algebra of finite presentation. $\square$

Remarks (2.4.8). — (i) Note that a morphism $f : X \rightarrow Y$ faithfully flat and quasi-compact need not be universally open, even when X and Y are noetherian. Take for example for Y a reduced irreducible algebraic curve (II, 7.4.2) of generic point y , and let X be the prescheme $\sum Y \amalg \text{Spec}(k(y))$, f being the canonical morphism; it is clear that f is flat and surjective, hence faithfully flat, and quasi-compact, but the image by f of the open part $\text{Spec}(k(y))$ of X is the singleton $\{y\}$, which is not open in Y (II, 7.4.3).

(ii) For every prescheme X , the canonical morphism $f : X_{\text{red}} \rightarrow X$ is a closed immersion and a universal homeomorphism (2.4.3) (vi); but when X is locally noetherian, f is not flat unless X is reduced, that is $f = 1_X$ (2.2.17).

Proposition (2.4.9). — *Let Y be a discrete prescheme. Then every morphism $f : X \rightarrow Y$ is universally open.*

Proof. The question being local on Y (2.4.3) (vii), we can reduce to the case where the underlying space of Y is a point; replacing f by f_{red} (2.4.3) (vi), we can suppose moreover that Y is the spectrum of a field k ; on the other hand, for every base change $Y' \rightarrow Y$, the open sets of $X' = X \times_Y Y'$ are the inverse images of open affine sets of X' , so we can suppose $X = \text{Spec}(B)$ affine, B being a k -algebra. It suffices then to prove that for every k -algebra A' , if we set $B' = B \otimes_k A'$, the image by $f' : \text{Spec}(B') \rightarrow \text{Spec}(A')$ of every open set of the form $U = D(t)$ ($t \in B'$) is open in $\text{Spec}(A')$. Now, B is the inductive limit of the family of its sub- k -algebras of finite type B_α , so $B' = \varinjlim (B_\alpha \otimes_k A')$; there exists an index α such that t is the image in B' of an element $t_\alpha \in B'_\alpha = B_\alpha \otimes_k A'$. But

since k is a field, B_α is a k -algebra of finite presentation and a flat k -module, hence B'_α is an A' -algebra of finite presentation and a flat A' -module; we conclude from (2.4.6) that the image by $f'_\alpha : \operatorname{Spec}(B'_\alpha) \rightarrow \operatorname{Spec}(A')$ is open in $\operatorname{Spec}(A')$. It suffices to check that for every point $y' \in f'_\alpha(U_\alpha)$, the intersection $V = U \cap f'^{-1}(y')$ is non-empty. But by definition of y' , $V_\alpha = U_\alpha \cap f'^{-1}_\alpha(y')$ is non-empty; that is, V_α is a non-empty open subset of $\operatorname{Spec}(B'_\alpha \otimes_k k(y'))$ and V is its inverse image by the morphism $v : \operatorname{Spec}(B \otimes_k k(y')) \rightarrow \operatorname{Spec}(B_\alpha \otimes_k k(y'))$. As B_α is a sub-algebra of B and k is a field, the homomorphism $B_\alpha \otimes_k k(y') \rightarrow B \otimes_k k(y')$ is injective, so the morphism v is dominant (I, 1.2.7), which concludes the proof. $\square$ IV-2 | 22

Corollary (2.4.10). — *Let k be a field, X, Y two k -preschemes; then the projection morphism $X \times_k Y \rightarrow X$ is universally open. In particular, for every extension K of k , the projection morphism $X_{(K)} \rightarrow X$ is universally open.*

Proof. It suffices to apply (2.4.9) to the structural morphism $Y \rightarrow \operatorname{Spec}(k)$. $\square$

Remark (2.4.11). — If $f : X \rightarrow Y$ is an open morphism, we know that, for every part E of Y , one has $f^{-1}(\overline{E}) = \overline{f^{-1}(E)}$ (Bourbaki, *Top. gén.*, chap. I, 3^e éd., §5, no. 4, prop. 7). This remark applies for example when f is a flat morphism locally of finite presentation (2.4.6), or a morphism projection $X \times_k Y \rightarrow X$ where X, Y are preschemes over a field k (2.4.10), and generalises then (2.3.10).

2.5. Permanence of properties of Modules by faithfully flat descent

Proposition (2.5.1). — *Let $f : X \rightarrow Y$ be a morphism, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module, $g : Y' \rightarrow Y$ a faithfully flat morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$, $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$. For $\mathcal{F}$ to be flat (resp. faithfully flat) relative to f , it is necessary and sufficient that $\mathcal{F}'$ be flat (resp. faithfully flat) relative to f' .*

Proof. It suffices to apply (2.2.10) replacing $X, Y, Z, \mathcal{F}, \mathcal{G}'$ by $Y', X, Y, \mathcal{O}_{Y'}, \mathcal{F}$ respectively; we conclude that for $\mathcal{F}$ to be flat (resp. faithfully flat) relative to f , it is necessary and sufficient that $\mathcal{F}'$ be flat (resp. faithfully flat) relative to $g \circ f'$. But since $g' : X' \rightarrow X$ is the canonical projection, g' is faithfully flat (2.2.13) and we have $g \circ f' = f \circ g'$; therefore for $\mathcal{F}'$ to be flat (resp. faithfully flat) relative to $f \circ g'$, it is necessary and sufficient that $\mathcal{F}$ be so relative to f (2.2.11) (iii). $\square$

Proposition (2.5.2). — *Let $f : X' \rightarrow X$ be a faithfully flat and quasi-compact morphism, $\mathcal{F}' = f^*(\mathcal{F})$. Consider, for a quasi-coherent Module, the property of being:*

- (i) of finite type;
- (i) of finite presentation;
- (i) locally free of finite type;
- (i) locally free of rank n .

Then, if $\mathbf{P}$ denotes one of the preceding properties, for $\mathcal{F}$ to have property $\mathbf{P}$, it is necessary and sufficient that $\mathcal{F}'$ have it.

Proof. Note that the question is local on X , and we can therefore suppose X affine; we then know (2.2.12) that there is an affine scheme X'' and a faithfully flat morphism $g : X'' \rightarrow X'$ which is a surjective local isomorphism; by hypothesis, the question is local on X , and taking into account (2.2.3) and (II, 6.1.4.1), it therefore suffices to prove the IV-2 | 23

Lemma (2.5.3). — *Let A be a ring, A' a faithfully flat A -algebra, M an A -module; $M' = M \otimes_A A'$. For M to be of finite type (resp. of finite presentation), it is necessary and sufficient that M' be so.*

The proof follows from Bourbaki, *Alg. comm.*, chap. I, §3, no. 6, prop. 11. $\square$

Remark (2.5.4). — The assertions of (2.5.2) for properties (i) and (ii) are still valid when one supposes only f quasi-faithfully flat (2.3.3) and quasi-compact. This is because one is reduced to Bourbaki, *loc. cit.* in this way.

Lemma (2.5.4.1). — *Let $\rho : A \rightarrow A'$ be a homomorphism of rings, such that the corresponding morphism $f : \operatorname{Spec}(A') \rightarrow \operatorname{Spec}(A)$ is surjective; suppose that there exists an A' -module N' of finite type, which is A -flat and has $\operatorname{Spec}(A')$ as support. Then, if $u : P \rightarrow Q$ is a homomorphism of A -modules such that $u \otimes 1 : P \otimes_A A' \rightarrow Q \otimes_A A'$ is surjective, u is surjective.*

Indeed, one deduces first that the homomorphism $u \otimes 1_{N'} : (P \otimes_A A') \otimes_{A'} N' \rightarrow (Q \otimes_A A') \otimes_{A'} N'$ is surjective. Let $\mathfrak{q}$ be a prime ideal of A' , and let $\mathfrak{p} = \rho^{-1}(\mathfrak{q})$; the corresponding homomorphism $(P \otimes_A N')_{\mathfrak{q}} \rightarrow (Q \otimes_A N')_{\mathfrak{q}}$ is surjective, and can be written $u_{\mathfrak{p}} \otimes 1 : P_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N'_{\mathfrak{q}} \rightarrow Q_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N'_{\mathfrak{q}}$ (**0I**, 1.5.4). By hypothesis we have $N'_{\mathfrak{q}} \neq 0$ and $N'_{\mathfrak{q}}$ is a flat $A_{\mathfrak{p}}$ -module (**0I**, 6.3.1); by virtue of Nakayama's lemma, we have $\mathfrak{q}N'_{\mathfrak{q}} \neq N'_{\mathfrak{q}}$, and a fortiori $\mathfrak{p}N'_{\mathfrak{q}} \neq N'_{\mathfrak{q}}$, hence $N'_{\mathfrak{q}}$ is a faithfully flat $A_{\mathfrak{p}}$ -module (**0I**, 6.4.1). It follows that $u_{\mathfrak{p}}$ is surjective (**0I**, 6.4.1), and this holding for every $\mathfrak{p} \in \text{Spec}(A)$, since f is surjective, one concludes finally that u is surjective (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, th. 1).

Proposition (2.5.5). — *Let $f : X \rightarrow Y$ be a morphism, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite type and f -flat, $\mathcal{E}$ an $\mathcal{O}_Y$ -Module quasi-coherent of finite type; for every $y \in Y$, set $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_y} k(y)$. For a point $x \in X$ to be a maximal point of $\text{Supp}(\mathcal{E} \otimes_Y \mathcal{F})$, it is necessary and sufficient that $y = f(x)$ be a maximal point of $\text{Supp}(\mathcal{E})$, and that x be a maximal point of $\text{Supp}(\mathcal{F}_y)$ in $f^{-1}(y)$. If this is so, then we have*

$$(5) \quad \text{long}((\mathcal{E} \otimes_Y \mathcal{F})_x) = \text{long}(\mathcal{E}_y) \cdot \text{long}((\mathcal{F}_y)_x).$$

Proof. It is clear that $f(\text{Supp}(\mathcal{E} \otimes_Y \mathcal{F})) \subset \text{Supp}(\mathcal{E})$ (**0I**, 5.2.2); the image by f of every irreducible component of $\text{Supp}(\mathcal{E} \otimes_Y \mathcal{F})$ is hence contained in an irreducible component of $\text{Supp}(\mathcal{E})$. One can restrict to the case where $\text{Supp}(\mathcal{F}) = X$. Indeed (**ErrIII**, 30), since $\mathcal{F}$ is of finite type, there exists a closed sub-prescheme X' of X having $\text{Supp}(\mathcal{F})$ as underlying space and an $\mathcal{O}_{X'}$ -Module quasi-coherent of finite type $\mathcal{G}$ such that, if $j : X' \rightarrow X$ is the canonical injection, we have $\mathcal{F} = j_*(\mathcal{G})$. If one sets $f' = f \circ j$, it is clear that $\mathcal{G}$ is f' -flat and that $\text{Supp}(\mathcal{E} \otimes_Y \mathcal{F}) = \text{Supp}(\mathcal{E} \otimes_Y \mathcal{G})$.

Suppose therefore $\text{Supp}(\mathcal{F}) = X$; if Z is an irreducible component of $\text{Supp}(\mathcal{E})$, one has $f^{-1}(Z) \subset \text{Supp}(\mathcal{E} \otimes_Y \mathcal{F})$ (**I**, 9.1.13), and it results from (2.3.4) that every irreducible component of $f^{-1}(Z)$ dominates Z ; in other words, if x is the generic point of an irreducible component of $\text{Supp}(\mathcal{E} \otimes_Y \mathcal{F})$ contained in $f^{-1}(Z)$, $y = f(x)$ is the generic point of Z ; moreover (**0I**, 2.1.8), x is the generic point of an irreducible component of $f^{-1}(y) = \text{Supp}(\mathcal{F}_y)$ (**I**, 9.1.13), and reciprocally every generic point of one of these components is the generic point of an irreducible component of $f^{-1}(Z)$.

It remains to prove (2.5.5.1); one has $(\mathcal{E} \otimes_Y \mathcal{F})_x = \mathcal{E}_y \otimes_{\mathcal{O}_y} \mathcal{F}_x$ (**I**, 9.1.12) and $(\mathcal{F}_y)_x = \mathcal{F}_x / \mathfrak{m}_y \mathcal{F}_x$; one is hence reduced to proving the

Lemma (2.5.5.2). — *Let A, B be two local rings, $\rho : A \rightarrow B$ a local homomorphism, N a B -module which is a faithfully flat A -module and such that $N / \mathfrak{m}N$ is a B -module of finite length; then one has*

$$(2.5.5.3) \quad \text{long}_B(M \otimes_A N) = \text{long}_A(M) \cdot \text{long}_B(N / \mathfrak{m}N).$$

If M is of infinite length, the same is true of $M \otimes_A N$, for every strictly increasing sequence of n submodules M_i of M ($1 \leq i \leq n$) yields n distinct submodules $M_i \otimes_A N$ of $M \otimes_A N$; since $N \neq \mathfrak{m}N$ since N is a faithfully flat A -module, the formula (2.5.5.3) is true in this case. Suppose therefore M of finite length. If $M = 0$ the two sides of (2.5.5.3) are zero. One can hence suppose $M \neq 0$; if $\mathfrak{m}^k M \neq 0$, it follows from Nakayama's lemma that $\mathfrak{m}^{k+1} M \neq \mathfrak{m}^k M$; since the submodules $\mathfrak{m}^k M$ are of finite length, there necessarily exists an integer r such that $\mathfrak{m}^r M = 0$. The $\mathfrak{m}^k M \otimes_A N$ are hence sub- B -modules of $M \otimes_A N$, $(\mathfrak{m}^k M \otimes_A N) / (\mathfrak{m}^{k+1} M \otimes_A N)$ being also isomorphic to $(\mathfrak{m}^k M / \mathfrak{m}^{k+1} M) \otimes_{A/\mathfrak{m}} (N / \mathfrak{m}N)$ ($0 \leq k \leq r-1$); the length of the latter, considered as a B -module, is hence the product of $\text{long}_B(N / \mathfrak{m}N)$ and of the rank of the $(A/\mathfrak{m})$ -vector space $\mathfrak{m}^k M / \mathfrak{m}^{k+1} M$, rank which is also the length of the A -module $\mathfrak{m}^k M / \mathfrak{m}^{k+1} M$. By addition (for $0 \leq k \leq r-1$), one deduces also the formula (2.5.5.3). $\square$

Remark (2.5.5.4). — We note that when N is an A -module of finite type, saying that N is a faithfully flat A -module is equivalent to saying that N is a flat A -module; indeed, Nakayama's lemma then shows that $\mathfrak{m}N \neq N$.

Lemma (2.5.6). — *Let B be a ring (not necessarily commutative), V, W two isomorphic left B -modules, C the ring $\text{End}_B(V)$, and let $M = \text{Hom}_B(V, W)$, equipped with its canonical structure of right C -module. Then M is a C -module isomorphic to C_d ; moreover, for every $u \in M$, the following conditions are equivalent:*

- a) $\{u\}$ is a basis of the C -module M .
- b) u is an isomorphism from V to W .

Proof. If u is an isomorphism from V to W , the map $v \mapsto u \circ v$ from C to M is evidently a bijection, hence b) implies a). Conversely, suppose that $\{u\}$ is a basis of the C -module M . By hypothesis, there

exists an isomorphism u' from V to W , and $\{u'\}$ is therefore a basis of M ; there is thus an invertible element w of C (that is, an automorphism of V) such that $u = u' \circ w$, which implies that u is an isomorphism from V to W . $\square$

Corollary (2.5.7). — *The hypotheses on B , V and W being those of (2.5.6), suppose moreover that one of the following conditions is satisfied:*

- (i) V and W are Noetherian B -modules;
- (ii) V and W are modules of finite presentation over a commutative sub-ring of B .

Then the conditions a) and b) of (2.5.6) are also equivalent to the following:

- a') u is a generator of the C -module M .
- b') u is an epimorphism from V to W .

Proof. One knows indeed that an epimorphism of an A -module E onto itself is *bijective* in the two following cases: 1° E is a Noetherian A -module (Bourbaki, *Alg.*, chap. VIII, §2, no. 2, lemma 3); 2° A is commutative and E is an A -module of finite presentation (8.9.3)¹; hence b) and b') are equivalent.

On the other hand, if u generates M and if $\{u'\}$ is a basis of M , there exists $v \in C$ such that $u' = u \circ v$, which proves that u is surjective; hence a') implies b'), and since a) evidently implies a'), this finishes proving the corollary. $\square$

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Proposition (2.5.8). — *Let A be a commutative semi-local ring, B an A -algebra (not necessarily commutative), V and W two B -modules. Let A' be a commutative A -algebra which is a faithfully flat A -module; set $B' = B \otimes_A A'$, $V' = V \otimes_A A'$, $W' = W \otimes_A A'$, so that B' is an A' -algebra, V' and W' are B' -modules. Suppose moreover that one of the following conditions is satisfied:*

- (i) A and A' are Noetherian, V and W are A -modules of finite type.
- (ii) B is an A -module of finite type, V is a projective B -module of finite type and an A -module of finite presentation.

Then, if V' and W' are isomorphic B' -modules, V and W are isomorphic B -modules.

Proof. We note that in case (ii), W' being A' -isomorphic to V' , is an A' -module of finite type, whence it results that W is an A -module of finite type (Bourbaki, *Alg. comm.*, chap. I, §3, no. 6, prop. 11); hence in all cases V and W are A -modules of finite type. Moreover:

(2.5.8.1) *Under either of the hypotheses (i), (ii), $\text{Hom}_B(V, W)$ is an A -module of finite type.*

This is evident in case (i), since $\text{Hom}_A(V, W)$ is then an A -module of finite type, and $\text{Hom}_B(V, W)$ is a sub- A -module of $\text{Hom}_A(V, W)$. In case (ii), V is a direct factor of a free B -module B^p , hence $\text{Hom}_B(V, W)$ is a direct factor of $\text{Hom}_B(B^p, W) = W^p$, and since W is an A -module of finite type, the same is true of $\text{Hom}_B(V, W)$.

Set $C = \text{End}_B(V)$, $M = \text{Hom}_B(V, W)$, which are A -modules of finite type in cases (i) and (ii). One knows that under either of the conditions (i), (ii), the canonical homomorphism

$$(2.5.8.2) \quad \text{Hom}_A(V, W) \otimes_A A' \longrightarrow \text{Hom}_{A'}(V', W')$$

is bijective (Bourbaki, *Alg. comm.*, chap. II, §2, no. 10, prop. 11). As A' is a flat A -module, $\text{Hom}_B(V, W) \otimes_A A'$ identifies canonically with a sub- A' -module of $\text{Hom}_A(V, W) \otimes_A A'$. The image of this submodule by the homomorphism (2.5.8.2) is contained in $\text{Hom}_{B'}(V', W')$, for if $u \in \text{Hom}_B(V, W)$ and $a' \in A'$, the image of $u \otimes a'$ by (2.5.8.2) is the homomorphism $u' : V' \rightarrow W'$ such that $u'(x \otimes 1) = u(x) \otimes a'$; for every $b \in B$, we therefore have $u'((b \otimes 1)(x \otimes 1)) = u'(bx \otimes 1) = u(bx) \otimes a' = bu(x) \otimes a' = (b \otimes 1)(u(x) \otimes a')$, whence our assertion. This being so:

(2.5.8.3) *Under either of the hypotheses (i), (ii), the homomorphism*

$$(2.5.8.4) \quad \text{Hom}_B(V, W) \otimes_A A' \longrightarrow \text{Hom}_{B'}(V', W')$$

is bijective.

For every $b \in B$, denote by $h(b)$ (resp. $h'(b)$) the homothety $x \mapsto bx$ in V (resp. W), which is an A -endomorphism. Let $(b_\alpha)_{\alpha \in I}$ be a system of generators of the A -algebra B ; the map

$$u \longmapsto (h'(b_\alpha) \circ u - u \circ h(b_\alpha))_{\alpha \in I}$$

¹The reader can verify that (2.5.7) and (2.5.8) are not used before §9.

from $\text{Hom}_A(V, W)$ to $(\text{Hom}_A(V, W))^I$ is A -linear, and by definition, its kernel is none other than $\text{Hom}_B(V, W)$; in other words, we have an exact sequence

$$0 \longrightarrow \text{Hom}_B(V, W) \longrightarrow \text{Hom}_A(V, W) \longrightarrow (\text{Hom}_A(V, W))^I.$$

The same reasoning applies by replacing A, B, V, W by A', B', V', W' ; moreover, we have a diagram

$$(2.5.8.5) \quad \begin{array}{ccccc} 0 \gg \text{Hom}_B(V, W) \otimes_A A' & \gg & \text{Hom}_A(V, W) \otimes_A A' & \gg & (\text{Hom}_A(V, W))^I \otimes_A A' \\ \downarrow r & & \downarrow s & & \downarrow t \\ 0 \longrightarrow \text{Hom}_{B'}(V', W') & \longrightarrow & \text{Hom}_{A'}(V', W') & \longrightarrow & (\text{Hom}_{A'}(V', W'))^I \end{array}$$

where r is the homomorphism (2.5.8.4), s the homomorphism (2.5.8.2), and t the composed homomorphism

$$(\text{Hom}_A(V, W))^I \otimes_A A' \xrightarrow{w} (\text{Hom}_A(V, W) \otimes_A A')^I \xrightarrow{s^I} (\text{Hom}_{A'}(V', W'))^I$$

w being the canonical homomorphism (Bourbaki, *Alg.*, chap. II, 3^e éd., §3, no. 7). One verifies immediately that the diagram (2.5.8.5) is commutative, and since A' is a flat A -module, its rows are exact. Finally, we have seen

that s is an isomorphism, and it is therefore the same for s^I ; in case (ii), one can take I finite, and one knows then that w is bijective (Bourbaki, *loc. cit.*, prop. 7); in case (i), one notes that if $\tilde{B}$ (resp. $\tilde{B}'$) is the image of B under h (resp. h') in $\text{End}_A(V)$ (resp. $\text{End}_A(W)$), $\tilde{B}$ and $\tilde{B}'$ are A -modules of finite type, hence one can again take I finite. Thus, in all cases, t is bijective, and one therefore concludes that the same is true of r .

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It follows therefore from (2.5.8.4) that if one sets $C' = C \otimes_A A'$, $M' = M \otimes_A A'$, one has canonical bijections

$$(2.5.8.6) \quad C' \xrightarrow{\sim} \text{End}_{B'}(V'), \quad M' \xrightarrow{\sim} \text{Hom}_{B'}(V', W')$$

of which the first is an isomorphism of A' -algebras, the second constituting with the first a di-isomorphism of right C' -modules.

(2.5.8.7) Reduction to the case $V = B_s$. The hypothesis that V' and W' are isomorphic B' -modules implies that C'_d and M' are isomorphic right C' -modules. Let us show that in order to prove (2.5.8), it suffices to find an element $u \in M$ that is a generator of M as a right C -module; indeed, in case (i), V' and W' are A' -modules of finite type, hence Noetherian since A' is Noetherian; in case (ii), V' and W' are (isomorphic) A' -modules of finite presentation. One can therefore apply (2.5.7) to A', B', V' and W' and one concludes that $u' = u \otimes 1$ is a B' -isomorphism from V' to W' . But since A' is faithfully flat over A , this implies that u is bijective (01, 6.4.1), that is, the conclusion of (2.5.8). If one notes that in case (i), C and M are A -modules of finite type, and that in case (ii), C is (as was seen in (2.5.8.1)) a direct factor of V^n , hence a projective A -module of finite type, one sees that one is reduced (by changing notations) to proving (2.5.8) in the particular case where $V = B_s$, and that it suffices to prove the existence of a generator of the B -module W . One notes that then B is an A -module of finite type.

(2.5.8.8) Reduction to the case where A and A' are fields, B a simple algebra with centre A . Let $\mathfrak{r}$ be the radical of the semi-local ring A ; it suffices to prove that $W/\mathfrak{r}W$ is a monogenic $(B/\mathfrak{r}B)$ -module, for if there exists a surjective homomorphism $B_s/\mathfrak{r}B_s \rightarrow W/\mathfrak{r}W$, it gives by composition a homomorphism $B_s \rightarrow B_s/\mathfrak{r}B_s \rightarrow W/\mathfrak{r}W$, which itself (since B_s is a free B -module) can be written $B_s \rightarrow W \rightarrow W/\mathfrak{r}W$, so that the surjective homomorphism considered is $f \otimes 1 : B_s \otimes_A (A/\mathfrak{r}) \rightarrow W \otimes_A (A/\mathfrak{r})$. As W is an A -module of finite type, Nakayama's lemma shows that f is surjective (Bourbaki, *Alg. comm.*, chap. II, §3, no. 2, cor. 1 of prop. 4). Setting $A_1 = A/\mathfrak{r}$, $A'_1 = A' \otimes_A A_1 = A'/\mathfrak{r}A'$, $B_1 = B/\mathfrak{r}B = B \otimes_A A_1$, $W_1 = W/\mathfrak{r}W = W \otimes_A A_1$, the hypotheses (i) (resp. (ii)) remain satisfied when one replaces A, A', B, V by $A_1, A'_1, B_1, V_1 = (B_1)_s, W_1$ respectively; moreover, $V'_1 = V' \otimes_A A_1 = V_1 \otimes_{A_1} A'_1$ and $W'_1 = W' \otimes_{A_1} A'_1$ are B'_1 -isomorphic (with $B'_1 = B' \otimes_A A_1 = B_1 \otimes_{A_1} A'_1$) and A'_1 is a faithfully flat A_1 -module. One can therefore suppose for the proof of (2.5.8) that A is a finite product of fields (commutative). As B is an A -module of finite type, it is an artinian ring; let $\mathfrak{N}$ be its radical. It will now suffice to prove that $W/\mathfrak{N}W$ is a monogenic $(B/\mathfrak{N})$ -module, for by using Nakayama's lemma,

this implies that W is a monogenic B -module; on the other hand, $B'/\mathfrak{N}B' = (B \otimes_A A') \otimes_B (B/\mathfrak{N}) = (B/\mathfrak{N}) \otimes_A A'$ and similarly $W'/\mathfrak{N}W' = (W/\mathfrak{N}W) \otimes_A A'$. One can therefore suppose moreover that $\mathfrak{N} = 0$, that is, that B is a *semi-simple* A -algebra.

Let us now note that since A is a finite product of fields k_i ($1 \leq i \leq n$), A' is a direct product of k_i -algebras A'_i ($1 \leq i \leq n$), each of the A'_i being annihilated by the k_j of index $j \neq i$; the hypothesis that A' is a faithfully flat A -module implies that the $A'_i \neq 0$; there is therefore a quotient A''_i of A'_i that is a field, and A'' , the direct product of the A''_i , is a faithfully flat A -module and a quotient of A' . Considering then the ring $B'' = B' \otimes_{A'} A'' = B \otimes_A A''$, $W'' = W' \otimes_{A'} A'' = W \otimes_A A''$ is a B'' -module isomorphic to B''_s ; one can therefore restrict to proving (2.5.8) by replacing A' by A'' , that is, one can also suppose that A' is a *finite product of fields*.

Let Z be the centre of B , which is a finite product of fields and an A -module of finite type; note that B and W are Z -modules of finite type, projective as any Z -module; moreover, $B' = B \otimes_Z Z'$ and $W' = W \otimes_Z Z'$ on setting $Z' = Z \otimes_A A'$, and Z' is a faithfully flat Z -module. One can therefore replace A by Z in the hypothesis, that is, suppose that A is the *centre* of B , B being semi-simple and A' a finite product of fields. If k_i ($1 \leq i \leq n$) are the component fields of A , B is a direct product of simple rings B_i , k_i being the centre of B_i , and W is a direct sum of submodules W_i ($1 \leq i \leq n$), W_i being annihilated by the B_j of index $j \neq i$; moreover, the reasoning made above shows that one can suppose that A' is a product of fields k'_i ($1 \leq i \leq n$), k'_i being an extension of k_i and being

annihilated by the k_j of index $j \neq i$. The hypothesis that $(B')_s$ and W' are isomorphic B' -modules implies then that, for every i , $(B_i \otimes_{k_i} k'_i)_s$ and $W_i \otimes_{k_i} k'_i$ are isomorphic $(B_i \otimes_{k_i} k'_i)$ -modules; it therefore suffices to prove (2.5.8) when $n = 1$, that is, in the case where A and A' are *fields*, B a *simple algebra with centre* A .

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(2.5.8.9) *End of the proof.* One knows then (Bourbaki, *Alg.*, chap. VIII, §5, prop. 6 and 8) that every B -module is a direct sum of modules isomorphic to a minimal ideal of B , and two B -modules of finite rank over A are therefore isomorphic if and only if they have *the same rank* over A . But by hypothesis, $[W' : A'] = [B'_s : A'] = [B_s : A]$; hence $[W : A] = [B_s : A]$, which finishes the proof. $\square$

2.6. Permanence of set-theoretic and topological properties of morphisms by faithfully flat descent

Proposition (2.6.1). — *Let $f : X \rightarrow Y$ be an S -morphism of S -preschemes, $g : S' \rightarrow S$ a surjective morphism, $X' = X_{(S')}$, $Y' = Y_{(S')}$, $f' = f_{(S')} : X' \rightarrow Y'$. Consider, for a morphism, the property of being:*

- (i) *surjective;*
- (i) *injective;*
- (i) *with finite fibres (as sets);*
- (i) *bijective;*
- (i) *radiciel.*

Then, if $\mathbf{P}$ denotes one of the preceding properties and if f' has property $\mathbf{P}$, the same is true of f .

Proposition (2.6.2). — *With the notations being those of (2.6.1), suppose the morphism $g : S' \rightarrow S$ faithfully flat and quasi-compact. Consider for a morphism the property of being:*

- (i) *open;*
- (i) *closed;*
- (i) *quasi-compact and universally bicontinuous onto its image;*
- (i) *a universal homeomorphism;*

Then, if $\mathbf{P}$ denotes one of the preceding properties and if f' has property $\mathbf{P}$, the same is true of f .

Proof. Since the morphism $Y' \rightarrow Y$ is faithfully flat and quasi-compact (2.2.13) and (1.1.2), one can, by virtue of (I, 3.3.11), reduce to the case where $Y = S$, $Y' = S'$. If g' is the projection $X' \rightarrow X$, we know that for every part M of X , $g^{-1}(f(M)) = f'(g'^{-1}(M))$ (I, 3.4.8). If f' is an open (resp. closed) morphism, then, for every open (resp. closed) part M of X , $f'(g'^{-1}(M))$ is open (resp. closed) in Y' , and since g is faithfully flat and quasi-compact, we conclude by virtue of (2.3.12) that $f(M)$ is open (resp. closed) in Y . This proves the proposition for cases (i) and (ii). Let us prove it in case (iii) (which will imply the conclusion in case (iv), taking into account (2.6.1) (iv)); by virtue of (2.6.1) (ii), f is injective, and it remains to prove that f , considered as a map from X onto $f(X)$, is a quasi-compact and *open* map. Since f' is quasi-compact, the same is true of f (1.1.4). It then suffices

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for every closed set Z of X to have $Z = f^{-1}(\overline{f(Z)})$; since g' is surjective, this relation is equivalent to $g'^{-1}(Z) = g'^{-1}(f^{-1}(\overline{f(Z)}))$ or to $g'^{-1}(Z) = f'^{-1}(g^{-1}(\overline{f(Z)}))$. Now, since f is quasi-compact, its composite with the canonical injection $Z \rightarrow X$ (where Z is the reduced sub-prescheme of X having Z as underlying space) is also quasi-compact, and one concludes that $f(Z)$ is pro-constructible. Applying (2.3.10) to $f(Z)$ in Y (image of the morphism $f|_Z$), we obtain $g^{-1}(\overline{f(Z)}) = \overline{g^{-1}(f(Z))}$; the formula to prove is therefore equivalent to: $Z' = f'^{-1}(\overline{f'(Z')})$, where $Z' = g'^{-1}(Z)$; but this last formula follows from the hypothesis that f' is a homeomorphism onto $f'(X')$. $\square$

Remark (2.6.3). — In cases (i) and (ii), the conclusions of (2.6.2) are still valid when one supposes only g quasi-faithfully flat (2.3.3) and quasi-compact; indeed, by virtue of (2.1.4), (I, 3.5.2) and (I, 9.1.13.1), one can still reduce to the case where $Y = S$, $Y' = S'$; the conclusion then follows from (2.3.12). In cases (iii) and (iv), the conclusions remain valid when one supposes only g quasi-faithfully flat; indeed, one then uses (2.3.10) and the fact that g is surjective. Finally, if g is faithfully flat and locally of finite presentation, or if g is surjective and S discrete, the conclusion of (2.6.2) is valid when $\mathbf{P}$ is the property:

(iii bis) of being a homeomorphism onto a sub-space image;

this follows indeed from the proof given in (2.6.2) and from Remark (2.4.11).

Corollary (2.6.4). — With the notations being those of (2.6.1), suppose the morphism $g : S' \rightarrow S$ faithfully flat and quasi-compact. Consider for a morphism the property of being:

- (i) universally open;
- (i) universally closed;
- (i) quasi-compact and universally bicontinuous;
- (i) a universal homeomorphism;
- (i) quasi-compact;
- (i) quasi-compact and dominant.

Then, if $\mathbf{P}$ denotes one of the preceding properties, for f to have property $\mathbf{P}$, it is necessary and sufficient that f' have it.

Proof. Properties (v) and (vi) are included for the record, being consequences of (1.1.4), (1.1.6) and (2.3.7). For the others, the condition is necessary by virtue of (2.4.3). Conversely, suppose for example that f' is universally open, and let $Y_1 \rightarrow Y$ be an arbitrary morphism; set $X_1 = X \times_Y Y_1$ and $f_1 = f_{(Y_1)}$. On the other hand, set $Y'_1 = Y_1 \times_Y Y'$, $X'_1 = X' \times_{Y'} Y'_1$, $Y'_1 = X_1 \times_{Y_1} Y'_1$, $f'_1 = f'_{(Y'_1)} = (f_1)_{(Y'_1)} : X'_1 \rightarrow Y'_1$; since $g' = g_{(Y'_1)} : Y'_1 \rightarrow Y'$ is faithfully flat and quasi-compact (2.2.13) and (1.1.2), and that f' is universally open, f'_1 is open, and it follows from (2.6.2) that f_1 is universally open. The same reasoning applies for the other cases. $\square$

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2.7. Permanence of various properties of morphisms by faithfully flat descent

Proposition (2.7.1). — Let $f : X \rightarrow Y$ be an S -morphism of S -preschemes, $g : S' \rightarrow S$ a faithfully flat and quasi-compact morphism, $X' = X_{(S')}$, $Y' = Y_{(S')}$, $f' = f_{(S')} : X' \rightarrow Y'$. Consider, for a morphism, the property of being:

- (i) separated;
- (i) quasi-separated;
- (i) locally of finite type;
- (i) locally of finite presentation;
- (i) of finite type;
- (i) of finite presentation;
- (i) proper;
- (i) an isomorphism;
- (i) a monomorphism;
- (i) an open immersion;
- (i) a quasi-compact immersion;
- (i) a closed immersion;
- (i) affine;
- (i) quasi-affine;

- (i) *finite*;
- (i) *quasi-finite*;
- (i) *integral*.

Then, if $\mathbf{P}$ denotes one of the preceding properties, for f to have property $\mathbf{P}$, it is necessary and sufficient that f' have it.

Proof. It has been shown in Chapters I, II and in Chapter IV, §1, that if f has one of the properties $\mathbf{P}$ above, the same is true of f' (without hypothesis on the morphism $g : S' \rightarrow S$). It therefore remains to prove the converse; since the projection $Y' \rightarrow Y$ is a faithfully flat and quasi-compact morphism (2.2.13) and (1.1.2), one can reduce to the case where $S = Y$, $S' = Y'$ by virtue of (I, 3.3.11).

(i) To say that f is separated means that the diagonal morphism $\Delta_f : X \rightarrow X \times_Y X$ is closed; since $\Delta_{f'} = (\Delta_f)_{(Y')}$ (I, 5.3.4), if $\Delta_{f'}$ is closed, the same is true of Δ_f by virtue of (2.6.2).

(ii) has already been proved under weaker hypotheses (1.2.5).

(iii) and (iv): The question is evidently local on X and Y , and, taking into account (2.2.12), it therefore suffices to prove the

Lemma (2.7.1.1). — *Let A be a ring, B an A -algebra, A' a faithfully flat A -algebra and $B' = B \otimes_A A'$. For B to be an A -algebra of finite type (resp. of finite presentation), it is necessary and sufficient that B' be an A' -algebra of finite type (resp. of finite presentation).*

We already know that the condition is necessary without hypothesis on A' (1.3.4), (1.3.6), (1.4.3) and (1.4.6). Suppose that B' is an A' -algebra of finite type; let $(B_\alpha)_{\alpha \in I}$ be the increasing filtering family of A -sub-algebras of B of finite type, so that $B = \varinjlim B_\alpha$, and consequently $B' = \varinjlim (B_\alpha \otimes_A A')$, the tensor product commuting with inductive limits; if (x'_i) is a finite system of generators of the A' -algebra B' , there exists an index α such that all the x'_i belong to the sub-algebra $B_\alpha \otimes_A A'$ of B' , whence $B' = B_\alpha \otimes_A A'$, and since A' is faithfully flat, $B = B_\alpha$ (0_I, 6.4.1).

Suppose now that B' is an A' -algebra of finite presentation; we already know from the preceding that B is an A -algebra of finite type, so there exists a surjective homomorphism $C = A[T_1, \dots, T_n] \rightarrow B$ of A -algebras and an A -homomorphism $C \rightarrow B$; let $\mathfrak{J}$ be the kernel of this homomorphism, so that one has an exact sequence $0 \rightarrow \mathfrak{J} \rightarrow C \rightarrow B' \rightarrow 0$ (since A' is A -flat), $\mathfrak{J}' = \mathfrak{J} \otimes_A A'$ (identified with an ideal of C'). Since B' is an A' -algebra of finite presentation, $\mathfrak{J}'$ is a C' -module of finite type (1.4.4); but $\mathfrak{J} = \mathfrak{J} \otimes_C C'$ and C' is a faithfully flat C -module (2.2.13) and (2.2.3); we know then that $\mathfrak{J}$ is a C -module of finite type (Bourbaki, *Alg. comm.*, chap. I, §3, no. 6, prop. 11), hence B is an A -algebra of finite presentation.

(v) follows from (iii) and from (2.6.2) (v) by virtue of (1.5.2).

(vi) follows similarly from (iv), (v) and (ii) by virtue of (1.6.1).

(vii) follows from (i), (v) and (2.6.4) (ii) (II, 5.4.1).

(viii) First note that since f' is an isomorphism, it is a universal homeomorphism, hence the same is true of f (2.6.4); we conclude already that f is quasi-compact and separated (2.4.4); it remains to prove that $\theta : \mathcal{O}_Y \rightarrow f_*(\mathcal{O}_X)$ is an isomorphism of $\mathcal{O}_Y$ -Modules. Now, if we set $f' = (\psi', \theta')$, $\theta' : \mathcal{O}_{Y'} \rightarrow f'_*(\mathcal{O}_{X'})$ is composed of the canonical homomorphism $g^*(f_*(\mathcal{O}_X)) \rightarrow f'_*(\mathcal{O}_{X'})$ and of $g^*(\theta)$ (2.3.2); but the first of these two homomorphisms is bijective by virtue of the hypothesis on g (2.3.1), so if θ' is bijective, θ is bijective (2.2.7), which proves (viii).

(ix) The proposition follows from (viii), from (I, 5.3.4) and from (I, 5.3.8) which reduces the monomorphisms to isomorphisms.

(x) If f' is an open immersion, $f'(X')$ is open in Y' , and we have $f'(X') = g^{-1}(f(X))$ (I, 3.4.8); it follows from (2.3.12) that $f(X)$ is open. One can then replace Y (resp. Y') by the sub-prescheme induced on the open $f(X)$ (resp. $f'(X')$), taking into account (1.1.2) and (2.2.13); then f' becomes an isomorphism, so the same is true of f by (viii), and this establishes (x).

(xi) If f' is a quasi-compact immersion, f' is a quasi-compact morphism and quasi-separated (II, 5.1.1), hence the same is true of f by (ii) and (2.6.2) (v). Let Z be the closed sub-prescheme of Y closed image of X by f (1.7.8), and set $f = j \circ g'$ with $j' = j_{(Y')}$, $g' = g_{(Y')}$, and we know that j' is the canonical closed immersion $Z' \rightarrow Y'$ of the sub-prescheme Z' of Y' , closed image of X' by f' (2.3.2). The hypothesis on f' means that g' is an open immersion; it follows by (x) that g is also an open immersion, and this shows that f is an immersion.

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(xii) To say that f (resp. f') is a closed immersion means that f (resp. f') is a quasi-compact immersion and a closed morphism; one sees therefore that (xii) follows from (xi) and from (2.6.2) (ii).

(xiii) and (xiv) Suppose f' affine (resp. quasi-affine); note then that f' is quasi-compact and quasi-separated (II, 5.1.1), hence the same is true of f by (ii) and (2.6.2) (v). Set $\mathcal{A} = f_*(\mathcal{O}_X)$, $\mathcal{A}' = f'_*(\mathcal{O}_{X'})$; by virtue of (2.3.1), the canonical homomorphism of $\mathcal{O}_{Y'}$ -Algebras $g^*(\mathcal{A}) \rightarrow \mathcal{A}'$ is bijective; consequently, if $h : Z = \text{Spec}(\mathcal{A}) \rightarrow Y$ is the structural morphism, the structural morphism $h' : Z' = \text{Spec}(\mathcal{A}') \rightarrow Y'$ is identified with $h_{(Y')}$ (II, 1.5.2). Let then $u : X \rightarrow Z$ (resp. $u' : X' \rightarrow Z'$) be the Y -morphism (resp. Y' -morphism) canonically corresponding to the identical homomorphism of $\mathcal{A}$ (resp. $\mathcal{A}'$) (II, 1.2.7); since we have the commutative diagram

$$\begin{array}{ccc} X & \longleftarrow & X' \\ u \uparrow & & \uparrow u' \\ Z & \longleftarrow & Z' \\ h \uparrow & & \uparrow h' \\ Y & \longleftarrow & Y' \end{array}$$

and that $h' \circ u' = f'$, it follows from (II, 1.2.7) that $u' = u_{(Z')}$. Moreover, g' is faithfully flat and quasi-compact (1.1.2) and (2.2.13). This being the case, the hypothesis on f' means that u' is an isomorphism (resp. an open immersion) (II, 5.1.6); it follows then from (viii) (resp. (x)) that u is an isomorphism (resp. an open immersion), whence (xiii) (resp. (xiv)).

(xv) If f' is finite, it is affine, hence so is f by (xiii); moreover, with the notations of the proof of (xiii), $\mathcal{A}'$ is an $\mathcal{O}_{Y'}$ -Module of finite type, and $\mathcal{A}'$ is isomorphic to $g^*(\mathcal{A})$; it follows from (2.5.2) that $\mathcal{A}$ is an $\mathcal{O}_Y$ -Module of finite type, hence f is a finite morphism.

(xvi) To say that f is quasi-finite means that f is a morphism of finite type and that for every $y \in Y$, $f^{-1}(y)$ is finite (II, 6.2.2) and (I, 6.4.4); the conclusion follows therefore from (v) and (1.5.2).

(xvii) One sees as in (xv) that f is affine. We can reduce to the case where $Y = \text{Spec}(A)$, $Y' = \text{Spec}(A')$, and then $X = \text{Spec}(B)$, $X' = \text{Spec}(B')$, where $B' = B \otimes_A A'$; B is equal to the inductive limit of its sub- A -algebras of finite type B_α , hence $B' = \varinjlim B'_\alpha$, where $B'_\alpha = B_\alpha \otimes_A A'$, and B'_α is an A' -algebra of finite type. But by hypothesis B' is integral over A' , hence B'_α is an A' -module of finite type, and B_α is therefore an A -module of finite type (2.5.2). $\square$

Corollary (2.7.2). — *With the hypotheses and notations being those of (2.7.1), suppose f quasi-compact; let $\mathcal{L}$ be a quasi-coherent $\mathcal{O}_X$ -Module invertible, $\mathcal{L}' = \mathcal{L} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$, its inverse image. For $\mathcal{L}$ to be ample (resp. very ample) for f , it is necessary and sufficient that $\mathcal{L}'$ be ample (resp. very ample) for f' .*

Remarks (2.7.3). — (i) It follows from (2.6.1), (2.6.4) and (2.5.4.1) that the conclusions of (2.7.1) are valid in cases (i), (iii), (v), (vii) and (xvi) when one supposes only that g is quasi-compact and quasi-faithfully flat (2.3.3); we have already remarked that (2.7.1) is valid in case (ii) only assuming g surjective and quasi-compact.

(ii) With the notations and hypotheses of (2.7.1), it can happen that f is proper and f' projective without f being quasi-projective. Indeed, Hironaka [?] has given an example of a proper non-projective morphism $f : X \rightarrow Y$, where X and Y are two regular algebraic schemes ($\mathbf{0_I}$, 4.1.4) over the same field k , Y being projective over k ; moreover Y is a union of two affine opens Y_i ($i = 1, 2$) such that $f_i : X \times_Y Y_i \rightarrow Y_i$ is projective for $i = 1, 2$. Let then $Y' = Y_1 \amalg Y_2$ be the sum prescheme; it is clear that the canonical morphism $g : Y' \rightarrow Y$ coinciding with the canonical injections into Y_1 and Y_2 , is faithfully flat and quasi-compact by virtue of (I, 5.5.10); however, although $f' : X \times_Y Y' \rightarrow Y'$ (coinciding with f_i on each of the Y_i) is projective (II, 5.5.6), it is not of the same type as f .

(iii) Under the hypotheses of (2.7.1), it can happen that f' is a local isomorphism without f being a local immersion.

2.8. Preschemes over a regular base of dimension 1; closure of a closed subscheme of the generic fibre

Proposition (2.8.1). — *Let Y be a locally Noetherian, regular, irreducible prescheme of dimension 1, with generic point η , $f : X \rightarrow Y$ a morphism, $X_\eta = f^{-1}(\eta)$ the fibre at the generic point, $i : X_\eta \rightarrow X$ the canonical morphism. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -Module, $\mathcal{F}_\eta = i^*(\mathcal{F})$, $\mathcal{G}'$ a quotient $\mathcal{O}_{X_\eta}$ -Module of $\mathcal{F}_\eta$, and let $\overline{\mathcal{G}'}$ be the $\mathcal{O}_X$ -Module image of $\mathcal{F}$ by the composed homomorphism (**0I**, 4.4.3.2)*

$$\mathcal{F} \xrightarrow{\rho_{\mathcal{F}}} i_*(i^*(\mathcal{F})) \longrightarrow i_*(\mathcal{G}').$$

Then $\overline{\mathcal{G}'}$ is a quotient $\mathcal{O}_X$ -Module of $\mathcal{F}$, quasi-coherent and f -flat, such that $i^(\overline{\mathcal{G}'}) = \mathcal{G}'$, and it is the only quotient $\mathcal{O}_X$ -Module of $\mathcal{F}$ having these properties.*

Proof. Since i_* is quasi-compact and quasi-separated (**1.1.2**) and (**1.2.2**), it follows from (**1.7.4**) that for every quasi-coherent $\mathcal{O}_{X_\eta}$ -Module $\mathcal{H}'$, $i_*(\mathcal{H}')$ is a quasi-coherent $\mathcal{O}_X$ -Module; moreover, for every open subset U of X , we have $(i_*(\mathcal{H}'))|_U = (i|_{U \cap X_\eta})_*(\mathcal{H}'|_{U \cap X_\eta})$ by definition (**0I**, 3.4.1). If we prove the proposition when X and Y are affine, it will follow by gluing in the general case, given the uniqueness assertion valid in the affine case. In other words, we are reduced to proving the following: IV-2 | 34

Lemma (2.8.1.1). — *Let A be a regular Noetherian ring (**0IV**, 17.3.6), integral of dimension 1, K its field of fractions, M an A -module, N' a quotient K -module of $M_{(K)} = M \otimes_A K$ by a sub- K -module P' , $\overline{N}'$ the image of M by the composed homomorphism $M \rightarrow M_{(K)} \rightarrow N'$. Then $\overline{N}'$ is a flat A -module, and is the unique quotient module N of M which is a flat A -module and such that the kernel of the surjective homomorphism $M_{(K)} \rightarrow N_{(K)}$ is equal to P' .*

Proof. Since for every maximal ideal $\mathfrak{m}$ of A , $A_{\mathfrak{m}}$ is a regular local ring of dimension 1, hence a discrete valuation ring, it amounts to the same to say that an A -module N is flat or that it is without torsion (**0I**, 6.3.4). Since N' is a K -vector space, it is an A -module without torsion, hence so is $\overline{N}'$, being a submodule of N' ; moreover, it is immediate to verify that $\overline{N}'_{(K)}$ identifies with N' . Conversely, if N is a quotient A -module of M having the properties stated, the fact that N is a flat A -module implies that the canonical homomorphism $N \rightarrow N_{(K)} = N \otimes_A K$ is *injective*. Since $N_{(K)}$ identifies with N' , the conclusion follows from the commutativity of the diagram

$$\begin{array}{ccc} M & \longrightarrow & N \\ \downarrow & & \downarrow \\ M_{(K)} & \longrightarrow & N_{(K)} \end{array}$$

□

Corollary (2.8.2). — *Under the conditions of (2.8.1), for $\mathcal{F}$ to be f -flat, it is necessary and sufficient that the canonical homomorphism $\mathcal{F} \rightarrow i_*(\mathcal{F}_\eta) = i_*(i^*(\mathcal{F}))$ be injective.*

(2.8.3). The formation of the $\mathcal{O}_X$ -Module $\overline{\mathcal{G}'}$ is functorial in $\mathcal{F}$ and $\mathcal{G}'$: more precisely, if $\mathcal{F}_1, \mathcal{F}_2$ are two quasi-coherent $\mathcal{O}_X$ -Modules, $u : \mathcal{F}_1 \rightarrow \mathcal{F}_2$ an $\mathcal{O}_X$ -homomorphism, $\mathcal{G}'_i$ a quotient $\mathcal{O}_{X_\eta}$ -Module of $(\mathcal{F}_i)_\eta$ ($i = 1, 2$) and $v : \mathcal{G}'_1 \rightarrow \mathcal{G}'_2$ a homomorphism making the diagram

$$\begin{array}{ccc} (\mathcal{F}_1)_\eta & \xrightarrow{i^*(u)} & (\mathcal{F}_2)_\eta \\ \downarrow & & \downarrow \\ \mathcal{G}'_1 & \xrightarrow{v} & \mathcal{G}'_2 \end{array}$$

commutative (homomorphism determined in a unique way (when it exists) by this property), then the diagram

$$\begin{array}{ccc} \mathcal{F}_1 & \xrightarrow{u} & \mathcal{F}_2 \\ \downarrow & & \downarrow \\ \overline{\mathcal{G}}'_1 & \xrightarrow{w} & \overline{\mathcal{G}}'_2 \end{array}$$

is commutative, and there is therefore a unique homomorphism $w : \overline{\mathcal{G}}'_1 \rightarrow \overline{\mathcal{G}}'_2$ making the diagram IV-2 | 35

$$\begin{array}{ccc} \mathcal{F}_1 & \xrightarrow{u} & \mathcal{F}_2 \\ \downarrow & & \downarrow \\ \overline{\mathcal{G}}'_1 & \xrightarrow{w} & \overline{\mathcal{G}}'_2 \end{array}$$

commutative.

Proposition (2.8.4). — *The hypotheses on Y being those of (2.8.1), let X_1, X_2 be two Y -preschemes, $\mathcal{F}_i$ a quasi-coherent $\mathcal{O}_{X_i}$ -Module, $\mathcal{G}'_i$ a quotient $\mathcal{O}_{(X_i)_\eta}$ -Module of $(\mathcal{F}_i)_\eta$ ($i = 1, 2$). Then we have*

$$(2.8.4.1) \quad \overline{\mathcal{G}}'_1 \otimes_{\mathcal{O}_Y} \overline{\mathcal{G}}'_2 = \overline{\mathcal{G}'_1 \otimes_{k(\eta)} \mathcal{G}'_2}.$$

Proof. Indeed, setting $X = X_1 \times_Y X_2$, the first member of (2.8.4.1) is a quasi-coherent $\mathcal{O}_X$ -Module which is Y -flat (0_I, 6.2.1), whose inverse image in X_η is $\mathcal{G}'_1 \otimes_{k(\eta)} \mathcal{G}'_2$ (I, 9.1.5) and which is a quotient of $\mathcal{F}_1 \otimes_{\mathcal{O}_Y} \mathcal{F}_2$; the conclusion therefore follows from the uniqueness property of (2.8.1). $\square$

Proposition (2.8.5). — *The hypotheses on X and Y being those of (2.8.1), let Z' be a closed subscheme of X_η . There then exists a unique closed subscheme $\overline{Z}'$ of X which is Y -flat and such that $i^{-1}(\overline{Z}') = Z'$.*

Proof. If $\mathcal{J}'$ is the quasi-coherent Ideal of $\mathcal{O}_{X_\eta}$ defining Z' , it suffices to apply (2.8.1) to the case where $\mathcal{F} = \mathcal{O}_X$ and $\mathcal{G}' = \mathcal{O}_{X_\eta} / \mathcal{J}'$; if $\overline{\mathcal{G}}' = \mathcal{O}_X / \mathcal{J}$, we have indeed $\mathcal{J}' = i^*(\mathcal{J})\mathcal{O}_{X_\eta}$, hence $i^{-1}(\overline{Z}') = Z'$ (I, 4.4.5). $\square$

We note that the prescheme $\overline{Z}'$ is the *closed image* of Z' by the composed morphism $Z' \rightarrow X_\eta \xrightarrow{i} X$, where the first arrow is the canonical injection (I, 9.5.3); its underlying space is the *closure* in X of Z' (I, 9.5.4), which justifies the notation adopted. We also say that $\overline{Z}'$ is the subscheme of X which is the *closure* of Z' .

Corollary (2.8.6). — *Let X_1, X_2 be two Y -preschemes, Z'_i a closed subscheme of $(X_i)_\eta$ ($i = 1, 2$). Then we have*

$$(2.8.6.1) \quad \overline{Z'_1 \times_{k(\eta)} Z'_2} = \overline{Z'_1} \times_Y \overline{Z'_2}.$$

Proof. This follows from (2.8.4) and (2.8.5). $\square$

§3. ASSOCIATED PRIME CYCLES AND PRIMARY DECOMPOSITION

We give in this section primarily the translation of results on modules presented in Bourbaki, Alg. comm., chap. IV, which we follow very closely. The notions that follow seem scarcely to be of interest except in the case of locally Noetherian preschemes. IV-2 | 36

3.1. Associated prime cycles of a Module

Definition (3.1.1). — Let X be a prescheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. We say that a point $x \in X$ is *associated* to $\mathcal{F}$ if the maximal ideal $\mathfrak{m}_x$ of $\mathcal{O}_x$ is associated to the $\mathcal{O}_x$ -Module $\mathcal{F}_x$ (in other words, if $\mathfrak{m}_x$ is the annihilator of an element of $\mathcal{F}_x$). We denote by $\text{Ass}(\mathcal{F})$ the set of $x \in X$ associated to $\mathcal{F}$. We say that an irreducible closed subset Z of X is an *associated prime cycle* of $\mathcal{F}$ if its generic point is associated to $\mathcal{F}$. When $\mathcal{F} = \mathcal{O}_X$, we also say that the points (resp. prime cycles) associated to $\mathcal{F}$ are *associated to the prescheme* X .

We say that an associated prime cycle of $\mathcal{F}$ (resp. X) is *immersed* if it is contained in another associated prime cycle of $\mathcal{F}$ (resp. X) (in other words, if it is not *maximal* among these cycles).

If X is locally Noetherian, the *irreducible components* of X are none other than the maximal (or *non-immersed*) prime cycles associated to X .

It is clear that if $x \in \text{Ass}(\mathcal{F})$, then $\mathcal{F}_x \neq 0$, in other words

$$(3.1.1.1) \quad \text{Ass}(\mathcal{F}) \subset \text{Supp}(\mathcal{F}).$$

If $x \in X$ is associated to $\mathcal{F}$, it is obviously associated to $\mathcal{F}|U$ for every open neighbourhood U of x , and conversely, if it is associated to $\mathcal{F}|U$ for one of these neighbourhoods, it is associated to $\mathcal{F}$.

We note finally that for a quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, the existence of immersed associated prime cycles is a *local* question, since if y and z are two points of $\text{Ass}(\mathcal{F})$ such that $y \in \{z\}$, every neighbourhood of y contains z .

Proposition (3.1.2). — Let A be a Noetherian ring, M an A -module, $X = \text{Spec}(A)$, $\mathcal{F} = \tilde{M}$; for a point $x \in X$ to be associated to $\mathcal{F}$, it is necessary and sufficient that the prime ideal $\mathfrak{j}_x$ of A be associated to the module M (in other words, be the annihilator of an element $f \in M$).

Proof. This follows from the definition (3.1.1) and from Bourbaki, *loc. cit.*, §1, no. 2, cor. of prop. 5, applied to $S = A - \mathfrak{j}_x$. $\square$

Proposition (3.1.3). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module, x a point of X , Z the reduced closed subscheme of X having $\{x\}$ as underlying space (I, 5.2.1). The following conditions are equivalent:

- a) $x \in \text{Ass}(\mathcal{F})$.
- b) There exists an open neighbourhood U of x and a section $f \in \Gamma(U, \mathcal{F})$ such that $U \cap Z$ is equal to $\text{Supp}(\mathcal{O}_U \cdot f)$.
- b') There exists an open neighbourhood U of x and a section $f \in \Gamma(U, \mathcal{F})$ such that $U \cap Z$ is an irreducible component of $\text{Supp}(\mathcal{O}_U \cdot f)$.
- c) There exists an open neighbourhood U of x and a sub-Module of $\mathcal{F}|U$ isomorphic to $\mathcal{O}_Z|U$ ($\mathcal{O}_Z$ being identified with a quotient of $\mathcal{O}_X$).
- c') There exists an open neighbourhood U of x and a coherent sub-Module $\mathcal{G}$ of $\mathcal{F}|U$ such that $U \cap Z$ is an irreducible component of $\text{Supp}(\mathcal{G})$.

Proof. It is clear that c) implies b), for it suffices to take for f the element of $\Gamma(U, \mathcal{F})$ corresponding to the unit section of $\mathcal{O}_Z|U$. Since $U \cap Z$ is irreducible (0_I, 2.1.6), b) implies b') and b') implies c') since $\mathcal{O}_X$ is coherent (0_I, 5.3.4). To see that c') implies a), we can restrict to the case $U = X = \text{Spec}(A)$ is affine, A being therefore Noetherian, and where $\mathcal{F} = \tilde{M}$, where M is an A -module, and $\mathcal{G} = \tilde{N}$, where N is a submodule of M , of finite type. We know then that the minimal elements of $\text{Supp}(\mathcal{G})$ are the maximal points of $V(\text{Ann}(N))$ (0_I, 1.7.4), and these are also the minimal elements of $\text{Ass}(N)$ (Bourbaki, *Alg. comm.*, chap. IV, §1, no. 3, cor. 1 of prop. 7); since $\text{Ass}(N) \subset \text{Ass}(M) = \text{Ass}(\mathcal{F})$, we see that c') implies a). Finally a) implies c) by virtue of (3.1.2), taking again X affine, $\mathcal{F} = \tilde{M}$, and Z defined by the ideal fA (with the notations of (3.1.2)). $\square$

Corollary (3.1.4). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Then the maximal associated prime cycles of $\mathcal{F}$ are the irreducible components of $\text{Supp}(\mathcal{F})$, and the generic points of these components are the points $x \in X$ such that $\mathcal{F}_x$ is an $\mathcal{O}_x$ -Module of finite length and $\neq 0$.

Proof. Indeed, if x is the generic point of one of the irreducible components Z of $\text{Supp}(\mathcal{F})$, it follows from the equivalence of a) and c') in (3.1.3) that x belongs to $\text{Ass}(\mathcal{F})$, and Z is an associated prime cycle of $\mathcal{F}$, necessarily maximal by virtue of (3.1.1.1); the converse follows trivially from (3.1.1.1); finally, the last assertion being obviously local follows from Bourbaki, *Alg. comm.*, chap. IV, §2, no. 5, cor. 2 of prop. 7. $\square$

Corollary (3.1.5). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. For $\mathcal{F} = 0$, it is necessary and sufficient that $\text{Ass}(\mathcal{F}) = \emptyset$.*

Proof. The question being local, we are reduced to the case where X is affine, and the conclusion follows immediately from (3.1.2) and from Bourbaki, *Alg. comm.*, chap. IV, §1, no. 1, cor. 1 of prop. 2. $\square$

Proposition (3.1.6). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module; then $\text{Ass}(\mathcal{F})$ is locally finite (that is to say, every point of X admits a neighbourhood whose intersection with $\text{Ass}(\mathcal{F})$ is finite).*

Proof. It suffices to consider the case where X is affine, hence Noetherian, and then the proposition follows from (3.1.2) and from Bourbaki, *Alg. comm.*, chap. IV, §1, no. 4, cor. of th. 2. $\square$

Proposition (3.1.7). — *Let X be a prescheme.*

- (i) *Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ be an exact sequence of quasi-coherent $\mathcal{O}_X$ -Modules. Then $\text{Ass}(\mathcal{F}') \subset \text{Ass}(\mathcal{F}) \subset \text{Ass}(\mathcal{F}') \cup \text{Ass}(\mathcal{F}'')$.*
- (ii) *Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$ -Module, $(\mathcal{F}_\alpha)$ a family of quasi-coherent sub- $\mathcal{O}_X$ -Modules of $\mathcal{F}$ such that $\mathcal{F}$ is the union of the $\mathcal{F}_\alpha$. Then $\text{Ass}(\mathcal{F}) = \bigcup_\alpha \text{Ass}(\mathcal{F}_\alpha)$.*
- (iii) *For every family $(\mathcal{F}_\alpha)$ of quasi-coherent $\mathcal{O}_X$ -Modules, we have $\text{Ass}(\bigoplus \mathcal{F}_\alpha) = \bigcup \text{Ass}(\mathcal{F}_\alpha)$.*

Proof. We are immediately reduced to the corresponding propositions for modules (Bourbaki, *loc. cit.*, IV-2 | 38 §1, no. 1, formula (1), prop. 3 and cor. 1 of prop. 3). $\square$

Proposition (3.1.8). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module, U an open subset of X , $\mathcal{J}$ a coherent Ideal of $\mathcal{O}_X$ defining a closed subscheme of X having $X - U$ as underlying space. The following conditions are equivalent:*

- a) $\text{Ass}(\mathcal{F}) \subset U$.
- b) *For every affine open V of X , every section of $\mathcal{F}$ over V whose restriction to $V \cap U$ is zero, is equal to 0.*
- c) *The canonical homomorphism $\mathcal{F} \rightarrow \mathcal{H}om_{\mathcal{O}_X}(\mathcal{J}, \mathcal{F})$ is injective.*

Proof. The question being local, we can suppose $X = \text{Spec}(A)$ affine, A being a Noetherian ring, $\mathcal{F} = \tilde{M}$, where M is an A -module, and $\mathcal{J} = \tilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A . The homomorphism $M \rightarrow \text{Hom}_A(\mathfrak{J}, M)$ sends $m \in M$ to the homomorphism $x \mapsto xm$ from $\mathfrak{J}$ to M ; to say that it is not injective means that there exists an $m \neq 0$ in M such that $\mathfrak{J}m = 0$.

Let us first show that c) implies a): indeed, if $\text{Ass}(\mathcal{F}) \not\subset U$, there would exist a prime ideal $\mathfrak{p} \in \text{Ass}(M)$ containing $\mathfrak{J}$, hence an element $m \neq 0$ of M such that $\mathfrak{J}m = 0$. Secondly, b) implies c): indeed, if $\mathfrak{J}m = 0$ for some $m \neq 0$ in M , for every prime ideal $\mathfrak{q} \not\supset \mathfrak{J}$, there exists an $a \in \mathfrak{J}$ not in $\mathfrak{q}$, hence the relation $am = 0$ implies that the canonical image of m in $M_{\mathfrak{q}}$ is 0, in other words m would be a section $\neq 0$ of $\mathcal{F}$ over X whose restriction to U would be zero. Finally a) implies b). To see this, note that the canonical homomorphism $M \rightarrow \prod_{\mathfrak{p} \in \text{Ass}(M)} M_{\mathfrak{p}}$ is injective: indeed, if N is the kernel of this homomorphism, we have $\text{Ass}(N) \subset \text{Ass}(M)$; if there existed a $\mathfrak{p} \in \text{Ass}(N)$, there would be an element $n \in N$ whose $\mathfrak{p}$ would be the annihilator; but by definition of N , there is an element $s \notin \mathfrak{p}$ such that $sn = 0$, which is absurd; we conclude that $\text{Ass}(N) = \emptyset$, whence $N = 0$. Now condition a) implies that if $m \in M$ is a section of $\mathcal{F}$ whose restriction to U is zero, the canonical image of m in $M_{\mathfrak{p}}$ is zero for every $\mathfrak{p} \in \text{Ass}(M)$, hence $m = 0$. $\square$

Corollary (3.1.9). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module, f a section of $\mathcal{O}_X$ over X . For f to be $\mathcal{F}$ -regular (0IV, 15.2.2), it is necessary and sufficient that $\text{Ass}(\mathcal{F}) \subset X_f$.*

Proof. Indeed, it is immediate that to say that f is $\mathcal{F}$ -regular means that the canonical homomorphism $\mathcal{F} \rightarrow \mathcal{H}om_{\mathcal{O}_X}(f\mathcal{O}_X, \mathcal{F})$ is injective, and it suffices to apply (3.1.8) to the Ideal $\mathcal{J} = f\mathcal{O}_X$. $\square$

Proposition (3.1.10). — *Let X, Y be locally Noetherian preschemes, $f : X \rightarrow Y$ an integral morphism. IV-2 | 39 Then, for every quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, we have $f(\text{Ass}(\mathcal{F})) = \text{Ass}(f_*(\mathcal{F}))$.*

Proof. The question being local on Y and the morphism f being affine, we are immediately reduced to the case where Y is affine, in other words to the following: $\square$

Lemma (3.1.10.1). — *Let A, B be two Noetherian rings, $\rho : A \rightarrow B$ a ring homomorphism making B an integral A -algebra, M a B -module. Then the prime ideals $\mathfrak{p} \in \text{Ass}(M_{[\rho]})$ are the inverse images by ρ of the prime ideals $\mathfrak{q} \in \text{Ass}(M)$.*

Proof. Indeed, if $\mathfrak{q} \in \text{Ass}(M)$, $\mathfrak{q}$ is the annihilator in B of an element $x \in M$, hence $\rho^{-1}(\mathfrak{q})$ is the annihilator in A of x . Conversely, let $\mathfrak{p} \in \text{Ass}(M_{[\rho]})$, so that $\mathfrak{p}$ is the inverse image by ρ of the annihilator $\mathfrak{b}$ in B of an element $x \in M$; it follows from the first theorem of Cohen-Seidenberg that there exists a prime ideal $\mathfrak{q}$ of B containing $\mathfrak{b}$ and whose inverse image is $\mathfrak{p}$ (Bourbaki, *Alg. comm.*, chap. V, §2, no. 1, cor. 2 of th. 1); by considering one of the minimal prime ideals among those contained in $\mathfrak{q}$ and containing $\mathfrak{b}$, we can obviously suppose that $\mathfrak{q}$ itself is one of these minimal ideals. But since $B \cdot x \subset M$ is isomorphic to $B/\mathfrak{b}$, we know that $\mathfrak{q} \in \text{Ass}(B/\mathfrak{b}) \subset \text{Ass}(M)$ (Bourbaki, *Alg. comm.*, chap. IV, §1, no. 4, th. 2). $\square$

Corollary (3.1.11). — *Under the hypotheses of (3.1.10), for $\mathcal{F}$ to be without immersed associated prime cycle, it suffices that the same be true of $f_*(\mathcal{F})$.*

Proof. Suppose indeed that $f_*(\mathcal{F})$ has no immersed associated prime cycle. Note that if A is an integral algebra over a field k , all the prime ideals of A are maximal (Bourbaki, *Alg. comm.*, chap. V, §2, no. 1, prop. 1); it therefore follows from (I, 6.2.2) that the fibres of f are discrete spaces; if x, x' are two distinct points of $\text{Ass}(\mathcal{F})$, neither of them can be adherent to the other if $f(x) = f(x')$; and if $f(x) \neq f(x')$, (3.1.10) and the hypothesis imply that neither of the two points $f(x), f(x')$ can be adherent to the other, hence the same is true of x and x' . $\square$

Remark (3.1.12). — Under the hypotheses of (3.1.10), it can on the contrary happen that $\mathcal{F}$ be without immersed associated prime cycle, but not $f_*(\mathcal{F})$. Take for example $Y = \text{Spec}(k[T])$ where k is a field (“affine line”), and X the sum of $X_1 = Y$ and $X_2 = \text{Spec}(k)$, the morphism $X_2 \rightarrow Y$ corresponding to the canonical homomorphism $k[T] \rightarrow k[T]/\mathfrak{m}$, where $\mathfrak{m}$ is the maximal ideal (T) . It is clear that the morphism $f : X \rightarrow Y$ is finite; if one takes $\mathcal{F} = \mathcal{O}_X$, $\mathcal{F}$ is without immersed associated prime cycle, but $f_*(\mathcal{F}) = \tilde{M}$, where M is the $k[T]$ -module direct sum of $k[T]$ and k , so $\text{Ass}(M)$ is formed by the generic point (0) of Y and the point $\mathfrak{m}$.

Proposition (3.1.13). — *Let X be a locally Noetherian prescheme, U an open subset of X , $i : U \rightarrow X$ the canonical injection. For every quasi-coherent $\mathcal{O}_U$ -Module $\mathcal{F}$, we have $\text{Ass}(i_*(\mathcal{F})) = \text{Ass}(\mathcal{F})$.*

Proof. Recall that $i_*(\mathcal{F})$ is a quasi-coherent $\mathcal{O}_X$ -Module (1.2.2) and (1.7.4); since $i_*(\mathcal{F})|_U = \mathcal{F}$, we have $(\text{Ass}(i_*(\mathcal{F}))) \cap U = \text{Ass}(\mathcal{F})$, and it therefore remains to prove that $\text{Ass}(i_*(\mathcal{F})) \subset U$. But by (3.1.8), this relation means that for every affine open V of X , every section of $i_*(\mathcal{F})$ over V which is zero in $U \cap V$, is zero, a condition trivially satisfied since $\Gamma(V, i_*(\mathcal{F})) = \Gamma(U \cap V, \mathcal{F}) = \Gamma(U \cap V, i_*(\mathcal{F}))$. $\square$

3.2. Irredundant decompositions

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Proposition (3.2.1). — *Let X be a locally Noetherian prescheme, U an open dense subset of X . The following conditions are equivalent:*

- a) X is reduced.
- b) The sub-prescheme induced on U is reduced and X is without immersed prime cycle.
- c) X is without immersed prime cycle and for every generic point x of an irreducible component of X , we have $\text{long}(\mathcal{O}_x) = 1$.

The prime cycles associated to X are then identical to the irreducible components of X .

Proof. It is clear that if X is reduced it is the same for the sub-prescheme induced on U . Moreover, the existence of immersed prime cycles being a local question, we can restrict to the case $X = \text{Spec}(A)$ is affine, A being Noetherian. If A is reduced, we know that the minimal prime ideals of A form a reduced primary decomposition of (0) (Bourbaki, *Alg. comm.*, chap. IV, §2, no. 5, prop. 10) and are

the elements of $\text{Ass}(A)$, so there are no immersed associated prime ideals of A , which shows that $a)$ implies $b)$. It is immediate that $b)$ implies $c)$, since a generic point x of an irreducible component of X belongs to U , hence $\mathcal{O}_x$ is a field. Finally, $c)$ implies $a)$: it suffices to remark that if $\mathcal{N}$ is the Nilradical of $\mathcal{O}_X$, which is a coherent Ideal, $\text{Supp}(\mathcal{N})$ cannot contain any of the generic points of the irreducible components of X by hypothesis; if $\text{Supp}(\mathcal{N})$ were not empty and x was one of the maximal points of this closed set, the criterion (3.1.3), $c')$ would show that $x \in \text{Ass}(\mathcal{O}_X)$, and $\{x\}$ would be an immersed prime cycle of X , contradicting the hypothesis; hence $\mathcal{N} = 0$. $\square$

We say that a locally Noetherian prescheme X is *irredundant* if $\mathcal{O}_X$ is irredundant (which implies that X is irreducible); for X to be *integral*, it is necessary and sufficient that the $\mathcal{O}_X$ -Module $\mathcal{O}_X$ be integral (I, 2.1.8) and (3.2.1). If $\mathcal{O}_X$ is integral, $\mathcal{O}_X$ is integral at every point x , that is to say $\mathcal{O}_x$ is an integral ring. We say that a closed subprescheme Y of X is *primary* in X if the Ideal $\mathcal{J}$ of $\mathcal{O}_X$ defining Y is primary in $\mathcal{O}_X$.

Definition (3.2.2). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. We say that $\mathcal{F}$ is *reduced* if it satisfies the two following conditions: 1° $\mathcal{F}$ is without immersed associated prime cycle; 2° for every maximal point x of $\text{Supp}(\mathcal{F})$, we have $\text{long}(\mathcal{F}_x) = 1$.

Condition 1° means that the associated prime cycles of $\mathcal{F}$ are the irreducible components of $\text{Supp}(\mathcal{F})$ (3.1.4), and condition 2° means that for every generic point x of such a component we have $\text{long}(\mathcal{F}_x) = 1$.

For an affine scheme X , this definition gives in particular the notion of *reduced module* over a Noetherian ring A ; an A -module M of finite type is said *reduced* if it has no immersed associated prime ideals and if, for every $\mathfrak{p} \in \text{Ass}(M)$, $\text{long}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = 1$. Returning to the case of a locally Noetherian prescheme X and a coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, we say that $\mathcal{F}$ is *reduced at a point* $x \in X$ if $\mathcal{F}_x$ is a reduced $\mathcal{O}_x$ -module; that means that, on the local scheme $\text{Spec}(\mathcal{O}_x)$, $\mathcal{F}_x$ is reduced; it amounts therefore to saying that x does not belong to any immersed associated prime cycle of $\mathcal{F}$ and that $\text{long}(\mathcal{F}_z) = 1$ for all maximal points z of $\text{Supp}(\mathcal{F})$ such that $x \in \overline{\{z\}}$. It is clear that if $\mathcal{F}$ is a coherent $\mathcal{O}_X$ -Module that is reduced, it is reduced at every point of X ; conversely, if $\mathcal{F}$ is *reduced at a point* x , there exists an open neighbourhood U of x such that $\mathcal{F}|_U$ is a reduced $\mathcal{O}_U$ -Module: it suffices in fact to take U not meeting any immersed associated prime cycle of $\mathcal{F}$ (such a neighbourhood exists since these cycles form a locally finite set of closed subsets of X). To say that $\mathcal{O}_X$ is reduced at a point x is equivalent to saying that X is *reduced* at the point x .

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Proposition (3.2.3). — Let X be a locally Noetherian prescheme, U an open subset of X , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module such that $U \cap \text{Supp}(\mathcal{F})$ is dense in $\text{Supp}(\mathcal{F})$. The following conditions are equivalent:

- a) $\mathcal{F}$ is reduced.
- b) $\mathcal{F}|_U$ is reduced, and $\mathcal{F}$ is without immersed associated prime cycle.
- c) There exists a reduced closed subprescheme X' of X and a coherent $\mathcal{O}_{X'}$ -Module $\mathcal{F}'$ without torsion and of rank 1 on every irreducible component of X' , such that if $j : X' \rightarrow X$ is the canonical injection, we have $j_*(\mathcal{F}') = \mathcal{F}$.

Moreover, when this holds, the subprescheme X' is defined by the Ideal $\mathcal{J}$ of $\mathcal{O}_X$ which is the annihilator of $\mathcal{F}$.

Proof. To see that $c)$ implies $a)$, we can, by virtue of (3.1.3), limit to the case where $X' = X$ is integral, with generic point x and one can further suppose $X = \text{Spec}(A)$ affine and $\mathcal{F} = \tilde{M}$, where A is integral and M is an A -module without torsion of rank 1; the annihilator of every element of M being then reduced to 0, we have $\text{Ass}(\mathcal{F}) = \{x\}$ and $\mathcal{F}_x$ is isomorphic to the field of fractions of A , so the conditions of the definition (3.2.2) are satisfied. It is clear that if $\mathcal{F}$ has no immersed associated prime cycles and if $U \cap \text{Supp}(\mathcal{F})$ is dense in $\text{Supp}(\mathcal{F})$, then $\text{Ass}(\mathcal{F}) = \text{Ass}(\mathcal{F}|_U)$, hence $a)$ and $b)$ are equivalent. If $a)$ is satisfied, take for X' the reduced closed subprescheme of X whose $\text{Supp}(\mathcal{F})$ is the underlying space (I, 5.2.1), and let $\mathcal{F}' = j^*(\mathcal{F})$; a point x of $\text{Ass}(\mathcal{F})$ is necessarily a maximal point of X' , and as $\text{long}(\mathcal{F}_x) = 1$, $\mathcal{F}'_x$ is isomorphic to $\mathcal{F}_x$, hence to the field $k(x) = \mathcal{O}_{X',x}$, which proves that $\mathcal{F}'$ is without torsion and of rank 1 (I, 7.4.6) and (I, 7.4.2). Finally, the last assertion is trivial, since for every $y \in X'$, the annihilator of the $\mathcal{O}_{X',y}$ -Module $\mathcal{F}'_y$ is zero. $\square$

Definition (3.2.4). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. We say that $\mathcal{F}$ is *irredundant* if $\text{Ass}(\mathcal{F})$ is reduced to a single element; if $\mathcal{F}$ is of finite type, we say that $\mathcal{F}$ is *integral* if moreover $\mathcal{F}$ is reduced (in other words if $\text{long}(\mathcal{F}_x) = 1$). We say that a quasi-coherent sub- $\mathcal{O}_X$ -Module $\mathcal{G}$ of a quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ is *primary* in $\mathcal{F}$ if $\mathcal{F}/\mathcal{G}$ is irredundant.

For an affine scheme X , this definition gives in particular the notion of *integral module* over a Noetherian ring A ; an A -module M of finite type is said *integral* if $\text{Ass}(M)$ is reduced to a single prime ideal $\mathfrak{p}$, and if moreover $\text{long}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = 1$. Returning to the case of a locally Noetherian prescheme X and a coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, we say that $\mathcal{F}$ is *integral at a point* $x \in X$ if $\mathcal{F}_x$ is an *integral $\mathcal{O}_x$ -module*: that means that x belongs to only a *single* associated prime cycle (necessarily non immersed) of $\mathcal{F}$ and that $\text{long}(\mathcal{F}_z) = 1$ at its generic point z . It is clear that if $\mathcal{F}$ is a coherent $\mathcal{O}_X$ -Module that is integral, it is integral at every point of X ; conversely, if $\mathcal{F}$ is *integral at a point* x , there exists an open neighbourhood U of x such that $\mathcal{F}|_U$ is an integral $\mathcal{O}_U$ -Module: it suffices in fact to take U meeting no immersed associated prime cycle of $\mathcal{F}$ (such a neighbourhood exists since these cycles form a locally finite set of closed subsets of X).

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Definition (3.2.5). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. A *decomposition* of $\mathcal{F}$ is called *irredundant* if it is a family $(\mathcal{F}_{\alpha})_{\alpha \in I}$ of quotient $\mathcal{O}_X$ -Modules of $\mathcal{F}$ such that the $\mathcal{F}_{\alpha}$ are irredundant, the family $(\text{Supp}(\mathcal{F}_{\alpha}))$ is locally finite, and the canonical homomorphism $\mathcal{F} \rightarrow \bigoplus_{\alpha \in I} \mathcal{F}_{\alpha}$ is injective. Such a decomposition is said to be *reduced* if the sets $\text{Ass}(\mathcal{F}_{\alpha})$ are pairwise distinct, and if there exists no subset $J \neq I$ such that the subfamily $(\mathcal{F}_{\alpha})_{\alpha \in J}$ is an irredundant decomposition of $\mathcal{F}$.

If $(\mathcal{F}_{\alpha})_{\alpha \in I}$ is an irredundant decomposition (resp. reduced irredundant decomposition) of $\mathcal{F}$, and if one sets $\mathcal{G}_{\alpha} = \mathcal{F}/\mathcal{F}_{\alpha}$, we say also that the family $(\mathcal{G}_{\alpha})_{\alpha \in I}$ of sub- $\mathcal{O}_X$ -Modules of $\mathcal{F}$ is a *primary decomposition* (resp. a *reduced primary decomposition*) of 0 in $\mathcal{F}$; one observes that the injectivity of the canonical homomorphism $\mathcal{F} \rightarrow \bigoplus_{\alpha \in I} \mathcal{F}_{\alpha}$ is equivalent to the condition $\bigcap_{\alpha \in I} \mathcal{G}_{\alpha} = 0$.

Proposition (3.2.6). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Then there exists a reduced irredundant decomposition $(\mathcal{F}^{(x)})_{x \in \text{Ass}(\mathcal{F})}$, formed of coherent $\mathcal{O}_X$ -Modules such that for every $x \in \text{Ass}(\mathcal{F})$, we have $\text{Ass}(\mathcal{F}^{(x)}) = \{x\}$. For every $x \in \text{Ass}(\mathcal{F})$ such that $\{x\}$ is not immersed, $\mathcal{F}^{(x)}$ is uniquely determined as the image of the canonical homomorphism $\mathcal{F} \rightarrow i_*(i^*(\mathcal{F}))$, where i is the canonical morphism $\text{Spec}(k(x)) \rightarrow X$.

Proof. For every $x \in \text{Ass}(\mathcal{F})$, let U be an affine open neighbourhood of x , with ring A , and let $\mathcal{F}|_U = \tilde{M}$, where M is an A -module of finite type. We know (Bourbaki, *Alg. comm.*, chap. IV, §2, no. 3, prop. 4) that there exists a submodule N of M such that, if we set $P = M/N$, we have $\text{Ass}(P) = \{x\}$ and $\text{Ass}(N) = \text{Ass}(M) - \{x\}$. Let $\mathcal{G} = \tilde{P}$, which is a quasi-coherent $\mathcal{O}_U$ -Module; let $u : j^*(\mathcal{F}) \rightarrow \mathcal{G}$ be the surjective homomorphism corresponding to $M \rightarrow P$; from this we derive by composition a homomorphism $j_*(u) : j_*(j^*(\mathcal{F})) \rightarrow j_*(\mathcal{G})$, from which by composition a homomorphism

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$$v : \mathcal{F} \xrightarrow{\rho_{\mathcal{F}}} j_*(j^*(\mathcal{F})) \xrightarrow{j_*(u)} j_*(\mathcal{G})$$

whose u is the restriction to U ; we denote by $\mathcal{F}^{(x)}$ the image of $\mathcal{F}$ by this homomorphism, which is a coherent $\mathcal{O}_X$ -Module (I, 6.1.1). We have $\text{Ass}(j_*(\mathcal{G})) = \{x\}$ by virtue of (3.1.13), and *a fortiori* (3.1.7) $\text{Ass}(\mathcal{F}^{(x)}) = \{x\}$, since $\mathcal{F}^{(x)} \neq 0$. Moreover, if $\mathcal{H}$ is the kernel of v , $\text{Ass}(\mathcal{H})$ is contained in the intersection of the $\text{Ass}(\mathcal{N}^{(x)})$, which is empty; by (3.1.5), $\mathcal{H} = 0$. In view of the fact that $\text{Ass}(\mathcal{F})$ is locally finite (3.1.6), it is clear that $(\mathcal{F}^{(x)})_{x \in \text{Ass}(\mathcal{F})}$ is a reduced irredundant decomposition of $\mathcal{F}$ satisfying the conditions of the statement. The characterisation of $\mathcal{F}^{(x)}$ when $\{x\}$ is not immersed follows from Bourbaki, *Alg. comm.*, chap. IV, §2, no. 3, prop. 5, the question being local, and taking into account (I, 1.6.7). $\square$

Corollary (3.2.7). — Under the hypotheses of (3.2.6), if $\mathcal{F}$ has no immersed associated prime cycle, there exists only one reduced irredundant decomposition of $\mathcal{F}$.

Corollary (3.2.8). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. There exists a finite filtration $(\mathcal{F}_i)_{0 \leq i \leq n}$ of $\mathcal{F}$ such that $\mathcal{F}_0 = \mathcal{F}$, $\mathcal{F}_n = 0$, formed of coherent $\mathcal{O}_X$ -Modules and such that the quotients $\mathcal{F}_i/\mathcal{F}_{i+1}$ are zero or irredundant, and $\text{Ass}(\mathcal{F}_i/\mathcal{F}_{i+1}) \subset \text{Ass}(\mathcal{F})$.

Proof. Indeed, $\mathcal{F}$ is isomorphic to a sub- $\mathcal{O}_X$ -Module of a finite direct sum $\bigoplus_{j=1}^n \mathcal{G}_j$, where the $\mathcal{G}_j$ are irredundant and coherent (3.2.6); since every quasi-coherent sub- $\mathcal{O}_X$ -Module of $\mathcal{G}_j$ is either zero or irredundant (3.1.7), the $\mathcal{F}_i = \mathcal{F} \cap (\bigoplus_{j=1}^{n-i} \mathcal{G}_j)$ answer the question, $\mathcal{F}_i / \mathcal{F}_{i+1}$ being isomorphic to a sub- $\mathcal{O}_X$ -Module coherent of $\mathcal{G}_{n-i}$. $\square$

3.3. Relations with flatness

Proposition (3.3.1). — *Let $f : X \rightarrow Y$ be a morphism, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module which is f -flat, $\mathcal{E}$ a quasi-coherent $\mathcal{O}_Y$ -Module. If, for every $y \in Y$, we set $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$, then*

$$(3.3.1.1) \quad \text{Ass}(\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F}) \supset \bigcup_{y \in \text{Ass}(\mathcal{E})} \text{Ass}(\mathcal{F}_y)$$

and the two members are equal if Y is locally Noetherian.

Proof. (It is understood that $\mathcal{F}_y$ is a sheaf on the fibre $f^{-1}(y)$, and we identify this fibre with a subspace of X (I, 3.6.1).) The question being local on X and on Y , we are reduced to the case where X and Y are affine, and the proposition is then proved in Bourbaki, *Alg. comm.*, chap. IV, §2, no. 6, th. 2. $\square$

Corollary (3.3.2). — *Let Y be a locally Noetherian prescheme without immersed associated prime cycles, $f : X \rightarrow Y$ a morphism, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module which is f -flat. Then, for every $x \in \text{Ass}(\mathcal{F})$, $f(x)$ is a maximal point of Y .*

Proof. It suffices to apply (3.3.1) with $\mathcal{E} = \mathcal{O}_Y$, since $\text{Ass}(\mathcal{O}_Y)$ is by hypothesis the set of maximal points of Y . $\square$

Corollary (3.3.3). — *Under the hypotheses of (3.3.1), suppose moreover that X and Y are locally Noetherian, $\mathcal{E}$ and $\mathcal{F}$ coherent. Then the following conditions are equivalent:*

- a) $\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F}$ is without immersed associated prime cycle.
- b) For every point $y \in \text{Ass}(\mathcal{E}) \cap f(\text{Supp}(\mathcal{F}))$, $\overline{\{y\}}$ is a non-immersed associated prime cycle of $\mathcal{E}$ and $\mathcal{F}_y$ is without immersed associated prime cycle.

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Proof. Suppose a) verified. The hypotheses imply that $\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F}$ is a coherent $\mathcal{O}_X$ -Module (0I, 5.3.11 and (I, 5.3.5)); its associated prime cycles are therefore the irreducible components of $\text{Supp}(\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F})$ (3.1.4) and for every maximal point x of $\text{Supp}(\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F})$, $f(x) = y$ is a maximal point of $\text{Supp}(\mathcal{E})$ and x is a maximal point of $\text{Supp}(\mathcal{F}_y)$ (2.5.5). Since, by virtue of (3.3.1) and of the hypothesis, $y \in f(\text{Supp}(\mathcal{F}))$, we see that $\mathcal{F}_y \neq 0$ (I, 9.1.13), every point of $\text{Ass}(\mathcal{E}) \cap f(\text{Supp}(\mathcal{F}))$ is the image by f of a maximal point of $\text{Supp}(\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F})$, whence condition b) is verified.

Conversely, suppose b) verified, and show that if z, z' are two distinct points of $\text{Ass}(\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F})$, neither of them can be adherent to the other. Firstly, if $f(z) = f(z') = y$, we have $y \in \text{Ass}(\mathcal{E})$ and z and z' belong to $\text{Ass}(\mathcal{F}_y)$ by (3.3.1), hence by hypothesis none of the two points z, z' is adherent to the other in $f^{-1}(y)$, so none of them can be adherent to the other in X . If $y = f(z)$ and $y' = f(z')$ are distinct, they belong to $\text{Ass}(\mathcal{E}) \cap f(\text{Supp}(\mathcal{F}))$, hence neither of them can be adherent to the other in Y ; it follows from the continuity of f that neither of the two points z, z' can be adherent to the other in X . $\square$

Proposition (3.3.4). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism, $\mathcal{E}$ a coherent $\mathcal{O}_Y$ -Module, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module which is f -flat. Then the following conditions are equivalent:*

- a) $\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F}$ is reduced (3.2.2).
- b) For every point $y \in \text{Ass}(\mathcal{E}) \cap f(\text{Supp}(\mathcal{F}))$, $\overline{\{y\}}$ is a non-immersed associated prime cycle of $\mathcal{E}$, $\text{long}(\mathcal{E}_y) = 1$ and $\mathcal{F}_y$ is reduced.

Proof. Suppose a) verified. One already knows from (3.3.3) that for every $y \in \text{Ass}(\mathcal{E}) \cap f(\text{Supp}(\mathcal{F}))$, $\overline{\{y\}}$ is a non-immersed associated prime cycle of $\mathcal{E}$ and $\mathcal{F}_y$ is without immersed associated prime cycle. Moreover (2.5.5), for every $x \in \text{Ass}(\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F}) \cap f^{-1}(y)$, we have $1 = \text{long}((\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F})_x) = \text{long}(\mathcal{E}_y) \cdot \text{long}((\mathcal{F}_y)_x)$, hence $\text{long}(\mathcal{E}_y) = 1$, which proves b).

Conversely, suppose $b)$ verified; one already knows that every $x \in \text{Ass}(\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F})$ is a maximal point of $\text{Supp}(\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F})$, that $y = f(x)$ is a maximal point of $\text{Supp}(\mathcal{E})$ and x a maximal point of $\text{Supp}(\mathcal{F}_y)$ (3.3.1) and (3.3.3); moreover it follows from the hypothesis and (2.5.5) that $\text{long}((\mathcal{E} \otimes_{\mathcal{O}_Y} \mathcal{F})_x) = 1$, which proves $a)$. $\square$

Corollary (3.3.5). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a flat morphism; if Y is reduced at the points of $f(X)$ and if $f^{-1}(y)$ is a reduced $k(y)$ -prescheme for every $y \in f(X)$, then X is reduced.*

Proof. Since the Nilradical $\mathcal{N}_Y$ is coherent, the set of points where Y is reduced is open (0_I, 5.2.2), and one can restrict to the case where Y is reduced. It suffices then to apply (3.3.4) with $\mathcal{E} = \mathcal{O}_Y$ and $\mathcal{F} = \mathcal{O}_X$. $\square$

Proposition (3.3.6). — *Let $f : X \rightarrow S, g : Y \rightarrow S$ be two morphisms, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module, $\mathcal{G}$ a quasi-coherent $\mathcal{O}_Y$ -Module. Suppose that: 1° $\mathcal{G}$ is g -flat; 2° X is locally Noetherian, and for every $s \in S$, $g^{-1}(s)$ is locally Noetherian (which will hold if Y is also locally Noetherian). Let $Z = X \times_S Y$; for every couple (x, y) such that $x \in X, y \in Y$ and $f(x) = g(y) = s$, let $T_{x,y}$ be the prescheme $\text{Spec}(k(x) \otimes_{k(s)} k(y))$, and let $I_{x,y}$ be the image of $\text{Ass}(\mathcal{O}_{T_{x,y}})$ by the canonical monomorphism $T_{x,y} \rightarrow Z$ (I, 3.4.9). Then*

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$$(3.3.6.1) \quad \text{Ass}(\mathcal{F} \otimes_S \mathcal{G}) = \bigcup_{x \in \text{Ass}(\mathcal{F})} \left(\bigcup_{y \in \text{Ass}(\mathcal{G}_{f(x)})} I_{x,y} \right)$$

where for every $s \in S, \mathcal{G}_s = \mathcal{G} \otimes_{\mathcal{O}_S} k(s)$.

Proof. Let $p : Z \rightarrow X, q : Z \rightarrow Y$ be the canonical projections, so that we have the commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{p} & Z \\ f \uparrow & & \uparrow q \\ S & \xleftarrow{g} & Y \end{array}$$

Setting $\mathcal{G}' = q^*(\mathcal{G})$, so that $\mathcal{F} \otimes_S \mathcal{G} = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}'$; since $\mathcal{G}'$ is p -flat (2.1.4), it follows from (3.3.1) that

$$(3.3.6.2) \quad \text{Ass}(\mathcal{F} \otimes_S \mathcal{G}) = \bigcup_{x \in \text{Ass}(\mathcal{F})} \text{Ass}(\mathcal{G}'_x)$$

with $\mathcal{G}'_x = \mathcal{G}' \otimes_{\mathcal{O}_X} k(x)$. If $s = f(x)$, we have $\mathcal{G}'_x = \mathcal{G}_s \otimes_{k(s)} k(x)$, and $p^{-1}(x) = g^{-1}(s) \otimes_{k(s)} k(x)$; moreover, since the field $k(x)$ is a flat $k(s)$ -module, the morphism $p^{-1}(x) \rightarrow g^{-1}(s)$ is flat (2.1.4); applying (3.3.1) to this morphism, it follows that

$$(3.3.6.3) \quad \text{Ass}(\mathcal{G}'_x) = \bigcup_{y \in \text{Ass}(\mathcal{G}_s)} \text{Ass}(\mathcal{O}_{T_{x,y}})$$

from which the proposition follows. $\square$

We note that if, in the statement, we drop hypothesis 2°, one can still deduce, by virtue of (3.3.1), the relation

$$(3.3.6.4) \quad \text{Ass}(\mathcal{F} \otimes_S \mathcal{G}) \supset \bigcup_{x \in \text{Ass}(\mathcal{F})} \left(\bigcup_{y \in \text{Ass}(\mathcal{G}_{f(x)})} I_{x,y} \right).$$

Corollary (3.3.7). — *Under the hypotheses of (3.3.6), suppose moreover that S is locally Noetherian and that $f(\text{Ass}(\mathcal{F})) \subset \text{Ass}(\mathcal{O}_S)$. Then*

$$(3.3.7.1) \quad \text{Ass}(\mathcal{F} \otimes_S \mathcal{G}) = \bigcup_{(x,y) \in C} I_{x,y}$$

where C is the set of couples (x, y) such that $x \in \text{Ass}(\mathcal{F}), y \in \text{Ass}(\mathcal{G})$ and $f(x) = g(y)$.

Proof. Since $\mathcal{G}$ is g -flat, it follows from (3.3.1) that the relation “ $s \in \text{Ass}(\mathcal{G}_s)$ ” is equivalent to $y \in \text{Ass}(\mathcal{G})$: the conclusion follows from (3.3.6.1). $\square$

Remarks (3.3.8). — We will see later (4.2.2) that under the hypotheses of (3.3.6), $T_{x,y}$ is a prescheme without immersed associated prime cycle; it will follow that if $\mathcal{F}$ and the $\mathcal{G}_s$ are without immersed associated prime cycle, the same is true of $\mathcal{F} \otimes_S \mathcal{G}$.

Corollary (3.3.9). — Under the conditions of (3.3.7), we have

$$(3.3.9.1) \quad q(\text{Ass}(\mathcal{F} \otimes_S \mathcal{G})) \subset \text{Ass}(\mathcal{G})$$

(where $q : X \times_S Y \rightarrow Y$ is the canonical projection).

Proof. Indeed, if $(x, y) \in Z$, we have $q(I_{x,y}) = \{y\} \subset \text{Ass}(\mathcal{G})$. □

3.4. Properties of the sheaves $\mathcal{F}/t\mathcal{F}$

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Proposition (3.4.1). — Let X be a locally Noetherian prescheme, t a section of $\mathcal{O}_X$ over X , Y the closed subprescheme of X defined by the Ideal $t\mathcal{O}_X$ of $\mathcal{O}_X$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, S the closed subprescheme having $\text{Supp}(\mathcal{F})$ as underlying space, (S_i) the family of the reduced closed subpreschemes of X having as underlying spaces the irreducible components of S ; we denote by s_i the generic point of S_i . Finally, let Z be an irreducible component of $\text{Supp}(\mathcal{F}/t\mathcal{F}) = S \cap Y$, z its generic point.

(i) For every i such that $Z \subset S_i$, Z is an irreducible component of $S_i \cap Y$.

(ii) If Z is not equal to any of the S_i , we have

$$(3.4.1.1) \quad \text{long}((\mathcal{F}/t\mathcal{F})_z) \geq \sum_i \text{long}(\mathcal{F}_{s_i})$$

where the sum of the second member is extended to all i such that $Z \subset S_i$.

(iii) Suppose that Z is not equal to any S_i . For the two members of (3.4.1.1) to be equal, it is necessary and sufficient that the two following conditions be satisfied:

α) t_z is $\mathcal{F}_z$ -regular (**0_{IV}**, 15.1.4).

α) For every i such that $M_{p_i} \neq 0$, the image of the germ t_z in $\mathcal{O}_{S_i,z}$ generates the maximal ideal of this ring (which implies that $\mathcal{O}_{S_i,z}$ is a discrete valuation ring whose t_z is a uniformiser).

Proof. Suppose that Z is not equal to any S_i . As in the demonstration of (3.4.1), we can reduce to the case where $X = \text{Spec}(\mathcal{O}_z)$; the proposition is then (taking into account (3.1.2)) a consequence of (**0_{IV}**, 16.4.6.3).

(i) If $j : Y \rightarrow X$ is the canonical injection, we have $\mathcal{F}/t\mathcal{F} = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_Y = j^*(\mathcal{F})$, hence $\text{Supp}(\mathcal{F}/t\mathcal{F}) = j^{-1}(S) = S \cap Y$ (**I**, 9.1.13), whence the assertion.

(ii) and (iii): Suppose that Z is not equal to any of the S_i . We can, to prove (ii) and (iii), replace X by $\text{Spec}(\mathcal{O}_z)$; and if $M = \mathcal{F}_z$, we can suppose $\mathcal{F} = \tilde{M}$, denoting by p_i the minimal ideals of $\mathcal{O}_z$. Moreover, since M is an $\mathcal{O}_z$ -module of finite type, on a $S = S'_{\text{red}}$, with $S' = \text{Spec}(\mathcal{O}_z/\mathfrak{a})$, where $\mathfrak{a}$ is the annihilator of M in $\mathcal{O}_z$ (**0_I**, 1.7.4), and the two members of (3.4.1.1) have the same values whether one considers M as an $\mathcal{O}_z$ -module or as an $(\mathcal{O}_z/\mathfrak{a})$ -module; one can therefore finally replace X by $S' = \text{Spec}(A)$, where A is a local Noetherian ring, M being a faithful A -module; since Z is closed in S , the hypothesis that $Z \neq S_i$ for all i means that $s_i \notin Z$, hence $\dim(A) > 0$; finally, to say that z is the generic point of Z , an irreducible component of $S_i \cap V(t)$, means that A/tA is an Artinian local ring. We are therefore reduced to proving the following statement: □

Lemma (3.4.1.2). — Let A be a local Noetherian ring of dimension > 0 , p_i the minimal prime ideals of A , $\mathfrak{m}$ its maximal ideal, t an element of $\mathfrak{m}$ such that A/tA is Artinian. Then, for every A -module M of finite type, we have

$$(3.4.1.3) \quad \text{long}(M/tM) \geq \sum_i \text{long}(M_{p_i});$$

moreover, for the two members of (3.4.1.3) to be equal, it is necessary and sufficient that the following conditions be satisfied:

α) t is M -regular;

α) for every i such that $M_{p_i} \neq 0$, the image of t in A/p_i generates the maximal ideal of this ring (which implies that A/p_i is a discrete valuation ring).

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Proof. Since A is not of dimension 0 and A/tA is Artinian, we necessarily have $\dim(A) = 1$ (0_{IV}, 16.3.4) and $t \notin \mathfrak{p}_i$ for every i : the principal ideal (t) is therefore an ideal of definition, and contains by consequence a power of its maximal ideal $\mathfrak{m}$. Let N be the submodule of elements of M annihilated by a power of t (or by a power of $\mathfrak{m}$, which amounts to the same since one has just seen this); if one sets $P = M/N$, t is P -regular, since for $x \in M$ the relation $tx \in N$ implies $t^k(tx) = 0$ for an integer k , hence $x \in N$. This being so, we have the

Lemma (3.4.1.4). — *Let A be a ring,*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

an exact sequence of A -modules. If $t \in A$ is M'' -regular, the sequence

$$0 \longrightarrow M'/tM' \longrightarrow M/tM \longrightarrow M''/tM'' \longrightarrow 0$$

is exact.

Proof. Since $M/tM = M \otimes_A (A/tA)$, it suffices to prove the exactness at M'/tM' ; now, if the image x' of an element x' of M' is such that $x' = ty$ with $y \in M$, we deduce, for the images x'', y'' in M'' , $x'' = ty''$; but as $x'' = 0$, the hypothesis implies $y'' = 0$, hence y is the image of an element $y' \in M'$, and the relation $x' = ty'$ proves that $M' \rightarrow M$ is injective. $\square$

This lemma being established, we derive the relation

$$(3.4.1.5) \quad \text{long}(M/tM) = \text{long}(N/tN) + \text{long}(P/tP).$$

On the other hand, for every i , we have $N_{\mathfrak{p}_i} = 0$ since $t \notin \mathfrak{p}_i$; hence $M_{\mathfrak{p}_i} = P_{\mathfrak{p}_i}$; to prove (3.4.1.3), it suffices to do so replacing M by P ; moreover, if the two members of (3.4.1.3) are equal, it follows from the same inequality for P and from (3.4.1.5) that necessarily $\text{long}(N/tN) = 0$, hence $N/tN = 0$ and finally $N = 0$, by the lemma of Nakayama, N being of finite type; now, $N = 0$ means that t is M -regular. $\text{Ass}(M) \subset \bigcup \{\mathfrak{p}_i\}$.

We proceed then by induction on $n = \sum \text{long}(M_{\mathfrak{p}_i})$. If $n = 0$, we have $M_{\mathfrak{p}_i} = 0$ for every i , hence $M = 0$ since none of the $\mathfrak{p}_i$ belongs to $\text{Ass}(M)$; the two members of (3.4.1.3) are then zero, and the assertion β) of (3.4.1.2) is trivially satisfied. If $n > 0$, the reasoning from the beginning of the proof of (3.4.1) allows us moreover to suppose that the A -module M is *faithful*: this implies $M_{\mathfrak{p}_i} \neq 0$ for every i (Bourbaki, *Alg. comm.*, chap. II, §2, no. 2, cor. 2 of prop. 4), and therefore $\text{Ass}(M) = \bigcup \{\mathfrak{p}_i\}$.

Suppose first $n = 1$; there is then only a single minimal prime ideal $\mathfrak{p}$ of A , and to say that $M_{\mathfrak{p}}$ is of length 1 means that $M_{\mathfrak{p}}$ is isomorphic to the residue field $k = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$, hence $\mathfrak{p}$ is the annihilator of M as an $A_{\mathfrak{p}}$ -module. By consequence $M_{\mathfrak{p}}$ is annihilated by $\mathfrak{p}A_{\mathfrak{p}}$, hence $\mathfrak{p}$ is the annihilator of M since M is supposed faithful. This being so, the hypothesis $M \neq 0$ implies $M/tM \neq 0$ by the lemma of Nakayama, hence $\text{long}(M/tM) \geq 1$, which proves (3.4.1.3) in this case. Moreover, if $\text{long}(M/tM) = 1$, M is necessarily *monogenic* (Bourbaki, *Alg. comm.*, chap. II, §3, no. 2, cor. 2 of prop. 4) hence isomorphic to a quotient $A/\mathfrak{b}$; since it is faithful, we necessarily have $\mathfrak{b} = 0$ and M is isomorphic to A ; since $\text{long}(A/tA) = 1$, tA is necessarily equal to the maximal ideal $\mathfrak{m}$, and since A is a local integral Noetherian ring, this proves that A is a discrete valuation ring (Bourbaki, *Alg. comm.*, chap. VI, §3, no. 6, prop. 9), whose t is the uniformiser. Conversely, if A is a discrete valuation ring, t its uniformiser, M is without torsion, hence isomorphic to a submodule of A (M being of finite type), and therefore isomorphic to A itself, whence $\text{long}(M/tA) = \text{long}(A/tA) = 1$.

Suppose now $n \geq 2$; there exists then an exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

with $M' \neq 0$, $M'' \neq 0$ and $\text{Ass}(M) = \text{Ass}(M') \cup \text{Ass}(M'')$; indeed, if $\text{Ass}(M)$ is not reduced to a single element, this follows from Bourbaki, *Alg. comm.*, chap. IV, §1, no. 1, prop. 4; if on the contrary $\text{Ass}(M)$ is reduced to a single prime ideal, the latter is necessarily the unique minimal prime ideal $\mathfrak{p}$ of A ; the hypothesis implies $\text{long}(M_{\mathfrak{p}}) \geq 2$ and it suffices to take for M' the inverse image of a submodule of $M_{\mathfrak{p}}$ distinct from 0 and from $M_{\mathfrak{p}}$. Since t is M -regular, t does not belong to any of the prime ideals of $\text{Ass}(M)$ (Bourbaki, *Alg. comm.*, chap. IV, §1, no. 1, cor. 2 of prop. 2), hence, for the same reason, t is M' -regular and M'' -regular. This last property implies by (3.4.1.4) that the sequence

$$0 \longrightarrow M'/tM' \longrightarrow M/tM \longrightarrow M''/tM'' \longrightarrow 0$$

is exact; as moreover it is the same for the sequence

$$0 \longrightarrow M'_{\mathfrak{p}_i} \longrightarrow M_{\mathfrak{p}_i} \longrightarrow M''_{\mathfrak{p}_i} \longrightarrow 0$$

for every i , we have

$$\begin{aligned} \text{long}(M/tM) &= \text{long}(M'/tM') + \text{long}(M''/tM'') \\ \text{long}(M_{\mathfrak{p}_i}) &= \text{long}(M'_{\mathfrak{p}_i}) + \text{long}(M''_{\mathfrak{p}_i}) \end{aligned}$$

and the induction hypothesis implies the inequality (3.4.1.3). Moreover, the two members can only be equal if the analogous inequalities for M' and M'' are also equalities. By virtue of the induction hypothesis, this is equivalent to the property β) for the $\mathfrak{p}_i$ such that $M'_{\mathfrak{p}_i} \neq 0$ or $M''_{\mathfrak{p}_i} \neq 0$; but these ideals are precisely those for which $M_{\mathfrak{p}_i} \neq 0$. $\square$

Corollary (3.4.2). — *Under the general hypotheses of (3.4.1), suppose that Z is not equal to any of the S_i and that $\text{long}((\mathcal{F}/t\mathcal{F})_z) = 1$. Then there exists a single S_i containing Z , and for this value of i , we have $\text{long}(\mathcal{F}_{S_i}) = 1$; moreover $\mathcal{O}_{S_i, z}$ is a discrete valuation ring whose t_z is a uniformiser, and t_z is $\mathcal{F}_z$ -regular.*

Proof. This follows from (3.4.1), the two members of (3.4.1.1) being then equal. $\square$

Proposition (3.4.3). — *Let X be a locally Noetherian prescheme, t a section of $\mathcal{O}_X$ over X , Y the closed subprescheme of X defined by the Ideal $t\mathcal{O}_X$ of $\mathcal{O}_X$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, T an associated prime cycle of $\mathcal{F}$, T' an irreducible component of $T \cap Y$, x the generic point of T' . Suppose that t_x is $\mathcal{F}_x$ -regular; then $x \in \text{Ass}(\mathcal{F}/t\mathcal{F})$.*

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Proof. As in the proof of (3.4.1), we can reduce to the case where $X = \text{Spec}(\mathcal{O}_x)$; the proposition is then (taking into account (3.1.2)) a consequence of (0_{IV}, 16.4.6.3). $\square$

Proposition (3.4.4). — *Let X be a locally Noetherian prescheme, t a section of $\mathcal{O}_X$ over X , Y the closed subprescheme of X defined by the Ideal $t\mathcal{O}_X$ of $\mathcal{O}_X$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, (S_i) the family of the irreducible components of $\text{Supp}(\mathcal{F})$. Let y be a point of Y such that t_y is $\mathcal{F}_y$ -regular and such that none of the associated prime cycles immersed of $\mathcal{F}/t\mathcal{F}$ contains y . Then the irreducible components of $\text{Supp}(\mathcal{F}/t\mathcal{F})$ containing y are exactly the irreducible components of the $S_i \cap Y$ containing y , and the prime cycles associated to $\mathcal{F}$ containing y are non-immersed.*

Proof. Let us first prove the last assertion. Let $T \supset T_1$ be two associated prime cycles of $\mathcal{F}$ containing y ; if x is the generic point of an irreducible component of $T \cap Y$ containing y , x is a generalisation of y , hence contained in every neighbourhood of y , and the hypothesis that t_y is $\mathcal{F}_y$ -regular implies that t_x is $\mathcal{F}_x$ -regular (0_{IV}, 15.2.4), hence, by virtue of (3.4.3), $x \in \text{Ass}(\mathcal{F}/t\mathcal{F})$. Let x_1 be the generic point of an irreducible component of $T_1 \cap Y$ containing y , and let x be the generic point of an irreducible component of $T \cap Y$ containing x_1 ; it follows from what precedes that x_1 and x both belong to $\text{Ass}(\mathcal{F}/t\mathcal{F})$, and since $x_1 \in \overline{\{x\}}$, the hypothesis implies that $x_1 = x$. Let us still designate by T and T_1 the integral closed subpreschemes of X having T and T_1 as underlying spaces respectively, and set $A = \mathcal{O}_{T, x}$, $A_1 = \mathcal{O}_{T_1, x}$; then $A_1 = A/\mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal of A . By definition of x and x_1 , A/tA and A_1/tA_1 are Artinian rings; on the other hand, it was seen above that t_x is $\mathcal{F}_x$ -regular, hence x cannot belong to $\text{Ass}(\mathcal{F})$, and consequently A_1 is not Artinian. We therefore have $\dim A = \dim A_1 = 1$; but this implies $\mathfrak{p} = 0$ and $A = A_1$ (0_{IV}, 16.1.2.2); since $\text{Spec}(A)$ and $\text{Spec}(A_1)$ are respectively dense in T and T_1 , we have indeed $T = T_1$.

The S_i containing y are therefore all the associated prime cycles of $\mathcal{F}$ containing y ; if x is the generic point of an irreducible component of $S_i \cap Y$ containing y , we deduce also from (0_{IV}, 15.2.4) that t_x is $\mathcal{F}_x$ -regular, hence, by (3.4.3), $x \in \text{Ass}(\mathcal{F}/t\mathcal{F})$; this demonstrates the first assertion of (3.4.4). $\square$

Proposition (3.4.5). — *Let X be a locally Noetherian prescheme, t a section of $\mathcal{O}_X$ over X , Y the closed subprescheme of X defined by the Ideal $t\mathcal{O}_X$ of $\mathcal{O}_X$. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, y a point of Y ; suppose that t_y is $\mathcal{F}_y$ -regular and that $\mathcal{F}/t\mathcal{F}$ is integral at the point y (3.2.4). Then $\mathcal{F}$ is integral at the point y .*

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Proof. In view of (3.4.4), it suffices to prove that y is contained in a single irreducible component of $\text{Supp}(\mathcal{F})$, and that if s is the generic point of this component, we have $\text{long}(\mathcal{F}_s) = 1$; the conclusion then follows from (3.4.2). $\square$

Proposition (3.4.6). — *The hypotheses being those of (3.4.1), let x be a point of Y . Suppose that Y contains none of the S_i containing x , and that $\mathcal{F}/t\mathcal{F}$ is reduced at the point x (3.2.2). Then t_z is $\mathcal{F}_z$ -regular and $\mathcal{F}$ is reduced at the point x . Moreover, if z_j is the generic point of an irreducible component of $\text{Supp}(\mathcal{F}/t\mathcal{F})$ containing x , z_j is contained in only one of the S_i , and $\mathcal{O}_{S_i, z_j}$ is a discrete valuation ring whose t_{z_j} is a uniformiser.*

Proof. The fact that t_z is $\mathcal{F}_z$ -regular follows from the following lemma applied to the ring $\mathcal{O}_z$:

Lemma (3.4.6.1). — *Let A be a Noetherian ring, M an A -module of finite type, $\mathfrak{p}_i$ the minimal elements of $\text{Supp}(M)$, t an element of A . Suppose that t does not belong to any of the $\mathfrak{p}_i$ and that M/tM is a reduced A -module (3.2.2). Then t is M -regular.*

Every prime ideal $\mathfrak{p} \in \text{Supp}(M)$ contains one of the $\mathfrak{p}_i$; since t does not belong to any of the $\mathfrak{p}_i$, the homothety of ratio t in $M_{\mathfrak{p}}$ is not nilpotent (Bourbaki, *Alg. comm.*, chap. IV, §1, no. 4, cor. of prop. 9). Denote by N the submodule of M formed by the elements annihilated by a power of t , and set $P = M/N$; we shall show that $N = 0$. Since t is P -regular, on an exact sequence (3.4.1.4)

$$0 \longrightarrow N/tN \longrightarrow M/tM \longrightarrow P/tP \longrightarrow 0.$$

Since N is of finite type, it is annihilated by a power of t , and it suffices therefore to prove that $N/tN = 0$ for every $\mathfrak{p} \in \text{Ass}(M/tM)$, or equivalently that the homomorphism $u_{\mathfrak{p}} : (M/tM)_{\mathfrak{p}} \rightarrow (P/tP)_{\mathfrak{p}}$ is bijective for every $\mathfrak{p} \in \text{Ass}(M/tM)$. Now, we have $(P/tP)_{\mathfrak{p}} \neq 0$; in effect, since $\mathfrak{p} \in \text{Supp}(M/tM) = \text{Supp}(M) \cap V(t)$ (0_{IV}, 1.7.5), the image of t in $A_{\mathfrak{p}}$ is contained in the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$, hence the hypothesis $P_{\mathfrak{p}} = tP_{\mathfrak{p}}$ would imply $P_{\mathfrak{p}} = 0$ by the lemma of Nakayama; one would then have $M_{\mathfrak{p}} = N_{\mathfrak{p}}$ and the homothety of ratio t in $M_{\mathfrak{p}}$ would be nilpotent, since $\mathfrak{p} \in \text{Supp}(M)$. This being so, the hypothesis that M/tM is reduced implies $\text{long}((M/tM)_{\mathfrak{p}}) = 1$, and since $(P/tP)_{\mathfrak{p}} \neq 0$, $u_{\mathfrak{p}}$ is necessarily bijective, which proves the lemma.

By hypothesis, none of the associated prime cycles immersed of $\mathcal{F}/t\mathcal{F}$ contains x , hence none of the associated prime cycles immersed of $\mathcal{F}$ contains x , by virtue of (3.4.4). On the other hand, applying (3.4.2) to an irreducible component of $\text{Supp}(\mathcal{F}/t\mathcal{F})$ containing x , we see that $\text{long}(\mathcal{F}_{S_i}) = 1$ for every S_i containing x , which completes the proof that $\mathcal{F}_x$ is reduced; the last assertions are also consequences of (3.4.2). $\square$

Corollary (3.4.7). — *Let A be a local Noetherian ring, $\mathfrak{m}$ its maximal ideal, M an A -module of finite type, $(x_i)_{1 \leq i \leq k}$ a family of elements of $\mathfrak{m}$ forming part of a system of parameters for M (0_{IV}, 16.3.6). If the A -module $N = M/(\sum_{i=1}^k x_i M)$ is integral (3.2.4), then M is integral and the sequence $(x_i)_{1 \leq i \leq k}$ is M -regular.*

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Proof. By induction on k , one is immediately reduced to the case $k = 1$; we write x instead of x_1 ; the hypothesis that x forms part of a system of parameters for M implies $\dim(M) = \dim(A) - 1$ (0_{IV}, 16.3.7). Setting $n = \dim(N)$; there is therefore an element $\mathfrak{p}$ of $\text{Supp}(M)$ such that $\dim(M/\mathfrak{p}M) = n + 1$ (0_{IV}, 16.3.5), and moreover x forms part of a system of parameters for $M/\mathfrak{p}M$ (0_{IV}, 16.3.5); in addition, the support of N' , being isomorphic to N , is integral; moreover, the support of N' (equal to the intersection of $\text{Supp}(M)$ and $V(x)$) cannot contain $\text{Supp}(M')$, and this last set is irreducible by construction. The hypothesis implies then that x is M' -regular by virtue of (3.4.6). One concludes that the kernel of the homothety $z \mapsto xz$ in M is contained in $\mathfrak{p}^j M \subset \mathfrak{m}^j M$ for every integer j , and this kernel is therefore reduced to 0 (0_{IV}, 7.3.5), which proves that x is M -regular. One can then apply (3.4.5). $\square$

Remark (3.4.8). — An analogous proposition to (3.4.7), where one replaces “integral” by “reduced”, is no longer necessarily exact. Consider for example the ring of polynomials $C = K[X, Y, Z]$ over a field K , the quotient ring $B = C/\mathfrak{p}\mathfrak{q}$, where $\mathfrak{p} = CZ$, $\mathfrak{q} = CX^2 + CY$; let A be the local ring of B corresponding to the maximal image of $CX + CY + CZ$ in B . If x, z are the canonical images of X, Z in A , it is clear that $xz \neq 0$ but $x^2z^2 = 0$; on the other hand, since A/xA is isomorphic to $K[Y, Z]/(YZ)$, we have $\dim(A/xA) = 1$ while $\dim(A) = 2$, hence x forms part of a system of parameters of A (0_{IV}, 16.3.4), A/xA is reduced, but A is not.

Proposition (3.4.9). — *Let A be a Noetherian ring, M an A -module of finite type, f an M -regular element such that M/fM has no immersed associated prime ideal. If $\mathfrak{p}_i$ ($1 \leq i \leq m$) are the prime ideals associated*

to M/fM , then, for every integer $n > 0$, $f^n M$ is the intersection of the inverse images of $f^n M_{\mathfrak{p}_i}$ by the canonical homomorphisms $M \rightarrow M_{\mathfrak{p}_i}$ ($1 \leq i \leq m$).

Proof. It amounts to showing that the $\mathfrak{p}_i$ are also the prime ideals associated to $M/f^n M$, for then the saturations of $f^n M$ for the $\mathfrak{p}_i$ are the submodules of the reduced primary decomposition (necessarily unique) of $f^n M$ in M . Now, we have

$$\text{Ass}(f^{n-1}M/f^n M) \subset \text{Ass}(M/f^n M) \subset \text{Ass}(M/f^{n-1}M) \cup \text{Ass}(f^{n-1}M/f^n M)$$

by (3.1.7), and since f is M -regular, $f^{n-1}M/f^n M$ is isomorphic to M/fM ; it therefore suffices to reason by induction on n . $\square$

§4. CHANGE OF BASE FIELD FOR ALGEBRAIC PRESCHMES

We will develop in §5 the general theory of the dimension of preschemes; but the theory of the dimension of *algebraic* preschemes can be developed in a more elementary way, and since it moreover presents numerous special features, we will give here a rapid exposition, independent of the general theory.

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If K is a field and L an extension of K , we will denote by $\deg.\text{tr}_K L$ the *transcendence degree* of L over K .

4.1. Dimension of algebraic preschemes

Definition (4.1.1). — Let k be a field, X a prescheme locally of finite type over k . The *dimension* of X is defined to be the number

$$(4.1.1.1) \quad \dim X = \sup_x \deg.\text{tr}_k k(x)$$

where x ranges over the set of maximal points of X .

We will see in §5 (5.2.2) that the definition (4.1.1) depends only in appearance on the base field k over which X is locally of finite type, and that the number $\dim(X)$ thus defined coincides with the dimension of the *underlying space* X (defined in (0_{IV}, 14.1.2)). We obviously have $\dim X = \dim X_{\text{red}}$.

We note that each $k(x)$ is a finitely generated extension of k (I, 6.3.3) and therefore has a finite transcendence degree over k . If X is of finite type over k and non-empty, it is Noetherian (I, 6.3.7), hence has only a finite number of irreducible components, and therefore $\dim(X)$ is finite and ≥ 0 ; the definition (4.1.1) gives

$$\dim(\emptyset) = -\infty.$$

It is clear that, if we denote by (X_α) the family of reduced closed subpreschemes of X having as underlying spaces the irreducible components of X (I, 5.2.1), then

$$(4.1.1.2) \quad \dim(X) = \sup_\alpha \dim(X_\alpha).$$

This reduces the calculation of the dimension to the case of *integral* preschemes (locally of finite type over k). Finally, we obviously have

$$(4.1.1.3) \quad \dim(X) = \dim(U)$$

for every sub-prescheme induced on an everywhere dense open subset U of X ; this finally reduces the notion of dimension to the case of an *affine scheme of finite type* over k .

Theorem (4.1.2). — Let X, Y be two preschemes locally of finite type over a field k , $f : X \rightarrow Y$ a k -morphism.

- (i) If f is quasi-compact and dominant, then $\dim(Y) \leq \dim(X)$.
- (ii) If f is quasi-finite, then $\dim(X) \leq \dim(Y)$.
- (iii) Suppose that X is of finite type over k . For $\dim(X) \geq n$ (resp. $\dim(X) \leq n$, $\dim(X) = n$), it is necessary and sufficient that there exist an open dense subset U of X , and a k -morphism $g : U \rightarrow \mathbf{V}(k^n)$ ($= \text{Spec}(k[T_1, \dots, T_n])$), which we will also denote $\mathbf{V}_k^n$ which is surjective (resp. finite, resp. finite and surjective). If X is an affine scheme, one can take $U = X$.

Proof. (i) Let y be a maximal point of Y ; we know that $f^{-1}(y)$ contains a maximal point x of X (1.1.5), hence $k(x)$ is an extension of $k(y)$, whence the inequality $\deg.\mathrm{tr}_k k(y) \leq \deg.\mathrm{tr}_k k(x) \leq \dim(X)$, which proves (i).

(ii) If x is a maximal point of X , $k(x)$ is a finite extension of $k(f(x))$ (II, 6.2.2), hence of the same transcendence degree over k . If one considers the reduced closed subprescheme of Y having $\overline{\{f(x)\}}$ as underlying space, one sees that one is reduced to proving the

Corollary (4.1.2.1). — *Let Y be a k -prescheme locally of finite type; for every subprescheme Z of Y , $\dim(Z) \leq \dim(Y)$. Suppose moreover that the irreducible components of Y all have the same dimension; then, for Z to be rare in Y , it is necessary and sufficient that $\dim(Z) < \dim(Y)$.*

Indeed, let z be a maximal point of Z ; by considering one of the reduced closed subpreschemes of Y having as underlying space one of the irreducible components of Y containing z , one reduces to the case where Y is integral with generic point y , then, in considering within Y an affine open containing z , to the case where Y is affine and of finite type over k , say $Y = \mathrm{Spec}(A)$, and Z closed in Y , say $Z = \mathrm{Spec}(A/\mathfrak{a})$ where $\mathfrak{a}$ is an integral ideal of A . By virtue of the Noether normalisation lemma (Bourbaki, *Alg. comm.*, chap. V, §3, no. 1, th. 1) there exists a finite sequence $(x_i)_{1 \leq i \leq n}$ of elements of A , algebraically independent over k , such that A is integral over the ring $B = k[x_1, \dots, x_n]$ and that $\mathfrak{a} \cap B$ is generated by a subfamily $(x_i)_{1 \leq i \leq p}$ (possibly empty) of A . Let $g : Y \rightarrow \mathrm{Spec}(B) = \mathbf{V}_k^n$ be the finite dominant morphism corresponding to the injection canonical $B \rightarrow A$; since $C = B/(\mathfrak{a} \cap B)$ is isomorphic to $k[x_{p+1}, \dots, x_n]$, g induces on Z a dominant morphism $Z \rightarrow \mathbf{V}_k^{n-p}$; we have therefore $\deg.\mathrm{tr}_k k(y) = n$ and $\deg.\mathrm{tr}_k k(z) = n - p$, hence $\deg.\mathrm{tr}_k k(z) \leq \deg.\mathrm{tr}_k k(y) = \dim(Y)$, which demonstrates (i). Moreover, if $p = 0$, Z contains an open non-empty subset of Y , hence Z is not rare; if on the contrary $p > 0$, we have $\overline{\{z\}}$ is rare in $\overline{\{y\}}$, which finishes proving (4.1.2.1).

(iii) The proof that the conditions stated are sufficient follows from (i) and (ii). To prove that they are necessary, one can consider an open dense subset U of X , union of affine opens U_i pairwise disjoint and irreducible ($1 \leq i \leq m$), each containing one of the maximal points of X (0I, 2.1.6); one can moreover suppose X reduced and if X is affine, one can of course take $U = X$. The same reasoning shows that if $\dim(U_i) = n_i$, there exists a k -morphism $g_i : U_i \rightarrow \mathbf{V}_k^{n_i}$ which is finite and dominant, hence surjective (II, 6.1.10); for each n_i there is then a morphism $h_i : \mathbf{V}_k^{n_i} \rightarrow \mathbf{V}_k^n$ which is a closed immersion corresponding to the canonical homomorphism $k[T_1, \dots, T_n] \rightarrow k[T_1, \dots, T_{n_i}]$, and which is the identity for $n_i = n$. Since U is the sum of the U_i , one takes for g the morphism which on each U_i coincides with $h_i \circ g_i$; it is obviously finite and surjective. When $n' \geq n$, one composes with the canonical closed immersion $\mathbf{V}_k^n \rightarrow \mathbf{V}_k^{n'}$; when $n' \leq n$, one composes with g the canonical morphism $p : \mathbf{V}_k^n \rightarrow \mathbf{V}_k^{n'}$ corresponding to the canonical injection $k[T_1, \dots, T_{n'}] \rightarrow k[T_1, \dots, T_n]$, noting that p is faithfully flat, hence surjective (I, 3.5.2) and (II, 6.1.5), whence the corollary. $\square$

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Remark (4.1.3). — The corollary (4.1.2.1) shows that in the formula (4.1.1.1), one can suppose that x ranges over the set of *all* points of X .

Corollary (4.1.4). — *Let X be a prescheme locally of finite type over a field k , K an extension of k ; then $\dim(X \otimes_k K) = \dim(X)$.*

Proof. One can restrict to the case where X is of finite type over k ; then the morphism $u : X \otimes_k K \rightarrow X$ is faithfully flat (2.2.13), (i), hence if U is an open dense subset in X , $u^{-1}(U) = U \otimes_k K$ is dense in $X \otimes_k K$ (2.3.10); if $g : U \rightarrow \mathbf{V}_k^n$ is finite and surjective, it is the same for $g_{(K)} : U \otimes_k K \rightarrow \mathbf{V}_k^n$ (I, 3.5.2) and (II, 6.1.5), whence the corollary. $\square$

Corollary (4.1.5). — *Let X and Y be two preschemes locally of finite type over a field k ; then $\dim(X \times_k Y) = \dim(X) + \dim(Y)$.*

Proof. It suffices to prove that if U (resp. V) is an affine neighbourhood of a point x (resp. y) of X (resp. Y) such that $\deg.\mathrm{tr}_k k(x) = m = \dim U$ and $\deg.\mathrm{tr}_k k(y) = n = \dim V$, then $\dim(U \times_k V) = m + n$, in other words, by virtue of (4.1.2) that there exist finite surjective k -morphisms $f : X \rightarrow \mathbf{V}_k^m$, $g : Y \rightarrow \mathbf{V}_k^n$; then $f \times g : X \times_k Y \rightarrow \mathbf{V}_k^{m+n}$ is finite and surjective (I, 3.5.2) and (II, 6.1.5), whence the corollary. $\square$

4.2. Associated prime cycles on algebraic preschemes

Proposition (4.2.1). — *Let K and L be two extensions of a field k , such that $K \otimes_k L$ is Noetherian. Then the prime ideals associated to $K \otimes_k L$ are minimal, and if E is the residue field of the local ring of such an ideal, we have*

$$(4.2.1.1) \quad \deg.\mathrm{tr}_K E = \deg.\mathrm{tr}_k L, \quad \deg.\mathrm{tr}_L E = \deg.\mathrm{tr}_k K$$

whence

$$(4.2.1.2) \quad \deg.\mathrm{tr}_k E = \deg.\mathrm{tr}_k K + \deg.\mathrm{tr}_k L.$$

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Proof. We know that K is an algebraic extension of a purely transcendental extension $K' = k(\mathbf{t})$, where $\mathbf{t} = (t_\alpha)_{\alpha \in I}$ is a family of indeterminates; the ring $k[\mathbf{t}] \otimes_k L = L[\mathbf{t}]$ is integral, hence the same is true of $K' \otimes_k L$, which is a ring of fractions, and the field of fractions of $K' \otimes_k L$ is $L(\mathbf{t})$; we have the commutative diagram of canonical homomorphisms

$$\begin{array}{ccccc} K & \longrightarrow & K \otimes_k L & \longrightarrow & K \otimes_{K'} L(\mathbf{t}) \\ \uparrow & & \uparrow & & \uparrow \\ K' & \longrightarrow & K' \otimes_k L & \longrightarrow & L(\mathbf{t}) \\ & \swarrow & \uparrow & \nwarrow & \uparrow \\ & & k & & L \end{array}$$

Since K is faithfully flat over K' , $K \otimes_k L = K \otimes_{K'} (K' \otimes_k L)$ is faithfully flat over $K' \otimes_k L$ (0_I, 6.5.2); moreover $K' \otimes_k L$ identifies with a subring of $K \otimes_k L$; the trace on $K' \otimes_k L$ of a prime ideal $\mathfrak{p}$ associated to $K \otimes_k L$ is the ideal 0, an element $\neq 0$ of $K' \otimes_k L$ not being a zero divisor in $K \otimes_k L$ (0_I, 6.3.4 and Bourbaki, *Alg. comm.*, chap. IV, §1, no. 1, cor. 3 of prop. 2). Since moreover K is algebraic over K' , $K \otimes_k L$ is an integral algebra over $K' \otimes_k L$, and the prime ideals of $K \otimes_k L$ inducing 0 on $K' \otimes_k L$ are necessarily without mutual inclusion relation (Bourbaki, *Alg. comm.*, chap. V, §2, no. 1, cor. 1 of prop. 1); this proves the first assertion (Bourbaki, *Alg. comm.*, chap. IV, §1, no. 3, cor. 1 of prop. 7). Furthermore, the residue field E of $\mathfrak{p}$ is algebraic over the residue field of the ideal (0) of $K' \otimes_k L$, that is to say $L(\mathbf{t})$; hence $\deg.\mathrm{tr}_k E = \mathrm{Card}(I) = \deg.\mathrm{tr}_k K$, in other words we have the first relation (4.2.1.1); exchanging the roles of K and L , we have the second relation (4.2.1.1), whence (4.2.1.2). $\square$

Corollary (4.2.2). — *Under the hypotheses of (3.3.6), if the preschemes $T_{x,y}$ are locally Noetherian, they have no immersed associated prime cycle.*

Corollary (4.2.3). — *Under the hypotheses of (3.3.6) (resp. (3.3.7)), if the $T_{x,y}$ are locally Noetherian and if $\mathcal{F}$ and the $\mathcal{G}_s$ (for $s \in S$) (resp. $\mathcal{F}$ and $\mathcal{G}$) are without immersed associated prime cycle, the same is true of $\mathcal{F} \otimes_S \mathcal{G}$.*

Proof. This follows from (4.2.2), (3.3.2) and the proof of (3.3.6). In particular, since every prescheme over a field k is flat over k , we have thus proved assertion (i) of the following: $\square$

Proposition (4.2.4). — *Let k be a field, X and Y two k -preschemes locally Noetherian such that $X \times_k Y$ is locally Noetherian. Suppose moreover X and Y integral. Then:*

- (i) $X \times_k Y$ is without immersed associated prime cycle; each of the irreducible components of $X \times_k Y$ dominates X and Y , and the set of these components is in bijective correspondence with the set of the irreducible components of $\mathrm{Spec}(R(X) \otimes_k R(Y))$ (in other words the set of the minimal prime ideals of $R(X) \otimes_k R(Y)$), where $R(X)$ and $R(Y)$ are the fields of rational functions of X and Y respectively.
- (ii) If a maximal point z of $X \times_k Y$ is identified with a minimal prime ideal $\mathfrak{p}$ of $R(X) \otimes_k R(Y)$, the local ring $\mathcal{O}_{X \times_k Y, z}$ is isomorphic to the ring of fractions $(R(X) \otimes_k R(Y))_{\mathfrak{p}}$. In particular, if $R(X)$ or $R(Y)$ is separable over k , $X \times_k Y$ is reduced.
- (iii) If moreover X and Y are locally of finite type over k , every irreducible component of $X \times_k Y$ is of dimension $\dim(X) + \dim(Y)$.

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Proof. Assertion (iii) follows from (4.1.2), taking into account (i) and (ii). To prove (ii), one can obviously restrict to the case where $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$ are affine, A and B being integral rings with fields of fractions $K = R(X)$ and $L = R(Y)$; assertion (i) shows that every minimal prime ideal $\mathfrak{q}$ of $A \otimes_k B$ is the trace on $A \otimes_k B$ of a minimal prime ideal $\mathfrak{p}$ of $K \otimes_k L$; since $K \otimes_k L$ is a ring of fractions of $A \otimes_k B$, the isomorphism of the rings of fractions $(A \otimes_k B)_{\mathfrak{q}}$ and $(K \otimes_k L)_{\mathfrak{p}}$ follows from (0_I, 1.2.5). Finally, if for example $R(Y)$ is separable over k , we know that the ring $R(X) \otimes_k R(Y)$ is reduced (Bourbaki, *Alg.*, chap. VIII, §7, no. 3, th. 1); so the same is true of the local rings at the maximal points of $X \times_k Y$. One deduces that $X \times_k Y$ is reduced (3.2.1). $\square$

Proposition (4.2.5). — *Let k be a field, X and Y two k -preschemes locally Noetherian, $\mathcal{F}$ (resp. $\mathcal{G}$) a quasi-coherent $\mathcal{O}_X$ -Module (resp. $\mathcal{O}_Y$ -Module). Let (Z'_λ) (resp. (Z''_μ)) be the family of the associated prime cycles of $\mathcal{F}$ (resp. $\mathcal{G}$), and denote also by Z'_λ (resp. Z''_μ) the reduced subprescheme of X (resp. Y) having Z'_λ (resp. Z''_μ) as underlying space. Then, if $Z'_\lambda \times_k Z''_\mu$ is locally Noetherian, the irreducible components $Z_{\lambda\mu\nu}$ of $Z'_\lambda \times_k Z''_\mu$ dominate Z'_λ and Z''_μ , and $(Z_{\lambda\mu\nu})$ is the family of the distinct associated prime cycles of $\mathcal{F} \otimes_k \mathcal{G}$.*

Proof. It suffices to apply (4.2.4) to the product $\operatorname{Spec}(k(x)) \times_k \operatorname{Spec}(k(y))$. $\square$

In particular:

Corollary (4.2.6). — *Let k be a field, X, Y two k -preschemes locally Noetherian such that $X \times_k Y$ is locally Noetherian. Let (Z'_λ) (resp. (Z''_μ)) be the family of reduced subpreschemes having as underlying spaces the irreducible components of X (resp. Y). Then the irreducible components $Z_{\lambda\mu\nu}$ of $Z'_\lambda \times_k Z''_\mu$ dominate Z'_λ and Z''_μ , and $(Z_{\lambda\mu\nu})$ is the family of the irreducible components of $X \times_k Y$.*

Proof. Indeed, one can restrict to the case where X and Y are reduced (I, 5.1.8); the irreducible components of X (resp. Y) are then the associated prime cycles of $\mathcal{O}_X$ (resp. $\mathcal{O}_Y$) (3.2.1). Applying (4.2.5) to $\mathcal{F} = \mathcal{O}_X$ and $\mathcal{G} = \mathcal{O}_Y$, and noting that by definition $\mathcal{O}_X \otimes_k \mathcal{O}_Y = \mathcal{O}_{X \times_k Y}$, the corollary follows from the fact that $\mathcal{O}_X$ and $\mathcal{O}_Y$ have no immersed associated prime cycles by hypothesis (4.2.3). $\square$

Let us apply the preceding results to the case where Y is the spectrum of an extension K of k :

Proposition (4.2.7). — *Let k be a field, X a k -prescheme, K an extension of k such that $X \otimes_k K$ is locally Noetherian, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module, x' a point of $X \otimes_k K$, x its image in X .*

- (i) *Let (Z_λ) be the family of reduced subpreschemes of X having as underlying spaces the associated prime cycles of $\mathcal{F}$; then the irreducible components $Z_{\lambda\mu}$ of the $Z_\lambda \otimes_k K$ are the cycles premiers associated to $\mathcal{F} \otimes_k K$, and $Z_{\lambda\mu}$ dominates Z_λ ; moreover, for $Z_{\lambda\mu}$ to be immersed, it is necessary and sufficient that Z_λ be so.*
- (ii) *For x to belong to an immersed associated prime cycle of $\mathcal{F}$, it is necessary and sufficient that x' belong to an immersed associated prime cycle of $\mathcal{F} \otimes_k K$; for $\mathcal{F}$ to be without immersed associated prime cycle, it is necessary and sufficient that the same hold for $\mathcal{F} \otimes_k K$.*
- (iii) *The indices λ such that $x \in Z_\lambda$ are the same as those for which there exists an index μ such that $x' \in Z_{\lambda\mu}$. In particular, if x' belongs to only one associated prime cycle of $\mathcal{F} \otimes_k K$, x belongs to only one associated prime cycle of $\mathcal{F}$.*
- (iv) *If X is locally of finite type over k , we have $\dim(Z_{\lambda\mu}) = \dim(Z_\lambda)$.*

Proof. One will note that the hypothesis implies that X itself is locally Noetherian (2.2.13) and (2.2.14); assertion (i) results from (4.2.5) and the proof of (4.2.3), for $\mathcal{G} = \mathcal{O}_Y$, with $Y = \operatorname{Spec}(K)$; (ii) and (iii) result from (i) and (2.3.5); finally, (iv) is a particular case of (4.2.4), (iii). $\square$

Corollary (4.2.8). — *If X is locally of finite type over k , the set of dimensions of associated prime cycles is the same for $\mathcal{F}$ and $\mathcal{F} \otimes_k K$; the set of dimensions of the irreducible components is the same for X and $X \otimes_k K$.*

Proposition (4.2.9). — *Suppose the hypotheses of (4.2.5) are satisfied, and suppose moreover that $\mathcal{F}$ and $\mathcal{G}$ are coherent. Let $(\mathcal{F}_\lambda)_{\lambda \in L}$ and $(\mathcal{G}_\mu)_{\mu \in M}$ be irredundant decompositions of $\mathcal{F}$ and $\mathcal{G}$ respectively; for every couple $(\lambda, \mu) \in L \times M$, let $(\mathcal{H}_{\lambda\mu\nu})_{\nu \in S(\lambda, \mu)} = \operatorname{Ass}(\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu)$ (3.2.5). Then $(\mathcal{H}_{\lambda\mu\nu})$ (for all triples (λ, μ, ν)) is an irredundant decomposition of $\mathcal{F} \otimes_k \mathcal{G}$; it is reduced if $(\mathcal{F}_\lambda)$ and $(\mathcal{G}_\mu)$ are.*

Proof. Note that $\mathcal{F} \otimes_k \mathcal{G}$ and each of the $\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu$ are coherent, and each of the $\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu$ identifies with a quotient of $\mathcal{F} \otimes_k \mathcal{G}$; each of the $\mathcal{H}_{\lambda\mu\nu}$ identifies by definition with a coherent quotient of

$\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu$, hence also with a coherent quotient of $\mathcal{F} \otimes_k \mathcal{G}$. The family of supports of the $\mathcal{H}_{\lambda\mu\nu}$ is locally finite, for $\text{Supp}(\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu)$ is contained in the underlying space of $\text{Supp}(\mathcal{F}_\lambda) \times_k \text{Supp}(\mathcal{G}_\mu)$ (where $\text{Supp}(\mathcal{F}_\lambda)$ and $\text{Supp}(\mathcal{G}_\mu)$ denote the corresponding reduced closed subschemes), and for λ and μ given, the family of supports of the $\mathcal{H}_{\lambda\mu\nu}$ is locally finite by hypothesis; our assertion therefore results from the fact that the families $(\text{Supp}(\mathcal{F}_\lambda))$ and $(\text{Supp}(\mathcal{G}_\mu))$ are locally finite. To show that $(\mathcal{H}_{\lambda\mu\nu})$ is an irredundant decomposition of $\mathcal{F} \otimes_k \mathcal{G}$, it suffices therefore to prove that the canonical homomorphism $\mathcal{F} \otimes_k \mathcal{G} \rightarrow \bigoplus_{\lambda,\mu,\nu} \mathcal{H}_{\lambda\mu\nu}$ is injective; but it is the composite of the homomorphisms $\mathcal{F} \otimes_k \mathcal{G} \rightarrow \bigoplus_{\lambda,\mu} (\mathcal{F}_\lambda \otimes_k \mathcal{G}_\mu) \rightarrow \bigoplus_{\lambda,\mu,\nu} \mathcal{H}_{\lambda\mu\nu}$; the second of these homomorphisms is injective by definition, and the same holds for the first, which is none other than the tensor product of the canonical injective homomorphisms $\mathcal{F} \rightarrow \bigoplus_\lambda \mathcal{F}_\lambda$, $\mathcal{G} \rightarrow \bigoplus_\mu \mathcal{G}_\mu$ (recall that $\mathcal{F}$ and $\mathcal{G}$ are flat over k). Finally, if $(\mathcal{F}_\lambda)$ and $(\mathcal{G}_\mu)$ are reduced, one can suppose that $M = \text{Ass}(\mathcal{F})$ and $M = \text{Ass}(\mathcal{G})$; the fact that the $\mathcal{H}_{\lambda\mu\nu}$ then form a partition of $\text{Ass}(\mathcal{F} \otimes_k \mathcal{G})$ follows from (4.2.2) (cf. (3.2.5)). $\square$

Corollary (4.2.10). — *Under the hypotheses of (4.2.7), suppose moreover $\mathcal{F}$ coherent, and let $(\mathcal{F}_\lambda)_{\lambda \in L}$ be an irredundant decomposition of $\mathcal{F}$; for every $\lambda \in L$, let $(\mathcal{F}_{\lambda\mu})_{\mu \in \text{Ass}(\mathcal{F}_\lambda \otimes_k K)}$ be a reduced irredundant decomposition of $\mathcal{F}_\lambda \otimes_k K$. Then $(\mathcal{F}_{\lambda\mu})$ is an irredundant decomposition of $\mathcal{F} \otimes_k K$; it is reduced if $(\mathcal{F}_\lambda)$ is.*

Proof. One applies (4.2.9) with $Y = \text{Spec}(K)$, $\mathcal{G} = \mathcal{O}_Y = K$, M reduced to a single element. $\square$

4.3. Reminders on tensor products of fields For the convenience of the reader, we recall here certain properties of tensor products of fields which we will use in the following sections; for proofs, we refer to [?].

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(4.3.1). Recall that an extension L of a field k is called *primary* if the largest algebraic separable extension of k contained in L is k itself.

Proposition (4.3.2). — *Let K, L be two extensions of a field k . If L is a primary extension of k , then $\text{Spec}(L \otimes_k K)$ is irreducible, and if ξ is its generic point, $k(\xi)$ is a primary extension of K . Conversely, if for every finite separable extension K' of k , $\text{Spec}(L \otimes_k K')$ is irreducible, L is a primary extension of k .*

Proof. See the proof in [?], p. 14-03 to 14-06. $\square$

Corollary (4.3.3). — *If k is separably closed (i.e. if its algebraic closure is purely inseparable over k), then, for any two extensions K, L of k , $\text{Spec}(L \otimes_k K)$ is irreducible, and conversely.*

Proof. This follows immediately from (4.3.2). $\square$

Corollary (4.3.4). — *Let L be an extension of a field k , L_s the largest algebraic separable extension of k contained in L (sometimes called the separable algebraic closure of k in L); suppose L_s of finite degree over k , and let K be a Galois extension of k containing L_s . Then $\text{Spec}(L \otimes_k K)$ has $[L_s : k]$ irreducible components, the sum of whose residue fields at the generic points are the primary extensions of K .*

Proof. Indeed, we have $L \otimes_k K = L \otimes_{L_s} (L_s \otimes_k K)$ and one knows that $L_s \otimes_k K$ is isomorphic to a product of $[L_s : k]$ fields isomorphic to K (Bourbaki, Alg., chap. VIII, §7, no. 3, cor. 2 of th. 1); hence $L \otimes_k K$ is isomorphic to a product of $[L_s : k]$ rings isomorphic to $L \otimes_{L_s} K$; since L is a primary extension of L_s , the conclusion follows from (4.3.2). $\square$

Proposition (4.3.5). — *Let K, L be two extensions of a field k . If L is a separable extension of k , the ring $L \otimes_k K$ is reduced. Conversely, if for every finite purely inseparable extension K' of k , the ring $L \otimes_k K'$ is reduced, L is a separable extension of k .*

Proof. For the proof, see Bourbaki, Alg., chap. VIII, §7, no. 3, th. 1. $\square$

Corollary (4.3.6). — *If k is perfect, then for any two extensions K, L of k , $K \otimes_k L$ is reduced, and conversely.*

Proof. This follows immediately from (4.3.5). $\square$

Corollary (4.3.7). — *Let L be a separable extension of k and let K be an extension of k such that K or L is finite over k ; then the residue fields of the semi-local ring $L \otimes_k K$ are separable extensions of K .*

Proof. One knows indeed that $L \otimes_k K$ is a direct product of these residue fields E_i ; for every extension K' of K , $L \otimes_k K'$ is therefore also isomorphic to the direct product of the rings $E_i \otimes_K K'$; since $L \otimes_k K'$ is reduced, the E_i are separable over K by virtue of (4.3.5). $\square$

Corollary (4.3.8). — *If K and L are two finite separable extensions of k , $K \otimes_k L$ is a product of a finite number of separable extensions of k .*

Proposition (4.3.9). — *If k is algebraically closed, then, for any two extensions K, L of k , $K \otimes_k L$ is integral, and conversely.*

Proof. This is a consequence of (4.3.3) and (4.3.6), a perfect and separably closed field being algebraically closed. $\square$

4.4. Irreducible and connected preschemes over an algebraically closed field

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(4.4.1). Let k be a field, X a k -prescheme, K an extension of k . Since the morphism $\text{Spec}(K) \rightarrow \text{Spec}(k)$ is faithfully flat, quasi-compact, and universally open (2.4.9), the same holds for the projection morphism $p : X \otimes_k K \rightarrow X$ (2.2.13). It therefore follows from (2.3.5) that every irreducible component of $X \otimes_k K$ dominates an irreducible component of X , and (since p is surjective) every irreducible component of X is dominated by an irreducible component of $X \otimes_k K$; one thus deduces from p a surjective application from the set of irreducible components of $X \otimes_k K$ to the set of irreducible components of X . In the same way, it follows from the fact that p is continuous and surjective that every connected component of X is the image by p of a union of connected components of $X \otimes_k K$, whence a surjective application from the set of connected components of $X \otimes_k K$ to the set of connected components of X . One concludes that every irreducible (resp. connected) component Z' of $X \otimes_k K$ is contained in a unique set of the form $p^{-1}(Z)$, where Z is an irreducible (resp. connected) component of X ; for every subprescheme Z of X having Z as underlying space (and still denoted Z), Z' is an irreducible (resp. connected) component of $Z \otimes_k K$. In particular, if $X \otimes_k K$ has a finite number n (resp. n') of irreducible (resp. connected) components (which is the case when $X \otimes_k K$ is of finite type over K , for then $X \otimes_k K$ is of finite type over K , hence Noetherian), the number of irreducible (resp. connected) components of X is $\leq n$ (resp. $\leq n'$); to say that it is equal to n (resp. n') means that for every reduced subprescheme Z of X having as underlying space an irreducible (resp. connected) component of X , $Z \otimes_k K$ is irreducible (resp. connected) (I, 5.1.8); the number of irreducible (resp. connected) components of $X \otimes_k K$ is therefore independent of K if and only if for every irreducible (resp. connected) component Z of X , $Z \otimes_k K$ is irreducible (resp. connected) for every extension K of k .

In particular, if $X \otimes_k K$ is irreducible (resp. connected), the same holds for X (which already follows from the fact that p is surjective). In the same way, if $X \otimes_k K$ is reduced (resp. integral), the same holds for X since p is faithfully flat (2.1.13).

In this section and the following, we shall examine more closely what one can say about the irreducible (resp. connected) components of $X \otimes_k K$ when K varies; what precedes already shows that the number of these components is an increasing function of K .

Lemma (4.4.2). — *Let X', X be two topological spaces, $f : X' \rightarrow X$ a continuous map. Suppose the following conditions are satisfied:*

- (i) *f is surjective and open (resp. such that the topological space X identifies canonically with the quotient of X' by the equivalence relation defined by f).*
- (ii) *For every $x \in X$, $f^{-1}(x)$ is irreducible (resp. connected).*

Then for X' to be irreducible (resp. connected), it is necessary and sufficient that X be so.

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Proof. The assertion relative to connectedness is none other than Bourbaki, *Top. gén.*, chap. I^{er}, 3^e éd., §11, no. 3, prop. 7. Let us prove the assertion relative to irreducibility; the condition being trivially necessary since f is surjective, let us show it is sufficient. Let X'_1, X'_2 be two closed parts of X' such that $X' = X'_1 \cup X'_2$, and let X_i ($i = 1, 2$) denote the set of $x \in X$ such that $f^{-1}(x) \subset X'_i$; since $X_i = X - f(X' - X'_i)$, and since f is open, we see that X_i is a closed part of X ($i = 1, 2$). On the other hand, for every $x \in X$, $f^{-1}(x)$ is irreducible by hypothesis, and is the union of the two closed parts $X'_1 \cap f^{-1}(x)$ and $X'_2 \cap f^{-1}(x)$; one of these two parts must therefore be equal to $f^{-1}(x)$, which means that $x \in X_1$ or $x \in X_2$; hence $X = X_1 \cup X_2$, and since X is irreducible, we deduce $X = X_1$ or $X = X_2$, and therefore $X' = X'_1$ or $X' = X'_2$. $\square$

Remark (4.4.3). — If the topology of X is not the quotient of that of X' by the equivalence relation defined by f , it can happen that X and all the fibres $f^{-1}(x)$ are irreducible, without X' being connected: one obtains an example by taking for X an irreducible scheme (for example the affine line $\text{Spec}(k[T])$) over a field k algebraically closed, considering a closed point x_0 of X and taking for X' the space that is the *sum* of $\{x_0\}$ and of the open subspace $X - \{x_0\}$ of X . The same if one replaces the hypothesis “ f open” by “ f closed” (and even “ f proper”) (which still implies that the topology of X is the quotient of that of X' by the relation defined by f), it can happen that X and all the fibres $f^{-1}(x)$ are irreducible without X' being so. Let us take for X the affine line over k , denote by S the product $X \times_k \mathbf{P}$, where $\mathbf{P}$ is the projective line over k ; if x_0 is a closed point of X , t_0 a closed point of $\mathbf{P}$, $p : S \rightarrow X$, $q : S \rightarrow \mathbf{P}$ the projections, we denote by X' the reduced subscheme having as underlying space the closed set $p^{-1}(x_0) \cup q^{-1}(t_0)$, and we take for f the restriction of p to X' ; f is proper but X' is not irreducible.

Theorem (4.4.4). — *Let k be an algebraically closed field, X a k -prescheme. If X is irreducible (resp. connected), then $X \otimes_k K$ is also irreducible (resp. connected) for every extension K of k .*

Proof. One saw in (4.4.1) that the morphism $p : X \otimes_k K \rightarrow X$ is faithfully flat and open; to be able to apply the lemma (4.4.2), it suffices therefore to verify that the fibres $p^{-1}(x)$ are irreducible for every $x \in X$. But the $k(x)$ -prescheme $p^{-1}(x)$ is isomorphic to $\text{Spec}(k(x) \otimes_k K)$ (I, 3.6.2), hence is integral by virtue of the hypothesis on k and of (4.3.9). $\square$

Corollary (4.4.5). — *Let k be an algebraically closed field, X a k -prescheme, K an extension of k , $p : X \otimes_k K \rightarrow X$ the canonical projection. If Z is an irreducible (resp. connected) part of X , $p^{-1}(Z)$ is irreducible (resp. connected); in particular, if X_0 is an irreducible (resp. connected) component of X containing Z , $p^{-1}(X_0)$ is an irreducible (resp. connected) component of $X \otimes_k K$ containing $p^{-1}(Z)$.*

Proof. The second assertion results from (4.4.1) and the first applied by replacing Z by X_0 ; to prove the first, let X' denote the sub-prescheme of $X \otimes_k K$ having X_0 as underlying space, $Z' = p^{-1}(Z)$; the equivalence relation R defined by p on X' is open, hence the same holds for the one it induces on the saturated part Z' , and Z identifies with the quotient space $Z' / R_{Z'}$ (Bourbaki, *Top. gén.*, chap. I^{er}, 3^e éd., §5, no. 2, prop. 4); one can therefore apply the lemma (4.4.2) to Z and Z' , whence the first assertion from (4.4.4). $\square$

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Corollary (4.4.6). — *Under the hypotheses and with the notation of (4.4.5), the map $Z \mapsto p^{-1}(Z)$ is a bijection from the set of irreducible (resp. connected) components of X to the set of irreducible (resp. connected) components of $X \otimes_k K$, whose inverse bijection is $Z' \mapsto p(Z')$.*

4.5. Geometrically irreducible and geometrically connected preschemes

Proposition (4.5.1). — *Let k be a field, X a k -prescheme, Ω an algebraically closed extension of k . Let n (resp. n') be the number of irreducible (resp. connected) components of $X \otimes_k \Omega$; this number is independent of the algebraically closed extension Ω chosen. Moreover, for every extension K of k , the number of irreducible (resp. connected) components of $X \otimes_k K$ is $\leq n$ (resp. $\leq n'$).*

Proof. Indeed, two algebraically closed extensions Ω, Ω' of k can always be considered as sub-extensions of a same extension E of k ; it suffices then to apply (4.4.6) to $X \otimes_k \Omega$ and $X \otimes_k \Omega'$ and to their common extension E of Ω and Ω' to show the first assertion. The second is obtained by taking for Ω an algebraically closed extension of K and using (4.4.1). $\square$

Definition (4.5.2). — The number n (resp. n') of (4.5.1) is called the *geometric number of irreducible* (resp. *connected*) *components* of X (relative to k). If $n = 1$ (resp. $n' \leq 1$), one says that X is a k -prescheme *geometrically irreducible* (resp. *geometrically connected*).

One thus recovers the definition given for a particular case in (III, 4.3.4). By virtue of (4.5.1), the following properties are equivalent:

- a) X is geometrically irreducible (resp. geometrically connected).
- b) For an algebraically closed extension Ω of k , $X \otimes_k \Omega$ is irreducible (resp. connected).
- c) For every extension K of k , $X \otimes_k K$ is irreducible (resp. connected).

(4.5.3). Let X be a k -prescheme, Z a locally closed part of X . If we still denote by Z any of the subpreschemes of X having Z as underlying space, it results from (I, 5.1.8) that for an extension K of k , the number of irreducible (resp. connected) components of $Z \otimes_k K$ is independent of the subprescheme with underlying space Z that one has chosen; one thus also defines the geometric number of irreducible (resp. connected) components of Z as that of any subprescheme of X having Z as underlying space.

Proposition (4.5.4). — *Let X, Y be two k -preschemes, $f : X \rightarrow Y$ a k -morphism surjective irreducible (resp. geometrically connected); it is the same for Y .*

Proof. Indeed, for every extension K of k , $f_{(K)} : X_{(K)} \rightarrow Y_{(K)}$ is surjective (I, 3.5.2), and $X_{(K)}$ being irreducible (resp. connected) by hypothesis, the same holds for $Y_{(K)}$. $\square$

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Definition (4.5.5). — One says that a morphism $f : X \rightarrow Y$ of preschemes is *irreducible* (resp. *connected*), if, for every $y \in Y$, the $k(y)$ -prescheme $f^{-1}(y)$ is geometrically irreducible (resp. geometrically connected).

Proposition (4.5.6). — (i) *Let X be a k -prescheme, K an extension of k . Then the geometric number of irreducible (resp. connected) components of $X_{(K)}$ relative to K is equal to the geometric number of irreducible (resp. connected) components of X relative to k . In particular, for the K -prescheme $X_{(K)}$ to be geometrically irreducible (resp. geometrically connected), it is necessary and sufficient that the k -prescheme X be so.*
(ii) *Let $f : X \rightarrow Y, g : Y' \rightarrow Y$ be two morphisms, and set $X' = X \times_Y Y', f' = f_{(Y')} : X' \rightarrow Y'$. If f is irreducible (resp. connected), the same holds for f' . The converse is true if g is surjective.*

Proof. (i) If Ω is an algebraically closed extension of K , we have $X \otimes_k \Omega = X_{(K)} \otimes_K \Omega$, whence the assertion.

(ii) For every $y' \in Y'$, if $y = g(y')$, we have $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$ (I, 3.6.4), hence the two assertions result from (i). $\square$

Proposition (4.5.7). — *Let $f : X \rightarrow Y$ be a surjective irreducible (resp. connected) morphism, $g : Y' \rightarrow Y$ a morphism, and set $X' = X \times_Y Y'$. If moreover Y' is irreducible (resp. connected), and f universally open (resp. flat and quasi-compact, or universally open, or universally closed), then X' is irreducible (resp. connected).*

Proof. The properties of f that one considers being all stable by base change, one can reduce to the case where $Y' = Y$; it then suffices to apply (4.4.2) (taking into account (2.3.12)). $\square$

Corollary (4.5.8). — *Let X, Y be two k -preschemes.*

- (i) *If X is geometrically irreducible (resp. geometrically connected) and Y is irreducible (resp. connected), then $X \times_k Y$ is irreducible (resp. connected).*
- (ii) *If X and Y are geometrically irreducible (resp. geometrically connected), the same holds for $X \times_k Y$.*

Proof. (i) The structural morphism $X \rightarrow \text{Spec}(k)$ is surjective and universally open (2.4.9) and in addition irreducible (resp. connected); it suffices therefore to apply (4.5.7).

(ii) If Ω is an algebraically closed extension of k , we have $(X \times_k Y)_{(\Omega)} = X_{(\Omega)} \times_{\Omega} Y_{(\Omega)}$ (III, 3.3.10); by hypothesis $X_{(\Omega)}$ and $Y_{(\Omega)}$ are geometrically irreducible (resp. geometrically connected) Ω -preschemes (4.5.6), and it suffices therefore to apply (i). $\square$

Proposition (4.5.9). — *Let X be a k -prescheme. The following conditions are equivalent:*

- a) *X is geometrically irreducible (that is to say, for every extension K of k , $X \otimes_k K$ is irreducible).*
- b) *For every finite separable extension K of k , $X \otimes_k K$ is irreducible.*
- c) *X is irreducible, and if x is its generic point, $k(x)$ is a primary extension of k .*

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Proof. It is clear that a) implies b). To see that b) implies c), consider a finite separable extension K of k ; if $p : X \otimes_k K \rightarrow X$ is the canonical projection, the maximal points of $X \otimes_k K$ are those of the fibre $p^{-1}(x) = \text{Spec}(k(x) \otimes_k K)$ (2.3.4) and (0_I, 2.1.8); to say that this fibre is irreducible for every

finite separable extension K of k amounts to saying that $k(y)$ is a primary extension of k , by virtue of (4.3.2). Conversely, if $c)$ is satisfied, the same reasoning (taking into account (4.3.2)) shows that for every extension K of k , $X \otimes_k K$ is irreducible, hence $c)$ implies $a)$. $\square$

Corollary (4.5.10). — *Let X be a k -prescheme irreducible, x its generic point, k' the algebraic separable closure of k in $k(x)$, k'' a Galois extension of k (of finite degree or not) containing k' . Suppose k' of finite degree over k ; then the irreducible components of $X \otimes_k k''$ are geometrically irreducible; their number, equal to $[k' : k]$ and is also the geometric number of irreducible components of X .*

Proof. As in (4.5.9), one is reduced to considering the maximal points of the fibre $\text{Spec}(k(x) \otimes_k k'')$; by virtue of (4.3.4), they are equal in number to $[k' : k]$ and the residue fields at these points are primary extensions of k'' , which, taking into account (4.5.9), proves the corollary. $\square$

Corollary (4.5.11). — *Let X be a k -prescheme. Suppose that X has only a finite number of maximal points x_i ($1 \leq i \leq r$) and that for each i , the algebraic separable closure k'_i of k in $k(x_i)$ is a finite extension of k . Then there exists a finite separable extension L of k such that the irreducible components of $X \otimes_k L$ are geometrically irreducible; their number, equal to $\sum_i [k'_i : k]$, is the geometric number of irreducible components of X .*

Proof. The maximal points of $X \otimes_k L$ are those of the various fibres $\text{Spec}(k(x_i) \otimes_k L)$ (2.3.4) and (0_I, 2.1.8), applied to the irreducible components of X and to their inverse images in $X \otimes_k L$. It suffices therefore to take for L a Galois extension finite of k containing all the k'_i and to apply (4.5.10). $\square$

One will note that (4.5.11) applies in particular to every k -prescheme X of finite type over k , for X is then Noetherian, hence has only a finite number of irreducible components, and each of the $k(x_i)$ is a finite type extension of k , hence the same holds for the algebraic separable closure of k in $k(x_i)$ ([?], p. 6-06, lemma 5).

Remarks (4.5.12). — (i) The notions defined in (4.5.2) depend on the base field k . For short, and whenever it is necessary to mention the base field, one says “ k -irreducible” (resp. “ k -connected”) instead of “geometrically irreducible (resp. connected) relative to k ”. One will note that this terminology, quite natural in the context of schemes, is exactly the *opposite* of that of Weil: for this author, “ k -irreducible” (resp. “ k -connected”, “ k -normal”, etc.) denotes an *intrinsic* property of a prescheme X , independent of the choice of the field k taken as “base field” (in other words, of the morphism $X \rightarrow \text{Spec}(k)$ chosen); we express these properties by saying simply that X is irreducible (resp. connected, normal, etc.). Weil says on the other hand “absolutely irreducible”, “absolutely connected”, “absolutely normal”, etc., for the corresponding notions relative to the prescheme $X \otimes_k \Omega$, where Ω is an algebraically closed suitable extension of k . We will avoid in what follows using the qualifier “absolute”, which may lend to confusion, and whenever we use the abbreviated terminology introduced above (in conflict with the accepted terminology, see above), we will refer to (4.5.12) to avoid all ambiguity.

(ii) The proposition (4.5.9) gives a “birational” criterion (that is, depending only on the residue field at the generic point) for a k -prescheme X to be geometrically irreducible. There is no analogous criterion for X to be geometrically connected.

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Proposition (4.5.13). — *Let X, Y be two k -preschemes, $f : Y \rightarrow X$ a k -morphism. Suppose Y geometrically connected and non-empty, and X connected. Then X is geometrically connected.*

Proof. Let Ω be an algebraic closure of k , $X' = X_{(\Omega)}$, $Y' = Y_{(\Omega)}$, $f' = f_{(\Omega)} : Y' \rightarrow X'$; note that the morphisms $p : X' \rightarrow X$ and $q : Y' \rightarrow Y$ are open (2.4.10) and closed since Ω is an algebraic extension of k (II, 6.1.10). Hence $U = p(U')$ is open and closed, non-empty in X , and hence is equal to X . Since Y is non-empty, one concludes (I, 3.4.7) that $V' = f'^{-1}(U')$ is non-empty; furthermore, it is an open and closed part of Y' , and since this last space is connected by hypothesis, $V' = Y'$. We deduce that $U' = X'$, without which the open and closed set $X' - U'$ would show, applying the same reasoning to $X' - U'$, that $f'^{-1}(X' - U') = Y'$, which is absurd since Y' is non-empty. $\square$

Corollary (4.5.13.1). — (i) *Let X, Y be two k -preschemes, $f : Y \rightarrow X$ a k -morphism. If Y is geometrically connected and non-empty, the connected component X_0 of X containing $f(Y)$ is geometrically connected.*

- (ii) Let X be a k -prescheme. If Y is an irreducible component of X which is geometrically irreducible, the connected component X_0 of X containing Y is geometrically connected.

Proof. (i) One can reduce to the case where Y is reduced, so that f factorizes as $f : Y \xrightarrow{g} X_0 \xrightarrow{j} X$, denoting X_0 as the reduced subprescheme having X_0 as underlying space (I, 5.2.2); it suffices to apply (4.5.13) to g .

(ii) Considering a prescheme having Y as underlying space, and remarking that Y is *a fortiori* geometrically connected, it suffices to apply (i) to the canonical injection $Y \rightarrow X$. $\square$

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Corollary (4.5.14). — Let X be a k -prescheme. If x is a point of X such that $k(x)$ is a primary extension of k (in particular if x is a rational point of X), the connected component of x in X is geometrically connected.

Proof. It suffices to apply (4.5.13.1) to $Y = \text{Spec}(k(x))$ and to the canonical morphism $Y \rightarrow X$, taking into account that Y is geometrically irreducible by (4.5.9). $\square$

Proposition (4.5.15). — Let X be a k -prescheme, x a point of X , k' the algebraic separable closure of k in $k(x)$. Suppose:

- (i) X is connected.
- (ii) k' is a finite extension of k .

Then the geometric number of connected components of X is $\leq [k' : k]$, and if k'' is a Galois extension finite of k containing k' , the connected components of $X \otimes_k k''$ are geometrically connected.

Proof. There exist finite Galois extensions k'' of k containing k' by virtue of condition (ii). Set $X'' = X \otimes_k k''$; the projection morphism $p : X'' \rightarrow X$ is faithfully flat and finite, hence closed (II, 6.1.10) and it follows (2.3.6), (ii) the image by p of every connected component X''_α of X'' is equal to X ; X''_α therefore contains a point $x''_\alpha \in p^{-1}(x)$. But $p^{-1}(x)$ has a number of points equal to the number of irreducible components of $\text{Spec}(k(x) \otimes_k k'')$ (I, 3.4.9), that is $[k' : k]$ (4.3.4), and *a fortiori* the number of connected components of X'' is $\leq [k' : k]$. For every $x''_\alpha \in p^{-1}(x)$, $k(x''_\alpha)$ is a primary extension of k'' by (4.3.5) and (I, 3.4.9). One can therefore apply the corollary (4.5.14) to x''_α and X'' , which proves that all the connected components of X'' are geometrically connected. $\square$

Corollary (4.5.16). — Let X be a k -prescheme, (X_α) the family of its connected components, and for every α , let $x_\alpha \in X_\alpha$. Suppose:

- (i) The family (X_α) is finite.
- (ii) For every α , the algebraic separable closure k'_α of k in $k(x_\alpha)$ is a finite extension of k .

Then the geometric number of connected components of X is at most $\sum_\alpha [k'_\alpha : k]$, and there exists a finite separable extension k'' of k such that all the connected components of $X \otimes_k k''$ are geometrically connected.

Proof. Noting that the connected components of X are open by virtue of condition (i), it suffices to apply to each of the subpreschemes of X induced on the open sets X_α the result of (4.5.15); for an extension k'' answering the question, it suffices to take a Galois extension finite of k containing all the k'_α . $\square$

Corollary (4.5.17). — Suppose that the k -prescheme X contains a point x such that the algebraic separable closure of k in $k(x)$ is finite over k . For X to be geometrically connected, it is necessary and sufficient that for every finite separable extension K of k , $X \otimes_k K$ be connected.

Remark (4.5.18). — We shall see in §8 (4.8.5) that the conclusion of (4.5.17) is still valid if, instead of supposing verified the condition of the statement, one supposes that X is *quasi-compact*.

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Proposition (4.5.19). — Let X be a k -prescheme, k' an algebraically closed extension of k , $X' = X \otimes_k k'$, $p : X' \rightarrow X$ the canonical projection. Suppose $p^{-1}(Z)$ is contained in a single irreducible component X'_0 of X' . Then $a = X_0 = p(X'_0)$ is an irreducible component of X containing Z , and moreover X_0 is geometrically irreducible. If moreover X'_0 is the only irreducible component of X' that meets Z' , then X_0 is the only irreducible component of X that meets Z .

Proof. To show that $X'_0 = p^{-1}(X_0)$, we will apply the following: $\square$

Lemma (4.5.19.1). — *Let T, T' be two preschemes, $p : T' \rightarrow T$ a morphism; consider the prescheme $T'' = T' \times_T T'$, and let $p_1 : T'' \rightarrow T', p_2 : T'' \rightarrow T'$ be the canonical projections. For an open subset U' of T' to be of the form $p^{-1}(U)$, where $U \subset T$, it is necessary and sufficient that $p_1^{-1}(U') = p_2^{-1}(U')$.*

Proof. The structural morphism $q : T'' \rightarrow T$ is equal to $p \circ p_1$ and to $p \circ p_2$ by definition, hence the relation is necessary. Conversely, suppose it is verified; if u' is a point of U' , t a point of $p^{-1}(p(u'))$; since $p(t') = p(u')$, there exists a point $t'' \in T''$ such that $p_1(t'') = u', p_2(t'') = t'$ (I, 3.4.7); hence $t'' \in p_1^{-1}(U') = p_2^{-1}(U')$ by hypothesis, hence $t' = p_2(t'') \in U'$, which proves the lemma. $\square$

This lemma being established, to show that $X'_0 = p^{-1}(X_0)$, let us form the product $X'' = X' \times_X X'$ (relative to the morphism $p : X' \rightarrow X$); if p_1 and p_2 are the canonical projections, it suffices to show that $p_1^{-1}(X'_0) = p_2^{-1}(X'_0)$. Now, if we set $S = \text{Spec}(k), S' = \text{Spec}(k')$, observe that if $S'' = S' \times_S S' = \text{Spec}(k' \otimes_k k')$, we can also write $X'' = X' \times_{S'} S''$. If $\varphi'' : X'' \rightarrow S''$ is the structural morphism, it suffices to prove that for every $s'' \in S'', p_1^{-1}(X'_0) \cap \varphi''^{-1}(s'') = p_2^{-1}(X'_0) \cap \varphi''^{-1}(s'')$. If one sets $T = \text{Spec}(k(s'')), U = X'' \times_{S''} T$, and if $v : U \rightarrow X''$ is the canonical projection, and since we have the diagram

$$\begin{array}{ccccccc} X & \xleftarrow{p_1} & X' & \xleftarrow{v} & X'' & \xleftarrow{\quad} & U \\ \uparrow & & \uparrow & & \uparrow & & \uparrow \\ S & \xleftarrow{\quad} & S' & \xleftarrow{\quad} & S'' & \xleftarrow{\quad} & T \end{array}$$

By construction, U_1 and U_2 contain both the set $w^{-1}(Z)$, where $w = p \circ p_1 \circ v$, and the two irreducible components of X' containing Z' ; they are therefore all equal to X'_0 by (4.4.5), and finally $U_1 = U_2$.

Since p is surjective and X'_0 dominates an irreducible component of X , $X_0 = p(X'_0)$ is an irreducible component of X containing Z , and since $p^{-1}(X_0)$ is an irreducible component of $X \otimes_k K$, X_0 is geometrically irreducible.

To prove the last assertion, let X_1 be a second irreducible component of X meeting Z . Taking $z \in Z \cap X_1$, one sees that there exists an irreducible component of X' meeting Z' and dominating X_1 (2.3.5); but since X'_0 is by hypothesis the only irreducible component of X' meeting Z' , we have necessarily $X_1 = X_0$.

Remark (4.5.20). — The fact that Z' is contained in a single irreducible component X'_0 of X' does *not* imply that $X_0 = p(X'_0)$ is the only irreducible component of X containing Z .

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Proposition (4.5.21). — *Let k be a separably closed field, and let k' be an algebraic closure of k . For every k -prescheme X , the canonical projection $X \otimes_k k' \rightarrow X$ is a universal homeomorphism. In particular the irreducible (resp. connected) components of X are geometrically irreducible (resp. geometrically connected).*

Proof. By definition, k' is a purely inseparable extension of k ; the morphism $\text{Spec}(k') \rightarrow \text{Spec}(k)$ is integral, surjective, and radical, hence the first assertion results from (2.4.5), (i)); the second follows from (4.5.1). $\square$

4.6. Geometrically reduced algebraic preschemes

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Proposition (4.6.1). — *Let k be a field, X a k -prescheme, Ω a perfect extension of k . The following conditions are equivalent:*

- a) *For every reduced k -prescheme S , $X \times_k S$ is reduced.*
- b) *For every extension K of k , $X \otimes_k K$ is reduced.*
- c) *$X \otimes_k \Omega$ is reduced.*
- d) *For every finite purely inseparable extension k' of k , $X \otimes_k k'$ is reduced.*
- e) *X is reduced, and for every irreducible component X_α of X , with generic point x_α , $k(x_\alpha)$ is a separable extension of k .*

Proof. It is trivial that $a) \Rightarrow b) \Rightarrow c)$; since k' can be considered as a sub-extension of Ω , we saw (4.4.1) that $c)$ implies $d)$. Let us show that $d)$ implies $e)$. Taking $k' = k$, one sees first that X is reduced; to prove that $k(x_\alpha)$ is separable over k , it suffices to show that $k(x_\alpha) \otimes_k k'$ is a reduced ring for every finite purely inseparable extension k' of k (4.3.5). One can restrict to the case where $X = \text{Spec}(A)$ is affine, A being a reduced k -algebra; then by hypothesis the ring $A \otimes_k k'$ is reduced (I, 5.1.4) and $k(x_\alpha) \otimes_k k'$ is a ring of fractions of this ring, hence is reduced (0I, 1.2.8). Finally, to prove that $e)$ implies $a)$, one can restrict to the case where $X = \text{Spec}(A)$, $S = \text{Spec}(B)$ are affine, A and B being k -algebras. The $k(x_\alpha)$ are the fields of fractions of the quotient rings $A/\mathfrak{p}_\alpha$, where $\mathfrak{p}_\alpha = \mathfrak{i}_{x_\alpha}$ are the minimal prime ideals of A ; since by hypothesis $\bigcap_\alpha \mathfrak{p}_\alpha = 0$, A is contained in the product $\prod A/\mathfrak{p}_\alpha$, hence in $\prod k(x_\alpha)$; one can similarly consider B as a subalgebra of a product $\prod L_\beta$ of extensions of k . It follows that $A \otimes_k B$ identifies with a subalgebra of $(\prod_\alpha k(x_\alpha)) \otimes_k (\prod_\beta L_\beta)$, and this tensor product itself identifies with a subalgebra of $\prod_{\alpha,\beta} (k(x_\alpha) \otimes_k L_\beta)$ (Bourbaki, Alg., chap. II, 3^e éd., §7, no. 7, prop. 15). Now by hypothesis the $k(x_\alpha) \otimes_k L_\beta$ are reduced (4.3.5), hence the same holds for $A \otimes_k B$. $\square$

Definition (4.6.2). — When the equivalent conditions a) to e) of the proposition (4.6.1) are satisfied, one says that X is *geometrically reduced* (or *separable*, or *universally reduced*) over k . One says that a k -prescheme X is *geometrically integral* over k if, for every extension K of k , $X \otimes_k K$ is integral; this amounts (by virtue of (4.6.1)) to saying that X is separable and geometrically irreducible over k .

One says that a k -algebra A (commutative) is *separable* if $\text{Spec}(A)$ is separable over k ; this means therefore that for every extension K of k , the ring $A_{(K)} = A \otimes_k K$ is reduced. One will note that this definition coincides with that of Bourbaki, Alg., chap. VIII, §7, no. 5, def. 1, when A is a ring of finite rank over k (loc. cit., cor. of prop. 7), but not in general, a k -algebra being able to have zero radical $\neq 0$ even if it is integral.

Corollary (4.6.3). — *Let X be a k -prescheme integral; for X to be geometrically reduced (resp. geometrically integral) over k , it is necessary and sufficient that its field of rational functions $R(X)$ be a separable (resp. separable and primary) extension of k .*

Proof. This follows immediately from (4.5.9) and (4.6.1). $\square$

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Corollary (4.6.4). — *Let X be a k -prescheme reduced. Then, for every separable extension k' of k , $X' = X \otimes_k k'$ is reduced.*

Proof. It suffices to apply the equivalence of a) and e) in (4.6.1), replacing X by $\text{Spec}(k')$ and S by X . $\square$

Proposition (4.6.5). — (i) *Let X be a k -prescheme, K an extension of k . For X to be geometrically reduced (resp. geometrically integral) over k , it is necessary and sufficient that $X \otimes_k K$ be geometrically reduced (resp. geometrically integral) over K .*
(ii) *Let X, Y be two k -preschemes. If X and Y are geometrically reduced (resp. geometrically integral) over k , the same holds for $X \times_k Y$.*

Proof. Assertion (i) is a trivial consequence of the definitions and of (4.4.1). For the part of assertion (ii) concerning separability, observe that if Ω is an algebraically closed extension of k , $(X \times_k Y) \otimes_k \Omega = X \times_k (Y \otimes_k \Omega)$; since $Y \otimes_k \Omega$ is reduced, the same holds for $X \times_k (Y \otimes_k \Omega)$ by virtue of (4.6.1), a). The rest of assertion (ii) results from what precedes and from (4.5.8), (ii). $\square$

Proposition (4.6.6). — *Let X be a k -prescheme of finite type. There exists a finite purely inseparable extension k' of k such that $(X_{(k')})_{\text{red}}$ is geometrically reduced over k' .*

Proof. Since X can be covered by a finite number of open affines U_i , one can restrict to showing that for each i , there is a finite purely inseparable extension k_i of k such that $(U_i \otimes_k k_i)_{\text{red}}$ is geometrically reduced over k_i ; one can then suppose that the k_i are contained in a same finite purely inseparable extension k' of k ; $(U_i \otimes_k k')_{\text{red}}$ identifies with $(U_i \otimes_k k_i)_{\text{red}} \otimes_{k_i} k'$ by virtue of (I, 5.1.8), hence is geometrically reduced over k' by hypothesis, and the same holds for $(X \otimes_k k')_{\text{red}}$. Suppose therefore $X = \text{Spec}(A)$, where A is a k -algebra of finite type, and let Ω be the algebraic closure of k . Denote by $\mathfrak{N}'$ the nilradical of $A \otimes_k \Omega$; since $A \otimes_k \Omega$ is a Noetherian ring, $\mathfrak{N}'$ is an ideal generated by a finite number of elements of the form $y_i = \sum_j x_{ij} \otimes \zeta_{ij}$ where $x_{ij} \in A$, $\zeta_{ij} \in \Omega$. Let K be a finite sub-extension of Ω containing the ζ_{ij} , and $\mathfrak{N}$ the ideal of $A \otimes_k K$ generated by the y_i ($A \otimes_k K$ being identified with a subring of $A \otimes_k \Omega$); it is clear that the y_i are nilpotent in $A \otimes_k K$; on the other hand, since $\mathfrak{N} \otimes_K \Omega = \mathfrak{N}'$ (and hence $\mathfrak{N} \cap (A \otimes_k K) = \mathfrak{N}$), $\mathfrak{N}$ contains the nilradical of $A \otimes_k K$, hence it is equal to this nilradical. Since $(A \otimes_k \Omega)/\mathfrak{N}' = ((A \otimes_k K)/\mathfrak{N}) \otimes_K \Omega$, one sees that $(X \otimes_k K)_{\text{red}} \otimes_K \Omega$ is reduced, and by (4.6.1), $(X_{(K)})_{\text{red}}$ is geometrically reduced over K . Replacing K by the quasi-Galois extension over k that it generates in Ω , one sees by (I, 5.1.8) that one can suppose moreover K quasi-Galois over k , hence a separable extension of a finite purely inseparable extension k' of k (Bourbaki, Alg., chap. V, §10, no. 9, prop. 14). By virtue of (I, 5.1.8), $(X_{(k')})_{\text{red}} \otimes_{k'} \bar{k}$ is isomorphic to $(X_{(K)})_{\text{red}}$, hence is geometrically reduced; the same holds for $(X_{(k')})_{\text{red}} \otimes_{k'} \bar{k} \cdot K$ by (4.6.5), (i); replacing K again we are done. $\square$

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Corollary (4.6.7). — *If K is an extension of finite type of k , there exists a finite purely inseparable extension k' of k such that the residue fields of the semi-local ring $K \otimes_k k'$ are separable over k' .*

Proof. It suffices to apply (4.6.6) to $X = \text{Spec}(A)$, where A is a k -algebra of finite type whose field of fractions is K . $\square$

Corollary (4.6.8). — *Let X be a k -prescheme of finite type. There exists a finite extension k' of k such that $(X_{(k')})_{\text{red}}$ is geometrically reduced over k' and the irreducible components of $X_{(k')}$ are geometrically irreducible and the connected components of $X_{(k')}$ are geometrically connected.*

Proof. This follows immediately from (4.6.6), (4.5.10) and (4.5.15). $\square$

Definition (4.6.9). — Let k be a field, X a k -prescheme, x a point of X . One says that X is *geometrically reduced* (or *separable*) (resp. *geometrically punctually integral*) at the point x over k , if, for every extension k' of k and every point x' of $X' = X \otimes_k k'$ above x , X' is reduced (resp. integral) at x' (i.e., $\mathcal{O}_{X',x'}$ is reduced (resp. integral)). One says that X is *geometrically punctually integral* if X is geometrically punctually integral at all its points, that is, if for every extension k' of k , all the local rings of $X \otimes_k k'$ are integral (in which case one says also that $X \otimes_k k'$ is *punctually integral*).

Note that, for X to be geometrically reduced over k (4.6.2), it is necessary and sufficient that it be geometrically reduced over k at every point $x \in X$.

Proposition (4.6.10). — *Let k be a field, X a k -prescheme, k' an extension of k , x' a point of $X' = X \otimes_k k'$, x its image in X . For X to be geometrically punctually reduced (resp. geometrically punctually integral) at x over k , it is necessary and sufficient that X' be so at x' over k' . In particular, for X to be geometrically punctually reduced (resp. geometrically punctually integral) over k , it is necessary and sufficient that X' be so over k' .*

Proof. It suffices to prove the first assertion. The condition being obviously necessary, let us prove it is sufficient. Suppose therefore X' geometrically punctually reduced (resp. geometrically punctually integral) at x' over k' , and let us prove that X is so at x over k , that is, for every extension k'' of k and every point x'' of $X'' = X \otimes_k k''$ above x , $\mathcal{O}_{X'',x''}$ is reduced (resp. integral). For this, note that there exists a point z of $X' \times_X X''$ projecting to x' and x'' (I, 3.4.7). Let s be the point of $\text{Spec}(k' \otimes_k k'')$ image of z (I, 3.4.9); set $K = k(s)$, $Z = X \otimes_k K$, so that K can be considered as a composite extension of k' and k'' , and z as a point of Z whose image in X' (resp. X'') is x' (resp. x''). Since X' is geometrically

punctually reduced (resp. geometrically punctually integral) at x' over k' , the ring $\mathcal{O}_{Z,z}$ is reduced (resp. integral), and since $\mathcal{O}_{Z,z}$ is an $\mathcal{O}_{X'',x''}$ -module faithfully flat, it follows that $\mathcal{O}_{X'',x''}$ is reduced (resp. integral) (0I, 6.5.1). $\square$

Corollary (4.6.11). — Suppose k perfect (resp. algebraically closed). For X to be geometrically punctually reduced (resp. geometrically punctually integral) at the point x over k , it is necessary and sufficient that $\mathcal{O}_{X,x}$ be reduced (resp. integral).

Proof. This is obviously necessary. Conversely, if this condition is satisfied, then, for every extension k' of k , $\text{Spec}(\mathcal{O}_{X,x}) \otimes_k k'$ is reduced (4.6.4) (resp. integral (4.4.4)); hence, for every point x' of $X' = X \otimes_k k'$ above x , $\mathcal{O}_{X',x'}$, which is isomorphic to a local ring of $\text{Spec}(\mathcal{O}_{X,x}) \otimes_k k'$, is reduced (resp. integral). $\square$

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Proposition (4.6.12). — Let k be a field, X a k -prescheme, x a point of X , Ω a perfect extension of k . The following conditions are equivalent:

- a) X is geometrically reduced at x over k , that is, for every extension k' of k and every point x' of $X' = X \otimes_k k'$ above x , $\mathcal{O}_{X',x'}$ is reduced.
- b) The prescheme $X \otimes_k \Omega$ is reduced at every point above x .
- c) For every finite purely inseparable extension k' of k , $X' = X \otimes_k k'$ is reduced at the unique point above x .
- d) $\text{Spec}(\mathcal{O}_{X,x})$ is geometrically reduced over k .
- e) $\mathcal{O}_{X,x}$ is reduced, and for every irreducible component Z of X containing x , with generic point z , $k(z)$ is a separable extension of k .

Proof. The implications $d) \Rightarrow a) \Rightarrow b) \Rightarrow c)$ are immediate, taking into account for the last that every purely inseparable extension of k is isomorphic to a sub-extension of Ω . Let us prove that $c)$ implies $d)$. It suffices to show that for every finite purely inseparable extension k' of k , $\mathcal{O}_{X,x} \otimes_k k'$ is reduced (4.6.1); now this ring is the local ring of $X' = X \otimes_k k'$ at the unique point x' of X' above x (I, 3.5.8) and (3.5.7); it is therefore reduced by hypothesis $c)$. Finally, the equivalence of $d)$ and $e)$ results from the equivalence of $b)$ and $e)$ in (4.6.1), since the irreducible components of X containing x correspond bijectively to the irreducible components of $\text{Spec}(\mathcal{O}_{X,x})$ (I, 2.4.2). $\square$

Corollary (4.6.13). — Under the hypotheses of (4.6.12), suppose moreover that X is locally Noetherian. Then conditions $a)$ to $e)$ of (4.6.12) are also equivalent to the following:

- f) There exists a neighbourhood U of x that is geometrically reduced over k .

Proof. Indeed, this condition trivially implies that X is geometrically reduced at x over k . Conversely, if this is so, since $\mathcal{O}_{X,x}$ is reduced, there exists a reduced open neighbourhood U of x in X (I, 6.1.13); taking U small enough, we can moreover suppose that U meets no irreducible components of X not containing x . Then the criterion (4.6.12, $e)$) proves that U is geometrically reduced over k , taking into account the criterion (4.6.1, $e)$). $\square$

It follows from (4.6.13) that when X is locally Noetherian, the set of $x \in X$ at which X is geometrically reduced over k is *open*, and it is the largest open subprescheme of X that is geometrically reduced over k .

(4.6.14). It results from (4.6.10) and (4.6.11) that if Ω is an algebraically closed extension of k , it amounts to the same to say that X is geometrically punctually integral at x , or that $X \otimes_k \Omega$ is integral at every point above x .

For X to be geometrically punctually integral at x , it is necessary that X be integral at x , and that if z is the generic point of the unique irreducible component of X containing x , $k(z)$ be a separable extension of k ; this follows from (4.6.10). But these conditions are *not sufficient*, as shown by the example in (4.5.12), (ii)). A sufficient but non-necessary condition is that X be geometrically reduced at x and that x belong to only a single irreducible component of X , and that this component be moreover geometrically irreducible; this follows from (4.6.1) and (4.5.9).

Recall that if a prescheme *locally Noetherian* is punctually integral, it is *locally integral* (I, 6.1.13). If a k -prescheme X , locally of finite type over k , is geometrically punctually integral over k , it follows

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from this remark that, for every extension k' of k , $X \otimes_k k'$ is *locally integral* (one says in this case that X is *geometrically locally integral*).

Proposition (4.6.15). — (i) Let k be a field, X a k -prescheme of finite type. For X to be geometrically punctually integral, it is necessary and sufficient that it be geometrically reduced and that the geometric number of its irreducible components be equal to the geometric number of its connected components.
 (ii) Let X be a prescheme locally of finite type over k . If X is geometrically punctually integral at a point x , there exists an open neighbourhood U of x that is geometrically punctually integral. In other words, the set of $x \in X$ at which X is geometrically punctually integral is open.

Proof. (i) Given an algebraically closed extension Ω of k , $X_{(\Omega)}$ is punctually integral (or locally integral, which amounts to the same since $X_{(\Omega)}$ is Noetherian) if and only if it is reduced and the number of its connected components is equal to the number of its irreducible components (I, 6.1.10). It then suffices to apply (4.5.1), (4.6.1) and the first remark of (4.6.14).

(ii) The question being local on X , one can restrict to the case where X is a prescheme of finite type over k . Moreover, one knows that there is an open neighbourhood of x in X which is geometrically reduced (4.6.13), so one can suppose X geometrically reduced. There then exists a finite extension k' of k such that $X' = X \otimes_k k'$ is reduced and its irreducible components are geometrically irreducible (4.6.8). Let $p : X' \rightarrow X$ be the canonical projection, which is a finite surjective morphism, and let x'_j ($1 \leq j \leq r$) be the points of $p^{-1}(x)$; by hypothesis, X' is integral at each x'_j , hence each x'_j has an open integral neighbourhood V'_j in X' . Since p is closed (II, 6.1.10), each of the $p(V'_j)$ is a neighbourhood of x , hence there exists an open neighbourhood U of x such that $p^{-1}(U)$ is contained in the union of the V'_j , and is therefore integral at each of its points, hence locally integral. The irreducible components of $p^{-1}(U)$ are therefore open and pairwise disjoint, and since they are geometrically irreducible (4.5.9), $p^{-1}(U) = U \otimes_k k'$ is geometrically punctually integral by (i), and the same holds for U by definition. $\square$

Proposition (4.6.16). — Let k be a field, X a k -prescheme locally Noetherian, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. The following conditions are equivalent:

- a) For every extension of finite type k' of k , setting $X' = X \otimes_k k'$, $\mathcal{F}' = \mathcal{F} \otimes_k k'$, $\mathcal{F}'$ is a reduced $\mathcal{O}_{X'}$ -Module (3.2.2).
- b) For every finite purely inseparable extension k' of k , $\mathcal{F}'$ is a reduced $\mathcal{O}_{X'}$ -Module.
- c) $\mathcal{F}$ is reduced, and if $\mathcal{J}$ is the Ideal of $\mathcal{O}_X$ annihilating $\mathcal{F}$, the closed subprescheme defined by $\mathcal{J}$ is geometrically reduced over k .

Moreover, if X is locally of finite type over k , these conditions are also equivalent to:

- d) For every extension k' of k (or for an algebraically closed extension k' of k), $\mathcal{F}'$ is a reduced $\mathcal{O}_{X'}$ -Module.

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Proof. It is clear that a) implies b). Condition b) implies first that $\mathcal{F}$ is reduced: this means (3.2.3) that if Y is the closed subprescheme of X defined by the coherent Ideal $\mathcal{J}$ of $\mathcal{O}_X$, annihilator of $\mathcal{F}$, Y is reduced, and there is an $\mathcal{O}_Y$ -Module coherent without torsion $\mathcal{G}$ of rank 1 on every irreducible component of Y , such that $j_*(\mathcal{G}) = \mathcal{F}$, $j : Y \rightarrow X$ being the canonical injection. Now, for every extension k' of k , the annihilator of $\mathcal{F}'$ is $\mathcal{J}' = \mathcal{J} \otimes_k k'$, the morphism projection $X' \rightarrow X$ being flat (2.1.11); the closed subprescheme of X' defined by $\mathcal{J}'$ is $Y' = Y \otimes_k k'$, and if $j' : Y' \rightarrow X'$ is the canonical injection, and $\mathcal{G}' = \mathcal{G} \otimes_k k'$, we have $\mathcal{F}' = j'_*(\mathcal{G}')$. Note also that, for every extension k' of k for which X' is locally Noetherian, $\mathcal{F}'$ is without immersed associated prime cycle (4.2.7). Moreover, every maximal point y' of Y' is above a maximal point y of Y ; since $\mathcal{G}_{y'}$ is isomorphic to $\mathcal{O}_{Y,y}$, $\mathcal{G}'_{y'}$ is isomorphic to $\mathcal{O}_{Y',y'}$. One sees therefore (by (3.2.3)) that, whether $\mathcal{F}'$ is reduced or the prescheme Y' is reduced, it amounts to the same thing. The fact that b) implies c) and that c) implies a) are therefore consequences of (4.6.1, d) and b)); in a), the hypothesis that k' is an extension of finite type of k is only used to ensure that X' is locally Noetherian. When X is locally of finite type over k , X' is locally Noetherian for every extension k' of k , which proves in this case the equivalence of d) and the other conditions. $\square$

Definition (4.6.17). — When $\mathcal{F}$ satisfies the equivalent conditions of (4.6.16), one says that $\mathcal{F}$ is *geometrically reduced* (or *separable*) over k .

Proposition (4.6.18). — Let k be a field, X a k -prescheme locally Noetherian, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. The following conditions are equivalent:

- a) For every extension of finite type k' of k , setting $X' = X \otimes_k k'$, $\mathcal{F}' = \mathcal{F} \otimes_k k'$, $\mathcal{F}'$ is an integral $\mathcal{O}_{X'}$ -Module (3.2.4).
- b) For every finite extension k' of k , $\mathcal{F}'$ is an integral $\mathcal{O}_{X'}$ -Module.
- c) $\mathcal{F}$ is reduced (or integral), and if $\mathcal{J}$ is the Ideal of $\mathcal{O}_X$ annihilating $\mathcal{F}$, the closed subprescheme defined by $\mathcal{J}$ is geometrically integral over k .

Moreover, if X is locally of finite type over k , these conditions are also equivalent to:

- d) For every extension k' of k (or for an algebraically closed extension k' of k), $\mathcal{F}'$ is an integral $\mathcal{O}_{X'}$ -Module.

Proof. Conditions a), b) and c) imply that $\mathcal{F}$ is integral, hence $\text{Ass}(\mathcal{F})$ reduces to a single point x , which is the generic point of $\text{Supp}(\mathcal{F})$. For every extension k' of k for which X' is locally Noetherian, the associated prime cycles of $\mathcal{F}'$ are therefore non-immersed, and are the maximal points of $\text{Supp}(\mathcal{F}')$, which are all above x . On the other hand, by virtue of (4.6.16), each of the conditions a), b), c) implies that $\mathcal{F}$ is geometrically reduced; it is trivial that a) implies b), and b) implies that $\text{Supp}(\mathcal{F})$ is geometrically irreducible (4.5.9), hence c), since the subprescheme Y defined by the annihilator $\mathcal{J}$ of $\mathcal{F}$ is then geometrically reduced and geometrically irreducible, hence geometrically integral; this shows that c) implies a), and c) implies d) when X is locally of finite type over k . $\square$

Definition (4.6.19). — When $\mathcal{F}$ satisfies the equivalent conditions of (4.6.18), one says that $\mathcal{F}$ is *geometrically integral* over k .

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Proposition (4.6.20). — Let k be a field, X a k -prescheme locally Noetherian, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Let K be an extension of k such that $X \otimes_k K$ is locally Noetherian. Then, if $\mathcal{F}$ is geometrically reduced (resp. geometrically integral) over k , $\mathcal{F} \otimes_k K$ is geometrically reduced (resp. geometrically integral) over K .

Proof. Indeed, for every finite extension K' of K , $(X \otimes_k K) \otimes_K K' = X \otimes_k K'$ is locally Noetherian, and one saw in the proofs of (4.6.16) and (4.6.18) that the hypothesis on $\mathcal{F}$ implies that $\mathcal{F} \otimes_k K'$ is reduced (resp. integral) over K , whence the conclusion. $\square$

Proposition (4.6.21). — Let k be a field, X, Y two k -preschemes locally Noetherian such that $X \times_k Y$ is locally Noetherian. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, $\mathcal{G}$ a coherent $\mathcal{O}_Y$ -Module.

- (i) If $\mathcal{F}$ is geometrically reduced (resp. geometrically integral) and $\mathcal{G}$ reduced (resp. integral), then $\mathcal{F} \otimes_k \mathcal{G}$ is reduced (resp. integral).
- (ii) If $\mathcal{F}$ and $\mathcal{G}$ are geometrically reduced (resp. geometrically integral), the same holds for $\mathcal{F} \otimes_k \mathcal{G}$.

Proof. (i) Using (3.2.3), (4.6.16) and (4.6.18), one can restrict to the case where $\text{Supp}(\mathcal{F}) = X$, $\text{Supp}(\mathcal{G}) = Y$, X being geometrically reduced (resp. geometrically integral), Y reduced (resp. integral), $\mathcal{F}$ and $\mathcal{G}$ without immersed associated prime cycles, and $\mathcal{F}_x$ isomorphic to $\mathcal{O}_x$ at every maximal point x of X , $\mathcal{G}_y$ isomorphic to $\mathcal{O}_y$ at every maximal point y of Y . One knows then (4.2.3) that $\mathcal{F} \otimes_k \mathcal{G}$ is without immersed associated prime cycle, and that the associated prime cycles of $\mathcal{F} \otimes_k \mathcal{G}$ are exactly the irreducible components of $X \times_k Y$ (4.2.5). One can therefore restrict to the case where X and Y are irreducible; then, if X is geometrically reduced and Y reduced, one knows (4.2.4), (ii) that $X \times_k Y$ is reduced; if X is geometrically integral and Y integral, one knows that $X \times_k Y$ is also irreducible (4.5.8), (i)). Finally, every maximal point z of $X \times_k Y$ being above x and y , the hypotheses on $\mathcal{F}$ and $\mathcal{G}$ imply, by virtue of (I, 9.1.12) that $(\mathcal{F} \otimes_k \mathcal{G})_z$ is isomorphic to $\mathcal{O}_z$, which finishes the proof in this case.

(ii) The reasoning is analogous, but here (after reduction to the case where $X = \text{Supp}(\mathcal{F})$, $Y = \text{Supp}(\mathcal{G})$), $X \times_k Y$ is geometrically reduced (resp. geometrically integral) by virtue of (4.6.5), (ii). $\square$

The definitions (4.6.18) and (4.6.19) localize:

Definition (4.6.22). — Let k be a field, X a k -prescheme locally Noetherian, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. One says that $\mathcal{F}$ is *geometrically reduced* (or *separable*) (resp. *geometrically punctually integral*) at a point $x \in X$ if, for every finite purely inseparable (resp. finite) extension k' of k , $\mathcal{F} \otimes_k k'$ is reduced (resp. integral) at each of the points x' of $X \otimes_k k'$ above x (cf. (3.2.2) and (3.2.4)).

If $\mathcal{F}$ is reduced at x , to say that $\mathcal{F}$ is geometrically reduced at x is equivalent to saying that $\mathcal{F}$ is reduced at x and that the closed subscheme Y defined by the annihilator $\mathcal{J}$ of $\mathcal{F}$ is geometrically reduced at x ; it follows that if X is locally of finite type over k , saying that $\mathcal{F}$ is geometrically punctually integral at x is equivalent to saying that $\mathcal{F}$ is reduced at x and that the subscheme Y is geometrically punctually integral at x .

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4.7. Multiplicities in the primary decomposition on an algebraic prescheme

Lemma (4.7.1). — Let A, B be two local rings, $\mathfrak{m}$ the maximal ideal of A , $\rho : A \rightarrow B$ a local homomorphism. Suppose that B is a flat A -module, and $B/\mathfrak{m}B$ a B -module of finite length. Then, for an A -module M of finite length, it is necessary and sufficient that $M_{(B)}$ be a B -module of finite length, and we have

$$(4.7.1.1) \quad \text{long}_B(M_{(B)}) = \text{long}_A(M) \cdot \text{long}_B(B/\mathfrak{m}B).$$

Proof. This is a particular case of (2.5.5.2). $\square$

Corollary (4.7.2). — Let A, B be two Noetherian rings, $\rho : A \rightarrow B$ a ring homomorphism. Let $\mathfrak{q}$ be a minimal prime ideal of B , $\mathfrak{p} = \rho^{-1}(\mathfrak{q})$. If B is a flat A -module, then $\mathfrak{p}$ is a minimal prime ideal of A , $M_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$ -module of finite length, and we have

$$(4.7.2.1) \quad \text{long}_{B_{\mathfrak{q}}}(M_{(B)})_{\mathfrak{q}} = (\text{long}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}))(\text{long}_{B_{\mathfrak{q}}}(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}})).$$

Proof. Indeed, $B_{\mathfrak{q}}$ is a flat $A_{\mathfrak{p}}$ -module (0_I, 6.3.2) and the homomorphism $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}$ is $A_{\mathfrak{p}}$ -flat, hence faithfully flat (0_I, 6.6.2); the fact that $\mathfrak{p}$ is minimal follows from (2.3.5). Moreover, $A_{\mathfrak{p}}$ is then an Artinian ring (Bourbaki, *Alg. comm.*, chap. IV, §2, no. 5, prop. 9) and $M_{\mathfrak{p}}$ is therefore an $A_{\mathfrak{p}}$ -module of finite length; the formula (4.7.2.1) is therefore a particular case of (4.7.1.1). $\square$

Proposition (4.7.3). — Let k be a field of characteristic exponent p , X an integral prescheme locally Noetherian, $K = R(X)$ the field of rational functions of X . Let k' be an extension of k , (X'_{α}) the family of irreducible components of $X' = X_{(k')}$, x'_{α} the generic point of X'_{α} .

- (i) Suppose that X' is locally Noetherian (which occurs if X is of finite type over k , or if k' is an extension of finite type of k). Then for k'' an extension of k' , (X''_{β}) the family of irreducible components of X'' , x''_{β} the generic point of X''_{β} . If x''_{β} is above x'_{α} , we have

$$(4.7.3.1) \quad \text{long}(\mathcal{O}_{x''_{\beta}}) = \text{long}(\mathcal{O}_{x'_{\alpha}}) \cdot \text{long}(\mathcal{O}_{Z'', x''_{\beta}})$$

where we have set $Z'' = (X'_{\text{red}}) \otimes_k k''$. In particular, if k'' is a separable extension of k' such that Z'' is locally Noetherian, we have $\text{long}(\mathcal{O}_{x''_{\beta}}) = \text{long}(\mathcal{O}_{x'_{\alpha}})$.

- (ii) If X is locally of finite type over k , or if k' is an extension of finite type of k , the numbers $\text{long}(\mathcal{O}_{x'_{\alpha}})$ are powers of p .
- (iii) Suppose X is locally of finite type over k . Then, if k' is perfect, all the numbers $\text{long}(\mathcal{O}_{x'_{\alpha}})$ are equal and do not depend on the particular perfect extension K of k considered. Moreover, there exists a finite purely inseparable extension k_1 of k such that, if x_1 is the generic point of $X_1 = X \otimes_k k_1$ (which is irreducible), we have $\text{long}(\mathcal{O}_{x'_{\alpha}}) = \text{long}(\mathcal{O}_{x_1})$ for every α .

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Proof. (i) Recall (4.2.4) that the irreducible components X'_{α} , which are finite in number, correspond bijectively to the minimal prime ideals $\mathfrak{p}_{\alpha}$ of $K \otimes_k k'$, and that $\mathcal{O}_{x'_{\alpha}}$ is an Artinian ring, isomorphic to $(K \otimes_k k')_{\mathfrak{p}_{\alpha}}$; these rings therefore do not depend on the extensions K and k' of k (and are therefore the same, k' being fixed, for all k -preschemes integral having the same fields of rational functions). The formula (4.7.3.1) results from (4.7.2.1) applied to the Noetherian ring A of an affine neighbourhood of x'_{α} in X' , and to the ring B of an affine neighbourhood small enough of x''_{β} in X'' , $\mathfrak{p}$ and $\mathfrak{q}$ being the minimal prime ideals corresponding to x'_{α} and x''_{β} respectively, and taking $M = A$: one notes that $A/\mathfrak{p}$ is the ring of an affine neighbourhood of x'_{α} in X'_{red} , and that $B = A \otimes_{k'} k''$, and that

$\mathcal{O}_{Z'', x''_\beta} = (B/pB)_q = B_q/pB_q$, whence (4.7.3.1). The last assertion of (i) results from the fact that in this case Z'' is reduced (4.2.4), hence $pB_q = qB_q$.

(ii) If k' is an extension of finite type of k , there is a purely transcendental extension k'_0 of k such that k' is a finite extension of k'_0 ; since k'_0 is separable over k , it results from (i) applied with k' replaced by k and k'' by k'_0 respectively, that we can replace X by $X \otimes_k k'_0$, in other words suppose k' is an extension *finite* of k . Then there is a finite quasi-Galois extension k'' of k containing k' , and the formula (4.7.3.1) shows that it suffices to prove the assertion for k'' . But k'' is a separable extension of a finite purely inseparable extension k_1 of k , hence, by virtue of (i), one is reduced to proving the assertion when $k' = k_1$ is *purely inseparable*. One knows that $K \otimes_k k_1$ is an Artinian ring having only a single minimal prime ideal (4.3.2); if $\bar{K}$ is the algebraic closure of K , the length of $K \otimes_k k_1$ divides that of the Artinian ring $\bar{K} \otimes_k k_1$ by virtue of (4.7.1.1). But $\bar{K} \otimes_k k_1$ is a $\bar{K}$ -algebra having only a single prime ideal (necessarily maximal), and its quotient by this ideal is necessarily a finite extension of $\bar{K}$, hence necessarily equal to $\bar{K}$; the length of $\bar{K} \otimes_k k_1$ is therefore equal to its rank over $\bar{K}$, that is, $[k_1 : k]$, which is a power of p .

Suppose next that X is locally of finite type over k ; then, X'' is locally Noetherian for every extension k'' of k' . Taking for k'' an algebraically closed extension, one is reduced, by virtue of (4.7.3.1), to proving the assertion when k' is the *algebraic closure* $\bar{k}$. Now, one knows (replacing X by an open affine of finite type over k) that there is a finite purely inseparable extension k_1 of k such that $(X \otimes_k k_1)_{\text{red}}$ is separable over k_1 (4.6.6); hence $(X \otimes_k k_1)_{\text{red}} \otimes_{k_1} \bar{k}$ is reduced (4.6.4), and the same holds for one and we conclude as above.

(iii) Note that two extensions k', k'' of k can always be contained in a same extension E of k ; it suffices then to compare the numbers for k' and for an algebraically closed extension E common to Ω and Ω' to show the first assertion. The second is obtained by taking for Ω an algebraically closed extension $\bar{k}$ and using the fact that the extension k_1 provided by (ii) determines a unique generic point of $X_1 = X \otimes_k k_1$, and the relation $\text{long}(\mathcal{O}_{x'_\alpha}) = \text{long}(\mathcal{O}_{x_1})$ for every α , follows from (4.7.3.1). $\square$

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Definition (4.7.4). — Let k be a field, X a k -prescheme integral, K the field of rational functions of X . The *purely inseparable multiplicity* of K (or of X) over k is defined to be the upper bound of the lengths of the Artinian rings $K \otimes_k k_1$, when k_1 ranges over the set of finite purely inseparable extensions of k . The *separable multiplicity* of K (or of X) over k is defined to be the geometric number of irreducible components of X if this number is finite, and $+\infty$ otherwise. The *total multiplicity* of K (or of X) over k is the product of the purely inseparable multiplicity and the separable multiplicity.

One will note that if p is the characteristic exponent of k and if the purely inseparable multiplicity of K over k is finite, it is a power $q = p^f$ of p , as follows from (4.7.3), (ii)); when $p \neq 1$, f is well-determined and is called the *exponent of inseparability* of K over k ; when $p = 1$, the purely inseparable multiplicity is always equal to 1.

To say that the purely inseparable multiplicity (resp. separable, resp. total) of X over k is 1 means that X is geometrically reduced over k (resp. geometrically irreducible, resp. geometrically integral) over k .

When X is locally of finite type over k , the purely inseparable multiplicity of X over k is also the common length of the local rings A_i at the minimal prime ideals of $A = K \otimes_k k'$, for every perfect extension k' of k ; the separable multiplicity of X over k is the number of minimal prime ideals of $A = K \otimes_k \Omega$ for every algebraically closed extension Ω of k , and finally the total multiplicity is the sum of the lengths of the local rings A_i of $A = K \otimes_k \Omega$ at its minimal prime ideals; these numbers are all three *finite* in this case.

Definition (4.7.5). — Let k be a field, X a k -prescheme locally of finite type over k , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, x a point of X such that $\mathcal{F}_x$ is an $\mathcal{O}_x$ -module of finite length. The *geometric length* of $\mathcal{F}$ at x (relative to k), or the *purely inseparable multiplicity* of x for $\mathcal{F}$, is the product $\lambda_x(\mathcal{F}) = (\text{long}_{\mathcal{O}_x}(\mathcal{F}_x))\mu_i(k(x) | k)$, where $\mu_i(k(x) | k)$ denotes the purely inseparable multiplicity of $k(x)$ over k .

When X is *integral*, and x the generic point of X , $\text{long}(\mathcal{O}_x) = 1$, hence the purely inseparable multiplicity of x for $\mathcal{O}_X$ is none other than the purely inseparable multiplicity of X over k , defined in (4.7.4).

Proposition (4.7.6). — *Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ be an exact sequence of coherent $\mathcal{O}_X$ -Modules. If at a point $x \in X$ the geometric length $\lambda_x(\mathcal{F})$ of $\mathcal{F}$ at this point is defined, the same holds for $\lambda_x(\mathcal{F}')$ and $\lambda_x(\mathcal{F}'')$, of $\mathcal{F}'$ and $\mathcal{F}''$ at the point x , and conversely; moreover, we have*

$$(4.7.6.1) \quad \lambda_x(\mathcal{F}) = \lambda_x(\mathcal{F}') + \lambda_x(\mathcal{F}'').$$

Proof. The proposition results immediately from the definition, for $\text{long}(\mathcal{F}_x)$ is finite if and only if $\text{long}(\mathcal{F}'_x)$ and $\text{long}(\mathcal{F}''_x)$ are finite, and we have $\text{long}(\mathcal{F}_x) = \text{long}(\mathcal{F}'_x) + \text{long}(\mathcal{F}''_x)$. $\square$

(4.7.7). Under the hypotheses of (4.7.5), the only points $x \in X$ such that $\mathcal{F}_x$ is a non-zero $\mathcal{O}_x$ -module of finite length are (by virtue of (3.1.2)) the *maximal points* of $\text{Supp}(\mathcal{F})$, as follows from Bourbaki, *Alg. comm.*, chap. IV, §2, no. 5, cor. 2 of prop. 7, the question being local on X . One knows that there exists a closed subscheme Y having $\text{Supp}(\mathcal{F})$ as underlying space, and a coherent $\mathcal{O}_Y$ -Module $\mathcal{G}$ such that $\mathcal{F} = j_*(\mathcal{G})$, $j : Y \rightarrow X$ being the canonical injection. If x is a maximal point of Y , we see that the geometric lengths of $\mathcal{F}$ and of $\mathcal{G}$ at x are equal, $\mathcal{O}_{X,x}$ and $\mathcal{O}_{Y,x}$ having the same residue fields.

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Proposition (4.7.8). — *Under the hypotheses of (4.7.5), let x be a maximal point of $\text{Supp}(\mathcal{F})$. Let k' be a perfect extension of k ; set $X' = X \otimes_k k'$, $\mathcal{F}' = \mathcal{F} \otimes_k k'$; then the geometric length $\lambda_x(\mathcal{F})$ of $\mathcal{F}$ at x is equal to the length of the $\mathcal{O}_{x'}$ -module $\mathcal{F}'_{x'}$, at every maximal point x' of $\text{Supp}(\mathcal{F}')$ above x . Moreover, there exists a finite purely inseparable extension k_1 of k such that, setting $X_1 = X \otimes_k k_1$, $\mathcal{F}_1 = \mathcal{F} \otimes_k k_1$, $\lambda_x(\mathcal{F})$ is also equal to the length of the $\mathcal{O}_{x_1}$ -module $(\mathcal{F}_1)_{x_1}$ at every maximal point x_1 of $\text{Supp}(\mathcal{F}_1)$ above x .*

Proof. The remark of (4.7.7) shows that one can restrict to the case where x is a maximal point of X and one can therefore suppose $X = \text{Spec}(A)$ affine Noetherian, $\mathcal{F} = \tilde{M}$, where M is an A -module of finite type; let $\mathfrak{p}$ (resp. $\mathfrak{q}$) be the minimal prime ideal of A (resp. B) corresponding to x (resp. x'). We have then, applying the formula (4.7.2.1):

$$\text{long}_{\mathcal{O}_{x'}}(\mathcal{F}'_{x'}) = \text{long}_{\mathcal{O}_x}(\mathcal{F}_x) \cdot \text{long}_{B_{\mathfrak{q}}}(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}).$$

But since $k(x) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$, the same reasoning as in (4.7.3) shows that $\text{long}_{B_{\mathfrak{q}}}(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}})$ is equal to the length of the local ring $(B/\mathfrak{p}B)_{\mathfrak{q}'}$, where $\mathfrak{q}' = \mathfrak{q}/\mathfrak{p}B$, and this length is by definition $\mu_i(k(x) | k)$, whence the first assertion. The second is proved similarly, replacing k' by k_1 and using (4.7.3), (iii). $\square$

One says, in the case where x is a maximal point of $\text{Supp}(\mathcal{F})$, that the geometric length of $\mathcal{F}$ at x is the *purely inseparable multiplicity of the maximal prime cycle $\{x\}$ associated to $\mathcal{F}$* .

Corollary (4.7.9). — *Under the hypotheses of (4.7.5), let k' be an extension of k , $X' = X \otimes_k k'$, $\mathcal{F}' = \mathcal{F} \otimes_k k'$; let Z be an irreducible component of $\text{Supp}(\mathcal{F})$, and Z' an irreducible component of $\text{Supp}(\mathcal{F}')$ that dominates Z (4.2.7). Then the purely inseparable multiplicities of Z and of Z' with respect to $\mathcal{F}$ and $\mathcal{F}'$ respectively are the same.*

Proof. It suffices to consider a perfect extension k'' of k' and, setting $X'' = X \otimes_k k''$, $\mathcal{F}'' = \mathcal{F} \otimes_k k''$, a maximal point x'' of $\text{Supp}(\mathcal{F}'')$ above the generic point of Z' ; then by (4.7.8) $\text{long}(\mathcal{F}''_{x''})$ is equal to the purely inseparable multiplicities of Z and of Z' . $\square$

Proposition (4.7.10). — *Let k be a field, X a k -prescheme locally of finite type over k , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, x a point of X , z_i the generic points of the irreducible components of $\text{Supp}(\mathcal{F})$ containing x (4.6.22). For $\mathcal{F}$ to be geometrically reduced at x over k , it is necessary and sufficient that x not belong to any immersed associated prime cycle of $\mathcal{F}$ and that, for every i , the geometric length $\lambda_{z_i}(\mathcal{F})$ of $\mathcal{F}$ at the point z_i be equal to 1.*

Proof. Let us first show that the conditions of the statement are sufficient. Indeed, if k' is a finite extension of k , x' a point of $X' = X \otimes_k k'$ above x , and $\mathcal{F}' = \mathcal{F} \otimes_k k'$, x' does not belong to any immersed associated prime cycle of $\mathcal{F}'$ by (4.2.7); moreover, by (4.7.8), the geometric length of $\mathcal{F}'$ is

equal to 1 at every generic point z'_j of an irreducible component of $\text{Supp}(\mathcal{F}')$ containing x' , such a point being above one of the z_i by (2.3.4); *a fortiori* $\text{long}(\mathcal{F}'_{z'_j}) = 1$, which proves that $\mathcal{F}'$ is reduced at x' .

Conversely, it results first from (4.2.7) that if, for one finite extension k' of k , $\mathcal{F}'$ is reduced at all points x' of X' above x , then x does not belong to any immersed associated prime cycle of $\mathcal{F}$. Moreover, taking for k' the field k_1 from the statement of (4.7.8) (where x is replaced by one of the z_i), the hypothesis that $\mathcal{F}$ is geometrically reduced at x implies $\lambda_{z_i}(\mathcal{F}) = 1$. $\square$

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Proposition (4.7.11). — *Let k be a field, X a prescheme locally of finite type over k , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, k' an extension of k , $X' = X \otimes_k k'$, $\mathcal{F}' = \mathcal{F} \otimes_k k'$, x a point of X , x' a point of X' above x . For $\mathcal{F}$ to be geometrically punctually reduced (resp. geometrically punctually integral) at x , it is necessary and sufficient that $\mathcal{F}'$ be geometrically punctually reduced (resp. geometrically punctually integral) at x' .*

Proof. The irreducible components of $\text{Supp}(\mathcal{F}')$ containing x' each dominate an irreducible component of $\text{Supp}(\mathcal{F})$ containing x , and conversely each of these latter is dominated by an irreducible component of $\text{Supp}(\mathcal{F}')$ containing x' ((4.2.7), (i)) and (2.3.4)). The assertion concerning separability results therefore from (4.2.7), (ii), (4.7.9) and (4.7.10). Let Y be the closed subscheme of X defined by the Ideal $\mathcal{J}$ annihilator of $\mathcal{F}$, so that $Y' = Y \otimes_k k'$ is the closed subscheme of X' defined by the Ideal $\mathcal{J}' = \mathcal{J} \otimes_k k'$ annihilator of $\mathcal{F}'$; to say that $\mathcal{F}$ (resp. $\mathcal{F}'$) is geometrically punctually reduced (resp. integral) at x (resp. x') is the same as saying that $\mathcal{F}$ (resp. $\mathcal{F}'$) is geometrically punctually reduced at x (resp. x') and that Y (resp. Y') is geometrically punctually integral at x (resp. x'). By virtue of the assertion of the statement concerning separability, the hypothesis that $\mathcal{F}$ is geometrically punctually integral at x , or the hypothesis that $\mathcal{F}'$ is geometrically punctually integral at x' , imply both that there exists in Y a neighbourhood of x that is separable, and the conclusion results from (4.6.10). $\square$

Definition (4.7.12). — *Let k be a field, X a k -prescheme locally of finite type over k , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, x a maximal point of the support of $\mathcal{F}$. The total multiplicity of x (or of the prime cycle $\{x\}$) for $\mathcal{F}$ (relative to k) is the product of the purely inseparable multiplicity of x for $\mathcal{F}$ by the separable multiplicity of $k(x)$ over k .*

It amounts to the same to say that the total multiplicity of x for $\mathcal{F}$ is the product of the length $\text{long}_{\mathcal{O}_x}(\mathcal{F}_x)$ by the total multiplicity of $k(x)$ over k (4.7.4). With the notation of (4.7.11), the total multiplicity of x for $\mathcal{F}$ is equal to the sum of the total multiplicities for $\mathcal{F}'$ of the maximal points x'_i of $\text{Supp}(\mathcal{F}')$ that are above x ; moreover, there exists a finite extension k' of k such that the total multiplicity of x for $\mathcal{F}$ be equal to $\sum_i \text{long}(\mathcal{F}'_{x'_i})$.

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Proposition (4.7.13). — *Let the hypotheses and notation be those of (4.7.10). For $\mathcal{F}$ to be geometrically punctually integral at the point x , it suffices that x belong to only one associated prime cycle (necessarily non-immersed) of $\mathcal{F}$ and that the total multiplicity for $\mathcal{F}$ of the generic point z of this cycle be equal to 1.*

Proof. Indeed, if k' is a finite extension of k , x' a point of $X' = X \otimes_k k'$ above x , and $\mathcal{F}' = \mathcal{F} \otimes_k k'$, x' can belong to only one associated prime cycle of $\mathcal{F}'$, since there is only one maximal point of $\text{Supp}(\mathcal{F}')$ above z by hypothesis; moreover, $\mathcal{F}$ is geometrically reduced by (4.7.10), whence the conclusion. $\square$

4.8. Fields of definition

(4.8.1). Given a prescheme X , we will denote, in this section, by $\mathfrak{S}(X)$ the set of subpreschemes of X , and as usual by $\mathfrak{P}(X)$ the set of subsets of the underlying space of X ; for every quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, we denote by $\Phi(\mathcal{F})$ the set of quasi-coherent sub- $\mathcal{O}_X$ -Modules of $\mathcal{F}$.

(4.8.2). Let k be a field, X, Y two k -preschemes, $\mathcal{F}, \mathcal{G}$ two quasi-coherent $\mathcal{O}_X$ -Modules, K, K' two extensions of k such that $K' \subset K$. We then have, corresponding to the morphism $\text{Spec}(K) \rightarrow \text{Spec}(K')$, canonical maps

$$(4.8.2.1) \quad \Phi(\mathcal{F} \otimes_k K') \longrightarrow \Phi(\mathcal{F} \otimes_k K)$$

$$(4.8.2.2) \quad \text{Hom}(\mathcal{F} \otimes_k K', \mathcal{G} \otimes_k K') \longrightarrow \text{Hom}(\mathcal{F} \otimes_k K, \mathcal{G} \otimes_k K)$$

$$(4.8.2.3) \quad \mathfrak{S}(X_{(K')}) \longrightarrow \mathfrak{S}(X_{(K)})$$

$$(4.8.2.4) \quad \text{Hom}_{K'}(X_{(K')}, Y_{(K')}) \longrightarrow \text{Hom}_K(X_{(K)}, Y_{(K)})$$

$$(4.8.2.5) \quad \mathfrak{P}(X_{(K')}) \longrightarrow \mathfrak{P}(X_{(K)}).$$

If $p_X : X_{(K)} \rightarrow X_{(K')}$ is the canonical projection, the map (4.8.2.5) is $M \mapsto p_X^{-1}(M)$, and likewise (4.8.2.3) is $Z \mapsto p_X^{-1}(Z) = Z \otimes_{K'} K$ (I, 4.4.1). The map (4.8.2.1) is $\mathcal{H} \mapsto p_X^*(\mathcal{H}) = \mathcal{H} \otimes_{\mathcal{O}_{X_{(K')}}} \mathcal{O}_{X_{(K)}}$; (4.8.2.2) is $u \mapsto p_X^*(u)$ and finally (4.8.2.4) is the map $f \mapsto f_{(K)}$.

If we denote by $\varepsilon_{K, K'}$ any one of these canonical maps, it is clear that if K'', K', K are three extensions of k such that $K'' \subset K' \subset K$, then we have

$$(4.8.2.6) \quad \varepsilon_{K, K''} = \varepsilon_{K, K'} \circ \varepsilon_{K', K''}.$$

Proposition (4.8.3). — *The canonical maps (4.8.2.1) to (4.8.2.5) are injective.*

Proof. Indeed, we know that p_X is faithfully flat and quasi-compact; the fact that (4.8.2.5) is injective thus follows from the surjectivity of p_X . The injectivity of (4.8.2.1) follows from (2.2.2), that of (4.8.2.2) from (2.2.7), that of (4.8.2.3) from (2.2.15), and finally that of (4.8.2.4) from (2.2.16). $\square$

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Definition (4.8.4). — The notations being those of (4.8.2), we say that a quasi-coherent sub- $\mathcal{O}_{X_{(K)}}$ -Module of $\mathcal{F} \otimes_k K$ (resp. a homomorphism $\mathcal{F} \otimes_k K \rightarrow \mathcal{G} \otimes_k K$, resp. a subprescheme of $X_{(K)}$, resp. a K -morphism $X_{(K)} \rightarrow Y_{(K)}$, resp. a subset of $X_{(K)}$) is *defined over K'* if it belongs to the image of the map (4.8.2.1) (resp. (4.8.2.2), resp. (4.8.2.3), resp. (4.8.2.4), resp. (4.8.2.5)). We then say that K' is a *field of definition* of the object considered.

It is clear that K itself is a field of definition of an element of any one of the five target sets of the maps (4.8.2.1) to (4.8.2.5). We note however that for a K -prescheme Z , to say for example that a quasi-coherent $\mathcal{O}_Z$ -Module $\mathcal{H}$ is “defined over a sub-corps K' of K ” has no meaning unless Z has been *given* in the form $X_{(K)}$ for a prescheme X over a sub-corps k of K , and if $\mathcal{H}$ is a sub- $\mathcal{O}_Z$ -Module which has been *given* in the form $\mathcal{F} \otimes_k K$, where $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -Module; analogous remarks apply for the other notions. It follows from (4.8.2.6) that if an element of one of the target sets of the maps of (4.8.2) is defined on K' , it is also defined on every field K'' such that $K' \subset K'' \subset K$. Finally, with the notations of (4.8.2.6), if K_1 is an extension of K , for an element of the target set of $\varepsilon_{K, K'}$ to be defined over K' , it suffices that its image by $\varepsilon_{K_1, K}$ be defined over K' , by virtue of the relation $\varepsilon_{K_1, K'} = \varepsilon_{K_1, K} \circ \varepsilon_{K, K'}$.

Proposition (4.8.5). — *With the notations of (4.8.2), let (X_α) be an open covering of X . For an element $\mathcal{H} \in \Phi(\mathcal{F} \otimes_k K)$ (resp. $u \in \text{Hom}(\mathcal{F} \otimes_k K, \mathcal{G} \otimes_k K)$, resp. $Z \in \mathfrak{S}(X_{(K)})$, resp. $M \in \mathfrak{P}(X_{(K)})$) to be defined over K' , it is necessary and sufficient that for all α , $\mathcal{H}|(X_\alpha)_{(K)}$ (resp. $u|(X_\alpha)_{(K)}$, $Z \cap (X_\alpha)_{(K)}$, $M \cap (X_\alpha)_{(K)}$) be defined over K' .*

Proof. This follows immediately from the injectivity of the maps (4.8.2.1), (4.8.2.2), (4.8.2.3), and (4.8.2.5): for example, if for all α , there is a quasi-coherent sub- $\mathcal{O}_{(X_\alpha)_{(K')}}$ -Module $\mathcal{H}'_\alpha$ of $(\mathcal{F}|X_\alpha) \otimes_k K'$ such that $\mathcal{H}|(X_\alpha)_{(K)} = \mathcal{H}'_\alpha \otimes_{K'} K$, for any two indices α, β , we will have $\mathcal{H}|(X_\alpha \cap X_\beta)_{(K)} = (\mathcal{H}'_\alpha|(X_\alpha \cap X_\beta)_{(K')}) \otimes_{K'} K = (\mathcal{H}'_\beta|(X_\alpha \cap X_\beta)_{(K')}) \otimes_{K'} K$, hence necessarily $\mathcal{H}'_\alpha|(X_\alpha \cap X_\beta)_{(K')} =$

$\mathcal{H}'_\beta|(X_\alpha \cap X_\beta)_{(K')}$ whatever α and β , and therefore there is a quasi-coherent sub- $\mathcal{O}_{X_{(K')}}$ -Module $\mathcal{H}'$ of $\mathcal{F} \otimes_k K'$ such that $\mathcal{H}'|(X_\alpha)_{(K')} = \mathcal{H}'_\alpha$ for all α , hence $\mathcal{H} = \mathcal{H}' \otimes_{K'} K$. The reasoning is the same in the other cases. $\square$

Lemma (4.8.6). — *Let k be a field, K an extension of k , A a k -algebra, M an A -module, $\bar{N}$ a sub- $A_{(K)}$ -module of $M_{(K)}$, K' a sub-extension of K . The following conditions are equivalent:*

- a) $\bar{N}$ is of the form $N' \otimes_{K'} K$, where N' is a sub- $A_{(K')}$ -module of $M_{(K')}$.
- b) If $X = \text{Spec}(A)$, the quasi-coherent $\mathcal{O}_{X_{(K)}}$ -Module $\tilde{N}$ on $X_{(K)} = \text{Spec}(A_{(K)})$ is defined over K' .
- c) $\bar{N}$ is of the form $N' \otimes_{K'} K$, where N' is a sub- K' -vector space of $M_{(K')}$.

Proof. The equivalence of a) and b) follows trivially from (I, 1.6.5); it is clear that a) implies c). Inversely, if c) is satisfied, we know that we can write (up to canonical identification) $N' = \bar{N} \cap M_{(K')}$. Since $\bar{N}$ and $M_{(K')}$ are by hypothesis $A_{(K')}$ -modules, the same is true of N' . $\square$

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Corollary (4.8.7). — *Under the hypotheses of (4.8.6), there exists a smallest sub-corps K' of K containing k such that the equivalent conditions of (4.8.6) are satisfied.*

Proof. It suffices to see this for condition c), which follows from Bourbaki, Alg., chap. II, 3rd ed., §8, n° 6, prop. 6. $\square$

Lemma (4.8.8). — *The notations being those of (4.8.2):*

- (i) Let $\bar{u} : \mathcal{F} \otimes_k K \rightarrow \mathcal{G} \otimes_k K$ be an $\mathcal{O}_{X_{(K)}}$ -homomorphism, and let $\mathcal{H} \subset (\mathcal{F} \otimes_k K) \oplus (\mathcal{G} \otimes_k K)$ be its graph. For $\bar{u}$ to be defined over K' , it is necessary and sufficient that $\mathcal{H}$ be so.
- (ii) Let $\bar{Z}$ be a closed subprescheme of $X_{(K)}$, and let $\bar{\mathcal{J}}$ be the quasi-coherent Ideal of $\mathcal{O}_{X_{(K)}}$ defining $\bar{Z}$. For $\bar{Z}$ to be defined over K' , it is necessary and sufficient that $\bar{\mathcal{J}}$ be so.
- (iii) Let $\bar{f}$ be a K -morphism $X_{(K)} \rightarrow Y_{(K)}$ and let $\bar{Z}$ be the subprescheme of $X_{(K)} \times_K Y_{(K)}$, graph of $\bar{f}$ (I, 5.3.11); for $\bar{f}$ to be defined over K' , it is necessary and sufficient that $\bar{Z}$ be so.

Proof. Assertion (ii) follows immediately from (I, 4.4.5). The necessity in assertion (iii) is evident; suppose conversely that $\bar{Z} = Z'_{(K')}$, where Z' is a subprescheme of $(X \times_k Y)_{(K')}$; if $p' : Z' \rightarrow X_{(K')}$ is the restriction to Z' of the first projection, we know that $p'_{(K)}$ is an isomorphism of preschemes, hence the same is true of p' (2.7.1, (viii)); but this means (I, 5.3.11) that Z' is the graph of a K' -morphism $f' : X_{(K')} \rightarrow Y_{(K')}$, and we then have $\bar{f} = f'_{(K)}$ (I, 5.3.12).

Finally, to prove (i), we can, by virtue of (4.8.5), suppose that $X = \text{Spec}(A)$ is affine, $\mathcal{F} = \tilde{M}$, $\mathcal{G} = \tilde{N}$, where M and N are A -modules, and $\bar{u} = (\bar{v})^\sim$, where $\bar{v} : M_{(K)} \rightarrow N_{(K)}$ is an $A_{(K)}$ -homomorphism. Suppose that the graph of $\bar{v}$ is of the form $P'_{(K')}$, where P' is a sub- $A_{(K')}$ -module of $M_{(K')} \oplus N_{(K')}$; if $p' : P' \rightarrow M_{(K')}$ is the restriction to P' of the first projection, we know that $p' \otimes 1_K$ is an isomorphism of $A_{(K')}$ -modules, hence, by faithful flatness (0_I, 6.4.1), p' is an isomorphism of $A_{(K')}$ -modules, which proves that P' is the graph of an $A_{(K')}$ -homomorphism $v' : M_{(K')} \rightarrow N_{(K')}$ and we obviously have $\bar{v} = v' \otimes 1_K$. $\square$

Proposition (4.8.9). — *Let k be a field, K an extension of k , X a k -prescheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. Let $\bar{\mathcal{H}}$ be a quasi-coherent sub- $\mathcal{O}_{X_{(K)}}$ -Module of $\mathcal{F} \otimes_k K$. Then there exists a smallest field of definition of $\bar{\mathcal{H}}$.*

Proof. When X is affine, the assertion is none other than (4.8.7). In the general case, consider an open covering (X_α) of X by affine opens; by virtue of what precedes, there exists for all α a smallest sub-corps K'_α of K containing k and such that $\bar{\mathcal{H}}|(X_\alpha)_{(K)}$ is defined over K'_α . By virtue of (4.8.5), the sub-corps of K generated by the K'_α is the smallest field of definition of $\bar{\mathcal{H}}$. $\square$

Corollary (4.8.10). — *The notations being those of (4.8.9), let $\mathcal{G}$ be a second quasi-coherent $\mathcal{O}_X$ -Module, and let $\bar{u} : \mathcal{F} \otimes_k K \rightarrow \mathcal{G} \otimes_k K$ be an $\mathcal{O}_{X_{(K)}}$ -homomorphism; then there exists a smallest field of definition of $\bar{u}$.*

Proof. Indeed, it follows from (4.8.8) that such a field is also a smallest field of definition of the graph of $\bar{u}$, and the existence of the latter follows from (4.8.9). $\square$

Corollary (4.8.11). — *Let k be a field, X a k -prescheme, K an extension of k , $\bar{Z}$ a closed subprescheme of $X_{(K)}$; then there exists a smallest field of definition of $\bar{Z}$.*

Proof. Indeed, if $\bar{\mathcal{I}}$ is the quasi-coherent Ideal of $\mathcal{O}_{X_{(K)}}$ defining $\bar{Z}$, it follows from (4.8.8) that this amounts to the existence of a smallest field of definition of $\bar{\mathcal{I}}$, which follows from (4.8.9). $\square$

Corollary (4.8.12). — *Let k be a field, X, Y two k -preschemes, Y being supposed separated, K an extension of k , $\bar{f} : X_{(K)} \rightarrow Y_{(K)}$ a K -morphism. There then exists a smallest field of definition of $\bar{f}$.*

Proof. Indeed, the existence of such a field, by virtue of (4.8.8), amounts to that of a smallest field of definition of the graph $\bar{Z}$ of $\bar{f}$; but since $\bar{Z}$ is a closed subprescheme of $(X \times_k Y)_{(K)}$ (I, 5.4.3), the existence of a smallest field of definition of $\bar{Z}$ follows from (4.8.11). $\square$

Proposition (4.8.13). — *Under the hypotheses of (4.8.11) (resp. (4.8.9), (4.8.10), (4.8.12)) suppose further that X is of finite type over k (resp. that X is of finite type over k and $\mathcal{F}$ coherent, resp. that X is of finite type over k and $\mathcal{F}$ and $\mathcal{G}$ coherent, resp. that X and Y are of finite type over k). Then the smallest field of definition of $\bar{Z}$ (resp. of $\bar{\mathcal{H}}$, of $\bar{u}$, of $\bar{f}$) is a finite type extension of k .*

Proof. We can restrict ourselves to treating the case of (4.8.9), and suppose X affine since X is a finite union of affine opens of finite type over k . With the notations of (4.8.6), it suffices to show that if A is a k -algebra of finite type and M an A -module of finite type, there exists a sub-extension K' of K which is of finite type over k (every sub-extension of a finite type extension being of finite type) and which satisfies conditions a), b), or c) of (4.8.6). Now, let $(m_i)_{1 \leq i \leq n}$ be a system of generators of the A -module M ; since A is Noetherian, the same is true of $A_{(K)}$, hence $\bar{M}$ is an $A_{(K)}$ -module of finite type, admitting a finite system of generators n_j ($1 \leq j \leq r$) such that $n_j = \sum_i (m_i \otimes 1) b_{ij}$, where $b_{ij} \in A_{(K)}$; each b_{ij} can be written $\sum_h a_{ijh} \otimes c_{ijh}$, where $a_{ijh} \in A$ and $c_{ijh} \in K$; it is clear that the extension K' of k generated by the c_{ijh} answers the question. $\square$

Proposition (4.8.14). — *Let k be a field, X a k -prescheme of finite type, K an extension of k containing a perfect extension of k . Then the smallest field of definition of the closed subprescheme $(X_{(K)})_{\text{red}}$ of $X_{(K)}$ is a finite radical extension of k .*

Proof. If p is the characteristic exponent of k , the hypothesis implies that $k' = k^{p^{-\infty}}$ is contained in K ; we know that $(X_{(k')})_{\text{red}}$ is geometrically reduced over k' (4.6.1), hence $(X_{(k')})_{\text{red}} \otimes_{k'} K$ is reduced, and is therefore equal to $(X_{(K)})_{\text{red}}$ (I, 5.1.8); in other words, k' is a field of definition of $(X_{(K)})_{\text{red}}$. Since k' is an algebraic extension of k , the conclusion follows from (4.8.13). $\square$

(4.8.15). Remarks.

- (i) The conclusion of (4.8.14) does not necessarily hold if one does not suppose that K contains a perfect extension of k . For example, let k_0 be a field of characteristic $p > 0$, k the field of rational fractions $k_0(s, t)$ in 2 indeterminates, L the field $k(s^{1/p}, t^{1/p})$, and $X = \text{Spec}(L)$. Let on the other hand u be a third indeterminate and K the field $k(u, s^{1/p} + ut^{1/p})$; one easily verifies that k is algebraically closed in K and that $L \otimes_k K$ has a non-zero nilradical, hence $X_{(K)}$ is not reduced; but since $(X_{(K)})_{\text{red}}$ is not defined over k , and K does not contain any sub-extension of finite degree over k other than k , the smallest field of definition of $(X_{(K)})_{\text{red}}$ cannot be finite over k .
- (ii) Given an element of one of the target sets of the maps (4.8.2), and an extension K_1 of K , it is equivalent to say that the object considered admits a smallest field of definition or that its image by $\varepsilon_{K_1, K}$ admits a smallest field of definition (the two sub-corps of definition being necessarily the same), as we have seen in (4.8.4).

4.9. Fields of definition of a part of a prescheme

Proposition (4.9.1). — *Let k be a field, X a k -prescheme, K an extension of k , $\bar{T}$ a closed part of $X_{(K)}$, K' a sub-extension of K . The following conditions are equivalent:*

- a) $\bar{T}$ is defined over K' .
- b) The open part $\bar{U} = X_{(K)} - \bar{T}$ of $X_{(K)}$ is defined over K' .
- b') The subprescheme of $X_{(K)}$ induced on the open $\bar{U}$ is defined over K' .
- c) There exists a closed subprescheme $\bar{Z}$ of $X_{(K)}$ having $\bar{T}$ as underlying space, and defined over K' .

Proof. If $g : X_{(K)} \rightarrow X_{(K')}$ is the canonical projection, to say that $\bar{T}$ is defined over K' means that there exists a closed part T' of $X_{(K')}$ such that $\bar{T} = g^{-1}(T')$; since $g^{-1}(X_{(K')} - T') = X_{(K)} - g^{-1}(T')$ for all closed parts T' of $X_{(K')}$, a), b), and b') are equivalent. If $\bar{Z}$, having $\bar{T}$ as underlying space, is defined over K' , we have $\bar{Z} = Z' \otimes_{K'} K$, where Z' is a closed subprescheme of $X_{(K')}$, and if T' is the underlying space of Z' , we have $\bar{T} = g^{-1}(T')$ (I, 4.4.1), hence $\bar{T}$ is defined over K' . Inversely, to see that a) implies c), it suffices to consider a closed subprescheme Z' of $X_{(K')}$ having T' as underlying space (I, 5.2.1) and to take $\bar{Z} = Z' \otimes_{K'} K$. $\square$

Corollary (4.9.2). — *With the notations of (4.9.1), suppose that K' is a field of definition of $\bar{T}$; every extension $K'' \subset K'$ of k , such that K' is a radical extension of K'' , is also a field of definition of $\bar{T}$.*

Proof. It suffices to observe that the canonical projection $g : X_{(K')} \rightarrow X_{(K'')}$ is a morphism that is integer, surjective, and radical, hence a homeomorphism (2.4.5). $\square$

(4.9.3). Remark. It follows from (4.9.2) that a closed part $\bar{T}$ of $X_{(K)}$ does not necessarily admit a *smallest* field of definition. For example, if $K = k(t)$, where t is an indeterminate and k a perfect field of characteristic $p > 0$, a closed part $\bar{F}$ of $X_{(K)}$ cannot have a smallest field of definition unless it is already defined over k : indeed, for every sub-corps K' of K containing k and $\neq k$, we have $K'^p \neq K'$, K'^p contains k and K' is a radical extension of K'^p , hence if K' is a field of definition of $\bar{F}$, so is K'^p . However:

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Proposition (4.9.4). — *With the notations of (4.9.1), suppose that K is a separable extension of K' ; for K' to be a field of definition of $\bar{T}$, it is necessary and sufficient that K' be a field of definition of the reduced subprescheme of $X_{(K)}$ having $\bar{T}$ as underlying space.*

Proof. The condition is evidently sufficient by virtue of (4.9.1); to see that it is necessary, note that if $\bar{T} = g^{-1}(T')$, where T' is a closed part of $X_{(K')}$, and if Z' is the reduced subprescheme of $X_{(K')}$ having T' as underlying space, then $\bar{Z} = Z' \otimes_{K'} K = \text{Spec}(K) \times_{K'} Z'$ is reduced by virtue of the criterion (4.6.1, e)) and the hypothesis on K , and has $\bar{T}$ as underlying space. $\square$

Corollary (4.9.5). — *Suppose one of the following hypotheses satisfied:*

- a) k is of characteristic 0.
- b) K is an algebraic extension of k and X is of finite type over k .

Then every closed part $\bar{T}$ of $X_{(K)}$ has a smallest field of definition, which is a finite separable extension of k (and finite over k in hypothesis b)).

Proof. In hypothesis a), K is separable on each of its sub-corps, and it follows from (4.9.4) that the fields of definition of $\bar{T}$ are the same as those of the reduced subprescheme of $X_{(K)}$ having $\bar{T}$ as underlying space. The corollary then follows from (4.8.11). In hypothesis b), we know that K is a radical extension of a separable extension K_1 of k . Then, for every sub-extension K' of K , K' is radical on $K' \cap K_1$, hence, by virtue of (4.9.2), K' is a field of definition of $\bar{T}$ if and only if $K' \cap K_1$ is. This reduces us to the case where $K' = K_1$, hence the case where K' is a separable algebraic extension of k , and one finishes the reasoning as in case a). $\square$

Proposition (4.9.6). — *Let k be a field, X a k -prescheme, K an extension of k , $\bar{U}$ a part that is both open and closed of $X_{(K)}$, $\bar{V}$ its complement. Let K' be a sub-extension of K ; then the following conditions are equivalent:*

- a) K' is a field of definition of the part $\bar{U}$ of $X_{(K)}$.

- a') K' is a field of definition of the closed subprescheme of $X_{(K)}$ induced on the open $\bar{U}$.
 b) K' is a field of definition of the part $\bar{V}$ of $X_{(K)}$.
 b') K' is a field of definition of the closed subprescheme of $X_{(K)}$ induced on the open $\bar{V}$.

Proof. It is clear that a') implies a), and a) implies b') by virtue of (4.9.1). Since $\bar{U}$ and $\bar{V}$ play symmetric roles, b') implies b) and b) implies a'), which completes the proof. $\square$

Corollary (4.9.7). — Under the hypotheses of (4.9.6), there exists a smallest field of definition K' for the part $\bar{U}$ of $X_{(K)}$, and K' is also the smallest field of definition of the part $\bar{V}$ of $X_{(K)}$ and of the closed subpreschemes of $X_{(K)}$ induced on the opens $\bar{U}$ and $\bar{V}$. If in addition X is of finite type over k , K' is a finite separable extension of k .

Proof. Taking into account (4.9.6) and (4.8.11), we only have to demonstrate the last assertion. By virtue of (4.8.15, , (ii)), one can restrict oneself to the case where K is algebraically closed. Let $\bar{k}$ be the algebraic closure of k contained in K , and denote by W_i the connected components of $X_{(\bar{k})}$ ($1 \leq i \leq m$); if $p : X_{(K)} \rightarrow X_{(\bar{k})}$ is the canonical projection, we know that the $p^{-1}(W_i)$ are the connected components of $X_{(K)}$ (4.4.6), hence $\bar{U}$ is a union of a certain number of these components, and is therefore defined over $\bar{k}$; the conclusion follows from (4.9.5, , b)). $\square$

§5. DIMENSION, DEPTH, AND REGULARITY IN LOCALLY NOETHERIAN PRESCHMES

This section merely restates, in geometric language, and with various complements of a technical nature, the notions and results of commutative Algebra exposed in chapter 0, §§16 and 17. IV-2 | 86

5.1. Dimension of preschemes

(5.1.1). Let A be a ring, $\mathfrak{J}$ an ideal of A . Recall (0, 16.1) that the definitions of the dimension of a ring (0, 16.1.1) and those of the theory of combinatorial dimension of topological spaces (0, 14.1.2) give the following relations:

$$(5.1.1.1) \quad \dim(\operatorname{Spec}(A)) = \dim(A)$$

$$(5.1.1.2) \quad \dim(V(\mathfrak{J})) = \dim(A/\mathfrak{J})$$

$$(5.1.1.3) \quad \operatorname{codim}(V(\mathfrak{J}), \operatorname{Spec}(A)) = \operatorname{ht}(\mathfrak{J})$$

(5.1.1.4) $\operatorname{Spec}(A)$ is a catenary space $\Leftrightarrow A$ is a catenary ring.

(5.1.1.5) $\operatorname{Spec}(A)$ is equidimensional $\Leftrightarrow A$ is equidimensional $\Leftrightarrow$ the minimal prime ideals $\mathfrak{p}_\alpha$ of A are such that the dimensions of the rings $A/\mathfrak{p}_\alpha$ are all equal.

(5.1.1.6) $\operatorname{Spec}(A)$ is equicodimensional $\Leftrightarrow A$ is equicodimensional $\Leftrightarrow$ all the maximal ideals have the same height.

Recall that a Noetherian ring A is called *biequidimensional* if $\operatorname{Spec}(A)$ is biequidimensional, that is if $\operatorname{Spec}(A)$ is Noetherian and A is equidimensional, catenary and of finite dimension.

Proposition (5.1.2). — Let X be a prescheme, Y a closed irreducible part of X , y the generic point of Y . We have

$$(5.1.2.1) \quad \operatorname{codim}(Y, X) = \dim(\mathcal{O}_{X,y}).$$

Proof. Indeed (I, 2.4.2) the closed irreducible parts of X containing y are canonically in bijective correspondence with the closed irreducible parts of $\operatorname{Spec}(\mathcal{O}_{X,y})$, hence in bijective correspondence with the prime ideals of $\mathcal{O}_{X,y}$, and (5.1.2.1) results from the definitions. $\square$

In particular, one recovers the fact that the generic points of the irreducible components of X are the *only* points $x \in X$ such that $\dim(\mathcal{O}_x) = 0$ (I, 1.1.14).

Corollary (5.1.3). — For every closed part Y of a prescheme X , we have

$$(5.1.3.1) \quad \operatorname{codim}(Y, X) = \inf_{y \in Y} \dim(\mathcal{O}_{X,y}).$$

Moreover, if X is locally Noetherian, then for every $x \in Y$

$$(5.1.3.2) \quad \text{codim}_x(Y, X) = \inf_{y \in Y, x \in \overline{\{y\}}} \dim(\mathcal{O}_{X,y}).$$

Proof. The relation (5.1.3.2) follows from (5.1.3.1) and (0, 14.2.6). $\square$

This corollary permits us to *define*, for an arbitrary part Y of a prescheme X , the *codimension* $\text{codim}(Y, X)$ of Y in X as equal to the second member of (5.1.3.1).

Proposition (5.1.4). — *For every prescheme X , we have*

$$(5.1.4.1) \quad \dim(X) = \sup_{x \in X} (\dim(\mathcal{O}_x)).$$

If every closed irreducible part of X contains a closed point, we also have

$$(5.1.4.2) \quad \dim(X) = \sup_{x \in F} (\dim(\mathcal{O}_x))$$

where F is the set of closed points of X .

Proof. It suffices to remark (by virtue of (I, 2.4.2)) that the chains of closed irreducible parts of X correspond bijectively to the chains of closed irreducible parts of a local scheme of X ; when every closed irreducible part of X contains a closed point, one can obviously restrict to the local schemes at the closed points of X . $\square$

One will note that every closed irreducible part of X contains a closed point when X is *quasi-compact* (0_I, 2.1.3); we shall see a little further (5.1.11) that this also holds when X is *locally Noetherian*.

Corollary (5.1.5). — *For a prescheme X to be catenary, it is necessary and sufficient that, for every $x \in X$, the local ring $\mathcal{O}_x$ be catenary. If moreover every closed irreducible part of X contains a closed point, for X to be catenary, it suffices, for every closed point x of X , that $\mathcal{O}_x$ be catenary.*

Proof. The reasoning is the same as in (5.1.4), since it involves comparing chains of closed irreducible parts having the same endpoints. $\square$

(5.1.6). We shall now examine properties more special linked to Noetherian hypotheses.

Recall (0, 16.2.3) that a *local Noetherian ring* $A \neq 0$ is of *finite dimension*, equal to the minimum number of generators of an ideal of definition of A ; for every prime ideal $\mathfrak{p}$ of A , the height of $\mathfrak{p}$, equal to $\dim(A_{\mathfrak{p}})$, is therefore also finite. These properties, combined with (5.1.2), (5.1.3) and (5.1.4), show that:

Proposition (5.1.7). — *For every non-empty closed part Y of a locally Noetherian prescheme X , $\text{codim}(Y, X)$ is finite. If X is Noetherian and affine and Y a closed irreducible part of X , $\text{codim}(Y, X)$ is equal to the minimum number of sections s_i of $\mathcal{O}_X$ above X such that Y is an irreducible component of the set of $x \in X$ such that $s_i(x) = 0$ for every i .*

Corollary (5.1.8). — *Let X be a locally Noetherian prescheme, $\mathcal{L}$ an $\mathcal{O}_X$ -Module invertible, f a section of $\mathcal{L}$ above X . Then every irreducible component of the set Z of $x \in X$ such that $f(x) = 0$ is of codimension ≤ 1 in X ; it is of codimension 1 if Z contains no irreducible component of X .*

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Proof. One can restrict to the case where $\mathcal{L} = \mathcal{O}_X$. If y is the generic point of an irreducible component of Z , the ideal (f_y) of $\mathcal{O}_{X,y}$ must be such that $\mathcal{O}_{X,y}/(f_y)$ has only a single prime ideal, which means that f_y generates an ideal of definition of the local Noetherian ring $\mathcal{O}_{X,y}$; we therefore have $\text{codim}(Y, X) \leq 1$ (5.1.7); if Z contains no irreducible component of X , we cannot have $\text{codim}(Y, X) = 0$ by virtue of (0, 14.2.1). $\square$

Proposition (5.1.9). — *Let X be a locally Noetherian prescheme, Y a closed part of X . If $x \in Y$ is such that $\mathcal{O}_{X,x}$ is a catenary ring, we have*

$$(5.1.9.1) \quad \text{codim}_x(Y, X) = \dim(\mathcal{O}_{X,x}) - \text{codim}(\overline{\{x\}}, Y) = \dim(\mathcal{O}_{X,x}) - \dim(\mathcal{O}_{Y,x}).$$

Proof. Let Y_i ($1 \leq i \leq n$) be the irreducible components of Y containing x (which are finite in number since Y is locally Noetherian), and let y_i be the generic point of Y_i . Setting $A = \mathcal{O}_{X,x}$, and $\mathfrak{p}_i$ being the prime ideal of A corresponding to $Y_i \cap \text{Spec}(A)$, the closed irreducible parts of Y containing x correspond bijectively to the prime ideals of A containing one of the $\mathfrak{p}_i$, hence $\dim(\mathcal{O}_{Y,x}) = \sup_i \dim(A/\mathfrak{p}_i)$; on the other hand, $\mathcal{O}_{X,y_i} = A_{\mathfrak{p}_i}$, and the hypothesis on A implies $\dim(A) = \dim(A/\mathfrak{p}_i) + \dim(A_{\mathfrak{p}_i})$ (0, 16.1.4); the conclusion results therefore from the relation $\text{codim}_x(Y, X) = \inf_i (\dim(\mathcal{O}_{X,y_i}))$ (5.1.2) and (0, 14.2.6). $\square$

Proposition (5.1.10). — (i) *In every prescheme X , every locally constructible non-empty part E contains a point x such that $\{x\}$ is a locally closed part of X (or, which amounts to the same, such that x is isolated in $\overline{\{x\}}$).*
 (ii) *Let X be a locally Noetherian prescheme, x a point of X such that $\{x\}$ is locally closed in X ; then $\dim(\overline{\{x\}}) \leq 1$; every point $y \neq x$ of $\overline{\{x\}}$ is therefore a closed point in X .*

Proof. (i) The result is a particular case of the following lemma: $\square$

Lemma (5.1.10.1). — *Let X be a topological space having the following property: for every locally closed non-empty part $Z \neq \emptyset$ of X , there exists a part Z' of Z , locally closed in X (or in Z , which amounts to the same), and a point $x \in Z'$, closed in Z' . Then every locally closed non-empty part $Z \neq \emptyset$ of X contains a point x isolated in $\overline{\{x\}}$.*

Proof. Indeed, let $Z' \subset Z$ be a locally closed part containing a point x such that x be closed in Z' . There is a neighbourhood U open of x in X such that $Z' \cap U$ is closed in U , hence x is also closed in U ; this means that $U \cap \overline{\{x\}} = \{x\}$, in other words x is isolated in $\overline{\{x\}}$. $\square$

The lemma applies to every space underlying a prescheme X , for Z is then also an underlying space of a prescheme (I, 5.2.1) and it suffices to take for Z' an open affine in Z , which is a Kolmogorov space quasi-compact, hence contains a closed point (0_I, 2.1.3).

(ii) Let Z be the reduced subscheme of X having $\overline{\{x\}}$ as underlying space; the hypothesis implies that $\{x\}$ is open in Z ; hence, for every $z \in Z$, the generic point x of $\text{Spec}(\mathcal{O}_{Z,z})$ is isolated in $\text{Spec}(\mathcal{O}_{Z,z})$; but $A = \mathcal{O}_{Z,z}$ is a local Noetherian integral ring, and the hypothesis implies that there exists an $f \in A$ such that A_f is a field; one knows (0, 16.3.3) that this implies $\dim(A) \leq 1$. Since $\dim(\mathcal{O}_{Z,z}) \leq 1$ for every $z \in Z$, we have $\dim(Z) \leq 1$.

Corollary (5.1.11). — *If X is a locally Noetherian prescheme non-empty and containing a closed point, every closed part of X contains a closed point.*

Proof. Indeed, every closed part of X is constructible (0_{III}, 9.1.1 and 9.1.5) and it suffices to apply (5.1.10). $\square$

(5.1.12). Let X be a prescheme, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite type, $S = \text{Supp}(\mathcal{F})$ its support, which is closed in X (0_I, 5.2.2). If, for every $x \in X$, one considers $\text{Supp}(\mathcal{F}_x)$ as a closed part of the local scheme $\text{Spec}(\mathcal{O}_x)$, by definition (0, 16.1.7) $\dim(\mathcal{F}_x) = \dim(\text{Supp}(\mathcal{F}_x))$; but we have

$$\text{Supp}(\mathcal{F}_x) = S \cap \text{Spec}(\mathcal{O}_{X,x}) = \text{Spec}(\mathcal{O}_{S,x})$$

where, in the last term, S denotes any closed subscheme of X having S as underlying space. Remarking that the irreducible components of $\text{Spec}(\mathcal{O}_{X,x})$ and the irreducible components of $\text{Spec}(\mathcal{O}_{S,x})$ are the intersections of the irreducible components of S containing x , and correspond bijectively to these latter, one sees that

$$(5.1.12.1) \quad \dim(\mathcal{F}_x) = \dim(\mathcal{O}_{S,x})$$

by virtue of (5.1.1.1); it also results from (5.1.2) that

$$(5.1.12.2) \quad \dim(\mathcal{F}_x) = \text{codim}(\overline{\{x\}}, S)$$

when X is locally Noetherian.

One says that $\mathcal{F}$ is *equidimensional at $x \in X$* if $\mathcal{F}_x$ is an equidimensional $\mathcal{O}_{X,x}$ -module (0, 16.1.7), that is if $\text{Supp}(\mathcal{F}_x)$ is equidimensional as a closed part of $\text{Spec}(\mathcal{O}_x)$; this amounts to saying that the

ring $\mathcal{O}_{S,x}$ is *equidimensional*. The *dimension* of $\mathcal{F}$ is denoted $\dim(\mathcal{F})$ and means the dimension of the support $\text{Supp}(\mathcal{F})$; it results from (5.1.4) and (5.1.12.1) that

$$(5.1.12.3) \quad \dim(\mathcal{F}) = \sup_{x \in X} \dim(\mathcal{F}_x).$$

If $X = \text{Spec}(A)$ is an affine scheme and $\mathcal{F} = \tilde{M}$, where M is an A -module of finite type, one has $\dim(\mathcal{F}) = \dim(M)$ by virtue of (0, 16.1.7) and (5.1.4).

Proposition (5.1.13). — *Let X be a prescheme, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite type, x a point of X , x' a generisation of x in X ; then*

$$(5.1.13.1) \quad \dim(\mathcal{F}_{x'}) \leq \dim(\mathcal{F}_x).$$

Proof. This follows immediately from (5.1.12.1) and the definitions. $\square$

5.2. Dimension of an algebraic prescheme

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Proposition (5.2.1). — *Let k be a field, X an irreducible prescheme locally of finite type over k , ξ its generic point. Then X is biequidimensional and we have $\dim(X) = \deg.\text{tr}_k k(\xi)$.*

Proof. Every local ring $\mathcal{O}_x$ is the local ring of a k -algebra of finite type, hence a quotient of a regular local ring (0, 17.3.9), and one knows (0, 16.5.12) that these rings are catenary; since X is irreducible, (5.1.5) shows that X is catenary, and it suffices to show that for every closed point $x \in X$, we have

$$(5.2.1.1) \quad \dim(\mathcal{O}_x) = \deg.\text{tr}_k k(\xi).$$

One can obviously suppose X reduced and affine, hence integral, with ring of fractions A , $k(\xi) = K$ the field of fractions of A . One knows (Bourbaki, *Alg. comm.*, chap. V, §3, no. 1, th. 1) that there exists a sub- k -algebra $B = k[t_1, \dots, t_n]$ of A , where the t_i are algebraically independent over k , such that A is a finite B -algebra. Let $\mathfrak{m} = \mathfrak{i}_x$, which by hypothesis is a maximal ideal of A ; $\mathfrak{n} = B \cap \mathfrak{m}$ is therefore also a maximal ideal of B (Bourbaki, *Alg. comm.*, chap. V, §2, no. 1, prop. 1), and $A_{\mathfrak{m}}$ is a local ring of the $B_{\mathfrak{n}}$ -algebra $\text{finite } S^{-1}A$, where $S = B - \mathfrak{n}$; since $B_{\mathfrak{n}}$ is integrally closed and $S^{-1}A$ integral, $\dim(A_{\mathfrak{m}}) = \dim(B_{\mathfrak{n}})$ (0, 16.1.6). One can therefore restrict to the case where $A = k[t_1, \dots, t_n]$; one knows then that $k' = k(x)$ is a finite extension of k (I, 6.4.2); let $A' = k'[t_1, \dots, t_n]$; taking into account (I, 2.4.6) and (3.3.14), there is a maximal ideal $\mathfrak{m}'$ of A' above $\mathfrak{m}$ and such that k' is the residue field of $A'_{\mathfrak{m}'}$. Since A' is integral and is a finite A -algebra, the same reasoning as above shows $\dim(A'_{\mathfrak{m}'}) = \dim(A_{\mathfrak{m}})$, so one is reduced to the case where $A/\mathfrak{m} = k$. Now in this case $\mathfrak{m}$ is generated by the polynomials $t_i - a_i$ (where $a_i \in k$, $1 \leq i \leq n$, the a_i being the images of the t_i in $A/\mathfrak{m}$). Replacing t_i by $t_i - a_i$, one sees that $\mathfrak{m} = \sum_{i=1}^n A t_i$. The completion of the local ring $A_{\mathfrak{m}}$ is then the ring of formal power series $k[[t_1, \dots, t_n]]$; one knows that it has the same dimension as $A_{\mathfrak{m}}$, and on the other hand, the dimension of $k[[t_1, \dots, t_n]]$ is equal to n (0, 17.1.4); whence the conclusion. $\square$

Corollary (5.2.2). — *For a prescheme X locally of finite type over a field k , $\dim(X)$ coincides with the number defined in (4.1.1).*

Proof. Indeed, if X_{α} are the reduced subpreschemes having as underlying spaces the irreducible components of X , we have $\dim(X) = \sup_{\alpha} (\dim(X_{\alpha}))$ (0, 14.1.2.1) and it suffices to apply (5.2.1) to the X_{α} . $\square$

Corollary (5.2.3). — *Let X be a prescheme locally of finite type over a field k , x a point of X . We have*

$$(5.2.3.1) \quad \dim_x(X) = \dim(\mathcal{O}_x) + \deg.\text{tr}_k k(x).$$

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Proof. One knows that there exists an open neighbourhood U of x in X such that $\dim_x(X) = \dim(U)$ (0, 14.1.4.1), and one can suppose that the irreducible components of U are the $X_i \cap U$, where the X_i are the irreducible components of X containing x ; since $U \cap X_i$ is dense in X_i , it follows from (4.1.1.3) that $\dim(X_i) = \dim(U \cap X_i)$, hence $\dim_x(X) = \sup_i (\dim(X_i))$. Moreover, the minimal prime ideals of $\mathcal{O}_x$ correspond to the generic points of the X_i , hence (0, 16.1.1.1), $\dim(\mathcal{O}_x) = \sup_i (\dim(\mathcal{O}_{X_i, x}))$. One is therefore reduced to the case where X is irreducible; by (5.2.1), X is biequidimensional, hence

$\dim(X) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X)$, and one knows that $\dim(\overline{\{x\}}) = \deg.\operatorname{tr}_k k(x)$ by (5.1.2) and $\operatorname{codim}(\overline{\{x\}}, X) = \dim(\mathcal{O}_x)$ by (5.1.2). $\square$

Corollary (5.2.4). — *Let X be a prescheme locally of finite type over a field k , $\mathcal{L}$ an $\mathcal{O}_X$ -Module invertible, f a section of $\mathcal{L}$ above X , Y the set of $x \in X$ such that $f(x) = 0$ and that is rare in X . Then $\dim(Y) \leq \dim(X) - 1$; the two members of this inequality are equal if Y meets every irreducible component of maximal dimension of X .*

If (X_α) is the family of irreducible components of X , we have

$$\dim(Y) = \sup_{\alpha} (\dim(Y \cap X_\alpha))$$

and one is therefore reduced to the case where X is irreducible ($Y \cap X_\alpha$ being rare in X_α since X_α has non-empty interior in X). One can restrict to the case $Y \neq \emptyset$; then, for every maximal point x of Y , we have (since X is biequidimensional) $\dim(\overline{\{x\}}) = \dim(X) - \operatorname{codim}(\overline{\{x\}}, X)$, and since Y is rare in X , we have $\operatorname{codim}(\overline{\{x\}}, X) = 1$ by (5.1.8); whence the corollary.

Remarks (5.2.5). — (i) Contrary to what occurs for algebraic k -preschemes, a locally Noetherian prescheme X is not necessarily catenary (cf. (5.6.11)); it is however if all its local rings $\mathcal{O}_x$ are quotients of regular rings (0, 16.5.12) and in particular if X is regular (I, 4.1.4). But even an integral scheme $X = \operatorname{Spec}(A)$, where A is a regular ring, is not necessarily biequidimensional, in other words the codimensions in X of the various closed points of X are not necessarily the same.

(ii) When X is a prescheme locally of finite type over a field k , one saw in (4.1.1.3) that $\dim(U) = \dim(X)$ for every open everywhere dense subset U of X . This result does not extend to Noetherian preschemes, even if they are biequidimensional.

5.3. Dimension of the support of a Module and Hilbert polynomial This section uses results from IV-2 | 92 chapter III; it will not be used later in this chapter.

Proposition (5.3.1). — *Let A be a local Artinian ring, X a projective scheme over $\operatorname{Spec}(A)$, $\mathcal{L}$ an $\mathcal{O}_X$ -Module invertible very ample relative to A , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module $\neq 0$; if we set $\mathcal{F}(n) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n}$ for $n \in \mathbf{Z}$, the degree of the Hilbert polynomial $P(n) = \chi_A(\mathcal{F}(n))$ of $\mathcal{F}$ with respect to A (III, 2.5.3) is equal to the dimension of $\operatorname{Supp}(\mathcal{F})$.*

Proof. We reason by induction on $d = \dim(\operatorname{Supp}(\mathcal{F}))$. One knows that there exists a closed subprescheme Y of X whose $\operatorname{Supp}(\mathcal{F})$ is the underlying space, and an $\mathcal{O}_Y$ -Module coherent $\mathcal{G}$ such that $\mathcal{F} = j_*(\mathcal{G})$, where $j : Y \rightarrow X$ is the canonical injection (ErrIII, 30). It is immediate that the Hilbert polynomials of $\mathcal{F}$ and of $\mathcal{G}$ are the same, so one can restrict to the case where $X = \operatorname{Supp}(\mathcal{F})$. Suppose first $d = 0$; all the points of X are closed, X is an Artinian scheme (I, 6.2.2), hence $\mathcal{F}(n) = \mathcal{F}$ for every integer n , and one has by (III, 2.5.3)

$$\chi_A(\mathcal{F}(n)) = \operatorname{long}_A(\Gamma(X, \mathcal{F}))$$

for n large enough, and the degree of the polynomial is therefore 0. Suppose now $d > 0$, and set $Z = \operatorname{Ass}(\mathcal{F})$, which is a finite set (3.1.6); there exists an integer $m > 0$ and a section $f \in \Gamma(X, \mathcal{L}^{\otimes m})$ such that X_f is a neighbourhood of Z (II, 4.5.4). Multiplication by f is a homomorphism

$$\mu_f : \mathcal{F} \longrightarrow \mathcal{F}(m)$$

which is injective (3.1.8); one therefore has an exact sequence

$$(5.3.1.1) \quad 0 \longrightarrow \mathcal{F} \xrightarrow{\mu_f} \mathcal{F}(m) \longrightarrow \mathcal{G} \longrightarrow 0$$

where $\mathcal{G}$ is coherent. By virtue of the Nakayama lemma, the points $x \in \operatorname{Supp}(\mathcal{G})$ are exactly those for which $f(x) = 0$. We shall now deduce that

$$(5.3.1.2) \quad \dim(\operatorname{Supp}(\mathcal{G})) = d - 1.$$

This will result from the following lemma: $\square$

Lemma (5.3.1.3). — *Let A be a local Artinian ring, X a projective scheme over $\operatorname{Spec}(A)$, $\mathcal{L}$ an $\mathcal{O}_X$ -Module invertible ample relative to A ; then, for every section g of $\mathcal{L}$ above X such that $g(x) = 0$ meets every irreducible component of X of dimension $\neq 0$.*

Proof. Indeed (II, 5.5.7), the set X_g is an open affine of X , and if it contains an irreducible component X' of X , the reduced closed subscheme of X having X' as underlying space is both projective and affine over A (III, 4.4.2), and is therefore finite over A , hence an Artinian scheme, therefore of dimension 0. $\square$

This lemma being established, note that since Z contains the maximal points of X , X_f is dense, hence $\text{Supp}(\mathcal{G})$ is rare in X , and the relation (5.3.1.2) results from the lemma and from (5.2.4).

From the exact sequence (5.3.1.1), we deduce, for every n , the exact sequence

$$0 \longrightarrow \mathcal{F}(n) \longrightarrow \mathcal{F}(n+m) \longrightarrow \mathcal{G}(n) \longrightarrow 0$$

and for n large enough, one therefore has the exact sequence (III, 2.2.3)

$$0 \longrightarrow \Gamma(X, \mathcal{F}(n)) \longrightarrow \Gamma(X, \mathcal{F}(n+m)) \longrightarrow \Gamma(X, \mathcal{G}(n)) \longrightarrow 0$$

whence, taking into account (III, 2.5.3), for n large enough,

$$\chi_A(\mathcal{G}(n)) = \chi_A(\mathcal{F}(n+m)) - \chi_A(\mathcal{F}(n)).$$

Since by virtue of (5.3.1.2) and the induction hypothesis, the degree of the polynomial $\chi_A(\mathcal{G}(n))$ is $d-1$, the preceding relation implies that the degree of $\chi_A(\mathcal{F}(n))$ is d , Q.E.D.

5.4. Dimension of the image of a morphism

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Proposition (5.4.1). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism.*

- (i) *If f is quasi-finite, we have $\dim(X) \leq \dim(\overline{f(X)}) \leq \dim(Y)$.*
- (ii) *If f is surjective and open (resp. surjective and closed), we have $\dim(X) \geq \dim(Y)$.*

Proof. (i) One can replace f by f_{red} (II, 6.2.4), hence suppose X and Y reduced; if Z is the closed reduced subscheme of Y having $\overline{f(X)}$ as underlying space, we have $f = j \circ g$, where $j : Z \rightarrow Y$ is the canonical injection and $g : X \rightarrow Z$ is a quasi-finite morphism (I, 5.2.2) and (II, 6.2.4). One can therefore restrict to the case where $f(X)$ is dense in Y . For every $x \in X$, $\mathcal{O}_x$ is then an $\mathcal{O}_{f(x)}$ -module quasi-finite (II, 6.2.2), and it follows that $\mathfrak{m}_{f(x)}\mathcal{O}_x$ is an ideal of definition of $\mathcal{O}_x$ (0I, 7.4.4); but one knows (0, 16.3.10) that if $A \rightarrow B$ is a local homomorphism of local Noetherian rings, such that, if $\mathfrak{m}$ is the maximal ideal of A , $\mathfrak{m}B$ is an ideal of definition of B , then $\dim(B) \leq \dim(A)$; this finishes proving (i) by virtue of (5.1.4).

(ii) The definition of dimension (0, 14.1.2) shows that it suffices to prove that for every suite $(y_i)_{0 \leq i \leq n}$ of distinct elements of Y such that $y_i \in \overline{\{y_{i+1}\}}$ for $0 \leq i \leq n-1$, there exists a suite $(x_i)_{0 \leq i \leq n}$ of points of X such that $x_i \in \overline{\{x_{i+1}\}}$ for $0 \leq i \leq n-1$ and $f(x_i) = y_i$ for every i . Suppose first f surjective and open, and let us show the existence of x_i by induction on i ; the existence of $x_0 \in X$ such that $f(x_0) = y_0$ results from the fact that f is surjective. If the x_i have been determined for $i \leq m$ in a way satisfying $f(x_i) = y_i$ for $i \leq m$ and $x_i \in \overline{\{x_{i+1}\}}$ for $i \leq m$, we note that since f is surjective and y_{m+1} a generisation of y_m (1.10.3), there exists $x_{m+1} \in f^{-1}(y_{m+1})$ which is a generisation of x_m and the induction can continue.

Suppose now f surjective and closed, and let us show the existence of x_i by descending induction on i ; the existence of x_n such that $f(x_n) = y_n$ results again from the fact that f is surjective. Suppose now f surjective and closed and let us show the existence of the x_i by descending induction on i : if the x_i have been determined satisfying the conditions sought for $i > m$, we note $f(\overline{\{x_{m+1}\}}) = \overline{\{y_{m+1}\}}$ since f is closed (Bourbaki, *Top. gén.*, chap. I^{er}, 3^e éd., §5, no. 4, prop. 9); there exists therefore $x_m \in \overline{\{x_{m+1}\}}$ such that $f(x_m) = y_m$ and the descending induction can continue. $\square$

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Corollary (5.4.2). — *If X, Y are two locally Noetherian preschemes, $f : X \rightarrow Y$ a finite morphism (hence closed), we have $\dim(X) = \dim(\overline{f(X)})$. If moreover f is surjective, $\dim(X) = \dim(Y)$.*

Remarks (5.4.3). — (i) One saw in (4.1.2) that if X and Y are two preschemes locally of finite type over a field k and f a k -morphism, the inequality $\dim(Y) \leq \dim(X)$ is already verified when f is quasi-compact and dominant. On the contrary, one can have $\dim(Y) > \dim(X)$ even when f is of finite type, bijective and a local immersion, if one only supposes X and Y locally Noetherian.

- (ii) If A and B are two Noetherian rings such that $A \subset B$ and B is a finite A -algebra, the corollary (5.4.2) shows again that $\dim(B) = \dim(A)$ (0, 16.1.5). Suppose moreover that A is a local Noetherian ring; then B is a Noetherian *semi-local* ring (Bourbaki, *Alg. comm.*, chap. IV, §2, no. 5, cor. 3 of prop. 9); if $\mathfrak{n}_i$ ($1 \leq i \leq r$) are the maximal ideals of B , we have therefore

$$(5.4.3.1) \quad \dim(A) = \sup_i \dim(B_{\mathfrak{n}_i}).$$

One will observe that, under these conditions, one does not necessarily have $\dim(B_{\mathfrak{n}_i}) = \dim(A)$ for every i .

5.5. Dimension formula for a morphism of finite type

(5.5.1). Recall (0, 16.3.9) that if A, B are two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, we have

$$(5.5.1.1) \quad \dim(B) \leq \dim(A) + \dim(B \otimes_A k).$$

We deduce:

Proposition (5.5.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism, x a point of X , $y = f(x)$. Then*

$$(5.5.2.1) \quad \dim(\mathcal{O}_x) \leq \dim(\mathcal{O}_y) + \dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)).$$

In particular, if x is a maximal point of the fibre $f^{-1}(y)$, we have

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$$(5.5.2.2) \quad \dim(\mathcal{O}_x) \leq \dim(\mathcal{O}_y)$$

since $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$ is the local ring of x in the prescheme $f^{-1}(y)$, and is of dimension 0 by hypothesis.

We shall now obtain a more precise formula than (5.5.2.1) when one supposes that f is a morphism of finite type.

Proposition (5.5.3). — *Let A be a Noetherian ring, T an indeterminate, $\mathfrak{p}$ a prime ideal of A ; then $\mathfrak{p}' = \mathfrak{p}A[T]$ is a prime ideal of $B = A[T]$ and $\mathfrak{p}' \cap A = \mathfrak{p}$. There exists an infinity of prime ideals of B distinct from $\mathfrak{p}'$, whose intersection with A is $\mathfrak{p}$; these ideals are pairwise without inclusion relation. Moreover, if $\mathfrak{q}$ is such an ideal, we have*

$$(5.5.3.1) \quad \dim(B_{\mathfrak{q}}) = \dim(B_{\mathfrak{p}'}) + 1 = \dim(A_{\mathfrak{p}}) + r$$

Proof. In the first assertions, we immediately reduce, by replacing A by $A/\mathfrak{p}$ and observing that $A[T]/\mathfrak{p}A[T] = (A/\mathfrak{p})[T]$, to the case $\mathfrak{p} = 0$; they then result from the fact that the prime ideals of $B = A[T]$ whose intersection with A reduces to 0 are exactly those which do not meet the multiplicative subset $S = A - \{0\}$ of the integral ring A ; now there is an increasing bijection from the set of these prime ideals onto the set of prime ideals of $S^{-1}A[T] = K[T]$, where K is the field of fractions of A (Bourbaki, *Alg. comm.*, chap. II, §2, no. 5, prop. 11). Moreover, by (5.5.1.2), $\dim(B_{\mathfrak{q}}) \leq \dim(A_{\mathfrak{p}}) + \dim(B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}})$, and if k is the field of fractions of $A/\mathfrak{p}$, $B_{\mathfrak{q}}/\mathfrak{p}B_{\mathfrak{q}}$ identifies canonically with $(k[T])_{\bar{\mathfrak{q}}}$, hence is a discrete valuation ring, hence of dimension 1. Finally, if $\mathfrak{p} = \mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_m$ is a chain of prime ideals of A of maximal length, the ideals $\mathfrak{p}_j B$ ($0 \leq j \leq m$) are prime, pairwise distinct in B , and contained in $\mathfrak{q}$; so $\dim(B_{\mathfrak{q}}) \geq m + 1 = \dim(A_{\mathfrak{p}}) + 1$ and hence $\dim(B_{\mathfrak{q}}) = \dim(A_{\mathfrak{p}}) + 1$. This relation also writes $\text{ht}(\mathfrak{q}) = \text{ht}(\mathfrak{p}) + 1$; since $\mathfrak{q} \neq \mathfrak{p}'$, we also have $\text{ht}(\mathfrak{q}) \geq \text{ht}(\mathfrak{p}') + 1 \geq \text{ht}(\mathfrak{p}) + 1$ by definition of the height of a prime ideal; this finishes proving (5.5.3.1). $\square$

Corollary (5.5.4). — *For every Noetherian ring A , we have*

$$(5.5.4.1) \quad \dim(A[T_1, \dots, T_r]) = \dim(A) + r$$

(T_i indeterminates).

Corollary (5.5.5). — *For every locally Noetherian prescheme X , the dimension of $X[T_1, \dots, T_r] = X \otimes_{\mathbb{Z}} \mathbb{Z}[T_1, \dots, T_r]$ (T_i indeterminates) is $\dim(X) + r$.*

Proof. This results from (5.5.4) and (0, 14.1.7). $\square$

Corollary (5.5.6). — *Under the hypotheses of (5.5.3), let $\mathfrak{q}$ be a prime ideal of $B = A[T]$ such that $\mathfrak{q} \cap A = \mathfrak{p}$; if k and k' are the residue fields of $A_{\mathfrak{p}}$ and $B_{\mathfrak{q}}$ respectively, we have*

$$(5.5.6.1) \quad \dim(A_{\mathfrak{p}}) + 1 = \dim(B_{\mathfrak{q}}) + \deg.\mathrm{tr}_k k'.$$

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Proof. If $\mathfrak{q} = \mathfrak{p}B$, by (5.5.3.1), $\dim(B_{\mathfrak{q}}) = \dim(A_{\mathfrak{p}})$, and in this case $k' = k(T)$, so formula (5.5.6.1) is indeed verified. In the contrary case, $\dim(B_{\mathfrak{q}}) = \dim(A_{\mathfrak{p}}) + 1$, and since $\mathfrak{q}$ corresponds to a nonzero prime ideal $\mathfrak{q}' \neq 0$ of $k[T]$, k' is an algebraic extension of k , so we again have formula (5.5.6.1). $\square$

Lemma (5.5.7). — *Let A be a local integral Noetherian ring, $\mathfrak{m}$ its maximal ideal, k its residue field.*

label=(i) For every integral ring B containing A , such that $B = A[x]$ (for some $x \in B$) and every prime ideal $\mathfrak{q}$ of B such that $\mathfrak{q} \cap A = \mathfrak{m}$, we have

$$(5.5.7.1) \quad \dim(A) + \deg.\mathrm{tr}_A B \geq \dim(B_{\mathfrak{q}}) + \deg.\mathrm{tr}_k k'$$

where k' denotes the residue field of $B_{\mathfrak{q}}$ and $\deg.\mathrm{tr}_A B$ the degree of transcendence of the field of fractions of B over that of A .

lbel=(ii) Suppose that for every maximal ideal $\mathfrak{n}$ of $A[T]$ such that $\mathfrak{n} \cap A = \mathfrak{m}$, the ring $(A[T])_{\mathfrak{n}}$ is catenary; then the two members of (5.5.7.1) are equal.

Proof. (i) If x is transcendental over the field of fractions of A , then $\deg.\mathrm{tr}_A B = 1$ and the two members of (5.5.7.1) are equal by virtue of (5.5.6). In the contrary case, $B = A[T]/\mathfrak{p}$, where $\mathfrak{p}$ is a nonzero prime ideal of $A[T]$ such that $\mathfrak{p} \cap A = 0$ since B contains A ; so $\mathrm{ht}(\mathfrak{p}) = 1$ by virtue of (5.5.3). The ideal $\mathfrak{q}$ of B is of the form $\mathfrak{n}/\mathfrak{p}$, where $\mathfrak{n} \supset \mathfrak{p}$ is a prime ideal of $A[T]$ such that $\mathfrak{n} \cap A = \mathfrak{m}$, and $B_{\mathfrak{q}} = (A[T])_{\mathfrak{n}}/\mathfrak{p}(A[T])_{\mathfrak{n}}$; formula (0, 16.1.4.1), applied to $X = \mathrm{Spec}((A[T])_{\mathfrak{n}})$ and $Y = \mathrm{Spec}(B_{\mathfrak{q}})$, gives

$$(5.5.7.2) \quad \dim((A[T])_{\mathfrak{n}}) \geq \mathrm{ht}(\mathfrak{p}(A[T])_{\mathfrak{n}}) + \dim(B_{\mathfrak{q}})$$

since $\mathrm{ht}(\mathfrak{p}(A[T])_{\mathfrak{n}}) = \mathrm{ht}(\mathfrak{p}) = 1$ by virtue of the bijective correspondence between prime ideals of $A[T]$ contained in $\mathfrak{n}$ and prime ideals of $(A[T])_{\mathfrak{n}}$. Finally, formula (5.5.6.1) gives

$$(5.5.7.3) \quad \dim((A[T])_{\mathfrak{n}}) = \dim(A) + 1 - \deg.\mathrm{tr}_k k'$$

since the residue fields of $B_{\mathfrak{q}}$ and of $(A[T])_{\mathfrak{n}}$ are the same; moreover, $\deg.\mathrm{tr}_A B = 0$, which finishes proving (5.5.7.1).

(ii) If $(A[T])_{\mathfrak{n}}$ is catenary, the two members of (5.5.7.2) are also equal (0, 16.1.4), hence so are the two members of (5.5.7.1). $\square$

Theorem (5.5.8). — (Dimension formula). *Let A be a local Noetherian integral ring, B an integral ring containing A and which is an A -algebra of finite type, $\mathfrak{q}$ a prime ideal of B such that $\mathfrak{q} \cap A$ is the maximal ideal $\mathfrak{m}$ of A , k and k' the residue fields of A and $B_{\mathfrak{q}}$ respectively. We then have the inequality*

$$(5.5.8.1) \quad \dim(A) + \deg.\mathrm{tr}_A B \geq \dim(B_{\mathfrak{q}}) + \deg.\mathrm{tr}_k k'.$$

Moreover, the two members are equal if, for every sub- A -algebra A' of finite type of $B[T]$ and every maximal ideal $\mathfrak{m}'$ of A' such that $\mathfrak{m}' \cap A = \mathfrak{m}$, $A'_{\mathfrak{m}'}$ is catenary.

Proof. Let $B = A[x_1, \dots, x_n]$ and argue by induction on n . Set $C = A[x_1, \dots, x_{n-1}]$ and $\mathfrak{r} = \mathfrak{q} \cap C$; $C_{\mathfrak{r}}$ is a local integral Noetherian ring, and if we set $S = C - \mathfrak{r}$, $B' = S^{-1}B$, $B' = C_{\mathfrak{r}}[x_n]$, and $\mathfrak{q}' = S^{-1}\mathfrak{q}$, we have $B_{\mathfrak{q}} = B'_{\mathfrak{q}'}$ and $\mathfrak{q}' \cap C_{\mathfrak{r}} = \mathfrak{r}C_{\mathfrak{r}}$; moreover the fields of fractions of B' and $C_{\mathfrak{r}}$ are those of B and C respectively. By (5.5.7.1), we thus have

$$(5.5.8.2) \quad \dim(C_{\mathfrak{r}}) + \deg.\mathrm{tr}_C B \geq \dim(B_{\mathfrak{q}}) + \deg.\mathrm{tr}_k k'.$$

On the other hand, the induction hypothesis gives

$$(5.5.8.3) \quad \dim(A) + \deg.\mathrm{tr}_A C \geq \dim(C_{\mathfrak{r}}) + \deg.\mathrm{tr}_k k_1$$

whence (5.5.8.1) by adding (5.5.8.2) and (5.5.8.3) member by member. To prove the second assertion, note first that every sub- A -algebra of finite type of $C[T]$ is also a sub- A -algebra of finite type of $B[T]$; so by induction on n , we can suppose that the two members of (5.5.8.3) are equal. On the other hand, to see that the two members of (5.5.8.2) are equal, it suffices, by virtue of (5.5.7), to verify that if $\mathfrak{n}$

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is a maximal ideal of $C_\tau[T]$ such that $\mathfrak{n} \cap C_\tau = \tau C_\tau$, then $(C_\tau[T])_\mathfrak{n}$ is catenary; but $C_\tau[T] = S'^{-1}C[T]$ where $S' = C - \tau$; the ideal $\mathfrak{n}$ is thus of the form $S'^{-1}\mathfrak{n}'$, where $\mathfrak{n}'$ is a prime ideal of $C[T]$ such that $\mathfrak{n}' \cap C = \tau$, whence $(C_\tau[T])_\mathfrak{n} = (C[T])_{\mathfrak{n}'}$; if $\mathfrak{m}'$ is a maximal ideal of $C[T]$ containing $\mathfrak{n}'$, $(C[T])_{\mathfrak{m}'}$ is thus a local ring of $(C[T])_{\mathfrak{n}'}$, and since by hypothesis this last ring is catenary, so is the former by (0, 16.1.4). $\square$

5.6. Dimension formula and universally catenary rings

Proposition (5.6.1). — *Let A be a Noetherian ring. The following properties are equivalent:*

label=a) Every polynomial ring $A[T_1, \dots, T_n]$ is catenary.

lbbel=b) Every A -algebra of finite type is catenary.

lcbel=c) A is catenary, and for every local integral A -algebra, essentially of finite type (1.3.8), B' , designating by $\mathfrak{p}$ the inverse image in A of the maximal ideal $\mathfrak{q}$ of B' , by A' the image of $A_\mathfrak{p}$ in B' , by k and k' the residue fields of A' and B' respectively,

$$(5.6.1.1) \quad \dim(A') + \deg.\mathrm{tr}_{A'}(B') = \dim(B') + \deg.\mathrm{tr}_k k'.$$

Proof. The fact that *b)* implies *c)* results from (5.5.8); indeed, B' is an A' -algebra essentially of finite type (1.3.10), hence of the form $B''_{\mathfrak{q}''}$, where B'' is a sub- A' -algebra of finite type of B' , $\mathfrak{q}''$ a prime ideal of B'' above the maximal ideal $\mathfrak{p}'$ of A' ; moreover the fields of fractions of B' and B'' are the same, hence $\deg.\mathrm{tr}_{A'}(B') = \deg.\mathrm{tr}_{A'}(B'')$. To prove (5.6.1.1), it suffices to show that (under hypothesis *b)*) every sub- A' -algebra A_1 of finite type of $B'[T]$ is catenary; indeed the two members of (5.5.8.1), where A, B , and $\mathfrak{q}$ are replaced by A', B'' , and $\mathfrak{q}''$, will then be equal, whence the equality (5.6.1.1). But hypothesis *b)* implies that every A -algebra essentially of finite type is catenary (0, 16.1.4), and A_1 is such an A -algebra (1.3.9).

It is trivial that *b)* implies *a)*; conversely, *a)* implies *b)*, since every A -algebra of finite type is a quotient of a polynomial algebra (0, 16.1.4).

It remains to prove that *c)* implies *b)*. Since every quotient by a prime ideal of an A -algebra of finite type is an integral A -algebra of finite type, if B is an integral A -algebra of finite type, $\mathfrak{q}$ and $\mathfrak{q}'$ two prime ideals of B such that $\mathfrak{q}' \subset \mathfrak{q}$, by (0, 16.1.4.2)

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$$(5.6.1.2) \quad \dim(B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}) + \dim(B_{\mathfrak{q}'}) = \dim(B_{\mathfrak{q}}).$$

Let $\mathfrak{p}, \mathfrak{p}'$ be the inverse images of $\mathfrak{q}, \mathfrak{q}'$ respectively in A , $\mathfrak{n}$ the kernel of the homomorphism $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}$. The image A' of $A_{\mathfrak{p}}$ in $B_{\mathfrak{q}}$ being isomorphic to $A_{\mathfrak{p}}/\mathfrak{n}$, formula (5.6.1.1) applied to A' and $B' = B_{\mathfrak{q}}$ writes

$$(5.6.1.3) \quad \dim(A_{\mathfrak{p}}/\mathfrak{n}) + \deg.\mathrm{tr}_{A_{\mathfrak{p}}/\mathfrak{n}}(B_{\mathfrak{q}}) = \dim(B_{\mathfrak{q}}) + \deg.\mathrm{tr}_{k(\mathfrak{p})} k(\mathfrak{q}).$$

On the other hand, the kernel of $A_{\mathfrak{p}'} \rightarrow B_{\mathfrak{q}'}$ is $\mathfrak{n}A_{\mathfrak{p}'}$, so, applying formula (5.6.1.1) where A' is replaced by $A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}$ and B' by $B_{\mathfrak{q}'}$, we have

$$(5.6.1.4) \quad \dim(A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}) + \deg.\mathrm{tr}_{A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}}(B_{\mathfrak{q}'}) = \dim(B_{\mathfrak{q}'}) + \deg.\mathrm{tr}_{k(\mathfrak{p}')} k(\mathfrak{q}').$$

Finally, $B/\mathfrak{q}'$ is an integral A -algebra of finite type, the inverse image of $\mathfrak{q}/\mathfrak{q}'$ in A is $\mathfrak{p}$, and the kernel of the homomorphism $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}$ is $\mathfrak{p}'A_{\mathfrak{p}}$, so, applying (5.6.1.1) where A' is replaced by $A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}$ and B' by $B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}$, we have

$$(5.6.1.5) \quad \dim(A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}) + \deg.\mathrm{tr}_{A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}}(B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}) = \dim(B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}) + \deg.\mathrm{tr}_{k(\mathfrak{p})} k(\mathfrak{q}).$$

Adding (5.6.1.4) and (5.6.1.5) member by member, and noting that $k(\mathfrak{p}')$ (resp. $k(\mathfrak{q}')$) is the field of fractions of $A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}$ (resp. $B_{\mathfrak{q}}/\mathfrak{q}'B_{\mathfrak{q}}$); moreover $A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}$ and $A_{\mathfrak{p}}/\mathfrak{n}$ have the same field of fractions, and $B_{\mathfrak{q}'}$ and $B_{\mathfrak{q}}$ have the same field of fractions; finally, since A is supposed catenary, by (0, 16.1.4.2)

$$\dim(A_{\mathfrak{p}}/\mathfrak{p}'A_{\mathfrak{p}}) + \dim(A_{\mathfrak{p}'}/\mathfrak{n}A_{\mathfrak{p}'}) = \dim(A_{\mathfrak{p}}/\mathfrak{n}).$$

Taking these remarks into account and (5.6.1.3), we obtain relation (5.6.1.2). $\square$

Definition (5.6.2). — A Noetherian ring A is said to be *universally catenary* if it satisfies the equivalent conditions *a), b), c)* of (5.6.1).

Remarks (5.6.3). — (i) If A is universally catenary, then so is $S^{-1}A$ for every multiplicative subset S of A , as follows immediately from ((5.6.1), a)) and the fact that every ring of fractions of a catenary ring is catenary. Conversely, if for every prime ideal $\mathfrak{p}$ of A , the ring $A_{\mathfrak{p}}$ is universally catenary, then A is universally catenary: this follows from ((5.6.1), c)), noting that by setting $S = A - \mathfrak{p}$, $S^{-1}A$ is an $A_{\mathfrak{p}}$ -algebra of finite type, and $B_{\mathfrak{q}} = (S^{-1}A)_{S^{-1}\mathfrak{q}}$.

(ii) A locally Noetherian prescheme X is said to be *universally catenary* if, for every $x \in X$, the ring $\mathcal{O}_x$ is universally catenary. It follows from (i) that for A to be a universally catenary ring, it is necessary and sufficient that the scheme $\text{Spec}(A)$ be universally catenary.

(iii) The criterion ((5.6.1), c)) only involves the images of A in integral A -algebras of finite type, hence *integral quotient rings* of A . We conclude that if $\mathfrak{p}_i$ ($1 \leq i \leq n$) are the minimal prime ideals of A , it amounts to the same to say that A is universally catenary or that each of the $A/\mathfrak{p}_i$ is. This also shows that a locally Noetherian prescheme X is universally catenary, or that X_{red} is.

(iv) The criterion ((5.6.1), b)) and remark (i) show that, if A is a universally catenary ring, then so is every A -algebra *essentially of finite type* (1.3.8).

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Proposition (5.6.4). — *Every quotient of a regular ring is universally catenary.*

Proof. It suffices to see that every regular ring A is universally catenary ((5.6.3), (iv)); or, we know that $A[T_1, \dots, T_n]$ is regular (0, 17.3.7), hence catenary (0, 16.5.12), and we conclude by ((5.6.1), a)). $\square$

In particular, it follows from the Cohen structure theorem (0, 19.8.8) that every *complete local Noetherian ring* is universally catenary. Similarly, every algebra of finite type over a field, every prescheme being a quotient of a regular prescheme, and every prescheme *locally of finite type over a field* is universally catenary.

We will see further on (6.3.7) that in (5.6.4), one can replace “regular ring” by “Cohen-Macaulay ring”.

Proposition (5.6.5). — *Let Y be an irreducible locally Noetherian prescheme, X an irreducible prescheme, $f : X \rightarrow Y$ a dominant morphism locally of finite type. Let ξ (resp. η) be the generic point of X (resp. Y), and let $e = \dim(f^{-1}(\eta)) = \deg.\text{tr}_{k(\eta)} k(\xi)$ (“dimension of the generic fibre”, cf. (0_I, 2.1.8) and (4.1.1)). For every $x \in X$, setting $y = f(x)$, we have*

$$(5.6.5.1) \quad e + \dim(\mathcal{O}_y) \geq \deg.\text{tr}_{k(y)} k(x) + \dim(\mathcal{O}_x)$$

$$(5.6.5.2) \quad \dim(\mathcal{O}_x) \leq \dim(\mathcal{O}_y) + \dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)) - \delta(x)$$

where $\delta(x) = \dim_x(f^{-1}(y)) - e$.

Moreover, if Y is universally catenary, the two members of (5.6.5.1) (resp. (5.6.5.2)) are equal. If furthermore x is closed in $f^{-1}(y)$, we have

$$(5.6.5.3) \quad \dim(\mathcal{O}_x) = \dim(\mathcal{O}_y) + e.$$

Proof. We can evidently restrict to the case where X and Y are affine (hence f of finite type) and reduced (1.5.4), hence integral. Since f is dominant, $\mathcal{O}_y$ identifies with a subring of $\mathcal{O}_x$, and the fields of fractions of $\mathcal{O}_y$ and of $\mathcal{O}_x$ are $k(\eta)$ and $k(\xi)$ respectively (I, 8.2.7); moreover, $\mathcal{O}_x$ is a local ring of an $\mathcal{O}_y$ -algebra integral of finite type B at a prime ideal $\mathfrak{q}$ (I, 3.6.5); applying (5.5.8.1) with $A = \mathcal{O}_y$, B , and $\mathfrak{q}$, we obtain (5.6.5.1). Moreover, $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y) = \mathcal{O}_x / \mathfrak{m}_y \mathcal{O}_x$ is none other than the local ring of the fibre $f^{-1}(y)$ at the point x (I, 3.6.1); since $f^{-1}(y)$ is a prescheme of finite type over $k(y)$, it follows from (5.2.3) that

$$\dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)) = \dim_x(f^{-1}(y)) - \deg.\text{tr}_{k(y)} k(x);$$

so the inequality (5.6.5.2) is just another form of (5.6.5.1).

The fact that the two members of (5.6.5.1) are equal when Y is universally catenary is none other than the equality (5.6.1.1), applied to $A = \mathcal{O}_y$ and $B' = \mathcal{O}_x$. To show that moreover (5.6.5.3) holds when x is closed in $f^{-1}(y)$, it suffices to note that, in general, $\deg.\text{tr}_{k(y)} k(x)$ is the dimension of $\overline{\{x\}} \cap f^{-1}(y)$, as follows from (5.2.1) applied to the closed reduced sub-prescheme of $f^{-1}(y)$ having this subspace as underlying space.

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We will show further on (13.1.1) that, under the conditions of (5.6.5), we *always* have $\delta(x) \geq 0$, and (5.6.5.2) makes precise in this case (5.2.1). $\square$

Corollary (5.6.6). — *Under the general hypotheses of (5.6.5), we have*

$$(5.6.6.1) \quad \dim(X) \leq \dim(Y) + e.$$

If we furthermore suppose Y universally catenary, then, for the two members of (5.6.6.1) to be equal, it is necessary and sufficient that

$$(5.6.6.2) \quad \dim(Y) = \sup_{y \in f(X)} \dim(\mathcal{O}_y).$$

In particular, the two members of (5.6.6.1) are equal when Y is locally of finite type over a field.

Proof. The inequality (5.6.6.1) results from (5.6.5.1) applied to a closed point x of X , taking into account (5.1.4.2) and (5.1.11). On the other hand, every nonempty fibre $f^{-1}(y)$ contains a closed point x in this fibre (5.1.11), and if we suppose Y universally catenary, at this point we have the relation (5.6.5.3). Taking the upper bounds of the two members of (5.6.5.3) when x runs through X , and noting that every closed point x of X is *a fortiori* closed in $f^{-1}(f(x))$, the second assertion of the corollary follows from (5.1.4.1) and (5.1.4.2).

We note that condition (5.6.6.2) is trivially satisfied when f is surjective. $\square$

Corollary (5.6.7). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, n an integer. If, for every $y \in Y$, $\dim(f^{-1}(y)) \leq n$, then*

$$(5.6.7.1) \quad \dim(X) \leq \dim(Y) + n.$$

Proof. By virtue of (0, 14.1.7) and of (1.5.4), we can restrict to the case where X and Y are affine, hence f of finite type, X and Y reduced; let X_i ($1 \leq i \leq m$) be the sub-preschemes (integral) of X having for underlying spaces the irreducible components of X ; we have $\dim(X) = \sup(\dim(X_i))$. If Z_i is the closed reduced sub-prescheme of Y having for underlying space $\overline{f(X_i)}$, Z_i is integral (0_I, 2.1.5); the restriction $X_i \rightarrow Y$ of f factors as $X_i \xrightarrow{g_i} Z_i \xrightarrow{j_i} Y$, where j_i is the canonical injection (I, 5.2.2), and g_i is dominant and of finite type (1.5.4). We have $\dim(Z_i) \leq \dim(Y)$ for every i ; on the other hand, if z_i is the generic point of Z_i , $\dim(g_i^{-1}(z_i)) \leq n$ for every i by hypothesis; so to prove (5.6.7.1), it suffices to show that $\dim(X_i) \leq \dim(Z_i) + n$ for every i , which follows from (5.6.6.1). $\square$

Corollary (5.6.8). — *Let Y be a locally Noetherian prescheme, X an irreducible prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, n an integer ≥ 0 . Suppose that Y is universally catenary, and that for every $y \in Y$, $\dim(f^{-1}(y)) \geq n$ (resp. $\dim(f^{-1}(y)) = n$). Then*

$$(5.6.8.1) \quad \dim(X) \geq \dim(Y) + n$$

(resp.

$$(5.6.8.2) \quad \dim(X) = \dim(Y) + n$$

).

Proof. Since $n \geq 0$, the hypothesis implies that f is *surjective*, hence Y irreducible; moreover, if η is the generic point of Y , $\dim(f^{-1}(\eta)) \geq n$ (resp. $\dim(f^{-1}(\eta)) = n$). The conclusion follows from (5.6.6). $\square$

Remarks (5.6.9). — (i) Even if Y is regular, X irreducible, $f : X \rightarrow Y$ dominant and of finite type, the two members of (5.6.6.1) are not necessarily equal, as shown by the example where $Y = \text{Spec}(A)$, A a discrete valuation ring, $X = \text{Spec}(K)$ where K is the field of fractions of A , f being the canonical morphism.

(ii) Example ((5.4.3), (i)) shows that in (5.6.6) and (5.6.8), one cannot suppress the hypothesis that X is irreducible, the other hypotheses being satisfied. However, we will see (10.6.1) that with additional hypotheses on Y , satisfied for example when Y is a prescheme locally of finite type over a field, or over a Dedekind ring having an infinity of prime ideals, such phenomena cannot occur.

Proposition (5.6.10). — *Let A be a local integral Noetherian universally catenary ring, B an integral ring containing A and which is an A -algebra of finite type. Then, for every maximal ideal $\mathfrak{n}$ of B , $\dim(B_{\mathfrak{n}}) = \dim(A)$.*

Proof. Indeed, $\deg .\mathrm{tr}_A B = 0$ and the residue field k' of $B_{\mathfrak{n}}$ is an algebraic extension of the field of fractions of A . The conclusion follows from formula (5.6.1.1), every maximal ideal being above that of A . $\square$

Example (5.6.11). — Let A be a local Noetherian integral ring and *integrally closed*; then, as we have seen (0, 16.1.6), the conclusion of (5.6.10) is valid for every integral finite A -algebra B containing A . On the other hand, we will construct an example of a local Noetherian integral *catenary* ring A for which the conclusion of (5.6.10) will fail. We use the construction of ((5.2.5), (i)). Let k_0 be a field, k a pure transcendental extension $k_0(S_i)_{i \in \mathbb{N}}$ of infinite transcendence degree, V the discrete valuation ring $k[S]_{(S)}$, local ring of the polynomial ring $k[S]$ at the prime ideal (S) ; finally let E be the polynomial ring $V[T]$; we have seen that in E the maximal ideal $\mathfrak{m} = (S) + (T)$ is of height 2 and the maximal ideal $\mathfrak{m}' = (ST - 1)$ of height 1; the residue fields are $k(\mathfrak{m}) = k$ and $k(\mathfrak{m}') = k(S)$, fields of fractions of V ; by choice of k , these fields are isomorphic. Let ε and ε' be the canonical homomorphisms of E onto $E/\mathfrak{m} = k(\mathfrak{m})$ and $E/\mathfrak{m}' = k(\mathfrak{m}')$ respectively; let σ be an isomorphism of $k(\mathfrak{m})$ onto $k(\mathfrak{m}')$, and consider the subring C of E formed by $x \in E$ such that $\varepsilon'(x) = \sigma(\varepsilon(x))$ (this construction is a particular case of “gluing procedures” which will be systematically studied in chap. V). It is immediate that $\mathfrak{n} = \mathfrak{m}\mathfrak{m}' = \mathfrak{m} \cap \mathfrak{m}'$ is a maximal ideal of C , $C/\mathfrak{n}$ identifying with the subfield of $(E/\mathfrak{m}) \times (E/\mathfrak{m}')$ formed by the pairs $(z, \sigma(z))$. We have $E = C + C(ST - 1)$, hence E is a C -algebra of finite type; we will now see that C is Noetherian: this will result from the following lemma:

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Lemma (5.6.11.1). — *Let R be a ring, S a subring, $\mathfrak{R} = \mathrm{Ann}_S(R/S)$ the conductor of S in R (largest ideal of S which is also an ideal of R).*

- label=(i) For every ideal $\mathfrak{J} \subset \mathfrak{R}$ of R , there exists a strictly increasing map from the set of ideals $\mathfrak{L}$ of S such that $R \cdot \mathfrak{L} = \mathfrak{J}$ onto the set of sub- $(S/\mathfrak{R})$ -modules of $\mathfrak{J}/\mathfrak{R}\mathfrak{J}$.*
- lbbel=(ii) If $S/\mathfrak{R}$ and R are Noetherian and if R is an S -module of finite type, then every increasing sequence of ideals of S contained in $\mathfrak{R}$ is stationary.*

Proof. (i) This is clear since $\mathfrak{L}$ is an ideal of S and $R \cdot \mathfrak{L} = \mathfrak{J}$ ($\mathfrak{L}$ being an ideal of S), whence the conclusion.

(ii) Let $(\mathfrak{L}_n)$ be an increasing sequence of ideals of S contained in $\mathfrak{R}$. The sequence of ideals $R \cdot \mathfrak{L}_n$ of R is stationary since R is Noetherian; we can thus suppose that all the ideals $R \cdot \mathfrak{L}_n$ are equal to the same ideal $\mathfrak{J}$ of R . Since R is Noetherian, $\mathfrak{J}/\mathfrak{R}\mathfrak{J}$ is also a finitely generated S -module, hence also an $(S/\mathfrak{R})$ -module of finite type. But since $S/\mathfrak{R}$ is Noetherian, $\mathfrak{J}/\mathfrak{R}\mathfrak{J}$ is an $(S/\mathfrak{R})$ -Noetherian module, and the conclusion follows from (i). $\square$

We apply this lemma with $R = E$, $S = C$; it is clear that $\mathfrak{n}$ is the conductor of C in E ; moreover, for every ideal $\mathfrak{a}$ of C , $\mathfrak{a}/(\mathfrak{n} \cap \mathfrak{a})$ is isomorphic to $(\mathfrak{a} + \mathfrak{n})/\mathfrak{n}$, hence a sub- C -module of $C/\mathfrak{n}$, which is a simple C -module; it therefore suffices to show that every ideal $\mathfrak{a} \cap \mathfrak{n}$ of C is of finite type, and this follows from (5.6.11.1). It follows from (0, 16.1.5) that $\dim(C) = \dim(E) = 2$. Let $A = C_{\mathfrak{n}}$; if $U = C - \mathfrak{n}$, the ring of fractions $B = U^{-1}E$ is thus the integral closure of A , and it is an A -module of finite type; moreover, since $\mathfrak{m}$ and $\mathfrak{m}'$ are the only prime ideals of E containing $\mathfrak{n}$, B is a semi-local ring whose local rings at the two maximal ideals are isomorphic to $E_{\mathfrak{m}}$ and $E_{\mathfrak{m}'}$ respectively, hence are of dimension 2 and 1 respectively. This shows that $\dim(B) = 2$, hence also $\dim(A) = 2$ (0, 16.1.5). Since A is a local integral ring, it is necessarily catenary (every prime ideal distinct from 0 and from the maximal ideal being necessarily of height 1); but it does not satisfy the conclusion of (5.6.10), and *a fortiori* is not universally catenary.

Let us also note that $\mathfrak{n}$ is a prime ideal of height 2 in C , and that for every prime ideal $\mathfrak{p}$ of height 1 in C , there exists a *single* prime ideal $\mathfrak{p}'$ of E above $\mathfrak{p}$, necessarily of height 1, and such that $C_{\mathfrak{p}} = E_{\mathfrak{p}'}$. Indeed, there is at least one prime ideal $\mathfrak{p}'$ of E above $\mathfrak{p}$, and it follows from the Cohen–Seidenberg

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theorem that such an ideal is necessarily of height 1; moreover, since $\mathfrak{p} \not\supset \mathfrak{n}$, $C_{\mathfrak{p}}$ is integrally closed (Bourbaki, *Alg. comm.*, chap. V, §1, no. 5, cor. 5 of prop. 16), hence $E_{\mathfrak{p}} = C_{\mathfrak{p}}$, which proves our assertions.

It would be interesting to know whether every local integral Noetherian ring satisfying the conclusion of (5.6.10) is universally catenary; this is what Nagata [33] has affirmed, but his proof does not appear to be complete.

5.7. Depth and property (S_k)

Definition (5.7.1). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent. We call *depth* (resp. *codepth*) of $\mathcal{F}$ at a point $x \in X$ the number $\text{prof}(\mathcal{F}_x)$ (resp. $\text{coprof}(\mathcal{F}_x)$) (0, 16.4.5 and 16.4.9). We call *codepth* of $\mathcal{F}$ the number

$$(5.7.1.1) \quad \text{coprof}(\mathcal{F}) = \sup_{x \in X} \text{coprof}(\mathcal{F}_x).$$

We say that $\mathcal{F}$ is an $\mathcal{O}_X$ -Module de Cohen-Macaulay at a point x if $\mathcal{F}_x$ is an $\mathcal{O}_x$ -module de Cohen-Macaulay, that is to say (0, 16.5.1) if $\text{coprof}(\mathcal{F}_x) = 0$. We say that $\mathcal{F}$ is an $\mathcal{O}_X$ -Module de Cohen-Macaulay if it is so at every point, that is if $\text{coprof}(\mathcal{F}) = 0$. A point $x \in X$ such that $\mathcal{O}_x$ is a Cohen-Macaulay ring is also called a *Cohen-Macaulay point* of X .

We call *codepth* of X and write $\text{coprof}(X)$ the number $\text{coprof}(\mathcal{O}_X)$. We say that X is a *Cohen-Macaulay prescheme* if $\mathcal{O}_X$ is an $\mathcal{O}_X$ -Module de Cohen-Macaulay, that is if $\text{coprof}(X) = 0$. Every locally Noetherian prescheme of dimension 0 is evidently a Cohen-Macaulay prescheme. To say that $\text{Spec}(A)$ is a Cohen-Macaulay scheme means that A is a Cohen-Macaulay ring (0, 16.5.13).

Definition (5.7.2). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent, k a positive or negative integer. We say that $\mathcal{F}$ possesses *property* (S_k) if, for every $x \in X$, we have

$$(5.7.2.1) \quad \text{prof}(\mathcal{F}_x) \geq \inf(k, \dim(\mathcal{F}_x)).$$

We say that $\mathcal{F}$ possesses property (S_k) at a point $x \in X$ if, for every generisation x' of x in X , we have

$$(5.7.2.2) \quad \text{prof}(\mathcal{F}_{x'}) \geq \inf(k, \dim(\mathcal{F}_{x'})).$$

We say that X satisfies property (S_k) at a point x (resp. satisfies (S_k)) if $\mathcal{O}_X$ satisfies (S_k) at the point x (resp. satisfies (S_k)).

For $\mathcal{F}$ to satisfy (S_k) , it is necessary and sufficient that it satisfy it at every point. If U is an open subset of X and if $\mathcal{F}$ satisfies (S_k) , then so does $\mathcal{F}|_U$; reciprocally, if (U_α) is an open covering of X and if $\mathcal{F}|_{U_\alpha}$ satisfies (S_k) for every α , then $\mathcal{F}$ satisfies (S_k) .

Remarks (5.7.3). — (i) Recall that one always has $\text{prof}(\mathcal{F}_x) \leq \dim(\mathcal{F}_x)$ (0, 16.4.5.1) if $\mathcal{F}_x \neq 0$. To IV-2 | 104

say that $\mathcal{F}$ possesses property (S_k) thus means that one has $\text{prof}(\mathcal{F}_x) \geq k$ except at points $x \in X$ such that $\dim(\mathcal{F}_x) < k$, and that at these latter points one has $\dim(\mathcal{F}_x) = \text{prof}(\mathcal{F}_x)$, that is to say (0, 16.5.1) that $\mathcal{F}$ is an $\mathcal{O}_X$ -Module de Cohen-Macaulay at these points; we note that only at points where $\dim(\mathcal{F}_x) = k$ does one have $\text{prof}(\mathcal{F}_x) = k$, hence $\mathcal{F}$ is an $\mathcal{O}_X$ -Module de Cohen-Macaulay at these points. To say that $\mathcal{F}$ possesses (S_k) for all k signifies that $\mathcal{F}$ is an $\mathcal{O}_X$ -Module de Cohen-Macaulay. It is clear that for $k' \geq k$, property $(S_{k'})$ implies (S_k) ; for $k \leq 0$, every $\mathcal{O}_X$ -Module coherent satisfies (S_k) .

(ii) To verify condition (5.7.2.1), one can restrict to the case $x \in \text{Supp}(\mathcal{F})$; in the contrary case indeed $\dim(\mathcal{F}_x) = -\infty$ (0, 14.1.2).

(iii) If $X = \text{Spec}(A)$, where A is a Noetherian ring, and $\mathcal{F} = \widetilde{M}$, where M is an A -module of finite type, one says that M possesses property (S_k) if $\mathcal{F}$ possesses this property. For a locally Noetherian prescheme X , to say that $\mathcal{F}$ possesses property (S_k) at a point $x \in X$ signifies that if one sets $Y = \text{Spec}(\mathcal{O}_x)$, the $\mathcal{O}_Y$ -Module $\widetilde{\mathcal{F}}_x$ possesses property (S_k) ; one notes that condition (5.7.2.1) in general does not entail (5.7.2.2) for every generisation x' of x , since there is no inequality relation between $\text{prof}(\mathcal{F}_x)$ and $\text{prof}(\mathcal{F}_{x'})$ (0, 16.4.6).

(iv) It follows immediately from the definition that if $\mathcal{F}$ satisfies (S_k) at a point x , it also satisfies (S_k) at every point x' which is a generisation of x .

(v) Property (S_k) is especially important for $k = 1$ and $k = 2$; it was introduced for $k = 2$ by Serre, to express his criterion of normality (cf. (5.8.5)).

(vi) Let X be a locally Noetherian prescheme, $j : Y \rightarrow X$ the canonical closed immersion, $\mathcal{G}$ an $\mathcal{O}_Y$ -Module coherent. It is clear that for every $x \in Y$, $\dim(\mathcal{G}_x) = \dim((j_*(\mathcal{G}))_x)$ and $\text{prof}(\mathcal{G}_x) = \text{prof}((j_*(\mathcal{G}))_x)$, whence $\text{coprof}(\mathcal{G}_x) = \text{coprof}((j_*(\mathcal{G}))_x)$. For $\mathcal{G}$ to satisfy (S_k) , it suffices that $j_*(\mathcal{G})$ satisfy (S_k) .

Proposition (5.7.4). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent. Set $S = \text{Supp}(\mathcal{F})$, and for every $n \geq 0$, denote by Z_n the set of $z \in X$ such that $\text{coprof}(\mathcal{F}_z) > n$, so that $Z_n \subset S$.*

label=(i) For $\mathcal{F}$ to satisfy property (S_k) , it is necessary and sufficient that, for every $n \geq 0$, we have

$$(5.7.4.1) \quad \text{codim}(Z_n, S) > n + k.$$

label=(ii) Suppose furthermore that the Z_n are closed in X . Then, for $\mathcal{F}$ to satisfy property (S_k) at a point $x \in S$, it is necessary and sufficient that, for every $n \geq 0$, we have

$$(5.7.4.2) \quad \text{codim}_x(Z_n, S) > n + k.$$

Proof. (i) By definition (5.1.3), $\text{codim}(Z_n, S) = \inf_{z \in Z_n} (\dim(\mathcal{O}_{S,z}))$ and the inequality (5.7.4.1) means that, for every $z \in X$, and every $n \geq 0$, the relation

$$\dim(\mathcal{F}_z) - \text{prof}(\mathcal{F}_z) \geq n + 1$$

implies the relation

$$\dim(\mathcal{F}_z) \geq n + k + 1.$$

But if one sets $a = \dim(\mathcal{F}_z)$, $b = \text{prof}(\mathcal{F}_z)$, we have $b \leq a$, and to say that for every $n \geq 0$, $b \leq a - n - 1$ implies $k \leq a - n - 1$ is equivalent to saying $b \geq \inf(k, a)$, whence the proposition.

(ii) The argument is the same as in (i), except that one restricts to $z \in X$ which are generisations of x , and takes account of (5.1.3.2). $\square$

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Proposition (5.7.5). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent. For $\mathcal{F}$ to satisfy (S_k) ($k \geq 1$), it is necessary and sufficient that for every integer r such that $0 \leq r < k$, every open subset U of X , and every $(\mathcal{F}|U)$ -regular suite $(f_i)_{1 \leq i \leq r}$ of sections of $\mathcal{O}_X$ above U , $(\mathcal{F}|U) / (\sum_{i=1}^r f_i(\mathcal{F}|U))$ be without embedded associated prime cycle.*

Proof. Let us first prove the proposition for $k = 1$; it then says that for $\mathcal{F}$ to satisfy (S_1) , it is necessary and sufficient that $\mathcal{F}$ be without embedded associated prime cycle. Indeed, to say that $\mathcal{F}$ satisfies (S_1) means that at the points $x \in \text{Supp}(\mathcal{F})$ such that $\dim(\mathcal{F}_x) > 0$, one has $\text{prof}(\mathcal{F}_x) \geq 1$ (since at the other points of $\text{Supp}(\mathcal{F})$ one has $\dim(\mathcal{F}_x) = 0$, hence $\text{prof}(\mathcal{F}_x) = 0$); but to say that $\text{prof}(\mathcal{F}_x) \geq 1$ means that $\mathfrak{m}_x$ is not associated to $\mathcal{F}$ (0, 16.4.6, (i)), or equivalently that x is not associated to $\mathcal{F}$ (3.1.1); on the other hand, if S is a sub-prescheme of X having for underlying space $\text{Supp}(\mathcal{F})$, by (5.1.12.1) $\dim(\mathcal{F}_x) = \dim(\mathcal{O}_{S,x})$; to say that $\dim(\mathcal{F}_x) > 0$ means that x is not a maximal point of $\text{Supp}(\mathcal{F})$ (5.1.2), whence the conclusion.

In the second place, let us show that, for k arbitrary > 1 , the condition of the statement is necessary. One can restrict to considering the case $U = X$, and our assertion will be proved (by induction on k and by virtue of the first part of the argument) if we show that when $\mathcal{F}$ satisfies (S_k) ($k > 1$) and f is an $\mathcal{F}$ -regular section of $\mathcal{O}_X$ above X , then $\mathcal{F}/f\mathcal{F}$ satisfies (S_{k-1}) (i.e. satisfies (5.7.2.1) where k is replaced by $k - 1$). Indeed, for every $x \in X$, f_x is an $\mathcal{F}_x$ -regular element; if it is invertible, $\mathcal{F}_x/f_x\mathcal{F}_x = 0$ and the conclusion is trivial. If on the contrary $f_x \in \mathfrak{m}_x$, one knows that $\text{prof}(\mathcal{F}_x/f_x\mathcal{F}_x) = \text{prof}(\mathcal{F}_x) - 1$ (0, 16.4.6, (i)), and $\dim(\mathcal{F}_x/f_x\mathcal{F}_x) = \dim(\mathcal{F}_x) - 1$ (0, 16.3.4), and our assertion follows.

Let us finally prove that for $k > 1$, the condition of the statement is sufficient. We proceed by induction on k ; let x be a point of X , and suppose first that $\dim(\mathcal{F}_x) \geq k$. The induction hypothesis implies that $\mathcal{F}$ satisfies (S_{k-1}) , hence $\text{prof}(\mathcal{F}_x) \geq k - 1$; taking account of (0, 15.2.4), there is thus an open neighbourhood U of x and an $(\mathcal{F}|U)$ -regular suite $(f_i)_{1 \leq i \leq k-1}$ of sections of $\mathcal{O}_X$ above U , such that $(f_i)_x \in \mathfrak{m}_x$ for $1 \leq i \leq k - 1$. The hypothesis implies that $\mathcal{G} = (\mathcal{F}|U) / (\sum_{i=1}^{k-1} f_i(\mathcal{F}|U))$ is without embedded associated prime cycle; but $\dim(\mathcal{G}_x) = \dim(\mathcal{F}_x) - (k - 1) \geq 1$ (0, 16.3.4), hence x is not associated to $\mathcal{G}$; so $\text{prof}(\mathcal{G}_x) \geq 1$ (0, 16.4.6), and since $\text{prof}(\mathcal{G}_x) = \text{prof}(\mathcal{F}_x) - (k - 1)$ (0, 16.4.6), we have $\text{prof}(\mathcal{F}_x) \geq k$. Suppose in the second place that $\dim(\mathcal{F}_x) = r < k$; since $\mathcal{F}$ satisfies (S_{k-1}) , we have $\text{prof}(\mathcal{F}_x) \geq \inf(k - 1, r) = r$, and this finishes the proof. $\square$

Corollary (5.7.6). — Suppose that $\mathcal{F}$ satisfies (S_k) ($k \geq 1$); if (f_i) ($1 \leq i \leq r$) is an $\mathcal{F}$ -regular suite of sections of $\mathcal{O}_X$ above X and if $r < k$, $\mathcal{F} / (\sum_{i=1}^r f_i \mathcal{F})$ satisfies (S_{k-r}) .

Proof. This results immediately from (5.7.5). $\square$

Corollary (5.7.7). — For $\mathcal{F}$ to satisfy (S_2) , it is necessary and sufficient that $\mathcal{F}$ be without embedded associated prime cycle and that for every open subset U of X , every $(\mathcal{F}|U)$ -regular section f of $\mathcal{O}_X$ above U , $(\mathcal{F}|U) / f(\mathcal{F}|U)$ be without embedded associated prime cycle.

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Remarks (5.7.8). — Let X be a locally Noetherian prescheme of dimension 1; it then amounts to the same to say that X is a Cohen-Macaulay prescheme, or that it satisfies (S_1) , or that it satisfies one of the properties (S_n) for $n \geq 1$, by virtue of definitions (5.7.1) and (5.7.2). By virtue of (5.7.5), it also amounts to the same, for preschemes of dimension 1, to say that X is a Cohen-Macaulay prescheme or that it has no embedded associated prime cycle. For example, a *reduced* locally Noetherian prescheme of dimension 1 is a Cohen-Macaulay prescheme.

Proposition (5.7.9). — Let A, B be two Noetherian rings, $\rho : A \rightarrow B$ a homomorphism of rings, M a B -module such that $M_{[\mathfrak{p}]}$ is an A -module of finite type. Let $\mathfrak{p}$ be a prime ideal of A ; the prime ideals of B above $\mathfrak{p}$ and belonging to $\text{Supp}(M)$ are finite in number, and if $(\mathfrak{q}_i)_{1 \leq i \leq n}$ is the family of these ideals, we have

$$(5.7.9.1) \quad \dim_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = \sup_i \dim_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i})$$

$$(5.7.9.2) \quad \text{prof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = \inf_i \text{prof}_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}).$$

Proof. If $\mathfrak{b}$ is the annihilator of M , $B/\mathfrak{b}$ identifies with a sub- A -module of $\text{End}_A(M_{[\mathfrak{p}]})$, hence is of finite type since A is Noetherian; so there are only a finite number of prime ideals of B above $\mathfrak{p}$ and containing $\mathfrak{b}$, and these are precisely those belonging to $\text{Supp}(M)$. Replacing B by $B/\mathfrak{b}$, which does not change the second members of (5.7.9.1) and (5.7.9.2) (by (0, 16.1.9) and (0, 16.4.8)), one can thus suppose that B is a finite A -algebra. Set $S = A - \mathfrak{p}$ and $B' = S^{-1}B$; B' is a semi-local Noetherian ring whose maximal ideals are $S^{-1}\mathfrak{q}_i$ ($1 \leq i \leq n$) and since $M_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$ -module of finite type, $\dim_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = \dim_{B'}(M_{\mathfrak{p}})$ (0, 16.1.9). This being so, relation (5.7.9.1) becomes a particular case of (0, 16.1.7.4). To prove relation (5.7.9.2), we immediately reduce as in (0, 16.4.8) to the case $\text{prof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) = 0$, and the same argument as in (0, 16.4.8) shows that $M_{\mathfrak{p}}$ contains a sub- B' -module of finite length, and hence also a simple sub- B' -module; but such a sub-module is necessarily isomorphic to the residue field of one of the $B_{\mathfrak{q}_i}$, hence there is at least one index i such that $\text{prof}_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}) = 0$ (0, 16.4.6), which finishes the proof. $\square$

Corollary (5.7.10). — Suppose verified the hypotheses of (5.7.9), and suppose in addition that A is a local ring; then, for M to be an A -module de Cohen-Macaulay, it is necessary and sufficient that, for all prime ideals $\mathfrak{q}_i$ of B above the maximal ideal of A , $M_{\mathfrak{q}_i}$ be a $B_{\mathfrak{q}_i}$ -module de Cohen-Macaulay, and moreover that all the $\dim_{B_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i})$ be equal.

Proof. It follows from (5.7.9.1) and (5.7.9.2) that these conditions are equivalent to $\dim_A(M) = \text{prof}_A(M)$. $\square$

Corollary (5.7.11). — Suppose verified the hypotheses of (5.7.9).

label=(i) If $M_{[\mathfrak{p}]}$ possesses property (S_k) , then so does M .

lbbel=(ii) Suppose that for every pair of prime ideals $\mathfrak{q}, \mathfrak{q}'$ of B such that $\rho^{-1}(\mathfrak{q}) = \rho^{-1}(\mathfrak{q}')$, $\dim_{B_{\mathfrak{q}}}(M_{\mathfrak{q}}) = \dim_{B_{\mathfrak{q}'}}(M_{\mathfrak{q}'})$. Then, if M possesses property (S_k) , so does $M_{[\mathfrak{p}]}$.

Proof. This results immediately from (5.7.9.1) and (5.7.9.2) and the definition of property (S_k) . $\square$

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(5.7.12). Conforming to definitions (5.7.1), given a Noetherian ring A and an A -module M of finite type, one defines $\text{coprof}_A(M)$ as $\sup_{x \in X} (\text{coprof}_{A_x}(M_x))$, where $X = \text{Spec}(A)$; we will see further on (6.11.5) that this definition coincides with that of (0, 16.4.9) when A is a local Noetherian ring.

Corollary (5.7.13). — *Let A, B be two Noetherian rings, $\rho : A \rightarrow B$ a homomorphism of rings, M a B -module such that $M_{[\mathfrak{p}]}$ is an A -module of finite type. Then*

$$(5.7.13.1) \quad \text{coprof}_A(M_{[\mathfrak{p}]}) \geq \text{coprof}_B(M).$$

Proof. This results from the preceding definition and from (5.7.9.1) and (5.7.9.2). $\square$

5.8. Regular preschemes and property (R_k) . Serre's normality criterion

(5.8.1). Recall (0_I, 4.1.4) that a ringed space $(X, \mathcal{O}_X)$ is called *regular* at a point $x \in X$ if $\mathcal{O}_x$ is a *regular local ring*. When dealing with preschemes, we will only use this terminology when X is *locally Noetherian*.

Definition (5.8.2). — A locally Noetherian prescheme X is said to be *regular in codimension $\leq k$* , or *possesses property (R_k)* if the set of points where X is *not* regular is of codimension $> k$ (that is to say (5.1.3), for every $x \in X$, the relation $\dim(\mathcal{O}_x) \leq k$ implies that $\mathcal{O}_x$ is regular).

To say that X is regular signifies that X possesses property (R_k) for every k . If $X = \text{Spec}(A)$, where A is a Noetherian ring, one says that A possesses property (R_k) if X possesses this property; to say that X is regular signifies that the ring A is *regular* (0, 17.3.6). For a locally Noetherian prescheme X , we say that X possesses property (R_k) *at a point $x \in X$* if the local ring $\mathcal{O}_x$ possesses property (R_k) ; this means that for every generisation x' of x in X , the relation $\dim(\mathcal{O}_{x'}) \leq k$ implies that $\mathcal{O}_{x'}$ is regular. To say that X is regular at a point x is equivalent to saying that X satisfies (R_n) for every $n \geq 0$ at the point x , by virtue of (0, 17.3.6).

Proposition (5.8.3). — *If k is a field, X a prescheme locally of finite type over k , then, for every $x \in X$, there exists an open neighbourhood of x in X isomorphic to a sub-scheme of a regular k -scheme.*

Proof. Indeed, there is an affine open neighbourhood U of x isomorphic to a k -scheme of the form $\text{Spec}(A)$, where A is a k -algebra of finite type; A is therefore isomorphic to a quotient of a polynomial algebra $k[T_1, \dots, T_n]$, and it is known that this last algebra is a regular ring (0, 17.3.7). $\square$

(5.8.4). By virtue of (5.8.2), to say that X possesses property (R_0) signifies that for every maximal point x of X , $\mathcal{O}_x$ is a *field* (0, 17.1.4), or equivalently that X is *reduced* at this point. Since the set U of $x \in X$ where X is reduced is *open* (the nilradical of X being an $\mathcal{O}_X$ -Module coherent (I, 6.1.1) and (0_I, 5.2.2)), it amounts to the same to say that X possesses property (R_0) or that the set U is *everywhere dense*. Consequently:

Proposition (5.8.5). — *For a locally Noetherian prescheme X to be reduced, it is necessary and sufficient that it satisfy properties (S_1) and (R_0) .*

Proof. Taking into account (5.7.5), this results from the preceding remark and from (3.2.1). $\square$

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Theorem (5.8.6). — (Serre's criterion). *Let X be a locally Noetherian prescheme. For X to be normal, it is necessary and sufficient that X satisfy properties (S_2) and (R_1) , that is to say that, for every $x \in X$, the following properties hold:*

- label=(i) *If $\dim(\mathcal{O}_x) \leq 1$, $\mathcal{O}_x$ is regular (that is, is a field or a discrete valuation ring).*
- lbbel=(ii) *If $\dim(\mathcal{O}_x) \geq 2$, then $\text{prof}(\mathcal{O}_x) \geq 2$.*

Proof. The conditions are *necessary*. Indeed, to say that X is normal signifies that for every $x \in X$, $\mathcal{O}_x$ is a local Noetherian integrally closed ring. If $\dim(\mathcal{O}_x) = 0$ (resp. $\dim(\mathcal{O}_x) = 1$), we conclude that $\mathcal{O}_x$ is a field since $\mathcal{O}_x$ is integral (resp. that $\mathcal{O}_x$ is a discrete valuation ring, by virtue of (II, 7.1.6)). On the other hand, for every nonzero element $f_x \neq 0$ of $\mathcal{O}_x$, one knows (Bourbaki, Alg. comm., chap. VII, §1, no. 4, prop. 8) that the prime ideals associated to $\mathcal{O}_x / f_x \mathcal{O}_x$ are non-immersed, hence $\mathcal{O}_x$ satisfies (S_2) (5.7.7).

The conditions are *sufficient*. The question being local, one can moreover suppose that $X = \text{Spec}(A)$, where A is a local Noetherian ring *reduced* (I, 5.1.4); if R is the total ring of fractions of A , R is a direct product of a finite number of fields, and (taking account of (II, 6.3.6)), it suffices to prove

that A is *integrally closed* in R . So let $h = f/g$ be an element of R integral over A , g and f being elements of A such that g is a non-zero-divisor of 0 . We have a relation of the form

$$(5.8.6.1) \quad f^n + \sum_{i=1}^n a_i f^{n-i} g^i = 0 \quad \text{with } a_i \in A \text{ for } 1 \leq i \leq n.$$

Let $\mathfrak{p}$ be a prime ideal of A such that $\dim(A_{\mathfrak{p}}) = 1$; if $f_{\mathfrak{p}}$ and $g_{\mathfrak{p}}$ are the images of f and g in $A_{\mathfrak{p}}$, it follows from (5.8.6.1) that $f_{\mathfrak{p}}/g_{\mathfrak{p}}$ (which belongs to the total ring of fractions of $A_{\mathfrak{p}}$, since $g_{\mathfrak{p}}$ is a non-zero-divisor of 0 in $A_{\mathfrak{p}}$ by flatness (0_I, 5.3.1)) is integral over $A_{\mathfrak{p}}$; but since $A_{\mathfrak{p}}$ is regular, hence integrally closed, we have $f_{\mathfrak{p}}/g_{\mathfrak{p}} \in A_{\mathfrak{p}}$. But hypothesis (S₂) implies (5.7.7) that g is a non-zero-divisor and that A/gA has no non-immersed associated prime ideals $\mathfrak{p}_i$ ($1 \leq i \leq n$); or, gA is the intersection of the primary ideals $\mathfrak{q}_i$ corresponding to the $\mathfrak{p}_i$, and from what we have just seen, the $\mathfrak{q}_i$ are the inverse images in A , by the homomorphisms $A \rightarrow A_{\mathfrak{p}_i}$, of the ideals $(gA)_{\mathfrak{p}_i}$ (Bourbaki, *Alg. comm.*, chap. IV, §2, no. 3, prop. 5). But by the Hauptidealsatz (0, 16.3.2) $\dim(A_{\mathfrak{p}_i}) = 1$ for $1 \leq i \leq n$, hence $(fA)_{\mathfrak{p}_i} \subset (gA)_{\mathfrak{p}_i}$ for every i by what precedes; since fA is contained in the intersection of the inverse images of $(fA)_{\mathfrak{p}_i}$ ($1 \leq i \leq n$), we have $fA \subset gA$, that is to say $f/g \in A$. $\square$

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5.9. Z-pure and Z-closed Modules Some of the notions and results of this section and the following are particular cases of notions and results developed in chapter III in the theory of local cohomology. For the convenience of the reader, we give here an independent treatment.

(5.9.1). Let X be a locally Noetherian prescheme, Z a subset of X *stable by specialisation*: this means that for every finite subset M of Z , the closure of M is contained in Z , and hence Z is the union of an increasing filtering family (Z_{α}) of closed subsets of X ; conversely, it is clear that such a union is stable by specialisation.

Set $U_{\alpha} = X - Z_{\alpha}$, so that $X - Z$ is the intersection of the filtering decreasing family of opens U_{α} ; let $i_{\alpha} : U_{\alpha} \rightarrow X$ be the canonical injection, and for $U_{\alpha} \supset U_{\beta}$, let $i_{\alpha\beta} : U_{\beta} \rightarrow U_{\alpha}$ be the canonical injection, so that $i_{\beta} = i_{\alpha} \circ i_{\alpha\beta}$. Let $\mathcal{F}$ be an $\mathcal{O}_X$ -Module (not necessarily quasi-coherent); from the canonical homomorphism (0_I, 4.4.3.2)

$$\mathcal{F}|U_{\alpha} \longrightarrow (i_{\alpha\beta})_*(\mathcal{F}|U_{\beta})$$

we deduce, by application of the functor $(i_{\alpha})_*$, a homomorphism

$$\rho_{\beta\alpha} : (i_{\alpha})_*(\mathcal{F}|U_{\alpha}) \longrightarrow (i_{\beta})_*(\mathcal{F}|U_{\beta})$$

and one verifies immediately that $\rho_{\gamma\alpha} = \rho_{\gamma\beta} \circ \rho_{\beta\alpha}$ for $U_{\alpha} \supset U_{\beta} \supset U_{\gamma}$; in other words, the $\mathcal{O}_X$ -Modules $(i_{\alpha})_*(\mathcal{F}|U_{\alpha})$ form an *inductive system* for the homomorphisms $\rho_{\beta\alpha}$. We set

$$(5.9.1.1) \quad \mathcal{H}_{X/Z}^0(\mathcal{F}) = \varinjlim_{\alpha} (i_{\alpha})_*(\mathcal{F}|U_{\alpha}).$$

This $\mathcal{O}_X$ -Module does not depend on the increasing family (Z_{α}) of closed sets whose union is Z : indeed, let V be a Noetherian open subset of X ; one knows (G, II, 3.10.1) that in the category of $\mathcal{O}_V$ -Modules, the functor $\mathcal{G} \mapsto \Gamma(V, \mathcal{G})$ commutes with inductive limits; so by virtue of (5.9.1.1)

$$(5.9.1.2) \quad \Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F})) = \varinjlim_{\alpha} \Gamma(V \cap U_{\alpha}, \mathcal{F}).$$

Let then (Z'_{λ}) be a second increasing filtering family of closed sets of X , with union Z ; $V \cap Z_{\alpha}$ is therefore the union of the $V \cap Z_{\alpha} \cap Z'_{\lambda}$; but $V \cap Z_{\alpha}$ is locally closed in X , hence every closed irreducible subset of $V \cap Z_{\alpha}$ admits a generic point; since the $V \cap Z_{\alpha} \cap Z'_{\lambda}$ are closed in $V \cap Z_{\alpha}$ and form (for α fixed) an increasing filtering family, there exists an index λ such that $V \cap Z_{\alpha} \cap Z'_{\lambda} = V \cap Z_{\alpha}$ (0_{III}, 9.2.4), in other words $V \cap Z_{\alpha} \subset V \cap Z'_{\lambda}$. This proves that the filtering decreasing families $V \cap U_{\alpha}$, $V \cap U'_{\lambda}$ (where $U'_{\lambda} = X - Z'_{\lambda}$) are cofinal in each other, whence our assertion, by virtue of (5.9.1.2).

We note that the set Z is not necessarily constructible: one has an example of this by taking $X = \text{Spec}(A)$, where A is an integral Noetherian ring having an infinity of maximal ideals, and Z the complement of the generic point of X .

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If Z is closed and if $i : X - Z \rightarrow X$ is the canonical injection, we have

$$(5.9.1.3) \quad \mathcal{H}_{X/Z}^0(\mathcal{F}) = i_*(\mathcal{F}|_{X-Z})$$

and in particular, for $Z = \emptyset$, $\mathcal{H}_{X/\emptyset}^0(\mathcal{F}) = \mathcal{F}$.

Proposition (5.9.2). — *label=(i) The functor $\mathcal{F} \mapsto \mathcal{H}_{X/Z}^0(\mathcal{F})$ is exact on the left.*
lbel=(ii) If $\mathcal{F}$ is quasi-coherent, then so is $\mathcal{H}_{X/Z}^0(\mathcal{F})$.

Proof. Assertion (i) results from the definition (5.9.1.1), from the fact that $(i_\alpha)_*$ is a left exact functor and from the fact that the inductive limit preserves exactness in the category of $\mathcal{O}_X$ -Modules. Assertion (ii) results from (I, 9.2.2) and the fact that an inductive limit of quasi-coherent $\mathcal{O}_X$ -Modules is quasi-coherent (I, 1.3.9). $\square$

Remark (5.9.3). — If $\mathcal{F}$ is an $\mathcal{O}_X$ -Algebra, then so is $\mathcal{H}_{X/Z}^0(\mathcal{F})$ (0I, 4.2.4); in particular $\mathcal{H}_{X/Z}^0(\mathcal{O}_X)$ is a quasi-coherent $\mathcal{O}_X$ -Algebra, and for every $\mathcal{O}_X$ -Module $\mathcal{F}$, $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is a $\mathcal{H}_{X/Z}^0(\mathcal{O}_X)$ -Module, which is quasi-coherent if $\mathcal{F}$ is quasi-coherent (I, 9.6.1). More particularly, suppose that $X = \text{Spec}(A)$, where A is integral and Noetherian; then $\mathcal{H}_{X/Z}^0(\mathcal{O}_X)$ is the $\mathcal{O}_X$ -Algebra $\tilde{B}$, where

$$(5.9.3.1) \quad B = \bigcap_{\mathfrak{p} \in X-Z} A_{\mathfrak{p}}.$$

This results indeed from (5.9.1.2) and (I, 8.2.1.1).

Proposition (5.9.4). — *Let X be a locally Noetherian prescheme, Z a subset of X stable by specialisation, X' a locally Noetherian prescheme, $f : X' \rightarrow X$ a flat morphism. Then $Z' = f^{-1}(Z)$ is stable by specialisation and for every quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, we have a canonical isomorphism*

$$(5.9.4.1) \quad f^*(\mathcal{H}_{X/Z}^0(\mathcal{F})) \xrightarrow{\sim} \mathcal{H}_{X'/Z'}^0(f^*(\mathcal{F})).$$

Proof. Indeed, with the notations of (5.9.1), $Z'_\alpha = f^{-1}(Z_\alpha)$ is closed in X' and Z' is the union of the Z'_α ; moreover, $(i_\alpha)_{(X')}$ is the canonical injection $i'_\alpha : U'_\alpha \rightarrow X'$, where $U'_\alpha = X' - Z'_\alpha = f^{-1}(U_\alpha)$. Since f is flat, one knows (2.3.1) that the canonical homomorphism $f^*((i_\alpha)_*(\mathcal{F}|_{U_\alpha})) \rightarrow (i'_\alpha)_*(f^*(\mathcal{F})|_{U'_\alpha})$ is bijective; since, for $\alpha \leq \beta$, the diagram

$$\begin{array}{ccc} f^*((i_\alpha)_*(\mathcal{F}|_{U_\alpha})) & \xrightarrow{\sim} & (i'_\alpha)_*(f^*(\mathcal{F})|_{U'_\alpha}) \\ \downarrow & & \downarrow \\ f^*((i_\beta)_*(\mathcal{F}|_{U_\beta})) & \xrightarrow{\sim} & (i'_\beta)_*(f^*(\mathcal{F})|_{U'_\beta}) \end{array}$$

is commutative, passing to the limit one obtains a canonical isomorphism $\varinjlim_\alpha (f^*((i_\alpha)_*(\mathcal{F}|_{U_\alpha}))) \xrightarrow{\sim} \mathcal{H}_{X'/Z'}^0(f^*(\mathcal{F}))$. But since the functor f^* commutes with inductive limits (0I, 4.3.2), this gives by definition the isomorphism (5.9.4.1) sought. $\square$

Corollary (5.9.5). — *Under the hypotheses of (5.9.4), if $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent, then so is $\mathcal{H}_{X'/Z'}^0(f^*(\mathcal{F}))$. The converse is true when f is a faithfully flat and quasi-compact morphism.*

Proof. The first assertion results from (5.9.4.1) and (0I, 5.3.11); the second amounts to saying that if $\mathcal{H}_{X'/Z'}^0(f^*(\mathcal{F}))$ is of finite type, then so is $\mathcal{H}_{X/Z}^0(\mathcal{F})$; this results from (5.9.4.1) and (2.5.2). $\square$

Corollary (5.9.6). — *Let X be a locally Noetherian prescheme, Z a subset of X stable by specialisation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent. For every $x \in X$, setting $X_x = \text{Spec}(\mathcal{O}_x)$, $Z_x = Z \cap X_x$, we have a canonical functorial isomorphism*

$$(5.9.6.1) \quad (\mathcal{H}_{X/Z}^0(\mathcal{F}))_x \xrightarrow{\sim} \mathcal{H}_{X_x/Z_x}^0(\mathcal{F}_x).$$

Proof. It suffices to apply (5.9.4) to the canonical morphism $X_x \rightarrow X$, which is flat, and (I, 1.6.5). $\square$

(5.9.7). With the notations of (5.9.1), for every α one has a canonical functorial homomorphism $\mathcal{F} \rightarrow (i_\alpha)^*(\mathcal{F}|_{U_\alpha})$ (0I, 4.4.3.2), and these homomorphisms form an inductive system; passing to the inductive limit, one deduces a canonical functorial homomorphism

$$(5.9.7.1) \quad \rho_{X/Z} : \mathcal{F} \longrightarrow \mathcal{H}_{X/Z}^0(\mathcal{F}).$$

Proposition (5.9.8). — Let X be a locally Noetherian prescheme, Z a subset of X stable by specialisation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module. The following properties are equivalent:

label=a) The homomorphism $\rho_{X/Z}$ (5.9.7.1) is injective (resp. bijective).

lbbel=b) For every Noetherian open subset V of X , the homomorphism

$$(\rho_{X/Z})_V : \Gamma(V, \mathcal{F}) \longrightarrow \varinjlim_{\alpha} \Gamma(V \cap U_{\alpha}, \mathcal{F})$$

is injective (resp. bijective).

a') For every closed subset $T \subset Z$ of X , the canonical homomorphism ($\mathbf{0}_I$, 4.4.3.2)

$$\mathcal{F} \longrightarrow i_*(\mathcal{F}|_{X-T})$$

(where $i : X - T \rightarrow X$ is the canonical injection) is injective (resp. bijective).

b') For every closed subset $T \subset Z$ of X and every Noetherian open subset V of X , the restriction homomorphism

$$\Gamma(V, \mathcal{F}) \longrightarrow \Gamma(V \cap (X - T), \mathcal{F})$$

is injective (resp. bijective).

Proof. Taking account of (5.9.1.2), the equivalence of $a)$ and $b)$ (resp. $a')$ and $b')$) results from the definition of the functor Γ and from the fact that it is left exact. Since the homomorphism $(\rho_{X/Z})_V$ is the composite

$$(5.9.8.1) \quad \Gamma(V, \mathcal{F}) \longrightarrow \Gamma(V \cap (X - T), \mathcal{F}) \longrightarrow \varinjlim_{\alpha} \Gamma(V \cap U_{\alpha}, \mathcal{F})$$

for every closed subset $T \subset Z$, if $(\rho_{X/Z})_V$ is injective, then so is $\Gamma(V, \mathcal{F}) \rightarrow \Gamma(V \cap (X - T), \mathcal{F})$; on the other hand, the fact that $b')$ implies $b)$ results from the definition of an inductive limit. It remains to show that if $\rho_{X/Z}$ is bijective, then so is $\Gamma(V, \mathcal{F}) \rightarrow \Gamma(V \cap (X - T), \mathcal{F})$, and for this it suffices, by virtue of (5.9.8.1), to see that if $U' \subset U$ are two open sets contained in V and containing $V \cap Z$, the restriction homomorphism $\Gamma(U, \mathcal{F}) \rightarrow \Gamma(U', \mathcal{F})$ is injective; but this results from the fact that $\rho_{X/Z}$ is injective, by replacing in what precedes V by U and $V \cap (X - T)$ by U' . The equivalence of $a)$ and $a')$ follows from (5.9.1.3) and the fact that the filtering families (U_{α}) and $(X - T)_{T \subset Z, T \text{ closed}}$ are cofinal. $\square$

Definition (5.9.9). — Under the hypotheses of (5.9.8), we say that $\mathcal{F}$ is Z -pure (resp. Z -closed) if the homomorphism $\rho_{X/Z}$ is injective (resp. bijective).

If $X = \text{Spec}(A)$ is affine, $\mathcal{F} = \tilde{M}$, where M is an A -module, we say that M is Z -pure (resp. Z -closed) when $\mathcal{F}$ is Z -pure (resp. Z -closed).

We say that $\mathcal{F}$ is Z -pure (resp. Z -closed) at a point $x \in X$ if (with the notations of (5.9.6)) $\mathcal{F}_x$ is Z_x -pure (resp. Z_x -closed); this amounts to the same, by virtue of (5.9.6), as saying that the canonical homomorphism $\mathcal{F}_x \rightarrow (\mathcal{H}_{X/Z}^0(\mathcal{F}))_x$ is injective (resp. bijective).

We note that for every $x \in X - Z$, $\mathcal{F}$ is Z -closed at the point x , by virtue of (5.9.8).

Corollary (5.9.10). — label=(i) Let (V_{λ}) be an open covering of X . For $\mathcal{F}$ to be Z -pure (resp. Z -closed), it is necessary and sufficient that for every λ , $\mathcal{F}|_{V_{\lambda}}$ be $(Z \cap V_{\lambda})$ -pure (resp. $(Z \cap V_{\lambda})$ -closed).

lbbel=(ii) Let Z' be a subset of Z stable under specialisation. If $\mathcal{F}$ is Z -pure (resp. Z -closed), it is Z' -pure (resp. Z' -closed).

Proof. This follows immediately from (5.9.8), $b')$. $\square$

Proposition (5.9.11). — Under the hypotheses of (5.9.8), the $\mathcal{O}_X$ -Modules $\text{Ker}(\rho_{X/Z})$ and $\text{Coker}(\rho_{X/Z})$ have their support contained in Z , and the $\mathcal{O}_X$ -Module $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is Z -closed. Furthermore, if $u : \mathcal{F} \rightarrow \mathcal{F}'$ is a homomorphism of $\mathcal{O}_X$ -Modules such that $\mathcal{F}'$ is Z -closed, u factors in a unique way as $\mathcal{F} \xrightarrow{\rho_{X/Z}} \mathcal{H}_{X/Z}^0(\mathcal{F}) \xrightarrow{v} \mathcal{F}'$. If in addition the supports of $\text{Ker}(u)$ and of $\text{Coker}(u)$ are contained in Z , v is an isomorphism.

Proof. The first assertion means that, for every $x \in X - Z$, we have

$$\text{Ker}(\rho_{X/Z})_x = \text{Coker}(\rho_{X/Z})_x = 0,$$

i.e. that $(\rho_{X/Z})_x$ is bijective, or in other words that $\mathcal{F}$ is Z -pure at x , as has been noted above (5.9.9).

To show that $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is Z -closed, it suffices to see that for every Noetherian open subset V of X , $\Gamma(V \cap U_\beta, \mathcal{H}_{X/Z}^0(\mathcal{F}))$ is equal to $\Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F}))$; but by definition

$$\Gamma(V \cap U_\beta, \mathcal{H}_{X/Z}^0(\mathcal{F})) = \varinjlim_\alpha \Gamma(V \cap U_\alpha \cap U_\beta, \mathcal{F}).$$

Now, the double family $(U_\alpha \cap U_\beta)$ is a decreasing filtering family, and our assertion results from (5.9.1) and the theorem of the double inductive limit.

Let us pass to the second part of the proposition. The existence and uniqueness of v result from **IV-2 | 113** the fact that for every α , $(i_\alpha)_*(u)$ is the unique homomorphism making the diagram

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{u} & \mathcal{F}' \\ \downarrow & & \downarrow \\ (i_\alpha)_*(\mathcal{F}|_{U_\alpha}) & \xrightarrow{(i_\alpha)_*(u)} & (i_\alpha)_*(\mathcal{F}'|_{U_\alpha}) \end{array}$$

commutative, from which there is a unique homomorphism w making all the diagrams

$$\begin{array}{ccc} (i_\alpha)_*(\mathcal{F}|_{U_\alpha}) & \xrightarrow{(i_\alpha)_*(u)} & (i_\alpha)_*(\mathcal{F}'|_{U_\alpha}) \\ \downarrow & & \downarrow \\ \mathcal{H}_{X/Z}^0(\mathcal{F}) & \xrightarrow{w} & \mathcal{H}_{X/Z}^0(\mathcal{F}') \end{array}$$

commutative, and finally that $\mathcal{F}'$ and $\mathcal{H}_{X/Z}^0(\mathcal{F}')$ identify canonically by hypothesis.

It remains to see that if the supports of $\text{Ker}(u)$ and of $\text{Coker}(u)$ are contained in Z , v is an isomorphism. It suffices to see that for every Noetherian open subset V , the corresponding homomorphism $\Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F})) \rightarrow \Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F}'))$ is then an isomorphism. Now, if a section $t \in \Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F}))$ has image 0 in $\Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F}'))$, note that for some index α , we have $t \in \Gamma(V \cap U_\alpha, \mathcal{F})$, and by virtue of the hypothesis on u , we have $t_y = 0$ for every $y \in V \cap (X - Z)$; there is thus an open set containing $V \cap (X - Z)$ such that the restriction of t to this open set is zero, hence by definition t is the element 0 of $\Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F}))$. Let us now prove that every section $s' \in \Gamma(V, \mathcal{H}_{X/Z}^0(\mathcal{F}'))$ is the image of a section of $\mathcal{H}_{X/Z}^0(\mathcal{F})$ above V . By hypothesis, for every $x \in V \cap (X - Z)$ there exists a section $s^{(x)}$ of $\mathcal{F}$ above a neighbourhood $W^{(x)}$ of x in X , whose image by u is $s'|_{W^{(x)}}$; $s'|_{W^{(x)}}$ is thus also the image by v of the section $t^{(x)}$ of $\mathcal{H}_{X/Z}^0(\mathcal{F})$, the canonical image of $s^{(x)}$. Furthermore, since as we have seen v is injective, the restrictions of $t^{(x)}$ and $t^{(x')}$ to $W^{(x)} \cap W^{(x')}$ are identical for any two points x, x' of $V \cap (X - Z)$; the $t^{(x)}$ are therefore restrictions of the same section t of $\mathcal{H}_{X/Z}^0(\mathcal{F})$ above an open neighbourhood U of $(X - Z) \cap V$. But since $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is Z -closed, t extends in a unique way to a section of $\mathcal{H}_{X/Z}^0(\mathcal{F})$ above V , whose image by v has the same restriction as s' to U , and coincides therefore with s' for the same reason.

We say that $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is the Z -closure of $\mathcal{F}$. □

Remarks (5.9.12). —

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- label=(i) Let $\mathcal{C}(X)$ be the category of $\mathcal{O}_X$ -Modules, and let $\mathcal{C}_Z(X)$ be the subcategory of $\mathcal{C}(X)$ formed by the $\mathcal{O}_X$ -Modules with support contained in Z ; this subcategory is localising in the sense of Gabriel, and the functor $\mathcal{H}_{X/Z}^0$ is none other than the localisation functor of Gabriel (cf. [?]); this would furnish another proof of (5.9.11)). When Z is closed, the functor $i^* : \mathcal{C}(X) \rightarrow \mathcal{C}(X - Z)$ (where $i : X - Z \rightarrow X$ is the canonical injection) defines an equivalence of categories $\mathcal{C}(X)/\mathcal{C}_Z(X) \approx \mathcal{C}(X - Z)$.
- lbel=(ii) It results from (5.9.11) that the condition $\mathcal{H}_{X/Z}^0(\mathcal{F}) = 0$ is equivalent to $\text{Supp}(\mathcal{F}) \subset Z$. It entails indeed this last condition since the kernel of $\rho_{X/Z}$ is then equal to $\mathcal{F}$. Conversely, if $\text{Supp}(\mathcal{F}) \subset Z$, it suffices to apply the second part of (5.9.11) to the unique homomorphism $u : \mathcal{F} \rightarrow 0$ to conclude that the corresponding morphism $v : \mathcal{H}_{X/Z}^0(\mathcal{F}) \rightarrow 0$ is an isomorphism.
- lbel=(iii) The above developments hold in their entirety for any ringed space which is locally Noetherian, whose every irreducible closed subset admits exactly one generic point.

In particular they apply, on a locally Noetherian prescheme, to *sheaves of abelian groups* (considered as Modules over the simple sheaf associated to the constant presheaf $\mathbf{Z}$). We have moreover for every $x \in X$ the canonical isomorphism (5.9.6.1), in which $\widetilde{\mathcal{F}}_x$ denotes the sheaf induced on the subspace X_x of X by the sheaf $\mathcal{F}$; the direct proof results immediately from the definition (5.9.1.2) and the theorem of the double inductive limit.

5.10. Property (S₂) and Z-closure

(5.10.1). Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module; for every subset T of X , we set

$$(5.10.1.1) \quad \text{prof}_T(\mathcal{F}) = \inf_{x \in T} \text{prof}(\mathcal{F}_x).$$

Proposition (5.10.2). — *Let X be a locally Noetherian prescheme, Z a subset of X stable under specialisation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module. The following conditions are equivalent:*

label=a) $\mathcal{F}$ is Z-pure.

lbel=b) $\text{Ass}(\mathcal{F})$ does not meet Z .

If furthermore $\mathcal{F}$ is coherent, these conditions are equivalent to the following:

resume,label=a) $\text{prof}_Z(\mathcal{F}) \geq 1$.

Proof. To say that $\mathcal{F}$ is Z-pure means that for every Noetherian open subset V of X , and every open set $U \supset X - Z$, the restriction homomorphism $\Gamma(V, \mathcal{F}) \rightarrow \Gamma(V \cap U, \mathcal{F})$ is injective (5.9.8); but by (3.1.8) this is equivalent to $V \cap \text{Ass}(\mathcal{F}) \subset U$, whence the equivalence of a) and b). Furthermore, to say that $x \in \text{Ass}(\mathcal{F})$ means that no element of $\mathfrak{m}_x$ is $\mathcal{F}_x$ -regular (3.1.2), hence, when $\mathcal{F}$ is coherent, this can be written $\text{prof}(\mathcal{F}_x) = 0$; from this we immediately deduce in this case the equivalence of b) and c). $\square$

Corollary (5.10.3). — *Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$ be an exact sequence of quasi-coherent $\mathcal{O}_X$ -Modules. If $\mathcal{F}$ is Z-pure, so is $\mathcal{F}'$; conversely, if $\mathcal{F}'$ and $\mathcal{F}''$ are Z-pure, so is $\mathcal{F}$.*

Proof. This results from the form (5.10.2), b) of the condition for a quasi-coherent $\mathcal{O}_X$ -Module to be Z-pure, and from (3.1.7). $\square$

Corollary (5.10.4). — *Suppose $\mathcal{F}$ coherent. For $\mathcal{F}$ to be Z-pure at a point $x \in X$, it is necessary and sufficient that $\text{prof}_{Z_x}(\widetilde{\mathcal{F}}_x) \geq 1$ (with the notations of (5.9.6)).*

Proof. This results immediately from (5.10.2) and (5.9.6). $\square$

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Theorem (5.10.5). — *Let X be a locally Noetherian prescheme, Z a subset stable under specialisation, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. For $\mathcal{F}$ to be Z-closed, it is necessary and sufficient that $\text{prof}_Z(\mathcal{F}) \geq 2$.*

Proof. By virtue of (5.10.2), we can limit ourselves to the case where $\mathcal{F}$ is Z-pure and $\text{prof}_Z(\mathcal{F}) \geq 1$. Furthermore, to say that $\text{prof}_Z(\mathcal{F}) \geq 2$ is equivalent to saying that for every closed subset Z_α of Z , $\text{prof}_{Z_\alpha}(\mathcal{F}) \geq 2$; and similarly, it results from (5.9.8) that to say that $\mathcal{F}$ is Z-closed for every α is equivalent to saying that $\mathcal{F}$ is Z-closed in the case where Z is closed. The question being local, it suffices, for every $x \in Z$, to prove the theorem for $\mathcal{F}|U$, where U is an affine neighbourhood of x , and one can therefore reduce to the case where $X = U$ is affine. We know then that $\text{Ass}(\mathcal{F})$ is finite (3.1.6), and since $\text{Ass}(\mathcal{F}) \subset X - Z$, there is a section f of $\mathcal{O}_X$ above X such that $\text{Ass}(\mathcal{F}) \subset X_f \subset X - Z$ (II, 4.5.4); we deduce from this that f is $\mathcal{F}$ -regular (3.1.9) and that for every $y \in Z$, we have $f_y \in \mathfrak{m}_y$, hence $\text{prof}(\mathcal{F}_y) = 1 + \text{prof}(\mathcal{F}_y / f_y \mathcal{F}_y)$ (0, 16.4.6). The condition $\text{prof}_Z(\mathcal{F}) \geq 2$ is thus equivalent to $\text{prof}_Z(\mathcal{F} / f\mathcal{F}) \geq 1$, or equivalently (5.10.2) to the fact that $\mathcal{F} / f\mathcal{F}$ is Z-pure, and it suffices to see that this last property is equivalent to the fact that $\mathcal{F}$ is Z-closed.

Consider the exact sequence $0 \rightarrow \mathcal{F} \xrightarrow{f} \mathcal{F} \rightarrow \mathcal{F}/f\mathcal{F} \rightarrow 0$ (the homothety of ratio $f : \mathcal{F} \xrightarrow{f} \mathcal{F}$ being injective by hypothesis); setting $W = X - Z$, we have the commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \Gamma(X, \mathcal{F}) & \xrightarrow{f} & \Gamma(X, \mathcal{F}) & \longrightarrow & \Gamma(X, \mathcal{F}/f\mathcal{F}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \Gamma(W, \mathcal{F}) & \xrightarrow{f} & \Gamma(W, \mathcal{F}) & \longrightarrow & \Gamma(W, \mathcal{F}/f\mathcal{F}) \end{array}$$

where the two rows are exact (X being affine). If the restriction homomorphism $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(W, \mathcal{F})$ is bijective, we deduce from this diagram that

$$\Gamma(X, \mathcal{F}/f\mathcal{F}) \longrightarrow \Gamma(W, \mathcal{F}/f\mathcal{F})$$

is injective, and this shows (5.9.8) that if $\mathcal{F}$ is Z -closed, $\mathcal{F}/f\mathcal{F}$ is Z -pure. Conversely, suppose that $\mathcal{F}/f\mathcal{F}$ is Z -pure, and let s be a section of $\mathcal{F}$ above W ; since $X_f \subset W$, as $f^n(s|_{X_f})$ extends to a section t of $\mathcal{F}$ above X (I, 1.4.1); moreover, the restrictions of t and of $f^n s$ to X_f being the same, it follows that the restriction of t to W is equal to $f^n s$ by virtue of the relation $\text{Ass}(\mathcal{F}) \subset X_f$ (5.10.2); since f is $\mathcal{F}$ -regular, it suffices to see that t is of the form $f^n t'$, where $t' \in \Gamma(X, \mathcal{F})$ to show that the homomorphism $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(W, \mathcal{F})$ is surjective, hence bijective. Now, to say that $t = f^n t'$ means that the image of t in $\Gamma(X, \mathcal{F}/f^n \mathcal{F})$ is zero. But since $f^k \mathcal{F}/f^{k+1} \mathcal{F}$ is isomorphic to $\mathcal{F}/f\mathcal{F}$, hence Z -pure by hypothesis, we deduce from (5.10.3) that $\mathcal{F}/f^n \mathcal{F}$ is Z -pure. But by definition the image of $t|_W = f^n s$ in $\Gamma(W, \mathcal{F}/f^n \mathcal{F})$ is zero, whence the conclusion. $\square$

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Corollary (5.10.6). — Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module. For $\mathcal{F}$ to be Z -closed at a point x , it is necessary and sufficient that $\text{prof}_{Z_x}(\mathcal{F}_x) \geq 2$.

Proof. This results from (5.9.6) and (5.10.5). $\square$

Theorem (5.10.7). — (Hartshorne). Let X be a locally Noetherian prescheme, Y a closed subset of X . Suppose that for every $y \in Y$, we have $\text{prof}(\mathcal{O}_{X,y}) \geq 2$; then for every connected component C of X , $C - (C \cap Y)$ is connected.

Proof. We can limit ourselves to the case where X is connected; it results then from (5.10.5) that the canonical homomorphism $\mathcal{O}_X \rightarrow i_*(\mathcal{O}_X|_{X-Y})$ (where $i : X - Y \rightarrow X$ is the canonical injection) is bijective. It follows that the restriction homomorphism $\Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(X - Y, \mathcal{O}_X)$ is also bijective. It now suffices to apply the lemma (III, 7.8.6.1). $\square$

Corollary (5.10.8). — Let X be a locally Noetherian prescheme, d an integer such that for every $x \in X$, the relation $\dim(\mathcal{O}_x) \geq d$ implies $\text{prof}(\mathcal{O}_x) \geq 2$. Suppose X connected; then, if X', X'' are two distinct irreducible components of X , there exists a suite $(X_i)_{0 \leq i \leq n}$ of irreducible components of X such that $X_0 = X'$, $X_n = X''$, and that, for $1 \leq i \leq n$, $\text{codim}(X_{i-1} \cap X_i, X) \leq d - 1$ (we say then that X is connected in codimension $\leq d - 1$).

If Y is a closed subset of X such that $\text{codim}(Y, X) \geq d$, we have $\dim(\mathcal{O}_{X,y}) \geq d$ for every $y \in Y$ (5.1.3), hence $\text{prof}(\mathcal{O}_{X,y}) \geq 2$ for every $y \in Y$, and it results from (5.10.7) that $X - Y$ is connected. On the other hand, for $\text{codim}(Y, X) \geq d$, it is necessary and sufficient that $\text{codim}(Y \cap V, V) \geq d$ for every open neighbourhood V of y in X (0, 14.2.3). We note finally that if $\mathfrak{Y}$ denotes the set of closed subsets Y of X of codimension $\geq d$, the union of two sets of $\mathfrak{Y}$ belongs to $\mathfrak{Y}$ (0, 14.2.5), and every closed set contained in a set of $\mathfrak{Y}$ belongs to $\mathfrak{Y}$; this is also expressed by saying that $\mathfrak{Y}$ is an antifilter of closed subsets of X . The corollary then results from the following topological lemma:

Lemma (5.10.8.1). — Let X be a connected locally Noetherian topological space, $\mathfrak{Y}$ an antifilter of closed subsets of X . Suppose that if Y is a closed subset of X such that for every $y \in Y$, there is a neighbourhood V of y in X and a $Y_y \in \mathfrak{Y}$ such that $V \cap Y = V \cap Y_y$, then $Y \in \mathfrak{Y}$. The following conditions are then equivalent:

label=a) For every $Y \in \mathfrak{Y}$, $X - Y$ is connected.

lbbel=b) If X' and X'' are two distinct irreducible components of X , there exists a suite $(X_i)_{0 \leq i \leq n}$ of irreducible components of X such that $X_0 = X'$, $X_n = X''$ and that, for $1 \leq i \leq n$, $X_{i-1} \cap X_i \notin \mathfrak{Y}$.

Proof. Suppose $b)$ verified and let us prove that $U = X - Y$ is connected for every $Y \in \mathfrak{Y}$. If U' and U'' are two distinct irreducible components of U , there exist two irreducible components X', X'' of X such that $X' \cap U = U'$, $X'' \cap U = U''$ (0I, 2.1.6); let us form for these two components a suite (X_i) having the property stated in $b)$ and set $U_i = X_i \cap U$ for $1 \leq i \leq n$; then U_i is an irreducible component of U (0I, 2.1.6), and moreover $U_i \cap U_{i-1} \neq \emptyset$ for $1 \leq i \leq n$; if we had $X_{i-1} \cap X_i \in \mathfrak{Y}$ for some i with $1 \leq i \leq n$, we would deduce $X_{i-1} \cap X_i \subset Y$ by definition of Y , whence $U_{i-1} \cap U_i = \emptyset$, contradicting the hypothesis. This completes the proof that U is connected.

Let us now show that $a)$ implies $b)$. Denote by Y the union of the family $(X_\alpha \cap X_\beta)$, where (X_α, X_β) ranges over the set of couples of *distinct* irreducible components of X such that $X_\alpha \cap X_\beta \in \mathfrak{Y}$. For every point $y \in Y$ in a neighbourhood V of y in X meeting only a finite number of irreducible components of X ; it follows on the one hand that Y is closed and on the other that $V \cap Y$ is the intersection of V and of a set of $\mathfrak{Y}$; by virtue of the hypothesis on $\mathfrak{Y}$, we have $Y \in \mathfrak{Y}$, hence $U = X - Y$ is connected; moreover Y is rare in X . Let X', X'' be two distinct irreducible components of X , U', U'' their respective traces on U ; these are distinct irreducible components of U (0I, 2.1.6). Now the union of the irreducible components W of U such that there exist irreducible components $(U_i)_{0 \leq i \leq n}$ of U for which $U_0 = U'$, $U_{i-1} \neq U_i$ and $U_{i-1} \cap U_i \neq \emptyset$ for $1 \leq i \leq n$ and $U_n = W$ is an *open and closed* set in U , since U is locally Noetherian and its irreducible components form a locally finite family of closed sets. There is thus such a suite (U_i) for which $U_n = U''$; let X_i ($0 \leq i \leq n$) be the irreducible component of X such that $X_i \cap U = U_i$ (0I, 2.1.6); since $U_{i-1} \neq U_i$, we have $X_{i-1} \neq X_i$ for $1 \leq i \leq n$; if we had $X_{i-1} \cap X_i \in \mathfrak{Y}$ for some i with $1 \leq i \leq n$, we would deduce $X_{i-1} \cap X_i \subset Y$ by definition of Y , whence $U_{i-1} \cap U_i = \emptyset$, contradicting the hypothesis. This completes the proof of the lemma.

We note that the hypothesis on X in (5.10.8) is verified when X is a *Cohen-Macaulay prescheme* and $d \geq 2$. $\square$

Corollary (5.10.9). — *If a Noetherian local ring A satisfies (S_2) and is catenary, it is equidimensional.*

Proof. The hypothesis of (5.10.8) is then verified by $X = \text{Spec}(A)$, with $d = 2$. To show that all the irreducible components of X have the same dimension, it suffices then, by virtue of (5.10.8), to show that two such components X', X'' have the same dimension when one further supposes that $\text{codim}(X' \cap X'', X) = 1$. There is then an irreducible component Z of $X' \cap X''$ such that $\text{codim}(Z, X) = 1$, hence $\text{codim}(Z, X') = 1$, since $\text{codim}(Z, X') \geq 1$; similarly $\text{codim}(Z, X'') = 1$, and as X is catenary, this implies $\dim(X') = \dim(X'')$. $\square$

Proposition (5.10.10). — *Let X be a locally Noetherian prescheme, Z a subset of X stable under specialisation, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module and suppose that the $\mathcal{O}_X$ -Module $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent. Then:*

- label=(i) *We have $\text{prof}_Z(\mathcal{H}_{X/Z}^0(\mathcal{F})) \geq 2$.*
- lbbel=(ii) *For every point $x \in \text{Ass}(\mathcal{F}) \cap (X - Z)$, we have $\text{codim}(Z \cap \overline{\{x\}}, \overline{\{x\}}) \geq 2$.*
- lcbel=(iii) *The set U of the $x \in X$ such that $\text{prof}_x(\widetilde{\mathcal{F}}_x) \geq 2$ (notations of (5.9.6)) is open in X ; we have $X - U \subset Z$, and U is the largest open subset of X such that $\mathcal{F}|_U$ is $(Z \cap U)$ -closed.*

Proof. Set $\mathcal{F}' = \mathcal{H}_{X/Z}^0(\mathcal{F})$ for short. We know (5.9.11) that $\mathcal{F}'$ is Z -closed, and assertion (i) thus results from (5.10.5) applied to $\mathcal{F}'$. Let $x \in \text{Ass}(\mathcal{F}) \cap (X - Z)$; since the restrictions of $\mathcal{F}$ and $\mathcal{F}'$ to $X - Z$ are canonically isomorphic (5.9.11), we have $x \in \text{Ass}(\mathcal{F}')$. Consider a point $y \in Z \cap \overline{\{x\}}$; the prime ideal $\mathfrak{p}$ of $\mathcal{O}_y$ corresponding to x is associated to the $\mathcal{O}_y$ -module $\mathcal{F}'_y$, hence we have, by virtue of (i) and (0, 16.4.6.2), $2 \leq \text{prof}(\mathcal{F}'_y) \leq \dim(\mathcal{O}_y/\mathfrak{p}) = \text{codim}(\{y\}, \overline{\{x\}})$, whence (ii). To prove (iii), note that U is the set of the $x \in X$ such that $\widetilde{\mathcal{F}}_x$ is Z_x -closed (5.10.5), or equivalently, by virtue of (5.9.6), the set of points where the canonical homomorphism $\widetilde{\mathcal{F}}_x \rightarrow \widetilde{\mathcal{F}}'_x$ is bijective; it is therefore the complement of the union of the supports of $\text{Ker}(\rho_{X/Z})$ and $\text{Coker}(\rho_{X/Z})$, and these last are coherent $\mathcal{O}_X$ -Modules by virtue of the hypothesis (0I, 5.3.4), hence have closed support (0I, 5.2.2); this shows that U is open, and U is clearly the largest open set such that $\mathcal{F}|_U$ is $(Z \cap U)$ -closed; finally the inclusion $X - U \subset Z$ results from (5.9.11).

We will see further on (5.11.1) that in the most important cases assertion (ii) conversely implies that $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent. $\square$

(5.10.11). Let X be a locally Noetherian prescheme, Z a subset of X stable under specialisation; we have seen that $\mathcal{A} = \mathcal{H}_{X/Z}^0(\mathcal{O}_X)$ is a quasi-coherent $\mathcal{O}_X$ -Algebra (5.9.3); the X -scheme $X' = \text{Spec}(\mathcal{H}_{X/Z}^0(\mathcal{O}_X))$ (II, 1.3.1) is called the Z -closure of X . Furthermore for every $\mathcal{O}_X$ -Module $\mathcal{F}$, $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is an $\mathcal{A}$ -Module, which is quasi-coherent; in this last case, there is thus a unique $\mathcal{O}_{X'}$ -Module $\mathcal{F}'$ such that, denoting by $g : X' \rightarrow X$ the structural morphism (II, 1.4.3)

$$(5.10.11.1) \quad \mathcal{H}_{X/Z}^0(\mathcal{F}) = g_*(\mathcal{F}').$$

Proposition (5.10.12). — *The notations being those of (5.10.11):*

- label=(i) Let x be a point of X . For the morphism $X'_x \rightarrow X_x (= \text{Spec}(\mathcal{O}_{X,x}))$ deduced from g (notations of (5.10.12)) to be an isomorphism, it is necessary and sufficient that $\mathcal{O}_X$ be Z -closed at the point x (which holds for every $x \in X - Z$).
- lbel=(ii) Suppose furthermore that g be a finite morphism (see in (5.11.2) sufficient conditions for this to hold). Then the set U of the points $x \in X$ where X satisfies (S_2) is open in X ((5.7.2)) and $\text{codim}(X - U, X) \geq 2$.
- lcbel=(iii) Under the same hypotheses as in (ii), X' satisfies (S_2) and for every $x' \in X'$ such that $\text{codim}(\overline{\{x'\}}, X') \leq 1$, the point $x = g(x')$ is such that $\text{codim}(\overline{\{x\}}, X) = \text{codim}(\overline{\{x'\}}, X')$.

(5.10.13). We will now apply the results above to the case where Z is one of the sets $Z^{(n)}(X)$ (or simply $Z^{(n)}$), defined as the set of the $x \in X$ such that $\dim(\mathcal{O}_x) \geq n$; it is clear that $Z^{(n)}$ is stable under specialisation; for a closed subset T of X to be contained in $Z^{(n)}$, it is necessary and sufficient that $\text{codim}(T, X) \geq n$. We shall be interested here in the case $n = 2$. IV-2 | 119

Proposition (5.10.14). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module with support equal to X .*

- label=(i) For $\mathcal{F}$ to satisfy (S_1) , it is necessary and sufficient that it be $Z^{(1)}$ -pure.
- lbel=(ii) For $\mathcal{F}$ to satisfy (S_2) , it is necessary and sufficient that it be $Z^{(2)}$ -closed and $Z^{(1)}$ -pure, or equivalently that it be $Z^{(2)}$ -closed and have no associated prime cycle of codimension 1.

Proof. label=(i) To say that $\mathcal{F}$ has the property (S_1) means that $\mathcal{F}$ has no immersed associated prime cycle (5.7.5), or equivalently that for every $x \in \text{Ass}(\mathcal{F})$, we have $\dim(\mathcal{F}_x) = 0$ (3.1.4), in other words $\dim(\mathcal{O}_x) = 0$ (5.1.12.1); but this is equivalent to saying that $\text{Ass}(\mathcal{F})$ does not meet $Z^{(1)}$, and the conclusion results from (5.10.2).

lbel=(ii) To say that $\mathcal{F}$ is $Z^{(2)}$ -closed means that $\text{prof}_{Z^{(2)}}(\mathcal{F}) \geq 2$ (5.10.5), or equivalently that, for every $x \in X$, $\dim(\mathcal{F}_x) \geq 2$ implies $\text{prof}(\mathcal{F}_x) \geq 2$; this shows that property (S_2) implies this condition; it implies in addition that $\mathcal{F}$ satisfies (S_1) , hence has no immersed associated prime cycle. Conversely, suppose that $\mathcal{F}$ is $Z^{(2)}$ -closed and has no associated prime cycle of codimension 1; to verify (S_2) , it remains to show that if $x \in X$ is such that $\dim(\mathcal{F}_x) = 1$ (or, what amounts to the same, $\dim(\mathcal{O}_x) = 1$), then $\text{prof}(\mathcal{F}_x) = 1$. But the hypothesis $\dim(\mathcal{O}_x) = 1$ implies $x \notin \text{Ass}(\mathcal{F})$, and this last relation is equivalent to $\text{prof}(\mathcal{F}_x) \neq 0$, i.e. $\text{prof}(\mathcal{F}_x) \geq 1$. If $\mathcal{F}$ is $Z^{(1)}$ -pure, it is noted above that it satisfies (S_1) , hence all the associated prime cycles of $\mathcal{F}$ are of codimension 0, hence the preceding applies.

We note that it can happen that $\mathcal{F}$ is $Z^{(2)}$ -closed and does not satisfy (S_1) : this is the case for example when X is of dimension 1 (for then $Z^{(2)} = \emptyset$, and every $\mathcal{O}_X$ -Module is $Z^{(2)}$ -closed) and has immersed associated prime cycles.

Recall that in Chapter III, in the study of local cohomology, a cohomological characterisation of property (S_n) for all $n \geq 1$, generalising (5.10.14), is given. □

Corollary (5.10.15). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module with support X . Suppose that $\mathcal{F}$ has no associated prime cycles of codimension 1 and that $\mathcal{F}' = \mathcal{H}_{X/Z^{(2)}}^0(\mathcal{F})$ is coherent. Then:* IV-2 | 120

- label=(i) $\mathcal{F}'$ satisfies the property (S_2) .
- lbel=(ii) The set U of the $x \in X$ such that $\mathcal{F}$ satisfies (S_2) at the point x (5.7.2) is open in X and $\text{codim}(X - U, X) \geq 2$.

Proof. label=(i) We know (5.9.11) that $\mathcal{F}'$ is $Z^{(2)}$ -closed, and furthermore $\text{Supp}(\mathcal{F}') = X$, since the maximal points of X belong to $X - Z^{(2)}$ and at these points $\mathcal{F}'_x = \mathcal{F}_x \neq 0$, hence the support of $\mathcal{F}'$ is dense in X , and as $\mathcal{F}'$ is coherent, $\text{Supp}(\mathcal{F}')$ is closed, hence $\text{Supp}(\mathcal{F}') = X$. It remains to see that $\mathcal{F}'$ has no associated prime cycles of codimension 1. But if $x \in \text{Ass}(\mathcal{F}')$ and $\dim(\mathcal{F}'_x) = \dim(\mathcal{O}_x) = 1$, we have $x \in X - Z^{(2)}$, hence since $\mathcal{F}'_x = \mathcal{F}_x$ we would have $x \in \text{Ass}(\mathcal{F})$, contrary to the hypothesis, which completes the proof of (i).
 lbbel=(ii) We have $Z^{(n)}(X_x) = Z^{(n)}(X) \cap X_x$, with the notations of (5.9.6), taking into account (I, 2.4.2); on the other hand, the hypothesis that $\mathcal{F}$ has no associated prime cycles of codimension 1 implies the same hypothesis for $\widetilde{\mathcal{F}}_x$; for $\widetilde{\mathcal{F}}_x$ to satisfy (S_2) , it is therefore necessary and sufficient, by virtue of (5.10.14), that $\widetilde{\mathcal{F}}_x$ be $Z^{(2)}(X_x)$ -closed; assertion (ii) therefore results from (5.10.6) and (5.10.10), (iii). $\square$

Proposition (5.10.16). — *Let X be a locally Noetherian prescheme,*

$$X' = \text{Spec}(\mathcal{H}_{X/Z^{(2)}}^0(\mathcal{O}_X))$$

its $Z^{(2)}$ -closure, $g : X' \rightarrow X$ the structural morphism. Suppose that X has no associated prime cycle of codimension 1.

- label=(i) *For X to satisfy (S_2) at a point $x \in X$, it is necessary and sufficient that the morphism $X'_x \rightarrow X_x$ deduced from g (notations of (5.10.12)) be an isomorphism. This condition is always satisfied if $\text{codim}(\overline{\{x\}}, X) \leq 1$.*
- lbbel=(ii) *Suppose furthermore that g be a finite morphism (see in (5.11.2) sufficient conditions for this to hold). Then the set U of the points where X satisfies (S_2) is open and $\text{codim}(X - U, X) \geq 2$; furthermore U is the largest open subset of X such that the restriction $g^{-1}(U) \rightarrow U$ of g is an isomorphism.*
- lcbel=(iii) *Under the same hypotheses as in (ii), X' satisfies (S_2) and for every $x' \in X'$ such that $\text{codim}(\overline{\{x'\}}, X') \leq 1$, the point $x = g(x')$ is such that $\text{codim}(\overline{\{x\}}, X) = \text{codim}(\overline{\{x'\}}, X')$.*
- ldbel=(iv) *The hypotheses being those of (ii), let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module with support X , such that $\mathcal{H}_{X/Z^{(2)}}^0(\mathcal{F})$ is coherent; then the $\mathcal{O}_{X'}$ -Module $\mathcal{F}'$ such that $g_*(\mathcal{F}') = \mathcal{H}_{X/Z^{(2)}}^0(\mathcal{F})$ is coherent and satisfies (S_2) , and its support is a union of irreducible components of X' .*

Proof. Assertions (i) and (ii) are included for reference, having already been proved in substance: (i) IV-2 | 121 results in fact from (5.10.12), (i) and (5.10.14), and (ii) is a particular case of (5.10.15), (ii).

Let us prove (iii); set $x = g(x')$; since g is finite, it is of the same for the morphism $X'_x \rightarrow X_x$, hence $\dim(\mathcal{O}_{X',x'}) \leq \dim(X_x) \leq \dim(\mathcal{O}_{X,x})$ by virtue of (5.4.1). Suppose first that $\dim(\mathcal{O}_{X',x'}) \leq 1$ and let us show that then $\dim(\mathcal{O}_{X,x}) \leq 1$. Otherwise, we would have $\dim(\mathcal{O}_{X,x}) \geq 2$, i.e. $x \in Z^{(2)}$; then we would have $\text{prof}(\mathcal{O}_{X',x'}) \geq 2$ by (5.10.12), which is absurd. On the other hand, $x \in X - Z^{(2)}$, and hence $\mathcal{O}_{X',x'}$ is isomorphic to $\mathcal{O}_{X,x}$ (5.10.12), (i), whence $\dim(\mathcal{O}_{X',x'}) = \dim(\mathcal{O}_{X,x})$. Furthermore, since X has no associated prime cycles of codimension 1, the hypothesis $\dim(\mathcal{O}_{X,x}) = 1$ implies $x \notin \text{Ass}(\mathcal{O}_X)$, hence $\text{prof}(\mathcal{O}_{X,x}) \neq 0$, and hence also $\text{prof}(\mathcal{O}_{X',x'}) = 1$. Suppose now $\dim(\mathcal{O}_{X',x'}) \geq 2$, hence $\dim(\mathcal{O}_{X,x}) \geq 2$, i.e. $x \in Z^{(2)}$; then we deduce from (5.10.12) that $\text{prof}(\mathcal{O}_{X',x'}) \geq 2$ by (5.10.12), which establishes the assertions of (iii).

To prove (iv) it suffices to replace $\mathcal{O}_X$ by $\mathcal{F}$ in the preceding reasoning, which establishes that $\mathcal{F}'$ satisfies (S_2) and that if $\dim(\mathcal{F}'_{x'}) \leq 1$, $\mathcal{F}_x$ and $\mathcal{F}'_{x'}$ are isomorphic; in particular if $\dim(\mathcal{F}'_{x'}) = 0$, we have $\dim(\mathcal{O}_{X,x}) = 0$, hence $\dim(\mathcal{F}_x) = 0$ since $\mathcal{F}$ has support X , and finally $\dim(\mathcal{O}_{X',x'}) = 0$; every irreducible component of $\text{Supp}(\mathcal{F}')$ is therefore an irreducible component of X' , since $\mathcal{F}'$ is coherent, hence $\text{Supp}(\mathcal{F}')$ is closed, hence $\text{Supp}(\mathcal{F}')$ is a union of irreducible components of X' . $\square$

Proposition (5.10.17). — *Let A be an integral Noetherian ring, and denote by $A^{(1)}$ the intersection of the local rings $A_{\mathfrak{p}}$, where $\mathfrak{p}$ ranges over the set of prime ideals of A of height 1. Suppose that $A^{(1)}$ is a finite A -algebra. Then:*

- label=(i) *The ring $A^{(1)}$ satisfies the condition (S_2) .*

lbel=(ii) The set U of the $\mathfrak{p} \in \operatorname{Spec}(A)$ such that the canonical homomorphism $A_{\mathfrak{p}} \rightarrow (A^{(1)})_{\mathfrak{p}}$ is bijective is equal to the set of the $\mathfrak{p}$ such that $A_{\mathfrak{p}}$ satisfies (S_2) ; U is open in $X = \operatorname{Spec}(A)$ and $\operatorname{codim}(X - U, X) \geq 2$.

lbel=(iii) For every multiplicative subset S of A , $(S^{-1}A)^{(1)}$ is a $(S^{-1}A)$ -algebra that is finite.

lbel=(iv) Let B be a finite A -algebra, integral and containing A . Then $B^{(1)}$ is a finite B -algebra. Furthermore, for every prime ideal $\mathfrak{q}$ of B of height 1, the prime ideal $\mathfrak{q} \cap A$ of A is of height 1.

Proof. Taking into account the formula (5.9.3.1), we see that $X' = \operatorname{Spec}(A^{(1)})$ is the $Z^{(2)}$ -closure of $X = \operatorname{Spec}(A)$; as A has no immersed prime ideals, properties (i) and (ii) are particular cases of (5.10.16), (i), (ii) and (iii). To prove (iii), it suffices to remark that $(S^{-1}A)^{(1)} = S^{-1}A^{(1)}$, which is a particular case of (5.9.4): indeed $S^{-1}A$ is a flat A -module, the prime ideals of $S^{-1}A$ are the $S^{-1}\mathfrak{p}$, where $\mathfrak{p} \in \operatorname{Spec}(A)$ does not meet S , and $\operatorname{ht}(S^{-1}\mathfrak{p}) = \operatorname{ht}(\mathfrak{p})$, whence (iii). As $A^{(1)}$ is a finite A -algebra, $S^{-1}A^{(1)}$ is a finite $S^{-1}A$ -algebra, whence (iii). To prove (iv), set $Y = \operatorname{Spec}(B)$, and let $f : Y \rightarrow X$ be the structural morphism; since it is finite, it results from (5.4.1) that for every $y \in Y$, $\dim(\mathcal{O}_{Y,y}) \leq \dim(\mathcal{O}_{X,f(y)})$; hence, if $T = f^{-1}(Z^{(2)}(X))$, we have $T \supset Z^{(2)}(Y)$. Let us show that $\mathcal{G} = \mathcal{H}_{Y/Z^{(2)}(Y)}^0(f_*(\mathcal{O}_Y))$ is coherent; indeed $f_*(\mathcal{O}_Y) = \tilde{B}$, B being considered as an A -module; but since B is an integral finite A -algebra, its field of fractions is finite over the field of fractions of A , and therefore B is contained in a free A -module of finite type, and by (5.9.2), (i) $\mathcal{G}$ is a quasi-coherent sub- $\mathcal{O}_X$ -Module of $(\mathcal{H}_{X/Z^{(2)}(X)}^0(\mathcal{O}_X))^n$ for a suitable n ; this last being coherent by hypothesis, so is $\mathcal{G}$. Now, it results from the definition (5.9.1.2) that $\mathcal{G}$ is isomorphic to $f_*(\mathcal{H}_{Y/T}^0(\mathcal{O}_Y))$; this proves a fortiori that $\mathcal{H}_{Y/T}^0(\mathcal{O}_Y)$ is a coherent $\mathcal{O}_Y$ -Module. It results then from (5.10.10), (ii), applied to $\mathcal{O}_Y$ and to codimension 2, that $Z^{(2)}(Y) \subset T$, and finally $T = Z^{(2)}(Y)$. This proves the two assertions of (iv). $\square$

5.11. Coherence criteria for the Modules $\mathcal{H}_{X/Z}^0(\mathcal{F})$

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Proposition (5.11.1). — Let X be a locally Noetherian prescheme, Z a subset of X stable under specialisation, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Denote by (x_α) the family of points of $\operatorname{Ass}(\mathcal{F}) \cap (X - Z)$ and, for every α , let Y_α be the reduced closed subscheme of X with underlying space $\overline{\{x_\alpha\}}$, and $Z_\alpha = Z \cap \overline{\{x_\alpha\}}$. Then the following two conditions are equivalent:

label=a) $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is a coherent $\mathcal{O}_X$ -Module.

lbel=b) For every α , $\mathcal{H}_{Y_\alpha/Z_\alpha}^0(\mathcal{O}_{Y_\alpha})$ is a coherent $\mathcal{O}_{Y_\alpha}$ -Module.

These two conditions imply the following:

resume,label=a) For every α , we have $\operatorname{codim}(Z_\alpha, Y_\alpha) \geq 2$.

Moreover, the three conditions a), b), c) are equivalent when in addition one of the following properties is satisfied:

label=(i) Every point of X admits an open neighbourhood isomorphic to a subscheme of a regular scheme (in which case we also say that X is locally immersible in a regular scheme).

lbel=(ii) For every α , Y_α is universally catenary (5.6.2) and its normalisation (II, 6.3.8) Y'_α is finite over Y_α .

Proof. All the properties under consideration are local on X , so we can limit ourselves to the case where $X = \operatorname{Spec}(A)$ is affine, and $\mathcal{F} = \tilde{M}$, where M is an A -module of finite type. Then, for every α , if h_α is the canonical injection $Y_\alpha \rightarrow X$, $(h_\alpha)_*(\mathcal{O}_{Y_\alpha}) = \mathcal{G}_\alpha$ is the $\mathcal{O}_X$ -Module corresponding to the A -module quotient $A/\mathfrak{I}_{x_\alpha}$, and, by definition of $\operatorname{Ass}(\mathcal{F})$, this A -module is isomorphic to a sub- A -module of M . Since $\mathcal{H}_{X/Z}^0(\mathcal{G}_\alpha)$ is a quasi-coherent sub- $\mathcal{O}_X$ -Module of $\mathcal{H}_{X/Z}^0(\mathcal{F})$ (5.9.2), the hypothesis that $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent implies that so is $\mathcal{H}_{X/Z}^0(\mathcal{G}_\alpha)$. On the other hand, it results from the definition (5.9.1.2) that $\mathcal{H}_{X/Z}^0(\mathcal{G}_\alpha)$ is isomorphic to $\mathcal{H}_{Y_\alpha/Z_\alpha}^0(\mathcal{O}_{Y_\alpha})$; this proves that a) implies b). To see that b) implies a), it suffices to show that there is a finite filtration $(\mathcal{F}_i)_{0 \leq i \leq n}$ of $\mathcal{F}$ formed by coherent $\mathcal{O}_X$ -Modules, with $\mathcal{F}_0 = \mathcal{F}$, $\mathcal{F}_n = 0$, and such that $\mathcal{H}_{X/Z}^0(\mathcal{F}_i/\mathcal{F}_{i+1})$ is coherent for every i : this results, by descending induction on i , from the exact sequences

$$0 \longrightarrow \mathcal{F}_{i+1} \longrightarrow \mathcal{F}_i \longrightarrow \mathcal{F}_i/\mathcal{F}_{i+1}$$

from the fact that $\mathcal{H}_{X/Z}^0$ is a left exact functor (5.9.2), and finally from (0_I, 5.3.3) and (I, 6.1.1). By virtue of (3.2.8), it therefore suffices to prove that $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent when $\mathcal{F}$ is irredundant; in other words, $\text{Ass}(M) = \{\mathfrak{p}\}$ is reduced to a single element. We now note the

Lemma (5.11.1.1). — *Let A be a Noetherian ring, M an A -module of finite type, such that $\text{Ass}(M) = \{\mathfrak{p}\}$. There exists a finite filtration $(M_h)_{0 \leq h \leq m}$ of M such that $M_0 = M$, $M_m = 0$ and that M_h/M_{h+1} is isomorphic to a submodule of $A/\mathfrak{p}$.*

Proof. First note that the canonical homomorphism $M \rightarrow M_{\mathfrak{p}} = N$ is injective (Bourbaki, *Alg. comm.*, chap. IV, § 1, no 2, prop. 6). Set $B = A_{\mathfrak{p}}$, $\mathfrak{m} = \mathfrak{p}A_{\mathfrak{p}}$, the maximal ideal of B ; we have $\text{Ass}(N) = \{\mathfrak{m}\}$ (*loc. cit.*, prop. 5), and since N is a B -module of finite type, there is an integer r such that $\mathfrak{m}^r N = 0$; if we set $N'_j = \mathfrak{m}^j N$ for $0 \leq j \leq r$, N'_j/N'_{j+1} is an $(A/\mathfrak{p})$ -module of finite type, hence a direct sum of a finite number of submodules isomorphic to $B/\mathfrak{m}$; in other words, there is a finite filtration $(N_h)_{0 \leq h \leq m}$ of N such that $N_0 = N$, $N_m = 0$ and that N_h/N_{h+1} is isomorphic to $B/\mathfrak{m}$, i.e. to the field of fractions of $A/\mathfrak{p}$; the filtration of the $M_h = M \cap N_h$ answers the question, since M_h/M_{h+1} is isomorphic to a sub- $(A/\mathfrak{p})$ -module of finite type of $N_h/N_{h+1} = B/\mathfrak{m}$; but one knows that such a submodule is isomorphic to a submodule of $A/\mathfrak{p}$. $\square$

The existence of the filtration (M_h) then shows, by the same reasoning as above, that to prove (granting *b*) that $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent, we can limit ourselves to showing that $\mathcal{H}_{X/Z}^0(\mathcal{P})$ is coherent, where $\mathcal{P} = (A/\mathfrak{q})^\sim$, $\mathfrak{q}$ being an ideal associated to M . But if $\mathfrak{q} = \mathfrak{I}_y$ with $y \in Z$, the support of $\mathcal{P}$ is a closed set contained in Z , since Z is stable under specialisation; the definition (5.9.1.2) then shows that $\mathcal{H}_{X/Z}^0(\mathcal{P}) = 0$. If on the contrary $\mathfrak{q} = \mathfrak{I}_{x_\alpha}$ for some α , we have $\mathcal{P} = (h_\alpha)_*(\mathcal{O}_{Y_\alpha})$ by definition, and we have seen above that $\mathcal{H}_{X/Z}^0(\mathcal{P})$ is isomorphic to $\mathcal{H}_{Y_\alpha/Z_\alpha}^0(\mathcal{O}_{Y_\alpha})$, hence is coherent by virtue of *b*).

We have already seen (5.10.10), (ii) that *a*) implies *c*). It remains to prove that, under one or the other of the hypotheses (i), (ii), *c*) implies *b*). Note that if X satisfies (i), so does each Y_α . Similarly (5.11.1); it therefore suffices to prove (5.9.3.1) the corollary below. $\square$

Corollary (5.11.2). — *Let A be a Noetherian integral ring, satisfying one of the two following hypotheses:*

label=(i) A is a quotient of a regular Noetherian ring.

lbbel=(ii) A is universally catenary, and its integral closure A' is a finite A -algebra.

Then the ring $A^{(1)}$, intersection of the $A_{\mathfrak{p}}$ where $\mathfrak{p}$ ranges over the set of prime ideals of A of height 1, is a finite A -algebra.

Proof. (i) Set $X = \text{Spec}(A)$. The set U of the points $x \in X$ where X satisfies (S_2) is open in X by (6.11.2)². Furthermore, if one sets $Z = X - U$, we have $\text{codim}(Z, X) \geq 2$: indeed, for every $x \in X$ such that $\dim(A_{x'}) \leq 1$, we have $\dim(A_{x'}) \leq 1$ for every generalisation x' of x , hence $\text{prof}(A_{x'}) \geq 1$, hence X satisfies (S_2) at the point x . We have thus $Z \subset Z^{(2)}$, and $\mathcal{H}_{X/Z^{(2)}}^0(\mathcal{O}_X) = j_*(\mathcal{H}_{U/Z^{(2)}}^0(\mathcal{O}_U))$, where $j: U \rightarrow X$ is the canonical injection (5.9.1.2). But since the prescheme U satisfies (S_2) , it results from (5.10.14) that $\mathcal{H}_{U/Z^{(2)}}^0(\mathcal{O}_U)$ is isomorphic to $\mathcal{O}_U$; on the other hand, since $\text{codim}(Z, X) \geq 2$, one knows, from Chapter III, § 9, that $j_*(\mathcal{O}_U)$ is a coherent $\mathcal{O}_X$ -Module, which proves the proposition in case (i).

(ii) The ring A' is, by virtue of hypothesis (ii), a Noetherian integral ring integrally closed, hence (Bourbaki, *Alg. comm.*, chap. VII, § 1, no 6, th. 4) the intersection of its local rings $A'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over the set of prime ideals of height 1 of A' . Now, for such a prime ideal $\mathfrak{p}'$, if one sets $\mathfrak{p} = \mathfrak{p}' \cap A$ and $S = A - \mathfrak{p}$, $A'_{\mathfrak{p}'}$ is a local ring at a prime ideal $S^{-1}\mathfrak{p}'$ of $S^{-1}A'$; since $S^{-1}\mathfrak{p}'$ lies above the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ of $A_{\mathfrak{p}}$, it is a maximal ideal of $S^{-1}A'$; and $S^{-1}A'$ is by hypothesis a finite $A_{\mathfrak{p}}$ -algebra; since $S^{-1}\mathfrak{p}'$ lies above the maximal ideal of $A_{\mathfrak{p}}$, $A_{\mathfrak{p}}$ is universally catenary. We deduce from (5.6.10) that $\dim(A_{\mathfrak{p}}) = \dim(A'_{\mathfrak{p}'}) = 1$. It results from this that $A^{(1)} \subset A'$, and as A' is a finite module of A by hypothesis, so is $A^{(1)}$ since A is Noetherian; the proposition is therefore proved in case (ii). $\square$

Remark (5.11.3). — The fact that $A^{(1)}$ is a finite A -algebra is not strictly necessary in hypothesis (ii) of (5.11.2): one only supposes that A is *catenary*. An example is given by the catenary local ring A

²The reader may verify that (5.11.2) is not used in the proof of (6.11.2).

constructed in (5.6.11), whose integral closure A' (denoted B in (5.6.11)) is a finite A -algebra; if $A^{(1)}$ were a finite A -algebra, it would be contained in the field of fractions of A , hence contained in A' . But on the other hand, with the notations of (5.6.11), every prime ideal $\mathfrak{r}$ of height 1 in A is of the form $\mathfrak{p}A$, where $\mathfrak{p}$ is a prime ideal of height 1 in C , and $A_{\mathfrak{p}\mathfrak{A}} = C_{\mathfrak{p}}$; we know (5.6.11) that $C_{\mathfrak{p}} = E_{\mathfrak{p}'}$, where $\mathfrak{p}'$ is the unique prime ideal of E above $\mathfrak{p}$, hence $A_{\mathfrak{p}\mathfrak{A}}$ is integrally closed and contains A' ; by definition, $A^{(1)} \subset A'$, and the hypothesis that $A^{(1)}$ is a finite A -algebra would imply finally $A^{(1)} = A'$. But this conclusion is absurd, since one of the two prime ideals of $\mathfrak{n}A$ of A is of height 1, while $\mathfrak{n}A$ is of height 2, which would contradict (5.10.17), (iv).

Corollary (5.11.4). — *Let X be a locally Noetherian prescheme, Z a closed subset of X , $U = X - Z$, $i : U \rightarrow X$ the canonical injection, $\mathcal{F}$ an $\mathcal{O}_U$ -Module coherent. For $i_*(\mathcal{F})$ to be a coherent $\mathcal{O}_X$ -Module, it is necessary that for every $x \in \text{Ass}(\mathcal{F})$, $\text{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geq 2$. This condition is sufficient in each of the two following cases:*

- label=(i) *The prescheme X is locally immersible in a regular scheme.*
- lbbel=(ii) *The prescheme X is universally catenary and universally Japanese (which means, by definition, that every point $x \in X$ admits an affine open neighbourhood whose ring is universally Japanese (0, 23.1.1)).*

Proof. One knows (I, 9.4.7) that there exists a quasi-coherent sub- $\mathcal{O}_X$ -Module $\mathcal{G}$ of $i_*(\mathcal{F})$ such that $\mathcal{G}|_U = \mathcal{F}$. We have clearly $\text{Ass}(\mathcal{F}) \subset \text{Ass}(\mathcal{G}) \subset \text{Ass}(i_*(\mathcal{F}))$, and since $\text{Ass}(i_*(\mathcal{F})) = \text{Ass}(\mathcal{F})$ (3.1.13), one has $\text{Ass}(\mathcal{G}) = \text{Ass}(\mathcal{F})$; it suffices then to apply (5.11.1) to the coherent $\mathcal{O}_X$ -Module $\mathcal{G}$, noting that $i_*(\mathcal{F}) = \mathcal{H}_{X/Z}^0(\mathcal{G})$ and that, when X is universally catenary and universally Japanese, hypothesis (ii) of (5.11.1) is satisfied by definition. $\square$

Corollary (5.11.5). — *Let X be a locally Noetherian prescheme, Z a subset of X stable under specialisation, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module such that $\text{Ass}(\mathcal{F}) \subset X - Z$. Then the condition:*

label=a) $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is a coherent $\mathcal{O}_X$ -Module

implies the following:

ldbel=d) *For every closed subset T of Z in X (or only for a filtering family*

of closed subsets T of Z , with union Z), $i_(\mathcal{F}|_{X-T})$ (where $i : X - T \rightarrow X$ is the canonical injection) is a coherent $\mathcal{O}_X$ -Module. IV-2 | 125*

Furthermore, the conditions a) and d) are satisfied in each of the two following cases:

- label=(i) *A is a quotient of a regular Noetherian ring.*
- lbbel=(ii) *A is universally catenary and universally Japanese.*

We then have the implications

$$a) \Leftrightarrow b) \Rightarrow c) \Leftrightarrow d)$$

Proof. When X satisfies one of the hypotheses (i), (ii) of (5.11.1), $a)$ and $d)$ are equivalent. Note that $\text{Ass}(i_*(\mathcal{F}|_{X-T})) = \text{Ass}(\mathcal{F})$ by virtue of the hypothesis and of (3.1.13); it results therefore from (5.10.2) that the canonical applications

$$i_*(\mathcal{F}|_{X-T}) \longrightarrow \mathcal{H}_{X/Z}^0(\mathcal{F})$$

are injective; the fact that $a)$ implies $d)$ is therefore a consequence of this remark. Conversely, the condition $d)$ implies, by virtue of (5.11.1) that $\text{codim}(T \cap Y_\alpha, Y_\alpha) \geq 2$ with the notations of (5.11.1); hence $\text{codim}(Z \cap Y_\alpha, Y_\alpha) \geq 2$ since Z is a union of its closed subsets in X , and the last assertion of the corollary follows from (5.11.1). $\square$

Corollary (5.11.6). — *Let A be a Noetherian integral ring, $X = \text{Spec}(A)$. Consider the following properties:*

- label=a) *For every integral quotient ring B of A , the ring $B^{(1)}$ (notation of (5.11.2)) is a finite B -algebra.*
- lbbel=b) *For every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ and every closed subset Z of X , stable under specialisation, and such that for every $x \in \text{Ass}(\mathcal{F}) \cap (X - Z)$ we have $\text{codim}(\overline{\{x\}} \cap Z, \overline{\{x\}}) \geq 2$, the $\mathcal{O}_X$ -Module $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent.*
- lcbel=c) *For every closed subset T of X and every coherent $\mathcal{O}_U$ -Module $\mathcal{G}$ (where $U = X - T$) such that for every $x \in \text{Ass}(\mathcal{G})$ we have $\text{codim}(\overline{\{x\}} \cap T, \overline{\{x\}}) \geq 2$, $i_*(\mathcal{G})$ (where $i : U \rightarrow X$ is the canonical injection) is a coherent $\mathcal{O}_X$ -Module.*

ldbel=d) For every integral quotient ring B of A and every ideal $\mathfrak{J}$ of height ≥ 2 in B , the ring $\bigcap_{\mathfrak{q} \not\supset \mathfrak{J}} B_{\mathfrak{q}}$ is a finite B -algebra.

We then have the implications

$$a) \Leftrightarrow b) \Rightarrow c) \Leftrightarrow d)$$

Furthermore, the conditions a), b), c) and d) are satisfied in each of the two following cases:

label=(i) A is a quotient of a regular Noetherian ring.

lbbel=(ii) A is universally catenary and universally Japanese.

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Proof. Let $B = A/\mathfrak{q}$, where $\mathfrak{q}$ is a prime ideal of A , so that $Y = \text{Spec}(B)$ is the closed subset $V(\mathfrak{q})$ of X ; set $Z = Z^{(2)}(Y)$, which is a subset of X stable under specialisation. If condition b) is satisfied, one can apply it to the $\mathcal{O}_X$ -Module coherent $\mathcal{F} = (A/\mathfrak{q})^\sim$ and to $Z = V(\mathfrak{J})$; inversely, one proves that d) implies c) by using the equivalence of a) and b) in (5.11.1). It is clear that c) is a particular case of b). Finally, the fact that a) (and hence each of the other conditions) is satisfied when A satisfies one of the hypotheses (i), (ii) results from (5.11.2), by the definition of universally catenary rings and of universally Japanese rings. $\square$

Remarks (5.11.7). — label=(i) If, in (5.11.5), condition d) implies a) without supplementary hypothesis on X ; we shall see further on (7.2.4) that this is indeed the case also when X is a local scheme. Similarly, we shall prove that when A is a local Noetherian ring the four properties a), b), c), d) of (5.11.6) are equivalent (7.2.4). If A satisfies property a) of (5.11.6), it is of the same for every ring of fractions $S^{-1}A$ (5.10.17), (iii) and (iv); in particular, it extends to all Noetherian rings.

lbbel=(ii) If A satisfies property a) of (5.11.6), it is of the same for every integral finite A -algebra C . Indeed every integral quotient ring of $S^{-1}A$ is of the form $S^{-1}(A/\mathfrak{q})$, where $\mathfrak{q}$ is a prime ideal of A not meeting S ; and on the other hand, if $\mathfrak{r}$ is a prime ideal of C , $\mathfrak{p}$ its inverse image in A , $C_{\mathfrak{r}}$ is an $(A/\mathfrak{p})$ -algebra that is finite integral with the field of fractions finite over that of $A/\mathfrak{p}$; our assertions are therefore consequences of (5.10.17), (iii) and (iv).

5.12. Relations between the properties of a Noetherian local ring A and of a quotient ring A/tA

We have already seen in (3.4) relations between the properties of A and of A/tA concerning the associated prime ideals, as well as the properties of being integral or reduced. In this section we give other relations between the properties of these rings, related to the notions of dimension and of depth.

Proposition (5.12.1). — Let A be a Noetherian local ring, $X = \text{Spec}(A)$, t an element of A belonging to a system of parameters (0, 16.3.6), X_0 the closed subspace $V(t)$ of X , X_1 the complementary open subset $X - X_0$. Let Z be a subset of X such that every specialisation of a point of Z belongs to Z . Suppose that X is biequidimensional (which will be the case if A is equidimensional and a quotient of a regular local ring (0, 16.5.12)). Then if we set $Z_0 = Z \cap X_0$, $Z_1 = Z \cap X_1$, we have

$$(5.12.1.1) \quad \text{codim}(Z_0, X_0) \leq \text{codim}(Z, X) \leq \text{codim}(Z_1, X_1).$$

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Proof. The second inequality results from the definition (5.1.3); to prove the first, we can limit ourselves to proving it when Z is closed: indeed, by hypothesis Z is a union of closed subsets Z_{α} , and if $\text{codim}(X_0 \cap Z_{\alpha}, X_0) \leq \text{codim}(Z_{\alpha}, X)$ for every α , we will have also $\text{codim}(Z_0, X_0) = \inf_{\alpha} (\text{codim}(X_0 \cap Z_{\alpha}, X_0)) \leq \inf_{\alpha} (\text{codim}(Z_{\alpha}, X)) = \text{codim}(Z, X)$. Suppose therefore Z closed, so that

$$\text{codim}(Z, X) = \dim(X) - \dim(Z).$$

On the other hand, X_0 is clearly catenary, equidimensional and of dimension $\dim(X) - 1$ by virtue of (0, 16.3.4); we thus also have

$$\text{codim}(Z_0, X_0) = \dim(X_0) - \dim(Z_0) = \dim(X) - 1 - \dim(Z_0).$$

But we have $\dim(Z_0) \geq \dim(Z) - 1$ (0, 16.3.4), which completes the proof of the first inequality (5.12.1.1). $\square$

Proposition (5.12.2). — *Let A be a Noetherian local ring, M an A -module of finite type, t an M -regular element belonging to the maximal ideal $\mathfrak{m}$ of A , k an integer ≥ 1 . Suppose that A is a catenary ring. Then, if M/tM is equidimensional and satisfies (S_k) , the same is true of M .*

Proof. Taking into account (Err_{III}, 30) and (5.7.3), (vi), we can limit ourselves to the case where $\text{Supp}(M) = \text{Spec}(A) = X$. Set $X_0 = V(t) = \text{Spec}(A/tA)$; since M/tM satisfies (S_1) by hypothesis, it has no immersed associated prime ideals (5.7.5). Applying (3.4.4) to the closed point of X (and of every closed subscheme of X), we see that the irreducible components of $\text{Supp}(M/tM)$ are exactly those of $Y_i \cap X_0$, where Y_i ($1 \leq i \leq r$) are the irreducible components of X . Since t is by hypothesis M -regular, $V(t)$ contains no maximal point of X (3.1.8); each of the irreducible components of $Y_i \cap X_0$ is therefore of codimension 1 in Y_i (5.1.8); since X is catenary and since all the irreducible components of $\text{Supp}(M/tM)$ have by hypothesis the same dimension, we conclude that the Y_i all have the same dimension, in other words X is equidimensional, hence *biequidimensional* since A is local and catenary. To see that M satisfies (S_k) , let us apply the criterion (5.7.4) (with the notations of (5.7.4)): we must verify that, for every integer $n \geq 0$, $\text{codim}(Z_n, X) > n + k$. But the hypothesis on M/tM and the criterion (5.7.4) show that $\text{codim}(Z_n \cap X_0, X_0) > n + k$ for every integer $n \geq 0$. But every specialisation of a point of Z_n still belongs to Z_n , as we shall see in (6.11.5)³; since X is biequidimensional and t belongs to a system of parameters for M (0, 16.4.1), and since t is a non-zero-divisor for A (the annihilator of M being nilpotent by virtue of the hypothesis $\text{Supp}(M) = \text{Spec}(A)$), the conclusion results from (5.12.1). $\square$

Remark (5.12.3). — If, in the statement of (5.12.2), one only supposes that M/tM is equidimensional, the conclusion does not necessarily follow that M is equidimensional.

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Corollary (5.12.4). — *Let A be a Noetherian local catenary ring, t a regular element belonging to the maximal ideal of A , k an integer ≥ 1 . If A/tA satisfies (S_k) , the same is true of A .*

Proof. If $k = 1$, the corollary results from (3.4.4), in view of the interpretation of condition (S_1) (5.7.5). If $k \geq 2$, it results from the criterion of Hartshorne (5.10.9) that A/tA is equidimensional, and one can apply (5.12.2). $\square$

Proposition (5.12.5). — *Let A be a Noetherian local catenary ring, t an element of A belonging to the maximal ideal of A , k an integer ≥ 0 . If the ring A/tA is reduced, equidimensional and satisfies the property (R_k) , the ring A also has these three properties.*

Proof. It is already known (3.4.6) that A is reduced, and it results from (5.12.2), applied for $k = 1$, that A is equidimensional. Set $X = \text{Spec}(A)$, $X_0 = V(t) = \text{Spec}(A/tA)$, and let Z be the set of points $x \in X$ where X is not regular; by virtue of (0, 17.3.2), every specialisation of a point of Z also belongs to Z . It results on the other hand from (0, 17.1.8) that for every point $x \in X_0$ where X is regular, X is also regular, hence the set Z'_0 of the points where X_0 is not regular contains $Z \cap X_0$; the hypothesis therefore implies that

$$k \leq \text{codim}(Z'_0, X_0) \leq \text{codim}(Z_0, X_0).$$

But, since A is equidimensional and catenary, X is biequidimensional, hence from (5.12.1)

$$\text{codim}(Z_0, X_0) \leq \text{codim}(Z, X)$$

which proves that X satisfies (R_k) . $\square$

Remark (5.12.6). — If, in the statement of (5.12.5), one only supposes that A/tA is reduced and possesses property (R_k) , it does not necessarily follow that A possesses property (R_k) .

Corollary (5.12.7). — *Let A be a Noetherian local catenary ring, t a regular element belonging to the maximal ideal of A . If A/tA is integral and integrally closed, the same is true of A .*

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Proof. By virtue of Serre's criterion (5.8.6) and of the fact that A is local, it suffices to show that $\text{Spec}(A)$ satisfies (S_2) and (R_1) . But the hypothesis on A/tA implies that A satisfies (R_1) by (5.12.5), and that A satisfies (S_2) by (5.12.4). $\square$

³The reader may verify that (5.12.2) is not used in the proof of (6.11.5).

Proposition (5.12.8). — (Hironaka’s lemma). *Let A be a Noetherian local ring, reduced, equidimensional and catenary. Suppose furthermore that for every minimal prime ideal $\mathfrak{q}_i$ of A , the ring $B_i = A/\mathfrak{q}_i$ is such that $B_i^{(1)}$ (notation of (5.10.17)) is a finite B_i -algebra (which occurs for instance when A possesses one of the properties (i) or (ii) of (5.11.6)). Let t be an element of A , non-zero-divisor in A , and such that:*

label=(i) The A -module A/tA has only one non-immersed associated prime ideal $\mathfrak{p}$ (necessarily of height 1 by the Hauptidealsatz (0, 16.3.2)), but one does not suppose that this is the only prime ideal associated to A/tA .

lbbel=(ii) The ideal $\mathfrak{p}A_{\mathfrak{p}}$ of $A_{\mathfrak{p}}$ is generated by $t/1$.

lcbel=(iii) The ring $A/\mathfrak{p}$ is integrally closed.

Under these conditions, the ring A is integral and integrally closed, and we have $\mathfrak{p} = tA$.

Proof. Since $Z = V(tA)$ is an irreducible closed subset of $X = \text{Spec}(A)$, whose generic point z is such that $\mathfrak{I}_z = \mathfrak{p}$, hypothesis (i) shows that $\mathfrak{p}$ cannot contain more than *one* minimal prime ideal $\mathfrak{q}$ of A , in other words $\mathfrak{p}$ belongs to a *single* irreducible component of X ; moreover $\mathfrak{q}$ is the inverse image of (0) by the canonical homomorphism $A \rightarrow A_{\mathfrak{p}}$, and $A_{\mathfrak{p}}$ is therefore isomorphic to $(A/\mathfrak{q})_{\mathfrak{p}\mathfrak{q}}$, which proves that the integral quotient $A/\mathfrak{q}$ is *integrally closed* as an intersection of integrally closed rings (I, 8.2.1.1). If A is *Noetherian and normal*, A is composed of a finite direct product of integral domains that are integrally closed (I, 5.1.4). Let us now note that since $A^{(1)}$ is a finite A -algebra, it is a semi-local Noetherian ring, and every non-immersed prime ideal associated to $A^{(1)}/tA^{(1)}$ is of height 1 by the Hauptidealsatz (0, 16.3.2); now, for such a prime ideal $\mathfrak{r}$, $\mathfrak{r} \cap A$ is of height 1 by virtue of (5.10.17), (iv), and as it contains tA , it cannot be $\mathfrak{p}$ by virtue of hypothesis (i). On the other hand, $A^{(1)}$ is contained in the integral closure A' of A ; if one sets $S = A - \mathfrak{p}$, the integral closure of $A_{\mathfrak{p}}$ is $S^{-1}A'$ (Bourbaki, *Alg. comm.*, chap. V, § 1, no 5, prop. 17), hence $S^{-1}A' = A_{\mathfrak{p}}$ since $A_{\mathfrak{p}}$ is a discrete valuation ring; it follows from this that there exists only *one* prime ideal $\mathfrak{p}'$ of A' above $\mathfrak{p}$, and *a fortiori* one prime ideal $\mathfrak{p}^{(1)}$ of $A^{(1)}$ above $\mathfrak{p}$, and we have $A_{\mathfrak{p}} = A'_{\mathfrak{p}'} = A^{(1)}_{\mathfrak{p}^{(1)}}$. Note now that $A^{(1)}$ satisfies (S_2) (5.10.17), (i) and since t is a non-zero-divisor in $A^{(1)}$, $A^{(1)}/tA^{(1)}$ has no immersed associated prime ideals (5.7.7). The prime ideal $\mathfrak{p}^{(1)}$ is therefore the unique associated prime ideal of $A^{(1)}/tA^{(1)}$, in other words $tA^{(1)}$ is a $\mathfrak{p}^{(1)}$ -primary ideal of $A^{(1)}$; this means also that the inverse image of $tA^{(1)}$ by the canonical projection $A^{(1)} \rightarrow A^{(1)}/\mathfrak{p}^{(1)}$ is the maximal ideal of $A_{\mathfrak{p}}$, from (ii); hence $tA^{(1)} = \mathfrak{p}^{(1)}$ is prime in $A^{(1)}$. On the other hand, $A^{(1)}/\mathfrak{p}^{(1)}$ is finite over $A/\mathfrak{p}$ and has the same field of fractions (since the residue field of $A^{(1)}_{\mathfrak{p}^{(1)}} = A_{\mathfrak{p}}$), hence it is identical to $A/\mathfrak{p}$ by virtue of hypothesis (iii). One can therefore write $A^{(1)} = A + \mathfrak{p}^{(1)} = A + tA^{(1)}$, and since t is contained in the radical of the semi-local Noetherian ring $A^{(1)}$, the lemma of Nakayama implies $A^{(1)} = A$, whence $\mathfrak{p}^{(1)} = \mathfrak{p} = tA$. But since A is catenary and A/tA is integral and integrally closed by virtue of hypothesis (iii), we conclude from (5.12.7) that A is integrally closed. $\square$

Corollary (5.12.10). — *Let A be an integral Noetherian local ring, satisfying one of the two following hypotheses:*

label=a) A is universally catenary and universally Japanese.

lbbel=b) A is a quotient of a regular ring.

Let $(x_i)_{1 \leq i \leq n}$ be a suite of n elements of A ; set $\mathfrak{J} = x_1A + \dots + x_nA$, and suppose the following conditions satisfied:

label=(i) There exists only one non-immersed prime ideal $\mathfrak{p}$ associated to $A/\mathfrak{J}$ and $\mathfrak{p}$ is of height n .

lbbel=(ii) The maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ of $A_{\mathfrak{p}}$ is generated by the $x_i/1$ (hence $A_{\mathfrak{p}}$ is a regular local ring of dimension n).

lcbel=(iii) The ring $A/\mathfrak{p}$ is integrally closed.

Under these conditions, for every integer i such that $0 \leq i \leq n$, the quotient ring $A/(x_1A + \dots + x_iA)$ is integrally closed and of dimension $\dim(A) - i$. In particular, $\mathfrak{J}$ is equal to $\mathfrak{p}$, A is integrally closed and $(x_i)_{1 \leq i \leq n}$ is an A -regular sequence.

Proof. We proceed by induction on n , the proposition being trivial for $n = 0$ and reducing to Hironaka’s lemma (5.12.8) for $n = 1$. We may therefore suppose $n \geq 2$. Set $\mathfrak{J}' = x_1A + \dots + x_{n-1}A$, and let $\mathfrak{q}$ be a minimal prime ideal among those containing $\mathfrak{J}'$. If $\mathfrak{q}$ is minimal among the prime

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ideals of A containing $\mathfrak{q} + x_n A$, $\mathfrak{p}'/\mathfrak{q}$ is of height 1 by the Hauptidealsatz (0, 16.3.1); let us show that $\mathfrak{q} \subset \mathfrak{p}$. Indeed, a prime ideal $\mathfrak{n}$ of B minimal among those containing $\mathfrak{q} + x_n A$ is of height $\leq n - 1$ by the Hauptidealsatz, and since $\mathfrak{q}$ is minimal, there is at most one such ideal. Since it contains $\mathfrak{J}$, it is necessarily equal to $\mathfrak{p}$, whence $\mathfrak{q} \subset \mathfrak{p}$.

Consider now the integral ring $B = A/\mathfrak{q}$, and let t be the image of x_n in B . It is immediate (cf. (5.6.3), (iv)) that if A satisfies hypothesis a) (resp. b)), then so does B , and B and t satisfy the hypotheses of Hironaka's lemma. Application of (5.12.8) proves that $B = A/\mathfrak{q}$ is integrally closed. Let us now use the induction hypothesis, which shows that $A/(x_1 A + \dots + x_i A)$ is integrally closed and of dimension $\dim(A) - i$ for $0 \leq i \leq n - 1$ and that $\mathfrak{q} = \mathfrak{J}'$, whence $\mathfrak{p} = \mathfrak{J}' + x_n A = \mathfrak{J}$. We conclude that $A/\mathfrak{J} = A/\mathfrak{p}$ is integrally closed by hypothesis (iii), and that $\dim(A/\mathfrak{p}) = n$ and that $\dim(A/\mathfrak{p}) = \dim(A) - n$ since A is biequidimensional, which completes the proof. $\square$

5.13. Permanence properties by passage to the inductive limit Throughout this section, $(A_\alpha, \varphi_{\beta\alpha})$ denotes an inductive system of rings whose index set is a pre-ordered filtering set; we set $A = \varinjlim A_\alpha$ and denote by φ_α the canonical homomorphism $A_\alpha \rightarrow A$.

Proposition (5.13.1). — *label=(i) For every index α , let $\mathfrak{a}_\alpha$ be an ideal of A_α such that, for $\alpha \leq \beta$, $\varphi_{\beta\alpha}(\mathfrak{a}_\alpha) \subset \mathfrak{a}_\beta$. Then the $\mathfrak{a}_\alpha$ form an inductive system for the restrictions of the $\varphi_{\beta\alpha}$, $\mathfrak{a} = \varinjlim \mathfrak{a}_\alpha$ identifies canonically with an ideal of A , $A/\mathfrak{a}$ with $\varinjlim (A_\alpha/\mathfrak{a}_\alpha)$. If furthermore $\mathfrak{a}_\alpha = \varphi_{\beta\alpha}^{-1}(\mathfrak{a}_\beta)$ for $\alpha \leq \beta$, then $\mathfrak{a}_\alpha = \varphi_\alpha^{-1}(\mathfrak{a})$ for every α .*

lbbel=(ii) Conversely, for every ideal $\mathfrak{a}$ of A , if we set $\mathfrak{a}_\alpha = \varphi_\alpha^{-1}(\mathfrak{a})$ for every α , the $\mathfrak{a}_\alpha$ form an inductive system such that $\mathfrak{a} = \varinjlim \mathfrak{a}_\alpha$.

Proof. It is clear that (ii) is a particular case of (i). In (i), the fact that $\mathfrak{a}$ identifies canonically with an ideal of A and $A/\mathfrak{a}$ with $\varinjlim (A_\alpha/\mathfrak{a}_\alpha)$ comes from the exactness of the functor $\varinjlim$ in the category of abelian groups. Furthermore, $\mathfrak{a}$ is the union of the increasing family of the $\varphi_\alpha(\mathfrak{a}_\alpha)$, hence, if $x_\alpha \in A_\alpha$ is such that $\varphi_\alpha(x_\alpha) \in \mathfrak{a}$, there exists $\beta \geq \alpha$ such that $\varphi_\alpha(x_\alpha) = \varphi_\beta(y_\beta)$ for some $y_\beta \in \mathfrak{a}_\beta$; but this implies the existence of a $\gamma \geq \beta$ such that $\varphi_{\gamma\alpha}(x_\alpha) = \varphi_{\gamma\beta}(y_\beta) \in \mathfrak{a}_\gamma$, and hence $x_\alpha \in \mathfrak{a}_\alpha$ if $\mathfrak{a}_\alpha = \varphi_{\gamma\alpha}^{-1}(\mathfrak{a}_\gamma)$. $\square$

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Corollary (5.13.2). — *If $\mathfrak{N}_\alpha$ is the nilradical of A_α , the nilradical of A identifies with $\varinjlim \mathfrak{N}_\alpha$, and A_{red} with $\varinjlim (A_\alpha)_{\text{red}}$. In particular, if the A_α are reduced, so is A .*

Proof. It is clear that $\varphi_{\beta\alpha}(\mathfrak{N}_\alpha) \subset \mathfrak{N}_\beta$ for $\alpha \leq \beta$, and that $\varinjlim \mathfrak{N}_\alpha$ is a nilpotent ideal of A , hence contained in the nilradical $\mathfrak{N}$ of A . Conversely, if $x \in \mathfrak{N}$, we have $x^h = 0$ for some integer h ; one can write $x = \varphi_\alpha(x_\alpha)$ for some index α and some $x_\alpha \in A_\alpha$; the relation $\varphi_\alpha(x_\alpha^h) = 0$ implies the existence of a $\beta \geq \alpha$ such that $\varphi_{\beta\alpha}(x_\alpha^h) = 0$, or equivalently $x_\beta^h = 0$ on setting $x_\beta = \varphi_{\beta\alpha}(x_\alpha)$; hence $x_\beta \in \mathfrak{N}_\beta$ and since $x = \varphi_\beta(x_\beta)$ this completes the proof, taking account of (5.13.1). $\square$

Proposition (5.13.3). — *label=(i) For every index α , let $\mathfrak{p}_\alpha$ be a prime ideal of A_α such that, for $\alpha \leq \beta$, $\varphi_{\beta\alpha}(\mathfrak{p}_\alpha) \subset \mathfrak{p}_\beta$. Then $\mathfrak{p} = \varinjlim \mathfrak{p}_\alpha$ is a prime ideal of A . In particular, if the A_α are integral, so is A .*

lbbel=(ii) If furthermore $\mathfrak{p}_\alpha = \varphi_{\beta\alpha}^{-1}(\mathfrak{p}_\beta)$ for $\alpha \leq \beta$, then $A_{\mathfrak{p}}$ identifies canonically with $\varinjlim (A_\alpha)_{\mathfrak{p}_\alpha}$.

lcbel=(iii) If A is a local ring, $\mathfrak{p}$ its maximal ideal, and if the homomorphisms φ_α are injective, then, setting $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, the homomorphisms $(A_\alpha)_{\mathfrak{p}_\alpha} \rightarrow A_{\mathfrak{p}}$ are injective.

Proof. The two assertions of (i) are in fact equivalent, since $A/\mathfrak{p} = \varinjlim (A_\alpha/\mathfrak{p}_\alpha)$. Suppose therefore all the A_α integral, and let x, y be two elements of A such that $xy = 0$. There exists thus an index α such that $x = \varphi_\alpha(x_\alpha)$, $y = \varphi_\alpha(y_\alpha)$ with x_α, y_α in A_α , and $\varphi_\alpha(x_\alpha y_\alpha) = 0$; but this implies the existence of an index $\beta \geq \alpha$ such that, setting $x_\beta = \varphi_{\beta\alpha}(x_\alpha)$, $y_\beta = \varphi_{\beta\alpha}(y_\alpha)$, $x_\beta y_\beta = 0$; since A_β is integral, we have $x_\beta = 0$ or $y_\beta = 0$, which completes the proof that A is integral.

Let us now prove (ii). The hypothesis implies that the $A_\alpha - \mathfrak{p}_\alpha$ form an inductive system of multiplicative subsets of the A_α ; hence the local rings $(A_\alpha)_{\mathfrak{p}_\alpha}$ form an inductive system (where the transition homomorphisms are *local*), and the canonical homomorphisms $(A_\alpha)_{\mathfrak{p}_\alpha} \rightarrow A_{\mathfrak{p}}$ an inductive system of homomorphisms, whose inductive limit $\psi : \varinjlim (A_\alpha)_{\mathfrak{p}_\alpha} \rightarrow A_{\mathfrak{p}}$ is a surjective

homomorphism. It remains to see that ψ is injective. Now, if $x_\alpha \in A_\alpha$ and $s_\alpha \in A_\alpha - \mathfrak{p}_\alpha$ are such that $\varphi_\alpha(x_\alpha)/\varphi_\alpha(s_\alpha) = 0$ in $A_{\mathfrak{p}}$, there exists $s' \in A - \mathfrak{p}$ such that $s'\varphi_\alpha(x_\alpha) = 0$, and, replacing α by a larger index, we can suppose $s' = \varphi_\alpha(s'_\alpha)$ with $s'_\alpha \in A_\alpha - \mathfrak{p}_\alpha$, whence $\varphi_\alpha(s'_\alpha x_\alpha) = 0$; there is thus a $\beta \geq \alpha$ such that $\varphi_{\beta\alpha}(s'_\alpha x_\alpha) = 0$ in $(A_\beta)_{\mathfrak{p}_\beta}$, which proves that the canonical image of x_α/s_α in $\varinjlim ((A_\alpha)_{\mathfrak{p}_\alpha})$ is zero.

Finally, the first assertion of (iii) results from (ii) and from the fact that the elements of $A_\alpha - \mathfrak{p}_\alpha$ are invertible in A , and that $A_{\mathfrak{p}_\alpha} = A$, and the fact that $(A_\alpha)_{\mathfrak{p}_\alpha} \rightarrow A$ is injective follows by flatness from the hypothesis that φ_α is injective. $\square$

Corollary (5.13.4). — Suppose that the A_α are all integral, and that the $\varphi_{\beta\alpha}$ are injective. Then, if A'_α is the integral closure of A_α , $A' = \varinjlim A'_\alpha$ is the integral closure of A . In particular, if A_α is integrally closed for every α , so is A . If the A_α are local rings that are unibranch (resp. geometrically unibranch) (0, 23.2.1) and the $\varphi_{\beta\alpha}$ are injective local homomorphisms, A is unibranch (resp. geometrically unibranch).

Proof. If K_α is the field of fractions of A_α , it results from (5.13.3), (ii) applied to $\mathfrak{p}_\alpha = 0$, that $K = \varinjlim K_\alpha$ is the field of fractions of A , and $A' = \varinjlim A'_\alpha$ identifies with a subring of K , clearly integral over A . Conversely, let $z \in K$ be an element satisfying an integral dependence equation $z^n + \sum_{i=1}^n c_i z^{n-i} = 0$, where the $c_i \in A$. There exists an index α , a $z_\alpha \in K_\alpha$ and $c_{i\alpha} \in A_\alpha$ such that z (resp. the c_i) are the images of z_α (resp. the $c_{i\alpha}$), which proves that $z \in A'$. It follows immediately that if the A_α are local and the $\varphi_{\beta\alpha}$ injective local homomorphisms, A' is local and the $\varphi_{\beta\alpha}$ local, hence if the A'_α are local, A' is a single extension of k_α for every α , $k' = \varinjlim k'_\alpha$, which is the residue field of A' , is a purely inseparable (resp. radical) extension of k , and A is therefore geometrically unibranch. $\square$

(5.13.5). We shall say that a ring A is *normal*, equivalently, if for every prime ideal $\mathfrak{p}$ of A , $A_{\mathfrak{p}}$ is *integral* and *integrally closed*. We note that this implies that A is *reduced* and that $\mathfrak{p}$ can contain only a *single* minimal prime ideal of A , in other words $\mathfrak{p}$ belongs to a single irreducible component of $\text{Spec}(A)$; moreover $\mathfrak{q}$ is the inverse image of (0) by the canonical homomorphism $A \rightarrow A_{\mathfrak{p}}$, and $A_{\mathfrak{p}}$ is therefore isomorphic to $(A/\mathfrak{q})_{\mathfrak{p}\mathfrak{q}}$, which proves that the integral quotient $A/\mathfrak{q}$ is *integrally closed* as an intersection of integrally closed rings (I, 8.2.1.1). If A is *Noetherian* and *normal*, A is a composed direct product of a finite number of integral domains that are integrally closed (I, 5.1.4).

Corollary (5.13.6). — Suppose that the A_β are normal and that for $\alpha \leq \beta$, every irreducible component of $\text{Spec}(A_\beta)$ dominates an irreducible component of $\text{Spec}(A_\alpha)$. Then $A = \varinjlim A_\alpha$ is normal.

Proof. Indeed, let $\mathfrak{p}$ be a prime ideal of A and for every α , set $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$. By hypothesis, for every α , $\mathfrak{p}_\alpha$ contains a single minimal ideal $\mathfrak{q}_\alpha$ of A_α , and for $\alpha \leq \beta$, $\varphi_{\beta\alpha}^{-1}(\mathfrak{q}_\beta)$ is a minimal ideal of A_α contained in $\mathfrak{p}_\alpha$, hence equal to $\mathfrak{q}_\alpha$. Setting $B_\alpha = A_\alpha/\mathfrak{q}_\alpha$, the integral rings B_α therefore form an inductive system for the homomorphisms $\varphi'_{\beta\alpha} : B_\alpha \rightarrow B_\beta$ deduced from the $\varphi_{\beta\alpha}$ by passage to quotients, which are injective; furthermore the B_α are integrally closed since they are the same by definition; as furthermore $(B_\alpha)_{\mathfrak{p}'_\alpha} = (A_\alpha)_{\mathfrak{p}_\alpha}$; since $\varinjlim (B_\alpha)_{\mathfrak{p}'_\alpha} = \varinjlim (A_\alpha)_{\mathfrak{p}_\alpha} = A_{\mathfrak{p}}$ is an integral ring integrally closed (5.13.3), this proves the corollary. $\square$

Proposition (5.13.7). — Suppose that the A_α are regular and that, for $\alpha \leq \beta$, $\varphi_{\beta\alpha}$ makes A_β a flat A_α -module; suppose furthermore that $A = \varinjlim A_\alpha$ is Noetherian. Then A is regular.

Proof. By definition, it suffices to prove that for every prime ideal $\mathfrak{p}$ of A , the local Noetherian ring $A_{\mathfrak{p}}$ is regular. Setting $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, we have $A_{\mathfrak{p}} = \varinjlim ((A_\alpha)_{\mathfrak{p}_\alpha})$ by (5.13.3); since by hypothesis the $(A_\alpha)_{\mathfrak{p}_\alpha}$ are regular and $(A_\beta)_{\mathfrak{p}_\beta}$ is a flat $(A_\alpha)_{\mathfrak{p}_\alpha}$ -module for $\alpha \leq \beta$ (0_I, 6.3.2), we see that we are reduced to the case where the A_α and A are *local* rings. It then suffices to show that, if M, N are two A -modules of finite type, $\text{Tor}_i^A(M, N) = 0$ for i sufficiently large (0, 17.3.1 and 17.2.6). But we have the following lemma (where the A_α are again arbitrary):

Lemma (5.13.7.1). — For every A -module M of finite presentation, there exists an index α and an A_α -module of finite presentation M_α such that M is isomorphic to $M_\alpha \otimes_{A_\alpha} A$.

Proof. Indeed, one can suppose that $M = A^r/P$, where P is a sub- A -module of finite type. Since $A' = \varinjlim (A'_\alpha)$ and since P is of finite type, there exists an α and a sub- A_α -module of finite type P_α of A_α^r such that P is the image canonique of $P_\alpha \otimes_{A_\alpha} A$; the exactness to the right of the tensor product then shows that M is isomorphic to $(A_\alpha^r/P_\alpha) \otimes_{A_\alpha} A$. $\square$

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This lemma being established, M and N are of finite presentation, since A is Noetherian; write therefore $M = M_\alpha \otimes_{A_\alpha} A$, $N = N_\alpha \otimes_{A_\alpha} A$; by virtue of the hypothesis, A is a flat A_α -module, hence $\mathrm{Tor}_i^A(M, N) = \mathrm{Tor}_i^{A_\alpha}(M_\alpha, N_\alpha) \otimes_{A_\alpha} A$ up to isomorphism (III, 6.3.9); since by hypothesis A_α is regular, there exists i_0 such that $\mathrm{Tor}_i^{A_\alpha}(M_\alpha, N_\alpha) = 0$ for $i \geq i_0$. $\square$

Corollary (5.13.8). — Suppose that the A_α are Noetherian and satisfy property (R_k) for an integer $k \geq 0$. Suppose furthermore that for $\alpha \leq \beta$, $\varphi_{\beta\alpha}$ makes A_β a flat A_α -module, that $A = \varinjlim A_\alpha$ is Noetherian and that, for every α , ${}^a\varphi_\alpha$ is a homeomorphism of $\mathrm{Spec}(A)$ onto $\mathrm{Spec}(A_\alpha)$. Then A satisfies (R_k) .

Proof. Indeed let $\mathfrak{p}$ be a prime ideal of A and, for every α , set $\mathfrak{p}_\alpha = \varphi_\alpha^{-1}(\mathfrak{p})$, so that $\mathfrak{p} = \varinjlim \mathfrak{p}_\alpha$ and $A_\mathfrak{p} = \varinjlim (A_\alpha)_{\mathfrak{p}_\alpha}$ (5.13.3). It results from (I, 2.4.2) and from the hypothesis that the morphism $\mathrm{Spec}(A_\mathfrak{p}) \rightarrow \mathrm{Spec}((A_\alpha)_{\mathfrak{p}_\alpha})$ is a surjective homeomorphism for every α ; we thus have $\dim(A_\mathfrak{p}) = \dim((A_\alpha)_{\mathfrak{p}_\alpha})$ for every α . Suppose then that $\dim(A_\mathfrak{p}) \leq k$; we have thus $\dim((A_\alpha)_{\mathfrak{p}_\alpha}) \leq k$, and the hypothesis implies that $(A_\alpha)_{\mathfrak{p}_\alpha}$ is a regular ring; since furthermore, for $\alpha \leq \beta$, $(A_\beta)_{\mathfrak{p}_\beta}$ is a flat $(A_\alpha)_{\mathfrak{p}_\alpha}$ -module (0I, 6.3.2), we conclude from (5.13.7) that $A_\mathfrak{p}$ is regular, which proves the corollary. $\square$

§6. FLAT MORPHISMS OF LOCALLY NOETHERIAN PRESCHMES

Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism. For every $y \in Y$, the fibre $f^{-1}(y) = X \times_Y \mathrm{Spec}(k(y))$ is also a locally Noetherian prescheme, since it suffices to see this when $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are spectra of Noetherian rings, and then $B \otimes_A k(y)$ is a ring of fractions of the quotient ring $B/\mathfrak{i}_y B$, and hence is Noetherian. We first propose in this section to relate the properties of X , of Y , and of the fibres $f^{-1}(y)$ (where y ranges over Y), under the hypothesis that the morphism f is flat. The questions treated will reduce to the study of the relations between the local Noetherian rings $\mathcal{O}_y, \mathcal{O}_x$, and $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$, where the homomorphism $\mathcal{O}_y \rightarrow \mathcal{O}_x$ is local and makes $\mathcal{O}_x$ a flat $\mathcal{O}_y$ -module. In (6.11) to (6.13) (which are fairly separate from the rest of the section, by virtue of their “absolute” rather than “relative” nature), we apply some of the preceding results to find criteria allowing us to assert that the singular locus (or certain analogous sets) of certain preschemes are closed sets; these criteria will play an important role in §7.

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6.1. Flatness and dimension

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Proposition (6.1.1). — Let A, B be two local Noetherian rings, $\mathfrak{m}$ the maximal ideal of A , $k = A/\mathfrak{m}$ its residue field, $\varphi : A \rightarrow B$ a local homomorphism. Suppose that for every prime ideal $\mathfrak{p}$ of A , distinct from $\mathfrak{m}$, and for every minimal element $\mathfrak{q}$ of the set of prime ideals of B containing $\mathfrak{p}B$, $\varphi^{-1}(\mathfrak{q})$ is distinct from $\mathfrak{m}$ (in other words, that no irreducible component of $\mathrm{Spec}(B/\mathfrak{p}B)$ is contained in the inverse image of the closed point $\mathfrak{m}$ of $\mathrm{Spec}(A)$ in $\mathrm{Spec}(B)$). Then we have

$$(6.1.1.1) \quad \dim(B) = \dim(A) + \dim(B \otimes_A k).$$

Proof. We argue by induction on $n = \dim(A)$; the assertion is trivial for $n = 0$, since $\mathfrak{m}B$ is then contained in the nilradical of B because $\mathfrak{m}$ is the nilradical of A . We can thus suppose $n > 0$. Let $\mathfrak{q}_i$ ($1 \leq i \leq m$) be the minimal prime ideals of B , and set $\mathfrak{p}_i = \varphi^{-1}(\mathfrak{q}_i)$; for every i , we have $\mathfrak{p}_i \neq \mathfrak{m}$; otherwise there would exist (since $n > 0$) a prime ideal $\mathfrak{p} \neq \mathfrak{m}$ contained in $\mathfrak{p}_i$ and since $\mathfrak{q}_i$ is minimal among the prime ideals of B containing $\mathfrak{p}B$, one would arrive at a contradiction with the hypothesis. It follows that $\mathfrak{m}$ is distinct from the union of the $\mathfrak{p}_i$ and the minimal prime ideals $\mathfrak{p}'_j$ ($1 \leq j \leq r$) of A (Bourbaki, *Alg. comm.*, chap. II, §1, n° 1, prop. 2) and there exists an $a \in \mathfrak{m}$ belonging to none of the $\mathfrak{p}_i$ nor the $\mathfrak{p}'_j$. Set $A' = A/aA$, $B' = B/aB$; we have (0, 16.3.4)

$$\dim(A') = \dim(A) - 1, \quad \dim(B') = \dim(B) - 1$$

by construction of a , and on the other hand $B' \otimes_{A'} k = B \otimes_A k = B/\mathfrak{m}B$, hence

$$\dim(B' \otimes_{A'} k) = \dim(B \otimes_A k)$$

and it therefore suffices to prove (6.1.1.1) for A' and B' ; but, by virtue of the bijective correspondence between ideals of A (resp. B) containing a and ideals of A' (resp. B'), the hypotheses of the statement are also valid for A' and B' ; one can thus apply the induction hypothesis, which completes the proof. $\square$

Corollary (6.1.2). — Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, $M \neq 0$ a finite type A -module, $N \neq 0$ a finite type B -module. If N is a flat A -module (resp. if B is a flat A -module), then we have

$$(6.1.2.1) \quad \dim_B(M \otimes_A N) = \dim_A(M) + \dim_{B \otimes_A k}(N \otimes_A k)$$

(resp. (6.1.1.1)).

Proof. It suffices to prove the assertion concerning N ; on the other hand, if $\mathfrak{b}$ is the annihilator of N , one can replace B by $B/\mathfrak{b}$, and hence suppose that $\text{Supp}(N) = \text{Spec}(B)$; the hypothesis then means that the morphism $f : \text{Spec}(B) \rightarrow \text{Spec}(A)$ corresponding to φ is quasi-flat (2.3.3); we conclude (2.3.4) that for every prime ideal $\mathfrak{p} \neq \mathfrak{m}$ of A , every irreducible component of $f^{-1}(V(\mathfrak{p}))$ dominates $V(\mathfrak{p})$, and hence the condition of (6.1.1) is satisfied, whence the conclusion. $\square$

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Corollary (6.1.3). — Let A, B be two local Noetherian rings, $\mathfrak{m}$ the maximal ideal of A , k its residue field, $\varphi : A \rightarrow B$ a local homomorphism. Suppose that $\dim(B \otimes_A k) = 0$ (that is to say (0, 16.2.1) that $\mathfrak{m}B$ is an ideal of definition of B). Then $\dim(B) \leq \dim(A)$. If in addition there exists a finite type B -module N that is a flat A -module and whose support equals $\text{Spec}(B)$ (which is the case in particular if B is a flat A -module), then $\dim(B) = \dim(A)$.

Proof. The first assertion follows from (5.5.1) and the second from (6.1.2). $\square$

Corollary (6.1.4). — Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a surjective morphism. For every closed subset Z of Y , we then have

$$(6.1.4.1) \quad \text{codim}(f^{-1}(Z), X) \leq \text{codim}(Z, Y).$$

Moreover, if f is quasi-flat (2.3.3), the two sides of (6.1.4.1) are equal.

Proof. Indeed, if y is a maximal point of Z and x a maximal point of the fibre $f^{-1}(y)$, we have $\dim(\mathcal{O}_{X,x}) \leq \dim(\mathcal{O}_{Y,y})$ by (5.5.2); the inequality (6.1.4.1) then follows from (5.1.2.1) and (5.1.3.1) and the fact that f is surjective. If in addition f is quasi-flat, we know (2.3.4) that every irreducible component of $f^{-1}(Z)$ dominates an irreducible component of Z ; the maximal points of $f^{-1}(Z)$ are thus the maximal points of the fibres $f^{-1}(y)$, where y ranges over the set of maximal points of Z (0_I, 2.1.8), and at each of these maximal points x we have $\dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)) = 0$; since in addition $\mathcal{O}_x$ is a flat $\mathcal{O}_y$ -module, we have $\dim(\mathcal{O}_x) = \dim(\mathcal{O}_y)$ by (6.1.1), whence the equality in (6.1.4.1), taking into account that f is surjective. $\square$

Proposition (6.1.1) admits the following partial converse:

Proposition (6.1.5). — Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, M a finite type B -module. Suppose that:

- 1° A is a regular ring.
- 2° M is a Cohen-Macaulay B -module.
- 3° We have $\dim_B(M) = \dim(A) + \dim_{B \otimes_A k}(M \otimes_A k)$.

Then M is a flat A -module.

Proof. We proceed by induction on $n = \dim(A)$. If $n = 0$, A is a field since it is regular and the assertion is trivial. Suppose $n > 0$; let $\mathfrak{m}$ be the maximal ideal of A , and let $x \in \mathfrak{m}$ be an element not belonging to $\mathfrak{m}^2$; we then know (0, 17.1.8) that $A' = A/xA$ is regular and $\dim(A') = \dim(A) - 1$. Set

$$B' = B/xB, \quad M' = M/xM = M \otimes_A A';$$

we thus have

$$B' \otimes_{A'} k = B \otimes_A k, \quad M' \otimes_{A'} k = M \otimes_A k$$

and by (5.5.1.2) we have

$$\dim_{B'}(M') \leq \dim(A') + \dim_{B' \otimes_{A'} k}(M' \otimes_{A'} k);$$

we thus conclude from (0, 16.3.4) that

$$\dim_B(M) \leq \dim_{B'}(M') + 1 \leq (\dim(A') + \dim_{B \otimes_A k}(M \otimes_A k)) + 1 = \dim(A) + \dim_{B \otimes_A k}(M \otimes_A k).$$

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Since by hypothesis the extreme members of these inequalities are equal, we necessarily have: (i) $\dim_{B'}(M') = \dim_B(M) - 1$, and since M is by hypothesis a Cohen-Macaulay B -module, this means that x is M -regular (0, 16.1.9 and 16.5.5); (ii) $\dim_{B'}(M') = \dim(A') + \dim_{B' \otimes_{A'} k}(M' \otimes_{A'} k)$; since M' is a Cohen-Macaulay B' -module by (i) and (0, 16.1.9 and 16.5.5) and A' is regular, the induction hypothesis shows that M' is a flat A' -module; we then deduce from (0_{III}, 10.2.7) that M is a flat A -module, since x is M -regular by (i). $\square$

6.2. Flatness and projective dimension

Proposition (6.2.1). — (i) Let A, B be two rings, $\varphi : A \rightarrow B$ a homomorphism such that B is a flat A -module. Then, for every A -module M , we have

$$(6.2.1.1) \quad \dim. \operatorname{proj}_B(M \otimes_A B) \leq \dim. \operatorname{proj}_A(M).$$

(ii) Suppose in addition that A is a Noetherian ring, B a faithfully flat A -module, and M a finite type A -module; then

$$(6.2.1.2) \quad \dim. \operatorname{proj}_B(M \otimes_A B) = \dim. \operatorname{proj}_A(M).$$

Proof. (i) We can restrict to the case where $n = \dim. \operatorname{proj}_A(M)$ is finite; there thus exists a left resolution

$$0 \rightarrow P_n \rightarrow P_{n-1} \rightarrow \dots \rightarrow P_0 \rightarrow M \rightarrow 0$$

of M by projective A -modules; since B is a flat A -module, the sequence

$$0 \rightarrow P_n \otimes_A B \rightarrow P_{n-1} \otimes_A B \rightarrow \dots \rightarrow P_0 \otimes_A B \rightarrow M \otimes_A B \rightarrow 0$$

is exact; on the other hand, the $P_i \otimes_A B$ are projective B -modules; whence the conclusion.

(ii) Suppose that $\dim. \operatorname{proj}_B(M \otimes_A B) = m$, and consider an exact sequence

$$0 \rightarrow R \rightarrow P_{m-1} \rightarrow P_{m-2} \rightarrow \dots \rightarrow P_0 \rightarrow M \rightarrow 0$$

where the P_i ($0 \leq i \leq m-1$) are projective A -modules of finite type; since A is Noetherian, R is also a finite type A -module. Since B is a flat A -module, we also have an exact sequence

$$0 \rightarrow R \otimes_A B \rightarrow P_{m-1} \otimes_A B \rightarrow \dots \rightarrow P_0 \otimes_A B \rightarrow M \otimes_A B \rightarrow 0$$

and the hypothesis on $M \otimes_A B$ implies that $R \otimes_A B$ is a projective B -module (0, 17.2.1). Since B is a faithfully flat A -module, we conclude that R is a projective A -module (Bourbaki, *Alg. comm.*, chap. I^{er}, §3, n° 6, prop. 12); hence $\dim. \operatorname{proj}_A(M) \leq m$, which completes the proof. $\square$

Corollary (6.2.2). — Let $f : X \rightarrow Y$ be a flat morphism of preschemes, $\mathcal{E}$ an $\mathcal{O}_Y$ -Module quasi-coherent. If $\dim. \operatorname{proj}(\mathcal{E}) \leq n$ (0, 17.2.14), then $\dim. \operatorname{proj}(f^*(\mathcal{E})) \leq n$.

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Proposition (6.2.3). — Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, N a finite type B -module. Suppose that B and N are flat A -modules. Then we have

$$(6.2.3.1) \quad \dim. \operatorname{proj}_B(N) = \dim. \operatorname{proj}_{B \otimes_A k}(N \otimes_A k).$$

Proof. Consider a left resolution of N by free B -modules of finite type

$$\dots \rightarrow L_i \rightarrow L_{i-1} \rightarrow \dots \rightarrow L_0 \rightarrow N \rightarrow 0.$$

Since the L_i and N are flat A -modules (B being a flat A -module) it follows from (2.1.10) that the $Z_i(L.)$ are flat A -modules, that $L. \otimes_A k$ is a left resolution of the $(B \otimes_A k)$ -module $N \otimes_A k$, and that $Z_i(L. \otimes_A k) = Z_i(L.) \otimes_A k$ for every i . Note that since B is Noetherian, the $Z_i(L.)$ are B -modules of finite type, hence the $Z_i(L. \otimes_A k)$ are $(B \otimes_A k)$ -modules of finite type. Now, to say that a B -module (resp. a $(B \otimes_A k)$ -module) of finite type is flat is equivalent to saying that it is *free* or equivalently *projective* (0_{III}, 10.1.3). Since the $Z_i(L.)$ are flat A -modules, it follows moreover from (0_{III}, 10.2.5) (taking $C = B$) that, for $Z_i(L.)$ to be a flat B -module, it is necessary and sufficient that $Z_i(L. \otimes_A k)$ be a flat $(B \otimes_A k)$ -module. The smallest integer n such that $Z_{n-1}(L.)$ is a free B -module is thus also the smallest integer such that $Z_{n-1}(L. \otimes_A k)$ is a free $(B \otimes_A k)$ -module, which proves the proposition (0, 17.2.1). $\square$

6.3. Flatness and depth

Proposition (6.3.1). — *Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, M a finite type A -module, N a finite type B -module. If N is a flat A -module $\neq 0$, then we have*

$$(6.3.1.1) \quad \text{prof}_B(M \otimes_A N) = \text{prof}_A(M) + \text{prof}_{B \otimes_A k}(N \otimes_A k).$$

Proof. We can restrict to the case where $M \neq 0$, since otherwise the two sides of (6.3.1.1) are both equal to $+\infty$. We proceed by induction on the integer n equal to the second member of (6.3.1.1) (which is finite by the hypotheses, (0, 16.4.6.2) and the Nakayama lemma applied to N). Suppose first that $n = 0$; then, if $\mathfrak{m}$ and $\mathfrak{n}$ denote the maximal ideals of A and B respectively, $\mathfrak{m}$ is an ideal premier associated to M and $\mathfrak{n}/\mathfrak{m}B$ an ideal premier associated to $N \otimes_A k$ (0, 16.4.6, (i)). We then conclude from (3.3.1) that $\mathfrak{n}$ is a prime ideal of B associated to $M \otimes_A N$, hence (0, 16.4.6, (i)) $\text{prof}_B(M \otimes_A N) = 0$. Suppose now $n > 0$, and distinguish two cases:

a) Suppose $\text{prof}_A(M) > 0$. Let $x \in \mathfrak{m}$ be an M -regular element, and set

$$A' = A/xA, \quad B' = B/xB, \quad M' = M/xM, \quad N' = N/xN;$$

we have

$$B' \otimes_{A'} k = B \otimes_A k, \quad N' \otimes_{A'} k = N \otimes_A k$$

and, since N is a flat A -module, the hypothesis that x is M -regular implies that x is also $(M \otimes_A N)$ -regular (0_I, 6.1.1). We have by (0, 16.4.6, (ii) and 16.4.8)

$$\text{prof}_{A'}(M') = \text{prof}_A(M) - 1, \quad \text{prof}_{B'}(M' \otimes_{A'} N') = \text{prof}_B(M \otimes_A N) - 1$$

and

$$\text{prof}_{B' \otimes_{A'} k}(N' \otimes_{A'} k) = \text{prof}_{B \otimes_A k}(N \otimes_A k).$$

The equality (6.3.1.1) is thus a consequence of the same relation for A', B', M' , and N' ; but since N is a flat A -module, $N' = N \otimes_A A'$ is a flat A' -module (0_I, 6.2.1); one can therefore apply the induction, which proves (6.3.1.1) in this case.

b) Suppose $\text{prof}_{B \otimes_A k}(N \otimes_A k) > 0$. Consider a $y \in \mathfrak{n}$ that is $(N \otimes_A k)$ -regular; it follows from (0_{III}, 10.2.4) that y is N -regular and that

$$N' = N/yN$$

is a flat A -module, since N is assumed to be a flat A -module. Applying (0_I, 6.1.2) to the exact sequence of A -modules

$$0 \longrightarrow N \xrightarrow{y} N \longrightarrow N' \longrightarrow 0$$

we obtain isomorphisms

$$(M \otimes_A N)/y(M \otimes_A N) \xrightarrow{\sim} M \otimes_A N' \quad \text{and} \quad N' \otimes_A k \xrightarrow{\sim} (N \otimes_A k)/y(N \otimes_A k)$$

and moreover y is $(M \otimes_A N)$ -regular. We thus have

$$\text{prof}_{B \otimes_A k}(N' \otimes_A k) = \text{prof}_{B \otimes_A k}(N \otimes_A k) - 1$$

and

$$\text{prof}_B(M \otimes_A N') = \text{prof}_B(M \otimes_A N) - 1;$$

the induction hypothesis shows that the relation (6.3.1.1) holds for A, B, M , and N' , and from what precedes, we deduce (6.3.1.1) for A, B, M , and N . $\square$

Corollary (6.3.2). — *Under the hypotheses of (6.3.1), suppose in addition that $M \neq 0$ and $N \neq 0$; then we have*

$$(6.3.2.1) \quad \text{coprof}_B(M \otimes_A N) = \text{coprof}_A(M) + \text{coprof}_{B \otimes_A k}(N \otimes_A k).$$

Proof. This follows immediately from (6.3.1.1) and (6.1.2) and from the definition of *coprofondéur* (0, 16.4.9). $\square$

Corollary (6.3.3). — *Under the hypotheses of (6.3.1), suppose in addition that $M \neq 0$ and $N \neq 0$; for $M \otimes_A N$ to be a Cohen-Macaulay B -module, it is necessary and sufficient that M be a Cohen-Macaulay A -module and that $N \otimes_A k$ be a $(B \otimes_A k)$ -module de Cohen-Macaulay.*

Proof. This follows from the corollary (6.3.2) and the definition of Cohen-Macaulay modules, by the fact that their coprofondéur is zero (0, 16.5.1), taking into account that the condition $N \neq 0$ is equivalent to $N \otimes_A k \neq 0$ by the Nakayama lemma. $\square$

Corollary (6.3.4). — *Under the hypotheses of (6.3.1), suppose in addition that $N \otimes_A k$ is a B -module of finite length; then we have*

$$(6.3.4.1) \quad \text{prof}_B(M \otimes_A N) = \text{prof}_A(M).$$

If in addition $M \neq 0$ and $N \neq 0$, then we have

$$(6.3.4.2) \quad \text{coprof}_B(M \otimes_A N) = \text{coprof}_A(M).$$

Proof. It amounts to saying that $N \otimes_A k$ is a B -module of finite length, or a $(B \otimes_A k)$ -module of finite length, and we know (0, 16.2.3) that modules of finite length $\neq 0$ are of dimension 0 and depth 0. The corollary (6.3.4) is applicable in particular when $B \otimes_A k$ is a ring of finite length, that is to say when $\mathfrak{m}B$ is an ideal of definition of the ring B . $\square$

Corollary (6.3.5). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a flat morphism.*

- (i) *If, at a point $x \in X$, $\text{coprof}(\mathcal{O}_x) \leq n$, then, setting $y = f(x)$, $\text{coprof}(\mathcal{O}_y) \leq n$ and $\text{coprof}(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)) \leq n$; in particular, if $\mathcal{O}_x$ is a Cohen-Macaulay ring, the same is true of $\mathcal{O}_y$ and of $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$.*
- (ii) *Suppose conversely that $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$ is a Cohen-Macaulay ring. Then, if $\mathcal{O}_y$ is a Cohen-Macaulay ring, $\text{coprof}(\mathcal{O}_x) \leq n$ and $\text{coprof}(\mathcal{O}_y) \leq n$ (resp. $\mathcal{O}_x$ is a Cohen-Macaulay ring).*

Proof. It suffices to apply (6.3.2) for $M = A = \mathcal{O}_y$ and $N = B = \mathcal{O}_x$. $\square$

Corollary (6.3.6). — *Let A be a Cohen-Macaulay ring. Then $A' = A[T_1, \dots, T_n]$ (T_i indeterminates) is a Cohen-Macaulay ring.*

Proof. Indeed, if we set $Y = \text{Spec}(A)$, $X = \text{Spec}(A')$ and $f : X \rightarrow Y$ is the canonical morphism, f is flat (since A' is a free A -module (2.1.2)) and for every $y \in Y$, $k(y)[T_1, \dots, T_n]$ is a regular ring (0, 17.3.7), hence a fortiori a Cohen-Macaulay ring; it therefore suffices to apply (6.3.5). $\square$

Corollary (6.3.7). — *Every quotient of a Cohen-Macaulay ring is universally catenary.*

Proof. This follows from (6.3.6), (5.6.1), and (0, 16.5.12). $\square$

Proposition (6.3.8). — *Let A be a local Noetherian ring; suppose that there exists a Cohen-Macaulay finite type A -module M whose support is equal to $\text{Spec}(A)$. Let B be a quotient ring of A , and let $f : \text{Spec}(\hat{B}) \rightarrow \text{Spec}(B)$ be the canonical morphism; then, for every $x \in \text{Spec}(B)$, $f^{-1}(x)$ is a Cohen-Macaulay prescheme.*

Proof. Since B is a finite type A -module, on a $\hat{B} = B \otimes_A \hat{A}$ (0I, 7.3.3); hence f is flat and finite, and the fibres of the two morphisms $\text{Spec}(\hat{B}) \rightarrow \text{Spec}(B)$ and $\text{Spec}(\hat{A}) \rightarrow \text{Spec}(A)$ at the same point are the same. One is therefore reduced to proving the proposition when $B = A$. Let $\mathfrak{p} = \mathfrak{i}_x$ be a prime ideal; since $\mathfrak{p} \in \text{Supp}(M)$, we know (0, 16.5.6 and 16.5.10) that $M_{\mathfrak{p}}$ is a Cohen-Macaulay $A_{\mathfrak{p}}$ -module. Taking into account (I, 3.6.5), we see that if we set $A'' = A' \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}}$ and $M'' = M' \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}}$, M'' is an A'' -module de Cohen-Macaulay; on the other hand, $M_{\mathfrak{p}}$ is by hypothesis a Cohen-Macaulay $A_{\mathfrak{p}}$ -module and $A''_{\mathfrak{p}''}$ is a flat $A_{\mathfrak{p}}$ -module (0I, 7.3.3 and 6.3.2). We conclude from (6.3.3) that, if k is the residue field of $A_{\mathfrak{p}}$, $k \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ is a Cohen-Macaulay ring. But since $A_{\mathfrak{p}}$ is of dimension 0, the prime ideals of A'' correspond bijectively to those of $k \otimes_{A_{\mathfrak{p}}} A''$ (I, 3.5.7), and if $\mathfrak{q}''$ is the prime ideal of $k \otimes_{A_{\mathfrak{p}}} A''$ corresponding to $\mathfrak{p}''$, the local rings $(k \otimes_{A_{\mathfrak{p}}} A'')_{\mathfrak{q}''}$ and $k \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ are isomorphic. It follows that $k \otimes_{A_{\mathfrak{p}}} A''_{\mathfrak{p}''}$ is a Cohen-Macaulay ring. Q.E.D. $\square$

6.4. Flatness and property (S_k)

Proposition (6.4.1). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism, $\mathcal{E}$ a coherent $\mathcal{O}_Y$ -Module, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module that is f -flat.*

- (i) *Let $x \in \text{Supp}(\mathcal{F})$ be such that $\mathcal{E} \otimes_Y \mathcal{F}$ has property (S_k) at the point x ; then $\mathcal{E}$ has property (S_k) at the point $y = f(x)$.*
- (ii) *Suppose that for every $y \in f(\text{Supp}(\mathcal{F}))$, $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_y} k(y)$ has property (S_k) ; if for a point $y \in f(\text{Supp}(\mathcal{F}))$, $\mathcal{E}$ has property (S_k) at the point y , then $\mathcal{E} \otimes_Y \mathcal{F}$ has property (S_k) at every point of $f^{-1}(y)$.*

Proof. (i) Setting $y = f(x)$ and let y' be a generisation of y ; as in the proof of (6.4.1) there exists a generic point x' of an irreducible component of $f^{-1}(y')$ that is a generisation of x ; hence on a $\dim(\mathcal{O}_{x'} \otimes_{\mathcal{O}_{y'}} k(y')) = 0$ and by the hypothesis and (6.1.2), $\dim(\mathcal{O}_{x'}) = \dim(\mathcal{O}_{y'})$. The hypothesis implies, either that $\dim(\mathcal{O}_{x'}) > k$, in which case $\dim(\mathcal{O}_{y'}) > k$, or that $\mathcal{O}_{x'}$ is a regular ring, and then $\mathcal{O}_{y'}$ is a regular ring by (6.5.1).

(ii) For every generisation x' of $x \in f^{-1}(y)$, $y' = f(x')$ is a generisation of y , we can restrict to verifying that if $x \in \text{Supp}(\mathcal{F})$ and $f(x) = y$, we have $\text{prof}(\mathcal{G}_x) \geq \inf(k, \dim(\mathcal{G}_x))$; it follows from (6.1.2) and (6.3.1) that

$$\begin{aligned} \dim(\mathcal{G}_x) &= \dim(\mathcal{E}_y) + \dim(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y)) \\ \text{prof}(\mathcal{G}_x) &= \text{prof}(\mathcal{E}_y) + \text{prof}(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y)). \end{aligned}$$

By hypothesis, we have

$$\begin{aligned} \text{prof}(\mathcal{E}_y) &\geq \inf(k, \dim(\mathcal{E}_y)) \\ \text{prof}(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y)) &\geq \inf(k, \dim(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y))) \end{aligned}$$

whence, by adding member by member,

$$\begin{aligned} \text{prof}(\mathcal{G}_x) &\geq \inf(k, \dim(\mathcal{E}_y)) + \inf(k, \dim(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y))) \geq \\ &\geq \inf(k, \dim(\mathcal{E}_y) + \dim(\mathcal{F}_x \otimes_{\mathcal{O}_y} k(y))) = \inf(k, \dim(\mathcal{G}_x)) \end{aligned}$$

which proves (ii). □

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Corollary (6.4.2). — *Let X, Y be two locally Noetherian preschemes, $\mathcal{E}$ a coherent $\mathcal{O}_Y$ -Module, $f : X \rightarrow Y$ a flat morphism. Suppose that for every $y \in Y$, the prescheme $f^{-1}(y)$ verifies (S_k) at every point x ; then, for $f^*(\mathcal{E})$ to verify (S_k) at a point x , it is necessary and sufficient that $\mathcal{E}$ verify (S_k) at the point $f(x)$.*

Remarks (6.4.3). — Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, M a finite type B -module that is a flat A -module, and suppose in addition that the A -module A and the $(B \otimes_A k)$ -module $M \otimes_A k$ both have property (S_n) ; can one then conclude that the B -module M has property (S_n) ? We do not know the answer even when $n = 1$, $M = B$ and $B = \hat{A}$, in other words, we do not know whether in general the completion of a local Noetherian ring satisfying (S_n) also satisfies (S_n) , even for $n = 1$. Setting $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$, and $\mathcal{F} = \hat{M}$, it would suffice, by (6.4.1), (ii), to show that for every $y \in Y$, $\mathcal{F}_y$ has property (S_k) (and not only when y is the closed point of Y); it would also suffice to show that the set U of $x \in X$ such that $\mathcal{F}_x \otimes_{\mathcal{O}_{f(x)}} k(f(x))$ satisfies (S_k) is open in X (since it contains the closed point of X by hypothesis). We shall show later (12.1.4) that U is open when B is a local ring of a finite type A -algebra, and that all the preschemes $f^{-1}(y)$ are Cohen-Macaulay preschemes (otherwise said satisfy (S_n) for all n) when the hypotheses of (6.3.8) are satisfied. On the other hand, for $B = \hat{A}$ and $M = B$, we have seen in (6.3.8) that the preschemes $f^{-1}(y)$ are Cohen-Macaulay preschemes for all n , for which it suffices that A satisfies (S_n) , it is necessary and sufficient that its completion $\hat{A}$ satisfies (S_n) . It remains to see whether this property holds without restriction on A .

6.5. Flatness and property (R_k)

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Proposition (6.5.1). — Let A, B be two local Noetherian rings, k the residue field of A , $\varphi : A \rightarrow B$ a local homomorphism, for which B is a flat A -module. Then:

- (i) If B is regular, A is regular.
- (ii) If A and $B \otimes_A k$ are regular, B is regular.

This proposition is given for the record, having already been proved in (0, 17.3.3).

Corollary (6.5.2). — Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a flat morphism.

- (i) If X is regular at a point x , Y is regular at the point $f(x)$.
- (ii) If for some $y \in f(X)$, Y is regular at the point y and if $f^{-1}(y)$ is a regular prescheme at a point x , then X is regular at the point x .

Proposition (6.5.3). — Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a flat morphism.

- (i) If X has property (R_k) at a point x (resp. if X has property (R_k)), Y has property (R_k) at the point $f(x)$ (resp. Y has property (R_k) at every point of $f(X)$).
- (ii) Suppose that for every $y \in f(X)$, the prescheme $f^{-1}(y)$ has property (R_k) ; then, if for a point $y \in f(X)$, Y has property (R_k) at the point y , X has property (R_k) at every point of $f^{-1}(y)$.

Proof. (i) Set $y = f(x)$ and let y' be a generisation of y ; as in the proof of (6.4.1), there exists a generic point x' of an irreducible component of $f^{-1}(y')$ that is a generisation of x ; hence $\dim(\mathcal{O}_{x'} \otimes_{\mathcal{O}_{y'}} k(y')) = 0$ and by the hypothesis and (6.1.2), $\dim(\mathcal{O}_{x'}) = \dim(\mathcal{O}_{y'})$. The hypothesis implies, either that $\dim(\mathcal{O}_{x'}) > k$, in which case $\dim(\mathcal{O}_{y'}) > k$, or that $\mathcal{O}_{x'}$ is a regular ring, and then $\mathcal{O}_{y'}$ is a regular ring by (6.5.1).

- (ii) For every generisation x' of $x \in f^{-1}(y)$, $y' = f(x')$ is a generisation of y ; we can restrict to verifying that if $\dim(\mathcal{O}_x) \leq k$, then $\mathcal{O}_x$ is a regular ring. Now, by (6.1.2)

$$\dim(\mathcal{O}_x) = \dim(\mathcal{O}_y) + \dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y))$$

hence if $\dim(\mathcal{O}_x) \leq k$, we have *a fortiori* $\dim(\mathcal{O}_y) \leq k$ and $\dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)) \leq k$ and the hypothesis implies that $\mathcal{O}_y$ and $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$ are regular rings. We then conclude from (6.5.1) that $\mathcal{O}_x$ is a regular ring. □

Corollary (6.5.4). — Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a flat morphism.

- (i) If X is normal at a point x , Y is normal at the point $f(x)$.
- (ii) If, for every $y \in f(X)$, $f^{-1}(y)$ is a normal prescheme and if Y is normal at a point $y \in f(X)$, then X is normal at every point of $f^{-1}(y)$.

Proof. This follows immediately from the Serre normality criterion (5.8.6) and from (6.4.1) applied for $k = 2$ and (6.5.3) applied for $k = 1$. □

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Remarks (6.5.5). — (i) Let A, B be two local Noetherian rings, $\varphi : A \rightarrow B$ a local homomorphism such that B is a flat A -module. Let k be the residue field of A , and suppose that the two rings A and $B \otimes_A k$ satisfy property (R_i) (5.8.2); it does *not* then necessarily follow that B satisfies (R_i) , even in the particular case where $i = 0$ or $i = 1$ and where B is the completion $\hat{A}$ of A . Nagata has indeed given an example where A is normal (hence satisfies (R_1)) but where $\hat{A}$ is not even reduced (hence does not satisfy (R_0)) [30]. One cannot apply the proposition (6.5.3) in this case because the fibres $f^{-1}(y)$ do not necessarily satisfy property (R_i) at the points distinct from the closed point of $Y = \text{Spec}(A)$. We shall show however later (7.8.3), (v) that such phenomena do not occur for the local Noetherian rings most commonly encountered in applications.

- (ii) The property for a prescheme of being *integral* does not at all behave like the properties we have just examined in this number and the preceding ones: it can happen that $f : X \rightarrow Y$ is a flat morphism of finite type, that all the fibres $f^{-1}(y)$ are regular (and even geometrically regular (6.7.6)) and that Y is integral, without X even being locally integral. For

example, let k be an algebraically closed field of characteristic 0, and let A be the integral ring $k[U, V]/(UV - (U + V)^3)$ (whose spectrum is therefore a “cubic with a double point”); A is not integrally closed, and if u, v are the canonical images of U, V in A , the integral closure of A is $C = A[t]$, where $t = (u - v)/(u + v)$, which satisfies the equation $t^2 = 1 + u + v$, whence $u = \frac{1}{2}(t^3 + t^2 - t)$, $v = \frac{1}{2}(-t^3 + t^2 + t)$ and hence $C = k[t]$, isomorphic to the polynomial ring in one indeterminate over k . If $\mathfrak{m} = Au + Av$, the maximal ideal of A (corresponding to the “double point” of the cubic), there are two maximal ideals $\mathfrak{n}_1, \mathfrak{n}_2$ of C , generated by $t - 1$ and $t + 1$ respectively. Then B is the subring of the product $C \times C$ consisting of the pairs of polynomials (f, g) such that $f(1) = g(-1)$ and $f(-1) = g(1)$ ($\text{Spec}(B)$ is the scheme obtained by “gluing” two copies of $\text{Spec}(C)$, the point $\mathfrak{n}_1$ (resp. $\mathfrak{n}_2$) of one being “glued” to the point $\mathfrak{n}_2$ (resp. $\mathfrak{n}_1$) of the other; cf. chap. V, where this operation will be discussed in a general fashion). There are thus two maximal ideals $\mathfrak{r}_1, \mathfrak{r}_2$ of B above the maximal ideal $\mathfrak{m}$ of A . Moreover, since the “gluing” process commutes with localisation and completion, one easily verifies that the canonical homomorphisms $\hat{A}_{\mathfrak{m}} \rightarrow \hat{B}_{\mathfrak{r}_1}$ and $\hat{A}_{\mathfrak{m}} \rightarrow \hat{B}_{\mathfrak{r}_2}$ are bijective, and hence $B_{\mathfrak{r}_1}$ and $B_{\mathfrak{r}_2}$ are flat $A_{\mathfrak{m}}$ -modules (Bourbaki, *Alg. comm.*, chap. III, §3, n° 5, prop. 10) having moreover the same residue field as $A_{\mathfrak{m}}$. For every other maximal ideal $\mathfrak{p}$ of A , it is immediate that there are two maximal ideals $\mathfrak{q}_1, \mathfrak{q}_2$ of B above $\mathfrak{p}$, and that the homomorphisms $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}_1}$ and $A_{\mathfrak{p}} \rightarrow B_{\mathfrak{q}_2}$ are bijective. One thus sees that the morphism $\text{Spec}(B) \rightarrow \text{Spec}(A)$ is flat and finite, and that all its fibres are geometrically regular (it is even *étale*, as we shall see later (17.6.3)); however it is immediate that B is not integral.

6.6. Transitivity properties

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Proposition (6.6.1). — *For a locally Noetherian prescheme Z , let us denote by $\mathbf{P}(Z)$ any one of the following properties:*

- a) Z is a Cohen-Macaulay prescheme.
- b) Z satisfies (S_n) .
- c) Z is regular.
- d) Z satisfies (R_n) .
- e) Z is normal.
- f) Z is reduced.

Let X, Y, Z be three locally Noetherian preschemes, $f : X \rightarrow Y, g : Y \rightarrow Z$ two morphisms.

- (i) Suppose that f and g are flat and that for every $y \in f(X)$ (resp. every $z \in g(Y)$), the prescheme $f^{-1}(y)$ (resp. $g^{-1}(z)$) satisfies $\mathbf{P}$. Then $h = g \circ f$ is flat and for every $z \in h(X)$, $h^{-1}(z)$ satisfies $\mathbf{P}$.
- (ii) Suppose the following conditions are satisfied: f is faithfully flat, $h = g \circ f$ is flat, for every $z \in h(X)$, the prescheme $f^{-1}(z)$ satisfies $\mathbf{P}$, and for every $z \in g(Y)$, the prescheme $h^{-1}(z)$ satisfies $\mathbf{P}$. Then g is flat and for every $z \in g(Y)$, $g^{-1}(z)$ satisfies $\mathbf{P}$.

Proof. (i) We already know that if f and g are flat, so is $h = g \circ f$, and that if f is flat (resp. faithfully flat, g is flat (2.1.6) and (2.2.13)). Furthermore, for every $z \in Z$, $f_z = f \otimes 1_{k(z)} : h^{-1}(z) \rightarrow g^{-1}(z)$ is flat (2.1.4), and for every $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (I, 3.6.4). The conclusion follows respectively from (6.3.5), (ii), (6.4.2), (6.5.2), (ii), (6.5.3), (ii), (6.5.4), (ii), and (3.3.5), (ii).

(ii) We already know that g is flat (2.2.13); moreover, for every $z \in Z$, f_z is faithfully flat (2.2.13). The conclusion follows respectively from (6.3.5), (i), (6.4.2), (i), (6.5.2), (i), (6.5.3), (i), (6.5.4), (i), and (2.1.13).

□

Remark (6.6.2). — Suppose h flat, f faithfully flat, and suppose that for every $z \in Z$, $h^{-1}(z)$ is of *coprofondéur* $\leq n$ (5.7.1); then it follows from the reasoning of (6.6.1), (ii), and from (6.3.2) that $g^{-1}(z)$ is of *coprofondéur* $\leq n$ for every $z \in g(Y)$ and that for every $y \in Y$, $f^{-1}(y)$ is of *coprofondéur* $\leq n$.

6.7. Application to base changes in algebraic preschemes

Proposition (6.7.1). — *Let k be a field, X a locally Noetherian k -prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Let k' be an extension of k ; set $X' = X \otimes_k k'$, $\mathcal{F}' = \mathcal{F} \otimes_k k'$ and let $p : X' \rightarrow X$ be the canonical projection. Suppose, either that X is locally of finite type over k , or that k' is a finite extension of k , so that in both cases X' is locally Noetherian. Let x' be a point of X' , $x = p(x')$. Then:*

- (i) *We have $\text{coprof}(\mathcal{F}_x) = \text{coprof}(\mathcal{F}'_{x'})$; in particular, for $\mathcal{F}_x$ to be a Cohen-Macaulay $\mathcal{O}_x$ -module, it is necessary and sufficient that $\mathcal{F}'_{x'}$ be a Cohen-Macaulay $\mathcal{O}_{x'}$ -module.*
- (ii) *For $\mathcal{F}$ to satisfy property (S_n) at the point x , it is necessary and sufficient that $\mathcal{F}'$ satisfy (S_n) at the point x' .*

Proof. We know that p is faithfully flat (2.2.13); the two assertions are therefore consequences respectively of (6.3.2) and (6.4.2), provided that we prove that $p^{-1}(x) = \text{Spec}(k(x) \otimes_k k')$ is a Cohen-Macaulay prescheme; since one of the two fields $k(x)$, k' is a finite extension of k by hypothesis, we are reduced to proving the following: $\square$

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Lemma (6.7.1.1). — *Let K, L be two extensions of a field k , one of which is of finite type (so that $K \otimes_k L$ is a Noetherian ring). Then $K \otimes_k L$ is a Cohen-Macaulay ring.*

Proof. Suppose for example that L is a finite type extension of k , so that L is a finite extension of a pure transcendental extension $k(\mathbf{t})$ of k ($\mathbf{t}$ being a finite system $(t_i)_{1 \leq i \leq m}$ of indeterminates). Then $A = K \otimes_k k(\mathbf{t})$, $B = K \otimes_k L$; B is a finite type A -module, and B is an A -module flat (0_I, 6.2.1) and of finite type; the morphism $h : \text{Spec}(B) \rightarrow \text{Spec}(A)$ is therefore finite, and since every Artinian prescheme is Cohen-Macaulay (6.3.5), or if S is the set of elements $\neq 0$ of $k[\mathbf{t}]$, we have $A' = S^{-1}A$, where $A' = K \otimes_k k[\mathbf{t}] = K[\mathbf{t}]$; by (0, 16.5.13), it suffices to prove that A' is a Cohen-Macaulay ring, and this follows from the fact that A' is regular (0, 17.3.7). $\square$

Corollary (6.7.2). — *For $\mathcal{O}_x$ to be a Cohen-Macaulay ring (resp. for X to satisfy (S_n) at the point x), it is necessary and sufficient that $\mathcal{O}_{x'}$ be a Cohen-Macaulay ring (resp. that X' satisfy (S_n) at the point x').*

Corollary (6.7.3). — *Let X, Y be two k -preschemes locally Noetherian, at least one of which is locally of finite type over k . Let $\mathcal{F}$ (resp. $\mathcal{G}$) be a coherent $\mathcal{O}_Y$ -Module (resp. a coherent $\mathcal{O}_X$ -Module). If $\mathcal{F}$ and $\mathcal{G}$ satisfy property (S_n) , then so does $\mathcal{F} \otimes_k \mathcal{G}$.*

Proof. The hypothesis implies that $X \times_k Y$ is locally Noetherian; let $p : X \times_k Y \rightarrow X$ and $q : X \times_k Y \rightarrow Y$ be the canonical projections, which are flat morphisms. Suppose for example that X is locally of finite type over k ; one can write $\mathcal{F} \otimes_k \mathcal{G} = p^*(\mathcal{F}) \otimes_k \mathcal{G}$, and since $\mathcal{F}$ is flat relative to the structural morphism $X \rightarrow \text{Spec}(k)$, $p^*(\mathcal{F})$ is q -flat (2.1.4); it suffices to see that for every $y \in Y$, $p^*(\mathcal{F}) \otimes_{\mathcal{O}_Y} k(y)$ has property (S_n) ; but $p^*(\mathcal{F}) \otimes_{\mathcal{O}_Y} k(y) = \mathcal{F} \otimes_k k(y)$, and since X is locally of finite type over k , the conclusion follows from (6.7.1), (ii). $\square$

Proposition (6.7.4). — *For a locally Noetherian prescheme Z , and a point $z \in Z$, let $\mathbf{P}(Z, z)$ be one of the following properties:*

- a) *Z is regular at the point z .*
- b) *Z satisfies (R_n) at the point z .*
- c) *Z is normal at the point z .*
- d) *Z is reduced at the point z .*

This being so, under the hypotheses and with the notations of (6.7.1), if $\mathbf{P}(X', x')$ is true, then so is $\mathbf{P}(X, x)$; the converse is true if k' is a separable extension of k .

Proof. The case where $\mathbf{P}$ is property d) has already been treated for the record, having already been treated (2.1.13) and (4.6.1). The same reasoning as at the beginning of the proof of (6.7.1) shows that the first assertion follows respectively from (6.5.2), (i), (6.5.3), (i), (6.5.4), (i); of the same kind, the second assertion follows from (6.5.2), (ii), (6.5.3), (ii), (6.5.4), (ii). $\square$

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Lemma (6.7.4.1). — *Let K, L be two extensions of a field k . If L is a separable extension of k , then $K \otimes_k L$ is a regular ring.*

Proof. We distinguish two cases:

A) L is a *finite type* extension of k . Then, with the notations of (6.7.1.1), we can suppose that L is a *finite separable* extension of $k(\mathfrak{t})$. For every $s \in \operatorname{Spec}(A)$, $k(s) \otimes_{k(\mathfrak{t})} L$ is composed of a direct sum of a finite number of fields, finite separable extensions of $k(s)$, hence a regular ring; it follows from (6.5.2), (ii) that it suffices to prove that A is a regular ring; as $A = S^{-1}A'$, it suffices to prove that A' is regular (0, 17.3.6), but one has seen that this was indeed the case in the proof of (6.7.1.1).

B) Let (L_λ) be the filtered family of sub-extensions of L that are of finite type over k ; by A), each of the rings $C_\lambda = K \otimes_k L_\lambda$ is regular; on the other hand, for $L_\lambda \subset L_\mu$, L_μ is a flat L_λ -module, hence C_μ is a flat C_λ -module (0_I, 6.2.1). Since $C = K \otimes_k L = \varinjlim (K \otimes_k L_\lambda)$ is Noetherian, we can apply (5.13.7), and C is therefore regular. $\square$

Remarks (6.7.5). — (i) In the proof, the fact that $\mathbf{P}(X, x)$ holds implies $\mathbf{P}(X', x')$, without the hypothesis that k' is a *separable* extension of k (4.3.1); but even when X is regular it can happen that X' is not regular without k' separable (6.7.6). We have already seen (4.6.1) that when $\mathbf{P}$ is property d), it can happen that X is regular without X' being regular; it is interesting to give an example of this in which k is *algebraically closed* in the field of rational functions K of X .

(ii) As recalled in (i), it follows from (4.6.1) that when X is a prescheme that is algebraic over k which is not geometrically reduced, there are finite radiciel extensions k' of k such that $X' = X \otimes_k k'$ is not reduced.

Definition (6.7.6). — Let k be a field, X a k -prescheme locally Noetherian. We say that X is *geometrically regular* at a point x (resp. has property (R_n) geometric, resp. is *geometrically normal* at a point x) if, for every finite extension k' of k , $X' = X_{(k')} = X \otimes_k k'$ is regular (resp. has property (R_n) , resp. is normal) at every point x' whose projection in X is x . We say that X is *geometrically regular* (or equivalently *geometrically regular in codimension $\leq n$*), resp. *geometrically normal*) if X is geometrically regular (resp. has the geometric property (R_n) , resp. is geometrically normal) at every point.

We say that an algebra A over k is a *geometrically regular* (resp. *geometrically normal*, resp. *geometrically reduced*, resp. *having geometric property (R_n)*) ring if $\operatorname{Spec}(A)$ has the same property.

If $X = \operatorname{Spec}(K)$, where K is an extension of k , it amounts to the same thing to say that X is geometrically regular, or geometrically normal, or geometrically reduced, or that K is a *separable* extension of k : this follows from (4.6.1) and from the fact that if K is separable from k and k' a finite extension, $K \otimes_k k'$ is composed of a direct sum of a finite number of fields (Bourbaki, *Alg.*, chap. VIII, §7, n° 3, cor. 1 of th. 1).

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Proposition (6.7.7). — Let k be a field, X a locally Noetherian k -prescheme, x a point of X ; let us denote by $\mathbf{Q}(k')$ the relation “ k' is an extension of k , and $\mathbf{P}(X_{(k')}, x')$ is true for every point x' of $X_{(k')}$ whose projection is x ”, where $\mathbf{P}(Z, z)$ is one of the properties a), b), c) of (6.7.4). Then the following properties are equivalent:

- a) $\mathbf{Q}(k')$ is true for every finite extension k' of k .
- b) $\mathbf{Q}(k')$ is true for every finite radiciel extension k' of k .
- c) $\mathbf{Q}(k')$ is true for every finite type extension k' of k .

Suppose in addition that X is locally of finite type over k ; then the three preceding properties are also equivalent to the following:

- d) $\mathbf{Q}(k')$ is true for every extension k' of k .
- e) $\mathbf{Q}(k')$ is true for every perfect extension k' of k .
- f) $\mathbf{Q}(k')$ is true for every extension $k' = k^{p^{-s}}$, where p is the characteristic exponent of k , and $s > 0$.

Proof. To prove the equivalence of a), b), and c), it clearly suffices to establish that b) implies c). Let K be a finite type extension of k , which is therefore the field of fractions of a k -algebra of finite type A ; set $Y = \operatorname{Spec}(A)$. We know (4.6.6) that there exists a finite radiciel extension k' of k such that $Y' = (Y \otimes_k k')_{\text{red}}$ is separable over k' ; if η' is the generic point of Y' , $K' = k(\eta')$ is a separable extension of k' by (4.6.1). Since, by hypothesis, $\mathbf{P}(X_{(k')}, x')$ is true for every point x' of $X_{(k')}$ above x , it follows from (6.7.4) that $\mathbf{P}(X_{(K')}, x'')$ is true for every point x'' of $X_{(K')}$ above x , since $X_{(K')} = (X_{(k')})_{(K')}$, and x'' is above a point x' of $X_{(k')}$ above x , and K' is separable over k' . But we

also have $X_{(K')} = (X_{(K)})_{(K')}$ and for every $x_0 \in X_{(K)}$ above x , there exists $x'' \in X_{(K')}$ above x_0 ; it follows from (6.7.4) that $\mathbf{P}(X_{(K)}, x_0)$ is true.

Suppose now X locally of finite type over k , so that for *every* extension K of k , $X_{(K)}$ is locally Noetherian. Since a radiciel extension of k is isomorphic to a sub-extension of a perfect extension of k , it follows immediately from (6.7.4) that $e)$ implies $f)$; similarly, $f)$ implies $b)$; $d)$ implies trivially $e)$, and finally $e)$ implies $d)$. We therefore need to prove that $b)$ implies $e)$. We take $k' = k^{p^{-\infty}}$ in $e)$. Since the question is local on X , one can moreover restrict to the case where X is affine and of finite type over k , by replacing X by a neighbourhood of x ; let thus $X = \text{Spec}(B)$, where B is a k -algebra of finite type; moreover, by definition of property $\mathbf{P}$ in all cases considered, one can also replace X by the local scheme of X at the point x , hence suppose that B is a *local* Noetherian ring. Set $B' = B \otimes_k k'$; k' is the inductive limit of its *finite* sub-extensions k'_λ and if we set $B'_\lambda = B \otimes_k k'_\lambda$, the morphism $f_\lambda : \text{Spec}(B') \rightarrow \text{Spec}(B'_\lambda)$ is a *homeomorphism* of $\text{Spec}(B')$ onto $\text{Spec}(B'_\lambda)$ for every λ ; indeed f_λ is bijective by (I, 3.5.2), (I, 3.5.7), and (I, 3.5.8); on the other hand, it is closed by (II, 6.1.10). One can now apply (5.13.6) which proves (by hypothesis $b)$) that $b)$ implies $e)$ when $\mathbf{P}(Z, z)$ is property (R_n) at the point z . The case where $\mathbf{P}(Z, z)$ is the property of being regular (i.e. of having property (R_n) for all n) follows trivially. Finally, taking into account the Serre normality criterion (5.8.6), $b)$ implies $e)$ when $\mathbf{P}(Z, z)$ is the property of being normal at the point z , since being normal at z is equivalent to having both properties (S_2) and (R_1) : one then applies what precedes for $n = 1$, and (6.7.2), (ii) for $n = 2$. $\square$

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Corollary (6.7.8). — *Let k be a field; for a k -prescheme locally Noetherian X and a point $x \in X$, let us denote by $\mathbf{P}(X, x)$ one of the following properties:*

- (i) $\text{coprof}(\mathcal{O}_x) \leq n$.
- (ii) $\mathcal{O}_x$ is a Cohen-Macaulay ring.
- (iii) X satisfies property (S_n) at the point x .
- (iv) X is geometrically regular at the point x .
- (v) X has property (R_n) geometric at the point x .
- (vi) X is geometrically normal at the point x .
- (vii) X is geometrically reduced (i.e. separable) at the point x .

Let k' be an extension of k ; suppose, either that k' is of finite type, or that X is locally of finite type over k , so that $X' = X \otimes_k k'$ is locally Noetherian. Let x' be a point of X' whose projection in X is x . Then, for $\mathbf{P}(X, x)$ to hold, it is necessary and sufficient that $\mathbf{P}(X', x')$ hold.

Proof. This has already been seen for property (vii) (4.6.11), and for properties (i), (ii), and (iii) (6.7.1). For (iv), (v), and (vi), this follows from (6.7.7): indeed, the fact that the condition is necessary follows from the equivalence of the criteria $a)$ and $c)$, and from the equivalence of $c)$ and $d)$ when X is locally of finite type over k . To see that the condition is sufficient, either k'' is a finite radiciel extension of k ; one can always consider k' and k'' as sub-extensions of an extension K of k ; set $X'' = X \otimes_k k''$, $X_0 = X \otimes_k K$, and note that since k'' is a radiciel extension of k , there is only *one* point x'' of X'' above x (I, 3.5.7) and (I, 3.5.8). Let x_0 be any point of X_0 above x ; if $\mathbf{P}(X', x')$ is true, then so is $\mathbf{P}(X_0, x_0)$ by (6.7.7), $c)$ and $d)$) (since k' is a finite type extension of k and K is an extension of k' , one can suppose that k' is of finite type, and if X is locally of finite type over k , X' is locally of finite type over k'). We deduce from (6.7.4) that the property $\mathbf{Q}(k'')$ is true (with the notations of (6.7.7)), hence $\mathbf{P}(X, x)$ is true by (6.7.7), $b)$). $\square$

6.8. Regular, normal, reduced, smooth morphisms

Definition (6.8.1). — Let X, Y be two preschemes, $f : X \rightarrow Y$ a morphism such that the fibre $f^{-1}(y)$ is a locally Noetherian prescheme for every $y \in Y$, x a point of X . We say respectively that f is a morphism:

- (i) of coprofondéur $\leq n$ at the point x ;
- (ii) of Cohen-Macaulay at the point x ;
- (iii) (S_n) at the point x ;
- (iv) regular at the point x ;
- (v) (R_n) at the point x ;
- (vi) normal at the point x ;
- (vii) reduced at the point x ;

if f is flat at the point x , and if in addition the corresponding property $\mathbf{P}(f^{-1}(f(x)), x)$ (notation of (6.7.8)) is true.

We say that f is *smooth* at the point x if it is locally of finite presentation in a neighbourhood of x in X , and if it is regular at the point x . We say that f is a morphism of coprofondéur $\leq n$, Cohen-Macaulay, (S_n) , regular, (R_n) , normal, reduced, smooth, if it has the corresponding property at every point of X .

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Proposition (6.8.2). — Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism, x a point of X . Let us denote by $\mathbf{M}(f, x)$ one of the properties (i) to (vii) of the definition (6.8.1), or the property for f to be smooth at the point x . Let Y' be a locally Noetherian prescheme, $g : Y' \rightarrow Y$ a morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. Suppose that f or g is locally of finite type. Then, for every $x' \in X'$ above x , the property $\mathbf{M}(f, x)$ implies $\mathbf{M}(f', x')$.

Proof. Set $y = f(x)$, $y' = f'(x')$; by transitivity of fibres (I, 3.6.4), on a $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$; since $f^{-1}(y)$ is locally of finite type over $k(y)$ or $k(y')$ is of finite type over $k(y)$ (I, 6.4.11), it follows from (6.7.8) that the properties $\mathbf{P}(f^{-1}(y), x)$ and $\mathbf{P}(f'^{-1}(y'), x')$ are equivalent; moreover, if f is flat at the point x , f' is flat at the point x' (2.1.4), which proves the proposition, taking into account (1.4.3), (iii) and (2.7.1), (iv). $\square$

Proposition (6.8.3). — For a morphism f of locally Noetherian preschemes, let $\mathbf{M}(f)$ be one of the following properties: to be Cohen-Macaulay, (S_n) , regular, (R_n) , normal, reduced.

- (i) Let X, Y, Z be three locally Noetherian preschemes, $f : X \rightarrow Y$, $g : Y \rightarrow Z$ two morphisms. If $\mathbf{M}(f)$ and $\mathbf{M}(g)$ are true, $\mathbf{M}(g \circ f)$ is true.
- (ii) Conversely, if f is surjective and if $\mathbf{M}(f)$ and $\mathbf{M}(g \circ f)$ are true, then $\mathbf{M}(g)$ is true.
- (iii) Let X, Y, Y' be three locally Noetherian preschemes, $f : X \rightarrow Y$, $h : Y' \rightarrow Y$ two morphisms; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. Suppose that f or h is locally of finite type. Then, if $\mathbf{M}(f)$ is true, so is $\mathbf{M}(f')$; the converse is true when h is faithfully flat.

The conclusions of (i) and (iii) are still true when $\mathbf{M}$ is the property of being smooth and (in (iii)) h is quasi-compact.

Proof. (i) We already know that if f and g are flat, so is $u = g \circ f$, and that if f is flat and f surjective, g is flat (2.1.6) and (2.2.13). Moreover, for every $z \in Z$, $f_z = f \otimes 1_{k(z)} : u^{-1}(z) \rightarrow g^{-1}(z)$ is flat (resp. faithfully flat if f is) and the morphism $f_z^{-1}(y) \otimes_{k(y)} k(y')$ is isomorphic to $f^{-1}(y)$ dans $g^{-1}(z)$ (I, 3.6.4). When $\mathbf{M}$ is the property of being regular, (R_n) , normal or reduced, the hypothesis implies that $\mathbf{M}(f)$ and $\mathbf{M}(g)$ hold, and each of the fibres $f_z^{-1}(y')$ has for $y' \in Y'_z$ the corresponding property c), d), e), f) of (6.6.1); we deduce from (6.6.1), (i) that X'_z has the same property, hence $\mathbf{M}(g \circ f)$ is true.

(ii) Conversely, the hypothesis that $\mathbf{M}(g \circ f)$ and $\mathbf{M}(f)$ are true and that f is surjective implies that Y'_z has the corresponding property for every z , by (6.6.1), (ii), f_z being surjective for every z .

(iii) The first assertion follows from (6.8.2). On the other hand, if h is faithfully flat, f is flat (2.4.1); moreover, with the notations of (6.8.2), the properties $\mathbf{P}(f^{-1}(y), x)$ and $\mathbf{P}(f'^{-1}(y'), x')$ are

equivalent (6.7.8), hence $\mathbf{M}(f)$ and $\mathbf{M}(f')$ are equivalent. Finally, the last assertion of the proposition follows from (1.4.3), (iii) and (2.7.1), (iv). $\square$

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Remarks (6.8.4). — (i) If f is faithfully flat, g flat, and $g \circ f$ is of coprofondur $\leq n$, then g is of coprofondur $\leq n$, as follows from (6.6.2).
(ii) When, in (6.8.1), one takes Y to be the spectrum of a field k , the notions (iv), (v), and (vi) reduce to those defined in (6.7.5). It is clear that these latter are the ones relative to the base field k . The definition (6.8.1) leads then to saying that a prescheme X is “regular (resp. (R_n) , resp. normal) over k ” instead of saying that it is “geometrically regular (resp. (R_n) , resp. normal)”; one will avoid confusing this notion with the property of being regular (resp. (R_n) , resp. normal) which is independent of k . One can make the same remarks in this regard as in (4.5.12).

Proposition (6.8.5). — Let k be a field, X, Y two k -preschemes locally Noetherian, at least one of which is locally of finite type over k . For a k -prescheme Z , and a point $z \in Z$, let us denote by $\mathbf{P}(Z)$ one of the properties c), d), e) (resp. f)) of (6.6.1); the property “ $\mathbf{P}(Z)$ geometric” is then defined in (6.7.6) (resp. (4.6.1)). Then:

- (i) If X has property $\mathbf{P}$ and Y has property $\mathbf{P}$ geometric, $X \times_k Y$ has the geometric property $\mathbf{P}$.
- (ii) If X and Y have property $\mathbf{P}$ geometric, so does $X \times_k Y$.

Proof. Let $f : X \times_k Y \rightarrow X$, $g : X \rightarrow \text{Spec}(k)$ be the structural morphisms, which are faithfully flat (2.2.13). The hypothesis that Y has the geometric property $\mathbf{P}$ implies by (6.8.2) that $\mathbf{M}(f)$ is true, $\mathbf{M}$ being the property corresponding to $\mathbf{P}$; under hypothesis (ii), $\mathbf{M}(g)$ is also true, hence assertion (ii) follows from (6.8.3), (i). As for assertion (i), it follows directly from (6.6.1), (i). $\square$

Theorem (6.8.6). — Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , $y = f(x)$. The following properties are equivalent:

- a) f is smooth at the point x .
- b) f is regular at the point x .
- c) $\mathcal{O}_x$ is a formally smooth $\mathcal{O}_y$ -algebra (0, 19.3.1) for the preadic topologies on $\mathcal{O}_x$ and $\mathcal{O}_y$.
- c') $\mathcal{O}_x$ is a formally smooth $\mathcal{O}_y$ -algebra (0, 19.3.1) for the discrete topologies on $\mathcal{O}_x$ and $\mathcal{O}_y$.

The equivalence of a) and b) follows from the definitions (6.8.1) and from the fact that for locally Noetherian preschemes, morphisms locally of finite type are locally of finite presentation (1.4.2). Secondly, for $\mathcal{O}_x$ to be an $\mathcal{O}_y$ -algebra smooth for the preadic topologies, it is necessary and sufficient that $\mathcal{O}_x$ be a flat $\mathcal{O}_y$ -module and that $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$ be a $k(y)$ -algebra formally smooth for its preadic topology (0, 19.7.1); but for $\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y)$ to be a $k(y)$ -algebra formally smooth for its preadic topology, it is necessary and sufficient that it be a $k(y)$ -algebra geometrically regular (0, 19.6.6); this shows the equivalence of b) and c). Finally, to prove the equivalence of c) and c'), we can restrict to the case where $Y = \text{Spec}(A)$ and $X = \text{Spec}(C)$ are affine, A being Noetherian and C a finite type A -algebra, and one can therefore write C in the form $A[T_1, \dots, T_n]/\mathfrak{J}$. Since here $\mathfrak{J}/\mathfrak{J}^2$ is a C -module of finite presentation, the equivalence of c) and c') follows from (0, 22.6.4) applied to A , $B = A[T_1, \dots, T_n]$ and $C = B/\mathfrak{J}$.

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Corollary (6.8.7). — Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism locally of finite type. Then the set of points $x \in X$ where f is smooth (or regular) is open in X .

Proof. It follows from (0, 22.6.5) that the set of $x \in X$ satisfying condition c') of (6.8.6) is open in X , and the conclusion follows from (6.8.6). $\square$

Remark (6.8.8). — In (17.5.1), we shall show the equivalence of b) and c') in (6.8.6), as well as the corollary (6.8.7), remain valid without the Noetherian hypothesis on X and Y , provided one restricts to morphisms locally of finite presentation.

6.9. The theorem of generic flatness

Theorem (6.9.1). — *Let Y be a locally Noetherian prescheme, integral, $u : X \rightarrow Y$ a morphism of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. There then exists a non-empty open subset U of Y such that $\mathcal{F}|_{u^{-1}(U)}$ is flat over U .*

Proof. One can clearly restrict to the case where $Y = \operatorname{Spec}(A)$ is affine; then X is a finite union of open affines X_i of finite type over Y ; if, for each i , there is an open non-empty U_i in Y such that $\mathcal{F}|_{(X_i \cap u^{-1}(U_i))}$ is flat over U_i , it is clear that taking U as the intersection of the U_i answers the question; one can therefore suppose that $X = \operatorname{Spec}(B)$, where B is a finite type A -algebra. We then have $\mathcal{F} = \tilde{M}$, where M is a finite type B -module; the theorem will therefore follow from the following: $\square$

Lemma (6.9.2). — *Let A be an integral Noetherian ring, B a finite type A -algebra, M a finite type B -module. Then there exists $f \neq 0$ in A such that M_f is a free A_f -module.*

Proof. Let K be the field of fractions of A ; then $B \otimes_A K$ is a finite type algebra over K and $M \otimes_A K$ a $(B \otimes_A K)$ -module of finite type. We argue by induction on the dimension n of the support of $M \otimes_A K$, which is $-\infty$ or finite and ≥ 0 . Suppose first $n = -\infty$, that is to say $M \otimes_A K = 0$; if $(m_i)_{1 \leq i \leq r}$ is a generating system of the B -module M , there exists $f \neq 0$ in A such that $fm_i = 0$ for $1 \leq i \leq r$; hence $M_f = 0$ and the lemma is true in this case. Suppose now $n \geq 0$. We know that there exists a composition series $M = M_1 \supset M_2 \supset \dots \supset M_q = 0$ of the B -module M such that each of the quotients $N_i = M_i/M_{i+1}$ is isomorphic to a B -module of the form $B/\mathfrak{p}_i$, where $\mathfrak{p}_i$ is a prime ideal of B (Bourbaki, *Alg. comm.*, chap. IV, §1, n° 4, th. 1). If the theorem is true for each of the N_i , there is for each i an $f_i \neq 0$ in A such that $(N_i)_{f_i}$ is a free A_{f_i} -module; setting $f = f_1 f_2 \cdots f_{q-1}$, it follows that $(N_i)_f$ is a free A_f -module for $1 \leq i \leq q-1$. But $(N_i)_f = (M_i)_f / (M_{i+1})_f$ (0_I, 1.3.2) and since an extension of free modules is free, we deduce that M_f is a free A_f -module. Replacing B by $B/\mathfrak{p}$ ($\mathfrak{p}$ a prime ideal of B), which is still of finite type over A , one can thus restrict to the case where $M = B$ and B is integral. We then know (Bourbaki, *Alg. comm.*, chap. V, §3, n° 1, cor. 1 of th. 1) that there exists an element $g \neq 0$ in A and elements t_i ($1 \leq i \leq m$) of B , algebraically independent over A and such that B_g is entire over $A_g[t_1, \dots, t_m]$. One can replace A by A_g , B by B_g and hence assume that B is entire over $C = A[t_1, \dots, t_m]$, hence a C -module of finite type and *without torsion*. We know moreover (4.1.2) that the dimension of $\operatorname{Spec}(B \otimes_A K)$ is equal to m , hence on a $m = n$.

This being so, if h is the rank of the torsion-free C -module B , there exists an exact sequence of C -modules

$$0 \longrightarrow C^h \longrightarrow B \longrightarrow M' \longrightarrow 0$$

where M' is a C -module of *torsion* of finite type; the support of M' therefore does not contain the generic point of $\operatorname{Spec}(C)$ (I, 7.4.6) and hence the support of $M' \otimes_A K$ does not contain the generic point of $\operatorname{Spec}(C \otimes_A K)$ (I, 9.1.13.1); we thus conclude (4.1.2.1) that its dimension is $< n$. By the induction hypothesis, there exists an $f \neq 0$ in A such that M'_f is a free A_f -module; on the other hand C_f^h is an A_f -module free; hence it follows that M_f is a free A_f -module. Q.E.D. $\square$

Corollary (6.9.3). — *Let S be a Noetherian prescheme, $u : X \rightarrow S$ a morphism of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Then there is a partition $(S_i)_{1 \leq i \leq n}$ of S in a finite number of locally closed subsets of S such that, if one denotes by S_i the reduced sub-prescheme having space S_i as underlying space, and if one sets $X_i = X \times_S S_i$, $\mathcal{F}_i = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S_i}$, the $\mathcal{O}_{X_i}$ -Module $\mathcal{F}_i$ is flat over S_i .*

Proof. We proceed by Noetherian induction (0_I, 2.2.2) on the set of closed subsets T of S such that the reduced sub-prescheme having T for underlying space satisfies the conclusion of (6.9.3). One can thus restrict to proving the corollary for S having for underlying space a closed subset $\neq S$. Since the morphism $S_{\text{red}} \rightarrow S$ is of finite type and surjective, one can replace S by S_{red} , hence suppose S reduced and non-empty. Since S is Noetherian, the interior T of an irreducible component of S is non-empty, and the prescheme induced on T is integral; there is therefore by (6.9.1) a non-empty open subset $U \subset T$ such that $\mathcal{F}|_{u^{-1}(U)}$ is flat over U . If Y is the sub-prescheme reduced by S having for underlying space $S - U$, it is clear by hypothesis that there is a partition (Y_i) of Y into locally closed sets in S , such that the Y_i and U form a partition answering the question. $\square$

6.10. Dimension and depth of a Module normally flat along a closed sub-prescheme

(6.10.1). Let X be a locally Noetherian prescheme, $\mathcal{I}$ a quasi-coherent Ideal of $\mathcal{O}_X$, Y the closed sub-prescheme of X defined by $\mathcal{I}$, $j : Y \rightarrow X$ the canonical injection. For every integer $k \geq 0$, the $\mathcal{O}_X$ -Module $\mathcal{I}^k \mathcal{F} / \mathcal{I}^{k+1} \mathcal{F}$ is annihilated by $\mathcal{I}$, hence is of the form $j_*(\mathcal{G}_k)$, where $\mathcal{G}_k = j^*(\mathcal{I}^k \mathcal{F} / \mathcal{I}^{k+1} \mathcal{F}) = j^*(\mathcal{I}^k \mathcal{F})$ is a coherent $\mathcal{O}_Y$ -Module. By abuse of language, we denote $\mathrm{gr}_{\mathcal{I}}^*(\mathcal{F})$ the graded $\mathcal{O}_Y$ -Module equal to the direct sum

$$\bigoplus_{k=0}^{\infty} \mathcal{G}_k = j^* \left(\bigoplus_{k=0}^{\infty} \mathcal{I}^k \mathcal{F} / \mathcal{I}^{k+1} \mathcal{F} \right);$$

in particular, we have $\mathrm{gr}_{\mathcal{I}}^0(\mathcal{F}) = \mathcal{G}_0 = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_Y = j^*(\mathcal{F})$. We shall say (following Hironaka) that $\mathcal{F}$ is *normally flat along* Y if $\mathrm{gr}_{\mathcal{I}}^*(\mathcal{F})$ is a flat $\mathcal{O}_Y$ -Module. This is the same as saying ((2.1.12) and (0_I, 6.1.2)) that each of the $\mathcal{O}_Y$ -Modules $\mathcal{G}_k$ is *locally free*.

Proposition (6.10.2). — *Let X be a locally Noetherian prescheme, Y a closed integral sub-prescheme of X . For every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, there exists an open subset U of X such that $Y \cap U \neq \emptyset$ and that $\mathcal{F}|_U$ is normally flat along $Y \cap U$.*

Proof. Let $\mathcal{I}$ be the coherent Ideal of $\mathcal{O}_X$ defining Y ; the $\mathcal{O}_Y$ -Algebra $\mathcal{B} = \mathrm{gr}_{\mathcal{I}}^*(\mathcal{O}_X)$, an image of $\mathcal{I} / \mathcal{I}^2$, is quasi-coherent and of finite type since it is generated by $\mathrm{gr}_{\mathcal{I}}^1(\mathcal{O}_X)$, image of $\mathcal{I} / \mathcal{I}^2$; as $\mathrm{gr}_{\mathcal{I}}^*(\mathcal{F})$ is a graded $\mathcal{B}$ -module quasi-coherent generated by $\mathrm{gr}_{\mathcal{I}}^1(\mathcal{F})$, it is a finite type $\mathcal{B}$ -Module. If we set $Z = \mathrm{Spec}(\mathcal{B})$, the structural morphism $u : Z \rightarrow Y$ is then of finite type, and if $\mathcal{H}$ is the coherent $\mathcal{O}_Z$ -Module such that $u_*(\mathcal{H}) = \mathrm{gr}_{\mathcal{I}}^*(\mathcal{F})$, it suffices to apply u and to $\mathcal{H}$ the theorem of generic flatness (6.9.1) to prove the proposition. $\square$

Proposition (6.10.3). — *With the notations of (6.10.1), suppose that $\mathcal{F}$ is normally flat along Y . Then:*

- (i) $Y \cap \mathrm{Supp}(\mathcal{F})$ is a part that is both open and closed in Y (in other words, a union of connected components of Y).
- (ii) If $\mathcal{I}$ is locally nilpotent, $\mathrm{Supp}(\mathcal{F})$ is an open and closed part of X , and for every $x \in \mathrm{Supp}(\mathcal{F})$, we have

$$(6.10.3.1) \quad \dim(\mathcal{F}_x) = \dim(\mathcal{O}_{Y,x})$$

$$(6.10.3.2) \quad \mathrm{prof}(\mathcal{F}_x) = \mathrm{prof}(\mathcal{O}_{Y,x})$$

$$(6.10.3.3) \quad \mathrm{coprof}(\mathcal{F}_x) = \mathrm{coprof}(\mathcal{O}_{Y,x}).$$

Corollary (6.10.4). — *Let $U_{S_n}(\mathcal{F})$ (resp. $U_{C_n}(\mathcal{F})$) be the set of $x \in X$ such that $\mathcal{F}$ satisfies (S_n) at the point x (resp. such that $\mathrm{coprof}(\mathcal{F}_x) \leq n$). If $\mathcal{F}$ is normally flat along Y , then $U_{S_n}(\mathcal{F}) = U_{S_n}(\mathcal{O}_Y)$ and $U_{C_n}(\mathcal{F}) = U_{C_n}(\mathcal{O}_Y)$.*

Proposition (6.10.5). — *With the notations of (6.10.1), suppose that Y is connected and non-empty, and that $\mathcal{F}$ is normally flat along Y . For every integer $n \geq 0$, we set*

$$(6.10.5.1) \quad r(n) = \mathrm{rg}_{\mathcal{O}_Y}(\mathrm{gr}_{\mathcal{I}}^n(\mathcal{F}))$$

(the $\mathcal{O}_Y$ -Module locally free $\mathrm{gr}_{\mathcal{I}}^n(\mathcal{F})$ being necessarily of constant rank). Then:

- (i) There exists a polynomial $P \in \mathbf{Q}[T]$ such that $P(n) = r(n)$ for n sufficiently large.
- (ii) We have $Y \cap \mathrm{Supp}(\mathcal{F}) = \emptyset$, or $Y \cap \mathrm{Supp}(\mathcal{F}) = Y$ (in other words $Y \subset \mathrm{Supp}(\mathcal{F})$).

In the second case, let $d - 1$ denote the degree of P ; for every maximal point y of Y , we have

$$(6.10.5.2) \quad \dim(\mathcal{F}_y) = d$$

and in particular

$$(6.10.5.3) \quad \mathrm{codim}(Y, \mathrm{Supp}(\mathcal{F})) = d.$$

- (iii) Suppose $Y \subset \mathrm{Supp}(\mathcal{F})$. For every $x \in Y$, we have

$$(6.10.5.4) \quad \dim(\mathcal{F}_x) = \dim(\mathcal{O}_{Y,x}) + d.$$

Proposition (6.10.6). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, Y a closed irreducible sub-prescheme of X , of generic point $y \in \text{Supp}(\mathcal{F})$. There then exists a neighbourhood U of y in X such that, for every $x \in U \cap Y$, we have IV-2 | 157

$$(6.10.6.1) \quad \dim(\mathcal{F}_x) = \dim(\mathcal{F}_y) + \dim(\mathcal{O}_{Y,x})$$

$$(6.10.6.2) \quad \text{prof}(\mathcal{F}_x) = \text{prof}(\mathcal{F}_y) + \text{prof}(\mathcal{O}_{Y,x})$$

$$(6.10.6.3) \quad \text{coprof}(\mathcal{F}_x) = \text{coprof}(\mathcal{F}_y) + \text{coprof}(\mathcal{O}_{Y,x}).$$

6.11. Criteria for the sets $U_{S_n}(\mathcal{F})$ or $U_{C_n}(\mathcal{F})$ to be open

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Lemma (6.11.1). — Let X be a locally Noetherian prescheme, locally immersible in a regular scheme (5.11.1); let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module. Then:

- (i) (M. Auslander) The function $x \mapsto \text{coprof}(\mathcal{F}_x)$ is semi-continuous superiorly in X (in other words, for every integer n , the set $U_{C_n}(\mathcal{F})$ of $x \in X$ such that $\text{coprof}(\mathcal{F}_x) \leq n$ is open).
- (ii) For every integer n , the set $U_{S_n}(\mathcal{F})$ of $x \in X$ such that $\mathcal{F}$ has property (S_n) at the point x is open.

Proposition (6.11.2). — Let X be a locally Noetherian prescheme, regular and locally immersible in a regular scheme (5.11.1); let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module. Then:

- (i) (M. Auslander) The function $x \mapsto \text{coprof}(\mathcal{F}_x)$ is semi-continuous superiorly in X (in other words, for every integer n , the set $U_{C_n}(\mathcal{F})$ of $x \in X$ such that $\text{coprof}(\mathcal{F}_x) \leq n$ is open).
- (ii) For every integer n , the set $U_{S_n}(\mathcal{F})$ of $x \in X$ such that $\mathcal{F}$ has property (S_n) at the point x is open.

Corollary (6.11.3). — Under the hypotheses of (6.11.2), the set $\text{CM}(\mathcal{F})$ of points $x \in X$ such that $\mathcal{F}_x$ is a Cohen-Macaulay module is open in X .

Proof. Indeed, it is the set $U_{C_0}(\mathcal{F})$. □

Remarks (6.11.4). — (i) The reasoning of (6.11.2), (ii) shows that (without hypothesis on X) when $U_{C_n}(\mathcal{F})$ is open for every integer n , then $U_{S_n}(\mathcal{F})$ is also open.

- (ii) We have $\text{CM}(\mathcal{F}) = \bigcap_n U_{S_n}(\mathcal{F})$. If every point $x \in X$ admits a neighbourhood V that is open such that V is of finite dimension, the decreasing sequence of intersections $V \cap U_{S_n}(\mathcal{F})$ is stationary since $\dim(\mathcal{F}_x)$ is finite for every x ; hence, if the $U_{S_n}(\mathcal{F})$ are open, $\text{CM}(\mathcal{F})$ is then also open in X .

- (iii) We write $U_{S_n}(X)$, $U_{C_n}(X)$, $\text{CM}(X)$ instead of $U_{S_n}(\mathcal{O}_X)$, $U_{C_n}(\mathcal{O}_X)$, $\text{CM}(\mathcal{O}_X)$.

Proposition (6.11.5). — Let A be a local Noetherian ring, M a finite type A -module. For every prime ideal $\mathfrak{p}$ of A , we have

$$(6.11.5.1) \quad \text{coprof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \leq \text{coprof}_A(M).$$

Proof. We can restrict to the case where $M_{\mathfrak{p}} \neq 0$. Since $\hat{A}$ is a faithfully flat A -module (0I, 6.5.1), there exists a prime ideal $\mathfrak{q}$ of $\hat{A}$ above $\mathfrak{p}$ (0I, 6.5.1); since $\hat{M}_{\mathfrak{q}} = (M \otimes_A \hat{A})_{\mathfrak{q}} = M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \hat{A}_{\mathfrak{q}}$ and $\hat{A}_{\mathfrak{q}}$ is a flat $A_{\mathfrak{p}}$ -module, hence a flat A -module flat (0I, 6.2.1), it follows from (6.3.2) that $\text{coprof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \leq \text{coprof}_{\hat{A}_{\mathfrak{q}}}(\hat{M}_{\mathfrak{q}})$. On the other hand, $X = \text{Spec}(\hat{A})$ is a sub-scheme of a regular scheme by the Cohen structure theorem (0, 19.8.8, (i)); on deduces from (6.11.2) that $\text{coprof}_{\hat{A}_{\mathfrak{q}}}(\hat{M}_{\mathfrak{q}}) \leq \text{coprof}_{\hat{A}}(\hat{M})$; finally, we know that $\text{coprof}_{\hat{A}}(\hat{M}) = \text{coprof}_A(M)$ (0, 16.4.10), which completes the proof of (6.11.5.1). □

Proposition (6.11.6). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, n an integer > 0 . Suppose that for every closed integral sub-prescheme Y of X , there exists a non-empty open subset W of Y such that the sub-prescheme of Y induced on the open W satisfies (S_n) . Then the set $U_{S_n}(\mathcal{F})$ is open in X . IV-2 | 160

Corollary (6.11.7). — With the notations of (6.11.6):

- (i) The set $U_{S_1}(\mathcal{F})$ is open in X .
- (ii) For $U_{S_n}(\mathcal{F})$ to be open, it suffices that every maximal point x of $\text{Supp}(\mathcal{F})$, belonging to $U_{S_n}(\mathcal{F})$, be interior to $U_{S_n}(\mathcal{F})$.

Proposition (6.11.8). — Let X be a locally Noetherian prescheme satisfying the following property:

(CMU) Every closed integral sub-prescheme Y of X contains a non-empty open W such that the prescheme induced by X on W is a Cohen-Macaulay prescheme.

Then, for every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, the function $x \mapsto \text{coprof}(\mathcal{F}_x)$ is locally constructible and semi-continuous superiorly; the sets $U_{C_n}(\mathcal{F})$ and $U_{S_n}(\mathcal{F})$ are open in X .

Proof. Indeed, let Y be a closed integral sub-prescheme of X , of generic point y ; by (6.10.6), there is a neighbourhood V of y in X such that, for every $x \in V \cap Y$, we have

$$(6.11.8.1) \quad \text{coprof}(\mathcal{F}_x) = \text{coprof}(\mathcal{F}_y) + \text{coprof}(\mathcal{O}_{Y,x}).$$

But by hypothesis there exists a non-empty open W of Y such that, for $x \in V \cap W$ we have $\text{coprof}(\mathcal{O}_{Y,x}) = 0$, hence $\text{coprof}(\mathcal{F}_x)$ is constant in a neighbourhood of y in Y , which proves that the function $x \mapsto \text{coprof}(\mathcal{F}_x)$ is locally constructible (0_{III}, 9.3.2); moreover it follows from (6.11.5) and from (0_{III}, 9.3.4) that this function is semi-continuous superiorly. The last assertion follows from (6.11.4), (i), or equivalently from (6.11.6). $\square$

Remarks (6.11.9). — (i) If X satisfies the hypothesis (CMU) of (6.11.8), then, for every morphism locally of finite type $u : X' \rightarrow X$, X' also satisfies (CMU).
 (ii) Suppose that there exists a coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ such that $\text{Supp}(\mathcal{F}) = X$ and that $\mathcal{F}$ is a Cohen-Macaulay $\mathcal{O}_X$ -Module. Then X satisfies condition (CMU).

6.12. Nagata's criteria for $\text{Reg}(X)$ to be open

(6.12.1). Given a locally Noetherian prescheme X , the *singular locus* of X is defined to be the set of points $x \in X$ such that X is not regular at the point x (in other words, such that the local ring $\mathcal{O}_x$ is not regular); it is denoted $\text{Sing}(X)$; the complementary set $X - \text{Sing}(X)$, consisting of the $x \in X$ where X is regular, is denoted $\text{Reg}(X)$. We propose in this number to give conditions under which $\text{Sing}(X)$ is closed (i.e. $\text{Reg}(X)$ is open).

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Proposition (6.12.2). — Let X be a locally Noetherian prescheme. The following conditions are equivalent:

- a) $\text{Reg}(X)$ is open in X .
- b) For every $x \in \text{Reg}(X)$, there exists a non-empty open subset of $\overline{\{x\}}$ contained in $\text{Reg}(X)$.

Moreover, these conditions are implied by the following:

- c) For every $x \in \text{Reg}(X)$, denoting by Y the closed reduced sub-prescheme of X having $\overline{\{x\}}$ as underlying space, $\text{Reg}(Y)$ is a neighbourhood of x in Y .

Proof. The equivalence of a) and b) follows from the fact that $\text{Reg}(X)$ is stable by generisation (0, 17.3.2) and from (0_{III}, 9.2.6). To prove that c) implies b), consider a closed integral sub-prescheme Z of Y of generic point z ; if Y' is the sub-prescheme integral of X having as underlying space $Y' = \overline{\{z\}}$ of z in X , Z is open in Y' and the sub-prescheme Z of Y' , being reduced, is induced by Y' on the open Z of the underlying space of Y' . The hypothesis implies that $\text{Reg}(Y')$ is a neighbourhood of z in Y' ; it therefore suffices to apply (6.12.2) by replacing X by Y and Y by Z . $\square$

Corollary (6.12.3). — Let X be a locally Noetherian prescheme. The following conditions are equivalent:

- a) For every sub-prescheme Y of X , $\text{Reg}(Y)$ is open in Y .
- b) For every closed integral sub-prescheme Y of X , $\text{Reg}(Y)$ contains a non-empty open part of Y .

Proof. It is clear that a) implies b), since if Y is integral and y is its generic point, $\mathcal{O}_{Y,y}$ is a field, hence $y \in \text{Reg}(Y)$. Conversely, to see that b) implies a), consider a closed integral sub-prescheme Z of Y of generic point z ; if Y' is the sub-prescheme integral of X having $\overline{\{z\}}$ as underlying space, $Y' = \overline{\{z\}}$ in X , and the sub-prescheme Z of Y' , being reduced, is induced by Y' on the open part of its underlying space. The hypothesis implies that $\text{Reg}(Y')$ is a neighbourhood of z in Y' ; it suffices then to apply (6.12.2) by replacing X and Y by Z . $\square$

Theorem (6.12.4). — (Nagata). — Let A be a Noetherian ring, $X = \text{Spec}(A)$. The following conditions are equivalent:

- a) For every prescheme X' locally of finite type over X , $\text{Reg}(X')$ is open in X' .
- b) For every finite integral A -algebra A' , there exists a non-empty open of $X' = \text{Spec}(A')$ contained in $\text{Reg}(X')$.

- c) For every prime ideal $\mathfrak{p}$ of A and every finite radical extension K' of the field of fractions K of $A/\mathfrak{p}$, finite on A , having K' for field of fractions, and such that in $X' = \text{Spec}(A')$, a non-empty open subset is contained in $\text{Reg}(X')$.

Corollary (6.12.5). — (Zariski). — Let k be a field; for every k -prescheme X locally of finite type over k , the set of $x \in X$ where X is regular (resp. geometrically regular over k) is open in X .

Corollary (6.12.6). — Let A be a ring having one of the following properties:

- (i) A is a Dedekind ring and its field of fractions K is of characteristic 0.
- (ii) A is a semi-local Noetherian ring of dimension ≤ 1 .

Then, for every prescheme X locally of finite type over A , $\text{Reg}(X)$ is open in X .

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Theorem (6.12.7). — (Nagata). — Let A be a local Noetherian complete ring, $X = \text{Spec}(A)$. Then $\text{Reg}(X)$ is open in X .

Proof. By (6.12.2), we are reduced to the case where A is moreover integral, and we must show that $\text{Reg}(X)$ contains a non-empty open subset of X . We distinguish two cases:

I) The field of fractions K of A is of characteristic 0. — We know then (0, 19.8.8) that there exists a complete discrete valuation ring C and a sub-ring B of A such that A is a finite B -algebra and such that B is isomorphic to a formal power series ring $C[[T_1, \dots, T_r]]$. As C is regular (II, 7.1.6), so is B (0, 17.3.8); moreover, the field of fractions L of B being of characteristic 0, K is a separable finite extension of L , so we can apply (6.12.4), I) to B , which proves the proposition in this case.

II) The field of fractions K of A is of characteristic $p > 0$. — Then A contains the prime field $\mathbf{F}_p$, and the theorem has already been proved in this case (0, 22.7.6). $\square$

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Corollary (6.12.8). — Let A be a local Noetherian complete ring. Then, for every prescheme X locally of finite type over A , $\text{Reg}(X)$ is open in X .

Proof. We verify condition c) of (6.12.4). If $\mathfrak{p}$ is prime in A , $A/\mathfrak{p}$ is again a local Noetherian complete ring; if K' is a finite extension of the field of fractions K of $A/\mathfrak{p}$, K' is the field of fractions of a sub- A -algebra A' of K' , generated by a finite system of generators of K' over K , integral over A ; A' is a semi-local integral ring of dimension ≤ 1 (0, 16.1.5), and therefore, in $X' = \text{Spec}(A')$, the set reduced to the generic point is open and evidently contained in $\text{Reg}(X')$, which proves condition c) of (6.12.4) in this case. This corollary applies in particular when $A = \mathbf{Z}$. $\square$

Proposition (6.12.9). — Let X be a locally Noetherian prescheme such that $\text{Reg}(X)$ is open; then, for every $n \geq 0$, the set $U_{R_n}(X)$ of the points $x \in X$ where X satisfies (R_n) is open.

Proof. Indeed (0, 8.2), $U_{R_n}(X)$ is the union of $\text{Reg}(X)$ and the set of $x \in \text{Sing}(X)$ such that the generic points z_i of the irreducible components of $\text{Sing}(X)$ containing x satisfy the relation $\dim(\mathcal{O}_{X,z_i}) \geq n + 1$ (since we have $z_i \notin \text{Reg}(X)$); in other words, these points $x \in \text{Sing}(X)$ are those for which $\text{codim}_x(\text{Sing}(X), X) \geq n + 1$ (0, 1.2); as the function $x \mapsto \text{codim}_x(\text{Sing}(X), X)$ is semi-continuous inferiorly (0, 14.2.6), the set of these points is open in $\text{Sing}(X)$, which completes the proof. $\square$

We also note the following elementary result:

Proposition (6.12.10). — Let X be a locally Noetherian prescheme. The set $U_{R_1}(X)$ is open in X . For $U_{R_0}(X)$ to be open in X , it is necessary and sufficient that every maximal point $x \in X$ such that $x \in U_{R_1}(X)$ (which means that X is reduced at x) be interior to $U_{R_1}(X)$.

Proof. By definition (0, 8.2), $U_{R_1}(X)$ is the set $X - \bigcup X'_\alpha$, where X'_α ranges over the set of irreducible components of X such that X is not reduced at their generic point; as the set of irreducible components of X is locally finite, $U_{R_1}(X)$ is open. Regarding $U_{R_0}(X)$, the condition of the statement being trivially necessary, let us prove that it is sufficient. Let X''_β be the irreducible components of X such that at their generic point x''_β the prescheme X is reduced; by hypothesis there exists an open $U \subset U_{R_1}(X)$ containing all the x''_β . Denote then by Z_λ the irreducible components of the closed set

$Z = X - U$ whose generic point z_λ is such that $\dim(\mathcal{O}_{X,z_\lambda}) \leq 1$ and that $\mathcal{O}_{X,z_\lambda}$ is non-regular. No point belonging to one of the Z_λ can belong to $U_{R_1}(X)$; but conversely, if $x \in Z$ does not belong to any of the Z_λ , then, for every generisation x' of x , either x' belongs to U , or $\dim(\mathcal{O}_{X,x'}) \geq 2$, or $\dim(\mathcal{O}_{X,x'}) = 1$ and, as x' does not belong to any of the Z_λ , $\overline{\{x'\}}$ is necessarily an irreducible component of Z , of codimension 1 in X , distinct from the Z_λ , hence $\mathcal{O}_{X,x'}$ is regular by definition. We conclude that $U_{R_1}(X) = X - \bigcup Z_\lambda$; as the set of the Z_λ is locally finite in Z , $\bigcup Z_\lambda$ is closed, which finishes proving that $U_{R_1}(X)$ is open in X . $\square$

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6.13. Criteria for $\text{Nor}(X)$ to be open

(6.13.1). Given a locally Noetherian prescheme X , we denote by $\text{Nor}(X)$ the set of $x \in X$ where X is *normal*; this set contains $\text{Reg}(X)$ and is contained in the (open) set of points x such that $\mathcal{O}_{X,x}$ is integral (i.e. the set of points x where X is reduced, and which belong to only a single irreducible component of X).

Proposition (6.13.2). — *Let X be a locally Noetherian prescheme, normalisé (II, 6.3.8), X' its normalisation. If the canonical morphism $f : X' \rightarrow X$ is finite, $\text{Nor}(X)$ is open in X .*

Proof. Indeed, on a $X' = \text{Spec}(f_*(\mathcal{O}_{X'}))$ and $f_*(\mathcal{O}_{X'})$ is a coherent $\mathcal{O}_X$ -Algebra by hypothesis (II, 6.1.3). To say that X is normal at a point x means that the canonical homomorphism $\mathcal{O}_x \rightarrow (f_*(\mathcal{O}_{X'}))_x$ is bijective (II, 6.3.4); but the set of such points is open since $\mathcal{O}_X$ and $f_*(\mathcal{O}_{X'})$ are coherent (0_I, 5.2.7). $\square$

Corollary (6.13.3). — *If A is an integral Noetherian Japanese ring, $\text{Nor}(\text{Spec}(A))$ is open in $\text{Spec}(A)$.*

Proposition (6.13.4). — *Let X be a locally Noetherian prescheme. For $\text{Nor}(X)$ to be open, it is necessary and sufficient that every maximal point x of X such that X is reduced at the point x (which means that $x \in \text{Nor}(X)$) be interior to $\text{Nor}(X)$.*

Proof. It suffices to prove the sufficiency of the condition. By virtue of the remark made in (6.13.1), we can restrict (by replacing X by an open subset of X in which X is reduced) to the case where X is *reduced*, i.e. suppose the maximal points of X belong to $U_{S_1}(X)$ and to $U_{R_1}(X)$; as $\text{Nor}(X) = U_{S_1}(X) \cap U_{R_1}(X)$ by Serre's criterion (0, 8.6), we deduce from (6.11.7) and (6.12.10) that these two sets are open; hence $\text{Nor}(X)$ is open in X . $\square$

Corollary (6.13.5). — *If X is such that $\text{Reg}(X)$ is open in X , then $\text{Nor}(X)$ is open in X .*

Proof. Indeed, every maximal point x is reduced (hence $\mathcal{O}_{X,x}$ is a field), hence belongs to $\text{Reg}(X)$, hence is interior to $\text{Reg}(X)$ by hypothesis, and *a fortiori* is interior to $\text{Nor}(X)$. $\square$

Proposition (6.13.6). — *Let A be an integral Noetherian ring, K its field of fractions, K' a finite extension of K , A' the integral closure of A in K' . For A' to be a finite A -algebra, it is necessary and sufficient that the two following conditions be satisfied:*

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- (i) *There exists $f \neq 0$ in A such that the integral closure A'_f of the ring of fractions A_f is a finite A_f -algebra.*
- (ii) *For every prime ideal $\mathfrak{p} \in \text{Spec}(A)$, the integral closure $A'_\mathfrak{p}$ of the local ring $A_\mathfrak{p}$ is a finite $A_\mathfrak{p}$ -algebra.*

Proof. The conditions are necessary, for if S is any multiplicative subset of A , $S^{-1}A'$ is the integral closure of $S^{-1}A$ in K' and is by hypothesis a $S^{-1}A$ -module of finite type. To see that the conditions are sufficient, consider the increasing filtrant family (B_λ) of sub- A -algebras of K' which are finite A -algebras and have K' as field of fractions. Set $Y = \text{Spec}(A)$, $X_\lambda = \text{Spec}(B_\lambda)$, and denote by u_λ the morphism $X_\lambda \rightarrow Y$, and let $S_\lambda = X_\lambda - \text{Nor}(X_\lambda)$, $T_\lambda = u_\lambda(S_\lambda)$. We can restrict to those B_λ which contain a finite system of generators of the A -module A'_f ; this ensures that $(B_\lambda)_f = A'_f$ for all λ , or equivalently that $u_\lambda^{-1}(D(f))$ is contained in $\text{Nor}(X_\lambda)$. By (6.13.2), S_λ is therefore *closed* in X_λ , and as u_λ is a finite morphism, T_λ is *closed* in Y . But for every $\mathfrak{p} \in \text{Spec}(A)$, there exists by hypothesis (ii) a λ such that $(B_\lambda)_\mathfrak{p} = A'_\mathfrak{p}$, and hence all the points of X_λ above $\mathfrak{p}$ belong to $\text{Nor}(X_\lambda)$; in other words on a $\bigcap_\lambda T_\lambda = \emptyset$; as Y is Noetherian and the T_λ are closed, there exists λ_0 such that $T_{\lambda_0} = \emptyset$, hence $B_{\lambda_0} = A'$. Q.E.D. $\square$

Proposition (6.13.7). — *Let A be a Noetherian ring, $X = \text{Spec}(A)$. The following conditions are equivalent:*

- a) For every prescheme X' locally of finite type over X , $\text{Nor}(X')$ is open in X' .
- b) For every finite integral A -algebra A' , $\text{Nor}(\text{Spec}(A'))$ is open in $\text{Spec}(A')$.
- c) For every prime ideal $\mathfrak{p}$ of A and every finite radical extension K' of the field of fractions K of $A/\mathfrak{p}$, there exists a sub- A -algebra A' of K' , finite over A , having K' as field of fractions and such that $\text{Nor}(\text{Spec}(A'))$ is open in $\text{Spec}(A')$.

Proof. The fact that a) implies b) and that b) implies c) is proved as in (6.12.4). To show that c) implies a), by (6.13.2), we are reduced as in (6.12.4) to proving that if A' is an integral A -algebra of finite type, the generic point of $\text{Spec}(A')$ is interior to $\text{Nor}(\text{Spec}(A'))$. We proceed as in (6.12.4) by first proving the following: $\square$

Lemma (6.13.7.1). — *Let A be an integral Noetherian ring, A' an integral A -algebra of finite type, containing A and such that the field of fractions K' of A' is a separable extension of the field of fractions K of A . If $\text{Nor}(\text{Spec}(A))$ is open in $\text{Spec}(A)$, then $\text{Nor}(\text{Spec}(A'))$ is open in $\text{Spec}(A')$.*

Proof. We must prove (by (6.13.2)) that the generic point of $\text{Spec}(A')$ is interior to $\text{Nor}(\text{Spec}(A'))$. The proof follows the same approach as that of (6.12.4), I), and we preserve the notations. The proof of (6.12.4), I) shows (by replacing A by an A_f with $f \neq 0$) that the fibres $g^{-1}(z)$ of the morphism $g : X' \rightarrow X$ are regular and *a fortiori* normal. Moreover g is flat and X is normal, hence (0, 4.2), (ii) X' is normal. This lemma being established, we pass to the general case as in (6.12.4), II), whose notations we still preserve; applying hypothesis c), we see each time that $\text{Nor}(X'_1)$ is open and we are reduced as before to the case where X' is normal and $g : X' \rightarrow X$ flat and surjective; we conclude each time that X' is normal with the aid of (0, 4.2), (i)). $\square$

6.14. Base change and integral closure

Proposition (6.14.1). — *Let Y, Y' be two locally Noetherian preschemes, $g : Y' \rightarrow Y$ a normal morphism (6.8.1). Then, for every normal Y -prescheme X , the prescheme $X' = X \times_Y Y'$ is normal.*

Proof sketch. We note that in this statement we do not suppose X locally Noetherian. The proof proceeds in several stages. $\square$

Lemma (6.14.1.1). — *Let R be a ring that is a finite direct product of a finite number of fields.*

- (i) *For a sub-ring A of R having R as its total ring of fractions to be normal, it is necessary and sufficient that it be integrally closed in R .*
- (ii) *Let (A_λ) be a family of normal sub-rings of R ; if $A = \bigcap_\lambda A_\lambda$ admits R as total ring of fractions, A is normal.*

Proof. (i) As A is a sub-ring of R , $A_{\mathfrak{p}}$ is a sub-ring of $R_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p}$ of A ; moreover $R_{\mathfrak{p}}$ is a ring of fractions of $A_{\mathfrak{p}}$, and is necessarily a finite direct product of fields, hence every element which is not a zero divisor of 0 in $R_{\mathfrak{p}}$ is invertible; this proves that $R_{\mathfrak{p}}$ is the total ring of fractions of $A_{\mathfrak{p}}$. If A is integrally closed in R , $A_{\mathfrak{p}}$ is hence integrally closed in $R_{\mathfrak{p}}$; but if $R_{\mathfrak{p}}$ is a finite direct product of 2 fields at least, the integral closure of $A_{\mathfrak{p}}$ in $R_{\mathfrak{p}}$ is a direct product of 2 rings at least which are not reduced to 0, which is absurd since $A_{\mathfrak{p}}$ is a local ring; hence $R_{\mathfrak{p}}$ is necessarily a field and $A_{\mathfrak{p}}$ is integral and integrally closed, which by definition means that A is normal. Conversely, if A is normal, $R_{\mathfrak{p}}$, total ring of fractions of an integral ring $A_{\mathfrak{p}}$, is a field and $A_{\mathfrak{p}}$ is integrally closed in $R_{\mathfrak{p}}$; if $x \in R$ is an integer over A , its image in each $R_{\mathfrak{p}}$ belongs to $A_{\mathfrak{p}}$, hence $x \in A$ (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 1, cor. 1 du th. 1), and hence A is integrally closed in R .

(ii) As $A \subset A_\lambda \subset R$ for each λ , R is also the total ring of fractions of each A_λ . In view of the characterisation of normal rings having R as total ring of fractions given in (i), assertion (ii) follows from Bourbaki, *Alg. comm.*, chap. V, § 1, n° 3, prop. 12. $\square$

The proof of (6.14.1) proceeds in several stages:

I) *Reduction to the case where $Y = \text{Spec}(A)$, $Y' = \text{Spec}(A')$, $X = \text{Spec}(B)$, A, A' are local Noetherian rings, A integral, B the integral closure of A .* — We must prove that for every $x' \in X'$, the local ring $\mathcal{O}_{X', x'}$ is integral and integrally closed; let x, y, y' be the images of x' in X, Y, Y' respectively; if we set $A = \mathcal{O}_{Y, y}$, $A' = \mathcal{O}_{Y', y'}$, $B = \mathcal{O}_{X, x}$, the rings $\mathcal{O}_{X', x'}$, for x, y, y' fixed, are the localisations of $B' = B \otimes_A A'$ at the prime ideals of B' (I, 3.6.5), and it suffices to prove that B' is a normal ring. Note

moreover that $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is a normal morphism by definition (6.8.1) and I, 3.6.5); one is hence reduced to the case of local schemes. Denote by (B_α) the family of *integral closures* of the sub- A -algebras of *finite type* of B ; it is clear that B is the union of the increasing filtrant family of the B_α . As $\varinjlim$ commutes with the tensor product, B' is hence isomorphic to $\varinjlim B'_\alpha$, where we have set $B'_\alpha = B_\alpha \otimes_A A'$. If we prove that C'_α is the integral closure of B'_α in C' , then it will follow that C' is the integral closure of A' in B' . We can hence restrict to the case where B is a A -algebra of finite type; as B is supposed *reduced*, it is sufficient to prove the claim when B is *integral* and is the *integral closure* of A .

II)-V) The remaining reductions and the completion of the proof are technically involved. In II), one reduces to the case where A is a local integral ring of dimension 1, B a discrete valuation ring, and the morphism $\text{Spec}(B) \rightarrow \text{Spec}(A)$ *radiciel*. In III), one establishes a key lemma:

Lemma (6.14.1.2). — *Let A be a local Noetherian integral ring of dimension 1, A' a local Noetherian ring, $A \rightarrow A'$ a local homomorphism such that the corresponding morphism $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is normal. Let K be the field of fractions of A , $\mathfrak{m}$ the maximal ideal of A , $k = A/\mathfrak{m}$ its residue field.*

- (i) *If $\mathfrak{q}'_j$ ($1 \leq j \leq r$) are the minimal ideals of A' , $K' = K \otimes_A A'$ is a finite direct product of integral and integrally closed rings $K'_j \supset A'/\mathfrak{q}'_j$ ($1 \leq j \leq r$), K_j having for field of fractions the field of fractions L'_j of $A'/\mathfrak{q}'_j$, so that the total ring of fractions L' of A' identifies to the direct product of the L'_j and A' to a sub-ring of K' .*
- (ii) *The ideal $\mathfrak{p}' = \mathfrak{m}A'$ of A' is prime; the ring $A'_{\mathfrak{p}'}$ is of dimension 1 and identifies to a sub-product of the fields L'_j for the indices j such that $\mathfrak{q}'_j \subset \mathfrak{p}'$.*
- (iii) *If $\overline{A'}_{\mathfrak{p}'}$ denotes the sub-ring of L' , product of $A'_{\mathfrak{p}'}$ and of the L'_j such that $\mathfrak{q}'_j \not\subset \mathfrak{p}'$, one has*

$$A' = K' \cap \overline{A'}_{\mathfrak{p}'}.$$

Lemma (6.14.1.4). — *Let R be a ring, $\mathfrak{a}$ and $\mathfrak{b}$ two ideals of R ; if $R/\mathfrak{a}$ and $R/\mathfrak{b}$ are Noetherian, so is $R/(\mathfrak{a} \cap \mathfrak{b})$.*

Proof. Indeed, let $\mathfrak{c}$ be an ideal of R such that $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{c}$; we have therefore $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{a} \cap \mathfrak{c} \subset \mathfrak{c}$; but $\mathfrak{c}/(\mathfrak{a} \cap \mathfrak{c})$ is an R -module isomorphic to $(\mathfrak{a} + \mathfrak{c})/\mathfrak{a}$, hence an ideal of $R/\mathfrak{a}$, and is therefore of finite type; on the other hand $(\mathfrak{a} \cap \mathfrak{c})/(\mathfrak{a} \cap \mathfrak{b})$ is a sub- R -module of $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$, and this last is isomorphic to $(\mathfrak{a} + \mathfrak{b})/\mathfrak{b}$, hence is of finite type as ideal of $R/\mathfrak{b}$; hence $(\mathfrak{a} \cap \mathfrak{c})/(\mathfrak{a} \cap \mathfrak{b})$ is of finite type, and it is so finally for $\mathfrak{c}/(\mathfrak{a} \cap \mathfrak{b})$. $\square$

The proof of (6.14.1) is completed using these lemmas along with the normality criteria of (6.8.2) and the results of (5.13.6). IV-2 | 173

Corollary (6.14.2). — *Let k be a field, X a normal k -prescheme (not necessarily locally Noetherian). Then, for every separable extension k' of k , $X_{(k')} = X \otimes_k k'$ is normal.*

Proof. We know indeed by (6.7.6) that the morphism $\text{Spec}(k') \rightarrow \text{Spec}(k)$ is then normal. $\square$

Corollary (6.14.3). — *Let k be a field, X, Y two integral k -preschemes, normal, whose fields of rational functions $K = R(X)$, $L = R(Y)$ are separable extensions of k . Then the prescheme $X \times_k Y$ is normal.*

Proof. The question being local on X and Y , we can suppose that $X = \text{Spec}(A)$, $Y = \text{Spec}(B)$, where A and B are two integral and integrally closed rings, of fields of fractions K and L respectively; suppose first that A is a k -algebra of finite type, hence K a finite type extension of k . By flatness, $A \otimes_k B$ is a sub- k -algebra of $K \otimes_k L$; but $K \otimes_k L$ is a Noetherian normal ring (by (6.7.4), I) or (6.14.2)), hence a finite direct product of integral rings C_i , so that if E_i is the field of fractions of C_i , the direct product E of the E_i is the total ring of fractions of $K \otimes_k L$ and also of $A \otimes_k B$. This shows that $A \otimes_k L$ and $K \otimes_k B$ are normal (as they are intersections of the C_i with appropriate direct factors), and it follows from (6.14.2) that $A \otimes_k B = (A \otimes_k L) \cap (K \otimes_k B)$, hence $A \otimes_k B$ is normal by (6.14.1.1).

In the general case, A is the union of the increasing filtrant family of its sub- k -algebras of finite type A_α , and $A \otimes_k B$ is the limit of the $A_\alpha \otimes_k B$. It follows from (0, 13.6) that to prove that every irreducible component of $\text{Spec}(A_\beta \otimes_k B)$ dominates an irreducible component of $\text{Spec}(A'_\alpha)$ for $\alpha \leq \beta$, it suffices to use (I, 3.7), (ii) and the fact that A_α and A_β are integral rings. $\square$

Proposition (6.14.4). — *Let A be a Noetherian ring, A' a Noetherian A -algebra such that the morphism $\mathrm{Spec}(A') \rightarrow \mathrm{Spec}(A)$ is normal. Let B be an A -algebra, C the integral closure of A in B ; set $B' = B \otimes_A A'$, $C' = C \otimes_A A'$, C' identifying to a sub-ring of B' ; then C' is the integral closure of A' in B' .*

Proof. The proof proceeds in several steps:

I) *Reduction to the case where B is reduced.* — Set $B_0 = B_{\mathrm{red}} = B/\mathfrak{N}$, where $\mathfrak{N}$ is the nilradical of B , and let C_0 be the integral closure of A in B_0 . By (6.14.4.1) below, the inverse image C of C_0 by the canonical homomorphism $\varphi : B \rightarrow B_0$ is the integral closure of A in B .

II) *Reduction to the case where B is an integral A -algebra of finite type containing A .* — Let (B_α) be the increasing filtrant family of sub- A -algebras of finite type of B , and let C_α be the integral closure of A in B_α . It follows from the definition of integral closure that C is the union of the C_α ; if we set $B'_\alpha = B_\alpha \otimes_A A'$, $C'_\alpha = C_\alpha \otimes_A A'$, C'_α is contained in B'_α and it suffices to show that C'_α is the integral closure of A' in B'_α .

III–V) The remaining cases reduce eventually to the case where A is a field and B is algebraically closed over A , where the conclusion follows from (6.14.1). $\square$

Lemma (6.14.4.1). — *Let A be a ring, B an A -algebra, $B_0 = B/\mathfrak{n}$ the quotient of B by a nilideal. If C_0 is the integral closure of A in B_0 , the inverse image C of C_0 by the canonical homomorphism $\varphi : B \rightarrow B_0$ is the integral closure of A in B .*

Proof. Indeed, if $x \in B$ is such that $\varphi(x)$ satisfies a monic integral dependence equation with coefficients in A , we deduce that x satisfies a relation of the form $x^n + a_1 x^{n-1} + \dots + a_n \in \mathfrak{n}$ with $a_i \in A$, whence, raising to a sufficiently large power, we obtain a monic integral dependence equation for x with coefficients in A . We have therefore an exact sequence of A -modules $0 \rightarrow C \rightarrow B \rightarrow B_0/C_0$, hence, by flatness, the exact sequence $0 \rightarrow C' \rightarrow B' \rightarrow B'_0/C'_0$, where we have set $B'_0 = B_0 \otimes_A A'$, $C'_0 = C_0 \otimes_A A'$. If we prove that C'_0 is the integral closure of A' in B'_0 , the lemma (6.14.4.1) will show that C' is the integral closure of A' in B' . $\square$

Corollary (6.14.5). — *Let A be a Noetherian ring, A' a Noetherian A -algebra such that the morphism $\mathrm{Spec}(A') \rightarrow \mathrm{Spec}(A)$ is normal. Let B, C be two A -algebras, $\varphi : B \rightarrow C$ an A -homomorphism making C a B -algebra, D the integral closure of B in C ; if we set $B' = B \otimes_A A'$, $C' = C \otimes_A A'$, $D' = D \otimes_A A'$, D' is the integral closure of B' in C' .*

Proof. Let (B_λ) be the increasing filtrant family of sub- A -algebras of finite type of B , and set $B'_\lambda = B'_\lambda \otimes_A A'$, so that B' is the union of the filtrant increasing family of the sub- A' -algebras of finite type B'_λ . Let D_λ be the integral closure of B_λ in C , and set $D'_\lambda = D_\lambda \otimes_A A'$; if we prove that D'_λ is the integral closure of B'_λ in C' , it will follow that D' is the integral closure of B' in C' . But B_λ and B'_λ are Noetherian and the morphism $\mathrm{Spec}(B'_\lambda) \rightarrow \mathrm{Spec}(B_\lambda)$ is normal (6.8.2); we can hence apply (6.14.4) by replacing A, A' and B by B_λ, B'_λ and C respectively, hence the corollary. $\square$

One can for example apply (6.14.5) when A is a local ring and A' its completion $\hat{A}$, a local excellent ring, since in this case $\mathrm{Spec}(A') \rightarrow \mathrm{Spec}(A)$ is a regular morphism (7.8.2).

6.15. Geometrically unibranch preschemes

(6.15.1). We say that a prescheme X is *unibranch* (resp. *geometrically unibranch*) at a point x , or that the point x is *unibranch* (resp. *geometrically unibranch*) in X if the local ring $\mathcal{O}_{X,x}$ is unibranch (resp. geometrically unibranch) (0, 23.2.1); we say that X is *unibranch* (resp. *geometrically unibranch*) if it is so at every point. It follows from this definition that, for X to be unibranch (resp. geometrically unibranch) at a point, it is necessary and sufficient that X_{red} be so at this point.

(6.15.2). It should be noted that with the definitions of (6.15.1), a local ring A can be unibranch (resp. geometrically unibranch) *without* $\text{Spec}(A)$ being so; in other words, it can happen that there exist prime ideals $\mathfrak{p}$ of A such that $A_{\mathfrak{p}}$ is not unibranch; this also amounts to saying that the notion of a geometrically unibranch point on a prescheme is *not stable by generisation*. We shall see this in the following example.

(6.15.3). We say, to abbreviate, that a morphism $f : X \rightarrow Y$ is *radiciel at a point* $y \in Y$, if $f^{-1}(y)$ is empty or reduced to a single point x and if $k(x)$ is a radiciel extension of $k(y)$. It amounts to the same to say that f is radiciel at all the points of Y , or that f is radiciel (I, 3.5.8). If $g : f^{-1}(y) \rightarrow \text{Spec}(k(y))$ is the morphism obtained from f by the base change $\text{Spec}(k(y)) \rightarrow Y$, it amounts to the same to say that f is radiciel at the point $y \in Y$, or that g is radiciel.

Lemma (6.15.3.1). — (i) Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms. If f is radiciel at a point y and g radiciel at the point $z = g(y)$, then $g \circ f$ is radiciel at the point $g(y)$. The converse is true if f is surjective.

(ii) Let $f : X \rightarrow Y$, $h : Y' \rightarrow Y$ be two morphisms, and let $f' = f_{(Y')} : X \times_Y Y' \rightarrow Y'$. Let $y' \in Y'$ and $y = h(y')$; then for f to be radiciel at the point y , it is necessary and sufficient that f' be radiciel at the point y' .

Proof. (i) If f is radiciel at y and g radiciel at $z = g(y)$, $g^{-1}(z)$ is reduced to y and $f^{-1}(g^{-1}(z)) = f^{-1}(y)$ is empty or reduced to a single point x ; moreover $k(x)$ is radiciel extension of $k(z)$, hence $k(x)$ is radiciel extension of $k(y)$. Conversely, if f is surjective and $g \circ f$ is radiciel at $z = g(y)$, $g^{-1}(z)$ is reduced to at most two points; but $f^{-1}(g^{-1}(z))$ would have at least two distinct points, a contradiction; furthermore, $f^{-1}(y) = f^{-1}(g^{-1}(z))$ is reduced to a single point x , and by hypothesis $k(z) \subset k(y) \subset k(x)$ with $k(x)$ radiciel over $k(z)$, hence $k(y)$ radiciel over $k(z)$ and $k(x)$ radiciel over $k(y)$.

(ii) Let $g : f^{-1}(y) \rightarrow \text{Spec}(k(y))$, $g' : f'^{-1}(y') \rightarrow \text{Spec}(k(y'))$ be the fibre morphisms of f and f' respectively; as $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$ (I, 3.6.4), g' is the base change of g , and the assertion follows from (I, 6.1), (v). $\square$

(6.15.4). Generalising the definition given in (I, 2.2.9), for two reduced preschemes X, Y , we say that a morphism $f : X \rightarrow Y$ is *birational* if the restriction of f to the set of maximal points of X is a bijection onto the set of maximal points of Y , and if, for every maximal point y of Y , the morphism $X \times_Y \text{Spec}(\mathcal{O}_y) \rightarrow \text{Spec}(\mathcal{O}_y)$ induced by f is an isomorphism (in other words, if the fibre $f^{-1}(y)$ is reduced to a single point x (maximal in X) and if the homomorphism $k(y) \rightarrow k(x)$ induced by f is bijective); this notion reduces to that of (I, 2.2.9) when X and Y have only a *finite* number of irreducible components.

Lemma (6.15.4.1). — Let $f : X \rightarrow Y$ be a morphism, $g : Y' \rightarrow Y$ a flat morphism. If f_{red} is birational, so is f'_{red} .

Proof. Indeed, if y' is a maximal point of Y' , we know (I, 3.4), (ii) that $y = g(y')$ is a maximal point of Y ; there exists a unique point x of X in $f^{-1}(y)$ by hypothesis; since moreover x is maximal in X and $k(x) = k(y)$. It follows therefore by (I, 3.4.9) that $f'^{-1}(y')$ is reduced to a single point x' and that $k(x') = k(y')$. We conclude by the remark that, by (I, 3.7), (ii), every irreducible component of X' dominates one irreducible component of Y' . $\square$

Proposition (6.15.5). — Let $f : X \rightarrow Y$ be a morphism such that f_{red} is entire and birational (6.15.4), hence surjective.

(i) For Y to be geometrically unibranch at a point y , it is necessary and sufficient that f be radiciel at the point y and that X be geometrically unibranch at the unique point x of $f^{-1}(y)$.

- (ii) For Y to be geometrically unibranch, it is necessary and sufficient that X be geometrically unibranch and that f be radiciel.

Proof. We can evidently restrict to the case where X and Y are *reduced*; the fact that f is surjective results from f being a closed morphism (II, 6.1.10) and from $f(X)$ containing all the maximal points of Y . This immediately shows that (i) implies (ii). Using (I, 3.6.5) and (II, 6.1.5), we can suppose, to prove (i), that $Y = \text{Spec}(\mathcal{O}_{X,y})$, and similarly $X = \text{Spec}(B)$. If A is geometrically unibranch, it is *integral*, and hence B is also integral (since f is birational). Conversely, if f is radiciel at y and X is geometrically unibranch at x , B has a single maximal ideal (since every ideal of A lies above the maximal of A and A is local), hence B is a local ring, *integral*. As f is dominant by hypothesis and A is reduced (I, 1.2.7), we conclude that A is also *integral*. Thus, to prove (i), it amounts to showing that A is integral and geometrically unibranch at point x means B is integral and B is the integral closure, i.e. the morphism $g : Y \rightarrow X$ is radiciel at the point x (6.15.3). $\square$

Proposition (6.15.6). — *Let k be a field, X a k -prescheme. If X is normal, then, for every extension k' of k , $X' = X \otimes_k k'$ is geometrically unibranch.*

Proof. We know that k' is the algebraic extension of a pure extension k_0 of k , and if k'' is the largest separable extension of k_0 contained in k' , k' is a radiciel extension of k'' . We know (6.14.2) that $X \otimes_k k''$ is normal; as $X'' \otimes_{k''} k' = X'' \otimes_{k''} k'$, we see that we can restrict to the case where the extension k' of k is *radiciel*. Moreover (I, 3.6.5), we can suppose that $X = \text{Spec}(A)$, where A is a local integral and integrally closed ring (since X is normal). The projection morphism $f : X' \rightarrow X$ is a universal homeomorphism, since $\text{Spec}(k') \rightarrow \text{Spec}(k)$ is a universal homeomorphism (I, 4.5); as $X' = \text{Spec}(A')$ where $A' = A \otimes_k k'$, we see that A' is a local ring whose nilradical $\mathfrak{N}'$ is the unique minimal prime ideal, hence $X'_{\text{red}} = \text{Spec}(A'_0)$, where $A'_0 = A' / \mathfrak{N}'$ is a local integral ring; moreover, if K is the field of fractions of A , the field of fractions K'_0 of A'_0 is radiciel over K , since the morphism f is radiciel. As A'_0 is entire over A , its integral closure B is also the integral closure of A in K'_0 ; moreover, every element of K'_0 whose p^m -th power (for some m) belongs to A belongs to K (for p the characteristic exponent of k); in particular B is an integral closure of A in K'_0 . But as A is integrally closed and K'_0 is radiciel over K , every element of K'_0 integral over A belongs to A (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 3, lemme 3), and *a fortiori* A'_0 is geometrically unibranch, and it is so also of X' . $\square$

Proposition (6.15.7). — *Let k be a field, X a k -prescheme, k' an extension of k , $X' = X \otimes_k k'$. Let x' be a point of X' , x its projection in X . For X to be geometrically unibranch at the point x , it is necessary and sufficient that X' be geometrically unibranch at the point x' . For X to be geometrically unibranch, it is necessary and sufficient that X' be so.*

Proof. The second assertion follows from the first and from the fact that the projection $f : X' \rightarrow X$ is surjective. To demonstrate the first assertion, we can, by (I, 5.1.8), restrict to the case where X is reduced, and by (I, 3.6.5), to the case where $X = \text{Spec}(A)$ (with $A = \mathcal{O}_{X,x}$) is a local scheme. Note that $A' = \mathcal{O}_{X',x'}$ is by hypothesis an A -module faithfully flat, hence containing A , and hence the hypothesis that X is geometrically unibranch at x , or the hypothesis that X' is geometrically unibranch at x' , both imply that A is a local integral ring (since A , being reduced, is also isomorphic to a sub-ring of A'_{red}). Let K be the field of fractions of A , B the integral closure of A , and set $Y = \text{Spec}(B)$; it amounts to saying that X is geometrically unibranch at the point x , or that the morphism $g : Y \rightarrow X$ is radiciel at the point x (6.15.3). Let $Y' = Y \otimes_k k' = Y \times_X X'$, so that we have the commutative diagram:

$$\begin{array}{ccc} Y & \xleftarrow{h} & Y' \\ \uparrow g & & \uparrow g' \\ X & \xleftarrow{f} & X' \end{array}$$

Note that f is a flat morphism; hence by (6.15.4.1) the entire morphism g'_{red} is birational. Moreover, as Y is normal, Y' is geometrically unibranch by (6.15.6). As f is entire and birational, f is radiciel at this point (6.15.5); on the other hand, x is the projection of x' in X and it is equivalent to say that g is radiciel at the point x or that g' is radiciel at the point x' (6.15.3.1); finally, for X to be geometrically unibranch at x , it suffices that g be radiciel at this point, hence the proposition. $\square$

Lemma (6.15.8). — *Let k be a separably closed field (in other words, such that the algebraic closure of k is radiciel over k), X a k -prescheme locally of finite type over k , x a closed point of X . If X is unibranch at the point x , it is geometrically unibranch at this point.*

Proof. Indeed, we know (I, 6.4.2) that $k(x)$ is an algebraic extension of k ; as the integral closure of $(\mathcal{O}_x)_{\text{red}}$ (integral closure which is by hypothesis a local ring) is an extension *algebraic* of $k(x)$, it is therefore an extension *radiciel* of $k(x)$, hence also of $k(x)$ by hypothesis, hence also of $k(x)$. $\square$

Corollary (6.15.9). — *Let k be a field, X a k -prescheme locally of finite type over k , k' a separably closed extension of k (in other words, such that the algebraic closure of k' is radiciel over k'); then, for X to be geometrically unibranch, it is necessary and sufficient that $X' = X \otimes_k k'$ be unibranch.*

Proof. By (6.15.7), we are reduced to proving that if X is unibranch and k separably closed, then X is geometrically unibranch. But X is geometrically unibranch at its closed points (6.15.8); we conclude from (10.4.9) that X is geometrically unibranch at all its points, relying on the fact (noted at the end of (6.15.2)) that the set of points where X is geometrically unibranch is constructible (the corollary (6.15.9) will of course not be used up to this point). $\square$

Proposition (6.15.10). — *Let Y, Y' be two locally Noetherian preschemes, $g : Y' \rightarrow Y$ a normal morphism (6.8.1), $f : X \rightarrow Y$ a morphism; set $X' = X \times_Y Y'$ and let $p : X' \rightarrow X$ be the canonical projection, x' a point of X' . Then, if X is reduced (resp. geometrically unibranch) at the point $x = p(x')$, X' is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch) at the point x' .*

Proof. By (I, 3.6.5), we can restrict to the case where $Y = \text{Spec}(A)$, $Y' = \text{Spec}(A')$, $X = \text{Spec}(B)$, where A, A' and B are local Noetherian rings, the homomorphisms $A \rightarrow A'$ and $A \rightarrow B$ being local, and x being the closed point of X ; it suffices to prove that $B' = B \otimes_A A'$ is reduced (resp. geometrically unibranch, resp. integral and geometrically unibranch at its points). Suppose first B reduced only; B is the union of the increasing filtrant family of its sub- A -algebras of finite type B_λ , and by flatness, B' is the union of the filtrant increasing family of the sub- A' -algebras of finite type $B'_\lambda = B_\lambda \otimes_A A'$. But, the morphism $\text{Spec}(B'_\lambda) \rightarrow \text{Spec}(B_\lambda)$ is normal (6.8.2), and *a fortiori* reduced; the fact that B'_λ is reduced follows then; one concludes that B' is itself reduced by (0, 13.2).

Suppose now B geometrically unibranch; by (I, 5.1.8), we can moreover suppose B reduced, hence integral. Let C be its integral closure, which is by hypothesis a local ring; set $Z = \text{Spec}(C)$, $C' = C \otimes_A A'$, $Z' = \text{Spec}(C') = Z \times_X X'$, so that we have the commutative diagram:

$$\begin{array}{ccc} Z & \xleftarrow{h} & Z' \\ \uparrow l & & \uparrow l' \\ X & \xleftarrow{p} & X' \end{array}$$

By the first part of the argument, X' is reduced; moreover, as $Z' = Z \times_Y Y'$, it follows from (6.14.1) that Z' is normal (and *a fortiori* geometrically unibranch). As f is entire and birational, so is f' by (6.15.4.1), since p is flat. Finally, the hypothesis that X is geometrically unibranch at x implies that f is radiciel at this point (6.15.5); we conclude by (6.15.3.1) that f' is radiciel at every point $x' \in p^{-1}(x)$, and it follows from (6.15.5) that X' is geometrically unibranch (hence integral since it is reduced) at each of these points. $\square$

Remarks (6.15.11). — (i) In the proof of (6.15.10), one cannot avoid appealing to (6.14.1), even when $X = Y$, as one makes use of the integral closure of the ring B , which is not necessarily a Noetherian ring even if B is.
 (ii) The example (0, 5.2), (ii) shows that in (6.15.10), one cannot replace “geometrically unibranch” by “integral” in the statement, even if the residue field $k(x)$ is algebraically closed and the morphism g is étale.

§7. RELATIONS BETWEEN A LOCAL NOETHERIAN RING AND ITS COMPLETION. EXCELLENT RINGS

7.1. Formal equidimensionality and formally catenary rings

Definition (7.1.1). — We say that a local Noetherian ring A is *formally equidimensional* if its completion $\hat{A}$ is equidimensional (0, 16.1.4).

Proposition (7.1.2). — Let A be a local Noetherian ring, $\mathfrak{p}_i$ ($1 \leq i \leq n$) its minimal prime ideals. For A to be formally equidimensional, it is necessary and sufficient that the $A/\mathfrak{p}_i$ be equidimensional (in other words that the $A/\mathfrak{p}_i$ have the same dimension).

Proof. Indeed, this follows from the fact that for each of the $\mathfrak{p}_i$, $\mathfrak{p}' \cap A$ contains one of the $\mathfrak{p}_i$, hence $\mathfrak{p}'$ contains one of the $\mathfrak{p}_i \hat{A}$, and $\hat{A}/\mathfrak{p}_i \hat{A} = (A/\mathfrak{p}_i) \otimes_A \hat{A}$ is the completion of the local ring $A/\mathfrak{p}_i$ (0I, 7.3.3), hence a maximal chain of prime ideals of $\hat{A}$ identifies therefore canonically to a chain of maximal prime ideals of one of the $\hat{A}/\mathfrak{p}_i \hat{A}$, and reciprocally, hence the conclusion. $\square$

Proposition (7.1.3). — Let A, A' be two local Noetherian rings, $\mathfrak{m}$ the maximal ideal of A , $\varphi : A \rightarrow A'$ a local homomorphism; suppose that A' is a flat A -module, and that A' is equidimensional and catenary. Then:

- (i) A is equidimensional and catenary.
- (ii) Suppose moreover that $\mathfrak{m}A'$ is an ideal of definition of A' . Then, if $\mathfrak{a}$ is an ideal of A , for $A'/\mathfrak{a}A'$ to be equidimensional, it is necessary and sufficient that $A/\mathfrak{a}$ be so. In particular, for every prime ideal $\mathfrak{p}$ of A , $A'/\mathfrak{p}A'$ is formally equidimensional.

Proof. Let us first prove the second assertion of (ii). Set $X = \text{Spec}(A)$, $X' = \text{Spec}(A')$, $Y = V(\mathfrak{p}) = \text{Spec}(A/\mathfrak{p})$, $Y' = V(\mathfrak{p}A') = \text{Spec}(A'/\mathfrak{p}A')$; if $f : X' \rightarrow X$ is the morphism corresponding to φ , Y' is also the closed sub-prescheme $f^{-1}(Y)$ of X' . As $\mathfrak{m}A'$ is an ideal of definition of A' , one has

$$(7.1.3.1) \quad \dim(X) = \dim(X')$$

by (6.1.3); similarly $\mathfrak{m}A'/\mathfrak{p}A'$ is an ideal of definition of $A'/\mathfrak{p}A'$ and Y' is flat over Y (I, 1.4), hence (6.1.3)

$$(7.1.3.2) \quad \dim(Y) = \dim(Y').$$

Let y be the generic point of Y , Y'_i ($1 \leq i \leq r$) the irreducible components of Y' , y'_i the generic point of Y'_i ($1 \leq i \leq r$); one has $f(y'_i) = y$ for all i (I, 3.4). On the other hand y'_i is a maximal point of the fibre $f^{-1}(y)$ (0I, 2.1.8), hence $\mathcal{O}_{X',y'_i} \otimes_{\mathcal{O}_{X,y}} k(y) = \mathcal{O}_{X',y'_i}/\mathfrak{m}_y \mathcal{O}_{X',y'_i}$ is of dimension 0, in other words $\mathfrak{m}_y \mathcal{O}_{X',y'_i}$ is an ideal of definition of $\mathcal{O}_{X',y'_i}$; one can again apply (6.1.3), which gives

$$(7.1.3.3) \quad \dim(\mathcal{O}_{X',y'_i}) = \dim(\mathcal{O}_{X,y})$$

that is to say (0, 1.2)

$$(7.1.3.4) \quad \text{codim}(Y'_i, X') = \text{codim}(Y, X)$$

for every i ; by (7.1.3.1), (7.1.3.2) and (7.1.3.4) and the inequality $\dim(Y'_i) \leq \dim(Y')$, one deduces **IV-2 | 184**

$$(7.1.3.6) \quad \dim(X) \leq \dim(Y) + \text{codim}(Y, X)$$

hence the two sides of this inequality are equal by (0, 14.2.2.2); moreover

$$(7.1.3.7) \quad \dim(Y'_i) = \dim(Y'_i) = \dim(Y)$$

for all i , which proves the second assertion of (ii).

Let us now demonstrate the first assertion of (ii). Let $\mathfrak{p}_j$ ($1 \leq j \leq n$) be the minimal prime ideals of A among those containing $\mathfrak{a}$, and let $\mathfrak{p}'_{jh}$ ($1 \leq h \leq m_j$) be the minimal prime ideals of A' among those containing $\mathfrak{p}_j A'$; we know then (I, 3.4) that the $\mathfrak{p}'_{jh}$ ($1 \leq j \leq n, 1 \leq h \leq m_j$ for every j) are also the minimal prime ideals of A' among those containing $\mathfrak{a}A'$. From what we have seen above, for every j , the dimensions of the rings $A'/\mathfrak{p}'_{jh}$ ($1 \leq h \leq m_j$) are all equal and are equal to $\dim(A'/\mathfrak{p}_j A')$, which itself is equal to $\dim(A/\mathfrak{p}_j)$ by (7.1.3.2). To say that $A'/\mathfrak{a}A'$ is equidimensional means therefore that the dimensions of the $A/\mathfrak{p}_j$ are all equal, that is to say that $A/\mathfrak{a}$ is equidimensional.

Let us now prove (i). Let $\mathfrak{r}'$ be a minimal prime ideal of A' among those containing $\mathfrak{m}A'$, and let $A'' = A'_{\mathfrak{r}'}$; as A'' is a flat A' -module, it is also a flat A -module; moreover A'' is equidimensional and catenary (0, 16.1.4), and $\mathfrak{m}A''$ is an ideal of definition of A'' . One can hence restrict to the case where $\mathfrak{m}A'$ is an ideal of definition of A' . If then $\mathfrak{p}$ and $\mathfrak{q} \subset \mathfrak{p}$ are two prime ideals of A , one can apply (ii) to $A/\mathfrak{q}$ and to $A'/\mathfrak{q}A'$, which is equidimensional and flat over $A/\mathfrak{q}$; if $Z = \text{Spec}(A/\mathfrak{q})$ and $Y = \text{Spec}(A/\mathfrak{p})$, one can apply (7.1.3.6), hence $\dim(Z) = \dim(Y) + \text{codim}(Y, Z)$; moreover one can

apply (7.1.3.6); as X is equicodimensional, these relations show that it is biequidimensional by (0, 14.3.3). $\square$

Corollary (7.1.4). — *Let A be a local Noetherian ring formally equidimensional. Then:*

- (i) *A is equidimensional and catenary (in other words, biequidimensional).*
- (ii) *For every ideal $\mathfrak{a}$ of A , for $A/\mathfrak{a}$ to be equidimensional, it is necessary and sufficient that $A/\mathfrak{a}$ be so. In particular, for every prime ideal $\mathfrak{p}$ of A , $A/\mathfrak{p}$ is formally equidimensional.*

Proof. If $\mathfrak{m}$ is the maximal ideal of A , $\mathfrak{m}\hat{A}$ is an ideal of definition of $\hat{A}$. By hypothesis $\hat{A}$ is equidimensional, and we know that it is catenary (0, 6.4); it therefore suffices to apply (7.1.3) to $A' = \hat{A}$. $\square$

Corollary (7.1.5). — *Let A be a local Noetherian ring such that there exists an A -module of finite type M which is an A -module of Cohen-Macaulay and for which $\text{Supp}(M) = \text{Spec}(A)$ (this is the case for example if A is a Cohen-Macaulay ring). Then A is formally equidimensional, hence (7.1.4) for every quotient ring B of A to be formally equidimensional, it is necessary and sufficient that it be equidimensional.*

Proof. Indeed, $\hat{M} = M \otimes_A \hat{A}$ is a $\hat{A}$ -module of Cohen-Macaulay (0, 16.5.2), of support equal to $\text{Spec}(\hat{A})$; hence (0, 16.5.4), $\hat{A}$ is equidimensional. $\square$

Remark (7.1.6). — We have seen (6.3.8) that under the hypotheses of (7.1.5), for every quotient ring B of A , the fibres of the canonical morphism $\text{Spec}(\hat{B}) \rightarrow \text{Spec}(B)$ are Cohen-Macaulay schemes; hence ((6.4.3) and (0, 7.5)), for B to have no immersed associated prime ideals, it is necessary and sufficient that it be so for $\hat{B}$.

Lemma (7.1.7). — *Let A be a local Noetherian ring, $\mathfrak{p}_i$ ($1 \leq i \leq r$) its minimal prime ideals. Suppose that each of the rings $A/\mathfrak{p}_i$ is formally equidimensional. Then, for every prime ideal $\mathfrak{q}$ of A and every i such that $\mathfrak{p}_i \subset \mathfrak{q}$, the ring $A_{\mathfrak{q}}/\mathfrak{p}_i A_{\mathfrak{q}}$ is formally equidimensional.*

Proof. As $A_{\mathfrak{q}}/\mathfrak{p}_i A_{\mathfrak{q}}$ is the local ring of $A/\mathfrak{p}_i$ at the prime ideal $\mathfrak{q}/\mathfrak{p}_i$, we can restrict to the case where A is integral and formally equidimensional. Set $A' = \hat{A}$, and let $\mathfrak{q}'$ be one of the minimal prime ideals of A' among those containing $\mathfrak{q}A'$; set $B = A_{\mathfrak{q}}$, $B' = A'_{\mathfrak{q}'}$. Set $C = \hat{B}$, $C' = \hat{B}'$; as B' is a B -module flat (0_I, 6.3.2), we know that C' is a C -module flat (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, prop. 4); as B' is catenary (0, 6.4) and equidimensional by hypothesis (by (7.1.3), (i)), it is the same for $B' = A'_{\mathfrak{q}'}$ (0, 16.1.4); moreover, as A' is isomorphic to a quotient of a regular ring by the theorem of Cohen (0, 19.8.8), it is the same of B' (0, 17.3.9); hence we conclude from (7.1.5) that C' is equidimensional. On the other hand C' is catenary (0, 6.4), hence C is equidimensional by (7.1.3), (i)). $\square$

Theorem (7.1.8). — *Let A be a local Noetherian ring. The following conditions are equivalent:*

- a) *Every integral quotient ring of A is formally equidimensional.*
- b) *The quotient rings of A by its minimal prime ideals are formally equidimensional.*
- c) *Every local ring B which is an A -algebra essentially of finite type (0, 3.8) and which is equidimensional, is formally equidimensional.*

Proof. It is trivial that c) implies a) and that a) implies b). On the other hand, as every integral prime ideal of A contains a minimal prime ideal of A , b) implies a) by (7.1.4), (ii). It remains to see that a) implies c).

It suffices to prove that the quotients of B by its minimal prime ideals are formally equidimensional (7.1.2); as every quotient ring of B is still an A -algebra essentially of finite type (0, 3.9), one can in the first instance suppose B integral. If $\mathfrak{q}$ is the inverse image in A of the maximal ideal of B , we know that B is also an $A_{\mathfrak{q}}$ -algebra essentially of finite type (0, 3.10), and it follows from a) and from (7.1.7) that every integral quotient of $A_{\mathfrak{q}}$ is formally equidimensional; one can hence suppose that the homomorphism $A \rightarrow B$ is a local homomorphism. The kernel $\mathfrak{r}$ of this homomorphism is then a prime ideal of A , and by a), every integral quotient of $A/\mathfrak{r}$ is formally equidimensional; as B is an $(A/\mathfrak{r})$ -algebra essentially of finite type, one sees that one can also suppose that A is integral and is a sub-ring of B .

Set $A' = \hat{A}$, and $C' = A'[T_1, \dots, T_n] = C \otimes_A A'$; there is a unique prime ideal $\mathfrak{p}'$ of C' above the maximal ideal $\mathfrak{m}A'$ of A' , hence above $\mathfrak{p}$; set $B' = C'_{\mathfrak{p}'}$; the homomorphism $B \rightarrow B'$ is local and

makes B' a B -module flat. We know then that $\widehat{B}'$ is *equidimensional*, from which we will deduce that $\widehat{B}$ is the same, which will complete the proof.

Indeed, A' is a quotient of a regular ring by the theorem of Cohen (0, 19.8.8), hence it is the same for B' (0, 17.3.9); by (7.1.5), in order to show that B' is *equidimensional*, it suffices to show that C' is *equidimensional*, since C' is a quotient of a regular ring. But the minimal prime ideals of C' are the ideals $\mathfrak{q}'C'$, where $\mathfrak{q}'$ ranges over the minimal prime ideals of A' (0, 5.3), and one has $C'/\mathfrak{q}'C' = (A'/\mathfrak{q}')[T_1, \dots, T_n]$. As A' is equidimensional by hypothesis (since A is integral), it is the same of C' by (0, 5.4). Q.E.D. $\square$

Definition (7.1.9). — When the equivalent conditions of (7.1.8) are satisfied, we say that A is a *formally catenary* ring. For a local Noetherian ring, to be formally equidimensional or formally catenary and equidimensional is therefore the same, by (7.1.2).

Proposition (7.1.10). — *Every quotient local ring of a local Cohen-Macaulay ring is formally catenary.*

Proof. This follows immediately from (7.1.8) and from (7.1.5) applied to the integral quotient rings of A . $\square$

Proposition (7.1.11). — (i) *A local Noetherian ring that is formally catenary is universally catenary.*
 (ii) *If A is a local Noetherian ring formally catenary, every local ring B which is an A -algebra essentially of finite type is formally catenary.*

Proof. Assertion (ii) follows immediately from condition c) of (7.1.8) and from the fact that if B is a local ring which is an A -algebra essentially of finite type, C is also an A -algebra essentially of finite type (0, 3.9). To prove (i), note first that, by (7.1.8), b), a local Noetherian ring formally catenary A is catenary, since the quotients of A by its minimal prime ideals are catenary (7.1.4). If now E is an A -algebra of finite type, it follows from what precedes and from (ii) that every local ring of E at a prime ideal is catenary, hence E is catenary. Hence A is universally catenary. $\square$

Remarks (7.1.12). — (i) It is not known whether, conversely, a local Noetherian ring which is universally catenary is always formally catenary.
 (ii) A local Noetherian ring A of dimension 1 is necessarily formally equidimensional, its maximal ideal not being able to be minimal (for in this case A would be of dimension 0). As $\dim(\widehat{A}) = 1$, this shows that A is even *formally equidimensional*, and *a fortiori* formally catenary (hence *universally catenary*). On the other hand, the local ring of dimension 2 defined in (0, 6.11), which is catenary but non-universally catenary, is *a fortiori* not formally catenary.

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Corollary (7.1.13). — *Let Y be a locally Noetherian prescheme irreducible of dimension 1, X an irreducible prescheme, $f : X \rightarrow Y$ a dominant morphism of finite type, ξ, η the generic points of X and Y respectively. Then, for every $y \in f(X)$, the dimensions of the irreducible components of $f^{-1}(y)$ are all equal to $\deg .\mathrm{tr}_{k(\eta)} k(\xi)$.*

Proof. By hypothesis, $f^{-1}(\eta)$ is irreducible of generic point ξ and of dimension $\deg .\mathrm{tr}_{k(\eta)} k(\xi)$ (0, 2.1). As Y is irreducible and of dimension 1, one has $\dim(\mathcal{O}_y) = 1$ for every $y \neq \eta$, and $\mathcal{O}_y$ is universally catenary by (7.1.12), (ii). If $y \in f(X)$ and if z is the generic point of an irreducible component Z of $f^{-1}(y)$, one has, by (0, 6.5)

$$\dim(Z) = 1 + \dim(f^{-1}(\eta)) - \dim(\mathcal{O}_{X,z}).$$

But one has $\dim(\mathcal{O}_{X,z}) \leq \dim(\mathcal{O}_y)$ (0, 16.3.9), and on the other hand, as z is not the generic point of X , $\dim(\mathcal{O}_{X,z}) > 0$; hence $\dim(\mathcal{O}_{X,z}) = 1$ and $\dim(Z) = \dim(f^{-1}(\eta))$. $\square$

7.2. Strictly formally catenary rings

(7.2.1). Notations (7.2.1). — For an *integral* Noetherian ring A , we denote by $A^{(1)}$ the intersection of the local rings $A_{\mathfrak{p}}$, where $\mathfrak{p}$ ranges over the set of prime ideals of A of height 1 (0, 10.17). If A is a local Noetherian integral ring, we denote by $A^{(\omega)}$ the intersection of the $A_{\mathfrak{p}}$ where this time $\mathfrak{p}$ ranges over the set of all prime ideals of A distinct from the maximal ideal.

We say that a local Noetherian ring A is *strictly equidimensional* if, for every associated prime ideal $\mathfrak{p} \in \text{Ass}(A)$, one has $\dim(A/\mathfrak{p}) = \dim(A)$; this amounts to saying that A is *equidimensional* and *without immersed associated prime ideals*.

(7.2.1.1). Example (7.2.1.1). — Let A be a local Noetherian ring of dimension 1. Then the following conditions are equivalent:

- a) A is without immersed associated prime ideals.
- a') $\hat{A}$ is without immersed associated prime ideals.
- b) A is strictly equidimensional.
- b') $\hat{A}$ is strictly equidimensional.
- c) A is a Cohen-Macaulay ring.

Indeed a) signifies that the maximal ideal $\mathfrak{m}$ of A is not associated to A . Since a) implies b). Inversely, b) implies that every associated prime ideal $\mathfrak{p} \in \text{Ass}(A)$ is distinct from $\mathfrak{m}$, hence minimal, and hence b) implies a). We already know that a) and c) are equivalent for a local ring of dimension 1 (0, 7.8), and that a') and b') are equivalent by the same reasoning. As $\dim(\hat{A}) = 1$, one sees also that a') and b') are equivalent to c).

Proposition (7.2.2). — Let A be a local Noetherian integral ring; set $A' = \hat{A}$, $X = \text{Spec}(A)$, $X' = \text{Spec}(A')$ and denote by $f : X' \rightarrow X$ the canonical morphism. Let a be the closed point of X , $j : X - \{a\} \rightarrow X$ the canonical injection. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, $\mathcal{F}' = f^*(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'}$; then the following conditions are equivalent:

- a) The $\mathcal{O}_X$ -Module $j_*(\mathcal{F}|_{X - \{a\}})$ is coherent.
- b) For every $x' \in \text{Ass}(\mathcal{F}')$, one has $\dim(\{x'\}) \geq 2$.

Proof. Let a' be the unique point of the fibre $f^{-1}(a)$, and let $j' : X' - \{a'\} \rightarrow X'$ be the canonical injection. As the morphism f is faithfully flat and quasi-compact, it is equivalent to say that $j_*(\mathcal{F}|_{X - \{a\}})$ is coherent and that $j'_*(\mathcal{F}'|_{X' - \{a'\}})$ is coherent (5.9.5). On the other hand, as A' is isomorphic to a quotient of a regular local ring by the theorem of Cohen (0, 19.8.8), it follows from (5.11.4) that condition b) of the statement is equivalent to the fact that $j'_*(\mathcal{F}'|_{X' - \{a'\}})$ is coherent. Hence the proposition. $\square$

Proposition (7.2.3). — Let A be a local Noetherian integral ring. The notations being those of (7.2.2), the following conditions are equivalent:

- a) $A^{(1)}$ is a finite A -algebra.
- b) For every closed part T of X of codimension ≥ 2 , if one denotes by $i : X - T \rightarrow X$ the canonical injection, $i_*(\mathcal{O}_T)$ is a coherent $\mathcal{O}_X$ -Module.
- c) For every $x' \in \text{Ass}(\mathcal{O}_{X'})$ and every closed part T of X of codimension ≥ 2 , one has $\text{codim}(f^{-1}(T) \cap \overline{\{x'\}}, \{x'\}) \geq 2$.

Proof. Set $Z = Z^{(2)}(X)$ (0, 10.13) and $Z' = f^{-1}(Z)$, which are stable by specialisation; the conditions a) and b) are equivalent respectively to the following properties: $a_1)$ $\mathcal{H}_{X/Z}^0(\mathcal{O}_X)$ is coherent; $b_1)$ $\mathcal{H}_{X/T}^0(\mathcal{O}_X)$ is coherent for every closed part T of X of codimension ≥ 2 . By (5.9.5), these two last properties are equivalent respectively to: $a'_1)$ $\mathcal{H}_{X'/Z'}^0(\mathcal{O}_{X'})$ is coherent; $b'_1)$ posing $T' = f^{-1}(T)$, $\mathcal{H}_{X'/T'}^0(\mathcal{O}_{X'})$ is coherent for every closed part T of X of codimension ≥ 2 . Since every point of $\text{Ass}(\mathcal{O}_{X'})$ projects to the generic point of X since f is a flat morphism (I, 3.2); since by definition this point does not belong to Z , we see that $\text{Ass}(\mathcal{O}_{X'})$ does not meet Z' ; the equivalence of $a'_1)$ and $b'_1)$ follows from (5.11.4), and this gives the equivalence of a), b) and c). $\square$

Proposition (7.2.4). — Let A be a local Noetherian ring and set $X = \text{Spec}(A)$. The following conditions are equivalent:

- a) For every integral quotient ring B of A , the ring $B^{(1)}$ is a finite B -algebra.
- b) For every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$ and every part Z , stable by specialisation, and such that for every $x \in \text{Ass}(\mathcal{F}) \cap (X - Z)$, one has $\text{codim}(\overline{\{x\}} \cap Z, \{x\}) \geq 2$, the $\mathcal{O}_X$ -Module $\mathcal{H}_{X/Z}^0(\mathcal{F})$ is coherent.
- c) For every coherent $\mathcal{O}_U$ -Module $\mathcal{G}$ (where $U = X - T$) such that, for every $x \in \text{Ass}(\mathcal{G})$, one has $\text{codim}(\{x\} \cap T, \{x\}) \geq 2$, $i_*(\mathcal{G})$ (where $i : U \rightarrow X$ is the canonical injection) is a coherent $\mathcal{O}_X$ -Module.
- d) For every integral quotient ring B of A and every ideal $\mathfrak{J}$ of height ≥ 2 in B , the ring $\bigcap_{\mathfrak{p} \not\supset \mathfrak{J}} B_{\mathfrak{p}}$ is a finite B -algebra.
- e) For every integral quotient ring B of A and every local integral ring $C = B_{\mathfrak{q}}$ and every prime ideal $\mathfrak{q}$ of B such that $\dim(C) \geq 2$, the ring $C^{(\omega)}$ is a finite C -algebra (or, what amounts to the same, if $Y = \text{Spec}(C)$, and $j : U \rightarrow Y$ is the complement in Y of the closed point of C and $j : U \rightarrow Y$ the canonical injection, $j_*(\mathcal{O}_U)$ is a coherent $\mathcal{O}_Y$ -Module).
- f) For every integral quotient ring B of A and every local ring $C = B_{\mathfrak{q}}$ and every prime ideal $\mathfrak{q}$ of B such that $\dim(C) \geq 2$, one has $\dim(\widehat{C}/\mathfrak{r}) \geq 2$ for every $\mathfrak{r} \in \text{Ass}(\widehat{C})$.

Theorem (7.2.5). — Let A be a local Noetherian ring. The following conditions are equivalent:

- a) For every integral quotient ring B of A , the completion $\widehat{B}$ is strictly equidimensional (7.2.1).
- b) A is formally catenary (7.1.9) and the fibres of the morphism $\text{Spec}(\widehat{A}) \rightarrow \text{Spec}(A)$ satisfy (S_1) (in other words, have no immersed associated prime cycle).
- c) A is universally catenary (0, 6.2) and for every integral quotient ring B of A , the ring $B^{(1)}$ is a finite B -algebra (cf. (7.2.4)).
- d) A is universally catenary and for every integral quotient ring B of A and every local integral ring $C = B_{\mathfrak{q}}$ and every prime ideal $\mathfrak{q}$ of B such that $\dim(C) \geq 2$, the ring $C^{(\omega)}$ is a finite C -algebra.
- e) A is universally catenary and for every integral quotient ring B of A and every local integral ring $C = B_{\mathfrak{q}}$ and every prime ideal $\mathfrak{q}$ of B such that $\dim(C) \geq 2$, the completion $\widehat{C}$ has no associated prime cycle of dimension 1.

Moreover, when these conditions are satisfied, then, for every quotient ring B of A which is strictly equidimensional, the completion $\widehat{B}$ is strictly equidimensional.

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Definition (7.2.6). — When a local Noetherian ring A satisfies the equivalent conditions of (7.2.5), we say that it is *strictly formally catenary*.

Corollary (7.2.7). — Let A be a local Noetherian ring such that there exists an A -module of finite type M which is a Cohen-Macaulay A -module and for which $\text{Supp}(M) = \text{Spec}(A)$; then A is strictly formally catenary.

Proof. Indeed, A is formally catenary (7.1.5) and (7.1.9), and on the other hand the fibres of $\text{Spec}(\widehat{A}) \rightarrow \text{Spec}(A)$ are Cohen-Macaulay preschemes (6.3.8), hence *a fortiori* satisfy (S_1) . $\square$

Corollary (7.2.8). — If A is a local Noetherian ring strictly formally catenary, every local ring B which is an A -algebra essentially of finite type is strictly formally catenary.

Proof. Indeed, B is formally catenary (7.1.11) and it follows from (7.4.4)⁴ that the fibres of $\text{Spec}(\widehat{B}) \rightarrow \text{Spec}(B)$ satisfy (S_1) , hence the conclusion. $\square$

Corollary (7.2.9). — Every local Noetherian ring of dimension 1 is strictly formally catenary.

Proof. Indeed, every integral quotient ring of such a ring A is of dimension 0 or 1, hence its completion is strictly equidimensional (7.2.1.1). $\square$

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Remarks (7.2.10). — (i) It follows from (7.2.7) and (7.2.8) that every quotient of a Cohen-Macaulay local ring is strictly formally catenary. A local Noetherian ring of dimension

⁴The corollary (7.2.8) is not used to demonstrate (7.4.4).

2 satisfying (S_2) is strictly formally catenary, since it is then a Cohen-Macaulay ring. Recall however that there are local Noetherian integral rings of dimension 2 which are not universally catenary, nor *a fortiori* strictly formally catenary (0, 6.11).

- (ii) It is not known whether a formally catenary ring (7.1.9) is strictly formally catenary; this is related to whether one knows whether for a local Noetherian integral ring, $\widehat{B}$ has no immersed associated prime ideals (6.4.3). We also do not know whether, in the equivalent conditions $c)$, $d)$, $e)$ of (7.2.5), one can replace the hypothesis that A is universally catenary by the hypothesis that A is catenary; one can show that this is so when A is henselian (18.9.6).

7.3. Formal fibres of local Noetherian rings

(7.3.1). We shall consider in this number and the next two properties $\mathbf{P}(Z, k)$ of the following form:

“ Z is a locally Noetherian prescheme on a field k , and for every $z \in Z$, one has $Q(\mathcal{O}_z, k)$ ”

where $Q(A, k)$ is a property of a local Noetherian ring A which is a k -algebra; one supposes that if k' is a field isomorphic to k , and if A' is a local Noetherian ring which is a k' -algebra di-isomorphic to A , the properties $Q(A, k)$ and $Q(A', k')$ are equivalent.

If X, Y are two locally Noetherian preschemes, we say that a morphism $f : X \rightarrow Y$ is a **P-morphism** if:

- 1° f is flat;
- 2° for every $y \in Y$, the property $\mathbf{P}(f^{-1}(y), k(y))$ is true.

Lemma (7.3.2). — *Let X, Y be two locally Noetherian preschemes, $f : X \rightarrow Y$ a morphism. The following properties are equivalent:*

- a) f is a **P-morphism**.
- b) For every $x \in X$, if one sets $y = f(x)$, then $\mathcal{O}_x$ is an $\mathcal{O}_y$ -module flat and the property $Q(\mathcal{O}_x \otimes_{\mathcal{O}_y} k(y), k(y))$ is true.
- c) For every $x \in X$, if one sets $y = f(x)$, the morphism $\text{Spec}(\mathcal{O}_x) \rightarrow \text{Spec}(\mathcal{O}_y)$ corresponding to $\mathcal{O}_y \rightarrow \mathcal{O}_x$ is a **P-morphism**.
- c') For every closed point $x \in X$, the morphism $\text{Spec}(\mathcal{O}_x) \rightarrow \text{Spec}(\mathcal{O}_y)$ is a **P-morphism**.

Proof. The equivalence of a) and b) follows from the definitions; it is the same for c), by (I, 2.4.2); finally the equivalence of b) and c') follows from the fact that for every $x \in X$, $\overline{\{x\}}$ contains a closed point (0, 1.11). $\square$

Corollary (7.3.3). — *Let A, B be two Noetherian rings, $\varphi : A \rightarrow B$ a homomorphism such that the corresponding morphism $\text{Spec}(B) \rightarrow \text{Spec}(A)$ is a **P-morphism**. Then, for every multiplicative subset S of A , the morphism $\text{Spec}(S^{-1}B) \rightarrow \text{Spec}(S^{-1}A)$ is a **P-morphism**.*

Proof. This follows immediately from (7.3.2) and from (I, 1.6.2). $\square$

(7.3.4). We shall always suppose in what follows that the property Q is such that the three following conditions are satisfied:

- (P_I) (*transitivity*). — If $f : X \rightarrow Y$ is a regular morphism (6.8.1) and $g : Y \rightarrow Z$ a **P-morphism**, then $g \circ f$ is a **P-morphism**.
- (P_{II}) (*descent*). — If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are morphisms of locally Noetherian preschemes such that f is faithfully flat and that $g \circ f$ is a **P-morphism**, then g is a **P-morphism**.
- (P_{III}) — For every field k , the property $\mathbf{P}(\text{Spec}(k), k)$ is true.

Remarks (7.3.5). — (i) Conditions (P_I) and (P_{III}) imply that every *regular morphism* is a **P-morphism**.

- (ii) Note that the hypotheses of (P_I) (resp. (P_{II})) imply that $h = g \circ f$ (resp. g) is flat (I, 2.13); the hypotheses of (P_I) or of (P_{II}) also imply that for every $z \in Z$, $f_z : h^{-1}(z) \rightarrow g^{-1}(z)$ is flat (I, 1.4); as for every $y \in g^{-1}(z)$, $f_z^{-1}(y)$ is isomorphic to $f^{-1}(y)$ (I, 3.6.4), this shows that it suffices, to verify conditions (P_I) and (P_{II}), to do so only when Z is the *spectrum of a field*.

- (iii) In certain cases, the property Q will be such that the following condition is also satisfied: (P'_I) — If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are two **P-morphisms**, then $g \circ f$ is a **P-morphism**.

(7.3.6). We shall say that the property $\mathbf{P}$ is *geometric* if it satisfies (in addition to (P_I) , (P_{II}) and (P_{III})) the condition:

(P_{IV}) (*finite extension of base field*). — If $\mathbf{P}(Z, k)$ is true, then for every extension k' of finite type of k , $\mathbf{P}(Z \otimes_k k', k')$ is true.

Lemma (7.3.7). — Let $f : X \rightarrow Y$ be a $\mathbf{P}$ -morphism of locally Noetherian preschemes, $g : Y' \rightarrow Y$ a morphism locally of finite type. Then, if $\mathbf{P}$ satisfies condition (P_{IV}) , the morphism $f' = f_{(Y')} : X \times_Y Y' \rightarrow Y'$ is a $\mathbf{P}$ -morphism.

Proof. Indeed, for every $y' \in Y'$, if one sets $y = g(y')$, $k(y')$ is an extension of finite type of $k(y)$ (I, 6.4.11) and $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$ (I, 3.6.4); it therefore suffices to apply (P_{IV}) . $\square$

(7.3.8). *Examples (7.3.8).* — The following properties $\mathbf{P}(Z, k)$ satisfy conditions (P_I) , (P_{II}) and (P_{III}) :

- (i) (also denoted (i_n)) Z is of coprofondéur $\leq n$.
- (ii) Z is a Cohen-Macaulay prescheme.
- (iii) (also denoted (iii_n)) Z satisfies (S_n) .
- (iv) Z is regular.
- (v) (also denoted (v_n)) Z satisfies (R_n) .
- (vi) Z is reduced.
- (vii) Z is normal.

For properties (ii) to (vii), this follows from (6.6.1) which, in fact, demonstrates the stronger condition (P'_I) . For (i), property (P_I) follows from (6.6.2); property (P_{II}) follows from (6.3.2) and from the fact that a regular prescheme is of coprofondéur 0.

Moreover, it follows from (6.7.1) that properties (i), (ii) and (iii) are *geometric*; and by (6.7.8), it is the same for the following:

- (iv') Z is geometrically regular.
- (v') (also denoted (v'_n)) Z possesses the geometric property (R_n) .
- (vi') Z is geometrically reduced.
- (vii') Z is geometrically normal.

Remarks (7.3.9). — (i) Each of properties (iv'), (v'_n) , (vi'), (vii') implies the corresponding property (iv), (v_n) , (vi), (vii). Property (iv) implies all of (i) through (vii), and property (iv') implies all the properties enumerated in (7.3.8). Let us also recall that (i_0) is equivalent to (ii); the conjunction of (iii_1) and of (v_0) (resp. of (iii_1) and (v'_0)) is equivalent to (vi) (resp. (vi')) (0, 8.5); finally the conjunction of (iii_2) and (v_1) (resp. (iii_2) and (v'_1)) is equivalent to (vii) (resp. (vii')) (0, 8.6).

- (ii) In all the examples of (7.3.8), the property $Q(\mathcal{O}_z, k)$ which serves to define $\mathbf{P}$ is such that, for every generisation z' of z in Z , $Q(\mathcal{O}_z, k)$ implies $Q(\mathcal{O}_{z'}, k)$: by (I, 3.4), it suffices to verify this for properties (i) to (vii) (and even (i) to (v) by the remark (7.3.9), (i)); this follows from (6.3.9) for (i), and finally from (0, 16.5.10 and 17.3.2) for (ii) and (iv).

(7.3.10). We shall say that a property $\mathbf{P}(Z, k)$ is of the *first type* if the property $Q(A, k)$ which serves to define $\mathbf{P}$ is a property $\mathbf{R}(A)$ not involving k and is true when A is a field. It is clear that in the examples of (7.3.8), properties (i) to (vii) are of the first type, properties (iv') to (vii') of the second type.

We shall say that $\mathbf{P}(Z, k)$ is of the *second type* if the property $Q(A, k)$ is of the form

“For every extension k' of finite type of k and every closed point z' of $\text{Spec}(A \otimes_k k')$ above the closed point of $\text{Spec}(A)$, one has $\mathbf{R}(\mathcal{O}_{z'})$ ”

$\mathbf{R}(A)$ being assumed true when A is a field.

(7.3.13). Given a semi-local Noetherian ring A , we call *formal fibres* of A the fibres of the canonical morphism $f : \text{Spec}(\hat{A}) \rightarrow \text{Spec}(A)$; for every prime ideal $\mathfrak{p}$ of A , the formal fibre at $\mathfrak{p}$ is then the prescheme $\text{Spec}(\hat{A} \otimes_A k(\mathfrak{p}))$; as the completion of the local ring $A/\mathfrak{p}$ is $\hat{A}/\mathfrak{p}\hat{A} = (A/\mathfrak{p}) \otimes_A \hat{A}$, one sees that the formal fibre of $A/\mathfrak{p}$ at the generic point (o) of $\text{Spec}(A/\mathfrak{p})$ is also the formal fibre of A at $\mathfrak{p}$.

The property **P** being defined as in (7.3.1), we shall say that *the formal fibres of A have property $\mathbf{P}$* , or that A is a **P**-ring, if, for every $x \in \operatorname{Spec}(A)$, $\mathbf{P}(f^{-1}(x), k(x))$ is true. As f is flat, this amounts to saying that f is a **P**-morphism.

Proposition (7.3.14). — *Let A be a semi-local Noetherian ring, $\mathfrak{m}_i$ ($1 \leq i \leq r$) its maximal ideals, and set $A_i = A_{\mathfrak{m}_i}$; for A to be a **P**-ring, it is necessary and sufficient that each of the A_i be so.*

Proof. Indeed, $\hat{A}$ is the direct product of the $\hat{A}_i$, hence the formal fibre of A at a point $x \in \operatorname{Spec}(A)$ is the sum of the formal fibres at x of those A_i such that $x \in \operatorname{Spec}(A_i)$. $\square$

Proposition (7.3.15). — *Let A be a semi-local Noetherian ring.*

- (i) *If A is a **P**-ring, every quotient ring of A is a **P**-ring.*
- (ii) *If $\mathbf{P}$ satisfies moreover condition $(\mathbf{P}_{\text{IV}})$ (7.3.6), then every finite A -algebra is a **P**-ring.*

Proof. For every ideal $\mathfrak{a}$ of A , $\hat{A} \otimes_A (A/\mathfrak{a})$ is the completion of $A/\mathfrak{a}$, and the formal fibres of $A/\mathfrak{a}$ are the formal fibres of A at the points of $V(\mathfrak{a})$, hence (i). On the other hand, if B is a finite A -algebra, $\hat{B} = B \otimes_A \hat{A}$, and (ii) follows from (7.3.7). $\square$

Proposition (7.3.16). — *Suppose that the property $\mathbf{P}$ is of the first (resp. second) type, defined from a property $\mathbf{R}$ (7.3.10). When $\mathbf{P}$ is of the second type, suppose in addition that the relation*

“for every finite extension k' of k , and every $z' \in Z \otimes_k k'$, $\mathbf{R}(\mathcal{O}_{z'})$ is true”

implies $\mathbf{P}(Z, k)$ for the examples (iv') to (vii') of (7.3.8), by (6.7.7) and (4.6.1).

Let A be a semi-local Noetherian ring. If the property $\mathbf{R}$ satisfies conditions $(\mathbf{R}_{\text{I}})$ and $(\mathbf{R}_{\text{II}})$, the following properties are equivalent:

- a) *A is a **P**-ring.*
- b) *For every integral quotient ring B of A (resp. every finite integral A -algebra B) and every prime ideal $\mathfrak{q}$ of $\hat{B}$ whose inverse image in B is 0, $\mathbf{R}(\hat{B}_{\mathfrak{q}})$ is true.*

If moreover $\mathbf{R}$ satisfies condition $(\mathbf{R}'_{\text{I}})$, properties a) and b) are also equivalent to:

- c) *For every integral quotient ring B of A (resp. every finite A -algebra B) if one sets $Y = \operatorname{Spec}(B)$, $Y' = \operatorname{Spec}(\hat{B})$, and if $g : Y' \rightarrow Y$ is the canonical morphism, one has (with the notations of (7.3.12))*

$$(7.3.16.1) \quad \mathbf{U}_{\mathbf{R}}(Y') = g^{-1}(\mathbf{U}_{\mathbf{R}}(Y)).$$

Proposition (7.3.17). — *Suppose that the property $\mathbf{R}$ satisfies conditions $(\mathbf{R}'_{\text{I}})$ and $(\mathbf{R}_{\text{II}})$ and that $\mathbf{P}$ is a property of the first (resp. second) type defined from $\mathbf{R}$ (7.3.10). Then, if A is a **P**-ring, $\mathbf{R}(A)$ and $\mathbf{R}(\hat{A})$ are equivalent.*

Proof. This follows from (7.3.12.1) applied to $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(\hat{A})$. $\square$

Proposition (7.3.18). — *Suppose that $\mathbf{P}$ is a property of the first (resp. second) type defined from a property $\mathbf{R}$ satisfying conditions $(\mathbf{R}'_{\text{I}})$ and $(\mathbf{R}_{\text{II}})$ as well as the following condition:*

- ($\mathbf{R}_{\text{III}}$) *For every local Noetherian complete ring C , if one sets $Z = \operatorname{Spec}(C)$, the set $\mathbf{U}_{\mathbf{R}}(Z)$ (7.3.12) is open in Z .*

*Then, if A is a **P**-ring and if one sets $X = \operatorname{Spec}(A)$, the set $\mathbf{U}_{\mathbf{R}}(X)$ is open in X .*

Proof. Indeed, if $X' = \operatorname{Spec}(\hat{A})$ and if $f : X' \rightarrow X$ is the canonical morphism, on a $\mathbf{U}_{\mathbf{R}}(X') = f^{-1}(\mathbf{U}_{\mathbf{R}}(X))$ (7.3.12). As f is faithfully flat and quasi-compact and by hypothesis $\mathbf{U}_{\mathbf{R}}(X')$ is open in X' , the conclusion follows from (I, 3.12). $\square$

Remarks (7.3.19). — (i) The property $\mathbf{R}$ which serves to define $\mathbf{P}$ satisfies condition $(\mathbf{R}_{\text{III}})$ in all the examples enumerated in (7.3.8). For (i), (ii) and (iii), this follows from (6.11.2) and from the theorem of Cohen (0, 19.8.8); when $\mathbf{P}$ is one of the properties (iv), (iv'), (v), (v'), this follows from (6.12.7) and (6.12.9), and when $\mathbf{P}$ is one of the properties (vii), (vii'), from (6.12.7) and (6.13.4); finally, for (vi) and (vi') the assertion of (7.3.18) is trivial, being true for every locally Noetherian prescheme.

- (ii) We have already noted (6.4.3) that when $\mathbf{P}$ is one of the properties (ii) or (iii_n) of (7.3.8), we do not know whether every local Noetherian ring is a **P**-ring; recall however that when A is a quotient of a Cohen-Macaulay ring, the formal fibres of A are Cohen-Macaulay preschemes (6.3.8).

- (iii) The most interesting case of the notion of **P**-ring is that which corresponds to the strongest property (iv') of (7.3.8), that is to say the rings whose *formal fibres are geometrically regular*. Fields, and more generally local Noetherian complete rings, trivially satisfy this property.
- (iv) Let A be a local Noetherian ring of *dimension* 1; $\text{Spec}(A)$ is then formed of the closed point a , corresponding to the maximal ideal $\mathfrak{m}$, and of the maximal points b_i ($1 \leq i \leq r$) corresponding to the minimal prime ideals of A . One has $\dim(\hat{A}) = 1$ and the maximal ideal of $\hat{A}$ is $\mathfrak{m}\hat{A}$; the formal fibre at a is then $\text{Spec}(k)$, where $k = A/\mathfrak{m}$ is the residue field; the formal fibre at each b_i is the spectrum of an Artinian ring whose residue fields are the residue fields L_{ij} of $\text{Spec}(\hat{A})$ at the maximal points b_{ij} above b_i . As an Artinian ring is a Cohen-Macaulay ring, one sees that A is a **P**-ring when **P** is property (ii) of (7.3.8). Moreover, as a reduced Artinian ring is a direct sum of fields, the following properties are equivalent (6.7.7):

- a) the formal fibres of A are geometrically reduced;
- b) the formal fibres of A are geometrically normal;
- c) the formal fibres of A are geometrically regular;

moreover, when A is reduced, they are also equivalent to:

- d) $\hat{A}$ is reduced and L_{ij} is a separable extension of K_i for every couple (i, j) (4.6.1).

In particular, if A is a *discrete valuation ring*, K its field of fractions, and if $\hat{K}$ is the completion of K for a valuation corresponding to A (field of fractions of $\hat{A}$), for the formal fibres of A to be geometrically regular, it is necessary and sufficient that $\hat{K}$ be a *separable extension* of K . This is always the case when K is of *characteristic* 0.

7.4. Permanence of properties of formal fibres

(7.4.1). In the remainder of this number, we suppose that the property **P** is of the form defined in (7.3.1), and *satisfies conditions* (P_I), (P_{II}) and (P_{III}) of (7.3.4). Moreover, we suppose that the property **Q** which serves to define **P** is such that, for every generisation z' of z in Z , $Q(\mathcal{O}_z, k)$ *implies* $Q(\mathcal{O}_{z'}, k)$.

Lemma (7.4.2). — *Let A, A' be local Noetherian rings, $\varphi : A \rightarrow A'$ a local homomorphism such that $f = {}^a\varphi : \text{Spec}(A') \rightarrow \text{Spec}(A)$ is a **P**-morphism. Then, if the formal fibres of A' are geometrically regular, A is a **P**-ring.*

Proof. Consider the completed homomorphism $\hat{\varphi} : \hat{A} \rightarrow \hat{A}'$ and the corresponding morphism $\hat{f} = {}^a\hat{\varphi}$; one has the commutative diagram

$$\begin{array}{ccc} \text{Spec}(\hat{A}) & \xleftarrow{\hat{f}} & \text{Spec}(\hat{A}') \\ g \uparrow & & g' \uparrow \\ \text{Spec}(A) & \xleftarrow{f} & \text{Spec}(A') \end{array}$$

where g and g' are the canonical morphisms. As by hypothesis f is a **P**-morphism and g' a regular morphism, it follows from (P_I) that $f \circ g' = g \circ \hat{f}$ is a **P**-morphism. On the other hand, the hypothesis that f is a **P**-morphism implies that f is flat, hence it is the same for $\hat{f}$ (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 4, cor. de la prop. 3), which is moreover a local homomorphism, hence faithfully flat (0_I, 6.6.2); it follows then from (P_{II}) that g is a **P**-morphism. $\square$

Corollary (7.4.3). — (i) *Let A be a **P**-ring local Noetherian, $A' = \hat{A}$ its completion. Suppose that for every prime ideal $\mathfrak{p}'$ of A' , the formal fibres of $A'_{\mathfrak{p}'}$ are geometrically regular; then, for every prime ideal $\mathfrak{p}$ of A , $A_{\mathfrak{p}}$ is a **P**-ring.*

- (ii) *Suppose that **P** satisfies condition (P_{IV}) (7.3.6). Let A be a **P**-ring local Noetherian, B a local A -algebra essentially of finite type, and such that the homomorphism $A \rightarrow B$ is local. Set $A' = \hat{A}$, and let $\mathfrak{n}'$ be the unique prime ideal of $B' = B \otimes_A A'$ above the maximal ideal of B and the maximal ideal of A' . If the formal fibres of $B'_{\mathfrak{n}'}$ are geometrically regular, then B is a **P**-ring.*

Theorem (7.4.4). — *The hypotheses on **P** being those of (7.4.1), let A be a **P**-ring local Noetherian.*

- (i) *For every prime ideal $\mathfrak{p}$ of A , $A_{\mathfrak{p}}$ is a **P**-ring.*

- (ii) Suppose moreover that $\mathbf{P}$ satisfies condition (P_{IV}) . Then every local ring which is an A -algebra essentially of finite type is a $\mathbf{P}$ -ring.

Corollary (7.4.5). — Suppose that the property $\mathbf{P}$ satisfies conditions (P_I) , (P_{II}) , (P_{III}) . The following conditions are equivalent:

- a) For every prime ideal $\mathfrak{p}$ of A , $A_{\mathfrak{p}}$ is a $\mathbf{P}$ -ring.
- b) For every maximal ideal $\mathfrak{m}$ of A , $A_{\mathfrak{m}}$ is a $\mathbf{P}$ -ring.

If moreover $\mathbf{P}$ satisfies (P_{IV}) , properties a) and b) are also equivalent to:

- c) For every A -algebra of finite type B and every prime ideal $\mathfrak{q}$ of B , $B_{\mathfrak{q}}$ is a $\mathbf{P}$ -ring.

The equivalence of a) and b) follows from (7.4.4), (i), and that of a) and c) from (7.4.4), (ii). When condition a) of (7.4.5) is satisfied, for a semi-local Noetherian ring, this definition coincides with that of (7.3.13). The ring $\mathbf{Z}$ is a $\mathbf{P}$ -ring when conditions (P_I) , (P_{II}) and (P_{III}) are satisfied. The ring $\mathbf{Z}$ is a $\mathbf{P}$ -ring (7.3.19), (iv)); every local Noetherian complete ring is a $\mathbf{P}$ -ring.

Proposition (7.4.6). — Suppose that the property $\mathbf{P}$ satisfies conditions (P_I) , (P_{II}) and (P_{III}) . Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A , $\hat{A}$ the separated completion of A for the $\mathfrak{J}$ -adic topology. Then, if A is a $\mathbf{P}$ -ring (7.4.5), the canonical morphism $\mathrm{Spec}(\hat{A}) \rightarrow \mathrm{Spec}(A)$ is a $\mathbf{P}$ -morphism.

Corollary (7.4.7). — Suppose that $\mathbf{P}$ satisfies conditions (P_I) , (P'_I) , (P_{II}) , (P_{III}) and (P_{IV}) . Then, if A is a $\mathbf{P}$ -ring (7.4.5), the canonical morphism $\mathrm{Spec}(A[[T_1, \dots, T_r]]) \rightarrow \mathrm{Spec}(A)$ is a $\mathbf{P}$ -morphism. In particular, if moreover A is integral, K its field of fractions, and if $\mathfrak{p}$ is a prime ideal of $B = A[[T_1, \dots, T_r]]$ such that $\mathfrak{p} \cap A = 0$, then the property $\mathbf{P}(B_{\mathfrak{p}}, K)$ is true.

Proof. Indeed, the canonical morphism $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ factorises as

$$\mathrm{Spec}(A[[T_1, \dots, T_r]]) \xrightarrow{f} \mathrm{Spec}(A[T_1, \dots, T_r]) \xrightarrow{g} \mathrm{Spec}(A).$$

It is clear that the morphism g is regular (0, 17.3.7); by (7.4.5), $A[T_1, \dots, T_r]$ is a $\mathbf{P}$ -ring, hence it follows from (7.4.6) that f is a $\mathbf{P}$ -morphism; as g is regular, hence also a $\mathbf{P}$ -morphism (7.3.5), (i), it is the same for $g \circ f$ by (P'_I) . $\square$

Remark (7.4.8). — The preceding results pose the following problems:

- A) Let A be a Zariski ring complete, and let $\mathfrak{J}$ be an ideal of definition of A ; if $A/\mathfrak{J}$ is a $\mathbf{P}$ -ring, is it the same for A ?
- B) Let k be a complete non-discrete valued field; we call the ring of restricted formal power series $k\{T_1, \dots, T_n\}$ the sub-ring of $k[[T_1, \dots, T_n]]$ formed by the series whose coefficients tend to 0. Is such a ring a $\mathbf{P}$ -ring?
- C) If A is a linearly topologised $\mathbf{P}$ -ring, S a multiplicative subset of A , are the rings $A\{S^{-1}\}$ (0_I, 7.6.1) and $A_{\{S\}}$ (0_I, 7.6.15) also $\mathbf{P}$ -rings?

7.5. A criterion for $\mathbf{P}$ -morphisms

(7.5.0). This number will not be used in the sequel of chapter IV, and can therefore be omitted on a first reading. We shall see later (7.9.8) that the results of this number can be considerably improved when one has the “resolution of singularities” at one’s disposal. IV-2 | 203

In the remainder of this number, we shall consider a property $\mathbf{R}(A)$, and we denote by $\mathbf{P}(Z, k)$ the property

“ Z is a locally Noetherian prescheme on a field k , and, for every finite extension k' of k , if one sets $Z' = Z \otimes_k k'$, the property $\mathbf{R}(\mathcal{O}_{Z'})$ is true for every $z' \in Z'$.”

Theorem (7.5.1). — Let A, B be two local Noetherian complete rings, $\mathfrak{m}$ the maximal ideal of A , $k = A/\mathfrak{m}$ its residue field, $\varphi: A \rightarrow B$ a local homomorphism such that:

- (i) The residue field of B is a finite extension of k .
- (ii) B is a flat A -module.

Let moreover $\mathbf{R}(C)$ be a property satisfying condition (R_{III}) (7.3.18) and the following condition:

- (R_{IV}) For every local ring C in a local Noetherian complete ring and every C -regular element t above the maximal ideal of C , the property $\mathbf{R}(C/tC)$ implies $\mathbf{R}(C)$.

Then, for every ideal $\mathfrak{p}$ of A , $\mathbf{R}(\widehat{B}_{\mathfrak{q}})$ is true for every prime ideal $\mathfrak{q}$ of $\widehat{B}$ above $\mathfrak{p}$, if and only if $\mathbf{R}(\widehat{A}_{\mathfrak{p}'})$ is true for every prime ideal $\mathfrak{p}'$ of $\widehat{A}$ above $\mathfrak{p}$ and the fibre of the morphism $\mathrm{Spec}(B) \rightarrow \mathrm{Spec}(A)$ over $\mathfrak{p}$ satisfies **P**.

7.6. Two applications to the finiteness of integral closure

(7.6.1). Let A be a local Noetherian integral ring. We shall say (following Nagata) that A is a *N-1 ring* (or a (N_1) -ring) if its integral closure A' is a finite A -module. We shall say that A is a *N-2 ring* (or a (N_2) -ring, or a *Japanese ring*) if, for every extension K' of finite degree of the field of fractions K of A , the integral closure A' of A in K' is a finite A -module.

Proposition (7.6.2). — *Let A be a local Noetherian integral ring whose formal fibres are geometrically reduced. Then A is a (N_1) -ring.*

Proposition (7.6.3). — *Let A be a local Noetherian integral ring whose formal fibres are geometrically normal. Then A is an (N_2) -ring.*

Corollary (7.6.4). — *Every local Noetherian complete integral ring is an (N_2) -ring (i.e. Japanese).*

Proof. Indeed, the formal fibres of a complete local ring are geometrically regular (7.3.19), (iii)), hence *a fortiori* geometrically normal. $\square$

Corollary (7.6.5). — *Let A be a Noetherian ring whose formal fibres are geometrically normal (in the sense of (7.4.5)). Then A is universally Japanese (0, 23.1.1).*

Proposition (7.6.6). — *Let A be a reduced local Noetherian ring, A' its normalisation, $f : X' = \mathrm{Spec}(A') \rightarrow X = \mathrm{Spec}(A)$ the corresponding morphism. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, U the complement of the closed point a in X , $U' = f^{-1}(U)$, $\mathcal{F}' = f^*(\mathcal{F})$; suppose that, for every point $x \in U'$, the $\mathcal{O}_x$ -module $\mathcal{F}'_x = \mathcal{F}_x \otimes A'_x$ is reflexive (i.e. that $\mathcal{F}'$ is reflexive on U'). Then the canonical homomorphism $\mathcal{F} \rightarrow f_*(f^*(\mathcal{F}))$ is bijective.*

Corollary (7.6.7). — *Under the hypotheses of (7.6.6), the canonical homomorphism $A \rightarrow \Gamma(U', \mathcal{O}_{X'})$ is bijective. Moreover, if A is integral and A' is finite over A , the normalisation A' of A identifies to $\Gamma(U, \mathcal{O}_X)$.*

7.7. Nagata rings

Definition (7.7.1). — We say that a local Noetherian ring A is a *Nagata ring* if the following two conditions are satisfied:

- (i) A is universally Japanese (0, 23.1.1).
- (ii) For every prime ideal $\mathfrak{p}$ of A and every finite extension of the field of fractions of $A/\mathfrak{p}$, the integral closure of $A/\mathfrak{p}$ in this extension has an open regular locus.

A Noetherian ring A is said to be a *Nagata ring* if, for every prime ideal $\mathfrak{p}$ of A , $A_{\mathfrak{p}}$ is a Nagata ring in the above sense.

Proposition (7.7.2). — *A local Noetherian ring A whose formal fibres are geometrically regular is a Nagata ring. In particular, every complete local Noetherian ring is a Nagata ring.*

Proposition (7.7.3). — *Let A be a Nagata ring. Then every A -algebra of finite type is a Nagata ring.*

7.8. Excellent rings

Definition (7.8.1). — A Noetherian ring A is said to be *quasi-excellent* if it satisfies the following conditions:

- (i) For every prime ideal $\mathfrak{p}$ of A , the formal fibres of $A_{\mathfrak{p}}$ are geometrically regular.
- (ii) For every A -algebra of finite type B , $\mathrm{Reg}(B)$ is open in $\mathrm{Spec}(B)$.

A Noetherian ring A is said to be *excellent* if it is quasi-excellent and universally catenary.

Proposition (7.8.2). — *A Noetherian ring A is quasi-excellent if and only if it satisfies the condition (i) of (7.8.1) and the following condition:*

- (ii') *For every integral quotient ring B of A , there exists a non-empty open subset of $\mathrm{Spec}(B)$ which is regular.*

Theorem (7.8.3). — (i) *Every quotient ring of a quasi-excellent (resp. excellent) ring is quasi-excellent (resp. excellent).*

- (ii) Every algebra of finite type over a quasi-excellent (resp. excellent) ring is quasi-excellent (resp. excellent).
- (iii) Every ring of fractions of a quasi-excellent (resp. excellent) ring is quasi-excellent (resp. excellent).
- (iv) If A is a quasi-excellent (resp. excellent) local ring, its completion $\hat{A}$ is quasi-excellent (resp. excellent).

Corollary (7.8.4). — Every complete Noetherian local ring is excellent. The ring $\mathbf{Z}$ is excellent. Every field is excellent. Every Dedekind ring whose field of fractions is of characteristic 0 is excellent.

Corollary (7.8.5). — Every algebra of finite type over an excellent ring is excellent. In particular, every algebra of finite type over $\mathbf{Z}$ or over a field is excellent.

Corollary (7.8.6). — Let A be an excellent ring. Then A is a Nagata ring (7.7.1). In particular, A is universally Japanese.

Proposition (7.8.7). — Let A be a quasi-excellent ring. Then, for every ideal $\mathfrak{J}$ of A , the $\mathfrak{J}$ -adic completion $\hat{A}$ is quasi-excellent.

Remark (7.8.8). — Let A be a quasi-excellent local ring, $\hat{A}$ its completion. Then the canonical morphism $\mathrm{Spec}(\hat{A}) \rightarrow \mathrm{Spec}(A)$ is a regular morphism (6.8.1). This is one of the key properties of excellent rings.

7.9. Excellent rings and resolution of singularities

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(7.9.1). Given a reduced locally Noetherian prescheme X , we call a morphism $f : X' \rightarrow X$ a *resolution morphism* for X if X' is regular and if f is proper and birational. When such a morphism exists, we say that one can *resolve the singularities* of X (or more simply that one can *resolve* X). For example, if A is a Japanese ring of dimension 1, one can resolve $X = \mathrm{Spec}(A)$, since the morphism $X' \rightarrow X$ (where X' is the normalisation of X) is finite (hence proper) and birational, and the local rings of X' are integrally closed rings of dimension 1, hence discrete valuation rings, and therefore regular.

(7.9.2). It is clear that if one can resolve X , then one can also resolve every prescheme induced on an open subset of X and every local prescheme $\mathrm{Spec}(\mathcal{O}_x)$ of X (II, 5.4.2). On the other hand, if one can resolve X , it is clear that there exists in X an everywhere dense open subset contained in $\mathrm{Reg}(X)$. These remarks show immediately that, if for every integral closed sub-prescheme Y of X and every integral Y -prescheme Y' finite and radicial over Y one can resolve Y' , then the affine open subsets of X satisfy condition (iii) of (7.8.2).

Proposition (7.9.3). — Let A be a reduced local Noetherian ring, and suppose that one can resolve $\mathrm{Spec}(A)$. Then the fibres of the canonical morphism $\mathrm{Spec}(\hat{A}) \rightarrow \mathrm{Spec}(A)$ at the maximal points of $\mathrm{Spec}(A)$ are regular.

Proof. Set $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(\hat{A})$, and let $g : X' \rightarrow X$ be the canonical morphism. Let $v : Y \rightarrow X$ be a resolution morphism; set $Y' = X' \times_X Y$, and let $f : Y' \rightarrow X'$ and $g' : Y' \rightarrow Y$ be the canonical projections, so that we have a commutative diagram

$$\begin{array}{ccc} X' & \xleftarrow{f} & Y' \\ g \uparrow & & \downarrow g' \\ X & \xleftarrow{v} & Y. \end{array}$$

It will suffice to prove that the prescheme Y' is regular. Indeed, as v is birational and of finite type, there is an everywhere dense open subset U of X such that the restriction $v^{-1}(U) \rightarrow U$ of v is an isomorphism and such that $v^{-1}(U)$ is everywhere dense in Y (I, 6.5.5); the preschemes induced respectively on the open subset $g^{-1}(U)$ of X' and on the open subset $g'^{-1}(v^{-1}(U))$ of Y' are therefore isomorphic. This proves that $g^{-1}(U)$ is regular, and *a fortiori* so are the fibres of g at the maximal points of X , which are contained in U .

Note that the morphism f is proper (II, 5.4.2), and in particular of finite type, hence Y' is Noetherian, since X' is. Let a' be the closed point of X' . To see that Y' is regular, it will suffice to prove that, for every $y' \in f^{-1}(a')$, the local ring $\mathcal{O}_{Y', y'}$ is regular. Indeed, then $f^{-1}(a') \subset \mathrm{Reg}(Y')$. But since X' is the spectrum of a complete local ring and f is of finite type, $\mathrm{Reg}(Y')$ is open in Y'

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(6.12.8); as f is closed and $f^{-1}(a') \subset \text{Reg}(Y')$, there exists a neighbourhood V' of a' in X' such that $f^{-1}(V') \subset \text{Reg}(Y')$. Since A is a local ring, necessarily $V' = X'$, hence $\text{Reg}(Y') = Y'$, and the proposition will be proved.

Now if a is the closed point of X , then $g^{-1}(a) = \{a'\}$ and the residue fields $k(a)$ and $k(a')$ are isomorphic; therefore $g'^{-1}(v^{-1}(a)) = f^{-1}(a')$, and the restriction $f^{-1}(a') \rightarrow v^{-1}(a)$ of g' is an isomorphism of these two fibres. Let $y = g'(y')$ and set $B = \mathcal{O}_{Y,y}$; if $\mathfrak{n}$ is the maximal ideal of B , what precedes shows that in the Noetherian ring $B' = B \otimes_A \hat{A}$ there is a unique maximal ideal $\mathfrak{n}'$ above $\mathfrak{n}$ and that $\mathcal{O}_{Y',y'} = B'_{\mathfrak{n}'}$. To prove that the local ring $B'_{\mathfrak{n}'}$ is regular, it suffices to show that its completion $(0, 17.1.5)$ is regular; but, by hypothesis, B is regular, hence so is its completion $\hat{B}$. The conclusion therefore follows from the following lemma. $\square$

Lemma (7.9.3.1). — *Let A, B be two local Noetherian rings, and let $\rho : A \rightarrow B$ be a local homomorphism such that the ring $B' = B \otimes_A \hat{A}$ is Noetherian. Then, for every maximal ideal $\mathfrak{n}'$ of B' above the maximal ideal $\mathfrak{n}$ of B and the maximal ideal of $\hat{A}$, the completions of B and of $B'_{\mathfrak{n}'}$ are isomorphic.*

Proof. Let $\mathfrak{m}$ be the maximal ideal of A and set $B'' = B'_{\mathfrak{n}'}$. For every integer $h > 0$, one has $B'/\mathfrak{m}^h B' = B \otimes_A (\hat{A}/\mathfrak{m}^h \hat{A})$; but $\hat{A}/\mathfrak{m}^h \hat{A}$ is isomorphic to $A/\mathfrak{m}^h$, so $B'/\mathfrak{m}^h B'$ is isomorphic to $B/\mathfrak{m}^h B$, and in particular is a local ring whose maximal ideal is $\mathfrak{n}'/\mathfrak{m}^h B'$. It follows that $B''/\mathfrak{m}^h B''$, isomorphic to $(B'/\mathfrak{m}^h B')_{\mathfrak{n}'}$, is B -isomorphic to $B'/\mathfrak{m}^h B'$, and finally to $B/\mathfrak{m}^h B$. In particular, the maximal ideal of B'' is equal to $\mathfrak{n}B''$, and $B''/\mathfrak{n}^h B''$ is isomorphic to $B/\mathfrak{n}^h$ by the preceding argument. Since $\hat{B}'' = \varprojlim (B''/\mathfrak{n}^h B'')$, the lemma is proved, and so is (7.9.3). $\square$

Corollary (7.9.4). — *Let A be a local Noetherian ring.*

- (i) *If, for every integral quotient ring B of A , one can resolve $\text{Spec}(B)$, then the formal fibres of A are regular.*
- (ii) *Suppose that, for every integral quotient ring B of A , and every integral ring B' containing B , finite as a B -algebra and whose field of fractions is radical over that of B , one can resolve $\text{Spec}(B')$; then the formal fibres of A are geometrically regular.*

Proof. Every formal fibre of A being a formal fibre of an integral quotient ring of A at the generic point of its spectrum, it is clear that (i) is an immediate consequence of (7.9.3), and (ii) follows from (i) and from (6.7.7). $\square$

Proposition (7.9.5). — *Let X be a locally Noetherian prescheme such that, for every integral prescheme Y finite over X , one can resolve Y . Then the rings of the affine open subsets of X satisfy conditions (ii) and (iii) of (7.8.2); if moreover X is universally catenary (5.6.3) (and in particular if X is locally immersible in a regular prescheme (5.6.4)), then X is an excellent prescheme (7.8.5).*

Proof. This follows immediately from (7.9.4) and (7.9.2). $\square$

Remark (7.9.6). — It is possible that the converse of (7.9.5) is true, that is, that the hypothesis that X is reduced and that the rings of the affine open subsets of X satisfy conditions (ii) and (iii) of (7.8.2) implies the possibility of resolving X (and hence also the possibility of resolving every reduced prescheme locally of finite type over X , by (7.4.4) and (6.12.4)). This is true, at least, if one restricts to reduced Noetherian preschemes whose residue fields have characteristic 0, as shown by recent results of Hironaka [35]. By contrast, in the general case, one so far knows how to resolve only algebraic schemes of dimension 2 over a perfect field. In view of the importance that resolution of singularities is coming to have in the topological study of schemes (notably as regards their homological and homotopical properties), this gives a special interest to the category of excellent rings or excellent preschemes.

One may also note in this connection that the less delicate part of Hironaka's arguments shows that, independently of any hypothesis on the characteristics of the residue fields, and using only properties (ii) and (iii) of (7.8.2) for the local rings of a prescheme X , the problem of resolving the singularities of X reduces to the case where $X = \text{Spec}(A)$, A being a complete local Noetherian integral ring. Consequently, if later results were to disprove the conjecture advanced above, and were therefore to lead to the formulation of more restrictive conditions on the local rings of X , this could hardly happen except through restrictive conditions imposed on complete local rings

(probably conditions concerning their residue fields, perhaps the condition $[k : k^p] < +\infty$ when p is the characteristic exponent of k).

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(7.9.7). Consider, on the one hand, a full subcategory $\mathcal{C}$ of the category of locally Noetherian preschemes and, on the other hand, a property $\mathbf{R}(A)$, subject to the following conditions:

- 1° For every $X \in \mathcal{C}$, every prescheme locally of finite type over X belongs to $\mathcal{C}$. For every Noetherian ring A such that $\text{Spec}(A) \in \mathcal{C}$ and every multiplicative subset S of A , one has $\text{Spec}(S^{-1}A) \in \mathcal{C}$.
- 2° For every $X \in \mathcal{C}$, the set $U_{\mathbf{R}}(X)$ of $x \in X$ such that $\mathbf{R}(\mathcal{O}_{X,x})$ is true is open in X .
- 3° For every local Noetherian ring A such that $\text{Spec}(A) \in \mathcal{C}$ and every regular element t in the maximal ideal of A , $\mathbf{R}(A/tA)$ implies $\mathbf{R}(A)$.

As in (7.5.0), denote by $\mathbf{P}(Z, k)$, for a field k and a k -prescheme $Z \in \mathcal{C}$, the following property:

“For every finite extension k' of k , all the local rings of $Z \otimes_k k'$ satisfy the property $\mathbf{R}$.”

We shall further suppose that $\mathbf{R}$ satisfies the following condition:

- 4° If $\mathbf{P}(Z, k)$ is true, then, for every extension k'' of finite type of k , all the local rings of $Z \otimes_k k''$ satisfy the property $\mathbf{R}$.

Note that these conditions are satisfied when one takes for $\mathcal{C}$ the category of excellent preschemes and for $\mathbf{R}$ one of the properties (i) to (viii) of (7.5.3). For condition 1°, this follows from (7.8.3), (ii), and for conditions 2° and 3°, from the arguments of (7.5.3), taking into account that every excellent ring is catenary. Finally, condition 4° follows, for (i), (ii) and (iii), from (6.7.1); for (iv), (v), (vi), from (6.7.7); and for (viii), from (4.6.1). As regards property (vii) of (7.5.3), the corresponding property $\mathbf{P}(Z, k)$ implies that Z is locally integral (being locally Noetherian) and that each of the integral sub-preschemes of which Z is the sum is geometrically integral, by (4.5.9) and (4.6.1); one again concludes condition 4° above in this case.

Proposition (7.9.8). — *Let A be a local Noetherian ring, k its residue field, B a local Noetherian ring such that $\text{Spec}(B) \in \mathcal{C}$, and $\varphi : A \rightarrow B$ a local homomorphism making B a flat A -module. Set $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$, and let $f : X \rightarrow Y$ be the morphism corresponding to φ . Suppose that:*

- 1° *the property $\mathbf{P}(\text{Spec}(B \otimes_A k), k)$ is true;*
- 2° *for every finite morphism $Y_1 \rightarrow Y$, one can resolve $(Y_1)_{\text{red}}$.*

Then, for every $y \in Y$, the property $\mathbf{P}(f^{-1}(y), k(y))$ is true.

Proof. Note that condition 2° of (7.9.8) is still satisfied when one replaces Y by the spectrum of a local ring at a maximal ideal of a finite A -algebra A' (7.9.2); on the other hand, by condition 1° of (7.9.7), the spectrum of a ring of fractions of $B \otimes_A A'$ also belongs to $\mathcal{C}$. The lemma (7.3.16.2) then shows (as in part I) of the proof of (7.5.2)) that it suffices to prove that, if Y is integral and if y is the generic point of Y , the local rings of the fibre $f^{-1}(y)$ satisfy $\mathbf{R}$.

This being so, by hypothesis there exists a regular prescheme Y' and a proper birational morphism $g : Y' \rightarrow Y$. As Y' is Noetherian and locally integral, the hypothesis that g is birational implies that Y' is integral. Let $X' = X \times_Y Y'$, so that we have a commutative diagram

$$(7.9.8.1) \quad \begin{array}{ccc} X & \xleftarrow{g'} & X' \\ & & \downarrow f' \\ Y & \xleftarrow{g} & Y' \end{array}$$

where f' and g' are the canonical projections. Taking condition 2° of (7.9.7) into account, the same argument as at the beginning of (7.9.3) shows that it is enough to prove that $U_{\mathbf{R}}(X') = X'$, then, denoting by a the closed point of X , that $g'^{-1}(a) \subset U_{\mathbf{R}}(X')$.

Let $x' \in g'^{-1}(a)$, and set $y' = f'(x')$, so that $b = g(y')$ is the closed point of Y . Then $f'^{-1}(y') = f^{-1}(b) \otimes_k k(y')$, and as $k(y')$ is an extension of finite type of k (since g is proper, hence of finite type), the hypothesis that $\mathbf{P}(f^{-1}(b), k)$ is true implies, by condition 4° of (7.9.7), that the same is true of

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$\mathbf{P}(f'^{-1}(y'), k(y'))$. But since $\mathcal{O}_{Y', y'}$ is a regular ring and $\mathrm{Spec}(\mathcal{O}_{X', x'}) \in \mathcal{C}$ by condition 1° of (7.9.7), the lemma (7.5.1.1) proves that $\mathbf{R}(\mathcal{O}_{X', x'})$ is true, which finishes the proof. $\square$

Corollary (7.9.9). — *Let A, B be two local Noetherian rings, and let $\varphi : A \rightarrow B$ be a local homomorphism making B a flat A -module. Suppose that A is integral and geometrically unibranch, and that for every finite integral A -algebra A' one can resolve $\mathrm{Spec}(A')$ (which will be the case, for example, if the formal fibres of A are geometrically regular and the residue field of A has characteristic 0, by Hironaka's results). Let k be the residue field of A , and suppose that $\mathrm{Spec}(B \otimes_A k)$ is geometrically pointwise integral (4.6.9). Then B is integral.*

Proof. We apply (7.9.8), taking for $\mathcal{C}$ the category of all locally Noetherian preschemes and for $\mathbf{R}$ the property of being integral. The arguments of (7.5.3) and (7.9.7) show that conditions 1°, 2°, 3° and 4° of (7.9.7) are satisfied in this case. The hypothesis on $B \otimes_A k$ and the proposition (7.9.8) show (with the notation of (7.9.8)) that the fibres $f^{-1}(y)$ are geometrically pointwise integral for $y \in Y$. *A fortiori* the $f^{-1}(y)$ for $y \in Y$ are reduced preschemes, and since A is integral (hence *a fortiori* reduced), we already see that B is reduced (3.3.5). It remains to prove that $X = \mathrm{Spec}(B)$ is irreducible. But the proof of (7.9.8) proves that X' is locally integral (being locally Noetherian). As the morphism $g' : X' \rightarrow X$ is surjective, it is enough to see that X' is irreducible, or (since X' is locally integral) that X' is connected. For this it suffices to prove that the fibre $g'^{-1}(a')$ is connected. Indeed, if this point is established, X' cannot be the sum of two non-empty open preschemes X'_1, X'_2 , for then one would have, say, $g'^{-1}(a) \cap X'_1 = \emptyset$; but the restriction $X'_1 \rightarrow X$ of g' is proper (since X'_1 is a closed sub-prescheme of X'), and $g'(X'_1)$ would be a non-empty closed subset of X not containing the closed point a , which is absurd. Now $g'^{-1}(a) = g^{-1}(b) \otimes_k k(a)$; but $g : Y' \rightarrow Y$ is proper and birational, so, in the Stein factorisation $Y' \rightarrow Y'' \rightarrow Y$ of g (III, 4.3.3), the finite morphism $Y'' \rightarrow Y$ is also birational and therefore Y'' is integral. Since A is assumed geometrically unibranch, it follows from (III, 4.3.4) that $g^{-1}(b)$ is geometrically connected, which finishes the proof. $\square$

Remarks (7.9.10). — (i) We shall show later (18.8.11) that, for a reduced local ring B to be geometrically unibranch, it is necessary and sufficient that, for every étale morphism $h : X' \rightarrow X = \mathrm{Spec}(B)$ and every point $x' \in X'$ above the closed point of X , $\mathcal{O}_{X', x'}$ be integral. It follows that if, in (7.9.9), one supposes not only that $\mathrm{Spec}(B \otimes_A k)$ is geometrically pointwise integral, but geometrically unibranch, then one can conclude that B is geometrically unibranch.

(ii) Let X, Y be two excellent preschemes, let $f : X \rightarrow Y$ be a flat morphism, and suppose that, for every finite morphism $Y_1 \rightarrow Y$, one can resolve $(Y_1)_{\mathrm{red}}$. It follows from (7.9.8), taking for $\mathbf{R}$ the property of being regular, that the set U of points $x \in X$ such that the fibre at the point $f(x)$ of the morphism

$$\mathrm{Spec}(\mathcal{O}_{X, x} \otimes_{\mathcal{O}_{Y, f(x)}} k(f(x))) \longrightarrow \mathrm{Spec}(k(f(x)))$$

is geometrically regular is stable under generisation. We do not know whether this set is open (or, equivalently (0, 9.2.5), whether it is constructible), even in the particular case where Y is the spectrum of $\mathbf{Z}$ or of a polynomial ring in one indeterminate $k[T]$ over a field k , and where consequently the “resolvability” condition imposed on Y is trivially satisfied.

§8. PROJECTIVE LIMITS OF PRESCHMES

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.8.1.1 The notion of a projective limit of preschemes was defined in (II, 2.4.1); if the ordered set $\mathbf{L}$ of indices is filtered and increasing, and if the morphisms of the projective system are *affine*, then the projective limit exists, and is a prescheme. The results of the present paragraph will essentially show that the algebraic geometry of $\mathcal{O}_y$ -preschemes of finite presentation (and the theory of Modules of finite presentation on these preschemes) is essentially equivalent to the algebraic geometry of preschemes of finite presentation over “sufficiently small” open neighborhoods of y . A particularly important case, and to a certain extent a classical one, is the one where Y is integral and where y is its generic point, so that $\mathcal{O}_y$ is nothing other than the field $\mathbf{R}(Y) = K$ of rational functions on Y . The results of the present paragraph thus allow reinterpreting algebraic geometry over K in terms of

algebraic geometry over non-empty open subsets “sufficiently small” of Y , that is to say, intuitively, in terms of “families” of geometric objects indexed by the points of such an open subset. This point of view has moreover been commonly used for a long time, not only in Algebraic Geometry over algebraically closed fields, but also in the study of arithmetic of varieties defined over a *number field* K (a finite extension of $\mathbf{Q}$), by considering the latter as the field of fractions of its ring of integers A (“the theory of *reduction modulo $\mathfrak{p}$* ” ($\mathfrak{p}$ a prime ideal of A); cf. (I, 3.7)). The results of §§ 8 and 9 thus provide, among other things, some of the foundations of the language of “reduction modulo $\mathfrak{p}$ ” in arithmetic.

We note that in the example considered here, the morphisms $S_\mu \rightarrow S_\lambda$ (for $\lambda \leq \mu$) are the *canonical open immersions* $U_\mu \rightarrow U_\lambda$, and *a fortiori* are *flat* morphisms (but not faithfully flat in general), which explains the interest of the statements that do not make such a restriction.

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. 8.1.2 b) Suppose that the A_λ are *fields*, so that $A = \varinjlim A_\lambda$ is also a field. This case arises generally when one starts from geometric data over a field K of any kind (for example k the prime subfield of K). There is then interest in considering K as an inductive limit of its sub-extensions that are of *finite type over k* , which allows in many cases reducing questions to the case where K is a finite type extension of k . Using also the method sketched in a), one can then reduce more generally to the case of a base ring A which is an *algebra of finite type over k* .

We note that in this example, the morphisms $S_\mu \rightarrow S_\lambda$ are *faithfully flat*.

c) Suppose one is interested in properties, local over Y , of preschemes of finite presentation Y over any prescheme Y , which one can then assume to be an affine scheme of ring A . There is then interest in considering A as an inductive limit of its sub-rings which are *$\mathbf{Z}$ -algebras of finite type*, which allows reducing many questions to the case where Y is the spectrum of such an algebra. This is the basis for the “Kroneckerian point of view” according to which Algebraic Geometry reduces to the Algebraic Geometry of preschemes of finite type over $\mathbf{Z}$ (which one sometimes qualifies as

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. 8.1.2.cont “Absolute Algebraic Geometry”). This example shows us in particular that for most “relative” questions on a prescheme of base Y , one can reduce to the case where Y is *noetherian*.

We note that in this example, in contrast to the preceding ones, the morphisms $S_\mu \rightarrow S_\lambda$ have in general no particular regularity property.

. 8.1.3 In example a) of (8.1.2), we saw that if Y is an integral prescheme of generic point y , $f : X \rightarrow Y$ a morphism of finite presentation, then, if the generic fibre $f^{-1}(y)$ is proper over $k(y)$, there is an open neighborhood U of y in Y such that $f^{-1}(U)$ is proper over U ; it follows *a fortiori* that for all $s \in U$, $f^{-1}(s)$ is proper over $k(s)$. One arrives at a need for a converse, ensuring that if $f^{-1}(s)$ is proper over $k(s)$ for “sufficiently many” points $s \in U$, then $f^{-1}(y)$ is proper over $k(y)$. For example, suppose that X and Y are preschemes algebraic over a field k algebraically closed (one can take for k the field $\mathbf{C}$ of complex numbers, to fix ideas); one sometimes needs to know that if, for all $s \in Y$, *rational over k* , the fibre $f^{-1}(s)$ is proper over $k(s)$, then $f^{-1}(y)$ is proper over $k(y)$, and consequently $f^{-1}(U)$ is proper over U for a neighborhood U of y . This statement will follow easily from the following: the set E of points $s \in Y$ such that $f^{-1}(s)$ is proper over $k(s)$ is *constructible* (and hence equal to all of Y if it contains the closed points of Y , by the Hilbert Nullstellensatz ((10.4.8))); this amounts to saying that if $f^{-1}(y)$ is *not proper* over $k(y)$, then there exists an open neighborhood U of y such that $f^{-1}(s)$ is not proper over $k(s)$ for all $s \in U$ (cf. ((9.6.1), (iv))). This example illustrates the interest of systematically developing *constructibility criteria* for the most important notions: this is what will be done in § 9.

8.1. Projective limits of preschemes

8.2.1 Let S_0 be a ringed space, $\mathbf{L}$ a preordered filtered increasing set, $(\mathcal{A}_\lambda, \varphi_{\mu\lambda})$ an inductive system of $\mathcal{O}_{S_0}$ -Algebras (not necessarily commutative), with $\mathbf{L}$ as index set. We know that, considered as an inductive system of $\mathcal{O}_{S_0}$ -Modules, $(\mathcal{A}_\lambda, \varphi_{\mu\lambda})$ admits an inductive limit $\mathcal{A}$; denote by $\varphi_\lambda : \mathcal{A}_\lambda \rightarrow \mathcal{A}$ the canonical homomorphism. Let $m_\lambda : \mathcal{A}_\lambda \otimes \mathcal{A}_\lambda \rightarrow \mathcal{A}_\lambda$ be the homomorphism of $\mathcal{O}_{S_0}$ -Modules that defines the multiplication in the $\mathcal{O}_{S_0}$ -Algebra $\mathcal{A}_\lambda$; the hypothesis on the $\varphi_{\mu\lambda}$ implies that the m_λ form an inductive system of homomorphisms, and as the functor $\varinjlim$ commutes with the tensor product, $m = \varinjlim m_\lambda$ is a homomorphism $\mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}$ of $\mathcal{O}_{S_0}$ -Modules; by passage to the limit on the commutative diagrams expressing the associativity of m_λ and the existence of the identity section in $\mathcal{A}_\lambda$, one sees that m defines on $\mathcal{A}$ a structure of $\mathcal{O}_{S_0}$ -Algebra and that φ_λ is a homomorphism of $\mathcal{O}_{S_0}$ -Algebras for all $\lambda \in \mathbf{L}$. Furthermore $\mathcal{A}$ is the inductive limit of the system $(\mathcal{A}_\lambda, \varphi_{\mu\lambda})$ in the category of $\mathcal{O}_{S_0}$ -Algebras, in other words, for every $\mathcal{O}_{S_0}$ -Algebra $\mathcal{B}$, the canonical map

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$$(8.2.1.1) \quad \mathrm{Hom}_{S_0\text{-Alg.}}(\mathcal{A}, \mathcal{B}) \longrightarrow \varprojlim_{\lambda} \mathrm{Hom}_{S_0\text{-Alg.}}(\mathcal{A}_\lambda, \mathcal{B})$$

which to every homomorphism $f : \mathcal{A} \rightarrow \mathcal{B}$ of $\mathcal{O}_{S_0}$ -Algebras associates the family $(f \circ \varphi_\lambda)$, is *bijective*. Indeed, one already knows that it is injective and identifies $\mathrm{Hom}_{S_0\text{-Alg.}}(\mathcal{A}, \mathcal{B})$ with a part of $\varprojlim \mathrm{Hom}_{S_0\text{-Mod.}}(\mathcal{A}_\lambda, \mathcal{B})$; it suffices to see that if (f_λ) is an inductive system of homomorphisms of $\mathcal{O}_{S_0}$ -Algebras, $f_\lambda : \mathcal{A}_\lambda \rightarrow \mathcal{B}$, its inductive limit $f : \mathcal{A} \rightarrow \mathcal{B}$, which is by definition a homomorphism of $\mathcal{O}_{S_0}$ -Modules, is also a homomorphism of $\mathcal{O}_{S_0}$ -Algebras; but this follows from passage to the inductive limit in the commutative diagram

$$\begin{array}{ccc} \mathcal{A}_\lambda \otimes \mathcal{A}_\lambda & \xrightarrow{f_\lambda \otimes f_\lambda} & \mathcal{B} \otimes \mathcal{B} \\ \uparrow m_\lambda & & \uparrow \\ \mathcal{A}_\lambda & \xrightarrow{f_\lambda} & \mathcal{B} \end{array}$$

and from the fact that the functor $\varinjlim$ commutes with tensor products. We note finally that if the $\mathcal{A}_\lambda$ are commutative $\mathcal{O}_{S_0}$ -Algebras, so is $\mathcal{A}$.

8.2.2 Suppose now that S_0 is a *prescheme*, and that the $\mathcal{A}_\lambda$ are *quasi-coherent* $\mathcal{O}_{S_0}$ -Algebras (commutative); we know then that $\mathcal{A} = \varinjlim \mathcal{A}_\lambda$ is a quasi-coherent $\mathcal{O}_{S_0}$ -Algebra (I, 4.1.1). Denote by S_λ (resp. $\mathcal{A}$) the spectrum of the $\mathcal{O}_{S_0}$ -Algebra $\mathcal{A}_\lambda$ (resp. $\mathcal{A}$) (II, 1.3.1), and let $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ (for $\lambda \leq \mu$) and $u_\lambda : S \rightarrow S_\lambda$ be the S_0 -morphisms corresponding to the homomorphisms $\varphi_{\mu\lambda}$ and φ_λ respectively (II, 1.2.7); it is clear that $(S_\lambda, u_{\lambda\mu})$ is a *projective system* in the category of S_0 -preschemes. We note that the $u_{\lambda\mu}$ and u_λ are affine morphisms (II, 1.6.2), hence *quasi-compact* and *separated*.

Proposition (8.2.3). — *With the notations of (8.2.2), the morphisms $u_\lambda : S \rightarrow S_\lambda$ make S a projective limit of the projective system $(S_\lambda, u_{\lambda\mu})$ in the category of preschemes. Furthermore, if $h : S_0 \rightarrow T$ is a morphism, making every S_0 -prescheme into a T -prescheme, S is also a projective limit of the system $(S_\lambda, u_{\lambda\mu})$ in the category of T -preschemes.*

Proof. It suffices to show that if X is any S_0 -prescheme, the canonical map

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$$(8.2.3.1) \quad \mathrm{Hom}_{S_0}(X, S) \longrightarrow \varprojlim_{\lambda} \mathrm{Hom}_{S_0}(X, S_\lambda)$$

which to every S_0 -morphism $v : X \rightarrow S$ associates the family $(u_\lambda \circ v)$, is *bijective*. Now, if $g : X \rightarrow S_0$ is the structural morphism and if one sets $\mathcal{B} = g_*(\mathcal{O}_X)$, which is an $\mathcal{O}_{S_0}$ -Algebra, the map (8.2.3.1) identifies canonically with (8.2.1.1) (II, 1.2.7), and the conclusion follows from what we saw in (8.2.1.1).

The other assertions of (8.1) are consequences of the following general lemma: □

Lemma (8.2.4). — *Let $\mathcal{C}$ be a category, T an object of $\mathcal{C}$, $\mathcal{C}_T$ the subcategory of objects of $\mathcal{C}$ above T . Let $(S_\lambda, u_{\lambda\mu})$ be a projective system in $\mathcal{C}_T$; then every projective limit of this system in $\mathcal{C}_T$ is also a projective limit in $\mathcal{C}$, and conversely.*

Proof. Let $f_\lambda : S_\lambda \rightarrow T$ be the structural morphism. Suppose that S is the projective limit of $(S_\lambda, u_{\lambda\mu})$ in $\mathcal{C}$, and denote by $u_\lambda : S \rightarrow S_\lambda$ the canonical corresponding morphisms. Consider then a projective

system of T -morphisms $w_\lambda : Y \rightarrow S_\lambda$, where $Y \in \mathcal{C}_T$. By hypothesis there exists a unique morphism (in $\mathcal{C}$) $w : Y \rightarrow S$ such that $w_\lambda = u_\lambda \circ w$ for all λ . The hypothesis that the u_λ are T -morphisms implies that the morphisms $f_\lambda \circ u_\lambda : S \rightarrow T$ are all the same, and this morphism f makes S into a T -object. If $g : Y \rightarrow T$ is the structural morphism of Y , one has $f \circ w = f_\lambda \circ u_\lambda \circ w = f_\lambda \circ w_\lambda = g$ for all λ , which proves that w is a T -morphism. Conversely, suppose (with the same notations) that S is a projective limit of $(S_\lambda, u_{\lambda\mu})$ in $\mathcal{C}_T$, and consider now a projective system of morphisms (of $\mathcal{C}$) $w_\lambda : Y \rightarrow S_\lambda$. The composed morphisms $f_\lambda \circ w_\lambda : Y \rightarrow T$ are then all the same: indeed, for any two indices λ, μ , there is an index ν such that $\lambda \leq \nu$ and $\mu \leq \nu$, hence $f_\nu = f_\lambda \circ u_{\lambda\nu} = f_\mu \circ u_{\mu\nu}$; as $w_\lambda = u_{\lambda\nu} \circ w_\nu$ and $w_\mu = u_{\mu\nu} \circ w_\nu$, one has $f_\lambda \circ w_\lambda = f_\lambda \circ u_{\lambda\nu} \circ w_\nu = f_\nu \circ w_\nu$ and similarly $f_\mu \circ w_\mu = f_\nu \circ w_\nu$. If $g : Y \rightarrow T$ is the unique morphism thus defined, g makes Y into a T -object, and the w_λ are T -morphisms; they thus have a projective limit $w : Y \rightarrow S$ which is a T -morphism, and *a fortiori* a morphism of $\mathcal{C}$; moreover the first part of the reasoning shows that every projective limit w' (in $\mathcal{C}$) of the projective system (w_λ) is necessarily also a T -morphism, hence equal to w , which finishes proving the lemma. $\square$

Proposition (8.2.5). — *Under the conditions of (8.2.2), let α be an element of $\mathbf{L}$, X_α an S_α -prescheme. For all $\lambda \geq \alpha$, set $X_\lambda = X_\alpha \times_{S_\alpha} S_\lambda$, and for $\alpha \leq \lambda \leq \mu$, set $v_{\lambda\mu} = 1_{X_\alpha} \times u_{\lambda\mu}$, so that $(X_\lambda, v_{\lambda\mu})$ is a projective system of X_α -preschemes, whose index set is formed of the $\lambda \geq \alpha$ in $\mathbf{L}$. Set also $X = X_\alpha \times_{S_\alpha} S$ and $v_\lambda = 1_{X_\alpha} \times u_\lambda$. Then the X_α -morphisms $v_\lambda : X \rightarrow X_\lambda$ make X a projective limit of the projective system $(X_\lambda, v_{\lambda\mu})$ in the category of X_α -preschemes, or in the category of all preschemes.*

Proof. This will follow from the following general lemma: $\square$

Lemma (8.2.6). — *Let $\mathcal{C}$ be a category in which fibered products exist, $q : T' \rightarrow T$ a morphism of $\mathcal{C}$, $\mathcal{C}_T$ (resp. $\mathcal{C}_{T'}$) the category of objects of $\mathcal{C}$ above T (resp. T'). Let $(S_\lambda, u_{\lambda\mu})$ be a projective system (not necessarily filtered) in $\mathcal{C}_T$, and set $S'_\lambda = S_\lambda \times_T T'$, $u'_{\lambda\mu} = u_{\lambda\mu} \times 1_{T'}$, so that $(S'_\lambda, u'_{\lambda\mu})$ is a projective system in $\mathcal{C}_{T'}$. Then, if (S, u_λ) is a projective limit of $(S_\lambda, u_{\lambda\mu})$ in $\mathcal{C}_T$, $(S \times_T T', u_\lambda \times 1_{T'})$ is a projective limit of $(S'_\lambda, u'_{\lambda\mu})$ in $\mathcal{C}_{T'}$.*

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Proof. By hypothesis, for all λ , one has a commutative diagram

$$\begin{array}{ccccc} S' & \xrightarrow{u'_\lambda} & S'_\lambda & \xrightarrow{h'_\lambda} & T' \\ p \downarrow & & p_\lambda \uparrow & & \uparrow q \\ S & \xrightarrow{u_\lambda} & S_\lambda & \xrightarrow{h_\lambda} & T \end{array}$$

where we have set $S' = S \times_T T'$, $u'_\lambda = u_\lambda \times 1_{T'}$, $h'_\lambda = h_\lambda \times 1_{T'}$. Let Y be a T' -object, and consider a projective system of T' -morphisms $w'_\lambda : Y \rightarrow S'_\lambda$, where $w'_\lambda = p_\lambda \circ w'_\lambda$ are T -morphisms from Y to S_λ . The hypothesis that the u_λ are T -morphisms implies that the morphisms $f_\lambda \circ u_\lambda : S \rightarrow T$ are all the same, and this morphism f makes S into a T -object. By definition of the fibered product, there exists therefore by hypothesis a unique T -morphism $w : Y \rightarrow S$ such that $u_\lambda \circ w = w_\lambda$ for all λ . By definition of the fibered product, there is a unique T' -morphism $w' : Y \rightarrow S'$ such that $p \circ w' = w$. One has then $u'_\lambda \circ w' = p_\lambda \circ w'_\lambda$ for each λ ; on the other hand, writing that w' is a T' -morphism and w'_λ a T' -morphism, one sees that $h'_\lambda \circ u'_\lambda \circ w' = g' = h'_\lambda \circ w'_\lambda$. The definition of S'_λ as fibered product $S_\lambda \times_T T'$, and the fact that w' is a T' -morphism, imply that w' is the unique T' -morphism verifying these relations, whence the lemma. $\square$

Remark (8.2.7). — Given any ringed space S , the inductive limits with respect to any preordered (not necessarily filtered) set $\mathbf{L}$ of (not necessarily commutative) $\mathcal{O}_S$ -Algebras exist by (8.2.1), and on the other hand, for two homomorphisms of $\mathcal{O}_S$ -Algebras $\mathcal{A} \rightarrow \mathcal{B}$, $\mathcal{A} \rightarrow \mathcal{C}$, the tensor product $\mathcal{B} \otimes_{\mathcal{A}} \mathcal{C}$ is the “amalgamated sum” in this category.

When S is a prescheme, the tensor product $\mathcal{B} \otimes_{\mathcal{A}} \mathcal{C}$ is a quasi-coherent $\mathcal{O}_S$ -Algebra whenever $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ are (I, 1.3.13); one concludes that, in the category of quasi-coherent $\mathcal{O}_S$ -Algebras, the inductive limits for any preordered index set always exist. This allows generalizing the definition of a projective limit of preschemes and the propositions (8.2.3) and (8.2.5) to the case where the preordered set $\mathbf{L}$ is not necessarily filtered.

. 8.2.8 With the notations of (8.2.2), set $u_{\lambda\mu} = (\psi_{\lambda\mu}, \theta_{\lambda\mu})$ and $u_\lambda = (\psi_\lambda, \theta_\lambda)$; $(S_\lambda, \psi_{\lambda\mu})$ is then a projective system of topological spaces and (ψ_λ) a projective system of continuous maps from the spaces underlying S and S_λ respectively.

Proposition (8.2.9). — *With the notations of (8.2.8), the projective limit of the projective system (ψ_λ) of continuous maps is a homeomorphism from the space underlying S to the projective limit of the projective system $(S_\lambda, \psi_{\lambda\mu})$ of topological spaces.*

Proof. Let T be the topological limit space of the system $(S_\lambda, \psi_{\lambda\mu})$ and set $\psi = \varprojlim \psi_\lambda : S \rightarrow T$. One can reduce to the case where $S_0 = S_\alpha$ for an $\alpha \in \mathbf{L}$, and $\lambda \geq \alpha$. IV-3 | 11

(i) Let us show first that the topology of S is the inverse image by ψ of the topology of T ; in other words, if $\pi_\lambda : T \rightarrow S_\lambda$ is the canonical map, the map $\psi^{-1}(\pi_\lambda^{-1}(U_\lambda))$, where U_λ runs through the open subsets of S_λ . Now every open subset of S is by definition a union of open subsets obtained as follows: one considers an affine open U_0 of S_0 , of ring A_0 , so that $\psi_\alpha^{-1}(U_0)$ is the open affine of S of ring $A = \Gamma(U_0, \mathcal{A})$, and one takes an element $f \in A$ and one considers in $\text{Spec}(A)$, identified with $\psi_\alpha^{-1}(U_0)$, the open subset $D(f)$. These open subsets $D(f)$ form a base of the topology of S (II, 1.3.1). Now, if one sets $A_\lambda = \Gamma(U_0, \mathcal{A}_\lambda)$, one has $A = \varinjlim A_\lambda$ (I, 1.3.9), hence there exists an index λ such that f is the canonical image of an element $f_\lambda \in A_\lambda$; one has then $D(f) = \psi_\lambda^{-1}(D(f_\lambda))$ (I, 1.2.2), and as $\psi_\lambda = \pi_\lambda \circ \psi$, our assertion is shown.

(ii) Let us show now that ψ is *bijective*, which will finish the proof. As S is a Kolmogoroff space, it follows already from (i) that ψ is injective, and it remains to show that ψ is surjective. One can evidently replace T by an open $\pi_\alpha^{-1}(U_0)$, where U_0 is open affine in $S_\alpha = S_0$, hence one can reduce to the case where the S_λ and S are affine, in other words $\mathcal{A}$ is the sheaf associated to an A_0 -algebra A_λ , and $A = \varinjlim A_\lambda$ the canonical homomorphisms. By definition, an element of T is a family $(\mathfrak{p}_\lambda)_{\lambda \in \mathbf{L}}$ where $\mathfrak{p}_\lambda$ is a prime ideal of A_λ and where $\mathfrak{p}_\lambda = \varphi_{\mu\lambda}^{-1}(\mathfrak{p}_\mu)$ for $\lambda \leq \mu$. One knows then ((5.13.3) and (5.13.1)) that there is a prime ideal $\mathfrak{p}$ of A such that $\mathfrak{p}_\lambda = \varphi_\lambda^{-1}(\mathfrak{p})$ for all $\lambda \in \mathbf{L}$, which finishes the proof. □

In particular, one has thus shown the

Corollary (8.2.10). — *Let $(A_\lambda)_{\lambda \in \mathbf{L}}$ be a filtered inductive system of rings, and let $A = \varinjlim A_\lambda$, $\varphi_\lambda : A_\lambda \rightarrow A$ the canonical homomorphisms. The canonical map $\mathfrak{p} \mapsto (\varphi_\lambda^{-1}(\mathfrak{p}))$ is a homeomorphism from $\text{Spec}(A)$ to the topological space $\varprojlim \text{Spec}(A_\lambda)$.*

Corollary (8.2.11). — *With the notations of (8.2.8), for every quasi-compact open subset U of S , there exists an index λ and a quasi-compact open subset U_λ of S_λ such that $U = \psi_\lambda^{-1}(U_\lambda)$.*

Proof. This follows from the fact that, by definition of the projective limit topology, the $\psi_\lambda^{-1}(U_\lambda)$ (U_λ open quasi-compact of S_λ) form a base of the topology of S , and since the index set $\mathbf{L}$ is filtered. □

Corollary (8.2.12). — *With the notations of (8.2.8), the inductive limit of the system of homomorphisms $\theta_\lambda^\# : \psi_\lambda^*(\mathcal{O}_{S_\lambda}) \rightarrow \mathcal{O}_S$ of sheaves of rings on S is an isomorphism*

$$(8.2.12.1) \quad \varinjlim_\lambda \psi_\lambda^*(\mathcal{O}_{S_\lambda}) \xrightarrow{\sim} \mathcal{O}_S.$$

Proof. One can evidently suppose the S_λ affine; with the notations of the proof of (8.2.9), it suffices to see that the inductive limit of the system of canonical maps $(A_\lambda)_{\mathfrak{p}_\lambda} \rightarrow A_{\mathfrak{p}}$ is an isomorphism, which is nothing other than (0_I, 5.13.3, (ii)). □

Proposition (8.2.13). — *Suppose that the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are open immersions, so that S_μ identifies with an open subset of S_λ for $\lambda \leq \mu$. Then, for all $\alpha \in \mathbf{L}$, the underlying space of the prescheme S identifies with the subspace of S_α which is the intersection of the S_λ for $\lambda \geq \alpha$, and the structural sheaf $\mathcal{O}_S$ with the sheaf induced by $\mathcal{O}_{S_\alpha}$ on this intersection; more generally, for every $\mathcal{O}_{S_\alpha}$ -Module $\mathcal{F}_\alpha$, $u_\alpha^*(\mathcal{F}_\alpha)$ identifies with the $\mathcal{O}_S$ -Module induced by $\mathcal{F}_\alpha$ on S .*

Proof. The first assertion follows from (8.2.9), the definition of a projective limit of topological spaces; moreover the $\psi_\lambda^*(\mathcal{O}_{S_\lambda})$ are all equal to the sheaf induced by $\mathcal{O}_{S_\alpha}$ on S ; with the notations of the proof of (8.2.9), the homomorphisms $(A_\lambda)_{\mathfrak{p}_\lambda} \rightarrow (A_\mu)_{\mathfrak{p}_\mu}$ are bijective for a system $(\mathfrak{p}_\lambda)$ of prime ideals corresponding to a single point of S ; the assertion relative to $\mathcal{O}_S$ follows from (8.2.12). The last assertion then follows from the definition of $u_\alpha^*(\mathcal{F}_\alpha)$ (0_I, 4.3.1). $\square$

Remark (8.2.14). — The results of (8.2.9) and (8.2.12) show that S is the projective limit of the projective system $(S_\lambda, u_{\lambda\mu})$ in the category of all ringed spaces (or of all locally ringed spaces). Indeed, let Y be a ringed space, and consider a projective system of morphisms of ringed spaces $w_\lambda : Y \rightarrow S_\lambda$. If one sets $w_\lambda = (\rho_\lambda, \omega_\lambda)$, the ρ_λ form a projective system of continuous maps, and the ω_λ form a system of homomorphisms of sheaves of rings; in view of (8.2.9), their limits projective give respectively a continuous map $\rho : Y \rightarrow S$ and, in view of (8.2.12), a homomorphism of sheaves of rings $\omega^\# : \rho^*(\mathcal{O}_S) \rightarrow \mathcal{O}_Y$ such that $\omega_\lambda^\# = \omega^\# \circ \rho^*(\theta_\lambda^\#)$, which shows our assertion.

8.2. Constructible parts in a projective limit of preschemes

. 8.3.1 Throughout this paragraph, we assume the conditions of (8.2.2) are satisfied, and we keep the same notations.

Theorem (8.3.2). — For all λ , let E_λ, F_λ be two subsets of S_λ . Set

$$(8.3.2.1) \quad E = \bigcap_{\lambda} u_\lambda^{-1}(E_\lambda), \quad F = \bigcup_{\lambda} u_\lambda^{-1}(F_\lambda).$$

Suppose the following conditions are satisfied:

- (i) For all λ , E_λ is pro-constructible and F_λ is ind-constructible (1.9.4).
- (ii) For $\lambda \leq \mu$, we have

$$E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda), \quad F_\mu \supset u_{\lambda\mu}^{-1}(F_\lambda).$$

- (iii) There exists $\alpha \in \mathbf{L}$ such that S_α is quasi-compact (which implies that S_λ is quasi-compact for $\lambda \geq \alpha$).

Then the following conditions are equivalent:

- a) $E \subset F$.
- b) There exists $\lambda \geq \alpha$ such that $u_\lambda^{-1}(E_\lambda) \subset u_\lambda^{-1}(F_\lambda)$ (and then $u_\mu^{-1}(E_\mu) \subset u_\mu^{-1}(F_\mu)$ for $\mu \geq \lambda$).
- c) There exists $\lambda \geq \alpha$ such that $E_\lambda \subset F_\lambda$ (and then $E_\mu \subset F_\mu$ for $\mu \geq \lambda$).

The remarks in parentheses in b) and c) follow from (ii). Set

$$G_\lambda = E_\lambda \cap (S_\lambda - F_\lambda), \quad G = E \cap (S - F).$$

Then G_λ is a pro-constructible part of S_λ (1.9.5, (i)), and by (8.3.2.1) and (ii), one has

$$G_\mu \subset u_{\lambda\mu}^{-1}(G_\lambda) \quad \text{for } \lambda \leq \mu$$

$$G = \bigcap_{\lambda} u_\lambda^{-1}(G_\lambda).$$

One is thus reduced to proving the particular case of (8.3.2) corresponding to $F_\lambda = \emptyset$ for all λ :

Corollary (8.3.3). — For all λ , let E_λ be a pro-constructible part of S_λ such that, for $\lambda \leq \mu$, $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$.

Suppose there exists $\alpha \in \mathbf{L}$ such that S_α is quasi-compact. Then the following conditions are equivalent:

- a) $E = \bigcap_{\lambda} u_\lambda^{-1}(E_\lambda) = \emptyset$.
- b) There exists λ such that $u_\lambda^{-1}(E_\lambda) = \emptyset$ (and then $u_\mu^{-1}(E_\mu) = \emptyset$ for $\mu \geq \lambda$).
- c) There exists λ such that $E_\lambda = \emptyset$ (and then $E_\mu = \emptyset$ for $\mu \geq \lambda$).

Proof. It is clear that c) implies b): since S_λ is quasi-compact, so is S (8.2.9); E_λ being pro-constructible, so is $u_\lambda^{-1}(E_\lambda)$ (1.9.5, (vi)); then the filtering decreasing family of pro-constructible sets $u_\lambda^{-1}(E_\lambda)$ has empty intersection, hence (1.9.9) one of them is empty.

Finally, let us show that b) implies c). As S_α is quasi-compact and $\mathbf{L}$ is filtered, one can replace S_α by an open affine, hence suppose (8.2.2) that S and the S_λ for $\lambda \geq \alpha$ are affine; one has then (1.9.2.1), for $\lambda \geq \alpha$,

$$u_\lambda(S) = \bigcap_{\mu \geq \lambda} u_{\lambda\mu}(S_\mu).$$

Hence $E_\lambda \cap u_\lambda(S) = \bigcap_{\mu \geq \lambda} (E_\lambda \cap u_{\lambda\mu}(S_\mu))$. Now, as u_λ and the $u_{\lambda\mu}$ are quasi-compact, $u_\lambda(S)$ and $u_{\lambda\mu}(S_\mu)$ are pro-constructible in S_λ (1.9.5, (vii)), hence the sets $E_\lambda \cap u_{\lambda\mu}(S_\mu)$ for $\mu \geq \lambda$ form a filtering decreasing family of pro-constructible parts of S_λ . As S_λ is quasi-compact, the hypothesis $b)$ implies that the intersection of this family is empty, hence (1.9.9) one of the sets $E_\lambda \cap u_{\lambda\mu}(S_\mu)$ is empty, hence $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$ is empty. Q.E.D. $\square$

Corollary (8.3.4). — For all λ , let F_λ be an ind-constructible part of S_λ such that for $\lambda \leq \mu$, $u_{\lambda\mu}^{-1}(F_\lambda) \subset F_\mu$. Suppose there exists $\alpha \in \mathbf{I}$ such that S_α is quasi-compact. Then the following conditions are equivalent:

- a) The set $F = \bigcup_\lambda u_\lambda^{-1}(F_\lambda)$ is equal to S .
- b) There exists λ such that $u_\lambda^{-1}(F_\lambda) = S$ (and then $u_\mu^{-1}(F_\mu) = S$ for $\mu \geq \lambda$).

This follows immediately from (8.3.3) by passage to complements.

Corollary (8.3.5). — For all λ , let E_λ, F_λ be two constructible parts of S_λ such that, for $\mu \geq \lambda$, $E_\mu \subset u_{\lambda\mu}^{-1}(E_\lambda)$ and $F_\mu \supset u_{\lambda\mu}^{-1}(F_\lambda)$. Suppose there exists an index α such that S_α is quasi-compact. Then, for $E \subset F$ (resp. $E = F$), it is necessary and sufficient that there exists λ such that $E_\lambda \subset F_\lambda$ (resp. $E_\lambda = F_\lambda$), in which case we also have $E_\mu \subset F_\mu$ (resp. $E_\mu = F_\mu$) for $\mu \geq \lambda$. This is a particular case of (8.3.2). IV-3 | 14

Corollary (8.3.6). — Suppose there exists α such that S_α is quasi-compact. For $S = \emptyset$, it is necessary and sufficient that there exists λ such that $S_\lambda = \emptyset$.

Corollary (8.3.7). — On a , for all λ ,

$$(8.3.7.1) \quad u_\lambda(S) = \bigcap_{\mu \geq \lambda} u_{\lambda\mu}(S_\mu).$$

Proof. It is clear that the first member of (8.3.7.1) is contained in the second. Let s be a point of S_λ and set $X_\lambda = \text{Spec}(k(s))$; consider the projective system $(X_\mu, v_{\mu\nu})$ where $X_\mu = X_\lambda \times_{S_\lambda} S_\mu$ and $v_{\mu\nu} = 1 \times u_{\mu\nu}$ for $\lambda \leq \mu \leq \nu$; its projective limit is $X = X_\lambda \times_{S_\lambda} S$ (8.2.5). If $s \in \bigcap_{\mu \geq \lambda} u_{\lambda\mu}(S_\mu)$, this implies that $X_\mu \neq \emptyset$ for all $\mu \geq \lambda$ (I, 3.4.8); it follows from (8.3.6) that $X \neq \emptyset$, hence $s \in u_\lambda(S)$, hence (I, 3.4.8). $\square$

Proposition (8.3.8). — (i) For the morphism $u_\lambda : S \rightarrow S_\lambda$ to be dominant (resp. surjective), it is necessary and sufficient that for $\mu \geq \lambda$ the morphism $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ be dominant (resp. surjective).
 (ii) If, for an index λ , the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are flat (resp. faithfully flat) for $\mu \geq \lambda$, then the morphism $u_\lambda : S \rightarrow S_\lambda$ is flat (resp. faithfully flat).
 (iii) Suppose that the morphisms $u_\mu : S \rightarrow S_\mu$ are surjective for $\mu \geq \lambda$. For u_λ to be an open (resp. universally open) morphism, it is necessary and sufficient that, for $\mu \geq \lambda$, $u_{\lambda\mu}$ be an open (resp. universally open) morphism.

Proof. (i) As $u_\lambda(S) \subset u_{\lambda\mu}(S_\mu)$ for $\mu \geq \lambda$, the necessity of the conditions is trivial, and it follows immediately from (8.3.7.1) that if the $u_{\lambda\mu}$ are surjective, so is u_λ . Suppose now the $u_{\lambda\mu}$ dominant for $\mu \geq \lambda$, and consider in S_λ an open quasi-compact non-empty subset U_λ ; then $u_{\lambda\mu}(S_\mu) \cap U_\lambda$ is non-empty for all μ , hence by (8.3.7.1), $u_\lambda(S) \cap U_\lambda$ is non-empty, hence u_λ is dominant. IV-3 | 15

We note that it can happen that all the $u_{\lambda\mu}$ are open without it being so for u_λ if the u_μ are not surjective.

(ii) By (i), it suffices to consider the case where the $u_{\lambda\mu}$ are flat; one can then reduce to the case where S_λ is affine, hence the S_μ for $\mu \geq \lambda$ and S are also affine, and the assertion follows from (0_I, 6.2.3).

(iii) By (8.2.5) and (I, 3.5.2), it suffices to treat the case of open morphisms. As $u_\lambda = u_{\lambda\mu} \circ u_\mu$ and u_μ is surjective, one knows that if u_λ is open it is the same of $u_{\lambda\mu}$ (Bourbaki, *Top. gén.*, chap. I^{er}, 3^e éd., § 5, n^o 1, prop. 1). $\square$

Conversely, to show that u_λ is open when all the $u_{\lambda\mu}$ are for $\mu \geq \lambda$, it suffices to see that for every open quasi-compact V of S , $u_\lambda(V)$ is open in S_λ ; but there exists then $\mu \geq \lambda$ such that $V = u_\mu^{-1}(V_\mu)$, where V_μ is open in S_μ (8.2.11); as u_μ is surjective, one has $V_\mu = u_\mu(V)$ and $u_\lambda(V) = u_{\lambda\mu}(u_\mu(V))$ is therefore open by hypothesis.

. 8.3.9 For any prescheme X , we will denote as usual by $\mathfrak{P}(X)$ the power set of the space underlying X , by $\mathfrak{C}(X)$ (resp. $\mathfrak{Dc}(X)$, $\mathfrak{Fc}(X)$, $\mathfrak{L}\mathfrak{Fc}(X)$) the set of constructible parts (resp. constructible and open, constructible and closed, locally constructible and closed) of X . It is clear that $(\mathfrak{P}(S_\lambda), u_{\lambda\mu}^{-1})$ is an *inductive system of sets* and that the maps $u_\lambda^{-1} : \mathfrak{P}(S_\lambda) \rightarrow \mathfrak{P}(S)$ form an inductive system of maps, whence, passing to the inductive limit, a canonical map

$$(8.3.9.1) \quad u_{\mathfrak{P}} : \varinjlim \mathfrak{P}(S_\lambda) \longrightarrow \mathfrak{P}(S).$$

Furthermore, it follows from (1.8.2) that $u_{\lambda\mu}^{-1}$ sends $\mathfrak{C}(S_\lambda)$ (resp. $\mathfrak{Dc}(S_\lambda)$, $\mathfrak{Fc}(S_\lambda)$, $\mathfrak{L}\mathfrak{Fc}(S_\lambda)$) into $\mathfrak{C}(S_\mu)$ (resp. $\mathfrak{Dc}(S_\mu)$, $\mathfrak{Fc}(S_\mu)$, $\mathfrak{L}\mathfrak{Fc}(S_\mu)$) and that u_λ^{-1} sends $\mathfrak{C}(S_\lambda)$ (resp. $\mathfrak{Dc}(S_\lambda)$, $\mathfrak{Fc}(S_\lambda)$, $\mathfrak{L}\mathfrak{Fc}(S_\lambda)$) into $\mathfrak{C}(S)$ (resp. $\mathfrak{Dc}(S)$, $\mathfrak{Fc}(S)$, $\mathfrak{L}\mathfrak{Fc}(S)$). Hence by restriction of (8.3.9.1), canonical maps

$$(8.3.9.2) \quad \varinjlim \mathfrak{C}(S_\lambda) \longrightarrow \mathfrak{C}(S)$$

$$(8.3.9.3) \quad \varinjlim \mathfrak{Dc}(S_\lambda) \longrightarrow \mathfrak{Dc}(S)$$

$$(8.3.9.4) \quad \varinjlim \mathfrak{Fc}(S_\lambda) \longrightarrow \mathfrak{Fc}(S)$$

$$(8.3.9.5) \quad \varinjlim \mathfrak{L}\mathfrak{Fc}(S_\lambda) \longrightarrow \mathfrak{L}\mathfrak{Fc}(S).$$

. 8.3.10 Let $g_\alpha : X_\alpha \rightarrow S_\alpha$ be a morphism; with the notations of (8.2.5) one has a canonical map $v_{\mathfrak{P}} : \varinjlim \mathfrak{P}(X_\lambda) \rightarrow \mathfrak{P}(X)$; on the other hand, for all $\lambda \geq \alpha$ and a morphism of projection $g_\lambda : X_\lambda \rightarrow S_\lambda$, it is clear that $g_\lambda^{-1} : \mathfrak{P}(S_\lambda) \rightarrow \mathfrak{P}(X_\lambda)$ form an inductive system of maps, and that the diagrams

$$(8.3.10.1) \quad \begin{array}{ccc} \varinjlim \mathfrak{P}(S_\lambda) & \xrightarrow{u_{\mathfrak{P}}} & \mathfrak{P}(S) \\ \varinjlim g_\lambda^{-1} \uparrow & & \uparrow g^{-1} \end{array}$$

$$\varinjlim \mathfrak{P}(X_\lambda) \xrightarrow{v_{\mathfrak{P}}} \mathfrak{P}(X)$$

are commutative, and one has analogous diagrams replacing $\mathfrak{P}$ by $\mathfrak{C}$, $\mathfrak{Dc}$, $\mathfrak{Fc}$, or $\mathfrak{L}\mathfrak{Fc}$.

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Theorem (8.3.11). — Suppose there exists $\alpha \in \mathbf{L}$ such that S_α is quasi-compact and quasi-separated. Then the canonical maps (8.3.9.2), (8.3.9.3), (8.3.9.4), and (8.3.9.5) are bijective.

Proof. By the preceding remark, it follows from (8.3.5) that under the hypothesis that for an $\alpha \in \mathbf{L}$, S_α is quasi-compact, the map (8.3.9.2) is *injective*. Furthermore:

It follows that these maps are *surjective*; as every constructible part of S is a finite union of sets of the form $U \cap V^c$, where U and V are open and constructible, it will suffice to prove that (8.3.9.3) is surjective (and also (8.3.9.4), by passage to complements). Now, as the morphisms $S_\lambda \rightarrow S_\alpha$ and $S \rightarrow S_\alpha$ are separated, the S_λ for $\lambda \geq \alpha$ and S are quasi-compact and quasi-separated, and one knows that the constructible open parts in such a prescheme are the quasi-compact open subsets (1.8.1). The conclusion follows hence from (8.2.11), except for the map (8.3.9.5). To prove that the latter is surjective, consider a locally closed and constructible part Z in S ; Z is then quasi-compact (0_{III}, 9.1.8). As every point $x \in Z$ admits by hypothesis a quasi-compact open neighborhood V_x in S such that $Z \cap V_x$ is closed in V_x , one can cover Z by a finite number of the V_x , hence there is a quasi-compact open U containing Z such that Z is closed in U ; as Z is also constructible in S , it is also constructible in U (0_{III}, 9.1.8). One knows (8.2.11) that there is an index λ and a quasi-compact open U_λ of S_λ such that $U = u_\lambda^{-1}(U_\lambda)$. Applying to U (which is a projective limit of the $U_\mu = u_{\lambda\mu}^{-1}(U_\lambda)$ for $\mu \geq \lambda$) the fact that (8.3.9.4) is surjective, one sees that there exists $\mu \geq \lambda$ and a closed constructible set Z_μ in U_μ such that $Z = u_\mu^{-1}(Z_\mu)$. But as the canonical immersion $U_\mu \rightarrow S_\mu$ is quasi-compact by hypothesis (1.2.7), each U_μ is open quasi-compact in S_μ , so that Z_μ is a constructible part of S_μ ; each of the $Z_{\mu\lambda}$ is of finite presentation on S_μ (1.6.2). This finishes the proof. $\square$

Corollary (8.3.12). — Suppose S_0 quasi-compact, and let, for all λ , Z_λ be a constructible part of S_λ such that $Z_\mu = u_{\lambda\mu}^{-1}(Z_\lambda)$ for $\mu \geq \lambda$. For Z to be open (resp. closed, resp. locally closed) in S , it is necessary and sufficient that there exists α such that Z_α is so in S_α .

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Proof. Cover S_α by a finite number of affine open subsets $U_\alpha^{(i)}$; then the $U^{(i)} = u_\alpha^{-1}(U_\alpha^{(i)})$ form a finite open affine covering of S , and for Z to be open (resp. closed) in S , it is necessary and sufficient that each $Z \cap U^{(i)}$ be open (resp. closed) in $U^{(i)}$. As $\mathbf{L}$ is filtered and S_α is affine, hence quasi-compact and quasi-separated, the result follows from (8.3.11). $\square$

Proposition (8.3.13). — *Suppose that the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are flat for $\lambda \leq \mu$, and that there exists α such that S_α is quasi-compact. For all λ , let Z'_λ, Z''_λ be two pro-constructible parts of S_λ , such that $Z'_\mu = u_{\lambda\mu}^{-1}(Z'_\lambda)$ and $Z''_\mu = u_{\lambda\mu}^{-1}(Z''_\lambda)$ for $\mu \geq \lambda$; suppose in addition that $\overline{Z''_\alpha}$ is constructible in S_α . Let $Z' = u_\alpha^{-1}(Z'_\alpha)$, $Z'' = u_\alpha^{-1}(Z''_\alpha)$; for $Z'' \subset \overline{Z'}$, it is necessary and sufficient that there exists $\lambda \geq \alpha$ such that $Z''_\lambda \subset \overline{Z'_\lambda}$.*

Proof. Indeed, one knows that $u_\lambda : S \rightarrow S_\lambda$ is also a flat morphism for all λ (8.3.8); as Z'_λ is pro-constructible, the closure of Z'_μ for $\mu \geq \lambda$ (resp. of Z') in S_μ (resp. S) is equal to $u_{\lambda\mu}^{-1}(\overline{Z'_\lambda})$ (resp. $u_\lambda^{-1}(\overline{Z'_\lambda})$) (2.3.10). As the $u_{\lambda\mu}^{-1}(\overline{Z'_\lambda})$ and $u_\lambda^{-1}(\overline{Z'_\lambda})$ are constructible (1.8.2), the conclusion follows from (8.3.2). $\square$

8.3. Criteria for irreducibility and connectedness for projective limits of preschemes

Proposition (8.4.1). — *Suppose there exists an index α such that S_α is quasi-compact.*

- (i) *If S is not irreducible and if moreover the underlying space of S is noetherian and S_α is quasi-separated, there exists $\lambda \geq \alpha$ such that, for $\mu \geq \lambda$, S_μ is not irreducible.*
- (ii) *If S is not connected, there exists $\lambda \geq \alpha$ such that, for $\mu \geq \lambda$, S_μ is not connected.*

Proof. Suppose that S is the union of two distinct closed sets S', S'' (resp. disjoint non-empty closed sets S', S'') in S . In case (i), S' and S'' are constructible since the space S is noetherian. By (8.3.11), there exist $\lambda \geq \alpha$ and two closed constructible sets S'_λ, S''_λ of S_λ such that $S' = u_\lambda^{-1}(S'_\lambda)$, $S'' = u_\lambda^{-1}(S''_\lambda)$; as $S = S' \cup S''$, it follows from (8.3.11) that one can suppose $S_\lambda = S'_\lambda \cup S''_\lambda$; as S'_λ and S''_λ are distinct from S_λ , this proves that S_λ is not irreducible.

In case (ii), S' and S'' are open quasi-compact, hence, by (8.2.11), there exist $\lambda \geq \alpha$ and two open quasi-compact subsets S'_λ, S''_λ of S_λ such that $S' = u_\lambda^{-1}(S'_\lambda)$, $S'' = u_\lambda^{-1}(S''_\lambda)$. Moreover, S' and S'' are at once pro-constructible and ind-constructible, hence constructible (1.9.11), and it follows from (8.3.5) that one can take λ such that $S_\lambda = S'_\lambda \cup S''_\lambda$ and $S'_\lambda \cap S''_\lambda = \emptyset$; this shows that S_λ is not connected. $\square$

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Proposition (8.4.2). — *Suppose that the underlying space of S is noetherian and that one of the two following conditions is satisfied:*

- a) *For $\lambda \leq \mu$, $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ is dominant, and there exists α such that S_α is quasi-compact.*
- b) *There exists α such that the underlying space of S_α is noetherian, and for $\mu \geq \lambda$, $u_{\lambda\mu}$ is a homeomorphism from S_μ to a subspace of S_λ .*

Under these conditions, there exists λ such that, for $\mu \geq \lambda$:

- (i) *For every irreducible component Y_i of S ($1 \leq i \leq m$), $\overline{u_\mu(Y_i)}$ is an irreducible component of S_μ , and the map $Y_i \mapsto \overline{u_\mu(Y_i)}$ is a bijection from the set of irreducible components of S to the set of irreducible components of S_μ .*
- (ii) *For every connected component C_j of S ($1 \leq j \leq n$), $\overline{u_\mu(C_j)}$ is a connected component of S_μ , and the map $C_j \mapsto \overline{u_\mu(C_j)}$ is a bijection from the set of connected components of S to the set of connected components of S_μ .*

Proof. We first establish the

Lemma (8.4.2.1). — *Under condition a) or b) of (8.4.2), there exists λ such that, for $\mu \geq \lambda$, $u_\mu : S \rightarrow S_\mu$ is dominant.*

In case a), this has already been proved without supposing S noetherian (8.3.8). In case b), set $Z_\alpha = \overline{u_\alpha(S)}$; as a closed part of the noetherian S_α , Z_α is constructible, and as $u_\alpha^{-1}(Z_\alpha) = S$, it follows from (8.3.5) that there exists $\lambda \geq \alpha$ such that $Z_\mu = u_{\lambda\mu}^{-1}(Z_\lambda) = S_\mu$. But as $u_{\lambda\mu}$ is a homeomorphism

from S_μ to a subspace and the composed map $S \rightarrow Z_\mu \rightarrow Z_\alpha$ is dominant, it is the same of $S \rightarrow Z_\mu = S_\mu$.

The lemma being established, we can suppose that for all λ , u_λ is dominant.

(i) Each S_λ is a union of the $\overline{u_\lambda(Y_i)}$, which are irreducible. On the other hand, if U_i is the complement of S of the union of the Y_j of index $j \neq i$ ($1 \leq i \leq m$), the U_i are pairwise disjoint and $Y_i = \overline{U_i}$ (0_I, 2.1.6). As the space S is noetherian, the U_i are quasi-compact, hence there exists an index λ and open subsets $U_{i\lambda}$ of S_λ such that $U_i = u_\lambda^{-1}(U_{i\lambda})$ for $1 \leq i \leq m$ (8.2.11). One concludes that if one sets $U_{i\mu} = u_{\lambda\mu}^{-1}(U_{i\lambda})$ for $\lambda \leq \mu$, the $U_{i\mu}$ are pairwise disjoint, the $U_i = u_\mu^{-1}(U_{i\mu})$ are the same and u_μ is dominant. Consequently, no $\overline{U_{i\mu}}$ is contained in another; hence $\overline{U_{i\mu}} = u_\mu(Y_i)$, which proves that the $\overline{U_{i\mu}}$ are the irreducible components of S_μ , and finishes the proof.

(ii) As the space S is noetherian, the C_j are open and closed in S and quasi-compact; the same reasoning as in (i) shows that there exist λ and open subsets $V_{j\lambda}$ of S_λ such that $C_j = u_\lambda^{-1}(V_{j\lambda})$ for $1 \leq j \leq n$. One sees also, as in (i), that if one sets $V_{j\mu} = u_{\lambda\mu}^{-1}(V_{j\lambda})$ for $\mu \geq \lambda$, the $V_{j\mu}$ are pairwise disjoint, and $u_\mu(C_j)$ is dense in $V_{j\mu}$. Furthermore, it follows from (8.3.4) that for μ large enough, the union of the $V_{j\mu}$ is S_μ , since every open subset in a prescheme is ind-constructible (1.9.6). The $V_{j\mu}$ are thus the connected components of S_μ . $\square$

Corollary (8.4.3). — Suppose one of the conditions a), b) of (8.4.2) is satisfied, the underlying space of S being noetherian; then, for S to be irreducible, it is necessary and sufficient that there exists λ such that S_μ is irreducible for all $\mu \geq \lambda$.

Proposition (8.4.4). — Suppose there exists α such that S_α is quasi-compact and that, for $\lambda \leq \mu$, $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ is dominant. Then, for S to be connected, it is necessary and sufficient that there exists λ_0 such that S_λ is connected for all $\lambda \geq \lambda_0$. IV-3 | 19

Proof. The condition is sufficient by (8.4.1); on the other hand, by (8.3.8), (i), we have seen that $u_\lambda : S \rightarrow S_\lambda$ is dominant for all λ sufficiently large, so if S is connected, since $u_\lambda(S)$ is dense in S_λ and connected, S_λ is connected. $\square$

Corollary (8.4.5). — Let k be a field, X a k -prescheme quasi-compact and connected. For X to be geometrically connected (4.5.2), it is necessary and sufficient that, for every finite and separable extension K of k , $X \otimes_k K$ be connected.

Proof. The condition is trivially necessary. To see that it is sufficient, we note that if Ω is an algebraic closure of k , $X \otimes_k \Omega$ is connected (4.5.1) and, for $k \subset k' \subset k'' \subset \Omega$, the morphism $X \otimes_k k'' \rightarrow X \otimes_k k'$ is surjective. One is thus reduced, by (8.4.4), to proving that $X \otimes_k k'$ is connected for every finite extension k' of k . But if K is the largest separable extension contained in k' , the morphism $X \otimes_k k' \rightarrow X \otimes_k K$ is finite, surjective and radiciel, hence (2.4.5) a homeomorphism, and as $X \otimes_k K$ is connected by hypothesis, so is $X \otimes_k k'$. $\square$

Remarks (8.4.6). — (i) The proof of (8.4.2) shows that the conclusion of the proposition is valid if one only supposes that the underlying space of S is noetherian, that there exists α such that S_α is quasi-compact, and finally that the $u_\lambda : S \rightarrow S_\lambda$ are dominant.

(ii) On the other hand, the conclusion of (8.4.2) can fail when the u_λ are not dominant for λ sufficiently large, even when the S_λ and S are noetherian, as the following example shows, taking $\mathbf{N}$ for the set of indices, all the S_n equal to $\text{Spec}(A \times K) = \text{Spec}(A) \amalg \text{Spec}(K)$, where K is a field, A a K -algebra of any kind, and all the morphisms $u_{n,n+1}$ equal to the same morphism corresponding to the homomorphism $(x, y) \mapsto (j(y), y)$ of $A \times K$ into itself, where $j : K \rightarrow A$ is the canonical homomorphism. One easily verifies that the inductive limit of this system of rings is K , the canonical homomorphism u_n corresponding to the second projection $A \times K \rightarrow K$. One sees then that $S = \text{Spec}(K)$ is irreducible even though none of the S_n are connected.

8.4. Modules of finite presentation on a projective limit of preschemes

8.5.1 We continue to use the notations and hypotheses of (8.2.2); we will moreover limit ourselves in this section to the case where S_0 is one of the S_λ , which one can always reduce to.

When, in this section, we consider a family $(\mathcal{F}_\lambda)$, where, for all $\lambda \in \mathbf{L}$, $\mathcal{F}_\lambda$ is an $\mathcal{O}_{S_\lambda}$ -Module, it will be understood that this family satisfies the condition

$$(8.5.1.1) \quad \mathcal{F}_\mu = u_{\lambda\mu}^*(\mathcal{F}_\lambda) \quad \text{for } \lambda \leq \mu.$$

We will then set

$$(8.5.1.2) \quad \mathcal{F} = u_\lambda^*(\mathcal{F}_\lambda)$$

which is an $\mathcal{O}_S$ -Module not depending on the index λ , by virtue of (8.5.1.1).

Let now $(\mathcal{F}_\lambda), (\mathcal{G}_\lambda)$ be two such families of $\mathcal{O}_{S_\lambda}$ -Modules. It is clear that the maps $f_\lambda \mapsto u_{\lambda\mu}^*(f_\lambda)$ from $\text{Hom}_{\mathcal{O}_{S_\lambda}}(\mathcal{F}_\lambda, \mathcal{G}_\lambda)$ into $\text{Hom}_{\mathcal{O}_{S_\mu}}(\mathcal{F}_\mu, \mathcal{G}_\mu)$ define an inductive system of abelian groups $(\text{Hom}_{\mathcal{O}_{S_\lambda}}(\mathcal{F}_\lambda, \mathcal{G}_\lambda), u_{\lambda\mu}^*)$, and that the maps $f_\lambda \mapsto u_\lambda^*(f_\lambda)$ form an *inductive system* of homomorphisms of abelian groups

$$u_\lambda^* : \text{Hom}_{\mathcal{O}_{S_\lambda}}(\mathcal{F}_\lambda, \mathcal{G}_\lambda) \longrightarrow \text{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{G});$$

whence, passing to the inductive limit, a canonical homomorphism of abelian groups

$$(8.5.1.3) \quad u_{\mathcal{F}, \mathcal{G}} : \varinjlim \text{Hom}_{\mathcal{O}_{S_\lambda}}(\mathcal{F}_\lambda, \mathcal{G}_\lambda) \longrightarrow \text{Hom}_{\mathcal{O}_S}(\mathcal{F}, \mathcal{G}).$$

We note that when $\mathcal{F}_\lambda = \mathcal{O}_{S_\lambda}$, the condition (8.5.1.1) is satisfied, and $\mathcal{F} = \mathcal{O}_S$; the homomorphism (8.5.1.3) gives then (0_I, 5.1.1) a canonical homomorphism of abelian groups

$$(8.5.1.4) \quad u_{\mathcal{G}} : \varinjlim \Gamma(S_\lambda, \mathcal{G}_\lambda) \longrightarrow \Gamma(S, \mathcal{G}).$$

Theorem (8.5.2). — (i) Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated) and that, for an $\lambda \in \mathbf{L}$, $\mathcal{F}_\lambda$ is quasi-coherent and of finite type (resp. of finite presentation) and $\mathcal{G}_\lambda$ quasi-coherent. Then the homomorphism $u_{\mathcal{F}, \mathcal{G}}$ is injective (resp. bijective).
 (ii) Suppose S_0 quasi-compact and quasi-separated. For every $\mathcal{O}_S$ -Module quasi-coherent $\mathcal{F}$ of finite presentation, there exist $\lambda \in \mathbf{L}$ and an $\mathcal{O}_{S_\lambda}$ -Module quasi-coherent $\mathcal{F}_\lambda$ of finite presentation such that $\mathcal{F}$ is isomorphic to $u_\lambda^*(\mathcal{F}_\lambda)$.

Proof. (i) One can evidently reduce to the case where $S_0 = S_\lambda$ and the morphisms $u_{0\lambda} : S_\lambda \rightarrow S_0$ are affine. Consider first the case where $S_0 = \text{Spec}(A_0)$ is affine. Then the assertion (i) is equivalent to the

Lemma (8.5.2.1). — Let A_0 be a ring, $(A_\lambda)_{\lambda \in \mathbf{L}}$ an inductive system of A_0 -algebras, $A = \varinjlim A_\lambda$; let M_0, N_0 be two A_0 -modules, and set $M_\lambda = M_0 \otimes_{A_0} A_\lambda$, $N_\lambda = N_0 \otimes_{A_0} A_\lambda$, $M = M_0 \otimes_{A_0} A = \varinjlim M_\lambda$, $N = N_0 \otimes_{A_0} A = \varinjlim N_\lambda$. If M_0 is of finite type (resp. of finite presentation), the canonical homomorphism

$$(8.5.2.2) \quad \varinjlim \text{Hom}_{A_\lambda}(M_\lambda, N_\lambda) \longrightarrow \text{Hom}_A(M, N)$$

is injective (resp. bijective).

One knows that there are canonical functorial isomorphisms

$$\text{Hom}_{A_\lambda}(M_\lambda, N_\lambda) \simeq \text{Hom}_{A_0}(M_0, N_\lambda), \quad \text{Hom}_A(M, N) \simeq \text{Hom}_{A_0}(M_0, N)$$

so that (8.5.2.2) is nothing other, up to canonical isomorphisms, than the canonical homomorphism

$$(8.5.2.3) \quad \varinjlim \text{Hom}_{A_0}(M_0, N_\lambda) \longrightarrow \text{Hom}_{A_0}(M_0, \varinjlim N_\lambda)$$

which to every inductive system of A_0 -module homomorphisms $\theta_\lambda : M_0 \rightarrow N_\lambda$ associates its inductive limit. Now, if M_0 is of finite type (resp. of finite presentation), one has an exact sequence $A_0^n \rightarrow M_0 \rightarrow 0$ (resp. $A_0^p \rightarrow A_0^n \rightarrow M_0 \rightarrow 0$); as it is clear that (8.5.2.3) is bijective when $M_0 = A_0^n$, it suffices to use the left exactness of the $\text{Hom}_{A_0}(M_0, P)$ functor and the exactness of the functor $\varinjlim$ (in the category of abelian groups) to conclude.

Let us now pass to the case where S_0 is quasi-compact, and let (U_i) be a finite open affine covering of S_0 ; for all λ , the $U_{i\lambda} = u_{0\lambda}^{-1}(U_i)$ form a finite open affine covering of S_λ , and the $V_i = u_0^{-1}(U_i)$ a finite open affine covering of S . To see that $u_{\mathcal{F}, \mathcal{G}}$ is injective, it suffices, if $f = u_\lambda^*(f_\lambda)$ is such that

$f = 0$, to show that there exists $\mu \geq \lambda$ such that $f_\mu = u_{\lambda\mu}^*(f_\lambda) = 0$. By the lemma (8.5.2.1), for each i there exists $\mu \geq \lambda$ such that $f_\mu|_{U_{i,\mu}} = 0$; it thus suffices to take μ larger than all the λ_i .

Suppose moreover S_0 quasi-separated and $\mathcal{F}_0$ of finite presentation, and let $f : \mathcal{F} \rightarrow \mathcal{G}$ be an $\mathcal{O}_S$ -homomorphism. By the lemma (8.5.2.1), for each i , there exist λ_i and a homomorphism $f_{\lambda_i}^{(i)} : \mathcal{F}_{\lambda_i}|_{U_{i,\lambda_i}} \rightarrow \mathcal{G}_{\lambda_i}|_{U_{i,\lambda_i}}$ such that $u_{\lambda_i}^*(f_{\lambda_i}^{(i)}) = f|_{V_i}$ (with the notations of the lemma). As $\mathbf{L}$ is filtered, one can suppose all the λ_i equal to a same λ . Note now that S_λ is quasi-separated (0_I, 5.2.5), and $\mathcal{F}_\lambda$ is a quasi-coherent $\mathcal{O}_{S_\lambda}$ -Module of finite presentation; as, for every pair of indices i, j , $U_{ij\lambda} = U_{i\lambda} \cap U_{j\lambda}$ is quasi-compact and quasi-separated (1.2.7), it follows from (8.2.5) that for every pair (i, j) , there exists an index $\lambda_{ij} \geq \lambda$ such that $u_{\lambda_{ij}}^*(f_{\lambda}^{(i)}|_{U_{ij\lambda}}) = u_{\lambda_{ij}}^*(f_{\lambda}^{(j)}|_{U_{ij\lambda}})$ for $\mu \geq \lambda_{ij}$; taking μ larger than all the λ_{ij} , the $u_{\lambda\mu}^*(f_{\lambda}^{(i)}|_{U_{ij\mu}})$ coincide in $U_{i\mu} \cap U_{j\mu}$ for all pairs (i, j) , and therefore define an S_μ -morphism $f_\mu : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$ such that $f = u_\mu^*(f_\mu)$.

Before passing to the proof of (ii), let us note the following corollaries of (i):

Corollary (8.5.2.4). — Suppose S_0 quasi-compact, $\mathcal{F}_\lambda$ quasi-coherent of finite type, $\mathcal{G}_\lambda$ quasi-coherent of finite presentation. Let $f_\lambda : \mathcal{F}_\lambda \rightarrow \mathcal{G}_\lambda$ be a homomorphism. For $f = u_\lambda^*(f_\lambda) : \mathcal{F} \rightarrow \mathcal{G}$ to be an isomorphism, it is necessary and sufficient that there exist $\mu \geq \lambda$ such that $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$ is an isomorphism.

Proof. One can always suppose $S_\lambda = S_0$; the question being local on S_0 , one can moreover (S_0 being quasi-compact and $\mathbf{L}$ filtered) reduce to the case where S_0 is affine, hence quasi-separated. The condition being trivially sufficient, it remains to show it is necessary: now, by hypothesis, there exists an $\mathcal{O}_S$ -homomorphism $g : \mathcal{G} \rightarrow \mathcal{F}$ such that $g \circ f = 1_{\mathcal{F}}$ and $f \circ g = 1_{\mathcal{G}}$. As $\mathcal{G}$ is of finite presentation, there exist $\nu \geq \lambda$ and a homomorphism $g_\nu : \mathcal{G}_\nu \rightarrow \mathcal{F}_\nu$ such that $g = u_\nu^*(g_\nu)$ by ((8.5.2), (i)); one thus has $u_\nu^*(g_\nu \circ f_\nu) = 1_{\mathcal{F}}$ and $u_\nu^*(f_\nu \circ g_\nu) = 1_{\mathcal{G}}$; taking into account that $\mathcal{F}_\nu$ and $\mathcal{G}_\nu$ are of finite type, one concludes by ((8.5.2), (i)) that there exists $\mu \geq \nu$ such that $g_\mu \circ f_\mu = 1_{\mathcal{F}_\mu}$ and $f_\mu \circ g_\mu = 1_{\mathcal{G}_\mu}$, whence the corollary. $\square$

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Corollary (8.5.2.5). — Suppose S_0 quasi-compact and quasi-separated. Suppose that $\mathcal{F}_\lambda, \mathcal{G}_\lambda$ are $\mathcal{O}_{S_\lambda}$ -Modules quasi-coherent of finite presentation. For $\mathcal{F}$ and $\mathcal{G}$ to be isomorphic, it is necessary and sufficient that there exist $\mu \geq \lambda$ such that $\mathcal{F}_\mu$ and $\mathcal{G}_\mu$ are isomorphic. Moreover, for every isomorphism $f : \mathcal{F} \xrightarrow{\sim} \mathcal{G}$, there exist $\nu \geq \mu$ and an isomorphism $f_\nu : \mathcal{F}_\nu \xrightarrow{\sim} \mathcal{G}_\nu$ such that $f = u_\nu^*(f_\nu)$.

Proof. This follows from (8.5.2.4) and from ((8.5.2), (i)) since every homomorphism $f : \mathcal{F} \rightarrow \mathcal{G}$ is of the form $u_\mu^*(f_\mu)$ for some $\mu \geq \lambda$ and some homomorphism $f_\mu : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$. $\square$

(ii) Consider first the case where $S_0 = \text{Spec}(A_0)$ is affine. Then the assertion is equivalent to the lemma (5.13.7.1). In the general case, starting from a finite open affine covering (U_i) of S_0 , one deduces (8.5.2.1) that for each i , there exist $\lambda(i)$ and an $\mathcal{O}_{U_{i,\lambda(i)}}$ -Module quasi-coherent of finite presentation $\mathcal{F}_i$ such that $u_{\lambda(i)}^*(\mathcal{F}_i) = \mathcal{F}|_{V_i}$ (with the notations of the proof of (i)). As $\mathbf{L}$ is filtered, one can suppose all the $\lambda(i)$ equal to a same λ . Note now that S_λ is quasi-separated (1.2.7), that $\mathcal{F}_\lambda$ is a quasi-coherent $\mathcal{O}_{S_\lambda}$ -Module of finite presentation; as, for every pair (i, j) , $U_{ij\lambda} = U_{i\lambda} \cap U_{j\lambda}$ is quasi-compact and quasi-separated (1.2.7), it follows from (8.1) that for every pair (i, j) , there exists an isomorphism $\theta_{ij} : \mathcal{F}_i|_{U_{ij\lambda}} \xrightarrow{\sim} u_{\lambda,\lambda_{ij}}^*(\mathcal{F}_j)|_{U_{ij\lambda}}$ such that $u_\lambda^*(\theta_{ij})$ is the identity automorphism of $\mathcal{F}|_{(V_i \cap V_j)}$; one can further suppose all the λ_{ij} equal to λ , so that $\theta_{ij} : \mathcal{F}_i|_{U_{ij\lambda}} \xrightarrow{\sim} \mathcal{F}_j|_{U_{ij\lambda}}$. Finally, for all triples i, j, k , if one sets $U_{ijk,\lambda} = U_{i\lambda} \cap U_{j\lambda} \cap U_{k\lambda}$, $U_{ijk,\lambda}$ is quasi-compact, and if $\theta'_{ij}, \theta'_{jk}$ and θ'_{ik} denote the restrictions of θ_{ij}, θ_{jk} and θ_{ik} to $U_{ijk,\lambda}$, one has $u_\lambda^*(\theta'_{ik}) = u_\lambda^*(\theta'_{ij} \circ \theta'_{jk})$. There is thus, by (i), an index $\mu \geq \lambda$ such that $u_{\lambda\mu}^*(\theta'_{ik}) = u_{\lambda\mu}^*(\theta'_{ij} \circ \theta'_{jk})$, hence the isomorphisms $u_{\lambda\mu}^*(\theta_{ij}) : u_{\lambda\mu}^*(\mathcal{F}_i)|_{U_{ij\mu}} \xrightarrow{\sim} u_{\lambda\mu}^*(\mathcal{F}_j)|_{U_{ij\mu}}$ satisfy the gluing condition, and define on S_μ an $\mathcal{O}_{S_\mu}$ -Module quasi-coherent $\mathcal{F}_\mu$ of finite presentation (0_I, 3.3.1) such that $\mathcal{F}$ is isomorphic to $u_\mu^*(\mathcal{F}_\mu)$. $\square$

Scholium (8.5.3). — The result of (8.5.2) can be expressed by saying that if S_0 is quasi-compact and quasi-separated, the category of $\mathcal{O}_S$ -Modules quasi-coherent of finite presentation is determined up to equivalence by the data of the categories of $\mathcal{O}_{S_\lambda}$ -Modules quasi-coherent of finite presentation, the functors $u_{\lambda\mu}^*$ between these categories, and the transitivity isomorphisms $u_{\mu\lambda} \circ u_{\lambda\mu} \xrightarrow{\sim} u_{\mu\lambda}$. In an image-like fashion, one can say that the data of an $\mathcal{O}_S$ -Module quasi-coherent of finite presentation $\mathcal{F}$ is “functorially” equivalent to the data of

an $\mathcal{O}_{S_\lambda}$ -Module quasi-coherent of finite presentation $\mathcal{F}_\lambda$ for λ sufficiently large, and that, if an $\mathcal{O}_{S_\mu}$ -Module quasi-coherent of finite presentation $\mathcal{F}_\mu''$ also has $\mathcal{F}$ as its inverse image, then $\mathcal{F}_\lambda'$ and $\mathcal{F}_\mu''$ have the same inverse image in an S_v for v large enough ($v \geq \lambda, v \geq \mu$).

Corollary (8.5.4). — Suppose S_0 quasi-compact and quasi-separated; then, for every quasi-coherent $\mathcal{O}_{S_\lambda}$ -Module $\mathcal{G}_\lambda$, the canonical homomorphism (8.5.1.4) is bijective. IV-3 | 23

Proof. Indeed, it suffices to apply (8.5.2), (i) to the case where $\mathcal{F}_\lambda = \mathcal{O}_{S_\lambda}$, which is of finite presentation. □

Proposition (8.5.5). — Suppose S_0 quasi-compact, and suppose that $\mathcal{F}_\lambda$ is an $\mathcal{O}_{S_\lambda}$ -Module quasi-coherent of finite presentation. For $\mathcal{F}$ to be locally free (resp. locally free of rank n), it is necessary and sufficient that there exists $\mu \geq \lambda$ such that $\mathcal{F}_\mu$ is locally free (resp. locally free of rank n).

Proof. The condition being trivially sufficient, let us prove it is necessary. If $\mathcal{F}$ is locally free (resp. locally free of rank n), there exists a finite open affine covering (V_i) of S such that $\mathcal{F}|_{V_i}$ is isomorphic to $\mathcal{O}_S^n|_{V_i}$ (resp. $\mathcal{O}_S^n|_{V_i}$) for all i . By (8.2.11) there exists $v \geq \lambda$ and for each i a quasi-compact open U_{iv} of S_v such that $V_i = u_v^{-1}(U_{iv})$. As S_v is quasi-compact, each U_{iv} is a finite union of open affines; one is thus reduced to the case where S_0 is affine and $\mathcal{F} = \mathcal{O}_S^n$; one knows then (8.2.5) that there exists $\mu \geq \lambda$ such that $\mathcal{F}_\mu$ is isomorphic to $\mathcal{O}_{S_\mu}^n$ (8.5.2). □

Proposition (8.5.6). — Suppose S_0 quasi-compact, and consider an exact sequence

$$\mathcal{F}_\lambda \longrightarrow \mathcal{G}_\lambda \longrightarrow \mathcal{H}_\lambda \longrightarrow 0$$

of homomorphisms of $\mathcal{O}_{S_\lambda}$ -Modules quasi-coherent, where $\mathcal{F}_\lambda$ and $\mathcal{G}_\lambda$ are of finite type and $\mathcal{H}_\lambda$ of finite presentation. For the corresponding sequence $\mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H} \rightarrow 0$ to be exact, it is necessary and sufficient that there exists $\mu \geq \lambda$ such that the sequence $\mathcal{F}_\mu \rightarrow \mathcal{G}_\mu \rightarrow \mathcal{H}_\mu \rightarrow 0$ is exact (in which case it is exact for $v \geq \mu$).

Proof. The fact that the condition is sufficient and the last assertion follow from the fact that the functor u_λ^* (resp. $u_{\lambda\mu}^*$) is right exact. To show that the condition is necessary, note that by ((8.5.2), (i)) there exists $v \geq \lambda$ such that the composed $\mathcal{F}_v \rightarrow \mathcal{G}_v \rightarrow \mathcal{H}_v$ is zero. Changing notations, one can therefore already suppose that $g_\lambda \circ f_\lambda = 0$. If one sets $\mathcal{H}'_\lambda = \text{Coker}(\mathcal{F}_\lambda \rightarrow \mathcal{G}_\lambda)$, one has then a homomorphism $f'_\lambda : \mathcal{H}'_\lambda \rightarrow \mathcal{H}_\lambda$; by hypothesis, $u_\lambda^*(f'_\lambda)$ is an isomorphism, and it follows from (8.5.2.4) that there exists $\mu \geq \lambda$ such that $u_{\lambda\mu}^*(f'_\lambda)$ is an isomorphism, which finishes the proof. □

Corollary (8.5.7). — Suppose S_0 quasi-compact, $\mathcal{F}_\lambda$ quasi-coherent of finite type, $\mathcal{G}_\lambda$ quasi-coherent. Let $f_\lambda : \mathcal{F}_\lambda \rightarrow \mathcal{G}_\lambda$ be a homomorphism. For $f = u_\lambda^*(f_\lambda) : \mathcal{F} \rightarrow \mathcal{G}$ to be surjective, it is necessary and sufficient that there exists $\mu \geq \lambda$ such that $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$ is surjective.

Proof. This is the particular case of (8.5.6) applied to the sequence $0 \rightarrow \mathcal{H}'_\lambda \rightarrow 0 \rightarrow 0$, where $\mathcal{H}'_\lambda = \text{Coker}(f_\lambda)$, which is quasi-coherent and of finite type (taking into account that $\text{Im } f_\lambda$ and $\text{Ker } g_\lambda$ are the reciprocal images of $\text{Im } f_\lambda$ and $\text{Ker } g_\lambda$ for $\mu \geq \lambda$), and in view of exactness to the right of the functors u_λ^* and $u_{\lambda\mu}^*$. □

Corollary (8.5.8). — Suppose S_0 quasi-compact and the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ flat. Then:

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- (i) Let $\mathcal{F}_\lambda \xrightarrow{f_\lambda} \mathcal{G}_\lambda \xrightarrow{g_\lambda} \mathcal{H}_\lambda$ be a sequence of homomorphisms of $\mathcal{O}_{S_\lambda}$ -Modules quasi-coherent, such that $\text{Im } f_\lambda$ and $\text{Ker } g_\lambda$ are of finite type. For the corresponding sequence $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$ to be exact, it is necessary and sufficient that there exists $\mu \geq \lambda$ such that the sequence $\mathcal{F}_\mu \xrightarrow{f_\mu} \mathcal{G}_\mu \xrightarrow{g_\mu} \mathcal{H}_\mu$ is exact.
- (ii) Let $f_\lambda : \mathcal{F}_\lambda \rightarrow \mathcal{G}_\lambda$ be a homomorphism of $\mathcal{O}_{S_\lambda}$ -Modules quasi-coherent such that $\text{Ker } f_\lambda$ is of finite type. For $f = u_\lambda^*(f_\lambda) : \mathcal{F} \rightarrow \mathcal{G}$ to be injective, it is necessary and sufficient that there exists $\mu \geq \lambda$ such that $f_\mu = u_{\lambda\mu}^*(f_\lambda) : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$ is injective.

Proof. (i) Taking into account (8.3.8), (ii), note that, by flatness, $\text{Im } f$ and $\text{Ker } g$ (resp. $\text{Im } f_\mu$ and $\text{Ker } g_\mu$) are the reciprocal images of $\text{Im } f_\lambda$ and $\text{Ker } g_\lambda$ for $\mu \geq \lambda$ (0_I, 6.7.2). Suppose the sequence $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$ is exact. Since $\text{Im } f_\lambda$ is of finite type, there exists $\mu \geq \lambda$ such that the composed $\mathcal{F}_\mu \rightarrow \mathcal{G}_\mu \rightarrow \mathcal{H}_\mu$ is zero, by (8.5.2), (i). Changing notations, one can therefore suppose that

$g_\lambda \circ f_\lambda = 0$. Then, as $\text{Ker } g_\lambda$ is of finite type, it follows from (8.5.7) that there exists $\mu \geq \lambda$ such that the homomorphism $\mathcal{F}_\mu \rightarrow \text{Ker } g_\mu$ is surjective, which finishes proving (i).

(ii) This is the particular case of (i) applied to the sequence $0 \rightarrow \mathcal{F}_\lambda \rightarrow \mathcal{G}_\lambda$. $\square$

Lemma (8.5.9). — Suppose S_0 quasi-compact and quasi-separated, $\mathcal{F}_\lambda$ quasi-coherent of finite type; let $\mathcal{G}'_\lambda, \mathcal{G}''_\lambda$ be two quasi-coherent quotients of $\mathcal{F}_\lambda$, $\mathcal{G}'_\lambda$ being in addition supposed of finite presentation. For $\mathcal{G}''$ to be a quotient of $\mathcal{G}'$, it is necessary and sufficient that there exists $\mu \geq \lambda$ such that $\mathcal{G}''_\mu$ is a quotient of $\mathcal{G}'_\mu$.

Proof. By hypothesis, there are two surjective homomorphisms $p'_\lambda : \mathcal{F}_\lambda \rightarrow \mathcal{G}'_\lambda, p''_\lambda : \mathcal{F}_\lambda \rightarrow \mathcal{G}''_\lambda$; by exactness to the right of u_λ^* and $u_{\lambda\mu}^*$, $p' = u_\lambda^*(p'_\lambda), p'' = u_\lambda^*(p''_\lambda), p'_\mu = u_{\lambda\mu}^*(p'_\lambda), p''_\mu = u_{\lambda\mu}^*(p''_\lambda)$ are also surjective; moreover, if there exists a homomorphism $f : \mathcal{G}' \rightarrow \mathcal{G}''$ (resp. $f_\mu : \mathcal{G}'_\mu \rightarrow \mathcal{G}''_\mu$) such that $p'' = f \circ p'$ (resp. $p''_\mu = f_\mu \circ p'_\mu$), this homomorphism is necessarily unique, which shows that the question is *local* on S_λ , and that one can therefore (as S_0 is quasi-compact and $\mathbf{L}$ filtered) suppose S_λ affine, hence *quasi-separated*. It is clear that the condition in the statement is sufficient. Conversely, if there exists a homomorphism $f : \mathcal{G}' \rightarrow \mathcal{G}''$ such that $p'' = f \circ p'$, it follows from (8.5.2), (i) that there exists $\mu \geq \lambda$ and a homomorphism $f_\mu : \mathcal{G}'_\mu \rightarrow \mathcal{G}''_\mu$ such that $f = u_\mu^*(f_\mu)$. Furthermore, by (8.5.7), one can suppose μ taken large enough for p_μ to be surjective, which finishes the proof. $\square$

. 8.5.10 To finish this section, for every quasi-coherent Module $\mathcal{F}$ on a prescheme, denote by $\Omega(\mathcal{F})$ the set of Modules which are quotients of $\mathcal{F}$ and are of finite presentation. If $\mathcal{F}_\lambda$ is quasi-coherent and $\mathcal{G}_\lambda \in \Omega(\mathcal{F}_\lambda)$, it follows from the fact that $u_{\lambda\mu}^*$ and u_λ^* are exact to the right that $\mathcal{G}_\mu \in \Omega(\mathcal{F}_\mu)$ for $\mu \geq \lambda$ and $\mathcal{G} \in \Omega(\mathcal{F})$; it is clear that $(\Omega(\mathcal{F}_\lambda), u_{\lambda\mu}^*)$ is an inductive system of sets, and that the $u_\lambda^* : \Omega(\mathcal{F}_\lambda) \rightarrow \Omega(\mathcal{F})$ form an inductive system of maps, whence, passing to the inductive limit, a canonical map

$$(8.5.10.1) \quad u_\Omega : \varinjlim \Omega(\mathcal{F}_\lambda) \longrightarrow \Omega(\mathcal{F}).$$

Furthermore, if $(\mathcal{F}'_\lambda)$ is a second family of quasi-coherent $\mathcal{O}_{S_\lambda}$ -Modules and if, for all λ , $\mathcal{F}'_\lambda$ is a quotient of $\mathcal{F}_\lambda$, then $\mathcal{F}'$ is a quotient of $\mathcal{F}$ and one has a commutative diagram

$$(8.5.10.2) \quad \begin{array}{ccc} \varinjlim \Omega(\mathcal{F}_\lambda) & \xrightarrow{u_\Omega} & \Omega(\mathcal{F}) \\ \uparrow & & \uparrow \\ \varinjlim \Omega(\mathcal{F}'_\lambda) & \xrightarrow{u_\Omega} & \Omega(\mathcal{F}') \end{array}$$

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Proposition (8.5.11). — Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated). Suppose that $\mathcal{F}_\lambda$ is quasi-coherent of finite type (resp. of finite presentation) for all λ ; then the canonical map (8.5.10.1) is injective (resp. bijective).

Proof. The first assertion follows more precisely from the lemma (8.5.9). For the second, consider an $\mathcal{O}_S$ -Module quotient $\mathcal{G}$ of $\mathcal{F}$ which is of finite presentation. It follows from (8.5.2), (ii) that there exist $\lambda' \in \mathbf{L}$ and an $\mathcal{O}_{S_{\lambda'}}$ -Module quasi-coherent of finite presentation $\mathcal{G}_{\lambda'}$ such that $\mathcal{G} = u_{\lambda'}^*(\mathcal{G}_{\lambda'})$; as $\mathbf{L}$ is filtered, one can replace λ and λ' by an index majoring them and suppose $\lambda' = \lambda$. Consider then the canonical homomorphism $p : \mathcal{F} \rightarrow \mathcal{G}$; it follows from (8.5.2), (i) that there exists $\mu \geq \lambda$ and a homomorphism $p_\mu : \mathcal{F}_\mu \rightarrow \mathcal{G}_\mu$ such that $p = u_\mu^*(p_\mu)$. Furthermore, by (8.5.7), one can suppose μ taken large enough for p_μ to be surjective, which finishes the proof. $\square$

8.5. Sub-preschemes of finite presentation of a projective limit of preschemes

. 8.6.1 Given a prescheme Y , we will denote in this section by $\mathfrak{S}\mathrm{pr}(Y)$ the ordered set (I, 4.1.10) of sub-preschemes of Y which are of *finite presentation* on Y (1.6.1), by $\mathfrak{S}\mathrm{pro}(Y)$ (resp. $\mathfrak{S}\mathrm{prf}(Y)$) the part of $\mathfrak{S}\mathrm{pr}(Y)$ formed of sub-preschemes induced on open subsets (resp. closed sub-preschemes) of Y , of finite presentation on Y ; recall that saying a sub-prescheme Z of Y belongs to $\mathfrak{S}\mathrm{pro}(Y)$ means it is induced on an open subset and that the underlying space Z is *retrocompact* in Y (resp. that it is closed and that the ideal $\mathcal{J}$ of $\mathcal{O}_Y$ which defines it is of *finite type*, which also means that $j_*(\mathcal{O}_Z) \in \Omega(\mathcal{O}_Y)$ if $j : Z \rightarrow Y$ is the canonical injection) (1.6.1 and 1.4.5).

. 8.6.2 We keep the notations of (8.2.2) and suppose that S_0 is one of the S_λ . Let Y_λ be a sub-prescheme of S_λ ; then $Y_\mu = u_{\lambda\mu}^{-1}(Y_\lambda)$ (resp. $Y = u_\lambda^{-1}(Y_\lambda)$) is a sub-prescheme of S_μ for $\mu \geq \lambda$ (resp. of S); it is induced on an open subset (resp. it is closed) if Y_λ is so, and is of finite presentation on S_μ (resp. S) if Y_λ is of finite presentation on S_λ (1.6.2, (iii)). Hence $(\mathfrak{S}\mathrm{pr}(S_\lambda), u_{\lambda\mu}^{-1})$ (resp. $(\mathfrak{S}\mathrm{pro}(S_\lambda), u_{\lambda\mu}^{-1})$, $(\mathfrak{S}\mathrm{prf}(S_\lambda), u_{\lambda\mu}^{-1})$) is an inductive system and the maps $u_\lambda^{-1} : \mathfrak{S}\mathrm{pr}(S_\lambda) \rightarrow \mathfrak{S}\mathrm{pr}(S)$ (resp. $\mathfrak{S}\mathrm{pro}(S_\lambda) \rightarrow \mathfrak{S}\mathrm{pro}(S)$, $\mathfrak{S}\mathrm{prf}(S_\lambda) \rightarrow \mathfrak{S}\mathrm{prf}(S)$) form an inductive system of maps; whence, passing to the inductive limit, canonical maps

$$(8.6.2.1) \quad \varinjlim \mathfrak{S}\mathrm{pr}(S_\lambda) \longrightarrow \mathfrak{S}\mathrm{pr}(S)$$

$$(8.6.2.2) \quad \varinjlim \mathfrak{S}\mathrm{pro}(S_\lambda) \longrightarrow \mathfrak{S}\mathrm{pro}(S)$$

$$(8.6.2.3) \quad \varinjlim \mathfrak{S}\mathrm{prf}(S_\lambda) \longrightarrow \mathfrak{S}\mathrm{prf}(S).$$

Recall (I, 4.1.10) that $\mathfrak{S}\mathrm{pr}(X)$, for any prescheme X , is an *ordered* set by the relation “ Z is a sub-prescheme of the sub-prescheme Y ” which we write $Z \leq Y$. The maps $u_{\lambda\mu}^{-1}$ and u_λ^{-1} are *increasing* for the corresponding order relations in $\mathfrak{S}\mathrm{pr}(S_\lambda)$, $\mathfrak{S}\mathrm{pr}(S_\mu)$, $\mathfrak{S}\mathrm{pr}(S)$. Furthermore, one defines an order relation on the set $\varinjlim \mathfrak{S}\mathrm{pr}(S_\lambda)$ by writing $\eta \leq \eta'$ for two elements of this set whenever there exist λ and two elements Y_λ, Y'_λ of $\mathfrak{S}\mathrm{pr}(S_\lambda)$, whose η and η' are the canonical images, and such that $Y_\lambda \leq Y'_\lambda$; one verifies easily that this does not depend on the choice of representatives, and that one obtains a *relation of order*. This being the case, the fact that the u_λ^{-1} are increasing implies immediately that the canonical map (8.6.2.1) is *increasing*; it is the same evidently for (8.6.2.2) and (8.6.2.3).

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Proposition (8.6.3). — Suppose S_0 quasi-compact (resp. quasi-compact and quasi-separated). Then the maps (8.6.2.1), (8.6.2.2), (8.6.2.3) are injective (resp. bijective).

Proof. Taking into account the remarks of (8.6.1), the assertions relative to (8.6.2.3) follow from (8.5.11) applied to $\mathcal{F}_\lambda = \mathcal{O}_{S_\lambda}$; similarly, the assertions relative to (8.6.2.2) are particular cases of (8.3.5) and (8.3.11), taking into account that the S_λ and S are quasi-compact. It remains to consider the map (8.6.2.1). Let us first prove that it is surjective when S_0 is quasi-compact and quasi-separated. Let Z be a sub-prescheme of S , of finite presentation on S ; as S is quasi-compact, it is the same of Z , hence there exists a quasi-compact open U of S such that Z is a sub-prescheme closed of U , of finite presentation on U . There exists then an index λ and a quasi-compact open U_λ of S_λ such that $U = u_\lambda^{-1}(U_\lambda)$ (8.2.11); as S_λ is quasi-separated, each U_λ is of the same type of U_λ (1.2.7), and by the same argument as in the case of (8.6.2.3), it is surjective.

Finally, to see that (8.6.2.1) is injective when S_0 is quasi-compact, it suffices to prove the following more precise result: □

Corollary (8.6.3.1). — Suppose S_0 quasi-compact, and let Z'_λ, Z''_λ be two sub-preschemes of S_λ , of finite presentation on S_λ . For $Z' = u_\lambda^{-1}(Z'_\lambda)$ to be majorized by $Z'' = u_\lambda^{-1}(Z''_\lambda)$ (I, 4.1.10), it is necessary and sufficient that there exists $\mu \geq \lambda$ such that $Z'_\mu = u_{\lambda\mu}^{-1}(Z'_\lambda)$ is majorized by $Z''_\mu = u_{\lambda\mu}^{-1}(Z''_\lambda)$.

Corollary (8.6.4). — Suppose S_0 quasi-compact, and let Z_λ be a sub-prescheme of S_λ , of finite presentation on S_λ . For $Z = u_\lambda^{-1}(Z_\lambda)$ to be a sub-prescheme induced on an open subset (resp. a closed sub-prescheme) of S , it is necessary and sufficient that there exists $\mu \geq \lambda$ such that $Z_\mu = u_{\lambda\mu}^{-1}(Z_\lambda)$ is induced on an open subset (resp. a closed sub-prescheme) of S_μ .

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Proof. Let $(U_\lambda^{(i)})$ be a finite open affine covering of S_λ , and set $U_\mu^{(i)} = u_{\lambda\mu}^{-1}(U_\lambda^{(i)})$ for $\mu \geq \lambda$ and $U^{(i)} = u_\lambda^{-1}(U_\lambda^{(i)})$. If Z is open (resp. closed) in S , $Z \cap U^{(i)}$ is open (resp. closed) in $U^{(i)}$, and conversely.

Since $\mathbf{L}$ is filtered, it suffices to show the corollary when S_λ is affine, hence quasi-separated; but then the result follows from the bijectivity of the maps (8.6.2.1), (8.6.2.2), and (8.6.2.3). $\square$

8.6. Criteria for a projective limit of preschemes to be reduced (resp. integral)

Proposition (8.7.1). — *Suppose that S is not reduced. Then there exists λ_0 such that for $\lambda \geq \lambda_0$, S_λ is not reduced.*

Proof. The question being local on S_0 , one can suppose $S_0 = \text{Spec}(A_0)$ affine, whence $S = \text{Spec}(A)$, where $A = \varinjlim A_\lambda$ is the inductive limit of an inductive system of A_0 -algebras (A_λ) . One knows then (5.13.2) that the nilradical of A is the inductive limit of the nilradicals of the A_λ ; so if it is not zero, one of the A_λ contains a nilpotent element $a_\lambda \neq 0$ whose image in A is non-zero, and the image of a_λ in the A_μ for $\mu \geq \lambda$ is therefore a nilpotent element $\neq 0$. $\square$

Proposition (8.7.2). — *Suppose one of the following hypotheses is satisfied:*

- a) S_0 is quasi-compact, the nilradical $\mathcal{N}_0$ of $\mathcal{O}_{S_0}$ is an ideal of finite type (which will be the case when S_0 is noetherian), and the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are open immersions.
- b) The morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are faithfully flat.

Under these conditions, for S to be reduced, it is necessary and sufficient that there exists λ_0 such that S_λ is reduced for $\lambda \geq \lambda_0$.

Furthermore, in case b), if S is reduced, all the S_λ are reduced.

Proof. The last assertion follows from the fact that the morphism $S \rightarrow S_\lambda$ is then faithfully flat for all λ (8.3.8) and from (2.1.13). On the other hand, (8.7.1) proves that the condition is necessary in hypothesis a); then (8.2.13), the underlying space of S identifies with the intersection of the underlying spaces of the S_λ (the S_λ being identified with sub-preschemes induced on open subsets of S_0), and the structural sheaf $\mathcal{O}_S$ identifies with the sheaf induced by all the $\mathcal{O}_{S_\lambda}$; in particular for all $s \in S$ and for all S_λ , the local ring $\mathcal{O}_s$ is the same for S and for all S_λ . If $\mathcal{N}_\lambda$ is the Nilradical of $\mathcal{O}_{S_\lambda}$, the Nilradical $\mathcal{N}$ of $\mathcal{O}_S$ has at each point of S the same fibre $\mathcal{N}_s$ (nilradical of $\mathcal{O}_s$) (induced on S); the hypothesis that S is reduced implies by $u_\lambda^*(\mathcal{N}_\lambda) = 0$; as $\mathcal{N}_0$ is supposed of finite type, it is the same of $\mathcal{N}_\lambda$, and the conclusion follows from (8.5.7). $\square$

Corollary (8.7.3). — *Suppose one of the following hypotheses is satisfied:*

- a) S_0 is a noetherian prescheme and the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are open immersions.
- b) The morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are faithfully flat.

Then, for S to be integral, it is necessary and sufficient that there exists λ_0 such that S_λ is integral for $\lambda \geq \lambda_0$.

Proof. Recall that saying a prescheme is integral means it is at once reduced and irreducible; the corollary thus follows from (8.7.2) and (8.4.3). $\square$

Remark (8.7.4). — If one does not make hypotheses on the $u_{\lambda\mu}$, it can happen that S is integral even though all the S_λ are not reduced and not connected, as the example (8.4.4) shows, where one takes A non-reduced.

8.7. Preschemes of finite presentation on a projective limit of preschemes

8.8.1 Keeping always the notations and hypotheses of (8.2.2), we will assume given in this section two S_α -preschemes X_α, Y_α , which defines (8.2.5) by setting $X_\lambda = X_\alpha \times_{S_\alpha} S_\lambda$, $Y_\lambda = Y_\alpha \times_{S_\alpha} S_\lambda$, $v_{\lambda\mu} = 1_{X_\alpha} \times u_{\lambda\mu}$, $w_{\lambda\mu} = 1_{Y_\alpha} \times u_{\lambda\mu}$ (for $\alpha \leq \lambda \leq \mu$), whose projective limits are respectively $X = X_\alpha \times_{S_\alpha} S$, $Y = Y_\alpha \times_{S_\alpha} S$, the canonical morphisms $v_\lambda : X \rightarrow X_\lambda$ and $w_\lambda : Y \rightarrow Y_\lambda$ being respectively equal to $1_{X_\alpha} \times u_\lambda$ and $1_{Y_\alpha} \times u_\lambda$. For $\alpha \leq \lambda \leq \mu$, one has a canonical map $e_{\mu\lambda} : \text{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \rightarrow \text{Hom}_{S_\mu}(X_\mu, Y_\mu)$, which to every S_λ -morphism $f_\lambda : X_\lambda \rightarrow Y_\lambda$ associates the morphism $f_\mu = f_\lambda \times 1_{S_\mu} : X_\lambda \times_{S_\lambda} S_\mu \rightarrow Y_\lambda \times_{S_\lambda} S_\mu$, and it is clear that $(\text{Hom}_{S_\lambda}(X_\lambda, Y_\lambda), e_{\lambda\mu})$ is an inductive system of sets. Similarly, one has a canonical map $e_\lambda : \text{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \rightarrow \text{Hom}_S(X, Y)$ which to f_λ associates $f = f_\lambda \times 1_S : X_\lambda \times_{S_\lambda} S \rightarrow Y_\lambda \times_{S_\lambda} S$ and (e_λ) is an inductive system of maps; whence, passing to the inductive limit, a canonical map, functorial in S_α, X_α and Y_α :

$$(8.8.1.1) \quad e : \varinjlim \text{Hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \text{Hom}_S(X, Y).$$

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Theorem (8.8.2). — (i) Suppose X_α quasi-compact (resp. quasi-compact and quasi-separated), and Y_α locally of finite type (resp. locally of finite presentation) on S_α . Then the map (8.8.1.1) is injective (resp. bijective).
 (ii) Suppose S_0 quasi-compact and quasi-separated. For every prescheme X of finite presentation on S , there exist $\lambda \in \mathbf{L}$, a prescheme X_λ of finite presentation on S_λ , and an S -isomorphism $X \xrightarrow{\sim} X_\lambda \times_{S_\lambda} S$.

Proof. (i) Consider first the case where $S_0 = \operatorname{Spec}(A_0)$, $X_\alpha = \operatorname{Spec}(B_\alpha)$ and $Y_\alpha = \operatorname{Spec}(C_\alpha)$ are affine; then the $S_\lambda = \operatorname{Spec}(A_\lambda)$ and $S = \operatorname{Spec}(A)$ are also affine, with $A = \varinjlim A_\lambda$, and the assertions of (i) are equivalent to the

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Lemma (8.8.2.1). — Let A_0 be a ring, $(A_\lambda)_{\lambda \in \mathbf{L}}$ an inductive system of A_0 -algebras, $A = \varinjlim A_\lambda$; let B_α be an A_α -algebra, C_α an A_α -algebra of finite type (resp. of finite presentation). Then the canonical homomorphism

$$(8.8.2.2) \quad \varinjlim \operatorname{Hom}_{A_\lambda\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A_\lambda, B_\alpha \otimes_{A_\alpha} A_\lambda) \longrightarrow \operatorname{Hom}_{A\text{-alg.}}(C_\alpha \otimes_{A_\alpha} A, B_\alpha \otimes_{A_\alpha} A)$$

is injective (resp. bijective).

Lemma (8.8.2.3). — Let E be a ring, G an E -algebra of finite type (resp. of finite presentation), (F_λ) an inductive system of E -algebras. Then the canonical homomorphism

$$\varinjlim \operatorname{Hom}_{E\text{-alg.}}(G, F_\lambda) \longrightarrow \operatorname{Hom}_{E\text{-alg.}}(G, \varinjlim F_\lambda)$$

which to every inductive system of homomorphisms $\theta_\lambda : G \rightarrow F_\lambda$ of E -algebras associates its inductive limit, is injective (resp. bijective).

These lemmas are proved by reduction to the case of polynomial algebras and quotients thereof, using the universal properties of polynomial algebras and the finite presentation hypothesis.

(ii) follows from (i) and the gluing arguments of (8.5.2). □

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Corollary (8.8.2.4). — Suppose S_0 quasi-compact and quasi-separated, X_α of finite presentation on S_α , Y_α of finite type on S_α and quasi-separated on S_α (which will be the case for example if Y_α is also of finite presentation on S_α). Let $f_\alpha : X_\alpha \rightarrow Y_\alpha$ be an S_α -morphism. For $f : X \rightarrow Y$ to be an isomorphism, it is necessary and sufficient that there exists $\lambda \geq \alpha$ such that $f_\lambda : X_\lambda \rightarrow Y_\lambda$ is an isomorphism.

Corollary (8.8.2.5). — Suppose S_0 quasi-compact and quasi-separated, X_α and Y_α of finite presentation on S_α . For X and Y to be S -isomorphic, it is necessary and sufficient that there exists $\lambda \geq \alpha$ such that X_λ and Y_λ are S_λ -isomorphic. Moreover, for every S -isomorphism $f : X \xrightarrow{\sim} Y$, there exists $\mu \geq \lambda$ and an S_μ -isomorphism $f_\mu : X_\mu \xrightarrow{\sim} Y_\mu$ such that $f = f_\mu \times 1_S$.

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Scholium (8.8.3). — The essential content of (8.8.2) can be expressed by saying that if S_0 is quasi-compact and quasi-separated, the category of S -preschemes of finite presentation is determined up to equivalence by the data of the categories of S_λ -preschemes of finite presentation, the functors $\rho_{\mu\lambda} : X_\lambda \mapsto X_\lambda \times_{S_\lambda} S_\mu$ (for $\lambda \leq \mu$) between these categories, and the transitivity isomorphisms $\rho_{\nu\mu} \circ \rho_{\mu\lambda} \xrightarrow{\sim} \rho_{\nu\lambda}$. In an image-like fashion, one can say that the data of an S -prescheme of finite presentation X is “functorially” equivalent to the data of an S_λ -prescheme of finite presentation X'_λ for λ sufficiently large; if an S_μ -prescheme of finite presentation X''_μ is such that X is S -isomorphic to $X''_\mu \times_{S_\mu} S$, there exists ν such that $\lambda \leq \nu$, $\mu \leq \nu$ and such that $X'_\lambda \times_{S_\lambda} S_\nu$ and $X''_\mu \times_{S_\mu} S_\nu$ are S_ν -isomorphic. The fact that $\mathbf{L}$ is filtered implies moreover that if one gives a finite family $(X^{(i)})_{i \in I}$ of S -preschemes of finite presentation, and a finite family $(f^{(j)})_{j \in J}$ of S -morphisms between these preschemes (thus $f^{(j)}$ being of the form $X^{(\sigma(j))} \rightarrow X^{(\tau(j))}$ where σ and τ are two maps of J into I), there is then an index $\lambda \in \mathbf{L}$, a family $(X^{(i)}_\lambda)_{i \in I}$ of S_λ -preschemes of finite presentation and a family $(f^{(j)}_\lambda)_{j \in J}$ of S_λ -morphisms such that $X^{(i)}$ identifies with $X^{(i)}_\lambda \times_{S_\lambda} S$ for all i and j ; moreover, if one has a relation $f^{(j)} = f^{(k)}$, one can suppose $f^{(j)}_\lambda = f^{(k)}_\lambda$.

8.8. First applications to the elimination of noetherian hypotheses

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Proposition (8.9.1). — *Let A be a ring, X an A -prescheme.*

- (i) *The following conditions are equivalent:*
 - a) *X is of finite presentation on A .*
 - b) *There exists a noetherian ring A_0 , a prescheme X_0 of finite type on A_0 , a ring homomorphism $A_0 \rightarrow A$, and an A -isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.*
 - c) *There exists a sub-ring A_0 of A , which is a $\mathbf{Z}$ -algebra of finite type, a prescheme X_0 of finite type on A_0 , and an A -isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.*
- (ii) *If $\mathcal{F}$ is an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation, one can suppose the sub-ring A_0 of A is such that there exists an $\mathcal{O}_{X_0}$ -Module coherent $\mathcal{F}_0$ such that $\mathcal{F}$ is isomorphic to $\mathcal{F}_0 \otimes_{A_0} A$; $\text{Supp}(\mathcal{F})$ is constructible and closed in X , and there is a closed sub-prescheme Z of X , having $\text{Supp}(\mathcal{F})$ as underlying space, such that the canonical immersion $Z \rightarrow X$ is of finite presentation.*
- (iii) *If Y is a second A -prescheme of finite presentation, and $f : X \rightarrow Y$ an A -morphism, one can suppose the sub-ring A_0 of A such that there exists a prescheme Y_0 of finite type on A_0 and an A_0 -morphism $f_0 : X_0 \rightarrow Y_0$ (necessarily of finite type) such that f identifies with $f_0 \otimes 1_A$.*

Proof. (i) As A is the inductive limit of its sub-rings that are of finite type over $\mathbf{Z}$, a) implies c) by (8.8.2), (ii); c) implies b) since a $\mathbf{Z}$ -algebra of finite type is a noetherian ring; finally, if A_0 is noetherian, an A_0 -prescheme of finite type is of finite presentation (1.6.1), hence b) implies a) by (1.6.2, (iii)).

The assertion (ii) follows immediately from (8.5.2), (ii), (8.3.11) and (1.8.2), and the assertion (iii) from (8.8.2), (i). $\square$

. 8.9.2 Proposition (8.9.1) and the results of (8.5.2) and (8.8.2) allow extending in many cases to morphisms of finite presentation $X \rightarrow Y$ results proved by noetherian techniques assuming Y is locally noetherian. We will see many examples in the sequel and will limit ourselves here to giving a few results of this type.

Proposition (8.9.3). — *Let A be a ring, M an A -module of finite presentation. Then every surjective A -endomorphism of M is bijective.* IV-3 | 35

Proof. Indeed, consider A as the inductive limit of its sub- $\mathbf{Z}$ -algebras of finite type. It follows from (8.5.2.6) that there is one of these sub-algebras A_0 and an A_0 -module M_0 of finite type such that M is A -isomorphic to $M_0 \otimes_{A_0} A$; moreover, if $f : M \rightarrow M$ is a surjective A -endomorphism, one can suppose (8.5.2), (i) that there exists an A_0 -endomorphism $f_0 : M_0 \rightarrow M_0$ such that $f = f_0 \otimes 1_A$; finally (8.5.7) one can suppose that f_0 is surjective. But as A_0 is noetherian and M_0 an A_0 -module of finite type (Bourbaki, Alg., chap. VIII, § 2, n° 2, lemma 3) f_0 is bijective, and it is the same of f . $\square$

Proposition (8.9.4). — (“generic flatness theorem”). *Let Y be an integral prescheme, $u : X \rightarrow Y$ a morphism of finite type and locally of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. Then there exists a non-empty open subset U of Y such that $\mathcal{F}|_{u^{-1}(U)}$ is flat over U .*

Proof. The reasoning from the beginning of (6.9.1) reduces to proving the

Lemma (8.9.4.1). — *Let A be an integral ring, B an A -algebra of finite presentation, M a B -module of finite presentation. Then there exists $f \neq 0$ in A such that M_f is an A_f -module libre.*

Indeed, by (8.9.1) there is a sub- $\mathbf{Z}$ -algebra of finite type A_0 of A , a A_0 -algebra of finite type B_0 and a B_0 -module of finite type M_0 such that B is A -isomorphic to $B_0 \otimes_{A_0} A$ and M is B -isomorphic to $M_0 \otimes_{A_0} A$; by (6.9.2), there exists $f_0 \neq 0$ in A_0 such that $(M_0)_{f_0}$ is an $(A_0)_{f_0}$ -module libre. As $M_{f_0} = (M_0)_{f_0} \otimes_{(A_0)_{f_0}} A_{f_0}$ and $A_{f_0} = (A_0)_{f_0} \otimes_{A_0} A$, M_{f_0} is an A_{f_0} -module libre. $\square$

Corollary (8.9.5). — *Let S be a prescheme quasi-compact and quasi-separated, $u : X \rightarrow S$ a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. Then there exists a partition $(S_i)_{1 \leq i \leq n}$ of S into a finite number of locally closed sets in S , having for each $1 \leq i \leq n$ a sub-prescheme S'_i of S of finite presentation, having S_i as underlying space, such that if one sets $X_i = X \times_S S'_i$, the $\mathcal{O}_{X_i}$ -Module $\mathcal{F}_i = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'_i}$ is flat over S'_i .*

8.9. Permanence of properties of morphisms by passage to the projective limit

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. 8.10.0 In this section we keep the general hypotheses and notations of (8.8.1).

Proposition (8.10.1). — *If there exists λ such that, for $\mu \geq \lambda$, f_μ is an open (resp. universally open) morphism, then f is an open (resp. universally open) morphism.*

Proof. As $X = X_\lambda \times_{Y_\lambda} Y$, the assertion relative to universally open morphisms is a particular case of (2.4.3, (iii)). Suppose f_μ open for $\mu \geq \lambda$; it suffices to see that for every open quasi-compact U of X , $f(U)$ is open in Y . Now, it exists $\mu \geq \lambda$ such that $U = v_\mu^{-1}(U_\mu)$, where U_μ is a quasi-compact open in X_μ (8.2.11); one has then $f(v_\mu^{-1}(U_\mu)) = w_\mu^{-1}(f_\mu(U_\mu))$ (I, 3.4.8), hence $f(U)$ is open in Y . $\square$

Corollary (8.10.2). — *Let $f : X \rightarrow Y$ be a morphism. For f to be universally open, it suffices that, for every integer $n > 0$, if one sets $Y_n = Y \otimes_{\mathbf{Z}} \mathbf{Z}[T_1, \dots, T_n](= \mathbf{V}_Y^n)$, and $X_n = X \times_Y Y_n$, the projection $f_n = f \times 1_{Y_n} : X_n \rightarrow Y_n$ is an open morphism.*

Proposition (8.10.3). — *Suppose there exists α such that: 1° S_α is quasi-compact; 2° the morphisms $X_\alpha \rightarrow S_\alpha$, $Y_\alpha \rightarrow S_\alpha$ are quasi-compact and the morphism $Y_\alpha \rightarrow S_\alpha$ is quasi-separated; 3° for $\alpha \leq \lambda \leq \mu$, the morphisms $u_{\lambda\mu} : S_\mu \rightarrow S_\lambda$ are flat; 4° $f_\alpha(X_\alpha)$ is constructible in Y_α . Then, for f to be dominant, it is necessary and sufficient that there exists $\lambda \geq \alpha$ such that f_λ is dominant.*

Proposition (8.10.4). — *Suppose there exists α such that Y_α is quasi-compact and of finite type on S_α . For the morphism f to be separated, it is necessary and sufficient that there exists $\lambda \geq \alpha$ such that f_λ is separated.*

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Theorem (8.10.5). — *Suppose S_0 quasi-compact, X_α and Y_α of finite presentation on S_α , and let $f_\alpha : X_\alpha \rightarrow Y_\alpha$ be an S_α -morphism. Consider, for a morphism, the property of being:*

- (i) an isomorphism;
- (i bis) a monomorphism;
- (ii) an immersion;
- (iii) an open immersion;
- (iv) a closed immersion;
- (v) separated;
- (vi) surjective;
- (vii) radiciel;
- (viii) affine;
- (ix) quasi-affine;
- (x) finite;
- (xi) quasi-finite;
- (xii) proper.

Then, if $\mathbf{P}$ denotes one of the preceding properties, for f to possess $\mathbf{P}$, it is necessary and sufficient that there exists $\lambda \geq \alpha$ such that f_λ possesses property $\mathbf{P}$ (in which case f_μ also possesses it for $\mu \geq \lambda$).

If S_0 is moreover supposed quasi-separated, the same conclusion is valid when $\mathbf{P}$ is the property of being:

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- (xiii) projective;
- (xiv) quasi-projective.

Proof (sketch). The case where $\mathbf{P}$ is one of properties (i) or (v) was not inserted in the statement since, for those cases, the theorem follows from what was already shown in (8.8.2.4) and (8.10.4) respectively. Moreover, taking into account (I, 5.4.1) and (5.3.4), (v) follows also from (iv). The case (i bis) follows immediately from (i), using (I, 5.3.8) and noting that the diagonal Δ_{f_λ} of $X_\lambda \times_{Y_\lambda} X_\lambda$ is deduced from Δ_f by base change $S \rightarrow S_\lambda$.

We note on the other hand that (vi), (vii) and (xi) are in fact conditions on the fibres $f^{-1}(y)$ of the morphisms considered, taking into account the transitivity of fibres by base change (I, 3.6.4): condition (vi) means that all fibres are non-empty; condition (vii) that they must be radiciel (I, 3.5.8); and condition (xi) that they must be finite (II, 6.2.2) and (I, 6.4.4), taking into account that f and the f_λ are morphisms of finite type by (1.5.4, (v)). The theorem in these three cases will thus be a consequence of a general result on a type of properties concerning only the fibres, which will be established in (9.3.3); we postpone the proof of the theorem in case (xi) until then, and note that

(8.11) and (8.12) will not use the theorem up to (9.3.3) and that (8.11) and (8.12) are not used before (9.3.3).

For the remaining cases, one can reduce to proving that the condition in the statement is necessary, all considered properties **P** being invariant by base change (see chap. I^{er} and II in the sections concerning each of these properties). One can moreover suppose $S_\alpha = S_0$ and $Y_\alpha = S_\alpha$, hence $Y_\lambda = S_\lambda$ for all $\lambda \geq \alpha$. Finally, properties (i) through (xii) are local on S_0 , hence one can further suppose $S_0 = \text{Spec}(A_0)$ affine (and quasi-separated). We designate by A_λ the ring of S_λ (resp. S).

Cases (ii), (iii), (iv): these follow from (8.6.3) applied to the sub-prescheme X' associated to f .

Cases (viii) and (ix): follow from (8.10.5), applied to the morphism $j : X \rightarrow \mathbf{V}_S^r = \text{Spec}(S[T_1, \dots, T_r])$ of S -preschemes of finite presentation.

Case (x): By hypothesis, $X = \text{Spec}(B)$, where B is an A -algebra which is an A -module of finite presentation (1.4.7); it follows from (8.5.2), (ii) that there is a λ and an A_λ -module of finite presentation B_λ such that the A -module B is isomorphic to $B_\lambda \otimes_{A_\lambda} A$. The A -algebra structure on B is defined by an A -homomorphism $m : B \otimes_A B \rightarrow B$; by (8.5.2), (i) there exists $\mu \geq \lambda$ an A_μ -homomorphism $m_\mu : B_\mu \otimes_{A_\mu} B_\mu \rightarrow B_\mu$ such that $m = m_\mu \otimes 1$. Considering the usual diagrams expressing the associativity and commutativity of m , one sees, applying (8.5.2), (i) again, that one can define on B_ν a multiplication making it into an A_ν -algebra for ν taken large enough, so that $X'_\nu = \text{Spec}(B_\nu)$ is S_ν -isomorphic to $X_\nu \times_{S_\nu} S$, by (i), hence there exist $\rho \geq \nu$ such that X_ρ and X'_ρ are S_ρ -isomorphic, which finishes the proof in this case.

Case (xii): Apply to the morphism $f_0 : X_0 \rightarrow S_0$ the proposition (8.10.5.1) (Chow's lemma for morphisms of finite presentation).

Cases (xiii) and (xiv): By (xii) and (II, 5.5.3) (which is applicable since the S_λ are quasi-compact and quasi-separated, taking into account (1.7.19)), it suffices to consider the case where f is quasi-projective. $\square$

Proposition (8.10.5.1). — (Chow's lemma for morphisms of finite presentation). Let A be a ring, X, Y two A -preschemes of finite presentation, $f : X \rightarrow Y$ a separated A -morphism. Then there exist two A -preschemes X', P of finite presentation and A -morphisms $p : P \rightarrow Y, j : X' \rightarrow P, g : X' \rightarrow X$, such that the diagram

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$$\begin{array}{ccc} X' & \xrightarrow{j} & P \\ \uparrow g & & \uparrow p \\ X & \xrightarrow{f} & Y \end{array}$$

is commutative, and such that: $1^\circ p$ is projective; $2^\circ g$ is projective and surjective; $3^\circ j$ is an open immersion.

8.10. Application to quasi-finite morphisms We propose to show in this section the two following theorems:

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Theorem (8.11.1). — Let $f : X \rightarrow Y$ be a morphism which is proper, locally of finite presentation, and quasi-finite. Then the morphism f is finite.

Theorem (8.11.2). — Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, quasi-finite and separated. Then the morphism f is quasi-affine, and a fortiori quasi-projective.

Remark (8.11.3). — We will see below that the proofs of (8.11.1) and (8.11.2), in the case where Y is locally noetherian, yield another proof of Chevalley's theorem (III, 4.4.2).

8.11.4 The hypotheses and conclusions of (8.11.1) and (8.11.2) are all local on Y (1.6.1, 1.2.6, (II, 5.1.1), (II, 5.4.1) and (II, 6.2.2)), hence one can suppose $Y = \text{Spec}(A)$ affine. One knows that there exists then a sub-ring A_0 of A , which is a $\mathbf{Z}$ -algebra of finite type, and a morphism of finite type $f_0 : X_0 \rightarrow \text{Spec}(A_0)$ such that X identifies with $X_0 \otimes_{A_0} A$ and f with $f_0 \times 1$ (8.9.1). Furthermore, A can be considered as an inductive limit of its sub-rings containing A_0 and which are $\mathbf{Z}$ -algebras of finite type; using (8.1.2, c)) and (8.10.5), (v), (xi) and (xii), one sees that it suffices to prove theorems (8.11.1) and (8.11.2) for f_0 ; using now the method of (8.1.2, a)) and (8.10.5), (v), (ix), (x), (xi) and (xii), one can replace Y by $\text{Spec}(\mathcal{O}_y)$, where y is any point of Y .

Proposition (8.11.5). — *Let $f : X \rightarrow Y$ be a morphism of finite presentation. The following properties are equivalent:*

- a) f is a closed immersion.
- b) f is a proper monomorphism.
- c) f is proper and for all $y \in Y$, $f^{-1}(y)$ is radiciel and geometrically reduced over $k(y)$ (that is to say either empty or $k(y)$ -isomorphic to $\text{Spec}(k(y))$).

Proof. It is clear that a) implies b). To see that b) implies c), note (I, 3.3.12) that for all $y \in Y$, the morphism $f^{-1}(y) \rightarrow \text{Spec}(k(y))$ deduced from f by base extension is a monomorphism, hence is injective, and in any case affine. Moreover, if A is the ring of $f^{-1}(y)$, the homomorphism canonical $A \otimes_k A \rightarrow A$ is bijective (I, 5.3.8). This evidently implies that $A = k$, unless $A = 0$.

It remains to prove that c) implies a). It follows first from (8.11.1) that f is finite, then a fortiori $X = \text{Spec}(\mathcal{A})$, where $\mathcal{A}$ is an $\mathcal{O}_Y$ -Algebra finite. It suffices then to prove that the canonical homomorphism $\mathcal{O}_Y \rightarrow \mathcal{A}_y$ is surjective (II, 1.4.10), or equivalently that for all $y \in Y$, the homomorphism $\mathcal{O}_y \rightarrow \mathcal{A}_y$ is surjective. But by hypothesis $f^{-1}(y) = \text{Spec}(\mathcal{A}_y/\mathfrak{m}_y\mathcal{A}_y)$ (II, 1.5.5) is such that the corresponding homomorphism $k(y) = \mathcal{O}_y/\mathfrak{m}_y \rightarrow \mathcal{A}_y/\mathfrak{m}_y\mathcal{A}_y$ is bijective; as $\mathcal{A}_y$ is an $\mathcal{O}_y$ -module of finite type, the lemma of Nakayama shows that $\mathcal{O}_y \rightarrow \mathcal{A}_y$ is surjective, which finishes the proof. $\square$

Proposition (8.11.6). — *If a morphism $f : X \rightarrow Y$ of finite presentation is a universal homeomorphism, it is finite, surjective, and radiciel (the converse being true by (2.4.5, (i))).*

Proof. Indeed, f being of finite type, universally closed, and separated by (2.4.4), is proper by definition (II, 5.4.1). As it is evidently quasi-finite (II, 6.2.3), it is finite by (8.11.1). One knows moreover that it is surjective (2.4.2), and evidently radiciel. $\square$

8.11. New proof and generalization of Zariski's "Main Theorem"

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Lemma (8.12.1). — *Let $f : X \rightarrow Y$ be a morphism quasi-compact and quasi-separated, $\mathcal{C}$ a quasi-coherent $\mathcal{O}_Y$ -Algebra, $Z = \text{Spec}(\mathcal{C})$, which is a Y -prescheme affine above Y . Let $g : X \rightarrow Z$ be a Y -morphism, $\varphi = \mathcal{A}(g) : \mathcal{C} \rightarrow f_*(\mathcal{O}_X)$ the corresponding $\mathcal{O}_Y$ -homomorphism of $\mathcal{O}_Y$ -Algebras (II, 1.2.7). Suppose that g is an immersion. Then, for the closed image of X by g to be equal to Z , it is necessary and sufficient that φ be injective; g is then an open immersion.*

Lemma (8.12.2). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \rightarrow Y$ a morphism quasi-separated and of finite type, $\mathcal{C}$ a quasi-coherent $\mathcal{O}_Y$ -Algebra, $Z = \text{Spec}(\mathcal{C})$, $\varphi : \mathcal{C} \rightarrow f_*(\mathcal{O}_X)$ the corresponding $\mathcal{O}_Y$ -homomorphism. Let $(\mathcal{C}_\lambda)$ be the filtering increasing family of quasi-coherent sub- $\mathcal{O}_Y$ -Algebras of finite type of $\mathcal{C}$ (whose $\mathcal{C}$ is the union (I, 9.6.6) and (1.7.9)); set $Z_\lambda = \text{Spec}(\mathcal{C}_\lambda)$ and let g_λ be the composed morphism $X \xrightarrow{g} Z \rightarrow Z_\lambda$. Then the following conditions are equivalent:*

- a) The morphism $g : X \rightarrow Z$ is an immersion.
- b) There exists λ such that $g_\lambda : X \rightarrow Z_\lambda$ is an immersion.
- c) There exists λ such that $g_\lambda : X \rightarrow Z_\lambda$ is an immersion.

Moreover, when g_λ is an immersion, it is the same of g_μ for $\mu \geq \lambda$, and there exists λ such that, for $\mu \geq \lambda$, g_μ is an open immersion.

Proposition (8.12.3). — *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \rightarrow Y$ a morphism separated and of finite type. Let $\mathcal{B} = f_*(\mathcal{O}_X)$, which is a quasi-coherent $\mathcal{O}_Y$ -Algebra (I, 9.2.2); let $\mathcal{C}$ be the integral closure of $\mathcal{O}_Y$ in $\mathcal{B}$, which is a quasi-coherent $\mathcal{O}_Y$ -Algebra (II, 6.3.4). Set $Z = \text{Spec}(\mathcal{C})$, and let $g : X \rightarrow Z$ be the Y -morphism corresponding to the canonical injection $\mathcal{C}_\lambda \rightarrow \mathcal{B} = f_*(\mathcal{O}_X)$ (II, 1.2.7). Let $(\mathcal{C}_\lambda)$ be the filtering increasing family of sub- $\mathcal{O}_Y$ -Algebras quasi-coherent of finite type of $\mathcal{C}$ (whose $\mathcal{C}$ is the union (I, 9.6.6) and (1.7.9)), and, for all λ , $g_\lambda : X \rightarrow Z_\lambda$ the Y -morphism corresponding to the injection $\mathcal{C}_\lambda \rightarrow \mathcal{B} = f_*(\mathcal{O}_X)$. Then the following conditions are equivalent:*

- a) There exists a factorization of f as

$$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.

- a') There exists a factorization $X \xrightarrow{f'} Y' \xrightarrow{u} Y$ of f , where f' is an open immersion and u a finite morphism.

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- b) The morphism $g : X \rightarrow Z$ is an immersion.
- c) There exists λ such that $g_\lambda : X \rightarrow Z_\lambda$ is an immersion.

Moreover, when g_λ is an immersion, so is g_μ for $\mu \geq \lambda$, and there exists λ such that, for $\mu \geq \lambda$, g_μ is an open immersion.

We will say that a morphism $f : X \rightarrow Y$, where Y is quasi-compact and quasi-separated, is *pseudo-finite*, if it is of finite type and satisfies condition a) of (8.12.3) (in which case it is necessarily separated).

Corollary (8.12.4). — Let Y be a prescheme quasi-compact and quasi-separated, $f : X \rightarrow Y$ a morphism. IV-3 | 45

- (i) Suppose f pseudo-finite. Then, for every morphism $Y' \rightarrow Y$, where Y' is quasi-compact and quasi-separated, $f_{(Y')} : X' = X_{(Y')} \rightarrow Y'$ is pseudo-finite.
- (ii) Let (U_λ) be a covering of Y by open quasi-compact subsets. For f to be pseudo-finite, it is necessary and sufficient that for all λ , the restriction $f_\lambda : f^{-1}(U_\lambda) \rightarrow U_\lambda$ of f is a pseudo-finite morphism.
- (iii) Suppose moreover Y noetherian, and f of finite type. Then, for f to be pseudo-finite, it is necessary and sufficient that, for every $y \in Y$, the morphism $f_y = f \times 1 : X_y = X \times_Y \text{Spec}(\mathcal{O}_y) \rightarrow \text{Spec}(\mathcal{O}_y)$ be pseudo-finite.

.8.12.5 We can now give a proof of Zariski's "Main theorem" (III, 4.4.3) not using the "global" cohomological results of chap. III, but appealing instead to finer properties of local noetherian rings; we will moreover generalize the statement of the theorem by eliminating the noetherian hypotheses:

Theorem (8.12.6). — ("Main theorem" of Zariski). Let Y be a prescheme quasi-compact and quasi-separated. If a morphism $f : X \rightarrow Y$ is quasi-finite, separated and of finite presentation, there exists a factorization of f

$$(8.12.6.1) \quad X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.

Proof. By virtue of (8.12.4) (ii) and the local character (on Y) of the notions of quasi-finite, separated, and finitely presented morphism, we can reduce to the case where $Y = \text{Spec}(A)$ is affine. Applying (8.9.1), we can suppose that there is a subring A_0 of A , which is a $\mathbf{Z}$ -algebra of finite type, and an A -isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$, f being identified via this isomorphism with $f_0 \times 1$, where $f_0 : X_0 \rightarrow \text{Spec}(A_0)$ is a morphism of finite type; moreover ((8.10.5), (v) and (xi)) we can suppose that f_0 is separated and quasi-finite; if we prove that f_0 is pseudo-finite, the same will hold for f by (8.12.4) (i). Since A_0 is then noetherian and the notions of morphism of finite type, separated, and quasi-finite are preserved by base change, it follows from (8.12.4) (iii) that we can even suppose that A is a local ring, *essentially of finite type over $\mathbf{Z}$* (I, 1.3.8). Set $n = \dim(A)$, and proceed by induction on n ; for $n = 0$, the theorem is evident, A being a field and the morphism f being already finite (II, 6.2.2). Set $B = \Gamma(X, \mathcal{O}_X)$; denote by C the integral closure of A in B , set $Z = \text{Spec}(C)$ and let $g : X \rightarrow Z$ be the Y -morphism corresponding to the injective A -homomorphism $C \rightarrow B$; by virtue of (8.12.3), it suffices to show that g is an open immersion. Let a be the closed point of Y , and let $U = Y - \{a\}$; U is noetherian and all its local rings are essentially of finite type over $\mathbf{Z}$ and of dimension $< n$; taking into account the induction hypothesis and (8.12.4) (iii), we see that the restriction $f^{-1}(U) \rightarrow U$ of f is a pseudo-finite morphism. We conclude from (8.12.3) that, if $h : Z \rightarrow Y$ is the structural morphism, the restriction $f^{-1}(U) \rightarrow h^{-1}(U)$ of g is an open immersion. Set $A' = \hat{A}$, $Y' = \text{Spec}(A')$, $X' = X_{(Y')}$, $f' = f_{(Y')} : X' \rightarrow Y'$. Since the canonical morphism $u : Y' \rightarrow Y$ is flat, it follows from (2.3.1) that $B' = \Gamma(X', \mathcal{O}_{X'})$ is identified with the A' -algebra $B \otimes_A A'$. On the other hand, since A is an excellent local ring (7.8.3), the morphism $u : Y' \rightarrow Y$ is regular, and *a fortiori* normal, and consequently (6.14.4) the integral closure of A' in B' is equal to $C' = C \otimes_A A'$. We thus see that $Z' = \text{Spec}(C')$ is equal to $Z_{(Y')}$ and the morphism $g' : X' \rightarrow Z'$ coming from the injection $C' \rightarrow B'$ is equal to $g_{(Y')}$. Since $u : Y' \rightarrow Y$ is faithfully flat and quasi-compact, to prove that g is an open immersion, it suffices to prove that g' is an open immersion (2.7.1) (x). Note now that $u^{-1}(a)$ is reduced to the closed point a' of Y' and consequently $U' = Y' - \{a'\} = u^{-1}(U)$. If $h' : Z' \rightarrow Y'$ is the canonical morphism, the fact that the restriction $f^{-1}(U) \rightarrow h^{-1}(U)$ of g is an open immersion implies that the same holds for the restriction $f'^{-1}(U') \rightarrow h'^{-1}(U')$ of g' . Note that f' is a separated and quasi-finite morphism (II, 6.2.4); since A' is complete, we deduce from (II, 6.2.6) that X' is Y' -isomorphic to a sum $X'_1 \sqcup X'_2$,

where the restriction $f'|_{X'_1} = f'_1 : X'_1 \rightarrow Y'$ is a *finite* morphism, and $X'_2 \subset f'^{-1}(U')$. It follows that B' is the direct product of the two A' -algebras $\Gamma(X'_1, \mathcal{O}_{X'_1}) = B'_1$ and $\Gamma(X'_2, \mathcal{O}_{X'_2}) = B'_2$; we immediately conclude that the integral closure C' of A' in B' is composed of the integral closures C'_1, C'_2 of A' in B'_1, B'_2 respectively, whence $Z' = Z'_1 \sqcup Z'_2$, where $Z'_i = \text{Spec}(C'_i)$ ($i = 1, 2$); and the canonical morphism $g' : X' \rightarrow Z'$ is such that $g'|_{X'_i}$ is the canonical morphism $g'_i : X'_i \rightarrow Z'_i$ ($i = 1, 2$). But since B'_1 is already a finite A' -algebra, we have $C'_1 = B'_1$, and g'_1 is therefore an isomorphism. On the other hand, since $X'_2 \subset f'^{-1}(U')$ and is open in $f'^{-1}(U')$, we already know that g'_2 is an open immersion. We conclude that g' is an open immersion. $\square$

Remark (8.12.7). — When, in (8.12.6), we suppose that Y is an affine *scheme*, the proof by reduction to the noetherian case shows that, in the factorization (8.12.6.1), the morphisms f' and u are also morphisms of *finite presentation* (I, 1.6.2).

Corollary (8.12.8). — Let Y be a quasi-compact scheme such that there exists an ample $\mathcal{O}_Y$ -Module (II, 4.5.3), $f : X \rightarrow Y$ a quasi-finite and quasi-projective morphism. Then there exists a factorization of f as

$$X \xrightarrow{f'} Y' \xrightarrow{u} Y$$

where f' is an open immersion and u a finite morphism.⁵

Proof. The hypothesis implies that X is identified with a quasi-compact sub- Y -scheme of a Y -scheme of the form $Z = \mathbf{P}_Y^r$ (II, 5.3.3). There is thus a quasi-compact open neighborhood U of X in Z such that X is closed in U ; since Z is a scheme, the canonical injection $U \rightarrow Z$ is a morphism of finite presentation ((I, 1.2.7) and (I, 1.6.2)), so the composite morphism $g : U \rightarrow Z \rightarrow Y$ is also a morphism of finite presentation (the fact that $\mathbf{P}_Y^r$ is of finite presentation over Y resulting immediately from the definition (II, 4.1.1)). Let $\mathcal{J}$ be the quasi-coherent Ideal of $\mathcal{O}_U$ defining the closed sub-prescheme X ; since U is a quasi-compact scheme, $\mathcal{J}$ is the inductive limit of its quasi-coherent sub-Ideals of finite type $\mathcal{J}_\lambda$ (I, 9.4.9). If X_λ is the closed sub-prescheme of U defined by $\mathcal{J}_\lambda$, we thus have $X = \bigcap_\lambda X_\lambda$. For all $y \in Y$, we thus have $f^{-1}(y) = \bigcap_\lambda (X_\lambda \cap g^{-1}(y))$, and since the sets $X_\lambda \cap g^{-1}(y)$ are closed in the noetherian space $g^{-1}(y)$, for every y there exists an index $\lambda(y)$ such that $f^{-1}(y) = X_{\lambda(y)} \cap g^{-1}(y)$. Denote by E_λ the set of $y \in Y$ such that the fibre $X_\lambda \cap g^{-1}(y)$ of the restriction of g to X_λ is a finite $k(y)$ -prescheme. The hypothesis that f is quasi-finite implies, by virtue of the above, that $Y = \bigcup E_\lambda$. Now, each of the X_λ is, by definition, of *finite presentation* over Y ; it thus follows from (9.2.3) and (9.2.6) that the E_λ are *constructible* subsets in the scheme Y ; since they form a filtered increasing collection, there exists an index λ such that $E_\lambda = Y$ (I, 1.9.9), and for this index λ , the morphism $f_\lambda : X_\lambda \rightarrow Y$, restriction of g to X_λ , is thus *quasi-finite*. Since it is of finite presentation and separated, we can apply (8.12.6) to it, and f_λ thus factorizes as

$$X_\lambda \xrightarrow{j_\lambda} Y'_\lambda \xrightarrow{u_\lambda} Y$$

where j_λ is an immersion and u_λ a finite morphism. Since X is a closed sub-prescheme of X_λ , we have thus proved that f possesses the property (8.12.3) a), whence the corollary by virtue of the equivalence of (8.12.3) a) and (8.12.3) a'). $\square$

Corollary (8.12.9). — Let $f : X \rightarrow Y$ be a locally quasi-finite morphism (Err_{III}, 20). For every $x \in X$ there exists an open neighborhood U of x in X , an open neighborhood V of $y = f(x)$ in Y , such that $f(U) \subset V$ and a factorization

$$U \xrightarrow{f'} V' \xrightarrow{u} V$$

of the restriction of f to U , where f' is an open immersion and u a finite morphism.

Proof. It suffices to take for V an affine neighborhood of y in Y , for U an affine neighborhood of x in X contained in $f^{-1}(V)$ and such that $f|_U$ is quasi-finite. The morphism $U \rightarrow V$ restriction of f being then affine (hence quasi-projective), we can apply (8.12.8) to it. $\square$

Corollary (8.12.10). — Let Y be an integral and normal prescheme, X an integral prescheme, $f : X \rightarrow Y$ a birational and locally quasi-finite morphism (Err_{III}, 20). Then f is a local isomorphism; for f to be an open immersion, it is necessary and sufficient that f be moreover separated.

⁵The reader will verify that Corollaries (8.12.8) to (8.12.11) are not used in § 9.

Proof. The second assertion follows immediately from the first and from (I, 8.2.8). To prove the first assertion, we can suppose X and Y affine and f quasi-finite; consider the factorization $f = u \circ f'$ of (8.12.8), which allows us to identify X by f' with a sub-prescheme induced on an open subset of Y' . Since X is integral, we can, by virtue of (I, 5.2.3), replace Y' by the reduced sub-prescheme of Y' having for underlying space $\bar{X}$, so we can suppose that Y' is also integral. Moreover, since f is birational, u is as well. The conclusion then follows from the following lemma: $\square$

Lemma (8.12.10.1). — *Let Y' be an integral prescheme, Y an integral and normal prescheme; then a morphism $u : Y' \rightarrow Y$ that is finite and birational is an isomorphism.*

Proof. Set $\mathcal{A} = u_*(\mathcal{O}_{Y'})$, so that $\mathcal{A}$ is a finite $\mathcal{O}_Y$ -Algebra, Y' being identified with $\text{Spec}(\mathcal{A})$ (II, 1.3.6). If $R(Y)$ is the field of rational functions of Y , we thus have, for all $y \in Y$, $\mathcal{O}_{Y,y} \subset \mathcal{A}_y \subset R(Y)$; but since the ring $\mathcal{O}_{Y,y}$ is by hypothesis integrally closed and has $R(Y)$ as its field of fractions, we necessarily have $\mathcal{A}_y = \mathcal{O}_{Y,y}$, whence the lemma. $\square$

Corollary (8.12.11). — *Let Y be an integral and normal prescheme, X an integral prescheme, $f : X \rightarrow Y$ a dominant and locally quasi-finite morphism. Let K and L be the fields of rational functions of Y and X respectively (extension of K); and let Y' be the integral closure of Y relative to L (II, 6.3.4); then f factorizes in a unique way as $f : X \xrightarrow{f'} Y' \xrightarrow{u} Y$, where f' is birational, and corresponds to the identity automorphism of L ; f' is then a local isomorphism, and for f' to be an open immersion, it is necessary and sufficient that f be separated.*

Proof. The existence and uniqueness of the factorization of f follow from (II, 6.3.9). It follows from ((II, 6.2.4), (v)), reducing to the affine case, that f' is locally quasi-finite; moreover, it follows from (I, 5.5.1) that, for f' to be separated, it is necessary and sufficient that f be so, since u is affine, hence separated; the two last assertions are thus consequences of (8.12.10) applied to f' . $\square$

8.12. Translation in terms of pro-objects

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Proposition (8.13.1). — *Let S be a prescheme, $(X_\lambda, v_{\lambda\mu})$ a filtered projective system of S -preschemes; suppose that there exists α such that $v_{\alpha\lambda}$ is an affine morphism for all $\lambda \geq \alpha$ (which implies (II, 1.6.2) that $v_{\lambda\mu}$ is affine for $\alpha \leq \lambda \leq \mu$), so that the projective limit $X = \varprojlim X_\lambda$ exists in the category of S -preschemes (8.2.3). Let Y be an S -prescheme, and, for all $\lambda \geq \alpha$, let $e_\lambda : \text{Hom}_S(X_\lambda, Y) \rightarrow \text{Hom}_S(X, Y)$ be the map which, to every S -morphism $f_\lambda : X_\lambda \rightarrow Y$, associates $f = f_\lambda \circ v_\lambda$, where $v_\lambda : X \rightarrow X_\lambda$ is the canonical morphism. The family (e_λ) is an inductive system of maps, which thus defines a canonical map*

$$(8.13.1.1) \quad \varinjlim \text{Hom}_S(X_\lambda, Y) \longrightarrow \text{Hom}_S(X, Y).$$

Suppose X_α quasi-compact (resp. quasi-compact and quasi-separated), and the structural morphism $Y \rightarrow S$ locally of finite type (resp. locally of finite presentation). Then the map (8.13.1.1) is injective (resp. bijective).

Proof. Set, for $\lambda \geq \alpha$, $Z_\lambda = Y \times_S X_\lambda$, so that $Z_\lambda = Z_\alpha \times_{X_\alpha} X_\lambda$. Set also $Z = Y \times_S X = Z_\alpha \times_{X_\alpha} X$; we know (8.2.5) that, if we also set $w_{\lambda\mu} = 1 \times v_{\lambda\mu} : Z_\mu \rightarrow Z_\lambda$ for $\alpha \leq \lambda \leq \mu$ and $w_\lambda = 1 \times v_\lambda : Z \rightarrow Z_\lambda$ for $\alpha \leq \lambda$, Z is the projective limit of the system $(Z_\lambda, w_{\lambda\mu})$ and the w_λ are the canonical corresponding morphisms. Note moreover that the morphism $Z_\alpha \rightarrow X_\alpha$ is locally of finite type (resp. locally of finite presentation (I, 1.3.4) and (I, 1.4.3)). Finally, we know that

$$\text{Hom}_S(X_\lambda, Y) = \text{Hom}_{X_\lambda}(X_\lambda, Z_\lambda) \quad \text{and} \quad \text{Hom}_S(X, Y) = \text{Hom}_X(X, Z)$$

(I, 3.3.14). It then suffices to apply ((8.8.2), (i)) replacing Y_λ by Z_λ and y by $X_\lambda = S_\lambda$. $\square$

Corollary (8.13.2). — *With the notations of (8.13.1), suppose X_α quasi-compact and quasi-separated, and the $v_{\alpha\lambda}$ affine for $\alpha \leq \lambda$; suppose moreover that $Y = \varprojlim Y_\rho$, where $(Y_\rho, t_{\rho\sigma})$ is a filtered projective system of S -preschemes such that, for each ρ , the structural morphism $Y_\rho \rightarrow S$ is locally of finite presentation. Then we have a canonical bijection*

$$(8.13.2.1) \quad \text{Hom}_S(X, Y) \xrightarrow{\sim} \varprojlim_{\rho} \left(\varinjlim_{\lambda} \text{Hom}_S(X_\lambda, Y_\rho) \right).$$

Proof. Indeed, the fact that Y is the projective limit of the Y_ρ implies in particular that the canonical map $\mathrm{Hom}_S(X, Y) \rightarrow \varprojlim_\rho \mathrm{Hom}_S(X, Y_\rho)$ is bijective; and on the other hand, the hypotheses imply, for each ρ , the existence of a canonical bijection $\mathrm{Hom}_S(X, Y_\rho) \xrightarrow{\sim} \varinjlim_\lambda \mathrm{Hom}_S(X_\lambda, Y_\rho)$ by virtue of (8.13.1); whence the conclusion. $\square$

.8.13.3 The preceding results allow us to interpret in the theory of preschemes the notions of “pro-variety” or “pro-scheme” that arise in certain applications (for example in the theory of the local class field following the ideas of Serre [?] or in the theory of Néron on the reduction of abelian varieties [?]). Let us recall briefly here the notion of *pro-object* of a category, referring to chap. V for more ample developments (we will not in fact use before chap. V the interpretation that follows, and the reader may thus omit until then the reading of the end of this number). Given a category $\mathcal{C}$, the category $\mathbf{Pro}(\mathcal{C})$ of *pro-objects* of $\mathcal{C}$ has for objects the *projective systems* (in the universe in which we place ourselves) $\mathbf{X} = (X_\mu)_{\mu \in M}$ of objects of $\mathcal{C}$ whose index sets (depending on the projective system considered) are supposed preordered and filtered. Given two such pro-objects $\mathbf{X} = (X_\mu)_{\mu \in M}$, $\mathbf{X}' = (X'_{\mu'})_{\mu' \in M'}$, the *morphisms* of $\mathbf{X}$ into $\mathbf{X}'$ are by definition the elements of the set $\varprojlim_{\mu'} (\varinjlim_\mu \mathrm{Hom}(X_\mu, X'_{\mu'}))$; the verification that one can take these sets for sets of morphisms is immediate, the composition of systems of morphisms $u_{\mu'}^\mu : X_\mu \rightarrow X'_{\mu'}$, $u_{\mu''}^{\mu'} : X'_{\mu'} \rightarrow X''_{\mu''}$, which are inductive in the upper index and projective in the lower index, being done “argument by argument”, in other words by considering the system of $u_{\mu''}^{\mu''} = u_{\mu''}^{\mu'} \circ u_{\mu'}^\mu$.

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.8.13.4 Consider a quasi-compact and quasi-separated prescheme S , and denote by $\mathcal{C}$ the full subcategory of the category $(\mathbf{Sch})_S$ of S -preschemes, formed by the S -preschemes X having the following property: the morphism $X \rightarrow S$ factorizes as $X \xrightarrow{g} X_0 \xrightarrow{f} S$, where $g : X \rightarrow X_0$ is affine and $f : X_0 \rightarrow S$ is of finite presentation; we will say for brevity that the preschemes of $\mathcal{C}$ are *essentially affine over S* .

Consider on the other hand the full subcategory $\mathcal{C}'$ of $(\mathbf{Sch})_S$ formed by the S -preschemes of finite presentation, and the category $\mathbf{Pro}(\mathcal{C}')$ of pro-objects of $\mathcal{C}'$. We will say that an object $\mathbf{X} = (X_\mu)_{\mu \in M}$ of $\mathbf{Pro}(\mathcal{C}')$ is *essentially affine* if there exists $\gamma \in M$ such that for all $\mu \geq \gamma$, the transition morphism $v_{\gamma\mu} : X_\mu \rightarrow X_\gamma$ is *affine* (which implies that for $\gamma \leq \mu \leq \nu$, $v_{\mu\nu} : X_\nu \rightarrow X_\mu$ is affine). Note that an object of $\mathbf{Pro}(\mathcal{C}')$ isomorphic to an essentially affine object is not necessarily itself essentially affine. We will denote by $\mathcal{C}'$ the full subcategory of $\mathbf{Pro}(\mathcal{C}')$ formed by the essentially affine pro-objects of $\mathcal{C}'$.

This being so, it follows from (8.2.2) and (8.2.3) that for every object $\mathbf{X} = (X_\mu)_{\mu \in M}$ of $\mathcal{C}'$, the S -prescheme $X = \varprojlim_\mu X_\mu$ exists; moreover, since, for μ large enough, the morphism $X \rightarrow X_\mu$ is affine (8.2.2), X is essentially affine over S by definition. Set $X = L(\mathbf{X})$; let us show that we have thus defined a *functor*

$$(8.13.4.1) \quad L : \mathcal{C}' \longrightarrow \mathcal{C}$$

Indeed, for any two objects $\mathbf{X} = (X_\mu)$, $\mathbf{X}' = (X'_{\mu'})$ of $\mathcal{C}'$, one has for each μ' a canonical map

$$\varinjlim_\mu \mathrm{Hom}_S(X_\mu, X'_{\mu'}) \longrightarrow \mathrm{Hom}_S(\varprojlim_\mu X_\mu, X'_{\mu'})$$

defined in (8.13.1.1), and by the definition of the projective limit, a canonical bijection

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$$\varprojlim_{\mu'} \mathrm{Hom}_S(\varinjlim_\mu X_\mu, X'_{\mu'}) \xrightarrow{\sim} \mathrm{Hom}_S(\varprojlim_\mu X_\mu, \varprojlim_{\mu'} X'_{\mu'})$$

whence a canonical map

$$(8.13.4.2) \quad \varprojlim_{\mu'} (\varinjlim_\mu \mathrm{Hom}_S(X_\mu, X'_{\mu'})) \longrightarrow \mathrm{Hom}(\varprojlim_\mu X_\mu, \varprojlim_{\mu'} X'_{\mu'})$$

evidently functorial in $\mathbf{X}$ and $\mathbf{X}'$, and which completes the definition of the functor L .

Proposition (8.13.5). — *The hypotheses and notations being those of (8.13.4), the functor L is fully faithful. If moreover S is a noetherian prescheme (which already implies that S is quasi-compact and quasi-separated (I, 1.2.8)), L is an equivalence of categories.*

Proof. To say that L is fully faithful means that the map (8.13.4.2) is bijective for all X, X' in $\mathcal{C}'$, which is a particular case of (8.13.2): indeed, the structural morphisms $X_\mu \rightarrow S$ being of finite presentation, are in particular quasi-compact and quasi-separated, hence the X_μ are quasi-compact and quasi-separated.

To show that when S is noetherian L is an equivalence of categories, it suffices, since we already know that L is fully faithful, to prove that every S -prescheme essentially affine X is S -isomorphic to an object of the form $L(X)$ where $X \in \mathcal{C}'$ (0_{III}, 8.1.5). Now, there is by hypothesis a factorization $X \xrightarrow{g} X_0 \xrightarrow{f} S$ of the structural morphism, f being of finite presentation and g affine. We can thus write $X = \text{Spec}(\mathcal{A})$, where $\mathcal{A}$ is a quasi-coherent $\mathcal{O}_{X_0}$ -Algebra (II, 1.3.1). Now, since X_0 is noetherian (since S is noetherian and f is of finite type), $\mathcal{A}$ is the filtered inductive limit of the family $(\mathcal{A}_\lambda)$ of its quasi-coherent sub- $\mathcal{O}_{X_0}$ -Algebras of finite type (I, 9.6.6). Set $X_\lambda = \text{Spec}(\mathcal{A}_\lambda)$; the morphisms $X_\lambda \rightarrow X_0$ are of finite type, hence of finite presentation since X_0 is noetherian, and consequently the same holds for the composite morphisms $X_\lambda \rightarrow X_0 \rightarrow S$ (I, 1.6.2); in other words, the X_λ belong to $\mathcal{C}'_0$ and since the morphisms $X_\lambda \rightarrow X_0$ are affine, $X = (X_\lambda)$ is an object of $\mathcal{C}'$ whose projective limit exists and is S -isomorphic to X by virtue of (8.2.2). This completes the proof. $\square$

Remark (8.13.6). — It follows from (I, 1.6.2) and (II, 1.6.2) that if X and Y are essentially affine over S , then so is $X \times_S Y$. We thus conclude for example (0_{III}, 8.2.5) that a $\mathcal{C}$ -group is nothing other than a $(\text{Sch})_S$ -group which is a prescheme essentially affine over S . On the other hand, finite products exist in the category $\mathcal{C}'$: indeed, if $X = (X_\mu)_{\mu \in M}$, $Y = (Y_\rho)_{\rho \in R}$ are two objects of $\mathcal{C}'$, the products $X_\mu \times_S Y_\rho$ are S -preschemes of finite presentation, and taking for transition morphisms $X_\nu \times_S Y_\sigma \rightarrow X_\mu \times_S Y_\rho$ the products of the transition morphisms $X_\nu \rightarrow X_\mu$ and $Y_\sigma \rightarrow Y_\rho$, we see immediately that $(X_\mu \times_S Y_\rho)$ is the product of X and Y in $\text{Pro}(\mathcal{C}'_0)$; moreover (II, 1.6.2) the transition morphisms thus defined are affine for μ and ρ large enough, so the product $X \times Y$ thus defined belongs to $\mathcal{C}'$. We conclude as above that a $\mathcal{C}'$ -group is a $\text{Pro}(\mathcal{C}'_0)$ -group that is essentially affine. We deduce from (8.13.5) that the categories of $\mathcal{C}'$ -groups and $\mathcal{C}$ -groups are *equivalent* when S is noetherian. It seems plausible that when S is the spectrum of a field k , the category of $\mathcal{C}$ -groups is *equivalent* to that of $\mathbf{K}$ -groups, where $\mathbf{K}$ is the category of S -preschemes *quasi-compact*; in other words, every prescheme in groups over k that is quasi-compact would be essentially affine. On the other hand, if we denote by $\mathcal{C}'_0\text{-gr}$ the category of $\mathcal{C}'_0$ -groups, it is plausible that the category of $\mathcal{C}'$ -groups is equivalent to the full subcategory of $\text{Pro}(\mathcal{C}'_0\text{-gr})$ formed by the “essentially affine proalgebraic groups”, that is to say by the projective systems $\mathbf{G} = (G_\mu)_{\mu \in M}$, where the G_μ are algebraic groups over k and the transition morphisms $G_\nu \rightarrow G_\mu$ are affine (which one can also express by saying that $\mathbf{G}$ is an extension of an algebraic group by an affine pro-algebraic group). The conjunction of these two conjectures amounts to the following: every prescheme in quasi-compact groups over k is an “extension” of an “algebraic group” (i.e. a prescheme in groups of finite type over k) by a prescheme in *affine* groups over k .

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The only pro-algebraic groups encountered in practice up to now being in fact essentially affine, there will thus undoubtedly be an advantage in substituting the study of general pro-algebraic groups (introduced and studied by Serre [?]) by that of schemes in quasi-compact groups over k , whose definition is conceptually simpler.

8.13. Characterization of a prescheme locally of finite presentation over another, in terms of the functor it represents

. 8.14.1 Given a prescheme S , we will say again, as in (8.13.4), that a filtered projective system $(X_\lambda, v_{\lambda\mu})$ of S -preschemes is *essentially affine* if there exists α such that $v_{\alpha\lambda}$ is an affine morphism for $\lambda \geq \alpha$.

The following statement, which will be especially useful in chap. V, gives a converse to ((8.8.2), (i)):

Proposition (8.14.2). — *Let S be a prescheme, $f : X \rightarrow S$ a morphism. For every S -prescheme T , set*

$$h_X(T) = \text{Hom}_S(T, X)$$

so that h_X is a contravariant functor from the category $(\text{Sch})_S$ of S -preschemes to the category $\mathbf{Ens}$ of sets (0_{III}, 8.1.1), and X an object representing the functor (0_{III}, 8.1.8). The following conditions are equivalent:

- a) f is locally of finite presentation.
 b) For every filtered projective system (Z_λ) of S -preschemes, essentially affine (8.13.4) and formed of quasi-compact and quasi-separated preschemes, the canonical map (8.13.1.1)

$$(8.14.2.1) \quad \varinjlim h_X(Z_\lambda) \longrightarrow h_X(\varprojlim Z_\lambda)$$

is bijective.

- c) For every filtered projective system (Z_λ) of S -preschemes, such that the Z_λ are affine schemes, the map (8.14.2.1) is bijective.
 c') For every affine open subset U of S and every filtered projective system (Z_λ) of U -preschemes, such that the Z_λ are affine schemes, the map (8.14.2.1) is bijective.

Proof. The fact that a) implies b) is nothing other than (8.13.1); it is trivial that b) implies c) and that c) implies c'). It remains to see that c') implies a), and since the property a) is local on S , we can reduce to the case where S is affine.

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Suppose first that X is also affine; the assertion to prove is then equivalent to the following: $\square$

Corollary (8.14.2.2). — Let A be a ring, B an A -algebra. In order that, for every filtered inductive system (C_λ) of A -algebras, the canonical map

$$(8.14.2.3) \quad \varinjlim \operatorname{Hom}_{A\text{-alg.}}(B, C_\lambda) \longrightarrow \operatorname{Hom}_{A\text{-alg.}}(B, \varinjlim C_\lambda)$$

be bijective, it is necessary and sufficient that B be an A -algebra of finite presentation.

Proof. It remains only to show that the condition is necessary. Let us first take for (C_λ) the filtered inductive system of sub- A -algebras of finite type of B , of B , so that $\varinjlim C_\lambda = B$. The fact that (8.14.2.3) is bijective implies in particular that the identity 1_B of B factorizes as $B \rightarrow C_\lambda \rightarrow B$ for some suitable λ , which implies $C_\lambda = B$, hence B is an A -algebra of finite type. Set then $B = C/\mathfrak{J}$, where $C = A[T_1, \dots, T_n]$ and $\mathfrak{J}$ is an ideal of C . Then $\mathfrak{J}$ is the filtered inductive limit of its ideals of finite type $\mathfrak{J}_\lambda$; setting $C_\lambda = C/\mathfrak{J}_\lambda$, and using the exactness of the inductive limit functor $\varinjlim$, we see that B is still isomorphic to the inductive limit of the filtered system (C_λ) . There thus exists a λ and an A -homomorphism $u : B \rightarrow C_\lambda$ such that the composite $B \xrightarrow{u} C_\lambda \xrightarrow{p_\lambda} B$ (where p_λ is the canonical homomorphism) is the identity. Set $q_\lambda : C \rightarrow C_\lambda$ the canonical homomorphism, and set $t_i = p_\lambda(q_\lambda(T_i))$; we thus have $p_\lambda(u(t_i)) = p_\lambda(q_\lambda(T_i))$, in other words $u(t_i) - q_\lambda(T_i) \in \mathfrak{J}/\mathfrak{J}_\lambda$ ($1 \leq i \leq n$); if $\rho_{\mu\lambda} : C_\lambda \rightarrow C_\mu$ is the canonical homomorphism, there thus exists $\mu \geq \lambda$ such that the n elements $u(t_i) - q_\lambda(T_i)$ belong to $\mathfrak{J}_\mu/\mathfrak{J}_\lambda$ ($1 \leq i \leq n$); we thus have $\rho_{\mu\lambda}(u(t_i)) = \rho_{\mu\lambda}(q_\lambda(T_i)) = q_\mu(T_i)$. Replacing λ by μ and u by $\rho_{\mu\lambda} \circ u$, we can thus suppose $u(t_i) = q_\lambda(T_i)$ for all i , and if $r = p_\lambda \circ q_\lambda$ is the canonical homomorphism $C \rightarrow C/\mathfrak{J} = B$, we can thus write $u(r(T_i)) = q_\lambda(T_i)$ for all i , from which $q_\lambda = u \circ r$. But this implies necessarily that $\mathfrak{J} = \mathfrak{J}_\lambda$, for if $z \in \mathfrak{J}$, then $r(z) = 0$; hence on $B = C_\lambda$.

Now pass to the case where S is affine and X arbitrary; it amounts, in virtue of what has just been shown, to proving that for every affine open subset V of X and every filtered projective system (Z_λ) of affine S -preschemes, the map

$$(8.14.2.4) \quad \varinjlim \operatorname{Hom}_S(Z_\lambda, V) \longrightarrow \operatorname{Hom}_S(\varprojlim Z_\lambda, V)$$

is bijective. It is immediate that this map is injective, for if $(v_\lambda), (v'_\lambda)$ are two inductive systems of S -homomorphisms $v_\lambda : Z_\lambda \rightarrow V, v'_\lambda : Z_\lambda \rightarrow V$ such that the morphisms

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$$Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v_\lambda} V, \quad Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v'_\lambda} V$$

are equal (u_λ being the canonical morphism), then the morphisms

$$Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v_\lambda} V \xrightarrow{j} X, \quad Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{v'_\lambda} V \xrightarrow{j} X$$

(where j is the canonical injection) are equal, which implies $j \circ v_\lambda = j \circ v'_\lambda$ by hypothesis for some suitable λ , hence $v_\lambda = v'_\lambda$.

It remains to prove that (8.14.2.4) is surjective. Let $v : Z \rightarrow V$ be an S -morphism; by hypothesis there exists λ and an S -morphism $w_\lambda : Z_\lambda \rightarrow X$ such that $j \circ v$ factorizes as $Z \xrightarrow{u_\lambda} Z_\lambda \xrightarrow{w_\lambda} X$, and it

suffices to prove that there exists $\mu \geq \lambda$ such that the morphism

$$Z_\mu \xrightarrow{u_{\lambda\mu}} Z_\lambda \xrightarrow{w_\lambda} X$$

(where $u_{\lambda\mu}$ is the transition morphism) factorizes as $Z_\mu \xrightarrow{v_\mu} V \xrightarrow{j} X$. Set, for $\mu \geq \lambda$, $U_\mu = u_{\lambda\mu}^{-1}(w_\lambda^{-1}(V))$. Since the Z_λ are quasi-compact and the U_μ , being open, are ind-constructible (I, 1.9.6), we deduce from (8.3.4) that there exists $\mu \geq \lambda$ such that $U_\mu = Z_\mu$. $\square$

Remark (8.14.3). — The fact that the map (8.14.2.1) is injective when f is locally of finite type ((8.8.2), (i)) naturally leads one to ask whether this result also admits a converse. This is not the case, even when S and X are affine, for there exist monomorphisms $X \rightarrow S$ that are not of finite type (I, 2.4.2), and which therefore disprove this conjecture.

§9. CONSTRUCTIBLE PROPERTIES

9.1. The principle of finite extension

Proposition (9.1.1). — (Principle of finite extension). Let k be a field, $\mathfrak{C}$ a set of extensions of k . Suppose the following conditions are satisfied: IV-3 | 54

label=(i) If $K \in \mathfrak{C}$ and if there exists a k -homomorphism $K \rightarrow K'$ (where K' belongs to the universe under consideration), then $K' \in \mathfrak{C}$.

lbbel=(ii) If $K \in \mathfrak{C}$, there exists a sub-extension K' of K , of finite type over k , such that $K' \in \mathfrak{C}$.

lcbel=(iii) If $K \in \mathfrak{C}$ is the field of fractions of an integral k -algebra A of finite type, there exists $f \in A - \{0\}$ such that for every maximal ideal $\mathfrak{m}$ of A we have $A_f/\mathfrak{m} \in \mathfrak{C}$.

Let Ω be an algebraically closed extension of k (in the universe under consideration). The following conditions are equivalent: IV-3 | 55

label=(a) $\mathfrak{C}$ is non-empty.

lbbel=(b) There exists $K' \in \mathfrak{C}$ which is a finite extension of k .

lcbel=(c) We have $\Omega \in \mathfrak{C}$.

Proof. The condition (i) implies that b entails c , and c trivially entails a ; let us prove that a entails b . By virtue of (ii) and (iii) there exists an extension $K' \in \mathfrak{C}$ which is of the form $A/\mathfrak{m}$, where A is a k -algebra of finite type on k and $\mathfrak{m}$ a maximal ideal of A . By the Hilbert Nullstellensatz (Bourbaki, Alg. comm., chap. V, §3, n° 1, cor. 2 of th. 1), K' is a finite extension of k . $\square$

Corollary (9.1.2). — Under the hypotheses (i), (ii), (iii) of (9.1.1), if k is algebraically closed and if $\mathfrak{C}$ is non-empty, $\mathfrak{C}$ contains all the extensions of k in the universe under consideration.

Proof. Indeed, as a entails c , we have $k \in \mathfrak{C}$, and the conclusion follows from hypothesis (i). $\square$

Remark (9.1.3). — In practice, one will verify condition (ii) of (9.1.1) by noting that K is the inductive limit of its sub-extensions of finite type K_α , and by applying the results of §8, taking into account the fact that for $K_\alpha \subset K_\beta$, K_β is faithfully flat over K_α . Frequently the set $\mathfrak{C}$ is formed by fields belonging to a set $\mathfrak{C}'$ of k -algebras satisfying the following condition:

(i bis) If $A \in \mathfrak{C}'$ and if there exists a homomorphism of k -algebras $A \rightarrow A'$ (where A' belongs to the universe under consideration), then $A' \in \mathfrak{C}'$.

When this is the case, condition (i) is trivially satisfied, and one will verify generally condition (iii) of (9.1.1) by noting that the field of fractions K of A is the inductive limit of the algebras A_f (for the relation $D(f) \supset D(g)$), and by applying the results of §8, taking into account eventually the fact that the morphisms $D(g) \rightarrow D(f)$ are open immersions.

Proposition (9.1.4). — Let k be a field, Ω an algebraically closed extension of k , X and Y two preschemes of finite type over k . The following conditions are equivalent:

label=(a) There exists an Ω -morphism $X_{(\Omega)} \rightarrow Y_{(\Omega)}$ (resp. an Ω -morphism having one (determined) of the properties (i) to (xiv) of (8.10.5)).

lbbel=(b) There exists a finite extension k' of k and a k' -morphism $X_{(k')} \rightarrow Y_{(k')}$ (resp. a k' -morphism having the considered property).

lcbel=(c) *There exists an extension K of k and a K -morphism $X_{(K)} \rightarrow Y_{(K)}$ (resp. a K -morphism having the considered property).*

Proof. We apply the remark (9.1.3), taking for $\mathfrak{C}'$ the set of all k -algebras A (of the universe under consideration) such that there exists an A -morphism $X \otimes_k A \rightarrow Y \otimes_k A$ (resp. a morphism having one of the properties of (8.10.5)). The condition (i bis) of (9.1.3) is then verified because the properties considered in (8.10.5) are all stable under base change. The condition (iii) of (9.1.1) is therefore satisfied by virtue of (8.8.2, , (i)) (resp. (8.10.5)), since $\text{Spec}(k)$ is quasi-compact and quasi-separated. It remains to verify condition (ii) of (9.1.1) which also follows from (8.8.2, , (i)) and (8.10.5), by considering K as an inductive limit of its sub-extensions of finite type. We conclude therefore by (9.1.1). In particular, if there exists an extension K of k and a K -isomorphism $X_{(K)} \xrightarrow{\sim} Y_{(K)}$, we say that X and Y are *geometrically isomorphic*. $\square$

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The following corollary generalizes (II, 6.6.5):

Corollary (9.1.5). — *Let k be a field, X a k -prescheme. If there exists an extension K of k such that $X_{(K)}$ is projective (resp. quasi-projective) over K , then X is projective (resp. quasi-projective) over k .*

Proof. The morphism $\text{Spec}(K) \rightarrow \text{Spec}(k)$ being faithfully flat and quasi-compact, and X being of finite type over k already by (II, 2.7.1, , (v)), the hypothesis means that there exists a closed immersion (resp. an immersion) $X_{(K)} \rightarrow \mathbf{P}_K^r = \mathbf{P}_k^r \otimes_k K$ ((II, 5.4.2), (ii) and (II, 5.5.3)); applying property (iv) (resp. (ii)) of (8.10.5), we deduce that there exists a finite extension k' of k and a closed immersion (resp. an immersion) $X_{(k')} \rightarrow \mathbf{P}_{k'}^r$, in other words $X_{(k')}$ is projective (resp. quasi-projective) over k' . We conclude by (II, 6.6.5). $\square$

Proposition (9.1.6). — *Let k be a field, Ω an algebraically closed extension of k , X a prescheme of finite type over k , $\mathcal{F}, \mathcal{G}$ two coherent $\mathcal{O}_X$ -Modules. Suppose that there exists an isomorphism $\mathcal{F} \otimes_k \Omega \xrightarrow{\sim} \mathcal{G} \otimes_k \Omega$. Then there exists a finite extension k' of k and an isomorphism $\mathcal{F} \otimes_k k' \xrightarrow{\sim} \mathcal{G} \otimes_k k'$.*

Proof. The reasoning is the same as in (9.1.4), applying (8.5.2, , (i)) (one uses in the proof the property (iii) of (9.1.1)), the fact that the morphisms $D(g) \rightarrow D(f)$ (with the notations of (9.1.3)) are open immersions, and *a fortiori* the morphisms are flat. $\square$

9.2. Constructible and ind-constructible properties

Definition (9.2.1). — Let $\mathbf{P}(X, \mathcal{F}, k)$ be a relation. We say that $\mathbf{P}$ is a *constructible* (resp. *ind-constructible*) *property of algebraic preschemes* if the two following conditions are satisfied:

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- label=(i) If k is a field, X an algebraic prescheme over k , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, k' an extension of k , for $\mathbf{P}(X, \mathcal{F}, k)$ to be true, it is necessary and sufficient that $\mathbf{P}(X_{(k')}, \mathcal{F} \otimes_k k', k')$ be true.
- lbbel=(ii) Let S be an integral noetherian prescheme, of generic point η , $u : X \rightarrow S$ a morphism of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. For every $s \in S$, set $X_s = u^{-1}(s) = X \times_S \text{Spec}(k(s))$, $\mathcal{F}_s = \mathcal{F} \otimes_{\mathcal{O}_S} k(s)$. Let E be the set of $s \in S$ such that $\mathbf{P}(X_s, \mathcal{F}_s, k(s))$ is true. Then one of the two sets $E, S - E$ contains an open non-empty set (and consequently is a neighborhood of η , or E is a neighborhood of η if it contains η).

Remarks (9.2.2). — (i) It is understood here a convention of language and not a mathematical definition in the strict sense. We have analogous “definitions” for relations between k , one or several k -preschemes, coherent Modules on these preschemes, or homomorphisms between these Modules.

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(ii) We shall also have to consider relations involving *constructible parts* of preschemes. For example, let $\mathbf{P}(X, Z, k)$ be a relation; we say (by abuse of language) that $\mathbf{P}$ is a *constructible* (resp. *ind-constructible*) *property of the constructible part Z of X* if the two following conditions are satisfied:

- 1° If k is a field, X an algebraic prescheme over k , Z a *constructible part* of X , k' an extension of k , then, for $\mathbf{P}(X, Z, k)$ to be true, it is necessary and sufficient that $\mathbf{P}(X_{(k')}, p^{-1}(Z), k')$ be true ($p : X_{(k')} \rightarrow X$ being the projection).
- 2° Let S be an integral noetherian prescheme, of generic point η , $u : X \rightarrow S$ a morphism of finite type, Z a constructible part of X . For every $s \in S$, set $X_s = u^{-1}(s)$, $Z_s = Z \cap X_s$. Let E

be the set of $s \in S$ such that $\mathbf{P}(X_s, Z_s, k(s))$ is true. Then one of the two sets $E, S - E$ contains an open non-empty set (resp. contains an open non-empty set if it contains η).

We note that in condition 2° one must assume that Z is a constructible part of X , and *not merely* a constructible part of X_s for every s ; the conjunction of these two properties entails the second (0, 1.8.2), but not conversely.

(iii) If $\mathbf{P}$ is a constructible property, it is clear that the same is true of “non $\mathbf{P}$ ”. If $\mathbf{P}, \mathbf{Q}$ are two constructible (resp. ind-constructible) properties, then so are the properties “ $\mathbf{P}$ or $\mathbf{Q}$ ” and “ $\mathbf{P}$ and $\mathbf{Q}$ ”; indeed, if, under the hypotheses of (9.2.1, , (ii)), E, E' are two parts of S and if E contains an open non-empty set, then so does $E \cup E'$, and if $S - E$ and $S - E'$ each contain an open non-empty set, then so does $(S - E) \cap (S - E') = (S - (E \cup E'))$.

(iv) Let $\mathbf{P}(X, \mathcal{F}, k)$ be a relation satisfying condition (9.2.1, , (i)); let S be a prescheme, $u : X \rightarrow S$ a morphism of type fini, with the notations of (9.2.1, , (ii)), and let E be the set of $s \in S$ such that $\mathbf{P}(X_s, \mathcal{F}_s, k(s))$ is true. Let moreover $g : S' \rightarrow S$ be any morphism, and set $X' = X_{(S')}$, $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$; it follows from the transitivity of the fibres (I, 3.6.4) and condition (9.2.1, , (i)) that the set of $s' \in S'$ such that $\mathbf{P}(X'_{s'}, \mathcal{F}'_{s'}, k(s'))$ is true is equal to $g^{-1}(E)$. This extends immediately to the case where $\mathbf{P}$ involves several preschemes, Modules, morphisms of preschemes or homomorphisms of Modules.

(v) As we shall see in the sequel of this paragraph and in the rest of chap. IV, most of the properties $\mathbf{P}$ satisfying condition (9.2.1, , (i)) also satisfy (9.2.1, , (ii)). As possible exceptions, note the property of being *projective*, or *affine*, or *quasi-affine* on the base field (for an algebraic prescheme); we shall see (9.6.2) that these properties are *ind-constructible*, but we shall give later an example where S is an open non-empty part of $\text{Spec}(\mathbf{Z})$ (or an open part of an elliptic curve over a finite field) and where all the fibres X_s except the generic fibre X_η are projective over $k(s)$ (all the preschemes X_s being of dimension 2).

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Proposition (9.2.3). — *Let $\mathbf{P}$ be a constructible (resp. ind-constructible) property of algebraic preschemes, S a prescheme, X a prescheme of finite presentation over S , $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. Then the set E of $s \in S$ such that $\mathbf{P}(X_s, \mathcal{F}_s, k(s))$ is true is locally constructible (resp. ind-constructible). Moreover, if S is irreducible of generic point η , one of the two sets $E, S - E$ is a neighborhood of η in S (resp. E is a neighborhood of η if it contains this point).*

Proof. To prove these assertions, one can reduce to the case where $S = \text{Spec}(A)$ is affine. One knows then that there exists a sub-ring A_0 of A which is a $\mathbf{Z}$ -algebra of finite type, an A_0 -prescheme of finite type X_0 and a coherent $\mathcal{O}_{X_0}$ -Module $\mathcal{F}_0$ such that X is isomorphic to $X_0 \otimes_{A_0} A$ and $\mathcal{F}$ to $\mathcal{F}_0 \otimes_{A_0} A$ (8.9.1). Let $p : S \rightarrow S_0 = \text{Spec}(A_0)$ be the morphism corresponding to the injection $A_0 \rightarrow A$, and let E_0 be the set of $s_0 \in S_0$ such that

$$\mathbf{P}((X_0)_{s_0}, (\mathcal{F}_0)_{s_0}, k(s_0))$$

is true; then, by virtue of (9.2.2, , (iv)), we have $E = p^{-1}(E_0)$; one can therefore reduce to the case where S is the spectrum of a $\mathbf{Z}$ -algebra of finite type, hence a noetherian scheme. We use the criterion of constructibility (0_{III}, 9.2.3) (resp. the criterion of ind-constructibility (I, 1.9.10)); we are then led, using as in (9.2.2, , (iv)) above and replacing S by a closed integral subscheme of S , to the case where S is noetherian and integral, and where it suffices to prove that E is rare in S or contains an open non-empty set of S (resp. that E contains an open non-empty set if it contains the generic point); but this is guaranteed by virtue of condition (9.2.1, , (ii)). $\square$

Corollary (9.2.4). — *Let $\mathbf{P}$ be a constructible (resp. ind-constructible) property of algebraic preschemes, X, Y two S -preschemes of finite presentation, $f : X \rightarrow Y$ an S -morphism. For every $s \in S$, set $X_s = X \times_S \text{Spec}(k(s))$, $Y_s = Y \times_S \text{Spec}(k(s))$, $f_s = f \times 1 : X_s \rightarrow Y_s$. Then the set E of $s \in S$ such that the property $\mathbf{P}(f_s^{-1}(y), k(y))$ is true for every $y \in Y_s$ is locally constructible (resp. ind-constructible).*

Proof. Indeed, let Z be the set of $y \in Y$ such that $\mathbf{P}(f^{-1}(y), k(y))$ is true. As the fibres $f^{-1}(y)$ and $f_s^{-1}(y)$ are isomorphic, we see that E is the set of $s \in S$ such that $Y_s \subset Z$; if $g : Y \rightarrow S$ is the structural morphism, we therefore have $E = S - g(Y - Z)$. Since f is of finite presentation (I, 1.6.2, , (v)), it follows from (9.2.3) that Z is locally constructible (resp. ind-constructible) in Y . As g is of finite presentation, $g(Y - Z)$ is locally constructible (resp. pro-constructible) in S , by virtue of

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the theorem of Chevalley (I, 1.8.4) (resp. of (I, 1.9.5, , (vii))); hence E is locally constructible (resp. ind-constructible) in S . $\square$

Remark (9.2.5). — We note that if $\mathbf{P}$ is a property of algebraic preschemes for which Proposition (9.2.3) is true, then $\mathbf{P}$ also satisfies the condition (9.2.1, , (ii)): this follows from the fact that in an irreducible noetherian space, a constructible set is either rare or contains an open non-empty set (0_{III}, 9.2.3).

Proposition (9.2.6). — Denote by $\mathbf{P}$ one of the following properties of a k -prescheme algebraic X :

- label=(i) X is empty.
- lbbel=(ii) X is finite over k .
- lcbel=(iii) X is radiciel over k .
- ldbel=(iv) $\dim(X)$ belongs to a given part Φ of the set $\bar{\mathbf{Z}} = \mathbf{Z} \cup \{-\infty\}$.

Then $\mathbf{P}$ is constructible.

Proof. It is clear that (i) and (ii) are particular cases of (iv), by taking Φ to be the set $\{-\infty\}$ and the set $\{-\infty, 0\}$ respectively. It thus suffices to prove (iii) and (iv). In each of the two cases, condition (i) of (9.2.1) is satisfied by virtue of (2.7.1, , (xv)) and (4.1.4). On the other hand, in case (iii), property $\mathbf{P}$ satisfies the conclusion of (9.2.3) by virtue of (1.8.7); it remains to see that the same holds in case (iv). This will follow from the following more precise proposition: $\square$

Proposition (9.2.6.1). — If $f : X \rightarrow S$ is a morphism of finite presentation, the function $s \mapsto \dim(f^{-1}(s))$ is locally constructible.

Proof. The question is local on S , so one can assume that $S = \text{Spec}(A)$ is affine and prove that for every n , the set E of $s \in S$ such that $\dim(X_s) = n$ is constructible. The same reasoning as in (9.2.3) reduces us to the case where A is noetherian and integral. $\square$

Corollary (9.2.6.2). — Let S be an integral noetherian prescheme of generic point η , $f : X \rightarrow S$ a morphism of type fini. Then there exists a neighborhood U of η in S such that the function $s \mapsto \dim(X_s)$ is constant in this neighborhood.

9.3. Constructible properties of morphisms of algebraic preschemes

Proposition (9.3.1). — Let $\mathbf{P}(X, k)$ be a constructible (resp. ind-constructible) property of algebraic preschemes. Denote by $\mathbf{P}'(f, X, Y, k)$ the following relation: $f : X \rightarrow Y$ is a k -morphism of k -preschemes algebraic such that for every $y \in Y$, we have the property $\mathbf{P}(f^{-1}(y), k(y))$. Then $\mathbf{P}'$ is constructible (resp. ind-constructible). IV-3 | 60

Proof. Indeed, as $\mathbf{P}$ satisfies condition (9.2.1, , (i)), so does $\mathbf{P}'$ by virtue of the transitivity of the fibres (I, 3.6.4); on the other hand, condition (9.2.1, , (ii)) follows from (9.2.4), by virtue of the remark (9.2.5). $\square$

Proposition (9.3.2). — Denote by $\mathbf{P}$ one of the following properties of a k -morphism of k -preschemes algebraic $f : X \rightarrow Y$:

- label=(i) f is surjective.
- lbbel=(ii) f is quasi-finite.
- lcbel=(iii) f is radiciel.
- ldbel=(iv) For every $y \in Y$, $\dim(f^{-1}(y))$ belongs to Φ (notation of (9.2.6)).

Then $\mathbf{P}$ is constructible.

Proof. This follows immediately from (9.3.1) and (9.2.6) if one takes into account that f is of finite type (I, 1.5.4, , (v)), the characterization of radiciel morphisms (I, 3.5.8), and that of quasi-finite morphisms (II, 6.2.2). $\square$

Proposition (9.3.3). — Suppose verified the hypotheses of (8.8.1), and let us moreover retain the notations; suppose moreover that S_α is quasi-compact, X_α and Y_α are of finite presentation over S_α , and let $f_\alpha : X_\alpha \rightarrow Y_\alpha$ be an S_α -morphism. Let $\mathbf{P}$ be an ind-constructible property of morphisms of algebraic preschemes. For every $s \in S$ (resp. $s_\alpha \in S_\alpha$) set $X_s = X \times_S \text{Spec}(k(s))$, $Y_s = Y \times_S \text{Spec}(k(s))$, $f_s = f_\alpha \times 1 : X_s \rightarrow Y_s$ (resp. $X_{\lambda, s_\alpha} = X_\lambda \times_{S_\lambda} \text{Spec}(k(s_\lambda))$, $Y_{\lambda, s_\alpha} = Y_\lambda \times_{S_\lambda} \text{Spec}(k(s_\lambda))$, $f_{\lambda, s_\alpha} = f_\lambda \times 1 : X_{\lambda, s_\alpha} \rightarrow Y_{\lambda, s_\alpha}$). Then, in

order that for every $s \in S$ we have $\mathbf{P}(f_s)$, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that for every $s_\lambda \in S_\lambda$ we have $\mathbf{P}(f_{\lambda, s_\lambda})$.

Proof. Indeed, let E (resp. E_λ) be the set of $s \in S$ (resp. $s_\lambda \in S_\lambda$) such that property $\mathbf{P}(f_s)$ (resp. $\mathbf{P}(f_{\lambda, s_\lambda})$) is true; it follows from (9.2.2, (iv)) that $E_\mu = u_{\lambda\mu}^{-1}(E_\lambda)$ for $\lambda \leq \mu$, and $E = u_\lambda^{-1}(E_\lambda)$; by virtue of (9.2.3), E (resp. E_λ) is ind-constructible in S (resp. S_λ); the proposition then follows from (8.3.4) applied to the ind-constructible part E of S . $\square$

Remark (9.3.4). — The combination of (9.3.3) and (9.3.2, (ii)) proves the assertion (8.10.5, (xi)).

Proposition (9.3.5). — Let $\mathbf{P}(X, Y, k)$ be the property: “ X and Y are two preschemes of finite type over the field k , and there exists an extension k' of k and a k' -morphism $g : X_{(k')} \rightarrow Y_{(k')}$ satisfying $\mathbf{Q}(g)$ ”, where $\mathbf{Q}$ is one of the properties (i) to (xiv) of (8.10.5). Then $\mathbf{P}$ is an ind-constructible property.

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Proof. The definition of $\mathbf{P}$ shows that condition (9.2.1, (i)) is satisfied, taking into account that the property $\mathbf{Q}(g)$ is stable under change of the base field, and that two extensions of k can always be considered as sub-extensions of a third extension. To verify (9.2.1, (ii)), one can reduce to the case where $S = \text{Spec}(A)$ is affine; if $K = k(\eta)$, the field of fractions of A , and a finite extension K' of K and a K' -morphism $g' : (X_\eta)_{(K')} \rightarrow (Y_\eta)_{(K')}$, satisfying $\mathbf{Q}(g')$, and K' is evidently the field of fractions of an integral A -algebra finite A' . If one sets $S' = \text{Spec}(A')$, it follows from (8.10.5) that there is a neighborhood U' of the generic point η' of S' such that, if one sets $X' = X \otimes_A A'$, $Y' = Y \otimes_A A'$, there exists for every $s' \in U'$, a morphism $X'_{s'} \rightarrow Y'_{s'}$ having property $\mathbf{Q}$. But the morphism $h : S' \rightarrow S$ is finite, hence closed, and since $h^{-1}(\eta) = \{\eta'\}$, $h(U')$ contains a neighborhood U of η in S ; for every $s \in U$, there is therefore an $s' \in U'$ such that $h(s') = s$, and as $X'_{s'} = X_s \otimes_{k(s)} k(s')$, $Y'_{s'} = Y_s \otimes_{k(s)} k(s')$, property $\mathbf{P}(X_s, Y_s, k(s))$ is true for every $s \in U$. $\square$

Example (9.3.6). — Take for example $\mathbf{Q}$ the property of being an isomorphism. Then, combining (9.3.5) and (9.3.3), one obtains the following property: with the notations and hypotheses being those of (8.8.1), S_α being quasi-compact, X_α and Y_α of finite presentation on S_α , in order that, for every $s \in S$, X_s and Y_s be geometrically isomorphic (9.1.4), it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that, for every $s_\lambda \in S_\lambda$, X_{λ, s_λ} and Y_{λ, s_λ} be geometrically isomorphic.

An analogous result holds when the preschemes considered are endowed with “composition laws” of a certain kind, for example “preschemes in groups”, “preschemes in rings”, etc. (0_{III}, 8.2.1). Then the preceding statement is also valid when one replaces “isomorphism” by “isomorphisms” which are also *homomorphisms* for the considered composition laws (0_{III}, 8.2.2); it suffices here to use not only (8.10.5) but also (8.8.2, (i)) by noting that the notion of homomorphism for a composition law is expressed by writing that diagrams of morphisms of preschemes are commutative.

One can also, instead of considering morphisms of preschemes as in (9.3.5), consider homomorphisms of Modules, using (9.1.6) instead of (9.1.5).

9.4. Constructibility of certain properties of modules

(9.4.1). Notations. In this number and the following ones until the end of §9, we shall systematically use the following notations: given a morphism $f : X \rightarrow S$, we set, for every $s \in S$, $X_s = f^{-1}(s) = X \times_S \text{Spec}(k(s))$; for every quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, $\mathcal{F}_s$ will denote the $\mathcal{O}_{X_s}$ -Module $\mathcal{F} \otimes_{\mathcal{O}_S} k(s)$, and for every homomorphism $u : \mathcal{F} \rightarrow \mathcal{G}$ of $\mathcal{O}_X$ -Modules quasi-coherent, $u_s : \mathcal{F}_s \rightarrow \mathcal{G}_s$ will be the morphism $p^*(u)$, p denoting the canonical projection $X_s \rightarrow X$. For every section g of $\mathcal{F}$ over X , we will denote by g_s the image of g by the canonical homomorphism $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(X_s, \mathcal{F}_s)$. For every closed part Z of X , we will denote by Z_s the preimage $p^{-1}(Z) = Z \cap X_s$ (I, 3.6.1). Finally, if Y is a second S -prescheme and $h : X \rightarrow Y$ an S -morphism, we will denote h_s the morphism $h \times 1 : X_s \rightarrow Y_s$.

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Proposition (9.4.2). — Let S be an integral noetherian prescheme of generic point η , $f : X \rightarrow S$ a morphism of finite presentation, $\mathcal{F}, \mathcal{G}, \mathcal{H}$ three quasi-coherent $\mathcal{O}_X$ -Modules of finite presentation. Let $u : \mathcal{F} \rightarrow \mathcal{G}$, $v : \mathcal{G} \rightarrow \mathcal{H}$ be two homomorphisms of $\mathcal{O}_X$ -Modules, and suppose that the sequence

$$\mathcal{F}_\eta \xrightarrow{u_\eta} \mathcal{G}_\eta \xrightarrow{v_\eta} \mathcal{H}_\eta$$

is exact. Then there exists a neighborhood U of η in S such that, for every $s \in U$, the sequence $\mathcal{F}_s \xrightarrow{u_s} \mathcal{G}_s \xrightarrow{v_s} \mathcal{H}_s$ is exact.

Proof. With the general notations of (9.2.1), it is here the relation $\mathbf{P}(X, \mathcal{F}, \mathcal{G}, \mathcal{H}, u, v, k)$: “ X is an algebraic prescheme over the field k , $\mathcal{F} \xrightarrow{u} \mathcal{G} \xrightarrow{v} \mathcal{H}$ is an exact sequence of coherent $\mathcal{O}_X$ -Modules”. As for every extension k' of k , the projection $X_{(k')} \rightarrow X$ is a faithfully flat morphism (2.2.13), the condition (9.2.1, , (i)) is satisfied (2.2.7). By virtue of (9.2.3), it suffices to reduce to the case where S is affine, noetherian, and integral, and where $\mathcal{F}, \mathcal{G}, \mathcal{H}$ are coherent $\mathcal{O}_X$ -Modules, and the hypothesis implies that there exists a neighborhood U of η in S such that the sequence is exact on $f^{-1}(U)$, applying the general method of (8.1.2, , a)) and the fact that it suffices to prove that $\text{Ker}(v_s) = (\text{Ker } v)_s$ and $\text{Im}(u_s) = (\text{Im } u)_s$ for s close to η in S ; by the flatness results (0_I, 5.3.4), one is led to prove the proposition in the particular case where the sequence $\mathcal{F} \xrightarrow{u} \mathcal{G} \xrightarrow{v} \mathcal{H} \rightarrow 0$ is exact. But then there is an open U in S containing η and such that $\mathcal{H}|_{f^{-1}(U)}$ is flat on U (6.9.1); it follows from (2.1.8) that for every $s \in U$, the sequence $0 \rightarrow \mathcal{F}_s \rightarrow \mathcal{G}_s \rightarrow \mathcal{H}_s \rightarrow 0$ is exact, which completes the proof. $\square$

Corollary (9.4.3). — *Let S be an integral noetherian prescheme, of generic point η , $f : X \rightarrow S$ a morphism of finite presentation. Let $\mathcal{L}^\bullet = (\mathcal{L}^i)_{i \in \mathbb{Z}}$ be a complex of $\mathcal{O}_X$ -Modules quasi-coherent of finite presentation. For every i , there exists a neighborhood U of η in S such that the canonical homomorphisms*

$$(\mathcal{H}^i(\mathcal{L}^\bullet))_s \longrightarrow \mathcal{H}^i(\mathcal{L}_s^\bullet)$$

are bijective for every $s \in U$.

Proof. One can evidently reduce to a complex of three terms of degrees $-1, 0, +1$: $\mathcal{M} \xrightarrow{u} \mathcal{N} \xrightarrow{v} \mathcal{P}$, and to $i = 0$; the homomorphism to consider is then the canonical homomorphism $(\text{Ker } v / \text{Im } u)_s \rightarrow \text{Ker}(v_s) / \text{Im}(u_s)$. Using (8.9.1) and (8.5.2, , (i)), one sees that one can reduce to the case where S (hence also X) is noetherian, and hence $\mathcal{M}, \mathcal{N}, \mathcal{P}$ are $\mathcal{O}_X$ -Modules coherent; then $\text{Im}(u)$ and $\text{Ker}(v)$ are also coherent (0_I, 5.3.4) and there exists moreover a neighborhood U of η such that for $s \in U$, we have $\text{Ker}(v_s) = (\text{Ker}(v))_s$ and $\text{Im}(u_s) = (\text{Im}(u))_s$; the conclusion then follows from (9.4.2) applied to the suite exact $0 \rightarrow (\text{Im } u)_\eta \rightarrow (\text{Ker } v)_\eta \rightarrow (\text{Ker } v / \text{Im } u)_\eta \rightarrow 0$, taking into account the fact that $\mathcal{O}_\eta = k(\eta)$ (since S is integral) and by the flatness of the suite exact on the right. $\square$

Proposition (9.4.4). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}, \mathcal{G}, \mathcal{H}$ three $\mathcal{O}_X$ -Modules quasi-coherent of finite presentation. Let $u : \mathcal{F} \rightarrow \mathcal{G}, v : \mathcal{G} \rightarrow \mathcal{H}$ be two homomorphisms of $\mathcal{O}_X$ -Modules. Then the set E of $s \in S$ such that the sequence $\mathcal{F}_s \xrightarrow{u_s} \mathcal{G}_s \xrightarrow{v_s} \mathcal{H}_s$ is exact is locally constructible.*

Proof. Taking into account (9.2.3), it suffices to establish that the property $\mathbf{P}$ considered in (9.4.2) is locally constructible. We have already remarked in the proof of (9.4.2) that condition (9.2.1, , (i)) is satisfied, and it remains to verify condition (9.2.1, , (ii)). Suppose therefore S integral noetherian of generic point η , and let us prove that E or $S - E$ is a neighborhood of η . If $\eta \in E$, our assertion follows from (9.4.2), and one can therefore assume $\eta \notin E$, that is, the sequence $\mathcal{F}_\eta \xrightarrow{u_\eta} \mathcal{G}_\eta \xrightarrow{v_\eta} \mathcal{H}_\eta$ is not exact. We distinguish two cases:

1° Set $w = v \circ u$, and suppose first that $w_\eta = v_\eta \circ u_\eta \neq 0$. As $\mathcal{F}, \mathcal{G}, \mathcal{H}$ are coherent, so is $\mathcal{N} = \text{Ker}(w)$ (0_I, 5.3.4); it follows from (9.4.2) that there is a neighborhood U of η in S such that, for $s \in S$, $\text{Ker}(w_s) = \mathcal{N}_s$; restricting S , one can therefore assume this relation is verified for every $s \in S$. Let j the canonical injection $\mathcal{N} \rightarrow \mathcal{F}$, and set $\mathcal{M} = \text{Coker}(j)$; the exactness on the right of the functor $\mathcal{F}_s$ entails that $M_s = \text{Coker}(j_s)$ for every $s \in S$; as M is coherent (0_I, 5.3.4), the hypothesis $w_\eta \neq 0$ means that $M_\eta \neq 0$; it follows from (1.8.6) that there is a neighborhood U of η in S such that $M_s \neq 0$ for $s \in U$, hence $w_s \neq 0$ for $s \in U$, and therefore $\mathcal{F}_s$ is not g_s -flat, i.e., $S - E$ is a neighborhood of η .

2° Suppose that $w_\eta = 0$; by virtue of (8.5.2, , (i)), applied following the general method of (8.1.2, , a)), there exists a neighborhood U of η such that $w|_{f^{-1}(U)} = 0$; replacing S by U , one can already suppose $w = 0$. Then $\mathcal{F} \xrightarrow{u} \mathcal{G} \xrightarrow{v} \mathcal{H}$ is a complex of three terms $\mathcal{L}^\bullet$, to which one can apply (9.4.3); by hypothesis we have $(\mathcal{H}^0(\mathcal{L}^\bullet))_\eta \neq 0$, and $\mathcal{H}^0(\mathcal{L}^\bullet)$ is coherent (0_I, 5.3.3 and 5.3.4), so it follows from (1.8.6) that there is a neighborhood U of η such that $(\mathcal{H}^0(\mathcal{L}^\bullet))_s \neq 0$ for every $s \in U$; but as one can assume that $\mathcal{H}^0(\mathcal{L}_s^\bullet) = (\mathcal{H}^0(\mathcal{L}^\bullet))_s$ for $s \in U$ by (9.4.3), one sees that $S - E$ is again a neighborhood of η in S . $\square$

Corollary (9.4.5). — Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}, \mathcal{G}$ two $\mathcal{O}_X$ -Modules quasi-coherent of finite presentation, $u : \mathcal{F} \rightarrow \mathcal{G}$ a homomorphism of $\mathcal{O}_X$ -Modules. Then the set of $s \in S$ such that u_s is injective (resp. surjective, bijective, zero) is locally constructible. IV-3 | 64

Proof. It suffices to apply (9.4.4) to the sequences $0 \rightarrow \mathcal{F} \xrightarrow{u} \mathcal{G}, \mathcal{F} \xrightarrow{u} \mathcal{G} \rightarrow 0, 0 \rightarrow \mathcal{F} \xrightarrow{u} \mathcal{G} \rightarrow 0, \mathcal{F} \xrightarrow{u} \mathcal{G} \xrightarrow{1_{\mathcal{G}}} \mathcal{G}$. □

Corollary (9.4.6). — Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. Let h be a section of $\mathcal{F}$ over X ; for every $s \in S$, let h_s be the corresponding section of $\mathcal{F}_s$ over X_s (for the projection morphism $X_s \rightarrow X$). Then, the set of $s \in S$ such that $h_s = 0$ is locally constructible.

Proof. It suffices to remark that h corresponds to a homomorphism $u : \mathcal{O}_X \rightarrow \mathcal{F}$ ($\mathbf{0}_I$, 5.1.1) and h_s to the homomorphism u_s . □

Proposition (9.4.7). — Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. The set E (resp. E') of $s \in S$ such that $\mathcal{F}_s$ is an $\mathcal{O}_{X_s}$ -Module locally free (resp. locally free of rank n) is locally constructible.

Proof. If X is an algebraic prescheme over a field k , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, k' an extension of k , then, for $\mathcal{F}$ to be locally free (resp. locally free of rank n), it is necessary and sufficient that $\mathcal{F} \otimes_k k'$ also be, since the projection $X_{(k')} \rightarrow X$ is a faithfully flat morphism (2.2.7). In other words, the condition (9.2.1, (i)) is verified for the properties considered. The constructibility then follows, with the structure of the proof considering four separate cases depending on the generic fiber. □

Lemma (9.4.7.1). — Let S be an integral noetherian prescheme of generic point η , X, Y two S -preschemes of type fini, $g : X \rightarrow Y$ an S -morphism, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. If $\mathcal{F}_\eta$ is not g_η -flat, then there exists a neighborhood U of η in S such that for every $s \in U$, $\mathcal{F}_s$ is not g_s -flat.

Proof. Taking into account (2.1.2) and Bourbaki, Alg. comm., chap. I^{er}, §2, n^o 3, Remark 1, the hypothesis means that there exists an open non-empty V of Y_η , and an $\mathcal{O}_\eta$ -Module coherent $\mathcal{G}$ and an $\mathcal{O}_Z$ -homomorphism $u : \mathcal{G} \rightarrow \mathcal{H}$ such that $(1 \otimes u)_s$ is not injective for $s \in U$. Using (8.5.2, (i) and (ii)) and the method of (8.1.2, a)), one can assume that these data are defined over a neighborhood of η ; the result follows from (9.4.5). □

Remark (9.4.7.2). — The lemma (9.4.7.1) will later be generalized and freed from noetherian hypotheses (11.2.8).

Proposition (9.4.8). — Let S be a prescheme locally noetherian, $f : X \rightarrow S$ a morphism of type fini, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Suppose that for every $s \in S$, X_s is a locally integral prescheme. Then the set E (resp. E') of $s \in S$ such that $\mathcal{F}_s$ is an $\mathcal{O}_{X_s}$ -Module of torsion (resp. a torsion-free $\mathcal{O}_{X_s}$ -Module) is locally constructible. IV-3 | 65

Proof. One can evidently assume $S = \text{Spec}(A)$ affine and noetherian and prove E (resp. E') is constructible; replacing S by the reduced closed subprescheme having as underlying space a closed part of S , and noting (I, 3.6.4) that for $s \in Y$, the fibre $(X_{(Y)})_s$ is canonically identified to X_s and the sheaf $(\mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_Y)_s$ to $\mathcal{F}_s$, one sees that one is brought to the case where S is integral of generic point η , and has to prove that E or $S - E$ (resp. E' or $S - E'$) is a neighborhood of η . Note moreover that X is a finite union of affine opens U_i , each of the $(U_i)_s$ is induced on an open of X_s , hence locally integral; moreover, if $(U_i)_\eta$ is empty (resp. non-empty), one knows (9.2.6) that it is also true (resp. non-empty) in a neighborhood of η . One can therefore assume that all the $(U_i)_\eta$ are non-empty and integral; saying that $\mathcal{F}_s$ is of torsion (resp. torsion-free) is equivalent to saying that each $\mathcal{F}_s|_{(U_i)_s}$ is of torsion (resp. torsion-free). One is therefore brought to the case where $X = \text{Spec}(B)$ is affine, and $\mathcal{F} = \tilde{M}$, where M is a B -module of finite type; one will set $B_s = B \otimes_A k(s)$, $M_s = M \otimes_A k(s)$ and one can assume B_s is integral. We have four cases to consider:

1^o $\eta \in E$; M_η is then a B_η -module of torsion of finite type, and there is therefore an element $h \neq 0$ in B_η such that $hM_\eta = 0$; by virtue of (8.5.2, (i)), applied following the method of (8.1.2, a)), one can (replacing S by a neighborhood of η) assume that $h \in \Gamma(X, \mathcal{O}_X)$ is of the form g_η , where $g \in \Gamma(X, \mathcal{O}_X)$; let $u : \mathcal{F} \rightarrow \mathcal{F}$ the endomorphism of $\mathcal{F}$ defined by g ; by hypothesis, $u_\eta = 0$, hence

(9.4.5) the endomorphism $u_s : \mathcal{F}_s \rightarrow \mathcal{F}_s$, which is the multiplication by g_s , is zero in a neighborhood of η . On the other hand, let $v : \mathcal{O}_X \rightarrow \mathcal{O}_X$ the endomorphism defined by the multiplication by g ; as B_η is integral and $h = g_\eta \neq 0$, v_η is injective, and it follows hence from (9.4.5) that v_s is an injective endomorphism of $\mathcal{O}_{X_s}$ for s close to η , i.e., g_s is an $\mathcal{O}_{X_s}$ -regular element for these values of s ; hence $\mathcal{F}_s$ is of torsion in a neighborhood of η .

2° $\eta \in S - E$. Say that a B_η -module of finite type M_η is not a module of torsion means that its quotient M_η/T by its torsion submodule is $\neq 0$, and as it is a torsion-free module of finite type, it is isomorphic to a submodule of B_η^n ; there exists therefore an injective homomorphism $w : M_\eta \rightarrow B_\eta^n$ which is $\neq 0$. Applying (8.5.2, (i)) and (9.4.5) as in 1°, there exists a homomorphism $v : \mathcal{F} \rightarrow \mathcal{O}_X^n$ such that $v_\eta = \tilde{w}$, and such that for s close to η , $v_s : \mathcal{F}_s \rightarrow \mathcal{O}_{X_s}^n$ is injective; for these values of s , $\mathcal{F}_s$ is therefore torsion-free.

3° As M_η is a B_η -module of finite type with torsion, i.e., it is not locally free, following from the analysis above there exists a homomorphism $v : \mathcal{F} \rightarrow \mathcal{O}_X^n$ such that v_η is injective, and such that for s in a neighborhood of η , $v_s : \mathcal{F}_s \rightarrow \mathcal{O}_{X_s}^n$ is injective; for these values of s , $\mathcal{F}_s$ is torsion-free.

4° $\eta \in S - E'$. Let T the torsion submodule of M_η ; by hypothesis $T \neq 0$, and T is of finite type since M_η is noetherian. Using this time (8.5.2, (i) and (ii)) a coherent $\mathcal{O}_X$ -Module $\mathcal{G}$ and an injective homomorphism $u : \mathcal{G} \rightarrow \mathcal{F}$ such that $\mathcal{G}_\eta = \tilde{T}$ and that u_η is the canonical injection $\tilde{T} \rightarrow \mathcal{F}_\eta$. It follows from 1° and from (1.8.6) that in a neighborhood of η , $\mathcal{G}_s$ is a torsion $\mathcal{O}_{X_s}$ -Module $\neq 0$, and on the other hand it follows from (9.4.5) that in a neighborhood of η , u_s is injective. We conclude that in a neighborhood of η , the torsion submodule of $\mathcal{F}_s$ is not zero. $\square$

Remark (9.4.9). — The property “ X is a k -prescheme algebraic locally integral” does not satisfy condition (9.2.1, (i)), and it is therefore not certain that the statement (9.4.8) remains valid when one makes no hypothesis on S and one only assumes that f is a morphism of finite presentation and $\mathcal{F}$ an $\mathcal{O}_X$ -Module of finite presentation. Let us consider however the following particular case: S_0 being a locally noetherian prescheme, let $f_0 : X_0 \rightarrow S_0$ a morphism of type fini, such that the fibres $(X_0)_{s_0}$ are locally integral (for every $s_0 \in S_0$), and $\mathcal{F}_0$ a coherent $\mathcal{O}_{X_0}$ -Module; let $g : S \rightarrow S_0$ be any morphism, set $X = X_0 \times_{S_0} S$, $F = \mathcal{F}_0 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_S$, $f = f_0 \times 1_S : X \rightarrow S$, and suppose that for every $s \in S$, the fibre X_s is again locally integral. Then the set E (resp. E') of $s \in S$ such that $\mathcal{F}_s$ is of torsion (resp. torsion-free) is again *locally constructible*. IV-3 | 66

9.5. Constructibility of topological properties

Proposition (9.5.1). — Let $f : X \rightarrow S$ be a morphism of finite presentation, Z a locally constructible part of X . Then the set E of $s \in S$ such that $Z_s \neq \emptyset$ is locally constructible. IV-3 | 67

Proof. Indeed, $E = f(Z)$, and it suffices to apply the theorem of Chevalley (I, 1.8.4). $\square$

Corollary (9.5.2). — If Z', Z'' are two locally constructible parts of X , the set of $s \in S$ such that $Z'_s \subset Z''_s$ (resp. $Z'_s = Z''_s$) is locally constructible.

Proof. Indeed, the relation $Z'_s \subset Z''_s$ is equivalent to $(Z' \cap (\mathbb{C}Z''))_s = \emptyset$ and $Z' \cap \mathbb{C}Z''$ is locally constructible. $\square$

Proposition (9.5.3). — Let $f : X \rightarrow S$ be a morphism of finite presentation, Z, Z' two locally constructible parts of X such that $Z \subset Z'$. Then the set E of $s \in S$ such that Z_s is dense in Z'_s is locally constructible in S .

Proof. We must verify the two conditions of (9.2.2, (ii)). For the first condition, concerning an algebraic prescheme X over a field k , two constructible parts $Z \subset Z'$, and an extension k' of k , the projection $p : X_{(k')} \rightarrow X$ is faithfully flat and quasi-compact, hence $p^{-1}(\overline{Z}) = \overline{p^{-1}(Z)}$ and $p^{-1}(\overline{Z'}) = \overline{p^{-1}(Z')}$ by virtue of (2.3.10); as p is surjective, the relation $\overline{Z} = \overline{Z'}$ is equivalent to $p^{-1}(\overline{Z}) = p^{-1}(\overline{Z'})$.

Let us now verify the second condition, supposing S affine, noetherian, integral, and of generic point η . We distinguish two cases:

1° $\eta \in S - E$, hence Z_η is not dense in Z'_η . Montrons first that one can assume Z' closed. We have $\overline{Z'_\eta} = \overline{Z_\eta}$ (closure in X_η); set $V_\eta = X_\eta - \overline{Z_\eta}$, which is open in X_η and does not meet Z'_η ; one can assume V_η of the form $V \cap X_\eta$, where V is open (hence constructible) in X , and the hypothesis

$V_\eta \cap Z'_\eta = \emptyset$ entails then by (9.5.1) that $V_s \cap Z'_s = \emptyset$ for $s \in U$; hence Z_s is not dense in Z'_s for $s \in U$, otherwise $U \subset S - E$ is a neighborhood of η .

2° $\eta \in E$, hence Z_η is dense in Z'_η . It suffices to show that one can assume $Z' = Z$, and this follows from the method of reducing to irreducible closed parts. $\square$

Corollary (9.5.4). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, Z a locally constructible part of X . The set E of $s \in S$ such that Z_s is open (resp. closed, resp. locally closed) in X_s is locally constructible in S .*

Proof. Saying that Z_s is open in X_s means that $(X - Z)_s = X_s - Z_s$ is closed in X_s , and as $X - Z$ is locally constructible, one can reduce to considering the set of $s \in S$ such that Z_s is closed, and the set of s such that Z_s is locally closed. We verify in each case the two conditions of (9.2.2, (ii)). The first condition is a consequence of the fact that $p : X_{(k')} \rightarrow X$ is faithfully flat and quasi-compact, and of (2.3.12). We then verify the second condition, assuming S affine, noetherian, and integral. $\square$

Proposition (9.5.5). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, Z a locally constructible part of X . For every $s \in S$, let $D(s) \subset \mathbf{Z} \cup \{-\infty\}$ be the set of dimensions of the irreducible components of Z_s . Then the function $s \mapsto D(s)$ is locally constructible in S .* IV-3 | 69

Corollary (9.5.6). — *Let S be a prescheme noetherian and irreducible of generic point η , X a prescheme irreducible, $f : X \rightarrow S$ a morphism dominant of type fini. Then there exists a neighborhood U of η in S such that, for $s \in U$, all the irreducible components of X_s are of dimension $n = \dim(X_\eta)$.*

Remark (9.5.7). — One must be careful that under the hypotheses of (9.5.6), it can happen that X_s is not irreducible for any $s \neq \eta$ in a neighborhood of η ; in other words, the property “ X is an irreducible k -prescheme algebraic” is not constructible. IV-3 | 70

9.6. Constructibility of certain properties of morphisms

Proposition (9.6.1). — *Let X, Y be two S -preschemes of finite presentation, $f : X \rightarrow Y$ an S -morphism. Let E be the set of $s \in S$ for which f_s has one of the following properties:* IV-3 | 71

- label=(i) being surjective;
- lbel=(ii) being dominant;
- lcbel=(iii) being separated;
- ldbel=(iv) being proper;
- lebel=(v) being radical;
- lfbel=(vi) being finite;
- lgbel=(vii) being quasi-finite;
- lhbel=(viii) being an immersion;
- libel=(ix) being a closed immersion;
- ljbel=(x) being an open immersion;
- lkbcl=(xi) being an isomorphism;
- llbel=(xii) being a monomorphism.

Then E is locally constructible in S .

Proof. The assertions for (i), (v) and (vii) are inserted here only as a reminder, having already been established in (9.3.2).

(ii): Note first that f is of finite presentation (I, 1.6.2, (v)), hence, by the theorem of Chevalley (I, 1.8.4), $f(X)$ is locally constructible in Y . On the other hand, $f_s(X_s) = (f(X))_s$ (I, 3.6.1); the set of s such that f_s is dominant is the set of s such that $(f(X))_s$ is dense in Y_s ; the conclusion follows from (9.5.3).

(iii): As $f : X \rightarrow Y$ is of finite presentation, the diagonal immersion $\Delta_f : X \rightarrow X \times_Y X$ is of finite presentation (I, 1.6.2, (iv) and (v)); it follows hence from (1.8.4.1) that $\Delta_f(X) = \Delta_X$ is a locally constructible part of $X \times_Y X$. Saying that f_s is separated means (taking into account (I, 5.3.4)) that $(\Delta_X)_s$ is closed in $X_s \times_{Y_s} X_s = (X \times_Y X)_s$; the conclusion follows from (9.5.4).

(iv): We show that the property “ X and Y are algebraic preschemes over a field k , and $f : X \rightarrow Y$ a k -morphism proper” is constructible. Condition (9.2.1, (i)) is verified by virtue of (2.7.1, (vii)). It thus remains to verify (9.2.1, (ii)), hence one can assume S is affine, noetherian, and integral of

generic point η , and one must prove that E or $S - E$ is a neighborhood of η . The proof considers separately the cases where $\eta \in E$ and $\eta \notin E$.

(vi): The property is the conjunction of the properties (iv) and (vii) (8.11.1), hence the proposition follows from what has already been proved.

(ix): Verification of condition (9.2.1, , (i)) follows from (2.7.1, , (xi)). One can reduce to the case where S is affine, noetherian, and integral of generic point η , and prove that E or $S - E$ is a neighborhood of η .

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(viii): The verification of condition (9.2.1, , (i)) is done as in (ix), using (2.7.1, , (x)) and the fact that every immersion of one prescheme noetherian into another is quasi-compact.

(x): Using this time (2.7.1, , (ix)) and (8.10.5, , (iii)), we are brought to the case where S is affine, noetherian, integral of generic point η , and where $\eta \in S - E$. We distinguish three cases: 1° $f_\eta(X_\eta)$ is not a locally closed part of Y_η . 2° $f_\eta(X_\eta)$ is locally closed in Y_η but f_η is not an immersion. 3° $f_\eta(X_\eta)$ is open in Y_η and f_η is an immersion.

(xi): The property is the conjunction of (i) and (x) and the conclusion follows from what has been proved.

(xii): By virtue of (I, 5.3.8), saying that f_s is a monomorphism means that $\Delta_{f_s} = (\Delta_f)_s : X_s \rightarrow X_s \times_{Y_s} X_s = (X \times_Y X)_s$ is an isomorphism, and as $X \times_Y X$ is an S -prescheme of finite presentation (I, 1.6.2, , (iv)), the conclusion follows from (xi). $\square$

Proposition (9.6.2). — *Let X, Y be two S -preschemes of finite presentation, $f : X \rightarrow Y$ an S -morphism.*

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label=I) Let E be the set of $s \in S$ for which f_s has one of the following properties:

label=(i) being affine;

lbbel=(ii) being quasi-affine;

lcbel=(iii) being projective;

ldbel=(iv) being quasi-projective.

Then E is ind-constructible in S .

label=II) Let $\mathcal{L}$ be an $\mathcal{O}_X$ -Module invertible. Then the set E' of $s \in S$ such that $\mathcal{L}_s$ is an $\mathcal{O}_{X_s}$ -Module ample (resp. very ample) relatively to f_s is ind-constructible in S .

Proof. I) We verify conditions (i) and (ii) of (9.2.1). For the condition (9.2.1, , (i)), it follows, for properties (i) and (ii), from (2.7.1, , (xiii) and (xiv)); for the properties (iii) and (iv), it follows from (9.1.5). We then verify (9.2.1, , (ii)), assuming S noetherian, integral of generic point η , and that $f_\eta : X_\eta \rightarrow Y_\eta$ has one of the properties (i) to (iv) of the statement. Applying (8.10.5, , (viii), (ix), (xiii) and (xiv)), one sees that there exists a neighborhood U of η such that, if V and W are the preimages of U in X and Y respectively by the structural morphisms, the restriction $V \rightarrow W$ of f has the same property. The conclusion follows from the stability of these properties under base change (II, 4.6.13) and (4.4.10).

II) One proceeds similarly. For condition (9.2.1, , (i)) it follows this time from (2.7.2). For the condition (9.2.1, , (ii)), with the same notations as in I), it follows from (8.10.5, , (vi) and (x)), applied following the method of (8.1.2, , a)), that there exists a neighborhood U of η in S such that the restriction $\mathcal{L}|_V$ is ample (resp. very ample) relatively to the restriction $V \rightarrow W$ of f . The conclusion follows from the stability of the properties considered under base change (II, 4.6.13) and the stability of amplitude by base change (III, 4.7.1) and (II, 4.4.10). $\square$

Proposition (9.6.3). — *Let X, Y be two S -preschemes of finite presentation, $f : X \rightarrow Y$ an S -morphism proper, $\mathcal{L}$ an $\mathcal{O}_X$ -Module invertible. Then the set E (resp. E') of $s \in S$ such that $\mathcal{L}_s$ is an $\mathcal{O}_{X_s}$ -Module ample (resp. very ample) relatively to $f_s : X_s \rightarrow Y_s$ is locally constructible in S .*

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Corollary (9.6.4). — *Let $f : X \rightarrow S$ be a morphism proper of finite presentation, $\mathcal{L}$ an $\mathcal{O}_X$ -Module invertible. Then the set E of $s \in S$ such that $\mathcal{L}_s$ is ample (relatively to f_s) is open in S , and $\mathcal{L}|_{f^{-1}(E)}$ is ample relatively to the restriction $f^{-1}(E) \rightarrow E$ of f .*

Corollary (9.6.5). — *Under the hypotheses of (9.6.4), for $\mathcal{L}$ to be ample relatively to f , it is necessary and sufficient that, for every $s \in S$, $\mathcal{L}_s$ be ample relatively to f_s .*

Remarks (9.6.7). — (i) For each of the properties **P** considered in (9.6.1) the proposition (9.3.3) is applicable, and these properties (for the morphisms f_s) are therefore “stable” under passage to the

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projective limit of a projective system of essentially affine (8.13.4) S -preschemes X_λ to a suitable one of them.

(ii) Let Z be a locally constructible part of X such that, for every $s \in S$, $Z \cap X_s$ is open in X_s , and denote by Z_s the sub-prescheme of X_s induced on the open $Z \cap X_s$. Then, in the propositions (9.6.1) and (9.6.2, (I)), one can everywhere replace f_s by its restriction $f_s|_{Z_s} : Z_s \rightarrow Y_s$ without changing the conclusions.

9.7. Constructibility of properties of separability, geometric irreducibility, and geometric connectedness

Lemma (9.7.1). — *Let S be an irreducible prescheme of generic point η , $f : X \rightarrow S$ a morphism of finite presentation. If X_η has n irreducible components (resp. n' connected components), there exists a neighborhood U of η in S such that for every $s \in U$, X_s has at least n irreducible components (resp. at least n' connected components).* IV-3 | 76

Proof. One can reduce to the case where S is affine, hence X quasi-compact and quasi-separated. Let V_i ($1 \leq i \leq n$) be the interiors of the irreducible components of X_η , which are two by two disjoint and quasi-compact, and whose reunion is dense in X_η ; by virtue of (8.2.11), applied following the method of (8.1.2, (a))), there exist for each i open quasi-compact W_i in X such that $W_i \cap X_\eta = V_i$; as X is quasi-separated, the intersections $W_i \cap W_j$ are quasi-compact (I, 1.2.7), hence, replacing S by a neighborhood of η , one can assume $W_i \cap W_j = \emptyset$ for $i \neq j$ by virtue of (8.3.3). Moreover, as the W_i are constructible, there is a neighborhood U of η in S such that for $s \in U$ the $(W_i)_s$ are non-empty (9.5.1) and (9.2.3) and that the reunion of the $(W_i)_s$ is dense (9.5.3) and (9.2.3). The rest of the proof follows by considering the components of X_s .

The case of connected components is treated similarly, replacing the V_i by the connected components C_j ($1 \leq j \leq n'$) of X_η . □

Lemma (9.7.2). — *Let S be an irreducible prescheme of generic point η , $f : X \rightarrow S$ a morphism of finite presentation. If X_η is not reduced, there exists a neighborhood U of η in S such that, for every $s \in U$, X_s is not reduced.* IV-3 | 77

Proof. Indeed, let $\mathcal{N}$ the Nilradical of $\mathcal{O}_X$; it follows from (8.2.13), applied following the method of (8.1.2, (a))) that $\mathcal{N}_\eta$ is the nilradical of $\mathcal{O}_{X_\eta}$, and the hypothesis is that $\mathcal{N}_\eta \neq 0$. We conclude from (9.4.5) and (9.2.3) that there is a neighborhood U of η such that, for every $s \in U$, $\mathcal{N}_s \neq 0$; and as $\mathcal{N}_s$ is evidently contained in the Nilradical of $\mathcal{O}_{X_s}$, one sees that X_s is not reduced for $s \in U$. □

Lemma (9.7.4). — *Let k be a field, Ω an algebraically closed extension of k , F a non-constant polynomial of $k[T_1, \dots, T_n]$. The following conditions are equivalent:*

label=(a) $X = \text{Spec}(k[T_1, \dots, T_n]/(F))$ is geometrically integral.

lbbel=(b) $F_{(K)}$ is irreducible for every extension K of k .

lcbel=(c) $F_{(\Omega)}$ is irreducible.

We say in this case that F is geometrically irreducible.

Lemma (9.7.5). — *Let A be an integral ring, K its field of fractions, F a non-constant polynomial of $A[T_1, \dots, T_n]$ of degree d such that $F_{(K)}$ is geometrically irreducible. Then there exists $f \neq 0$ in A such that for every $x \in D(f)$, $F_{(k(x))}$ is geometrically irreducible.* IV-3 | 78

Lemma (9.7.5.1). — *Let A be an integral ring, $S = \text{Spec}(A)$, η the generic point of S , B the polynomial ring $A[T_1, \dots, T_m]$, $(P_i)_{i \in I}$ a finite family of elements of B , $\mathfrak{a}$ the ideal of B generated by the P_i . For every $s \in S$, let $\mathfrak{a}_s$ the ideal of $k(s)[T_1, \dots, T_m]$ generated by the polynomials $(P_i)_{(k(s))}$ ($i \in I$). Then, if $V(\mathfrak{a}_\eta) = \emptyset$, there exists a neighborhood U of η in S such that $V(\mathfrak{a}_s) = \emptyset$ for every $s \in U$.*

Proof. Indeed, let $Y = \text{Spec}(B)$, and let Z the closed part $V(\mathfrak{a})$ of Y ; as $\mathfrak{a}$ is an ideal of finite type, Z is constructible in Y (0_{III}, 9.1.5) and with the notations introduced in (9.4.1), $V(\mathfrak{a}_s) = Z_s$ for every $s \in S$; as the structural morphism $Y \rightarrow S$ is of finite presentation, the conclusion follows from (9.5.1) and (9.2.3). □

Proposition (9.7.6). — *Let S be a noetherian prescheme integral of generic point η , $f : X \rightarrow S$ a morphism of type fini, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. If $\mathcal{F}_\eta$ is without first associated immersed cycle, then there exists a neighborhood U of η in S such that, for every $s \in U$, $\mathcal{F}_s$ is without first associated immersed cycle.*

Theorem (9.7.7). — Let $f : X \rightarrow S$ be a morphism of finite presentation, and let E be the set of $s \in S$ for which X_s has one of the following properties: IV-3 | 79

- label=(i) being geometrically irreducible;
- lbbel=(ii) being geometrically connected;
- lcbel=(iii) being geometrically reduced;
- ldbel=(iv) being geometrically integral.

Then E is locally constructible in S .

Proof. To see that the properties considered are constructible, remark first that they trivially verify condition (9.2.1, (i)), by virtue of (4.5.6, (i)) and (4.6.5, (i)). It thus remains to verify (9.2.1, (ii)), hence one can assume that $S = \text{Spec}(A)$ is affine noetherian and integral of generic point η . By virtue of (4.6.8), there exists a finite extension K' of $K = k(\eta)$ such that $(X_\eta)_{(K')}$ has its connected components (resp. irreducible components) geometrically irreducible (resp. geometrically connected) and that $((X_\eta)_{(K')})_{\text{red}}$ is geometrically reduced.

Suppose first that $\eta \in S - E$ for one of the properties (i) to (iv). The proof proceeds by showing that X_s is not geometrically irreducible (resp. connected, reduced, integral) in a neighborhood of η , using the lemmas (9.7.1) and (9.7.2). IV-3 | 80

Now suppose that $\eta \in E$ and examine each of the four properties separately.

1° Suppose X_η geometrically integral. The proof uses the function field and the lemma on geometrically irreducible polynomials (9.7.4) and (9.7.5).

2° Suppose X_η geometrically irreducible. One replaces X by X_{red} (I, 1.5.8) and concludes from 1°.

3° Suppose X_η geometrically connected. The proof uses the surjection (5.10.8.1) applied to $\mathcal{F} = \{\emptyset\}$ and the characterization of geometric connectedness (4.5.13.1). IV-3 | 81

4° Suppose X_η geometrically reduced. The proof uses the results on geometrically irreducible components and (9.7.6). □

Proposition (9.7.8). — Let S be an irreducible prescheme of generic point η , and $f : X \rightarrow S$ a morphism of finite presentation. Let n (resp. n') be the geometric number of irreducible components (resp. connected components) of X_η (4.5.2). Then there exists a neighborhood U of η in S such that for every $s \in U$, the geometric number of irreducible components (resp. connected components) of X_s is equal to n (resp. n'). IV-3 | 82

Corollary (9.7.9). — Let $f : X \rightarrow S$ be a morphism of finite presentation. For every $s \in S$, let $n(s)$ (resp. $n'(s)$) be the geometric number of irreducible components (resp. connected components) of $f^{-1}(s)$ (4.5.2). Then the function $s \mapsto n(s)$ (resp. $s \mapsto n'(s)$) is locally constructible in S .

Corollary (9.7.10). — Let X be a locally noetherian prescheme such that, if X' is the normalization of X_{red} , the canonical morphism $f : X' \rightarrow X$ is finite. Then the set of points $x \in X$ such that X is geometrically unibranch at the point x is locally constructible in X .

Proof. Indeed, this set is by definition the set of points $x \in X$ such that the number of geometric points of $f^{-1}(x)$ is equal to 1. But as f is finite, this number is also the number of geometric irreducible components of the discrete space $f^{-1}(x)$ (II, 6.3.8) and of (4.5.11); the conclusion follows from (9.7.9). □

Remark (9.7.11). — Let Z be a locally constructible part of X such that, for every $s \in S$, $Z \cap X_s$ is open in X_s . Then, in theorem (9.7.7), one can everywhere replace X_s by Z_s without changing the conclusion: one sees it by repeating the reasoning made in (9.6.7, (ii)).

Proposition (9.7.12). — Let $f : X \rightarrow S$ be a morphism of finite presentation, and $g : S \rightarrow X$ a S -section of X (I, 2.5.5). For every $s \in S$, let X_s^0 be the connected component of $f^{-1}(s)$ which contains $g(s)$; then, the reunion X^0 of the X_s^0 for $s \in S$ is a locally constructible part of X .

Proof. We first show that one can always reduce to the case where $S = \text{Spec}(A)$ is affine; with the notations of (9.2.3), we have $f = (f_0)_{(S)}$, where $f_0 : X_0 \rightarrow S_0$ is a morphism of type fini, and one can moreover assume that there exists a S_0 -section $g_0 : S_0 \rightarrow X_0$ such that $g = (g_0)_{(S)}$ (8.9.1). Let x be a point of X , Y the sub-prescheme reduced of S having $\overline{\{f(x)\}}$ as underlying space, and Y' the sub-prescheme reduced of S having $\overline{f(Z)}$ as underlying space; as the restriction of f to Z factorizes in $Z \rightarrow Y \rightarrow S$ (I, 5.2.2), one can assume, replacing S by Y , that $f(x) = \eta$, the generic point of the IV-3 | 83

integral prescheme S . By hypothesis, X_η is the sum of two sub-preschemes X_η^0, X_η^1 , induced on open complementary sets of X_η . As $g : S \rightarrow X$ is continuous and injective, one of these two opens is empty, and as $g(\eta) \in X_\eta^0$ by definition, we have $g^{-1}(X^1) = \emptyset$. In other words, g is a S -section of X^0 ; on the other hand, as $(X^0)_\eta$ is geometrically connected (4.5.13), it follows that for s close to η by (9.7.7), $(X^0)_s = X_s^0$. $\square$

9.8. Primary decomposition in a neighborhood of a generic fibre

Proposition (9.8.1). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. Then the set E of $s \in S$ such that $\mathcal{F}_s$ is an $\mathcal{O}_{X_s}$ -Module without first associated immersed cycle is locally constructible.*

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Proof. One knows that for quasi-coherent Modules on algebraic preschemes, the property of being without first associated immersed cycle is invariant by extension of the base field (4.2.7); one can therefore reduce to the case where S is affine, noetherian, and integral of generic point η , and has to prove that E or $S - E$ is a neighborhood of η . One has seen in (9.7.6) that if $\eta \in E$, E is a neighborhood of η ; it remains to consider the case where $\mathcal{F}_\eta$ has associated immersed cycles. One can reduce to the case where X is affine, and where there is a coherent sub- $\mathcal{O}_{X_\eta}$ -Module $\mathcal{H}$ of $\mathcal{F}_\eta$ whose support Z' is non-empty and *rare* with respect to the support Z of $\mathcal{F}$ (3.1.3); then, by virtue of ((8.5.2), (i) and (ii)) and (8.5.8), applied following the method of ((8.1.2), a)), there exists a coherent sub- $\mathcal{O}_X$ -Module $\mathcal{G}$ of $\mathcal{F}$ such that $\mathcal{G}_\eta = \mathcal{H}$; if $Y = \text{Supp}(\mathcal{F})$, $Y' = \text{Supp}(\mathcal{G})$, $Y'_s = \text{Supp}(\mathcal{G}_s)$ for all $s \in S$ (I, 9.1.13) and in particular $Z = Y_\eta$; since $Z - Z'$ is dense in Z and $Z' \neq \emptyset$, there is a neighborhood U of η such that for $s \in U$, $Y_s - Y'_s$ is dense in Y_s , and $Y'_s \neq \emptyset$ ((9.5.1) and (9.5.3)); by considering a neighborhood of an irreducible component of Y'_s and a sufficiently small neighborhood of this point in X_s , we immediately deduce from (3.1.3) that $\mathcal{F}_s$ admits first associated immersed cycles. $\square$

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. 9.8.2 Let S be a noetherian integral prescheme, with generic point η , $f : X \rightarrow S$ a morphism of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Consider an irredundant reduced decomposition $(\mathcal{G}_i)_{i \in I}$ of $\mathcal{F}_\eta$ (3.2.6); the $\mathcal{G}_i$ are thus quotients of $\mathcal{F}_\eta$ and there is an injective homomorphism $h : \mathcal{F}_\eta \rightarrow \bigoplus \mathcal{G}_i$; moreover $\text{Ass}(\mathcal{G}_i)$ is reduced to a single point x_i . Let Z_i be the closure of $\{x_i\}$ in X , so that $(Z_i)_\eta = \text{Supp}(\mathcal{G}_i)$. Denote by $\mathcal{J}_i$ the coherent Ideal of $\mathcal{O}_X$ defining the closed reduced sub-prescheme of X with underlying space Z_i (sub-prescheme also denoted Z_i). By virtue of (8.5.2) and (8.5.8), applied following the method of ((8.1.2), a)), we can (restricting S to a neighborhood of η if necessary) suppose that there exist quotients $\mathcal{F}_i$ of $\mathcal{F}$ ($i \in I$) such that $\mathcal{G}_i = (\mathcal{F}_i)_\eta$ for every i , and a homomorphism $g : \mathcal{F} \rightarrow \bigoplus \mathcal{F}_i$ such that $h = g_\eta$. Moreover (I, 9.3.5), there exists an integer m such that $(\mathcal{J}_i^m \mathcal{F}_i)_\eta = (\mathcal{J}_i)_\eta^m \mathcal{G}_i = 0$ and, restricting S again, we can therefore assume that $\mathcal{J}_i^m \mathcal{F}_i = 0$ (8.5.2.5), hence that the support of $\mathcal{F}_i$ is contained in Z_i .

Proposition (9.8.3). — *Under the conditions of (9.8.2), for every $s \in S$ and every $i \in I$, denote by $x_{i,s\alpha}$ ($\alpha \in J_{s,i}$) the maximal points of $(Z_i)_s$. There exists a neighborhood U of η in S such that, for every $s \in U$, the $x_{i,s\alpha}$ (for $i \in I$ and $\alpha \in J_{s,i}$) are two by two distinct and that $\text{Ass}(\mathcal{F}_s)$ is the set $\{x_{i,s\alpha}\}$ (in other words, the first associated cycles of $\mathcal{F}_s$ are the irreducible components of $(Z_i)_s$). Moreover, one can shrink U such that the closure of $\{x_{i,s\alpha}\}$ in X_s is a first associated cycle of $\mathcal{F}_s$, and it is necessary and sufficient that $(Z_i)_\eta$ be a maximal first associated cycle of $\mathcal{F}_\eta$.*

Corollary (9.8.4). — *With the notations and hypotheses of (9.8.2), there exists a neighborhood U of η in S such that, for every $s \in U$, each of the $(\mathcal{F}_i)_s$ is without first associated immersed cycle, and the family $(\mathcal{F}_{i,s\alpha})_{\alpha \in J_{s,i}}$ is the unique irredundant reduced decomposition of $(\mathcal{F}_i)_s$, while the family $(\mathcal{F}_{i,s\alpha})_{i \in I, \alpha \in J_{s,i}}$ is an irredundant reduced decomposition of $\mathcal{F}_s$.*

Proposition (9.8.5). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. For every $s \in S$, let $E(s)$ (resp. $E'(s)$) be the set (finite part of $\mathbf{Z} \cup \{-\infty\}$) of dimensions of the first associated (resp. maximal first associated, i.e., the irreducible components of $\text{Supp}(\mathcal{F}_s)$) cycles of $\mathcal{F}_s$. Then the functions $s \mapsto E(s)$ and $s \mapsto E'(s)$ are locally constructible in S .*

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Proposition (9.8.6). — *With the hypotheses and notations of (9.8.2), let $i \in I$ be such that $(Z_i)_\eta$ is maximal (in other words, an irreducible component of $\text{Supp}(\mathcal{F}_\eta)$). Then there exists a neighborhood U of η in S such that for every $s \in U$ and every $\alpha \in J_{s,i}$, the geometric length of $\mathcal{F}_s$ in $x_{i,s\alpha}$ (relative to $k(s)$) (4.7.5) is equal to the geometric length of $\mathcal{F}_\eta$ in x_i (relative to $k(\eta)$).*

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Corollary (9.8.7). — Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. For every $s \in S$, let $M(s)$ be the set of couples (d, m) such that there exists an irreducible component of $\text{Supp}(\mathcal{F}_s)$ of dimension d and of radiciel multiplicity m for $\mathcal{F}_s$ (4.7.8). Then the function $s \mapsto M(s)$ is locally constructible in S . IV-3 | 87

Proposition (9.8.8). — Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. For every $s \in S$, let $l(s)$ be the sum of the total multiplicities for $\mathcal{F}_s$ (relative to $k(s)$) of the generic points of the irreducible components of $\text{Supp}(\mathcal{F}_s)$ (4.7.12). Then the function $s \mapsto l(s)$ is locally constructible in S .

Remark (9.8.9). — One can make more precise the preceding propositions in various ways; let us limit ourselves to one statement, given as an example. Say that a finite part P of a k -prescheme algebraic X is *saturated* if, for every couple of points x, y of P , the generic points of the irreducible components of $\{x\} \cap \{y\}$ also belong to P ; for every finite part Q of X , there exists a smallest finite part P of X containing Q and saturated; we say that P is the *saturate* of Q . For every coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, we call *primary skeleton* of $\mathcal{F}$ the system (P, Q, ω, d, m) where $Q = \text{Ass}(\mathcal{F})$, P is the saturate of Q , ω is the order relation $\{x\} \subset \overline{\{y\}}$ on P , d the function $x \mapsto \dim \overline{\{x\}}$ on P , m the application of the set of maximal elements of Q for the relation ω . We define in an evident manner the notion of *isomorphism* of two virtual skeletons, and finally, with the previous notations, we call *geometric primary type* of $\mathcal{F}$ the class (for the relation of isomorphism of virtual skeletons) of the primary skeleton of $\mathcal{F}$.

(9.8.9.1). Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation. For every $s \in S$, let $T(s)$ be the geometric primary type of $\mathcal{F}_s$. Then the function $s \mapsto T(s)$ is locally constructible in S . IV-3 | 88

9.9. Constructibility of local properties of fibres

Proposition (9.9.1). — Let $f : X \rightarrow S$ be a morphism locally of finite presentation, Z a locally constructible part of X such that for every $s \in S$, Z_s is closed in X_s , Φ a finite part of $\mathbf{Z} \cup \{\pm\infty\}$. The following parts of X are locally constructible: IV-3 | 88

- label=(i) The set of $x \in X$ such that $\dim_x(Z_{f(x)})$ belongs to Φ .
- lbel=(ii) The set of $x \in X$ such that $\text{codim}_x(Z_{f(x)}, X_{f(x)})$ belongs to Φ .
- lcbel=(iii) The set of $x \in X$ such that the local ring $\mathcal{O}_{X_{f(x)}, x}$ is equidimensional.

We note that properties (i) and (ii) express respectively that the functions $x \mapsto \dim_x(Z_{f(x)})$ and $x \mapsto \text{codim}_x(Z_{f(x)}, X_{f(x)})$ are locally constructible in X (0_{III}, 9.3.1).

Proposition (9.9.2). — Let $f : X \rightarrow S$ be a morphism locally of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation, Φ a finite part of $\mathbf{Z} \cup \{\pm\infty\}$. The following parts of X are locally constructible: IV-3 | 90

- label=(i) The set of points $x \in X$ such that $\mathcal{F}_{f(x)}$ is equidimensional at the point x (5.1.12).
- lbel=(ii) The set of points $x \in X$ such that the geometric lengths of $\mathcal{F}_{f(x)}$, relative to $k(f(x))$, of the generic points of the irreducible components of $\text{Supp}(\mathcal{F}_{f(x)})$ containing x (4.7.5), belong to Φ .
- lcbel=(iii) The set of points $x \in X$ such that the dimensions of the first associated cycles of $\mathcal{F}_{f(x)}$ and containing x belong to Φ .
- ldbel=(iv) The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is geometrically reduced at the point x (4.6.17).
- lebel=(v) The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is geometrically integral at the point x (4.6.22).
- lfbel=(vi) The set of $x \in X$ such that $\dim. \text{proj}((\mathcal{F}_{f(x)})_x) \in \Phi$.
- lgbel=(vii) The set of $x \in X$ such that $\text{coprof}((\mathcal{F}_{f(x)})_x) \in \Phi$.
- lhbel=(viii) The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ possesses the property (S_k) at the point x (5.7.2).
- libel=(ix) The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is a Cohen–Macaulay $\mathcal{O}_{X_{f(x)}}$ -Module at the point x (5.7.1).

Corollary (9.9.3). — Let $f : X \rightarrow S$ be a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module of finite presentation, $\mathbf{P}$ one of the properties (i) to (ix) of (9.9.2). Then the set of $s \in S$ such that the property $\mathbf{P}$ is true for all points $x \in X_s$ is locally constructible in S . IV-3 | 93

Proof. Indeed, its complement is the image by f of the complement of E , the set of points where $\mathbf{P}$ holds. As E is locally constructible in X , $X - E$ is also, hence $f(X - E)$ is locally constructible in S , by virtue of the theorem of Chevalley (I, 1.8.4). $\square$

Proposition (9.9.4). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation. The set of $x \in X$ such that $X_{f(x)}$ has at the point x one (determined) of the following properties:*

- label=(i) being geometrically regular;*
 - lbel=(ii) possessing the geometric property (R_k) ;*
 - lbel=(iii) being geometrically normal;*
 - lbel=(iv) being geometrically reduced (i.e. separable);*
 - lbel=(v) being geometrically punctually integral;*
- is a locally constructible part of X .*

Proof. For properties (iv) and (v), this follows from (9.9.2, (iv) and (v)) applied to $\mathcal{F} = \mathcal{O}_X$. For the other properties, taking into account (6.7.8), one reduces, as at the beginning of (9.9.2), to the case where S is noetherian and integral of generic point $\eta = f(x)$. Moreover, by the theorem of generic flatness (6.9.1), one can, by replacing S by a neighborhood of η , assume that f is a *flat* morphism. Then, saying that $X_{f(x)}$ is geometrically regular at the point y means that f is *regular* at the point y (6.8.1), and one knows that the set L of these points is *open* in X (6.8.7), which proves the proposition in case (i). $\square$

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Corollary (9.9.5). — *Let $f : X \rightarrow S$ be a morphism of finite presentation, and denote by $\mathbf{P}$ one of the properties (i) to (v) of (9.9.4). Then the set of $s \in S$ such that the property $\mathbf{P}$ is true for all points $x \in X_s$ is locally constructible in S .*

Proof. The proof is the same as that of (9.9.3) starting from (9.9.4). $\square$

Proposition (9.9.6). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, $\mathcal{L}_\bullet$ a complex formed of $\mathcal{O}_X$ -Modules quasi-coherent of finite presentation; for every $s \in S$, let $(\mathcal{L}_\bullet)_s$ the complex $\mathcal{L}_\bullet \otimes_{\mathcal{O}_S} k(s)$ of $\mathcal{O}_{X_s}$ -Modules coherent of finite type. For a given integer n , the set of $x \in X$ such that $(\mathcal{H}_n((\mathcal{L}_\bullet)_{f(x)}))_x = 0$ is locally constructible in X .*

Proof. One can reduce to the case where $\mathcal{L}_i = 0$ except for $i = 0, 1$ or 2 , and where $n = 1$. Moreover, the question being local on X , one can reduce to the case where $S = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affine, B being a finitely presented A -algebra, and the complex $\mathcal{L}_\bullet$ is a complex of $\mathcal{O}_X$ -Modules of finite presentation, zero except in degrees $0, 1$ and 2 . There exists then a sub-ring A_0 of A which is a $\mathbf{Z}$ -algebra of finite type, an A_0 -prescheme of finite type X_0 and a complex $\mathcal{L}_\bullet^{(0)}$ of $\mathcal{O}_{X_0}$ -Modules coherent, zero except in degrees $0, 1$ and 2 , such that X is isomorphic to $X_0 \otimes_{A_0} A$ and $\mathcal{L}_\bullet \simeq \mathcal{L}_\bullet^{(0)} \otimes_{A_0} A$. For every $s \in S$, if s_0 is the projection of s in $S_0 = \text{Spec}(A_0)$, the fibres of X and X_0 at s and s_0 are the same, and $(\mathcal{H}_n((\mathcal{L}_\bullet)_{f(x)}))_x = (\mathcal{H}_n((\mathcal{L}_\bullet^{(0)})_{f_0(x)}))_x$, hence one can reduce to the case where A is noetherian. Then, by (1.8.2), one is led to show that, for every $x \in X$, there is a neighborhood V of x in $\overline{\{x\}}$ such that, for every $x' \in V$, the set of dimensions of the first cycles associated to $\mathcal{F}_{f(x')}$ containing x' is the same. But this follows from (9.5.1) applied to the support Z of $\mathcal{H}_n(\mathcal{L}_\bullet)$, and $Z \cap \overline{\{x\}}$ is closed in the noetherian space X ; hence $Z_s \cap \overline{\{x\}} = \emptyset$ (resp. $\neq \emptyset$) for s in a neighborhood of η by (9.5.1). The set $V = f^{-1}(U) \cap \overline{\{x\}}$ is then the desired neighborhood. $\square$

§10. JACOBSON PRESCHMES

We have already had occasion to observe (5.2.5) that even *excellent* preschemes (7.8.5) do not always behave like the “varieties” of classical Algebraic Geometry, in particular concerning dimension questions; thus if X is the spectrum of a complete discrete valuation ring, the set of closed points of X (reduced to a single element) is not everywhere dense in X , and its complement (also reduced to a single element) is an open set everywhere dense in X whose dimension is zero, hence $< \dim(X)$. We examine in this paragraph general conditions under which such phenomena do not occur; it turns out that, under such conditions, the behaviour is much more satisfactory in many respects for the relations between dimension, codimension, depth, and coprofundity in these preschemes ((10.6) and (10.8)). Furthermore, in the preschemes considered, the fact that the set of closed points

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is everywhere dense (and even “very dense” (10.1.3)) allows many demonstrations to be simplified; one thus recovers the classical viewpoint of “algebraic varieties” which, from our point of view, are the sets of closed points of *algebraic preschemes* over a field, and one makes the link between the language of schemes and that of “varieties” or “algebraic spaces” of Serre ((10.9) and (10.10)).

10.1. Very dense subsets of a topological space

(10.1.1). We say that a subset of a topological space X is *quasi-constructible* if it is a finite union of locally closed parts of X . We say that a subset T of X is *locally quasi-constructible* if for every $x \in X$, there exists an open neighbourhood V of x such that $T \cap V$ is quasi-constructible in V . It is clear that every quasi-constructible part of X is locally quasi-constructible: the converse is true if X is quasi-compact. The reasoning of (0_{III}, 9.1.3), where one omits the word “retrocompact”, shows that the set of quasi-constructible (resp. locally quasi-constructible) parts of X , which we will denote by $\Omega c(X)$ (resp. $\Omega qc(X)$), is stable under finite union, finite intersection, and passage to complements. If $f : X \rightarrow Y$ is a continuous map, it follows immediately from these definitions that, for every quasi-constructible (resp. locally quasi-constructible) part Z of Y , $f^{-1}(Z)$ is quasi-constructible (resp. locally quasi-constructible) in X .

The constructible (resp. locally constructible) parts of X are evidently quasi-constructible (resp. locally quasi-constructible); the converse is true when X is Noetherian (resp. locally Noetherian).

In what follows, we will denote respectively by $\mathfrak{D}(X)$, $\mathfrak{F}(X)$, $\Omega f(X)$, $\mathfrak{C}(X)$, $\Omega c(X)$ the set of subsets of X that are respectively open, closed, locally closed, constructible, locally constructible.

Proposition (10.1.2). — *Let X be a topological space, X_0 a subset of X . The following conditions are equivalent:* IV-3 | 96

- label=a) *For every non-empty locally closed part Z of X , we have $Z \cap X_0 \neq \emptyset$.*
 - a') *For every closed part Z of X , we have $Z = \overline{Z \cap X_0}$.*
- lbel=b) *For every non-empty locally quasi-constructible part Z of X , we have $\overline{Z} \cap X_0 \neq \emptyset$.*
 - b') *For every locally quasi-constructible part Z of X , we have $Z \subset \overline{Z \cap X_0}$ (equivalently, $Z \cap X_0$ is dense in Z).*
- lbel=c) *The map $U \mapsto U \cap X_0$ from $\mathfrak{D}(X)$ to $\mathfrak{D}(X_0)$ is injective (hence bijective).*
 - c') *The map $F \mapsto F \cap X_0$ from $\mathfrak{F}(X)$ to $\mathfrak{F}(X_0)$ is injective (hence bijective).*
 - c'') *The map $Z \mapsto Z \cap X_0$ from $\Omega c(X)$ to $\Omega c(X_0)$ is injective (hence bijective).*
 - c''') *The map $Z \mapsto Z \cap X_0$ from $\Omega qc(X)$ to $\Omega qc(X_0)$ is injective.*

Furthermore, when these conditions are satisfied, the map $Z \mapsto Z \cap X_0$ from $\Omega c(X)$ to $\Omega qc(X_0)$ is bijective.

Proof. Note that the surjectivity assertions in $c)$ and $c')$ are trivial; they imply that every locally closed part of X_0 is the trace on X_0 of a locally closed part of X , and hence the map $\Omega c(X) \rightarrow \Omega c(X_0)$ defined in c'') is also surjective.

We will prove the implications

$$c''') \Rightarrow c'') \Rightarrow c') \Rightarrow c) \Rightarrow b) \Rightarrow b') \Rightarrow a') \Rightarrow a) \Rightarrow c''').$$

The first three are trivial. To see that $c)$ implies $b)$, note first that $c)$ implies that X_0 is dense in X . Replacing X and X_0 by a suitable open set U of X and $U \cap X_0$ respectively, we can reduce to the case where Z is locally closed in X , hence $Z = V \cap W$, where V and W are open in X ; the hypothesis $Z \neq \emptyset$ means that $V \not\subset W$, equivalently $V \cup W \neq W$. By virtue of $c)$, we have $(V \cup W) \cap X_0 \neq W \cap X_0$, hence $V \cap X_0 \not\subset W$, and therefore $(V \cap W) \cap X_0 \neq \emptyset$.

To see that $b)$ implies $b')$, it suffices to apply $b)$ to $Z \cap U$, where U is an arbitrary open neighbourhood of a point of Z . As $\overline{Z \cap X_0} \subset \overline{Z}$, it is trivial that $b')$ implies $a')$. To show that $a')$ implies $a)$, note that if Z is locally closed in X , we can write $Z = F - F'$ where F and F' are closed in X and $F' \subset F$; hence $Z \cap X_0 = (F \cap X_0) - (F' \cap X_0)$. If $Z \cap X_0 = \emptyset$, we would deduce that $F \cap X_0 = F' \cap X_0$, hence $F = F'$ by $a')$, that is to say $Z = \emptyset$.

To see that $a)$ implies $c''')$, it suffices to show that if $Z \neq \emptyset$ is locally quasi-constructible, the relation $Z' \cap X_0 = Z'' \cap X_0$ is equivalent to $((Z' \cup Z'') - (Z' \cap Z'')) \cap X_0 = \emptyset$. In other words, it suffices to prove that $a)$ implies $b)$; moreover, replacing X and X_0 by an open set U of X and $U \cap X_0$ respectively, we reduce to the case where Z is locally closed in X , whence the conclusion.

It remains to show that the map $\Omega_{qc}(X) \rightarrow \Omega_{qc}(X_0)$ is surjective. Let Z_0 be a locally quasi-constructible part of X_0 : there is a covering (V_α) of X_0 by open sets of X_0 such that $Z_0 \cap V_\alpha$ is quasi-constructible in V_α (and hence also in X_0). By virtue of c), for each α there is a single open set U_α of X such that $X_0 \cap U_\alpha = V_\alpha$, and by virtue of c'') a single quasi-constructible set Z_α in X such that $Z_0 \cap V_\alpha = X_0 \cap Z_\alpha$. If α and β are any two indices, we have $Z_\beta \cap U_\alpha \cap Z_0 \cap V_\beta = V_\alpha \cap Z_0 \cap V_\beta = Z_\alpha \cap X_0 \cap V_\beta \subset Z_\alpha \cap X_0$; as $Z_\beta \cap U_\alpha$ and Z_α are quasi-constructible in X , it follows from c'') that $Z_\beta \cap U_\alpha \subset Z_\alpha$. Setting $Z = \bigcup Z_\alpha$, since the V_α cover X_0 , it follows from c) that the U_α cover X , and we see that Z is locally quasi-constructible in X and $Z_0 = Z \cap X_0$. $\square$

Definition (10.1.3). — When a subset X_0 of a topological space satisfies the equivalent conditions of (10.1.2), we say that X_0 is *very dense* in X .

We have already seen in the course of the proof of (10.1.2) that X_0 is then *dense* in X .

Corollary (10.1.4). — If X_0 is *very dense* in X and U an open subset of X , $U \cap X_0$ is *very dense* in U . Conversely, if (U_α) is an open covering of X such that $U_\alpha \cap X_0$ is *very dense* in U_α for every α , then X_0 is *very dense* in X .

Proof. As every locally closed part of U is locally closed in X , the first assertion follows from criterion a) of (10.1.2); similarly for the second, since if $Z \neq \emptyset$ is locally closed in X , $Z \cap U_\alpha$ is locally closed in U_α for every α , and $Z \cap U_\alpha \neq \emptyset$ for at least one α . $\square$

10.2. Quasi-homeomorphisms

Proposition (10.2.1). — Let X_0, X be two topological spaces, $f : X_0 \rightarrow X$ a continuous map. The following conditions are equivalent:

- label=a) The map $U \mapsto f^{-1}(U)$ from $\mathfrak{D}(X)$ to $\mathfrak{D}(X_0)$ is bijective.
- a') The map $F \mapsto f^{-1}(F)$ from $\mathfrak{F}(X)$ to $\mathfrak{F}(X_0)$ is bijective.
- lbbel=b) The topology of X_0 is the inverse image by f of that of X , and $f(X_0)$ is *very dense* (10.1.3) in X .
- lcbel=c) The functor $\mathcal{F} \mapsto f^*(\mathcal{F})$ from the category of sheaves of sets (resp. sheaves of abelian groups) on X to the category of sheaves of sets (resp. sheaves of abelian groups) on X_0 is an equivalence of categories.

Proof. It is clear that a) and a') are equivalent, and a) implies that the topology of X_0 is the inverse image by f of that of X . On the other hand, if $f(X_0)$ is not *very dense* in X , there exist two distinct open sets U_1, U_2 of X such that $U_1 \cap f(X_0) = U_2 \cap f(X_0)$, and hence $f^{-1}(U_1) = f^{-1}(U_2)$, which shows that a) implies b). Conversely, condition b) implies that the maps $U \mapsto U \cap f(X_0)$ from $\mathfrak{D}(X)$ to $\mathfrak{D}(f(X_0))$ and $V \mapsto f^{-1}(V)$ from $\mathfrak{D}(f(X_0))$ to $\mathfrak{D}(X_0)$ are bijective, hence so is their composite $U \mapsto f^{-1}(U)$.

To see that a) implies c), it suffices to apply the definition of $f^*(\mathcal{F})$ (0I, 3.7.1) and the axioms of sheaves: a) implies that for every open set U of X , the canonical map $\Gamma(U, \mathcal{F}) \rightarrow \Gamma(f^{-1}(U), f^*(\mathcal{F}))$ is a functorial bijection in $\mathcal{F}$, whence c). It remains to show that c) implies b).

Suppose first that $f(X_0)$ is not *very dense* in X ; there then exist two distinct closed parts $Y \supset Y'$ of X such that $Y \cap f(X_0) = Y' \cap f(X_0)$. Let $\mathcal{F}$ (resp. $\mathcal{F}'$) be the sheaf of abelian groups on X that is the direct image by the canonical injection $Y \rightarrow X$ (resp. $Y' \rightarrow X$) of the simple sheaf associated to the constant presheaf $\mathbf{Z}$ on Y (resp. Y'); the definition of f^* (0I, 3.7.1) shows that $f^*(\mathcal{F})$ is isomorphic to $f^*(\mathcal{F}')$ but $\mathcal{F}$ is not isomorphic to $\mathcal{F}'$, so condition c) is not satisfied. (One notes that the functor f^* is not even *faithful*, for if u and v are the identity automorphism and the zero endomorphism of $\mathcal{F}|_{\mathcal{F}'}$, $f^*(u)$ and $f^*(v)$ are equal.)

Let us now show that c) implies that the topology of X_0 is the inverse image of that of X by f . Note that if condition c) is satisfied for the category of sheaves of sets, it is also satisfied for the category of sheaves of abelian groups, since it follows immediately from the definitions (0III, 8.2.5) that the latter is none other than the category of *group objects* in the category of sheaves of sets. It suffices therefore to prove our assertion assuming c) verified for the categories of sheaves of abelian groups on X and X_0 . Now, if $\mathbf{Z}_X$ denotes the simple sheaf on X associated to the constant presheaf equal to $\mathbf{Z}$, we have a canonical isomorphism $\Gamma(X, \mathcal{F}) \xrightarrow{\sim} \text{Hom}(\mathbf{Z}_X, \mathcal{F})$ functorial in $\mathcal{F}$ (by the same reasoning as in (0I, 5.1.1)); as it is clear by definition (0I, 3.7.1) that $f^*(\mathbf{Z}_X) = \mathbf{Z}_{X_0}$, the hypothesis that f^* is an equivalence implies that the canonical homomorphism $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(X_0, f^*(\mathcal{F}))$ is

bijective for every sheaf of abelian groups $\mathcal{F}$ on X . Now, let Y_0 be a closed part of X_0 ; let $\mathcal{F}_0$ be the sheaf that is the direct image by the canonical injection $Y_0 \rightarrow X_0$ of the simple sheaf associated to the constant presheaf $\mathbf{Z}$ on Y_0 . As f^* is an equivalence, $\mathcal{F}_0$ is isomorphic to a sheaf of the form $f^*(\mathcal{F})$, where $\mathcal{F}$ is a sheaf of abelian groups on X . Consider then the section s_0 of $\mathcal{F}_0$ over X_0 such that $(s_0)_{x_0} = 1_{x_0}$ if $x_0 \in Y_0$, $(s_0)_{x_0} = 0_{x_0}$ if $x_0 \notin Y_0$. The preceding remarks show that there exists a section s of $\mathcal{F}$ over X such that $(s_0)_{x_0} = s_{f(x_0)}$ for every $x_0 \in X_0$ (the fibres $(\mathcal{F}_0)_{x_0}$ and $\mathcal{F}_{f(x_0)}$ being canonically identified (0I, 3.7.1)); this implies that Y_0 is the inverse image by f of the set of $x \in X$ such that $s_x \neq 0_x$, and y is closed in X . Q.E.D. $\square$

Definition (10.2.2). — When a continuous map $f : X_0 \rightarrow X$ satisfies the equivalent conditions of (10.2.1), we say that f is a *quasi-homeomorphism* of X_0 into X .

By virtue of (10.2.1, b)), to say that a subset X_0 of a topological space X is very dense in X means therefore that the canonical injection $X_0 \rightarrow X$ is a quasi-homeomorphism.

Corollary (10.2.3). — *The composite of two quasi-homeomorphisms is a quasi-homeomorphism.*

Proof. This follows immediately from (10.2.1, a)). $\square$

Corollary (10.2.4). — *Let $f : X \rightarrow Y$ be a quasi-homeomorphism, Y' a locally quasi-constructible part of Y , $X' = f^{-1}(Y')$; then the restriction $f' = f|_{X'} : X' \rightarrow Y'$ is a quasi-homeomorphism.*

Proof. It is clear by virtue of (10.2.1, b)) that the topology induced on X' by that of X is the inverse image of the topology induced on Y' by that of Y . Let $Z \neq \emptyset$ be a closed part in Y' , and $z \in Z$; there is an open neighbourhood U of z in Y such that $U \cap Y'$ is the union of a finite number of parts Y'_i closed in U ; if j is an index such that $z \in Y'_j$, $Z \cap Y'_j$ is closed in U . Since $f(X) \cap U$ is very dense in U (10.1.4), $Z \cap Y'_j \cap f(X)$ is non-empty (10.1.2, a)), and a fortiori $Z \cap f(X)$; but as $Z \subset Y'$, we have $Z \cap f(X) = Z \cap f'(X')$; the criterion (10.1.2, a)) shows that $f'(X')$ is very dense in Y' , and we conclude with the help of (10.2.1, b)). $\square$

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Corollary (10.2.5). — *Let $f : X \rightarrow Y$ be a continuous map, (V_α) an open covering of Y . If, for every α , the restriction $f^{-1}(V_\alpha) \rightarrow V_\alpha$ of f is a quasi-homeomorphism, then f is a quasi-homeomorphism.*

Proof. This follows immediately from the criterion (10.2.1, b)) and (10.1.4). $\square$

Corollary (10.2.6). — *Let $f : X \rightarrow Y$ be a quasi-homeomorphism, Y' a locally quasi-constructible part of Y , $X' = f^{-1}(Y')$. For Y' to be quasi-compact (resp. Noetherian, resp. retrocompact in Y), it is necessary and sufficient that X' be quasi-compact (resp. Noetherian, resp. retrocompact in X).*

Proof. Let us prove the first two assertions; by virtue of (10.2.4), we can restrict to the case where $Y' = Y$. To say that X is quasi-compact (resp. Noetherian) means that for every increasing filtered family (U_α) in $\mathfrak{D}(X)$ having X as its greatest element (resp. for every increasing filtered family (U_α) in $\mathfrak{D}(X)$), there exists γ such that $U_\alpha = U_\gamma$ for $\alpha \geq \gamma$. As $U \mapsto f^{-1}(U)$ is a bijection from $\mathfrak{D}(Y)$ to $\mathfrak{D}(X)$, our assertion follows immediately from the preceding remark.

The quasi-compact open sets of X are therefore the sets of the form $f^{-1}(U)$ where U is a quasi-compact open set in Y , by virtue of (10.2.1, a)) and of what precedes. For X' to be retrocompact in X , it is necessary and sufficient that for every quasi-compact open set U in Y , $f^{-1}(U) \cap X' = f^{-1}(U \cap Y')$ be quasi-compact (0III, 9.1.1); the first part of the proof shows that this is equivalent to saying that $U \cap Y'$ is quasi-compact for every quasi-compact open set U , that is to say that Y' is retrocompact in Y . $\square$

Proposition (10.2.7). — Let $f : X \rightarrow Y$ be a quasi-homeomorphism. Then the map $Z \mapsto f^{-1}(Z)$ from $\mathfrak{P}(Y)$ to $\mathfrak{P}(X)$ defines by restriction the following bijections (cf. (10.1.1) for the notation):

$$\begin{aligned} \mathfrak{D}(Y) &\xrightarrow{\sim} \mathfrak{D}(X) \\ \mathfrak{F}(Y) &\xrightarrow{\sim} \mathfrak{F}(X) \\ \mathfrak{Q}f(Y) &\xrightarrow{\sim} \mathfrak{Q}f(X) \\ \mathfrak{Q}c(Y) &\xrightarrow{\sim} \mathfrak{Q}c(X) \\ \mathfrak{Q}qc(Y) &\xrightarrow{\sim} \mathfrak{Q}qc(X) \\ \mathfrak{C}(Y) &\xrightarrow{\sim} \mathfrak{C}(X) \\ \mathfrak{L}c(Y) &\xrightarrow{\sim} \mathfrak{L}c(X). \end{aligned}$$

Proof. For the first two, this is the definition (10.2.2); as the topology of X is the inverse image by f of that of $f(X)$, the five remaining maps, where one replaces Y by $f(X)$, are bijective. One can therefore (by (10.2.1, b)) restrict to the case where X is a very dense subspace of Y , and the fact that the maps $\mathfrak{Q}f(Y) \rightarrow \mathfrak{Q}f(X)$, $\mathfrak{Q}c(Y) \rightarrow \mathfrak{Q}c(X)$, $\mathfrak{Q}qc(Y) \rightarrow \mathfrak{Q}qc(X)$ are bijective has already been proved (10.1.2). Every locally constructible part being locally quasi-constructible, the maps $\mathfrak{C}(Y) \rightarrow \mathfrak{C}(X)$ and $\mathfrak{L}c(Y) \rightarrow \mathfrak{L}c(X)$ are injective; moreover, for every open set $U \subset Y$, the restriction $f^{-1}(U) \rightarrow U$ of f is a quasi-homeomorphism (10.2.4), hence if one shows that $\mathfrak{C}(Y) \rightarrow \mathfrak{C}(X)$ is surjective, the same will be true of $\mathfrak{L}c(Y) \rightarrow \mathfrak{L}c(X)$. But by virtue of (10.2.6), every retrocompact open part Z in X is of the form $f^{-1}(T)$, where T is a retrocompact open set in Y ; this clearly proves the surjectivity of $\mathfrak{C}(Y) \rightarrow \mathfrak{C}(X)$. $\square$

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Remarks (10.2.8). — (i) One has seen in the proof of (10.2.1) that if f is a quasi-homeomorphism, the canonical map

$$(10.2.8.1) \quad \Gamma(Y, \mathcal{F}) \longrightarrow \Gamma(X, f^*(\mathcal{F}))$$

is an isomorphism of abelian groups, functorial in $\mathcal{F}$, in the category of sheaves of abelian groups on X . As f^* is exact in this category ($\mathbf{0_I}$, 3.7.2), this implies that the canonical homomorphisms

$$H^i(Y, \mathcal{F}) \xrightarrow{\sim} H^i(X, f^*(\mathcal{F}))$$

are bijective for all i .

(ii) If $f : X \rightarrow Y$ is a continuous map and $\mathcal{F}, \mathcal{G}$ two sheaves of abelian groups on Y , we have $f^*(\mathcal{F} \otimes_Y \mathcal{G}) = f^*(\mathcal{F}) \otimes_{\mathbf{Z}_X} f^*(\mathcal{G})$ up to canonical isomorphism ($\mathbf{0_I}$, 4.3.3). We conclude that if f is a quasi-homeomorphism, f^* is again an equivalence of the category of sheaves of rings on Y and the category of sheaves of rings on X ; the datum of a structure of ringed space on Y is therefore equivalent to that of a structure of ringed space on X .

Given two ringed spaces $(X, \mathcal{O}_X), (Y, \mathcal{O}_Y)$, we say that a morphism of ringed spaces $f = (\psi, \theta) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a *quasi-isomorphism* if ψ is a quasi-homeomorphism of X into Y and $\theta^\# : \psi^*(\mathcal{O}_Y) \rightarrow \mathcal{O}_X$ is an isomorphism of sheaves of rings; in this case, the ringed space $(X, \mathcal{O}_X)$ is entirely determined, up to isomorphism, by $(Y, \mathcal{O}_Y)$, the space X , and the quasi-homeomorphism ψ (which one can take arbitrarily). When f is a quasi-isomorphism of ringed spaces, the functor

$$\mathcal{F} \longmapsto f^*(\mathcal{F})$$

is an equivalence of the category of $\mathcal{O}_Y$ -Modules and that of $\mathcal{O}_X$ -Modules, since $f^*(\mathcal{F})$ is canonically identified here with $\psi^*(\mathcal{F})$. We conclude for example with isomorphisms of bi- δ -functors

$$\mathrm{Ext}_{\mathcal{O}_Y}^i(\mathcal{F}, \mathcal{G}) \xrightarrow{\sim} \mathrm{Ext}_{\mathcal{O}_X}^i(f^*(\mathcal{F}), f^*(\mathcal{G})).$$

In general, the usual constructions of the theory of sheaves and homological algebra, carried out on the ringed space Y or the ringed space X , are equivalent.

10.3. Jacobson spaces

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Definition (10.3.1). — We say that a topological space X is a *Jacobson space* if the set X_0 of *closed points* of X is very dense in X (equivalently, if the canonical injection $X_0 \rightarrow X$ is a *quasi-homeomorphism*).

This means therefore (10.1.2) that every non-empty locally closed (or even merely locally quasi-constructible) part Z of X contains a closed point of X , or equivalently that *every closed part of X is the closure of the set of its closed points*.

Proposition (10.3.2). — *Let X be a Jacobson space, Z a locally quasi-constructible part of X ; then the subspace Z of X is a Jacobson space, and for a point of Z to be closed in Z , it is necessary and sufficient that it be closed in X .*

Proof. If X_0 is the set of closed points of X , $Z \cap X_0$ is very dense in Z by virtue of (10.2.4) applied to the injection $i : X_0 \rightarrow X$; as the set Z_0 of closed points of Z evidently contains $Z \cap X_0$, it is *a fortiori* very dense in Z , hence Z is a Jacobson space. Let us now show that in fact $Z_0 = Z \cap X_0$; let $x \in Z$ be a closed point of Z ; let $\{x\}$ be its closure in X ; $\overline{\{x\}} \cap Z = \{x\}$ is a locally quasi-constructible part, and as its intersection with X_0 is non-empty (10.1.2), we have $x \in X_0$. $\square$

Proposition (10.3.3). — *Let X be a topological space, (U_α) an open covering of X . For X to be a Jacobson space, it is necessary and sufficient that each of the subspaces U_α be a Jacobson space.*

Proof. The condition is necessary by virtue of (10.3.2). Conversely, let us show that the hypothesis that the U_α are Jacobson spaces implies that for a closed point $x \in X$ to be closed in X , it suffices that it be closed in U_α . It suffices indeed to see that x is also closed in each of the U_β that contain it; but $U_\alpha \cap U_\beta$ is open in U_α , hence x is closed in $U_\alpha \cap U_\beta$, and by virtue of (10.3.2), x is also closed in U_β , which completes the proof. $\square$

10.4. Jacobson preschemes and Jacobson rings

Definition (10.4.1). — We say that a prescheme X is a *Jacobson prescheme* if the underlying topological space of X is a Jacobson space. We say that a ring A is a *Jacobson ring* if $\text{Spec}(A)$ is a Jacobson prescheme.

Every closed part of $X = \text{Spec}(A)$ is of the form $Z = V(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal equal to its radical, and the set X_0 of closed points of X is the set of maximal ideals of A ; to say that $Z \cap X_0$ is dense in Z therefore means that $\mathfrak{a}$ is an *intersection of maximal ideals* (I, 1.1.4); as $\mathfrak{a}$ is an intersection of prime ideals, this is equivalent to saying that every prime ideal of A is an intersection of maximal ideals; by virtue of (10.3.1) and (10.1.2), the usual definition of Jacobson rings (Bourbaki, *Alg. comm.*, chap. V, §3, n° 4, déf. 1) therefore coincides with the definition (10.4.1).

Proposition (10.4.2). — *Let X be a prescheme, (U_α) a covering of X by affine open sets. For X to be a Jacobson prescheme, it is necessary and sufficient that the rings $\Gamma(U_\alpha, \mathcal{O}_X)$ be Jacobson rings.*

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Proof. This follows from (10.3.3) and the definition (10.4.1). $\square$

(10.4.3). A discrete prescheme is a Jacobson prescheme; an Artinian ring is therefore a Jacobson ring. Every principal ideal domain having an infinity of maximal ideals (for example $\mathbf{Z}$) is a Jacobson ring; a local Noetherian ring A is a Jacobson ring if and only if its maximal ideal is its only prime ideal, that is to say if A is Artinian. Every sub-prescheme of a Jacobson prescheme is a Jacobson prescheme by virtue of (10.3.2).

Proposition (10.4.4). — *Let B be an integral ring. The following conditions are equivalent:*

- label=a) *There exists $f \neq 0$ in B such that B_f is a field.*
- lbel=b) *The field of fractions of B is a B -algebra of finite type.*
- lbel=c) *There exists a field K containing B which is a B -algebra of finite type.*
- ldbel=d) *The generic point of $\text{Spec}(B)$ is isolated in $\text{Spec}(B)$.*

Proof. It is clear that $d)$ is equivalent to $a)$, since $d)$ means that there exists $f \neq 0$ in B such that $D(f)$ is reduced to the generic point of $\text{Spec}(B)$. It is trivial that $a)$ implies $b)$ and $b)$ implies $c)$. Finally, $c)$ implies $a)$, by virtue of Bourbaki, *Alg. comm.*, chap. V, §3, n° 1, cor. 2 of th. 1. $\square$

Proposition (10.4.5). — *Let A be a ring. The following conditions are equivalent:*

label=a) A is a Jacobson ring.

lbbel=b) For every non-maximal prime ideal $\mathfrak{p}$ of A and every $f \neq 0$ in $B = A/\mathfrak{p}$, B_f is not a field.

b') Every A -algebra of finite type K that is a field is a finite A -algebra.

Proof. We know that *a)* implies *b')* (Bourbaki, *Alg. comm.*, chap. V, §3, n° 4, cor. 3 of th. 3). Moreover, the kernel of the homomorphism $A \rightarrow K$ is then a maximal ideal $\mathfrak{m}$ of A , and K is a finite extension of $A/\mathfrak{m}$ (*loc. cit.*). It is trivial that *b')* implies *b)*, since $A/\mathfrak{p}$ is not a field, every B -algebra of finite type that is a field is an A -algebra of finite type, and B_f is a B -algebra of finite type. It remains to see that *b)* implies *a)*. We will use the following lemma: $\square$

Lemma (10.4.5.1). — *Let X be a topological space having the following property: for every non-empty locally closed part Z of X , there exists a part Z' of Z , locally closed in X (or in Z , which amounts to the same), and a point $x \in Z'$, closed in Z' . Then, for X to be a Jacobson space, it is necessary and sufficient that no non-closed point x of X be isolated in $\overline{\{x\}}$.*

Proof. If X is a Jacobson space, a non-closed point x of X cannot be isolated in $\overline{\{x\}}$, for this would mean that there exists an open set U of X such that $U \cap \overline{\{x\}} = \{x\}$; but $U \cap \overline{\{x\}}$ is locally closed in X and would contain no closed point of X , which is contrary to the hypothesis. Conversely, suppose the condition of the statement is satisfied; then the set of closed points X_0 of X is identical to the set of $x \in X$ that are isolated in $\overline{\{x\}}$; but it follows from (5.1.10.1) that this set is very dense in X , hence X is a Jacobson space by definition. IV-3 | 103

Recall (5.1.10) that the hypothesis made on X in (10.4.5.1) is always satisfied when the underlying space of X is a prescheme.

Returning to the proof of (10.4.5), condition *b)* implies that for every non-closed point x of $\text{Spec}(A)$, the generic point x of $\text{Spec}(A/\mathfrak{j}_x) = V(\mathfrak{j}_x) = \overline{\{x\}}$ is not isolated in $\overline{\{x\}}$, hence *b)* implies *a)* by virtue of the lemma (10.4.5.1) and (10.4.4). $\square$

Corollary (10.4.6). — *Every algebra of finite type B over a Jacobson ring A is a Jacobson ring, and the inverse image in A of every maximal ideal of B is a maximal ideal of A . In particular, every algebra of finite type over a field or over $\mathbf{Z}$ is a Jacobson ring.*

Proof. A B -algebra of finite type K which is a field is also an A -algebra of finite type, hence an A -algebra of finite type and *a fortiori* a B -module of finite type, whence the first assertion; the second was demonstrated in the course of the proof of (10.4.5), applied to $K = B/\mathfrak{m}$. $\square$

Corollary (10.4.7). — *If X is a Jacobson prescheme, $f : Y \rightarrow X$ a morphism locally of finite type, then Y is a Jacobson prescheme and the image by f of every closed point in Y is a closed point in X .*

Proof. The question being local on X and on Y , we reduce to the case where X and Y are affine, and the corollary then follows from (10.4.6). $\square$

Corollary (10.4.8). — *If k is an algebraically closed field, X a k -prescheme locally of finite type, the set of rational points of X over k is very dense in X .*

Proof. Indeed, X is a Jacobson prescheme (10.4.7) and the closed points of X are exactly the rational points over k (I, 6.4.2). $\square$

(10.4.9). The fact that preschemes locally of finite type over a field or over $\mathbf{Z}$ are Jacobson preschemes is particularly important, because of the possibility of reducing to this case in many questions of Algebraic Geometry (8.1.2, c)). We will give two examples:

(10.4.10). *Applications I: Proof of (6.15.9).* Let k be a separably closed field, X a k -prescheme locally of finite type over k and *unibranch*. We know that the integral closure of a k -algebra of finite type A in a finite extension of its field of fractions is a finite A -algebra (Bourbaki, *Alg. comm.*, chap. V, §3, n° 2, th. 2), hence every k -algebra of finite type is a universally Japanese ring; we conclude that the set of points $x \in X$ where X is geometrically unibranch is locally constructible (9.7.10). But the hypothesis and the lemma (6.15.8) imply that this set contains all the closed points of X . The conclusion follows therefore from (10.4.6), (10.3.1), and the bijectivity of the map $\mathcal{L}c(X) \rightarrow \mathcal{L}c(X_0)$ (where X_0 is the set of closed points of X) (10.2.7).

(10.4.11). Applications II:

Proposition (10.4.11). — *Let S be a prescheme, X an S -prescheme of finite type. Every S -endomorphism of X which is radicial is surjective (hence bijective).*

Proof. Let $f : X \rightarrow S$ be the structural morphism, $g : X \rightarrow X$ the S -endomorphism considered, and, for every $s \in S$, let g_s be the morphism deduced from g by the base change $\mathrm{Spec}(k(s)) \rightarrow S$, which is a $k(s)$ -endomorphism of the fibre $f^{-1}(s) = X \times_S \mathrm{Spec}(k(s))$. IV-3 | 104

To prove that g is surjective, it suffices to prove that g_s is surjective for every $s \in S$, hence one can (by virtue of (I, 3.5.7)) restrict to the case of a field k , in which case f is a morphism of finite presentation, since S is Noetherian. Applying (8.9.1) and (8.10.5, vii), we reduce to the case where $S = \mathrm{Spec}(A)$, where A is a sub- $\mathbf{Z}$ -algebra of finite type of k . But X is then a Jacobson prescheme (10.4.7), and $g(X)$ is constructible in X (1.8.5), hence, to prove that $g(X) = X$, it suffices to show that $g(X)$ contains all the closed points of X (10.3.1). $\square$

Lemma (10.4.11.1). — *Let Y be a $\mathbf{Z}$ -prescheme of finite type.*

label=(i) For a point $y \in Y$ to be closed in Y , it is necessary and sufficient that $k(y)$ be a finite field.

lbel=(ii) For every prime number p and every integer $d \geq 1$, the set of points $y \in Y$ such that $k(y)$ is an extension of $\mathbf{F}_p$ whose degree divides d is finite.

Proof. Assertion (i) follows from the fact that the image of a closed point $y \in Y$ in $\mathrm{Spec}(\mathbf{Z})$ is a closed point (10.4.7), hence a prime number, and from (I, 6.4.2). On the other hand, as Y is a finite union of affine open sets of finite type over $\mathbf{Z}$, we can restrict to proving (ii) in the case where $Y = \mathrm{Spec}(C)$, where C is a $\mathbf{Z}$ -algebra of finite type. Now, the points $y \in Y$ such that the degree of $k(y)$ over $\mathbf{F}_p$ divides d correspond bijectively to the homomorphisms $C \rightarrow \mathbf{F}_{p^d}$; but if (t_i) is a system of generators of the $\mathbf{Z}$ -algebra C , every homomorphism of C is determined by its values on the elements t_i , and hence there are at most a finite number of homomorphisms of C into a finite field. $\square$

Since X is a $\mathbf{Z}$ -prescheme of finite type, let $T_{p,d}$ be the set of closed points $z \in X$ such that $k(z)$ is an extension of $\mathbf{F}_p$ of degree dividing d ; it follows from (10.4.11.1) that the set $T_{p,d}$ is finite and that the set of closed points of X is the union of the $T_{p,d}$. Moreover, if $z \in T_{p,d}$ and if h is any endomorphism of X , $h(z) \in T_{p,d}$, that is to say $T_{p,d}$ is stable under every endomorphism of X . As g is by hypothesis injective, its restriction to $T_{p,d}$ is bijective since $T_{p,d}$ is finite, which completes the proof of the proposition.

We will see later (17.9.6) that when one furthermore supposes that X is an S -prescheme of finite presentation, and moreover that g is a monomorphism, then one can affirm that g is an automorphism of X .

10.5. Noetherian Jacobson preschemes

Proposition (10.5.1). — *Let B be an integral Noetherian ring. The conditions equivalent a) to d) of (10.4.4) are then also equivalent to the following:* IV-3 | 105

label=e) $\mathrm{Spec}(B)$ is finite.

lbel=f) B is a semi-local ring of dimension ≤ 1 .

Proof. It follows from the Artin–Tate theorem (0, 16.3.3) that conditions a) and f) are equivalent. Condition f) implies that $\mathrm{Spec}(B)$ is the union of the finite set of its closed points and its generic point, hence f) implies e) without supposing A Noetherian. Finally e) implies d) without supposing A Noetherian, for the generic point x of X is the complement of the union of the closures $\overline{\{y\}}$, where y runs through the set of points $y \neq x$, and as these points are finite in number, $\{x\}$ is the complement of a closed set in X . $\square$

Corollary (10.5.2). — *Let A be a Noetherian ring. For A to be a Jacobson ring, it is necessary and sufficient that there exist no prime ideal $\mathfrak{p}$ of A such that $A/\mathfrak{p}$ is a semi-local ring of dimension 1.*

Proof. This follows immediately from (10.5.1) and condition b) of (10.4.5), the prime ideals $\mathfrak{p}$ such that $A/\mathfrak{p}$ is semi-local of dimension 0 being the maximal ideals of A . $\square$

Corollary (10.5.3). — *Let X be a Noetherian irreducible prescheme. The following conditions are equivalent:*

label=a) The generic point of X is isolated.

lbbel=b) X is finite.

lcbel=c) X is of dimension ≤ 1 and the set of its closed points is finite.

Proof. There exists a finite number of irreducible affine open sets U_i ($1 \leq i \leq n$) covering X , each of which contains the generic point of X ; it suffices to prove the equivalence of a), b), and c) for each of the $(U_i)_{\text{red}}$ (taking into account (0, 14.1.7)). But this equivalence follows from (10.5.1). $\square$

Remark (10.5.4). — A Noetherian prescheme X satisfying the equivalent conditions of (10.5.3) is not necessarily an affine scheme; in fact it can even be *non-separated*. One obtains an example by replacing, in the example (I, 5.5.11) of the “affine line with doubled point”, X_1 and X_2 by the spectra of the valuation ring of a discrete valuation, U_{12} and U_{21} by the open sets reduced to the generic point in X_1 and X_2 respectively; the non-separated prescheme X thus obtained has exactly 3 points.

Proposition (10.5.5). — *Let X be a Noetherian prescheme.*

label=(i) The set X_0 of points $x \in X$ such that $\overline{\{x\}}$ is finite is very dense in X .

lbbel=(ii) For X to be a Jacobson prescheme, it is necessary and sufficient that there exist no sub-prescheme of X isomorphic to the spectrum of an integral semi-local ring of dimension 1.

Proof. The condition that $\overline{\{x\}}$ is finite is equivalent to the fact that x is isolated in $\overline{\{x\}}$ (10.5.3), hence $\overline{\{x\}}$, being the underlying space of a sub-prescheme (Noetherian) of X , and assertion (i) follows from (5.1.10.1); taking into account (10.4.5.1), for a non-closed point x of X to belong to X_0 , it is necessary and sufficient that the sub-prescheme closed and integral Y of X having $\overline{\{x\}}$ as underlying space be of dimension 1 and finite, hence a finite union of sub-preschemes that are spectra of integral semi-local rings of dimension 1. Conversely, if there exists a sub-prescheme Z of X which is the spectrum of an integral semi-local ring of dimension 1, Z is not a Jacobson prescheme (10.5.2), hence neither is X (10.3.2). $\square$

Remark (10.5.6). — Assertion (ii) of (10.5.5) is still valid when X is *locally Noetherian*: indeed, if (V_α) is a covering of X by affine Noetherian open sets, every sub-prescheme of one of the V_α is a sub-prescheme of X ; conversely, if a sub-prescheme Z of X is isomorphic to the spectrum of an integral semi-local ring of dimension 1, there is an α such that $V_\alpha \cap Z$ contains an affine open set U of Z not reduced to the generic point of Z , which is therefore also the spectrum of an integral semi-local ring of dimension 1. We conclude with the help of (10.3.3).

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Proposition (10.5.7). — *Let X be a locally Noetherian prescheme, Y a closed part of X such that every non-empty closed part of X meets Y . Then the prescheme induced on the open set $X - Y$ is a Jacobson prescheme.*

Proof. Let us apply the criterion (10.5.5, ii), and suppose that there is a sub-prescheme Z of $X - Y$ which is the spectrum of an integral semi-local ring of dimension 1, with generic z of Z being therefore isolated in Z (or in the closure $\overline{Z}$ of Z in X), and Z being distinct from $\{z\}$. Let $y \neq z$ be a point of Z ; as it does not belong to Y , it is not a closed point of X , and its closure $\overline{\{y\}}$ in X meets Y at a point $x \neq y$, which is therefore not in Z . The existence of the chain $\{x\} \subset \overline{\{y\}} \subset \overline{\{z\}}$ shows that the dimension of $\overline{\{z\}}$ would be ≥ 2 , and it would be the same for the dimension of $U \cap \overline{\{z\}}$, where U is an affine open neighbourhood (hence Noetherian) of x in X . But this contradicts the fact that the generic point z of $U \cap \overline{\{z\}}$ is isolated in $U \cap \overline{\{z\}}$ (10.5.3). $\square$

Corollary (10.5.8). — *Let A be a Noetherian ring; for every element f of the radical $\mathfrak{R}$ of A , the ring A_f is a Jacobson ring, and the open set $\text{Spec}(A) - V(\mathfrak{R})$ is a Jacobson scheme.*

Proof. Setting $X = \text{Spec}(A)$, $Y = V(\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of A , to say that every non-empty closed part of X meets Y is equivalent (0_I, 2.1.3) to saying that Y contains all the closed points of X , or equivalently that $\mathfrak{J}$ is contained in the radical $\mathfrak{R}$ of A . If $f \in \mathfrak{R}$, the open set $D(f)$ does not meet $V(\mathfrak{R})$, and is a Jacobson space by virtue of (10.5.7). $\square$

Corollary (10.5.9). — *Let A be a local Noetherian ring, $\mathfrak{m}$ its maximal ideal, $X = \text{Spec}(A)$, $Y = X - \{\mathfrak{m}\}$; then Y is a Jacobson scheme, whose closed points are the prime ideals $\mathfrak{p} \in \text{Spec}(A)$ such that $\dim(A/\mathfrak{p}) = 1$.*

Proof. The first assertion is a particular case of (10.5.8); on the other hand, the closed points of Y are the prime ideals $\mathfrak{p}$ of A which are maximal elements of the set of prime ideals $\neq \mathfrak{m}$, which, by definition of the dimension, means that $\dim(A/\mathfrak{p}) = 1$. $\square$

Proposition (10.5.10). — *Let A be a local Noetherian ring, reduced and complete, and which is not a field. Then the finite intersections of the kernels of the local homomorphisms of A into discrete valuation rings V , which make V a finite A -algebra, form a filter base tending towards 0 for the adic topology of A .*

Proof. It suffices (Bourbaki, *Alg. comm.*, chap. III, §2, n° 7, prop. 8) to prove that the intersection of the kernels in question is reduced to 0. Suppose first that A is integral and of dimension 1; by virtue of the theorem of Nagata (0, 23.1.5 and 23.1.6), the integral closure A' of A is a complete local integral ring and a finite A -algebra; it is therefore a discrete valuation ring, and the proposition follows immediately in this case.

Let us pass to the general case; let x be the closed point of $X = \operatorname{Spec}(A)$, and set $Y = X - \{x\}$; we know (10.5.9) that Y is a Jacobson prescheme. Let us show that the intersection of the prime ideals $\mathfrak{p}$ of A such that $\dim(A/\mathfrak{p}) = 1$ is reduced to 0; the proposition will follow since, for each such prime $\mathfrak{p}$, the intersection of the kernels of the local homomorphisms of $A/\mathfrak{p}$ into discrete valuation rings V , making V an $(A/\mathfrak{p})$ -algebra of finite type, is reduced to 0. But to say that the intersection of these prime ideals is reduced to 0 means that the set of these ideals is dense in X , or equivalently that Y is dense in X , and this also follows from (10.5.9). $\square$

10.6. Dimension in Jacobson preschemes The results of this subsection refine in certain cases and generalize results of §5.

Proposition (10.6.1). — *Let S be a locally Noetherian prescheme, satisfying moreover the following conditions: 1° S is a Jacobson prescheme; 2° for every $s \in S$, $\mathcal{O}_{S,s}$ is universally catenary (5.6.2); 3° every irreducible component S' of S is equicodimensional (equivalently, for every closed point s of S' and every sub-prescheme of S having S' as underlying space, we have $\dim(S') = \dim(\mathcal{O}_{S',s})$). We then have the following properties:*

label=(i) For every morphism locally of finite type $g : X \rightarrow S$, X satisfies the conditions 1°, 2°, and 3° above. In particular, if X is equidimensional (for example if X is irreducible), X is catenary and for every closed point x of X , we have $\dim(\mathcal{O}_{X,x}) = \dim(X)$ (0, 14.3.3).

lbbel=(ii) Let X, Y be two S -preschemes locally of finite type over S , $f : X \rightarrow Y$ an S -morphism; suppose X irreducible and f dominant. If ξ (resp. η) is the generic point of X (resp. Y) and $e = \dim(f^{-1}(\eta)) = \deg.\operatorname{tr}_{k(\eta)}k(\xi)$, we have

$$(10.6.1.1) \quad \dim(X) = \dim(Y) + e.$$

lcbel=(iii) Let X, Y be two S -preschemes locally of finite type over S , $f : X \rightarrow Y$ an S -morphism, n an integer ≥ 0 such that $\dim(f^{-1}(y)) \geq n$ (resp. $\dim(f^{-1}(y)) \leq n$) for every $y \in Y$. Then we have

$$(10.6.1.2) \quad \dim(X) \geq \dim(Y) + n$$

(resp.

$$(10.6.1.3) \quad \dim(X) \leq \dim(Y) + n).$$

Proof. (i) Property 1° for X follows from (10.4.7). For every $x \in X$, $\mathcal{O}_{X,x}$ is a local ring of an $\mathcal{O}_{S,g(x)}$ -algebra of finite type and a prime ideal, and the homomorphism $\mathcal{O}_{S,g(x)} \rightarrow \mathcal{O}_{X,x}$ is local; hence (5.6.3, iv) $\mathcal{O}_{X,x}$ is universally catenary. To show that X satisfies condition 3°, we consider several cases:

a) X is a closed irreducible sub-prescheme of S ; let S' be an irreducible component of S containing X , ξ the generic point of X , x a closed point of X ; for every $s \in S'$, $\mathcal{O}_{S',s}$, quotient of $\mathcal{O}_{S,s}$, is catenary (5.6.1), hence the conditions 2° and 3° imply that S' is biequidimensional (0, 14.3.3). By virtue of (5.1.2) and (0, 14.3.3.2), we have

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{S',x}) - \dim(\mathcal{O}_{S',\xi}) = \dim(S') - \dim(\mathcal{O}_{S',\xi})$$

which proves that $\dim(\mathcal{O}_{X,x})$ does not depend on the closed point x considered, whence the assertion in this case (5.1.4).

b) X is irreducible and g dominant. Then, for every closed point x of X , $g(x)$ is a closed point of S by (10.4.7); as $\mathcal{O}_{S,g(x)}$ is universally catenary, it follows from (5.6.5.3) that

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{S,g(x)}) + e$$

where $e = \dim(g^{-1}(\zeta))$, ζ being the generic point of S . As $\dim(S) = \dim(\mathcal{O}_{S,g(x)})$ by virtue of condition 3° for S , we have $\dim(\mathcal{O}_{X,x}) = \dim(S) + e$ for every closed point $x \in X$; this proves condition 3° for X (5.1.4), and at the same time the formula

$$(10.6.1.4) \quad \dim(X) = \dim(S) + e.$$

c) General case. Considering a reduced sub-prescheme X' of X having as underlying space an irreducible component of X , we reduce to the case where X is integral; using (I, 5.2.2) and the case a) proved above, one can then replace S by the sub-prescheme reduced having $g(X)$ as underlying space; we are then reduced to case b), and this completes the proof of (i).

(ii) The morphism f being locally of finite type (I, 1.3.4), we can apply the results of (i) by replacing S by Y ; moreover, as X is irreducible and f dominant, (10.6.1.4) gives (10.6.1.1).

(iii) The assertion relative to the case $\dim(f^{-1}(y)) \leq n$ for every $y \in Y$ has already been proved under more general hypotheses in (5.6.7). Suppose that $\dim(f^{-1}(y)) \geq n$ for every $y \in Y$, and consider a generic point η of an irreducible component Y' of Y ; there exists at least one irreducible component Z of $f^{-1}(\eta)$ of dimension $\geq n$; if ξ is the generic point of Z , ξ is also the generic point of an irreducible component X' of X such that $f(X')$ is dense in Y' (0I, 2.1.8). Considering the reduced sub-preschemes of X, Y having as underlying spaces X', Y' respectively, and the restriction $X' \rightarrow Y'$ of f (I, 5.2.2); it follows from (ii) that $\dim(X') \geq \dim(Y') + n$; this being true for every irreducible component Y' of Y , we conclude the inequality (10.6.1.2). $\square$

Corollary (10.6.2). — Suppose that X satisfies conditions $1^\circ, 2^\circ$, and 3° of (10.6.1). Then, for every dense open set U in X , we have $\dim(U) = \dim(X)$. IV-3 | 109

Proof. One knows that $\dim(U) \leq \dim(X)$ (0, 14.1.4); moreover, as X is a Jacobson prescheme, U contains a closed point of every irreducible component of X , hence $\dim(U) = \dim(X)$ by virtue of (10.6.1) and (0, 14.1.2.1). $\square$

Proposition (10.6.3). — Suppose that X satisfies conditions $1^\circ, 2^\circ$, and 3° of (10.6.1), and let Y be a closed part of X . Then, for every $x \in Y$, and every open neighbourhood U of x in X not meeting the irreducible components of Y that do not contain x , we have

$$(10.6.3.1) \quad \dim_x(Y) = \dim(U \cap Y) = \dim \overline{\{x\}} + \operatorname{codim}(\overline{\{x\}}, Y) = \sup_i \dim(Y_i)$$

where Y_i ($1 \leq i \leq m$) are the irreducible components of Y containing x .

Proof. Considering a closed sub-prescheme of X having Y as underlying space, one can restrict, by virtue of (10.6.1, i)), to the case where $Y = X$. By virtue of the choice of U , U is the union of the $U \cap X_i$, hence $\dim(U) = \sup \dim(U \cap X_i)$; but by (10.6.2) applied to a closed sub-prescheme of X having X_i as underlying space, one has (taking into account (10.6.1, i))), $\dim(U \cap X_i) = \dim(X_i)$; this proves that the second and fourth terms of (10.6.3.1) are equal. As $U \cap X_i$ is biequidimensional (10.6.1, i)), we have

$$(10.6.3.2) \quad \dim(X_i) = \dim(U \cap X_i) = \dim(U \cap \overline{\{x\}}) + \operatorname{codim}(U \cap \overline{\{x\}}, U \cap X_i)$$

applied to a closed sub-prescheme of X having $\overline{\{x\}}$ as underlying space, one has $\dim(U \cap \overline{\{x\}}) = \dim(\overline{\{x\}})$; as one also has, for the same reasons,

$$(10.6.3.3) \quad \dim(X_i) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X_i)$$

we obtain

$$\dim(U) = \dim(\overline{\{x\}}) + \sup_i \operatorname{codim}(\overline{\{x\}}, X_i) = \dim(\overline{\{x\}}) + \operatorname{codim}(\overline{\{x\}}, X)$$

by definition of the codimension (0, 14.2.1). This shows that $\dim(U)$ of the open neighbourhood U of x satisfying the conditions of the statement is independent of the choice of U , hence is equal to $\dim_x(X)$, by (0, 14.1.4.1). $\square$

Corollary (10.6.4). — *Under the hypotheses of (10.6.3), let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -Module, Y its support. For every $x \in Y$, we have*

$$(10.6.4.1) \quad \dim(\overline{\{x\}}) + \dim(\mathcal{F}_x) = \dim_x Y.$$

Proof. This follows from (10.6.3.1) and the formula $\dim(\mathcal{F}_x) = \operatorname{codim}(\overline{\{x\}}, Y)$ (5.1.12.2). $\square$

10.7. Examples and counter-examples

(10.7.1). Let S be a locally Noetherian prescheme of *dimension* ≤ 1 and suppose that S is a Jacobson prescheme; when S is Noetherian, this amounts to saying that the irreducible components of S of dimension 1 are infinite, since every $x \in S$ which is not a closed point is the generic point of such a component (10.4.5) and (10.5.4). Then S also satisfies conditions 2° and 3° of (10.6.1): indeed, every local ring $\mathcal{O}_{S,s}$ is of dimension 0 or 1, and is hence necessarily catenary, and by (5.6.4), universally catenary (7.2.9); on the other hand, an irreducible component S' of S , either is reduced to a point, or is of dimension 1, and for every closed point $s \in S'$, $\mathcal{O}_{S',s}$ is necessarily of dimension 1. We deduce from these remarks and from (10.6.1) that every prescheme locally of finite type over S satisfies also the properties 1° , 2° , and 3° of (10.6.1): this includes in particular preschemes *locally of finite type over a field or over $\mathbb{Z}$* .

(10.7.2). Let A be a local Noetherian ring *universally catenary* and let S be the complement of the closed point in $X = \operatorname{Spec}(A)$. Then S satisfies conditions 1° , 2° , and 3° of (10.6.1): indeed, we have already seen that S is a Jacobson prescheme (10.5.9); similarly, for the same reason as for the local rings $A_{\mathfrak{p}}$, it is a Jacobson prescheme, and moreover the local rings $\mathcal{O}_{S,s} = \mathcal{O}_{X,s}$ are universally catenary. On the other hand, an irreducible component X' of X is the complement of $\mathfrak{a}$ in X ; for every closed point x of S , the adherence of x is $\{x, \mathfrak{a}\}$, and X' is by hypothesis biequidimensional (0, 14.3.3), hence $\dim(\mathcal{O}_{X',x}) = \dim(X') - 1$ (0, 14.3.2), which proves that S' is equicodimensional (5.1.4).

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(10.7.3). Let A be a discrete valuation ring; let us show that in the ring $B = A[T_1, \dots, T_n]$ of polynomials in n indeterminates, there exist two maximal ideals $\mathfrak{m}, \mathfrak{n}$ of heights n and $n + 1$ respectively. We have already seen in (5.2.5, i) (for $n = 1$; the proof is by induction on n). Let $A[T_1, \dots, T_{n+1}]$ be a faithfully flat free $A[T_1, \dots, T_n]$ -module; hence there are in $A[T_1, \dots, T_{n+1}]$ two maximal ideals $\mathfrak{m}', \mathfrak{n}'$ of heights $n + 1$ and $n + 2$ respectively (5.5.1); moreover, by (5.5.3), these ideals are necessarily of heights $n + 1$ and $n + 2$. Suppose $n \geq 2$; let $R = 1 + 3$ which is a multiplicative part of B ; setting $B' = R^{-1}B$, the ideal $\mathfrak{J}' = R^{-1}\mathfrak{J}$ is contained in the *radical* of B' (Bourbaki, *Alg. comm.*, chap. III, §3, n° 5, prop. 12); so the topological subspace $X' = \operatorname{Spec}(B')$ is identified with a subspace of X , and the local rings $\mathcal{O}_{S,x}$ and $\mathcal{O}_{X',x}$ are the same by (I, 1.6.2). Considering then in X' the closed set $Y' = V(\mathfrak{J}')$, and setting $S = X' - Y'$; we know (10.5.7) that S is a Jacobson prescheme, evidently irreducible and Noetherian; moreover the local rings $\mathcal{O}_{S,s} = \mathcal{O}_{X',s}$ are universally catenary, being the same as the local rings of A .

However, there exist two closed points a, b of S such that $\mathcal{O}_{S,a}$ and $\mathcal{O}_{S,b}$ do not have the same dimension, that is to say S does not satisfy condition 3° of (10.6.1). To see this, consider the two maximal ideals $\mathfrak{m}' = R^{-1}\mathfrak{m}$ and $\mathfrak{n}' = R^{-1}\mathfrak{n}$, of heights n and $n + 1$ respectively; the corresponding points a', b' are the only closed points of X' , corresponding to $\mathfrak{a}$ and $\mathfrak{b}$. Since $B_{\mathfrak{m}'}$ is biequidimensional, $\mathfrak{p}'$ is then a prime ideal of height $n - 1$. Similarly $\mathfrak{n}'$ corresponds to a point b of X' such that the adherence of b in X' is $\{b, b'\}$. Since a and b are closed points of S , and are in S , this proves our claim.

10.8. Rectified depth

Definition (10.8.1). — Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. For every $x \in X$, we call the *rectified depth* of $\mathcal{F}$ at the point x and denote by $\text{prof}_x^*(\mathcal{F})$ the number (an integer ≥ 0 or $+\infty$) equal to

$$(10.8.1.1) \quad \text{prof}_x^*(\mathcal{F}) = \text{prof}(\mathcal{F}_x) + \dim(\overline{\{x\}})$$

where $\overline{\{x\}}$ is the closure of the point x in X . For any part Z of X , we call the *rectified depth* of $\mathcal{F}$ along Z and denote by $\text{prof}_Z^*(\mathcal{F})$ the number

$$(10.8.1.2) \quad \text{prof}_Z^*(\mathcal{F}) = \inf_{x \in Z} \text{prof}_x^*(\mathcal{F}).$$

In other terms, for every integer n , the relation $\text{prof}_Z^*(\mathcal{F}) \geq n$ is equivalent to $\text{prof}_x^*(\mathcal{F}) \geq n$ for every $x \in Z$. If $Z = X$, one writes $\text{prof}^*(\mathcal{F})$ instead of $\text{prof}_X^*(\mathcal{F})$.

Remarks (10.8.2). — (i) At every closed point $x \in X$, the rectified depth is equal to the depth. IV-3 | 111

(ii) Let Y be a closed sub-prescheme of X , $j : Y \rightarrow X$ the canonical injection, $\mathcal{G}$ a coherent $\mathcal{O}_Y$ -Module. We know (5.7.3, vi) that $\text{prof}(\mathcal{G}_x) = \text{prof}(j_*(\mathcal{G})_x)$ for every $x \in Y$; we deduce that $\text{prof}_x^*(\mathcal{G}) = \text{prof}_{j_*(x)}^*(j_*(\mathcal{G}))$ for every $x \in Y$.

(iii) The notion of rectified depth is only of interest when it is of *local* character, that is to say when it does not change upon replacing X by an arbitrary open neighbourhood U of x . This requires evidently that x not be isolated in $\overline{\{x\}}$, hence x is not a closed point and hence X must be a *Jacobson prescheme* (10.4.5.1); most often, it will also be necessary to know that $\dim(U) = \dim(X)$ for every dense open U in X , and one will therefore need to suppose that X also satisfies conditions 2° and 3° of (10.6.1).

Lemma (10.8.3). — Let X be a regular and biequidimensional prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Then, for every $x \in X$,

$$(10.8.3.1) \quad \text{prof}_x^*(\mathcal{F}) = \dim(X) - \dim.\text{proj}(\mathcal{F}_x).$$

Proof. Indeed, as X is biequidimensional, we have (0, 14.3.5.1)

$$\dim(\overline{\{x\}}) = \dim(X) - \text{codim}(\overline{\{x\}}, X) = \dim(X) - \dim(\mathcal{O}_{X,x})$$

by virtue of (5.1.2). On the other hand, as X is regular, we have by (0, 17.3.4)

$$\text{prof}(\mathcal{F}_x) = \dim(\mathcal{O}_{X,x}) - \dim.\text{proj}(\mathcal{F}_x)$$

whence the lemma. □

Corollary (10.8.4). — Under the hypotheses of (10.8.3), the function $x \mapsto \text{prof}_x^*(\mathcal{F})$ is lower semi-continuous.

Proof. This follows from (10.8.3.1), since $x \mapsto \dim.\text{proj}(\mathcal{F}_x)$ is upper semi-continuous (6.11.1). □

Proposition (10.8.5). — Let S be a locally Noetherian prescheme, X a prescheme locally of finite type over S , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Suppose that S satisfies the following conditions: 1° S is a Jacobson prescheme; 2° S is regular; 3° the irreducible components of S are equicodimensional. Then the function $x \mapsto \text{prof}_x^*(\mathcal{F})$ is lower semi-continuous in X ; in other words, for every integer n , the set U_n of $x \in X$ such that $\text{prof}_x^*(\mathcal{F}) \geq n$ is open.

Proof. As the local rings $\mathcal{O}_{S,s}$ of S are regular, they are universally catenary (0, 16.5.12), hence S satisfies conditions 1°, 2°, and 3° of (10.6.1), and X does too (10.6.1, i). The notion of rectified depth being of local character (10.8.2, iii), we can restrict to the case where $S = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affine, A being a regular ring and B an A -algebra of finite type, hence a quotient of a polynomial ring $C = A[T_1, \dots, T_n]$, and the latter is regular (0, 17.3.7). One can therefore suppose that X is a closed sub-prescheme of a regular prescheme Y also satisfying conditions 1°, 2°, and 3° of (10.6.1); taking into account the remark (10.8.2, ii), we are thus reduced to the case where X is moreover regular and Noetherian. But then, as the local rings of X are integral, the irreducible components of X are also irreducible. Then, as the local rings of X are catenary (0, 16.5.12), the hypothesis 3° of (10.6.1) implies that X is biequidimensional ((5.1.5) and (0, 14.3.3)); it therefore suffices to apply (10.8.4). □

Corollary (10.8.6). — *Under the hypotheses of (10.8.5), for every $x \in X$, the number $\text{prof}_x^*(\mathcal{F})$ is the unique integer n having the following property: there exists an open neighbourhood U of x in X such that for every point $x' \in U \cap \overline{\{x\}}$, closed in U , we have $\text{prof}(\mathcal{F}_{x'}) = n$. In particular, for $\text{prof}_x^*(\mathcal{F}) \geq m$ (resp. $\text{prof}_x^*(\mathcal{F}) \leq m$), it is necessary and sufficient that there exist an open neighbourhood V of x such that, for every point $x' \in V \cap \overline{\{x\}}$, closed in V , we have $\text{prof}(\mathcal{F}_{x'}) \geq m$ (resp. $\text{prof}(\mathcal{F}_{x'}) \leq m$).*

Proof. Indeed, one can restrict to the case where x is not a closed point; if $\text{prof}_y^*(\mathcal{F}) \leq n$, the set of $y \in X$ such that $\text{prof}_y^*(\mathcal{F}) \leq n$ is closed by virtue of (10.8.5), hence contains $\overline{\{x\}}$, and by virtue of the lower semi-continuity of $y \mapsto \text{prof}_y^*(\mathcal{F})$, there exists an open neighbourhood U of x such that $\text{prof}(\mathcal{F}_{x'}) = n$ for every $y \in U \cap \overline{\{x\}}$, hence $\text{prof}(\mathcal{F}_{x'}) = n$ if $x' \in U \cap \overline{\{x\}}$ is closed in U (since the rectified depth is local).

For preschemes satisfying the hypotheses of (10.8.5), the notion of rectified depth can therefore be defined with the aid of the depth at the *closed points* only (these forming a set that is very dense in every closed part of X). $\square$

Proposition (10.8.7). — *Let S be a locally Noetherian prescheme satisfying conditions 1^0 , 2^0 , and 3^0 of (10.6.1). Let X be a prescheme locally of finite type over S , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, $Y = \text{Supp}(\mathcal{F})$. Then, for every $x \in Y$, we have*

$$(10.8.7.1) \quad \text{prof}_x^*(\mathcal{F}) = \dim_x(Y) - \text{coprof}(\mathcal{F}_x).$$

Proof. Indeed, by definition, $\text{coprof}(\mathcal{F}_x) = \dim(\mathcal{F}_x) - \text{prof}(\mathcal{F}_x)$, and it follows from (10.6.4) that $\dim_x(Y) = \dim(\overline{\{x\}}) + \dim(\mathcal{F}_x)$; whence (10.8.7.1) by the definition of $\text{prof}_x^*(\mathcal{F})$. $\square$

Corollary (10.8.8). — *Under the hypotheses on f and $\mathcal{F}$ of (9.9.1), the function $x \mapsto \text{prof}_x^*(\mathcal{F}_{f(x)})$ is constructible.*

Proof. Note that for every $s \in S$, the fibre X_s , being a prescheme locally of finite type over $k(s)$, is a Jacobson prescheme (10.4.7); as moreover $\text{Spec}(k(s))$ satisfies the conditions of (10.6.1), we have $\text{prof}_x^*(\mathcal{F}_{f(x)}) = \dim_x(\text{Supp}(\mathcal{F}_{f(x)})) - \text{coprof}((\mathcal{F}_{f(x)})_x)$ (10.8.7). Now, if $Z = \text{Supp}(\mathcal{F})$, we have $Z_{f(x)} = \text{Supp}(\mathcal{F}_{f(x)})$ (I, 9.1.13) and Z is locally constructible (8.9.1); hence the functions $x \mapsto \dim_x(\text{Supp}(\mathcal{F}_{f(x)}))$ and $x \mapsto \text{coprof}((\mathcal{F}_{f(x)})_x)$ are locally constructible ((9.9.1) and (9.9.3)), which proves the proposition. $\square$

10.9. Maximal spectra and ultrapreschemes

(10.9.1). Let X be a Jacobson prescheme, and let $S(X)$ be the *ringed space* whose underlying space is the subspace of closed points of X , and whose sheaf of rings is the sheaf induced on this subspace by $\mathcal{O}_X$, in other words the sheaf $\psi^*(\mathcal{O}_X)$, with $\psi : S(X) \rightarrow X$ denoting the canonical injection. As ψ is a quasi-homeomorphism, one has seen in (10.2.8, ii)) that if $\theta : \mathcal{O}_X \rightarrow \psi_*(\mathcal{O}_{S(X)})$ is the homomorphism of sheaves of rings such that $\theta^\#$ is the identity, then $j_X = (\psi, \theta)$ is a *quasi-isomorphism* of ringed spaces, and $\mathcal{F} \mapsto j_X^*(\mathcal{F})$ is an equivalence of the category of $\mathcal{O}_X$ -Modules locally free (resp. coherent) and that of $\mathcal{O}_{S(X)}$ -Modules locally free (resp. coherent); moreover, if $\mathcal{L}$ is an $\mathcal{O}_X$ -Module locally free and if U is an open set of X , such that $\mathcal{L}|_U$ is isomorphic to $(\mathcal{O}_X|_U)^n$, $j_X^*(\mathcal{L})|(U \cap S(X))$ is isomorphic to $(\mathcal{O}_{S(X)}|(U \cap S(X)))^n$.

(10.9.2). Let X, Y be two Jacobson preschemes and $f = (\rho, \lambda) : X \rightarrow Y$ a morphism *locally of finite type*. We have seen in (10.4.7) that $\rho(S(X)) \subset S(Y)$ and that ρ restricted to $S(X)$ gives a continuous map $S(X) \rightarrow S(Y)$. On the other hand, for every open set V of Y , one defines by composition a homomorphism of rings

$$\Gamma(V \cap S(Y), \mathcal{O}_{S(Y)}) \longrightarrow \Gamma(f^{-1}(V), \mathcal{O}_X) \longrightarrow \Gamma(f^{-1}(V) \cap S(X), \mathcal{O}_{S(X)})$$

where the two extreme isomorphisms have been defined in (10.9.1); it is clear that this defines a homomorphism of sheaves of rings $S(\lambda) : \mathcal{O}_{S(Y)} \rightarrow \rho_*(\mathcal{O}_{S(X)})$ (recalling that the open sets of $S(Y)$ correspond bijectively to those of Y (resp. $S(X)$) (10.2.1)); we thus obtain a morphism of ringed

spaces $S(f) = (S(\rho), S(\lambda)) : (S(X), \mathcal{O}_{S(X)}) \rightarrow (S(Y), \mathcal{O}_{S(Y)})$ such that the diagram

$$\begin{array}{ccc} S(X) & \xrightarrow{S(f)} & S(Y) \\ j_X \downarrow & & \downarrow j_Y \\ X & \xrightarrow{f} & Y \end{array}$$

is commutative; moreover, if Z is a third Jacobson prescheme and $g : Y \rightarrow Z$ a morphism locally of finite type, it is clear that $S(g \circ f) = S(g) \circ S(f)$. We have thus defined a *covariant functor* $S : C \rightarrow C'$, where C' is the category of *ringed spaces with local rings* and C the category whose objects are the *Jacobson preschemes* and whose morphisms are the morphisms *locally of finite type* between Jacobson preschemes.

(10.9.3). Let us determine the sub-category C'' of C' consisting of the ringed spaces isomorphic to $S(X)$ and whose morphisms come from $S(f)$. Suppose first that $X = \text{Spec}(A)$, where A is a Jacobson ring; then $S(X)$ is the set of *maximal ideals* of A , equipped with: 1° the topology induced by that of X , so that a base of this topology is formed by the $D^m(h) = D(h) \cap S(X)$, the set of maximal ideals $\mathfrak{m}$ such that $h \notin \mathfrak{m}$, where h runs through A ; 2° the sheaf of rings $\mathcal{O}_{S(X)}$ such that $\Gamma(D^m(h), \mathcal{O}_{S(X)}) = A_f$. We call this ringed space the *maximal spectrum* of A and denote it $\text{Spm}(A)$.

One notes that if $j : D(h) \rightarrow X$ is the canonical injection, the ringed space induced by $S(X)$ on $S(D(h))$ is $S(D(h))$.

Let B be a second Jacobson ring, $Y = \text{Spec}(B)$, $\varphi : B \rightarrow A$ a homomorphism of rings making A a B -algebra of finite type, $f = ({}^a\varphi, \tilde{\varphi}) : X \rightarrow Y$ the corresponding morphism of preschemes, and $S(f) : \text{Spm}(A) \rightarrow \text{Spm}(B)$ the morphism of ringed spaces corresponding to f . It is clear that $S(f) = (\psi, \theta)$ is a morphism of *ringed spaces with local rings*, that is (Err_{II}, 1.8.2) for every $x \in \text{Spm}(A)$, $\theta_x^\#$ is a local homomorphism. Reciprocally:

Proposition (10.9.4). — *Let A, B be two Jacobson rings. If $u = (\psi, \theta) : \text{Spm}(A) \rightarrow \text{Spm}(B)$ is a morphism of ringed spaces with local rings such that $\Gamma(\theta) : B \rightarrow A$ makes A a B -algebra of finite type, there exists one and only one morphism of preschemes $f : \text{Spec}(A) \rightarrow \text{Spec}(B)$ such that $u = S(f)$.*

Proof. The uniqueness of f is evident, since if $f = ({}^a\varphi, \tilde{\varphi})$, one must have $\varphi = \Gamma(\theta)$; it remains to see that $S(f)$ is defined and equal to $u = S(f)$. Now, the first assertion follows from the fact that φ is supposed to make A a B -algebra of finite type, and the fact that $\theta_x^\#$ is a local homomorphism for every $x \in \text{Spm}(A)$ allows us to repeat the reasoning of (I, 1.7.3) restricting to the maximal points of $\text{Spm}(A)$: one shows thus successively that $\psi(x) = {}^a\varphi(x)$ for every $x \in \text{Spm}(A)$, then that $\theta_x^\# = \varphi_x : B_{\psi(x)} \rightarrow A_x$, which completes the proof that $u = S(f)$. $\square$

(10.9.5). Let us now consider a ringed space $(X, \mathcal{O}_X)$; we will say that an open part U of X is *ultra-affine* if the ringed space $(U, \mathcal{O}_X|_U)$ is isomorphic to a maximal spectrum $\text{Spm}(A)$, where A is a Jacobson ring. We will say that X is an *ultra-prescheme* if every point of X admits an ultra-affine open neighbourhood. One shows, as in (I, 2.1.3) and (I, 2.1.4), that the ultra-affine open sets form a *base* of the topology of X and that X is a Kolmogoroff space. If Y is a second ultra-prescheme, we will say that a morphism of ringed spaces $f : X \rightarrow Y$ is a *morphism of ultra-preschemes* if it satisfies the following conditions: 1° f is a morphism of *ringed spaces with local rings*; 2° for every $x \in X$, there is an ultra-affine open neighbourhood V of $f(x)$ in Y and an ultra-affine open neighbourhood U of x in X such that $f(U) \subset V$ and the homomorphism of rings $\Gamma(V, \mathcal{O}_Y) \rightarrow \Gamma(U, \mathcal{O}_X)$ corresponding to f makes $\Gamma(U, \mathcal{O}_X)$ a $\Gamma(V, \mathcal{O}_Y)$ -algebra of finite type.

It is immediate that one thus composes morphisms well, the composite of two morphisms of ultra-preschemes being an ultra-prescheme morphism thanks to the final remark of (10.9.3). It is clear that the category C''_0 thus defined is a sub-category of C' that contains C'' ; we now propose to show that $C'' = C''_0$, in other words:

Proposition (10.9.6). — *The functor $X \mapsto S(X)$ from C to C''_0 is fully faithful, that is for X, Y in C , the canonical map*

$$\text{Hom}_C(X, Y) \longrightarrow \text{Hom}_{C''_0}(S(X), S(Y))$$

is bijective. It is an equivalence of categories.

Proof. Let us first show that it is *injective*: let f, g be two morphisms locally of finite type from X to Y and suppose that $S(f) = S(g)$. This implies first that for every open set V of Y , $f^{-1}(V) \cap S(X) = g^{-1}(V) \cap S(X)$, hence (10.3.1) and (10.2.7) $f^{-1}(V) = g^{-1}(V)$; it therefore suffices to prove that for every affine open V of Y , f and g coincide on $f^{-1}(V) = g^{-1}(V)$, which reduces to the case where X and Y are affine. It suffices (I, 2.2.4) to prove that the two homomorphisms of rings $B \rightarrow \Gamma(X, \mathcal{O}_X)$ corresponding to f and g are the same; now, for every $s \in B$, the images of s by these homomorphisms are two sections of $\mathcal{O}_X$ over X which, by hypothesis, induce the *same* section over $S(X)$; we say therefore (10.2.8) that these sections are identical, whence our assertion.

Let us show in the second place that every morphism $h : S(X) \rightarrow S(Y)$ (in the category C_0'') is of the form $S(f)$, where $f : X \rightarrow Y$ is a morphism locally of finite type. By hypothesis, there is an ultra-affine open covering (U') of $S(X)$ (resp. (V'_μ)) and for each λ , it corresponds under h to an ultra-affine open of $S(Y)$. By (10.9.4) and the remark following, we can determine the unique morphism $f : X \rightarrow Y$ and conclude that $h = S(f)$.

2° It remains to prove that every ultra-prescheme X' is of the form $S(X)$ for some Jacobson prescheme X (which will necessarily be unique up to isomorphism, by 1°). There are ultra-affine open sets covering X' , each isomorphic to $S(U_\alpha)$ by open sets of X' , U_α being the spectrum of a Jacobson ring; for every pair of indices α, β , consider the unique open $U_{\alpha\beta}$ of U_α whose trace on $S(U_\alpha)$ is $S(U_\alpha) \cap S(U_\beta)$; by virtue of 1°, the identity of $U_\alpha^\beta \cap U_\beta^\alpha$ is of the form $S(\theta_{\alpha\beta})$, where $\theta_{\alpha\beta} : U_{\alpha\beta} \rightarrow U_{\beta\alpha}$ is an isomorphism of preschemes. One checks immediately (by virtue of 1°) that the family $(\theta_{\alpha\beta})$ satisfies the recollement condition (0_I, 4.1.7), and that this family defines a prescheme X , such that the U_α are identified with affine open sets of X ; it is clear that $X' = S(X)$, which completes the proof. $\square$

10.10. Serre's algebraic spaces

(10.10.1). The language introduced by Serre in (FAC) is sometimes convenient, in particular in the questions where the principal interest is attached to the rational points over a base field k (algebraically closed by hypothesis) of the “algebraic varieties” over k that one considers. We will briefly outline here this language, to allow the reader to translate the results of Serre into the language of schemes by connecting them to the preceding considerations. It is moreover possible to develop the language of Serre equally for preschemes over a non-algebraically-closed field (and even over an Artinian ring); but this introduces considerable technical complications, and moreover, over an arbitrary base field, the advantages (mainly psychological) of the Serre viewpoint disappear; thus we will confine ourselves to the framework fixed by Serre. The present subsection, like the preceding one, will not be used in the rest of this Treatise, and we will therefore limit ourselves to brief indications.

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(10.10.2). Given a fixed ring R , one can naturally (as in every category) define the notion of R -ultra-prescheme. Consider in particular an *algebraically closed* field k ; $\text{Spm}(k)$ is then identical to $\text{Spec}(k)$; we say that an $\text{Spm}(k)$ -ultra-prescheme X' is a *prealgebraic space* over k : it is therefore a k -ringed space with local rings whose every point has an open neighbourhood isomorphic to the maximal spectrum of a k -algebra of finite type; it amounts to the same to say (by (10.9.6)) that $X' = S(X)$, where X is a k -prescheme locally of finite type. If X, Y, Z are three k -preschemes locally of finite type over k , $X \times_k Y$ is also a k -prescheme locally of finite type (I, 1.3.4), hence by virtue of (10.9.6), the notion of product exists in the category of prealgebraic spaces over k (the “fibre product” $X' \times_{k'} Y'$, where X', Y', Z' are three prealgebraic spaces over k). One can also define the diagonal morphism $\Delta_{X'} : X' \rightarrow X' \times_{k'} X'$ (which is none other than $S(\Delta_X)$); we say that X' is an *algebraic space* over k if X is a scheme, and this amounts to the same as saying that the image of $\Delta_{X'}$ is a *closed part* of $X' \times_{k'} X'$.

(10.10.3). The simplifications which follow from the hypothesis that k is algebraically closed stem from the fact that, for a k -prescheme X locally of finite type, there is a bijective correspondence between *closed points* of X , *points of X with values in k* (I, 3.4.4), and *rational points of X* (I, 3.4.5), in virtue of (I, 6.4.2). This shows in particular (in virtue of (I, 4.3.1)) that for two k -prealgebraic spaces X', Y' , the underlying set of the product $X' \times_{k'} Y'$ is *identical to the set-theoretic product* $X' \times Y'$ (although of course the topology of the underlying space of $X' \times_{k'} Y'$ is *not* the product topology of the topologies of the underlying spaces of X' and Y' ; it is in general strictly finer than the latter).

On the other hand, the local rings $\mathcal{O}_{X'}$ at the points of a prealgebraic space X' over k are the k -algebras whose residue field is *isomorphic* to k ; if A and B are two such local k -algebras, every k -homomorphism $\varphi : A \rightarrow B$ is necessarily *local*: indeed, if an element x of the maximal ideal of A were such that $\varphi(x)$ is invertible, there would exist $\lambda \in k$ non-zero such that $\varphi(x - \lambda \cdot 1)$ belongs to the maximal ideal of B , which is absurd since $x - \lambda \cdot 1$ is invertible in A . We conclude immediately that if $X' = S(X)$, $Y' = S(Y)$ are two k -prealgebraic spaces, every morphism $X' \rightarrow Y'$ of *k -ringed spaces* is also a morphism of *ringed spaces with local rings* (I, 1.8.2); moreover, if A and B are two k -algebras of finite type, φ makes A a B -algebra of finite type; hence *every* morphism of k -prealgebraic spaces over k is, by virtue of (10.9.6), of the form $S(f)$, where $f : X \rightarrow Y$ is a *morphism of k -preschemes*.

(10.10.4). We say that a k -prealgebraic space $X' = S(X)$ is *reduced* if X is reduced; as the set of $x \in X$ where X is reduced is open (0_I, 5.2.2), its complement contains at least one closed point if it is non-empty (5.1.11), and it amounts to the same to say that X' is reduced or that each of its local rings $\mathcal{O}_{X'}$ is reduced. We have just seen in (10.10.3) that one can define $\mathcal{O}_{X'}$ as a *subsheaf* of $\mathcal{A}(X')$ (in (FAC), Serre limits himself in fact to *reduced* prealgebraic spaces). One notes that if X' and Y' are two reduced k -prealgebraic spaces, the same is true of $X' \times_{k'} Y'$: indeed, it suffices to see that if A and B are two reduced k -algebras of finite type, the same is true of $A \otimes_k B$; but we have already seen that the radicals of A and B are reduced to 0, and as k is algebraically closed, A and B are “separable” algebras in the sense of Bourbaki (*Alg.*, chap. VIII, §7, n° 5, prop. 5); hence $A \otimes_k B$ is without radical (*loc. cit.*, n° 6, cor. 3 of th. 3). However, if Z' is a third prealgebraic space over k , the “fibre product” $X' \times_{k'} Y'$ of two reduced prealgebraic spaces over Z' is not in general reduced, which implies that the category of these spaces is insufficient for many questions (notably in the theory of algebraic groups). But, as one has just seen above, one can keep the language of Serre without limiting oneself, as in (FAC), to considering only *quasi-compact* prealgebraic spaces, to the case of reduced prealgebraic spaces.

(10.10.5). Finally, one can also consider ultra-preschemes over an *arbitrary* field k while preserving a language close to that of Serre, and by introducing, as Weil did, an algebraically closed extension K of k (chosen large enough, for example of transcendence degree infinite over k , to have sufficiently many “generic points” in the sense of Weil). For every prescheme X locally of finite type over k , one associates the set $S_K(X) = S(X \otimes_k K)$ of points of X with values in K ; one has a canonical map $j : S_K(X) \rightarrow X$ which one shows to be a quasi-homeomorphism when one equips $S_K(X)$ with the topology that is the inverse image of that of X by j ; one equips $S_K(X)$ with the sheaf of k -algebras $j^*(\mathcal{O}_X)$, and one thus obtains a sub-category of the category of (k, K) -prealgebraic spaces. One can show that one can again define products and generalize the results of (10.10.3) and (10.10.4) ($\mathcal{A}(X')$ being replaced here by the sheaf $\mathcal{A}_K(X')$ of germs of maps from X' into K). However, this point of view makes an algebraically closed sub-field of k play an artificial role, and we note it only to set it aside.

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§11. TOPOLOGICAL PROPERTIES OF FINITELY PRESENTED FLAT MORPHISMS. FLATNESS CRITERIA

Recall that in §2 we considered results concerning flatness that do not depend on any finiteness hypothesis, and that §6 studies the notion of flatness in the context of locally Noetherian preschemes (but without finiteness hypothesis on morphisms); the present paragraph is devoted to the notion of f -flatness in the case where the morphism $f : X \rightarrow Y$ is *locally of finite presentation*. The interest of the notion of flat morphism of finite presentation lies in the fact that it is this notion which seems to express in the most technically adequate way the intuitive notion of a “family of algebraic preschemes parametrized by a scheme Y ”, the study of which is one of the principal objects of Algebraic Geometry. Moreover, even if one were only interested at the outset in the case of a Noetherian base scheme, it is indispensable, for certain technical reasons (for example for certain applications of the theory of “descent”, which leads to introducing non-Noetherian schemes), not to limit oneself to this case, as soon as it is a question of problems essentially related to morphisms locally of finite presentation. We follow this principle systematically, already supported by the results of §§8 and 9, throughout the rest of this Chapter, and even in the rest of our Treatise, at

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the cost of occasionally sacrificing the simplicity of certain proofs, that Noetherian hypotheses sometimes allow to simplify. In the present paragraph, this principle leads us to take up again, in the context of “finite presentation” (notably in n° 3), certain results of flatness, already obtained in the Noetherian context. The essential technical tool for the reduction to the Noetherian case is the theorem of compatibility of flatness with projective limits of preschemes (11.2.6), complementing the general results of § 8. We also prove in passing (11.3.1) a result often used in the sequel, implying that the set of points of flatness of a morphism locally of finite presentation is open.

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In n°s 4 to 8, we study the question of the “descent” of flatness, consisting of finding useful conditions on a base change morphism $Y' \rightarrow Y$ (not flat in general) to be able to conclude that if $X \times_Y Y'$ is flat over Y' , then X is flat over Y . These results, more technical than those of n°s 1 to 3, are of less frequent use in the sequel; they will however play an important role in certain non-projective construction techniques, in the following chapter. The only result from n°s 4 to 8 used in the rest of chap. IV is the valuative criterion of flatness (11.8), which will be applied in (15.2).

Finally, n°s 9 and 10 are devoted to the study of the notion that refines, in scheme theory, the notion of density in the topological sense, namely the notion of a family of sub-preschemes *schematically dense* in a given prescheme, and in particular the study of the behaviour of this notion under (flat or arbitrary) base change. This notion is mainly used, for the moment, in the study of group schemes.

11.1. Flatness loci (Noetherian case)

Theorem (11.1.1). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Then the set U of $x \in X$ such that $\mathcal{F}$ is f -flat at the point x is open in X .*

Proof. The question being evidently local on X and Y , we can suppose $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$, A being Noetherian and B an A -algebra of finite type. One has then $\mathcal{F} = \tilde{M}$, where M is a B -module of finite type. Let us apply the criterion (0_{III}, 9.2.6): it suffices therefore to prove the following assertion: □

(11.1.1.1). Let A be a Noetherian ring, B an A -algebra of finite type, M a B -module of finite type, $\mathfrak{q}$ a prime ideal of B , $\mathfrak{p}$ its inverse image in A . Suppose that $M_{\mathfrak{q}}$ is an $A_{\mathfrak{p}}$ -module flat. Then there exists $g \in B - \mathfrak{q}$ such that, for every prime ideal $\mathfrak{q}' \supset \mathfrak{q}$ of B such that $g \notin \mathfrak{q}'$, $M_{\mathfrak{q}'}$ is an $A_{\mathfrak{p}'}$ -module flat, denoting by $\mathfrak{p}'$ the inverse image of $\mathfrak{q}'$ in A (which amounts to saying (0_I, 6.3.1) that $M_{\mathfrak{q}'}$ is an A -module flat).

Proof of (11.1.1.1). Consider $B_{\mathfrak{q}'}$ as an A -algebra; evidently $\mathfrak{p}B_{\mathfrak{q}'} \subset \mathfrak{q}'B_{\mathfrak{q}'}$. We know then (0_{III}, 10.2.2) that for $M_{\mathfrak{q}'}$ to be an A -module flat, it is necessary and sufficient that $M_{\mathfrak{q}'}/\mathfrak{p}M_{\mathfrak{q}'}$ be an $(A/\mathfrak{p})$ -module flat and that $\text{Tor}_1^A(M_{\mathfrak{q}'}, A/\mathfrak{p}) = 0$. Now, we have $M_{\mathfrak{q}'} = M \otimes_B B_{\mathfrak{q}'}$; $B_{\mathfrak{q}'}$ being flat over B , $\text{Tor}_1^A(M_{\mathfrak{q}'}, A/\mathfrak{p}) = (\text{Tor}_1^A(M, A/\mathfrak{p}))_{\mathfrak{q}'} = (M/\mathfrak{p}M)_{\mathfrak{q}'}$ ($M/\mathfrak{p}M$ being considered in the second member as an A -module); moreover $(M/\mathfrak{p}M)_h$ being an A -module of finite type for a suitable h , $(M/\mathfrak{p}M)_h = 0$ if and only if there is a suitable element g and $(M/\mathfrak{p}M)_h = 0$ is equivalent to $(M/\mathfrak{p}M)_{\mathfrak{q}'} = 0$. Since A and B are Noetherian, $\text{Tor}_1^A(M, A/\mathfrak{p})$ is a B -module of finite type, hence (0_I, 5.2.2) there exists $g \in B - \mathfrak{q}$ such that $(\text{Tor}_1^A(M, A/\mathfrak{p}))_g = 0$. Moreover, $M/\mathfrak{p}M$ is a $B/\mathfrak{p}B$ -module of finite type, and it suffices to apply (0_I, 6.3.2). On the other hand, using the theorem of generic flatness (6.9.1) applied to the integral ring $A/\mathfrak{p}$, the $(A/\mathfrak{p})$ -algebra of finite type $B/\mathfrak{p}B$, and the $(B/\mathfrak{p}B)$ -module of finite type $M/\mathfrak{p}M$, there exists $h \in A - \mathfrak{p}$ such that, if $\bar{h}$ is its canonical image in $A/\mathfrak{p}$, $(M/\mathfrak{p}M)_{\bar{h}}$ is an $(A/\mathfrak{p})$ -module flat. On the other hand, as $M_{\mathfrak{q}}$ is an $A_{\mathfrak{p}}$ -module flat, and hence an A -module flat (0_I, 6.3.1), we also have $\text{Tor}_1^A(M_{\mathfrak{q}}, A/\mathfrak{p}) = 0$, which writes $(\text{Tor}_1^A(M, A/\mathfrak{p}))_{\mathfrak{q}} = 0$. Moreover, on has $M_{\mathfrak{q}'}/\mathfrak{p}M_{\mathfrak{q}'} = (M/\mathfrak{p}M)_h$ being considered as B -modules; finally, on has $(\text{Tor}_1^A(M, A/\mathfrak{p}))_{hg} = 0$ and $(M/\mathfrak{p}M)_{hg}$ is an $(A/\mathfrak{p})$ -module flat; hence M_{hg} is an A -module flat by (0_I, 6.3.2), and the conclusion follows since $hg \notin \mathfrak{q}$. C.Q.F.D. □

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Corollary (11.1.2). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}'$, $\mathcal{F}$ two coherent $\mathcal{O}_X$ -Modules, $u : \mathcal{F}' \rightarrow \mathcal{F}$ a homomorphism of $\mathcal{O}_X$ -Modules. Suppose that $\mathcal{F}$ is f -flat. Then the set U of $x \in X$ such that, setting $y = f(x)$, the homomorphism $u_x \otimes 1 : \mathcal{F}'_x \otimes_{\mathcal{O}_y} k(y) \rightarrow \mathcal{F}_x \otimes_{\mathcal{O}_y} k(y)$ is injective, is open in X .*

Proof. Indeed, let $\mathcal{N}$ (resp. $\mathcal{P}$) be the kernel (resp. the cokernel) of u ; applying (0III, 10.2.4)⁽¹⁾ to the local rings $\mathcal{O}_y$ and the $\mathcal{O}_x$ -modules $\mathcal{F}'_x$ and $\mathcal{F}_x$: to say that $u_x \otimes 1$ is injective is equivalent to saying that $\mathcal{N}_x = 0$ and that $\mathcal{P}$ is f -flat at the point x . Now $\mathcal{F}$ and $\mathcal{G}$ are coherent, hence the corollary follows from (11.1.1) and (0I, 5.2.2).⁶ $\square$

Corollary (11.1.3). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, g a section of $\mathcal{O}_X$ over X . The set of $x \in X$ such that $g_x \otimes 1$ is not a zero divisor in $\mathcal{O}_x \otimes_{\mathcal{O}_{f(x)}} k(f(x))$ is open in X .*

Proof. It suffices to apply (11.1.2) to the endomorphism of $\mathcal{O}_X$ defined by g . $\square$

Corollary (11.1.4). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, and f -flat. Let $(g_i)_{1 \leq i \leq n}$ be a suite of sections of $\mathcal{O}_X$ over X . Then the set U of $x \in X$ such that the suite $((g_i)_x \otimes 1)$ is $\mathcal{F}_x$ -regular and the $\mathcal{O}_x$ -module $\mathcal{G}_x$ (where $\mathcal{G} = \mathcal{F} / (\sum_{i=1}^n g_i \mathcal{F})$) is f -flat, is open in X .*

Proof. As $\mathcal{F}$ and $\mathcal{G}$ are coherent, the corollary follows from (11.1.1) and (0, 15.2.4). $\square$

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Corollary (11.1.5). — *Let X, Y, Z be three locally Noetherian preschemes, $f : X \rightarrow Y, g : Y \rightarrow Z$ two morphisms of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Then the set U of $y \in Y$ such that, for every generisation y' of y , $\mathcal{F}$ is $(g \circ f)$ -flat at all the points of $f^{-1}(y')$ (i.e. such that $\mathcal{F} \otimes_Y \text{Spec}(\mathcal{O}_{Y,y})$ is flat relative to the morphism $X \times_Y \text{Spec}(\mathcal{O}_{Y,y}) \rightarrow Z$) is open in Y .*

Proof. If V is the set of $x \in X$ where $\mathcal{F}$ is $(g \circ f)$ -flat, U is the set of $y \in Y$ such that for every generisation y' of y , one has $f^{-1}(y') \subset V$. Now V is locally open (11.1.1), hence locally constructible in X , and the set W of $y \in Y$ such that $f^{-1}(y) \subset V$ is also locally constructible in Y by virtue of Chevalley's theorem (1.8.4). But it follows then from (0III, 9.2.5) that the points of U are the interior points of W , whence the conclusion. $\square$

Corollary (11.1.6). — *Let A be a Noetherian ring, B an A -algebra of finite type, C a B -algebra of finite type, M a C -module of finite type. Then the set of $\mathfrak{q} \in \text{Spec}(B)$ such that $M_{\mathfrak{q}}$ is an A -module flat is open in $\text{Spec}(B)$.*

Proof. Taking into account (2.1.2), this is a consequence of (11.1.5) applied to $Z = \text{Spec}(A), Y = \text{Spec}(B), X = \text{Spec}(C), \mathcal{F} = \tilde{M}$.

The results of this subsection will be freed of the Noetherian hypotheses in (11.3). $\square$

11.2. Flatness of a projective limit of preschemes

(11.2.1). Let A be a ring, M, N two A -modules, A' an A -algebra; set $M' = M \otimes_A A', N' = N \otimes_A A'$. Recall (III, 6.3.8) that one defines, for every i , a canonical homomorphism

$$(11.2.1.1) \quad \psi_i : \text{Tor}_i^A(M, N) \otimes_A A' \longrightarrow \text{Tor}_i^{A'}(M', N')$$

in the following way: one considers a left resolution of M by free A -modules

$$(11.2.1.2) \quad \cdots \longrightarrow L_{i+1} \xrightarrow{f_i} L_i \longrightarrow \cdots \longrightarrow L_0 \xrightarrow{\varepsilon} M \longrightarrow 0$$

from which one deduces by tensorisation with A' a complex of A' -modules

$$(11.2.1.3) \quad \cdots \longrightarrow L'_{i+1} \xrightarrow{f'_i} L'_i \longrightarrow \cdots \longrightarrow L'_0 \xrightarrow{\varepsilon'} M' \longrightarrow 0$$

where $L'_i = L_i \otimes_A A', f'_i = f_i \otimes 1_{A'}, \varepsilon' = \varepsilon \otimes 1_{A'}$. Considering on the other hand a left resolution of M' by free A' -modules

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$$(11.2.1.4) \quad \cdots \longrightarrow L''_{i+1} \xrightarrow{f''_i} L''_i \longrightarrow \cdots \longrightarrow L''_0 \xrightarrow{\varepsilon''} M' \longrightarrow 0$$

⁶Here is a proof of (0III, 10.2.4), which has not been published in Bourbaki's *Commutative Algebra*. Taking into account (0I, 6.1.2), it suffices to see that $b)$ implies $a)$. Let $P = \text{Im}(u), Q = \text{Coker}(u), R = \text{Ker}(u)$. The composed map $M \otimes k \rightarrow P \otimes k \rightarrow N \otimes k$ is injective, and $M \otimes k \rightarrow P \otimes k$ is surjective, hence $P \otimes k \rightarrow N \otimes k$ is injective and $M \otimes k \rightarrow P \otimes k$ is bijective. The exact sequence $0 \rightarrow P \rightarrow N \rightarrow Q \rightarrow 0$ gives the exact sequence $0 = \text{Tor}_1^A(N, k) \rightarrow \text{Tor}_1^A(Q, k) \rightarrow P \otimes k \rightarrow N \otimes k \rightarrow Q \otimes k \rightarrow 0$ and since $P \otimes k \rightarrow N \otimes k$ is injective, one has $\text{Tor}_1^A(Q, k) = 0$; as Q is a B -module of finite type, (0III, 10.2.2) shows that Q is also an A -module flat; also $R \otimes k = 0$ by exact sequence, hence $R = 0$ by Nakayama.

as the L'_i are free A' -modules, one knows (M, V, 1.1) that there are A' -homomorphisms u_i forming a commutative diagram

$$\begin{array}{ccccccc} \cdots & \longrightarrow & L'_{i+1} & \xrightarrow{f'_i} & L'_i & \longrightarrow & \cdots \longrightarrow L'_0 \xrightarrow{\epsilon'} M' \longrightarrow 0 \\ & & \downarrow u_{i+1} & & \downarrow u_i & & \downarrow u_0 \\ \cdots & \longrightarrow & L''_{i+1} & \xrightarrow{f''_i} & L''_i & \longrightarrow & \cdots \longrightarrow L''_0 \xrightarrow{\epsilon''} M' \longrightarrow 0 \end{array}$$

Composing the homomorphism $u_\bullet : L'_\bullet \rightarrow L''_\bullet$ of complexes thus defined with the canonical homomorphism of A -module complexes $L_\bullet \rightarrow L'_\bullet$; noting that $(L_i \otimes_A N) \otimes_A A' = L'_i \otimes_{A'} N'$, one deduces a homomorphism of A -module complexes $L_\bullet \otimes_A N \rightarrow L''_\bullet \otimes_{A'} N'$, whence, passing to homology, the canonical homomorphisms (11.2.1.1). As the u_i are well-determined up to homotopy (M, V, 1.1), the homomorphisms (11.2.1.1) do not depend on the choice of the u_i nor on the choice of the free resolutions $L_\bullet$ and $L''_\bullet$.

As the $\text{Tor}_i^{A'}(M', N')$ are A' -modules, one canonically deduces from (11.2.1.1) A' -homomorphisms (11.2.1.5)

$$\varphi_i : \text{Tor}_i^A(M, N) \otimes_A A' \longrightarrow \text{Tor}_i^{A'}(M', N').$$

Consider now two homomorphisms of rings

$$\rho : A \longrightarrow A^{(1)}, \quad \sigma : A^{(1)} \longrightarrow A^{(2)},$$

and their composite $\sigma \circ \rho : A \rightarrow A^{(2)}$; setting $M^{(j)} = M \otimes_A A^{(j)}$, $N^{(j)} = N \otimes_A A^{(j)}$ for $j = 1, 2$. Then the canonical composite homomorphism

$$\text{Tor}_i^A(M, N) \longrightarrow \text{Tor}_i^{A^{(1)}}(M^{(1)}, N^{(1)}) \longrightarrow \text{Tor}_i^{A^{(2)}}(M^{(2)}, N^{(2)})$$

is the same as the canonical homomorphism deduced from $\sigma \circ \rho$; this follows from the fact that, if $L_\bullet^{(j)}$ is a free resolution of $M^{(j)}$, the diagram

$$\begin{array}{ccccc} L_\bullet & \longrightarrow & L_\bullet \otimes_A A^{(1)} & \longrightarrow & L_\bullet^{(1)} \\ \downarrow & & \downarrow & & \\ L_\bullet \otimes_A A^{(2)} & \longrightarrow & L_\bullet^{(1)} \otimes_{A^{(1)}} A^{(2)} & \longrightarrow & L_\bullet^{(2)} \end{array}$$

is commutative.

(11.2.2). With the notations of (11.2.1), consider now an inductive system of A -algebras $(A_\alpha, g_{\beta\alpha})$, and for every index α , set $M_\alpha = M \otimes_A A_\alpha$, $N_\alpha = N \otimes_A A_\alpha$; it follows from (11.2.1) that for every i , $(\text{Tor}_i^{A_\alpha}(M_\alpha, N_\alpha), \psi_{i\beta\alpha})$, where $\psi_{i\beta\alpha}$ is the canonical homomorphism (11.2.1.1) corresponding to $g_{\beta\alpha}$, is (always by virtue of (11.2.1)) an inductive system of homomorphisms; we propose to complete the result of (M, V, 9.4*) by showing that the

$$(11.2.2.1) \quad \psi_i = \varinjlim_\alpha \psi_{i\alpha} : \varinjlim_\alpha \text{Tor}_i^{A_\alpha}(M_\alpha, N_\alpha) \longrightarrow \text{Tor}_i^{A'}(M', N')$$

are isomorphisms of A' -modules. For this, associating to each M_α its free canonical resolution, everything amounts (taking into account the exactitude of the functor $\varinjlim$) to proving the

Lemma (11.2.2.2). — Let $(A_\alpha, g_{\beta\alpha})$ be a filtered inductive system of rings, $(M_\alpha, h_{\beta\alpha})$ a filtered inductive system of sets, $A' = \varinjlim_\alpha A_\alpha$, $M' = \varinjlim_\alpha M_\alpha$, $g_\alpha : A_\alpha \rightarrow A'$, $h_\alpha : M_\alpha \rightarrow M'$ the canonical maps. For every α , let $F(M_\alpha)$ be the A_α -module of formal linear combinations of elements of M_α ; let similarly $F(M')$ be the A' -module of formal linear combinations of elements of M' . If $h'_{\beta\alpha} : F(M_\alpha) \rightarrow F(M_\beta)$ (for $\alpha \leq \beta$) and $h'_\alpha : F(M_\alpha) \rightarrow F(M')$ are the A_α -homomorphisms deduced from $h_{\beta\alpha}$ and h_α respectively, $(F(M_\alpha), h'_{\beta\alpha})$ is an inductive system of A_α -modules and (h'_α) an inductive system of homomorphisms. Then

$$h'' = \varinjlim_\alpha h'_\alpha : \varinjlim_\alpha F(M_\alpha) \longrightarrow F(\varinjlim_\alpha M_\alpha)$$

is an isomorphism.

Proof. For the proof, see Bourbaki, *Alg.*, chap. II, 3^e éd., § 6, n^o 6, cor. of prop. 10. □

(11.2.3). Let us take up again the notations of (11.2.1) and consider particularly the case $i = 1$; set $T = \operatorname{Tor}_1^A(M, N)$, $T' = T \otimes_A A'$, $T'' = \operatorname{Tor}_1^{A'}(M', N')$; then ψ_1 is the homomorphism $\operatorname{Ker}(f_0 \otimes 1_N) / \operatorname{Im}(f_1 \otimes 1_N) \rightarrow \operatorname{Ker}(f'_0 \otimes 1_{N'}) / \operatorname{Im}(f'_1 \otimes 1_{N'})$ which is deduced by passage to quotients of the restriction $\operatorname{Ker}(f_0 \otimes 1_N) \rightarrow \operatorname{Ker}(f'_0 \otimes 1_{N'})$ of

$$u_1 \otimes 1 : L_0 \otimes_A N \longrightarrow L'_0 \otimes_{A'} N' = L''_0 \otimes_{A'} N'.$$

Set $R = \operatorname{Ker}(\varepsilon)$, $R'' = \operatorname{Ker}(\varepsilon'')$, so that one has the exact sequences

$$0 \longrightarrow R \xrightarrow{j} L_0 \xrightarrow{\varepsilon} M \longrightarrow 0 \quad \text{and} \quad 0 \longrightarrow R'' \xrightarrow{j''} L''_0 \xrightarrow{\varepsilon''} M' \longrightarrow 0,$$

whence one deduces the exact sequences of homology

$$(11.2.3.1) \quad 0 = \operatorname{Tor}_1^A(L_0, N) \longrightarrow T \xrightarrow{\partial} R \otimes_A N \xrightarrow{j \otimes 1} L_0 \otimes_A N \xrightarrow{\varepsilon \otimes 1} M \otimes_A N \longrightarrow 0$$

$$(11.2.3.2) \quad 0 = \operatorname{Tor}_1^{A'}(L''_0, N') \longrightarrow T'' \xrightarrow{\partial''} R'' \otimes_{A'} N' \xrightarrow{j'' \otimes 1} L''_0 \otimes_{A'} N' \xrightarrow{\varepsilon'' \otimes 1} M' \otimes_{A'} N' \longrightarrow 0.$$

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Lemma (11.2.4). — Let A be a ring, C an A -algebra, M an A -module, N a C -module. Set $C' = C \otimes_A A'$, $M' = M \otimes_A A'$, $N' = N \otimes_A A' = N \otimes_C C'$. Suppose that $M \otimes_A N$ is a C -module flat. Then the canonical homomorphism

$$\varphi_1 : \operatorname{Tor}_1^A(M, N) \otimes_A A' \longrightarrow \operatorname{Tor}_1^{A'}(M', N')$$

(cf. (11.2.1.5)) is surjective.

Proof. Let us keep the notations of (11.2.3); the right-exactness of the tensor product shows that the sequence $L'_1 \xrightarrow{f'_1} L'_0 \xrightarrow{\varepsilon'} M' \rightarrow 0$ is exact; as L'_0 and L'_1 are A' -modules free, one can take $L''_0 = L'_0$, $L''_1 = L'_1$, u_0 and u_1 being the identity maps and $f''_0 = f'_0$. As $R = \operatorname{Im}(f_0)$ and $R'' = \operatorname{Im}(f'_0)$, the homomorphism v is surjective, and it is also the case for $v \otimes 1$. The proof will be finished if one proves that the first row of (11.2.3.3) is exact, $v \otimes 1$ being surjective and $u_0 \otimes 1$ injective (Bourbaki, *Alg. comm.*, chap. I^{er}, § 1, n° 4, cor. 2 of prop. 2). Let us use for this the hypothesis that $M \otimes_A N$ is a C -module flat. Setting $Q = \operatorname{Ker}(\varepsilon \otimes 1) = \operatorname{Im}(j \otimes 1)$ in the exact sequence (11.2.3.1), one has the two exact sequences $0 \rightarrow Q \rightarrow L_0 \otimes_A N \rightarrow M \otimes_A N \rightarrow 0$ and $T \rightarrow R \otimes_A N \rightarrow Q \rightarrow 0$, where the homomorphisms are homomorphisms of C -modules; using the hypothesis of flatness, one deduces from (0_I, 6.1.2) the exact sequence

$$0 \longrightarrow Q \otimes_C C' \longrightarrow (L_0 \otimes_A N) \otimes_C C' \longrightarrow (M \otimes_A N) \otimes_C C' \longrightarrow 0$$

and on the other hand, the tensor product being right-exact, one has the exact sequence

$$T \otimes_C C' \longrightarrow (R \otimes_A N) \otimes_C C' \longrightarrow Q \otimes_C C' \longrightarrow 0$$

whence finally the exact sequence

$$T \otimes_C C' \longrightarrow (R \otimes_A N) \otimes_C C' \longrightarrow (L_0 \otimes_A N) \otimes_C C' \longrightarrow (M \otimes_A N) \otimes_C C' \longrightarrow 0.$$

But by definition, for every C -module P , $P \otimes_C C' = P \otimes_A A'$, hence the conclusion. $\square$

Lemma (11.2.5). — Let A be a ring, $\mathfrak{J}$ an ideal of A , B an A -algebra, M a B -module. Set $\mathfrak{J}' = \mathfrak{J}A'$, $B' = B \otimes_A A'$, $M' = M \otimes_A A' = M \otimes_B B'$. Let $\mathfrak{p}'$ be a prime ideal of B' containing $\mathfrak{J}B'$. Suppose verified one of the hypotheses:

- a) $\mathfrak{J}$ is nilpotent.
- b) A' is Noetherian, B' is an A' -algebra of finite type, M' a B' -module of finite type.

Under these conditions, suppose the two following properties verified:

- (i) $M \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$ -module flat.
- (ii) The composed canonical homomorphism

$$\operatorname{Tor}_1^A(M, A/\mathfrak{J}) \xrightarrow{\psi_1} \operatorname{Tor}_1^{A'}(M', A'/\mathfrak{J}') \xrightarrow{\theta} (\operatorname{Tor}_1^{A'}(M', A'/\mathfrak{J}'))_{\mathfrak{p}'}$$

(where ψ_1 is the canonical homomorphism (11.2.1.1) and θ the localisation of a B' -module at $\mathfrak{p}'$) is zero.

Then $M'_{\mathfrak{p}'}$ is an A' -module flat.

Proof. Let us note that under hypothesis b), $M'_{\mathfrak{p}'}$ is a $B'_{\mathfrak{p}'}$ -module of finite type, $B'_{\mathfrak{p}'}$ an A' -algebra noethérienne, and $\mathfrak{J}'B'_{\mathfrak{p}'}$ is contained in the radical $\mathfrak{p}'B'_{\mathfrak{p}'}$ of $B'_{\mathfrak{p}'}$; under hypothesis a), $\mathfrak{J}'$ is nilpotent; one can therefore apply the flatness criterion (0_{III}, 10.2.1) or (0_{III}, 10.2.2) as appropriate. First, $M'_{\mathfrak{p}'} \otimes_{A'} (A'/\mathfrak{J}') = (M' \otimes_{A'} (A'/\mathfrak{J}'))_{\mathfrak{p}'} = ((M \otimes_A (A/\mathfrak{J})) \otimes_{A/\mathfrak{J}} (A'/\mathfrak{J}'))_{\mathfrak{p}'}$; hypothesis (i) then implies that $M'_{\mathfrak{p}'} \otimes_{A'} (A'/\mathfrak{J}')$ is an $(A'/\mathfrak{J}')$ -module flat, taking into account (0_I, 6.2.1 and 6.3.2). It remains to see that $\mathrm{Tor}_1^{A'}(M'_{\mathfrak{p}'}, A'/\mathfrak{J}') = 0$; but this $B'_{\mathfrak{p}'}$ -module is equal to $(\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{J}'))_{\mathfrak{p}'}$ by flatness of $B'_{\mathfrak{p}'}$ over B' . But by virtue of hypothesis (ii), (11.2.4) applied to $C = A/\mathfrak{J}$ and $N = C$ shows that φ_1 is surjective (since $A'/\mathfrak{J}' = C \otimes_A A'$); hence the homomorphism θ of $\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{J}')$ engenders the $B'_{\mathfrak{p}'}$ -module $(\mathrm{Tor}_1^{A'}(M', A'/\mathfrak{J}'))_{\mathfrak{p}'}$, and the latter is zero. Q.E.D. $\square$

Theorem (11.2.6). — *The notations being those of (8.5.1) and (8.8.1), suppose S_α quasi-compact, X_α and Y_α of finite presentation over S_α ; let $f_\alpha : X_\alpha \rightarrow Y_\alpha$ be an S_α -morphism, $\mathcal{F}_\alpha$ an $\mathcal{O}_{X_\alpha}$ -Module quasi-coherent of finite presentation.*

- (i) *Let x be a point of X , x_α its canonical projection in X_α . For $\mathcal{F}$ to be f -flat at the point x , it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that $\mathcal{F}_\lambda$ be f_λ -flat at the point x_λ .*
- (ii) *For $\mathcal{F}$ to be f -flat, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that $\mathcal{F}_\lambda$ be f_λ -flat.*

One need only prove the sufficiency of the conditions, the necessity being (2.1.4). We proceed in several steps. IV-3 | 124

(11.2.6.1). Corollary. Let A_0 be a ring, $(A_\lambda)_{\lambda \in L}$ a filtered inductive system of A_0 -algebras of finite presentation, B_0 an A_0 -algebra of finite presentation, M_0 a B_0 -module of finite presentation. Set $B_\lambda = B_0 \otimes_{A_0} A_\lambda$, $M_\lambda = M_0 \otimes_{A_0} A_\lambda = M_0 \otimes_{B_0} B_\lambda$, $A = \varinjlim A_\lambda$, $B = \varinjlim B_\lambda = B_0 \otimes_{A_0} A$, $M = \varinjlim M_\lambda = M_0 \otimes_{A_0} A = M_0 \otimes_{B_0} B$.

- (i) Let $\mathfrak{p}$ be a prime ideal of B , and $\mathfrak{p}_\lambda$ its preimage in B_λ . For $M_\mathfrak{p}$ to be an A -module flat, it is necessary and sufficient that there exist λ such that $(M_\lambda)_{\mathfrak{p}_\lambda}$ be an A_λ -module flat.
- (ii) For M to be an A -module flat, it is necessary and sufficient that there exist λ such that M_λ be an A_λ -module flat.

Corollary (11.2.7). — *Let $S = \mathrm{Spec}(A)$ be an affine scheme, $f : X \rightarrow S$ a morphism, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent, x a point of X . The following conditions are equivalent:*

- a) f is a morphism of finite presentation, $\mathcal{F}$ is an $\mathcal{O}_X$ -Module of finite presentation and $\mathcal{F}$ is f -flat at the point x (resp. f -flat).*
- b) There exists an affine Noetherian scheme $S_0 = \mathrm{Spec}(A_0)$, a morphism of finite type $f_0 : X_0 \rightarrow S_0$, an $\mathcal{O}_{X_0}$ -Module coherent $\mathcal{F}_0$, a morphism $S \rightarrow S_0$ such that X is S -isomorphic to $X_0 \otimes_{S_0} S$, $\mathcal{F}$ is isomorphic to $\mathcal{F}_0 \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_S$, and $\mathcal{F}_0$ is f_0 -flat at the point x_0 projection of x in X_0 (resp. f_0 -flat).*
- c) The conditions of b) are verified and moreover A_0 is a sub- $\mathbf{Z}$ -algebra of finite type of A , corresponding to the injection canonical $A_0 \rightarrow A$.*

It is clear that c) implies b) and b) implies a) by (2.1.4). On the other hand, a) implies b): one can consider A as inductive limit of its sub- $\mathbf{Z}$ -algebras of finite type, and one knows by (8.9.1) that there is such a sub-algebra A_0 , an A'_0 -algebra of finite type B'_0 and a B'_0 -module of finite type M'_0 such that $B_0 = B'_0 \otimes_{A'_0} A_0$ and $M_0 = M'_0 \otimes_{A'_0} A_0$; considering the A_λ as sub- A_0 -algebras of finite type of A , the conclusion then results from (11.2.6) applied with c) to verify. IV-3 | 126

Proposition (11.2.8). — *Let $g : X \rightarrow S$, $h : Y \rightarrow S$ be two morphisms of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation; for every $s \in S$, let $X_s = g^{-1}(s) = X \times_S \mathrm{Spec}(k(s))$, $Y_s = h^{-1}(s) = Y \times_S \mathrm{Spec}(k(s))$, f_s the morphism $f \times 1 : X_s \rightarrow Y_s$, $\mathcal{F}_s = \mathcal{F} \otimes_{\mathcal{O}_S} k(s)$. Then the set E of $s \in S$ such that $\mathcal{F}_s$ is f_s -flat is locally constructible.*

Proof. The property in question verifies the constructibility condition (9.2.1, (i)), by virtue of (2.2.13) and (2.5.1). Taking into account (9.2.3), one can reduce to the case where S is affine, Noetherian and integral of generic point η and to prove that E is an open neighbourhood of η in S . If $\eta \in S - E$, it results first from (11.2.6), applying the method of (8.1.2, a), that there is a neighbourhood open U of η in S such that $\mathcal{F}|_{g^{-1}(U)}$ is $h^{-1}(U)$ -flat; a fortiori (2.1.4) $\mathcal{F}_s$ is f_s -flat for every $s \in U$, which finishes the proof. $\square$

The following theorem generalises (11.2.6.1), (ii):

Theorem (11.2.9). — (Raynaud) *Let A_0 be a ring, $(A_\lambda)_{\lambda \in L}$ a filtered inductive system of A_0 -algebras, B_0 an A_0 -algebra of finite presentation, $\mathfrak{J}_0$ an ideal of finite type of B_0 , M_0 a B_0 -module of finite presentation. Set $B_\lambda = B_0 \otimes_{A_0} A_\lambda$, $M_\lambda = M_0 \otimes_{A_0} A_\lambda = M_0 \otimes_{B_0} B_\lambda$, $A = \varinjlim A_\lambda$, $B = \varinjlim B_\lambda = B_0 \otimes_{A_0} A$, $\mathfrak{J} = \mathfrak{J}_0 B$, $M = \varinjlim M_\lambda = M_0 \otimes_{A_0} A = M_0 \otimes_{B_0} B$. For $\mathrm{gr}_{\mathfrak{J}}^*(M)$ to be an A -module flat, it is necessary and sufficient that there exist λ such that $\mathrm{gr}_{\mathfrak{J}_\lambda}^*(M_\lambda)$ be an A_λ -module flat; the canonical homomorphisms*

$$(11.2.9.1) \quad \begin{cases} \mathrm{gr}_{\mathfrak{J}_\lambda}^*(M_\lambda) \otimes_{A_\lambda} A \rightarrow \mathrm{gr}_{\mathfrak{J}}^*(M) \\ \mathrm{gr}_{\mathfrak{J}_\lambda}^*(M_\lambda) \otimes_{A_\lambda} A_\mu \rightarrow \mathrm{gr}_{\mathfrak{J}_\mu}^*(M_\mu) \end{cases} \quad \text{for } \lambda \leq \mu$$

are then bijective and $\mathrm{gr}_{\mathfrak{J}_\mu}^*(M_\mu)$ is an A_μ -module flat for $\lambda \leq \mu$.

The proof of this theorem proceeds in several steps (I–VI) and occupies the following pages. We only state the key intermediate results:

Lemma (11.2.9.2). — *Let A be a ring, B an A -algebra, M a B -module, $\mathfrak{J}$ an ideal of B . Let A' be an A -algebra; set $B' = B \otimes_A A'$, $M' = M \otimes_A A' = M \otimes_B B'$, $\mathfrak{J}' = \mathfrak{J}B'$. If $\mathrm{gr}_{\mathfrak{J}}^*(M)$ is an A -module flat, the canonical homomorphism*

$$(\mathrm{gr}_{\mathfrak{J}}^*(M)) \otimes_A A' \longrightarrow \mathrm{gr}_{\mathfrak{J}'}^*(M')$$

is bijective.

Proof. By induction on n , one proves that the canonical homomorphism $\mathfrak{J}^n M \otimes_A A' \rightarrow \mathfrak{J}'^n M'$ is bijective. For $n = 0$, this is the identity $M \otimes_A A' = M'$. For the inductive step, consider the exact sequence $0 \rightarrow \mathfrak{J}^{n+1} M \rightarrow \mathfrak{J}^n M \rightarrow \mathrm{gr}_{\mathfrak{J}}^n(M) \rightarrow 0$. Since $\mathrm{gr}_{\mathfrak{J}}^n(M)$ is A -flat, tensoring with A' preserves exactness, giving

$$0 \longrightarrow \mathfrak{J}^{n+1} M \otimes_A A' \longrightarrow \mathfrak{J}^n M \otimes_A A' \longrightarrow \mathrm{gr}_{\mathfrak{J}}^n(M) \otimes_A A' \longrightarrow 0.$$

By the induction hypothesis, $\mathfrak{J}^n M \otimes_A A' = \mathfrak{J}'^n M'$, and one identifies the quotient $\mathfrak{J}'^n M' / (\mathfrak{J}'^{n+1} M' \otimes_A A')$ as $\mathrm{gr}_{\mathfrak{J}'}^n(M')$. On the other hand, $\mathfrak{J}'^{n+1} M' \subset \mathfrak{J}'^n M'$ and $\mathfrak{J}'^n M' / \mathfrak{J}'^{n+1} M' = \mathrm{gr}_{\mathfrak{J}'}^n(M')$; since the canonical map $\mathfrak{J}^{n+1} M \otimes_A A' \rightarrow \mathfrak{J}'^{n+1} M'$ is surjective and both quotients $\mathrm{gr}_{\mathfrak{J}}^n(M) \otimes_A A'$ and $\mathrm{gr}_{\mathfrak{J}'}^n(M')$ embed into $\mathfrak{J}'^n M'$, one concludes that $\mathfrak{J}^{n+1} M \otimes_A A' \xrightarrow{\sim} \mathfrak{J}'^{n+1} M'$ and $\mathrm{gr}_{\mathfrak{J}}^n(M) \otimes_A A' \xrightarrow{\sim} \mathrm{gr}_{\mathfrak{J}'}^n(M')$. $\square$

Lemma (11.2.9.3). — *Let A be a ring, $B = \bigoplus_{i \geq 0} B_i$ a graded A -algebra with positive degrees, $M = \bigoplus_{i \in \mathbb{Z}} M_i$ a graded B -module. Suppose that B_0 is a local ring, and that each of the M_i is a B_0 -module of finite type. For M to be a B -module of finite type, it is necessary and sufficient that, if $\mathfrak{q}$ is the inverse image in A of the maximal ideal $\mathfrak{m}$ of B_0 , $M \otimes_A k(\mathfrak{q})$ be a $(B \otimes_A k(\mathfrak{q}))$ -module of finite type.*

Proof. The condition being evidently necessary, let us prove that it is sufficient. As B is an A_0 -algebra, one can replace A by $A_{\mathfrak{q}}$, in other words suppose that A is also a local ring with $\mathfrak{q}$ as maximal ideal. By hypothesis, there exists an integer N such that the canonical homomorphism

$$\bigoplus_{-N \leq i \leq N} (M_i \otimes_{B_0} B) \otimes_A k(\mathfrak{q}) \longrightarrow M \otimes_A k(\mathfrak{q})$$

is surjective; this also means that, for every integer n , the canonical homomorphism of $k(\mathfrak{q})$ -modules

$$\bigoplus_{-N \leq i \leq N} (M_i \otimes_{B_0} B_{n-i}) \otimes_A k(\mathfrak{q}) \longrightarrow M_n \otimes_A k(\mathfrak{q})$$

is surjective. Now, M_n is a B_0 -module of finite type, B_0 a local ring and $\mathfrak{q}B_0 \subset \mathfrak{m}$; hence Nakayama's lemma shows that each of the canonical homomorphisms

$$\bigoplus_{-N \leq i \leq N} (M_i \otimes_{B_0} B_{n-i}) \longrightarrow M_n$$

is surjective, whence the conclusion. $\square$

Corollary (11.2.9.4). — *Under the hypotheses of (11.2.9.3), suppose moreover that each of the B_i and each of the M_i is a B_0 -module of finite presentation, and that M is an A -module flat. For M to be a B -module of finite presentation, it is necessary and sufficient that $M \otimes_A k(\mathfrak{q})$ be a $(B \otimes_A k(\mathfrak{q}))$ -module of finite type.*

Proof. The condition being evidently necessary, let us prove that it is sufficient. By (11.2.9.3), M is a B -module of finite type, so there exists a free graded B -module L of finite type and a surjective graded homomorphism of degree 0, $u : L \rightarrow M$; let $R = \text{Ker}(u)$, which is a graded B -module. There is then a finite number of integers m_j ($1 \leq j \leq r$) such that for each integer i , R_i is the kernel of a surjective homomorphism $\bigoplus_{1 \leq j \leq r} B_{i+m_j} \rightarrow M_i$; one concludes from the hypothesis on the M_i and the B_i that R_i is a B_0 -module of finite type (Bourbaki, *Alg. comm.*, chap. I, § 2, n° 8, lemme 9). To prove that R is a B -module of finite type, it suffices by (11.2.9.3) to prove that $R \otimes_A k(q)$ is a $(B \otimes_A k(q))$ -module of finite type. Since M is A -flat, the sequence $0 \rightarrow R \otimes_A k(q) \rightarrow L \otimes_A k(q) \rightarrow M \otimes_A k(q) \rightarrow 0$ is exact (0_I, 6.1.2); as $L \otimes_A k(q)$ and $M \otimes_A k(q)$ are $(B \otimes_A k(q))$ -modules of finite type, so is $R \otimes_A k(q)$; it therefore suffices to apply (11.2.9.3). $\square$

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Lemma (11.2.9.5). — *Let A_0 be a Noetherian ring, B_0 an A_0 -algebra of finite type, $\mathfrak{J}_0$ an ideal of B_0 , M_0 a B_0 -module of finite type. There exists then a suite $(S_{0i})_{1 \leq i \leq m}$ of subschemes of $S_0 = \text{Spec}(A_0)$ having the following properties:*

- 1° *The underlying spaces of the S_{0i} are pairwise disjoint and form a covering of S_0 .*
- 2° *For each i , the set $\bigcup_{j \geq i} S_{0j}$ is open in S_0 .*
- 3° *Each scheme S_{0i} is affine.*
- 4° *If A_{0i} is the ring of S_{0i} and if one sets $B_{0i} = B_0 \otimes_{A_0} A_{0i}$, $\mathfrak{J}_{0i} = \mathfrak{J}_0 B_{0i}$, then $\text{gr}_{\mathfrak{J}_{0i}}^*(M_0 \otimes_{A_0} A_{0i})$ is an A_{0i} -module flat.*

Proof. One proceeds as in (6.9.3) by Noetherian induction, supposing the lemma true when one replaces A_0 by $A'_0 = A_0/\mathfrak{a}_0$, where $\mathfrak{a}_0$ is an ideal such that $V(\mathfrak{a}_0) \neq S_0$, B_0 by $B'_0 = B_0 \otimes_{A_0} A'_0$, $\mathfrak{J}_0$ by $\mathfrak{J}'_0 = \mathfrak{J}_0 B'_0$ and M_0 by $M'_0 = M_0 \otimes_{A_0} A'_0$. One considers the nilradical $\mathfrak{N}_0$ of A_0 . In other words, one can suppose that A_0 is *reduced*; on the other hand, $\text{gr}_{\mathfrak{J}_0}^*(B_0)$ is an A_0 -algebra of finite type and $\text{gr}_{\mathfrak{J}_0}^*(M_0)$ a $\text{gr}_{\mathfrak{J}_0}^*(B_0)$ -module of finite type; by the generic flatness theorem (6.9.1), there exists an open set $D(t_0) \subset S_0$ such that if one sets $A_{01} = (A_0)_{t_0}$, $\text{gr}_{\mathfrak{J}_0}^*(M_0) \otimes_{A_0} A_{01}$ is an A_{01} -module flat; but since A_{01} is a flat A_0 -module, $\text{gr}_{\mathfrak{J}_0}^*(M_0) \otimes_{A_0} A_{01}$ identifies with $\text{gr}_{\mathfrak{J}_{01}}^*(M_0 \otimes_{A_0} A_{01})$, where $B_{01} = B_0 \otimes_{A_0} A_{01}$ and $\mathfrak{J}_{01} = \mathfrak{J}_0 B_{01}$. The complement of $D(t_0)$ in S_0 is then of the form $V(\mathfrak{a}_0)$, and one concludes by applying the Noetherian induction hypothesis. $\square$

Lemma (11.2.9.6). — *Under the preceding hypotheses, let $\mathfrak{R}_0$ be an ideal of A_0 ; set $A'_0 = A_0/\mathfrak{R}_0$, $B'_0 = B_0 \otimes_{A_0} A'_0 = B_0/\mathfrak{R}_0 B_0$, $\mathfrak{J}'_0 = \mathfrak{J}_0 B'_0$, $\mathfrak{R} = \mathfrak{R}_0 A$, $A' = A/\mathfrak{R}$, and suppose moreover that $\text{gr}_{\mathfrak{J}'_0}^*(M_0 \otimes_{A_0} A'_0)$ is an A'_0 -module flat. Then there exist a finite number of elements q_h ($1 \leq h \leq s$) of Q whose images in $Q_{\mathfrak{p}}$ engender the $C_{\mathfrak{p}}$ -module $Q_{\mathfrak{p}}$, for every prime ideal $\mathfrak{p} \supset \mathfrak{R}B$ of B .*

Proof. Within the proof of (11.2.9), let L (resp. L_0) denote the free graded C -module (resp. C_0 -module) which is the source of the epimorphism u (resp. u_0), so that $Q = \text{Ker}(u : L \rightarrow \text{gr}_{\mathfrak{J}}^*(M))$. Set $M' = M \otimes_A A'$ and $\mathfrak{J}' = \mathfrak{J}(B/\mathfrak{R}B)$. By hypothesis, $\text{gr}_{\mathfrak{J}}^*(M)$ is an A -module flat, so by (11.2.9.2), $\text{gr}_{\mathfrak{J}'}^*(M')$ identifies with $\text{gr}_{\mathfrak{J}}^*(M) \otimes_A A'$. Tensoring with A' over A the exact sequence $0 \rightarrow Q \rightarrow L \xrightarrow{u} \text{gr}_{\mathfrak{J}}^*(M) \rightarrow 0$, we obtain by (0_I, 6.1.2) the exact sequence

$$(11.2.9.7) \quad 0 \rightarrow Q \otimes_A A' \rightarrow L \otimes_A A' \xrightarrow{u \otimes 1_{A'}} \text{gr}_{\mathfrak{J}'}^*(M') \rightarrow 0.$$

Set $C'_0 = C_0 \otimes_{A_0} A'_0 = C_0/\mathfrak{R}_0 C_0$ and let Q_0 be the kernel of $u_0 \otimes 1_{A'_0} : L_0 \otimes_{A_0} A'_0 \rightarrow \text{gr}_{\mathfrak{J}'_0}^*(M_0 \otimes_{A_0} A'_0)$; since $\text{gr}_{\mathfrak{J}'_0}^*(M_0 \otimes_{A_0} A'_0)$ is an A'_0 -module flat by hypothesis, (11.2.9.2) shows that $\text{gr}_{\mathfrak{J}'_0}^*(M_0 \otimes_{A_0} A'_0) \otimes_{A'_0} A'$ identifies with $\text{gr}_{\mathfrak{J}'}^*(M')$, and (0_I, 6.1.2) gives the exact sequence

$$(11.2.9.8) \quad 0 \rightarrow Q_0 \otimes_{A'_0} A' \rightarrow L \otimes_A A' \xrightarrow{u \otimes 1_{A'}} \text{gr}_{\mathfrak{J}'}^*(M') \rightarrow 0,$$

whence, by comparison with (11.2.9.7), an isomorphism of graded $(C \otimes_A A')$ -modules $Q \otimes_A A' \xrightarrow{\sim} Q_0 \otimes_{A'_0} A'$.

Since C'_0 is Noetherian, Q_0 is a C'_0 -module of finite type, hence $Q_0 \otimes_{A'_0} A'$ is a graded $(C \otimes_A A')$ -module of finite type, and consequently so is $Q \otimes_A A'$; let q_h ($1 \leq h \leq s$) be homogeneous elements of Q whose images in $Q \otimes_A A'$ generate this module.

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Consider now an arbitrary prime ideal $\mathfrak{p} \supset \mathfrak{A}B$ of B ; we have $C_{\mathfrak{p}} = \mathrm{gr}_{\mathfrak{J}_{\mathfrak{p}}}^*(B_{\mathfrak{p}})$ by flatness, and $\mathrm{gr}_{\mathfrak{J}_{\mathfrak{p}}}^0(B_{\mathfrak{p}}) = B_{\mathfrak{p}}/\mathfrak{J}_{\mathfrak{p}}$ is either 0 or a local ring; to prove the lemma, we may restrict to the second case. It is clear that each $(B_{\mathfrak{p}}/\mathfrak{J}_{\mathfrak{p}})$ -module $\mathrm{gr}_{\mathfrak{J}_{\mathfrak{p}}}^i(B_{\mathfrak{p}})$ is of finite presentation. On the other hand, for every index i , $M/\mathfrak{J}^i M$ identifies with $(M_0/\mathfrak{J}_0^i M_0) \otimes_{A_0} A$, hence is a B -module of finite presentation. By induction on i , the hypothesis that $\mathrm{gr}_{\mathfrak{J}}^*(M)$ is A -flat implies, by (0_I, 6.1.2), that $M/\mathfrak{J}^i M$ is A -flat. Applying (11.2.6.1) with M_0 replaced by $M_0/\mathfrak{J}_0^k M_0$ for $k \leq i$ shows that there exists an index λ_i such that, for $\mu \geq \lambda_i$, each $M_{\mu}/\mathfrak{J}_{\mu}^{k+1} M_{\mu}$ is A_{μ} -flat, and consequently (0_I, 6.1.2) each $\mathrm{gr}_{\mathfrak{J}_{\mu}}^k(M_{\mu})$ is A_{μ} -flat for $k \leq i$. It follows from (11.2.9.2) that each $(B/\mathfrak{J})$ -module $\mathrm{gr}_{\mathfrak{J}}^i(M)$ is of finite presentation, so $\mathrm{gr}_{\mathfrak{J}_{\mathfrak{p}}}^*(M_{\mathfrak{p}})$ is a $(B_{\mathfrak{p}}/\mathfrak{J}_{\mathfrak{p}})$ -module of finite presentation. The images of the q_h in $Q_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} k(\mathfrak{p}) = (Q \otimes_A A')_{\mathfrak{p}} \otimes_{A'} k(\mathfrak{p})$ generate this $(C_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} k(\mathfrak{p}))$ -module (since $C_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} k(\mathfrak{p}) = (C \otimes_A A')_{\mathfrak{p}} \otimes_{A'} k(\mathfrak{p})$); as we have the exact sequence

$$Q_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} k(\mathfrak{p}) \longrightarrow (L \otimes_A A')_{\mathfrak{p}} \otimes_{A'} k(\mathfrak{p}) \longrightarrow \mathrm{gr}_{\mathfrak{J}_{\mathfrak{p}}}^*(M_{\mathfrak{p}}) \otimes_{B_{\mathfrak{p}}} k(\mathfrak{p}) \longrightarrow 0,$$

we conclude that $\mathrm{gr}_{\mathfrak{J}_{\mathfrak{p}}}^*(M_{\mathfrak{p}}) \otimes_{B_{\mathfrak{p}}} k(\mathfrak{p})$ is a $(C_{\mathfrak{p}} \otimes_{B_{\mathfrak{p}}} k(\mathfrak{p}))$ -module of finite presentation. Applying (11.2.9.4) with A, B, M replaced by $B_{\mathfrak{p}}, C_{\mathfrak{p}}, \mathrm{gr}_{\mathfrak{J}_{\mathfrak{p}}}^*(M_{\mathfrak{p}})$ respectively, we conclude that $Q_{\mathfrak{p}}$ is a $C_{\mathfrak{p}}$ -module of finite type. By Nakayama's lemma (noting that $C = B$), the images of the q_h in $Q_{\mathfrak{p}}$ generate this $C_{\mathfrak{p}}$ -module. $\square$

Remark (11.2.10). — We do not know whether the generalisation of (11.2.6), (i) analogous to the theorem of Raynaud is valid.

11.3. Application to the elimination of Noetherian hypotheses

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Theorem (11.3.1). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. Then the set U of $x \in X$ such that $\mathcal{F}$ is f -flat at the point x is open in X . If moreover $\mathrm{Supp}(\mathcal{F}) = X$, $f|_U$ is an open morphism.*

Proof. The second assertion has already been proved (2.4.6) and has been included here only for reference.

The question being local on X and Y , we can suppose that X and Y are affine, and that f is a morphism of finite presentation. Let $x \in X$ be a point such that $\mathcal{F}$ is f -flat at the point x . Applying (11.2.7), one can suppose that $X = X_0 \times_{Y_0} Y$, $f = f_0 \times 1$, where Y_0 is Noetherian, $f_0 : X_0 \rightarrow Y_0$ a morphism of finite type, $\mathcal{F}_0$ a coherent $\mathcal{O}_{X_0}$ -Module; moreover, if x_0 is the projection of x in X_0 , one can suppose that $\mathcal{F}_0$ is f_0 -flat at the point x_0 . Then, by virtue of (11.1.1), the set U_0 of points of X_0 where $\mathcal{F}_0$ is f_0 -flat is an open neighbourhood of x_0 ; hence $\mathcal{F}$ is f -flat at the points of the inverse image of U_0 in X (2.1.4), and this proves that U is a neighbourhood of x . $\square$

Corollary (11.3.2). — *Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. Then the set U of $y \in Y$ such that, for every generisation y' of y , $\mathcal{F}$ is $(g \circ f)$ -flat at all the points of $f^{-1}(y')$ (i.e. such that $\mathcal{F} \otimes_Y \mathrm{Spec}(\mathcal{O}_{Y,y})$ is flat relative to the morphism $X \times_Y \mathrm{Spec}(\mathcal{O}_{Y,y}) \rightarrow Z$) is open in Y .*

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Proof. The same reasoning as in (11.1.5) shows that if V is the set of $x \in X$ where $\mathcal{F}$ is $(g \circ f)$ -flat, U is the set of $y \in Y$ whose every generisation belongs to $E = Y - f(X - V)$. As $g \circ f$ is of finite presentation, V is open in X by virtue of (11.3.1), hence ind-constructible; and $f(X - V)$ is pro-constructible in Y (1.9.5, vii), hence E is ind-constructible in Y . But it follows then from (0_I, 10.1) that U is the interior of E , hence open in Y . $\square$

Corollary (11.3.3). — *Let A be a ring, B an A -algebra of finite presentation, C a B -algebra of finite type, M a C -module of finite type. Then the set of $\mathfrak{q} \in \mathrm{Spec}(B)$ such that $M_{\mathfrak{q}}$ is an A -module flat is open in $\mathrm{Spec}(B)$.*

Proof. Taking into account (2.1.2), this is a consequence of (11.3.2) applied to $Z = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, $X = \mathrm{Spec}(C)$, $\mathcal{F} = \tilde{M}$. $\square$

Proposition (11.3.4). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, $\mathcal{I}$ a quasi-coherent ideal of finite type of $\mathcal{O}_X$, Y the closed sub-prescheme of X defined by $\mathcal{I}$, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. Suppose $\mathrm{gr}_{\mathcal{I}}(\mathcal{O}_X)$ is f -flat. Under these conditions:*

- label=(i) $\mathcal{F}$ is f -flat at the points of Y .
- lbbel=(ii) If $\mathrm{gr}_{\mathcal{J}}(\mathcal{O}_X)$ is f -flat, $\mathrm{gr}_{\mathcal{J}}(\mathcal{O}_X)$ is an $(\mathcal{O}_X/\mathcal{J})$ -Algebra of finite presentation and $\mathrm{gr}_{\mathcal{J}}(\mathcal{F})$ is an $(\mathrm{gr}_{\mathcal{J}}(\mathcal{O}_X))$ -Module of finite presentation.
- lcbel=(iii) Suppose $\mathrm{gr}_{\mathcal{J}}(\mathcal{O}_X)$ is f -flat (which implies that $\mathcal{O}_X/\mathcal{J}$ is f -flat). Then the set W of $y \in Y$ such that $(\mathrm{gr}_{\mathcal{J}}(\mathcal{F}))_y$ is an $(\mathcal{O}_X/\mathcal{J})_y$ -module flat is open in Y .
- ldbel=(iv) Suppose $\mathrm{gr}_{\mathcal{J}}(\mathcal{O}_X)$ is f -flat; let $S' \rightarrow S$ be any morphism, let $X' = X \times_S S'$, $p : X' \rightarrow X$ the canonical projection, $Y' = p^{-1}(Y)$ the closed sub-prescheme of X' , defined by $\mathcal{J}' = p^*(\mathcal{J})_{\mathcal{O}_{X'}}$, $\mathcal{F}' = p^*(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$; then, if $W' = p^{-1}(W)|_{Y'}$, one has $\mathrm{gr}_{\mathcal{J}'}(\mathcal{F}')|_{W'} \cong p^*(\mathrm{gr}_{\mathcal{J}}(\mathcal{F}))|_W$, and $\mathrm{gr}_{\mathcal{J}'}(\mathcal{F}')$ is an $(\mathcal{O}_{X'}/\mathcal{J}')$ -Module flat at the points of W' .

Proof. The questions being local on X and S , one can suppose that $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, B being an A -algebra of finite presentation, and $\mathcal{F} = \tilde{M}$, where M is a B -module of finite presentation, $\mathcal{J} = \tilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of finite type of B ; by virtue of (8.9.1) and (8.5.11), there exists a Noetherian sub-ring A_0 of A , an A_0 -algebra of finite type B_0 , an ideal $\mathfrak{J}_0$ of B_0 of finite type, a B_0 -module M_0 such that $B = B_0 \otimes_{A_0} A$, $\mathfrak{J} = \mathfrak{J}_0 B$, $M = M_0 \otimes_{A_0} A = M_0 \otimes_{B_0} B$. Moreover, A is the inductive limit of its sub- A_0 -algebras of finite type A_λ ; setting $B_\lambda = B_0 \otimes_{A_0} A_\lambda$, $M_\lambda = M_0 \otimes_{A_0} A_\lambda = M_0 \otimes_{B_0} B_\lambda$, $\mathfrak{J}_\lambda = \mathfrak{J}_0 B_\lambda$, $M = \varinjlim M_\lambda$. Since, by hypothesis, $\mathrm{gr}_{\mathfrak{J}}^*(M)$ is an A -module flat, it follows from (11.2.9) that there exists λ such that $\mathrm{gr}_{\mathfrak{J}_\lambda}^*(M_\lambda)$ is an A_λ -module flat. To prove that $\mathcal{F}$ is f -flat at the points of Y , one can therefore restrict to the case where S is Noetherian; but then (0_I, 6.1.2), applied by induction on n , proves that the $\mathcal{F}/\mathcal{J}^{n+1}\mathcal{F}$ are f -flat, and one concludes by (0_{III}, 10.2.6).

The proof of (ii) reduces in the same way to the case where S is Noetherian, as does (iii) and (iv). $\square$

Generalising the definition of (6.10.1), we say that for a prescheme X , a closed sub-prescheme Y of X defined by a quasi-coherent Ideal $\mathcal{J}$ of $\mathcal{O}_X$, and a quasi-coherent $\mathcal{O}_X$ -Module $\mathcal{F}$, $\mathcal{F}$ is *normally flat along Y at a point y* if $(\mathrm{gr}_{\mathcal{J}}^*(\mathcal{F}))_y$ is a flat $\mathcal{O}_{X,y}$ -module. We say that $\mathcal{F}$ is *normally flat along Y* if it is so at all points of Y .

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Corollary (11.3.5). — Under the general hypotheses of (11.3.4), suppose that $\mathrm{gr}_{\mathcal{J}}^*(\mathcal{O}_X)$ and $\mathrm{gr}_{\mathcal{J}}^*(\mathcal{F})$ are f -flat. Then the set W of $y \in Y$ such that $\mathcal{F}$ is normally flat along Y at the point y is open in Y , and (with the notations of (11.3.4, (iv))) $\mathcal{F}'$ is normally flat along Y' at all points of $W' = p^{-1}(W)$.

Proof. This follows from (11.3.4), (iii) and (iv) applied to the graded module $\mathrm{gr}_{\mathcal{J}}^*(\mathcal{F})$ regarded as a module over the graded $\mathcal{O}_Y$ -algebra $\mathrm{gr}_{\mathcal{J}}^*(\mathcal{O}_X)$, which is f -flat by hypothesis. $\square$

Proposition (11.3.6). — Let the notations and hypotheses be those of (8.5.1) and (8.8.1), and suppose S_α quasi-compact, X_α of finite presentation over S_α , Y_α a closed sub-prescheme of X_α defined by an Ideal $\mathcal{J}_\alpha$ of finite type of $\mathcal{O}_{X_\alpha}$ such that $\mathrm{gr}_{\mathcal{J}_\alpha}^*(\mathcal{O}_{X_\alpha})$ is flat over S_α ; finally suppose that $\mathcal{F}_\alpha$ is an $\mathcal{O}_{X_\alpha}$ -Module of finite presentation. For $\mathcal{F}$ to be normally flat along Y , it is necessary and sufficient that there exist λ such that $\mathcal{F}_\lambda$ is normally flat along Y_λ .

Proof. This follows from (11.3.4) applied to the graded modules $\mathrm{gr}_{\mathcal{J}_\lambda}^*(\mathcal{F}_\lambda)$ and $\mathrm{gr}_{\mathcal{J}}^*(\mathcal{F})$, together with the limit arguments of (8.5.1) and (8.8.1). $\square$

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Proposition (11.3.7). — Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, $\mathcal{F}, \mathcal{F}'$ two quasi-coherent $\mathcal{O}_X$ -Modules of finite presentation, $u : \mathcal{F}' \rightarrow \mathcal{F}$ an $\mathcal{O}_X$ -homomorphism, x a point of X and $y = f(x)$; suppose that $\mathcal{F}$ is f -flat at the point x . The following conditions are equivalent:

- label=a) $(\mathrm{Ker} u)_x = 0$ and $\mathrm{Coker} u$ is an $\mathcal{O}_X$ -Module f -flat at the point x .
- lbbel=b) The homomorphism $(u \otimes 1)_x : (\mathcal{F}' \otimes_{\mathcal{O}_y} k(y))_x \rightarrow (\mathcal{F} \otimes_{\mathcal{O}_y} k(y))_x$ is injective.

If moreover $\mathcal{F}'$ is f -flat, the set of points x satisfying the equivalent conditions above is open in X .

Proof. Condition a) implies b) by virtue of (0_I, 6.1.2), without hypothesis on $\mathcal{F}$. To prove the converse, one can restrict to the case where X and Y are affine, then, by virtue of (8.9.1) and (11.2.7), suppose that $X = X_0 \times_{Y_0} Y$, $f = f_0 \times 1$, $\mathcal{F}' = \mathcal{F}'_0 \otimes_{\mathcal{O}_{Y_0}} \mathcal{O}_Y$, $\mathcal{F} = \mathcal{F}_0 \otimes_{\mathcal{O}_{Y_0}} \mathcal{O}_Y$, where Y_0 is Noetherian,

$f_0 : X_0 \rightarrow Y_0$ a morphism of finite type, $\mathcal{F}_0, \mathcal{F}'_0$ two coherent $\mathcal{O}_{X_0}$ -Modules, $u_0 : \mathcal{F}'_0 \rightarrow \mathcal{F}_0$ a homomorphism; moreover, if x_0 is the projection of x in X_0 , one can suppose that $\mathcal{F}_0$ is f_0 -flat at the point x_0 . Setting $y_0 = f_0(x_0)$; by the transitivity of fibres (I, 3.6.4), the fibre morphism $f^{-1}(y) \rightarrow f_0^{-1}(y_0)$ is faithfully flat (2.2.13), and as $(u \otimes 1)_x$ is injective, this implies that $(u_0 \otimes 1)_{x_0}$ is also injective (2.2.7). This reduces, by (0_{III}, 10.2.4) applied to the Noetherian local rings $\mathcal{O}_{y_0}$ and $\mathcal{O}_{x_0}$ and to the $\mathcal{O}_{x_0}$ -modules $(\mathcal{F}'_0)_{x_0}$ and $(\mathcal{F}_0)_{x_0}$, to concluding that $(\text{Ker } u_0)_{x_0} = 0$ and that $\text{Coker } u_0$ is f_0 -flat at the point x_0 . One deduces from this that $\text{Coker } u$ is f -flat at the point x (2.1.4); applying (0_I, 6.1.2) to the exact sequence

$$0 \longrightarrow (\mathcal{F}'_0)_{x_0} \longrightarrow (\mathcal{F}_0)_{x_0} \longrightarrow (\text{Coker } u_0)_{x_0} \longrightarrow 0$$

one deduces that u_x is an injective homomorphism. Finally, it follows from (11.1.2) that the set U_0 of $z_0 \in X_0$ such that $(u_0 \otimes 1)_{z_0}$ is injective is open; by flatness, for every $z \in X$ lying over $z_0 \in U_0$, the morphism $(u \otimes 1)_z$ is injective, hence the inverse image U of U_0 in X is an open neighbourhood of x , which concludes the proof. $\square$

Theorem (11.3.8). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation, $(g_i)_{1 \leq i \leq n}$ a suite of sections of $\mathcal{O}_X$ over X ; set $\mathcal{F}_i = \mathcal{F} / \sum_{j=1}^i g_j \mathcal{F}$ for $0 \leq i \leq n$ (with $\mathcal{F}_0 = \mathcal{F}$). Let x be a point of X , $y = f(x)$, and set $X_y = f^{-1}(y) = X \times_Y \text{Spec}(k(y))$, $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$, which is an $\mathcal{O}_{X_y}$ -Module. Suppose that the $(g_i)_x$ belong to the maximal ideal $\mathfrak{m}_x$. The following conditions are equivalent:*

- label=a) *The suite $((g_i)_x)$ is $\mathcal{F}_x$ -regular and the $\mathcal{F}_i$ ($0 \leq i \leq n$) are f -flat at the point x .*
- lbbel=b) *The suite $((g_i)_x)$ is $\mathcal{F}_x$ -regular and $\mathcal{F}_n$ is f -flat at the point x .*
- lbbel=b') *There exists a neighbourhood U of x such that the suite $(g_i|_U)$ is $(\mathcal{F}|_U)$ -quasi-regular, and $\mathcal{F}_n$ is f -flat at the point x .*
- lcbel=c) *$\mathcal{F}$ is f -flat at the point x , and the suite $((g_i \otimes 1)_x)$ of elements of $\mathcal{O}_{X_y, x}$, images of the $(g_i)_x$, is $(\mathcal{F}_y)_x$ -regular.*
- ldbel=d) *$\mathcal{F}$ is f -flat at the point x , and for every morphism $Y' \rightarrow Y$ and every point $x' \in X' = X \times_Y Y'$ lying over x , if one sets $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$, the suite $((g_i \otimes 1)_{x'})$ of elements of $\mathcal{O}_{X', x'}$ is $\mathcal{F}'_{x'}$ -regular.*

Moreover the set of $x \in X$ satisfying these conditions is open in the set Z of x such that $g_i(x) = 0$ for all i .

Proof. That a) implies d) follows from (0, 15.1.15), and c) is a particular case of d); moreover a) implies b) trivially. That b) or c) implies a) follows from (11.3.7) by recurrence on n , and this shows that the set of $x \in X$ satisfying c) is open in Z . To show that b) implies a), one reduces by recurrence on n to the case $n = 1$; writing g instead of g_1 . The question being local, one can suppose $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$ affine, $\mathcal{F} = \tilde{M}$, where M is a B -module of finite presentation. By (8.9.1), write $B = B_0 \otimes_{A_0} A$, $M = M_0 \otimes_{A_0} A$ with A_0 Noetherian, B_0 an A_0 -algebra of finite type, M_0 a B_0 -module of finite type. It then suffices to show that there is a $\mu \geq \lambda$ such that at the point $x_\mu \in X_\mu$, $(g_\mu)_{x_\mu}$ is $(\mathcal{F}_\mu)_{x_\mu}$ -regular and $\mathcal{F}_\mu / g_\mu \mathcal{F}_\mu$ is flat on Y_μ at the point x_μ . The fact that b) implies a) follows from the following proposition: $\square$

Proposition (11.3.9). — *Let the notations and hypotheses be those of (8.5.1) and (8.8.1), $f_\alpha : X_\alpha \rightarrow Y_\alpha$ a morphism locally of finite presentation. Let $\mathcal{F}_\alpha, \mathcal{G}_\alpha$ be two quasi-coherent $\mathcal{O}_{X_\alpha}$ -Modules of finite presentation, $h_\alpha : \mathcal{F}_\alpha \rightarrow \mathcal{G}_\alpha$ an $\mathcal{O}_{X_\alpha}$ -homomorphism, $\mathcal{H}_\alpha$ its cokernel. Let x be a point of X , x_α its projection in X_α . For h_x to be injective and $\mathcal{H} = \text{Coker } h$ to be f -flat at the point x , it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that $(h_\lambda)_{x_\lambda}$ be injective and $\mathcal{H}_\lambda = \text{Coker } h_\lambda$ be f_λ -flat at the point x_λ . Moreover, the set of $x \in X$ having the above properties is open in X .*

Proof. The condition is necessary, as $\mathcal{H}$ is of finite presentation; the question being local, the injectivity condition follows from (0_I, 5.2.7). The sufficiency follows from (11.3.7), as does the openness assertion. $\square$

Lemma (11.3.9.1). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, $\mathcal{G}, \mathcal{H}$ two quasi-coherent $\mathcal{O}_X$ -Modules of finite presentation such that $\mathcal{H}$ is f -flat, $p : \mathcal{G} \rightarrow \mathcal{H}$ a surjective homomorphism. Then $\text{Ker}(p)$ is an $\mathcal{O}_X$ -Module of finite presentation.*

Proof. One can suppose X, Y affine, and that there exists a morphism $Y \rightarrow Y_0$ where Y_0 is Noetherian, a morphism of finite type $f_0 : X_0 \rightarrow Y_0$, two coherent $\mathcal{O}_{X_0}$ -Modules $\mathcal{G}_0, \mathcal{H}_0$ and a surjective homomorphism $p_0 : \mathcal{G}_0 \rightarrow \mathcal{H}_0$ such that $\mathcal{G}, \mathcal{H}$ and p are deduced by base change. Moreover, one can suppose $\mathcal{H}_0$ is f_0 -flat (11.2.7). Then, if $\mathcal{K}_0 = \text{Ker}(p_0)$, $\mathcal{K}_0$ is a coherent $\mathcal{O}_{X_0}$ -Module (0_I, 5.3.4), and by (0_I, 6.1.2), $\mathcal{K} = \text{Ker}(\mathcal{G} \rightarrow \mathcal{H})$ is deduced from $\mathcal{K}_0$ by base change, hence is of finite presentation. $\square$

Theorem (11.3.10). — (Flatness criterion by fibres). *Let S be a prescheme, $g : X \rightarrow S, h : Y \rightarrow S$ two morphisms, $f : X \rightarrow Y$ an S -morphism, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module, x a point of $X, y = f(x), s = h(y) = g(x)$. Suppose satisfied one of the two following hypotheses:*

- 1° S, X and Y are locally Noetherian and $\mathcal{F}$ is coherent.
- 2° g and h are locally of finite presentation and $\mathcal{F}$ is of finite presentation.

Then, with the notations of (9.4.1), if $\mathcal{F}_x \neq 0$, the following conditions are equivalent:

label=a) $\mathcal{F}$ is g -flat at the point x and $\mathcal{F}_s$ is f_s -flat at the point x .

lbbel=b) h is flat at the point y and $\mathcal{F}$ is f -flat at the point x .

Moreover, under hypothesis 2°, the set of $x \in X$ satisfying condition b) is open in X .

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Proof. The last assertion follows from (11.3.1) applied to $\mathcal{O}_Y$ and the morphism h , and to $\mathcal{F}$ and the morphism f . As $g = h \circ f$, b) implies a) without hypotheses 1° or 2° (2.1.6) and (2.1.4). To prove a) implies b), one can restrict to the case where S, X and Y are affine; under hypothesis 2°, by (11.2.7), one reduces to the case where S is Noetherian, i.e. hypothesis 1° is satisfied. The assertion then amounts to the following lemma, which improves (0_{III}, 10.2.5): $\square$

Lemma (11.3.10.1). — *Let $\rho : A \rightarrow B, \sigma : B \rightarrow C$ be homomorphisms of Noetherian local rings, k the residue field of A, M a C -module of finite type $\neq 0$. The following conditions are equivalent:*

label=a) M is an A -module flat and $M \otimes_A k$ is a $(B \otimes_A k)$ -module flat.

lbbel=b) B is an A -module flat and M is a B -module flat.

Proof. We first establish the more general: $\square$

Lemma (11.3.10.2). — *Let A be a commutative ring, B an A -algebra, $\mathfrak{J}$ an ideal of A, M a B -module. Consider the following conditions:*

label=(i) $\mathfrak{J}$ is nilpotent.

lbbel=(ii) B is Noetherian and M is ideally separated for the $\mathfrak{J}B$ -preadic topology (0_{III}, 10.2.1).

lcbel=(iii) $\mathfrak{J}B$ is contained in the radical of B .

Consider on the other hand the four properties:

label=a) M is a B -module flat.

lbbel=b) B is an A -module flat.

lcbel=c) M is an A -module flat and $M/\mathfrak{J}M$ is a $(B/\mathfrak{J}B)$ -module flat.

ldbel=d) $B/\mathfrak{J}B$ is an $(A/\mathfrak{J})$ -module flat and for every maximal ideal $\mathfrak{m} \supset \mathfrak{J}B$ of B one has $\mathfrak{m}M \neq M$.

Then:

1° If (i) or (ii) is satisfied, the conjunction of a) and b) implies c), and c) implies a).

2° If (i), or the conjunction of (ii) and (iii), is satisfied, the conjunction of c) and d) implies the conjunction of a) and b).

Proof. 1° The first assertion is immediate (0_I, 6.2.1). Suppose c) satisfied. Consider the graded rings $\text{gr}_{\mathfrak{J}}^*(A), \text{gr}_{\mathfrak{J}}^*(B)$ and the graded module $\text{gr}_{\mathfrak{J}}^*(M)$ relative to the $\mathfrak{J}$ -preadic filtrations, along with the surjective canonical maps (0_{III}, 10.1.2)

$$u : \text{gr}^0(B) \otimes_{\text{gr}^0(A)} \text{gr}^*(A) \longrightarrow \text{gr}^*(B)$$

$$v : \text{gr}^0(M) \otimes_{\text{gr}^0(A)} \text{gr}^*(A) \longrightarrow \text{gr}^*(M)$$

$$w : \text{gr}^0(M) \otimes_{\text{gr}^0(B)} \text{gr}^*(B) \longrightarrow \text{gr}^*(M).$$

Hypothesis c) implies that v is bijective (0_{III}, 10.2.1); as the two other maps are surjective, they are also bijective. But as $M/\mathfrak{J}M$ is a $(B/\mathfrak{J}B)$ -module flat by c), it follows from (0_{III}, 10.2.1) that M is a B -module flat.

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2° Either of (i) or (iii) implies that every maximal ideal of B contains $\mathfrak{J}B$. It follows from 1° and the conjunction of c) and d) that M is a B -module faithfully flat, and hence $\mathrm{gr}^0(M)$ is a $\mathrm{gr}^0(B)$ -module faithfully flat ($\mathbf{0_I}$, 6.2.1). Since c) implies that v is bijective, u is also bijective ($\mathbf{0_I}$, 6.4.1). Conditions (ii) and (iii) imply that B is ideally separated for the $\mathfrak{J}$ -preadic filtration (Bourbaki, *Alg. comm.*, chap. III, §5, n° 4, prop. 2); one deduces from ($\mathbf{0_{III}}$, 10.2.1) that B is an A -module flat. This also finishes the proof of (11.3.10). $\square$

Corollary (11.3.11). — *Let $g : X \rightarrow S$, $h : Y \rightarrow S$ be two morphisms locally of finite presentation, $f : X \rightarrow Y$ an S -morphism. The following conditions are equivalent:*

label=a) g is flat and $f_s : X_s \rightarrow Y_s$ is flat for every $s \in S$.

lbbel=b) h is flat at all points of $f(X)$ and f is flat.

Proof. It suffices to apply (11.3.10) with $\mathcal{F} = \mathcal{O}_X$. $\square$

Remarks (11.3.12). — It would be interesting to give proofs of (11.3.2) and (11.3.10) not using passage to the inductive limit; it would suffice for this to prove the following criterion: Let A be a ring, B an A -algebra of finite presentation, M a B -module of finite presentation, $\mathfrak{J}$ an ideal of A , $\mathfrak{p}$ a prime ideal of B containing $\mathfrak{J}B$. Then the following conditions are equivalent:

label=a) $M_{\mathfrak{p}}$ is an A -module flat and $\mathrm{Tor}_1^A(A/\mathfrak{J}, M_{\mathfrak{p}}) = 0$.

lbbel=b) $M_{\mathfrak{p}}/\mathfrak{J}M_{\mathfrak{p}}$ is an $(A/\mathfrak{J})$ -module flat and $\mathrm{Tor}_1^A(A/\mathfrak{J}, M_{\mathfrak{p}}) = 0$.

When A is Noetherian, this is a consequence of ($\mathbf{0_{III}}$, 10.2.2) applied to the Noetherian A -algebra $B_{\mathfrak{p}}$.

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Proposition (11.3.13). — *Let $f : X \rightarrow Y$ be a morphism of preschemes, x a point of X such that f is flat at the point x ; set $y = f(x)$.*

label=(i) *If X is reduced (resp. normal) at the point x , then Y is reduced (resp. normal) at the point y .*

lbbel=(ii) *Suppose moreover that f is of finite presentation at the point x . Then, if Y is reduced (resp. normal) at the point y and if the morphism f is reduced (resp. normal) at the point x (6.8.1), X is reduced (resp. normal) at the point x .*

Proof. Assertion (i) has been proved in (2.1.13). To prove (ii), one can restrict to the case $X = \mathrm{Spec}(B)$, $Y = \mathrm{Spec}(A)$, with B an A -algebra of finite presentation. One writes $B = B_0 \otimes_{A_0} A$ (8.9.1), where A_0 is a sub- $\mathbb{Z}$ -algebra of finite type of A , B_0 an A_0 -algebra of finite type. Let (C_{α}) be the family of sub- A_0 -algebras of finite type of A ; one has $A = \varinjlim C_{\alpha}$.

1° Suppose A reduced and f reduced at x . Let $\mathfrak{p}_{\alpha} = \mathfrak{m} \cap C_{\alpha}$, $A'_{\alpha} = (C_{\alpha})_{\mathfrak{p}_{\alpha}}$, so $A = \varinjlim A'_{\alpha}$ (5.13.3). Since f is reduced at x , there exists α_0 such that f_{α} is reduced at x_{α} for $\alpha \geq \alpha_0$ (6.7.8) and (11.2.6); as Y_{α} and X_{α} are Noetherian and A'_{α} reduced, X_{α} is reduced at x_{α} by (3.3.5). Since $\mathcal{O}_{X,x} = \varinjlim \mathcal{O}_{X_{\alpha},x_{\alpha}}$ (5.13.3), $\mathcal{O}_{X,x}$ is reduced (5.13.2).

2° Suppose A integrally closed and f normal at x . As C_{α} is universally Japanese (7.7.4), its integral closure C'_{α} is a finite C_{α} -algebra contained in A . Let $A''_{\alpha} = (C'_{\alpha})_{\mathfrak{p}'_{\alpha}}$ where $\mathfrak{p}'_{\alpha} = \mathfrak{m} \cap C'_{\alpha}$; then A''_{α} is Noetherian local integral and integrally closed, $A = \varinjlim A''_{\alpha}$. For $\alpha \geq \alpha_0$, f'_{α} is normal at x'_{α} (6.7.8) and (11.2.6), so X'_{α} is normal at x'_{α} by (6.5.4). One concludes that $\mathcal{O}_{X,x}$ is integrally closed (5.13.4). $\square$

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Corollary (11.3.14). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, x a point of X , $y = f(x)$. If Y is geometrically unibranch at the point y (6.15.1) and if the morphism f is geometrically normal at the point x (6.8.1), then X is geometrically unibranch at the point x .*

Proof. One can restrict to $Y = \mathrm{Spec}(A)$, A an integral local integrally closed ring. Let A' be the integral closure of A , $Y' = \mathrm{Spec}(A')$, y' the closed point of Y' , so that $g : Y' \rightarrow Y$ is radiciel at y (6.15.3), entire and birational. Setting $X' = X \times_Y Y'$ with projections $f' : X' \rightarrow Y'$, $g' : X' \rightarrow X$, then g' is entire and birational. Since f is of finite presentation and normal at the point x' (the unique point of $g'^{-1}(x)$), f' is of finite presentation and normal at x' (6.7.8). Hence $\mathcal{O}_{X',x'}$ is integrally closed (11.3.13); since $\mathcal{O}_{X,x}$ is reduced (11.3.13) and g' is birational (6.15.4.1), by (6.15.5) X is integral and geometrically unibranch at x . $\square$

Proposition (11.3.15). — *Let A be a ring, B an A -algebra of finite presentation, M a B -module of finite presentation, which is flat as an A -module. Then there exists a finite sequence $(f_i)_{1 \leq i \leq n}$ of elements of A , such that the ideal generated by the f_i is equal to A , and that, for $0 \leq i < n$, $M_{f_{i+1}} / (\sum_{j=1}^i f_j M_{f_{i+1}})$ is an $(A_{f_{i+1}} / (\sum_{j=1}^i f_j A_{f_{i+1}}))$ -module free.*

Proof. One can also say that the $D(f_i)$ form a covering of $X = \operatorname{Spec}(A)$, and that, setting $Z_1 = D(f_1)$ and by induction $Z_{i+1} = D(f_{i+1}) \cap V(f_1 A + \cdots + f_i A)$, the Z_i form a partition of X into locally closed sets, $A_{f_{i+1}} / (\sum_{j=1}^i f_j A_{f_{i+1}})$ being the ring of an affine sub-scheme of X having Z_{i+1} as underlying space.

To prove the proposition, one can first, by virtue of (11.2.7), suppose that there exist a Noetherian ring A_0 , an A_0 -algebra of finite type B_0 , a B_0 -module of finite type M_0 , flat over A_0 , such that $B = B_0 \otimes_{A_0} A$, $M = M_0 \otimes_{A_0} A$; it is clear that it suffices to prove the proposition for A_0, B_0, M_0 , for if the $f_i \in A_0$ satisfy the conditions of the statement, they will satisfy them also for A, B, M . One can therefore restrict to the case where A is Noetherian.

Note now that if C is a Noetherian ring, $\mathfrak{R}$ its nilradical and P a flat C -module, then for P to be a C -module free, it suffices that $P \otimes_C (C/\mathfrak{R})$ be a $(C/\mathfrak{R})$ -module free. Note also that if C is a Noetherian reduced ring, there exists $g \in C$ such that C_g is integral. Using now (6.9.2): one can define by induction a sequence $(f_i)_{i \geq 1}$ of elements of A :

- 1° f_1 is such that $A_1 = (A_{\text{red}})_{f_1}$ is integral and $M \otimes_A A_1$ is an A_1 -module free;
- 2° if the ideal $\mathfrak{J}_i$ generated by $f_1, \dots, f_i$ is A , $f_{i+1} = f_i$;
- 3° otherwise $\mathfrak{J}_i \neq A$, f_{i+1} is an element not belonging to $\mathfrak{J}_i$ such that $A_{i+1} = ((A/\mathfrak{J}_i)_{\text{red}})_{f_{i+1}}$ is integral and $M \otimes_A A_{i+1}$ is an A_{i+1} -module free.

As A is Noetherian, $(\mathfrak{J}_i)$ is stationary, hence there exists n such that $f_1, \dots, f_n$ generate the ideal A , and the f_i answer the question since $((A/\mathfrak{J}_i)_{\text{red}})_{f_{i+1}} = ((A/\mathfrak{J}_i)_{f_{i+1}})_{\text{red}}$. $\square$

Proposition (11.3.16). — *Let $f : X \rightarrow Y$ be a faithfully flat morphism of finite presentation, $g : Y \rightarrow Z$ a morphism of finite type such that $g \circ f : X \rightarrow Z$ is a morphism of finite type (resp. of finite presentation). Then g is a morphism of finite type (resp. of finite presentation).*

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Proof. As f is surjective and $g \circ f$ quasi-compact, g is quasi-compact (1.1.3). Let us show that g is quasi-separated: for an affine open $W \subset Z$, and a finite affine covering (V_i) of $g^{-1}(W)$, the $V_i \cap V_j$ are quasi-compact since f is of finite presentation and surjective (1.2.6) and (1.2.7). The question being local on Y and Z , one can suppose $Y = \operatorname{Spec}(B)$, $Z = \operatorname{Spec}(A)$ affine. If X' is the prescheme sum of the affine pieces of X , $f \circ p$ is faithfully flat of finite presentation and $g \circ f \circ p$ is of finite type (resp. of finite presentation), so one can suppose $X = \operatorname{Spec}(C)$ affine. The first assertion (g of finite type) then follows by faithful flatness and the fact that the application $B_\lambda \rightarrow B$ is injective. For the second (finite presentation), suppose $g \circ f$ is of finite presentation; then C is an A -algebra of finite presentation. Let (B_λ) be the filtered family of sub- A -algebras of finite type of B ; by (8.8.2), there exists α such that $C = C_\alpha \otimes_{B_\alpha} B$ with C_α a B_α -algebra of finite presentation faithfully flat (8.10.5) and (11.2.6). By faithful flatness $B = B_\lambda$ for some λ , hence B is an A -algebra of finite presentation. $\square$

Corollary (11.3.17). — *Let A be a ring, B an A -algebra, C a B -algebra of finite presentation which is a faithfully flat B -module. Then, if C is an A -algebra of finite type (resp. of finite presentation), B is an A -algebra of finite type (resp. of finite presentation).*

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Proof. This is the affine translation of (11.3.16). $\square$

11.4. Descent of flatness by arbitrary morphisms: case of an Artinian base prescheme

Theorem (11.4.1). — *Let A be a ring, $\mathfrak{J}$ a nilpotent ideal of A , $u_\lambda : A \rightarrow B_\lambda$ ($\lambda \in L$) a family of ring homomorphisms such that the intersection of the kernels of the u_λ is reduced to 0. Let M be an A -module such that for every $\lambda \in L$, $M \otimes_A B_\lambda$ is a B_λ -module free and that $M \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$ -module free. Then M is an A -module free. If moreover the index set L is finite, one can replace “free” by “flat” in the preceding statement.*

Proof. In both cases it suffices to prove that M is an A -module flat: when $M \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$ -module free, it follows that M is an A -module free by (0III, 10.1.2). We use the following lemma, which generalises (0III, 10.2.1): $\square$

Lemma (11.4.1.1). — *Let A be a ring equipped with a finite filtration $(\mathfrak{J}_n)_{0 \leq n \leq N+1}$ with $\mathfrak{J}_0 = A$, $\mathfrak{J}_{N+1} = 0$. Let M be an A -module equipped with the filtration $(\mathfrak{J}_n M)_{0 \leq n \leq N+1}$, and denote by $\text{gr}(A)$ and $\text{gr}(M)$ the corresponding graded ring and graded module. Suppose that $M \otimes_A (A/\mathfrak{J}_1)$ is an $(A/\mathfrak{J}_1)$ -module flat and that the canonical homomorphism*

$$(11.4.1.2) \quad \text{gr}^0(M) \otimes_{\text{gr}^0(A)} \text{gr}(A) \longrightarrow \text{gr}(M)$$

is injective. Then M is an A -module flat.

Proof. The conditions imply, by recurrence, that $M \otimes_A (A/\mathfrak{J}_N)$ is an $(A/\mathfrak{J}_N)$ -module flat. The canonical homomorphism $(M/\mathfrak{J}_N M) \otimes \mathfrak{J}_N \rightarrow \mathfrak{J}_N M$ is injective. Applying (0III, 10.2.1) to the $\mathfrak{J}_N$ -preadic filtration, M is an A -module flat. IV-3 | 144

To apply the lemma to (11.4.1), we denote by $\mathfrak{J}_n$ the intersection of the inverse images $u_\lambda^{-1}(\mathfrak{J}^n B_\lambda)$: one has $\mathfrak{J}_0 = A$, $\mathfrak{J}_m \mathfrak{J}_n \subset \mathfrak{J}_{m+n}$ for $m \geq 0$, $n \geq 0$; moreover, since $\mathfrak{J}^{N+1} = 0$, one also has $\mathfrak{J}_{N+1} = 0$ since the intersection of the kernels of the u_λ is reduced to 0. Equipping A with the filtration $(\mathfrak{J}_n)$, M with the filtration $(\mathfrak{J}_n M)$, and B_λ with the $\mathfrak{J}$ -preadic filtration; for each λ , one considers the commutative diagram

$$\begin{array}{ccc} \text{gr}^0(M) \otimes_{\text{gr}^0(A)} \text{gr}(A) & \xrightarrow{\varphi} & \text{gr}(M) \\ \downarrow f_\lambda & & \downarrow \theta_\lambda \\ \text{gr}_{\mathfrak{J}}^0(N_\lambda) \otimes_{\text{gr}_{\mathfrak{J}}^0(B_\lambda)} \text{gr}_{\mathfrak{J}}^*(B_\lambda) & \xrightarrow{\varphi_\lambda} & \text{gr}_{\mathfrak{J}}^*(N_\lambda) \end{array}$$

where $N_\lambda = M \otimes_A B_\lambda$. The hypothesis that N_λ is a B_λ -module flat implies that φ_λ is injective (0III, 10.2.1), hence $\text{Ker}(\varphi) \subset \text{Ker}(f_\lambda)$. Setting $A_0 = A/\mathfrak{J}_1$, $M_0 = M \otimes_A A_0$, one notes that

$$\text{gr}_{\mathfrak{J}}^0(N_\lambda) = N_\lambda / \mathfrak{J} N_\lambda = (M/\mathfrak{J} M) \otimes_A B_\lambda = (M/\mathfrak{J}_1 M) \otimes_A (B_\lambda / \mathfrak{J} B_\lambda)$$

and hence $\text{gr}_{\mathfrak{J}}^0(N_\lambda) \otimes_{\text{gr}_{\mathfrak{J}}^0(B_\lambda)} \text{gr}_{\mathfrak{J}}^*(B_\lambda) = M_0 \otimes_{A_0} \text{gr}_{\mathfrak{J}}^*(B_\lambda)$. The homomorphism f_λ can thus be written $1 \otimes \text{gr}(u_\lambda) : M_0 \otimes_{A_0} \text{gr}(A) \rightarrow M_0 \otimes_{A_0} \text{gr}_{\mathfrak{J}}^*(B_\lambda)$, where R_λ is the kernel of $\text{gr}(u_\lambda) : \text{gr}(A) \rightarrow \text{gr}_{\mathfrak{J}}^*(B_\lambda)$. As M_0 is by hypothesis a flat A_0 -module, the kernel of f_λ equals $M_0 \otimes_{A_0} R_\lambda$, whence the proposition in this case. When L is finite, the formula remains true under the sole hypothesis that M_0 is a flat A_0 -module (0I, 6.1.3), which terminates the proof. $\square$

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Remarks (11.4.2). — The conclusion of (11.4.1) can be inexact if L is infinite and one only supposes $M \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$ -module flat. For example, let V be a discrete valuation ring, K its field of fractions, $A = V[T]/(T^2)$; take $\mathfrak{J}$ the image of (T) in A , so $A/\mathfrak{J} = V$, and $M = K$; then M is an $(A/\mathfrak{J})$ -module flat but not an A -module flat, since $\mathfrak{J}$ is nilpotent and the intersection of the maximal ideals of A is the maximal ideal $\mathfrak{m}$ of the local Noetherian ring A ; one has $\mathfrak{m}K = K$, hence $M \otimes_A (A/\mathfrak{m}^n) = 0$ for all n , and yet M is not A -flat.

Corollary (11.4.3). — *Let A be a semi-local ring whose radical $\mathfrak{J}$ is nilpotent (for example an Artinian ring), $u_\lambda : A \rightarrow B_\lambda$ ($\lambda \in L$) a family of ring homomorphisms such that the intersection of the kernels of the u_λ is reduced to 0. For an A -module M to be flat, it is necessary and sufficient that for every $\lambda \in L$, $M \otimes_A B_\lambda$ be a B_λ -module flat.*

Proof. As $A/\mathfrak{J}$ is a direct product of a finite number of fields (Bourbaki, *Alg. comm.*, chap. II, §3, n° 5, prop. 16), A is a direct product of local rings A_i with nilpotent radical. One reduces to the case where A is local; then $A/\mathfrak{J}$ is a field, so $M \otimes_A (A/\mathfrak{J})$ is free, and the result follows from (11.4.1). $\square$

Corollary (11.4.4). — *Let A be a ring, M an A -module; suppose that there exists a nilpotent ideal $\mathfrak{J}$ of A such that $M \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$ -module free. Then the set of ideals $\mathfrak{R}$ of A such that $M \otimes_A (A/\mathfrak{R})$ is an $(A/\mathfrak{R})$ -module free admits a smallest element $\mathfrak{R}_0$ (which is also the smallest of the ideals $\mathfrak{R}$ such that $M \otimes_A (A/\mathfrak{R})$ is an $(A/\mathfrak{R})$ -module flat).*

Proof. The intersection $\mathfrak{R}_0$ of the ideals $\mathfrak{R}$ for which $M \otimes_A (A/\mathfrak{R})$ is $(A/\mathfrak{R})$ -free is the smallest such ideal, by (11.4.1) applied to $A/\mathfrak{R}_0$, its nilpotent ideal $\mathfrak{J}/\mathfrak{R}_0$, and the homomorphisms $A/\mathfrak{R}_0 \rightarrow A/\mathfrak{R}$. The assertion about “flat” follows from (0_{III}, 10.1.2). $\square$

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Proposition (11.4.5). — *Let Y be an irreducible prescheme, $f : X \rightarrow Y$ a morphism of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module of finite presentation. There exist then a non-empty open U in Y and a closed sub-scheme Z of U , finite over U , having the following property: for every base change $U' \rightarrow U$, if one sets $X' = X \times_Y U'$, $f' = f|_{U'}$ and $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{U'}$, then, for $\mathcal{F}'$ to be f' -flat, it is necessary and sufficient that the morphism $U' \rightarrow U$ factor through $U' \rightarrow Z \rightarrow U$. Such a Z has the same underlying space as U . Moreover, if Y is affine and (W_i) a finite affine covering of X , for $U' = \text{Spec}(A')$ affine with $U' \rightarrow U$ factoring through Z , each $\Gamma(W_i \times_Y U', \mathcal{F}')$ is an A' -module free.*

Proof. One reduces via (8.9.1) to the case where Y is Noetherian. Let $\mathfrak{R}$ be its nilradical; by (6.9.2), replacing Y by a non-empty open, one can suppose the M_{i0} are A_0 -modules free. By (11.4.4), there is an ideal $\mathfrak{J}_i \subset \mathfrak{R}$ such that $M_i \otimes_A A'$ is A' -free if and only if $\mathfrak{J}_i A' = 0$. Taking $\mathfrak{J} = \sum_i \mathfrak{J}_i$ and $Z = V(\mathfrak{J})$ gives the result. $\square$

Corollary (11.4.6). — *Let A be a ring such that $\text{Spec}(A)$ is irreducible, $\mathfrak{p}$ its unique minimal prime, B an A -algebra of finite presentation, M a B -module of finite presentation. Suppose $M_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$ -module flat; then there exists $t \in A - \mathfrak{p}$ such that M_t is an A_t -module free.*

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Proof. By (11.4.5), there is an ideal of finite type $\mathfrak{J}$ in A such that $M \otimes_A A'$ is A' -free iff $\mathfrak{J}A' = 0$. The hypothesis $M_{\mathfrak{p}}$ flat gives $\mathfrak{J}_{\mathfrak{p}} = 0$; as $\mathfrak{J}$ is of finite type, $\mathfrak{J}_t = 0$ for some $t \in A - \mathfrak{p}$. $\square$

Proposition (11.4.7). — *Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A , M an A -module ideally separated (0_{III}, 10.2.1) for the $\mathfrak{J}$ -preadic topology. Let $(\mathfrak{p}_i)_{1 \leq i \leq r}$ be a finite family of prime ideals of A containing $\mathfrak{J}$; for every $n \geq 0$, let $\mathfrak{p}_i^{(n)}$ be the n -th symbolic power; set $\mathfrak{J}_n = \bigcap_{i=1}^r \mathfrak{p}_i^{(n)}$, so $\mathfrak{J}_n \supset \mathfrak{J}^n$. Suppose the topology defined by $(\mathfrak{J}_n)$ is the $\mathfrak{J}$ -preadic topology. For M to be A -flat, it is necessary and sufficient that $M \otimes_A (A/\mathfrak{J})$ be $(A/\mathfrak{J})$ -flat and that $M_{\mathfrak{p}_i}$ be $A_{\mathfrak{p}_i}$ -flat for every i .*

Proof. As M is ideally separated, by (0_{III}, 10.2.1) it suffices to show that $M \otimes_A (A/\mathfrak{J}^n)$ is $(A/\mathfrak{J}^n)$ -flat for every n . In $A/\mathfrak{J}_n$, $\mathfrak{J}_1/\mathfrak{J}_n$ is nilpotent and the intersection of the kernels of $A/\mathfrak{J}_n \rightarrow A_{\mathfrak{p}_i}/\mathfrak{p}_i^n A_{\mathfrak{p}_i}$ is 0. By (11.4.1), it suffices that $M \otimes_A (A_{\mathfrak{p}_i}/\mathfrak{p}_i^n A_{\mathfrak{p}_i})$ be flat, which follows from $M_{\mathfrak{p}_i}$ being $A_{\mathfrak{p}_i}$ -flat. $\square$

Remarks (11.4.8). — The hypothesis in (11.4.7) on the topology defined by $(\mathfrak{J}_n)$ is satisfied if, for n sufficiently large, $\text{Ass}(A/\mathfrak{J}^n)$ is contained in $\{\mathfrak{p}_1, \dots, \mathfrak{p}_r\}$.

Corollary (11.4.9). — *Let A be a Noetherian ring, $\mathfrak{J}$ a nilpotent ideal of A , M an A -module. For M to be A -flat (resp. free), it is necessary and sufficient that $M \otimes_A (A/\mathfrak{J})$ be $(A/\mathfrak{J})$ -flat (resp. free) and that $M_{\mathfrak{p}}$ be $A_{\mathfrak{p}}$ -flat for every $\mathfrak{p} \in \text{Ass}(A/\mathfrak{J})$.*

Proof. The assertion relative to the case where $M \otimes_A (A/\mathfrak{J})$ is free reduces to the assertion relative to the case where it is flat by (0_{III}, 10.1.2). For the flat case, since $\mathfrak{J}$ is nilpotent, $\mathfrak{J}^n = 0$ for n sufficiently large; the result then follows from (11.4.7) applied to the decreasing sequence $\mathfrak{J}_n = \mathfrak{J}^n$. $\square$

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Corollary (11.4.10). — *Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A , M an A -module. Suppose M is ideally separated for the $\mathfrak{J}$ -preadic topology and $\text{gr}_{\mathfrak{J}}(A)$ is $(A/\mathfrak{J})$ -flat. For M to be A -flat, it is necessary and sufficient that $M \otimes_A (A/\mathfrak{J})$ be $(A/\mathfrak{J})$ -flat and that $M_{\mathfrak{p}}$ be $A_{\mathfrak{p}}$ -flat for every $\mathfrak{p} \in \text{Ass}(A/\mathfrak{J})$.*

Proof. By (11.4.8), it reduces to showing that $\text{Ass}(A/\mathfrak{J}^n)$ is contained in $\text{Ass}(A/\mathfrak{J})$ for every n . Now, if $a \in A$ does not belong to any $\mathfrak{p} \in \text{Ass}(A/\mathfrak{J})$, the multiplication by a is injective in $A/\mathfrak{J}$; since each $\mathfrak{J}^k/\mathfrak{J}^{k+1}$ is an $(A/\mathfrak{J})$ -module, the element $(\mathfrak{J}^k/\mathfrak{J}^{k+1})$ -regular, hence a is $(A/\mathfrak{J}^n)$ -regular for every n (Bourbaki, *Alg. comm.*, chap. III, § 2, no. 8, cor. 1 of th. 1), and therefore does not belong to any associated prime of $A/\mathfrak{J}^n$ (Bourbaki, *Alg. comm.*, chap. II, § 1, no. 1, prop. 2). $\square$

Proposition (11.4.11). — *Let A be an Artinian local ring, $\mathfrak{m}$ its maximal ideal, $Y = \text{Spec}(A)$, y its unique point, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Let (A'_α) be a family of local rings, $u_\alpha : A \rightarrow A'_\alpha$ local homomorphisms. Set $Y'_\alpha = \text{Spec}(A'_\alpha)$, $X'_\alpha = X \times_Y Y'_\alpha$, $\mathcal{F}'_\alpha = \mathcal{F} \otimes_A A'_\alpha$. Let x be a point of X , and suppose:*

- label=(i) *The intersection of the kernels of the u_α is reduced to 0.*
- lbbel=(ii) *The residue field extension $k(x)/k$ is primary (4.3.1) (automatically satisfied if k is separably closed).*
- lcbel=(iii) *For every α , there exists $x'_\alpha \in X'_\alpha$ whose projections in X and Y'_α are x and y'_α respectively, such that $\mathcal{F}'_\alpha$ is Y'_α -flat at x'_α .*

Then $\mathcal{F}$ is f -flat at x .

Proof. One reduces in three steps. I) Reduction to A'_α a local ring of a Y -prescheme of finite type with residue field a finite extension of k : using (11.2.6) and the fact that A is Artinian/Jacobson (10.4.7). II) Reduction to finitely many finite A -algebras: as A is Artinian, a finite number of the $u_\alpha^{-1}(\mathfrak{m}_\alpha^{n_i})$ have intersection 0; and $A'_\alpha/\mathfrak{m}_\alpha^{n_i}$ is finite over A . III) With the A'_i finite, take the prescheme sum $Y' = \coprod Y'_i$, $X' = X \times_Y Y'$; $\mathcal{F}'$ is Y' -flat near the inverse image of x (11.1.1); as $Y' \rightarrow Y$ is finite hence closed, one finds an open U of x in X where $\mathcal{F}$ is f -flat by (11.4.3). $\square$

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Proposition (11.4.12). — *Let A be an Artinian local ring of residue field k , $Y = \text{Spec}(A)$, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Let (A'_α) be a family of local rings, $u_\alpha : A \rightarrow A'_\alpha$ homomorphisms. Set $Y'_\alpha = \text{Spec}(A'_\alpha)$, $X'_\alpha = X \times_Y Y'_\alpha$, $\mathcal{F}'_\alpha = \mathcal{F} \otimes_A A'_\alpha$. Let $x \in X$, and suppose:*

- label=(i) *The intersection of the kernels of the u_α is reduced to 0.*
- lbbel=(ii) *For every α , $\mathcal{F}'_\alpha$ is f'_α -flat at all $x'_\alpha \in X'_\alpha$ whose projection in X is x and in Y'_α is y'_α , and $k(x)$ is a primary extension of $k(y)$.*

Then $\mathcal{F}$ is f -flat at x .

Proof. One reduces to the case of finitely many A'_α (as A is Artinian). Passing to the algebraic closure k' of k via a flat local homomorphism $A \rightarrow B$ with $B \otimes_A k \cong k'$ (0III, 10.3.1), the hypotheses of (11.4.11) are satisfied for B , hence the result by (11.4.3). $\square$

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Corollary (11.4.13). — *Let A be an Artinian local ring, $Y = \text{Spec}(A)$, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module. Let (Y'_α) be a family of A -preschemes, $X'_\alpha = X \times_Y Y'_\alpha$, $\mathcal{F}'_\alpha = \mathcal{F} \otimes_Y \mathcal{O}_{Y'_\alpha}$. Let $x \in X$ and suppose:*

- label=(i) *The intersection of the kernels of the structural homomorphisms $A \rightarrow \Gamma(Y'_\alpha, \mathcal{O}_{Y'_\alpha})$ is reduced to 0.*
- lbbel=(ii) *For every α , $\mathcal{F}'_\alpha$ is f'_α -flat at all $x'_\alpha \in X'_\alpha$ whose projection in X is x and in Y'_α is a closed point.*

Then $\mathcal{F}$ is f -flat at x .

Proof. Replacing the family (Y'_α) by the local schemes at the closed points reduces to (11.4.12). $\square$

11.5. Descent of flatness by arbitrary morphisms: general case

Theorem (11.5.1). — Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, $x \in X$, $y = f(x)$. Let (Y'_α) be a family of Y -preschemes, $Y'_\alpha = \text{Spec}(A'_\alpha)$ local, with closed points y'_α mapping to y ; let $\mathfrak{m}'_\alpha$ be the maximal ideal of A'_α and $u_\alpha : \mathcal{O}_y \rightarrow A'_\alpha$ the canonical homomorphism. Suppose the finite intersections of ideals $u_\alpha^{-1}(\mathfrak{m}'_\alpha)$ form a fundamental system of neighbourhoods of 0 in $\mathcal{O}_y$. Set $X'_\alpha = X \times_Y Y'_\alpha$, $\mathcal{F}'_\alpha = \mathcal{F} \otimes_Y \mathcal{O}_{Y'_\alpha}$. Suppose one of:

- label=(i) For every α , $\mathcal{F}'_\alpha$ is f'_α -flat at all points with projection x in X and y'_α in Y'_α .
 lbbel=(ii) For every α , there exists $x'_\alpha \in X'_\alpha$ projecting to x and y'_α , with $\mathcal{F}'_\alpha$ flat at x'_α , and $k(x)$ is a primary extension of $k(y)$.

Then $\mathcal{F}$ is f -flat at x .

Proof. By (0III, 10.2.2), it suffices to show $\mathcal{F}_x / \mathfrak{m}_y^n \mathcal{F}_x$ is $(\mathcal{O}_y / \mathfrak{m}_y^n)$ -flat for every $n > 0$. By hypothesis, **IV-3** | 151 there exists λ with $\mathfrak{J}_\lambda \subset \mathfrak{m}_y^n$; passing to quotients gives homomorphisms $u'_\alpha : \mathcal{O}_y / \mathfrak{J}_\lambda \rightarrow A'_\alpha / \mathfrak{m}'_\alpha$ with intersection of kernels 0. One applies (11.4.11) in case (i) and (11.4.12) in case (ii). $\square$

Corollary (11.5.2). — Let Y be locally Noetherian, $f : X \rightarrow Y$ locally of finite type, $\mathcal{F}$ coherent, $x \in X$, $y = f(x)$; set $A = \mathcal{O}_y$. Let $u : A \rightarrow A'$ be a homomorphism into a Zariski A -ring A' such that $u^{-1}(\text{rad}(A')) = \mathfrak{m}$ and $\hat{u} : \hat{A} \rightarrow \hat{A}'$ is injective. Set $Y' = \text{Spec}(A')$, $X' = X \times_Y Y'$, $\mathcal{F}' = \mathcal{F} \otimes_Y \mathcal{O}_{Y'}$. For $\mathcal{F}$ to be f -flat at x , it is necessary and sufficient that $\mathcal{F}'$ be f' -flat at every point projecting to x in X and to a closed point of Y' .

If A' is a finite A -algebra, one can replace the injectivity of $\hat{u}$ by the injectivity of u .

Proof. As A (resp. A') embeds into $\hat{A}$ (resp. $\hat{A}'$), u is injective. The result follows from (11.5.1, (i)) applied to the family of localisations of A' at its maximal ideals. $\square$

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Proposition (11.5.3). — Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation, $x \in X$, $y = f(x)$, $g = (\psi, \theta) : Y' \rightarrow Y$ a proper morphism of finite presentation. Suppose:

- label=(i) $\theta_y : \mathcal{O}_y \rightarrow (g_*(\mathcal{O}_{Y'}))_y$ is injective.
 lbbel=(ii) For every $x' \in X' = X \times_Y Y'$ projecting to x , $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$ is Y' -flat at x' .

Then $\mathcal{F}$ is f -flat at x .

Proof. By (11.3.1), the f' -flatness locus U' is open and contains $p^{-1}(x)$; as p is proper hence closed, U' contains a neighbourhood. One reduces to Y local Noetherian, then applies (11.5.1). $\square$

Corollary (11.5.4). — Let $f = (\psi, \theta) : X \rightarrow Y$ be a proper morphism of finite presentation, $p : X \times_Y X \rightarrow X$ the first projection. Suppose $\theta : \mathcal{O}_Y \rightarrow f_*(\mathcal{O}_X)$ is injective. Then, for f to be flat, it is necessary and sufficient that p be flat.

Proof. Apply (11.5.3) with $\mathcal{F} = \mathcal{O}_X$. $\square$

Proposition (11.5.5). — Let A be a ring, $Y = \text{Spec}(A)$, $f : X \rightarrow Y$ a morphism locally of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation, $x \in X$. Let $A \rightarrow A'$ be an injective homomorphism making A' an integral A -algebra. Set $Y' = \text{Spec}(A')$, $X' = X \times_Y Y'$, $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$. If $\mathcal{F}'$ is f' -flat at every point of X' projecting to x in X , then $\mathcal{F}$ is f -flat at x . **IV-3** | 153

Proof. One reduces to f of finite presentation, then to Y local; by (5.13.3) and (8.9.1), one can suppose A Noetherian. Setting $B = \hat{A}$, $Z = \text{Spec}(B)$; $\mathcal{F}$ is f -flat iff $\mathcal{F} \otimes_Y Z$ is Z -flat (2.1.4). As A' is integral over A , $Y' \rightarrow Y$ is proper (II, 6.1.10). One applies (11.5.3). $\square$

11.6. Descent of flatness by arbitrary morphisms: case of a unibranch base

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Lemma (11.5.5.2). — *Let A be a ring, A' an A -algebra, (A'_λ) an inductive system of A -algebras such that $A' = \varinjlim A'_\lambda$; set $Y = \operatorname{Spec}(A)$, $Y' = \operatorname{Spec}(A')$, $Y'_\lambda = \operatorname{Spec}(A'_\lambda)$. Let $f : X \rightarrow Y$ be a morphism of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation; set $X' = X \times_Y Y'$, $\mathcal{F}' = \mathcal{F} \otimes_Y \mathcal{O}_{Y'}$, $X'_\lambda = X \times_Y Y'_\lambda$, $f'_\lambda = f|_{Y'_\lambda}$, $\mathcal{F}'_\lambda = \mathcal{F} \otimes_Y \mathcal{O}_{Y'_\lambda}$. Let x be a point of X ; if $\mathcal{F}'$ is f' -flat at all points $x' \in X'$ lying over x , then there exists λ such that $\mathcal{F}'_\lambda$ is f'_λ -flat at all points of X'_λ lying over x .*

Proof. Let U' be the set of $x' \in X'$ such that $\mathcal{F}'$ is f' -flat at x' ; by (11.3.1), U' is open in X' since f' is of finite presentation (1.6.2); likewise U'_λ the flatness locus of $\mathcal{F}'_\lambda$ is open in X'_λ , and $U' = \bigcup v_\lambda^{-1}(U'_\lambda)$ where $v_\lambda : X' \rightarrow X'_\lambda$ is the canonical projection. Consider $T = \operatorname{Spec}(k(x))$, $T'_\lambda = T \times_Y Y'_\lambda$, and $V' = g'^{-1}(U')$, $V'_\lambda = g'^{-1}_\lambda(U'_\lambda)$. By hypothesis (I, 6.1.1)) $V' = T'$; as T' is quasi-compact, there exists λ such that $w_\lambda^{-1}(V'_\lambda) = T'$, hence $T'_\mu = V'_\mu$ for $\mu \geq \lambda$. By (8.3.3), applied to the closed complement $T'_\lambda - V'_\lambda$ of T'_λ , there exists $\mu \geq \lambda$ such that $T'_\mu = V'_\mu$, i.e. $\mathcal{F}'_\mu$ is f'_μ -flat at all points of X'_μ whose projection in X is x . C.Q.F.D. $\square$

Theorem (11.6.1). — *Let A be a local integral ring geometrically unibranch (0, 23.2.1), $Y = \operatorname{Spec}(A)$, $f : X \rightarrow Y$ a morphism locally of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. Let A' be a local ring, $u : A \rightarrow A'$ a local injective homomorphism; set $Y' = \operatorname{Spec}(A')$, $X' = X \times_Y Y'$, $f' = f|_{Y'}$, $\mathcal{F}' = \mathcal{F} \otimes_Y \mathcal{O}_{Y'}$. Let x be a point of X with $f(x) = y$ the closed point of Y , x' a point of X' whose projections in X and Y' are x and the closed point y' of Y' . If $\mathcal{F}'$ is f' -flat at the point x' , then $\mathcal{F}$ is f -flat at the point x .*

Proof. One reduces to f of finite presentation. We proceed in several steps. IV-3 | 155

I) *Reduction to A and A' integrally closed.* As u is injective and A integral, one can suppose A' integrally closed. Let A_1 be the integral closure of A ; since A is geometrically unibranch, the morphism $\operatorname{Spec}(k(y_1)) \rightarrow \operatorname{Spec}(k(y))$ is radiciel, and hence x_1 is the sole point of X_1 projecting to x in X and y_1 in Y_1 . One can apply (11.5.5), and this reduces to the case where A is also integrally closed.

II) *Reduction to A and A' local $\mathbf{Z}$ -algebras of finite type integrally closed.* One considers A' as an inductive limit of its sub- $\mathbf{Z}$ -algebras of finite type B'_λ integrally closed; since A' is integrally closed and the integral closure of a $\mathbf{Z}$ -algebra of finite type is also a finite $\mathbf{Z}$ -algebra (7.8.3), $A' = \varinjlim (B'_\lambda)_{\mathfrak{m}'_\lambda} = A'_\lambda$. By (11.5.5.2) and (8.9.1), there exists a pair (α, λ) sufficiently large and a coherent $\mathcal{O}_{X_{\alpha\lambda}}$ -Module $\mathcal{F}_{\alpha\lambda}$ such that it suffices to show $\mathcal{F}_{\alpha\lambda}$ is $f_{\alpha\lambda}$ -flat at $x_{\alpha\lambda}$. By (11.2.6), one can take $\mu \geq \lambda$ large enough so that $\mathcal{F}'_\mu$ is Y'_μ -flat at the appropriate point.

III) *Reduction to k' a finite and radiciel extension of k .* Using the part I) argument applied to (11.4.11), and the fact that $\mathbf{Z}$ is a Jacobson ring, one reduces to the case where k' is a finite extension of k . Let k'' be the largest separable extension of k in k' , k_1 a finite Galois extension containing k'' . Since k' is a radiciel extension of k'' , $k' \otimes_k k_1$ is a direct product of extensions radiciel of k_1 ; hence the canonical homomorphism $A'_1 \rightarrow \prod (A'_1)_{\mathfrak{n}_i}$ is injective. Since A_1 is integrally closed, A'_1 is an A' -module free of finite type; hence A_1 is an A -module flat, and the homomorphism is injective. Setting $Y'_1 = \operatorname{Spec}(B_1)$, $X'_1 = X' \times_{Y'} Y'_1$, $\mathcal{F}'_1 = \mathcal{F}' \otimes_{Y'} \mathcal{O}_{Y'_1}$; $\mathcal{F}'_1$ is Y'_1 -flat at the points over x' ; and as $A_1 \rightarrow B_1$ is injective and local, if $(\mathcal{F}_1)_{x_1}$ is an $\mathcal{O}_{y_1}$ -module flat, $\mathcal{F}_x$ is an $\mathcal{O}_y$ -module flat (0_I, 6.2.1).

IV) *End of proof.* One is finally reduced to proving (11.6.1) under the additional hypotheses: (i) A and A' are local $\mathbf{Z}$ -algebras of finite type (hence excellent rings (7.8.3)); (ii) A is integrally closed; (iii) k' is a finite and radiciel extension of k . Under (i) and (ii), the $\mathfrak{m}$ -adic and $\mathfrak{m}'$ -adic topologies on A and A' are induced by the completions. The completed $\hat{u} : \hat{A} \rightarrow \hat{A}'$ is an injective homomorphism. Since $\operatorname{Spec}(k') \rightarrow \operatorname{Spec}(k)$ is radiciel, there is only one point x' of X' projecting to x and y' . One can therefore apply (11.5.2). C.Q.F.D. $\square$

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Corollary (11.6.2). — *Let A be a local integral ring unibranch, $Y = \operatorname{Spec}(A)$, $f : X \rightarrow Y$ a morphism locally of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation. Let A' be a local ring, $A \rightarrow A'$ a local injective homomorphism; set $Y' = \operatorname{Spec}(A')$, $X' = X \times_Y Y'$, $f' = f|_{Y'}$, $\mathcal{F}' = \mathcal{F} \otimes_Y \mathcal{O}_{Y'}$. Let x be a point of X with $f(x) = y$ the closed point of Y ; suppose $\mathcal{F}'$ is f' -flat at all points x' of X' whose projections in X and Y' are x and y' respectively. Then $\mathcal{F}$ is f -flat at x .*

Proof. One reprises part I) of the proof of (11.6.1), which shows (with the same notations) that if $\mathcal{F}_1$ is f_1 -flat at *all* points x_1 of X_1 whose projections in X and Y_1 are x and y_1 , then $\mathcal{F}$ is f -flat at x ; one is thus reduced to the case where A is integrally closed, hence geometrically unibranch, and the conclusion follows from (11.6.1). $\square$

11.7. Counter-examples

(11.7.1). Consider first the case where A is an Artinian local ring and where the hypotheses of (11.4.11) are satisfied *except* the condition (ii) on the residue field. We shall see that the conclusion can then fail. Let k be a field admitting a Galois extension k' of degree $[k' : k] > 1$, and denote by Γ the Galois group. Let A be a k -algebra having a base of 3 elements $1, a, b$ with the multiplication $a^2 = ab = ba = b^2 = 0$, so that A is an Artinian local ring with maximal ideal $\mathfrak{m} = ka + kb$ of square zero. Set $A' = A \otimes_k k'$, which is a local Artinian ring with maximal ideal $\mathfrak{m}A'$ of square zero; A identifies canonically with a sub-ring of A' . Let $\mathfrak{J}$ be the sub- k' -space of $\mathfrak{m}'$ generated by $a + \gamma b$, where $\gamma \in k'$ does not belong to k ; it is clear that $\mathfrak{J}$ is an ideal of A' . Set $B = A'/\mathfrak{J}$; this is an Artinian ring which is an A -module *not flat*; for otherwise (Bourbaki, *Alg. comm.*, chap. II, §3, n° 2, cor. 2 of prop. 5), B would be an A -module free, and since the canonical homomorphism $A'/\mathfrak{m}A' \rightarrow B/\mathfrak{m}B$ is bijective, the canonical homomorphism $A' \rightarrow B = A'/\mathfrak{J}$ would also be bijective (*loc. cit.*, prop. 6), which is absurd.

(11.7.2). We now show that the result of (11.5.5) loses its validity when one no longer supposes f a morphism *proper* (and it suffices not to suppose g of finite type). Let $X = \text{Spec}(B)$, e_x the generic point, $Y = \text{Spec}(A)$, and take the curve Y reducible formed by two “coordinate axes” in the affine plane $V_y^2 = \text{Spec}(A_0)$. The ring A admits two minimal primes $\mathfrak{p}_1, \mathfrak{p}_2$; setting $B = B_1 \oplus B_2$ where $B_i = A/\mathfrak{p}_i$, then B_1 identifies with $k[T]$ and B_2 with $k[S]$, and the sum $Z = \text{Spec}(B)$ is the *normalised* of Y with respect to Y (II, 6.3.8). The corresponding morphism $f : X \rightarrow Y$ is not flat, even though for each branch one can show flatness after base change to a valuation ring dominating one of the branches; moreover the first projection $p : X \times_Y X \rightarrow X$ is an *isomorphism*, hence p is flat, yet f is not flat at z_1 .

(11.7.3). The examples in (11.7.2) and (11.7.3) explain the restriction to *unibranch* local rings in (11.6.1) and (11.6.2), and one cannot in general replace the hypothesis that A is unibranch but not geometrically unibranch. Consider the complete local integral ring $A = \mathbf{R}[[U, V]]/(U^2 + V^2)$ which is unibranch but not geometrically unibranch (6.15.11). The integral closure $\bar{A} = A[t]$ with $t = n/u$, $t^2 = -1$, is isomorphic to $\mathbf{C}[[U]]$. The two ideals $\mathfrak{p}', \mathfrak{p}''$ generated by $u + iv$ and $u - iv$ are the minimal primes of the local complete ring B . B has two minimal primes $\mathfrak{p}', \mathfrak{p}''$ and the ideal $\mathfrak{n} = \mathfrak{p}' + \mathfrak{p}''$ is the maximal ideal of B . The morphism f is not flat, although after base change to $A' = \bar{A}$, which is an integrally closed local ring with $A \rightarrow A'$ injective and local, the base-changed module is flat.

(11.7.4). One notes that in the preceding example, the homomorphism u is *injective* when Y is *irreducible* (one can take $Y = \text{Spec}(k[S, T]/(S(S^2 + T^2) - (S^2 - T^2)))$, a “cubic with a double point”); one can then replace Y by an affine neighbourhood of y so that $Y = \text{Spec}(A)$, where A is integral, whence $Z = \text{Spec}(B)$, where B is the integral closure of A ; as $B, \hat{\mathcal{O}}_y$ and $\hat{\mathcal{O}}_{z_1}$ identify with sub-rings of the field of fractions.

(11.7.5). The examples in (11.7.2) and (11.7.3) explain the restriction to *unibranch* local rings in (11.6.1) and (11.6.2). We now show that in (11.6.1), one cannot weaken the hypothesis on A to merely *unibranche* (rather than geometrically unibranch). Consider the complete local integral ring $A = \mathbf{R}[[U, V]]/(U^2 + V^2)$, which is unibranche (6.15.11). The integral closure is $\bar{A} \cong \mathbf{C}[[U]]$; setting $X = \text{Spec}(\bar{A})$, $Y = \text{Spec}(A)$, $A' = \mathbf{C}[[U]]$: the base change $Y' = \text{Spec}(A') \rightarrow Y$ satisfies all hypotheses of (11.6.1), yet one can verify that $B = \bar{A} \otimes_{\mathbf{R}} \mathbf{C}$ is not A -flat.

11.8. A valuative criterion of flatness

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Theorem (11.8.1). — Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module of finite presentation, x a point of X , $y = f(x)$. Suppose $\mathcal{O}_y$ integral (resp. reduced and Noetherian). For $\mathcal{F}$ to be f -flat at the point x , it is necessary and sufficient that, for every valuation ring (resp. every discrete valuation ring) A' and every local homomorphism $\mathcal{O}_y \rightarrow A'$, the following condition be satisfied: setting $Y' = \text{Spec}(A')$, $X' = X \times_Y Y'$, $f' = f_{(Y')}$, $\mathcal{F}' = \mathcal{F} \otimes_Y \mathcal{O}_{Y'}$, the $\mathcal{O}_{X'}$ -Module $\mathcal{F}'$ is f' -flat at all points x' of X' whose projections in X and Y' are x and the closed point y' of Y' .

Proof. The condition is evidently necessary (2.1.4), it remains to prove it is sufficient. One can evidently ((I, 3.6.5) and (I, 2.4.4)) restrict to the case where $Y = \text{Spec}(A)$ and y the closed point of Y .

(i) *Case where A is integral.* Let K be the field of fractions of A , A_1 the integral closure of A ; setting $Y_1 = \text{Spec}(A_1)$, $X_1 = X \times_Y Y_1$, $f_1 = f_{(Y_1)}$, by virtue of (11.5.5), it suffices to show that $\mathcal{F}_1 = \mathcal{F} \otimes_Y \mathcal{O}_{Y_1}$ is f_1 -flat at all points x_1 of X_1 projecting to x in X and to y_1 in Y_1 . Now, if $f_1(x_1) = y_1$, where $\mathfrak{m}_1$ is an ideal prime of A_1 above $\mathfrak{m} = \mathfrak{i}_y$ (which is necessarily a maximal ideal), $\mathcal{O}_{y_1} = (A_1)_{\mathfrak{m}_1}$ is integrally closed. Let A' be a valuation ring for K dominating $\mathcal{O}_{y_1}$; the homomorphism $A \rightarrow \mathcal{O}_{y_1}$ being local, $A \rightarrow A'$ is also local. There exists then at least one point x' of X' whose projections in X_1 and Y' are x_1 and y' respectively (I, 3.4.7); as by hypothesis $\mathcal{F}'$ is f' -flat at x' , and as $\mathcal{O}_{y'}$ is integrally closed, one can apply (11.6.1), and one deduces that $\mathcal{F}_1$ is f_1 -flat at the point x_1 , whence the theorem in this case.

(ii) *Case where A is reduced and Noetherian.* Let $\mathfrak{p}_i$ ($1 \leq i \leq m$) be the minimal primes of A ; as A is reduced, it identifies canonically with a sub-ring of the product $\prod A_i = \prod A/\mathfrak{p}_i$, A_i being local Noetherian integral. Setting $Y_i = \text{Spec}(A_i)$, $X_i = X \times_Y Y_i$, $\mathcal{F}_i = \mathcal{F} \otimes_Y \mathcal{O}_{Y_i}$, $f_i = f_{(Y_i)}$: by (11.5.2), it suffices to show that for each i , $\mathcal{F}_i$ is f_i -flat at every point x_i whose projections in X and Y_i are x and the closed point y_i . Now as A_i is Noetherian integral, there exists in its field of fractions K_i a sub-ring A'_i which is a semi-local Noetherian ring (hence a A_i -algebra finite) and whose local rings are geometrically unibranch (6.15.5) and (0, 23.2.5). As the maximal ideals of A'_i are necessarily above the maximal ideal of A_i , by (11.5.2) one deduces that for each i , one can apply (11.6.1) (using (I, 3.4.7) and (11.5.2)) to deduce $\mathcal{F}_i$ is f_i -flat at x_i , then the result by (11.5.2). $\square$

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Remarks (11.8.2). — label=(i) In the statement of (11.8.1), one can restrict to valuation rings A' that are *complete* and whose residue field is *algebraically closed*. Indeed, one knows that every such valuation ring A' is dominated by a valuation ring A'' (II, 7.1.2), and that if A' is a discrete valuation ring, one can suppose A'' is also a discrete valuation ring (0III, 10.3.1).

lbel=(ii) The proof of (11.8.1) simplifies when one supposes not only that A is integral and Noetherian, but that its completion $\hat{A}$ is also *integral*. Replacing X by $X \otimes_A \hat{A}$ and reasoning as in the proof of (11.5.3), one can indeed suppose $A = \mathcal{O}_y$ integral, Noetherian and *complete*. Now, one knows (II, 7.1.7) that every such ring A is dominated by a discrete valuation ring; the conclusion then follows directly from (11.5.2).

11.9. Separating and universally separating families of homomorphisms of sheaves of modules

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(11.9.1). Let X be a prescheme, $(f_\lambda)_{\lambda \in L}$ a family of morphisms $f_\lambda : Z_\lambda \rightarrow X$, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -Module; for every $\lambda \in L$, let $\mathcal{G}_\lambda$ be a quasi-coherent $\mathcal{O}_{Z_\lambda}$ -Module and

$$(11.9.1.1) \quad u_\lambda : \mathcal{F} \longrightarrow (f_\lambda)_*(\mathcal{G}_\lambda)$$

a homomorphism. We say that the family (u_λ) (or the corresponding family of $u_\lambda^* : (f_\lambda)^*(\mathcal{F}) \rightarrow \mathcal{G}_\lambda$) is *separating* if the intersection of the kernels of the u_λ is zero. In other words, for every open U of X , and every section t of $\mathcal{F}$ over U , such that for every λ , the section $u_\lambda(t)$ (which, by definition, is a section of $\mathcal{G}_\lambda$ over $f_\lambda^{-1}(U)$) is zero, then t is itself zero.

(11.9.2). With the notations of (11.9.1), let M be a second index set; for every $\lambda \in L$, let $(g_{\lambda\mu})_{\mu \in M}$ be a family of morphisms $g_{\lambda\mu} : Z'_{\lambda\mu} \rightarrow Z_\lambda$; for every pair (λ, μ) , let $\mathcal{H}_{\lambda\mu}$ be a quasi-coherent $\mathcal{O}_{Z'_{\lambda\mu}}$ -Module and $v_{\lambda\mu} : \mathcal{G}_\lambda \rightarrow (g_{\lambda\mu})_*(\mathcal{H}_{\lambda\mu})$ a homomorphism; set $h_{\lambda\mu} = f_\lambda \circ g_{\lambda\mu} : Z'_{\lambda\mu} \rightarrow X$ and consider the composed homomorphism

$$w_{\lambda\mu} : \mathcal{F} \xrightarrow{u_\lambda} (f_\lambda)_*(\mathcal{G}_\lambda) \xrightarrow{(f_\lambda)_*(v_{\lambda\mu})} (h_{\lambda\mu})_*(\mathcal{H}_{\lambda\mu}).$$

Suppose that, for every $\lambda \in L$, the family $(v_{\lambda\mu})_{\mu \in M}$ is separating: then, as is immediately seen, the family $(w_{\lambda\mu})_{(\lambda,\mu) \in L \times M}$ is also separating, and conversely if (u_λ) is separating then it suffices for the family of the $(f_\lambda)_*(v_{\lambda\mu})$ to be separating.

(11.9.3). To verify that the family (u_λ) is separating (with the notations of (11.9.1)), one can evidently first reduce to the case where X is *affine*, and then assume $Z_\lambda = X$ for every $\lambda \in L$. Moreover, by (11.9.2) and (11.9.5), if L is *finite* and the f_λ *quasi-compact*, one can, in the preceding reduction, also take the M_λ to be *finite*.

(11.9.4). Consider the case where $Z_\lambda = X$ for every λ , and $X = \text{Spec}(A)$ is affine; then $\mathcal{F} = \tilde{M}$ and $\mathcal{G}_\lambda = \tilde{N}_\lambda$, where M and N_λ are A -modules, and $u_\lambda = \tilde{\varphi}_\lambda$, where the $\varphi_\lambda : M \rightarrow N_\lambda$ are A -homomorphisms. To say that the family (u_λ) is separating then means that, for every $s \in A$, the intersection of the kernels of the $(\varphi_\lambda)_s : M_s \rightarrow (N_\lambda)_s$ is reduced to 0. We say that the family (φ_λ) is also *separating*. Note that if L is *finite*, this is the same as saying that $\bigcap_{\lambda \in L} \text{Ker}(\varphi_\lambda) = 0$, since $(\bigcap \text{Ker}(\varphi_\lambda))_s = \bigcap \text{Ker}((\varphi_\lambda)_s)$ (0I, 1.3.2). However, this relation is no longer exact in general when L is *infinite*.

(11.9.5). We now principally consider the problem of *base change* for separating families. With the notations of (11.9.1), consider a morphism $g : X' \rightarrow X$ and set $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} = g^*(\mathcal{F})$, and, for every λ , $Z'_\lambda = Z_\lambda \times_X X'$, $f'_\lambda = f_\lambda \times 1$, $\mathcal{G}'_\lambda = \mathcal{G}_\lambda \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} = g^*(\mathcal{G}_\lambda)$, where $g'_\lambda : Z'_\lambda \rightarrow Z_\lambda$ is the projection canonique. For every λ , denote by $u'_\lambda : \mathcal{F}' \rightarrow (f'_\lambda)_*(\mathcal{G}'_\lambda)$ the homomorphism obtained from u_λ by base change (0I, 4.4.3.2) corresponding to j_λ . Then u'_λ is defined as the composite

$$g^*(\mathcal{F}) \xrightarrow{g^*(u_\lambda)} g^*((f_\lambda)_*(\mathcal{G}_\lambda)) \xrightarrow{g^*(\psi)} g_*(((f'_\lambda)_*(\mathcal{G}'_\lambda))) = (f'_\lambda)_*(\mathcal{G}'_\lambda)$$

where $\rho_{g'_\lambda} : \mathcal{G}'_\lambda \rightarrow (j_{\lambda\mu})_*(j_{\lambda\mu}^*(\mathcal{G}'_\lambda))$ is the canonical homomorphism (0I, 4.4.3.2) corresponding to j_λ . When $f_\lambda : Z_\lambda \rightarrow X$ is an affine morphism, in the case where $\mathcal{G}_\lambda = \mathcal{O}_{Z_\lambda}$ and u_λ is the homomorphism $\rho_{\mathcal{F}} : \mathcal{F} \rightarrow (j_\lambda)_*((j_\lambda)^*(\mathcal{F})) = (j_\lambda)_*(\mathcal{G}_\lambda)$ corresponding to the injection canonical j_λ . When Z_λ is a sub-prescheme of X , $\mathcal{G}_\lambda = j_\lambda^*(\mathcal{F})$, and u_λ is the homomorphism $\rho_{\mathcal{F}} : \mathcal{F} \rightarrow (j_\lambda)_*((j_\lambda)^*(\mathcal{F})) = (j_\lambda)_*(\mathcal{G}_\lambda)$ corresponding to j_λ .

(11.9.6). Under the conditions of (11.9.5), suppose X and X' affine, and that one wants to prove that for every section t' of $\mathcal{F}'$ over X' whose images by all the u'_λ are zero, t' is itself zero. Then, one can restrict to the case where $Z_\lambda = X$ for every $\lambda \in L$. Indeed, with the notations of (11.9.3) and (11.9.5), if one sets $Y'_{\lambda\mu} = g_\lambda^{-1}(Y_{\lambda\mu})$, $j'_{\lambda\mu} : Y'_{\lambda\mu} \rightarrow Z'_\lambda$ the injection canonical and $v'_{\lambda\mu} : \mathcal{G}'_\lambda \rightarrow (j'_{\lambda\mu})_*((j'_{\lambda\mu})^*(\mathcal{G}'_\lambda))$ the homomorphism deduced from $v_{\lambda\mu}$ by base change; by (11.9.2), it is immediate that for each $\lambda \in L$, the family $(v_{\lambda\mu})_{\mu \in M_\lambda}$ is separating. One is thus reduced to proving that the family of the composed homomorphisms $((f_\lambda)_*(v_{\lambda\mu})) \circ u_\lambda$ is separating, which, since the $(f_\lambda)_*(\mathcal{G}_\lambda)$ are quasi-coherent $\mathcal{O}_X$ -Modules (I, 1.6.2) and by the definition, one can replace Z_λ by X and $\mathcal{G}_\lambda$ by $(f_\lambda)_*(\mathcal{G}_\lambda)$, whence our assertion.

Note that if L is *finite* and the f_λ *quasi-compact*, one can also take the M_λ finite, hence one reduces to the case of verifying that a *finite* family of homomorphisms of $\mathcal{F}$ into quasi-coherent $\mathcal{O}_X$ -Modules is separating.

Proposition (11.9.7). — *With the notations of (11.9.1) and (11.9.5), suppose $X = \text{Spec}(A)$ and $X' = \text{Spec}(A')$ affine, and further that A' is an A -module projective. Then, if the family (u_λ) is separating, every section t' of $\mathcal{F}'$ over X' whose images by all the u'_λ are zero is itself zero.*

Proof. By (11.9.6) one can restrict to $Z_\lambda = X$ for every λ . The proposition is then a consequence of the following: □

Lemma (11.9.7.1). — *Let A be a ring, $(M_\lambda)_{\lambda \in L}$ a family of A -modules, M an A -module and for each λ , $u_\lambda : M \rightarrow M_\lambda$ an A -homomorphism. Suppose the intersection of the kernels of the u_λ is reduced to 0. Then, for every projective A -module N , the intersection of the kernels of the homomorphisms $u_\lambda \otimes 1 : M \otimes_A N \rightarrow M_\lambda \otimes_A N$ is reduced to 0.*

Proof. N being a direct factor of a free A -module L , it suffices to prove the result for $N = L$, which is trivial since the restriction of v_λ is trivially the kernel of $u_\lambda \otimes 1_L$. The assertion then follows trivially from the hypothesis. □

Remarks (11.9.8). — Under the hypotheses of (11.9.7), we do not know if the family (u'_λ) is *separating*: one would need to prove that a section t' of $\mathcal{F}'$ over an open $D(h') \subset X'$ (where $h' \in A'$) whose images by all u'_λ are zero, is zero. However, one cannot apply (11.9.4) to $D(h') = \text{Spec}(A'_{h'})$, since A' being projective does not imply $A'_{h'}$ is projective (or even free) as an A -module.

Corollary (11.9.9). — Let X be an Artinian prescheme, $g : X' \rightarrow X$ a flat morphism (one notes that these two conditions are satisfied when X is the spectrum of a field and g an arbitrary morphism). Then, with the notations of (11.9.1) and (11.9.5), if the family (u_λ) is separating, so is the family (u'_λ) .

Proof. One can restrict to $X = \text{Spec}(A)$, the spectrum of an Artinian local ring A (I, 6.2.2); one notes then that for every affine open $U' = \text{Spec}(A')$ of X' , A' is an A -module flat, hence *projective* (0III, 10.1.3). It suffices then to apply (11.9.7). $\square$

Theorem (11.9.10). — Let X be a prescheme, (u_λ) a family of homomorphisms (11.9.1.1), (u'_λ) the family of homomorphisms deduced from (u_λ) by the base change g (11.9.5).

label=(i) If g is a faithfully flat morphism and if the family (u'_λ) is separating, then the family (u_λ) is separating.

lbbel=(ii) Conversely, suppose that $g : X' \rightarrow X$ is a flat morphism, that $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -Module of finite type, and that the family (u_λ) is separating. Then the family (u'_λ) is separating.

Proof. Assertion (i) follows immediately from the faithful flatness of g (0I, 6.3.2). For (ii), one can restrict to X and X' affine, and then to the case $Z_\lambda = X$ for every λ (11.9.6). Then $\mathcal{F} = \tilde{M}$, $\mathcal{G}_\lambda = \tilde{N}_\lambda$, $u_\lambda = \tilde{\varphi}_\lambda$ with M of finite type. As g is flat, $M' = M \otimes_A A'$ is a flat A -module, and $(\varphi_\lambda \otimes 1)$ have intersection of kernels equal to $(\cap \text{Ker } \varphi_\lambda) \otimes_A A' = 0$. $\square$

Remarks (11.9.11). — It should be noted that if the family of homomorphisms $u_\lambda : \mathcal{F} \rightarrow \mathcal{G}_\lambda$ is separating, it does not necessarily follow that, for every $x \in X$, the intersection of the kernels of the homomorphisms $(u_\lambda)_x : \mathcal{F}_x \rightarrow (\mathcal{G}_\lambda)_x$ is reduced to 0. For example, let X be a locally Noetherian prescheme, of Jacobson dimension ≥ 1 , and $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module; for every closed point x of X , and every integer $n \geq 0$, $\mathfrak{m}_x^n \mathcal{F}$ is a coherent $\mathcal{O}_X$ -Module of support contained in $\{x\}$. The family of canonical homomorphisms $\mathcal{F} \rightarrow \mathcal{F}/\mathfrak{m}_x^n \mathcal{F}$ (where x ranges over the set X_0 of closed points of X and n the set of integers ≥ 0) is *separating*: indeed, if t is a section of $\mathcal{F}$ over an open U of X whose images in all the $\Gamma(U, \mathcal{F}/\mathfrak{m}_x^n \mathcal{F})$ are zero, it follows immediately that for every closed point $x \in U$, we have $t_x = 0$; as the set of closed points contained in U is very dense in U , this implies $t = 0$ (10.2.1). However, if $\mathcal{F} = \mathcal{O}_X$, and if $y \in X$ is a non-closed point of X , we take $\mathcal{O}_{X,y} \neq 0$, but $(\mathcal{O}_X/\mathfrak{m}_x^n \mathcal{O}_X)_y = 0$ for every closed point x of X , and the intersection of the kernels of the homomorphisms $\mathcal{O}_{X,y} \rightarrow (\mathcal{O}_X/\mathfrak{m}_x^n \mathcal{O}_X)_y$ is equal to $\mathcal{O}_{X,y}$.

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Lemma (11.9.12). — The notations being those of (11.9.1) and (11.9.5), suppose the family (u_λ) separating; suppose in addition that X is an S -prescheme, where $S = \text{Spec}(A)$, and that $X' = X \times_S S'$, where $S' = \text{Spec}(A')$, A' being an A -algebra; suppose finally that the $\mathcal{G}_\lambda$ are S -flat. Let $(A'_\alpha)_{\alpha \in I}$ be the filtering family of sub- A -algebras of A' of finite type; for every $\alpha \in I$, set $S''_\alpha = \text{Spec}(A'_\alpha)$, $X''_\alpha = X \times_S S''_\alpha$, $Z''_{\alpha\lambda} = Z_\lambda \times_X X''_\alpha$, and let $f''_{\alpha\lambda}$, $\mathcal{F}''_\alpha$, $\mathcal{G}''_{\alpha\lambda}$ and $u''_{\alpha\lambda}$ be the morphisms, Modules and homomorphisms of Modules deduced from f_λ , $\mathcal{F}$, $\mathcal{G}_\lambda$ and u_λ by the base change $X''_\alpha \rightarrow X$. Then, if, for every $\alpha \in I$, the family $(u''_{\alpha\lambda})_{\lambda \in L}$ is separating, so is the family (u'_λ) .

Proof. It suffices to prove that if a section t' of $\mathcal{F}'$ over an affine open U' of X' is such that $u'_\lambda(t') = 0$ for every $\lambda \in L$, then $t' = 0$. If $h_\alpha : X' \rightarrow X''_\alpha$ is the projection, by (8.2.11) there exist an index α and an open quasi-compact $U''_\alpha \subset X''_\alpha$ such that $U' = h_\alpha^{-1}(U''_\alpha)$; in addition, by (8.5.2, (i)), one can suppose that there is a section t''_α of $\mathcal{F}''_\alpha$ over U''_α such that t' is the canonical image of t''_α . Replacing S by S''_α , Z_λ by $Z''_{\alpha\lambda}$, $\mathcal{F}$ by $\mathcal{F}''_\alpha$ and u_λ by $u''_{\alpha\lambda}$, $f''_{\alpha\lambda}^{-1}(U''_\alpha)$ and the restrictions of the corresponding modules and homomorphisms, one can thus suppose that $U' = X'$, that t' is the canonical image of a section t of $\mathcal{F}$ over X and that the homomorphism $A \rightarrow A'$ is *injective*. But the composed homomorphism

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$$\Gamma(X, \mathcal{F}) \xrightarrow{\Gamma(u_\lambda)} \Gamma(Z_\lambda, \mathcal{G}_\lambda) \xrightarrow{\Gamma(\rho)} \Gamma(Z'_\lambda, \mathcal{G}'_\lambda)$$

is equal by definition to the composed homomorphism

$$\Gamma(X, \mathcal{F}) \xrightarrow{\Gamma(\rho)} \Gamma(X', \mathcal{F}') \xrightarrow{\Gamma(u'_\lambda)} \Gamma(Z'_\lambda, \mathcal{G}'_\lambda)$$

hence the image of t by these composed homomorphisms is $u'_\lambda(t') = 0$; by the injectivity of the homomorphism $\Gamma(Z_\lambda, \mathcal{G}_\lambda) \xrightarrow{\Gamma(\rho)} \Gamma(Z'_\lambda, \mathcal{G}'_\lambda)$ one concludes that $u_\lambda(t) = 0$ for every $\lambda \in L$, hence $t = 0$, and finally $t' = 0$. $\square$

Proposition (11.9.13). — *The notations being those of (11.9.1) and (11.9.5), suppose that X is a prescheme over a field k , and that, setting $S = \operatorname{Spec}(k)$, one has $X' = X \times_S S'$, where S' is an arbitrary k -prescheme. Then, if the family $(u_\lambda)_{\lambda \in L}$ is separating, so is the family (u'_λ) .*

Proof. One can restrict to the case where $S' = \operatorname{Spec}(A')$ is affine. If A' is a k -algebra of finite type, the morphism $g : X' \rightarrow X$ is flat and of finite presentation, so one is in the conditions of application of (11.9.10), (ii), b). In the general case, one considers A' as the inductive limit of its sub- k -algebras A'_α of finite type, and one applies to each A'_α the result of (11.9.10), (ii), b); one then concludes with the aid of the lemma (11.9.12), since the $\mathcal{G}_\lambda$ are S -flat. $\square$

(11.9.14). Let us keep the notations of (11.9.1) and (11.9.5) and suppose that X is an S -prescheme. If for every base change $g : X \times_S S' \rightarrow X$, where $S' \rightarrow S$ is an arbitrary morphism, the corresponding family (u'_λ) is separating, we say that the family (u_λ) is *universally separating relative to S* . When the family (u_λ) is reduced to a single element u , we also say that u is *universally injective*, relative to S . It is clear that for every morphism $h : S' \rightarrow S$, the corresponding family (u'_λ) is universally separating relative to S' ; conversely, if h is *faithfully flat* and if (u'_λ) is universally separating relative to S' , then (u_λ) is universally separating relative to S , as follows immediately from (11.9.10), (i)) and the fact that for every morphism $S'' \rightarrow S$, the corresponding morphism $S'' \times_S S' \rightarrow S''$ is faithfully flat.

Proposition (11.9.15). — *The notations being those of (11.9.1), suppose that X is an S -prescheme, the $\mathcal{G}_\lambda$ being S -flat. Let S_0 be a closed sub-prescheme of S defined by a quasi-coherent nilpotent ideal $\mathcal{J}$ of $\mathcal{O}_S$, such that the $(\mathcal{O}_S / \mathcal{J})$ -Modules $\mathcal{J}^k / \mathcal{J}^{k+1}$ are all locally free. Let $(u_{\lambda 0})$ be the family of homomorphisms obtained from the base change $X \leftarrow X_0 = X \times_S S_0$. Then, if the family $(u_{\lambda 0})$ is separating (resp. universally separating relative to S_0), the family (u_λ) is separating (resp. universally separating relative to S).*

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Proof. The assertion relative to universally separating families is a consequence of the assertion relative to separating families. To prove the latter, one can (11.9.3) reduce to the case where $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$ are affine, $Z_\lambda = X$ for every λ , $\mathcal{F} = \tilde{M}$, $\mathcal{G}_\lambda = \tilde{N}_\lambda$, where M and the N_λ are B -modules, the N_λ being A -modules flat. The question being local on X and S , it suffices to verify that if $t \in M$ is such that $u_\lambda(t) = 0$ for every λ , then $t = 0$. One has $\mathcal{J} = \tilde{\mathfrak{J}}$, where $\mathfrak{J}$ is a nilpotent ideal of A . Suppose $\mathfrak{J}^{n+1} = 0$ and argue by induction on n , the assertion being trivially true for $n = 0$. If $\bar{t}$ is the class of t in $M / \mathfrak{J}^n M$, then, by the induction hypothesis, the class of $u_\lambda(t)$ in $N_\lambda / \mathfrak{J}^n N_\lambda$ is zero for every λ , hence $\bar{t} = 0$, that is $t \in \mathfrak{J}^n M$. But $\mathfrak{J}^n / \mathfrak{J}^{n+1}$ is a free $(A / \mathfrak{J})$ -module; if (e_α) is a base of this module, one can write $t = \sum_\alpha e_\alpha t_\alpha^0$, with $t_\alpha^0 \in A / \mathfrak{J}$. Since N_λ is an A -module flat, $\mathfrak{J}^n N_\lambda$ identifies with $\mathfrak{J}^n \otimes_{A/\mathfrak{J}} (N_\lambda / \mathfrak{J} N_\lambda)$ and one can write $u_\lambda(t) = \sum_\alpha e_\alpha \otimes u_{\lambda 0}(t_\alpha^0)$. Since by hypothesis $u_\lambda(t) = 0$, one deduces that $u_{\lambda 0}(t_\alpha^0) = 0$ for every α and every λ ; hence $t_\alpha^0 = 0$ since the family $(u_{\lambda 0})$ is separating, and consequently $t = 0$. $\square$

Theorem (11.9.16). — *The notations being those of (11.9.1), suppose that X is an S -prescheme locally Noetherian and that the $\mathcal{G}_\lambda$ are S -flat. For every $s \in S$, let $((u_\lambda)_s)_{\lambda \in L}$ be the family obtained from (u_λ) by the base change $X_s = X \times_S \operatorname{Spec}(\kappa(s)) \rightarrow X$. Then, for the family (u_λ) to be universally separating relative to S , it is necessary and sufficient that for every $s \in S$, the family $((u_\lambda)_s)$ is separating.*

Proof. The necessity of the condition follows trivially from the definitions. Conversely, suppose the condition of the statement verified, and let us prove first that the family (u_λ) is separating. One can (11.9.3) reduce to the case where $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$ are affine, $Z_\lambda = X$ for every λ , $\mathcal{F} = \tilde{M}$, $\mathcal{G}_\lambda = \tilde{N}_\lambda$, where B is Noetherian, M is a B -module of finite type and the N_λ are A -modules flat, and to show that, if $t \in M$ is such that $u_\lambda(t) = 0$ for every λ , then $t = 0$. To show that $t = 0$, it suffices to prove that for every maximal ideal $\mathfrak{p}$ of B , the image $t_{\mathfrak{p}}$ of t in $M_{\mathfrak{p}}$ is zero (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, cor. 1 du th. 1). One can thus also suppose that A is a local ring, and that M is a B -module of finite type, the intersection of $\mathfrak{m}^n M$ being reduced to 0 (**01**, 7.3.5), hence it suffices to prove that the family deduced from (u_λ) by the base change $\operatorname{Spec}(B / \mathfrak{m}^{n+1} B) \rightarrow \operatorname{Spec}(B)$ is separating, which since the maximal ideal $\mathfrak{m}$ is nilpotent, is precisely the conditions of application of (11.9.15), hence the conclusion.

Now let $h : S' \rightarrow S$ be a morphism base change, and (u'_λ) the family obtained from (u_λ) by the base change $h' : X \times_S S' \rightarrow X$; let us prove that (u'_λ) is also separating. Suppose first that h is *locally of finite type*; then $X \times_S S'$ is locally Noetherian; moreover, if $s' \in S'$ is above $s \in S$, then $X'_{s'} = X_s \otimes_{\kappa(s)} \kappa(s')$, and by (11.9.13) applied to X_s and to $X'_{s'} = X_s \times_{S_s} S'_{s'}$, the family $(u'_\lambda)_{s'}$ is separating; one can thus conclude by the first part of the proof that (u'_λ) is separating.

Finally, if h is arbitrary, one can evidently restrict to the case where $S = \operatorname{Spec}(A)$ and $S' = \operatorname{Spec}(A')$ are affine, and consider A' as the inductive limit of its sub- A -algebras of finite type. As the $\mathcal{G}_\lambda$ are S -flat, it suffices to apply the preceding argument and the lemma (11.9.12) to conclude. $\square$

Proposition (11.9.17). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation and f -flat, U an open of X , $j : U \rightarrow X$ the canonical injection, $u : \mathcal{F} \rightarrow j_*(j^*(\mathcal{F}))$ the canonical homomorphism (01, 4.4.3.2). For every $s \in S$, let X_s be the fibre $X \times_S \operatorname{Spec}(\kappa(s))$, U_s the open $U \cap X_s$ of X_s , $\mathcal{F}_s = \mathcal{F} \otimes_{\mathcal{O}_S} \kappa(s)$, $j_s : U_s \rightarrow X_s$ the canonical injection, $u_s : \mathcal{F}_s \rightarrow (j_s)_*((j_s)^*(\mathcal{F}_s))$ the canonical homomorphism. For u to be universally injective relative to S , it is necessary and sufficient that u_s be injective for every $s \in S$.*

Proof. It suffices to prove the sufficiency of the condition. When X is locally Noetherian, the proposition is an immediate corollary of (11.9.16). We shall reduce to this case in two steps, restricting ourselves, as one evidently can, to the case where $S = \operatorname{Spec}(A)$ and X are affine.

A) *Case where U is quasi-compact.* — We shall use the following lemma: $\square$

Lemma (11.9.17.1). — *Under the general hypotheses of (11.9.17), and in addition assuming S and X affine and U quasi-compact, the set E of $s \in S$ such that u_s is injective is constructible.*

Proof. Indeed, the fibres X_s are locally Noetherian preschemes, hence E can also, by (5.10.2), be defined as the set of $s \in S$ such that $\operatorname{Ass}(\mathcal{F}_s) \subset U_s$. Note in addition that U , being quasi-compact in an affine scheme, the verification of condition (9.2.1, (i)) follows also from (4.2.7); on the other hand, the verification of condition (9.2.1, (ii)) is done easily using the study of prime cycles associated to the neighbourhood of the generic fibre (9.8.3), as well as (9.5.2) and (9.5.3).

This lemma established, one can, by (8.9.1) and (8.2.11), suppose that there exists a Noetherian sub-ring A_0 of A , a prescheme of finite type X_0 over $S_0 = \operatorname{Spec}(A_0)$, an open U_0 of X_0 and an $\mathcal{O}_{X_0}$ -Module coherent $\mathcal{F}_0$ such that $X = X_0 \times_{S_0} S$ and $\mathcal{F} = p_0^*(\mathcal{F}_0)$, where $p_0 : X \rightarrow X_0$ is the projection. Let (A_α) be the filtering family of sub- A -algebras of A which are A_0 -algebras of finite type, and set $X_\alpha = X_0 \times_{S_0} S_\alpha$, $U_\alpha = U_0 \times_{S_0} S_\alpha$, $\mathcal{F}_\alpha = \mathcal{F}_0 \otimes_{S_0} S_\alpha$; let u_α the canonical homomorphism relative to $\mathcal{F}_\alpha$ and U_α , defined as in (11.9.17). For every $s \in S$, the hypothesis that u_s is injective implies also (11.9.10, (i)), where $s_\alpha = q_\alpha(s)$ is the image of s by the morphism $q_\alpha : S \rightarrow S_\alpha$. If E_α is the set of $s_\alpha \in S_\alpha$ such that $(u_\lambda)_{s_\alpha}$ is injective, one has $S = q_\alpha^{-1}(E_\alpha)$, and the E_α form a projective system of sets. But the lemma (11.9.17.1) applied to u_α , shows that E_α is constructible in S_α ; one deduces hence (8.3.4) that there exists an index α such that $E_\alpha = S_\alpha$. But then, since X_α is Noetherian, u_α is universally injective by (11.9.16), hence so is u . $\square$

(11.9.17.2). B) *General case.* — The open U is the union of a filtering increasing family of quasi-compact opens U_λ ; if $j_\lambda : U_\lambda \rightarrow X$ is the canonical injection and $u_\lambda : \mathcal{F} \rightarrow (j_\lambda)_*((j_\lambda)^*(\mathcal{F}))$ the corresponding homomorphism, the set E_λ of $s \in S$ such that $(u_\lambda)_s$ is injective is constructible. On the other hand, for a given $s \in S$, saying that u_s (resp. $(u_\lambda)_s$) is injective means that $\operatorname{Ass}(\mathcal{F}_s) \subset U \cap X_s$ (resp. $\operatorname{Ass}(\mathcal{F}_s) \subset U_\lambda \cap X_s$) (5.10.2). Since $\operatorname{Ass}(\mathcal{F}_s)$ is finite (3.1.6), saying that u_s is injective thus means that there exists λ such that $s \in E_\lambda$; the hypothesis thus implies $S = \bigcup_\lambda E_\lambda$. By (1.9.9), there exists an index λ such that $S = E_\lambda$, from which one concludes by the first part of the argument that u_λ is also universally injective. It follows then from the factorisation of u_λ :

$$\mathcal{F} \xrightarrow{u} j_*(\mathcal{F}|U) \longrightarrow (j_\lambda)_*(\mathcal{F}|U_\lambda)$$

that u is also universally injective.

Remarks (11.9.18). — Let X be a prescheme, $\mathcal{F}, \mathcal{G}$ two quasi-coherent $\mathcal{O}_X$ -Modules, $\mathcal{G}$ being in addition supposed *locally free*, $u : \mathcal{F} \rightarrow \mathcal{G}$ a homomorphism. The following conditions are equivalent:

label=a) for every morphism $g : X' \rightarrow X$, the homomorphism $g^*(u) : g^*(\mathcal{F}) \rightarrow g^*(\mathcal{G})$ is injective;

lbel=b) for every $x \in X$, the homomorphism $u \otimes 1_{\kappa(x)} : \mathcal{F} \otimes_{\mathcal{O}_X} \kappa(x) \rightarrow \mathcal{G} \otimes_{\mathcal{O}_X} \kappa(x)$ is injective;
 lbel=c) for every $x \in X$, there exists an open neighbourhood U of x such that $u|_U : \mathcal{F}|_U \rightarrow \mathcal{G}|_U$ is an isomorphism of $\mathcal{F}|_U$ onto a direct factor of $\mathcal{G}|_U$.

Indeed, it is clear that c) implies a) and that a) implies b). The fact that b) implies c) follows from (0, 19.1.12), (0_I, 5.2.5) and (0_I, 5.5.5). When the preceding equivalent conditions are verified, we say that u is *universally injective*.

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Proposition (11.10.1). — *Let X be a prescheme, $(Z_\lambda)_{\lambda \in L}$ a family of preschemes, and for every $\lambda \in L$, let $f_\lambda = (\psi_\lambda, \theta_\lambda) : Z_\lambda \rightarrow X$ be a morphism. The following conditions are equivalent:*

label=a) *The family of homomorphisms $\theta_\lambda : \mathcal{O}_X \rightarrow (f_\lambda)_*(\mathcal{O}_{Z_\lambda})$ is separating (11.9.1) (in other words the intersection of the kernels of the θ_λ is zero).*

lbel=b) *For every open U of X , every section t of $\mathcal{O}_X$ over U whose images by all the canonical homomorphisms*

$$(11.10.1.1) \quad (\theta_\lambda)_U : \Gamma(U, \mathcal{O}_X) \longrightarrow \Gamma(f_\lambda^{-1}(U), \mathcal{O}_{Z_\lambda})$$

are zero, is itself zero.

lbel=c) *For every open U of X , and every closed sub-prescheme Y of U such that for every $\lambda \in L$, there exists a factorisation*

$$(11.10.1.2) \quad f_\lambda^{-1}(U) \xrightarrow{g_\lambda} Y \xrightarrow{j} U$$

of the restriction $f_\lambda^{-1}(U) \rightarrow U$ of f_λ (where j is the canonical injection), one has $Y = U$.

If moreover X is an S -prescheme, these conditions are also equivalent to the following:

lbel=d) *For every separated morphism $g : X' \rightarrow S$ and every couple of S -morphisms u, v of an open U of X into X' , such that for every λ , the composites of u and v with the morphism $f_\lambda^{-1}(U) \rightarrow U$, restriction of f_λ , are equal, one has $u = v$.*

Proof. The equivalence of a) and b) follows from the definitions. To see that b) implies c), it suffices to consider the quasi-coherent Ideal $\mathcal{J}$ of $\mathcal{O}_U$ defining Y , and to note that, for every open $V \subset U$, the hypothesis implies that the morphism $(\theta_\lambda)_V$ factorises through

$$\Gamma(V, \mathcal{O}_U) \longrightarrow \Gamma(V, \mathcal{O}_U / \mathcal{J}) \longrightarrow \Gamma(f_\lambda^{-1}(V), \mathcal{O}_{Z_\lambda}).$$

One concludes that every section t of $\mathcal{J}$ over V has image 0 in all the $\Gamma(f_\lambda^{-1}(V), \mathcal{O}_{Z_\lambda})$, hence, by b), $\mathcal{J} = 0$ and $Y = U$. Conversely, if c) is verified, for proving b), it suffices, for every section t of $\mathcal{O}_U$ whose images by all the $(\theta_\lambda)_U$ are zero, to apply c) to the closed sub-prescheme Y of U defined by the Ideal $t\mathcal{O}_U$: the hypothesis that the images of t by the $(\theta_\lambda)_U$ are all zero implies that one has factorisations (11.10.1.2) for every λ (I, 4.1.9). To prove that c) implies d), it suffices to apply c) to the closed sub-prescheme Y of U , the image of the diagonal of $X' \times_S X'$ by $(u, v)_S$ and to use (I, 4.4.1). Conversely, to deduce b) from d), one considers the S -scheme $X' = S \otimes_{\mathbb{Z}} \mathbb{Z}[T]$ (T indeterminate) and one notes that the sections of $\mathcal{O}_U$ over U correspond bijectively to S -morphisms $U \rightarrow S[T]$ (I, 3.3.15); to say that two sections of $\mathcal{O}_U$ have the same images by all the $(\theta_\lambda)_U$ is equivalent to saying that the composites of the two corresponding morphisms with all the morphisms $f_\lambda^{-1}(U) \rightarrow U$ are equal. $\square$

Definition (11.10.2). — When the equivalent conditions of (11.10.1) are verified, we say that the family (f_λ) is *schematically dominant*. When the f_λ are the canonical immersions of a family (Z_λ) of sub-preschemes of X , we also say that the family (Z_λ) is *schematically dense*. IV-3 | 171

Remarks (11.10.3). — label=(i) The notion of schematically dominant family over X is *local* on X , as follows for example from condition b) in (11.10.1): if (W_α) is an open covering of X , the family (f_λ) is schematically dominant if and only if each of the families formed by the morphisms $f_\lambda^{-1}(W_\alpha) \rightarrow W_\alpha$, restrictions of f_λ , is.

lbel=(ii) If Z is the prescheme coproduct of the Z_λ , $f : Z \rightarrow X$ the morphism coinciding with f_λ on each Z_λ , saying that the family (f_λ) is schematically dominant is equivalent to saying that the family reduced to the single element f is schematically dominant.

- lcbel=(iii) Let M be a second index set and, for every $\lambda \in L$, let $(g_{\lambda\mu})_{\mu \in M}$ be a family of morphisms $g_{\lambda\mu} : Z'_{\lambda\mu} \rightarrow Z_\lambda$; if, for each λ , the family $(g_{\lambda\mu})_{\mu \in M}$ is schematically dominant, then it amounts to the same to say that the family $(f_\lambda)_{\lambda \in L}$ is schematically dominant, or that the family $(f_\lambda \circ g_{\lambda\mu})_{(\lambda,\mu) \in L \times M}$ is (11.9.2).
- ldbel=(iv) Let $f : Z \rightarrow X$ be a morphism such that $f_*(\mathcal{O}_Z)$ is a quasi-coherent $\mathcal{O}_X$ -Module (for example a quasi-compact and quasi-separated morphism (1.7.4)). Then, saying that f is schematically dominant means that the *scheme-theoretic image* of Z by f (I, 9.5.3) is identical to X .

Proposition (11.10.4). — *If the family of morphisms $f_\lambda : Z_\lambda \rightarrow X$ is schematically dominant, the union of the $f_\lambda(Z_\lambda)$ is dense in X . Conversely, if this union is dense in X and if X is reduced, the family (f_λ) is schematically dominant.*

Proof. The first assertion follows immediately from (11.10.1, b)). On the other hand, if X is reduced, and if the union of the $f_\lambda(Z_\lambda)$ is dense in X , then, if one has for every $\lambda \in L$ factorisations (11.10.1.2) for every $\lambda \in L$, one also has factorisations

$$(f_\lambda^{-1}(U))_{\text{red}} \xrightarrow{(g_\lambda)_{\text{red}}} Y_{\text{red}} \xrightarrow{j_{\text{red}}} U,$$

and the hypothesis implies that the underlying space of Y is identical to U , hence $Y = Y_{\text{red}} = U$ since U is reduced. $\square$

The results of n° 11.9 on separating families translate into results on schematically dominant families:

Theorem (11.10.5). — *Let (f_λ) be a family of morphisms $f_\lambda : Z_\lambda \rightarrow X$, $g : X' \rightarrow X$ a morphism, and set $Z'_\lambda = Z_\lambda \times_X X'$, $f'_\lambda = (f_\lambda)_{(X')} : Z'_\lambda \rightarrow X'$.*

- label=(i) *If g is faithfully flat and if the family (f'_λ) is schematically dominant, then the family (f_λ) is schematically dominant.*
- lbbel=(ii) *Suppose that g is a flat morphism and in addition that one of the two following conditions is verified:*
 label=a) *L is finite and the f_λ are quasi-compact.*
 lbbel=b) *The morphism g is locally of finite presentation.*
Then, if the family (f_λ) is schematically dominant, so is the family (f'_λ) .

Proof. This follows from (11.9.1) applied to the corresponding separating families, via the equivalence established in (11.10.1) and (11.10.3). $\square$

Proposition (11.10.6). — *The notations being those of (11.10.5), suppose that X is a prescheme over a field k , and that, setting $S = \text{Spec}(k)$, one has $X' = X \times_S S'$, where S' is an arbitrary k -prescheme. Then, if the family (f_λ) is schematically dominant, so is the family (f'_λ) .*

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Proof. This follows from (11.9.2) by the translation established in (11.10.1) and (11.10.3). $\square$

Corollary (11.10.7). — *Let X be a prescheme over a field k , $(Z_\lambda)_{\lambda \in L}$ a family of k -preschemes geometrically reduced over k , and for each $\lambda \in L$, let $f_\lambda : Z_\lambda \rightarrow X$ be a k -morphism. Let Y be the sub-prescheme reduced of X having for underlying space the adherence of $\bigcup_{\lambda \in L} f_\lambda(Z_\lambda)$. Let k' be an extension of k , $X', Z'_\lambda, f'_\lambda : Z'_\lambda \rightarrow X'$ the preschemes and morphisms deduced from X, Z_λ and f_λ by extension of the base to k' . Then, if Y' is the sub-prescheme reduced of X' having for underlying space the adherence of $\bigcup_{\lambda \in L} f'_\lambda(Z'_\lambda)$, one has $Y' = Y \otimes_k k'$. In particular, Y is geometrically reduced over k .*

Proof. As the Z_λ are reduced, the morphisms f_λ factorise as $Z_\lambda \xrightarrow{g_\lambda} Y \xrightarrow{j} X$, where j is the canonical injection (I, 5.2.2). It follows from (11.10.4) that (g_λ) is schematically dominant. Moreover, by hypothesis the Z'_λ are reduced; it follows from (11.10.6) that the family (g'_λ) is schematically dominant. Setting $Y'_1 = Y \otimes_k k'$, the closed sub-prescheme of X' , and since f'_λ factorises as $Z'_\lambda \rightarrow Y'_1 \rightarrow X'$, and since the g'_λ factorise through $Z'_\lambda \xrightarrow{g'_\lambda} Y'_1 \xrightarrow{j'} X'$ by (I, 5.2.2), we have $Y' \subset Y'_1$. Since the family (g'_λ) is schematically dominant, $Y' = Y'_1$, establishing the corollary. $\square$

Definition (11.10.8). — The notations being those of (11.10.5), suppose that X is an S -prescheme. We say that the family (f_λ) is *universally schematically dominant* (relative to S) if, for every base change

$S' \rightarrow S$, the family (f'_λ) corresponding to the base change $X' = X \times_S S' \rightarrow X$ is schematically dominant. When the f_λ are canonical immersions $Z_\lambda \hookrightarrow X$, we also say that the family of sub-preschemes Z_λ is *universally schematically dense in X (relative to S)*.

When S is the spectrum of a field, (11.10.6) thus means that a schematically dominant family is *universally schematically dominant (relative to S)*.

Theorem (11.10.9). — *The notations being those of (11.10.5), suppose that X is an S -prescheme locally Noetherian and that the Z_λ are S -flat. For every $s \in S$, let $X_s = X \times_S \text{Spec}(\kappa(s))$, $(Z_\lambda)_s = Z_\lambda \times_S \text{Spec}(\kappa(s))$, $(f_\lambda)_s = f_\lambda \times 1 : (Z_\lambda)_s \rightarrow X_s$. Then, for the family (f_λ) to be universally schematically dominant relative to S , it is necessary and sufficient that for every $s \in S$, the family $((f_\lambda)_s)$ is schematically dominant.*

Proof. This follows by translation from (11.9.6), via the equivalence of (11.10.1) and the characterisation of universally schematically dominant families given in (11.10.8). $\square$

Proposition (11.10.10). — *Let $f : X \rightarrow S$ be a morphism flat locally of finite presentation, U an open of X . For U to be universally schematically dense in X relative to S , it is necessary and sufficient that, for every $s \in S$, $U_s = U \cap X_s$ is schematically dense in X_s (or, what amounts to the same, that $\text{Ass}(\mathcal{O}_{X_s}) \subset U_s$).*

Proof. This is a particular case of (11.10.9) applied to the family consisting of the single canonical immersion $U \hookrightarrow X$, noting that U is S -flat since f is flat and U is open in X . $\square$

§12. STUDY OF FIBRES OF FINITELY PRESENTED FLAT MORPHISMS

12.1. Introduction We shall use throughout this paragraph the general notations of (9.4.1).

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(12.0.1). Given a morphism *locally of finite presentation* $f : X \rightarrow Y$, we have seen (9.9) that for certain *local* properties $\mathbf{P}$ of preschemes on corps or of Modules on these preschemes, the set E of $x \in X$ such that the property $\mathbf{P}$ holds, for the fibre $X_{f(x)}$, at the point x of this fibre, is *locally constructible in X* . We propose to show that, for most of these properties, if one furthermore supposes that the morphism f is *flat*, then E is even *open* in X . Similarly ((9.2) to (9.8)) we have shown that if f is of *finite presentation*, and if this time $\mathbf{P}$ denotes certain *global* properties of preschemes on corps or of Modules on these preschemes, the set F of $y \in Y$ such that the property $\mathbf{P}$ holds for the fibre X_y is *locally constructible in Y* . We shall show that if one furthermore supposes the morphism f to be *proper* and *flat*, F is even *open* in Y .

(12.0.2). The general method of proof of the properties in question comprises three steps. One first reduces to the case where Y is *affine* and X of finite presentation; then:

label=A) Using (8.9.1) (and possibly other results of § 8) and of (11.2.6), one reduces to the case where X and Y are *Noetherian*.

lbbel=B) One applies the results of § 9 recalled in (12.0.1) to prove that E is *constructible*.

lcbel=C) To see that E is open, it suffices, by (0_{III}, 9.2.5) to show that if $x \in E$, then every *generisation* x' of x also belongs to E . Using (II, 7.1.7), one sees, since Y is Noetherian, that there exists a spectrum of a *discrete valuation ring* Y_1 and a morphism $h : Y_1 \rightarrow X$ such that, if y_1 (resp. y'_1) is the closed point (resp. the generic point) of Y_1 , one has $h(y_1) = x$ and $h(y'_1) = x'$. One then performs the base change $g = f \circ h : Y_1 \rightarrow Y$; taking into account the results of §§ 4 and 6 on preschemes locally Noetherian on corps and on base changes of the base field, one is reduced to proving the assertion in question for a point x_1 of $X_1 = X \times_Y Y_1$ above y_1 and for a generisation x'_1 of x_1 above y'_1 . As $Y_1 = \text{Spec}(A)$, where A is a discrete valuation ring, y the closed point and y' the generic point of Y , the hypothesis that $\mathcal{F}$ is f -flat implies that t is $\mathcal{F}$ -regular (0_I, 6.3.4). One uses the results of (3.4) and (5.12) for this.

One proceeds similarly for the set $F \subset Y$.

12.2. Local properties of fibres of a morphism that is flat and locally of finite presentation

Theorem (12.1.1). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module f -flat and of finite presentation, Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$, k an integer. The following subsets of X are open:* IV-3 | 174

- label=(i) The set of $x \in X$ such that the dimensions of the prime cycles associated to $\mathcal{F}_{f(x)}$ and containing x are elements of Φ .*
- lbbel=(ii) The set of $x \in X$ such that the dimensions of the prime cycles associated to $\mathcal{F}_{f(x)}$ containing x all have dimension k .*
- lcbel=(iii) The set of $x \in X$ such that x does not belong to any immersed prime cycle associated to $\mathcal{F}_{f(x)}$.*
- ldbel=(iv) The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is equidimensional at the point x and possesses the property (S_k) at x (5.7.2).*
- lebel=(v) The set of $x \in X$ such that $\text{coprof}((\mathcal{F}_{f(x)})_x) \leq k$ (0, 16.4.9).*
- lfbel=(vi) The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is a Cohen-Macaulay $\mathcal{O}_{X_{f(x)}}$ -Module at the point x (5.7.1).*
- lgbel=(vii) The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is geometrically reduced at the point x (4.6.22).*
- lhbel=(viii) The set of $x \in X$ such that $\mathcal{F}_{f(x)}$ is geometrically punctually integral at the point x (4.6.22).*

Proof. The questions being local on X , one first reduces to the case where $X = \text{Spec}(A)$ and $Y = \text{Spec}(B)$ are affine, f a morphism of finite presentation. There is then a Noetherian sub-ring A_0 of A , a A_0 -prescheme of finite type X_0 , an $\mathcal{O}_{X_0}$ -Module coherent $\mathcal{F}_0$ such that $\mathcal{F}_0 \otimes_{A_0} A$ is isomorphic to $\mathcal{F}$ (8.9.1); in addition, by (11.2.6), one can suppose $\mathcal{F}_0$ is Y_0 -flat (with $Y_0 = \text{Spec}(A_0)$). If $h : X \rightarrow X_0$ is the canonical projection, the set E of points $x \in X$ where one of the properties (i) to (viii) is verified corresponds to the set E_0 where the corresponding property holds for $\mathcal{F}_0$: this follows for properties (i) to (iii) from (4.2.7); for properties (iv) to (vi) from (6.7.1); for properties (vii) and (viii) from (4.7.11).

One is thus reduced to the case where X and Y are Noetherian, and $\mathcal{F} = \tilde{M}$, where M is a B -module of finite type and the N_λ are A -modules flat. By (9.9.2), the set E is constructible for all the properties considered, and it remains in each case to verify step C) of (12.0.2).

(12.1.1.1) Let us begin with case (iii), the simplest. Replacing Y by the integral sub-scheme having $\{y'\}$ for underlying space, one can restrict to the case where Y is the spectrum of a discrete valuation ring, y the closed point and y' the generic point of Y . The hypothesis that $\mathcal{F}$ is f -flat implies that t is $\mathcal{F}$ -regular (0_I, 6.3.4). By hypothesis, none of the immersed prime cycles of $\mathcal{F}/t\mathcal{F}$ contains x . Then, it follows from (3.4.4) that x does not belong to any immersed prime cycle associated to $\mathcal{F}$, and it follows likewise for every generisation x' of x . In particular, x' does not belong to any of the immersed cycles associated to $\mathcal{F}_{y'}$ ($X_{y'}$ being a sub-prescheme induced on an open of X).

(12.1.1.2) Let us consider the cases (iv) and (v) ((vi) is a particular case of (v)). One proceeds as above (using (6.7.1) instead of (4.2.7)) and is again reduced to the case where Y is the spectrum of a discrete valuation ring. The ring $C = \mathcal{O}_{X,x}$ is then a localisation of an A -algebra of finite type, hence catenary (5.6.4), and by hypothesis the C -module M_x/tM_x verifies (S_k) ; one deduces from (5.12.2) (resp. from (0, 16.4.10)) that M_x verifies (S_k) and is equidimensional (resp. $\text{coprof}(M_x) \leq k$) since t is M_x -regular and belongs to the maximal ideal of C . Hence the conclusions since x' is a generisation of x .

(12.1.1.3) Let us pass to cases (vii) and (viii). One can evidently replace Y by the integral sub-scheme having $\{y'\}$ for underlying space, and X by $f^{-1}(\{y'\})$, without changing the fibres in y and y' , hence one can suppose A integral, of field of fractions $K = \kappa(y')$. There exists a finite extension K' of K such that, for the $(\mathcal{O}_X \otimes_A K')$ -Module $\mathcal{F}' = \mathcal{F} \otimes_A K'$, the prime cycles associated are geometrically irreducible and the geometric lengths of its support are respectively equal to the lengths of $\mathcal{F}'_{z'_i}$ at its maximal points z'_i of $\text{Supp}(\mathcal{F}')$ which are closed. It follows from (4.7.11) that $\mathcal{F} \otimes_A A'$ has property (vii) (resp. (viii)). Replacing A by A' and X by $X \otimes_A A'$, one sees that one can restrict to the case where A is a discrete valuation ring, the uniformiser t of A being M -regular. By hypothesis $\mathcal{F}/t\mathcal{F}$ is reduced (resp. integral) at the point x , it follows from (3.4.6) (resp. (3.4.5)) that $\mathcal{F}$ is reduced (resp. integral) at the point x , hence also in a neighbourhood of the point x , and in particular at the point x' , which completes the proof in cases (vii) and (viii).

(12.1.1.4) It remains to examine cases (i) and (ii). Replacing Y by the integral sub-scheme having $\{y'\}$ for underlying space, one can restrict to the case where Y is irreducible and y' its generic point. The detailed proof proceeds by reducing via (4.2.7) to the case where Y is a discrete valuation ring, and then applying the dimension theory from (5.1.2), (4.5.6), (4.7.9), and the flatness hypothesis on $\mathcal{F}$. We refer to the original for the complete argument. $\square$

Proposition (12.1.1.5). — *Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent and f -flat; for every $y \in Y$, set $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} \kappa(y)$. Let z' be a point of X , $y' = f(z')$, and suppose that $z' \in \text{Ass}(\mathcal{F}_{y'})$. Let Z' (resp. Y') be the sub-prescheme reduced of X (resp. Y) having for underlying space the adherence of $\{z'\}$ (resp. $\{y'\}$) in X (resp. Y). Then, for every $y \in f(Z')$, the dimensions of all the irreducible components of Z'_y are equal to $\dim(Z'_{y'})$ (dimension of the adherence of z' in $X_{y'}$) (which we shall express later (13.2.2) by saying that the restriction $Z' \rightarrow Y'$ of f is an equidimensional morphism); moreover, at every maximal point z of Z'_y , one has $z \in \text{Ass}(\mathcal{F}_y)$.*

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Proof. We shall reduce to the case where Y is the spectrum of a discrete valuation ring. Take Y_1 a spectrum of discrete valuation ring $h : Y_1 \rightarrow X$ a morphism such that, if y_1 and y'_1 are the closed point and the generic point of Y_1 respectively (II, 7.1.7), one has $h(y_1) = z'$, $h(y'_1) = y'$. Set $g = f \circ h : Y_1 \rightarrow Y$, $X_1 = X \times_Y Y_1$; let $f_1 : X_1 \rightarrow Y_1$, $g_1 : X_1 \rightarrow X$ be the projections canoniques; there is a Y_1 -section $h_1 : Y_1 \rightarrow X_1$ such that $u = g_1 \circ h_1$ (I, 3.3.14). If $Z'_1 = Z'_y \otimes_{\kappa(y')} \kappa(y'_1)$, one says (4.2.6) that the irreducible components of Z'_1 dominate $Z'_{y'}$; let z'_1 be the generic point of one of these components which contains $h_1(y'_1)$. As $h_1(y'_1)$ is a specialisation of z'_1 , it is *a fortiori* a specialisation of z'_1 ; let z_1 be a generic point of one of the irreducible components of $\{z'_1\} \cap (X_1)_{y_1}$ containing $h_1(y_1)$. Then $g_1(z_1)$ is a specialisation of $g_1(z'_1) = z'$ in X , and $g_1(z_1) \in X_{y'}$; but since $f(g_1(z_1)) = y$ and z is a generic point of Z'_y , we have necessarily $z = g_1(z_1)$.

Now suppose that $z_1 \in \text{Ass}((\mathcal{F}_1)_{y_1})$, where $\mathcal{F}_1 = \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_1}$. Since $\mathcal{F}$ is f -flat, t is $\mathcal{F}_1$ -regular by (0I, 6.3.4). By (3.4.3) we deduce that $z'_1 \in \text{Ass}(\mathcal{F}_{y'})$. As the points of $\text{Ass}(\mathcal{F}_y)$ are the z_i , this concludes the proof. $\square$

Remarks (12.1.2). — label=(i) Under the conditions of (12.1.1.5) with $\mathcal{F} = \mathcal{O}_X$ (so that f is flat), one cannot in general affirm that the restriction $Z' \rightarrow Y'$ of f is a flat morphism or even an open morphism. This is what the example ((6.5.5), (ii)) shows, where one takes for Z' one of the two irreducible components of $X = \text{Spec}(B)$. It is immediate that this restriction morphism is not open at the points of Z' above the “double point” of $Y' = Y$.

lbbel=(ii) Under the hypotheses of (12.1.1.5), with $\mathcal{F} = \mathcal{O}_X$, one cannot weaken the condition “ f is flat” to “ f is universally open” (cf. (2.4.6)), as we shall see later on an example ((14.4.10), (i)).

Corollary (12.1.3). — *Under the hypotheses of (12.1.1), the function $x \mapsto \text{coprof}((\mathcal{F}_{f(x)})_x)$ is upper semi-continuous in X and the function $x \mapsto \text{prof}_x^*(\mathcal{F}_{f(x)})$ (10.8.1) is lower semi-continuous in X .*

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Proof. The first assertion is none other than an equivalent formulation of ((12.1.1), (v)). To prove the second, one can first, taking into account (10.8.7) and (10.8.8), reduce as in (12.1.1) to the case where Y is the spectrum of a discrete valuation ring A with closed point y and generic point y' , $X = \text{Spec}(B)$ being affine, $\mathcal{F} = \tilde{M}$, where M is a B -module of finite type. Since $\mathcal{F}$ is f -flat, every irreducible component of $\text{Supp}(\mathcal{F})$ containing $x \in X_y$ dominates Y (2.3.4), that is to say its generic point z' belongs to $X_{y'}$; the irreducible components Z of $\text{Supp}(\mathcal{F})$ containing a generisation x' of x belonging to $X_{y'}$ are thus exactly those containing x , and moreover, by (12.1.1.5), we have $\dim(Z_{y'}) = \dim(Z_y)$, whence, if $T = \text{Supp}(\mathcal{F})$, $\dim_x(T_y) = \dim_{x'}(T_{y'})$. Moreover, for a uniformiser t of A , we saw in ((12.1.1), (v)) that $\text{coprof}(M_x) = \text{coprof}(M_x/tM_x)$, and since on the other hand $\text{coprof}(M_{x'}) \leq \text{coprof}(M_x)$ (6.11.5), we see that $\text{coprof}((\mathcal{F}_y)_x) \leq \text{coprof}((\mathcal{F}_{y'})_{z'})$, and the relation $\text{prof}_x^*(\mathcal{F}_y) \leq \text{prof}_{x'}^*(\mathcal{F}_{y'})$ then follows from (10.8.7). $\square$

Remark (12.1.4). — One also deduces from ((12.1.1), (i)) that the function $x \mapsto \dim_x(X_{f(x)})$ is upper semi-continuous in X , but we shall see later (13.1.3) that this property is true even without assuming f flat.

Corollary (12.1.5). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module and f -flat, $\mathcal{G}$ a coherent $\mathcal{O}_Y$ -Module, x a point of X , $y = f(x)$. Suppose that $\mathcal{G}$ has the property (S_k) at the point y , and that $\mathcal{F}_y$ has the property (S_k) at the point x and is equidimensional at the point x . Then:*

- label=(i) $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ possesses the property (S_k) at the point x .*
lbbel=(ii) If moreover Y is locally immersible in a regular scheme, or is an excellent prescheme (7.8.5), there exists a neighbourhood of x in X such that $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ has the property (S_k) in this neighbourhood.

Proof. Indeed, by ((12.1.1), (iv)), there exists an open neighbourhood V of x such that for every $x' \in V$, $\mathcal{F}_{f(x')}$ verifies (S_k) at the point x' . On the other hand, $\mathcal{G}$ verifies (S_k) at all points of $Y' = \text{Spec}(\mathcal{O}_y)$ (5.7.2). Replacing Y by Y' and X by $V \times_Y Y'$, one is reduced (taking into account (I, 3.6.5)) to the case where $\mathcal{G}$ verifies (S_k) in all of Y and where, for every $y \in f(X)$, $\mathcal{F}_y$ possesses the property (S_k) at all points of $f^{-1}(y)$; since $\mathcal{F}$ is f -flat, one then knows (6.4.1) that $\mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{G}$ possesses the property (S_k) in X , which proves (i).

To prove (ii), it suffices to observe that, under the hypotheses made, $\mathcal{G}$ possesses the property (S_k) in a neighbourhood U of y in Y ((6.11.2) and ((7.8.3), (iv))); one then concludes in the same way by replacing this time Y by U and X by $V \times_Y U$. $\square$

Theorem (12.1.6). — *Let $f : X \rightarrow Y$ be a morphism flat and locally of finite presentation, k an integer. The following subsets of X are open:* IV-3 | 178

- label=(i) The set of $x \in X$ such that $X_{f(x)}$ verifies the property (S_k) at the point x .*
lbbel=(ii) The set of $x \in X$ such that $X_{f(x)}$ verifies the property (R_k) geometric at the point x , is equidimensional and such that x does not belong to any immersed prime cycle associated to $X_{f(x)}$.
lcbel=(iii) The set of $x \in X$ such that $X_{f(x)}$ is geometrically regular (i.e. smooth over $\kappa(y)$) at the point x .
ldbel=(iv) The set of $x \in X$ such that $X_{f(x)}$ is geometrically normal at the point x .

Proof. Steps A) and B) of (12.0.2) are performed as in (12.1.1); for step A), one uses (6.7.8), as well as (4.2.7); for step B), one uses (9.9.2) and (9.9.4). It remains to examine step C) in each case.

For (iii), step C is unnecessary; as f is flat, one knows that (6.8.7) when Y is locally Noetherian and f locally of finite type, the set of $x \in X$ such that $X_{f(x)}$ is geometrically regular at the point x is open in X .

For (i), as in (12.1.1.2), one reduces to the case where Y is the spectrum of a discrete valuation ring A , where B is an A -algebra of finite type, x, x' two points of X such that $f(x) = y$ the closed point, $f(x') = y'$ the generic point of Y . The hypothesis that $\mathcal{F}$ is f -flat implies t is B -regular by flatness, and B/tB verifies (S_k) , hence by (5.12.4) B verifies (S_k) . As A is a universally catenary ring, this suffices to conclude (5.12.5).

For (ii), as in (12.1.1.3), to prove that the property considered is stable by generisation, one can, by considering a finite extension K' of $\kappa(y')$, and the invariance by change of the base field of the two other properties in (ii), replace in (ii) the property (R_k) geometric by the property (R_k) . Proceeding then as in (i), one reduces to the case where Y is the spectrum of a discrete valuation ring, and since A is catenary, it suffices to apply (5.12.5) to conclude.

For (iv), the set of points $x \in X$ such that $X_{f(x)}$ is geometrically normal at the point x is contained in the set of $x \in X$ such that $X_{f(x)}$ is geometrically punctually integral and verifies (S_2) at x , and this latter is open in X by (12.1.1, (viii)). One can thus already suppose that, for every $x \in X$, $X_{f(x)}$ is geometrically punctually integral and verifies (S_2) at x . By the Serre criterion (5.8.6), saying that $X_{f(x)}$ is geometrically normal at x amounts to $X_{f(x)}$ verifying (S_2) and the property (R_1) geometric at x ; by (ii) and the remarks preceding, this ensemble is intersection of two opens of X , hence is open in X . C.Q.F.D. $\square$

Corollary (12.1.7). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation. Then the set of points $x \in X$ where f possesses one of the following properties (6.8.1):* IV-3 | 179

- label=(i) to verify the property (S_k) ;*
lbbel=(ii) to be of coprofondueur $\leq n$;
lcbel=(iii) to be Cohen-Macaulay;
ldbel=(iv) to be regular (or smooth, which amounts to the same);

label=(v) to be normal;
lfbel=(vi) to be reduced;
 is open in X .

Proof. Indeed, it follows from (11.3.1) that the set of $x \in X$ where f is flat is open. One can thus restrict to the case where f is flat, and then the corollary results from (12.1.1, (iv), (vi) and (vii)) and from (12.1.6, (i), (ii) and (iv)). $\square$

Remarks (12.1.8). — *label*=(i) In (12.1.6, (ii)), one cannot suppress the hypothesis that x does not belong to any immersed prime cycle of $X_{f(x)}$. This is shown by the example (5.12.3), where one takes $X = \text{Spec}(A)$, Y the spectrum of the local ring A_0 of $k[T]$ at the prime ideal (T) and the morphism $X \rightarrow Y$ corresponding to the injection $A_0 \rightarrow A$, deduced by localisation and passage to the quotient of the injection $k[t_0, t_2] \rightarrow k[t_0, t_1, t_2, t_3]$.

lbbel=(ii) In the hypotheses of (12.1.1.5), with $\mathcal{F} = \mathcal{O}_X$ (so that f is flat), one cannot in general weaken the condition “ f is flat” to “ f is universally open” (cf. (2.4.6)), as we shall see later in an example (14.4.10, (i)).

12.3. Local and global properties of fibres of a proper, flat morphism of finite presentation

Theorem (12.2.1). — *Let $f : X \rightarrow Y$ be a morphism proper, flat and of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module f -flat and of finite presentation, Φ a finite subset of $\mathbf{Z} \cup \{\pm\infty\}$, k an integer. The following subsets of Y are open:*

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- label*=(i) The set of $y \in Y$ such that the dimensions of the prime cycles associated to $\mathcal{F}_y$ and containing x are elements of Φ .
- lbbel*=(ii) The set of $y \in Y$ such that the dimensions of the prime cycles associated to $\mathcal{F}_y$ containing x all have the same dimension k .
- lcbel*=(iii) The set of $y \in Y$ such that $\mathcal{F}_y$ has no immersed prime cycle.
- ldbel*=(iv) The set of $y \in Y$ such that $\mathcal{F}_y$ is equidimensional at every point x of X_y and possesses property (S_k) at x (5.7.2).
- lebel*=(v) The set of $y \in Y$ such that $\mathcal{F}_y$ possesses the property (S_k) and is equidimensional at every point of X_y .
- lfbel*=(vi) The set of $y \in Y$ such that $\text{coprof}(\mathcal{F}_y) \leq k$.
- lgbel*=(vii) The set of $y \in Y$ such that $\mathcal{F}_y$ is a Cohen-Macaulay Module.
- lhbel*=(viii) The set of $y \in Y$ such that $\mathcal{F}_y$ is geometrically reduced.
- libel*=(ix) The set of $y \in Y$ such that $\mathcal{F}_y$ is geometrically punctually integral at every point of X_y .
- ljbel*=(x) The set of $y \in Y$ such that $\mathcal{F}_y$ is geometrically integral.
- lkbel*=(xi) The set of $y \in Y$ such that $\mathcal{F}_y$ has no immersed prime cycle associated and that the sum $l(y)$ of the total multiplicities (4.7.12) of $\mathcal{F}_y$ at the maximal points of $\text{Supp}(\mathcal{F}_y)$ is $\leq k$.

Proof. With the exception of (ii), (iii), (iv), (x) and (xi), the properties $\mathbf{P}(y)$ considered are of the form: “for every $x \in X_y$, $\mathcal{F}_y$ has the property $\mathbf{Q}(x)$ ”, where $\mathbf{Q}(x)$ is one of the properties (i) to (viii) of (12.1.1). It follows from (12.1.1) that the set U of $x \in X$ such that $\mathbf{Q}(x)$ holds is open, and the set V of $y \in Y$ such that $\mathbf{P}(y)$ holds is none other than $Y - f(X - U)$; in all these cases, the theorem is thus already true by only supposing the morphism f closed (in particular proper).

For (iii), (i) with Φ reduced to a single element, it follows from (12.1.1) that the set U of $x \in X$ such that $\mathbf{Q}(x)$ holds is open, and by (10.8.1) the function $x \mapsto \dim(\text{Supp}(\mathcal{F}_{f(x)}))$ is semi-continuous inferiorly in X . The proof for the remaining cases (ii), (iv), (x), (xi) proceeds by reducing to the case where Y is Noetherian using steps A), B), C) of (12.0.2).

(12.2.1.1) Case (ii): The same reasoning as at the beginning shows, using (12.1.1, (iii)) that one can already suppose that for every $y \in Y$, $\mathcal{F}_y$ has no immersed associated prime cycle. It follows from the assertions in cases (i) and (ii).

(12.2.1.2) Case (iv): The same reasoning as at the beginning shows, using (12.1.1, (iii)) that one can already suppose (replacing Y by an open of Y) that for every $y \in Y$, $\mathcal{F}_y$ has no immersed associated prime cycle. The assertion of case (iv) is then an immediate consequence of cases (i) and (ii).

(12.2.1.3) Case (xi): For step A) of the reasoning, one uses (8.9.1), (11.2.6), (8.10.5, (xii)) (to preserve the hypothesis that f is proper), as well as (4.2.7), (4.5.6) and (4.7.9). For step B), one sees, as in case

(iv), that one can suppose that for every $y \in Y$, $\mathcal{F}_y$ has no immersed associated prime cycle, and one applies (9.8.8) which proves that the function $z \mapsto l(z)$ is constructible. It remains to verify (assuming Y Noetherian and f proper) that for every generisation y' of a point $y \in Y$, one has $l(y') \leq l(y)$. The proof uses a key lemma: $\square$

Lemma (12.2.1.4). — *Let Y be the spectrum of a discrete valuation ring, y its closed point, y' its generic point, $f : X \rightarrow Y$ a proper morphism, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent and f -flat, z_i (resp. z'_i) the maximal points of $\text{Supp}(\mathcal{F}_y)$ (resp. $\text{Supp}(\mathcal{F}_{y'})$). Suppose that $\mathcal{F}_{y'}$ has no immersed prime cycle associated. Then one has*

$$\sum_j \text{long}((\mathcal{F}_{y'})_{z'_j}) \leq \sum_i \text{long}((\mathcal{F}_y)_{z_i}).$$

Proof. We have $Y = \text{Spec}(A)$, where A is a discrete valuation ring; let t be a uniformiser, so that $\mathcal{F}_y = \mathcal{F}/t\mathcal{F}$. As $\mathcal{F}$ is f -flat, the z'_j are also the maximal points of $\text{Supp}(\mathcal{F})$ (2.3.4); for every z_i , denote by z'_{ik} those of the z'_j that are generisations of z_i ; it follows from (3.4.1.1) that

$$\text{long}((\mathcal{F}_y)_{z_i}) \geq \sum_k \text{long}((\mathcal{F}_{y'})_{z'_{ik}})$$

hence, summing over i ,

$$\sum_i \text{long}((\mathcal{F}_y)_{z_i}) \geq \sum_i \sum_k \text{long}((\mathcal{F}_{y'})_{z'_{ik}}).$$

The lemma will therefore be proved if we establish that for every z'_j , there is at least one index i such that z'_j is one of the z'_{ik} . But, as f is proper (hence closed) and $f(z'_j) = y'$, there exists $x \in X$ that is a specialisation of z'_j and such that $f(x) = y$, in other words $\overline{\{z'_j\}} \cap X_y$ is non-empty. As t is $\mathcal{F}$ -regular by flatness, it follows from (3.4.3) that there is at least one point of $\text{Ass}(\mathcal{F}_y)$ of which z'_j is a generisation; but as the points of $\text{Ass}(\mathcal{F}_y)$ are by hypothesis the z_i , this finishes the proof. $\square$

Corollary (12.2.2). — *Under the hypotheses of (12.2.1):*

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- label=(i) The function $y \mapsto \dim(\text{Supp}(\mathcal{F}_y))$ is continuous (hence locally constant) in Y .
- lbbel=(ii) The function $y \mapsto \text{coprof}(\mathcal{F}_y)$ (5.7.1) is semi-continuous superiorly in Y .
- lcbel=(iii) The function $y \mapsto l(y)$ (12.2.1), (xi) is semi-continuous superiorly in Y when the $\mathcal{F}_y$ have no immersed prime cycle associated.
- ldbel=(iv) The function $y \mapsto \text{prof}^*(\mathcal{F}_y)$ (10.8.1) is semi-continuous inferiorly in Y .

Proof. (i) follows from (12.2.1) applied to Φ equal to the smallest interval of $\mathbf{N}$ containing the dimensions of the prime cycles associated to $\mathcal{F}_y$, and noting that $z \mapsto \dim(\text{Supp}(\mathcal{F}_z))$ is semi-continuous superiorly at the point y ; it also follows from (12.2.1), (ii) applied to the set of dimensions of the irreducible components of $\text{Supp}(\mathcal{F}_y)$, that this function is also semi-continuous inferiorly at the point y , hence the conclusion. Assertions (ii) and (iii) are other formulations of (12.2.1), (vi) and (xi). Finally, from (12.1.3), the set of $x \in X$ such that $\text{prof}_x^*(\mathcal{F}_{f(x)}) \leq k$ is open, and the conclusion of (iv) follows by the same reasoning as at the beginning of (12.2.1). $\square$

Remarks (12.2.3). —

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- label=(i) For (12.2.1), (ii), one can dispense from the hypothesis that f is proper.
- lbbel=(ii) In (12.2.1), (ii), one cannot replace the condition that the set $E(y)$ of dimensions of irreducible components of $\text{Supp}(\mathcal{F}_y)$ be contained in Φ by the condition that it be equal to Φ .
- lcbel=(iii) The Remarks (ii) and (iii) show the necessity of including in (12.2.1), (iv)) the condition that $\mathcal{F}_y$ has no immersed prime cycle associated.
- ldbel=(iv) In (12.2.1), (xi), one cannot suppress the hypothesis that f be proper.
- lebel=(v) In (12.2.1), (xi), one cannot suppress the hypothesis that $\mathcal{F}_y$ has no immersed associated cycle.
- lfbel=(vi) We do not know if, in (12.2.1), (xi)), one can replace the hypothesis that $\mathcal{F}$ is f -flat by that $\text{Supp}(\mathcal{F}) = X$ and that f is universally open.

Theorem (12.2.4). — *Let $f : X \rightarrow Y$ be a morphism proper, flat and of finite presentation, k an integer ≥ 1 . The following subsets of Y are open:*

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- label=(i) The set of $y \in Y$ such that X_y possesses the property (S_k) .
 lbbel=(ii) The set of $y \in Y$ such that X_y verifies the property (R_k) geometric, is equidimensional at every point and has no immersed prime cycle associated to X_y .
 lcbel=(iii) The set of $y \in Y$ such that X_y is geometrically regular (i.e. smooth over $\kappa(y)$).
 ldbel=(iv) The set of $y \in Y$ such that X_y is geometrically normal.
 lebel=(v) The set of $y \in Y$ such that X_y is geometrically reduced.
 lfbel=(vi) The set of $y \in Y$ such that X_y is geometrically reduced and the number of geometric connected components of X_y over $\kappa(y)$ is equal to k .
 lgbel=(vii) The set of $y \in Y$ such that X_y is geometrically punctually integral (4.6.9).
 lhbel=(viii) The set of $y \in Y$ such that X_y is geometrically integral.
 libel=(ix) The set of $y \in Y$ such that X_y has no immersed prime cycle associated and that the total multiplicity (4.7.4) of X_y over $\kappa(y)$ is $\leq k$.

Proof. Except for (vi), these assertions are particular cases of assertions of (12.2.1), or are deduced from assertions of (12.1.6) as at the beginning of the proof of (12.2.1). For (vi), one reduces as always (taking into account the invariance of the geometric number of connected components by base change (4.5.6)) to the case where Y is Noetherian. The set U of $y \in Y$ such that X_y is geometrically reduced is open in Y by (v); it follows then from (III, 7.8.7 and 7.8.6) that there is a neighbourhood $V \subset U$ of y and an integer m such that, for every $z \in V$, $\Gamma(X_z, \mathcal{O}_{X_z})$ is isomorphic to $(\kappa(z))^m$; but by (III, 4.3.4), m is then the number of geometric connected components of X_z , hence the conclusion. One will give another proof of (vi) in (15.5.9). $\square$

12.4. Local cohomological properties of fibres of a flat morphism locally of finite presentation

Lemma (12.3.1). — *Let A be a ring, B a A -algebra of finite presentation which is an A -module flat, M a B -module of finite presentation, which is an A -module flat. Then the following conditions are equivalent:* IV-3 | 183

- label=a) M is a B -module projective.
 lbbel=b) M is a B -module flat.
 lcbel=c) For every $y \in \text{Spec}(A)$, $M \otimes_A \kappa(y)$ is a $(B \otimes_A \kappa(y))$ -module projective.

Proof. The equivalence of a) and b) follows from Bourbaki, *Alg. comm.*, chap. II, § 5, n° 2, cor. 2 du th. 1. As a) implies c) trivially, it remains to prove that c) implies b), which follows from the criterion of flatness by fibres (11.3.10), applied with $g = h$, $f = 1_X$. $\square$

Proposition (12.3.2). — *Let A be a ring, B a A -algebra of finite presentation which is an A -module flat, M a B -module of finite presentation which is an A -module flat. Then:* IV-3 | 184

- label=(i) There exists a left resolution of M by B -modules free of finite type.
 lbbel=(ii) One has

$$(12.3.2.1) \quad \dim. \text{proj}_B(M) = \sup_{y \in \text{Spec}(A)} \dim. \text{proj}_{B \otimes_A \kappa(y)}(M \otimes_A \kappa(y)) = \text{Tor. dim}_B(M)$$

where $\text{Tor. dim}_B(M)$ is the smallest integer i such that $\text{Tor}_j^B(M, N) = 0$ for every $j \geq i$ and every B -module N (and $+\infty$ if no such integer i exists).

Proof. (i) By (8.9.1), there exist a Noetherian sub-ring A_0 of A , an A_0 -algebra of finite type B_0 and a B_0 -module of finite type M_0 such that B is isomorphic to $B_0 \otimes_{A_0} A$ and M to $M_0 \otimes_{A_0} A$. Moreover by (11.2.7) one can suppose that B_0 and M_0 are A_0 -modules flat. There is then a left resolution $(L_0)_*$ of M_0 by B_0 -modules free of finite type, $(L_0)_* \otimes_{A_0} A$ is also a resolution of M formed by B -modules free of finite type (2.1.10).

(ii) By (0, 17.2.2, (ii)), on a $\text{Tor. dim}_B(M) \leq \dim. \text{proj}_B(M)$, and the definition of projective dimension proves immediately that, for every $y \in \text{Spec}(A)$, one has $\dim. \text{proj}_{B \otimes_A \kappa(y)}(M \otimes_A \kappa(y)) \leq \dim. \text{proj}_B(M)$. To prove the inverse inequalities, consider a left resolution (L_i) of M by B -modules free of finite type, and suppose that $\text{Tor. dim}_B(M) = n$ (resp. $\dim. \text{proj}_{B \otimes_A \kappa(y)}(M \otimes_A \kappa(y)) \leq n$ for every $y \in \text{Spec}(A)$). Then $R = \text{Im}(L_n \rightarrow L_{n-1}) = \text{Ker}(L_{n-1} \rightarrow L_{n-2})$ is also a B -module of finite type which is also an A -module flat, by the hypothesis on M and by (2.1.10). Moreover, one has $\text{Tor}_{n+1}^B(M, N) = \text{Tor}_1^B(R, N)$ for every B -module N (M, V, 7). The hypothesis $\text{Tor. dim}_B(M) = n$ implies $\text{Tor}_1^B(R, N) = 0$ for every B -module N , i.e. R is a B -module flat, hence projective by (12.3.1);

this establishes that $\dim. \text{proj}_B(M) \leq n$. Similarly, the hypothesis $\dim. \text{proj}_{B \otimes_A \kappa(y)}(M \otimes_A \kappa(y)) \leq n$ for every $y \in \text{Spec}(A)$ implies, by (2.1.10), that in each of the suites (exact by the flatness of A on M , of the L_i and of R)

$$0 \longrightarrow R \otimes_A \kappa(y) \longrightarrow L_{n-1} \otimes_A \kappa(y) \longrightarrow \dots \longrightarrow L_0 \otimes_A \kappa(y) \longrightarrow M \otimes_A \kappa(y) \longrightarrow 0$$

$R \otimes_A \kappa(y)$ is a $(B \otimes_A \kappa(y))$ -module projective (for every $y \in \text{Spec}(A)$). One concludes then also that R is a B -module projective, hence $\dim. \text{proj}_B(M) \leq n$. $\square$

Proposition (12.3.3). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, $\mathcal{L}_\bullet$ a complex of $\mathcal{O}_X$ -Modules quasi-coherent of finite presentation; for every $s \in S$, let $(\mathcal{L}_\bullet)_s$ the complex $\mathcal{L}_\bullet \otimes_{\mathcal{O}_S} \kappa(s)$ of $\mathcal{O}_{X_s}$ -Modules of finite type. Suppose that $\mathcal{L}_{n-1}$ is f -flat. Then the set U of $x \in X$ such that $(\mathcal{H}_n((\mathcal{L}_\bullet)_{f(x)}))_x = 0$ is open in X . If moreover $\mathcal{L}_n$ is f -flat, then on a $\mathcal{H}_n(\mathcal{L}_\bullet)|U = 0$.*

Proof. One can evidently restrict to the case where $n = 1$ and where the complex $\mathcal{L}_\bullet$ reduces to $0 \rightarrow \mathcal{L}_2 \xrightarrow{u} \mathcal{L}_1 \xrightarrow{v} \mathcal{L}_0 \rightarrow 0$. One can first reduce to the case where $S = \text{Spec}(A)$ and X are affine, then to the case where S is Noetherian; indeed, by (8.9.1) and (8.5.2), one knows that there exists a Noetherian prescheme $S' = \text{Spec}(A')$, where A' is a sub-ring of A , a morphism of finite type $f' : X' \rightarrow S'$ and three coherent $\mathcal{O}_{X'}$ -Modules $\mathcal{L}'_i$ ($0 \leq i \leq 2$), $f' = f'_{(S')}$, $\mathcal{L}'_i = \mathcal{L}'_i \otimes_{S'} S$, and two homomorphisms u', v' such that $u = u' \otimes 1$, $v = v' \otimes 1$ and $v' \circ u' = 0$. One can also suppose that $\mathcal{L}'_0$ is f' -flat (11.2.7), and that $\mathcal{L}'_1$ is f' -flat. By faithful flatness, the hypothesis $(\mathcal{H}_1((\mathcal{L}_\bullet)_{f(x)}))_x = 0$ is equivalent to $(\mathcal{H}_1((\mathcal{L}'_\bullet)_{f'(x')}))_{x'} = 0$, where x' is the projection of x into X' ; if U' is the set of the $x' \in X'$ such that $(\mathcal{H}_1((\mathcal{L}'_\bullet)_{f'(x')}))_{x'} = 0$, U is thus the reciprocal image of U' by the projection $X \rightarrow X'$, which reduces us to the first assertion for the case where S is Noetherian.

For the second assertion, one notes also that A is the inductive limit of its sub- A' -algebras of finite type; setting $X_\lambda = X' \times_{S'} S_\lambda$, $\mathcal{L}'_i^{(\lambda)} = \mathcal{L}'_i \otimes_{S'} S_\lambda$, and u_λ, v_λ the homomorphisms obtained from $u_\lambda = u' \otimes 1 : \mathcal{L}'_2^{(\lambda)} \rightarrow \mathcal{L}'_1^{(\lambda)}$, $v_\lambda = v' \otimes 1 : \mathcal{L}'_1^{(\lambda)} \rightarrow \mathcal{L}'_0^{(\lambda)}$, so that one has $\mathcal{H}_1(\mathcal{L}_\bullet^{(\lambda)}) = \text{Ker}(v_\lambda) / \text{Im}(u_\lambda)$. Since in the category of commutative groups, the $\varinjlim$ functor is exact, one has $\text{Ker}(v) = \varinjlim (\text{Ker}(v_\lambda))$, $\text{Im}(u) = \varinjlim (\text{Im}(u_\lambda))$, and $\mathcal{H}_1(\mathcal{L}_\bullet) = \varinjlim \mathcal{H}_1(\mathcal{L}_\bullet^{(\lambda)})$. If one has supposed that $\mathcal{L}_1$ is f -flat and (reducing to the case $X = U$) that $\mathcal{H}_1(\mathcal{L}_\bullet^{(\lambda)}) = 0$ for every λ , one deduces $\mathcal{H}_1(\mathcal{L}_\bullet) = 0$.

I) Now suppose S Noetherian. One knows (without flatness hypothesis on $\mathcal{L}_0$) that the set U is constructible in X (9.9.6). Using now (12.0.2), the method described in (12.0.2) also applies without change (taking into account the fact that for a prescheme Z over a corps k and an extension k' of k , the projection $Z \otimes_k k' \rightarrow Z$ is faithfully flat). One can thus suppose that S is the spectrum of a discrete valuation ring A , of which we designate by t a uniformiser, x being above the closed point of S and x' above the generic point of S . The hypothesis that $\mathcal{L}_0$ is f -flat implies that t is $\mathcal{L}_0$ -regular ($\mathbf{0}_I$, 6.3.4). One is then reduced to proving the following lemma (where B, M, N, P are replaced by $\mathcal{O}_x, (\mathcal{L}_2)_x, (\mathcal{L}_1)_x$ and $(\mathcal{L}_0)_x$ respectively): $\square$

Lemma (12.3.3.1). — *Let B be a local Noetherian ring, M, N, P three B -modules, N being supposed of finite type, $u : M \rightarrow N$, $v : N \rightarrow P$ two homomorphisms such that $v \circ u = 0$. Let t be an element of the maximal ideal of B such that t is P -regular and such that the sequence*

$$(12.3.3.2) \quad M/tM \xrightarrow{u \otimes 1_{B/tB}} N/tN \xrightarrow{v \otimes 1_{B/tB}} P/tP$$

is exact. Then the sequence $M \xrightarrow{u} N \xrightarrow{v} P$ is exact.

Proof. Note first that if one replaces M by its image Q in N and u by the injection $j : Q \rightarrow N$, the image of $j \otimes 1$ is the same as that of $u \otimes 1$, so one can, without changing the hypothesis or the conclusion, suppose u injective. On the other hand, if R is the image of v and $p : N \rightarrow R$ the canonical surjection, t is evidently R -regular, and the kernel of $p \otimes 1$ is contained in that of $v \otimes 1$; it is therefore equal to it if the sequence (12.3.3.2) is exact; as on the other hand one evidently has $\text{Ker}(p) = \text{Ker}(v)$, one sees that one can, for proving the lemma, suppose in addition v surjective. The lemma will then be a consequence of the following: $\square$

Lemma (12.3.3.3). — *Let B be a ring, M, N, P three B -modules, $u : M \rightarrow N$, $v : N \rightarrow P$ two homomorphisms such that u is injective, v surjective and $v \circ u = 0$. Let t be an element of B such that t is P -regular. Then on a*

$$(12.3.3.4) \quad \text{Ker}(v \otimes 1) / \text{Im}(u \otimes 1) = (\text{Ker}(v) / \text{Im}(u)) \otimes_B (B/tB)$$

as a canonical isomorphism.

Proof. Indeed, the hypothesis that the sequence (12.3.3.2) is exact will then imply that $(\text{Ker } v / \text{Im } u) \otimes_B (B/tB) = 0$, and as t belongs to the maximal ideal of B and $\text{Ker}(v) / \text{Im}(u)$ is a B -module of finite type (since B is supposed Noetherian in (12.3.3.1)), Nakayama's lemma proves that $\text{Ker}(v) = \text{Im}(u)$. It remains therefore to prove (12.3.3.3). Set $u' = u \otimes 1$, $v' = v \otimes 1$, $H = Z / \text{Im}(u)$, $Z = \text{Ker}(v)$, so that one has the exact sequences

$$0 \longrightarrow M \longrightarrow Z \longrightarrow H \longrightarrow 0$$

$$0 \longrightarrow Z \longrightarrow N \xrightarrow{v} P \longrightarrow 0$$

hence, tensoring by B/tB and using the fact that t is P -regular, the exact sequences

$$M/tM \xrightarrow{w'} Z/tZ \longrightarrow H/tH \longrightarrow 0$$

$$0 \longrightarrow Z/tZ \longrightarrow N/tN \xrightarrow{v'} P/tP \longrightarrow 0$$

hence $\text{Ker}(v') = Z/tZ$, and as $\text{Im}(w') = \text{Im}(u')$, one obtains (12.3.3.4). $\square$

(12.3.3.5). II) Suppose now also that $\mathcal{L}_1$ is f -flat; replacing eventually X by U , one can suppose $X = U$. It suffices to show that for every $x \in X$, one has $(\mathcal{H}_1(\mathcal{L}_\bullet))_x = 0$. Let $s = f(x)$, and let $\mathfrak{n}$ be the ideal $\mathfrak{m}_s \mathcal{O}_{X,x}$, which is contained in the maximal ideal of $\mathcal{O}_{X,x}$; as $(\mathcal{H}_1(\mathcal{L}_\bullet))_x$ is an $\mathcal{O}_{X,x}$ -module of finite type (X being locally Noetherian), it is separated for the $\mathfrak{n}$ -adic topology (0_I, 7.3.5), hence it suffices to show that its separated completion for this topology is 0. Now, by (III, 7.4.7.2), this separated completion is equal to $\varprojlim \mathcal{H}_1(\mathcal{L}_\bullet \otimes_{\mathcal{O}_S} (\mathcal{O}_{S,s} / \mathfrak{m}_s^{k+1}))$, and it suffices hence to prove that each of the terms of this projective system is zero. But this is true by hypothesis for $k = 0$; the conclusion follows by induction on k . The conclusion will follow from the following more general lemma (which one will apply for $A = \mathcal{O}_{S,s} / \mathfrak{m}_s^{k+1}$ and $\mathfrak{J}$ equal to the maximal ideal $\mathfrak{m}_s / \mathfrak{m}_s^{k+1}$ of A):

Lemma (12.3.3.5). — *Let A be a ring, $\mathfrak{J}$ a nilpotent ideal of A , $L_\bullet : P \xrightarrow{u} Q \xrightarrow{v} R$ a complex of A -modules. Set $A_0 = A / \mathfrak{J}$, $L_\bullet^{(0)} = L_\bullet \otimes_A A_0 : P_0 \xrightarrow{u_0} Q_0 \xrightarrow{v_0} R_0$, and suppose that $\text{gr}_{\mathfrak{J}}^*(A)$ is an A_0 -module flat, and that Q and R are A -modules flat. Then the relation $H_1(L_\bullet^{(0)}) = 0$ implies $H_1(L_\bullet) = 0$.*

Proof. Set $L_\bullet^{(k)} = L_\bullet \otimes_A (A / \mathfrak{J}^{k+1}) : P_k \xrightarrow{u_k} Q_k \xrightarrow{v_k} R_k$; as there exists an integer $k \geq 1$ such that $L_\bullet^{(k)} = L_\bullet$, it will suffice to prove, by induction on k , that the sequence $P_k \xrightarrow{u_k} Q_k \xrightarrow{v_k} R_k$ is exact (which one applies for $A = \mathcal{O}_{S,s} / \mathfrak{m}_s^{k+1}$ and $\mathfrak{J}$ equal to the maximal ideal $\mathfrak{m}_s / \mathfrak{m}_s^{k+1}$ of A). Let $x \in Q_k$ be an element such that $v_k(x) = 0$. By the induction hypothesis, the canonical image of x in $Q_{k-1} = Q_k / (\mathfrak{J}^k Q / \mathfrak{J}^{k+1} Q)$ belongs to $\text{Im}(u_{k-1})$, hence there exists $z \in P_k$ such that $x = u_k(z) + x'$ with $x' \in \mathfrak{J}^k Q / \mathfrak{J}^{k+1} Q$. Since $v_k \circ u_k = 0$, the relation $v_k(x) = 0$ implies $v_k(x') = 0$, and the $v_k(x') \in \mathfrak{J}^k R / \mathfrak{J}^{k+1} R$; but by hypothesis, $\mathfrak{J}^k / \mathfrak{J}^{k+1}$ is an $(A / \mathfrak{J})$ -module flat, hence the suite

$$\mathfrak{J}^k P / \mathfrak{J}^{k+1} P \xrightarrow{u''} \mathfrak{J}^k Q / \mathfrak{J}^{k+1} Q \xrightarrow{v''} \mathfrak{J}^k R / \mathfrak{J}^{k+1} R$$

(where u' and v' come from u and v by restriction and passage to quotients) is exact. Now, by hypothesis, $\mathfrak{J}^k / \mathfrak{J}^{k+1}$ is an $(A / \mathfrak{J})$ -module flat, hence the suite $(Q / \mathfrak{J} Q) \otimes_{A / \mathfrak{J}} (\mathfrak{J}^k / \mathfrak{J}^{k+1})$ identifies with $\mathfrak{J}^k Q / \mathfrak{J}^{k+1} Q$ by the flatness of Q on A , and similarly $(R / \mathfrak{J} R) \otimes_{A / \mathfrak{J}} (\mathfrak{J}^k / \mathfrak{J}^{k+1})$ identifies with $\mathfrak{J}^k R / \mathfrak{J}^{k+1} R$. The image of u' then identifies with that of u'' , and this concludes the proof. $\square$

Corollary (12.3.4). — *Let $f : X \rightarrow S$ be a morphism flat locally of finite presentation, $\mathcal{F}, \mathcal{G}$ two $\mathcal{O}_X$ -Modules of finite presentation and f -flat. Let n be an integer ≥ 0 , U the set of $x \in X$ such that $(\text{Ext}_{\mathcal{O}_{X,f(x)}}^n(\mathcal{F}_{f(x)}, \mathcal{G}_{f(x)}))_x = 0$ (resp. $(\text{Tor}_n^{\mathcal{O}_{X,f(x)}}(\mathcal{F}_{f(x)}, \mathcal{G}_{f(x)}))_x = 0$). Then U is open, and one has $\text{Ext}_{\mathcal{O}_X}^n(\mathcal{F}, \mathcal{G})|_U = 0$ (resp. $\text{Tor}_n^{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})|_U = 0$).*

Proof. One can evidently restrict to the case where S and X are affine. Then (12.3.2) gives a left resolution $\mathcal{L}_\bullet = (\mathcal{L}_i)$ of $\mathcal{F}$ formed by $\mathcal{O}_X$ -Modules free of finite type. If one sets $\mathcal{M}^\bullet = \mathcal{H}om(\mathcal{L}_\bullet, \mathcal{G})$ (resp. $\mathcal{N}_\bullet = \mathcal{L}_\bullet \otimes_{\mathcal{O}_X} \mathcal{G}$), one deduces that each $\mathcal{M}^i$ (resp. $\mathcal{N}_i$) is isomorphic to an $\mathcal{O}_X$ -Module of the form $\mathcal{G}^m$, hence the $\mathcal{M}^i$ and $\mathcal{N}_i$ are $\mathcal{O}_X$ -Modules of finite presentation and f -flat. Moreover, for every base change $S' \rightarrow S$, if one sets $X' = X \times_S S'$, $\mathcal{F}' = \mathcal{F} \otimes_S S'$, $\mathcal{G}' = \mathcal{G} \otimes_S S'$, $\mathcal{L}'_\bullet = \mathcal{L}_\bullet \otimes_S S'$ is still a left resolution of $\mathcal{F}'$ by $\mathcal{O}_{X'}$ -Modules free of finite type (2.1.10), and $\mathcal{M}'^\bullet = \mathcal{M}^\bullet \otimes_S S'$ is equal to $\mathcal{H}om(\mathcal{L}'_\bullet, \mathcal{G}')$ (resp. $\mathcal{N}'_\bullet = \mathcal{N}_\bullet \otimes_S S'$ is equal to $\mathcal{L}'_\bullet \otimes_{\mathcal{O}_{X'}} \mathcal{G}'$) from what precedes. In particular, for every $s \in S$, $\mathcal{E}xt_{X_s}^n(\mathcal{F}_s, \mathcal{G}_s) = \mathcal{H}^n((\mathcal{M}^\bullet)_s)$ (resp. $\mathcal{T}or_n^{X_s}(\mathcal{F}_s, \mathcal{G}_s) = \mathcal{H}_n((\mathcal{N}_\bullet)_s)$). Applying (12.3.3) to the f -flat complexes $\mathcal{M}^\bullet$ and $\mathcal{N}_\bullet$, one deduces immediately the corollary. $\square$

§13. EQUIDIMENSIONAL MORPHISMS

This paragraph is devoted to the study of the variation of the *dimension* of the fibres of a morphism locally of finite type $f : X \rightarrow Y$ (which has already appeared in connection with the “dimension formula” in 5.5 and 5.6). One first proves (13.1.3) that the function $x \mapsto \dim_x f^{-1}(f(x))$ is always *semi-continuous superiorly* in X (Chevalley’s theorem of semi-continuity). One then studies more particularly the morphisms called “equidimensional”, for which this function is *locally constant*. Unfortunately, the notion of equidimensional morphism is not stable by base change; this is why in many questions it is more convenient to work with the notion of universally open morphism, which is the object of §§ 14 and 15.

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13.1. Chevalley’s theorem of semi-continuity

Lemma (13.1.1). — *Let Y be an irreducible locally Noetherian prescheme, X an irreducible prescheme, $f : X \rightarrow Y$ a dominant morphism of finite type. Let ξ (resp. η) be the generic point of X (resp. Y) and let $e = \dim(f^{-1}(\eta))$. Then, for every $x \in X$, all the irreducible components of $f^{-1}(f(x))$ are of dimension $\geq e$.*

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Proof. The proposition is immediate when, for every $y \in f(X)$, $\mathcal{O}_y$ is a *universally catenary* ring: indeed, if x is the generic point of an irreducible component Z of $f^{-1}(y)$, it follows from (5.6.5), joined with (5.2.1), that $e + \dim(\mathcal{O}_y) = \dim(Z) + \dim(\mathcal{O}_x)$, but by (0, 16.3.9), one has $\dim(\mathcal{O}_x) \leq \dim(\mathcal{O}_y)$, hence the conclusion in this case.

We shall reduce the general case to this particular case. The question is evidently local on Y and, taking into account (4.1.1.3), it is also local on X ; one can thus restrict to the case where $Y = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affine and irreducible, B being an A -algebra of finite type. Moreover (1.5.4), one can suppose X and Y reduced, hence A and B integral. Consider A as inductive limit of its sub- $\mathbf{Z}$ -algebras of finite type; it follows then from (8.9.1) that there exist such a sub-algebra A_0 and an A_0 -algebra of finite type B_0 such that $B = B_0 \otimes_{A_0} A$. Set $Y_0 = \text{Spec}(A_0)$, $X_0 = \text{Spec}(B_0)$, and let $f_0 : X_0 \rightarrow Y_0$ be the morphism corresponding to $A_0 \rightarrow B_0$, so that $f = f_0 \times 1_Y$. It is not *a priori* evident that the prescheme X_0 be *integral*, but we shall now see that one can reduce to this case. $\square$

Lemma (13.1.2). — *Let Y_0, Y be two preschemes integral of generic points η_0, η , $g : Y \rightarrow Y_0$ a dominant morphism, $f_0 : X_0 \rightarrow Y_0$ a dominant morphism such that $f_0^{-1}(\eta_0)$ is irreducible. Let X'_0 be the unique irreducible component of X_0 meeting $f_0^{-1}(\eta_0)$ (0I, 2.1.8), and designate again by X'_0 the reduced closed sub-prescheme of X_0 having X'_0 for underlying space. Suppose that the prescheme $X = X'_0 \times_{Y_0} Y$ is integral; then X is isomorphic to $X'_0 \times_{Y_0} Y$.*

Proof. Indeed, if $j_0 : X'_0 \rightarrow X_0$ is the canonical injection, which is a closed immersion, $j = j_0 \times 1_Y : X'_0 \times_{Y_0} Y \rightarrow X_0 \times_{Y_0} Y = X$ is a closed immersion. On the other hand, X'_0 contains the fibre $f_0^{-1}(\eta)$; note on the other hand that $f^{-1}(\eta)$ is not empty (I, 3.4.7), hence contains the generic point of X ; hence the image of j necessarily contains X entirely. But as X is integral, the only closed sub-prescheme of X having X for underlying space is X itself, hence j is an isomorphism. $\square$

This lemma being established, one can thus suppose that X_0 is integral; for every $y_0 \in Y_0$, $\mathcal{O}_{y_0}$ is a $\mathbf{Z}$ -algebra essentially of finite type, hence a *universally catenary* ring (5.6.4); hence, for every $y_0 \in f_0(X_0)$, all the irreducible components of $f_0^{-1}(y_0)$ have dimension $\geq e$ by transitivity of the fibres (4.1.4). For every $y \in f(X)$, one has $y_0 = g(y) \in f_0(X_0)$ and by transitivity of the fibres and (4.2.7), the irreducible components of $f^{-1}(y)$ are the irreducible components of $f_0^{-1}(y_0)$, where

Z_0 ranges over the irreducible components of $f_0^{-1}(y_0)$, and each of the irreducible components of $p^{-1}(Z_0)$ dominates Z_0 and has dimension equal to $\dim(Z_0)$. Taking into account (0, 14.1.5), one sees that $\dim_x(f^{-1}(f(x))) = \dim_{x_0}(f_0^{-1}(f_0(x_0)))$, where $F_n(X) = g'^{-1}(F_n(X_0))$, and one is thus reduced to proving the theorem when Y is *Noetherian*, which we shall now suppose.

Theorem (13.1.3). — (Chevalley). *Let $f : X \rightarrow Y$ be a morphism locally of finite type. For every integer n , the set $F_n(X)$ of $x \in X$ such that $\dim_x(f^{-1}(f(x))) \geq n$ is closed; in other words, the function $x \mapsto \dim_x(f^{-1}(f(x)))$ is semi-continuous superiorly in X .*

Proof. I) Suppose first that f is locally of finite presentation. The question is evidently local on X and on Y , and one can thus suppose $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$ affine, B being an A -algebra of finite presentation. Then by (8.9.1) there is a Noetherian sub-ring A_0 of A , and an A_0 -algebra of finite type B_0 such that $B = B_0 \otimes_{A_0} A$, so $X = X_0 \times_{Y_0} Y$, $f = f_0 \times 1_Y$. Let $g : Y \rightarrow Y_0$ be the morphism corresponding to $A_0 \rightarrow A$, y a point of Y , $y_0 = g(y)$; one knows that $f^{-1}(y) = f_0^{-1}(y_0) \otimes_{\kappa(y_0)} \kappa(y)$, and it follows from (4.2.7) that if p is the canonical projection $f^{-1}(y) \rightarrow f_0^{-1}(y_0)$, the irreducible components of $f^{-1}(y)$ are the irreducible components of $p^{-1}(Z_0)$, where Z_0 ranges over the irreducible components of $f_0^{-1}(y_0)$, and each has dimension $\dim(Z_0)$. Taking into account (0, 14.1.5), one sees thus that if $g' : X \rightarrow X_0$ is the canonical projection, x a point of X and $x_0 = g'(x)$, one has $\dim_x(f^{-1}(f(x))) = \dim_{x_0}(f_0^{-1}(f_0(x_0)))$, hence $F_n(X) = g'^{-1}(F_n(X_0))$, and one is thus reduced to proving the theorem when Y is *Noetherian*, which we shall now suppose.

One can evidently suppose X and Y reduced (1.5.4). Considering the set of closed parts Y' of Y such that the theorem holds for the closed sub-prescheme having Y' for underlying space and for $X' = f^{-1}(Y')$, one can argue by Noetherian induction (0I, 2.2.2), and suppose that the theorem is true for every closed part $Y' \neq Y$ of Y . Let X_i ($1 \leq i \leq m$) be the closed sub-preschemes which are the reduced irreducible components of X ; by (0, 14.1.5), $F_n(X) = \bigcup_i F_n(X_i)$ and one can thus restrict to the case where X is *irreducible*. If Z is the closed sub-prescheme of Y having $\overline{f(X)}$ for underlying space, f factorises as $X \xrightarrow{f'} Z \xrightarrow{j} Y$, where j is the canonical injection, and f' is of finite type (1.5.4), hence it suffices to prove the theorem for Z and g ; by the induction hypothesis, one is thus reduced to the case where $Z = Y$, i.e. Y is irreducible and f dominant. Let η be the generic point of Y and set $e = \dim(f^{-1}(\eta))$; it follows from (13.1.1) that for $n \leq e$ one has $F_n(X) = X$, and one can thus restrict to the case $n > e$. But then (9.5.6), there is a neighbourhood U of η in Y such that $F_n(X) \subset f^{-1}(Y - U)$; as $Y - U \neq Y$, the induction hypothesis implies that $F_n(X)$ is closed.

II) Let us now pass to the general case, still supposing $S = \text{Spec}(A)$ and $X = \text{Spec}(B)$ affine, B being an A -algebra of finite type, hence of the form $B = A[T_1, \dots, T_n]/\mathfrak{J}$. Let $(\mathfrak{J}_\lambda)$ be the family of ideals of finite type of $A[T_1, \dots, T_n]$ contained in $\mathfrak{J}$; if X and the $X_\lambda = \text{Spec}(A[T_1, \dots, T_n]/\mathfrak{J}_\lambda)$ are considered as closed sub-preschemes of $Z = \text{Spec}(A[T_1, \dots, T_n])$, $X = \bigcap_\lambda X_\lambda$ for the underlying spaces. If $f_\lambda : X_\lambda \rightarrow S$ is the structural morphism, one deduces that $f^{-1}(s) = \bigcap_\lambda f_\lambda^{-1}(s)$ for every $s \in S$, and as the sets $f_\lambda^{-1}(s)$ are closed in the Noetherian space Z_s , it exists for every $x \in X$, a λ (depending on s) such that $f^{-1}(s) = f_\lambda^{-1}(s)$. Then, for every $x \in X$, $d(x) = \dim_x(f^{-1}(f(x)))$, $d_\lambda(x) = \dim_x(f_\lambda^{-1}(f_\lambda(x)))$, $d(x) = \inf_\lambda d_\lambda(x)$; as the functions d_λ are semi-continuous superiorly by the first part of the proof, so is d . C.Q.F.D. $\square$

Corollary (13.1.4). — *Under the hypotheses of (13.1.3), the set of $x \in X$ such that x is isolated in $f^{-1}(f(x))$ is open in X .*

Proof. Indeed, this is the complementary of $F_1(X)$ (0, 14.1.10). One notes that one thus recovers, under more general hypotheses, the consequence of the “Main theorem” of Zariski (III, 4.4.10). $\square$

Corollary (13.1.5). — *Under the hypotheses of (13.1.3), suppose also that f is a closed morphism. Then, for every integer n , the set of $y \in Y$ such that $\dim(f^{-1}(y)) \geq n$ is closed; in other words, the function $y \mapsto \dim(f^{-1}(y))$ is semi-continuous superiorly; in particular if y is a specialisation of y' , one has $\dim(f^{-1}(y)) \geq \dim(f^{-1}(y'))$.*

Proof. Indeed, to say that $\dim(f^{-1}(y)) \geq n$ means that $y \in f(F_n(X))$, where $F_n(X)$ is the set of $x \in X$ such that $\dim_x(f^{-1}(f(x))) \geq n$ (0, 14.1.6). As $F_n(X)$ is closed in X by (13.1.3) and f is a closed morphism, $f(F_n(X))$ is closed in Y . $\square$

Corollary (13.1.6). — *Let X, Y be two irreducible preschemes, $f : X \rightarrow Y$ a dominant morphism locally of finite type. Let ξ (resp. η) be the generic point of X (resp. Y) and let $e = \dim(f^{-1}(\eta))$. Then, for every $x \in X$, one has $\dim_x(f^{-1}(f(x))) \geq e$.*

Proof. Indeed, the set of x such that $\dim_x(f^{-1}(f(x))) < e$ is open by (13.1.3), and as it cannot contain ξ , it is empty. $\square$

Proposition (13.1.7). — *Let Y be a prescheme quasi-compact, $f : X \rightarrow Y$ a morphism of finite type. There exists an integer n such that, for every $y \in Y$, $\dim(f^{-1}(y)) \leq n$.*

Proof. As there is a finite covering by affine open sets (V_i) of Y such that each $f^{-1}(V_i)$ is a finite union of affine opens, one is immediately reduced to the case where $Y = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affine, B being an A -algebra of finite type. If B admits a system of n generators, then $B \otimes_A \kappa(y)$ is a $\kappa(y)$ -algebra admitting n generators, hence $\dim(f^{-1}(y)) \leq n$ by (4.1.1). $\square$

13.2. Equidimensional morphisms: case of dominant morphisms of irreducible preschemes

(13.2.1). Let Y be a prescheme irreducible, X a prescheme irreducible, $f : X \rightarrow Y$ a dominant morphism locally of finite type; let η be the generic point of Y . One knows (13.1.6) that for every $x \in X$, one has $\dim_x(f^{-1}(f(x))) \geq \dim(f^{-1}(\eta))$. IV-3 | 190

Definition (13.2.2). — Under the hypotheses of (13.2.1), we say that f is equidimensional at the point x (or that X is equidimensional over Y at the point x) if

$$\dim_x(f^{-1}(f(x))) = \dim(f^{-1}(\eta)).$$

We say that f is equidimensional (or that X is equidimensional over Y) if f is equidimensional at every point $x \in X$.

It follows from Chevalley's theorem (13.1.3) that the set of $x \in X$ where f is equidimensional is open and non-empty. Moreover, if f is equidimensional at the point x , all the irreducible components of $f^{-1}(f(x))$ containing x have the same dimension, which is $\geq \dim(f^{-1}(\eta))$ (13.1.6) and $\leq \dim_x(f^{-1}(f(x)))$ by (0, 14.1.5).

Proposition (13.2.3). — *Let Y be a prescheme irreducible locally Noetherian, X a prescheme irreducible, $f : X \rightarrow Y$ a dominant morphism locally of finite type; let η be the generic point of Y , x a point of X , $y = f(x)$, and suppose that one has*

$$(13.2.3.1) \quad \dim(\mathcal{O}_x) = \dim(\mathcal{O}_y) + \dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} \kappa(y)).$$

Then f is equidimensional at the point x . The converse is true if the two members of the inequality (5.6.5.2) are equal, in particular if $\mathcal{O}_y$ is a universally catenary ring.

Proof. This follows immediately from (5.6.5.2) and the inequality $\delta(x) \geq 0$ (13.1.1). $\square$

(13.2.4). Let now, more generally, Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism of finite type, x a point of X , X' a closed irreducible part of X containing x , $Y' = \overline{f(X')}$, η' the generic point of Y' . Designating again by X' and Y' the reduced closed sub-preschemes of X and Y respectively having X' and Y' for underlying spaces, the restriction $X' \rightarrow Y$ of f factorises as $X' \xrightarrow{f'} Y' \xrightarrow{j} Y$, where j is the canonical injection, and f' is of finite type (1.5.4). Set for short

$$(13.2.4.1) \quad A = \mathcal{O}_{Y,y}, \quad B = \mathcal{O}_{X,x}, \quad A' = \mathcal{O}_{Y',y}, \quad B' = \mathcal{O}_{X',x}.$$

Lemma (13.2.5). — *With the notations of (13.2.4), the following conditions are equivalent:*

label=a) $\dim(B') = \dim(A) + \dim(B \otimes_A \kappa(y))$.

lbbel=b) The following relations hold simultaneously:

label=(i) $\dim(A') = \dim(A)$.

lbbel=(ii) $\dim(B' \otimes_{A'} \kappa(y)) = \dim(B \otimes_A \kappa(y))$, in other words $\dim_x(X' \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$.

lcbel=(iii) $\dim_x(f'^{-1}(y)) = \dim_x(f'^{-1}(\eta'))$, in other words $f' : X' \rightarrow Y'$ is equidimensional (13.2.2) at the point x .

lcbel=c) One has simultaneously the relations:

label=(i) $\dim(A') = \dim(A)$.

lbbel=(ii) $\dim(B') = \dim(A) + \dim(B' \otimes_{A'} \kappa(y))$.

ldbel=d) One has the equality

$$\dim(B') = \dim(A') + \dim(B' \otimes_{A'} \kappa(y)) - (\dim_x(f'^{-1}(y)) - \dim_x(f'^{-1}(\eta')))$$

(a relation that is always verified when $\mathcal{O}_{Y,y}$ is a universally catenary ring, by (5.6.5)).

Proof. The formula (5.6.5.2) applied to the dominant morphism $f' : X' \rightarrow Y'$ gives

$$\dim(B') \leq \dim(A') + \dim(B' \otimes_{A'} \kappa(y)) - (\dim_x(f'^{-1}(y)) - \dim_x(f'^{-1}(\eta'))).$$

As A' , B' , $B' \otimes_{A'} \kappa(y)$ are quotient rings of A , B , $B \otimes_A \kappa(y)$ respectively, one has by (0, 16.1.2.1)

$$\dim(A') \leq \dim(A), \quad \dim(B') \leq \dim(B), \quad \dim(B' \otimes_{A'} \kappa(y)) \leq \dim(B \otimes_A \kappa(y)).$$

From this and (0, 16.3.9), one deduces

$$\dim(B') \leq \dim(A') + \dim(B' \otimes_{A'} \kappa(y)) \leq \dim(A) + \dim(B \otimes_A \kappa(y))$$

and

$$\dim(B') \leq \dim(B) \leq \dim(A) + \dim(B \otimes_A \kappa(y)).$$

The equivalence of conditions a) through d) follows from comparing these chains of inequalities. $\square$

(13.2.6). Recall now that the irreducible components X_i containing x are in finite number in X ((5.1.1) and (0, 14.2.1.1)) and that

$$(13.2.6.1) \quad \dim(\mathcal{O}_{X,x}) = \sup_i \dim(\mathcal{O}_{X_i,x}).$$

Proposition (13.2.7). — *Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism of finite type, x a point of X , $y = f(x)$; one has*

$$(13.2.7.1) \quad \dim(\mathcal{O}_{X,x}) \leq \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} \kappa(y)).$$

For the two members of (13.2.7.1) to be equal, it is necessary and sufficient that there exist a closed irreducible part X' of X containing x and verifying simultaneously the following conditions:

- label=(i) If $Y' = \overline{f(X')}$, one has $\dim(\mathcal{O}_{Y',y}) = \dim(\mathcal{O}_{Y,y})$.
- lbbel=(ii) $\dim_x(X' \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$ (in other words, X' contains one of the irreducible components of $f^{-1}(y)$ of maximal dimension among all the components). It amounts to the same to say that $\dim(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} \kappa(y)) = \dim(\mathcal{O}_{X',x} \otimes_{\mathcal{O}_{Y',y}} \kappa(y))$.
- lcbel=(iii) X' is equidimensional over Y' at the point x , in other words, one has

$$\dim_x(X' \cap f^{-1}(y)) = \dim(X' \cap f^{-1}(\eta')),$$

where η' is the generic point of Y' (and hence all the irreducible components of $X' \cap f^{-1}(y)$ have the same dimension (13.2.2)).

Moreover, one has the equality

$$\dim(\mathcal{O}_{X',x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X',x} \otimes_{\mathcal{O}_{Y',y}} \kappa(y))$$

(condition always implied by (iii) when $\mathcal{O}_{Y,y}$ is a universally catenary ring). Moreover, X' is then a irreducible component of X and Y' a irreducible component of Y .

Proof. The inequality (13.2.7.1) follows from (13.2.6.1), since for each irreducible component X_i of X containing x , one has by (13.2.5) (applied with $X' = X_i$) and (0, 16.3.9):

$$\dim(\mathcal{O}_{X_i,x}) \leq \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X_i,x} \otimes_{\mathcal{O}_{Y,y}} \kappa(y)) \leq \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} \kappa(y)).$$

For the two members of (13.2.7.1) to be equal, it is necessary and sufficient by (13.2.6.1) that there exist an irreducible component X' of X containing x with $\dim(\mathcal{O}_{X',x}) = \dim(\mathcal{O}_{X,x})$, hence verifying condition a) of (13.2.5), which by (13.2.5) is equivalent to conditions (i), (ii), (iii). The “moreover” statement on the equality $\dim(\mathcal{O}_{X',x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X',x} \otimes_{\mathcal{O}_{Y',y}} \kappa(y))$ follows from condition d) of (13.2.5). Finally, X' is an irreducible component of X since $\dim(\mathcal{O}_{X',x}) = \dim(\mathcal{O}_{X,x})$, and Y' is an irreducible component of Y by condition (i). $\square$

Corollary (13.2.8). — *If the two members of (13.2.7.1) are equal, the closed irreducible parts X' of X containing x and verifying conditions (i) to (iv) of (13.2.7) are exactly those for which $\dim(\mathcal{O}_{X',x}) = \dim(\mathcal{O}_{X,x})$ (which implies necessarily that X' is a irreducible component of X).*

Proof. This follows immediately from (13.2.7) and (13.2.5), since conditions (i) to (iv) of (13.2.7) for X' are equivalent by (13.2.5) to $\dim(\mathcal{O}_{X',x}) = \dim(\mathcal{O}_{X,x})$. $\square$

Corollary (13.2.9). — *Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism of finite type, x a point of X , $y = f(x)$. Suppose that $\mathcal{O}_{X,y}$ is a universally catenary ring. Then the following conditions are equivalent:*

label=a) The two members of (13.2.7.1) are equal.

lbel=b) There exists an irreducible component Z of $f^{-1}(y)$ containing x , of dimension $\dim_x(f^{-1}(y))$, such that, for every x' in a neighbourhood of x in Z , one has

$$(13.2.9.1) \quad \dim(\mathcal{O}_{X,x'}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X,x'} \otimes_{\mathcal{O}_{Y,y}} \kappa(y)).$$

lbel=c) There exists an irreducible component Z of $f^{-1}(y)$ containing x , of dimension $\dim_x(f^{-1}(y))$, such that for the generic point z of Z , one has

$$(13.2.9.2) \quad \dim(\mathcal{O}_{X,z}) = \dim(\mathcal{O}_{Y,y}).$$

Proof. Let us show that *a)* implies *b)*. Let $\dim_x(f^{-1}(y)) = n$; by (13.2.7), there exists an irreducible component X' of X verifying conditions (i) to (iv) of (13.2.7); let Z be an irreducible component of dimension n of $f^{-1}(y)$, containing x and contained in X' . As $\mathcal{O}_{X,y}$ is universally catenary, since $f^{-1}(y)$ is locally Noetherian, there exists an open neighbourhood U of x in X meeting no irreducible component of $f^{-1}(y)$ other than those containing x , hence (4.1.1.3) $\dim_{x'}(f^{-1}(y)) = n$ for every $x' \in U$; it is clear that conditions (i) to (iii) of (13.2.7) are satisfied when one replaces x by any $x' \in U$, and the same is true of condition (iv) since $\mathcal{O}_{Y,y}$ is universally catenary; hence the conclusion by (13.2.7). Condition *b)* implies trivially *c)* by (5.1.2). Finally, if *c)* is verified and if X'' is an irreducible component of X containing Z and such that conditions (i) to (iv) of (13.2.7) are verified, it is clear that these conditions are also verified when one replaces X' by X'' and x by z , hence the conditions of (13.2.7) are satisfied, hence *c)* implies *a)*. $\square$

Proposition (13.2.10). — *Let Y be a prescheme irreducible locally Noetherian, η its generic point, $f : X \rightarrow Y$ a morphism of finite type, y a point of $f(X)$. Let Z_i be the irreducible components of $f^{-1}(y)$, z_i the generic point of Z_i , and consider the following conditions:*

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label=a) For $x \in f^{-1}(y)$, one has the relation

$$(13.2.10.1) \quad \dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} \kappa(y)).$$

lbel=b) For every i , one has

$$(13.2.10.2) \quad \dim(\mathcal{O}_{X,z_i}) = \dim(\mathcal{O}_{Y,y}).$$

lbel=c) For every i , there exists an irreducible component X_i of X containing Z_i and such that the reduced closed sub-prescheme X_i of X is equidimensional over Y at the point z_i (in other words $\dim(X_i \cap f^{-1}(y)) = \dim(f_i^{-1}(\eta))$, where $f_i = f|X_i$).

Then a) implies b) and b) implies c); moreover, if $\mathcal{O}_{Y,y}$ is universally catenary, the three conditions a), b), c) are equivalent.

Proof. The ring $\mathcal{O}_{X,z_i} \otimes_{\mathcal{O}_{Y,y}} \kappa(y)$ being zero (5.1.2), *a)* evidently implies *b)*; *b)* implies *c)* by (13.2.7) applied at the point z_i . Inversely, suppose that $\mathcal{O}_{Y,y}$ is universally catenary; as condition *c)* implies that $f(X_i)$ is dense in Y , it follows from *c)* that the conditions (i) to (iv) of (13.2.7) are verified when one replaces X' by X_i and x by z_i , hence *c)* implies *b)*; finally *b)* implies *a)* by (13.2.9). $\square$

Corollary (13.2.11). — *The notations being those of (13.2.10), suppose that $\mathcal{O}_{Y,y}$ is universally catenary. For every $y' \in Y$, let $E(y')$ be the set of dimensions of the irreducible components of $f^{-1}(y')$ and set $d(y') = \dim(f^{-1}(y')) = \sup(E(y'))$. Then, if the conditions a), b), c) of (13.2.10) are verified, one has $E(y) \subset E(\eta)$, hence $d(y) \leq d(\eta)$.*

Proof. Indeed, with the notations of (13.2.10), $X_i \cap f^{-1}(\eta)$ is not empty and is by construction an irreducible component of $f^{-1}(\eta)$ ($\mathbf{0_I}$, 2.1.8). $\square$

Remarks (13.2.12). — *label=i) Recall (6.1.2) that the relation (13.2.10.1) is always verified when the morphism f is flat at the point x .*

- lbbel=(ii) With the notations of (13.2.10), suppose that $\mathcal{O}_{Y,y}$ is universally catenary and that the morphism f is proper; then, if the equivalent conditions a), b), c) of (13.2.10) are satisfied, for the same $d(y) = d(\eta)$, for it follows from (13.1.5) that one has $d(y) \geq d(\eta)$.
- lcbel=(iii) The morphism of (12.2.3, (ii)) is proper and flat and all the local rings are universally catenary; moreover, the two irreducible components X_1, X_2 of X are equidimensional over Y at every point; but $E(y)$ has two elements for every $y \neq y_0$, hence $E(y)$ is not constant in Y .

13.3. Equidimensional morphisms: general case

Proposition (13.3.1). — Let Y be a prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , $y = f(x)$. Designate by Y_α the irreducible components of Y containing y . Then the following conditions are equivalent: IV-3 | 194

- label=a) There exist an integer e and an open neighbourhood U of x such that the image by f of every irreducible component of U is dense in some Y_α , and that, for every $x' \in U$, the space $U \cap f^{-1}(f(x'))$ is equidimensional and of dimension e .
- lbbel=b) There exist an integer e and a Y -morphism quasi-finite $g : U \rightarrow Y[T_1, \dots, T_e]$ (prescheme that we shall also denote $Y[T_1, \dots, T_e]$ for short) such that the image by g of every irreducible component of U is dense in some irreducible component of $Y[T_1, \dots, T_e]$.

Proof. It is immediate that $a'')$ implies $a)$, for the irreducible components of each of the $U_\lambda \cap f^{-1}(f(x))$ are each irreducible components of one of the $U_\lambda \cap f^{-1}(f(x))$, hence the condition $a)$ is trivially $a'')$. Let us show next that $a')$ implies $a'')$; one can restrict to the case where f is of finite type and X and Y reduced (1.5.4). Let U_λ be an irreducible component of U , and suppose that $f(U_\lambda)$ is dense in Y_α ; let x_λ the generic point of U_λ ; by (0_I, 2.1.8), $U_\lambda \cap f^{-1}(y_\alpha)$ is the unique irreducible component of $U \cap f^{-1}(y_\alpha)$ containing x_λ and is by hypothesis of dimension e , equal to the dimensions of all the irreducible components of $U_\lambda \cap f^{-1}(y_\alpha)$, by the hypothesis and (13.1.1). But by Chevalley's theorem (13.1.3) and (13.1.1), the set V_λ of $x' \in U_\lambda$ such that $\dim_{x'}(f_\lambda^{-1}(f_\lambda(x))) = e$ is open and contains x , and it suffices to take the union of the V_λ for an open satisfying the conditions of $a'')$.

Let us now prove that $a)$ implies $b)$; one can restrict to the case where $Y = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affine and let us prove the following lemma first: □

Lemma (13.3.1.1). — Let $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$ two affine schemes, $f : X \rightarrow Y$ a morphism of finite type, y a point of $f(X)$, and set $e = \dim(f^{-1}(y))$. Then there exists a Y -morphism $g : X \rightarrow Y[T_1, \dots, T_e]$ such that (if one sets $X_y = X \times_Y \text{Spec}(\kappa(y))$, $g_y : X_y \rightarrow \text{Spec}(\kappa(y)[T_1, \dots, T_e])$ is finite. Moreover, for every such morphism g , g_y is necessarily surjective and the set U of $x \in X$ which are isolated in their fibre $g^{-1}(g(x))$ is open in X and contains $f^{-1}(y)$, and by definition the restriction $g|_U$ is a quasi-finite morphism.

Proof. Let $\mathfrak{p} = \mathfrak{i}_y$; the ring $B \otimes_A \kappa(\mathfrak{p})$ is a $\kappa(\mathfrak{p})$ -algebra of finite type, hence by the Noether normalisation lemma (Bourbaki, Alg. comm., chap. V, § 3, n° 1, th. 1) there are r elements algebraically independent over $\kappa(\mathfrak{p})$, $(t_i)_{1 \leq i \leq r}$, of $B \otimes_A \kappa(\mathfrak{p})$ such that if one sets $C' = \kappa(\mathfrak{p})[t_1, \dots, t_r]$, $B \otimes_A \kappa(\mathfrak{p})$ is a C' -algebra finite; one thus has $\dim(B \otimes_A \kappa(\mathfrak{p})) = \dim(C')$ (0_I, 16.1.5) and as $\dim(C') = r$ (5.2.1), one has $r = e$. As $B \otimes_A \kappa(\mathfrak{p}) = (B/\mathfrak{p}B) \otimes_{A/\mathfrak{p}} \kappa(\mathfrak{p})$, one can, multiplying the t_i by a suitable non-zero element $s_i \in B$, suppose that each t_i is the canonical image in $B \otimes_A \kappa(\mathfrak{p})$ of an element of B . Let then $u : A[T_1, \dots, T_e] \rightarrow B$ be the homomorphism such that $u(T_i) = s_i$ for every i , and let $g : X \rightarrow Y[T_1, \dots, T_e]$ be the corresponding morphism. It is clear that by reason of the choice of the s_i , g_y is finite. For every such morphism g , it follows from (5.4.2) and (4.1.2.1) that g_y is necessarily surjective. On the other hand, the set U of $x \in X$ isolated in their fibre $g^{-1}(g(x))$ is open in X by (13.1.4), and contains $f^{-1}(y)$ and by definition the restriction $g|_U$ is a quasi-finite morphism. □

Definition (13.3.2). — Let Y be a prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X . We say that f is equidimensional at the point x (or that X is equidimensional over Y at the point x) if the equivalent conditions of (13.3.1) are verified. We say that f is equidimensional (or that X is equidimensional over Y) if f is equidimensional at every point $x \in X$. IV-3 | 196

It is clear that, for f to be equidimensional at a point x , it suffices that f_{red} be. Moreover, the conditions of (13.3.1) show that the set of points where f is equidimensional is open in X .

One notes that when X and Y are *irreducible*, saying that f is equidimensional at the point x , by (13.3.1), a''), means that f is dominant and that $\dim_x(f^{-1}(f(x))) = \dim(f^{-1}(\eta))$ (where η is the generic point of Y); the definition (13.3.2) thus coincides with (13.2.2).

Proposition (13.3.3). — *Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , X_j the irreducible components of X (in finite number) containing x . For f to be equidimensional at the point x , it is necessary and sufficient that, for every j , $Y_j = \overline{f(X_j)}$ be a irreducible component of Y and, designating by X_j and Y_j the reduced closed sub-preschemes of X and Y having X_j and Y_j for underlying spaces respectively, and by $f_j : X_j \rightarrow Y_j$ the morphism deduced from f (I, 5.2.2), by y_j the generic point of Y_j , that f_j be equidimensional at the point x and that all the numbers $\dim(f_j^{-1}(y_j))$ be equal.*

Proof. This follows immediately from (13.3.1), a'). $\square$

Corollary (13.3.4). — *With the notations of (13.3.3), set $y = f(x)$. If $\mathcal{O}_{Y,y}$ is universally catenary and if f is equidimensional at the point x , one has*

$$(13.3.4.1) \quad \dim(\mathcal{O}_{X_j,x}) = \dim(\mathcal{O}_{Y_j,y}) + e - \deg \cdot \mathrm{tr}_{\kappa(y)} \kappa(x)$$

where e is the common value of the numbers $\dim(f_j^{-1}(y_j))$.

Proof. As each $\mathcal{O}_{Y_j,y}$, being a quotient of $\mathcal{O}_{Y,y}$, is a universally catenary ring (5.6.1), the equality follows from (5.6.5) applied to f_j . $\square$

Corollary (13.3.5). — *With the notations of (13.3.4), suppose that f be equidimensional at the point x and that $\mathcal{O}_{Y,y}$ be an universally catenary ring. If the ring $\mathcal{O}_{Y,y}$ is equidimensional, so is $\mathcal{O}_{X,x}$; the converse is true if the image by f of the union of the X_j is dense in a neighbourhood of y .*

Proof. Indeed, to say that $\mathcal{O}_{Y,y}$ is equidimensional means that the numbers $\dim(\mathcal{O}_{Y_j,y})$ are equal for all the irreducible components Y_j of Y containing y , as follows from (5.1.1.5) applied to the local scheme $\mathrm{Spec}(\mathcal{O}_{Y,y})$; as one has the relation (13.3.4.1), it follows by the same reasoning that $\mathcal{O}_{X,x}$ is then equidimensional. Inversely, if $\mathcal{O}_{X,x}$ is equidimensional, all the numbers $\dim(\mathcal{O}_{Y_j,y})$ are equal by (13.3.4.1); one deduces that $\mathcal{O}_{Y,y}$ is equidimensional if the Y_j are all the irreducible components of Y containing y , which follows from the supplementary hypothesis. $\square$

Proposition (13.3.6). — *Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , $y = f(x)$. Suppose that the ring $\mathcal{O}_y$ be equidimensional. Then, if $\mathcal{O}_x$ is equidimensional and if one has the equality*

$$(13.3.6.1) \quad \dim(\mathcal{O}_x) = \dim(\mathcal{O}_y) + \dim(\mathcal{O}_x \otimes_{\mathcal{O}_y} \kappa(y))$$

(cf. (13.2.7.1)), f is equidimensional at the point x , and the converse is true if $\mathcal{O}_y$ is a universally catenary ring.

Proof. Keep the notations of (13.3.3); it follows from (13.2.8) and the hypothesis that $\mathcal{O}_x$ is equidimensional that each Y_j is an irreducible component of Y and that each f_j is equidimensional at the point x ; moreover, by (13.2.8), one has (taking into account (5.6.5))

$$(13.3.6.2) \quad \dim_x(X_j \cap f^{-1}(y)) = \dim(\mathcal{O}_{X_j,x}) - \dim(\mathcal{O}_{Y_j,y}) + \deg \cdot \mathrm{tr}_{\kappa(y)} \kappa(x).$$

The first member of (13.3.6.2) is therefore independent of j ; but as f_j is equidimensional at the point x , one has $\dim_x(X_j \cap f^{-1}(y)) = \dim(f_j^{-1}(y_j))$, hence the criterion (13.3.3) shows that f is equidimensional at the point x .

Inversely, suppose that $\mathcal{O}_{Y,y}$ is a universally catenary ring and that f is equidimensional at the point x ; then by (13.3.5) $\mathcal{O}_{X,x}$ is equidimensional, and it follows from (13.3.4) that one has the relation

$$\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + e - \deg \cdot \mathrm{tr}_{\kappa(y)} \kappa(x)$$

where e is the common value of the numbers $\dim(f_j^{-1}(y_j)) = \dim_x(X_j \cap f^{-1}(y))$; but by definition e is also equal to $\dim_x(f^{-1}(y))$, hence the relation (13.3.6.1), taking into account (5.6.5.2). $\square$

Proposition (13.3.7). — Let Y be a prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, Z a closed part of X . Then the function IV-3 | 198

$$(13.3.7.1) \quad x \longmapsto \operatorname{codim}_x(Z \cap f^{-1}(f(x)), f^{-1}(f(x)))$$

is semi-continuous inferiorly in X .

Proof. As all the irreducible components of $f^{-1}(f(x))$ containing x have by hypothesis the same dimension (13.3.1), one has, by (5.2.1),

$$\operatorname{codim}_x(Z \cap f^{-1}(f(x)), f^{-1}(f(x))) = \dim_x(f^{-1}(f(x))) - \dim_x(Z \cap f^{-1}(f(x))).$$

But by hypothesis the first term of the right-hand side is *continuous* at x (13.3.1) and the second is semi-continuous superiorly by Chevalley's theorem (13.1.3); hence the conclusion. $\square$

Proposition (13.3.8). — Let Y be a prescheme, Y' a prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $g : Y' \rightarrow Y$ a flat morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$; if f is equidimensional at the point x , then f' is equidimensional at every point $x' \in X'$ above x .

Proof. The question being local on X and Y , one can restrict to the case where every irreducible component of X (resp. Y) contains x (resp. y); as the image by f of every irreducible component of X is then dense in an irreducible component of Y (13.3.1, a'), one knows (2.3.5) that the image by f' of every irreducible component of X' is dense in an irreducible component of Y' . On the other hand, by transitivity of the fibres (I, 3.6.4), it follows from (4.2.8) and (13.3.1) that, for every $z' \in X'$, the set of dimensions of the irreducible components of $f'^{-1}(f'(z'))$ is the same as the set of dimensions of the irreducible components of $f^{-1}(f(z))$, where z is the projection of z' in X ; hence the conclusion by (13.3.1, a). $\square$

Remarks (13.3.9). — The property of equidimensionality of a morphism $f : X \rightarrow Y$ is *not* stable by arbitrary base change $g : Y' \rightarrow Y$, even when g is the canonical injection of an irreducible component Y' of Y into Y (Y' being considered as a reduced closed sub-prescheme of Y). For example, let k be a corps, $A_0 = k[S, T]$ the ring of polynomials in two indeterminates, $\mathfrak{p}_1$ and $\mathfrak{p}_2$ the ideals A_0S and A_0T of A_0 ; let $A = A_0/\mathfrak{a}$ and $Y = \operatorname{Spec}(A)$, which has two irreducible components $Y_1 = \operatorname{Spec}(A/\mathfrak{p}_1)$, $Y_2 = \operatorname{Spec}(A/\mathfrak{p}_2)$; take $X = Y_1$, $f : X \rightarrow Y$ the canonical injection, which is evidently an equidimensional morphism. Take on the other hand $Y' = Y_2$, $g : Y' \rightarrow Y$ the canonical injection; then $X' = X \times_Y Y' = \operatorname{Spec}((A/\mathfrak{p}_1) \otimes_A (A/\mathfrak{p}_2)) = \operatorname{Spec}(A/(\mathfrak{p}_1 + \mathfrak{p}_2))$, and $\mathfrak{p}_1 + \mathfrak{p}_2$ is a maximal ideal of A , hence X' is reduced to a point; the morphism $f' : X' \rightarrow Y'$ is *not dominant*, hence f' is not equidimensional.

One can also give a counter-example where X and Y are *integral*, f *finite* and *birational* (and *a fortiori* equidimensional by (13.3.1, b)), $g : Y' \rightarrow Y$ *finite* and *dominant*.

In this example, the ring A is not geometrically unibranch; we shall see in the following paragraph (14.4.6) that such phenomena cannot occur when the points of Y are *geometrically unibranch*. The lack of stability of the notion of equidimensional morphism greatly restricts its interest, in favour of the notion of universally open morphism, which will be studied in detail in the following paragraph.

§14. UNIVERSALLY OPEN MORPHISMS

Sections 14 and 15 are devoted to the study of the notion of *universally open* morphism (2.4.2). IV-3 | 199
We have already seen (2.4.6) that a *flat* morphism locally of finite presentation is universally open, the converse being inexact. In § 14, we examine first the properties of the fibres of a universally open morphism $f : X \rightarrow Y$; when X and Y are locally Noetherian, f behaves in this respect (14.2.1) like a flat morphism (cf. 6.1.2), and is in particular *equidimensional* when it is locally of finite type and dominant and X is irreducible (14.2.2). Conversely, an equidimensional morphism locally of finite type $f : X \rightarrow Y$ is universally open when one supposes that Y is *geometrically unibranch*, and in particular when Y is *normal* (Chevalley criterion, 14.4.4). We also show that universally open morphisms locally of finite type $f : X \rightarrow Y$ (when X and Y are locally Noetherian and Y irreducible) admit “quasi-sections”, i.e. in a neighbourhood of a closed point x of a fibre $f^{-1}(y)$, there is a closed sub-prescheme X' of X containing x such that the restriction $X' \rightarrow Y$ of f is a morphism *quasi-finite* (hence with discrete fibres) and dominant (14.5.3).

In § 15 we study various properties of the fibres of universally open morphisms $f : X \rightarrow Y$, locally of finite type, notably when X and Y are locally Noetherian. One thus obtains in particular a criterion for a point $x \in X$ to belong to a *single* irreducible component of the fibre $f^{-1}(f(x))$ containing x : it suffices that Y be geometrically unibranch (for example normal) at $f(x)$ and that $f^{-1}(f(x))$ be geometrically punctually integral at the point x (15.5.3); if moreover Y is locally integral at $f(x)$, f flat at x and X locally integral at x , we also show (15.5.7) that the number of *geometric connected components* of a fibre $f^{-1}(y)$ is *locally constant* (15.5.7). Finally, when f admits a *section* g (which will be the case in the context of § 15.7), and the fibres of f are geometrically reduced, then X^0 (the union of the X_y^0 for $y \in Y$, where X_y^0 designates the connected component of $X_y = f^{-1}(y)$ at the point $g(y)$ (“neutral component” in the group case)) is studied, and one shows (15.6.4) that if f is universally open and the fibres X_y geometrically reduced, then X^0 is open in X .

14.1. Open morphisms

(14.1.1). Recall (1.10.2) that a continuous map $\psi : X \rightarrow Y$ is said to be *open at a point* $x \in X$ if the image by ψ of *every* neighbourhood of x in X is a neighbourhood of $\psi(x)$ in Y . One notes that this does *not imply* that there exists a fundamental system of neighbourhoods of x whose images are open in Y .

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Proposition (14.1.2). — Let X, Y be two topological spaces, $\psi : X \rightarrow Y$ a continuous map, x a point of X , $y = \psi(x)$.

label=(i) If ψ is open at x , then for every part Y' of Y containing $\psi(x)$ the restriction $\psi^{-1}(Y') \rightarrow Y'$ of ψ is open at the point x .

lbbel=(ii) Suppose that Y is a union of a locally finite family of closed parts (Y_i) and that for every i such that $\psi(x) \in Y_i$, the restriction $\psi^{-1}(Y_i) \rightarrow Y_i$ of ψ is open at the point x ; then ψ is open at the point x .

lcbel=(iii) Let $\gamma : X' \rightarrow X$ be a continuous map, x' a point of X' ; if the composed map $\psi \circ \gamma : X' \rightarrow Y$ is open at the point x' , ψ is open at the point $\gamma(x')$.

Proof. Assertion (i) follows from the criterion (1.10.3). For (ii), let U be an open neighbourhood of x in X ; for each i such that $\psi(x) \in Y_i$, $U \cap \psi^{-1}(Y_i)$ is a neighbourhood of x in $\psi^{-1}(Y_i)$, hence by hypothesis $\psi(U \cap \psi^{-1}(Y_i))$ contains a neighbourhood V_i of $\psi(x)$ in Y_i ; one has $V_i \subset \psi(U) \cap Y_i$, and since the reunion of the V_i for these indices is a neighbourhood V of $\psi(x)$ in Y (as the Y_i form a locally finite closed cover), one has $V \subset \psi(U)$, hence $\psi(U)$ is a neighbourhood of $\psi(x)$ in Y . Assertion (iii) is trivial. $\square$

Remarks (14.1.3). — label=(i) The set Z of points $x \in X$ at which a morphism is open is not necessarily open.

lbbel=(ii) Let X, Y be two Noetherian irreducible preschemes of points generics ξ, η respectively, $f : X \rightarrow Y$ a dominant morphism locally of finite type at the point ξ ; then f is *open at the point* ξ . Indeed, one can evidently restrict to the case where X and Y are reduced (hence integral) and affine; by the generic flatness theorem (6.9.1), there exists a non-empty open V of Y such that the restriction $f^{-1}(V) \rightarrow V$ of f is flat; it follows from (2.4.6) that this restriction is an open morphism. However, f is not necessarily universally open (not even *a fortiori* flat), as is shown by (11.7.5).

lcbel=(iii) We do not know if, when X and Y are locally Noetherian and $f : X \rightarrow Y$ a morphism locally of finite type, the set of points of X where f is open or not is *locally constructible*.

Proposition (14.1.4). — Let X, Y be two topological spaces, $\psi : X \rightarrow Y$ a continuous map. For every $y \in Y$, the set of $x \in \psi^{-1}(y)$ where ψ is open is a closed part of $\psi^{-1}(y)$.

Proof. Indeed, suppose that ψ is not open at a point $x \in \psi^{-1}(y)$; there exists then a neighbourhood V of x in X such that $\psi(V)$ does not contain a neighbourhood of y ; it follows that for every point $x' \in V \cap \psi^{-1}(y)$, ψ is not open at the point x' . $\square$

Remarks (14.1.5). — Even if X and Y are locally Noetherian and f a finite morphism, it may happen that f is open at all points of a fibre $f^{-1}(y)$, without there existing a *neighbourhood* of $f^{-1}(y)$ in which f is open.

(14.1.6). In what follows, the essential role will be played by the criterion (1.10.3) characterising the morphisms locally of finite presentation $f : X \rightarrow Y$ that are open at a point x by the “lifting of generisations y' of $y = f(x)$ ”: for every generisation y' of $y = f(x)$, there exists $x' \in X$, generisation of x , such that $y' = f(x')$. IV-3 | 201

14.2. Open morphisms and dimension of fibres

Proposition (14.2.1). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type. Let y be a point of Y and x a point of $f^{-1}(y)$.* IV-3 | 201

- label=(i) Suppose that f is open at x . Then, for every irreducible component Y_0 of Y containing y with generic point η , there exists an irreducible component Z of X containing x , dominating Y_0 and such that $\dim_x(Z \cap X_y) = \dim(Z \cap X_\eta)$.*
- lbbel=(ii) Suppose f universally open at x ; then for every irreducible component Y_0 of Y containing y , for every irreducible component Z of X containing x and dominating Y_0 , one has $\dim_x(Z \cap X_y) = \dim(Z \cap X_\eta)$.*

Proof. Let y' be any generisation of y distinct from y , and consider the closed reduced sub-prescheme Y' of Y with underlying space $\{\overline{y'}\}$; then no irreducible component X' of $f^{-1}(Y')$ containing x can be contained in $f^{-1}(y)$: indeed, if x' is the generic point of X' , one would have $f(x') = y \neq y'$, and since x' is its sole generisation in $f^{-1}(Y')$, this would contradict the hypothesis that f is open at the point x' , by virtue of the criterion (1.10.3). One can therefore apply (6.1.2) to the local homomorphism of Noetherian local rings $\mathcal{O}_y \rightarrow \mathcal{O}_x$, whence the conclusion. $\square$

Corollary (14.2.2). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ an open morphism of finite type, y a point of Y , x a maximal point of the fibre X_y . Then there exists an irreducible component Z of X containing x and equidimensional over Y (13.2.2) at the point x .*

Proof. This follows from (14.2.1) and (13.2.3). $\square$

(14.2.4). With the notations of (14.2.1), define $E(y) = \{\dim(Z_i) : Z_i \text{ irreducible component of } f^{-1}(y)\}$ and $d(y) = \dim(f^{-1}(y)) = \sup(E(y))$.

Corollary (14.2.5). — *Under the hypotheses of (14.2.4), suppose f is a proper morphism and Y is irreducible with generic point η . Then $d(y) = d(\eta)$ for every $y \in Y$.*

Proof. Let Y_i ($1 \leq i \leq n$) be the closed reduced sub-preschemes of Y having for underlying spaces the irreducible components of Y containing y ; if $f_i : f^{-1}(Y_i) \rightarrow Y_i$ is the restriction of f , each of the f_i is proper (II, 5.4.5) and open at all points of $f^{-1}(y)$ (14.1.2). Since the union of the Y_i is a neighbourhood of y in Y , one reduces to the case where Y is irreducible; let η be its generic point. It follows from (14.2.4) that $d(y) \leq d(\eta)$; on the other hand, since f is proper, one deduces from (13.1.5) that $z \mapsto \dim(f^{-1}(z))$ is upper semi-continuous; in particular $d(y) \geq d(\eta)$, hence $d(y) = d(\eta)$. Moreover, there is a neighbourhood V of y such that $d(z) \leq d(y)$ for every $z \in V$; but since z is a specialisation of η , one has $d(z) \geq d(\eta)$, whence $d(z) = d(y)$ for $z \in V$. $\square$

Corollary (14.2.6). — *Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ an open morphism of finite type, y a point of Y . Let y_i ($1 \leq i \leq n$) be the generic points of the irreducible components of Y containing y . Then, with the notations of (14.2.4):* IV-3 | 203

- label=(i) If for $1 \leq i \leq n$, one has $E(y) = E(y_i)$ (resp. $d(y) = d(y_i)$), the function E (resp. d) is constant in a neighbourhood of y .*
- lbbel=(ii) There exists an open neighbourhood U of $f^{-1}(y)$ such that the function $z \mapsto \dim(U \cap f^{-1}(z))$ is constant in a neighbourhood of y .*

Proof. (i) The same reasoning as in (14.2.5) shows that one can restrict to the case where Y is irreducible, of generic point η . If y' is any generisation of y , one has (applying (14.2.4) to the restriction $f^{-1}(Y') \rightarrow Y'$ of f , where $Y' = \{\overline{y'}\}$, and using (14.1.2)) that $E(y) \subset E(y') \subset E(\eta)$ and $d(y) \leq d(y') \leq d(\eta)$; this shows that $E(y) = E(\eta)$ (resp. $d(y) = d(\eta)$) implies $E(y) = E(y')$ (resp. $d(y) = d(y')$) for every generisation y' of y . Now, by (9.5.5), the functions E and d are locally constructible, hence it follows from (0III, 9.2.5) applied to the set of z such that $E(z) = E(y)$ (resp. $d(z) = d(y)$) that this set is a neighbourhood of y .

(ii) For each y_i , $f^{-1}(y_i)$ is a Noetherian space since f is of type fini; let x_{ij} ($1 \leq j \leq m_i$) be the generic points of its irreducible components, and set $X'_{ij} = \{x_{ij}\}$; the complementary U in X of the union of the X'_{ij} that do not meet $f^{-1}(y)$ is an open neighbourhood of $f^{-1}(y)$. Indeed, the restriction $f|_U : U \rightarrow Y$ of f is an open morphism of finite type (I, 6.3.5); for every (i, j) such that $x_{ij} \in U$, x_{ij} is a maximal point of $U \cap f^{-1}(y_i)$, and it follows from (13.1.1) applied to the restriction $X'_{ij} \rightarrow Y_i = \{y_i\}$ of f (by (I, 5.2.2) and (1.5.4)) that all irreducible components of $X'_{ij} \cap f^{-1}(y_i)$ have dimension $\geq \dim(X'_{ij} \cap f^{-1}(y_i))$. By (14.2.4), $\dim(f^{-1}(y)) \leq \dim(U \cap f^{-1}(y_i))$; but as $f|_U$ is open, one also has $\dim(f^{-1}(y)) \leq \dim(f^{-1}(y_i) \cap U)$ by (14.2.4); one can thus apply (i) to $f|_U$, whence the conclusion. $\square$

Remarks (14.2.7). — The example (13.2.12, (iii)) shows that under the hypotheses of (14.2.5) or (14.2.6), (ii), one cannot, in the conclusion, replace the function $z \mapsto \dim(f^{-1}(z))$ by the function $z \mapsto E(z)$; indeed, in this example, f is proper and flat (hence universally open (2.4.6)) and every neighbourhood of the unique point of $f^{-1}(y_0)$ contains the generic points of the irreducible components of $f^{-1}(\eta)$.

14.3. Universally open morphisms

(14.3.1). Recall (2.4.2) that to say a morphism $f : X \rightarrow Y$ is universally open means that for every morphism $g : Y' \rightarrow Y$, $f_{(Y')} : X_{(Y')} \rightarrow Y'$ is open; it suffices moreover that this be so when $Y' = Y[T_1, \dots, T_e]$, for every e (8.10.2) (and if Y is locally Noetherian, it suffices that it be so for every Y' locally Noetherian).

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Proposition (14.3.2). — *Let $f : X \rightarrow Y$ be a morphism of preschemes. If f is universally open, then, for every morphism $g : Y' \rightarrow Y$, where Y' is irreducible, the image by $f' = f_{(Y')} : X_{(Y')} \rightarrow Y'$ of every irreducible component of $X_{(Y')}$ is dense in Y' . Conversely, if this condition is verified for every irreducible Y' and every morphism of finite type $g : Y' \rightarrow Y$, and if moreover f is locally of finite presentation, then f is universally open.*

Proof. Indeed, it follows from (1.10.4) that if f is universally open, it satisfies the condition of the statement. Conversely, suppose that f is locally of finite presentation, and let us show that for every integer e , if one sets $Y'' = Y[T_1, \dots, T_e]$, $f_{(Y'')}$ is open. Indeed, let Y' be a closed sub-prescheme of Y'' having for underlying space an irreducible closed part of Y'' ; the composed morphism $Y' \rightarrow Y'' \rightarrow Y$ is of finite type, hence every irreducible component of $X_{(Y')}$ dominates Y' by hypothesis; one deduces from (1.10.4) that $f_{(Y'')}$ is an open morphism. $\square$

Definition (14.3.3). — Let $f : X \rightarrow Y$ be a morphism of preschemes, x a point of X . We say that f is *universally open at the point x* if, for every morphism $g : Y' \rightarrow Y$, setting $X' = X \times_Y Y'$, the morphism $f' = f_{(Y')} : X' \rightarrow Y'$ is open at every point x' of X' whose projection in X is x .

Remarks (14.3.3.1). — (i) The reasoning of (8.10.1) shows (with the same notations) that if x is a point of X , x_λ its projection in X_λ , then, if f_μ is open at x_μ for $\mu \geq \lambda$, f is open at x ; it suffices thus to restrict to open sets U of X containing x and to remark that the hypothesis implies that $f_\mu(U_\mu)$ is a neighbourhood of $f(x)$. One deduces that the statement (8.10.2) is still exact when one replaces “universally open” by “universally open at the point x ”, and “open morphism” by “morphism open at every point x_n of X_n whose projection in X is x ”: it suffices in the proof to restrict to opens V containing x_n .

(ii) The result of (14.1.4) is still true for a morphism $f : X \rightarrow Y$, replacing “open” by “universally open”.

Proposition (14.3.4). —

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- (i) Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ two morphisms, x a point of X . If f is universally open at x and g is universally open at $f(x)$, then $g \circ f$ is universally open at x . Conversely, if $g \circ f$ is universally open at x , g is universally open at $f(x)$.
- (ii) If $f : X \rightarrow Y$ is an S -morphism universally open at $x \in X$, then, for every base change $S' \rightarrow S$, $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is universally open at every point of $X_{(S')}$ above x .
- (iii) For $f : X \rightarrow Y$ to be universally open at $x \in X$, it suffices that f_{red} be.

ldbel=(iv) Let $f : X \rightarrow Y$ a morphism locally of finite presentation, x a point of X , $y = f(x)$; set $Y_1 = \text{Spec}(\mathcal{O}_y)$, $X_1 = X \times_Y Y_1$, $f_1 = f_{(Y_1)}$. For f to be universally open at x , it suffices that f_1 be (recall (I, 3.6.5) that X_1 is canonically identified with a sub-space of X).

Proof. Indeed, (ii) is an evident consequence of the definition (14.3.3); it also follows from the definition that to prove assertion (i), it suffices to do so when one suppresses everywhere the word “universally”, and this then follows from (14.1.2). Assertion (iii) follows from (ii) and from the fact that the canonical morphism $X_{\text{red}} \rightarrow X$ is surjective. Finally, condition (iv) is trivially necessary. On the other hand, if it is satisfied, and if $g : Y' \rightarrow Y$ is any morphism, $X' = X_{(Y')}$, $f' = f_{(Y')}$, x' a point of X' above x , then f' is locally of finite presentation, and to see that it is open at the point x' , it suffices to apply the criterion (1.10.3, b)). Let $y' = f'(x')$, $Y_1 = \text{Spec}(\mathcal{O}_{y'})$, $X'_1 = X' \times_{Y'} Y_1$, $f'_1 = f'_{(Y_1)}$; since $g(y') = y$, the composed morphism $Y_1 \rightarrow Y' \rightarrow Y$ factorises through $Y_1 \rightarrow Y_1 \rightarrow Y$ (I, 2.4.4), hence $X'_1 = X_1 \times_{Y_1} Y_1$ and $f'_1 = (f_1)_{(Y_1)}$; the conclusion then follows from the hypothesis that f_1 is open at the point x' and from (1.10.3). $\square$

Proposition (14.3.5). — Let X, Y two preschemes, $f : X \rightarrow Y$ a morphism, x a point of X . Let $(Y_i)_{1 \leq i \leq n}$ be a finite family of closed sub-preschemes of Y such that Y is a union of the Y_i , and suppose that for every i such that $f(x) \in Y_i$, the restriction $f^{-1}(Y_i) \rightarrow Y_i$ of f is universally open at x ; then f is universally open at x .

Proof. Taking into account the definition, this follows from the analogous proposition (14.1.2), (ii) for morphisms open at a point. $\square$

Proposition (14.3.6). — Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism locally of finite type. For f to be universally open at a point $x \in X$, it is necessary and sufficient that the following condition be verified: for every morphism $g : Y' \rightarrow Y$, where $Y' = \text{Spec}(A)$ is the spectrum of a discrete valuation ring, such that the image $g(y')$ of the closed point y' of Y' is equal to $y = f(x)$, and for every point $x' \in X' = X \times_Y Y'$ whose projections on X and Y' are x and y' respectively, there exists a generisation z' of x' in X' whose projection in Y' is the generic point of Y' (in other words, there exists an irreducible component of X' containing x' and dominating Y').

Moreover, one can, in the preceding condition, restrict to the case where A is complete, with algebraically closed residue field, and where x' is rational over $\kappa(y')$.

Proof. The necessity of the condition follows from the fact that f' must be open at the point x' , and from the criterion (1.10.3). To see that the condition is sufficient, consider a finite type morphism $g : Y'' \rightarrow Y$, and let $X'' = X \times_Y Y''$, $f'' = f_{(Y'')} : X'' \rightarrow Y''$, and x'' a point of X'' above x . Set $y'' = f''(x'')$, and let t be a generisation of y'' in Y'' , distinct from y'' . Since Y'' is locally Noetherian, it follows from (II, 7.1.9) and (0_{III}, 10.3.1) that there exists a discrete valuation ring A , complete with algebraically closed residue field of $k(x'')$, and a morphism $g : Y' = \text{Spec}(A) \rightarrow Y''$ such that, if s and y' are the generic point and the closed point of Y' , $g(s) = t$ and $g(y') = y''$. There is then a point $x' \in X' = X'' \times_{Y''} Y'$ whose projections in X'' and Y' are x'' and y' respectively, and which is rational over $\kappa(y')$ (I, 3.4.9). The hypothesis implies that there is a generisation z' of x' in X' whose projection in Y' is s ; if z'' is the projection of z' in X'' , z'' is a generisation of x'' in X'' , and its projection in Y'' is t ; one concludes that f'' is open at the point x'' (1.10.3), hence that f is universally open at the point x (14.3.3.1, (i)). $\square$

Corollary (14.3.7). — The notations being those of (14.3.6):

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label=(i) Given a point $y \in Y$, for f to be universally open at all points of $f^{-1}(y)$, it is necessary and sufficient that for every morphism $g : Y' \rightarrow Y$, where Y' is the spectrum of a discrete valuation ring, such that the image $g(y')$ of the closed point y' of Y' is y , every irreducible component of $X' = X \times_Y Y'$ dominate Y' .

lbbel=(ii) For f to be universally open, it is necessary and sufficient that for every morphism $g : Y' \rightarrow Y$, where Y' is the spectrum of a discrete valuation ring, every irreducible component of $X' = X \times_Y Y'$ dominate Y' .

Proof. It is clear that it suffices to prove (i); the necessity follows from (14.3.2) and its sufficiency from (14.3.6). $\square$

Proposition (14.3.8). — *Let Y be a prescheme locally Noetherian, irreducible, regular and of dimension 1 (for example the spectrum of a Dedekind ring), $f : X \rightarrow Y$ a morphism locally of finite type, y a point of Y . The following conditions are equivalent:*

label=a) f_{red} is flat at every point of $f^{-1}(y)$.

lbbel=b) f is universally open in a neighbourhood of $f^{-1}(y)$.

lcbel=c) f is open in a neighbourhood of $f^{-1}(y)$.

ldbel=d) Every irreducible component of X meeting $f^{-1}(y)$ dominates Y .

Proof. Since f_{red} is locally of finite type (1.3.4), a) implies that f_{red} is flat in a neighbourhood of $f^{-1}(y)$ (11.1.1), and it suffices to apply (2.4.6) in such a neighbourhood to see that a) implies b). The implication b) $\Rightarrow$ c) is trivial, and the implication c) $\Rightarrow$ d) follows from (1.10.4) applied to a neighbourhood of $f^{-1}(y)$. It remains to see that d) implies a). One can, by (1.3.4), restrict to the case where X is reduced. The question being moreover local on X and on Y , one can suppose $Y = \text{Spec}(A)$ and $X = \text{Spec}(B)$, where A is a regular local ring of dimension 1, i.e. a discrete valuation ring (II, 7.1.6). Let X_i ($1 \leq i \leq n$) be the closed integral sub-preschemes of X having for underlying spaces the irreducible components of X ; then, for every $x \in X$, $\mathcal{O}_{X,x}$ is a sub-ring of the composed direct of the $\mathcal{O}_{X_i,x}$; if $y = f(x)$, it will suffice to show that each $\mathcal{O}_{X_i,x}$ is an $\mathcal{O}_y$ -module without torsion, for then the same will be true of $\mathcal{O}_{X,x}$; since by hypothesis $\mathcal{O}_y$ is a discrete valuation ring (II, 7.1.6), it will follow from (0I, 6.3.4) that $\mathcal{O}_{X,x}$ is a flat $\mathcal{O}_y$ -module. But if $X_i = \text{Spec}(B_i)$, where B_i is an integral ring, hypothesis d) implies that the homomorphism $A \rightarrow B_i$ is injective (I, 1.2.7); hence B_i is an A -module without torsion, and a fortiori $\mathcal{O}_{X_i,x}$ is an $\mathcal{O}_y$ -module without torsion. $\square$

Remarks (14.3.9). — label=(i) In (14.3.8), one cannot dispense from the hypothesis that Y is regular.

lbbel=(ii) The example of (14.1.3), (i) shows that the set of points of X where a morphism is universally open is not necessarily open.

Lemma (14.3.10). — *Let A be a valuation ring (not necessarily discrete), k its residue field, K its field of fractions, and set $S = \text{Spec}(A)$. Let X be a prescheme irreducible, $f : X \rightarrow S$ a dominant morphism of finite type; let $X_0 = X \otimes_A k$ (resp. $X_1 = X \otimes_A K$) be the fibre of f at the closed point (resp. the generic point) of S . Then, if $X_0 \neq \emptyset$, one has $\dim(X_0) = \dim(X_1)$.*

Proof. One can restrict to the case where X is affine, replacing if necessary X by an affine open containing a generic point of an irreducible component of X_0 of maximal dimension, and using (4.1.1.3). Let $n = \dim(X_0) \geq 0$; it follows from (13.3.1.1) that there exists a neighbourhood U of X_0 in X and an S -morphism quasi-finite $g : U \rightarrow S[T_1, \dots, T_n] = Z$ such that the restriction $g_0 : U_0 = U \cap X_0 \rightarrow Z_0 = \text{Spec}(k[T_1, \dots, T_n])$ is finite and surjective. By base change $\text{Spec}(K) \rightarrow S$ and restriction to the open $U_1 = U \cap X_1$ of X_1 , one deduces from g a quasi-finite morphism $g_1 : U_1 \rightarrow Z_1 = \text{Spec}(K[T_1, \dots, T_n])$. Since U_1 is dense in X_1 , one has $\dim(U_1) = \dim(X_1)$ (4.1.1.3); the proposition will be established, by (4.1.2), if we prove that the morphism g_1 is dominant. Suppose the contrary; there would exist a polynomial $F_1 \neq 0$ in $K[T_1, \dots, T_n]$ such that $g_1(U_1) \subset V(F_1)$. If ω is a valuation on K associated to A , and if (a_α) is the family of coefficients of F_1 , one can, after multiplying F_1 by an element $\neq 0$ of K , suppose that $\inf(\omega(a_\alpha)) = 0$; in other words, F_1 comes from an $F \in A[T_1, \dots, T_n]$ such that the image F_0 of F in $k[T_1, \dots, T_n]$ is non-zero. Consider then in Z the closed subset $V(F)$; one has $V(F) \cap Z_0 = V(F_0)$, $V(F) \cap Z_1 = V(F_1)$, and since U_1 is dense in U (since it contains the generic point of X), $g(U) \subset V(F)$; in particular, $g_0(U_0) \subset V(F) \cap Z_0 = V(F_0)$; but since $F_0 \neq 0$, $V(F_0)$ is a proper closed part of Z_0 , contradicting the surjectivity of g_0 . $\square$

Proposition (14.3.11). — *Let $f : X \rightarrow S$ be a morphism of finite type, $(g_\lambda)_{\lambda \in L}$ a family of universally open morphisms $g_\lambda : Y_\lambda \rightarrow S$, and for every $\lambda \in L$, let $u_\lambda : Y_\lambda \rightarrow X$ be an S -morphism. For every $s \in S$, set $X_s = X \times_S \text{Spec}(\kappa(s))$, $(Y_\lambda)_s = Y_\lambda \times_S \text{Spec}(\kappa(s))$, and let $(u_\lambda)_s : (Y_\lambda)_s \rightarrow X_s$ be the morphism $u_\lambda \times 1$ deduced from u_λ by base change. Let $Z(s)$ be the adherence in X_s of the union of the sets $(u_\lambda)_s((Y_\lambda)_s)$, and set $d(s) = \dim(Z(s))$. Then, for every generisation s' of a point $s \in S$, one has $d(s) \leq d(s')$.*

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Proof. One knows (II, 7.1.4) that there exists a valuation ring A' and a morphism $h : S' = \text{Spec}(A') \rightarrow S$ such that if a (resp. b) is the closed point (resp. the generic point) of S' , $h(a) = s$ and $h(b) = s'$. Moreover, the morphism projection $p : X_s \otimes_{k(s)} k(a) \rightarrow X_s$ is a universally open equivalence relation

(2.4.10), hence for every part M of X_s , $p^{-1}(\overline{M})$ is equal to the adherence $\overline{p^{-1}(M)}$ (Bourbaki, *Top. gén.*, chap. I^{er}, 4^e éd., § 5, no. 3, prop. 7); one reasons similarly for $X_{s'}$, and taking into account (I, 3.4.8), (4.2.7) and the fact that the g_λ are universally open, one can reduce to proving the proposition in the situation obtained after base change $S' \rightarrow S$. One can therefore suppose $S' = S$, s the closed point and s' the generic point of S . The hypothesis that g_λ is open implies that every irreducible component of Y_λ dominates S (1.10.4), hence its generic point is a maximal point of $(Y_\lambda)_{s'}$; if Z denotes the adherence in X of $Z(s')$, one has $u_\lambda(Y_\lambda) \subset Z$, and hence $(u_\lambda)_s((Y_\lambda)_s) \subset Z_s = Z \cap X_s$; in other words, $Z(s) \subset Z_s$, whence $\dim(Z(s)) \leq \dim(Z_s)$. But applying (14.3.10) to a reduced sub-prescheme of X having for underlying space an irreducible component of Z_s , one obtains $\dim(Z(s)) \leq \dim(Z_{s'})$ (the inequality arising from the fact that there can be irreducible components of Z not meeting X_s). Since $Z(s')$ is by definition closed in $X_{s'}$, one has $Z_{s'} = Z(s')$, which completes the proof. $\square$

Remarks (14.3.12). — The case considered by M. Artin was that where $Y_\lambda = S$ for every λ , i.e. where (u_λ) is a family of S -sections of X . Another useful case is that where the family (u_λ) is reduced to a single element; one can always reduce to this case by considering the prescheme Y which is the coproduct of the Y_λ and the morphisms $g : Y \rightarrow S$ and $u : Y \rightarrow X$ whose restrictions to each Y_λ are respectively g_λ and u_λ .

Proposition (14.3.13). — *Let $f : X \rightarrow Y$ be a morphism locally of finite type, y a point of Y , x a maximal point of the fibre $X_y = X \times_Y \text{Spec}(\kappa(y))$. Consider the following conditions:*

label=a) f is universally open at the point x (or also at every point of the irreducible component of X_y of generic point x (14.3.3.1, (ii))).

lbel=b) For every irreducible component Y_0 of Y containing y , there exists an irreducible component Z of X containing x , dominant Y_0 and such that $\dim_x(Z \cap X_y) = \dim_x(Z \cap X_\eta) = \dim(Z \cap X_\eta)$, where η is the generic point of Y_0 (which means that Z is equidimensional over Y_0 at the point x (13.2.2)).

b') For every open neighbourhood U of x in X and every generisation y' of y , one has $\dim(U \cap X_{y'}) \geq \dim_x(U \cap X_y)$.

Then one has the implications $a) \Rightarrow b) \Leftrightarrow b')$.

Proof. The implication $a) \Rightarrow b)$ follows from (14.2.1) applied at the point x (which is a generic point of X_y) and from (13.2.7). The equivalence $b) \Leftrightarrow b')$ follows from (13.2.3). $\square$

Remarks (14.3.14). — We do not know if in (14.3.13), the conclusion subsists when one replaces hypothesis $a)$ by the weaker hypothesis that f be open at the point x .

14.4. The Chevalley criterion for universally open morphisms

Theorem (14.4.1). — *Let $f : X \rightarrow Y$ be a morphism locally of finite type, y a point of Y , x a maximal point of the fibre $X_y = X \times_Y \text{Spec}(\kappa(y))$. Suppose y geometrically unibranch. Then the following conditions are equivalent:*

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label=a) f is universally open at the point x (or also at every point of the irreducible component of X_y of generic point x (14.3.3.1, (ii))).

lbel=b) If Y_0 is the unique irreducible component of Y containing y , η its generic point, there exists an irreducible component Z of X containing x , dominating Y_0 and such that $\dim_x(Z \cap X_y) = \dim(Z \cap X_\eta)$ (i.e. Z is equidimensional over Y_0 at the point x (13.2.2)).

b') For every open neighbourhood U of x in X and every generisation y' of y , one has $\dim(U \cap X_{y'}) \geq \dim_x(U \cap X_y)$.

If moreover Y is locally Noetherian, these conditions are also equivalent to the following:

lbel=c) f is open at the point x .

Proof. Note first that since $(\mathcal{O}_y)_{\text{red}}$ is integral, y belongs to a single irreducible component Y_0 of Y . The fact that $b)$ and $b')$ are equivalent and that $a)$ implies $b')$ follows from (14.3.13); on the other hand, if Y is locally Noetherian, one has seen in (14.2.3) that $c)$ implies $b)$. It remains therefore to show that when y is geometrically unibranch, $b)$ implies $a)$.

Lemma (14.4.1.1). — *Let Y be a prescheme, $Y' = Y[T_1, \dots, T_n]$, y' a point of Y' , y its image in Y . If Y is geometrically unibranch at the point y , then Y' is geometrically unibranch at the point y' .*

Indeed, as for every $y \in Y$, $\text{Spec}(k(y)[T_1, \dots, T_n])$ is geometrically regular over $k(y)$ (0, 17.3.7), the structural morphism $Y' \rightarrow Y$ is smooth (6.8.1), and it suffices to apply (11.3.14).

Lemma (14.4.1.2). — *Let A be an integral local ring unibranch, B an integral ring over A , $\mathfrak{n}$ a prime ideal of B above the maximal ideal $\mathfrak{m}$ of A . Then the morphism $\text{Spec}(B_{\mathfrak{n}}) \rightarrow \text{Spec}(A)$ is surjective, i.e. for every prime ideal $\mathfrak{p}$ of A , there exists a prime ideal $\mathfrak{q}$ of B such that $\mathfrak{q} \subset \mathfrak{n}$ and $\mathfrak{q} \cap A = \mathfrak{p}$.*

Let K (resp. L) be the field of fractions of A (resp. B), A' the integral closure of A , B' the sub-ring of L generated by A' and B , so that one has a commutative diagram of canonical injections. Since B' is integral over B , there exists a prime ideal $\mathfrak{n}'$ of B' such that $\mathfrak{n}' \cap B = \mathfrak{n}$ (Bourbaki, *Alg. comm.*, chap. V, § 2, no. 1, th. 1), and (for the same reason) $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is surjective. On the other hand, since A is unibranch, A' is a *local* ring, hence $\mathfrak{n}' \cap A'$ is necessarily equal to the unique maximal ideal $\mathfrak{m}'$ of A' . By the second theorem of Cohen–Seidenberg (*loc. cit.*, § 2, no. 4, th. 3) the morphism $\text{Spec}(B'_{\mathfrak{n}'}) \rightarrow \text{Spec}(A')$ is surjective, hence the same is true of the composition $\text{Spec}(B'_{\mathfrak{n}'}) \rightarrow \text{Spec}(A') \rightarrow \text{Spec}(A)$; but this morphism is also the composition $\text{Spec}(B'_{\mathfrak{n}'}) \rightarrow \text{Spec}(B_{\mathfrak{n}}) \rightarrow \text{Spec}(A)$, hence the morphism $\text{Spec}(B_{\mathfrak{n}}) \rightarrow \text{Spec}(A)$ is surjective.

These lemmas being established, let us return to the proof of the implication $b) \Rightarrow a)$ in (14.4.1). By (14.3.3.1, (i)), it suffices to prove that, for every integer $n \geq 0$ and every point x' of $X' = X[T_1, \dots, T_n]$ above x , the morphism $f' : X' \rightarrow Y' = Y[T_1, \dots, T_n]$, deduced from f by base change, is open at the point x' . By the lemma (14.4.1.1), (2.3.4) and (4.2.7), one is therefore reduced to proving that f is *open* at the point x ; moreover, it suffices evidently ((14.1.2), (iii)) to show that the restriction of f to a closed sub-prescheme Z of X having Z as underlying space is open at the point x , so one can restrict to the case where $X = Z$ is irreducible and $V \cap X_y$ is irreducible. By (13.3.1), one can suppose that the morphism f factorises as $X \xrightarrow{g} Y'' = Y[T_1, \dots, T_m] \rightarrow Y$, where g is quasi-finite, dominant and locally of finite type. Since the structural morphism $Y'' \rightarrow Y$ is open (2.4.6), one is reduced to proving that $g : X \rightarrow Y''$ is open at the point x . By (14.4.1.1), $g(x)$ is a geometrically unibranch point of Y'' . One is therefore reduced to the following:

Lemma (14.4.1.3). — *Let X, Y be two irreducible preschemes, $f : X \rightarrow Y$ a morphism locally quasi-finite and dominant. If $x \in X$ is such that $y = f(x)$ is unibranch, then f is open at the point x .*

It suffices to prove that $f(\text{Spec}(\mathcal{O}_{X,x})) \rightarrow \text{Spec}(\mathcal{O}_{Y,y})$ is surjective (1.10.3). One can therefore suppose that X and Y are reduced (hence integral) and affine. By (8.12.9), the morphism f factorises as $X \xrightarrow{j} X_1 \xrightarrow{g} Y$, where j is an open immersion and g a *finite* dominant morphism; since X and X_1 are affine, j is affine, hence separated and quasi-compact, and thus factorises as $X \xrightarrow{h} X_2 \xrightarrow{u} X_1$, where X_2 is the schematic image of X by j , h the canonical injection and u an open immersion (I, 9.5.3). In other words, one can suppose that $X_1 = \text{Spec}(B)$, where B is a finite integral A -algebra containing A , and if $\mathfrak{n}$ is the prime ideal of B corresponding to the point x , the hypothesis that A is unibranch implies by (14.4.1.2) that $\text{Spec}(B_{\mathfrak{n}}) \rightarrow \text{Spec}(A)$ is surjective, i.e. $\text{Spec}(\mathcal{O}_{X,x}) \rightarrow \text{Spec}(\mathcal{O}_{Y,y})$ is surjective. $\square$

Corollary (14.4.2). — *Let $f : X \rightarrow Y$ be a morphism locally of finite type, y a point geometrically unibranch of Y , η the generic point of the unique irreducible component Y_0 of Y containing y . The following conditions are equivalent:*

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label=a) *f is universally open at all points of X_y (or, which amounts to the same (14.3.3.1, (ii)), at the maximal points of X_y).*

lbbel=b) *For every $x \in X_y$, there exists an irreducible component Z of X containing x and equidimensional over Y at the point x (13.2.2).*

b') *For every $x \in X_y$ and every open neighbourhood U of x in X , one has $\dim(U \cap X_{\eta}) \geq \dim_x(U \cap X_y)$.*

b'') *For every open U of X , one has $\dim(U \cap X_{\eta}) \geq \dim(U \cap X_y)$.*

When moreover Y is locally Noetherian, these conditions are also equivalent to the following:

lcbel=c) *f is open at all points of X_y (or, which amounts to the same (14.3.3.1, (ii)), at the maximal points of X_y).*

Proof. The equivalence of $a)$ and $c)$ when Y is locally Noetherian follows from (14.4.1); the conditions $b)$ or $b')$, applied to the *maximal* points of X_y , imply $a)$ by virtue also of (14.3.3.1, (ii)); finally, $b')$ and $b'')$ are equivalent, since $\dim(U \cap X_y) = \sup_{x \in X_y} (\dim_x(U \cap X_y))$.

It remains to see that condition $a)$ implies $b)$ and $b')$ at every point $x \in X_y$. Set $d = \dim_x(X_y)$, and let x' be the generic point of an irreducible component of X_y containing x and of dimension d . By $a)$ and (14.4.1), there is an irreducible component Z of X containing x' and equidimensional over Y at the point x' , so that $\dim_{x'}(Z \cap X_y) = \dim(Z \cap X_{\eta})$. But by construction $\dim_{x'}(Z \cap X_y) = \dim_{x'}(X_y) = d$, and $\dim_x(Z \cap X_y) \leq \dim(Z \cap X_y) \leq d$; taking into account (13.1.6), this shows that Z is equidimensional over Y at the point x ; hence $a)$ implies $b)$. Moreover, $\dim(X_{\eta}) \geq \dim(Z \cap X_{\eta}) = d = \dim_x(X_y)$. Replacing X by an open neighbourhood U of x , one sees that $a)$ implies $b')$. $\square$

Corollary (14.4.3). — *Let Y be a prescheme geometrically unibranch, $f : X \rightarrow Y$ a morphism locally of finite type. The following conditions are equivalent:*

label=a) f is universally open.

lbbel=b) For every open U of X , every $y \in Y$ and every generisation y' of y one has $\dim(U \cap X_{y'}) \geq \dim(U \cap X_y)$.

If moreover Y is locally Noetherian, these conditions are also equivalent to the following:

lcbel=c) f is open.

Proof. This follows from (14.4.2) applied to each point $y \in Y$. $\square$

Corollary (14.4.4). — (Chevalley criterion). *Let $f : X \rightarrow Y$ be a morphism locally of finite type.*

label=(i) If f is equidimensional at a point $x \in X$ (13.3.2) and if $y = f(x)$ is a point geometrically unibranch of Y , f is universally open at the point x .

lbbel=(ii) If Y is geometrically unibranch, f is universally open at all points of X where f is equidimensional, and the set of these points is open in X . In particular, if f is equidimensional, it is universally open.

Proof. Assertion (ii) is a trivial consequence of (i), since the set of points where f is equidimensional is already open (13.3.2). As to assertion (i), it follows from the fact that condition $b)$ of (14.4.1) is verified at the generic point of an irreducible component of X_y containing x , taking into account (13.3.1); it thus suffices to apply (14.4.1). $\square$

Remarks (14.4.5). — One can prove that if Y is locally Noetherian, and if all the generisations of $y = f(x)$ are geometrically unibranch points of Y (cf. (6.15.2)), then, if f is equidimensional at the point x , it is universally open in a neighbourhood of x .

Corollary (14.4.6). — *Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism locally of finite type. Let y be a point geometrically unibranch of Y , and suppose that for every $x \in f^{-1}(y)$, the ring $\mathcal{O}_{X,x}$ is equidimensional. Then the following conditions are equivalent:*

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label=a) f is equidimensional (13.3.2) at all points of $f^{-1}(y)$.

lbbel=b) f is open at all points of $f^{-1}(y)$.

lcbel=c) f is universally open at all points of $f^{-1}(y)$.

Proof. Indeed, $c)$ trivially implies $b)$, and $a)$ implies $c)$ by virtue of (14.4.4); finally, by the hypotheses on $\mathcal{O}_y$ and $\mathcal{O}_x$, $b)$ implies $a)$ by (14.2.2). $\square$

Proposition (14.4.7). — *Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , such that $y = f(x)$ is geometrically unibranch. The following conditions are equivalent:*

label=a) f is equidimensional (13.3.2) at the point x .

lbbel=b) The ring $\mathcal{O}_{X,x}$ is equidimensional and f is open at the generic points of the irreducible components of $f^{-1}(y)$ containing x (hence also at every point of such a component).

lcbel=c) The ring $\mathcal{O}_{X,x}$ is equidimensional, and f is universally open at the generic points of the irreducible components of $f^{-1}(y)$ containing x (hence also at every point of such a component).

Proof. Condition $a)$ implies that f is equidimensional at all the generic points of the irreducible components of $f^{-1}(y)$ containing x (13.3.1), and hence (by (14.4.4)) universally open at these points;

the same reasoning applied to each f_i (taking into account (13.3.3)) proves the last assertion of the proposition, by (14.3.3.1, (ii)). Moreover, by (14.2.1), one has the relations

$$(14.4.7.1) \quad \dim(\mathcal{O}_{X_i, x}) = \dim(\mathcal{O}_y) + \dim(\mathcal{O}_{X_i, x} \otimes_{\mathcal{O}_y} k(y))$$

$$(14.4.7.2) \quad \dim(\mathcal{O}_{X, x}) = \dim(\mathcal{O}_y) + \dim(\mathcal{O}_{X, x} \otimes_{\mathcal{O}_y} k(y))$$

and since f is equidimensional at the point x , it follows from (13.3.1) that $\dim_x(X_i \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$, hence (5.2.3) $\dim(\mathcal{O}_{X_i, x} \otimes k(y)) = \dim(\mathcal{O}_{X, x} \otimes k(y))$. One concludes that $\dim(\mathcal{O}_{X_i, x}) = \dim(\mathcal{O}_{X, x})$ for every i , i.e. $\mathcal{O}_{X, x}$ is equidimensional, which shows that a) implies c). It is clear that c) implies b); finally, b) implies the relation (14.4.7.2) by virtue of (14.2.1); it then follows from (14.3.6) that b) implies a). $\square$

Proposition (14.4.8). — *Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism locally of finite type, y a point of Y , x a point maximal of $X_y = f^{-1}(y)$. The following conditions are equivalent:*

label=a) The morphism f is universally open at the point x , i.e. for every base change $g : Y' \rightarrow Y$, the property $\mathbf{P}(Y')$ holds: for every point x' of $X' = X \times_Y Y'$ above x , $f' = f_{(Y')} : X' \rightarrow Y'$ is open at x' .

a') The property $\mathbf{P}(Y')$ is true for every finite morphism $g : Y' \rightarrow Y$.

a'') The property $\mathbf{P}(Y'')$ is true for the normalised Y'' of Y_{red} (II, 6.3.8).

lbbel=b) For every point x'' of $X'' = X \times_Y Y''$ above x , there exists an irreducible component Z'' of X'' containing x'' and equidimensional over Y'' at the point x'' .

Proof. It is trivial that a) implies $a')$. To show that $a')$ implies $a'')$, note that one can write $Y'' = \text{Spec}(\mathcal{B})$, where $\mathcal{B}$ is an $\mathcal{O}_Y$ -Algebra quasi-coherent, integral over $\mathcal{O}_Y$; since Y is Noetherian, $\mathcal{B}$ is an inductive limit of its sub- $\mathcal{O}_Y$ -Algebras $\mathcal{B}_\lambda$, which are quasi-coherent and of finite type (I, 9.6.6); but then the $\mathcal{B}_\lambda$ are finite $\mathcal{O}_Y$ -Algebras (II, 6.1.2); one can therefore write $Y'' = \varprojlim Y'_\lambda$, where $Y'_\lambda = \text{Spec}(\mathcal{B}_\lambda)$, with $X'_\lambda = X \times_Y Y'_\lambda$. By $a')$, the morphisms $f'_\lambda : X'_\lambda \rightarrow Y'_\lambda$ are open at all points above x ; one concludes by (8.10.1) and (14.3.3.1, (i)) that f'' is open at all points of X'' above x .

Since Y'' is normal by definition, the fact that b) implies $a'')$ follows by (14.4.4) applied to the equidimensional irreducible component of the statement. It remains therefore to show that $a'')$ implies a) and b). By (1.10.3), one can restrict to the case where $Y = \text{Spec}(\mathcal{O}_y)$, and noting that the canonical morphism $\text{Spec}(\mathcal{O}_y) \rightarrow Y$ is universally bicontinuous (I, 3.6.5), and that $Y'' \times_Y \text{Spec}(\mathcal{O}_y)$ is the normalised of $(\text{Spec}(\mathcal{O}_y))_{\text{red}}$ as it follows from the permutability of integral closure and localisation (Bourbaki, *Alg. comm.*, chap. V, § 1, no. 5, prop. 16). Suppose therefore $Y = \text{Spec}(A)$, where A is a local Noetherian ring, and y the closed point of Y ; one knows (0, 23.2.5) that there exists a factorisation

$$Y'' \xrightarrow{u} Y_1 \xrightarrow{v} Y$$

of the structural morphism, such that v is a morphism finite surjective, u a morphism entire, radiciel et dominant, hence by suite un homéomorphisme universel (2.4.5). If one sets $X_1 = X \times_Y Y_1$, $f_1 = f_{(Y_1)}$, the projection $X'' \rightarrow X_1$ is therefore a homeomorphism, and hypothesis $a'')$ implies that $f_1 = f_{(Y_1)} : X_1 \rightarrow Y_1$ is open at all points of X_1 above x . Moreover, this shows that to prove property b), it suffices to prove the same property where one replaces Y'' , X'' and x'' by Y_1 , X_1 and a point x_1 of X_1 above x . But Y_1 is Noetherian and moreover it is normal and radiciel over the normalised (6.15.1); the property follows then from (14.4.1). It remains to show that f is universally open at the point x , which follows from (14.4.8.1) applied to the entire surjective morphism $v : Y_1 \rightarrow Y$. $\square$

Lemma (14.4.8.1). — *Let $v : Y_1 \rightarrow Y$ be a morphism closed (resp. universally closed) and surjective. For a morphism $f : X \rightarrow Y$ to be open (resp. universally open) at a point $x \in X$, it suffices that $f_1 = f_{(Y_1)} : X_1 = X \times_Y Y_1 \rightarrow Y_1$ be open (resp. universally open) at all points of X_1 above x .*

Proof. The second assertion follows trivially from the first and from the fact that for every base change $Y' \rightarrow Y$, the morphism $v_{(Y')} : Y_1 \times_Y Y' \rightarrow Y'$ is again surjective and closed if v is universally closed. To prove the first assertion, consider an open neighbourhood U of x in X ; since v is closed and surjective, for $f(U)$ to be a neighbourhood of $y = f(x)$, it is necessary and sufficient that $v^{-1}(f(U))$ be a neighbourhood of $v^{-1}(y)$. But if $p : X_1 \rightarrow X$ is the canonical projection, one has

$v^{-1}(f(U)) = f_1(p^{-1}(U))$ (I, 3.4.8), and the hypothesis implies that $f_1(p^{-1}(U))$ is a neighbourhood of $f_1(p^{-1}(x)) = v^{-1}(y)$ (I, 3.4.8). $\square$

Corollary (14.4.9). — Under the hypotheses of (14.4.8), suppose moreover that Y is excellent. Then the equivalent conditions of (14.4.8) are also equivalent to the following: IV-3 | 215

label=a') The property $\mathbf{P}(Y')$ is true for any faithfully flat morphism $g : Y' \rightarrow Y$.

In particular, f is universally open at x if and only if f is open at the maximal points of X_y above x .

Proof. This follows immediately from (14.4.8) and (14.1.4). $\square$

Remarks (14.4.10). — label=(i) In (12.1.8), (ii), we claimed that for $\mathcal{F} = \mathcal{O}_X$ (so f flat), one cannot weaken the condition “ f is flat” to “ f is universally open” in (12.1.1.5). Here is the example: take $A_0 = k[S, T]$, $\mathfrak{a} = (T_1) + (T_2 - T_3)$, $X = \text{Spec}(A/\mathfrak{a})$ with two irreducible components, $Y = X_1 = \text{Spec}(A/\mathfrak{b})$ for suitable $\mathfrak{b}$.

lbbel=(ii) With the same notations, this is also an example where f is proper, equidimensional and flat, but $E(y)$ has two elements for $y \neq y_0$, and the fibre $X_{y'}$ at the generic point has two irreducible components of dimensions 0 and 1, whereas X_{y_0} has a single irreducible component of dimension 1.

14.5. Quasi-sections of flat morphisms of finite type

(14.5.1). Let $f : X \rightarrow Y$ be a morphism locally of finite type, y a point of Y , x a closed point of $X_y = f^{-1}(y)$. By a *quasi-section* of f at the point x (relative to Y , or to y), we shall mean a closed sub-prescheme X' of an open neighbourhood U of x in X , such that $x \in X'$ and that the restriction $f|_{X'} : X' \rightarrow Y$ is a morphism *quasi-finite* and *dominant*. IV-3 | 216

Proposition (14.5.2). — Let Y be a prescheme locally Noetherian, irreducible of generic point η , $f : X \rightarrow Y$ a morphism locally of finite type, y a point of Y , x a closed point of X_y .

label=(i) For there to exist a quasi-section of f at x , it is necessary and sufficient that there exist an irreducible component Z of X containing x and dominating Y .

lbbel=(ii) When f is universally open at x (in particular when f is flat at x), such a quasi-section always exists.

Proof. We can restrict to the case where $X = U$ is an affine open neighbourhood of x such that all the irreducible components of $f^{-1}(y)$ contain x . These components correspond to the minimal prime ideals of $\mathcal{O}_x/\mathfrak{m}_y\mathcal{O}_x$, and by hypothesis these ideals are distinct from $\mathfrak{m}_x/\mathfrak{m}_y\mathcal{O}_x$ (Bourbaki, *Alg. comm.*, chap. II, §1, n° 1, prop. 2); to obtain a $g \in \Gamma(U, \mathcal{O}_X)$ satisfying the conditions of the statement, it suffices to take $g \in \mathfrak{I}_x$ such that the image of g in $\mathfrak{m}_x$ does not belong to any of the preceding prime ideals. Moreover, we have $\text{codim}(X', X) \leq 1$ (5.1.8), and since X' contains none of the irreducible components of $f^{-1}(y)$ and these are of dimension e , we have (0, 14.2.2.2) $\dim_x(X' \cap f^{-1}(y)) \leq e - 1$. It then suffices to apply (14.5.1). $\square$ IV-3 | 217

Proposition (14.5.3). — Let Y be an irreducible prescheme, locally Noetherian, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X . Suppose that f is equidimensional at the point x and that x is closed in $f^{-1}(f(x))$. Then there exists an irreducible closed part X' of X , locally closed in X , containing x , and such that the restriction $X' \rightarrow Y$ of f (where X' denotes the reduced subprescheme of X having X' as underlying space) is a quasi-finite and dominant morphism.

Proof. Indeed, with the notations of (14.5.2), the hypothesis that x is closed in $f^{-1}(y)$ implies that x is not a maximal point of $X' \cap f^{-1}(y)$ as long as $e - 1 \geq 1$. It suffices to apply (14.5.2) by descending induction on $e = \dim_x(f^{-1}(f(x)))$ until one arrives at $e = 1$; applying (14.5.2) in this last case gives an X' such that $X' \cap f^{-1}(f(x))$ is of dimension 0, hence finite and discrete; since X' is equidimensional over Y , $X' \rightarrow Y$ de f is the reduced subprescheme of X having X' as underlying space, which is quasi-finite and dominant. $\square$

Corollary (14.5.4). — Under the hypotheses of (14.5.3), suppose further that $Y = \text{Spec}(A)$, where A is a Noetherian local integral ring, complete, and that $y = f(x)$ is the unique closed point of Y . Then there exists an integral local ring A' , containing A , which is a finite A -algebra and has the following property: if one sets

$Y' = \operatorname{Spec}(A')$ and $X' = X \times_Y Y'$, there exists a Y' -section $h : Y' \rightarrow X'$ such that the composite morphism $Y' \xrightarrow{h} X' \rightarrow X$ is an immersion whose image contains x .

Proof. Replacing if necessary X by an irreducible reduced subprescheme of X , we can, by virtue of (14.5.3), restrict to the case where the morphism f is already quasi-finite and dominant. Using (II, 6.2.5), we deduce that $\mathcal{O}_x = A'$ is an integral ring which is a finite A -algebra, and that X is the disjoint sum of the closed subprescheme $Y' = \operatorname{Spec}(A')$ and of a subprescheme Y'' ; the prescheme Y' answers the question, the composite morphism $Y' \rightarrow X' \rightarrow X$ being none other than the canonical morphism $\operatorname{Spec}(\mathcal{O}_x) \rightarrow X$. $\square$

(14.5.5). Remark. If, in the statement of (14.5.4), one does not require that the morphism $Y' \rightarrow X$ be an immersion, one can suppose that A' is moreover *integrally closed*: it suffices to replace A' by its integral closure A_1 , since one knows (0, 23.1.5) that A_1 is an A -module of finite type.

Proposition (14.5.6). — *Let Y be a locally Noetherian prescheme, irreducible, regular, and of dimension 1, $f : X \rightarrow Y$ a morphism locally of finite type, y a point of Y . The following conditions are equivalent:*

- a) f_{red} is flat at every point of $f^{-1}(y)$.
- b) f is universally open in a neighbourhood of $f^{-1}(y)$.
- c) f is open in a neighbourhood of $f^{-1}(y)$.
- d) Every irreducible component of X meeting $f^{-1}(y)$ dominates Y .
- e) For every point $x \in f^{-1}(y)$, closed in $f^{-1}(y)$, and every irreducible component X_i of X containing x , there exists an irreducible closed part X' of X , locally closed in X , containing x , contained in X_i , and such that the restriction $X' \rightarrow Y$ of f is a quasi-finite and dominant morphism.

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Proof. The equivalence of a), b), c), and d) has already been proved (14.3.8). It is clear that e) implies d). Finally, to prove that c) implies e), we can restrict to a morphism of finite type; consider a closed point x of $f^{-1}(y)$ and let X_i be an irreducible component of X containing x . Since the restriction $f_i : X_i \rightarrow Y$ of f to X_i is a dominant morphism, it follows from (14.2.2) that f_i is open at the generic points of $X_i \cap f^{-1}(y)$, hence f_i is equidimensional at the point x , and one concludes with the aid of (14.5.3). $\square$

(14.5.7). Remark. If, in the statement of (14.5.5), one supposes that $Y = \operatorname{Spec}(A)$, where A is a discrete valuation ring, complete, and that y is the closed point of Y , one can moreover suppose that $X' = \operatorname{Spec}(A')$, where A' is a discrete valuation ring which is a finite A -algebra, as the proof shows from (14.5.4) and the fact that a local integral ring, regular and of dimension 1, is a discrete valuation ring (II, 7.1.6).

Proposition (14.5.8). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, y a point of Y . For every Y -prescheme Y' , set $X(Y') = \operatorname{Hom}_Y(Y', X)$; let x be a point of $f^{-1}(y)$; in order that f be universally open at the point x , it is necessary and sufficient that the following condition be satisfied:*

For every discrete valuation ring A , complete, of algebraically closed residue field, every morphism $g : Y' = \operatorname{Spec}(A) \rightarrow Y$ such that the image by g of the closed point y' of Y' is equal to y , and every element $u_0 \in X(\operatorname{Spec}(k(y')))$ such that $u_0(y') = x$, there exists a discrete valuation ring B , a local A -homomorphism $A \rightarrow B$ making B a finite A -algebra, and, if one sets $Z' = \operatorname{Spec}(B)$, an element $u \in X(Z')$ such that, if z' is the closed point of Z' , the diagram

$$\begin{array}{ccc} \operatorname{Spec}(k(z')) & \longrightarrow & Z' \\ \downarrow & & \downarrow u \\ \operatorname{Spec}(k(y')) & \xrightarrow{u_0} & X \end{array}$$

is commutative.

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Proof. Note that if A and B satisfy the conditions of the statement, B is a complete discrete valuation ring (Bourbaki, *Alg. comm.*, chap. III, §3, n° 3, prop. 7, and chap. IV, §2, n° 2, cor. 3 of prop. 9) of residue

field isomorphic to that of A , hence algebraically closed; the condition of the statement is therefore *necessary*. Let us now prove that the condition is *sufficient*. Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. By hypothesis, there exists a point $x' \in X'$ above x and y' that is rational over $k(y')$; moreover, since $u(z') = x''$, there is in $X'' = X \times_Y Z'$ a point x'' above x' and the closed point z' of Z' ; the projections in X and Z' are x and z' respectively, and if T' is the underlying space of Z' , the morphism composed $Z' \xrightarrow{v} X'' \rightarrow X$ shows that x' lies in the closure of the image. By virtue of (14.3.6), the condition of the statement implies that $\bar{T} = g^{-1}(T')$ for all closed parts T' of $X_{(K')}$; applying (14.4.1) to the quasi-finite and dominant morphism f_i , one sees that f_i is universally open; since f is open at all the points of $X_i \cap f^{-1}(y)$ and since the closed points of X_y are dense in X_y (10.3.1), it follows from (14.1.4) that f is universally open at the point x . $\square$

Proposition (14.5.9). — *Let Y be a locally Noetherian irreducible prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, y a point of Y that is geometrically unibranch. The following conditions are equivalent:*

- a) f is universally open at every point of $f^{-1}(y)$.
- b) f is open at every point of $f^{-1}(y)$.
- c) For every irreducible component Z of $f^{-1}(y)$, with generic point z , there exists an irreducible component X_i of X containing z and equidimensional over Y at the point z .
- d) For every closed point x of $f^{-1}(y)$, there exists an irreducible closed part X' of X , locally closed in X , containing x , and such that the restriction $X' \rightarrow Y$ of f is a quasi-finite and dominant morphism.

Proof. The equivalence of a), b), and c) has already been proved (14.4.2). To prove that a) implies d), note that by virtue of (14.4.2), a) implies that there exists an irreducible component containing x and equidimensional on Y at the point x ; the existence of X' then follows from (14.5.3). Inversely, suppose d) satisfied; by virtue of the Chevalley criterion (14.4.4), the restriction $X' \rightarrow Y$ of f is universally open at the point x , and *a fortiori* f is open at the point x . Hence f is open at all the closed points of $X_y = f^{-1}(y)$. But X_y is a $k(y)$ -prescheme locally of finite type, hence a Jacobson prescheme; the set of closed points of X_y is therefore dense in X_y (10.3.1), and it follows from (14.1.4) that f is universally open at every point of $f^{-1}(y)$, which proves d) implies b). $\square$

Proposition (14.5.10). — *Let Y be a Noetherian prescheme, $f : X \rightarrow Y$ a morphism universally open, surjective, and locally of finite type. Then there exists a finite surjective morphism $g : Y' \rightarrow Y$ such that, if one sets $X' = X \times_Y Y'$ and $f' = f_{(Y')} : X' \rightarrow Y'$, every point $y' \in Y'$ admits a neighbourhood U' such that there exists a U' -section of $f'^{-1}(U')$.*

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Proof. We will prove the proposition in several steps.

I) *Reduction to the case where Y is integral.* If one has proved the proposition for each of the reduced subpreschemes Y_i having for underlying space an irreducible component of Y , and for the inverse images $f^{-1}(Y_i)$, it is clear that the prescheme Y' which is the sum of the corresponding Y'_i answers the question. We can therefore suppose Y integral. Then, in the conclusion, we can also take Y' integral (it suffices to replace it by an irreducible component suitable).

II) *Local character on Y .* We will now show that if one can cover Y by finitely many open sets U_j such that, for every j , the conclusion of the proposition is true for the morphism $f^{-1}(U_j) \rightarrow U_j$, the conclusion is also true for f . Indeed, one can evidently suppose U_j affine, so that $U_j = \text{Spec}(A_j)$, where A_j is an integral Noetherian ring whose field of fractions $K = R(Y)$ is the field of rational functions on Y . For every j , by hypothesis there is a finite integral A_j -algebra A'_j such that the morphism $g_j : U'_j = \text{Spec}(A'_j) \rightarrow \text{Spec}(A_j) = U_j$ is injective and that the morphism corresponding $g_j : U'_j = \text{Spec}(A'_j) \rightarrow \text{Spec}(A_j) = U_j$ satisfies the conditions of the proposition for U_j and $f^{-1}(U_j)$. Let then K' be a finite extension of K containing the field of fractions of all the A'_j (which are finite extensions of K). Consider the *normalisé* Y'' of Y in K' (II, 6.3.8), which is of the form $\text{Spec}(\mathcal{B})$, where $\mathcal{B}$ is an $\mathcal{O}_Y$ -Algebra coherent of $\mathcal{B}|U_j$. These definitions show that for every j , $A'_j = \Gamma(U_j, \mathcal{O}_{U'_j}) = \Gamma(U'_j, \mathcal{O}_{U'_j})$ identifies with a sub- A_j -algebra finite of $\Gamma(U_j, \mathcal{B})$; in other words, $(g_j)_*(\mathcal{O}_{U'_j})$ is a sub- $\mathcal{O}_{U_j}$ -Algebra coherent of $\mathcal{B}|U_j$. There exists therefore a sub- $\mathcal{O}_Y$ -Module coherent $\mathcal{C}'_j$ of $\mathcal{B}$ such that $\mathcal{C}'_j|U_j = \mathcal{A}'_j$ (I, 9.4.7). If one sets $\mathcal{C}' = \sum_j \mathcal{C}'_j$, the sub- $\mathcal{O}_Y$ -Algebra $\mathcal{B}'$ of $\mathcal{B}$

generated by $\mathcal{C}'$ is a *coherent* $\mathcal{O}_Y$ -Algebra. Let us then show that $Y' = \text{Spec}(\mathcal{B}')$ answers the question. Indeed, it is clear that the morphism $g : Y' \rightarrow Y$ is finite surjective and that Y' is integral; moreover, for every j , $\Gamma(U_j, \mathcal{B}')$ is a sub- A_j -algebra finite, and since the restriction of g to $g^{-1}(U_j) \rightarrow U_j$ factors as $g^{-1}(U_j) \rightarrow U'_j \rightarrow U_j$, and since the existence of local sections is stable under base change, this establishes our assertion for every $y \in Y$ belonging to some U_j .

III) *Reduction to the case where Y is integral, local and geometrically unibranch.* Suppose first that the proposition has been proved when Y is integral and *local* (with Y' integral), and let us show that it is valid when Y is integral (noetherian) affine of arbitrary dimension. Indeed, by virtue of reduction II), it suffices to prove for every point $y \in Y$ that there exists an affine open V of y in Y for which the proposition holds. Let $Y_1 = \text{Spec}(A_{\mathfrak{p}})$, where $\mathfrak{p} = \mathfrak{p}_y$; by hypothesis, there exists a finite morphism $g_1 : Y'_1 \rightarrow Y_1$, surjective, with $Y'_1 = \text{Spec}(B_1)$, B_1 being a finite $A_{\mathfrak{p}}$ -algebra integral. The $A_{\mathfrak{p}}$ -module B_1 admits a finite system of generators of the form z_i/s (with $s \in A - \mathfrak{p}$, z_i integer over A_s), which one can, by multiplying by a suitable element of A , take to be elements of the field of fractions of B_1 integer over A_s ; it follows that if V is the affine open $D(s) \subset Y$, Y'_1 identifies with $Y_1 \times_Y V'$, where V' is the affine open V satisfies the conditions of the proposition. Reasoning as at the start of reduction III), we can therefore suppose A local, integral and geometrically unibranch.

IV) *Reduction to the case where X is integral, affine, f quasi-finite, surjective, birational, and universally open.* Suppose therefore $Y = \text{Spec}(A)$ integral, local and geometrically unibranch. There exists then an irreducible subprescheme X_0 of X such that the restriction $f_0 : X_0 \rightarrow Y$ of f is a quasi-finite and dominant morphism and that $f_0(X_0)$ contains the closed point y of Y (14.5.9); since Y is geometrically unibranch, it follows from (14.4.1) that f_0 is still *universally open*. Since moreover one can suppose X_0 reduced, hence integral, and on the other hand, we can, by replacing X by X_0 , suppose that X is *integral* and f *quasi-finite and dominant*; since f is open and surjective, every open of Y containing y is equal to Y , f is *surjective*.

Let ξ, η be the generic points of X and Y respectively; $k(\xi)$ is therefore a finite extension of $K = k(\eta)$. By (4.6.8), there is a finite extension K' of K such that $(\text{Spec}(k(\xi) \otimes_K K'))_{\text{red}}$ is geometrically reduced over K and that its irreducible components are geometrically irreducible; it follows ((4.5.9) and (4.6.1)) that the residue fields of $\text{Spec}(k(\xi) \otimes_K K')$ are finite extensions, primary and separable, of K' , hence are *equal* to K' . Applying again (0, 23.2.5) and (6.15.5), there exists a finite A -algebra integral B , having K' as field of fractions, containing A and such that $\text{Spec}(B)$ is geometrically unibranch; since the morphism $\text{Spec}(B) \rightarrow \text{Spec}(A)$ is surjective, one can evidently replace Y by $\text{Spec}(B)$ and X by $X \otimes_A B$ to prove the proposition. Reasoning as at the beginning of the reduction III), one can therefore suppose A local, integral and $Y = \text{Spec}(A)$ geometrically unibranch, closed point y . Moreover, f is then a quasi-finite morphism, *surjective and universally open*; each irreducible component X_i of X ($1 \leq i \leq m$) dominates Y (1.10.4) and is *birational* over Y . Let then x be a point of $f^{-1}(y)$ and let X_i be an irreducible component of X containing x , and let X_i be the reduced subprescheme (hence integral) of X having X_i as underlying space, and let $f_i : X_i \rightarrow Y$ be the restriction of f . Applying again (14.4.1) to the quasi-finite and dominant morphism f_i , one sees that f_i is *universally open*; if U is an affine open of X_i containing x , $f_i(U)$ is therefore an open of Y containing x .

V) *End of the proof.* Suppose therefore Y local, integral and geometrically unibranch, X integral and affine, f quasi-finite, surjective, birational and universally open; it follows that f is automatically *separated*. By virtue of the Main Theorem (8.12.6), there exists a factorisation $X \xrightarrow{j} Z \xrightarrow{u} Y$, where j is an open immersion and u a finite morphism. Replacing moreover Z by the closed image of j (I, 9.5.10), one can suppose that Z is *integral*; since u is finite and birational and Y geometrically unibranch, it follows from (III, 4.3.5 and 4.3.4) that u (and hence f) is a *universally radiciel homeomorphism*. By (2.4.5, , (ii)) f is a finite morphism. But then one responds to the conditions of the statement of Y' *integral* by simply taking $Y' = X$ and $g = f$, the Y' -section being the diagonal morphism Δ_f . Q.E.D. $\square$

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This result can be used to develop criteria of “descent” for the properties of universally open and surjective morphisms. We mention in particular the following criterion, due to D. Mumford:

Corollary (14.5.11). — Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $g : Y' \rightarrow Y$ a morphism locally of finite type, universally open and surjective; set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. Then, for f to be affine, it is necessary and sufficient that f' be so.

Proof. One need only prove that the condition is sufficient. The question being local on Y , we can suppose Y affine, hence Noetherian. By virtue of (14.5.10), there exists a finite surjective morphism $h : Y_1 \rightarrow Y$ such that, if one sets $Y'_1 = Y' \times_Y Y_1$ and $g' = g_{(Y_1)} : Y'_1 \rightarrow Y_1$, every point of Y_1 admits a neighbourhood U_1 such that there exists a U_1 -section of $g'^{-1}(U_1)$. If one sets $X_1 = X \times_Y Y_1$, the projection $p : X_1 \rightarrow X$ is a finite surjective morphism; hence, if one proves that X_1 is a scheme, then X_1 is affine if and only if X is affine, by the theorem of Chevalley (II, 6.7.1). It then suffices to prove that the morphism $f_1 = f_{(Y_1)} : X_1 \rightarrow Y_1$ is affine. Now, if one sets $f'_1 = (f_1)_{(Y'_1)} : X'_1 \rightarrow Y'_1$, f'_1 is affine by virtue of the hypothesis. The question being local on Y_1 , it suffices to prove that f is affine when Y admits a Y -section s of Y' . Now, by the lemma: $\square$

Lemma (14.5.11.1). — Let $f : X \rightarrow S$, $g : Y \rightarrow S$ be two morphisms, $p_1 : X \times_S Y \rightarrow X$, $p_2 : X \times_S Y \rightarrow Y$ the canonical projections. If $s : S \rightarrow Y$ is a section of g , then $s' = (1, s \circ f)_S$ is a section of the morphisms $f : X \rightarrow S$ and X , equipped with the morphisms f and $s' : X \rightarrow X \times_S Y$, identifies with the product of the Y -preschemes S and $X \times_S Y$ for the morphisms $s : S \rightarrow Y$ and $p_2 : X \times_S Y \rightarrow Y$.

Proof. This is a particular case of (I, 3.3.11), where one replaces the diagram by

$$\begin{array}{ccccc} X & \xrightarrow{p_1} & X \times_S Y & \xrightarrow{s'} & X \\ f \downarrow & & \downarrow p_2 & & \downarrow f \\ S & \xrightarrow{s} & Y & \xrightarrow{g} & S \end{array}$$

$\square$

Applying this lemma with S, Y replaced by Y, Y' , one sees that one can write $f = (f')_{(Y)}$ for the base change $s : Y \rightarrow Y'$, hence f is affine since f' is. Q.E.D.

A variant of this criterion is the following:

Corollary (14.5.12). — With the notations and general hypotheses of (14.5.11), let $\mathcal{L}$ be an $\mathcal{O}_X$ -Module IV-3 | 223
invertible. Suppose that there exists a closed part Z of X , proper over Y (II, 5.4.10) and such that the prescheme induced by X on $X - Z$ is affine; it then suffices that $\mathcal{L}' = \mathcal{L} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$ be ample relative to f' .

Proof. Keeping the notations of the proof of (14.5.11), set $\mathcal{L}_1 = \mathcal{L} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_1}$; it suffices to prove that $\mathcal{L}_1$ is ample relative to f_1 , and one can again suppose that g' admits a section s . If $\mathcal{L}'_1 = \mathcal{L}' \otimes_{\mathcal{O}_{Y'}} \mathcal{O}_{Y'_1}$, one can then write, by virtue of the lemma (14.5.11.1), $\mathcal{L}_1 = \mathcal{L}'_1 \otimes_{\mathcal{O}_{Y_1}} \mathcal{O}_{Y_1}$ for the base change $s : Y_1 \rightarrow Y'_1$; the conclusion follows therefore from two applications of (II, 4.6.13, (iii)). $\square$

§15. STUDY OF FIBRES OF A UNIVERSALLY OPEN MORPHISM

15.1. Multiplicities of fibres of a universally open morphism

Proposition (15.1.1). — Let Y be an irreducible locally Noetherian prescheme, of generic point η , $f : X \rightarrow Y$ IV-3 | 223
a morphism locally of finite type, y a point of Y , $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module; set $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$ (sheaf of modules on the fibre $f^{-1}(y)$). Let z be the generic point of an irreducible component of $\text{Supp}(\mathcal{F})$, and let X_i ($1 \leq i \leq n$) be the closed reduced sub-preschemes of X whose underlying spaces are the irreducible components of $\text{Supp}(\mathcal{F})$ containing z , and which are such that the restriction $X_i \rightarrow Y$ of f is a universally open morphism at the point z ; let z_i be the generic point of X_i ($1 \leq i \leq n$). If $\lambda_z(\mathcal{F}_y)$ (resp. $\lambda_i(\mathcal{F}_\eta)$) denotes the geometric multiplicity of $\mathcal{F}_y$ at a point $x \in f^{-1}(y)$ (resp. that of $\mathcal{F}_\eta$ at a point $t \in f^{-1}(\eta)$) (IV, 4.7.5), then

$$(15.1.1.1) \quad \lambda_z(\mathcal{F}_y) \geq \sum_{i=1}^n \lambda_{z_i}(\mathcal{F}_\eta).$$

If moreover the ring $\mathcal{O}_y$ is regular, then

$$(15.1.1.2) \quad \text{long}((\mathcal{F}_y)_z) \geq \sum_{i=1}^n \text{long}((\mathcal{F}_\eta)_{z_i}).$$

Proof. The fibres $f^{-1}(y')$ of f (for all $y' \in Y$) do not change when we replace f by f_{red} , so we may suppose that X and Y are reduced (2.4.3, (vi)), hence Y integral; we set $K = k(\eta)$. We may moreover replace f by the restriction $\text{Supp}(\mathcal{F}) \rightarrow Y$ of f to the closed sub-prescheme having $\text{Supp}(\mathcal{F})$ as underlying space, and thus reduce to the case where $X = \text{Supp}(\mathcal{F})$. Finally, if $y = \eta$, there is only one irreducible component X_i of X containing z (since $z_i = z$), and (15.1.1.1) and (15.1.1.2) are trivial. We may thus restrict to the case $y \neq \eta$; it then follows that the z_i belong to $f^{-1}(\eta)$, and so the second members of (15.1.1.1) and (15.1.1.2) are well-defined. Let Z_i be the closed reduced sub-prescheme of $f^{-1}(\eta)$ having $X_i \cap f^{-1}(\eta)$ as underlying space. Since for each i , the K' -prescheme $(Z_i \otimes_K K')_{\text{red}}$ is geometrically reduced over K' , there exists a finite radiciel extension K' of K such that the projection $f^{-1}(\eta) \otimes_K K' \rightarrow f^{-1}(\eta)$ is finite, dominant, and radiciel ((I, 3.5.7)), hence a homeomorphism (2.4.5). So z'_i is the unique point of $f^{-1}(\eta) \otimes_K K'$ above z_i . It follows from (4.7.9) and (4.7.5) that

$$(15.1.1.3) \quad \lambda_{z_i}(\mathcal{F}_\eta) = \lambda_{z'_i}(\mathcal{F}_\eta \otimes_K K')_{z'_i} = \text{long}((\mathcal{F}_\eta \otimes_K K')_{z'_i}).$$

Let V be a discrete valuation ring with field of fractions K' , dominating $\mathcal{O}_y$ (II, 7.1.7); set $Y' = \text{Spec}(V)$, $X' = X \times_Y Y'$, $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$; if η' is the generic point of Y' and y' its closed point, then $g(y') = y$, $g(\eta') = \eta$; the morphism $\text{Spec}(k(\eta')) \rightarrow \text{Spec}(k(\eta))$ being faithfully flat, hence (2.2.13) and (2.3.4), there is a generic point z' of an irreducible component of $f'^{-1}(y')$ such that $g'(z') = z$. Par construction, $k(\eta') = K'$, hence $f'^{-1}(\eta') = f^{-1}(\eta) \otimes_K K'$; on the other hand, if we set $X'_i = X_i \times_Y Y'$, the restriction $X'_i \rightarrow Y'$ of f' is a universally open morphism at the point z' , hence (1.10.3) there exists a generisation z'_i of z' , belonging to an irreducible component of $X'_i \cap f'^{-1}(\eta')$, so we may suppose $z'_i = z'_i$. By (4.7.9) and (4.7.5),

$$(15.1.1.4) \quad \lambda_z(\mathcal{F}_y) = \lambda_{z'}(\mathcal{F}_{y'}) \geq \text{long}((\mathcal{F}'_{y'})_z)$$

and it follows from this inequality and (15.1.1.3) that to establish (15.1.1.1), it suffices to show the inequality

$$(15.1.1.5) \quad \text{long}((\mathcal{F}'_{y'})_z) \geq \sum_{i=1}^n \text{long}((\mathcal{F}'_{\eta'})_{z'_i}).$$

Now consider the second inequality (15.1.1.2); we use the following lemma:

Lemma (15.1.1.6). — *Let A be a local regular ring that is not a field, k its residue field, K its field of fractions. There exists a discrete valuation ring V dominating A , having K for field of fractions, and whose residue field is a pure transcendental extension of k .*

Set $S = \text{Spec}(A)$, and consider the S -scheme P obtained by blowing up the closed point s of S (II, 8.1.3); if $\mathfrak{m}$ is the maximal ideal of A , then by definition $P = \text{Proj}(B)$, where B is the graded A -algebra $\bigoplus_{n \geq 0} \mathfrak{m}^n$. If $h : P \rightarrow S$ is the structural morphism, the fibre $h^{-1}(s)$ is isomorphic to $\text{Proj}(B \otimes_A k)$ (II, 2.8.10), and by definition $B \otimes_A k$ is the graded algebra $\bigoplus_{n \geq 0} \mathfrak{m}^n / \mathfrak{m}^{n+1}$; but since A is regular, this algebra is isomorphic to a polynomial algebra $k[t_1, \dots, t_e]$ (t_i indeterminates) (0, 17.1.1) and hence $h^{-1}(s) \cong \mathbf{P}_k^{e-1}$. Let ζ be the generic point of $h^{-1}(s)$; if t'_i ($1 \leq i \leq e$) are the classes of generators of $\mathfrak{m}$ modulo $\mathfrak{m}^2$, the local ring $B_{(t'_i)}$ is an affine neighbourhood of ζ in P , and we have $k(\zeta) = k(t_1, \dots, t_{e-1})$; on the other hand, since A is integrally closed, $B_{(t'_i)}$ is also integrally closed, and hence the local ring $\mathcal{O}_{P, \zeta}$ is regular. Finally, as $A = \mathcal{O}_s$, $\dim(\mathcal{O}_{P, \zeta}) = \dim(\mathcal{O}_s) - (e - 1)$, since h is a birational morphism, that $\dim(h^{-1}(s)) = e - 1$, and the local ring of P at its generic point ζ is a discrete valuation ring of dimension 1, and $\mathcal{O}_{P, \zeta} = V$ is a discrete valuation ring that answers the question (II, 7.1.6).

This lemma being established, we can resume the reasoning for the inequality (15.1.1.1) with $Y' = \text{Spec}(V)$; this time $k(y')$ is a *separable* extension of $k(y)$, and by (4.7.9), $\text{long}((\mathcal{F}_y)_z) = \text{long}((\mathcal{F}'_{y'})_{z'})$; as moreover $f'^{-1}(\eta') = f^{-1}(\eta)$ and $\mathcal{F}'_{\eta'} = \mathcal{F}_\eta$, we are again reduced to proving (15.1.1.5). But if t is a uniformiser of $\mathcal{O}_{y'}$, $f'^{-1}(y')$ is the closed sub-prescheme of X' defined by the ideal $t\mathcal{F}'$; on the other hand, (15.1.1.5) is then just a special case of (3.4.1.1). $\square$

Corollary (15.1.2). — *Under the hypotheses of (15.1.1), suppose moreover that $\lambda_z(\mathcal{F}_y) = 1$ (resp. that $\mathcal{O}_y$ is regular and $\text{long}((\mathcal{F}_y)_z) = 1$). Then there is at most one irreducible component X_i of $\text{Supp}(\mathcal{F})$ containing*

z and such that the restriction $X_i \rightarrow Y$ of f is universally open at the point z . Moreover, if z_i is the generic point of this component, then $\lambda_{z_i}(\mathcal{F}_\eta) = 1$ (resp. $\text{long}((\mathcal{F}_\eta)_{z_i}) = 1$).

This follows immediately from (15.1.1), noting that one necessarily has $(\mathcal{F}_\eta)_{z_i} \neq 0$ by definition of X_i ((I, 9.1.13)).

Corollary (15.1.3). — *Let Y be an irreducible locally Noetherian prescheme of generic point η , $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module of support X , x a point of X , $y = f(x)$. Suppose that: 1° x belongs to a single irreducible component Z of $f^{-1}(y)$; 2° $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$ is geometrically reduced on $k(y)$, i.e. if z is the generic point of Z , $\lambda_z(\mathcal{F}_y) = 1$ (resp. $\mathcal{O}_y$ is regular and $\text{long}((\mathcal{F}_y)_z) = 1$). Then there is at most one irreducible component X' of X containing x , such that $\dim_x(X' \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$ and such that the restriction $X' \rightarrow Y$ of f is universally open at the generic points of the irreducible components of $f^{-1}(y)$ containing x . Moreover, if such a component X' exists, and if z' is its generic point, then $\lambda_{z'}(\mathcal{F}_\eta) = 1$ (resp. $\text{long}((\mathcal{F}_\eta)_{z'}) = 1$).*

Indeed, an irreducible component X' of X containing x and such that $\dim_x(X' \cap f^{-1}(y)) = \dim_x(f^{-1}(y))$ necessarily contains Z , since an irreducible component of $f^{-1}(y)$ containing x must be of the same dimension as the sole irreducible component of $f^{-1}(y)$ containing x . One may then apply (15.1.2) to z .

Remarks (15.1.4). — label=(i) In the statement of (15.1.1), one cannot replace “universally open” by “equidimensional”, as the example (14.4.10, (ii)) where one takes $\mathcal{F} = \mathcal{O}_X$ shows; the fibres of f are then reduced Artinian schemes, so $\lambda_x(\mathcal{F}_y) = 1$, but there are two irreducible components X_1, X_2 of X containing x' , and the second member of (15.1.1.1) is therefore equal to 2, even though the restriction of f to each of the components X_1, X_2 is an equidimensional morphism at every point. This same example also shows that in (15.1.2) and (15.1.3), one cannot replace the hypothesis “universally open” by “equidimensional”.

lbbel=(ii) It seems likely that for the validity of the inequality (15.1.1.2), one cannot suppress the hypothesis that $\mathcal{O}_y$ is regular.

15.2. Flatness of universally open morphisms with geometrically reduced fibres

(15.2.1). We have seen (2.4.6) that a morphism *flat* and locally of finite presentation is universally open. One has a partial converse of this result, subject to supplementary hypotheses: IV-3 | 226

Theorem (15.2.2). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, x a point of X , $y = f(x)$. Suppose the following conditions are satisfied:*

- label=(i) $\text{Supp}(\mathcal{F}) = X$.
- lbbel=(ii) f is universally open at the generic points of the irreducible components of $f^{-1}(y)$ containing x .
- lcbel=(iii) $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$ is geometrically reduced at the point x (4.6.22).
- ldbel=(iv) The ring $\mathcal{O}_y$ is reduced.

Then $\mathcal{F}$ is f -flat at the point x .

Proof. Since $\mathcal{O}_y$ is reduced and Noetherian, we shall apply the valuative criterion of flatness (11.8.1). Consider therefore a local homomorphism $\mathcal{O}_y \rightarrow A'$, where A' is a discrete valuation ring; set $Y' = \text{Spec}(A')$, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$; let x' be a point of X' whose projections in X and Y' are respectively x and the closed point of Y' ; if $\mathcal{F}' = \mathcal{F} \otimes_{\mathcal{O}_Y} Y'$, then $\text{Supp}(\mathcal{F}') = X'$ by ((I, 9.1.13)); every generic point z' of an irreducible component of $f'^{-1}(y')$ containing x' has for projection in $f^{-1}(y)$ a generic point z of an irreducible component of $f^{-1}(y)$ containing x , by (4.2.6); hence f' is universally open at z' . Finally, it follows from (4.7.11) that $\mathcal{F}'_{y'}$ is geometrically reduced at the point x' . One thus sees that one is reduced to showing the following: □

Lemma (15.2.2.1). — *Let $Y = \text{Spec}(A)$ be the spectrum of a discrete valuation ring, y its closed point, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of $f^{-1}(y)$, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent. Suppose the following conditions are satisfied:*

- label=(i) $\text{Supp}(\mathcal{F}) = X$.
- lbbel=(ii) f is open at the generic points of the irreducible components of $f^{-1}(y)$ containing x .
- lcbel=(iii) $\mathcal{F}_y = \mathcal{F} \otimes_{\mathcal{O}_Y} k(y)$ is reduced at the point x (3.2.2).

Then $\mathcal{F}$ is f -flat at the point x .

Proof. Let t be a uniformiser of A ; set $B = \mathcal{O}_x$ and $\mathcal{F}_x = M$; condition (i) implies that $\text{Supp}(M) = \text{Spec}(B)$ ((I, 9.1.13)). Condition (ii) implies, by (14.3.2), that every irreducible component of X containing x dominates Y ; in other words, the inverse image in A of every minimal prime ideal $\mathfrak{p}_i$ of B is 0, which means that $t \notin \mathfrak{p}_i$ for all i . Finally, (iii) means that M/tM is a (B/tB) -module *reduced* (3.2.2). It suffices to show that under these conditions, M is an A -module *without torsion* ($\mathbf{0_I}$, 6.3.4), and for this it evidently suffices to show that t is an M -regular element; but this follows from (3.4.6.1). $\square$

Corollary (15.2.3). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , $y = f(x)$. Suppose the following conditions are satisfied:*

- label=(i) f is universally open at the generic points of the irreducible components of $f^{-1}(y)$ containing x .*
- lbbel=(ii) $f^{-1}(y)$ is geometrically reduced (on $k(y)$) at the point x (4.6.9).*
- lcbel=(iii) The ring $\mathcal{O}_y$ is reduced.*

Under these conditions, f is flat at the point x .

Proof. It suffices to apply (15.2.2) with $\mathcal{F} = \mathcal{O}_X$ (4.6.22). $\square$

Remarks (15.2.4). — *label=(i) The first three conditions of (15.2.2) do not change when one replaces f by f_{red} ; but if $\mathfrak{N}$ is the nilradical of a local Noetherian ring A , $A/\mathfrak{N}$ is flat over $A/\mathfrak{N}$ but not over A itself when $\mathfrak{N} \neq 0$ (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. 2 of prop. 5). One thus sees that one cannot suppress condition (iv) in (15.2.2).*

lbbel=(ii) It follows from (2.4.6) that if the conclusion of (15.2.2) holds, then f is universally open in a neighbourhood of x , thus in particular universally open at the points of the generic points of the irreducible components of $f^{-1}(y)$ containing x . Even when (i), (iii), and (iv) are satisfied, one cannot replace (ii) by the weaker hypothesis that f is universally open at the point x only. This is what the example (14.1.3, (i)) shows, where f is evidently universally open at the point x , intersection of X_1 and X_2 ; the conditions (ii) and (iii) of (15.2.3) are also satisfied by this example. However, $\mathcal{O}_x$ is not an $\mathcal{O}_y$ -module without torsion, $\mathcal{O}_y$ identifying with a subring of $\mathcal{O}_x$ that contains the divisors of zero; hence f is not flat at the point x .

lcbel=(iii) In (15.2.3), the conclusion is no longer necessarily valid when one replaces hypothesis (ii) by the weaker hypothesis that $f^{-1}(y)$, considered as a $k(y)$ -prescheme, has no associated embedded prime cycles. An example is furnished by taking for Y a curve having a cusp at a point, say $Y = \text{Spec}(K[S, T]/(S^3 - T^2))$, K an algebraically closed field, and for X its normalisation; the morphism $f : X \rightarrow Y$ is finite, surjective, and radiciel, hence a universal homeomorphism (2.4.5), but $f^{-1}(y)$ is not a reduced $k(y)$ -prescheme. Here f is a finite morphism, surjective, hence $f : X \rightarrow Y$ is universally open and a radiciel morphism. But f is not flat at the point x , for if $\mathcal{O}_x$ were an $\mathcal{O}_y$ -module flat, it would be an $\mathcal{O}_y$ -module free of rank 1, which is absurd.

ldbel=(iv) Let us show finally that in (15.2.2) or (15.2.3), one cannot replace “geometrically reduced” by “reduced”. We shall define two rings A, A' having the following properties: IV-3 | 228

label=4) A' is the integral closure of A , an A -algebra finite whose residue field k' is a non-trivial finite radiciel extension of the residue field $k = A/\mathfrak{m}$ of A .

*lbbel=4) One can take $Y = \text{Spec}(A)$, $X = \text{Spec}(A')$, $\mathcal{F} = \mathcal{O}_x$, y and x being the closed points of Y and X respectively; the hypotheses (i) and (iv) of (15.2.2) are trivially satisfied; since A' is a discrete valuation ring, $f : X \rightarrow Y$ is radiciel, finite, and surjective, hence a universal homeomorphism (2.4.5), which proves hypothesis (i) of (15.2.3). Finally it is clear that $f^{-1}(y)$ is reduced, but f is not flat at the point x , for if A' were an A -module flat, it would be an A -module free of rank 1 (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, prop. 5), which is absurd.*

lebel=(v) The proof of (15.2.2) shows that one can weaken condition (iii) by replacing “geometrically reduced” by “reduced” provided one only uses discrete valuation rings A' whose residue field k' is separable over $k(y)$; in effect, if x' and z' have the same meanings as in (15.2.2), the radiciel multiplicities $\mu_s(k(z')|k')$ and $\mu_s(k(z)|k(y))$ are the same, by (4.7.3), hence

$\text{long}((\mathcal{F}_{y'})_z)$ and $\text{long}((\mathcal{F}_{y'})_{z'})$ are equal, and one can get the result still in the case Y is the spectrum of a discrete valuation ring y and y its closed point. For example, one can obtain this result still when $\mathcal{O}_y$ is *unibranch* and dominated by a discrete valuation ring whose residue field is a *separable* extension of $k(y)$, by (11.6.4). This is the case when $\mathcal{O}_y$ is a *regular* ring. But as Hironaka has pointed out, there are local rings of algebraic schemes, of dimension 2 over imperfect fields, that do not satisfy this condition.

15.3. Applications: criteria for reduction and irreducibility

Proposition (15.3.1). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, x a point of X , $y = f(x)$. Suppose that the conditions (i), (ii), and (iii) of (15.2.2) are satisfied. Then:*

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label=(i) *There exists a neighbourhood U of x in X such that $f|_U$ is universally open and, for every $x' \in U$, $\mathcal{F}_{f(x')}$ is geometrically reduced on $k(f(x'))$ at the point x' .*

lbbel=(ii) *If moreover Y is reduced at the point y , there exists a neighbourhood $U_1 \subset U$ of x such that $\mathcal{F}|_{U_1}$ is f -flat and reduced.*

Proof. By ((I, 5.1.8)), (4.2.7), and (4.7.11), the hypotheses do not change, nor the conclusion (i) of the statement, if one replaces Y by Y_{red} and X by $X \times_Y Y_{\text{red}}$; one may thus suppose Y *reduced*. It then follows from (15.2.2) that $\mathcal{F}$ is f -flat at the point x , hence also (11.1.1) in a neighbourhood of x , and *a fortiori* f is universally open in this neighbourhood (2.4.6). The fact that $\mathcal{F}_{f(x')}$ is geometrically reduced at every point x' of a neighbourhood of x follows from (12.1.1, (vii)). Assertion (ii) follows from (3.3.4). $\square$

Proposition (15.3.2). — *The notation being that of (15.3.1), suppose that the conditions (i), (ii), (iii) of (15.2.2) are satisfied, and suppose moreover that $f^{-1}(y)$ is equidimensional at the point x . Then f is equidimensional at the point x .*

Proof. For every $y' \in Y$, $\text{Supp}(\mathcal{F}_{y'}) = f^{-1}(y')$. By hypothesis, $\mathcal{F}_{y'}$ is reduced (hence satisfies condition (S₁)) and is equidimensional at x . By (12.1.1, (ii)), we may suppose $\mathcal{F}_{f(x')}$ is equidimensional and of constant dimension $e = \dim_x(f^{-1}(y))$ at the points x' for all $x' \in U$, which means that $f^{-1}(f(x'))$ is equidimensional at x' . Since moreover $\mathcal{F}|_U$ is f -flat, it follows from (2.3.4) and (13.3.1, a) that f is equidimensional at x . $\square$

Corollary (15.3.3). — *The notation being that of (15.3.1), suppose the following conditions satisfied:*

label=(i) $\text{Supp}(\mathcal{F}) = X$.

lbbel=(ii) f is universally open at the generic points of the irreducible components of $f^{-1}(y)$ containing x .

lcbel=(iii) $\mathcal{F}_y$ is geometrically punctually integral on $k(y)$ at the point x (4.6.22).

ldbel=(iv) Y is geometrically unibranch at the point y .

Then x belongs to a single irreducible component of X .

If moreover Y is integral at the point y and $\mathcal{F} = \mathcal{O}_X$, then X is integral at the point x and f is flat at the point x .

Proof. Note first that (i) and (iii) imply that x belongs to only a single irreducible component Z of $f^{-1}(y)$. It follows then from (14.4.7) and (15.3.2) that for every irreducible component X_i of X containing x (hence Z), the restriction f_i of f to X_i is equidimensional at the point x , and universally open at the generic point of Z . The first assertion of the statement then follows from (15.1.3) and the second from (15.3.1). $\square$

Remarks (15.3.4). — label=(i) The example (14.4.10, (ii)) shows that in (15.3.3), one cannot suppress the hypothesis that Y is geometrically unibranch at the point y .

lbbel=(ii) The remark (15.2.4, (vi)) shows that the conclusion of (15.3.3) is still valid under the following hypotheses: $\mathcal{O}_y$ is regular, $\text{Supp}(\mathcal{F}) = X$, and $\mathcal{F}_y$ is *integral* at the point x . We do not know whether in this statement one can replace the hypothesis that $\mathcal{O}_y$ is regular by that of being geometrically unibranch, or even integral and integrally closed.

15.4. Complements on Cohen–Macaulay morphisms

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Proposition (15.4.1). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module such that $\text{Supp}(\mathcal{F}) = X$, x a point of X , $y = f(x)$. For every $z \in X$, set $X_{f(z)} = f^{-1}(f(z))$, $\mathcal{F}_{f(z)} = \mathcal{F} \otimes_{\mathcal{O}_{f(z)}} k(f(z))$. Suppose that $\mathcal{F}$ is f -flat at the point x and that $\mathcal{F}_y$ is an $\mathcal{O}_{X_y}$ -Module de Cohen–Macaulay at the point x . Then f is equidimensional at the point x .*

Proof. Indeed ((11.1.1) and (12.1.1, (vi))), there exists a neighbourhood U of x such that $\mathcal{F}|_U$ is f -flat and that for every $x' \in U$, $\mathcal{F}_{f(x')}$ is an $\mathcal{O}_{X_{f(x')}}$ -Module de Cohen–Macaulay at the point x' ; this last property implies that $\mathcal{F}_{f(x')}$ is equidimensional at x' , and that x' does not belong to any embedded associated prime cycle of $\mathcal{F}_{f(x')}$ (0, 16.5.4). Moreover, by (12.1.1, (ii)), one can suppose that the dimensions of the irreducible components of $\text{Supp}(\mathcal{F}_{f(x')})|(U \cap X_{f(x')})$ take a (same) value independent of x' . As $\text{Supp}(\mathcal{F}_{f(x')}) = X_{f(x')}$ by hypothesis, it follows from (2.3.4) and (13.3.1, a) that f is equidimensional at the point x . $\square$

Proposition (15.4.2). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module such that $\text{Supp}(\mathcal{F}) = X$, x a point of X , $y = f(x)$. Suppose that $\mathcal{O}_y$ is regular and that $\mathcal{F}$ is an $\mathcal{O}_X$ -Module de Cohen–Macaulay at the point x . Then (with the notation of (15.4.1)) the following conditions are equivalent:*

label=a) $\mathcal{F}$ is f -flat at the point x and $\mathcal{F}_y$ is an $\mathcal{O}_{X_y}$ -Module de Cohen–Macaulay at the point x .

lbbel=b) $\mathcal{F}$ is f -flat at the point x .

lcbel=c) f is universally open in a neighbourhood of x .

ldbel=d) f is open at the generic points of the irreducible components of X_y containing x .

lebel=e) $\dim(\mathcal{F}_x) = \dim(\mathcal{O}_y) + \dim((\mathcal{F}_y)_x)$.

e') $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_y) + \dim(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_y} k(y))$.

Proof. As $\text{Supp}(\mathcal{F}) = X$, the conditions e) and e') are equivalent by definition (5.1.12). Condition a) implies trivially b); b) implies that $\mathcal{F}$ is f -flat at neighbouring points of x (11.1.1), hence b) implies c) (2.4.6); c) implies trivially d); and d) implies e') by (14.2.1). Finally, since $\mathcal{O}_y$ is regular and $\mathcal{F}_x$ is Cohen–Macaulay, it follows from (6.1.4) that e) implies a). $\square$

Proposition (15.4.3). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, $\mathcal{F}$ an $\mathcal{O}_X$ -Module of finite presentation such that $\text{Supp}(\mathcal{F}) = X$, x a point of X , $y = f(x)$. Suppose (with the notation of (15.4.1)) that the following conditions are satisfied:*

label=a) $\mathcal{F}$ is f -flat at the point x and $\mathcal{F}_y$ is an $\mathcal{O}_{X_y}$ -Module de Cohen–Macaulay at the point x .

Then:

label=a) There exists a neighbourhood U of x in X , and a Y -morphism quasi-finite $g : U \rightarrow Y' = Y[T_1, \dots, T_n]$ (T_i indeterminates), such that $\mathcal{F}|_U$ is g -flat at x .

Proof. Let us show first that b) implies a). Set $h : Y' \rightarrow Y$; let $Y'_y = h^{-1}(y)$, the $k(y)$ -prescheme regular (0, 17.3.7); let A be the local ring of Y'_y at the point $y' = g(x)$, k its residue field, and B the local ring of X_y at the point x ; set moreover $(\mathcal{F}_y)_x = M$, which is a B -module of finite type. The hypothesis that the morphism g is quasi-finite implies in the same way that $g_y : X_y \rightarrow Y'_y$ (II, 6.2.4), hence $M \otimes_A k$ is a $(B \otimes_A k)$ -module of finite length (II, 6.2.2); since by hypothesis M is an A -module flat and A is a Cohen–Macaulay ring (6.3.3), the fact that $\mathcal{F}$ is f -flat at x results from that it is g -flat and the morphism h is flat (2.1.6).

Let us now show that a) implies b). The question being local on X and Y , using (8.9.1) and (11.2.7), one reduces to the case where X and Y are Noetherian, using (6.7.1). The hypothesis implies that f is equidimensional at the point x ((15.4.1)), whence the existence of a quasi-finite morphism $g : U \rightarrow Y'$ such that $g_y : U_y \rightarrow Y'_y$ is finite and surjective (13.3.1, b); it suffices to prove that $\mathcal{F}|_U$ is g -flat at x . But by hypothesis $\mathcal{F}_y$ is g_y -flat at the point x . But as g_y is finite and surjective, it satisfies condition e') of (15.4.2) by (5.6.10); one may thus conclude by applying (15.4.2) to g_y and $\mathcal{F}_y$. $\square$

15.5. Separable rank of fibres of a separated quasi-finite morphism and universally open morphism. Application to geometric connected components of fibres of a proper morphism.

Proposition (15.5.1). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a separated and quasi-finite morphism. For every $z \in Y$, let $n(z)$ be the number of geometric points of $f^{-1}(z)$ (or the separable rank of $f^{-1}(z)$ on $k(z)$; cf. (I, 6.4.8)). One has the following properties:* IV-3 | 231

- label=(i) *The function $z \mapsto n(z)$ is semi-continuous inferiorly at every point $y \in Y$ such that f is universally open at the points of $f^{-1}(y)$.*
- lbbel=(ii) *Let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$; if the function $z \mapsto n(z)$ is constant in a neighbourhood of y , then f is proper at the point y (15.7.1).*
- lcbel=(iii) *Suppose that f is universally open and that every point of Y is geometrically unibranch; then, if the function $z \mapsto n(z)$ is locally constant in Y , the irreducible components of X are pairwise disjoint.*

Corollary (15.5.2). — *Under the general hypotheses of (15.5.1), let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$. Then, if y' is a generisation of y , one has*

$$(15.5.2.1) \quad n(y) \leq n(y').$$

Lemma (15.5.2.2). — *Under the general hypotheses of (15.5.1), if y is a point of Y , y' a generisation of y distinct from y , then there exists the spectrum of a discrete valuation ring Z , of closed point z and generic point z' , and a morphism $g : Z \rightarrow Y$ such that: $1^\circ g(z) = y$, $g(z') = y'$; 2° if one sets $f' = f|_Z$, $n(y)$ is the number of points of $f'^{-1}(z)$ and $n(y')$ is the number of points of $f'^{-1}(z')$.*

Proposition (15.5.3). — *Let Y be an irreducible locally Noetherian prescheme, $f : X \rightarrow Y$ a proper morphism. For every $z \in Y$, let $n(z)$ be the number of geometric connected components of $f^{-1}(z)$. Suppose that the restriction of f to every irreducible component of X is a dominant morphism. Then, if y_0 is a point that is geometrically unibranch in Y , the function $y \mapsto n(y)$ is semi-continuous inferiorly at the point y_0 .*

Proof. Consider the Stein factorisation $f = g \circ f'$ of f (III, 4.3.3), where $g : Y' \rightarrow Y$ is a finite morphism and $f' : X \rightarrow Y'$ a surjective morphism whose fibres are geometrically connected (III, 4.3.4). Since f' is surjective, the inverse image by f' of every irreducible component of X contains at least an irreducible component of Y' ; hence each of the irreducible components of Y' dominates Y . One may apply the Chevalley criterion (14.4.4) to the restrictions of g to each of these irreducible components, and one concludes that g is *universally open at every point of $g^{-1}(y_0)$* . If then y is any point, since the $X_i \cap f^{-1}(y)$ are pairwise disjoint, $\sum_i n_i(y') = n(y')$; but by hypothesis the extremal members are equal, hence $n(y) = \sum_i n_i(y)$, which implies that the $X_i \cap f^{-1}(y)$ are pairwise disjoint. The conclusion then follows from (15.5.2) and from the fact that the number of geometric connected components of $f^{-1}(y)$ equals the number of geometric points of $g^{-1}(y)$ (III, 4.3.4). $\square$

Corollary (15.5.4). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a proper morphism; let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$; then, with the notation of (15.5.3), the function $z \mapsto n(z)$ is semi-continuous inferiorly at the point y .*

Remarks (15.5.5). — label=(i) Even if Y is integral and normal, X integral, f finite and surjective (hence universally open by the Chevalley criterion (14.4.4)), the function $y \mapsto n(y)$ is not necessarily locally constant.

lbbel=(ii) The example (14.4.10, (ii)) shows that in (15.5.1), (iii), one cannot suppress the hypothesis that the points of Y are geometrically unibranch.

lcbel=(iii) The example (11.7.5) shows that the conclusion of (15.5.1), (i) is no longer valid if one only supposes the morphism f is *open* (even if f is finite and surjective).

ldbel=(iv) Finally, in (15.5.1), one cannot dispense with the hypothesis that the morphism f is *separated*, as the example of the affine line over a field with the origin “doubled” ((I, 5.5.11)) shows: here f is a local isomorphism (hence flat) and Y is the affine line, but $z \mapsto n(z)$ is not semi-continuous inferiorly.

Lemma (15.5.6). — *Let Y be the spectrum of a discrete valuation ring, a its closed point, b its generic point. Let $f : X \rightarrow Y$ be a morphism locally of finite type and open. Suppose that X is connected and that the fibre $f^{-1}(a)$ is a reduced prescheme. Then $f^{-1}(b)$ is connected.*

Proof. To prove (15.5.6), we apply the lemma (15.5.6.1) to X and to its closed part $Z = f^{-1}(a)$ which is rare since f is open. Note that by hypothesis, taking into account (15.2.2.1), f is flat at the points of $f^{-1}(a)$. For such a point x , $\mathcal{O}_x$ is an $\mathcal{O}_a$ -module without torsion (0_I, 6.3.4), and in particular, if t is a uniformiser of $\mathcal{O}_a$, t is $\mathcal{O}_x$ -regular. Moreover $\mathcal{O}_x/t\mathcal{O}_x$ is reduced by hypothesis; if it is of depth 0, it is therefore a field (0, 16.4.7), and $t\mathcal{O}_x$ is then the maximal ideal of $\mathcal{O}_x$, hence $\text{Spec}(\mathcal{O}_x) - \{x\}$ is reduced to a point, and *a fortiori* connected. If on the contrary $\text{prof}(\mathcal{O}_x/t\mathcal{O}_x) \geq 1$, then $\text{prof}(\mathcal{O}_x) \geq 2$, hence by the theorem of Hartshorne (5.10.7) $\text{Spec}(\mathcal{O}_x) - \{x\}$ is connected, which finishes the proof. $\square$

Lemma (15.5.6.1). — *Let X be a locally Noetherian topological space, irreducible. Let Z be a closed rare part of X having the following property: for every $x \in Z$, if one denotes by T_x the set of generisations of x in X , then $T_x - \{x\}$ is connected. Under these conditions, if X is connected, $X - Z$ is connected.*

Proposition (15.5.7). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a proper morphism. For every $z \in Y$, let $n(z)$ be the number of geometric connected components of $f^{-1}(z)$. Let y be a point of Y such that f is universally open at the points of $f^{-1}(y)$ and that $f^{-1}(y)$ is geometrically reduced on $k(y)$ (4.6.2). Then the function $z \mapsto n(z)$ is constant in a neighbourhood of y .*

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Proof. We already know that under the conditions of the statement, $z \mapsto n(z)$ is semi-continuous inferiorly superiorly, and by the same reasoning as at the beginning of the proof of (15.5.2), it suffices to show that if y' is a generisation of y , then

$$(15.5.7.1) \quad n(y') \leq n(y).$$

Using the reasoning from (15.5.2) and the lemma (15.5.2.2) applied to the Stein factorisation of f (recalling that the number of geometric connected components of $f^{-1}(y)$ equals the number of geometric points of $g^{-1}(y)$ (III, 4.3.4)), one reduces to the case where $n(y)$ and $n(y')$ are the numbers of points of $f^{-1}(y)$ and $f^{-1}(y')$ respectively, and where Y is the spectrum of a discrete valuation ring, y its closed point and y' its generic point. We may moreover replace X by any connected component of its connected components, and these being both open and closed in X , proper on Y ; the hypothesis f is open at the points of $f^{-1}(y)$, the lemma (15.5.6) implies that $n(y') = 1$, which completes the proof of (15.5.7.1). $\square$

Remark (15.5.8). — The relation (15.5.7.1) is no longer necessarily valid if f is not supposed proper, even if the other hypotheses of (15.5.7) are satisfied. With the notation of (15.5.5, (i)), it suffices to consider the restriction of f to $X - \{x'\}$, where x' is one of the two points of X above a point y distinct from y' and y'' ; then $n(y) = 1$, while if η is the generic point of Y , $n(\eta) = 2$.

Proposition (15.5.9). — *Let $f : X \rightarrow Y$ be a morphism flat and locally of finite presentation; for every $y \in Y$, let $n(y)$ be the number of geometric connected components of $f^{-1}(y)$.*

label=(i) If f is separated and quasi-finite, the function $y \mapsto n(y)$ (equal then to the number of geometric points of $f^{-1}(y)$) is semi-continuous inferiorly in Y ; if it is constant in a neighbourhood of y_0 , f is proper at the point y_0 .

lbbel=(ii) If f is proper, the function $y \mapsto n(y)$ is semi-continuous inferiorly in Y ; if moreover $f^{-1}(y_0)$ is geometrically reduced on $k(y_0)$, $n(y)$ is constant in a neighbourhood of y_0 .

Proof. In all cases, the questions are local on Y , so one may suppose Y affine and f of finite presentation; using (8.9.1), (8.10.5), and (11.2.7), one reduces to the case where Y is Noetherian (using also (8.2.11) for the second assertion of (i)). Since moreover f is then universally open (2.4.6), it suffices to apply (15.5.1), (15.5.4), and (15.5.7) to conclude. $\square$

Remark (15.5.10). — One will note that another proof has thus been obtained of (12.2.4, (vi)). Conversely, one can deduce (15.5.7) from (12.2.4, (vi)): in effect, one can restrict to the case where Y is reduced and f flat, and it follows then from the hypotheses of (15.5.7) and from (15.2.3) that f is flat in a neighbourhood open of $f^{-1}(y)$; as moreover f is proper, one can suppose this neighbourhood is of the form $f^{-1}(U)$, where U is an open neighbourhood of y in Y . The proof of (15.5.7) given above has the advantage of putting in evidence the result (15.5.6), which is of independent interest.

15.6. Connected components of fibres along a section

Proposition (15.6.1). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, $g : Y \rightarrow X$ a Y -section of X ((I, 2.5.5)). For every $y \in Y$, denote by X_y^0 the connected component of $f^{-1}(y)$ containing the point $g(y)$. Let y be a point of Y , y' a generisation of y . Then:* IV-3 | 236

label=(i) If there exists an irreducible component Z of X_y^0 , of dimension equal to $\dim(X_y^0)$ and containing $g(y')$ (which will be the case if X_y^0 is irreducible), then

$$(15.6.1.1) \quad \dim(X_y^0) \geq \dim(X_{y'}^0).$$

lbbel=(ii) If moreover X_y^0 is irreducible and the two members of (15.6.1.1) are equal, then $X_{y'}^0 \subset \overline{X_y^0}$ (closure in X).

Proof. (i) Let $\overline{Z}$ be the closure of Z in X ; since y is adherent to y' and g continuous, $g(y) \in \overline{Z}$. As $Z = \overline{Z} \cap f^{-1}(y')$, it follows from Chevalley's semi-continuity theorem (13.1.3) applied to the restriction of f to the closed reduced sub-prescheme of X having $\overline{Z}$ as underlying space, that $\dim_{g(y)}(\overline{Z} \cap f^{-1}(y)) \geq \dim(Z)$; but the irreducible components of $\overline{Z} \cap f^{-1}(y)$ containing $g(y)$ are evidently contained in X_y^0 , whence the inequality (15.6.1.1).

(ii) The reasoning of (i) shows moreover that if the two members of (15.6.1.1) are equal, $\dim_{g(y)}(\overline{Z} \cap f^{-1}(y')) = \dim_{g(y)}(X_y^0)$; if X_y^0 is irreducible, then $\overline{Z} = X_y^0$, hence $X_{y'}^0$ is contained in $\overline{Z}$, and *a fortiori* in $\overline{X_y^0}$. □

Corollary (15.6.2). — *With the notation of (15.6.1), suppose moreover that for every $y \in Y$, X_y^0 is irreducible. Then the function $z \mapsto \dim(X_z^0)$ is semi-continuous superiorly in Y . If moreover this function is constant in a neighbourhood of a point $y \in Y$, then $X_{y'}^0 \subset \overline{X_y^0}$ for every generisation y' of y .*

Proof. Let y be a point of Y , and let U be an open affine neighbourhood of $g(y)$ in X ; then for every $z \in U$, $U \cap X_z^0$ is dense in X_z^0 , hence $\dim(X_z^0) = \dim(U \cap X_z^0)$ (4.1.3). Replacing X by U , one can thus suppose f is of finite type. One then knows (9.7.10) that the function $z \mapsto \dim(X_z^0)$ is locally constructible, and the assertions of the corollary follow from (15.6.1) and (0III, 9.3.4). □

Proposition (15.6.3). — *With the notation of (15.6.1), suppose that for every $y \in Y$, X_y^0 is geometrically irreducible (4.5.2). Then, for every $y \in Y$ such that the function $z \mapsto \dim(X_z^0)$ is constant in a neighbourhood of y , f is universally open at X_y^0 .*

Proof. Apply the criterion (14.3.7). Let $h : Y' \rightarrow Y$ be a morphism, Y' being the spectrum of a discrete valuation ring, and denote by t, t' the closed point and the generic point respectively, and suppose $h(t) = y$, and hence $y' = h(t')$ is a generisation of y . Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$, $g' = g_{(Y')} : Y' \rightarrow X'$. With the same notation as in (15.6.1), the hypothesis on the X_y^0 implies that $X_t'^0 = X_y^0 \otimes_{k(y)} k(t)$ and $X_{t'}'^0 = X_{y'}^0 \otimes_{k(y')} k(t')$ are irreducible (4.4.1); since $\dim(X_y^0) = \dim(X_{y'}^0)$ and $\dim(X_y^0) = \dim(X_t'^0)$ (4.1.4), we see that one is reduced to proving the proposition for f' and $X_t'^0$, i.e. one can restrict to the case where Y is the spectrum of a discrete valuation ring, y its closed point and y' its generic point; if x' is the generic point of $X_{y'}^0$, it follows then from (15.6.2) that the point of $X_{y'}^0$ is a specialisation of x' , whence the conclusion. □

Proposition (15.6.4). — *Under the general hypotheses of (15.6.1), let X^0 be the union of the X_y^0 when y ranges over Y . Suppose that for every $y \in Y$, X_y^0 is geometrically reduced on $k(y)$ (4.6.2), and that f is universally open at every point of X_y^0 . Then X^0 is a neighbourhood of X_y^0 in X . In particular, if f is universally open and if, for every $y \in Y$, X_y^0 is geometrically reduced on $k(y)$, then X^0 is open in X .*

Proof. One can first reduce to the case where f is a morphism of finite type. Suppose then that f is of finite type. Then X^0 is locally constructible (9.7.12); it suffices to prove that X^0 contains every generisation x' of x (0III, 9.2.5). Set $y' = f(x')$; there exists a discrete valuation ring A and a morphism u of $Y' = \text{Spec}(A)$ to Y such that if z is the closed point and z' the generic point, $u(z) = x$ and $u(z') = x'$ (II, 7.1.9); if $h = f \circ u$, $h(z) = y$ and $h(z') = y'$. Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$, $g' = g_{(Y')} : Y' \rightarrow X'$. If $p_y : f'^{-1}(z) \rightarrow f^{-1}(y)$ (resp. $p_{y'} : f'^{-1}(z') \rightarrow f^{-1}(y')$)

is the canonical projection, $p_y^{-1}(X_y^0)$ (resp. $p_{y'}^{-1}(X_{y'}^0)$) is the connected component of $f'^{-1}(z)$ (resp. $f'^{-1}(z')$) containing $g'(z)$ (resp. $g'(z')$), since the X_i^0 are geometrically connected (4.5.13), and it suffices to apply (4.4.1). Moreover, the morphism u corresponds to a Y' -section v of X' such that $p_y(v(y)) = x$, $p_{y'}(v(y')) = x'$; as $v(y')$ is a generisation of $v(y)$, and $f'^{-1}(y) - X_y^0 = f^{-1}(y)$ is then closed in $f^{-1}(y)$, we can, replacing X by the open $X - (f^{-1}(y) - X_y^0)$, suppose $X^0 = f^{-1}(y)$, i.e. $X^0 = X$; in other words, the proposition will be established if one shows that $X^0 = X$, or yet again that $X_{y'}$ is connected; but $f^{-1}(y')$ is reduced; as moreover f is open at the points of $f^{-1}(y)$, the lemma (15.5.6) allows us to conclude. $\square$

Corollary (15.6.5). — *Let $f : X \rightarrow Y$ be a morphism flat and of finite presentation, g a Y -section of X ; for every $y \in Y$, let X_y^0 be the connected component of $f^{-1}(y)$ containing $g(y)$, and let X^0 be the union of the X_y^0 when y ranges over Y . Then, for every $y \in Y$ such that X_y^0 is geometrically reduced on $k(y)$, X^0 is a neighbourhood of X_y^0 in X . In particular, if f is reduced (6.8.1), X^0 is open in X .*

Proof. The question being local on Y , one can suppose $Y = \text{Spec}(A)$ affine. There exists then a sub-Noetherian ring A_0 and a morphism of finite type $f_0 : X_0 \rightarrow Y_0 = \text{Spec}(A_0)$ such that $X = X_0 \times_{Y_0} Y$ and $f = (f_0)_{(Y)}$ ((8.9.1) and (11.2.7)); moreover, one can suppose there exists a Y_0 -section g_0 of X_0 such that $g = (g_0)_{(Y)}$ (8.8.2). For every $y \in Y$, let y_0 be its projection in Y_0 ; it follows from (4.5.13) that $(X_0)_{y_0}^0$ is geometrically connected, hence if $p : f^{-1}(y) \rightarrow f_0^{-1}(y_0)$ is the canonical projection, $X_y^0 = p^{-1}((X_0)_{y_0}^0)$ (4.4.1); moreover, if $f^{-1}(y)$ is geometrically reduced on $k(y)$, it is also geometrically reduced on $k(y_0)$ (4.6.10). On thus reduces to the case where Y is Noetherian and f of finite type. Since moreover f is then universally open (2.4.6), (11.1.1), it suffices to apply (15.6.4). $\square$

Proposition (15.6.6). — *Let $f : X \rightarrow Y$ be a morphism such that Y is locally Noetherian and f locally of type finite, or that f is locally of finite presentation. Let g be a Y -section of X ; for every $y \in Y$, let X_y^0 be the connected component of $f^{-1}(y)$ containing $g(y)$; finally, let X^0 be the union of the X_y^0 when y ranges over Y .*

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Suppose that for every $y \in Y$, X_y^0 is geometrically irreducible, and that f is universally open at X_y^0 . Then:

- label=(i) *The function $y \mapsto \dim(X_y^0)$ is semi-continuous superiorly in Y .*
- lbbel=(ii) *Let $x_0 \in X^0$, $y_0 = f(x_0)$. If the function $y \mapsto \dim(X_y^0)$ is constant in a neighbourhood of y_0 , then there exists a neighbourhood open U of y_0 in Y such that f is universally open at the points of $X^0 \cap f^{-1}(U)$, and f is flat at the point x_0 .*
- lcbel=(iii) *Conversely, suppose that f is universally open at $X_{y_0}^0$, and that moreover one of the following conditions is satisfied:*
 - label= α *The fibre $f^{-1}(y_0)$ is geometrically reduced on $k(y_0)$ at the point $g(y_0)$ and the ring $\mathcal{O}_{Y, y_0}$ is Noetherian.*
 - lbbel= α *For the generic point y' of an irreducible component of Y containing y_0 , one has $\dim(f^{-1}(y')) = \dim(X_{y'}^0)$.*
- lbbel=(iv) *Then the function $y \mapsto \dim(X_y^0)$ is constant in a neighbourhood of y_0 .*
- lbbel=(iv) *The set W of the $x \in X^0$ such that $y \mapsto \dim(X_y^0)$ is constant in a neighbourhood of $f(x)$ and that $X_{f(x)}^0$ is geometrically reduced on $k(f(x))$, is open in X .*

Corollary (15.6.7). — *Suppose the preliminary conditions of (15.6.6) on f satisfied, and suppose moreover that for every $y \in Y$, X_y^0 is geometrically integral on $k(y)$. Then the following conditions are equivalent:*

- label=a) *The function $y \mapsto \dim(X_y^0)$ is locally constant in Y .*
- lbbel=b) *The morphism $f_{(Y_{\text{red}})} : X \times_Y Y_{\text{red}} \rightarrow Y_{\text{red}}$ deduced from f by base change is flat at the points of X^0 .*

Moreover, these conditions imply that X^0 is open in X . Finally, when Y is locally Noetherian, a) and b) are also equivalent to:

- b') *f is universally open at the points of X^0 .*

Proposition (15.6.8). — *Let $f : X \rightarrow Y$ be a separated morphism of finite presentation, g a Y -section of X ; for every $y \in Y$, let X_y^0 be the connected component of $f^{-1}(y)$ containing $g(y)$, and let X^0 be the union of the X_y^0 when y ranges over Y . If for every $y \in Y$ X_y^0 is proper on $k(y)$ (i.e. (15.7.1), there exists a neighbourhood open U of y in Y such that $X^0 \cap f^{-1}(U)$ is closed in $f^{-1}(U)$ and proper on U).*

Proceeding as at the beginning of the proof of (15.6.6), and using (8.10.5, (v)), and replacing (9.7.11) by (9.7.7), one reduces to the case where Y is Noetherian and f of finite type. One then knows (9.7.12), (9.7.11) that X^0 is locally constructible in X ; on the other hand, the X_y^0 are geometrically connected (for every $z \in Y$) by (4.5.13); finally, X^0 is universally submersive over Y (15.7.8). It suffices then to apply the criterion (15.7.9).

Corollary (15.6.9). — Let $f : X \rightarrow Y$ be a separated morphism such that Y is locally Noetherian and f locally of finite type, or that f is of finite presentation. Let g be a Y -section of X , and for every $y \in Y$, let X_y^0 be the connected component of $f^{-1}(y)$ containing $g(y)$. Let y be a point of Y such that X_y^0 is proper on $k(y)$. Then, for every generisation y' of y , one has $\dim(X_{y'}^0) \geq \dim(X_y^0)$.

Proof. Suppose first f of finite presentation; then replacing Y by a neighbourhood of y if necessary, one can suppose that $f|X^0$ is proper (15.6.8), and it suffices to apply (13.1.5) to this restriction. If Y is locally Noetherian and f locally of finite type, X is locally Noetherian; moreover the hypothesis implies that X_y^0 is Noetherian; therefore there is an open Noetherian $V \subset X$ containing X_y^0 ; the restriction f to $V \cap f^{-1}(W)$ is a morphism of type finite, and $g|W$ is a W -section of X , and by applying the above result to this morphism, one gets $\dim(X_{y'}^0) \geq \dim(X_y^0)$. $\square$

Remarks (15.6.10). — label=(i) The supplementary hypothesis on the existence of an irreducible component of $X_{y'}^0$ containing $g(y')$ and of dimension equal to $\dim(X_{y'}^0)$, made in (15.6.1), is not superfluous, as the following example shows. Let A be a discrete valuation ring, π a uniformiser of A , K the field of fractions of A , k its residue field. Consider the two Y -schemes $X_1 = \text{Spec}(A[t])$, $X_2 = \text{Spec}(K[u, v])$, where t, u, v are indeterminates. The projection of X_1 in Y has for image $\{y\}$ (i.e. the projection of X_2 to Y is the point $\{y'\}$). One can glue the two sub-preschemes X_1 and X_2 along their common open Y -isomorphism $Z_1 \cong Z_2$, getting a Y -scheme X ; $f^{-1}(y)$ is the disjoint union of $(X_1)_y$ and $(X_2)_y$, whilst $f^{-1}(y')$ has two irreducible components, isomorphic respectively to $(X_1)_{y'}$ and $(X_2)_{y'}$, having a unique common point x , which projects to the generic point y' of Y and such that $k(x_1) = K$. The “null section” g of X_1 (II, 1.7.9) is then a Y -section of X ; $f^{-1}(y)$ equals $(X_1)_y$, while $f^{-1}(y')$ has two irreducible components isomorphic to $(X_1)_{y'}$ and $(X_2)_{y'}$, with a single common point; even though $\dim(X_{y'}^0) = 1$ whilst X_y^0 is not a neighbourhood of $X_{y_0}^0$ in X (one cannot suppress the hypotheses α) and β) in (15.6.6)). Moreover, X^0 is not open in X , which shows that in (15.6.4) one cannot dispense with the hypothesis that X' is reduced.

lbel=(ii) Consider the morphism f defined in (12.2.3, (iii)), which is proper and flat; moreover the reciprocal $g : Y \rightarrow X$ is a Y -section of X . It satisfies $X_1 \cap f_0^{-1}(Y)$ an isomorphism, hence the reciprocal $g : Y \rightarrow X$ is a Y -section of X . We have $\dim(X_y) = 1$ whilst $\dim(X^0) = 0$ for $y \neq y_0$, even though X_y^0 is not a neighbourhood of $X_{y_0}^0$ in X . Moreover, X^0 is not open in X , which shows that in (15.6.4), one cannot dispense with the hypothesis that X' is reduced.

lbel=(iii) At chap. VI, we shall apply the preceding results to Y -preschemes that are groups locally of finite type. If G is such a prescheme, there is always a Y -section canonical g , the “unit section” and one shows that G_y is always geometrically irreducible and that one has $\dim(G_y^0) = \dim(G_y)$, by setting $G_y^0 = f^{-1}(y)$. It follows then from (15.6.2) and (15.6.3) that the function $z \mapsto \dim(G_z)$ is semi-continuous superiorly, and that if it is constant in a neighbourhood of a point y , f is universally open at all the points of G_y^0 . If moreover G_y is geometrically reduced on $k(y)$ at all its points, one shows that G is smooth (6.8.1) at all the points of G_y^0 ; it will follow then from (15.6.6) that if moreover $\mathcal{O}_y$ is reduced, then f is smooth at all the points of G_y^0 . These results apply for example in the theory of Picard schemes, where one will have at our disposal a general theorem assuring that, under certain conditions, the function $z \mapsto \dim(G_z)$ is locally constant. We also note that in the case of a Y -prescheme in groups G , the hypothesis β) of (15.6.6) is always satisfied.

One will note also that in the case of a Y -group prescheme G , the hypothesis β) of (15.6.6) is always satisfied.

15.7. Appendix: Valuative criteria for local properness

Definition (15.7.1). — Let $f : X \rightarrow Y$ be a morphism of preschemes, y a point of Y . One says that f is *proper at the point y* if there exists a neighbourhood open U of y in Y such that the restriction $f^{-1}(U) \rightarrow U$ of f is a proper morphism. One says that a part Z of X is *proper over Y at the point y* if there exists a neighbourhood open U of y such that $Z \cap f^{-1}(U)$ is closed in $f^{-1}(U)$ and proper on U (II, 5.4.10) (which amounts to saying that for every sub-prescheme having Z as closed underlying space, the restriction $Z \rightarrow Y$ of f to Z is proper at the point y).

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Let $g : Y' \rightarrow Y$ be any morphism; set $X' = X \times_Y Y'$, $f' = f_{(Y')}$, and, if $p : X' \rightarrow X$ is the canonical projection, $Z' = p^{-1}(Z)$. Then if Z is proper on Y at the point y , Z' is proper on Y' at every point y' above y (II, 5.4.2).

Proposition (15.7.2). — Let Y be an integral locally Noetherian prescheme, X an integral prescheme, $f : X \rightarrow Y$ a separated morphism, dominant, and of finite type, y a point of Y . The following conditions are equivalent:

label=a) f is proper at the point y .

lbbel=b) If $Y_1 = \text{Spec}(\mathcal{O}_y)$, $X_1 = X \times_Y Y_1$, the morphism $f_1 = f_{(Y_1)} : X_1 \rightarrow Y_1$ is proper.

lcbel=c) Every discrete valuation ring having $R(X)$ for field of fractions and dominating a local ring $\mathcal{O}_x$ of X (in which case one has necessarily $f(x) = y$).

Proof. The fact that *a*) implies *b*) follows from (II, 5.4.2), and the implication *b*) $\Rightarrow$ *c*) follows from (II, 7.3.10). It remains to show that *c*) implies *a*). The question being local on Y , one may suppose Y affine, hence Noetherian. By the lemma of Chow (II, 5.6.1), there exists an integral prescheme X' , a projective morphism $p : P \rightarrow Y$, an open immersion $j : X' \rightarrow P$ dominant, and a projective morphism birational (hence surjective) $g : X' \rightarrow X$ such that the diagram

$$\begin{array}{ccc} P & \xleftarrow{j} & X' \\ & & \downarrow g \\ Y & \xleftarrow{\quad} & X \end{array}$$

is commutative. Since X' is integral and j dominant, P is irreducible, and one may, replacing p by p_{red} , suppose P integral. It suffices to prove that there exists an open U of y in Y such that $f(X) \cap U$ is closed in U , and proper on U . But the image $f(X)$ is locally closed in Y , since f is of finite type, and open in its closure. Since f is of finite type, it factorises as $f = j_0 \circ g_0$, where $j_0 : Z' \rightarrow Y$ is the canonical injection and g_0 is the restriction $g_0(X) = Z''$ (which is open from Z'); the fact that g is proper implies *d*). Finally, by the Chow lemma (II, 7.2.2 and II, 7.3.5, (vi)), the uniqueness of the prolongation of a Y' -section follows from the hypothesis; by (15.7.2), X is proper over Y , which proves the proposition. $\square$

Remark (15.7.3). — One can, in (15.7.2), suppress the hypothesis that Y is locally Noetherian, in replacing in the condition *c*) the discrete valuation rings by any valuation rings; the proof is then unchanged, using (II, 7.3.10).

Corollary (15.7.4). — Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a separated morphism, Z a part of X such that the maximal points of its closure $\overline{Z}$ belong to Z (which is the case when Z is finite, or when Z is constructible (0III, 9.2.2)). Let y be a point of Y . The following conditions are equivalent:

label=a) The set $\overline{Z}$ is proper over Y at the point y .

lbbel=b) If $Y_1 = \text{Spec}(\mathcal{O}_y)$, $X_1 = X \times_Y Y_1$, and if $g_1 : X_1 \rightarrow X$ is the canonical projection and $Z_1 = g_1^{-1}(Z)$, the closure $\overline{Z}_1$ of Z_1 in X_1 is proper on Y_1 .

lcbel=c) For every scheme $Y' = \text{Spec}(A')$, where A' is a discrete valuation ring, and every morphism $g : Y' \rightarrow Y$, such that the image $g(y')$ of the closed point y' of Y' is y , one has the following property: setting $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$, $g' = g_{(X)} : X' \rightarrow X$, $Z' = g'^{-1}(Z)$, and denoting by η' the generic point of Y' , then for every $x' \in Z' \cap f'^{-1}(\eta')$, rational on $k(\eta')$, there exists a Y' -section of X' containing x' .

Proof. The fact that *a*) implies *b*) follows from the fact that $\overline{Z}_1 \subset g_1^{-1}(\overline{Z})$ and that $g_1^{-1}(Z)$ is proper on Y_1 (15.7.1). The fact that *b*) implies *c*) follows from (II, 7.3.3). It remains to show that *c*) implies

a). It suffices to prove that every irreducible component of $\bar{Z}$ is proper over Y at the point y , and the hypothesis thus allows us to restrict to the case where Z is reduced to a single point z . One can also, by (I, 5.2.2) and (II, 5.4.6), suppose that X and Y are reduced, then (by definition of a proper part (II, 5.4.10)) suppose that $\bar{Z} = X = \{\bar{z}\}$ and (applying again (I, 5.2.2)) that f is dominant, so X and Y are integral and Noetherian; one must then prove that f is proper at the point y of Y . Let us show for this that f satisfies condition c) of (15.7.2). Let A' be a discrete valuation ring having $R(X) = k(z)$ for field of fractions and dominating $\mathcal{O}_y$; with the notations of c), one has $g(y') = y$, and $g(\eta')$ is the generic point $\eta = f(z)$ of Y . Since by definition $k(\eta') = k(z)$, there exists a $k(\eta')$ -morphism $\text{Spec}(k(\eta')) \rightarrow \text{Spec}(k(z))$ sending η' to z , hence a $k(\eta')$ -section of $f'^{-1}(\eta')$ (I, 3.3.14); if z' is the image of η' by this section, z' is rational over $k(\eta')$ and $g'(z') = z$. One concludes from hypothesis c) that there exists a Y' -section h' of X' such that $h'(\eta') = z'$; let $h = g' \circ h' : Y' \rightarrow X$ be the corresponding Y -morphism, and set $x = h(y')$; it is clear that $f(x) = y$ and that $A' = \mathcal{O}_{y'}$ dominates $\mathcal{O}_z$, which completes the proof. $\square$

Corollary (15.7.5). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a separated morphism of finite type, y a point of Y . The following conditions are equivalent:*

label=a) *f is proper at the point y .*

lbbel=b) *If $Y_1 = \text{Spec}(\mathcal{O}_y)$, $X_1 = X \times_Y Y_1$, the morphism $f_1 = f_{(Y_1)} : X_1 \rightarrow Y_1$ is proper.*

lcbel=c) *For every scheme $Y' = \text{Spec}(A')$, where A' is a discrete valuation ring, and every morphism $g : Y' \rightarrow Y$, such that the image $g(y')$ of the closed point y' of Y' is y , one has the following property: setting $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$, and denoting by η' the generic point of Y' , for every $x' \in f'^{-1}(\eta')$, rational on $k(\eta')$, there exists a Y' -section of X' containing x' .*

Proof. This is a special case of (15.7.4) applied to $Z = X$. $\square$

Corollary (15.7.6). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism of finite type. The following conditions are equivalent:*

label=a) *There exists a proper morphism $g : X \rightarrow Z$ and an immersion $h : Z \rightarrow Y$ such that $f = h \circ g$.*

lbbel=b) *The sub-space $f(X)$ is locally closed in Y , and if Z' is the sub-prescheme of Y induced by the closure $\bar{f}(X)$ of $f(X)$ on the open $f(X)$ of Z' , and Z'' the prescheme induced by Z' on $f(X)$, then f factorises uniquely as $f = j_0 \circ g_0$, where $j_0 : Z'' \rightarrow Y$ is the canonical injection, and $g_0 : X \rightarrow Z''$ is proper.*

lcbel=c) *For every $y \in f(X)$, f is proper at the point y .*

lbbel=d) *For every scheme $Y' = \text{Spec}(A')$, where A' is a discrete valuation ring, and every morphism $g : Y' \rightarrow Y$, such that $g(Y') \subset f(X)$, there exists a Y' -section of $X' = X \times_Y Y'$, and every Y' -section h' of X' extends uniquely to a Y' -section of X' .*

Proof. It is clear that b) implies a). To see that a) implies c), note first that a) implies that $h(Z)$ is locally closed in Y and $g(X)$ is closed in Z , hence $f(X)$ is locally closed in Y ; for every $y \in f(X)$, there is thus an open neighbourhood U of y in Y such that $f(X) \cap U$ is closed in U , hence $f(X)$ is locally closed, and therefore open in its closure. Since this closure is the underlying space of Z' , and f factorises as $f = j_0 \circ g_0$, where $j_0 : Z' \rightarrow Y$ is the canonical injection and g_0 is a morphism from X to Z' , the fact that $g_0(X) = Z''$ (which is open in Z') implies the factorisation of the statement of b); the fact that g is proper then follows from the local character (on Z'') of this property, and from (II, 5.4.3). It is clear that c) implies d) by virtue of (15.7.5), the uniqueness of the extension of a Y' -section resulting from the hypothesis that f is separated (I, 7.2.2). Finally, let us prove that d) implies c); first, it follows from d), by (II, 7.2.3 and 7.2.4), that f is separated; one can then apply the criterion ((15.7.5), c)), which shows that f is proper at every point of $f(X)$. $\square$

Remarks (15.7.7). — label=(i) In (15.7.4, c) and (15.7.5, c), one can restrict to the case where the discrete valuation ring A' is *complete* and has an algebraically closed residue field. Indeed, in the proof of (15.7.4), if one considers a complete discrete valuation ring A'' , dominant A' , and having an algebraically closed residue field, the hypothesis c) is then satisfied for $Y'' = \text{Spec}(A'')$, $X'' = X' \times_{Y'} Y''$, and $f'' = f'_{(Y'')}$; but $X'' = X' \times_{Y'} Y''$, and it follows then from the remark (II, 7.3.9, (i)) that f' is proper; hence Y' , X' , and f' also satisfy hypothesis c) (II, 7.3.8).

lbel=(ii) By (II, 7.3.8), the condition c) of (15.7.5) may be replaced by the condition that f' is proper.

(15.7.8). One says that a morphism of preschemes $f : X \rightarrow Y$ is *submersive* if it is surjective and if the topology of Y is equal to the quotient of the topology of X by the equivalence relation defined by f . One says that f is *universally submersive* if for every base change $Y' \rightarrow Y$, the morphism $f' = f_{(Y')} : X \times_Y Y' \rightarrow Y'$ is submersive. Every surjective morphism *universally open*, or *universally closed*, or *faithfully flat* and quasi-compact is universally submersive (2.3.12). Given a morphism $f : X \rightarrow Y$, one says that a sub-set Z of X is *submersive on $f(Z)$* if the topology of the sub-space $f(Z)$ of Y is the quotient of the topology of the sub-space Z of X by the equivalence relation defined by $f|_Z$. One says that Z is *universally submersive on $f(Z)$* if, for every base change $g : Y' \rightarrow Y$, the image $Z' = p^{-1}(Z)$ of Z by the projection $p : X \times_Y Y' \rightarrow Y'$ is a submersive set on $f'(Z')$. One notes that if the morphism f admits a Y -section $h : Y \rightarrow X$, then every set Z containing $h(Y)$ is universally submersive on Y (15.7.8).

Proposition (15.7.9). — *Let Y be a locally Noetherian prescheme, y a point of Y , $f : X \rightarrow Y$ a separated morphism of finite type. Let Z be a locally constructible part of X having the following properties:*

label=(i) *For every generisation y' of y , $Z_{y'} = Z \cap f^{-1}(y')$ is a connected component of $f^{-1}(y')$ (4.5.2).*

lbel=(ii) *Z_y is proper on $\text{Spec}(k(y))$.*

lbel=(iii) *Z is universally submersive on $f(Z)$ (15.7.8).*

Then Z is proper at y (in the sense of (15.7.1)), i.e. there exists a neighbourhood open U of y such that $Z \cap f^{-1}(U)$ is closed in $f^{-1}(U)$ and proper on U .

Proof. Using the method of (8.1.2, a)) and (8.10.5, (xii)), one reduces to proving the result when $Y = \text{Spec}(A)$ is the spectrum of a local Noetherian ring, and hypotheses (i), (ii), (iii) imply that Z is proper on Y .

Note first that if A' is a local Noetherian ring, $\varphi : A \rightarrow A'$ a local homomorphism, $Y' = \text{Spec}(A')$ and $g : Y' \rightarrow Y$ the morphism corresponding to φ , the inverse image Z' of Z by the canonical projection $p : X \times_Y Y' \rightarrow X$ satisfies properties (i), (ii), (iii) of the statement, by (1.8.2) and (2.7.1, (vii)); it therefore suffices to prove that Z' is proper on Y' , provided that $A' = \hat{A}$.

We use this remark by taking $A' = \hat{A}$ (0_I, 7.3.5), i.e. we may suppose that A is *complete*. By (III, 5.5.2), X is the *sum* of two sub-preschemes X_0, X_1 induced on open and closed parts of X , such that $X_0 \cap f^{-1}(y) = Z_y$. Set $Z_0 = X_0 \cap Z$, $Z_1 = X_1 \cap Z$; these sets form a partition of Z into two open and closed parts in Z . Since the Z_y are connected, one necessarily has $Z_y \subset Z_0$ or $Z_y \subset Z_1$, hence $f(Z_0) \cap f(Z_1) = \emptyset$. But since Z_0 and Z_1 are saturated for the equivalence relation defined by $f|_Z$, it follows from (iii) that $f(Z_0)$ and $f(Z_1)$ are open in $f(Z)$. Since $f(Z_0)$ contains y , and every neighbourhood of y in Y is equal to Y , one has $f(Z_0) = f(Z)$, and therefore $f(Z_1) = \emptyset$, i.e. $Z_1 = \emptyset$ and $Z \subset X_0$.

One can now suppose that f is *proper*, and it remains to prove that a constructible part Z of a prescheme X proper over Y satisfying only conditions (i) and (iii) is *closed* in X . By (0_{III}, 9.2.5), it suffices to prove that for every $x' \in Z$ and every specialisation x of x' in X , one has $x \in Z$. Let A_1 be a *complete* discrete valuation ring, u a morphism of $Y_1 = \text{Spec}(A_1)$ into X such that, if y_1 and y'_1 are the closed point and generic point, $u(y_1) = x$ and $u(y'_1) = x'$ (II, 7.1.7); set $g = f \circ u : Y_1 \rightarrow Y$ and $X_1 = X \times_Y Y_1$, $f_1 = f_{(Y_1)}$, where $p : X_1 \rightarrow X$ is the canonical projection. If $x_1 = u_1(y_1)$, $x'_1 = u_1(y'_1)$, one has $p(x_1) = x$ and $p(x'_1) = x'$, and $x'_1 \in Z_1 = p^{-1}(Z)$; since conditions (i) and (iii) are stable under all base changes, one reduces to the following situation: Y is the spectrum of a complete discrete valuation ring, X is proper over Y , there exists a Y -section h of X such that $h(y) = x$, $h(y') = x'$, and $x' \in Z$; one must prove that $x \in Z$.

Now, Z_y is a connected component of $f^{-1}(y)$, proper on $\text{Spec}(k(y))$ since X is proper over Y . Applying (III, 5.5.2) again, there is an open and closed part X' of X such that $X' \cap f^{-1}(y) = Z_y$; since Z_y is connected (replacing X' by the connected component containing Z_y if needed), one necessarily has $Z \subset X'$; since $h(Y)$ is connected and contains $x' \in Z \subset X'$, it is necessarily contained in X' , hence $x = h(y) \in X' \cap f^{-1}(y) = Z_y$. $\square$

Corollary (15.7.10). — Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a separated morphism of finite type, universally submersive (15.7.8) and such that all the fibres $X_y = f^{-1}(y)$ are geometrically connected. Then, if for every $y \in Y$ X_y is proper on $k(y)$, f is proper at the point y .

Proof. By (15.7.5), one is reduced to the case where $Y = \operatorname{Spec}(\mathcal{O}_y)$, hence the hypotheses are stable by base change. It suffices then to apply (15.7.9) to $Z = X$. $\square$

Corollary (15.7.11). — Let $f : X \rightarrow Y$ be a separated morphism of finite presentation; suppose that there exists a surjective morphism of finite presentation $g : Y' \rightarrow Y$, which is moreover either proper or flat, and such that if one sets $X' = X \times_Y Y'$, there exists a Y' -section of X' (or even a Y' -morphism $Y' \rightarrow X$ ((I, 3.3.14))). Suppose finally that the fibres $X_y = f^{-1}(y)$ are geometrically connected. Then the set U of the $y \in Y$ such that X_y is proper on $k(y)$ is open in Y , and the restriction $f^{-1}(U) \rightarrow U$ of f is proper.

Proof.

Lemma (15.7.11.1). — Let $f : X \rightarrow S$, $g : Y \rightarrow S$ be two morphisms, $p_1 : X \times_S Y \rightarrow X$, $p_2 : X \times_S Y \rightarrow Y$ the canonical projections. If $s : S \rightarrow Y$ is a section of g , then $s' = (1, s \circ f)_S$ and $s' : X \rightarrow X \times_S Y$, equipped with the morphisms $f : X \rightarrow S$ and $s' : X \rightarrow X \times_S Y$, identifies with the fibre product of the Y -preschemes S and $X \times_S Y$ for the morphisms $s : S \rightarrow Y$ and $p_2 : X \times_S Y \rightarrow Y$.

This is a special case of ((I, 3.3.11)), replacing the diagram by

$$\begin{array}{ccccc} X & \xrightarrow{p_1} & X \times_S Y & \xrightarrow{s'} & X \\ \downarrow & & & & \downarrow f \\ S & \xrightarrow{s} & Y & \xrightarrow{g} & S \end{array}$$

Applying this lemma by replacing S, Y, Y' by Y, Y' respectively, one sees that $f = (f')_{(Y)}$ for the base change $s : Y \rightarrow Y'$, hence f is affine since f' is. C.Q.F.D. $\square$

§16. DIFFERENTIAL INVARIANTS. DIFFERENTIALLY SMOOTH MORPHISMS

In this paragraph we will present, in global form, some notions of differential calculus particularly useful in algebraic geometry. We will ignore many classic developments in differential geometry (connections, infinitesimal transformations associated to vector fields, jets, etc.), although these notions are translated in a particularly natural way for schemes. We will similarly ignore phenomena exclusive to characteristic $p > 0$ (some of which are seen, in the affine case, in (0, 21). For certain complements to the differential formalism for preschemes the reader may consult Exposés II and VII of [eAG64] as well as subsequent chapters of this treatise.

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16.1. Normal invariants of an immersion

(16.1.1). Let $(X, \mathcal{O}_X), (Y, \mathcal{O}_Y)$ be two ringed spaces and $f = (\psi, \theta) : Y \rightarrow X$ a morphism of ringed spaces (0, 4.1.1) such that the homomorphism

$$\theta^\# : \psi^*(\mathcal{O}_X) \longrightarrow \mathcal{O}_Y$$

is surjective, so that $\mathcal{O}_Y$ is identified with a sheaf of quotient rings $\psi^*(\mathcal{O}_X)/\mathcal{I}_f$. We can then endow $\psi^*(\mathcal{O}_X)$ with the $\mathcal{I}_f$ -preadic filtration.

Definition (16.1.2). — The $\mathcal{O}_Y$ -augmented sheaf of rings $\psi^*(\mathcal{O}_X)/\mathcal{I}_f^{n+1}$ is called the n 'th normal invariant of f ; the ringed space $(Y, \psi^*(\mathcal{O}_X)/\mathcal{I}_f^{n+1})$ is called the n 'th infinitesimal neighbourhood of Y along f and is denoted by $Y_f^{(n)}$ or simply $Y^{(n)}$. The sheaf of graded rings associated to the sheaf of filtered rings $\psi^*(\mathcal{O}_X)$

$$(16.1.2.1) \quad \mathcal{G}_\bullet(f) = \bigoplus_{n \geq 0} (\mathcal{I}_f^n / \mathcal{I}_f^{n+1})$$

is called the sheaf of graded rings associated to f . The sheaf $\mathcal{G}_1(f) = \mathcal{I}_f / \mathcal{I}_f^2$ is called the conormal sheaf of f (that will be denoted by $\mathcal{N}_{Y/X}$ when there is no risk of confusion).

It is clear that the $\mathcal{O}_{Y^{(n)}} = \psi^*(\mathcal{O}_X) / \mathcal{I}_f^{n+1}$ (that we also denote $\mathcal{O}_{Y_f^{(n)}}$) form a projective system of sheaves of rings on Y , the transition homomorphism $\phi_{nm} : \mathcal{O}_{Y^{(m)}} \rightarrow \mathcal{O}_{Y^{(n)}}$ for $n \leq m$ identifies $\mathcal{O}_{Y^{(n)}}$ with the quotient of $\mathcal{O}_{Y^{(m)}}$ by the power $(\mathcal{I}_f / \mathcal{I}_f^{n+1})^m$ of the *augmentation ideal* of $\mathcal{O}_{Y^{(n)}}$, kernel of $\phi_{0n} : \mathcal{O}_{Y^{(n)}} \rightarrow \mathcal{O}_Y$. The $Y^{(n)}$ therefore form an inductive system of ringed spaces, all having underlying space Y , and we have canonical morphisms of ringed spaces $h_n : Y^{(n)} \rightarrow X$ equal to (ψ, θ_n) , where $\theta_n^\#$ is the canonical morphism $\psi^*(\mathcal{O}_X) \rightarrow \psi^*(\mathcal{O}_X) / \mathcal{I}_f^{n+1}$. It is clear that the sheaf $\mathcal{G}_\bullet(f)$ is a sheaf of graded algebras over the sheaf of rings $\mathcal{O}_Y = \mathcal{G}_0(f)$ and the $\mathcal{G}_k(f)$ of $\mathcal{O}_Y$ -modules.

As with every sheaf of filtered rings, we have a *canonical surjective homomorphism* of graded $\mathcal{O}_Y$ -algebras

$$(16.1.2.2) \quad \mathbf{S}_{\mathcal{O}_Y}^\bullet(\mathcal{G}_1(f)) \longrightarrow \mathcal{G}_\bullet(f)$$

which coincide in degrees 0 and 1 with the identities.

Examples (16.1.3). —

- (i) Suppose that X is a locally ringed space, Y is reduced to a single point y (endowed with a ring $\mathcal{O}_y$) and that, if $x = \psi(y)$, $\theta^\# : \mathcal{O}_x \rightarrow \mathcal{O}_y$ is a *surjective* homomorphism of rings having as kernel the maximal ideal $\mathfrak{m}_x$ of $\mathcal{O}_x$. So the $\mathcal{O}_{Y^{(n)}}$ are identified with the rings $\mathcal{O}_x / \mathfrak{m}_x^{n+1}$ and $\mathcal{G}_\bullet(f)$ with the graded ring associated with the local ring $\mathcal{O}_x$ endowed with the $\mathfrak{m}_x$ -preadic filtration.
- (ii) Suppose that Y is a closed subset of an open subspace U of X and that the $\mathcal{O}_Y$ is induced on Y by a quotient sheaf $\mathcal{O}_U / \mathcal{I}$, where $\mathcal{I}$ is an ideal of $\mathcal{O}_U$ such that $\mathcal{I}_x = \mathcal{O}_x$ for every $x \notin Y$; if X is a locally ringed space we also suppose that $\mathcal{I}_x \neq \mathcal{O}_x$ for $y \in Y$ so that $(Y, \mathcal{O}_Y)$ is a locally ringed space.

Let $\psi_0 : Y \rightarrow U$ be the canonical injection and denote by $\theta_0 : \mathcal{O}_U \rightarrow (\psi_0)_*(\mathcal{O}_Y)$ the homomorphism such that $\theta_0^\#$ is the canonical homomorphism $\psi_0^*(\mathcal{O}_U) = \mathcal{O}_U|_Y \rightarrow (\mathcal{O}_U / \mathcal{I})|_Y$, so that $j_0 = (\psi_0, \theta_0) : Y \rightarrow U$ is a morphism of ringed spaces (and of locally ringed spaces if X is a locally ringed space); if $i : U \rightarrow X$ is the canonical injection (morphism of ringed spaces), $j = i \circ j_0$ is the morphism (ψ, θ) of Y to X where $\psi : Y \rightarrow X$ is the canonical injection and $\theta : \mathcal{O}_X \rightarrow \psi_*(\mathcal{O}_Y)$ is the homomorphism such that $\theta^\# = \theta_0^\#$. Since $\theta^\#$ is surjective we can apply the previous definitions; $\mathcal{O}_{Y^{(n)}}$ is equal to $\psi_0^*(\mathcal{O}_U / \mathcal{I}^{n+1})$, and we have $(\psi_0)_*(\mathcal{O}_{Y^{(n)}}) = \mathcal{O}_U / \mathcal{I}^{n+1}$, and $\mathcal{G}_n(j) = \mathcal{G}_n(j_0) = \psi_0^*(\mathcal{I}^n / \mathcal{I}^{n+1}) = j_0^*(\mathcal{I}^n / \mathcal{I}^{n+1})$.

(16.1.4). The example (16.1.3, (ii)) shows that in general the $\mathcal{O}_{Y^{(n)}}$ are *not canonically endowed with a structure of an $\mathcal{O}_Y$ -module*, or *a fortiori* with a structure of an $\mathcal{O}_Y$ -algebra. The data of such structure is equivalent to the data of a homomorphism of sheaves of rings $\lambda_n : \mathcal{O}_Y \rightarrow \mathcal{O}_{Y^{(n)}}$, right inverse to the augmentation morphism ϕ_{0n} ; it is also equivalent to the data of a morphism of ringed spaces $(1_Y, \lambda_n) : Y^{(n)} \rightarrow Y$ left inverse to the canonical morphism $(1_Y, \phi_{0n}) : Y \rightarrow Y^{(n)}$.

Proposition (16.1.5). — *Let $f = (\psi, \theta) : Y \rightarrow X$ be an immersion of preschemes. Then:*

- (i) $\mathcal{G}_\bullet(f)$ is a *quasi-coherent graded $\mathcal{O}_Y$ -algebra*.
- (ii) The $Y^{(n)}$ are *preschemes, canonically isomorphic to subpreschemes of X* .
- (iii) Every homomorphism of sheaves of rings $\lambda_n : \mathcal{O}_Y \rightarrow \mathcal{O}_{Y^{(n)}}$, right inverse to the augmentation homomorphism ϕ_{0n} , makes the $\mathcal{O}_{Y^{(n)}}$ and $\mathcal{O}_{Y^{(k)}}$ for $k \leq n$ *quasi-coherent $\mathcal{O}_Y$ -algebras*; the $\mathcal{O}_Y$ -module structures induced from the above structures on the $\mathcal{G}_k(f)$ for $k \leq n$ coincide with the ones defined in (16.1.2).

Proof. (i) Since the question is local on X and Y , we can reduce to the case where Y is a closed subpreschemes of X defined by a quasi-coherent ideal $\mathcal{I}$ of $\mathcal{O}_X$; since $\mathcal{O}_Y$ is the restriction to Y of $\mathcal{O}_X / \mathcal{I}$ the assertion (i) is evident, and $Y^{(n)}$ is the closed subprescheme of X defined by the quasi-coherent ideal $\mathcal{I}^{n+1}$ of $\mathcal{O}_X$. Finally, to prove (iii) we notice that the data of λ_n makes the ideal $\mathcal{I} / \mathcal{I}^n$ of the augmentation ϕ_{0n} and their quotients $\mathcal{I} / \mathcal{I}^{k+1}$ ($1 \leq k \leq n$) $\mathcal{O}_Y$ -modules, and it suffices to prove by induction on k that the $\mathcal{I} / \mathcal{I}^{k+1}$ are quasi-coherent $\mathcal{O}_Y$ -modules and the structure of quotient $\mathcal{O}_Y$ -module induced on $\mathcal{I}^k / \mathcal{I}^{k+1}$ is the same as defined on (16.1.2). The second assertion is immediate, $\mathcal{I}^k / \mathcal{I}^{k+1}$ being killed by $\mathcal{I} / \mathcal{I}^{n+1}$; the first result, by induction on k , is trivial for $k = 1$ and for $\mathcal{I} / \mathcal{I}^{k+1}$ being an extension of $\mathcal{I} / \mathcal{I}^k$ by $\mathcal{I}^k / \mathcal{I}^{k+1}$ (1.4.17). $\square$

Corollary (16.1.6). — Under the general hypotheses of (16.1.5), if the immersion f is locally of finite presentation then the $\mathcal{G}_n(f)$ are quasi-coherent $\mathcal{O}_Y$ -modules of finite type.

Proof. Indeed, with the notation from the proof of (16.1.5), $\mathcal{I}$ is an ideal of finite type of $\mathcal{O}_X$ (1.4.7), therefore the $\mathcal{I}^n / \mathcal{I}^{n+1}$ are $\mathcal{O}_Y$ -modules of finite type, hence the conclusion. $\square$

Corollary (16.1.7). — Under the general hypotheses of (16.1.5), let $g : X \rightarrow Y$ be a morphism of preschemes, left inverse to f . Therefore, for every n , the composite morphism $(1, \lambda_n) : Y^{(n)} \xrightarrow{h_n} X \xrightarrow{g} Y$ defines a homomorphism of sheaves of rings $\lambda_n : \mathcal{O}_Y \rightarrow \mathcal{O}_{Y^{(n)}}$ right inverse to the augmentation ϕ_{0n} , making $\mathcal{O}_{Y^{(n)}}$ a quasi-coherent $\mathcal{O}_Y$ -algebra; via these homomorphisms, the transition homomorphism $\phi_{nm} : \mathcal{O}_{Y^{(m)}} \rightarrow \mathcal{O}_{Y^{(n)}}$ ($n \leq m$) are homomorphisms of $\mathcal{O}_Y$ -algebras. Also, if g is locally of finite type, then the $\mathcal{O}_{Y^{(n)}}$ are quasi-coherent $\mathcal{O}_Y$ -modules of finite type.

Proof. The first assertion is an immediate result from the definitions and (16.1.5). On the other hand, if g is locally of finite type, then f is locally of finite presentation (1.4.3, (v)); the $\mathcal{G}_n(f)$ being then quasi-coherent $\mathcal{O}_Y$ -modules of finite type by (16.1.6), the same goes for the $\mathcal{O}_Y$ -modules $\mathcal{I} / \mathcal{I}^{n+1}$, being extensions of a finite number of the $\mathcal{G}_k(f)$ (III, 1.4.17). $\square$

Proposition (16.1.8). — Let X be a locally Noetherian prescheme, $j : Y \rightarrow X$ an immersion; Then the $Y^{(n)}$ are locally Noetherian preschemes, the $\mathcal{G}_n(j)$ are coherent $\mathcal{O}_Y$ -modules and the $\mathcal{G}_\bullet(j)$ is a coherent sheaf of rings over the space Y .

Proof. Everything is local on X and Y , so we reduce to the case where X is affine and j is a closed immersion and therefore all the assertions are evident except for the last, which follows from the fact that if A is a Noetherian ring and $\mathfrak{J}$ is an ideal of A , then $\text{gr}_\bullet^\mathfrak{J}(A)$ is a Noetherian ring, taking into account the exactness of the functor ψ^* and (0, 5.3.7). $\square$

Proposition (16.1.9). — Let X be a prescheme, $j : Y \rightarrow X$ an immersion locally of finite presentation, y a point of Y . The following conditions are equivalent:

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- (a) There exists an open neighbourhood U of y in Y such that $j|_U$ is a homeomorphism of U onto an open set of X .
- (b) There is an integer $n > 0$ such that the canonical homomorphism

$$(\phi_{n-1,n})_y : \mathcal{O}_{Y^{(n)},y} \longrightarrow \mathcal{O}_{Y^{(n-1)},y}$$

is bijective.

- (c) There is an integer $n > 0$ such that $(\mathcal{G}_n(j))_y = 0$.

In addition, if the integer n satisfies (b) or (c), then there is a neighbourhood V of y in Y such that $\mathcal{G}_m(j)|_V = 0$ for $m \geq n$ and that $\phi_{nm}|_V : \mathcal{O}_{Y^{(m)}}|_V \rightarrow \mathcal{O}_{Y^{(n)}}|_V$ is bijective for $m \geq n$.

Proof. Since the questions is local on Y , we can restrict ourselves to the case where j is a closed immersion, Y being defined by a quasi-coherent ideal of finite type $\mathfrak{J}$ of $\mathcal{O}_X$. The equivalence of (b) and (c), for a given n , is immediate; also, since $\mathcal{I}^n / \mathcal{I}^{n+1}$ is an $\mathcal{O}_X$ -module of finite type, there is an open neighbourhood U of y in X such that $\mathcal{I}^n|_U = \mathcal{I}^{n+1}|_U$ (0, 5.2.2), so we also have $\mathcal{I}^n|_U = \mathcal{I}^m|_U$ for $m \geq n$ proving the last assertions. To prove that (a) implies (b), we can restrict ourselves to the cases where the underlying space of Y is equal to the underlying space of X and where $\mathcal{I}$ is generated by a finite number of sections over X : since $\mathcal{I}$ is contained in the nilradical $\mathcal{N}$ of $\mathcal{O}_X$ (I, 5.1.2), it is now nilpotent which proves b). Finally, to prove that (b) implies (a), we can restrict ourselves to the case where $\mathcal{I}^n = \mathcal{I}^m$; therefore, for every $y \in Y$, since $\mathcal{I}_y \subset \mathfrak{m}_y$, maximal ideal of $\mathcal{O}_{X,y}$, we must have $\mathcal{I}_y^n = 0$ because of Nakayama's lemma, since $\mathcal{I}_y$ is an ideal of finite type. The set of $x \in X$ such that $\mathcal{I}_x^n = 0$ is an open U of X contained in Y (0, 5.2.2); since on the other hand $\mathcal{I}_x \neq 0$ for $x \notin Y$, we must have $U = Y$. $\square$

Corollary (16.1.10). — For a restriction of the immersion j to an open neighbourhood of y in Y to be an open immersion (in other words, for j to be a local isomorphism on the point y), it is necessary and sufficient that $(\mathcal{G}_1(j))_y = (\mathcal{N}_{Y/X})_y = 0$.

Proof. The condition is clearly necessary, and the previous reasoning applied to $n = 1$ proves that it is sufficient. $\square$

Remark (16.1.11). —

- (i) Under the conditions of the definition (16.1.1), the projective limit of the projective system $(\mathcal{O}_{Y^{(n)}}, \phi_{nm})$ of sheaves of rings over Y is called the *normal invariant of infinite order* of f , and sometimes denoted by $\mathcal{O}_{Y^{(\infty)}}$. When X is a locally noetherian prescheme, $j : Y \rightarrow X$ a closed immersion, Y then is a closed subscheme of X defined by a coherent ideal $\mathcal{I}$ and $\mathcal{O}_{Y^{(\infty)}}$ is exactly the *formal completion* of $\mathcal{O}_X$ along Y (I, 10.8.4), and $Y^{(\infty)} = (Y, \mathcal{O}_{Y^{(\infty)}})$ is the formal prescheme that is the *completion* of X along Y (I, 10.8.5). In all cases, we could say that $Y^{(\infty)}$ is the *formal neighbourhood* of Y in X (via the morphism f). In the particular case we have just considered, it is the formal prescheme that is the inductive limit of the infinitesimal neighbourhoods of order n .
- (ii) Note that for a morphism of preschemes $f = (\psi, \theta) : Y \rightarrow X$, it can happen that the homomorphism $\theta^\# : \psi^*(\mathcal{O}_X) \rightarrow \mathcal{O}_Y$ is surjective without f being a local immersion and without f being injective. We have an example by taking Y to be a sum of preschemes Y_λ all isomorphic to $\text{Spec}(\mathcal{O}_x)$, where $x \in X$, and taking f to be the morphism equal to the canonical morphism in each of the Y_λ .

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16.2. Functorial properties of the normal invariants of an immersion

(16.2.1). Let $f = (\psi, \theta) : Y \rightarrow X$ and $f' = (\psi', \theta') : Y' \rightarrow X'$ by two morphisms of ringed spaces such that $\theta^\#$ and $\theta'^\#$ are surjective; consider a commutative diagram of morphisms of ringed spaces

$$(16.2.1.1) \quad \begin{array}{ccc} Y & \xrightarrow{f} & X \\ u \uparrow & & \uparrow v \\ Y' & \xrightarrow{f'} & X' \end{array}$$

Let $u = (\rho, \lambda), v = (\sigma, \mu)$. We have $\rho^*(\psi^*(\mathcal{O}_X)) = \psi'^*(\sigma^*(\mathcal{O}_{X'}))$ and as a result a commutative diagram of homomorphisms of sheaves of rings over Y'

$$\begin{array}{ccc} \rho^*(\psi^*(\mathcal{O}_X)) = \psi'^*(\sigma^*(\mathcal{O}_{X'})) & \xrightarrow{\psi'^*(\mu^\#)} & \psi'^*(\mathcal{O}_{X'}) \\ \rho^*(\theta^\#) \downarrow & & \downarrow \theta'^\# \\ \rho^*(\mathcal{O}_Y) & \xrightarrow{\lambda^\#} & \mathcal{O}_{Y'} \end{array}$$

from which we conclude, if $\mathcal{I}$ and $\mathcal{I}'$ are the kernels of $\theta^\#$ and $\theta'^\#$, that we have $\psi'^*(\mu^\#)(\rho^*(\mathcal{I})) \subset \mathcal{I}'$, having in mind the exactness of the functor ρ^* . We deduce that, for every integer n , $\psi'^*(\mu^\#)(\rho^*(\mathcal{I}^n)) \subset \mathcal{I}'^n$, which shows that $\psi'^*(\mu^\#)$ defines, passing to the quotients, a homomorphism of sheaves of rings

$$(16.2.1.2) \quad v_n : \rho^*(\psi^*(\mathcal{O}_X)/\mathcal{I}^{n+1}) \longrightarrow \psi'^*(\mathcal{O}_{X'})/\mathcal{I}'^{n+1}$$

and therefore a morphism of ringed spaces $w_n = (\rho, v_n) : Y'^{(n)} \rightarrow Y^{(n)}$ (which, for $n = 0$, is none other than u). It follows immediately from this definition that the diagrams

$$\begin{array}{ccccc} Y^{(n)} & \xrightarrow{h_{mn}} & Y^{(m)} & \xrightarrow{h_m} & X \\ w_n \uparrow & & \uparrow w_m & & \uparrow v \\ Y'^{(n)} & \xrightarrow{h'_{mn}} & Y'^{(m)} & \xrightarrow{h'_m} & X' \end{array} \quad (n \leq m)$$

(where the horizontal arrows are the canonical morphisms (16.1.2)) are commutative.

By passage to the quotients via the morphisms (16.2.1.2), and taking into account the exactness of the functor ρ^* , we obtain a di-homomorphism of graded algebras (relative to the morphism $\lambda^\# : \rho^*(\mathcal{O}_Y) \rightarrow \mathcal{O}_{Y'}$)

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$$(16.2.1.3) \quad \text{gr}(u) : \rho^*(\mathcal{G}_\bullet(f)) \longrightarrow \mathcal{G}_\bullet(f')$$

(or, if you like, a ρ -morphism (0, 3.5.1) $\mathcal{G}_\bullet(f) \rightarrow \mathcal{G}_\bullet(f')$), and in particular a di-homomorphism of conormal sheaves

$$\mathrm{gr}_1(u) : \rho^*(\mathcal{G}_1(f)) \longrightarrow \mathcal{G}_1(f').$$

It is also immediate that these homomorphisms give rise to a commutative diagram

$$(16.2.1.4) \quad \begin{array}{ccc} \rho^*(\mathbf{S}_{\mathcal{O}_Y}^\bullet(\mathcal{G}_1(f))) & \longrightarrow & \rho^*(\mathcal{G}_\bullet(f)) \\ \mathrm{S}(\mathrm{gr}_1(u)) \downarrow & & \downarrow \mathrm{gr}(u) \\ \mathbf{S}_{\mathcal{O}_Y}^\bullet(\mathcal{G}_1(f')) & \longrightarrow & \mathcal{G}_\bullet(f') \end{array}$$

where the horizontal arrow are the canonical morphisms (16.1.2.2).

Finally, if we have a commutative diagram of morphisms of ringed spaces

$$\begin{array}{ccc} Y & \xrightarrow{f} & X \\ u \uparrow & & \uparrow v \\ Y' & \xrightarrow{f'} & X' \\ u' \uparrow & & \uparrow v' \\ Y'' & \xrightarrow{f''} & X'' \end{array}$$

where $f'' = (\psi'', \theta'')$ is such that $\theta''^\#$ is surjective, and if w'_n and w''_n are defined from u', v' for one and $u'' = u \circ u', v'' = v \circ v'$ for the other, we have $w''_n = w'_n \circ w'_n$, which follows immediately from the definitions and from (0, 3.5.5); we have also $\mathrm{gr}(u'') = \mathrm{gr}(u') \circ \rho'^*(\mathrm{gr}(u))$ if $u' = (\rho', \lambda')$. Therefore we can say that $Y^{(n)}$ and $\mathcal{G}_\bullet(f)$ depend functorially on f .

Proposition (16.2.2). — *With the notation and hypotheses of (16.2.1), suppose also that f, f', u , and v are morphisms of preschemes. We have:*

- (i) *The morphisms $w_n : Y^{(n)} \rightarrow Y^{(n)}$ are morphisms of preschemes.*
- (ii) *If $Y' = Y \times_X X'$, u and f' the canonical projections, and if f is an immersion or if v is flat, we have $Y'^{(n)} = Y^{(n)} \times_X X'$.*
- (iii) *If $Y' = Y \times_X X'$ and if v is flat (resp. if f is an immersion), the homomorphism*

$$\mathrm{Gr}(u) = \mathrm{gr}(u) \otimes I : \mathcal{G}_\bullet(f) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \longrightarrow \mathcal{G}_\bullet(f')$$

is bijective (resp. surjective).

Proof.

- (i) The hypotheses immediately imply that, for every $y' \in Y'$, $\rho_{y'}^*(\theta_{\psi'(y')}^\#)$ is a local homomorphism (I, 1.6.2), so w_n is a morphism of preschemes (I, 2.2.1).
- (ii) and (iii) If f is an immersion, we can restrict ourselves to the case where f is a closed immersion, Y being defined by a quasi-coherent ideal $\mathcal{I}$ of $\mathcal{O}_X$ and $Y^{(n)}$ by the ideal $\mathcal{I}^{n+1}$; the assertions follows from (I, 4.4.5).

Second, suppose that v is flat; we can restrict ourselves to the case where $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(A')$ are affines, A' being a flat A -module; so $Y' = \mathrm{Spec}(B')$ where $B' = B \otimes_A A'$; in addition, if $\mathfrak{I}$ is the kernel of the homomorphism $A \rightarrow B$, the kernel $\mathfrak{I}'$ of $A' \rightarrow B'$ is identified with $\mathfrak{I} \otimes_A A'$ by flatness, and $\mathcal{I}^n / \mathcal{I}^{n+1}$ is equal to

$$\begin{aligned} \psi'^*(\sigma^*((\mathfrak{I}^n / \mathfrak{I}^{n+1})^\sim) \otimes_{\sigma^*(\mathcal{O}_X)} \mathcal{O}_{X'}) &= \\ \psi'^*(\sigma^*((\mathfrak{I}^n / \mathfrak{I}^{n+1})^\sim) \otimes_{\psi'^*(\sigma^*(\mathcal{O}_X))} \psi'^*(\mathcal{O}_{X'})) &= \rho^*(\mathcal{I}^n / \mathcal{I}^{n+1}) \otimes_{\rho^*(\psi^*(\mathcal{O}_X))} \psi'^*(\mathcal{O}_{X'}) \end{aligned}$$

and in particular for $n = 0$, we have

$$\mathcal{O}_{Y'} = \rho^*(\mathcal{O}_Y) \otimes_{\rho^*(\psi^*(\mathcal{O}_X))} \psi'^*(\mathcal{O}_{X'})$$

from which we have canonical isomorphism of $\mathcal{I}^n / \mathcal{I}^{n+1}$ with

$$\rho^*(\mathcal{I}^n / \mathcal{I}^{n+1}) \otimes_{\rho^*(\mathcal{O}_Y)} \mathcal{O}_{Y'} = (\mathcal{I}^n / \mathcal{I}^{n+1}) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$$

which proves (iii). Let now $C_n = \Gamma(Y, \mathcal{O}_{Y^{(n)}})$, $C'_n = \Gamma(Y', \mathcal{O}_{Y'^{(n)}})$. As $Y^{(n)}$ and $Y'^{(n)}$ are affine schemes (16.1.5), the kernel $\mathfrak{K}_n$ (resp. $\mathfrak{K}'_n$) of the homomorphism $C_n \rightarrow C_{n-1}$ (resp. $C'_n \rightarrow C'_{n-1}$) is $\Gamma(Y, \mathcal{I}^n / \mathcal{I}^{n+1})$ (resp. $\Gamma(Y', \mathcal{I}'^n / \mathcal{I}'^{n+1})$); therefore we can deduce from the above results that $\mathfrak{K}'_n = \mathfrak{K}_n \otimes_A A'$. Now, we have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathfrak{K}_n \otimes_A A' & \longrightarrow & C_n \otimes_A A' & \longrightarrow & C_{n-1} \otimes_A A' \longrightarrow 0 \\ & & \downarrow r & & \downarrow s_n & & \downarrow s_{n-1} \\ 0 & \longrightarrow & \mathfrak{K}'_n & \longrightarrow & C'_n & \longrightarrow & C'_{n-1} \longrightarrow 0 \end{array}$$

where the vertical arrow of the left is bijective and the two lines are exact (A' being a flat A -module). We deduce by induction that s_n is bijective for every n , because it is true by hypothesis for $n = 0$, and is deduced by application of the five lemma for all n . That proves the second assertion of (ii). $\square$

Corollary (16.2.3). — Let $g : X \rightarrow Y$, $u : Y' \rightarrow Y$ be two morphisms of preschemes, $X' = X \times_Y Y'$, $g' : X' \rightarrow Y'$ and $v : X' \rightarrow X$ by the canonical projections. Let $f : Y \rightarrow X$ by a Y -section of X (and therefore an immersion), $f' = f_{(Y')} : Y' \rightarrow X'$ the Y' -section of X' deduced from f by the base change u . We have:

- (i) The morphism $w_n : Y'_{f'}^{(n)} \rightarrow Y_f^{(n)}$ corresponding to f, f', u, v (16.2.1) and the canonical morphism $h'_n : Y'_{f'}^{(n)} \rightarrow X'$ identifies $Y'_{f'}^{(n)}$ with the product $Y_f^{(n)} \times_X X'$.
- (ii) If we endow $\mathcal{O}_{Y_f^{(n)}}$ (resp. $\mathcal{O}_{Y'_{f'}^{(n)}}$) with the structure of an $\mathcal{O}_Y$ -algebra defined by g (resp. with the structure of an $\mathcal{O}_{Y'}$ -algebra defined by g') (16.1.5, (iii)), then the homomorphism of $\mathcal{O}_{Y'}$ -algebras

$$(16.2.3.1) \quad \rho^*(\mathcal{O}_{Y_f^{(n)}}) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \longrightarrow \mathcal{O}_{Y'_{f'}^{(n)}}$$

induced by the homomorphism v_n (16.2.1.2) is bijective. Also, the homomorphism of $\mathcal{O}_{Y'}$ -modules

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$$(16.2.3.2) \quad \mathrm{Gr}_1(u) : \mathcal{G}_1(f) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \longrightarrow \mathcal{G}_1(f')$$

is bijective.

Proof.

- (i) Let us first note that $f' : Y' \rightarrow X'$ and $u : Y' \rightarrow Y$ identifies Y' with the product $Y \times_X X'$ (via the structure morphisms $f : Y \rightarrow X$ and $v : X' \rightarrow X$) (14.5.12.1). The conclusion of (i) now follows from (16.2.2, (ii)), the morphism g being an immersion.
- (ii) The commutative diagram

$$\begin{array}{ccc} Y_f^{(n)} & \xleftarrow{w_n} & Y'_{f'}^{(n)} \\ \downarrow h_n & & \downarrow h'_n \\ X & \xleftarrow{v} & X' \\ \downarrow g & & \downarrow g' \\ Y & \xleftarrow{u} & Y' \end{array}$$

identifies $Y'_{f'}^{(n)}$ with the product $Y_f^{(n)} \times_X X'$, so (I, 3.3.9) it identifies (via the morphisms $g' \circ h'_n$ and w_n) $Y'_{f'}^{(n)}$ to the product $Y_f^{(n)} \times_Y Y'$. Since $Y_f^{(n)}$ (resp. $Y'_{f'}^{(n)}$) is the affine prescheme over Y (resp. over Y') associated with the $\mathcal{O}_Y$ -algebra $\mathcal{O}_{Y_f^{(n)}}$ (resp. to the $\mathcal{O}_{Y'}$ -algebra $\mathcal{O}_{Y'_{f'}^{(n)}}$), the fact that the canonical homomorphism (16.2.3.1) is bijective follows from (II, 1.5.2). Finally, the canonical homomorphism (16.2.3.1) is compatible with the augmentations $\mathcal{O}_{Y_f^{(n)}} \rightarrow \mathcal{O}_Y$ and $\mathcal{O}_{Y'_{f'}^{(n)}} \rightarrow \mathcal{O}_{Y'}$; since $\mathcal{O}_{Y_f^{(n)}}$ is a direct sum (as an $\mathcal{O}_Y$ -module) of

$\mathcal{O}_Y$ and the augmentation ideal $\mathcal{I} / \mathcal{I}^{n+1}$, we can therefore see that the canonical homomorphism (16.2.3.1), restricted to $\mathcal{I} / \mathcal{I}^{n+1} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'}$, is a bijection of the latter onto $\mathcal{I}' / \mathcal{I}'^{n+1}$. For $n = 1$ this shows that $\mathrm{Gr}_1(u)$ is bijective.

□

We note that, under the hypotheses of (16.2.3), the homomorphisms $\mathrm{Gr}_n(u)$ are *surjective* in view of the above, but are not bijective in general for $n \geq 2$. However:

Corollary (16.2.4). — *Under the hypotheses of (16.2.3), suppose that $u : Y' \rightarrow Y$ is a flat morphism (resp. that the $\mathcal{G}_n(f)$ are flat $\mathcal{O}_Y$ -modules for $n \leq m$). Then the homomorphism*

$$\mathrm{Gr}_n(u) : \mathcal{G}_n(f) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \longrightarrow \mathcal{G}_n(f')$$

is bijective for all n (resp. for $n \leq m$).

Proof. If u is flat, then we deduce by base change that the same is true for $v : X' \rightarrow X$, and we already know in this case that $\mathrm{Gr}(u)$ is bijective (16.2.2, (iii)). If the $\mathcal{G}_n(f)$ are flat for $n \leq m$, then we first see by induction on n that the same holds for $\mathcal{I} / \mathcal{I}^{n+1}$ for $n \leq m$, because of the exact sequences

$$0 \longrightarrow \mathcal{I}^n / \mathcal{I}^{n+1} \longrightarrow \mathcal{I} / \mathcal{I}^{n+1} \longrightarrow \mathcal{I} / \mathcal{I}^n \longrightarrow 0$$

(0, 6.1.2); in addition, we have the commutative diagram

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$$\begin{array}{ccccccc} 0 & \longrightarrow & (\mathcal{I}^n / \mathcal{I}^{n+1}) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} & \longrightarrow & (\mathcal{I} / \mathcal{I}^{n+1}) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} & \longrightarrow & (\mathcal{I} / \mathcal{I}^n) \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y'} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathcal{I}^m / \mathcal{I}^{m+1} & \longrightarrow & \mathcal{I}' / \mathcal{I}'^{m+1} & \longrightarrow & \mathcal{I}' / \mathcal{I}'^m \longrightarrow 0 \end{array}$$

in which the lines are exact (the first by flatness (0, 6.1.2)) and the two last vertical arrows are bijective by virtue of (16.2.2, (ii)); hence the conclusion. □

Remarks (16.2.5). —

- (i) The reasoning of (16.2.2, (i)) still applies to (16.2.1.1) when these are morphisms of *locally ringed spaces* (I, 1.8.2).
- (ii) In (16.2.2, (ii)), the conclusion is no longer necessarily valid if we only suppose that v and f are morphisms of preschemes (f satisfying the condition of (16.1.1)). For example (with the notation of the proof of (16.2.2, (ii))), it can happen that $\mathfrak{I} = 0$ but the kernel $\mathfrak{I}'$ of $A' \rightarrow B' = B \otimes_A A'$ is not zero and that $B' \neq 0$, in which case we have $Y^{(n)} = Y$ for all n , but $Y'^{(n)} \neq Y'$. We have an example of this by taking $A = \mathbf{Z}$, $B = \mathbf{Q}$, $A' = \prod_{h=1}^{\infty} (\mathbf{Z} / m^h \mathbf{Z})$ where $m > 1$.

(16.2.6). Consider the particular case of the diagram (16.2.1.1) where $X' = X$, v is the identity, X a prescheme, Y a subscheme of X , Y' a subscheme of Y , f , u , and $f' = f \circ u$ the canonical injections; the di-homomorphism (16.2.1.3) gives us, by tensoring with $\mathcal{O}_{Y'}$ over $\rho^*(\mathcal{O}_Y)$, a homomorphism of graded $\mathcal{O}_{Y'}$ -algebras

$$(16.2.6.1) \quad u^*(\mathcal{G}_\bullet(f)) \longrightarrow \mathcal{G}_\bullet(f').$$

On the other hand, we identify $\mathcal{O}_Y$ to $\psi^*(\mathcal{O}_X) / \mathcal{I}_f$ and $\mathcal{O}_{Y'}$ to $\rho^*(\mathcal{O}_Y) / \mathcal{I}_u$; since ρ^* is an exact functor, we have $\rho^*(\mathcal{O}_Y) = \rho^*(\psi^*(\mathcal{O}_X)) / \rho^*(\mathcal{I}_f) = \psi'^*(\mathcal{O}_X) / \rho^*(\mathcal{I}_f)$, and since $\mathcal{O}_{Y'}$ is moreover identified with $\psi'^*(\mathcal{O}_X) / \mathcal{I}_{f'}$, we see that $\mathcal{I}_u = \mathcal{I}_{f'} / \rho^*(\mathcal{I}_f)$. We deduce that for every integer n there is a canonical homomorphism $\mathcal{I}_{f'}^n / \mathcal{I}_{f'}^{n+1} \rightarrow \mathcal{I}_u^n / \mathcal{I}_u^{n+1}$, from which we have a canonical morphism of graded $\mathcal{O}_{Y'}$ -algebras

$$(16.2.6.2) \quad \mathcal{G}_\bullet(f') \longrightarrow \mathcal{G}_\bullet(u).$$

Proposition (16.2.7). — *Let X be a prescheme, Y a subscheme of X , Y' a subscheme of Y , $j : Y' \rightarrow Y$ the canonical injection. We then have an exact sequence of conormal sheaves ($\mathcal{O}_{Y'}$ -modules)*

$$(16.2.7.1) \quad j^*(\mathcal{N}_{Y/X}) \longrightarrow \mathcal{N}_{Y'/X} \longrightarrow \mathcal{N}_{Y'/Y} \longrightarrow 0$$

where the arrows are the degree 1 components of the canonical homomorphisms (16.2.6.1) and (16.2.6.2).

Proof. The problem being local, we can restrict to the case where $X = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(A/\mathfrak{I})$ and $Y' = \operatorname{Spec}(A/\mathfrak{K})$, $\mathfrak{I}$ and $\mathfrak{K}$ being ideals of A such that $\mathfrak{I} \subset \mathfrak{K}$; everything reduces to seeing that the sequence of canonical morphisms $\mathfrak{I}/\mathfrak{K}\mathfrak{I} \rightarrow \mathfrak{K}/\mathfrak{K}^2 \rightarrow (\mathfrak{K}/\mathfrak{I})/(\mathfrak{K}/\mathfrak{I})^2 \rightarrow 0$ is exact, which is immediate given that the image of $\mathfrak{I}/\mathfrak{K}\mathfrak{I}$ in $\mathfrak{K}/\mathfrak{K}^2$ is $(\mathfrak{I} + \mathfrak{K}^2)/\mathfrak{K}^2$ and that $(\mathfrak{K}/\mathfrak{I})/(\mathfrak{K}/\mathfrak{I})^2$ is identified with $\mathfrak{K}/(\mathfrak{I} + \mathfrak{K}^2)$. $\square$

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It is easy to give examples where the sequence (16.2.7.1) extended on the left by 0 is not exact; with the above notation, it suffices to take $A = k[T]$, $\mathfrak{I} = AT^2$, $\mathfrak{K} = AT$, because then $(\mathfrak{I} + \mathfrak{K}^2)/\mathfrak{K}^2 = 0$ and $\mathfrak{I}/\mathfrak{K}\mathfrak{I} \neq 0$. See however (16.9.13) and (19.1.5) for some cases where the extended sequence is indeed exact.

16.3. Fundamental differential invariants of morphisms of preschemes

Definition (16.3.1). — Let $f : X \rightarrow S$ be a morphism of preschemes, $\Delta_f : X \rightarrow X \times_S X$ the corresponding diagonal morphism, which is an immersion (I, 5.3.9). We denote by $\mathcal{P}_f^n$ or $\mathcal{P}_{X/S}^n$, and call the *sheaf of principal parts of order n of the S -prescheme X* , the $\mathcal{O}_X$ -augmented sheaf of rings, n -th normal invariant of Δ_f (16.1.2). We will also write $\mathcal{P}_f^\infty = \mathcal{P}_{X/S}^\infty = \varprojlim_n \mathcal{P}_{X/S}^n$, $\mathcal{G}_n(\mathcal{P}_f) = \mathcal{G}_n(\mathcal{P}_{X/S}) = \mathcal{G}_n(\Delta_f)$ (16.1.2); the $\mathcal{O}_X$ -module $\mathcal{G}_1(\Delta_f)$, augmentation sheaf of ideals of $\mathcal{P}_{X/S}^1$, is denoted by Ω_f^1 or $\Omega_{X/S}^1$, and is called the $\mathcal{O}_X$ -module of 1-differentials of f , or of X with respect to S , or of the S -prescheme X .

It follows from this definition that $\mathcal{P}_{X/S}^0$ is canonically identified with $\mathcal{O}_X$ (16.1.2).

We have (16.1.2.2) a canonical surjective morphism of graded $\mathcal{O}_X$ -algebras

$$(16.3.1.1) \quad \mathbf{S}_{\mathcal{O}_X}^\bullet(\Omega_{X/S}^1) \longrightarrow \mathcal{G}_\bullet(\mathcal{P}_{X/S}).$$

And it follows from Definition (16.3.1) that for every open U of X we have $\mathcal{P}_{f|U}^n = \mathcal{P}_f^n|_U$, $\mathcal{P}_{f|U}^\infty = \mathcal{P}_f^\infty|_U$, $\mathcal{G}_n(\mathcal{P}_{f|U}) = \mathcal{G}_n(\mathcal{P}_f)|_U$, $\Omega_{f|U}^1 = \Omega_f^1|_U$ (in other words, the notions introduced are *local* on X).

(16.3.2). Denote by p_1, p_2 the two canonical projections of the product $X \times_S X$; since Δ_f is an X -section of $X \times_S X$ for both p_1 and p_2 , each of these morphisms define, for all n , a homomorphism of sheaves of rings $\mathcal{O}_X \rightarrow \mathcal{P}_{X/S}^n$, right inverse of the augmentation $\mathcal{P}_{X/S}^n \rightarrow \mathcal{O}_X$ (16.1.7); we can also say that we thus define on $\mathcal{P}_{X/S}^n$ two quasi-coherent augmented $\mathcal{O}_X$ -algebra structures; the corresponding $\mathcal{O}_X$ -module structures on $\mathcal{G}_n(\mathcal{P}_{X/S}^n)$ are the same. We also have, by passing to the limit, two $\mathcal{O}_X$ -algebra structures on $\mathcal{P}_{X/S}^\infty$.

(16.3.3). The morphism $s = (p_2, p_1)_S : X \times_S X \rightarrow X \times_S X$ is an involutive automorphism of $X \times_S X$, called the *canonical symmetry*, such that

$$(16.3.3.1) \quad p_1 \circ s = p_2, \quad p_2 \circ s = p_1, \quad s \circ \Delta_f = \Delta_f.$$

If we put $s = (\rho, \lambda)$, $p_i = (\pi_i, \mu_i)$ ($i = 1, 2$), $\Delta_f = (\delta, \nu)$, $\lambda^\#$ is then an isomorphism of $\rho^*(\pi_1^*(\mathcal{O}_X))$ onto $\pi_2^*(\mathcal{O}_X)$, and $\delta^*(\lambda^\#)$ fixes $\delta^*(\mathcal{O}_{X \times_S X})$ and the kernel $\mathcal{I}$ of the homomorphism $\nu^\# : \delta^*(\mathcal{O}_{X \times_S X}) \rightarrow \mathcal{O}_X$. Therefore:

Proposition (16.3.4). — The homomorphism $\sigma = \delta^*(\lambda^\#)$ induced from s (and also called the canonical symmetry) is an involutive automorphism of the projective system $(\mathcal{P}_{X/S}^n)$ of $\mathcal{O}_X$ -augmented sheaves of rings, and as a result also of the projective limit $\mathcal{P}_{X/S}^\infty$. This automorphism permutes the $\mathcal{O}_X$ -algebra structure on $\mathcal{P}_{X/S}^n$ and on $\mathcal{P}_{X/S}^\infty$.

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(16.3.5). In what follows, the two $\mathcal{O}_X$ -algebra structures defined on the $\mathcal{P}_{X/S}^n$ and on $\mathcal{P}_{X/S}^\infty$ will play very different roles: we will now agree, unless said otherwise, that when $\mathcal{P}_{X/S}^n$ or $\mathcal{P}_{X/S}^\infty$ is considered as an $\mathcal{O}_X$ -algebra, it is the algebra structure induced by p_1 .

For every open U of X and every section $t \in \Gamma(U, \mathcal{O}_X)$, we will simply denote by $t.1$ or even t the image of t under the structure morphism $\Gamma(U, \mathcal{O}_X) \rightarrow \Gamma(U, \mathcal{P}_{X/S}^n)$ (resp. $\Gamma(U, \mathcal{O}_X) \rightarrow \Gamma(U, \mathcal{P}_{X/S}^\infty)$) (that is to say, the homomorphism corresponding to p_1).

Definition (16.3.6). — We denote by d_f^n , or $d_{X/S}^n$ (resp. d_f^∞ , or $d_{X/S}^\infty$), or simply d^n (resp. d^∞), the homomorphism of sheaves of rings $\mathcal{O}_X \rightarrow \mathcal{P}_f^n = \mathcal{P}_{X/S}^n$ (resp. $\mathcal{O}_X \rightarrow \mathcal{P}_f^\infty = \mathcal{P}_{X/S}^\infty$) induced by p_2 . For every open U of X , and every $t \in \Gamma(U, \mathcal{O}_X)$, $d^n t$ (resp. $d^\infty t$) is called the *principal part of order n* (resp. *principal part of infinite order*) of t . We set $dt = d^1 t - t$, and we say that dt is the differential of t (an element of $\Gamma(U, \Omega_{X/S}^1)$, also denoted $d_{X/S}(t)$).

It follows immediately⁷ from this definition that we have

$$(16.3.6.1) \quad d(t_1 t_2) = t_1 dt_2 + t_2 dt_1$$

for every t_1, t_2 in $\Gamma(U, \mathcal{O}_X)$, that is, d is a *derivation* of the ring $\Gamma(U, \mathcal{O}_X)$ in the $\Gamma(U, \mathcal{O}_X)$ -module $\Gamma(U, \Omega_{X/S}^1)$.

In all notation introduced in (16.3.1) and (16.3.6), we will sometimes replace S by A when $S = \text{Spec}(A)$.

(16.3.7). Suppose in particular that $S = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affine schemes, B then being an A -algebra. Then Δ_f corresponds to the canonical surjective homomorphism $\pi : B \otimes_A B \rightarrow B$ such that $\pi(b \otimes b') = bb'$, with kernel $\mathfrak{I} = \mathfrak{I}_{B/A}$ (0, 20.4.1); $\mathcal{P}_f^n$ is the structure sheaf of the prescheme $\text{Spec}(P_{B/A}^n)$, where

$$P_{B/A}^n = (B \otimes_A B) / \mathfrak{I}^{n+1};$$

$\mathcal{G}_\bullet(\mathcal{P}_f)$ is the quasi-coherent $\mathcal{O}_X$ -module corresponding to the graded B -module

$$\text{gr}_\bullet^{\mathfrak{I}}(B \otimes_A B) = \bigoplus_{n \geq 0} (\mathfrak{I}^n / \mathfrak{I}^{n+1});$$

in particular $\Omega_f^1 = \Omega_{X/S}^1$ is the quasi-coherent $\mathcal{O}_X$ -module corresponding to the B -module of 1-differentials of B over A , $\Omega_{B/A}^1$ (0, 20.4.3). The projection morphisms $p_1 : X \times_S X \rightarrow X$, $p_2 : X \times_S X \rightarrow X$ corresponding to the two homomorphisms of rings $j_1 : B \rightarrow B \otimes_A B$, $j_2 : B \rightarrow B \otimes_A B$ such that $j_1(b) = b \otimes 1$, $j_2(b) = 1 \otimes b$, so that (by the convention of (16.3.5)), $P_{B/A}^n$ is always considered as a B -algebra via the composite homomorphism $B \xrightarrow{j_1} B \otimes_A B \rightarrow P_{B/A}^n$; the ring homomorphism $B \xrightarrow{j_2} B \otimes_A B \rightarrow P_{B/A}^n$ is denoted by $d_{B/A}^n$ and corresponds to $d_{X/S}^n$ acting on $\Gamma(X, \mathcal{O}_X)$; for every $t \in B$, dt is equal to $d_{B/A}^1 t$, defined in (0, 20.4.6).

If $\pi_n : B \otimes_A B \rightarrow P_{B/A}^n$ is the canonical homomorphism, so we have, in light of the preceding definitions,

$$(16.3.7.1) \quad \pi_n(b \otimes b') = b \cdot \pi_n(1 \otimes b') = b \cdot d_{B/A}^n(b') \quad \text{for } b \in B, b' \in B.$$

Proposition (16.3.8). — The image of the canonical homomorphism $d_{X/S}^n : \mathcal{O}_X \rightarrow \mathcal{P}_{X/S}^n$ generates the $\mathcal{O}_X$ -module $\mathcal{P}_{X/S}^n$. IV-4 | 16

Proof. We immediately reduce to the case where $X = \text{Spec}(B)$ and $S = \text{Spec}(A)$ are affine and the proposition follows from (16.3.7.1) since π_n is surjective. We note that in general $d_{X/S}^n$ is not surjective (even for $n = 1$). □

Proposition (16.3.9). — Suppose that $f : X \rightarrow S$ is a morphism locally of finite type. Then the $\mathcal{P}_f^n$ and the $\mathcal{G}_n(\mathcal{P}_f)$ are quasi-coherent $\mathcal{O}_X$ -modules of finite type.

Proof. This follows from (16.1.6) and from the fact that Δ_f is locally of finite presentation (I, 4.3.1). □

⁷[Trans.] This is, locally we have (0, 20.1.1).

16.4. Functorial properties of differential invariants

(16.4.1). Consider a commutative diagram of morphisms of preschemes

(16.4.1.1)

$$\begin{array}{ccc} X & \xleftarrow{u} & X' \\ f \downarrow & & \downarrow f' \\ S & \xleftarrow{w} & S' \end{array}$$

We deduce a commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{u} & X' \\ \Delta_f \downarrow & & \downarrow \Delta_{f'} \\ X \times_S X & \xleftarrow{v} & X' \times_{S'} X' \end{array}$$

where v is the composite homomorphism (I, 5.3.5) and (I, 5.3.15).

$$(16.4.1.2) \quad X' \times_{S'} X' \xrightarrow{(p'_1, p'_2)_S} X' \times_S X' \xrightarrow{u \times_S u} X \times_S X.$$

So we induce from u and v , as explained in (16.2.1), homomorphisms of augmented sheaves of rings

$$(16.4.1.3) \quad \nu_n : \rho^*(\mathcal{P}_{X/S}^n) \longrightarrow \mathcal{P}_{X'/S'}^n$$

(where we put $u = (\rho, \lambda)$); these homomorphisms form a projective system, and therefore give at the limit a homomorphism of sheaves of graded rings

$$(16.4.1.4) \quad \nu_\infty : \rho^*(\mathcal{P}_{X/S}^\infty) \longrightarrow \mathcal{P}_{X'/S'}^\infty;$$

on the other hand, by passing to the quotient, the homomorphisms ν_n give rise to a di-homomorphism of graded algebras (relative to $\lambda^\#$):

$$(16.4.1.5) \quad \text{gr}(u) : \rho^*(\mathcal{G}_\bullet(\mathcal{P}_{X/S})) \longrightarrow \mathcal{G}_\bullet(\mathcal{P}_{X'/S'}).$$

(16.4.2). If we have a commutative diagram

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$$\begin{array}{ccccc} X & \xleftarrow{u} & X' & \xleftarrow{u'} & X'' \\ f \downarrow & & \downarrow f' & & \downarrow f'' \\ S & \xleftarrow{w} & S' & \xleftarrow{w'} & S'' \end{array}$$

we deduce a commutative diagram

$$\begin{array}{ccccc} X & \xleftarrow{u} & X' & \xleftarrow{u'} & X'' \\ \Delta_f \downarrow & & \downarrow \Delta_{f'} & & \downarrow \Delta_{f''} \\ X \times_S X & \xleftarrow{v} & X' \times_{S'} X' & \xleftarrow{v'} & X'' \times_{S''} X'' \end{array}$$

where v' is defined from u', w', f', f'' as v is from u, w, f, f' . We verify immediately that if $u'' = u \circ u'$, $w'' = w \circ w'$, then the composite homomorphism $v \circ v'$ is equal to the homomorphism v'' deduced from u'', v'', f, f'' as v is from u, w, f, f' . If we put $u' = (\rho', \lambda')$, $u'' = (\rho'', \lambda'')$ it follows (16.2.1) that the homomorphism $\nu''_n : \rho''^*(\mathcal{P}_{X''/S''}^n) \rightarrow \mathcal{P}_{X''/S''}^n$ is equal to the composite

$$\rho'^*(\rho^*(\mathcal{P}_{X/S}^n)) \xrightarrow{\rho'^*(\nu_n^\#)} \rho'^*(\mathcal{P}_{X'/S'}^n) \xrightarrow{\nu''_n} \mathcal{P}_{X''/S''}^n,$$

and we have analogous transitivity properties for the homomorphisms (16.4.1.4) and (16.4.1.5), which lets us say that the $\mathcal{P}_{X/S}^n$, $\mathcal{P}_{X/S}^\infty$ and $\mathcal{G}_\bullet(\mathcal{P}_{X/S})$ depend functorially on f .

(16.4.3). We verify immediately (for example, by restricting ourselves to the affine case with help of (16.3.7)) that with the notation of (16.4.1), the diagram

$$(16.4.3.1) \quad \begin{array}{ccc} \rho^*(\mathcal{O}_X) & \xrightarrow{\lambda^\#} & \mathcal{O}_{X'} \\ \downarrow & & \downarrow \\ \rho^*(\mathcal{P}_{X/S}^n) & \xrightarrow{v_n} & \mathcal{P}_{X'/S'}^n \end{array}$$

where the vertical arrows are the ones defining the algebra structure chosen in (16.3.5) (that is to say, the ones coming from the first projections) is commutative; the same goes for the diagram

$$(16.4.3.2) \quad \begin{array}{ccc} \rho^*(\mathcal{O}_X) & \xrightarrow{\lambda^\#} & \mathcal{O}_{X'} \\ \rho^*(d_{X/S}^n) \downarrow & & \downarrow d_{X'/S'}^n \\ \rho^*(\mathcal{P}_{X/S}^n) & \xrightarrow{v_n} & \mathcal{P}_{X'/S'}^n \end{array}$$

the vertical arrows defining here the algebra structure from the second projection; besides, if σ and σ' are the canonical symmetries corresponding to f and f' (16.3.4), we have IV-4 | 18

$$v_n \circ \rho^*(\sigma) = \sigma' \circ v_n$$

which switches one diagram with the other. We deduce from (16.4.3.1) a canonical homomorphism of *augmented* $\mathcal{O}_{X'}$ -algebras

$$(16.4.3.3) \quad P^n(u) : u^*(\mathcal{P}_{X/S}^n) = \mathcal{P}_{X/S}^n \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} \longrightarrow \mathcal{P}_{X'/S'}^n$$

and it follows from (16.4.3.2) that the diagram

$$(16.4.3.4) \quad \begin{array}{ccc} \mathcal{O}_{X'} & \xrightarrow{\text{id}} & \mathcal{O}_{X'} \\ u^*(d_{X/S}^n) \downarrow & & \downarrow d_{X'/S'}^n \\ u^*(\mathcal{P}_{X/S}^n) & \xrightarrow{P^n(u)} & \mathcal{P}_{X'/S'}^n \end{array}$$

is commutative. We deduce a homomorphism of *graded* $\mathcal{O}_{X'}$ -algebras

$$(16.4.3.5) \quad \text{Gr}_\bullet(u) : u^*(\text{Gr}_\bullet(\mathcal{P}_{X/S})) \longrightarrow \text{Gr}_\bullet(\mathcal{P}_{X'/S'})$$

and in particular a homomorphism of $\mathcal{O}_{X'}$ -modules

$$(16.4.3.6) \quad \text{Gr}_1(u) : \Omega_{X/S}^1 \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} \longrightarrow \Omega_{X'/S'}^1$$

giving rise to a commutative diagram

$$(16.4.3.7) \quad \begin{array}{ccc} \mathcal{O}_{X'} & \xrightarrow{\text{id}} & \mathcal{O}_{X'} \\ d_{X/S} \otimes 1 \downarrow & & \downarrow d_{X'/S'} \\ \Omega_{X/S}^1 \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} & \longrightarrow & \Omega_{X'/S'}^1 \end{array}$$

(16.4.4). When $S = \text{Spec}(A)$, $S' = \text{Spec}(A')$, $X = \text{Spec}(B)$, $X' = \text{Spec}(B')$ are affine, so that we have a commutative diagram of ring homomorphisms

$$\begin{array}{ccc} B & \longrightarrow & B' \\ \uparrow & & \uparrow \\ A & \longrightarrow & A' \end{array}$$

the image of $\mathcal{I}_{B/A}$ in $B' \otimes_{A'} B'$ is contained in $\mathcal{I}_{B'/A'}$, and the homomorphism v_n corresponds to the homomorphism of rings $P_{B/A}^n \rightarrow P_{B'/A'}^n$ induced from the homomorphism $B \otimes_A B \rightarrow B' \otimes_{A'} B'$ by passing to quotients. The homomorphism (16.4.3.6) corresponds to the homomorphism defined in (0, 20.5.4.1), and the commutative diagram (16.4.3.7) to the diagram (0, 20.5.4.2).

Proposition (16.4.5). — Suppose that $X' = X \times_S S'$, f' and u the canonical homomorphisms $P^n(u)$ (16.4.3.3) and $\text{Gr}_1(u)$ (16.4.3.6) are bijective. IV-4 | 19

Proof. We have $X' \times_{S'} X' = (X \times_S X) \times_S S'$, and it suffices to apply (16.2.3, (ii)) replacing g by the first $p_1 : X \times_S X \rightarrow X$ and f by the diagonal Δ_f . $\square$

We note that under the hypotheses of (16.4.5) the homomorphism $\text{Gr}_\bullet(u)$ (16.4.3.5) is surjective, but not bijective in general. However (16.2.4):

Corollary (16.4.6). — Under the hypotheses of (16.4.5), suppose in addition that $w : S \rightarrow S'$ is flat (resp. that $\mathcal{G}_n(\mathcal{P}_{X/S}^n)$ are flat $\mathcal{O}_X$ -modules for $n \leq m$); then the homomorphism

$$\text{Gr}_n(u) : u^*(\mathcal{G}(\mathcal{P}_{X/S}^n)) \longrightarrow \mathcal{G}(\mathcal{P}_{X'/S'}^n)$$

is bijective for each n (resp. for $n \leq m$).

Proof. Indeed, if w is flat, then so is $v : X' \times_{S'} X' \rightarrow X \times_S X$, so the conclusion follows from (16.2.4). $\square$

(16.4.7). Let S be a prescheme, $\mathcal{E}$ a quasi-coherent $\mathcal{O}_S$ -Module, and set $X = \mathbf{V}(\mathcal{E})$ (II, 1.7.8), the vector bundle associated to $\mathcal{E}$, equal to $\text{Spec}(\mathbf{S}_{\mathcal{O}_S}(\mathcal{E}))$. Let $f : X \rightarrow S$ be the structure morphism. For every open U of S and every section $t \in \Gamma(U, \mathcal{E})$, t is identified with a section of $\mathbf{S}_{\mathcal{O}_S}(\mathcal{E})$ over U ; let t' be its image in $\Gamma(f^{-1}(U), \mathcal{O}_X) = \Gamma(U, f_*(\mathcal{O}_X)) = \Gamma(U, \mathbf{S}_{\mathcal{O}_S}(\mathcal{E}))$, and set

$$(16.4.7.1) \quad \delta(t) = d_{X/S}^n(t') - t' \in \Gamma(f^{-1}(U), \mathcal{P}_{X/S}^n);$$

it is clear that δ is a di-homomorphism of modules (corresponding to the homomorphism of rings $\Gamma(U, \mathcal{O}_S) \rightarrow \Gamma(f^{-1}(U), \mathcal{O}_X)$ of $\Gamma(U, \mathcal{E})$ into $\Gamma(f^{-1}(U), \mathcal{P}_{X/S}^n)$, and therefore the image belongs to the augmentation ideal of $\Gamma(f^{-1}(U), \mathcal{P}_{X/S}^n)$. We deduce (by varying U) a canonical homomorphism of $\mathcal{O}_X$ -algebras

$$(16.4.7.2) \quad f^*(\mathbf{S}_{\mathcal{O}_S}(\mathcal{E})) \longrightarrow \mathcal{P}_{X/S}^n$$

and in view of the above remark, if $\mathcal{K}$ is the ideal kernel of augmentation $\mathbf{S}_{\mathcal{O}_S}(\mathcal{E}) \rightarrow \mathcal{O}_S$, the image of $\mathcal{K}^{n+1}$ by (16.4.7.2) is zero, so that by factoring by $\mathcal{K}^{n+1}$, we finally have a canonical homomorphism

$$(16.4.7.3) \quad \delta_n : f^*(\mathbf{S}_{\mathcal{O}_S}(\mathcal{E})/\mathcal{K}^{n+1}) \longrightarrow \mathcal{P}_{X/S}^n.$$

Proposition (16.4.8). — Under the conditions of (16.4.7), the homomorphisms δ_n are bijective and form a projective system of isomorphisms; we deduce an isomorphism of graded $\mathcal{O}_S$ -algebras

$$(16.4.8.1) \quad f^*(\mathbf{S}_{\mathcal{O}_S}^\bullet(\mathcal{E})) \longrightarrow \mathcal{G}_\bullet(\mathcal{P}_{X/S}).$$

Proof. The fact that homomorphisms (16.4.7.3) form a projective system follows immediately from their definition. To prove they are isomorphisms, it suffices to prove that (16.4.8.1) is an isomorphism, since both filtrations involved in (16.4.7.3) are finite (Bourbaki, Alg. comm., chap. III, §2, no 8, cor. 3 of th. 1). To do this, consider the split exact sequence of $\mathcal{O}_S$ -modules

$$(16.4.8.2) \quad 0 \longrightarrow \mathcal{E} \xrightarrow{u} \mathcal{E} \oplus \mathcal{E} \xrightarrow{v} \mathcal{E} \longrightarrow 0$$

where, for every pair of sections s, t of $\mathcal{E}$ over an open U of S , we take $u(s) = (-s, s)$ and $v(s, t) = s + t$. We have

$$X \times_S X = \text{Spec}(\mathbf{S}_{\mathcal{O}_S}(\mathcal{E}) \otimes_{\mathcal{O}_S} \mathbf{S}_{\mathcal{O}_S}(\mathcal{E})) = \text{Spec}(\mathbf{S}_{\mathcal{O}_S}(\mathcal{E} \oplus \mathcal{E}))$$

((II, 1.4.6) and (II, 1.7.11)), and the diagonal morphism $X \rightarrow X \times_S X$ corresponds (II, 1.2.7) to the homomorphism of $\mathcal{O}_X$ -algebras $\mathbf{S}(v) : \mathbf{S}_{\mathcal{O}_S}(\mathcal{E} \oplus \mathcal{E}) \rightarrow \mathbf{S}_{\mathcal{O}_S}(\mathcal{E})$ (II, 1.7.4), such that if $\mathcal{I}$ is the kernel of this homomorphism, then we have

$$\mathcal{P}_{X/S}^n = f^*(\mathbf{S}_{\mathcal{O}_S}(\mathcal{E} \oplus \mathcal{E})/\mathcal{I}^{n+1}).$$

The proposition now will be a consequence of the following lemma: $\square$

Lemma (16.4.8.3). — Let Y be a ringed space, $0 \rightarrow \mathcal{F}' \xrightarrow{u} \mathcal{F} \xrightarrow{v} \mathcal{F}'' \rightarrow 0$ an exact sequence of $\mathcal{O}_Y$ modules such that each point $y \in Y$ has an open neighbourhood V such that the sequence $0 \rightarrow \mathcal{F}'|_V \rightarrow \mathcal{F}|_V \rightarrow \mathcal{F}''|_V \rightarrow 0$ is split. Let $\mathcal{I}$ be the kernel ideal of $\mathbf{S}(v)$:

$$\mathbf{S}_{\mathcal{O}_Y}(\mathcal{F}) \longrightarrow \mathbf{S}_{\mathcal{O}_Y}(\mathcal{F}''),$$

and let $\mathrm{gr}_{\mathcal{I}}^{\bullet}(\mathbf{S}_{\mathcal{O}_Y}(\mathcal{F}))$ be the graded $\mathcal{O}_Y$ -algebra associated to the $\mathcal{O}_Y$ -algebra $\mathbf{S}_{\mathcal{O}_Y}(\mathcal{F})$ endowed with the $\mathcal{I}$ -preadic filtration. Then the homomorphism of graded $\mathcal{O}_Y$ -algebras

$$(16.4.8.4) \quad \mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}') \otimes_{\mathcal{O}_Y} \mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}'') \longrightarrow \mathrm{gr}_{\mathcal{I}}^{\bullet}(\mathbf{S}_{\mathcal{O}_Y}(\mathcal{F}))$$

(where the first member is the graded tensor product of symmetric $\mathcal{O}_Y$ -algebras endowed with the canonical gradation (II, 1.7.4) and (II, 2.1.2)), induced by the canonical injection

$$\mathcal{F}' \longrightarrow \mathcal{I} = \mathrm{gr}_{\mathcal{I}}^1(\mathbf{S}_{\mathcal{O}_Y}(\mathcal{F})),$$

is bijective.

Proof. The injection $\mathcal{F}' \rightarrow \mathcal{I}$ indeed canonically gives a homomorphism of graded $\mathcal{O}_Y$ -algebras $\mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}') \rightarrow \mathrm{gr}_{\mathcal{I}}^{\bullet}(\mathbf{S}_{\mathcal{O}_Y}(\mathcal{F}))$, and since the second member is by definition a graded $\mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}'')$ -algebra, we induce the canonical homomorphism (16.4.8.4) by tensoring the above with $\mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}'')$. To prove the lemma we can, being a local problem, restrict to the case where $\mathcal{F} = \mathcal{F}' \oplus \mathcal{F}''$, u and v the canonical homomorphisms. Then the graded algebra $\mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F})$ is canonically identified with the graded tensor product $\mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}') \otimes_{\mathcal{O}_Y} \mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}'')$ (II, 1.7.4), and it is immediate that $\mathcal{I}$ is therefore the ideal $\mathcal{I}' \otimes_{\mathcal{O}_Y} \mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}'')$, where $\mathcal{I}'$ is the augmentation ideal of $\mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}')$, that is to say the (direct) sum of the $\mathbf{S}_{\mathcal{O}_Y}^m(\mathcal{F}')$ for $m \geq 1$. We conclude that $\mathcal{I}^n = \mathcal{I}'^n \otimes_{\mathcal{O}_Y} \mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}'')$, where this time $\mathcal{I}'^n$ is the direct sum of the $\mathbf{S}_{\mathcal{O}_Y}^m(\mathcal{F}')$ for $m \geq n$; we have therefore $\mathcal{I}^n / \mathcal{I}^{n+1} = \mathbf{S}_{\mathcal{O}_Y}^n(\mathcal{F}') \otimes_{\mathcal{O}_Y} \mathbf{S}_{\mathcal{O}_Y}^{\bullet}(\mathcal{F}'')$, which proves that (16.4.8.4) is bijective. $\square$

Having proved the lemma, it remains to see that the homomorphism (16.4.8.1) is the image by f^* of the homomorphism (16.4.8.4) corresponding to the exact sequence (16.4.8.2); we can easily see that it follows from the definition of u (16.4.8.2) and of δ (16.4.7.1), given the definition of the $\mathcal{O}_X$ -algebra structures of $\mathcal{P}_{X/S}^n$ and of the $d_{X/S}^n$ (16.3.5) and (16.4.3.6).

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In particular:

Corollary (16.4.9). — Under the conditions of (16.4.7), we have a canonical isomorphism

$$(16.4.9.1) \quad \mathrm{gr}_1(\delta) : f^*(\mathcal{E}) \simeq \Omega_{X/S}^1.$$

Corollary (16.4.10). — If $S = \mathrm{Spec}(A)$, $\mathcal{E} = \mathcal{O}_S^m$, so that

$$X = \mathrm{Spec}(A[T_1, \dots, T_m]),$$

then $\mathcal{P}_{X/S}^n$ is canonically identified with the $\mathcal{O}_X$ -algebra corresponding to the quotient $A[T_1, \dots, T_m]$ -algebra $A[T_1, \dots, T_m, U_1, \dots, U_m] / \mathfrak{K}^{n+1}$, where the U_i ($1 \leq i \leq m$) are m new indeterminates and $\mathfrak{K}$ is the ideal generated by $U_1, \dots, U_m$.

We thus recover in particular the structure of $\Omega_{X/S}^1$ in this case (0, 20.5.13).

In addition, note that the $d_{X/S}^n$ then corresponds to a polynomial $F(T_1, \dots, T_m)$, the class modulo $\mathfrak{K}^{n+1}$ of $F(T_1 + U_1, \dots, T_m + U_m)$, which follows from the definition (16.4.7.1).

Proposition (16.4.11). — Let $f : X \rightarrow S$ be a morphism, $g : S \rightarrow X$ a S -section of X , $S^{(n)}$ the n -th infinitesimal neighbourhood of S by the immersion g (16.1.2). Then there exists a unique isomorphism of $\mathcal{O}_S$ -algebras

$$(16.4.11.1) \quad \omega_n : g^*(\mathcal{P}_{X/S}^n) \longrightarrow \mathcal{O}_{S_g^{(n)}}$$

(via the $\mathcal{O}_S$ -algebra structure on $\mathcal{O}_{S_g^{(n)}}$ defined by f (16.1.7)), making the diagram

$$(16.4.11.2) \quad \begin{array}{ccc} \mathcal{O}_S = g^*(\mathcal{O}_X) & \xrightarrow{\lambda_n} & \mathcal{O}_{S_g^{(n)}} \\ & \searrow g^*(d_{X/S}^n) & \nearrow \omega_n \\ & g^*(\mathcal{P}_{X/S}^n) & \end{array}$$

commutative (where λ_n is the structure morphism).

Proof. In light of (I, 5.3.7), where we replace X, Y, S by S, X, S respectively and f by g , the diagrams

$$(16.4.11.3) \quad \begin{array}{ccc} S & \xrightarrow{g} & X \\ \downarrow g & & \downarrow \Delta_f \\ X & \xrightarrow{(g \circ f, 1_X)_S} & X \times_S X \end{array} \quad \begin{array}{ccc} S & \xrightarrow{g} & X \\ \downarrow g & & \downarrow \Delta_f \\ X & \xrightarrow{(1_X, g \circ f)_S} & X \times_S X \end{array}$$

identifies S with the product of the $(X \times_S X)$ -preschemes X and X by the morphisms Δ_f and $(g \circ f, 1_X)_S$ (resp. $(1_X, g \circ f)_S$). On the other hand, the diagrams

$$(16.4.11.4) \quad \begin{array}{ccc} X & \xrightarrow{(g \circ f, 1_X)_S} & X \\ \downarrow f & & \downarrow p_1 \\ S & \xrightarrow{g} & X \end{array} \quad \begin{array}{ccc} X & \xrightarrow{(1_X, g \circ f)_S} & X \\ \downarrow f & & \downarrow p_2 \\ S & \xrightarrow{g} & X \end{array}$$

identify X to the product of X -preschemes S and $X \times_S X$ via the morphisms g and p_1 (resp. p_2) (particular case of the associativity formula (I, 3.3.9.1)). We can say that Δ_f , considered as an X -section of $X \times_S X$ (relative to p_1 or p_2) plays the role of a *universal section* for the S -sections of X : each of these sections g in fact are deduced by *base change* $(g \circ f, 1_X)_S : X \rightarrow X \times_S X$. The definition of the homomorphism ω_n and the fact that it is bijective follows from the remarks of (16.2.3, (ii)) applied to the first diagram (16.4.11.4). The commutativity of the first diagram (16.4.11.4) follows also from (16.2.3, (ii)) this time applied to the second diagram (16.11.1.4). To explain ω_n , we can restrict ourselves to the case where g is a closed immersion: Indeed, for every $s \in S$, there is an open neighbourhood W of s in S such that $g(W)$ is closed in an open set U of X , and it is clear that $g|_W$ is a W -section of the morphism $U \cap f^{-1}(W)$. We can then suppose that S is a closed subscheme of X defined by a quasi-coherent ideal $\mathcal{K}$. Then the preceding definitions show that if W is an open of S , t is a section of $\mathcal{O}_X$ over $f^{-1}(W)$, $\omega_n(d^n t|_W)$ is equal to the canonical image of t in $\Gamma(W, (\mathcal{O}_X/\mathcal{K}^{n+1})|_W)$. The uniqueness of ω_n then follows since the image of $\mathcal{O}_X$ under $d_{X/S}^n$ generates the $\mathcal{O}_X$ -module $\mathcal{P}_{X/S}^n$ (16.3.8). $\square$

Corollary (16.4.12). — *Let k be a field, X a k -prescheme, x a point of X rational over k . Then $(\mathcal{P}_{X/S}^n)_x \otimes_{\mathcal{O}_x} k(x)$ is canonically isomorphic (as an augmented $k(x)$ -algebra) to $\mathcal{O}_x/\mathfrak{m}_x^{n+1}$.*

Proof. It suffices to consider the unique k -section g of X such that $g(\text{Spec}(k)) = \{x\}$. $\square$

Corollary (16.4.13). — *Let $f : X \rightarrow S$ be a morphism, s a point of S , $X_s = X \times_S \text{Spec}(k(s))$ the fibre of f in s . If $x \in X_s$ is rational over $k(s)$, $(\mathcal{P}_{X/S}^n)_x \otimes_{\mathcal{O}_s} k(s)$ is canonically isomorphic to $\mathcal{O}_{X_s, x}/\mathfrak{m}_x'^{n+1}$, is the maximal ideal of $\mathcal{O}_{X_s, x}$; more precisely, this isomorphism sends $(d^n t)_x \otimes 1$ (where t is a section of $\mathcal{O}_X$ over an open neighbourhood of x in X) to the class of $t_x \otimes 1$ modulo $\mathfrak{m}_x'^{n+1}$.*

Proof. This follows from (16.4.5) and (16.4.12). $\square$

The preceding corollaries justify the terminology “sheaf of principal parts of order n ”.

Proposition (16.4.14). — *Let $\rho : A \rightarrow B$ be a morphism of rings, S a multiplicative subset of B . Then the canonical homomorphisms*

$$(16.4.14.1) \quad S^{-1}P_{B/A}^n \longrightarrow P_{S^{-1}B/A}^n$$

deduced from the canonical homomorphisms $P_{B/A}^n \rightarrow P_{S^{-1}B/A}^n$ (16.4.4), form a projective system and are bijective. IV-4 | 23

Proof. It suffices to remark that $S^{-1}((B \otimes_A B)/\mathfrak{I}^{n+1}) = S^{-1}(B \otimes_A B)/(S^{-1}\mathfrak{I})^{n+1}$ by flatness, and that $S^{-1}(B \otimes_A B) = (S^{-1}B) \otimes_A (S^{-1}B)$ (I, 1.3.4). $\square$

Corollary (16.4.15). — *The notation being that of (16.4.14), let R be a multiplicative subset of A such that $\rho(R) \subset S$. Then we have canonical isomorphisms*

$$(16.4.15.1) \quad S^{-1}P_{B/A}^n \simeq P_{S^{-1}B/R^{-1}A}^n$$

forming a projective system.

Proof. It evidently suffices to define canonical isomorphisms

$$(16.4.15.2) \quad P_{S^{-1}B/A}^n \simeq P_{S^{-1}B/R^{-1}A}^n$$

that is to say that we reduce to the case there $\rho(R)$ consists of invertible elements of B . But then the isomorphism (16.4.15.2) is simply induced by the canonical isomorphism $B \otimes_A B \rightarrow B \otimes_{R^{-1}A} B$ by passing to quotients (I, 1.5.3). $\square$

Corollary (16.4.16). — *Let $f : X \rightarrow S$ be a morphism of preschemes, x a point of X , $s = f(x)$. Then we have canonical isomorphisms*

$$(16.4.16.1) \quad (\mathcal{P}_{X/S}^n)_x \simeq P_{\mathcal{O}_x/\mathcal{O}_s}^n$$

forming a projective system.

We deduce from these isomorphisms of the associated graded rings, and in particular a canonical isomorphism

$$(16.4.16.2) \quad (\Omega_{X/S}^1)_x \simeq \Omega_{\mathcal{O}_x/\mathcal{O}_s}^1.$$

Corollary (16.4.17). — *Let k be a field, K the field of rational functions $k(T_1, \dots, T_r)$. Then, for every integer n , the homomorphism of $K[U_1, \dots, U_r]$ (U_i indeterminates) into $P_{K/k}^n$ which sends U_i to $d^n T_i - T_i \cdot 1$ is surjective and defines an isomorphism from the quotient $K[U_1, \dots, U_n]/\mathfrak{m}^{n+1}$ (where $\mathfrak{m}$ is the ideal generated by the U_i) to $P_{K/k}^n$.*

Proof. This follows from (16.4.8), (16.4.10) and (16.4.14), where we take $A = k$, $B = k[T_1, \dots, T_r]$ and $S = B - \{0\}$. $\square$

We thus recover the fact that the dT_i form a basis of the K -vector space $\Omega_{K/k}^1$ (0, 20.5.10).

(16.4.18). Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two morphisms of preschemes, and consider the canonical homomorphism of augmented $\mathcal{O}_X$ -algebras (16.4.3.3)

$$(16.4.18.1) \quad g_{X/Y/Z} : \mathcal{P}_{X/Z}^n \longrightarrow \mathcal{P}_{X/Y}^n$$

$$(16.4.18.2) \quad f_{X/Y/Z} : f^*(\mathcal{P}_{Y/Z}^n) \longrightarrow \mathcal{P}_{X/Z}^n.$$

Then $g_{X/Y/Z}$ is surjective, and its kernel is the sheaf of ideals generated by the image under $f_{X/Y/Z}$ of the augmentation ideal of $f^*(\mathcal{P}_{Y/Z}^n)$.

Proof. First note that $g_{X/Y/Z}$ corresponds to the case in (16.4.3.3) where $X' = X$, $S' = Y$ and $S = Z$, IV-4 | 24
 $u = 1_X$, and $f_{X/Y/Z}$ to the case where we replace X', X, S, S' by X, Y, Z, Z respectively and u, f by f, g respectively.

We have a commutative diagram (I, 3.5)

$$(16.4.18.3) \quad \begin{array}{ccccc} X & \xrightarrow{\Delta_f} & X \times_Y X & \xrightarrow{j} & X \times_Z X \\ & \searrow f & \downarrow p & & \downarrow f \times_Z f \\ & & Y & \xrightarrow{\Delta_g} & Y \times_Z Y \end{array}$$

where $j = (1_X, 1_X)_Z$ is an immersion, $j \circ \Delta_f = \Delta_{g \circ f}$, and p is the structure morphism. Since we can restrict ourselves to the case where X, Y and Z are affine, we can suppose that the immersions Δ_f, Δ_g and j are closed, so that $\mathcal{O}_X$ and $\mathcal{O}_{X \times_Y X}$ are identified respectively with $\mathcal{O}_{X \times_Z X} / \mathcal{I}$ and $\mathcal{O}_{X \times_Z X} / \mathcal{L}$, where $\mathcal{L} \supset \mathcal{I}$ are two quasi-coherent ideals corresponding respectively to the immersions $\Delta_{g \circ f}$ and j . The $\mathcal{O}_X$ -algebra $\mathcal{P}_{X/Z}^n$ is identified with $\mathcal{O}_{X \times_Z X} / \mathcal{I}^{n+1}$, and $\mathcal{P}_{X/Y}^n$ is identified with $\mathcal{O}_{X \times_Y X} / (\mathcal{I} / \mathcal{L})^{n+1}$, which is to say with $\mathcal{O}_{X \times_Z X} / (\mathcal{I}^{n+1} + \mathcal{L})$, and therefore with the quotient of $\mathcal{P}_{X/Z}^n$ by $(\mathcal{I}^{n+1} + \mathcal{L}) / \mathcal{I}^{n+1}$. But we know (*loc. cit.*) that if p and j make $X \times_Y X$ the product of the $(Y \times_Z Y)$ -preschemes Y and $X \times_Z X$, so if $\mathcal{O}_Y$ is identified to $\mathcal{O}_{Y \times_Z Y} / \mathcal{K}$, where $\mathcal{K}$ is the ideal corresponding to Δ_g , $\mathcal{L}$ is equal to $(f \times_Z f)^*(\mathcal{K}) \cdot \mathcal{O}_{X \times_Z X}$ (I, 4.4.5). Since $(\mathcal{I}^{n+1} + \mathcal{L}) / \mathcal{I}^{n+1}$ is the ideal of $\mathcal{P}_{X/Z}^n$ generated by the image of $\mathcal{L}$, we deduce the proposition. $\square$

Corollary (16.4.19). — *With the notation of (16.4.18), we have an exact sequence of quasi-coherent $\mathcal{O}_X$ -modules*

$$(16.4.19.1) \quad f^*(\Omega_{Y/Z}^1) \xrightarrow{f_{X/Y/Z}} \Omega_{X/Z}^1 \xrightarrow{g_{X/Y/Z}} \Omega_{X/Y}^1 \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (0, 5.7.1).

Proposition (16.4.20). — *Let $f : Y \rightarrow Z$ be a morphism, $j : X \rightarrow Y$ a closed immersion, $\mathcal{K}$ the quasi-coherent sheaf of ideals of $\mathcal{O}_Y$ corresponding to j . It follows that $\mathcal{P}_{X/Y}^n = \mathcal{O}_X = \mathcal{O}_Y / \mathcal{K}$, the canonical homomorphism $j_{X/Y/Z} : j^*(\mathcal{P}_{Y/Z}^n) \rightarrow \mathcal{P}_{X/Z}^n$ is surjective, and its kernel is the ideal of $j^*(\mathcal{P}_{Y/Z}^n)$ generated by $j^*(\mathcal{O}_Y \cdot d_{Y/Z}^n(\mathcal{K}))$ (it should be noted that $d_{Y/Z}^n(\mathcal{K})$ is a subsheaf of abelian groups of $\mathcal{P}_{X/Z}^n$, but not an $\mathcal{O}_Y$ -module in general).*

Proof. We know (I, 5.3.8) that the diagonal $\Delta_j : X \rightarrow X \times_Y X$ is an isomorphism, from which the first assertion follows. If ω_1 and ω_2 are the two canonical homomorphisms of algebras $\mathcal{O}_Y \rightarrow \mathcal{P}_{Y/Z}^n$ corresponding respectively to the two canonical projections p_1, p_2 of $Y \times_Z Y \rightarrow Y$, recall that by definition ((16.3.5) and (16.3.6)) ω_1 is the structure homomorphism of the $\mathcal{O}_Y$ -algebra $\mathcal{P}_{Y/Z}^n$ and $\omega_2 = d_{Y/Z}^n$. The $\mathcal{O}_X$ -algebra $j^*(\mathcal{P}_{Y/Z}^n)$ is therefore identified with $\mathcal{P}_{Y/Z}^n / \omega_1(\mathcal{K}) \mathcal{P}_{Y/Z}^n$ and its quotient by the ideal generated by $j^*(d_{Y/Z}^n(\mathcal{K}))$ to $\mathcal{P}_{Y/Z}^n / (\omega_1(\mathcal{K}) + \omega_2(\mathcal{K})) \mathcal{P}_{Y/Z}^n$. Now note that we have a commutative diagram

$$\begin{array}{ccc} Y & \xleftarrow{j} & X \\ \Delta_f \downarrow & & \downarrow \Delta_{f \circ j} \\ Y \times_Z Y & \xleftarrow{j \times_Z j} & X \times_Z X \end{array}$$

identifying X with the product of the $(Y \times_Z Y)$ -preschemes Y and $X \times_Z X$ (I, 5.3.7). Since $j \times_Z j$ is an immersion, we therefore deduce from this remark and from (16.2.2) that if $\Delta_{Y/Z}^n$ and $\Delta_{X/Z}^n$ denote the infinitesimal neighbourhoods of order n of Y and X by the canonical immersions Δ_f and $\Delta_{f \circ j}$ respectively, then we have a diagram

$$\begin{array}{ccc} \Delta_{Y/Z}^n & \xleftarrow{\quad} & \Delta_{X/Z}^n \\ \downarrow & & \downarrow \\ Y \times_Z Y & \xleftarrow{j \times_Z j} & X \times_Z X \end{array}$$

making $\Delta_{X/Z}^n$ the product of the $(Y \times_Z Y)$ -preschemes $\Delta_{Y/Z}^n$ and $X \times_Z X$. We can also say that $\mathcal{P}_{X/Z}^n$ is identified with the sheaf of rings $\mathcal{P}_{Y/Z}^n \otimes_{\mathcal{O}_{Y \times_Z Y}} \mathcal{O}_{X \times_Z X}$. But we see immediately that

(for example, by restricting to the affine case) that $\mathcal{O}_{X \times_Z X} = \mathcal{O}_{Y \times_Z Y} / (p_1^*(\mathcal{K}) + p_2^*(\mathcal{K})) \mathcal{O}_{Y \times_Z Y}$. Therefore $\mathcal{P}_{X/Z}^n$ is identified with the quotient of $\mathcal{P}_{Y/Z}^n$ by the ideal generated by the image in $\mathcal{P}_{Y/Z}^n$ of $p_1^*(\mathcal{K}) + p_2^*(\mathcal{K})$. But by definition this ideal is generated by $\omega_1(\mathcal{K}) + \omega_2(\mathcal{K})$. $\square$

Corollary (16.4.21). — *Let $f : Y \rightarrow Z$ be a morphism, $j : X \rightarrow Y$ an immersion. We have an exact sequence of quasi-coherent $\mathcal{O}_X$ -modules*

$$(16.4.21.1) \quad \mathcal{N}_{X/Y} \longrightarrow j^*(\Omega_{Y/Z}^1) \longrightarrow \Omega_{X/Z}^1 \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (0, 20.5.12.1).

Corollary (16.4.22). — *If $f : X \rightarrow S$ is a morphism locally of finite presentation, $\mathcal{P}_{X/S}^n$ and $\Omega_{X/S}^1$ are quasi-coherent $\mathcal{O}_X$ -modules of finite presentation.*

Proof. We immediately reduce to the case where $S = \text{Spec}(A)$ is affine, $X = \text{Spec}(B)$, where $B = A[T_1, \dots, T_r]/\mathfrak{K}$, $\mathfrak{K}$ being an ideal of finite type of $C = A[T_1, \dots, T_r]$. Applying (16.4.20) where $Z = S$, $Y = \text{Spec}(C)$ and $\mathcal{K} = \mathfrak{K}$. Then $j^*(\mathcal{P}_{Y/Z}^n)$ is a free $\mathcal{O}_X$ -module of finite rank (16.4.10) and the hypothesis on $\mathfrak{K}$ implies that $j^*(\mathcal{O}_{Y \times_Z Y} \cdot d_{Y/Z}^n(\mathcal{K}))$ generates a quasi-coherent $\mathcal{O}_X$ -module of finite type; hence the conclusion. $\square$

Proposition (16.4.23). — *Let X, Y be two S -preschemes, $Z = X \times_S Y$ their product, $p : X \times_S Y \rightarrow X$ and $q : X \times_S Y \rightarrow Y$ the canonical projections. Then the canonical homomorphism*

$$(16.4.23.1) \quad p_{Z/X/S} \oplus q_{Z/Y/S} : p^*(\Omega_{X/S}^1) \oplus q^*(\Omega_{Y/S}^1) \longrightarrow \Omega_{(X \times_S Y)/S}^1$$

is bijective.

Proof. The commutative diagram

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$$\begin{array}{ccccc} Y & \xleftarrow{q} & X \times_S Y & \xleftarrow{\text{id}} & X \times_S Y \\ g \downarrow & & h \downarrow & & \downarrow p \\ S & \xleftarrow{\text{id}} & S & \xleftarrow{f} & X \end{array}$$

gives us a factorization of the canonical isomorphism $P^n(p)$ (16.4.5)

$$p^*(\mathcal{P}_{X/S}^n) \longrightarrow \mathcal{P}_{Z/S}^n \longrightarrow \mathcal{P}_{Z/Y}^n$$

and similarly, switching X with Y , we have a factorization of the isomorphism $P^n(q)$

$$q^*(\mathcal{P}_{Y/S}^n) \longrightarrow \mathcal{P}_{Z/S}^n \longrightarrow \mathcal{P}_{Z/X}^n.$$

This proves that the canonical homomorphism (16.4.18.1)

$$p_{Z/X/S} : p^*(\mathcal{P}_{X/S}^n) \longrightarrow \mathcal{P}_{Z/S}^n \quad (\text{resp. } q_{Z/Y/S} : q^*(\mathcal{P}_{Y/S}^n) \longrightarrow \mathcal{P}_{Z/S}^n)$$

is injective, and that the kernel of the canonical surjective homomorphism (16.4.18.2)

$$\mathcal{P}_{Z/S}^n \longrightarrow \mathcal{P}_{Z/Y}^n \quad (\text{resp. } \mathcal{P}_{Z/S}^n \longrightarrow \mathcal{P}_{Z/X}^n)$$

is direct summand of the image $p_{Z/X/S}$ (resp. $q_{Z/Y/S}$). On the other hand, this kernel is, by virtue of (16.4.18), generated by the image by $q_{Z/Y/S}$ (resp. $p_{Z/X/S}$) of the augmentation ideal of $q^*(\mathcal{P}_{Y/S}^n)$ (resp. $p^*(\mathcal{P}_{X/S}^n)$). We conclude the proposition by considering the case $n = 1$. $\square$

We immediately generalize (16.4.23) to the case of a product of any finite number of S -preschemes.

Remarks (16.4.24). —

- (i) We will see (17.2.3) that when the morphism $f : X \rightarrow Y$ in (16.4.18) is smooth, the homomorphism $f_{X/Y/Z}$ in (16.4.19.1) is locally left invertible and in particular injective. Similarly, when the morphism $f \circ j : X \rightarrow Z$ of (16.4.20) is smooth, the homomorphism on the left in (16.4.21.1) is locally left invertible and a fortiori injective (17.2.5). In Chapter V, we will also give a variant, in the case of modules over a prescheme, of the “imperfection modules” studied in (0, 20.6), and the exact sequences where they occur.

- (ii) Let X be a topological space, $\mathcal{A}$ a sheaf of rings over X and $\mathcal{B}$ a $\mathcal{A}$ -algebra over X . Then it is clear that

$$U \longmapsto P_{\Gamma(U, \mathcal{B})/\Gamma(U, \mathcal{A})}^n \quad (U \text{ open in } X)$$

is a presheaf of augmented $\Gamma(U, \mathcal{B})$ -algebras, and therefore the associated sheaf $\mathcal{P}_{\mathcal{B}/\mathcal{A}}^n$ is an augmented $\mathcal{B}$ -algebra. In the particular case where X is a prescheme, $f = (\psi, \theta) : X \rightarrow S$ a morphism of preschemes, it follows easily from (16.4.16) and from the exactness of the functor $\varinjlim$ that $\mathcal{P}_{X/S}^n$ is canonically isomorphic to $\mathcal{P}_{\mathcal{O}_X/\psi^*(\mathcal{O}_S)}^n$. It follows that the formalism developed in the present paragraph could be considered as a particular case of a differential formalism for ringed spaces endowed with a sheaf of algebras over the structure sheaf. However, we did not start with this point of view, which is less intuitive and less convenient for applications. It also seems that, for various kinds of “varieties”, the “global” constructions of the $\mathcal{P}^n$ analogous to those we have used here are also better suited for applications.

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16.5. Relative tangent sheaves and bundles; derivations.

(16.5.1). Let $f = (\psi, \theta) : X \rightarrow S$ be a morphism of ringed spaces. For every $\mathcal{O}_X$ -module $\mathcal{F}$, we say S -derivation (or (X/S) -derivation, or f -derivation) of $\mathcal{O}_X$ to $\mathcal{F}$ for every homomorphism of sheaves of additive groups $D : \mathcal{O}_X \rightarrow \mathcal{F}$ satisfying the following conditions:

- (a) for every open V of X , and all pair of sections (t_1, t_2) of $\mathcal{O}_X$ over V , we have

$$(16.5.1.1) \quad D(t_1 t_2) = t_1 D(t_2) + D(t_1) t_2;$$

- (b) for every open V of X , every section t of $\mathcal{O}_X$ over V , and every section s of $\mathcal{O}_S$ over an open U of S such that $V \subset f^{-1}(U)$, we have

$$(16.5.1.2) \quad D((s|V)t) = (s|V)D(t).$$

It is clear that this amounts to saying that, for all $x \in X$, the homomorphism of additive groups $D_x : \mathcal{O}_x \rightarrow \mathcal{F}_x$ is an $\mathcal{O}_{f(x)}$ -derivation.

Another interpretation consists of considering the $\mathcal{O}_X$ -algebra $\mathcal{D}_{\mathcal{O}_X}(\mathcal{F})$ as equal to $\mathcal{O}_X \oplus \mathcal{F}$, the algebra structure being defined by the condition that for every open V of X , the product of two sections of $\mathcal{O}_X$ (resp. of a section of $\mathcal{O}_X$ and a section of $\mathcal{F}$) over V is defined by the ring structure of $\Gamma(V, \mathcal{O}_X)$ (resp. the $\Gamma(V, \mathcal{O}_X)$ -module structure on $\Gamma(V, \mathcal{F})$), and the product of two sections of $\mathcal{F}$ over V is chosen to be zero; then $\mathcal{F}$ is an ideal of $\mathcal{D}_{\mathcal{O}_X}(\mathcal{F})$, the kernel of the canonical augmentation $\mathcal{D}_{\mathcal{O}_X}(\mathcal{F}) \rightarrow \mathcal{O}_X$, and to say that D is an S -derivation of $\mathcal{O}_X$ to $\mathcal{F}$ means that $1_{\mathcal{O}_X} + D$ is an $\mathcal{O}_S$ -homomorphism of algebras from $\mathcal{O}_X$ to $\mathcal{D}_{\mathcal{O}_X}(\mathcal{F})$, which, composed with the augmentation, gives $1_{\mathcal{O}_X}$.

The S -derivations of $\mathcal{O}_X$ to $\mathcal{F}$ clearly form a $\Gamma(X, \mathcal{O}_X)$ -module $\text{Der}_{\mathcal{O}_S}(\mathcal{O}_X, \mathcal{F})$.

When $\mathcal{F} = \mathcal{O}_X$, an S -derivation of $\mathcal{O}_X$ to itself is simply called an S -derivation of $\mathcal{O}_X$.

Proposition (16.5.2). — Let A be a ring, B an A -algebra, L a B -module; let $S = \text{Spec}(A)$, $X = \text{Spec}(B)$, $\mathcal{F} = \tilde{L}$. Then the map $D \mapsto \Gamma(D)$ which sends every S -derivation D of $\mathcal{O}_X$ to $\mathcal{F}$ to the map $\Gamma(D) : t \mapsto D(t)$ of B to L , is an isomorphism of B -modules from $\text{Der}_S(\mathcal{O}_X, \mathcal{F})$ to $\text{Der}_A(B, L)$ (cf. (0, 20.1.2)).

Proof. This follows immediately from the given interpretation of S -derivations in terms of homomorphisms of algebras, analogous to the interpretation given in (0, 20.1.6), and from the canonical correspondence between homomorphisms of $\mathcal{O}_X$ -algebras and homomorphisms of B -algebras ((I, 1.3.13) and (I, 1.3.8)). $\square$

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Proposition (16.5.3). — Let $f = (\psi, \theta) : X \rightarrow S$ be a morphism of preschemes.

- (i) The differential $d_{X/S} : \mathcal{O}_X \rightarrow \Omega_{X/S}^1$ (16.3.6) is an S -derivation.
 (ii) For every $\mathcal{O}_X$ -module $\mathcal{F}$, the map $u \mapsto u \circ d_{X/S}$ is an isomorphism of $\Gamma(X, \mathcal{O}_X)$ -modules

$$(16.5.3.1) \quad \text{Hom}_{\mathcal{O}_X}(\Omega_{X/S}^1, \mathcal{F}) \simeq \text{Der}_S(\mathcal{O}_X, \mathcal{F}).$$

Proof. The assertion (i) has already been written (16.3.6). On the other hand, it is immediate (in light of (0, 20.4.8)) that $u \mapsto u \circ d_{X/S}$ is injective, considering the restrictions to a fibre $\mathcal{O}_x$ of the two sides and using (16.4.16.2). To see that the homomorphism (16.5.3.1) is surjective, consider an S -derivation $D : \mathcal{O}_X \rightarrow \mathcal{F}$; for every affine open $V = \text{Spec}(B)$ of X , such that $f(V)$ is contained in an affine open $U = \text{Spec}(A)$ of S , $D_V : B \rightarrow \Gamma(V, \mathcal{F})$ is an A -derivation, and therefore there exists a

unique B -homomorphism $u_V : \Omega_{B/A}^1 \rightarrow \Gamma(V, \mathcal{F})$ such that $D_V = u_V \circ d_{B/A}$ (0, 20.4.8); in addition, the uniqueness of u_V shows immediately that for an affine open $W \subset V$ we have $u_W = u_V|_W$, and therefore the u_V define a homomorphism of $\mathcal{O}_X$ -modules $u : \mathcal{O}_X \rightarrow \mathcal{F}$ answering the question. $\square$

(16.5.4). With the notation of (16.5.1), for every open U of X , $\text{Der}_S(\mathcal{O}_U, \mathcal{F}|_U)$ is a $\Gamma(U, \mathcal{O}_X)$ -module and it is clear that the map $U \mapsto \text{Der}_S(\mathcal{O}_U, \mathcal{F}|_U)$ is a presheaf; in fact, it is even a *sheaf* (and therefore an $\mathcal{O}_X$ -module), in light of the pointwise characterization of S -derivations, seen in (16.5.1). This $\mathcal{O}_X$ -module is denoted by $\mathcal{D}er_S(\mathcal{O}_X, \mathcal{F})$ and is called the *sheaf of S -derivations of $\mathcal{O}_X$ in $\mathcal{F}$* , and what we have seen is further expressed in the following corollary:

Corollary (16.5.5). — For every $\mathcal{O}_X$ -module $\mathcal{F}$, the homomorphism of $\mathcal{O}_X$ -modules induced by $u \mapsto u \circ d_{X/S}$

$$(16.5.5.1) \quad \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/S}^1, \mathcal{F}) \longrightarrow \mathcal{D}er_S(\mathcal{O}_X, \mathcal{F})$$

is bijective.

Corollary (16.5.6). — (i) If the morphism $f : X \rightarrow S$ is locally of finite presentation and if $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -module, then $\mathcal{D}er_S(\mathcal{O}_X, \mathcal{F})$ is a quasi-coherent $\mathcal{O}_X$ -module.
 (ii) If in addition S is locally Noetherian and if $\mathcal{F}$ is coherent, then $\mathcal{D}er_S(\mathcal{O}_X, \mathcal{F})$ is a coherent $\mathcal{O}_X$ -module.

Proof. The assertion (i) follows from the isomorphism (16.5.5.1), from (16.4.22), and (I, 1.3.12); the assertion (ii) follows from (0, 5.3.5). $\square$

(16.5.7). We set

$$(16.5.7.1) \quad \mathfrak{G}_{X/S} = \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/S}^1, \mathcal{O}_X) = \mathcal{D}er_S(\mathcal{O}_X, \mathcal{O}_X),$$

and say that it is the *sheaf of S -derivations of $\mathcal{O}_X$* , or even the *tangent sheaf of X relative to S* : it is therefore the dual of the $\mathcal{O}_X$ -module $\Omega_{X/S}^1$. If f is locally of finite presentation, $\mathfrak{G}_{X/S}$ is a quasi-coherent $\mathcal{O}_X$ -module; if in addition S is locally Noetherian, then $\mathfrak{G}_{X/S}$ is coherent (16.5.6). IV-4 | 29

(16.5.8). Suppose in particular that $\Omega_{X/S}^1$ is a *locally free* $\mathcal{O}_X$ -module (of finite rank) (which will be the case then f is *smooth* (17.2.3)); then $\mathfrak{G}_{X/S}$ is locally free $\mathcal{O}_X$ -module of the same rank as $\Omega_{X/S}^1$ at each point. More specifically, suppose that $\Omega_{X/S}^1$ is of rank n at a point x ; then there are n sections s_i ($1 \leq i \leq n$) of $\mathcal{O}_X$ over an affine neighbourhood U of x such that the canonical images of the ds_i in $\Omega_{X/S}^1 \otimes_{\mathcal{O}_X} k(x)$ form a basis of this $k(x)$ -vector space; by virtue of Nakayama's lemma, the germs $(ds_i)_x = d(s_i)_x$ of the ds_i at the point x form a basis of the $\mathcal{O}_x$ -module $(\Omega_{X/S}^1)_x$, and therefore, by restricting U , we can suppose that the ds_i form a *basis* of the $\Gamma(U, \mathcal{O}_X)$ -module $\Gamma(U, \Omega_{X/S}^1)$. So the $\Gamma(U, \mathcal{O}_X)$ -module $\Gamma(U, \mathfrak{G}_{X/S})$ is dual to the above; we denote by $(D_i)_{1 \leq i \leq n}$ or $\left(\frac{\partial}{\partial s_i}\right)_{1 \leq i \leq n}$ the dual basis of $(ds_i)_{1 \leq i \leq n}$, so that, by (16.5.3), we have

$$(16.5.8.1) \quad D_i s_j = \langle D_i, ds_j \rangle = \left\langle \frac{\partial}{\partial s_i}, ds_j \right\rangle = \delta_{ij} \quad (\text{Kronecker's symbol}).$$

Every $\Gamma(S, \mathcal{O}_S)$ -derivation of the $\Gamma(S, \mathcal{O}_S)$ -algebra $\Gamma(U, \mathcal{O}_X)$ is therefore written in an unique way as

$$D = \sum_{i=1}^n a_i D_i = \sum_{i=1}^n a_i \frac{\partial}{\partial s_i},$$

where the a_i ($1 \leq i \leq n$) are sections of $\mathcal{O}_X$ over U . For every section $g \in \Gamma(U, \mathcal{O}_X)$, if we put $dg = \sum_{i=1}^n c_i ds_i$, then we have $c_i = \langle D_i, dg \rangle = D_i g$ by virtue of (16.5.8.1), in other words,

$$(16.5.8.2) \quad dg = \sum_{i=1}^n (D_i g) ds_i = \sum_{i=1}^n \frac{\partial g}{\partial s_i} ds_i.$$

(16.5.9). Let D_1, D_2 be two S -derivations of $\mathcal{O}_X$. For every open U of X , if D_1^U, D_2^U are the corresponding derivations of the ring $\Gamma(U, \mathcal{O}_X)$, the *bracket*

$$[D_1^U, D_2^U] = D_1^U \circ D_2^U - D_2^U \circ D_1^U$$

is also a derivation in this ring, and therefore the $\psi^*(\mathcal{O}_S)$ -endomorphism of $\mathcal{O}_X$

$$(16.5.9.1) \quad [D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1$$

is also an S -derivation; as we immediately check that this bracket satisfies the Jacobi identity, we have thus defined on $\text{Der}_S(\mathcal{O}_X, \mathcal{O}_X)$ a $\Gamma(S, \mathcal{O}_S)$ -Lie algebra structure. Since the definition of this structure commutes with the restriction to an open of X , we thus see that $\mathfrak{G}_{X/S}$ is canonically equipped with a $\psi^*(\mathcal{O}_S)$ -Lie algebra structure. Note that the mapping $(D_1, D_2) \mapsto [D_1, D_2]$ is not $\Gamma(X, \mathcal{O}_X)$ -bilinear.

(16.5.10). For every base change $g : S' \rightarrow S$, if we set $X' = X \times_S S'$, then we see (16.4.5) that we have a canonical isomorphism

$$(16.5.10.1) \quad \Omega_{X/S}^1 \otimes_S S' \simeq \Omega_{X'/S'}^1,$$

from which we deduce, by (16.5.10.1), a canonical homomorphism (Bourbaki, *Alg.*, chap. II, 3rd ed., IV-4 | 30 §5, n°3)

$$(16.5.10.2) \quad \mathfrak{G}_{X/S} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} \longrightarrow \mathfrak{G}_{X'/S'},$$

which is neither injective nor surjective in general. However:

Proposition (16.5.11). — (i) If $g : S' \rightarrow S$ is a flat morphism and if f is locally of finite type (resp. locally of finite presentation), then the homomorphism (16.5.10.2) is injective (resp. bijective).
 (ii) If $\Omega_{X/S}^1$ is a locally free $\mathcal{O}_X$ -module of finite type, then the homomorphism (16.5.10.2) is bijective.

Proof. The assertion (ii) follows from Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n°3, prop. 7. The assertion (i) follows similarly from Bourbaki, *Alg. Comm.*, chap. I, §2, n°10, prop. 11 and from the fact that if f is locally of finite type (resp. locally of finite presentation), then $\Omega_{X/S}^1$ is an $\mathcal{O}_X$ -module of finite type (resp. of finite presentation) ((16.3.9) (16.4.22)). $\square$

(16.5.12). Since $\Omega_{X/S}^1$ is a quasi-coherent $\mathcal{O}_X$ -module, we can consider the vector bundle over X defined by $\Omega_{X/S}^1$ (II, 1.7.8)

$$(16.5.12.1) \quad T_{X/S} = \mathbf{V}(\Omega_{X/S}^1)$$

which is called the *tangent bundle of X relative to S* . We have therefore a canonical bijection (II, 1.7.9)

$$\Gamma(T_{X/S}/S) \simeq \text{Hom}_{\mathcal{O}_X}(\Omega_{X/S}^1, \mathcal{O}_X) = \Gamma(X, \mathfrak{G}_{X/S})$$

by definition of $\mathfrak{G}_{X/S}$, and we can replace X by an open set U of X in this isomorphism; so we can say that the *tangent sheaf* of X relative to S is isomorphic to the *sheaf of germs of S -sections* of the tangent bundle of X relative to S . If $f : X \rightarrow Y$ is an S -morphism, we saw (16.4.19) that we have a canonical homomorphism $f_{X/Y/S} : f^*(\Omega_{Y/S}^1) \rightarrow \Omega_{X/S}^1$, which, having in mind that

$$\mathbf{V}(f^*(\Omega_{Y/S}^1)) = \mathbf{V}(\Omega_{Y/S}^1) \times_Y X \quad (\text{II, 1.7.11}),$$

gives us an X -morphism $T_{X/S}(f) : T_{X/S} \rightarrow T_{Y/S} \times_Y X$. If $g : Y \rightarrow Z$ is a second S -morphism, we have $T_{X/S}(g \circ f) = (T_{Y/S}(g) \times 1_X) \circ T_{X/S}(f)$ (0, 20.5.4.1).

It follows from (16.5.10.1) and from (II, 1.7.11) that for every base change $g : S' \rightarrow S$ we have a canonical isomorphism

$$(16.5.12.2) \quad T_{X'/S'} \simeq T_{X/S} \times_S S' = T_{X/S} \times_X X'.$$

(16.5.13). For every point $x \in X$, we define the *tangent space of X at the point x* (relative to S) to be the set of points in the fibre $T_{X/S} \times_X \text{Spec}(k(x))$ that are *rational over $k(x)$* , that is, the set

$$(16.5.13.1) \quad T_{X/S}(x) = \text{Hom}_{k(x)}(\Omega_{X/S}^1 \otimes_{\mathcal{O}_x} k(x), k(x)),$$

which is the *dual* of the $k(x)$ -vector space $\Omega_{\mathcal{O}_x/\mathcal{O}_S}^1 / \mathfrak{m}_x \cdot \Omega_{\mathcal{O}_x/\mathcal{O}_S}^1$. When $\Omega_{X/S}^1$ is an $\mathcal{O}_X$ -module of finite type, then $T_{X/S}(x)$ is a vector space of finite rank over $k(x)$, and for every base change $g : S \rightarrow S'$, IV-4 | 31 and every point $x' \in X' = X \times_S S'$ over x , we have a canonical isomorphism

$$(16.5.13.2) \quad T_{X'/S'}(x') \simeq T_{X/S}(x) \otimes_{k(x)} k(x').$$

If x is *rational over* $k(s)$, where $s = f(x)$ (so that $k(s) \rightarrow k(x)$ is an isomorphism), it follows from (16.4.13) that we have a canonical isomorphism

$$(16.5.13.3) \quad T_{X/S}(x) = T_{X_s/k(s)}(x) = \text{Hom}_{k(s)}(\mathfrak{m}'_x/\mathfrak{m}_x'^2, k(x)),$$

where $\mathfrak{m}'_x$ is the maximal ideal of $\mathcal{O}_{X_s, x} = \mathcal{O}_{X, x}/\mathfrak{m}_s \mathcal{O}_{X, x}$. In the case where S is the spectrum of a field k , we recover the definition of the Zariski tangent space of a point $x \in X$ *rational over* k , as the dual of $\mathfrak{m}_x/\mathfrak{m}_x^2$.

Let Y be a second S -prescheme and let $g : Y \rightarrow X$ be an S -morphism; then we have a canonical homomorphism of $\mathcal{O}_Y$ -modules (16.4.19)

$$(16.5.13.4) \quad g^*(\Omega_{X/S}^1) \rightarrow \Omega_{Y/S}^1.$$

Now note that if $y \in Y$ and $x = g(y)$, then we have

$$g^*(\Omega_{X/S}^1) \otimes_{\mathcal{O}_Y} k(y) = (\Omega_{X/S}^1 \otimes_{\mathcal{O}_X} k(x)) \otimes_{k(x)} k(y)$$

and consequently, if $\Omega_{X/S}^1$ is an $\mathcal{O}_X$ -module of *finite type*, then we can identify

$$\text{Hom}_{k(y)}(g^*(\Omega_{X/S}^1) \otimes_{\mathcal{O}_Y} k(y), k(y))$$

with $T_{X/S}(x) \otimes_{k(x)} k(y)$. We therefore deduce from the homomorphism (16.5.13.4) a homomorphism of $k(y)$ -vector spaces

$$(16.5.13.5) \quad T_y(g) : T_{Y/S}(y) \rightarrow T_{X/S}(x) \otimes_{k(x)} k(y)$$

called the *linear map tangent to g at the point y* . When y is *rational over* $k(s)$, we can identify $k(s)$, $k(y)$, and $k(x)$, and $T_y(g)$ is then a homomorphism of $k(s)$ -vector spaces $T_{Y/S}(y) \rightarrow T_{X/S}(x)$; also note that in this case, $g^*(\Omega_{X/S}^1) \otimes_{\mathcal{O}_Y} k(y)$ is identified with $\Omega_{X/S}^1 \otimes_{\mathcal{O}_X} k(x)$, and the above homomorphism is therefore defined without any finiteness conditions on $\Omega_{X/S}^1$ and it is none other than the homomorphism $T_{Y/S}(g)$ (16.5.12) restricted to the fibre of $T_{Y/S}$ at the point y .

(16.5.14). The interpretation of derivations of an A -algebra B to a B -module L , given in (0, 20.1.1), translates to the language of preschemes in the following way.

Consider two morphisms of preschemes $f : X \rightarrow S$, $g : Y \rightarrow S$, and a closed subprescheme Y_0 of Y defined by a *zero-square ideal* $\mathcal{J}$ of $\mathcal{O}_Y$ (so that Y and Y_0 have the *same underlying subspace*). Suppose we are given an S -morphism $u_0 : Y_0 \rightarrow X$, so that we have a commutative diagram

$$(16.5.14.1) \quad \begin{array}{ccc} X & \xleftarrow{u_0} & Y_0 \\ f \downarrow & \nwarrow u & \downarrow j \\ S & \xleftarrow{g} & Y \end{array}$$

and we suggest looking for an S -morphism $u : Y \rightarrow X$ such that $u_0 = u \circ j$ (in other words, if it is possible to complete the diagram above by the dotted arrow u , keeping it *commutative*). IV-4 | 32

For that, consider an affine open $U = \text{Spec}(C)$ of Y ; its inverse image $j^{-1}(U)$ is the affine open $U_0 = \text{Spec}(C/\mathcal{L})$, where $\mathcal{L} = \Gamma(U, \mathcal{J})$, a *zero-square ideal* in C ; suppose that U is small enough so that $u_0(U_0)$ is contained in an affine open $V = \text{Spec}(B)$ of X and that $g(U) = f(u_0(U_0))$ is contained in an affine open $W = \text{Spec}(A)$ of S , so that B and C are A -algebras and $u_0|_{U_0}$ corresponds to an A -homomorphism ψ from B to $C/\mathcal{L}$; Let $P(U_0)$ be the set of restrictions $u|_{U_0}$ of the sought homomorphisms, which corresponds canonically to A -homomorphisms of algebras $\phi : B \rightarrow C$ such that the *composite* $B \xrightarrow{\phi} C \rightarrow C/\mathcal{L}$ is equal to ψ . So we know (0, 20.1.1) that the set of such homomorphisms is either empty or of the form $\phi_1 + \text{Der}_A(B, \mathcal{L})$; when $P(U_0)$ is not empty, the additive group $\text{Der}_A(B, \mathcal{L})$ acts by addition on $P(U_0)$, which is therefore an *affine space* for the additive group $\text{Der}_A(B, \mathcal{L})$ (or even a *principal homogeneous space* (or *torsor*) under $\text{Der}_A(B, \mathcal{L})$).

Now notice that, since $\mathcal{L}$ is equipped with a B -module structure via ψ , we have an *isomorphism* $v \mapsto v \circ d_{B/A}$ of $\text{Hom}_B(\Omega_{B/A}^1, \mathcal{L})$ onto $\text{Der}_A(B, \mathcal{L})$ (0, 20.4.8). Besides, as $\mathcal{L}$ is square-zero, therefore a $(C/\mathcal{L})$ -module, every B -homomorphism $v : \Omega_{B/A}^1 \rightarrow \mathcal{L}$ can be considered as a $(C/\mathcal{L})$ -homomorphism $\Omega_{B/A}^1 \otimes_B (C/\mathcal{L}) \rightarrow \mathcal{L}$. As $\mathcal{J}$ is square-zero, it can be considered as a quasi-coherent

$\mathcal{O}_{Y_0}$ -module; let's introduce the $\mathcal{O}_{Y_0}$ -module

$$(16.5.14.2) \quad \mathcal{G} = \mathcal{H}om(u_0^*(\Omega_{X/S}^1), \mathcal{I});$$

it follows from the fact that $\Omega_{B/A}^1 = \Gamma(V, \Omega_{X/S}^1)$ (16.3.7) that we can write $\mathrm{Der}_A(B, \mathfrak{L}) = \Gamma(U_0, \mathcal{G})$.

As $P(U_0)$ is defined as a set of S -morphisms $U \rightarrow X$, it is clear that $U_0 \mapsto P(U_0)$ is a *sheaf of sets* $\mathcal{P}$ on Y_0 . We can use this fact to prove that the map $h : \Gamma(U_0, \mathcal{G}) \times P(U_0) \rightarrow P(U_0)$ defining the torsor structure on $P(U_0)$ is independent of choice of V and W and also that, if $U' \subset U$ is a second affine open of Y , U'_0 its inverse image in Y_0 , then the diagram

$$(16.5.14.3) \quad \begin{array}{ccc} \Gamma(U_0, \mathcal{G}) \times P(U_0) & \xrightarrow{h} & P(U_0) \\ \downarrow & & \downarrow \\ \Gamma(U'_0, \mathcal{G}) \times P(U'_0) & \xrightarrow{h'} & P(U'_0) \end{array}$$

is commutative (the vertical arrows being the restrictions). In light of the above remark, we reduce to proving the commutativity of the above diagram when h is defined as such from affine opens V , W and h' from affine opens $V' \subset V$ and $W' \subset W$. But because of the preceding description of h , this follows from the commutativity of the diagram (0, 20.5.4.2).

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The mapping $\Gamma(U_0, \mathcal{G}) \times P(U_0) \rightarrow P(U_0)$ therefore define a homomorphism of *sheaf of sets* $m : \mathcal{G} \times \mathcal{P} \times \mathcal{P} \rightarrow \mathcal{P}$ such that, for all open sets U_0 for which $\Gamma(U_0, \mathcal{P}) \neq \emptyset$, $m_{U_0} : \Gamma(U_0, \mathcal{G}) \times \Gamma(U_0, \mathcal{P}) \rightarrow \Gamma(U_0, \mathcal{P})$ is an external law defining in $\Gamma(U_0, \mathcal{P})$ a torsor structure for the group $\Gamma(U_0, \mathcal{G})$.

(16.5.15). In general, when we are given a sheaf of sets $\mathcal{P}$ over a topological space Z , a sheaf of groups $\mathcal{G}$ (not necessarily commutative), and a homomorphism of sheaves of sets $m : \mathcal{G} \times \mathcal{P} \rightarrow \mathcal{P}$ such that, for every open $U \subset Z$ such that $\Gamma(U, \mathcal{P}) \neq \emptyset$, $m_U : \Gamma(U, \mathcal{G}) \times \Gamma(U, \mathcal{P}) \rightarrow \Gamma(U, \mathcal{P})$ makes $\Gamma(U, \mathcal{P})$ a *torsor* under the group $\Gamma(U, \mathcal{G})$, then we say that $\mathcal{P}$ is a *pseudo-torsor* (or *formally principal homogeneous sheaf*) under the sheaf of groups $\mathcal{G}$. We say that $\mathcal{P}$ is a *torsor* (or *principal homogeneous sheaf*) under $\mathcal{G}$ ⁸ if in addition $\Gamma(U, \mathcal{P}) \neq \emptyset$ for every open $U \neq \emptyset$ in a suitable basis for the topology of Z .

For the general theory of torsors, we refer to [eAG64]; we will limit ourselves to recalling the canonical correspondence between isomorphism classes of torsors (for a *given* $\mathcal{G}$) and elements from the cohomology set $H^1(Z, \mathcal{G})$. Consider a torsor $\mathcal{P}$ under $\mathcal{G}$ and an open cover (U_λ) of Z such that $\Gamma(U_\lambda, \mathcal{P}) \neq \emptyset$ for every λ ; denote by p_λ an element of $\Gamma(U_\lambda, \mathcal{P})$. For every pair of indices λ, μ such that $U_\lambda \cap U_\mu \neq \emptyset$, there then exists a unique element $\gamma_{\lambda\mu}$ of $\Gamma(U_\lambda \cap U_\mu, \mathcal{G})$ such that $\gamma_{\lambda\mu} \cdot (p_\mu|_{U_\lambda \cap U_\mu}) = p_\lambda|_{U_\lambda \cap U_\mu}$; in addition, if λ, μ, ν are three indices such that $U_\lambda \cap U_\mu \cap U_\nu \neq \emptyset$, then the restrictions $\gamma'_{\lambda\mu}, \gamma'_{\mu\nu}, \gamma'_{\lambda\nu}$ of $\gamma_{\lambda\mu}, \gamma_{\mu\nu}, \gamma_{\lambda\nu}$ to $U_\lambda \cap U_\mu \cap U_\nu$ satisfy the condition $\gamma'_{\lambda\nu} = \gamma'_{\lambda\mu} \gamma'_{\mu\nu}$; in other words, $(\lambda, \mu) \mapsto \gamma_{\lambda\mu}$ is a 1-cocycle of the cover (U_λ) with values in $\mathcal{G}$. If, for every λ , p'_λ is a second element of $\Gamma(U_\lambda, \mathcal{P})$, then there exists a unique element $\beta_\lambda \in \Gamma(U_\lambda, \mathcal{G})$ such that $p'_\lambda = \beta_\lambda \cdot p_\lambda$, and the 1-cocycle $(\gamma'_{\lambda\mu})$ corresponding to the family (p'_λ) is given by $\gamma'_{\lambda\mu} = \beta_\lambda \gamma_{\lambda\mu} \beta_\mu^{-1}$, that is, it is *cohomologous* to $\gamma_{\lambda\mu}$. Conversely, the data of a 1-cocycle $(\gamma_{\lambda\mu})$ defines, for every pair (λ, μ) , an automorphism $\theta_{\lambda\mu}$ of the sheaf of sets $\mathcal{G}|_{U_\lambda \cap U_\mu}$, namely the right translation by $\gamma_{\lambda\mu}$, and the fact that it is a cocycle shows that we can *glue* the sheaves of sets $\mathcal{G}|_{U_\lambda}$ via the automorphisms $\theta_{\lambda\mu}$ (0, 3.3.1); we thus obtain a torsor under $\mathcal{G}$, denoted $\mathcal{P}$, and if we take for p_λ the unit section over U_λ , then the corresponding 1-cocycle is none other than the given 1-cocycle $(\gamma_{\lambda\mu})$; in addition, if we replace $(\gamma_{\lambda\mu})$ by a 1-cocycle $\gamma'_{\lambda\mu} = \beta_\lambda \gamma_{\lambda\mu} \beta_\mu^{-1}$ cohomologous to it, then we check immediately that the torsor obtained is isomorphic to $\mathcal{P}$.

In particular, if $(\gamma_{\lambda\mu})$ is a 1-coboundary, in other words of the form $\gamma_{\lambda\mu} = \beta_\lambda \beta_\mu^{-1}$, then the torsor $\mathcal{P}$ obtained is *isomorphic to* $\mathcal{G}$ (considered as a torsor under itself by left translations); we say in this case that $\mathcal{P}$ is *trivial*, and the converse is evident.

In particular, it follows from (III, 1.3.1) that we have:

Proposition (16.5.16). — *Let Z be an affine scheme, $\mathcal{G}$ a quasi-coherent $\mathcal{O}_Z$ -module; then every torsor over $\mathcal{G}$ is trivial.*

⁸[Trans.] This is nowadays more commonly called a $\mathcal{G}$ -torsor rather than a torsor under $\mathcal{G}$.

Returning to the problem considered in (16.5.13), we thus obtain:

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Proposition (16.5.17). — Let X, Y be two S -preschemes, Y_0 a closed subprescheme of Y defined by a quasi-coherent ideal $\mathcal{I}$ of $\mathcal{O}_Y$ such that $\mathcal{I}^2 = 0$, $j : Y_0 \rightarrow Y$ the canonical injection. Let $u_0 : Y_0 \rightarrow X$ be an S -morphism, and $\mathcal{P}$ the sheaf of sets on Y such that, for every open U of Y , $\Gamma(U, \mathcal{P})$ is the set of S -morphisms $u : U \rightarrow X$ such that $u_0|_{U_0} = u \circ (j|_{U_0})$, where $U_0 = j^{-1}(U)$. Then there exists on $\mathcal{P}$ the structure of a pseudo-torsor over the $\mathcal{O}_{Y_0}$ -module $\mathcal{G} = \mathcal{H}om_{\mathcal{O}_{Y_0}}(u_0^*(\Omega_{X/S}^1), \mathcal{I})$.

In particular:

Corollary (16.5.18). — With the notation of (16.5.16), suppose that Y is affine and $\Omega_{X/S}^1$ is of finite presentation; if there is a open cover (U_α) of Y , and, for every index α , an S -morphism $v_\alpha : U_\alpha \rightarrow X$ such that, if $U_\alpha^0 = j^{-1}(U_\alpha)$, we have $v_\alpha \circ (j|_{U_\alpha^0}) = u_0|_{U_\alpha^0}$, then there is an S -morphism $u : Y \rightarrow X$ such that $u \circ j = u_0$.

Proof. Indeed, $\mathcal{G}$ is a quasi-coherent $\mathcal{O}_{Y_0}$ -module (I, 1.3.12); by (16.5.16) and the fact that Y_0 is then affine, the sheaf $\mathcal{P}$, which is by hypothesis a torsor over $\mathcal{G}$, and not only a pseudo-torsor, is trivial; but if w is an isomorphism from $\mathcal{G}$ to $\mathcal{P}$ (as it is a torsor over $\mathcal{G}$), the image under w of the zero section of $\mathcal{G}$ is the S -morphism we want. $\square$

16.6. Sheaf of p -differentials and exterior differentials.

(16.6.1). Let $f : X \rightarrow S$ be a morphism of preschemes. We define the sheaf of p -differentials of X relative to S (p integer) to be the p 'th exterior power (0, 4.1.5) of the $\mathcal{O}_X$ -module $\Omega_{X/S}^1$, denoted by

$$(16.6.1.1) \quad \Omega_{X/S}^p = \bigwedge^p (\Omega_{X/S}^1).$$

So we have $\Omega_{X/S}^0 = \mathcal{O}_X$, and $\Omega_{X/S}^p = 0$ for $p < 0$; the $\Omega_{X/S}^p$ are the homogeneous components of the exterior algebra of $\Omega_{X/S}^1$

$$(16.6.1.2) \quad \Omega_{X/S}^\bullet = \bigwedge (\Omega_{X/S}^1) = \bigoplus_{p \in \mathbb{Z}} \bigwedge^p (\Omega_{X/S}^1),$$

which is therefore a quasi-coherent graded anti-commutative $\mathcal{O}_X$ -algebra whose elements of degree 1 are square-zero. For every affine U of X , we have $\Gamma(U, \Omega_{X/S}^\bullet) = \bigwedge (\Gamma(U, \Omega_{X/S}^1))$, where $\Gamma(U, \Omega_{X/S}^1)$ is considered as a $\Gamma(U, \mathcal{O}_X)$ -module.

When $S = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affines, B being then an A -algebra, we have (0, 4.1.5) $\Omega_{X/S}^p = (\Omega_{B/A}^p)^\sim$, by putting $\Omega_{B/A}^p = \bigwedge^p \Omega_{B/A}^1$.

Theorem (16.6.2). — There is one and only one endomorphism d of the sheaf of additive groups $\Omega_{X/S}^\bullet$ with the following properties:

- (i) $d \circ d = 0$.
- (ii) For every open set U of X and every section $f \in \Gamma(U, \mathcal{O}_X)$ we have $df = d_{X/S}f$.
- (iii) For every open set U of X , every pair of integers p, q and every pair of sections $\omega_p' \in \Gamma(U, \Omega_{X/S}^p)$, $\omega_q'' \in \Gamma(U, \Omega_{X/S}^q)$, we have

$$(16.6.2.1) \quad d(\omega_p' \wedge \omega_q'') = (d\omega_p') \wedge \omega_q'' + (-1)^p \omega_p' \wedge d\omega_q''.$$

Also, d is an endomorphism of graded $\psi^*(\mathcal{O}_X)$ -modules of degree $+1$.

Proof. Suppose that we have proved the existence of an endomorphism d . For every affine open U of X , every section of $\Omega_{X/S}^p$ over U is (because of (ii)) a linear combination of a finite number of elements of the form $g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)$, where g and the f_i are sections of $\mathcal{O}_X$ over U (0, 20.4.7). The conditions (i) and (iii) then show, by induction on p , that we necessarily have

$$(16.6.2.2) \quad d(g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)) = dg \wedge df_1 \wedge df_2 \wedge \cdots \wedge df_p.$$

This therefore proves the uniqueness of d and the last claim of the theorem. By virtue of this uniqueness property, to show the existence of d , we can restrict ourselves to the case where $S = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affines. Now (Bourbaki, *Alg.*, chap. III, 3rded., §10) to define an

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A -derivation D of degree $+1$ of an exterior algebra $\Lambda(M)$ (where M is a B -module and an A -algebra), such derivation taking its values in a *graded anti-commutative* A -algebra $C = \bigoplus_{n=0}^{\infty} C_n$, whose elements of degree 1 are square-zero, it suffices to give arbitrarily an A -derivation D_0 of B in C_1 and an A -homomorphism D_1 of M in C_2 ; then it exists one and only one A -anti-derivation D of $\Lambda(M)$ in C coinciding with D_0 in B and D_1 in M .

In the present case, D_0 is necessarily equal to $d_{B/A}$ by (ii); we reduce to seeing, having (16.6.2.2) in mind, that there is an A -homomorphism u of $\Omega_{B/A}^1$ in $\Omega_{B/A}^2$ such that

$$(16.6.2.3) \quad u(g.df) = dg \wedge df$$

for whichever f, g in A ; it suffices to show that there is an A -homomorphism $v : B \otimes_A \Omega_{B/A}^1 \rightarrow \Omega_{B/A}^2$ such that

$$(16.6.2.4) \quad v(g.\omega) = dg \wedge \omega$$

for $g \in B$ and $\omega \in \Omega_{B/A}^1$. Finally, since $\Omega_{B/A}^1 = \mathcal{I}/\mathcal{I}^2$ (where $\mathcal{I} = \mathcal{I}_{B/A}$ is the kernel of the canonical homomorphism $B \times_A B \rightarrow B$) and that $\Omega_{B/A}^1$ is generated by elements of the form $g.df$, it is enough to define an A -homomorphism $w : B \otimes_A (B \otimes_A B) \rightarrow \Omega_{B/A}^2$ such that

$$(16.6.2.5) \quad w(g' \otimes g \otimes f) = dg' \wedge (g.df)$$

and such that w is zero on the image of $B \otimes_A \mathcal{I}^2$. Or, since the second member of (16.6.2.5) is A -trilinear in g', g and f , the existence of w verifying (16.6.2.5) is immediate. Since, on the other hand, $\mathcal{I}$ is generated by elements of the form $1 \otimes x - x \otimes 1$ ($x \in B$), we reduce to checking that when $z = (1 \otimes x - x \otimes 1)(1 \otimes y - y \otimes 1)$ we have $w(g' \otimes z) = 0$. Or, since $z = 1 \otimes (xy) + (xy) \otimes 1 - x \otimes y - y \otimes x$, the formula (16.6.2.4) shows that it is enough to see that we have $d(xy) - x.dy - y.dx = 0$, which is to say that d is a derivation. IV-4 | 36

It remains to be shown that d verifies the condition (i). Now, the square of an anti-derivation is a derivation (Bourbaki, *loc. cit.*), and since $\Omega_{B/A}^\bullet$ is generated by $\Omega_{B/A}^1$ as a B -algebra, it is enough to verify that $d(dz) = 0$ for $z \in B$ and $x \in \Omega_{B/A}^1$; in the first case, this follows from the formula (16.6.2.3) when $g = 1$; for the second, we can restrict ourselves to the case where $z = g.df$ with f, g in B , and then we have, because of (16.6.2.1) and (16.6.2.3),

$$d(d(g.df)) = d(dg \wedge df) = (d(dg)) \wedge (df) - (dg) \wedge (d(df)) = 0.$$

□

Definition (16.6.3). — The anti-derivation d defined in (16.6.2) (also denoted by $d_{X/S}$) is called the *exterior differential* on X (relative to S).

Proposition (16.6.4). — For every base change $g : S' \rightarrow S$, if we put $X' = X \times_S S'$, the canonical morphism

$$(16.6.4.1) \quad \Omega_{X'/S'}^\bullet \otimes_{S'} S' \longrightarrow \Omega_{X'/S}^\bullet$$

deduced from the isomorphism (16.5.10.1) is bijective. Also, if s is a section of $\Omega_{X/S}^\bullet$ over an open set U of X , $s \otimes 1$ its inverse image, section of $\Omega_{X'/S'}^\bullet$ over the inverse image U' of U in X' , we have $d_{X'/S'}(s \otimes 1) = d_{X/S}(s) \otimes 1$.

Proof. The first claim is immediate, the formation of the exterior algebra of a module commutes with extending the scalar ring. To prove the second, we can, because of (16.6.2.2), restrict ourselves to the case where $s \in \Gamma(U, \mathcal{O}_X)$, and in this case the claim has already been proven (16.4.3.7). □

(16.6.5). Suppose that $\Omega_{X/S}^1$ is an locally free $\mathcal{O}_X$ -module of rank n in a point x , so that we have n sections $s_i \in \Gamma(U, \mathcal{O}_X)$ such that the ds_i form a basis for the $\Gamma(U, \mathcal{O}_X)$ -module $\Gamma(U, \Omega_{X/S}^1)$ (16.5.8). Then, for every integer $p \geq 1$, the p -differentials $ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}$ (for $i_1 \leq i_2 \leq \cdots \leq i_p$ elements of $[1, n]$) form a basis of $\binom{n}{p}$ elements of $\Gamma(U, \Omega_{X/S}^p)$ over $\Gamma(U, \mathcal{O}_X)$. Also the formula (16.6.2.2) shows that for every section $g \in \Gamma(U, \mathcal{O}_X)$, we have

$$(16.6.5.1) \quad d(g.ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}) = \sum_k (-1)^r \frac{\partial g}{\partial s_k} ds_{i_1} \wedge \cdots \wedge ds_{i_r} \wedge ds_k \wedge ds_{i_{r+1}} \wedge \cdots \wedge ds_{i_p}$$

where, in the second member, k varies in the set of the $n - p$ indexes different from the i_h, i_r being the biggest index $< k$.

We note that the relation $d(dg) = 0$ for every section $g \in \Gamma(U, \mathcal{O}_X)$ expresses itself in the form

$$D_i(D_j g) = D_j(D_i g) \quad \text{for } i \neq j;$$

in other words, the derivations D_i defined in (16.5.7) commute with each other.

16.7. The $\mathcal{P}_{X/S}^n(\mathcal{F})$.

(16.7.1). Let $f : X \rightarrow S$ be a morphism of preschemes, $\mathcal{F}$ an $\mathcal{O}_X$ -module. We denote by $X_{\Delta_f}^{(n)}$ the n 'th infinitesimal neighbourhood of X via the diagonal morphism $\Delta_f : X \rightarrow X \times_S X$, by $h_n : X_{\Delta_f}^{(n)} \rightarrow X \times_S X$ the canonical morphism (16.1.2), and consider the two composite morphisms

$$p_1^{(n)} : X_{\Delta_f}^{(n)} \xrightarrow{h_n} X \times_S X \xrightarrow{p_1} X, \quad p_2^{(n)} : X_{\Delta_f}^{(n)} \xrightarrow{h_n} X \times_S X \xrightarrow{p_2} X$$

so that, by definition, $p_1^{(n)}$ corresponds to the homomorphism of sheaves of rings $\mathcal{O}_X \rightarrow \mathcal{P}_{X/S}^n$ which we have chosen to define the $\mathcal{O}_X$ -algebra structure on $\mathcal{P}_{X/S}^n$ (16.3.5), and $p_2^{(n)}$ to the homomorphism of sheaves of rings $d_{X/S}^n : \mathcal{O}_X \rightarrow \mathcal{P}_{X/S}^n$ (16.3.6). Since $X_{\Delta_f}^{(n)}$ and X have the same underlying subspace, we can write

$$(16.7.1.1) \quad \mathcal{P}_{X/S}^n = (p_1^{(n)})_*((p_2^{(n)})^*(\mathcal{O}_X)).$$

More generally, we define

$$(16.7.1.2) \quad \mathcal{P}_{X/S}^n(\mathcal{F}) = (p_1^{(n)})_*((p_2^{(n)})^*(\mathcal{F})).$$

so that $\mathcal{P}_{X/S}^n = \mathcal{P}_{X/S}^n(\mathcal{O}_X)$; by definition, $\mathcal{P}_{X/S}^n(\mathcal{F})$ is an $\mathcal{O}_X$ -module.

(16.7.2). If we come back to the definition of the inverse image of modules on ringed spaces (0, 4.3.1) and having in mind that $X_{\Delta_f}^{(n)}$ and X have the same underlying space, we see that we can write the definition (16.7.1.2) in the form

$$(16.7.2.1) \quad \mathcal{P}_{X/S}^n(\mathcal{F}) = \mathcal{P}_{X/S}^n \otimes_{\mathcal{O}_X} \mathcal{F},$$

but where you have to be careful that, in the interpretation of the symbol $\otimes$, $\mathcal{P}_{X/S}^n$ is endowed with the structure of $\mathcal{O}_X$ -module defined by the homomorphism of sheaves of rings $d_{X/S}^n : \mathcal{O}_X \rightarrow \mathcal{P}_{X/S}^n$. It follows immediately from such formula (or directly from (16.7.1.2)) that $\mathcal{P}_{X/S}^n(\mathcal{F})$ is canonically equipped with a $\mathcal{P}_{X/S}^n$ -module structure.

Proposition (16.7.3). — (i) The functor $\mathcal{F} \mapsto \mathcal{P}_{X/S}^n(\mathcal{F})$ from the category of $\mathcal{O}_X$ -modules to the category of $\mathcal{P}_{X/S}^n$ -modules is right exact, and commutes with arbitrary inductive limits; it is exact when $\mathcal{P}_{X/S}^n$ is flat.

(ii) If $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -module (resp. of finite type, resp. of finite presentation), then $\mathcal{P}_{X/S}^n(\mathcal{F})$ is quasi-coherent (resp. of finite type, resp. of finite presentation).

Proof. The claims from (i) follow from (16.7.2.1) and the consideration of the symmetry of $\mathcal{P}_{X/S}^n$ (16.3.4). The claims from (ii) follow from the right exactness of the functor $\mathcal{F} \mapsto \mathcal{P}_{X/S}^n(\mathcal{F})$. $\square$

(16.7.4). The two structures of $\mathcal{O}_X$ -module on $\mathcal{P}_{X/S}^n$ define in $\mathcal{P}_{X/S}^n(\mathcal{F})$ two structures of $\mathcal{O}_X$ -modules, which happen to be permutable, and therefore an $\mathcal{O}_X$ -bimodule structure. It is convenient to denote the structure coming from the structure homomorphism $\mathcal{O}_X \rightarrow \mathcal{P}_{X/S}^n$ (chosen in (16.3.5)) on the left and the one coming from the homomorphism $d_{X/S}^n : \mathcal{O}_X \rightarrow \mathcal{P}_{X/S}^n$ on the right. On other words, for every open U of X , and every triplet $a \in \Gamma(U, \mathcal{O}_X)$, $b \in \Gamma(U, \mathcal{P}_{X/S}^n)$, $t \in \Gamma(U, \mathcal{F})$, we have by definition

$$(16.7.4.1) \quad a(b \otimes t) = (ab) \otimes t, \quad (b \otimes t)a = (b \cdot d^n a) \otimes t = b \otimes (at) = (d^n a) \cdot (b \otimes t).$$

The $\mathcal{O}_X$ -module structure coming from the definition (16.7.1.2) is therefore, under these conventions, the left $\mathcal{O}_X$ -module structure. If $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$ -module, then the same is true for $\mathcal{P}_{X/S}^n(\mathcal{F})$ for any one of its $\mathcal{O}_X$ -module structures. If also $\mathcal{F}$ is of finite type (resp. of finite presentation) and $f : X \rightarrow S$ is locally of finite type (resp. locally of finite presentation), $\mathcal{P}_{X/S}^n(\mathcal{F})$ is (for any one of its $\mathcal{O}_X$ -module structures) of finite type (resp. of finite presentation), which is a consequence of (16.3.9) and (16.4.22).

(16.7.5). The definition (16.7.2.1) entails the existence of a homomorphism of sheaves of commutative groups

$$(16.7.5.1) \quad d_{X/S, \mathcal{F}}^n : \mathcal{F} \longrightarrow \mathcal{P}_{X/S}^n(\mathcal{F}) \quad (\text{also denoted } d_{X/S}^n)$$

such that, in the notations of (16.7.4), we have

$$(16.7.5.2) \quad d_{X/S, \mathcal{F}}^n(t) = 1 \otimes t$$

and consequently, because of (16.7.4.1)

$$(16.7.5.3) \quad d_{X/S, \mathcal{F}}^n(at) = (1 \otimes t)a = (d_{X/S, \mathcal{F}}^n(t)).a$$

$$(IV.16.7.5.4) \quad d_{X/S, \mathcal{F}}^n(at) = (d_{X/S, \mathcal{F}}^n(a)).(1 \otimes t) = (d_{X/S, \mathcal{F}}^n(a))(d_{X/S, \mathcal{F}}^n(t)).$$

Therefore it is $\mathcal{O}_X$ -linear for the structure of *right* $\mathcal{O}_X$ -module on $\mathcal{P}_{X/S}^n(\mathcal{F})$, and *semilinear* (relative to the homomorphism σ (16.3.4)) for the *left* $\mathcal{O}_X$ -module structure.

Proposition (16.7.6). — *The right $\mathcal{O}_X$ -module $\mathcal{P}_{X/S}^n(\mathcal{F})$ is generated by the image of $\mathcal{F}$ by the homomorphism $d_{X/S, \mathcal{F}}^n$.*

Proof. This is an immediate consequence of (16.7.5.3) and of the particular case $\mathcal{F} = \mathcal{O}_X$ (16.3.8). $\square$

(16.7.7). The canonical homomorphisms of sheaves of rings

$$\phi_{nm} : \mathcal{P}_{X/S}^m \longrightarrow \mathcal{P}_{X/S}^n$$

for $n \leq m$ (16.1.2) define, because of (16.7.2.1), canonical homomorphisms

$$\mathcal{P}_{X/S}^m(\mathcal{F}) \longrightarrow \mathcal{P}_{X/S}^n(\mathcal{F}) \quad (n \leq m)$$

which are homomorphisms of $\mathcal{O}_X$ -bimodules in light of (16.1.6) and (7.4.1); also we have commutative diagrams

$$\begin{array}{ccc} \mathcal{P}_{X/S}^m(\mathcal{F}) & \longrightarrow & \mathcal{P}_{X/S}^n(\mathcal{F}) \\ & \nwarrow d_{X/S, \mathcal{F}}^n & \nearrow d_{X/S, \mathcal{F}}^n \\ & \mathcal{F} & \end{array}$$

We have therefore a projective system of $\mathcal{O}_X$ -bimodules $(\mathcal{P}_{X/S}^n(\mathcal{F}))$, and we define

$$(16.7.7.1) \quad \mathcal{P}_{X/S}^\infty(\mathcal{F}) = \varprojlim \mathcal{P}_{X/S}^n(\mathcal{F}).$$

Also, this shows that the homomorphisms (16.7.5.1) form a projective system of homomorphisms, and therefore define a canonical homomorphism

$$(16.7.7.2) \quad d_{X/S, \mathcal{F}}^\infty : \mathcal{F} \rightarrow \mathcal{P}_{X/S}^\infty(\mathcal{F}).$$

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(16.7.8). Let $\mathcal{F}, \mathcal{G}$ be two $\mathcal{O}_X$ -modules; it follows immediately from the definition (16.7.2.1) that we have a canonical isomorphism of $\mathcal{P}_{X/S}^n$ -modules

$$(16.7.8.1) \quad \mathcal{P}_{X/S}^n(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}) \simeq \mathcal{P}_{X/S}^n(\mathcal{F}) \otimes_{\mathcal{P}_{X/S}^n} \mathcal{P}_{X/S}^n(\mathcal{G})$$

(Bourbaki, *Alg.*, chap. II, 3rded., §5, n.1, prop. 3).

We conclude in particular (or we see directly from the definition (16.7.2.1)) that if $\mathcal{F}$ has an $\mathcal{O}_X$ -algebra structure (not necessarily associative), $\mathcal{P}_{X/S}^n(\mathcal{F})$ has a canonical $\mathcal{O}_X$ -algebra structure; the latter is associative (resp. commutative, res. unital, resp. a Lie algebra) if $\mathcal{F}$ is so. Also the canonical homomorphisms $\mathcal{P}_{X/S}^m(\mathcal{F}) \rightarrow \mathcal{P}_{X/S}^n(\mathcal{F})$ for $n \leq m$ (16.7.7) are then algebra di-homomorphisms; similarly, (16.7.5.1) is then an $\mathcal{O}_X$ -algebra homomorphisms when $\mathcal{P}_{X/S}^n(\mathcal{F})$ is equipped with the $\mathcal{O}_X$ -algebra structure from its structure of *right* $\mathcal{O}_X$ -module.

With the same notations, we equally have a canonical homomorphisms of $\mathcal{P}_{X/S}^n$ -modules

$$(16.7.8.2) \quad \mathcal{P}_{X/S}^n(\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})) \longrightarrow \mathcal{H}om_{\mathcal{P}_{X/S}^n}(\mathcal{P}_{X/S}^n(\mathcal{F}), \mathcal{P}_{X/S}^n(\mathcal{G}))$$

(Bourbaki, *Alg.*, chap. II, 3rded., §5, n.3), which is bijective when $\mathcal{P}_{X/S}^n$ is *locally free of finite type* (*loc. cit.*, prop. 7).

(16.7.9). Suppose we are in the situation described in (16.4.1); then from the canonical homomorphism $P^n(u)$ (16.4.3.3) we deduce immediately a canonical homomorphism of $\mathcal{O}_{X'}$ -bimodules

$$(16.7.9.1) \quad u^*(\mathcal{P}_{X/S}^n(\mathcal{F})) \longrightarrow \mathcal{P}_{X'/S'}^n(u^*(\mathcal{F})).$$

We leave it to the reader to extend the properties seen in (16.4) in the case $\mathcal{F} = \mathcal{O}_X$.

Remark (16.7.10). — The definition of $\mathcal{P}_{X/S}^n(\mathcal{F})$ in the form (16.7.1.2) still makes sense when $\mathcal{F}$ is a sheaf of sets (the inverse image of a sheaf of sets by $p_2^{(n)}$ being defined in (0, 3.7.1)); a variant of this definition allows us to define the “jet schemes” (relatively to S) for any prescheme X .

16.8. Differential operators.⁹

Definition (16.8.1). — Let $f = (\psi, \theta) : X \rightarrow S$ be a morphism of preschemes, $\mathcal{F}, \mathcal{G}$ two $\mathcal{O}_X$ -modules, n an integer ≥ 0 . We say that a morphism $D : \mathcal{F} \rightarrow \mathcal{G}$ of sheaves of additive groups is a differential operator of order $\leq n$ (relative to S) if there is a homomorphism of $\mathcal{O}_X$ -modules $u : \mathcal{P}_{X/S}^n(\mathcal{F}) \rightarrow \mathcal{G}$ (where $\mathcal{P}_{X/S}^n(\mathcal{F})$ is equipped with the structure of left $\mathcal{O}_X$ -modules (16.7.4)) such that we have $D = u \circ d_{X/S, \mathcal{F}}^n$.

It is clear, because of the existence of canonical morphisms

$$\mathcal{P}_{X/S}^m(\mathcal{F}) \longrightarrow \mathcal{P}_{X/S}^n(\mathcal{F})$$

for $n \leq m$ (16.7.7), that a differential operator of order $\leq n$ is a differential operator of order $\leq m$ for $n \leq m$. If $D : \mathcal{F} \rightarrow \mathcal{G}$ is a differential operator of order $\leq n$, then, for every open set U of X , $D|U : \mathcal{F}|U \rightarrow \mathcal{G}|U$ is a differential operator of order $\leq n$. IV-4 | 40

We say that a homomorphism $D : \mathcal{F} \rightarrow \mathcal{G}$ is a *differential operator* (relative to S) if, for every $x \in X$, there is an open neighbourhood U of x and an integer $n \geq 0$ such that $D|U : \mathcal{F}|U \rightarrow \mathcal{G}|U$ is a differential operator of order $\leq n$. The *order* of a differential operator is the least upper bound of all integers n so that D is a differential operator of order $\leq n$ (and therefore $+\infty$ if there is no such integer); such order is always finite if X is *quasi-compact*. The differential operator of order 0 are exactly the homomorphisms of $\mathcal{O}_X$ -modules $\mathcal{F} \rightarrow \mathcal{G}$; the operators of order < 0 are zero by convention. For $n \geq 0$ a differential operator of order n is not in general a homomorphism of $\mathcal{O}_X$ -modules but always a homomorphism of $\psi^*(\mathcal{O}_S)$ -modules.

When $\mathcal{F} = \mathcal{O}_X$, a differential operator of order ≤ 1 of $\mathcal{O}_X$ to $\mathcal{G}$ can be put in the form of $v + D$, where $v : \mathcal{O}_X \rightarrow \mathcal{G}$ is an $\mathcal{O}_X$ -homomorphism, and D is an S -derivation (16.5.1) of $\mathcal{O}_X$ to $\mathcal{G}$: this results from the structure of $P_{B/A}$ (0, 20.4.8).

(16.8.2). To describe in a more precise manner a differential operator of order $\leq n$, $D : \mathcal{F} \rightarrow \mathcal{G}$, it suffices, for every open set U of X whose image in S is contained in an affine open set V , to characterize the homomorphism $D = D_U : \Gamma(U, \mathcal{F}) \rightarrow \Gamma(U, \mathcal{G})$. If we put $\Gamma(V, \mathcal{O}_S) = A$, $\Gamma(U, \mathcal{O}_X) = B$, so that B is an A -algebra, we have $\Gamma(U, \mathcal{P}_{X/S}^n(\mathcal{F})) = (B \otimes_A B) / \mathcal{I}^{n+1}$, where we abbreviate $\mathcal{I} = \mathcal{I}_{B/A}$. Also put $M = \Gamma(U, \mathcal{F})$, $N = \Gamma(U, \mathcal{G})$; then the definition of D means that for every pair (U, V) satisfying the above, the A -homomorphism $D : M \rightarrow N$ factors through

$$M \longrightarrow ((B \otimes_A B) / \mathcal{I}^{n+1}) \otimes_B M \xrightarrow{v} N$$

where the first arrow is the canonical morphism $t \mapsto 1 \otimes t$, and v is a B -homomorphism, the structure of B -module coming from the first factor (whereas we recall that in the formation of the tensor product over B , the structure of B -module on $(B \otimes_A B) / \mathcal{I}^{n+1}$ comes from the second factor B). Note also that the B -module $((B \otimes_A B) / \mathcal{I}^{n+1}) \otimes_B M$ is isomorphic to $(B \otimes_A M) / \mathcal{I}^{n+1}(B \otimes_A M)$, where $(B \otimes_A M)$ is considered as a $(B \otimes_A B)$ module and its structure of B -module comes from $t \mapsto 1 \otimes t$ of B in $B \otimes_A B$. Let then D' be the B -homomorphism of $B \otimes_A M$ to N such that $D'(b \otimes t) = bD(t)$; then condition of factorization of D is to say that D' must be zero on the B -module $\mathcal{I}^{n+1}(B \otimes_A M)$.

(16.8.3). It is clear that the set of differential operators of order $\leq n$ from $\mathcal{F}$ to $\mathcal{G}$ forms an additive group, denoted by $\text{Diff}_{X/S}^n(\mathcal{F}, \mathcal{G})$; when $\mathcal{F} = \mathcal{G} = \mathcal{O}_X$, we also write $\text{Diff}_{X/S}^n$ instead of $\text{Diff}_{X/S}^n(\mathcal{O}_X, \mathcal{O}_X)$.

⁹For a more general formalism, see the exposé VII of [eAG64] (due to P. Gabriel).

We have seen (16.8.1), that given two open sets $U \supset V$ of X , we have a restriction homomorphism

$$\mathrm{Diff}_{X/S}^n(\mathcal{F}|U, \mathcal{G}|U) \longrightarrow \mathrm{Diff}_{X/S}^n(\mathcal{F}|V, \mathcal{G}|V)$$

from which we deduce that $U \mapsto \mathrm{Diff}_{U/S}^n(\mathcal{F}|U, \mathcal{G}|U)$ is a presheaf of additive groups; in fact, it is actually a *sheaf*, since for an open set U varying in X , the homomorphisms $u \mapsto u \circ d_{U/S, \mathcal{F}|U}^n$ are isomorphisms of sheaves of additive groups

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$$(16.8.3.1) \quad \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{P}_{U/S}^n(\mathcal{F}|U), \mathcal{G}|U) \simeq \mathrm{Diff}_{U/S}^n(\mathcal{F}|U, \mathcal{G}|U),$$

because of the fact that the image of $\mathcal{F}$ by $d_{X/S, \mathcal{F}}^n$ generates $\mathcal{P}_{X/S}^n(\mathcal{F})$ (16.7.6). We denote this sheaf by $\mathcal{D}\mathrm{iff}_{X/S}^n(\mathcal{F}, \mathcal{G})$, and we have:

Proposition (16.8.4). — *The isomorphisms (16.8.3.1) define an isomorphism of sheaves of additive groups*

$$(18.4.1) \quad \mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{P}_{X/S}^n(\mathcal{F}), \mathcal{G}) \simeq \mathcal{D}\mathrm{iff}_{X/S}^n(\mathcal{F}, \mathcal{G}).$$

When $\mathcal{F} = \mathcal{G} = \mathcal{O}_X$, we also write $\mathcal{D}\mathrm{iff}_{X/S}^n$ instead of $\mathcal{D}\mathrm{iff}_{X/S}^n(\mathcal{O}_X, \mathcal{O}_X)$; it results from (16.6.4) that $\mathcal{D}\mathrm{iff}_{X/S}^n$ is the dual of $\mathcal{P}_{X/S}^n$; so we also write $\langle t, D \rangle$ instead of $u(t)$ if t is a section of $\mathcal{P}_{X/S}^n$ over an open set and if u is a homomorphism from $\mathcal{P}_{X/S}^n$ to $\mathcal{O}_X$ corresponding to D .

(16.8.5). When $\mathcal{P}_{X/S}^n(\mathcal{F})$ has an $\mathcal{O}_X$ -bimodule structure (16.7.4), we deduce canonically an $\mathcal{O}_X$ -bimodule structure on $\mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{P}_{X/S}^n(\mathcal{F}), \mathcal{G})$, and therefore on $\mathcal{D}\mathrm{iff}_{X/S}^n(\mathcal{F}, \mathcal{G})$ because of (16.8.4.1). More precisely, to the left $\mathcal{O}_X$ -module structure on $\mathcal{P}_{X/S}^n(\mathcal{F})$ corresponds, because of (16.8.1), the left $\mathcal{O}_X$ -module structure on $\mathcal{D}\mathrm{iff}_{X/S}^n(\mathcal{F}, \mathcal{G})$ explained as follows: for every open set U of X , every section $a \in \Gamma(U, \mathcal{O}_X)$ and every differential operator $D : \mathcal{F}|U \rightarrow \mathcal{G}|U$, aD is the differential operator which, for every section $t \in \Gamma(U, \mathcal{F})$, makes correspond the section

$$(16.8.5.1) \quad (aD)(t) = a(D(t))$$

of $\Gamma(U, \mathcal{G})$. Similarly, the right $\mathcal{O}_X$ -module structure on $\mathcal{D}\mathrm{iff}_{X/S}^n(\mathcal{F}, \mathcal{G})$ is made explicit as follows: under the same notations as above, Da is the operator which, to every $t \in \Gamma(U, \mathcal{F})$, makes correspond the section

$$(16.8.5.2) \quad (Da)(t) = D(at).$$

Proposition (16.8.6). — *If $f : X \rightarrow S$ is a morphism locally of finite presentation, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module of finite presentation and $\mathcal{G}$ a quasi-coherent $\mathcal{O}_X$ -module, then $\mathcal{D}\mathrm{iff}_{X/S}^n(\mathcal{F}, \mathcal{G})$ is a quasi-coherent $\mathcal{O}_X$ -module for any of the structures defined in (16.8.5).*

Proof. The proposition follows from the fact that, under these hypothesis, $\mathcal{P}_{X/S}^n$ is a quasi-coherent $\mathcal{O}_X$ -module of finite presentation (16.7.4) and of (I, 1.3.12) $\square$

(16.8.7). The set of differential operators (of unspecified order (16.8.1)) is denoted by $\mathrm{Diff}_{X/S}(\mathcal{F}, \mathcal{G})$; we also see as in (16.8.3) that $U \mapsto \mathrm{Diff}_{U/S}(\mathcal{F}|U, \mathcal{G}|U)$ is a sheaf of additive groups, which we will denote by $\mathcal{D}\mathrm{iff}_{X/S}(\mathcal{F}, \mathcal{G})$. It is immediate that $\mathcal{D}\mathrm{iff}_{X/S}(\mathcal{F}, \mathcal{G})$ is the reunion of the increasing filtered family of its subsheaves $\mathcal{D}\mathrm{iff}_{X/S}^n(\mathcal{F}, \mathcal{G})$; if X is quasi-compact, $\mathrm{Diff}_{X/S}(\mathcal{F}, \mathcal{G})$ is similarly the union of its subgroups $\mathrm{Diff}_{X/S}^n(\mathcal{F}, \mathcal{G})$ (16.8.1). The $\mathcal{O}_X$ -bimodule structure on the $\mathcal{D}\mathrm{iff}_{X/S}^n(\mathcal{F}, \mathcal{G})$ induce therefore an $\mathcal{O}_X$ -bimodule structure on $\mathcal{D}\mathrm{iff}_{X/S}(\mathcal{F}, \mathcal{G})$, further explained in (16.8.5.1) and (16.8.5.2).

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Note that, for $n \leq m$, we have a commutative diagram

$$(16.8.7.1) \quad \begin{array}{ccc} \mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{P}_{X/S}^n(\mathcal{F}), \mathcal{G}) & \xrightarrow{\sim} & \mathcal{D}\mathrm{iff}_{X/S}^n(\mathcal{F}, \mathcal{G}) \\ \downarrow & & \downarrow \\ \mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{P}_{X/S}^m(\mathcal{F}), \mathcal{G}) & \xrightarrow{\sim} & \mathcal{D}\mathrm{iff}_{X/S}^m(\mathcal{F}, \mathcal{G}) \end{array}$$

where the horizontal arrows are the isomorphisms (16.8.4.1) and the horizontal arrow on the left comes from the canonical morphism $\mathcal{P}_{X/S}^m(\mathcal{F}) \rightarrow \mathcal{P}_{X/S}^n(\mathcal{F})$ (16.7.7). For every open set U of X , we then endow $\Gamma(U, \mathcal{P}_{X/S}^\infty(\mathcal{F})) = \varprojlim \Gamma(U, \mathcal{P}_{X/S}^n(\mathcal{F}))$ of the projective limit topology of the discrete topologies on $\Gamma(U, \mathcal{P}_{X/S}^n(\mathcal{F}))$, which defines on $\Gamma(U, \mathcal{P}_{X/S}^\infty(\mathcal{F}))$ a topological $\Gamma(U, \mathcal{O}_X)$ -bimodule structure, so that $\mathcal{P}_{X/S}^\infty(\mathcal{F})$ shows itself as a sheaf valued in the category of topological commutative

groups (0, 3.2.6). So [God58, II, 1.11], the limit of the inductive system of sheaves of commutative groups $(\mathcal{H}om_{\mathcal{O}_X}(\mathcal{P}_{X/S}^n(\mathcal{F}), \mathcal{G}))$ is precisely the sheaf of *continuous* germs of homomorphisms from $\mathcal{P}_{X/S}^\infty(\mathcal{F})$ to $\mathcal{G}$ (the latter equipped with the discrete topology): the continuous homomorphisms $\Gamma(U, \mathcal{P}_{X/S}^\infty(\mathcal{F}))$ into the discrete group $\mathcal{G}$ indeed correspond bijectively to the inductive systems of group homomorphisms $\Gamma(U, \mathcal{P}_{X/S}^n(\mathcal{F})) \rightarrow \Gamma(U, \mathcal{G})$. We can furthermore express (16.8.4) by saying there is a canonical isomorphism

$$\mathcal{H}om.\text{cont}_{\mathcal{O}_X}(\mathcal{P}_{X/S}^\infty(\mathcal{F}), \mathcal{G}) \simeq \mathcal{D}iff_{X/S}(\mathcal{F}, \mathcal{G})$$

where the first member denotes the sheaf of germs of continuous homomorphisms from $\mathcal{P}_{X/S}^\infty(\mathcal{F})$ to $\mathcal{G}$.

Proposition (16.8.8). — *Let $\mathcal{F}, \mathcal{G}$ be two $\mathcal{O}_X$ -modules, $D : \mathcal{F} \rightarrow \mathcal{G}$ a homomorphism of $\psi^*(\mathcal{O}_S)$ -modules, n an integer ≥ 0 . The following conditions are equivalent:*

- (a) *D is a differential operator of order $\leq n$.*
- (b) *For all sections a of $\mathcal{O}_X$ over an open set U , the homomorphism $D_a : \mathcal{F}|_U \rightarrow \mathcal{G}|_U$ such that, for every section t of $\mathcal{F}$ over an open set $V \subset U$, we have*

$$(16.8.8.1) \quad D_a(t) = D(at) - aD(t)$$

is a differential operator of order $\leq n - 1$.

- (c) *For every open set U of X , every family $(a_i)_{1 \leq i \leq n+1}$ of $n + 1$ sections of $\mathcal{O}_X$ over U and every section t of $\mathcal{F}$ over U , we have the identity*

$$(16.8.8.2) \quad \sum_{H \subset I_{n+1}} (-1)^{\text{Card}(H)} \left(\prod_{i \in H} a_i \right) D \left(\left(\prod_{i \notin H} a_i \right) t \right) = 0$$

(where I_{n+1} is the interval $1 \leq i \leq n + 1$ of $\mathbb{N}$).

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Proof. Lets prove first the equivalence of (a) and (b). By definition, to prove that D is a differential operator of order $\leq n$, it suffices to prove so for the restriction $D|_U : \mathcal{F}|_U \rightarrow \mathcal{G}|_U$ to any affine open set U of X , and on the other hand the property (c) is valid for every open set U if it is so on every affine open set. We can therefore restrict ourselves to the case where $S = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affines. Because of (16.8.2) (where we use the same notations), the condition (a) means that the A -homomorphism $D' : B \otimes_A M \rightarrow N$ such that $D'(b \otimes t) = bD(t)$ is zero on $\mathfrak{I}^{n+1}(B \otimes_A M)$, which is, because of (0, 20.4.4), equivalent to saying that D' annihilates every element of the form

$$\left(\prod_{i=1}^{n+1} (a_i \otimes 1 - 1 \otimes a_i) \right) \cdot (1 \otimes t)$$

where $a_i \in B$ and $t \in M$. Now, this element can be written as $\sum_{H \subset I_{n+1}} (\prod_{i \in H} a_i) \otimes ((\prod_{i \notin H} a_i) t)$, and the image under D' of this element is the first member of (16.8.8.2), which proves the equivalence of (a) and (c).

Let us prove now the equivalence of (b) and (c). Let us reason by induction on n , the statement being trivial for $n = 0$. Writing a_{n+1} instead of a in the condition (b), we see, by the induction hypothesis, that condition (b) means that for every family $(a_i)_{1 \leq i \leq n}$ of n sections of $\mathcal{O}_X$ over U and every section t of $\mathcal{F}$ over U , that

$$\sum_{H' \subset I_{n+1}} (-1)^{\text{Card}(H')} \left(\prod_{i \in H'} a_i \right) D_{a_{n+1}} \left(\left(\prod_{i \notin H'} a_i \right) t \right) = 0.$$

But if we replace on this relation $D_{a_{n+1}}$ by the definition (16.8.8.1), we check immediately that we have, up to sign, the first member of (16.8.8.2); from which we conclude. $\square$

Proposition (16.8.9). — *If $D : \mathcal{F} \rightarrow \mathcal{G}$ is a differential operator of order $\leq n$, and $D' : \mathcal{G} \rightarrow \mathcal{H}$ a differential operator of order $\leq n'$, then $D' \circ D : \mathcal{F} \rightarrow \mathcal{H}$ is a differential operator of order $\leq n + n'$.*

Proof. By hypothesis, we can write $D = u \circ d_{X/S, \mathcal{F}}^n$ and $D' = v \circ d_{X/S, \mathcal{G}}^{n'}$, where $u : \mathcal{P}_{X/S}^n \otimes_{\mathcal{O}_X} \mathcal{F} \rightarrow \mathcal{G}$ and $v : \mathcal{P}_{X/S}^{n'} \otimes_{\mathcal{O}_X} \mathcal{G} \rightarrow \mathcal{H}$ are $\mathcal{O}_X$ -homomorphisms. It all comes down to showing that the

composite homomorphism of sheaves of additive groups

$$\mathcal{F} \xrightarrow{d_{X/S, \mathcal{F}}^n} \mathcal{P}_{X/S}^n \otimes_{\mathcal{O}_X} \mathcal{F} \xrightarrow{u} \mathcal{G} \xrightarrow{d_{X/S, \mathcal{G}}^{n'}} \mathcal{P}_{X/S}^{n'} \otimes_{\mathcal{O}_X} \mathcal{G}$$

factors as

$$\mathcal{F} \xrightarrow{d_{X/S, \mathcal{F}}^{n+n'}} \mathcal{P}_{X/S}^{n+n'} \otimes_{\mathcal{O}_X} \mathcal{F} \xrightarrow{w} \mathcal{P}_{X/S}^{n'} \otimes_{\mathcal{O}_X} \mathcal{G}$$

where w is an $\mathcal{O}_X$ -homomorphism. It suffices to prove the

Lemma (16.8.9.1). — *There is one and only one $\mathcal{O}_X$ -homomorphism*

$$(16.8.9.2) \quad \delta : \mathcal{P}_{X/S}^{n+n'} \longrightarrow \mathcal{P}_{X/S}^{n'}(\mathcal{P}_{X/S}^n) = \mathcal{P}_{X/S}^{n'} \otimes_{\mathcal{O}_X} \mathcal{P}_{X/S}^n$$

making the following diagram commute

$$(16.8.9.3) \quad \begin{array}{ccc} \mathcal{O}_X & \xrightarrow{d_{X/S}^{n+n'}} & \mathcal{P}_{X/S}^{n+n'} \\ d_{X/S}^n \downarrow & & \downarrow \delta \\ \mathcal{P}_{X/S}^n & \xrightarrow{d_{X/S, \mathcal{P}_{X/S}^n}^{n'}} & \mathcal{P}_{X/S}^{n'}(\mathcal{P}_{X/S}^n) \end{array}$$

We will then have, indeed, a commutative diagram deduced from (16.8.9.3) by tensorization with $\mathcal{F}$

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{d_{X/S, \mathcal{F}}^{n+n'}} & \mathcal{P}_{X/S}^{n+n'}(\mathcal{F}) \\ d_{X/S, \mathcal{F}}^n \downarrow & & \downarrow \delta \otimes 1 \\ \mathcal{P}_{X/S}^n(\mathcal{F}) & \xrightarrow{d_{X/S, \mathcal{P}_{X/S}^n(\mathcal{F})}^{n'}} & \mathcal{P}_{X/S}^{n'}(\mathcal{P}_{X/S}^n(\mathcal{F})) \end{array}$$

and on the other hand, we verify immediately the from definition (16.7.5) that the diagram

$$\begin{array}{ccc} \mathcal{P}_{X/S}^n(\mathcal{F}) & \xrightarrow{u} & \mathcal{G} \\ d_{X/S, \mathcal{P}_{X/S}^n(\mathcal{F})}^{n'} \downarrow & & \downarrow d_{X/S, \mathcal{G}}^{n'} \\ \mathcal{P}_{X/S}^{n'}(\mathcal{P}_{X/S}^n(\mathcal{F})) & \xrightarrow{1 \otimes u} & \mathcal{P}_{X/S}^{n'}(\mathcal{G}) \end{array}$$

is commutative. We finish the proof by taking w to be the composite $\mathcal{O}_X$ -homomorphism

$$\mathcal{P}_{X/S}^{n+n'}(\mathcal{F}) \xrightarrow{\delta \otimes 1} \mathcal{P}_{X/S}^{n'}(\mathcal{P}_{X/S}^n(\mathcal{F})) \xrightarrow{1 \otimes u} \mathcal{P}_{X/S}^{n'}(\mathcal{G}).$$

□

Proof (16.8.9.3). It remains to prove the lemma (16.8.9.3). Considering (16.7.6), which proves the uniqueness of δ , we are brought back to the case where $S = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affines; letting $\mathfrak{I} = \mathfrak{I}_{B/A}$, it suffices to define a canonical homomorphism of B -modules

$$\phi : (B \otimes_A B) / \mathfrak{I}^{n+n'+1} \longrightarrow ((B \otimes_A B) / \mathfrak{I}^{n'+1}) \otimes_B ((B \otimes_A B) / \mathfrak{I}^{n+1})$$

the B -module structure of the two members coming from the first B factor; recall that on tensor product of the second member, $(B \otimes_A B) / \mathfrak{I}^{n'+1}$ must be considered as a right B -module by its second B factor, and $(B \otimes_A B) / \mathfrak{I}^{n+1}$ as a left B -module by its first B factor (16.7.2). It is the same to define a homomorphism of B -modules

$$\phi_0 : B \otimes_A B \longrightarrow ((B \otimes_A B) / \mathfrak{I}^{n'+1}) \otimes_B ((B \otimes_A B) / \mathfrak{I}^{n+1})$$

and prove it is zero on $\mathfrak{I}^{n+n'+1}$. Now, we immediately define a homomorphism by the condition that

$$\phi_0(b \otimes b') = \pi_{n'}(b \otimes 1) \otimes \pi_n(1 \otimes b') \quad \text{for } b, b' \text{ in } B$$

under the notations of (16.3.7). Also, it is immediate that ϕ_0 is a homomorphism of *rings*. Now, we can write

$$\phi_0(b \otimes 1 - 1 \otimes b) = \pi_{n'}(b \otimes 1 - 1 \otimes b) \otimes \pi_n(1 \otimes 1) + \pi_{n'}(1 \otimes b) \otimes \pi_n(1 \otimes 1) - \pi_{n'}(1 \otimes 1) \otimes \pi_n(1 \otimes b)$$

and we have

$$\pi_{n'}(1 \otimes b) \otimes \pi_n(1 \otimes 1) = \pi_{n'}(1 \otimes 1)b \otimes \pi_n(1 \otimes 1) = \pi_{n'}(1 \otimes 1) \otimes b\pi_n(1 \otimes 1) = \pi_{n'}(1 \otimes 1) \otimes \pi_n(b \otimes 1)$$

from which, finally

$$(16.8.9.4) \quad \phi_0(b \otimes 1 - 1 \otimes b) = \pi_{n'}(b \otimes 1 - 1 \otimes b) \otimes \pi_n(1 \otimes 1) + \pi_{n'}(1 \otimes 1) \otimes \pi_n(b \otimes 1 - 1 \otimes b).$$

A product of $n + n' + 1$ of terms of the form (16.8.9.4) is therefore necessarily zero, because the same is true for the product of $n + 1$ terms of the form $\pi_n(b \otimes 1 - 1 \otimes b)$ and of $n' + 1$ terms of the form $\pi_{n'}(b \otimes 1 - 1 \otimes b)$. The conclusion therefore results from (0, 20.4.4). $\square$

Corollary (16.8.10). — *The sheaf $\mathcal{D}iff_{X/S}(\mathcal{O}_X, \mathcal{O}_X)$ (also denoted $\mathcal{D}iff_{X/S}$) is canonically endowed with the structure of sheaf of rings, and the $\mathcal{D}iff_{X/S}^n$ form an increasing filtration compatible with such structure.*

In particular, $\mathcal{D}iff_{X/S}^0$ is a sheaf of subrings of $\mathcal{D}iff_{X/S}$, which is canonically identified with $\mathcal{O}_X$ (16.8.1). The formulas (16.8.5.1) and (16.8.5.2) show that the structure of $\mathcal{O}_X$ -bimodule of $\mathcal{D}iff_{X/S}$ comes from the multiplication on the left and on the right by sections of $\mathcal{O}_X$ considered as a sheaf of subrings of $\mathcal{D}iff_{X/S}$.

Remarks (16.8.11). — (i) Suppose that $\mathcal{F} = \bigoplus_{\lambda \in L} \mathcal{F}_\lambda$; then it is clear (16.7.2.1) that $\mathcal{P}_{X/S}^n(\mathcal{F}) = \bigoplus_{\lambda \in L} \mathcal{P}_{X/S}^n(\mathcal{F}_\lambda)$; since the functor $U \mapsto \Gamma(U, \mathcal{F})$ commutes with the formation of arbitrary direct sums, $d_{X/S, \mathcal{F}}^n$ is the homomorphism whose restriction to each $\mathcal{F}_\lambda$ is $d_{X/S, \mathcal{F}_\lambda}^n : \mathcal{F}_\lambda \rightarrow \mathcal{P}_{X/S}^n(\mathcal{F}_\lambda)$; then we conclude immediately that we have

$$\text{Diff}_{X/S}^n(\mathcal{F}, \mathcal{G}) = \prod_{\lambda \in L} \text{Diff}_{X/S}^n(\mathcal{F}_\lambda, \mathcal{G}),$$

and therefore also (0, 3.2.6)

$$\mathcal{D}iff_{X/S}^n(\mathcal{F}, \mathcal{G}) = \prod_{\lambda \in L} \mathcal{D}iff_{X/S}^n(\mathcal{F}_\lambda, \mathcal{G}).$$

Moreover, if $\mathcal{G} = \prod_{\mu \in M} \mathcal{G}_\mu$ (0, 3.2.6), we have

$$\text{Hom}_{\mathcal{O}_X}(\mathcal{P}_{X/S}^n(\mathcal{F}), \mathcal{G}) = \prod_{\mu \in M} \text{Hom}_{\mathcal{O}_X}(\mathcal{P}_{X/S}^n(\mathcal{F}), \mathcal{G}_\mu),$$

every homomorphism u from $\mathcal{P}_{X/S}^n(\mathcal{F})$ to $\mathcal{G}$ corresponds bijectively to the family of its composites $u_\mu : \mathcal{P}_{X/S}^n(\mathcal{F}) \rightarrow \mathcal{G} \rightarrow \mathcal{G}_\mu$. We have therefore IV-4 | 46

$$\text{Diff}_{X/S}^n(\mathcal{F}, \mathcal{G}) = \prod_{\mu \in M} \text{Diff}_{X/S}^n(\mathcal{F}, \mathcal{G}_\mu),$$

and consequently also

$$\mathcal{D}iff_{X/S}^n(\mathcal{F}, \mathcal{G}) = \prod_{\mu \in M} \mathcal{D}iff_{X/S}^n(\mathcal{F}, \mathcal{G}_\mu).$$

(ii) So far, we have hardly encountered differential operators $\mathcal{F} \rightarrow \mathcal{G}$ where $\mathcal{F}$ and $\mathcal{G}$ are not locally free of finite rank, in which case the structure is reduced locally, because of (i), to the case of the sheaf $\mathcal{D}iff_{X/S}$; the latter will be studied later (16.11) in a particular case.

16.9. Regular and quasi-regular immersions

Definition (16.9.1). — Let X be a ringed space. We say that an ideal $\mathcal{I}$ of $\mathcal{O}_X$ is regular (resp. quasi-regular) if, for every point $x \in \text{Supp}(\mathcal{O}_X/\mathcal{I})$, there is an open neighbourhood U of x in X and a regular sequence (0, 15.2.2) (resp. quasi-regular (0, 15.2.2)) of elements of $\Gamma(U, \mathcal{O}_X)$ which generates $\mathcal{I}|_U$.

We say that a regular (resp. quasi-regular) sequence of sections of $\mathcal{O}_X$ over U which generates $\mathcal{I}|_U$ is called a *regular system* (resp. *quasi-regular system*) of generators of $\mathcal{I}|_U$.

Definition (16.9.2). — Let $j : Y \rightarrow X$ be an immersion of preschemes and let U be an open set such that $j(Y) \subset U$ and that j is a closed immersion of Y in U . We say that j is regular (resp. quasi-regular) if the closed subscheme $j(Y)$ of U associated to j is defined by a regular (resp. quasi-regular) ideal of $\mathcal{O}_U$ (condition independent of the choice of U).

We say that a subscheme Y of a prescheme X is *regularly immersed* (resp. *quasi-regularly immersed*) if the canonical injection $j : Y \rightarrow X$ is a regular immersion (resp. quasi-regular immersion). If Y is a closed subscheme and $\mathcal{I}$ is the ideal of $\mathcal{O}_X$ that defines Y , it is the same as asking $\mathcal{I}$ to be *regular* (resp. *quasi-regular*).

For example, if A is an integral ring, f and element $\neq 0$ of A , the closed subscheme $V(f)$ of $\text{Spec}(A)$ (isomorphic to $\text{Spec}(A/(f))$) is *regularly immersed* in $\text{Spec}(A)$.

Every regular ideal is quasi-regular (0, 15.2.2); every regular immersion is quasi-regular (cf. (16.9.11) for a converse).

Proposition (16.9.3). — Let X be a ringed space, $\mathcal{I}$ an ideal of $\mathcal{O}_X$, $(f_i)_{1 \leq i \leq m}$ a finite sequence of sections of $\mathcal{O}_X$ over X generating $\mathcal{I}$. For the (f_i) to be a quasi-regular sequence (0, 15.2.2), it is necessary and sufficient that the following conditions are verified:

- (i) The canonical images of f_i in $\mathcal{I}/\mathcal{I}^2$ form a basis of this $\mathcal{O}_X/\mathcal{I}$ -module.
- (ii) The canonical surjective homomorphism (16.1.2.2)

$$\mathbf{S}_{\mathcal{O}_X/\mathcal{I}}^\bullet(\mathcal{I}/\mathcal{I}^2) \longrightarrow \mathcal{G}_{\mathcal{I}}^\bullet(\mathcal{O}_X)$$

is bijective.

Also, if this is true, then every sequence $(f'_i)_{1 \leq i \leq n}$ of n sections of $\mathcal{I}$ over X which generates $\mathcal{I}$ is quasi-regular.

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Proof. The two conditions stated above only translate the ones in (0, 15.2.2), given the definition of the canonical homomorphisms (16.2.1.1). The last claim follows from the fact that, if a module M over a commutative ring A admits a basis of n elements, every system of n generators of M is a basis (Bourbaki, *Alg. comm.*, chap. II, §3, cor. 5 of th. 1). $\square$

Corollary (16.9.4). — Let X be a locally ringed space, $\mathcal{I}$ an ideal of $\mathcal{O}_X$. For $\mathcal{I}$ to be quasi-regular, it is necessary and sufficient that it verifies the following conditions:

- (i) $\mathcal{I}$ is of finite type.
- (ii) $\mathcal{I}/\mathcal{I}^2$ is a locally free $\mathcal{O}_X/\mathcal{I}$ -module.
- (iii) The canonical homomorphism

$$(16.9.4.1) \quad \mathbf{S}_{\mathcal{O}_X/\mathcal{I}}^\bullet(\mathcal{I}/\mathcal{I}^2) \longrightarrow \mathcal{G}_{\mathcal{I}}^\bullet(\mathcal{O}_X)$$

is bijective.

Proof. The necessity of the conditions follows immediately from (16.9.3). To see that they are sufficient, it is enough to show, because of (16.9.3), that if, in a point $x \in \text{Supp}(\mathcal{O}_X/\mathcal{I})$, there is a neighbourhood U of x in X and n -sections f_i ($1 \leq i \leq n$) of $\mathcal{I}$ over U such that the image in $\mathcal{I}/\mathcal{I}^2$ form a basis of $(\mathcal{I}/\mathcal{I}^2)|_U$ over $(\mathcal{O}_X/\mathcal{I})|_U$, so there will be a neighbourhood $V \subset U$ of x such that the $f_i|_V$ generate $\mathcal{I}|_V$. Now, by hypothesis, we have $\mathcal{I}_x \neq \mathcal{O}_x$, so that $\mathcal{I}_x$ is contained in the maximal ideal of $\mathcal{O}_x$; since $\mathcal{I}_x$ is an $\mathcal{O}_x$ module of finite type and the classes of $(f_i)_x$ in $\mathcal{I}_x/\mathcal{I}_x^2$ generate this $(\mathcal{O}_x/\mathcal{I}_x)$ -module, Nakayama's lemma shows that the $(f_i)_x$ generate $\mathcal{I}_x$. Since $\mathcal{I}$ is of finite type, we conclude by (0, 5.2.2). $\square$

Corollary (16.9.5). — Let X be a locally ringed space, $\mathcal{I}$ a quasi-regular ideal of $\mathcal{O}_X$, $(f_i)_{1 \leq i \leq n}$ a sequence of sections of $\mathcal{I}$ over X , x a point of $\text{Supp}(\mathcal{O}_X/\mathcal{I})$. The following conditions are equivalent:

- (a) There is a neighbourhood of x in X such that $f_i|_U$ form a quasi-regular sequence of elements $\Gamma(U, \mathcal{O}_X)$ generating $\mathcal{I}|_U$.
- (b) The $(f_i)_x$ form a system of generators of $\mathcal{I}_x$ whose size is as small as possible.
- (b') The $(f_i)_x$ form a minimal set of generators of $\mathcal{I}_x$.
- (c) If $\bar{f}_i$ is the canonical image of f_i in $\Gamma(X, \mathcal{I}/\mathcal{I}^2)$, the $(\bar{f}_i)_x$ form a basis for the $(\mathcal{O}_x/\mathcal{I}_x)$ -module $\mathcal{I}_x/\mathcal{I}_x^2$.

Proof. By hypothesis, $\mathcal{O}_x$ is a local ring, $\mathcal{I}_x$ an ideal of finite type of $\mathcal{O}_x$ contained in the maximal ideal of $\mathcal{O}_x$; the equivalence of (b), (b') and (c) results from Nakayama's lemma (Bourbaki, *Alg. comm.*, chap. II, §3, n°2, prop. 5). It is clear that (a) implies (c) because of (16.9.3); on the other hand, from (0, 5.2.2) it follows that if condition (c) is verified (and therefore so is (b)), there is a neighbourhood U of x in X such that $(\mathcal{I}/\mathcal{I}^2)|_U$ has constant rank equal to n , and that the $f_i|_U$ generate $\mathcal{I}|_U$; it suffices now to apply the last assertion of (16.9.3) to U . $\square$

Remarks (16.9.6). — (i) Under the general hypothesis of (16.9.5), for the sequence (f_i) to generate $\mathcal{I}$, it is not enough that the $(\bar{f}_i)_y$ form a basis of the $(\mathcal{O}_y/\mathcal{I}_y)$ -module $(\mathcal{I}_y/\mathcal{I}_y^2)$ for all $y \in X$. We have an example by taking $X = \text{Spec}(A)$, where A is a Dedekind ring, and $\mathcal{I} = \tilde{\mathfrak{I}}$, where $\mathfrak{I}$ is a *non-principal* ideal of A ; then indeed $\mathcal{I}_y/\mathcal{I}_y^2 = 0$ in every point y different from $x \in X$ corresponding to $\mathfrak{I}$, and $\mathcal{I}_x/\mathcal{I}_x^2$ has rank 1 over the field $\mathcal{O}_x/\mathcal{I}_x$; also, $\mathcal{I}$ is evidently a regular ideal.

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- (ii) In (16.9.5), one cannot replace “quasi-regular” by “regular”, even when X is a prescheme (cf. (16.9.12)). Indeed, denote by B the ring of germs of infinitely differentiable functions on the point 0 on $\mathbb{R}$; it has a maximal ideal $\mathfrak{m}$ generated by the germ of t of the identity mapping to 0, and the intersection $\mathfrak{n}$ of the $\mathfrak{m}^k$ for $k > 0$ is not reduced to 0. Now let A be the quotient ring $B[T]/\mathfrak{n}TB[T]$, and let f_1, f_2 be the canonical images in A of the elements t and T of $B[T]$. The sequence (f_1, f_2) is *regular* in A : indeed, f_1 is not a zero divisor in A , because the relation $tP[T] \in \mathfrak{n}TB[T]$, for a polynomial $P \in B[T]$, implies that the products of t by the coefficients of P belongs to the ideal $\mathfrak{n}$, and it results immediately that the coefficients are the same in $\mathfrak{n}$, so $P[T] \in \mathfrak{n}TB[T]$. Since B/tB is isomorphic to $\mathbb{R}$, A/f_1A is isomorphic to the ring of polynomials $\mathbb{R}[T]$, therefore integral, and the image of f_2 in A/f_1A , being equal to T , is not a zero divisor, so that our claim is true. However, f_2 is a zero divisor in A , since for every non-zero element $x \in \mathfrak{n}$, the image of x in A is $\neq 0$, but the image of xT is zero. We conclude that the sequence (f_2, f_1) is *not regular* in A ; on the other hand, the ideal $\mathfrak{I} = f_1A + f_2A$ is distinct from A , so the conditions (b), (b') and (c) of (16.9.5) do not imply the condition (a) when we replace “quasi-regular” by “regular”.

(16.9.7). If $X = \text{Spec}(A)$ is an affine scheme, we'll say that the ideal $\mathfrak{I}$ of A is *regular* (resp. *quasi-regular*) if the ideal $\mathcal{I} = \tilde{\mathfrak{I}}$ of $\mathcal{O}_X$ is regular (resp. quasi-regular); we note that this notion is *local* and does not imply the existence of a *system of generators* of $\mathfrak{I}$ forming in A a regular (resp. quasi-regular) sequence as the example (16.9.5) shows; however this is true if A is *local* (16.9.5).

The proposition (16.9.4) can be translated in terms of quasi-regular immersions in the following manner:

Proposition (16.9.8). — Let $j : Y \rightarrow X$ be a morphism of preschemes; for j to be a quasi-regular immersion, it is necessary and sufficient that j satisfies the following conditions:

- (i) j is an immersion locally of finite presentation.
- (ii) The conormal sheaf $\mathcal{G}^1(j) = \mathcal{N}_{Y/X}$ (16.1.2) is a locally free $\mathcal{O}_Y$ -module.
- (iii) The canonical homomorphism

$$\mathbf{S}_{\mathcal{O}_Y}^\bullet(\mathcal{G}^1(j)) \longrightarrow \mathcal{G}^\bullet(j)$$

(16.1.2.2) is bijective.

Proof. The problem being local on Y , we can restrict ourselves to the case where j is the canonical injection of a closed subprescheme Y of X , so the translation of (16.9.4) into (16.9.8) results from

the description of $\mathcal{G}^1(j)$ and $\mathcal{G}^\bullet(j)$ in terms of the ideal $\mathcal{I}$ of $\mathcal{O}_X$ defining the subprescheme Y (16.1.3, (ii)). $\square$

Corollary (16.9.9). — *Let Y be a prescheme, X an Y -prescheme, $j : Y \rightarrow X$ a Y -section of X , so that the n 'th normal invariant $\mathcal{A}^{(n)}$ of j (16.1.2) is an augmented $\mathcal{O}_Y$ -algebra (16.1.7); take $\mathcal{A}^{(\infty)} = \varprojlim \mathcal{A}^{(n)}$. For j to be a quasi-regular immersion it is necessary and sufficient that j is locally of finite presentation, and that every $y \in Y$ admits an affine open neighbourhood U of ring C such that $\mathcal{A}^{(\infty)}|_U$ is isomorphic as an augmented $\mathcal{O}_U$ -algebra to $\mathcal{O}_U[[T_1, \dots, T_n]]$.*

Proof. We can reduce to the case where j is a closed immersion by restricting to a small neighbourhood of y (see the argument on (16.4.11)), and therefore $\mathcal{O}_Y$ is identified with the quotient algebra $\mathcal{O}_X/\mathcal{I}$ and the canonical surjective homomorphism $\mathcal{O}_X \rightarrow \mathcal{O}_Y$ admits a right inverse (16.1.7). We can therefore suppose $X = \text{Spec}(B)$, $Y = \text{Spec}(A)$ are affines, B being then an augmented A -algebra, and the augmentation ideal $\mathfrak{J}$ being of finite type. Since $\mathcal{A}^{(n)}$ is identified with $(B/\mathfrak{J}^{n+1})^\sim$, the corollary follows from the equivalence of (b) and (c) in (0, 19.5.4) when $B/\mathfrak{J} = A$. $\square$

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We note that, on the affine case considered, the fact that j is a quasi-regular immersion is further equivalent, because of (0, 19.5.4), to saying that B is a *formally smooth* A -algebra for the $\mathfrak{J}$ -preadic topology.

We also note that the condition that j is locally of finite presentation is always satisfied when $X \rightarrow Y$ is locally of finite type (1.4.3, (v)).

Proposition (16.9.10). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X , $j : Y \rightarrow X$ the canonical injection, y a point of Y .*

- (i) *For there to exist an open neighbourhood U of y in X such that the restriction $Y \cap U \rightarrow U$ of j be a regular immersion, it is necessary and sufficient that the kernel $\mathcal{I}_y$ of the surjective homomorphism $\mathcal{O}_{X,y} \rightarrow \mathcal{O}_{Y,y}$ is generated by a regular sequence of elements of $\mathcal{O}_{X,y}$.*
- (ii) *For the immersion j to be regular, it is necessary and sufficient that it is quasi-regular.*

Proof. (i) We can reduce to the case where Y is a subprescheme defined by a coherent ideal $\mathcal{I}$ of $\mathcal{O}_X$. The condition is clearly necessary. Conversely, if $\mathcal{I}_y$ is generated by a regular sequence $(s_i)_y$, where the s_i are sections of $\mathcal{I}$ over an open neighbourhood U of y in X , we can suppose that the s_i generate $\mathcal{I}|_U$ (0, 5.5.2) and form a regular sequence (0, 15.2.4), from which the claim follows.

- (ii) The fact that a quasi-regular immersion is regular follows from (i) and the identification of quasi-regular and regular sequences of $\mathcal{O}_{X,y}$, formed from elements of the maximal ideal (0, 15.1.11) $\square$

Corollary (16.9.11). — *Let X be a locally Noetherian prescheme; then every quasi-regular ideal of $\mathcal{O}_X$ is regular.*

Remarks (16.9.12). — (i) We note that a regular immersion is not in general a flat morphism, and therefore *a fortiori* neither are quasi-regular morphisms in the sense of (6.8.1).

- (ii) Let A be a local Noetherian ring; it follows immediately from (16.9.4) and from (0, 17.1.1) that for A to be *regular*, it is necessary and sufficient that its maximal ideal $\mathfrak{m}$ is *quasi-regular* (or regular, which amounts to the same thing given that A is Noetherian). For an affine Noetherian scheme X to be *regular*, it is necessary and sufficient that for every closed point $x \in X$, the canonical injection $\text{Spec}(k(x)) \rightarrow X$ to be a *regular immersion*.

Proposition (16.9.13). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X , Y' a subprescheme of Y , such that the canonical injection $j : Y' \rightarrow Y$ is regular. Then the sequence of $\mathcal{O}_{Y'}$ -modules*

$$(16.9.13.1) \quad 0 \longrightarrow j^*(\mathcal{N}_{Y/X}) \longrightarrow \mathcal{N}_{Y'/X} \longrightarrow \mathcal{N}_{Y'/Y} \longrightarrow 0$$

is exact; furthermore, for every $x \in X$, there is an open neighbourhood U of x such that the restrictions to U of the homomorphisms of (16.9.13.1) form a split exact sequence.

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Let us first prove the following lemma:

Lemma (16.9.13.2). — Let A be a ring, $\mathfrak{I}$ an ideal of A , $A' = A/\mathfrak{I}$, $(f_i)_{1 \leq i \leq r}$ a sequence of elements of A which is A' -regular, $\mathfrak{K} = \sum_i f_i A$, $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, $\mathfrak{K}' = \sum_i f_i A'$, so that $C = A/\mathfrak{L}$ is isomorphic to $A'/\mathfrak{K}'$. For every integer $n > 0$, and every integer $N \geq n$, we have the relation

$$(16.9.13.3) \quad \mathfrak{I} \cap \mathfrak{K}^n = \mathfrak{I} \mathfrak{K}^n + \mathfrak{I} \mathfrak{K}^N.$$

Proof. It is clearly sufficient to prove that every element of the first is contained in the second, and, by induction on n , we reduce to the case $N = n + 1$. An element of the first member of (16.9.13.3), being in $\mathfrak{K}^n$, is written as $P(f_1, \dots, f_r)$, where $P \in A[T_1, \dots, T_r]$ is homogeneous of degree n . If f'_i is the canonical image of f_i in A' , the hypothesis $P(f_1, \dots, f_r) \in \mathfrak{I}$ means that $P(f'_1, \dots, f'_r) = 0$. But $P(f'_1, \dots, f'_r) \in \mathfrak{K}'^n$, so the canonical image of $P(f'_1, \dots, f'_r)$ in $\mathfrak{K}'^n / \mathfrak{K}'^{n+1}$ is zero. Now the hypothesis that (f_i) is A' -regular implies that the canonical homomorphism $S_{\mathbb{C}}^n(\mathfrak{K}' / \mathfrak{K}'^2) \rightarrow \mathfrak{K}'^n / \mathfrak{K}'^{n+1}$ is bijective (0, 15.1.9); we conclude that the coefficients of P are in $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$. It follows immediately that $P(f_1, \dots, f_r) \in \mathfrak{I} \mathfrak{K}^n + \mathfrak{K}^{n+1}$, and since $P(f_1, \dots, f_r) \in \mathfrak{I}$, we finally have $P(f_1, \dots, f_r) \in \mathfrak{I} \mathfrak{K}^n + \mathfrak{I} \mathfrak{K}^{n+1}$ which proves the lemma. $\square$

By taking the quotient of both side of (16.9.13.3) by $\mathfrak{I} \mathfrak{K}^n$, we see that the relations (16.9.13.3) for $N \geq n$ entail

$$(16.9.13.4) \quad (\mathfrak{I} \cap \mathfrak{K}^n) / \mathfrak{I} \mathfrak{K}^n \subset \bigcap_{N \geq n} \mathfrak{K}^N \cdot (A / (\mathfrak{I} \mathfrak{K}^n)).$$

We deduce the

Corollary (16.9.13.5). — Suppose that the hypothesis of (16.9.13.2) are verified and also that the ring A is Noetherian and $\mathfrak{K}$ is contained in the radical of A . Then for every integer $n > 0$,

$$(16.9.13.6) \quad \mathfrak{I} \cap \mathfrak{K}^n = \mathfrak{I} \mathfrak{K}^n.$$

Proof. Indeed, the second member of (16.9.13.4) is then zero, given that $A/\mathfrak{I} \mathfrak{K}^n$ is an A -module of finite type (Bourbaki, *Alg. comm.*, chap. III, §3, n°3, prop. 6). $\square$

Taking in particular $n = 2$ in (16.9.13.6), and remarking that we have $\mathfrak{L}^2 = \mathfrak{I}^2 + \mathfrak{I} \mathfrak{K} + \mathfrak{K}^2 = \mathfrak{I} \mathfrak{L} + \mathfrak{K}^2$; since $\mathfrak{I} \mathfrak{L} \subset \mathfrak{L}^2$, we deduce that

$$\mathfrak{I} \cap \mathfrak{L}^2 = \mathfrak{I} \mathfrak{L} + (\mathfrak{I} \cap \mathfrak{K}^2) = \mathfrak{I} \mathfrak{L} + \mathfrak{I} \mathfrak{K}^2 = \mathfrak{I} \mathfrak{L},$$

in other words,

$$(16.9.13.7) \quad \mathfrak{I} \cap \mathfrak{L}^2 = \mathfrak{I} \mathfrak{L},$$

which we can also express in saying that the canonical homomorphism

$$\mathfrak{I} / \mathfrak{I} \mathfrak{L} \longrightarrow (\mathfrak{I} + \mathfrak{I}^2) / \mathfrak{I}^2$$

is bijective.

Proof (16.9.13). Having demonstrated the lemmas, let us prove the first claim of (16.9.13): It is clearly enough to prove that the sequence of stalks of the sheaves appearing in (16.9.13.1), in a point $x \in Y'$, is exact. Now, if we take $A = \mathcal{O}_{X,x}$, we can write $\mathcal{O}_{Y',x} = A' = A/\mathfrak{I}$, where $\mathfrak{I}$ is an ideal contained in the maximal ideal of A , then $\mathcal{O}_{Y',x} = A'/\mathfrak{K}'$, where $\mathfrak{K}'$ is generated by an A' -regular sequence of elements of A' , which themselves are images of elements from a A' -regular sequence of elements of A belonging to the maximal ideal of A . If $\mathfrak{K}$ is the ideal generated by the latter and $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, we have $\mathcal{O}_{Y',x} = A/\mathfrak{L}$, and since we are in the situation of (16.9.13.5), the canonical homomorphism $\mathfrak{I} / \mathfrak{I} \mathfrak{L} \rightarrow (\mathfrak{I} + \mathfrak{I}^2) / \mathfrak{I}^2$ is bijective. But this shows that the sequence

$$0 \longrightarrow \mathfrak{I} / \mathfrak{I} \mathfrak{L} \longrightarrow \mathfrak{L} / \mathfrak{L}^2 \longrightarrow (\mathfrak{I} / \mathfrak{L}) / (\mathfrak{I} / \mathfrak{L})^2 \longrightarrow 0$$

is exact (see the demonstration of (16.2.7)), and the modules making up this sequence are precisely the stalks in x of the sheaves of (16.9.13.1). The second claim follows from the fact that $\mathcal{N}_{X/Y}$ is a locally free $\mathcal{O}_{Y'}$ -module (16.9.8) and Bourbaki, *Alg.*, chap. II, 3rded., §1, n°11, prop. 21. $\square$

16.10. Differentially smooth morphisms.

Definition (16.10.1). — We say that a morphism of preschemes $f : X \rightarrow S$ is differentially smooth (or that X is differentially smooth over S) if it satisfies the following conditions:

- (i) $\Omega_{X/S}^1$ is a locally projective $\mathcal{O}_X$ -module, which is to say that every point $x \in X$ admits an affine open neighbourhood U such that $\Gamma(U, \Omega_{X/S}^1)$ is a projective $\Gamma(U, \mathcal{O}_X)$ -module (not necessarily of finite type).
- (ii) The canonical morphism (16.3.1.1)

$$\mathbf{S}_{\mathcal{O}_X}^\bullet(\Omega_{X/S}^1) \longrightarrow \mathcal{G}_\bullet(\mathcal{P}_{X/S})$$

is bijective.

In particular, if $\Omega_{X/S}^1$ is locally free of finite rank, the $\mathcal{P}_{X/S}^n$ are locally free of finite rank $\mathcal{O}_X$ -modules (being extensions of such modules).

We say that f is *differentially smooth at a point* $x \in X$ if there is an open neighbourhood U of x in X such that $f|_U$ is differentially smooth.

We will see later (17.12.4) that a smooth morphism is differentially smooth, which justifies the terminology; but the converse is not true; indeed, a *monomorphism* $f : X \rightarrow S$ is differentially smooth, since $\Omega_{X/S}^1 = 0$ because of (I, 5.3.8), and consequently the surjective homomorphism (16.3.1.1) is clearly bijective; however a monomorphism is not even necessarily flat, neither *a fortiori* smooth. Let's limit ourselves to proving the following proposition:

Proposition (16.10.2). — Let A be a ring, B a formally smooth A -algebra for the discrete topology (0, 19.3.1). Then $\text{Spec}(B)$ is differentially smooth over $\text{Spec}(A)$.

Proof. Indeed, $B \otimes_A B$ is then (with the discrete topology) a formally smooth B -algebra (by on or the other canonical homomorphisms $b \mapsto b \otimes 1, b \mapsto 1 \otimes b$ of B to $B \otimes_A B$) (0, 19.3.5, (iii)); therefore $B \otimes_A B$ is a formally smooth A -algebra with the discrete topology (0, 19.3.5, (ii)). Taking $\mathfrak{I} = \mathfrak{I}_{B/A}$, it follows that $B \otimes_A B$ is a formally smooth A -algebra for the $\mathfrak{I}$ -preadic topology (0, 19.3.8); since by hypothesis $B = (B \otimes_A B)/\mathfrak{I}$ is a formally smooth A -algebra for the discrete topology, the proposition follows from the equivalence of (a) and (b) in (0, 19.5.4). $\square$

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Proposition (16.10.3). — For a morphism $f : X \rightarrow S$ to be differentially smooth, it is necessary and sufficient that for every $x \in X$, there is an affine open neighbourhood of x , of ring A , such that $\Gamma(U, \mathcal{P}_{X/S}^\infty)$ is an augmented topological A -algebra isomorphic to the completion $\widehat{B}$, where $B = \mathbf{S}_A(V)$, V being a projective A -module and B being endowed with the B^+ -preadic topology (where B^+ is the augmentation ideal). If $\Omega_{X/S}^1$ is locally free of finite rank, we can replace $\widehat{B}$ with the ring of formal series $A[[T_1, \dots, T_n]]$.

Proof. The notion of a differentially smooth morphism is clearly local on X , so we reduce to the case where $S = \text{Spec}(B)$, $X = \text{Spec}(C)$. Consider $C \otimes_B C$ as a C -algebra (by the first factor); take $\mathfrak{I} = \mathfrak{I}_{C/B}$ and endow C with the $\mathfrak{I}$ -preadic topology; we can apply to the topological C algebra $C \otimes_B C$ and to the ideal $\mathfrak{I}$ of $C \otimes_B C$ the equivalence of (b) and (c) of (0, 19.5.4), given that $(C \otimes_B C)/\mathfrak{I} = C$ is clearly a formally smooth C -algebra for the discrete topologies. The topology of $\Gamma(U, \mathcal{P}_{X/S}^\infty)$ is clearly the projective limit topology of this ring (16.1.11). $\square$

We note that the integer n of the proposition (16.10.3) is the *rank* of $\Omega_{X/S}^1$ in the point x . We shall see (17.13.5) that when f is differentially smooth and locally of finite type, n is equal to the dimension of the fiber $f^{-1}(f(x))$.

Proposition (16.10.4). — Let $f : X \rightarrow S$, $g : S' \rightarrow S$ be two morphisms, and take $X' = X \times_S S'$, $f' = f_{(S')} : X' \rightarrow S'$.

- (i) If f is differentially smooth, the same is true for f' .
- (ii) Conversely, if g is faithfully flat and quasi-compact, and if f' is differentially smooth and $\Omega_{X'/S'}^1$ is a finite type $\mathcal{O}_{X'}$ -module, f is differentially smooth and $\Omega_{X/S}^1$ is a finite type $\mathcal{O}_X$ -module.

Proof. Indeed, if f is differentially smooth, the $\mathcal{G}_n(\mathcal{P}_{X/S}^n)$ are flat $\mathcal{O}_X$ -modules; therefore by (16.4.6), the homomorphism $\mathcal{G}_n(\mathcal{P}_{X/S}^n) \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} \rightarrow \mathcal{G}_n(\mathcal{P}_{X'/S'}^n)$ are bijective for every n , because of the commutative of the diagram (16.2.1.3), it follows from the definition (16.10.1) that f' is differentially

smooth. On the other hand, if g is faithfully flat and quasi-compact, it follows also from (16.4.6) that $\mathcal{G}_n(\mathcal{P}_{X/S}^n) \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} \rightarrow \mathcal{G}_n(\mathcal{P}_{X'/S'}^n)$ is bijective for every n . Suppose also that f' is differentially smooth and $\Omega_{X'/S'}^1$ is of finite rank. Since the canonical projection $X' \rightarrow X$ is a faithfully flat and quasi-compact morphism, it results first from (2.5.2) that $\Omega_{X/S}^1$ is an $\mathcal{O}_X$ -module locally free of finite rank, then from (2.2.7) that the canonical homomorphism (16.3.1.1) is bijective, and therefore f is differentially smooth. $\square$

Proposition (16.10.5). — *For a morphism locally of finite type $f : X \rightarrow S$ to be differentially smooth, it is necessary and sufficient that the diagonal immersion $\Delta_f : X \rightarrow X \times_S X$ to be quasi-regular.*

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Proof. Being a local problem, we can reduce to the case where S and X are affines, and therefore the diagonal subscheme of $X \times_S X$ is closed. The hypothesis that f is locally of finite type implies that Δ_f is locally of finite presentation (I, 4.3.1), therefore the diagonal prescheme of $X \times_S X$ is defined by an ideal $\mathcal{I}$ of finite type, and $\Omega_{X/S}^1 = \mathcal{I} / \mathcal{I}^2$ is an $\mathcal{O}_X$ -module of finite type. The proposition is now immediate from the comparison of the conditions of (16.10.1) and (16.9.4). $\square$

Remark (16.10.6). — Let $f : X \rightarrow S$ be a morphism such that the $\mathcal{O}_X$ -module $\Omega_{X/S}^1$ is locally free of finite rank. It results from (0, 20.4.7) that every $x \in X$ has an open neighbourhood such that there is a finite family $(z_\lambda)_{\lambda \in L}$ of sections of $\mathcal{O}_X$ over U for which $(dz_\lambda)_{\lambda \in L}$ forms a *basis* of the $\Gamma(U, \mathcal{O}_X)$ -module $\Gamma(U, \Omega_{X/S}^1)$.

16.11. Differential operators on a differentially smooth S -prescheme

(16.11.1). Let $f : X \rightarrow S$ be a morphism, U an open set of X , $(z_\lambda)_{\lambda \in L}$ a family of sections of $\mathcal{O}_X$ over U such that the dz_λ form system of generators of $\Omega_{X/S}^1|_U = \Omega_{U/S}^1$. Let m be an integer or the symbol ∞ , and take, for every λ

$$(16.11.1.1) \quad \zeta_\lambda = \delta z_\lambda = d^m z_\lambda - z_\lambda \in \Gamma(U, \mathcal{P}_{X/S}^m).$$

We will also use the usual notations from analysis; for every $\mathbf{p} = (p_\lambda) \in \mathbb{N}^{(L)}$ (so that $p_\lambda = 0$ except for a finite number of indices), we put

$$(16.11.1.2) \quad |\mathbf{p}| = \sum_\lambda p_\lambda, \quad \mathbf{p}! = \prod_\lambda (p_\lambda!)$$

$$(16.11.1.3) \quad \binom{\mathbf{p}}{\mathbf{q}} = \mathbf{p}! / (\mathbf{q}!(\mathbf{p} - \mathbf{q})!), \quad \text{for } \mathbf{p}, \mathbf{q} \text{ in } \mathbb{N}^{(L)}, \mathbf{q} \leq \mathbf{p}$$

and we adopt the convention that $\binom{\mathbf{p}}{\mathbf{q}} = 0$ when $\mathbf{q} \not\leq \mathbf{p}$,

$$(16.11.1.4) \quad \mathbf{z}^{\mathbf{p}} = \prod_\lambda (z_\lambda)^{p_\lambda}, \quad \boldsymbol{\zeta}^{\mathbf{p}} = \prod_\lambda (\zeta_\lambda)^{p_\lambda}.$$

We therefore have, with this notation

$$(16.11.1.5) \quad d^m(\mathbf{z}^{\mathbf{p}}) = (d^m(\mathbf{z}))^{\mathbf{p}} = (\boldsymbol{\zeta} + \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q} \leq \mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}} \mathbf{z}^{\mathbf{p}-\mathbf{q}} \boldsymbol{\zeta}^{\mathbf{q}}$$

$$(16.11.1.6) \quad \boldsymbol{\zeta}^{\mathbf{p}} = (d^m \mathbf{z} - \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q} \leq \mathbf{p}} (-1)^{|\mathbf{p}-\mathbf{q}|} \binom{\mathbf{p}}{\mathbf{q}} \mathbf{z}^{\mathbf{p}-\mathbf{q}} d^m(\mathbf{z}^{\mathbf{q}}).$$

Since the dz_λ generate $\Omega_{X/S}^1$, and are the images of δz_λ , and as the canonical morphism (16.3.1.1) is surjective, we conclude that for finite m , the δz_λ generate the $\mathcal{O}_U$ -algebra $\mathcal{P}_{U/S}^m$ (Bourbaki, *Alg. comm.*, chap. III, §2, n°8, cor. 2 du th. 1). Therefore the $\boldsymbol{\zeta}^{\mathbf{p}}$ (for $|\mathbf{p}| \leq m$) generate the $\mathcal{O}_U$ -module $\mathcal{P}_{U/S}^m$. A differential operator $D \in \text{Diff}_{U/S}^m$ is consequently entirely determined by the values of $\langle \boldsymbol{\zeta}^{\mathbf{p}}, D \rangle$ for $|\mathbf{p}| \leq m$, or, which amounts to the same by (16.11.1.5) and (16.11.1.6), by the values of the $\langle d^m(\mathbf{z}), D \rangle = D(\mathbf{z}^{\mathbf{p}})$ for $|\mathbf{p}| \leq m$; more precisely, it follows from (16.11.1.5) that we have

$$(16.11.1.7) \quad D(\mathbf{z}^{\mathbf{p}}) = \langle d^m(\mathbf{z}^{\mathbf{p}}), D \rangle = \sum_{\mathbf{q} \leq \mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}} \langle \boldsymbol{\zeta}^{\mathbf{q}}, D \rangle \mathbf{z}^{\mathbf{p}-\mathbf{q}}.$$

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Theorem (16.11.2). — Let $f : X \rightarrow S$ be a morphism, U an open set of X , $(z_\lambda)_{\lambda \in L}$ a family of sections of $\mathcal{O}_X$ over U such that the family $(dz_\lambda)_{\lambda \in L}$ generates $\Omega_{X/S}^1|_U = \Omega_{U/S}^1$. The following conditions are equivalent:

- (a) $f|_U$ is differentially smooth and (dz_λ) is a basis of the $\mathcal{O}_U$ -module $\Omega_{U/S}^1$.
- (b) There is a family $(D_{\mathbf{p}})_{\mathbf{p} \in \mathbb{N}^{(L)}}$ of differential operators of $\mathcal{O}_U$ to itself verifying the conditions

$$(16.11.2.1) \quad D_{\mathbf{p}}(z^{\mathbf{q}}) = \binom{\mathbf{q}}{\mathbf{p}} z^{\mathbf{q}-\mathbf{p}}, \quad (\mathbf{p}, \mathbf{q} \text{ in } \mathbb{N}^{(L)}).$$

Also, when these conditions are verified, the family $(D_{\mathbf{p}})$ is uniquely determined by the conditions (16.11.2.1) and satisfies the relations

$$(16.11.2.2) \quad D_{\mathbf{q}} \circ D_{\mathbf{p}} = D_{\mathbf{p}} \circ D_{\mathbf{q}} = \frac{(\mathbf{p} + \mathbf{q})!}{\mathbf{p}! \mathbf{q}!} D_{\mathbf{p} + \mathbf{q}} \quad (\mathbf{p}, \mathbf{q} \text{ in } \mathbb{N}^{(L)}).$$

Finally, if L is finite, for every integer m , the $D_{\mathbf{p}}$ such that $|\mathbf{p}| \leq m$ form a basis of the $\mathcal{O}_U$ -module $\mathcal{D}iff_{U/S}^m$, in other words, every differential operator of order $\leq m$ on U can be written uniquely as

$$D = \sum_{|\mathbf{p}| \leq m} a_{\mathbf{p}} D_{\mathbf{p}}$$

where the $a_{\mathbf{p}}$ are sections of $\mathcal{O}_X$ over U .

Proof. Note first that because of (16.11.1.6) and (16.11.1.5), we verify immediately that the conditions (16.11.2.1) are equivalent to

$$(16.11.2.3) \quad \langle \zeta^{\mathbf{p}}, D_{\mathbf{q}} \rangle = \delta_{\mathbf{p}\mathbf{q}} \quad (\text{Kronecker's symbol}).$$

The existence of the family $(D_{\mathbf{p}})$ verifying these conditions implies first (by taking $|\mathbf{p}| = 1$) that the dz_λ are linearly independent, and therefore form a basis of the $\mathcal{O}_U$ -module $\Omega_{U/S}^1$. Then, for every integer $m \geq 1$, we deduce similarly from (16.11.2.3) that the $\zeta^{\mathbf{p}}$ such that $|\mathbf{p}| \leq m$ are linearly independent; it follows that the canonical homomorphism (16.3.1.1) is injective, and therefore bijective, which proves that (b) implies (a). The converse follows immediately from the definition (16.10.1), the fact that $\zeta^{\mathbf{p}}$ form a basis of $\mathcal{P}_{U/S}^n$ for $|\mathbf{p}| \leq m$ implies the existence and uniqueness of a family of homomorphisms $u_{\mathbf{q},m} : \mathcal{P}_{U/S}^n \rightarrow \mathcal{O}_U$ ($|\mathbf{q}| \leq m$) such that $\langle \zeta^{\mathbf{p}}, u_{\mathbf{q},m} \rangle = \delta_{\mathbf{p}\mathbf{q}}$ for $|\mathbf{p}| \leq m$, $|\mathbf{q}| \leq m$. For a given value of $\mathbf{q}$, the differential operators corresponding to $u_{\mathbf{q},m}$ for $m \geq |\mathbf{q}|$ are identified with the same operator $D_{\mathbf{q}}$. This proves that (a) implies (b), and also that the family $(D_{\mathbf{q}})$ is uniquely determined, and that, if L is finite, for $|\mathbf{p}| \leq m$, the $D_{\mathbf{p}}$ form a basis of the dual $\mathcal{D}iff_{U/S}^m$ of $\mathcal{P}_{U/S}^n$. Finally, the relations (16.11.2.2) follows immediately from the expression of the values of the three operators considered on the $z^{\mathbf{r}}$, and of the fact that the $\zeta^{\mathbf{r}}$ for $|\mathbf{r}| \leq m$ generate $\mathcal{P}_{U/S}^n$. $\square$

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- Remarks (16.11.3).** —
- (i) The fact that, because of (16.11.2.2), the $D_{\mathbf{p}}$ are pairwise permutable naturally do not imply that the $\mathcal{O}_U$ -algebra $\mathcal{D}iff_{U/S}$ is commutative, the $D_{\mathbf{p}}$ do not commute with the sections of $\mathcal{O}_U$ unless $n = 0$.
 - (ii) The indices $\mathbf{p}$ such that $|\mathbf{p}| = 1$ are the $\epsilon_\lambda = (\epsilon_{\lambda\mu})_{\mu \in L}$ where $\epsilon_{\lambda\mu} = 0$ if $\mu \neq \lambda$ and $\epsilon_{\lambda\lambda} = 1$; when L is finite, the operators D_{ϵ_λ} are exactly the S -derivations D_i introduced in (16.5.7). We note that in general (contrary to what happens in classical analysis), it is not true that a differential operator of any order can be written as a linear combination of powers of D_i (cf. (16.12)).
 - (iii) For every integer $r \geq 1$, we can define the notion of *differentially smooth up to order r* by replacing in (16.10.1) the condition (ii) by the condition that the homomorphisms

$$\mathbf{S}_{\mathcal{O}_X}^m(\Omega_{X/S}^1) \longrightarrow \mathcal{G}_m(\mathcal{P}_{X/S}^n)$$

are bijective for every $m \leq r$. The argument of (16.11.2) proves also that if, in the condition (a), we replace “differentially smooth” by “differentially smooth up to order r ”, this condition is equivalent to (b) by restricting ourselves to $\mathbf{p} \in \mathbb{N}^{(L)}$, $\mathbf{q} \in \mathbb{N}^{(L)}$ such that $|\mathbf{p}| \leq r$, $|\mathbf{q}| \leq r$.

16.12. Case of characteristic zero: Jacobian criterion for differentially smooth morphisms.

(16.12.1). We say that a prescheme X is of characteristic p (p equal to zero or a prime number) if, for every affine open set U of X , the ring $\Gamma(U, \mathcal{O}_X)$ is of characteristic p (0, 21.1.1). It follows from (0, 21.1.3) that for X to be of characteristic 0, it is necessary and sufficient that for every closed point x of X , the residue field $k(x)$ is of characteristic 0, or even that X can be given a structure of $\mathbb{Q}$ -prescheme (necessarily unique).

Theorem (16.12.2). — *Let X be a scheme of characteristic 0, $f : X \rightarrow S$ a morphism. If $\Omega_{X/S}^1$ is a locally free $\mathcal{O}_X$ -module (not necessarily of finite type), f is differentially smooth.*

Proof. The problem being local on X , we can suppose there is a family (z_λ) of sections of $\mathcal{O}_X$ over X such that the (dz_λ) is a basis for the $\mathcal{O}_X$ -module $\Omega_{X/S}^1$. Applying the criterion (16.11.2), it is enough for the operators

$$D_{\mathbf{p}} = (\mathbf{p}!)^{-1} \prod_{\lambda} D_{\lambda}^{p_{\lambda}}$$

(where the D_{λ} are the coordinate forms corresponding to the basis (dz_{λ})) to verify the relations (16.11.2.1), which is a consequence of the fact that the D_{λ} are derivations. $\square$

(16.12.3). The theorem above is not true if we discard the hypothesis that X is of characteristic 0. For example, if $S = \text{Spec}(k)$, where k is a field of characteristic $p > 0$, $X = \text{Spec}(K)$ where $K = k(\alpha)$ where $\alpha \notin k$, $\alpha^p \in k$, we verify immediately that $\Omega_{X/S}^1$ has rank 1, and that the morphism $X \rightarrow S$ has rank 1, and that the morphism $X \rightarrow S$ is differentially smooth up to order $p - 1$ (16.11.3, (iii)), but not of order p . However, the proof of (16.12.2) proves that if $\Omega_{X/S}^1$ is locally free, and if $n! \mathcal{O}_X$ is invertible in $\Gamma(X, \mathcal{O}_X)$, then X is differentially smooth over S up to order n .

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§17. SMOOTH MORPHISMS, UNRAMIFIED (OR NET) MORPHISMS, AND ÉTALE MORPHISMS.

In this paragraph, we revisit the concepts studied in (0_{III}, 9), expressed in the geometric language of schemes from a global point of view, for preschemes locally of finite presentation over a given base.

Most of the results (except 17.7, 17.8, 17.9, 17.13, and 17.16) are reduced to various properties already encountered in (0_{III}, 9).

For more specific results on étale morphisms, the reader should consult §18.

17.1. Formally smooth morphisms, formally unramified morphisms, formally étale morphisms.

Definition (17.1.1). — Let $f : X \rightarrow Y$ be a morphism of preschemes. We say that f is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) if, for all affine schemes Y' , all closed subschemes Y'_0 of Y' defined by a nilpotent ideal $\mathcal{J}$ of $\mathcal{O}_{Y'}$, and every morphism $Y' \rightarrow Y$, the map

$$(17.1.1.1) \quad \text{Hom}_Y(Y', X) \longrightarrow \text{Hom}_Y(Y'_0, X)$$

induced by the canonical map $Y'_0 \rightarrow Y'$, is *surjective* (resp. *injective*, resp. *bijective*).

One also says that X is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) over Y .

It is clear that for f to be formally étale, it is necessary and sufficient for f to be formally smooth and formally unramified.

Remark (17.1.2). —

- (i) Suppose that $Y = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affine, so that f comes from a homomorphism of rings $\phi : A \rightarrow B$. According to (0, 19.3.1) and (0, 19.10.1), saying that f is formally smooth (resp. formally unramified, resp. formally étale) means that, via ϕ , B is a *formally smooth* (resp. *formally unramified*, resp. *formally étale*) A -algebra, for the discrete topologies on A and B .
- (ii) To verify that f is formally smooth (resp. formally unramified, resp. formally étale), we can, in Definition (17.1.1), restrict to the case where $\mathcal{J}^2 = 0$. To see this, if f satisfies the corresponding condition of Definition (17.1.1) in the particular case $\mathcal{J}^2 = 0$, and if we have $\mathcal{J}^n = 0$, then we consider the closed subscheme Y'_j of Y' defined by the sheaf of ideals

$\mathcal{I}^{j+1}$ for $0 \leq j \leq n-1$, so that Y'_j is a closed subscheme of Y'_{j+1} defined by a square-zero sheaf of ideals; the hypotheses imply that each of the maps

$$\mathrm{Hom}_Y(Y'_{j+1}, X) \longrightarrow \mathrm{Hom}_Y(Y'_j, X) \quad (0 \leq j \leq n-1)$$

is surjective (resp. injective, resp. bijective); by composition, we conclude that the same holds for (17.1.1.1). IV | 57

- (iii) Note that the properties of the morphism f defined in (17.1.1) are properties of the *representable functor* (0III, 8.1.8)

$$Y' \longmapsto \mathrm{Hom}_Y(Y', X)$$

from the category of Y -preschemes to the category of sets; they keep a meaning for *any* contravariant functor with the same domain and codomain, representable or not.

- (iv) Assume that the morphism f is formally unramified (resp. formally étale); consider an arbitrary Y -prescheme Z and a closed subscheme Z_0 of Z defined by a *locally nilpotent* sheaf of ideals $\mathcal{I}$ of $\mathcal{O}_Z$. Then the map

$$(17.1.2.1) \quad \mathrm{Hom}_Y(Z, X) \longrightarrow \mathrm{Hom}_Y(Z_0, X)$$

induced by the canonical injection $Z_0 \rightarrow Z$, is still injective (resp. bijective). To see this, let (U_α) be an affine open covering of Z such that the sheaves of ideals $\mathcal{I}|_{U_\alpha}$ are nilpotent, and for each α , let U_α^0 be the inverse image of U_α in Z_0 , which is the closed subscheme of U_α defined by $\mathcal{I}|_{U_\alpha}$. Let $f_0 : Z_0 \rightarrow X$ be a Y -morphism; by hypothesis, for each α , there is at most one (resp. one and only one) Y -morphism $f_\alpha : U_\alpha \rightarrow X$ whose restriction to Z_0 coincides with $f_0|_{U_\alpha}$. We immediately conclude that if f_α and f_β are defined, then, for each affine open $V \subset U_\alpha \cap U_\beta$, we have $f_\alpha|_V = f_\beta|_V$, as the restrictions of these morphisms to the inverse image V_0 of V in Z_0 coincide. There is therefore at most one (resp. one and only one) Y -morphism $f : Z \rightarrow X$ whose restriction to Z_0 coincides with f_0 .

Proposition (17.1.3). —

- (i) A monomorphism of preschemes is formally unramified; an open immersion is formally étale.
- (ii) The composition of two formally smooth (resp. formally unramified, resp. formally étale) morphisms is formally smooth (resp. formally unramified, resp. formally étale).
- (iii) If $f : X \rightarrow Y$ is a formally smooth (resp. formally unramified, resp. formally étale) S -morphism, then so is $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ for any base extension $S' \rightarrow S$.
- (iv) If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are two formally smooth (resp. formally unramified, resp. formally étale) S -morphisms, then so is $f \times_S g : X \times_S Y \rightarrow X' \times_S Y'$.
- (v) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms; if $g \circ f$ is formally unramified, then so is f .
- (vi) If $f : X \rightarrow Y$ is a formally unramified morphism, then so is $f_{\mathrm{red}} : X_{\mathrm{red}} \rightarrow Y_{\mathrm{red}}$.

Proof. According to (I, 5.5.12), it suffices to prove (i), (ii), and (iii). The assertions in (i) are both trivial. To prove (ii), consider two morphisms $f : X \rightarrow Y$, $g : Y \rightarrow Z$, an affine scheme Z' , a closed subscheme Z'_0 of Z defined by a nilpotent ideal and a morphism $Z' \rightarrow Z$. Suppose that f and g are formally smooth, and consider a Z -morphism $u_0 : Z'_0 \rightarrow X$; the hypothesis on g implies that there exists a Z -morphism $v : Z' \rightarrow Y$ such that $f \circ u_0 = v \circ j$ (where $j : Z'_0 \rightarrow Z$ is the canonical injection); the hypothesis on f then implies that there exists a morphism $u : Z' \rightarrow X$ such that $f \circ u = v$ and $u \circ j = u_0$, therefore $(g \circ f) \circ u$ is equal to the given morphism $Z' \rightarrow Z$ and $u \circ j = u_0$, which proves that $g \circ f$ is formally smooth; we argue the same way when we suppose that f and g are formally unramified. IV | 58

Finally, to prove (iii), let $X' = X_{S'}$, $Y' = Y_{S'}$, $f' = f_{S'}$; consider an affine scheme Y'' , a closed subscheme Y''_0 defined by a nilpotent sheaf of ideals, and a morphism $g : Y'' \rightarrow Y'$ making Y'' a Y' -prescheme; we then know by (I, 3.3.8) that $\mathrm{Hom}_{Y'}(Y'', X')$ is canonically identified with $\mathrm{Hom}_Y(Y'', X)$, and $\mathrm{Hom}_{Y'}(Y''_0, X')$ with $\mathrm{Hom}_Y(Y''_0, X)$, and the conclusion follows immediately from Definition (17.1.1). □

We note that a *closed immersion* is not necessarily formally smooth.

Proposition (17.1.4). — Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms, and suppose that g is formally unramified. Then, if $g \circ f$ is formally smooth (resp. formally étale), so is f .

Proof. Let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent sheaf of ideals, $h : Y' \rightarrow Y$ a morphism, $j : Y'_0 \rightarrow Y'$ the canonical injection, $u_0 : Y'_0 \rightarrow Y$ a Y -morphism, such that $f \circ u_0 = h \circ j$. Suppose that $g \circ f$ is formally smooth; then there exists a morphism $u : Y' \rightarrow X$ such that $u \circ j = u_0$ and $(g \circ f) \circ u = g \circ h$. But these two relations imply that $f \circ u$ and h are Z -morphisms from Y' to Y such that $(f \circ u) \circ j = h \circ j$; by virtue of the hypothesis that g is formally unramified, we get that $f \circ u = h$, in other words that u is a Y -morphism; thus f is formally smooth. Taking into account (17.1.3, (v)), this proves the proposition. $\square$

Corollary (17.1.5). — Suppose that g is formally étale; then, for $g \circ f$ to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that f is.

Proof. This follows from (17.1.4) and (17.1.3, (ii) and (iv)). $\square$

Proposition (17.1.6). — Let $f : X \rightarrow Y$ be a morphism of preschemes.

- (i) Let (U_α) be an open covering of X and, for each α , let $i_\alpha : U_\alpha \rightarrow X$ be the canonical injection. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each $f \circ i_\alpha$ is.
- (ii) Let (V_λ) be an open covering of Y . For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each of the restrictions $f^{-1}(V_\lambda) \rightarrow V_\lambda$ of f is.

Proof. First note that (ii) is a consequence of (i): if $j_\lambda : V_\lambda \rightarrow Y$ and $i_\lambda : f^{-1}(V_\lambda) \rightarrow X$ are the canonical injections, then the restriction $f_\lambda : f^{-1}(V_\lambda) \rightarrow V_\lambda$ of f is such that $j_\lambda \circ f_\lambda = f \circ i_\lambda$; if f is formally smooth (resp. formally unramified), then so is $f \circ i_\lambda$ since i_λ is formally étale (17.1.3); but since j_λ is formally étale, this means that f_λ is formally smooth (resp. formally unramified), by virtue of (17.1.5). Conversely, if all the f_λ are formally smooth (resp. formally unramified), the same applies to $j_\lambda \circ f_\lambda$ (17.1.3), so also to f in virtue of (i).

If we take into account that the i_α are formally étale, everything comes down to proving that if $f \circ i_\alpha$ are formally smooth (resp. formally unramified), then the same applies to f . IV | 59

Therefore let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent ideal $\mathcal{J}$, which we may assume to satisfy $\mathcal{J}^2 = 0$ (17.1.2, (ii)), and finally let $g : Y' \rightarrow Y$ be a morphism. Suppose we are given a Y -morphism $u_0 : Y'_0 \rightarrow Y$; denote by W_α (resp. W_α^0) the prescheme induced by Y' (resp. Y'_0) on the open subset $u_0^{-1}(U_\alpha)$ (we recall that Y' and Y'_0 share the same underlying topological space). Let us first suppose that the $f \circ i_\alpha$ are formally unramified, and show that, if u' and u'' are two Y -morphisms from Y' to X whose restrictions to Y'_0 coincide, then we have $u' = u''$. Indeed, taking into account (17.1.2, (iv)), the hypothesis that the $f \circ i_\alpha$ are formally unramified implies that for all α , we have $u'|_{W_\alpha} = u''|_{W_\alpha}$, since the restrictions of both Y -morphisms to W_α^0 coincide. Hence the conclusion follows.

Now suppose that the $f \circ i_\alpha$ are formally smooth and prove the existence of a Y -morphism $u : Y' \rightarrow X$ whose restriction to Y'_0 is u_0 . Now, since Y' is an affine scheme, we can apply (16.5.17), the hypotheses of which are satisfied, and the conclusion of which precisely proves the existence of u . $\square$

We can therefore say that the notions introduced in (17.1.1) are *local* on X and Y , which always allows, in virtue of (17.1.2, (i)), to be reduced to the study of formally smooth (resp. formally unramified, resp. formally étale) *algebras*.

17.2. General properties of differentials

Proposition (17.2.1). — *For a morphism $f : X \rightarrow Y$ to be formally unramified, it is necessary and sufficient that $\Omega_f^1 = 0$ (what we still write $\Omega_{X/Y}^1 = 0$ (16.3.1)).*

Proof. Taking into account (17.1.6), we reduce to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and the conclusion then follows from (0, 20.7.4) and the interpretation of $\Omega_{X/Y}^1$ in this case (16.3.7). $\square$

Corollary (17.2.2). — *Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms. For f being formally unramified, it is necessary and sufficient that the canonical morphism (16.4.19)*

$$f^*(\Omega_{Y/Z}^1) \longrightarrow \Omega_{X/Z}^1$$

is surjective.

Proof. This is an immediate consequence of (17.2.1) and the exact sequence (16.4.19.1). $\square$

Proposition (17.2.3). — *Let $f : X \rightarrow Y$ be a formally smooth morphism.*

- (i) *The $\mathcal{O}_X$ -module $\Omega_{X/Y}^1$ is locally projective (16.10.1). If f is locally of finite type, then $\Omega_{X/Y}^1$ is locally free and of finite type.*
- (ii) *For all morphisms $g : Y \rightarrow Z$, the sequence (16.4.19) of $\mathcal{O}_X$ -modules*

$$(17.2.3.1) \quad 0 \longrightarrow f^*(\Omega_{Y/Z}^1) \longrightarrow \Omega_{X/Z}^1 \longrightarrow \Omega_{X/Y}^1 \longrightarrow 0$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.3.1) form a split exact sequence.

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Proof.

- (i) We know (16.3.9) that if f is locally of finite type, then Ω_f^1 is an $\mathcal{O}_X$ -module of finite type. To prove that, in all cases, it is locally projective, we can reduce, by virtue of (17.1.6), to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and the result follows from the hypothesis on f and from (0, 20.4.9) and (0, 19.2.1).
- (ii) Again, we can restrict to the case where X , Y , and Z are affine (17.1.6), and the conclusion in this case follows from the interpretation of the sheaves of modules in the sequence (17.2.3.1) and from (0, 20.5.7). $\square$

Corollary (17.2.4). — *If $f : X \rightarrow Y$ is formally étale, then, for all morphisms $g : Y \rightarrow Z$, the canonical homomorphism of $\mathcal{O}_X$ -modules*

$$f^*(\Omega_{Y/Z}^1) \longrightarrow \Omega_{X/Z}^1$$

is bijective.

Proof. This follows from the exactness of the sequence (17.2.3.1) and from the fact that we then have $\Omega_{X/Y}^1 = 0$ (17.2.1). $\square$

Proposition (17.2.5). — *Let $f : X \rightarrow Y$ be a morphism, X' a subprescheme of X such that the composite morphism $X' \xrightarrow{j} X \xrightarrow{f} Y$ (where j is the canonical injection) is formally smooth. Then the sequence of $\mathcal{O}_X$ -modules (16.4.21)*

$$(17.2.5.1) \quad 0 \longrightarrow \mathcal{N}_{X'/X} \longrightarrow \Omega_{X/Y}^1 \otimes_{\mathcal{O}_X} \mathcal{O}_{X'} \longrightarrow \Omega_{X'/Y}^1 \longrightarrow 0$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.5.1) form a split exact sequence.

Proof. By virtue of (17.1.6), we reduce to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and $X' = \operatorname{Spec}(B/\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of B . The conormal sheaf $\mathcal{N}_{X'/X}$ then corresponds to the B -module $\mathfrak{J}/\mathfrak{J}^2$ (16.1.3), and the conclusion follows from (0, 20.5.14). $\square$

Proposition (17.2.6). — *Let X and Y be two preschemes, $f : X \rightarrow Y$ a morphism locally of finite type. The following conditions are equivalent:*

- (a) f is a monomorphism.
- (b) f is radicial and formally unramified.
- (c) For each $y \in Y$, the fibre $f^{-1}(y)$ is empty or $k(y)$ -isomorphic to $\operatorname{Spec}(k(y))$ (in other words, it is reduced to a single point z such that $k(y) \rightarrow \mathcal{O}_z/\mathfrak{m}_y\mathcal{O}_z$ is an isomorphism).

Proof. The fact that (a) implies (c) follows from (8.11.5.1). It is clear that (c) implies that f is radicial; let us prove that it also follows from (c) that $\Omega_{X/Y}^1 = 0$, which will prove that (c) implies (b) (17.2.1). Note that the $\mathcal{O}_X$ -module $\Omega_{X/Y}^1$ is quasi-coherent of finite type (16.3.9). It follows from (I, 9.1.13.1) that, for $(\Omega_{X/Y}^1)_x = 0$, it is necessary and sufficient that if we set $Y_1 = \operatorname{Spec}(k(y))$, $X_1 = f^{-1}(y) = X \times_Y Y_1$, then we have $(\Omega_{X_1/Y_1}^1)_x = 0$; but as the morphism $f_1 : X_1 \rightarrow Y_1$ induced by f is formally unramified by virtue of the hypothesis (c) (17.1.3), the conclusion follows from (17.2.1). Finally, let us prove that (b) implies (a); for this, consider the diagonal morphism $g = \Delta_f : X \rightarrow X \times_Y X$; since f is radicial, g is surjective (1.8.7.1); on the other hand, $\Omega_{X/Y}^1$ is by definition the conormal sheaf $\mathcal{G}_1(g)$ of the immersion g (16.3.1), and to say that f is formally unramified therefore means that $\mathcal{G}_1(g) = 0$ (17.2.1). In addition, g is locally of finite presentation (1.4.3.1); therefore the hypothesis $\mathcal{G}_1(g) = 0$ implies that g is an open immersion (16.1.10); being surjective, this immersion is an isomorphism, hence f is a monomorphism (I, 5.3.8). $\square$

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17.3. Smooth morphisms, unramified morphisms, étale morphisms

Definition (17.3.1). — We say that a morphism $f : X \rightarrow Y$ is *smooth* (resp. *unramified*, or *net*¹⁰ resp. *étale*) if it is locally of finite presentation and formally smooth (resp. formally unramified, resp. formally étale).

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) over Y .

We will see later (17.5.2) that this definition of a smooth morphism coincides with the definition already given in (6.8.1); until then, we will exclusively use definition (17.3.1).

It is clear that saying that f is étale means that it is *both* smooth and unramified.

Remark (17.3.2). —

- (i) Note that definition (17.3.1) can be phrased using only the functor

$$Y' \longmapsto \operatorname{Hom}_Y(Y', X)$$

considered in (17.1.2, (iii)) because to say that f is locally of finite presentation is equivalent to saying that the preceding functor *commutes with projective limits of affine schemes* (8.14.2).

- (ii) Let A be a ring and B an A -algebra. We say that B is a *smooth* (resp. *unramified*, resp. *étale*) A -algebra if the corresponding morphism $\operatorname{Spec}(B) \rightarrow \operatorname{Spec}(A)$ is smooth (resp. unramified, resp. étale). It is equivalent to say that B is an A -algebra of *finite presentation* (1.4.6) that is furthermore formally smooth (resp. formally unramified, resp. formally étale) for the discrete topologies.
- (iii) It follows from (17.1.6) and the definition of a morphism locally of finite presentation (1.4.2) that the notion of a smooth (resp. unramified, resp. étale) morphism is *local on X and on Y* .

Proposition (17.3.3). —

- (i) An open immersion is étale. For an immersion to be unramified, it is necessary and sufficient to it be locally of finite presentation.
- (ii) The composition of two smooth (resp. unramified, resp. étale) morphisms is smooth (resp. unramified, resp. étale).
- (iii) If $f : X \rightarrow Y$ is a smooth (resp. unramified, resp. étale) S -morphism, then so is $f_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ for any base extension $S' \rightarrow S$.
- (iv) If $f : X \rightarrow X'$ and $g : Y \rightarrow Y'$ are smooth (resp. unramified, resp. étale) S -morphisms, then so is $f \times_S g : X \times_S Y \rightarrow X' \times_S Y'$.

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¹⁰The words “net” and “formally net” seem more preferable to the terminology used in “unramified” (resp. formally unramified”) and will be used almost exclusively in Chapter V. In this chapter, we have kept the old terminology so as not to conflict with 0, 19.10.

(v) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms; if g is locally of finite type and if $g \circ f$ is unramified, then f is unramified.

Proof. This follows from (1.4.3) and (17.1.3). $\square$

Proposition (17.3.4). — Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms, and suppose that g is unramified. Then, if $g \circ f$ is smooth (resp. unramified, resp. étale), so is f .

Proof. As g and $g \circ f$ are locally of finite presentation, so is f (1.4.3, (v)); the conclusion thus follows from (17.1.4) and (17.1.3, (v)). $\square$

Corollary (17.3.5). — Suppose that g is étale; then, for f to be smooth (resp. unramified, resp. étale) it is necessary and sufficient that $g \circ f$ is.

Proof. This follows from (17.3.4) and (17.3.3, (ii)). $\square$

Proposition (17.3.6). — Let $g : Y \rightarrow S$ and $h : X \rightarrow S$ be two morphisms locally of finite presentation. For an S -morphism $f : X \rightarrow Y$ to be unramified, it is necessary and sufficient that the canonical homomorphism (16.4.19)

$$f^*(\Omega_{Y/S}^1) \longrightarrow \Omega_{X/S}^1$$

is surjective.

Proof. As f is locally of finite presentation (1.4.3, (v)), the proposition follows from (17.2.2). $\square$

Definition (17.3.7). — Let $f : X \rightarrow Y$ be a morphism. We say that f is smooth (resp. unramified, resp. étale) at a point $x \in X$, if there exists an open neighbourhood U of x in X such that the restriction $f|_U$ is a smooth (resp. unramified, resp. étale) morphism from U to Y .

We then also say that X is smooth (resp. unramified, resp. étale) over Y at the point x .

Taking into account remark (17.3.2, (iii)), it is equivalent to say that f is smooth (resp. unramified, resp. étale) and to say that f is smooth (resp. unramified, resp. étale) at all points of X .

It is clear that the set of points of X at which the morphism $f : X \rightarrow Y$ is smooth (resp. unramified, resp. étale) is open in X .

Proposition (17.3.8). — For all preschemes Y and all locally free $\mathcal{O}_Y$ -modules $\mathcal{E}$ of finite type, the vector bundle prescheme $\mathbf{V}(\mathcal{E})$ (II, 1.7.8) associated to $\mathcal{E}$ is a smooth Y -prescheme.

Proof. Indeed (17.3.2, (iii)), we can restrict ourselves to the case where $Y = \text{Spec}(A)$ is affine and $\mathbf{V}(\mathcal{E}) = \text{Spec}(A[T_1, \dots, T_r])$; as $A[T_1, \dots, T_r]$ is a formally smooth A -algebra for the discrete topologies (0, 19.3.2), and of finite presentation, this proves the proposition (17.3.2, (ii)) $\square$

Corollary (17.3.9). — Under the hypotheses of (17.3.8), the projective prescheme $\mathbf{P}(\mathcal{E})$ (II, 4.1.1) is a smooth Y -prescheme.

Proof. We can still restrict to the case where $Y = \text{Spec}(A)$ is affine and $\mathbf{P}(\mathcal{E}) = \mathbf{P}_A^r$. We then know (II, 2.3.14) that we have a finite open cover of $\mathbf{P}_A^r$ by the $D_+(T_i)$ ($0 \leq i \leq r$) respectively equal to the spectrum of the ring $S_{(f)}$, where we wrote S for $A[T_1, \dots, T_r]$ and f for T_i ; but it follows immediately from the definition of $S_{(f)}$ (II, 2.2.1), that this ring, in this case, is isomorphic to $A[T_0, \dots, T_{i-1}, T_{i+1}, \dots, T_r]$; hence the corollary follows by (17.3.8). $\square$

17.4. Characterizations of unramified morphisms.

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Theorem (17.4.1). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, x a point of X . The following properties are equivalent:*

label=(a) *f is unramified at the point x .*

lbbel=(b) *The diagonal morphism $\Delta_f : X \rightarrow X \times_Y X$ is an isomorphism local at the point x .*

b') *If one sets $\Delta_f = (\psi, \theta)$, $Z = X \times_Y X$ and $z = \psi(x)$, the homomorphism $\theta_z^\# : \mathcal{O}_{Z,z} \rightarrow \mathcal{O}_{X,x}$ is bijective.*

b'') *For every morphism $g : Y' \rightarrow Y$, and every point $y' \in Y'$ above $y = f(x)$, every Y' -section s' of $X' = X \times_Y Y'$ such that $x' = s'(y')$ is above x is an isomorphism local at the point y' .*

lcbel=(c) *$(\Omega_{X/Y}^1)_x = 0$.*

ldbel=(d) *The $k(y)$ -prescheme $f^{-1}(y)$ is unramified on $k(y)$ at the point x .*

d') *The point x is isolated in $X_y = f^{-1}(y)$ (equivalently (ErrIII, 20), the morphism f is quasi-finite at the point x) and the ring $\mathcal{O}_{X_y,x}$ is a field, extension finite and separable of $k(y)$.*

d'') *The ring $\mathcal{O}_{X_y,x} = \mathcal{O}_{X,x} / \mathfrak{m}_y \mathcal{O}_{X,x}$ is a field, extension finite separable of $k(y)$.*

lebel=(e) *The ring $\mathcal{O}_{X,x}$ is an $\mathcal{O}_{Y,y}$ -algebra formally unramified for the discrete topologies.*

Proof. Since f is locally of finite type, the $\mathcal{O}_X$ -Module $\Omega_{X/Y}^1$ is of finite type (16.3.9), hence it amounts to the same to say that $(\Omega_{X/Y}^1)_x = 0$ or that there exists a neighbourhood open U of x such that $\Omega_{X/Y}^1|_U = 0$. This proves the equivalence of a) and c). On the other hand, if one sets $A = \mathcal{O}_{Y,y}$, $B = \mathcal{O}_{X,x}$, then $(\Omega_{X/Y}^1)_x = \Omega_{B/A}^1$ (16.4.15), and the equivalence of c) and e) follows from (0, 20.7.4).

Since d') only involves properties of the morphism $X_y \rightarrow \text{Spec}(k(y))$, the equivalence of a) and d') will imply that of d) and d'). On the other hand, d') and d'') are equivalent, since it amounts to the same to say that $\mathcal{O}_{X_y,x}$ is a $k(y)$ -algebra *finite*, or that x is an isolated point of X_y , since X_y is a $k(y)$ -prescheme locally of finite type ((I, 6.4.4)).

We now prove the equivalence of b) and b'). We may restrict to the case where $Y = \text{Spec}(R)$ and $X = \text{Spec}(S)$ are affine and f of finite presentation; then $Z = \text{Spec}(S \otimes_R S)$ and Δ_f corresponds to the canonical surjective homomorphism $S \otimes_R S \rightarrow S$, and the hypothesis that $\theta_z^\# : \mathcal{O}_{Z,z} \rightarrow \mathcal{O}_{X,x}$ is bijective implies that, replacing X by a suitable open neighbourhood of x , the homomorphism $\theta : \mathcal{O}_Z \rightarrow \psi_*(\mathcal{O}_X)$ is itself bijective (0I, 5.2.7). This shows that b') implies b); the converse is obvious.

On the other hand, the equivalence of b) and b'') follows from ((I, 5.3.7)) without hypothesis of finitude on f : the data of a Y' -section $s' : Y' \rightarrow X'$ is equivalent to that of a Y -morphism $h = g' \circ s' : Y' \rightarrow X$ (where $g' : X' \rightarrow X$ is the canonical projection), so that $s' = (1_{Y'}, h)_X$, and then the diagram

$$(17.4.1.1) \quad \begin{array}{ccc} Y' & \xrightarrow{s'} & X' = Y' \times_Y X \\ h \downarrow & & \downarrow h \times 1_X \\ X & \xrightarrow{\Delta_f} & X \times_Y X \end{array}$$

identifies Y' with the $(X \times_Y X)$ -prescheme product of X and X' . Hence ((I, 4.3.2)) if Δ_f is an isomorphism local at the point x , s' is an isomorphism local at the point y' (since $x = h(y')$), which proves that b) implies b''). The converse is obtained by applying b'') to the case $Y' = X$, $y' = x$, $g = f$, and $s' = \Delta_f$.

To finish the proof of (17.4.1), it suffices to prove the implications $d'') \Rightarrow c) \Rightarrow b) \Rightarrow d'')$.

$d'') \Rightarrow c)$: Since $\Omega_{X/Y}^1$ is an $\mathcal{O}_X$ -Module of finite type, it follows from the lemma of Nakayama that condition c) is equivalent to $(\Omega_{X/Y}^1)_x / \mathfrak{m}_y (\Omega_{X/Y}^1)_x = 0$, that is to say (16.4.5) to $(\Omega_{X_y / \text{Spec}(k(y))}^1)_x = 0$. One is thus reduced to the case where Y is the spectrum of a field k and X a k -prescheme algebraic. The hypothesis that $\mathcal{O}_{X,x}$ is a field k' , finite extension of k , implies first that x is a *closed* point in X ((I, 6.4.2)), then that x is a *maximal* point, hence is an isolated point of X . Replacing X by the open $\{x\}$ of X , one may suppose $X = \text{Spec}(k')$; but then the hypothesis that k' is a finite separable extension of k implies $\Omega_{k'/k}^1 = 0$ (0, 20.6.20), which proves c).

$c) \Rightarrow b)$: We saw above that one then has $\Omega_{X/Y}^1|_U = 0$ for a neighbourhood open U of x in X ; the assertion $b)$ follows then from the definition of $\Omega_{X/Y}^1$ (16.3.1) and (16.1.9).

$b) \Rightarrow d'')$: Replacing X by a neighbourhood open of x , one may suppose that Δ_f is an open immersion; if one denotes by $f_y : X_y \rightarrow \text{Spec}(k(y))$ the morphism deduced by base change, Δ_{f_y} is then also an open immersion ((I, 5.3.4)), and since condition $d'')$ only concerns the prescheme X_y , one can reduce to the case where Y is the spectrum of a field k , X the spectrum of a k -algebra A of finite type; the property to be established is that A is a k -algebra *finite and separable*; such an algebra is a composed direct of finite separable extensions of k . Let K be an algebraically closed extension of k . Since $k(x) = k$ since k is algebraically closed, there is a Y -section s of X such that $s(Y) = \{x\}$, and by (17.4.1), $\{x\}$ is the inverse image of the diagonal $\Delta_X(X)$ by a morphism $X \rightarrow X \times_Y X$, hence $\{x\}$ is open in X by the hypothesis $b)$. One has thus shown that A is a *finite* k -algebra. To express that Δ_f is an open immersion, one can thus reduce to the case where A is a *local* finite k -algebra; the residue field of A , being a finite extension of k , is necessarily identical to k , and hence ((I, 3.4.9)) $X \times_k X$ is reduced to a single point and Δ_f is necessarily an *isomorphism*. But since A is a k -algebra, the canonical homomorphism $A \otimes_k A \rightarrow A$ can only be bijective if $A = k$. $\square$

Remark (17.4.2). — Suppose only that f is *locally of finite type*. Then $\Omega_{X/Y}^1$ is still an $\mathcal{O}_X$ -Module of finite type (16.3.9), and Δ_f a locally of finite presentation immersion (1.4.3.1). The entire demonstration of (17.4.1) remains valid, subject to replacing $a)$ by: *the restriction of f to a suitable neighbourhood of x is a morphism formally unramified*. One sees moreover that in this case the restriction of f to a suitable neighbourhood of x is a morphism *locally quasi-finite*.

Corollary (17.4.2). — Let $f : X \rightarrow Y$ be a morphism locally of finite presentation. The following properties are equivalent:

label=(a) f is unramified.

lbbel=(b) The diagonal morphism $\Delta_f : X \rightarrow X \times_Y X$ is an open immersion.

b') For every morphism $Y' \rightarrow Y$, every Y' -section of $X' = X \times_Y Y'$ is an open immersion.

lcbel=(c) $\Omega_{X/Y}^1 = 0$.

ldbel=(d) For every $y \in Y$, the $k(y)$ -prescheme $f^{-1}(y)$ is unramified on $k(y)$.

d') For every $y \in Y$, the $k(y)$ -prescheme $f^{-1}(y)$ is isomorphic to a prescheme of the form $\coprod_{\lambda \in L} \text{Spec}(K_\lambda)$, where for every $\lambda \in L$, K_λ is a finite extension finite and separable of $k(y)$.

lebel=(e) For every $x \in X$, the ring $\mathcal{O}_{X,x}$ is an $\mathcal{O}_{Y,f(x)}$ -algebra formally unramified for the discrete topologies.

Corollary (17.4.3). — If $f : X \rightarrow Y$ is unramified, then f is locally quasi-finite (Err_{III}, 20).

Proof. Indeed, by ((17.4.1), d')), for every $y \in Y$, the fibre $f^{-1}(y)$ is isomorphic to a prescheme of the form $\coprod_{\lambda \in L} \text{Spec}(K_\lambda)$, where the K_λ are finite separable extensions of $k(y)$; hence $f^{-1}(y)$ is a discrete topological space. $\square$

Proposition (17.4.4). — Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , $y = f(x)$. Set $A = \mathcal{O}_{Y,y}$, $B = \mathcal{O}_{X,x}$, which are local Noetherian rings, and let k be the residue field of A . Then the conditions equivalent $a)$ to $e)$ of theorem (17.4.1) are also equivalent to each of the following:

f) $\hat{B} \otimes_{\hat{A}} k$ is a field, finite separable extension of k (which implies that $\hat{B}$ is an $\hat{A}$ -algebra finite).

f') $\hat{B}$ is an $\hat{A}$ -algebra formally unramified for the adic topologies.

If moreover $k(x) = k(y)$, or if k is separably closed, these conditions are also equivalent to:

f'') The homomorphism $\hat{A} \rightarrow \hat{B}$ is surjective.

(17.4.5). Given an S -prescheme Y and two S -morphisms $f : X \rightarrow Y$, $g : X \rightarrow Y$, one canonically deduces an S -morphism $(f, g)_S : X \rightarrow Y \times_S Y$. We call the *prescheme of coincidences* of f and g the inverse image by $(f, g)_S$ of the diagonal $\Delta_{Y|S}$; it is therefore a sub-prescheme of X , which is a closed S -scheme when Y is an S -scheme ((I, 5.4.1)).

Proposition (17.4.6). — Let $h : Y \rightarrow S$ be a unramified morphism, and let $f : X \rightarrow Y$, $g : X \rightarrow Y$ be two S -morphisms. Then the prescheme C of coincidences of f and g is a sub-prescheme induced on an open of X ; if moreover Y is an S -scheme ((I, 5.4.1)), C is a closed sub-prescheme of X .

Proof. Indeed, since $\Delta_h : Y \rightarrow Y \times_S Y$ is an open immersion (17.4.2), the inverse image by $(f, g)_S$ of $\Delta_{Y|S}$ is a sub-prescheme induced on an open of X ((I, 4.4.1)). The last assertion follows from (17.4.5). $\square$

Corollary (17.4.7). — *Under the hypotheses of (17.4.6), let x be a point of X such that the two composed morphisms $\text{Spec}(k(x)) \rightarrow X \xrightarrow{f} Y$ and $\text{Spec}(k(x)) \rightarrow X \xrightarrow{g} Y$ are equal. Then there exists an open neighbourhood U of x such that $f|_U = g|_U$. If moreover Y is an S -scheme, there exists a closed open neighbourhood X' of x in X such that $f|_{X'} = g|_{X'}$. If finally X is connected, then $f = g$.*

This follows from (17.4.6) and ((I, 5.3.17)).

Corollary (17.4.8). — *Under the hypotheses of (17.4.6), suppose that the structural morphism $\varphi = h \circ f = h \circ g$ of X in S is closed. Let s be a point of S ; let X_s be the $k(s)$ -prescheme $\varphi^{-1}(s)$ and suppose that the two morphisms composed $X_s \rightarrow X \xrightarrow{f} Y$ and $X_s \rightarrow X \xrightarrow{g} Y$ are equal. Then there exists a neighbourhood open V of s in S such that $f|_{\varphi^{-1}(V)} = g|_{\varphi^{-1}(V)}$. If moreover φ is open, one can take V open and closed. If finally S is connected, $f = g$.*

Proposition (17.4.9). — *Let Y be a connected prescheme, $f : X \rightarrow Y$ a unramified and separated morphism. Then every Y -section g of X is an isomorphism of Y onto a connected component open of X , and the application $g \mapsto g(Y)$ is a bijection of $\Gamma(X/Y)$ onto the set of connected components Z of X (necessarily open in X) such that the restriction of f to Z is an isomorphism. In particular, if g' and g'' are two Y -sections of X such that $g'(y) = g''(y)$ for some $y \in Y$, then $g' = g''$.*

Proof. It follows from (17.4.1, b'') that a Y -section s of X is also an open immersion, and since X is a Y -scheme, s is also a closed immersion ((I, 5.4.7)); it follows that s is an isomorphism of Y onto a sub-prescheme of X that is both open and closed, hence a connected component open of X , and since $s(Y)$ is connected, it is necessarily a connected component. The rest of the proposition is immediate. $\square$

Remark (17.4.10). — In view of the remark (17.4.1.2), one sees that, in the statements (17.4.6) to (17.4.9), one can replace the words “unramified” everywhere by “formally unramified and locally of finite type”.

17.5. Characterizations of smooth morphisms.

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Theorem (17.5.1). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, x a point of X , $y = f(x)$. The following conditions are equivalent:*

label=(a) *f is smooth at the point x .*

lbbel=(b) *f is flat at the point x and the $k(y)$ -prescheme $f^{-1}(y)$ is smooth on $k(y)$ at the point x (6.8.1).*

b') *f is regular at the point x (6.8.1).*

lcbel=(c) *$\mathcal{O}_{X,x}$ is an $\mathcal{O}_{Y,y}$ -algebra formally smooth for the discrete topologies.*

Proof. One may reduce to the case where $Y = \text{Spec}(A)$, $X = \text{Spec}(C)$, where $C = B/\mathfrak{J}$, $B = A[T_1, \dots, T_n]$ being a polynomial algebra and $\mathfrak{J}$ an ideal of finite type. The equivalence of a) and c) follows from (0, 22.6.4). On the other hand, applying this result to the morphism locally of finite type $f^{-1}(y) \rightarrow \text{Spec}(k(y))$, one sees that the equivalence of b) and b') follows from the equivalence of a) and b) in (6.8.6). It remains to show the equivalence of a) and b).

Let us show first that a) implies b). The hypothesis a) implies first that $f^{-1}(y)$ is smooth over $k(y)$ at the point x by (17.3.3), and it suffices to show that $C_{\mathfrak{p}} = \mathcal{O}_{X,x}$ is an $A_{\mathfrak{r}}$ -module flat. The condition (17.5.1.1) implies that the canonical images v_i of the u_i in the maximal ideal $\mathfrak{m}$ of $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$ are linearly independent mod $\mathfrak{m}^2$: otherwise there would exist polynomials $w_i \in B$ not all in $\mathfrak{q}$ such that $\sum w_i(\partial u_i / \partial T_{j_k}) \in \mathfrak{q}$ for $1 \leq k \leq r$, which contradicts (17.5.1.1) since $\mathfrak{q}$ is prime. One concludes from (0, 17.1.7) that (v_i) is a regular sequence in $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$. But since B is a morphism locally of finite presentation, it follows from (11.3.8) that the canonical images u'_i of the u_i in $B_{\mathfrak{q}}$ also form a regular sequence and that $B_{\mathfrak{q}}/(\sum_i u'_i B_{\mathfrak{q}})$ is then an $A_{\mathfrak{r}}$ -module flat. It follows from Nakayama's lemma that $\sum_i u'_i B_{\mathfrak{q}} = \mathfrak{J}_{\mathfrak{q}}$, and $C_{\mathfrak{p}} = B_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}$ is thus indeed an $A_{\mathfrak{r}}$ -module flat.

Let us prove finally that b) implies a). With the same notation, the hypothesis that $B_{\mathfrak{q}}$ is an $A_{\mathfrak{r}}$ -module flat implies that the canonical injection $\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}} \rightarrow B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$ is injective (0I, 6.1.2), so that

$\mathfrak{J}_q/\tau\mathfrak{J}_q$ identifies with an ideal of $B_q/\tau B_q$. Since $B_q/\tau B_q$ is formally smooth, the Jacobian criterion (0, 22.6.4), together with (0, 19.1.12), proves the existence of r polynomials $v_i \in k(y)[T_1, \dots, T_n]$ and r indices j_i ($1 \leq i \leq r$) such that their images in $\mathfrak{J}_q/\mathfrak{J}_q^2$ engender this B_q -module and $\det(\partial v_i/\partial T_{j_k}) \notin q$. If one designates by u_i an element of $\mathfrak{J}$ whose image is v_i , by Nakayama the u_i' in $\mathfrak{J}_q$ engender $\mathfrak{J}_q$, and $C_p = B_q/\mathfrak{J}_q$; the Jacobian criterion (0, 22.6.4) together with (0, 19.1.12), proves that $C_p = B_q/\mathfrak{J}_q$ is an A_τ -algebra formally smooth for the discrete topologies. C.Q.F.D. $\square$

Corollary (17.5.2). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation. For f to be smooth (in the sense of (17.3.1)), it is necessary and sufficient that f be regular (6.8.1), i.e. that f be flat and that for every $y \in Y$, $f^{-1}(y)$ be a $k(y)$ -prescheme geometrically regular (6.7.6).*

One has thus established the equivalence of the two definitions of “smooth morphism” given in (6.8.1) and (17.3.1).

Proposition (17.5.3). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , $y = f(x)$. Set $A = \mathcal{O}_{Y,y}$, $B = \mathcal{O}_{X,x}$, which are local Noetherian rings. Then the conditions equivalent a) to c) of (17.5.1) are also equivalent to each of the following:*

d) *B is an A -algebra formally smooth for the preadic topologies.*

d') *$\widehat{B}$ is an $\widehat{A}$ -algebra formally smooth for the adic topologies.*

If moreover $k(x) = k(y)$, these conditions are also equivalent to:

d'') *$\widehat{B}$ is an $\widehat{A}$ -algebra isomorphic to a formal power series algebra $\widehat{A}[[T_1, \dots, T_n]]$.*

Remark (17.5.4). — Suppose that Y is a locally Noetherian prescheme, and $f : X \rightarrow Y$ a morphism locally of finite type. The criterion (17.5.3, d)), together with (0, 22.1.4) shows that for proving f is smooth, one can apply the definition (17.1.1), restricting to the case where the affine scheme Y' is the spectrum of a local Artinian ring.

Proposition (17.5.5). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , $y = f(x)$. Suppose that Y is reduced at the point y . Then, for f to be smooth at the point y , it is necessary and sufficient that f be universally open in a neighbourhood of x in $f^{-1}(y)$ and that $f^{-1}(y)$ be a $k(y)$ -prescheme geometrically regular at the point x .*

Proof. By (17.5.1), everything reduces to seeing that, if $f^{-1}(y)$ is a $k(y)$ -prescheme geometrically regular at the point x , it is equivalent to say that f is flat or universally open in a neighbourhood of x in $f^{-1}(y)$. But if f is flat at the point x , it is in a neighbourhood of x in X (11.1.1) and hence universally open in this neighbourhood (2.4.6). Reciprocally, the hypothesis that f is universally open in a neighbourhood of x in $f^{-1}(y)$ and that $f^{-1}(y)$ is a $k(y)$ -prescheme geometrically regular at the point x implies that f is flat at the point x , since $\mathcal{O}_{Y,y}$ is reduced (15.2.2). $\square$

Corollary (17.5.6). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , $y = f(x)$. Suppose that Y is reduced and geometrically unibranch (6.15.1) at the point y . Then, for f to be smooth at the point x , it is necessary and sufficient that f be equidimensional at the point x and that $f^{-1}(y)$ be a $k(y)$ -prescheme geometrically regular at the point x .*

This follows from (17.5.5) and the Chevalley criterion (14.4.4).

Proposition (17.5.7). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, smooth at a point $x \in X$; set $y = f(x)$. Then, for $\mathcal{O}_{X,x}$ to be reduced (resp. integrally closed, resp. geometrically unibranch), it is necessary and sufficient that $\mathcal{O}_{Y,y}$ be so.*

This has been proved in (11.3.13) and (11.3.14) completed by ErrIV, 30.

Proposition (17.5.8). — *Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, smooth at a point $x \in X$; set $y = f(x)$. Then:*

- label=(i) $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y}) + \dim(\mathcal{O}_{X,x} \otimes_{\mathcal{O}_{Y,y}} k(y))$.
- lbbel=(ii) $\text{coprof}(\mathcal{O}_{X,x}) = \text{coprof}(\mathcal{O}_{Y,y})$.
- lcbel=(iii) For $\mathcal{O}_{X,x}$ to have the property (S_n) (5.7.2) (resp. (R_n) (5.8.2)), it is necessary and sufficient that $\mathcal{O}_{Y,y}$ have it. In particular, for $\mathcal{O}_{X,x}$ to be regular, it is necessary and sufficient that $\mathcal{O}_{Y,y}$ be so.

These are special cases of (6.1.2), (6.3.2), (6.4.1) and (6.5.3).

17.6. Characterizations of étale morphisms.

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Theorem (17.6.1). — Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, x a point of X , $y = f(x)$. The following conditions are equivalent:

label=(a) f is étale at the point x .

a') f is smooth at the point x and unramified at the point x .

lbbel=(b) f is smooth at the point x and locally quasi-finite (**Err_{III}**, 20).

lcbel=(c) f is flat at the point x and unramified at the point x .

c') f is flat at the point x and the ring $\mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ is a field, extension finite separable of $k(y)$.

c'') f is flat, and for every $y \in Y$ and every algebraically closed extension k' of $k(y)$, the “geometric fibre” $f^{-1}(y) \otimes_{k(y)} k'$ is a sum of spectra of fields isomorphic to k' .

ldbel=(d) $\mathcal{O}_{X,x}$ is an $\mathcal{O}_{Y,y}$ -algebra formally étale for the discrete topologies.

Proof. The equivalence of $a)$ and $a')$ follows immediately from the definitions; that of $a)$ and $d)$ follows from the equivalence of $a)$ and $d)$ in (17.4.1) and (17.5.1). The equivalence of $c)$ and $c')$ follows from $c)$ in (17.4.1); inversely, if $c')$ is verified, f is regular (hence smooth by (17.5.1)) at the point x , for if K is any finite extension of a field k , $\text{Spec}(K \otimes_k k')$ is regular, being a finite sum of spectra of fields. The fact that $a')$ implies $b)$ follows from (17.4.1, $d')$) and (17.5.1, $b)$). It remains to show that $b)$ implies $c)$, and since one already knows that f is flat at the point x (by (17.5.1)), it suffices to show that $f^{-1}(y)$ is a $k(y)$ -prescheme unramified on $k(y)$; in other words, one is reduced to proving that $b)$ implies $c)$ when $Y = \text{Spec}(k)$ is the spectrum of a field k , X the spectrum of a k -algebra A of finite type. By hypothesis $b)$, A is a k -algebra formally smooth for the discrete topologies (for the topologies preadic, which coincide with the discrete topologies here); hence (0, 19.6.5), A is a regular local ring, hence a field since it is Artinian, and it then follows from (0, 19.6.5.1) that A must be a finite separable extension of k , which is not possible unless A is a field, finite separable extension of $k(y)$ (4.6.1). $\square$

Corollary (17.6.2). — Let $f : X \rightarrow Y$ be a morphism locally of finite presentation. The following conditions are equivalent:

label=(a) f is étale.

a') f is smooth and unramified.

lbbel=(b) f is smooth and locally quasi-finite (**Err_{III}**, 20).

lcbel=(c) f is flat and unramified.

c') f is flat, and every fibre $f^{-1}(y)$ is a $k(y)$ -scheme sum of spectra of fields, extensions finite and separable of $k(y)$.

c'') f is flat, and for every $y \in Y$ and every algebraically closed extension k' of $k(y)$, the “geometric fibre” $f^{-1}(y) \otimes_{k(y)} k'$ is a sum of spectra of fields isomorphic to k' .

Proposition (17.6.3). — Let Y be a locally Noetherian prescheme, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , $y = f(x)$. Set $A = \mathcal{O}_{Y,y}$, $B = \mathcal{O}_{X,x}$, which are local Noetherian rings, and let k be the residue field of A . Then the conditions equivalent $a)$ to $d)$ of (17.6.1) are also equivalent to each of the following:

e) $\widehat{B}$ is an $\widehat{A}$ -algebra formally étale for the adic topologies.

e') $\widehat{B}$ is an $\widehat{A}$ -module free and $\widehat{B} \otimes_{\widehat{A}} k$ is a field, extension finite and separable of k (which implies that $\widehat{B}$ is an $\widehat{A}$ -algebra finite).

If moreover $k(x) = k(y)$, or if k is separably closed, these conditions are also equivalent to:

e'') The canonical homomorphism $\widehat{A} \rightarrow \widehat{B}$ is bijective.

Proposition (17.6.4). — Under the hypotheses of (17.6.3), if f is étale at the point x , then $\dim(\mathcal{O}_{X,x}) = \dim(\mathcal{O}_{Y,y})$.

This is a particular case of (17.5.8, (i)) since x is isolated in its fibre $f^{-1}(y)$.

17.7. Properties of descent, passage to the limit, and constructibility.

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Proposition (17.7.1). — Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, $g : Y' \rightarrow Y$ a morphism, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$ and $g' : X' \rightarrow X$ the canonical projections. Let x' be a point of X' and set $x = g'(x')$, $y' = f'(x')$.

label=(i) If f' is unramified at the point x' , then f is unramified at the point x .

lbbel=(ii) Suppose moreover that g is flat at the point y' . Then, if f' is smooth (resp. étale) at the point x' , f is smooth (resp. étale) at the point x .

Proof. Set $y = f(x) = g(y')$, so that $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$; note that f' is locally of finite presentation.

(i) Since the property for a morphism (locally of finite presentation) of being unramified at a point only involves the fibre of the morphism at that point (17.4.1, d)), one may reduce to the case where Y and Y' are spectra of fields. But then (17.4.1, d')) it suffices to say that if (f') is unramified at x (resp. x'), or that it is étale at x (resp. x').

(ii) Since f' is flat at x' (17.5.1), the hypothesis that g is flat at y' implies f is flat at x (by transitivity, 2.2.11, (iv)). The fact that f is smooth (resp. étale) at x then follows from (17.5.1) (resp. (17.6.1)) since the fibre $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$ and being smooth (resp. étale) at x' implies $f^{-1}(y)$ is geometrically regular (resp. a sum of spectra of fields) at x (6.7.8) and this implies that X is a $k(y)$ -prescheme geometrically regular at x , hence f is smooth at x by (17.5.1). $\square$

Corollary (17.7.2). — label=(i) With the notation of (17.7.1), let U (resp. U') be the set of points of X (resp. X') where f (resp. f') is unramified; then $U' = g'^{-1}(U)$.

lbbel=(ii) Suppose moreover that g is flat, and let V (resp. V') be the set of points of X (resp. X') where f (resp. f') is smooth; then $V' = g'^{-1}(V)$.

Proof. (i). Since the property of being unramified only involves the fibre ((17.4.1), d)), and the fibre of f' at x' is $f^{-1}(y) \otimes_{k(y)} k(y')$, the result follows from the fact that non-ramification is stable under base change of fields ((6.7.8)).

(ii). By (i), one has $U' \cap V' = g'^{-1}(U \cap V)$. Since f' is smooth at x' means f' is flat at x' and f' is unramified at x' on the fibre, the result follows from the fact that flatness descends by flat morphisms ((2.2.11), (iv)) and from ((17.7.1), (ii)). $\square$

Corollary (17.7.3). — label=(i) Suppose g surjective; then, for f to be unramified, it is necessary and sufficient that f' be so.

lbbel=(ii) Let $f : X \rightarrow Y$ be an S -morphism, $g : S' \rightarrow S$ a morphism faithfully flat, or that g be quasi-compact. Suppose that f is locally of finite presentation, or that g is locally of finite presentation. Then, for f to be smooth (resp. unramified, resp. étale), it is necessary and sufficient that f' be so.

Proof. (i). The sufficiency follows from base change ((17.3.3), (ii)). For the necessity, by ((17.7.2), (i)), $U' = g'^{-1}(U)$; if f' is unramified, then $U' = X'$, and since g is surjective, g' is surjective, hence $U = X$.

(ii). The sufficiency is again ((17.3.3), (ii)). For the necessity, the base change of a smooth (resp. unramified, resp. étale) morphism is smooth (resp. unramified, resp. étale) ((17.3.3), (ii)); for the converse, use ((17.7.1)) and the fact that a faithfully flat quasi-compact morphism is surjective ((2.2.13)). $\square$

Proposition (17.7.4). — Let $f : X \rightarrow Y$ be a morphism, $g : Y' \rightarrow Y$ a morphism flat and locally of finite presentation, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$ and $g' : X' \rightarrow X$ the canonical projections. Let V (resp. V') be the set of $x \in X$ (resp. $x' \in X'$) where f (resp. f') has one of the following properties:

label=(i) locally of finite type;

lbbel=(ii) locally of finite presentation;

lcbel=(iii) flat;

ldbel=(iv) unramified;

lebel=(v) smooth;

lfbel=(vi) étale.

Then $V' = g'^{-1}(V)$ (in other words, for f' to have the property considered at a point x' , it is necessary and sufficient that f have this property at x).

Lemma (17.7.5). — *Let $f : X \rightarrow Y$ be a faithfully flat morphism and locally of finite presentation, $g : Y \rightarrow Z$ a morphism such that $g \circ f : X \rightarrow Z$ has one of the following properties: being*

- label=(i) locally of finite type;*
- lbbel=(ii) locally of finite presentation;*
- lcbel=(iii) of finite type.*

Then g has the same property. If moreover f is quasi-compact or g quasi-separated, the same conclusion holds for the property:

- (iv) of finite presentation.*

Remark (17.7.6). — Regarding the assertion relative to property (iv), one cannot suppress the hypothesis that f is quasi-compact; without this, it would imply that every morphism $g : Y \rightarrow Z$ quasi-compact and locally of finite presentation would be of finite presentation, a conclusion known to be erroneous (I, 6.4). Indeed, one can only restrict to the case where Z is affine, hence Y quasi-compact; there is then a finite covering of Y by affine opens U_i such that the restrictions $g|_{U_i}$ are of finite presentation; it would then suffice to take for X the prescheme *sum* of the U_i , for $f : X \rightarrow Y$ the canonical morphism, which is evidently faithfully flat and locally of finite presentation (I, 6.5); $g \circ f$ would be of finite presentation by virtue of the choice of the U_i and of (I, 6.5), whence our assertion.

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Proposition (17.7.7). — *Let $f : Y \rightarrow S$ and $h : X \rightarrow S$ be two morphisms locally of finite presentation, $g : X \rightarrow Y$ an S -morphism, x a point of X , $y = g(x)$. Suppose that g is flat at the point x . Then, if h is smooth (resp. unramified, resp. étale) at the point x , f is smooth (resp. unramified, resp. étale) at the point y .*

Proof. Set $s = h(x) = f(y)$. To say that h is unramified at the point x (resp. that f is unramified at the point y) means that $h^{-1}(s)$ is étale on $k(s)$ at the point x (resp. that $f^{-1}(s)$ is étale on $k(s)$ at the point y) (17.4.1) and (17.6.1); as the morphism $g_s : h^{-1}(s) \rightarrow f^{-1}(s)$ deduced from g is flat at the point x , one can reduce to proving the proposition when h is smooth or étale. Moreover, as h is then flat at x (17.5.1), f is flat at the point y (by transitivity, (2.2.11), (iv)). It therefore suffices to say that $f^{-1}(s)$ is smooth (resp. étale) on $k(s)$ at y , and one is thus reduced to the case where $S = \text{Spec}(k)$ is the spectrum of a field.

(i) *Case of smooth morphisms.* As g is a morphism locally of finite presentation (I, 4.3), (v)), there is a neighbourhood U of x in X in which g is flat (11.3.1) and h smooth. Moreover $g(U)$ is an open neighbourhood of y in Y (2.4.6); replacing X by U and Y by $g(U)$, one can suppose that g is *faithfully flat and smooth*, and one is reduced to proving that f is then smooth. If k' is an algebraically closed extension of k , $X \otimes_k k'$ is then smooth on k' (17.7.3), (ii)); it suffices to prove that $Y \otimes_k k'$ is smooth on k' (since $\text{Spec}(k') \rightarrow \text{Spec}(k)$ is faithfully flat); one can therefore restrict to the case where k is *algebraically closed*. The set of points of Y rational on k is then very dense in Y (10.4.8) and the set of points of Y where Y is smooth on k is open, so it suffices to prove that Y is smooth on k at each rational point y ; but at such a point, saying that Y is smooth on k is equivalent to saying that Y is *regular* at y (17.5.1) and (6.7.8)). One has $y = g(x)$ for some $x \in X$ and by hypothesis (17.5.1) X is regular at the point x ; as X and Y are then locally Noetherian and g is flat, Y is indeed regular at the point y (6.5.1), (i)).

(ii) *Case of étale morphisms.* By (i) one already knows that f is étale at the point y ; by virtue of (17.6.1), it suffices to show that f is *quasi-finite* at the point y . Or, as $\mathcal{O}_{X,x}$ is an $\mathcal{O}_{Y,y}$ -module faithfully flat (01, 6.6.2), $\mathcal{O}_{Y,y}$ is identified with a sub- k -module of $\mathcal{O}_{X,x}$ and as $\mathcal{O}_{X,x}$ is by hypothesis a k -algebra finite, so is $\mathcal{O}_{Y,y}$. □

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Proposition (17.7.8). — *With the notations being those of (8.8.1), suppose X_α and Y_α locally of finite presentation on S_α . Let $f_\alpha : X_\alpha \rightarrow Y_\alpha$ be an S_α -morphism, $f : X \rightarrow Y$ the corresponding S -morphism.*

- label=(i) Let x be a point of X , x_λ its canonical projection in X_λ . For f to be smooth (resp. unramified, resp. étale) at the point x , it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that f_λ is smooth (resp. unramified, resp. étale) at the point x_λ .*
- lbbel=(ii) Suppose moreover that X_α is quasi-compact. For f to be smooth (resp. unramified, resp. étale), it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that f_λ is smooth (resp. unramified, resp. étale).*

Proof. (i) By hypothesis, for every λ , $f^{-1}(y) = f_{\lambda}^{-1}(y_{\lambda}) \otimes_{k(y_{\lambda})} k(y)$; the part of the statement concerning unramified morphisms is thus clear, and it suffices to consider the case of smooth morphisms. As f and f_{λ} are locally of finite presentation, it reduces to the same by (17.5.1) to saying that $f^{-1}(y)$ is geometrically regular at the point x or that $f_{\lambda}^{-1}(y_{\lambda})$ is geometrically regular at the point x_{λ} (6.7.8). The proposition then results from (17.5.1) and (11.2.6).

(ii) For every λ , let U_{λ} be the open set of $x_{\lambda} \in X_{\lambda}$ where f_{λ} is smooth (resp. unramified, resp. étale); let V_{λ} be its inverse image in X . By hypothesis, for every $x \in X$, there exists a λ such that f_{λ} is smooth (resp. unramified, resp. étale) at the point x_{λ} . As, by (17.3.3), for $\lambda \leq \mu$ we have $V_{\lambda} \subset V_{\mu}$, and as X is quasi-compact, there exists an index μ such that $X = V_{\mu}$. Furthermore (17.7.3), (ii), one concludes that for some $\nu \geq \mu$, $U_{\nu} = X_{\nu}$, which means f_{ν} is smooth (resp. unramified, resp. étale) by (17.3.3). $\square$

Corollary (17.7.9). — *Let $S = \operatorname{Spec}(A)$ be an affine scheme, $f : X \rightarrow S$ a morphism. The following conditions are equivalent:*

- a) *f is a morphism of finite presentation and smooth (resp. unramified, resp. étale).*
- b) *There exists an affine Noetherian scheme $S_0 = \operatorname{Spec}(A_0)$, a morphism of finite type $f_0 : X_0 \rightarrow S_0$ and a morphism $S \rightarrow S_0$ such that the S -prescheme $X_0 \otimes_{S_0} S$ is S -isomorphic to X and that f_0 is smooth (resp. unramified, resp. étale).*
- c) *The conditions of b) are satisfied, and moreover A_0 is a sub- $\mathbf{Z}$ -algebra of finite type of A , and the morphism $S \rightarrow S_0$ corresponds to the canonical injection $A_0 \rightarrow A$.*

Proof. The proof from (17.7.8) is the same as that of (11.2.7) from (11.2.6). $\square$

Proposition (17.7.10). — *Let $f : Y \rightarrow S$, $h : X \rightarrow S$ be two morphisms locally of finite presentation, $g : X \rightarrow Y$ an S -morphism, x a point of X , $y = g(x)$. Suppose that h is flat at the point x and that f is unramified at the point y . Then f is étale at the point y and g is flat at the point x .* IV | 76

Proof. (One will see further on (18.4.9) that one can in fact dispense with the hypothesis that h is locally of finite presentation.)

The question being local on S , X and Y , one can suppose that S , X and Y are affine, f , g , h are morphisms of finite presentation, h flat and f unramified; taking into account (11.2.7) and (17.7.9), one can furthermore suppose that S , X and Y are Noetherian; finally one can reduce to the case where $S = \operatorname{Spec}(\mathcal{O}_{S,s})$ (where $s = f(y) = h(x)$). The ring $A = \mathcal{O}_{S,s}$ being a local Noetherian ring, there exists a complete local Noetherian ring B , of residue field algebraically closed, and a local homomorphism $A \rightarrow B$ making B an A -module faithfully flat (0III, (10.3.1)). Replacing X and Y by $X \times_S \operatorname{Spec}(B)$ and $Y \times_S \operatorname{Spec}(B)$, one concludes by (2.5.1) and (17.7.1) that one can reduce to the case where A is complete and with residue field algebraically closed. The hypothesis that f is unramified at the point y implies (17.4.4) that $\mathcal{O}_{Y,y}$ is an $\mathcal{O}_{S,s}$ -algebra finite, and as the residue field of $\mathcal{O}_{S,s}$ is algebraically closed, the homomorphism $\mathcal{O}_{S,s} \rightarrow \mathcal{O}_{Y,y}$ is surjective (17.4.4). But on the other hand, the hypothesis that h is flat at the point x implies that the composed homomorphism $\mathcal{O}_{S,s} \rightarrow \mathcal{O}_{Y,y} \rightarrow \mathcal{O}_{X,x}$ is injective (0I, 6.5.1), whence $\mathcal{O}_{S,s} \rightarrow \mathcal{O}_{Y,y}$ is bijective; this shows that f is étale at the point y (17.6.3); furthermore, $\mathcal{O}_{X,x}$ is an $\mathcal{O}_{Y,y}$ -module flat, and g is flat at the point x . $\square$

Proposition (17.7.11). — *Let S be a prescheme, X, Y two S -preschemes locally of finite presentation on S . For every $s \in S$, denote X_s, Y_s, f_s the preschemes and morphism deduced from X, Y, f by the change of base $\operatorname{Spec}(k(s)) \rightarrow S$. Then:*

- label=(i) *The set of $x \in X$ such that, if s is the image of x in S , $f_s : X_s \rightarrow Y_s$ is smooth (resp. unramified, resp. étale, resp. differentially smooth) at the point x , is locally constructible.*
- lbbel=(ii) *Suppose moreover that f is of finite presentation. The set of $y \in Y$ such that, if s is the image of y in S , f_s is smooth (resp. unramified, resp. étale, resp. differentially smooth) at all the points of $f_s^{-1}(y)$, is locally constructible.*
- lcbel=(iii) *Suppose X and Y of finite presentation on S . The set of $s \in S$ such that f_s is smooth (resp. unramified, resp. étale, resp. differentially smooth) is locally constructible.*

Proof. Let E be the set of $x \in X$ satisfying the property considered in (i). Denoting by $g : X \rightarrow S$ and $h : Y \rightarrow S$ the structural morphisms, one can first replace S by the reduced sub-prescheme S' of S

having $\overline{\{g(x)\}}$ as underlying space, X by $g^{-1}(S')$, Y by $h^{-1}(S')$; the fibres of X and X' (resp. Y and Y') at the points of S' being the same. Now let us restrict to the case where S is *integral* and where $\eta = g(x) = h(y)$ (where $y = f(x)$) is its generic point.

1° Suppose first that $x \in E$. The local rings $\mathcal{O}_{X_\eta, x}$ and $\mathcal{O}_{Y_\eta, y}$ are respectively equal to $\mathcal{O}_{X, x}$ and $\mathcal{O}_{Y, y}$; as the property of smoothness of a morphism of finite presentation at a point depends only on the local ring of the morphism at that point and on the local ring of its image (17.5.1), one sees that the hypothesis $x \in E$ means the morphism f is smooth at the point x ; it then has this property at the points of an open neighbourhood of x in X , and it suffices to apply (17.3.3), (iii)) to obtain the conclusion.

2° Suppose second that $x \in X - E$, and that the morphism f_η is *not flat* at the point x . The conclusion results from the following lemma which specifies (11.2.8): $\square$

Lemma (17.7.11.1). — *Let $g : X \rightarrow S$, $h : Y \rightarrow S$ be two morphisms locally of finite presentation, $f : X \rightarrow Y$ an S -morphism, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent. Then the set E of $x \in X$ such that $\mathcal{F}_{g(x)}$ is $f_{g(x)}$ -flat at the point x is locally constructible.*

Proof of Lemma (17.7.11.1). Reasoning again as at the beginning of (9.9.1) and using (2.5.1), one reduces to the case where S , X , and Y are Noetherian; then one reduces as above to the case where S is *integre*, where $\eta = g(x) = h(f(x))$ is its generic point, and it is a matter of showing that if $x \in E$ (resp. $x \notin E$) there exists a neighbourhood V of x in $\overline{\{x\}}$ contained in E (resp. in $X - E$). The case where $x \in E$ is an immediate consequence of (11.1.1). To treat the case where $x \notin E$, one reasons as in (9.4.7.1), keeping the same notations, so that one can suppose, by replacing Y and X by neighbourhoods of $f(x)$ and x respectively, that there exist two coherent $\mathcal{O}_Y$ -Modules $\mathcal{G}$, $\mathcal{H}$ and an $\mathcal{O}_Y$ -homomorphism $u : \mathcal{G} \rightarrow \mathcal{H}$ such that for every $s \in S$, $u_s : \mathcal{G}_s \rightarrow \mathcal{H}_s$ is injective, but that the homomorphism $1 \otimes u_\eta : \mathcal{F}_\eta \otimes_{\mathcal{O}_{Y_\eta}} \mathcal{G}_\eta \rightarrow \mathcal{F}_\eta \otimes_{\mathcal{O}_{Y_\eta}} \mathcal{H}_\eta$ is not injective at the point x , in other words $x \in \text{Supp}(\text{Ker}(1 \otimes u_\eta))$; setting $T = \overline{\{x\}}$, we therefore have $\text{Supp}(\text{Ker}(1 \otimes u_\eta)) \supset T_\eta$. But for s in a neighbourhood of η , we have $\text{Ker}(1 \otimes u_s) = (\text{Ker}(1 \otimes u))_s$ (9.4.2), hence $\text{Supp}(\text{Ker}(1 \otimes u_s)) = (\text{Supp}(\text{Ker}(1 \otimes u)))_s$ (I, 9.1.13.1); it follows finally from (9.5.2) that for s in a neighbourhood of η , $(\text{Supp}(\text{Ker}(1 \otimes u)))_s \supset T_s$, which establishes the lemma. $\square$

Proof of Proposition (17.7.11), continued. 3° Suppose now that $x \in X - E$, that the morphism f_η is *flat* at the point x , but that f_η does not smooth at the point x . Note that to say f_η is flat at the point x means that f itself is flat at the point x and replacing X by a neighbourhood of x , (11.1.1); one concludes that it is also true of f_s for every $s \in S$, and since for every $y \in Y$, $f^{-1}(y) = f_s^{-1}(y)$, it comes back to the same to say that $f_{g(x)}$ is smooth at the point x' or to say that f is smooth at the point x' . But the set of $x' \in X$ where f is smooth is open (17.5.1), and since it does not contain x by hypothesis, it also does not contain $\overline{\{x\}}$, which achieves the proof for the first property considered in (i).

We prove secondly in (i) when the property in question is that of being *étale*. Note that this property for f_s at the point x means that f_s is at the same time *smooth* and *quasi-finite* at the point x (17.6.1). Or, it comes back to the same to say that $f_{g(x)}$ is quasi-finite, or that f is itself quasi-finite at this point; it results from (13.1.4) that the set of points x such that $f_{g(x)}$ is quasi-finite at the point x is open in X , and *a fortiori* locally constructible; the conclusion thus results from the fact that the set of x such that $f_{g(x)}$ is smooth is also locally constructible.

Let us pass to the proof of (i) when the property in question is that of being *differentially smooth*. Let $p : X \times_Y X \rightarrow X$ be the second canonical projection; the second projection $X_s \times_{Y_s} X_s \rightarrow X_s$ is just p_s for every $s \in S$, and it results from (17.12.5) that for $f_{g(x)}$ to be differentially smooth at the point x , it is necessary and sufficient that $p_{g(x)}$ be smooth at $\Delta_f(x)$. As p is locally of finite presentation, the set of points $z \in X \times_Y X$ such that $p_{g(p(z))}$ is smooth at the point z is locally constructible, and the intersection of this set with the locally closed set $\Delta_f(X)$ is also locally constructible; the restriction of p to $\Delta_f(X)$ being an isomorphism on X , one sees that the set of $x \in X$ such that $f_{g(x)}$ is differentially smooth at the point x is locally constructible.

Finally, consider the property of being unramified; note that the diagonal morphism $\Delta_f : X \rightarrow X \times_Y X$ is an immersion locally of finite presentation (I, 4.3.1) and for every $s \in S$, the diagonal morphism $\Delta_{f_s} : X_s \rightarrow X_s \times_{Y_s} X_s$ is just $(\Delta_f)_s$; saying that $f_{g(x)}$ is unramified at the point x means

that $(\Delta_f)_{g(x)}$ is an isomorphism local at the point x (17.4.1), and since this is an immersion locally of finite presentation, it suffices from (17.9.1) to say that $(\Delta_f)_{g(x)}$ is étale at the point x ; it suffices therefore to apply what has been seen above for the property of being étale.

The assertions (ii) and (iii) of (17.7.11) follow from (i), the theorem of Chevalley on images of constructible sets (1.8.4), and (when h is the structural morphism) similar arguments. $\square$

17.8. Criteria for smoothness and non-ramification by fibres.

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Proposition (17.8.1). — *Let $g : Y \rightarrow S$, $h : X \rightarrow S$ be two morphisms locally of finite presentation. For an S -morphism $f : X \rightarrow Y$ to be unramified, it is necessary and sufficient that for every $s \in S$, the morphism $f_s : h^{-1}(s) \rightarrow g^{-1}(s)$ deduced from f by the change of base $\text{Spec}(k(s)) \rightarrow S$, be unramified.*

Proof. This follows immediately from the fact that, for a morphism locally of finite presentation, being unramified is a property of the fibres of this morphism (17.4.1, d)), and from the fact that for every $y \in Y$, one has $f^{-1}(y) = f_s^{-1}(y)$ where $s = g(y)$. $\square$

Proposition (17.8.2). — *Let $g : Y \rightarrow S$, $h : X \rightarrow S$ be two morphisms locally of finite presentation; suppose moreover that h is flat. For an S -morphism $f : X \rightarrow Y$ to be smooth (resp. étale), it is necessary and sufficient that for every $s \in S$, the morphism $f_s : h^{-1}(s) \rightarrow g^{-1}(s)$ be smooth (resp. étale). When this is the case, the morphism g is étale and flat at all points of $f(X)$.*

Proof. One knows in fact (11.3.10) that for every $s \in S$, f_s is flat at all points of $f(X)$, and it suffices that for every $s \in S$, f_s be smooth at all points of $f(X)$; but for a flat morphism locally of finite presentation, the fact of being smooth is a property of the fibres of this morphism (17.5.1, b)), and for every $y \in Y$, $f^{-1}(y) = f_s^{-1}(y)$ where $s = g(y)$. $\square$

Remark (17.8.3). — The preceding proofs show (taking into account (11.3.10)) that if the hypotheses on g and h are the same as above, then, for f to be unramified (resp. smooth, resp. étale) at a point $x \in X$, it suffices that, if one sets $s = h(x)$, f_s be unramified (resp. smooth, resp. étale) at the point x .

17.9. Étale morphisms and open immersions.

Theorem (17.9.1). — *Let $f : X \rightarrow Y$ be a morphism. The following properties are equivalent:*

- a) *f is an open immersion.*
- b) *f is a flat monomorphism locally of finite presentation.*
- c) *f is étale and radiciel.*

Proof. It results from (I, 4.3), (i) that a) implies b). The condition b) implies that for every $y \in Y$, the fibre $f^{-1}(y)$ is either empty or isomorphic to $\text{Spec}(k(y))$ (8.11.5.1), hence b) implies c), by virtue of (17.6.2, c)). It remains to show that c) implies a). The question being local on X and on Y (since f is injective), one can reduce to the case where Y is affine and f of finite presentation. As f is flat, this is an open morphism (2.4.6), hence, replacing Y by $f(X)$, one can suppose that f is surjective. For every morphism $Y' \rightarrow Y$, $f' = f_{(Y')} : X_{(Y')} \rightarrow Y'$ is again étale, radiciel, surjective and of finite presentation; in other words f is a universal homeomorphism, that is to say f is of finite type and is separated (I, 8.7.1, (ii)). The hypothesis that f is radiciel and of finite type implies that f is quasi-finite; hence (8.11.1) f is a finite morphism. To prove that f is an isomorphism, one can reduce to the case where $Y = \text{Spec}(A)$, A being a local ring. As f is of finite presentation, one has $X = \text{Spec}(B)$, where B is an A -module flat of finite presentation (I, 4.7), hence free (Bourbaki, Alg. comm., chap. II, § 5, n° 2, cor. 2 of th. 1). Moreover, if $\mathfrak{m}$ is the maximal ideal of A and k its residue field, $B/\mathfrak{m}B$ is by hypothesis a field, which is both radiciel extension and étale extension of k (17.6.1); hence $B/\mathfrak{m}B$ is isomorphic to k , and as B is a free A -module, B is isomorphic to A . C.Q.F.D. $\square$

Corollary (17.9.2). — *Let X be a connected prescheme; if $f : X \rightarrow Y$ is a closed étale immersion, then f is an isomorphism of X onto a connected component of Y .*

Proof. Indeed f is étale and radiciel, hence an isomorphism from X onto a sub-prescheme of Y ; but by hypothesis $f(X)$ is closed in Y , hence since open and closed, and since $f(X)$ is connected, it is a connected component of Y . $\square$

Corollary (17.9.3). — *Let $f : X \rightarrow Y$ be a morphism étale (resp. étale and separated). Then every Y -section $g : Y \rightarrow X$ of X is an open immersion (resp. open and closed); moreover the application $g \mapsto g(Y)$ is a bijection of the set $\Gamma(X/Y)$ of Y -sections of X to the set of open (resp. open and closed) parts Z of X such that the restriction of f to Z is a surjective morphism and radiciel of Z onto Y .*

Proof. Indeed, the fact that g is an open immersion follows already from the fact that f is unramified (17.4.1, b'')), and the restriction of f to the open $g(Y)$ of X is an isomorphism. Inversely, if Z is an open of X such that $f|_Z$ is a surjective morphism and radiciel, since f is étale, $f|_Z$ is an isomorphism by virtue of (17.9.1), which proves the corollary. $\square$

Corollary (17.9.4). — *Let Y be a connected prescheme and separated, $f : X \rightarrow Y$ a morphism étale. Then every Y -section g of X is an isomorphism of Y onto a connected component of X , and the application $g \mapsto g(Y)$ is a bijection of $\Gamma(X/Y)$ to the set of connected components of X such that the restriction of f to Z is a surjective morphism and radiciel of Z onto Y .*

Corollary (17.9.5). — *Let $g : Y \rightarrow S, h : X \rightarrow S$ be two morphisms locally of finite presentation; suppose moreover that h is flat. For an S -morphism $f : X \rightarrow Y$ to be an open immersion (resp. isomorphism), it is necessary and sufficient that for every $s \in S$, the morphism $f_s : h^{-1}(s) \rightarrow g^{-1}(s)$ deduced from f by the change of base $\text{Spec}(k(s)) \rightarrow S$, be an open immersion (resp. isomorphism).*

Proof. Indeed, if f_s is an open immersion for every $s \in S$, it results from (17.8.3) that f is étale, and as for every $y \in Y, f^{-1}(y) = f_s^{-1}(y)$ with $s = g(y)$, f is radiciel; hence f is an open immersion by virtue of (17.9.1). If moreover f_s is surjective for every $s \in S$, f is surjective, hence an isomorphism. $\square$

The following proposition specifies (10.4.11):

Proposition (17.9.6). — *Let S be a prescheme, X an S -prescheme of finite presentation. Every S -endomorphism of X which is a monomorphism is an automorphism of X .*

Proof. Let $f : X \rightarrow S$ be the structural morphism, g the S -endomorphism considered. The question being local on S , one can suppose S affine. Using (8.9.1) and (8.10.5), (i bis)), one is reduced to the case where $S = \text{Spec}(A)$, where A is a $\mathbf{Z}$ -algebra of finite type, and hence X is a $\mathbf{Z}$ -prescheme of finite type. It follows already from (10.4.11) that g is a bijective morphism, since a monomorphism is radiciel ((8.11.5.1)); it will thus suffice to show that g is an open immersion, and since g is radiciel, it will suffice, by virtue of (17.9.1), to prove that g is étale. Moreover, as the set of points where g is étale is open and X is a Jacobson prescheme ((10.4.7)), it suffices to show that g is étale at every closed point z of X ((10.3.1)). Set $z' = g(z)$; it follows from (10.4.11), (i) that z' is also a closed point of X , and since g is a monomorphism, the map $k(z') \rightarrow k(z)$ deduced from g is an isomorphism. For g to be étale at the point z , it is thus necessary and sufficient that the canonical homomorphism $\hat{\mathcal{O}}_{X,z'} \rightarrow \hat{\mathcal{O}}_{X,z}$ be bijective ((17.6.3), e'')). We will show for this purpose that for every integer n , the canonical homomorphism $\mathcal{O}_{X,z'}/\mathfrak{m}_{z'}^{n+1} \rightarrow \mathcal{O}_{X,z}/\mathfrak{m}_z^{n+1}$ is bijective, from which the conclusion will immediately follow. One can suppose that z belongs to the finite set $T_{p,d}$ of closed points t of X such that $k(t)$ is an extension of $\mathbf{F}_p$ whose degree divides d ; in which case the same holds for z' . Denote by $\mathcal{J}$ the coherent ideal of $\mathcal{O}_X$ such that $\mathcal{J}_x = \mathcal{O}_x$ for $x \notin T_{p,d}$, and $\mathcal{J}_x = \mathfrak{m}_x^{n+1}$ for $x \in T_{p,d}$. Let $T_{p,d,n}$ be the closed sub-prescheme of X defined by $\mathcal{J}$, $j : T_{p,d,n} \rightarrow X$ the canonical injection; the composed morphism $g \circ j : T_{p,d,n} \rightarrow X$ maps the set $T_{p,d}$ into itself, and for every $x \in T_{p,d,n}$, the homomorphism $\mathcal{O}_{X,g(x)} \rightarrow \mathcal{O}_{X,x}/\mathfrak{m}_x^{n+1}$ factors as

$$\mathcal{O}_{X,g(x)} \longrightarrow \mathcal{O}_{X,g(x)}/\mathfrak{m}_{g(x)}^{n+1} \longrightarrow \mathcal{O}_{X,x}/\mathfrak{m}_x^{n+1}.$$

By ((I, 4.9)), there therefore exists a unique endomorphism g' of $T_{p,d,n}$ such that $j \circ g' = g \circ j$. As j is a monomorphism, and g is also one by hypothesis, one deduces that g' is also a monomorphism, and the proposition will be proven if one shows that g' is an automorphism of $T_{p,d,n}$. But, for every $x \in T_{p,d,n}$, we have seen that $k(x)$ is a finite field, and as $\mathcal{O}_{X,x}$ is Noetherian, each of the $\mathfrak{m}_x^h/\mathfrak{m}_x^{n+1}$ is a $k(x)$ -vector space of finite rank; from which one concludes immediately that the local ring $\mathcal{O}_{X,x}/\mathfrak{m}_x^{n+1}$ at the point x has a finite number of elements, and a fortiori is a $\mathbf{Z}$ -module of finite type. As $T_{p,d,n}$ is the sum of the preschemes $\text{Spec}(\mathcal{O}_{X,x}/\mathfrak{m}_x^{n+1})$ in finite number, it is a finite $\mathbf{Z}$ -prescheme; the endomorphism g' is therefore also finite (II, 6.1.5, (v)), hence proper. But a proper

monomorphism is a closed immersion ((8.11.5)); if $T_{p,d,n} = \text{Spec}(B)$, g' corresponds to a surjective endomorphism $\varphi : B \rightarrow B$ of the ring B . But the set B is finite, hence φ is necessarily bijective. $\square$

17.10. Relative dimension of a smooth prescheme over another.

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Definition (17.10.1). — Let $f : X \rightarrow Y$ be a morphism locally of finite type. The *relative dimension* of f at a point $x \in X$ (or the *dimension of X relative to Y at the point x*) is the positive integer $\dim_x f = \dim_x(f^{-1}(f(x)))$.

To say that f is *quasi-finite* at the point x (Err_{III}, 20) is equivalent to saying that $\dim_x f = 0$. One has seen (13.1.3) that the function $x \mapsto \dim_x f$ is upper semi-continuous. One notes that, even when the morphism f has the property (S_1) (that is to say (6.8.1) is flat and such that its fibres have no embedded associated prime cycle), the function $x \mapsto \dim_x f$ is not necessarily continuous, as shown by the example where $Y = \text{Spec}(k)$, where k is a field, and $X = \text{Spec}(k[U, V, W]/\mathfrak{p}\mathfrak{q})$, where $\mathfrak{p} = (W)$ and $\mathfrak{q} = (U) + (V - W)$, the prime ideals of $k[U, V, W]$ (X being thus the reunion, in the affine space of 3 dimensions, of a plane and of a line non parallel to this plane).

To say that a morphism locally of finite presentation $f : X \rightarrow Y$ is *étale* at a point $x \in X$ means that f is smooth at the point x and that $\dim_x f = 0$ (17.6.1).

Proposition (17.10.2). — Let $f : X \rightarrow Y$ be a smooth morphism. For every $x \in X$, the $\mathcal{O}_{X,x}$ -Module locally free Ω_f^1 (17.2.3) has rank equal to $\dim_x f$ in x (which implies that $x \mapsto \dim_x f$ is a continuous function on X).

Proof. Indeed, if $y = f(x)$, $X_y = f^{-1}(y)$ is smooth over $k(y)$, and if $f_y : X_y \rightarrow \text{Spec}(k(y))$ is the structural morphism, one has $\Omega_{f_y}^1 = \Omega_f^1 \otimes_{\mathcal{O}_x} k(y)$ (16.4.5); one can thus reduce to the case where $Y = \text{Spec}(k)$ is the spectrum of a field. But (16.4.12) $(\Omega_{X/k}^1)_x \otimes_{\mathcal{O}_x} k(x)$ is then k -isomorphic to $\mathfrak{m}_x/\mathfrak{m}_x^2$, and since $\mathcal{O}_x$ is a regular local ring (17.5.1), $\text{rg}_k(\mathfrak{m}_x/\mathfrak{m}_x^2) = \dim(\mathcal{O}_x)$ (0, 17.1.1); hence $\text{rg}_{\mathcal{O}_x}(\Omega_{X/k}^1)_x = \dim(\mathcal{O}_x)$. But since $k(x) = k$, one has $\dim_x f = \dim(\mathcal{O}_x)$ (5.2.3). C.Q.F.D. $\square$

We will prove a reciprocal of this result further on (17.15.5).

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Corollary (17.10.3). — Let $f : X \rightarrow Y$, $g : Y \rightarrow Z$ be two smooth morphisms. Then for every $x \in X$, one has

$$\dim_x(g \circ f) = \dim_x f + \dim_{f(x)} g.$$

Proof. Indeed, $g \circ f$ is smooth (17.3.3), hence the three $\mathcal{O}_X$ -Modules $\Omega_{X/Y}^1$, $\Omega_{X/Z}^1$ and $f^*(\Omega_{Y/Z}^1)$ are locally free ((17.2.3) and (0_I, 5.4.5)); moreover the rank at x of $f^*(\Omega_{Y/Z}^1)$ is equal to the rank at y of $\Omega_{Y/Z}^1$. The equality (17.10.3) is thus a consequence of (17.10.2) and of the exactness of the suite (17.2.3). $\square$

Corollary (17.10.4). — Let $f : X \rightarrow Y$ be a smooth morphism, X' a sub-prescheme of X such that the composed morphism $X' \xrightarrow{j} X \xrightarrow{f} Y$ (where j is the canonical injection) be smooth. Then the conormal sheaf $\mathcal{N}_{X'/X}$ is an $\mathcal{O}_{X'}$ -Module locally free, and for every $x \in X'$, one has

$$(17.10.4.1) \quad \dim_x f = \dim_x(f \circ j) + \text{rg}_{\mathcal{O}_{X',x}}(\mathcal{N}_{X'/X})_x.$$

Proof. Indeed, $\Omega_{X/Y}^1 \otimes_{\mathcal{O}_x} \mathcal{O}_{X'}$ and $\Omega_{X'/Y}^1$ are then locally free and the exact sequence (17.2.5.1) is locally split in a neighbourhood of each point of X' , hence $\mathcal{N}_{X'/X}$ is also locally free, and the relation (17.10.4.1) results from the exactness of the suite (17.2.5.1). $\square$

17.11. Smooth morphisms of smooth preschemes.

Theorem (17.11.1). — Let $f : Y \rightarrow S$ and $h : X \rightarrow S$ be two morphisms locally of finite presentation, $g : X \rightarrow Y$ an S -morphism, x a point of X ; set $y = g(x)$, $s = f(y) = h(x)$. The following conditions are equivalent:

- a) f is smooth at the point y and g is smooth at the point x .
- b) g and h are smooth at the point x .
- c) h is smooth at the point x , and the canonical homomorphism (16.4.18)

$$(17.11.1.1) \quad (g^*(\Omega_{Y/S}^1))_x \longrightarrow (\Omega_{X/S}^1)_x$$

is left invertible (in other words is an isomorphism onto a direct factor of $(\Omega_{X/S}^1)_x$).

- c') h is smooth at the point x , and the canonical homomorphism

$$(17.11.1.2) \quad ((\Omega_{Y/S}^1)_y \otimes_{\mathcal{O}_{Y,y}} k(y)) \otimes_{k(y)} k(x) \longrightarrow (\Omega_{X/S}^1)_x \otimes_{\mathcal{O}_{X,x}} k(x)$$

is injective.

Suppose moreover that the homomorphism $k(y) \rightarrow k(x)$ is bijective. Then the preceding conditions are also equivalent to the following:

- d) h is smooth at the point x , and the canonical application $T_{X/S}(x) \rightarrow T_{Y/S}(y)$ (16.5.12) from the tangent vector space at x to the tangent vector space at y to Y is surjective.

Proof. The fact that a) implies b) results trivially from (17.3.3), (ii)); b) implies c) by application of (17.2.3), (ii)). To see that c) is equivalent to c'), note that by virtue of (17.2.3), (i)), $\Omega_{X/S}^1$ is an $\mathcal{O}_X$ -Module locally free of finite type; it suffices to apply (0, 19.1.12) to the ring $\mathcal{O}_{X,x}$ and the homomorphism (17.11.1.1) of $\mathcal{O}_{X,x}$ -modules of finite type. When $k(x) = k(y)$, the tangent linear application $T_{X/S}(x) \rightarrow T_{Y/S}(y)$ is the transpose of (17.11.1.2), by (16.5.12), hence the equivalence of c') and d) in this case.

It remains to prove that c) implies a). One can restrict to the case where $S = \text{Spec}(A)$, $Y = \text{Spec}(B)$, $X = \text{Spec}(C)$ are affine. The hypothesis c) implies ((17.2.3), (i)) that $(\Omega_{X/S}^1)_x = \Omega_{C_m/A}^1$ is a C_m -module locally free: indeed (16.10.6) there exist elements t_i ($1 \leq i \leq r$) of $B_p = \mathcal{O}_{Y,y}$ such that the differentials $d_{B/A}(t_i)$ engender the B_p -module Ω_{B/A_p}^1 and their images in $\Omega_{C_m/A}^1$ are part of a base of this C_m -module. As $\Omega_{Y/S}^1$ (resp. $\Omega_{X/S}^1$) is an $\mathcal{O}_Y$ -Module (resp. $\mathcal{O}_X$ -Module) of finite presentation (16.4.22), one can, replacing when needed X and Y by suitable open affine neighbourhoods of x and y respectively, suppose that $\Omega_{C/A}^1$ is a C -module libre and that the t_i are images of elements s_i ($1 \leq i \leq r$) of B such that the $d_{B/A}(s_i)$ are part of a base of the C -module $\Omega_{C/A}^1$. Let φ be the A -homomorphism from $B' = A[T_1, \dots, T_r]$ to B such that $\varphi(T_i) = s_i$ for every i ; the di-homomorphism corresponding $\Omega_{B'/A}^1 \rightarrow \Omega_{B/A}^1$ (0, 20.5.2) transforms the $d_{B'/A}(T_i)$, which form a base of $\Omega_{B'/A}^1$ (0, 20.4.13), into the $d_{B/A}(s_i)$ and is hence *surjective*; if $Y' = \text{Spec}(B')$ and if $u : Y \rightarrow Y'$ is the S -morphism corresponding to φ , one concludes that u is *unramified* (17.2.2). If one proves that the composed morphism $u \circ g : X \rightarrow Y'$ is *étale* at the point y , then from (17.3.5) that g is smooth at the point x ; finally, as the structural morphism $f' : Y' \rightarrow S$ is smooth (17.3.8), $f = f' \circ u$ will be smooth at the point y by (17.3.5). That for the proof that c) implies a), one sees thus that one can restrict to the case where $Y' = Y$, and hence suppose that f is a smooth morphism *smooth*.

By virtue of (17.8.2), one can reduce to the case where $S = \text{Spec}(k)$ is the spectrum of a field; moreover, by (16.4.5) and (17.7.1), (ii)), one can suppose that k is algebraically closed, and one can suppose that f and h are both smooth, and that the canonical homomorphism

$$g^*(\Omega_{Y/S}^1) \longrightarrow \Omega_{X/S}^1$$

is left invertible (0, 19.1.12). Then the set of points of X rational on k is very dense in X (10.4.8), hence, to prove that g is smooth, it suffices to prove that g is smooth at every point $x \in X$ rational on k , or equivalently that at such a point, $\mathcal{O}_{X,x}$ is an $\mathcal{O}_{Y,y}$ -module flat and $\mathcal{O}_{X,x}/\mathfrak{m}_y \mathcal{O}_{X,x}$ a regular ring (17.5.1). Or, $y = g(x)$ is the image of x , $B = \mathcal{O}_{Y,y}$, the residue field of B is equal to k , it is the same as that of A , hence x (resp. y) is a closed point of X (resp. Y) (I, 6.4.2), and one has $\dim(A) = \dim(B) = n$. Since f is unramified (17.7.1), the homomorphism $\hat{A} \rightarrow \hat{B}$ is surjective (17.4.4, f''); but as $\dim(\hat{A}) = \dim(\hat{B}) = n$, and $\hat{A}$ being a regular ring, integral, the homomorphism

$\widehat{A} \rightarrow \widehat{B}$ is also *injective* (0, 16.3.10). This homomorphism is thus bijective, which achieves the proof that f is étale at the point x (17.6.3), e''). $\square$

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Corollary (17.11.2). — *Under the general hypotheses of (17.11.1), the following conditions are equivalent:*

- a) f is smooth at the point y and g is étale at the point x .
- b) h is smooth at the point x and g is étale at the point x .
- c) h is smooth at the point x , and the canonical homomorphism

$$(g^*(\Omega_{X/S}^1))_x \longrightarrow (\Omega_{X/S}^1)_x$$

is bijective.

- c') h is smooth at the point x , and the canonical homomorphism

$$((\Omega_{Y/S}^1)_y \otimes_{\mathcal{O}_{Y,y}} k(y)) \otimes_{k(y)} k(x) \longrightarrow (\Omega_{X/S}^1)_x \otimes_{\mathcal{O}_{X,x}} k(x)$$

is bijective.

Suppose moreover that the homomorphism $k(y) \rightarrow k(x)$ is bijective. Then the preceding conditions are also equivalent to the following:

- d) h is smooth at the point x , and the canonical application $T_{X/S}(x) \rightarrow T_{Y/S}(y)$ (16.5.12) is bijective.

Proof. Each of conditions a), b), c) of (17.11.2) is equivalent to the conjunction of the corresponding condition of (17.11.1) and of the fact that g is unramified at the point x , taking into account (17.2.2) in what concerns condition c); whence the equivalence of a), b) and c). The equivalence of c) and c') results from the equivalence of the corresponding conditions of (17.11.1) and from Nakayama's lemma; the equivalence of c') and d) when $k(x) = k(y)$ is immediate by transposition (16.5.12). $\square$

Corollary (17.11.3). — *Let $h : X \rightarrow S$ be a smooth morphism, s_i ($1 \leq i \leq r$) sections of $\mathcal{O}_X$ above X (which are also sections of $h_*(\mathcal{O}_X)$ above S), $g : X \rightarrow S[T_1, \dots, T_r] = \mathbf{V}_S^r$ the S -morphism corresponding to the homomorphism of $\mathcal{O}_S$ -Modules $\mathcal{O}_S^r \rightarrow h_*(\mathcal{O}_X)$ defined by these sections (II, (1.2.7)). For g to be smooth (resp. étale) at a point $x \in X$, it is necessary and sufficient that the $(d_{X/S}(s_i))_x$ form part of a base (resp. constitute a base) of the $\mathcal{O}_{X,x}$ -module $(\Omega_{X/S}^1)_x$.*

Proof. It suffices to apply (17.11.1) (resp. (17.11.2)) taking for f the structural morphism $S[T_1, \dots, T_r] \rightarrow S$. $\square$

Corollary (17.11.4). — *For a morphism $h : X \rightarrow S$ to be smooth at a point $x \in X$, it is necessary and sufficient that there exist an open neighbourhood U of x , an integer r , and an S -morphism étale $U \rightarrow S[T_1, \dots, T_n]$.*

Proof. The condition is evidently sufficient since the structural morphism $S[T_1, \dots, T_n] \rightarrow S$ is smooth (17.3.8). To show that it is necessary, note that since h is smooth at the point x , there is an open neighbourhood U of x such that $\Omega_{X/S}^1|_U$ is locally free (16.10.6) for obtaining (restricting to U) sections s_i of $\mathcal{O}_X$ above U such that the $d_{X/S}(s_i)$ form a base of $\Omega_{X/S}^1|_U$; one concludes by applying (17.11.3). $\square$

Proposition (17.11.5). — *Let $f : Y \rightarrow S$, $h : X \rightarrow S$ be two smooth morphisms. For an S -morphism $g : X \rightarrow Y$ to be an open immersion, it is necessary and sufficient that g be a monomorphism of preschemes and that for every $x \in X$, if one sets $y = g(x)$, one has $\dim_y(f) = \dim_x(h)$ (for a generalisation of this proposition, see (18.10.5)).*

Proof. The condition is evidently necessary; let us prove that it is sufficient. By virtue of (17.9.5), one is immediately reduced to the case where $S = \text{Spec}(k)$ is the spectrum of a field, and by virtue of (2.7.1), (x)), one can suppose k algebraically closed. By virtue of (17.9.1), it suffices to prove that g is étale at the closed points (or equivalently, rational on k) of X (10.4.8). So let x be such a point, and set $y = g(x)$, which is also rational on k and hence closed. Set $A = \mathcal{O}_{Y,y}$, $B = \mathcal{O}_{X,x}$, which are regular local rings, and let k be the residue field of A . Let $d = \dim(A)$ and $(t_i)_{1 \leq i \leq d}$ a regular system of parameters for A . Set $B = \mathcal{O}_{X,x}$, $C = B/\mathfrak{m}_y B$. As g is a monomorphism, so is the morphism $\text{Spec}(C) \rightarrow \text{Spec}(k(y)) = \text{Spec}(k)$ deduced from g by base change (I, 3.3.12); but this means the corresponding homomorphism $u : k \rightarrow C$ is surjective (hence bijective), since u admits a left inverse

$v : C \rightarrow k$, and $u \circ v$ and the identity of $k \rightarrow C$ give the same morphism, which, composed with u , gives the identity of $k \rightarrow C$. As by hypothesis B is a regular ring of dimension d , the images of the t_i in B form a regular system of parameters for B (0, 17.1.7); the condition (17.6.3), e'') is thus verified by g at the point x , which achieves the proof. $\square$

17.12. Smooth subpreschemes of a smooth prescheme. Smooth morphisms and differentially smooth morphisms. IV | 85

Theorem (17.12.1). — *Let $f : X \rightarrow S$, $h : Y \rightarrow S$ be two morphisms locally of finite presentation, $j : Y \rightarrow X$ an immersion, y a point of Y , $x = j(y)$. The following conditions are equivalent:*

- a) h is smooth at the point y and f is smooth at the point x .
- b) f is smooth at the point x , and the canonical homomorphism (16.4.21)

$$(\mathcal{N}_{Y/X})_y \longrightarrow (j^*(\Omega_{X/S}^1))_y$$

is left invertible.

- c) h is smooth at the point y and there exists an open neighbourhood U of y in Y such that $j|_U : U \rightarrow X$ is a regular immersion (16.9.2).

Corollary (17.12.2). — *With the notations of (17.12.1), suppose that f is smooth at the point x , and let Y be a closed sub-prescheme of X defined by an Ideal $\mathcal{J}$ of $\mathcal{O}_X$, and S -smooth at the point x . Let $(g_i)_{1 \leq i \leq r}$ be sections of $\mathcal{J}$ above X . The following conditions are equivalent:*

- a) The $(g_i)_x$ form a system of generators of the $\mathcal{O}_{X,x}$ -module $\mathcal{J}_x$ whose number of elements is the smallest possible.
- b) If g'_i is the canonical image of g_i in $\Gamma(X, \mathcal{J} / \mathcal{J}^2)$, the $(g'_i)_x$ form a base of the $(\mathcal{O}_{X,x} / \mathcal{J}_x)$ -module $\mathcal{J}_x / \mathcal{J}_x^2$.
- c) The images of the $d_{X/S}g_i$ in $(\Omega_{X/S}^1)_x \otimes_{\mathcal{O}_{X,x}} k(x)$ are linearly independent over $k(x)$, and the $(g_i)_x$ engender $\mathcal{J}_x$.
- d) There exists an open neighbourhood U of x in X and sections g_j ($r+1 \leq j \leq n$) of $\mathcal{O}_X$ above U such that the S -morphism $u : U \rightarrow S[T_1, \dots, T_n]$ corresponding to the homomorphism $\mathcal{O}_S^n \rightarrow f_*(\mathcal{O}_U)$ defined by the sections $g_i|_U$ ($1 \leq i \leq r$) and g_j ($r+1 \leq j \leq n$) is an étale morphism, and such that the reciprocal image of the sub-prescheme $Y' = S[T_{r+1}, \dots, T_n]$ is the open $U \cap j(Y)$.

Proof. As $(g'_i)_x$ is the canonical image of $(g_i)_x$ and b) results from (17.10.4) (Bourbaki, Alg. comm., chap. II, § 3, n° 2, prop. 5). By virtue of (17.12.1), b)), $\mathcal{J}_x / \mathcal{J}_x^2$ is a free $(\mathcal{O}_{X,x} / \mathcal{J}_x)$ -module libre of rank r ; and the equivalence of b) and c) results from Bourbaki, loc. cit. Moreover, if a) is verified, $(g'_i)_x$ is the image of $(g_i)_x$ in $\mathcal{O}_{X,x} / \mathcal{J}_x$; the equivalence of a) and b) follows from the fact that $\mathcal{J}_x$ is of finite type and $\mathcal{J}_x / \mathcal{J}_x^2$ free, by virtue of the Nakayama lemma.

Moreover, if a) is verified, the images of the $d_{X/k}g_i$ engender $(\Omega_{X/k}^1)_x \otimes 1$ for $1 \leq i \leq r$; the d_i for $r+1 \leq i \leq n$ already engender $\Omega_{k(x)/k}^1$. As $(\Omega_{X/S}^1)_x$ is free of rank n , it results from Bourbaki, loc. cit., that replacing X by an open neighbourhood of x , one can suppose that there exist $n-r$ sections g_i ($r+1 \leq i \leq n$) of $\mathcal{O}_X$ above X such that the $d_{X/S}g_i$ for $1 \leq i \leq n$ form a base of $(\Omega_{X/S}^1)_x$; the fact that the corresponding étale morphism u is étale at the point x results from (17.11.3); it is then étale in a neighbourhood of x ; replacing X by this neighbourhood, u can be supposed étale. Moreover, it is immediate that Y (identified, by restriction of $j(Y)$ to X) is a closed sub-prescheme of $u^{-1}(Y')$, and by virtue of (17.2.5) and (17.11.2), the restriction to Y of u can be supposed étale. One deduces that for every $y' \in Y'$, $Y \cap u^{-1}(y')$ is an open immersion (17.9.6), whence $Y \rightarrow u^{-1}(Y')$ is an open immersion, and we conclude by (17.11.3) that this proves that a) implies d). The reciprocal d) $\Rightarrow$ a) follows from (17.11.3). $\square$

Corollary (17.12.3). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, $u : S \rightarrow X$ an S -section of X . For f to be smooth at a point $x \in X$, it is necessary and sufficient that u be a quasi-regular immersion in a neighbourhood of $s = f(x)$.*

Proof. The conclusion results from (17.12.1), since $f \circ u = 1_S$ is smooth. $\square$

Proposition (17.12.4). — *Every smooth morphism $f : X \rightarrow S$ is differentially smooth (in other words (16.10.5) $\Delta_f : X \rightarrow X \times_S X$ is a quasi-regular immersion). In particular, the $\mathcal{P}_f^n = \mathcal{P}_{X/S}^n$ and the $\text{gr}_n(\mathcal{P}_{X/S})$ are $\mathcal{O}_X$ -Modules locally free of finite type.*

Proof. It suffices to note that the structural morphism $p_1 : X \times_S X \rightarrow X$ is smooth (17.3.3), (iii)) and that Δ_f is an X -section for p_1 ; the conclusion results from (17.12.3). $\square$

Proposition (17.12.5). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation. The following conditions are equivalent:*

- a) *f is differentially smooth.*
- b) *For every morphism $g : Y' \rightarrow Y$, if one sets $X' = X \times_Y Y'$ and $f' = f_{(Y')} : X' \rightarrow Y'$, for every Y' -section s' of X' , f' is smooth at all points of $s'(Y')$.*
- c) *The second projection $p_2 : X \times_Y X \rightarrow X$ is smooth at all the points of the diagonal $\Delta_f(X)$.*

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Proof. The condition c) is a particular case of b): it suffices to take $Y' = X$ in b), f' is then the second projection p_2 , and Δ_f is an X -section for p_2 . In second place, c) implies a), since p_2 is a morphism locally of finite presentation, and if p_2 is smooth at the points of $\Delta_f(X)$, Δ_f is a quasi-regular immersion (17.12.3), hence f is differentially smooth. It remains to show that a) implies b). One can suppose that $Y' = Y$, and $f' = f$, and hence show that if f is differentially smooth, then for every Y -section s of X , f is smooth at the points of $s(Y)$. Or, if $m : G \times_S G \rightarrow G$ is the morphism defining the group prescheme structure, $m \circ (s \times 1_G) : S \times_S G \rightarrow G \times_S G \rightarrow G$ is an S -isomorphism of G , “left translation” τ_s , transforming the identity section of G into the points of $s(S)$; hence one concludes that G is smooth on S at the points of $s(S)$. $\square$

Corollary (17.12.6). — *With the notations being those of (8.8.1), suppose X_α quasi-compact, and $f_\alpha : X_\alpha \rightarrow S_\alpha$ locally of finite presentation. For $f : X \rightarrow S$ to be differentially smooth, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that $f_\lambda : X_\lambda \rightarrow S_\lambda$ be differentially smooth (in which case $f_\mu : X_\mu \rightarrow S_\mu$ is differentially smooth for $\mu \geq \lambda$).*

Proof. The sufficiency of the condition and the last assertion result from (16.10.4). To prove that the condition is necessary, note that $\Delta_f : X \rightarrow X \times_S X$ and $p_2 : X \times_S X \rightarrow X$ are deduced by change of base from $\Delta_{f_\lambda} : X_\lambda \rightarrow X_\lambda \times_{S_\lambda} X_\lambda$ and $p_{2,\lambda} : X_\lambda \times_{S_\lambda} X_\lambda \rightarrow X_\lambda$. By virtue of (17.12.5), for every $z \in X \times_S X$ in $\Delta_f(X)$, there exists an affine open $U(z)$ of z in $X \times_S X$ such that p_2 is smooth in $U(z)$; by virtue of (8.2.11) and (17.7.8), there exists an index $\lambda(z)$ and a neighbourhood $U_{\lambda(z)}$ of z in $X_{\lambda(z)} \times_{S_{\lambda(z)}} X_{\lambda(z)}$ such that $p_{2,\lambda(z)}$ is smooth in $U_{\lambda(z)}$. As X is quasi-compact, one can cover $\Delta_f(X)$ by a finite number of such neighbourhoods; by virtue of (8.3.4), there exists a λ superior to all the $\lambda(z_i)$ such that the inverse images of the $U_{\lambda(z_i)}$ in $X_\lambda \times_{S_\lambda} X_\lambda$ form a covering; it results then from (17.12.5) that this index λ answers the question. $\square$

Example (17.12.7). — Let S be a prescheme, G an S -prescheme in groups, locally of finite presentation on S . For G to be differentially smooth on S , it is necessary and sufficient that G be smooth on S at all the points of the “identity section” e of G . The condition is indeed necessary by (17.12.5). Inversely, suppose it is fulfilled; for every morphism $S' \rightarrow S$, $G' = G \times_S S'$ is then an S' -prescheme in groups locally of finite presentation on S' ; one can therefore, by virtue of (17.12.5), restrict to proving that for every S -section s of G , f is smooth at the points of $s(S)$. But if $m : G \times_S G \rightarrow G$ is the morphism that defines the group prescheme structure, $m \circ (s \times 1_G) : S \times_S G \rightarrow G \times_S G \rightarrow G$ is an S -isomorphism of G , the “left translation” τ_s , transforming the identity section of G into the points of $s(S)$; whence one concludes that G is smooth on S at the points of $s(S)$.

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Remark (17.12.8). — An S -prescheme in groups G , of finite type (and even finite) on a locally Noetherian prescheme S , can be differentially smooth on S without being smooth (or even flat) on S . Consider for example for S the spectrum of the dual number ring $D = k[T]/T^2k[T]$ over a field k , for G the spectrum of the D -algebra direct composition $E = D \oplus k$. One defines on G a structure of S -scheme in groups by defining a “diagonal application” δ , homomorphism of E in $E \otimes_D E$: if e' , e'' are the idempotents, canonical images of 1 of D in the factors D and k of E , one easily verifies that such an application is obtained by taking $\delta(e') = e' \otimes e' + e'' \otimes e''$ and $\delta(e'') = e' \otimes e'' + e'' \otimes e'$. The identity section of the group scheme G corresponds to the homomorphism $E \rightarrow D$ which is the

identity in D and 0 in k ; it is an isomorphism from S onto a connected component of G , and *a fortiori* G is differentially smooth on S (17.12.6), but it is clear that G is not S -flat.

17.13. Transversal morphisms.

(17.13.1). Let S be a prescheme, X, Y, X' three S -preschemes, $i : Y \rightarrow X$ an S -immersion, $f : X' \rightarrow X$ an S -morphism. Set $Y' = Y \times_X X'$, and let $g : Y' \rightarrow Y, j : Y' \rightarrow X'$ be the canonical projections, so that one has the commutative diagram

$$(17.13.1.1) \quad \begin{array}{ccc} Y & \xleftarrow{g} & Y' \\ i \downarrow & & \downarrow j \\ X & \xleftarrow{f} & X' \end{array}$$

and that j is an S -immersion. One has then a commutative diagram of $\mathcal{O}_{Y'}$ -Modules quasi-coherent

$$(17.13.1.2) \quad \begin{array}{ccccccc} g^*(\mathcal{N}_{Y/X}) & \longrightarrow & f^*(\Omega_{X/S}^1) \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{Y'} & \longrightarrow & g^*(\Omega_{Y/S}^1) & \longrightarrow & 0 \\ \text{gr}_1(g) \uparrow & & \uparrow & & \uparrow & & \\ \mathcal{N}_{Y'/X'} & \longrightarrow & \Omega_{X'/S}^1 \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{Y'} & \longrightarrow & \Omega_{Y'/S}^1 & \longrightarrow & 0 \end{array}$$

where the lower line is the exact suite (16.4.21) applied to j , the upper line comes from the same exact suite for i , by application of the exact functor to the right g^* (which thus leaves it exact); $\text{gr}_1(g)$ is defined in (16.2.1), and the commutativity of the two squares results from (0, 20.5.7.3) and (0, 20.5.11.3). We have seen moreover ((16.2.2), (iii)) that $\text{gr}_1(g)$ is here surjective, from which one deduces from (17.13.1.2) the exact sequence

$$(17.13.1.3) \quad g^*(\mathcal{N}_{Y/X}) \xrightarrow{\alpha} \Omega_{X'/S}^1 \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{Y'} \longrightarrow \Omega_{Y'/S}^1 \longrightarrow 0.$$

Proposition (17.13.2). — With the notations of (17.13.1), let x' be a point of Y' , $x = g(x')$ its image in Y ; suppose X and Y smooth on S at the point x , X' smooth on S at the point x' . Let $m, m - c$ be the relative dimensions of X and Y on S at the point x (17.10.1), n that of X' on S at the point x' . The following conditions are equivalent:

- a) Y' is smooth on S at the point x' and of relative dimension $n - c$.
- b) The homomorphism $\alpha \otimes 1 : g^*(\mathcal{N}_{Y/X}) \otimes_{\mathcal{O}_{Y'}} k(x') \rightarrow \Omega_{X'/S}^1 \otimes_{\mathcal{O}_{X'}} k(x')$ deduced from the homomorphism α of (17.13.1.3) is injective.

Let s be the image of x in S , and suppose that x' is rational on $k(s)$ (s being the image of x' in S). Then the conditions a) and b) are also equivalent to the following:

- b') The transposed homomorphism of $\alpha \otimes 1$

$$T_{X'/S}(x') \longrightarrow (\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} k(x'))^\vee = T_{X/S}(x)/T_{Y/S}(x)$$

(cf. (16.5.12)) is surjective.

Definition (17.13.3). — With the notations of (17.13.1), we say that the morphism f is *transversal* to Y at the point x' , relatively to S , if X and Y are smooth on S at the point x , X' smooth on S at the point x' , and the equivalent conditions a), b) of (17.13.2) are satisfied. When there is no confusion to fear, we suppress the mention of the prescheme S and simply say that f is transversal to Y at the point x' .

Remarks (17.13.4). — (i) Suppose that X, Y and X' are flat and locally of finite presentation on S . For every $s \in S$, denote X_s, Y_s, X'_s, Y'_s the fibres of X, Y, X', Y' at the point s , $f_s : X'_s \rightarrow X_s$ the morphism deduced from f by change of base. It results then from (17.5.1) that for f to be a transversal morphism to Y at the point x' , it is necessary and sufficient that, if s is the image of x in S , f_s be transversal to Y_s at the point x' , relatively to $k(s)$.¹¹

¹¹[Trans.] The original reference (17.5.9) is an erratum in the original; there is no item 17.5.9. The correct reference is (17.5.1).

(ii) One has seen in (17.13.2) that the set of points $x' \in Y'$ where f is transversal is open in Y' ; if Y' is moreover *proper* on S , one deduces that the set of $s \in S$ such that f_s is transversal to Y at *all* the points of Y'_s , relatively to S , is *open* in S .

(iii) The property of being smooth in a point, as well as the notion of relative dimension on S in a point, being stable by every change of base $S' \rightarrow S$ ((17.3.3) and (4.2.7)), the same holds for the property of a morphism $f : X' \rightarrow X$ of being transversal to a sub-prescheme of X at a point, relatively to S .

(iv) The condition *b*) of (17.13.2) also expresses that the homomorphism $\alpha : g^*(\mathcal{N}_{Y/X}) \rightarrow \Omega_{X/S}^1 \otimes_{\mathcal{O}_X} \mathcal{O}_{Y'}$ is *universally injective* with respect to Y' (11.9.18).

(17.13.5). Consider now a prescheme S , three S -preschemes X, Y, Z , two S -morphisms $f : Y \rightarrow X$, $g : Z \rightarrow X$; set $T = Y \times_X Z$; one then knows (I, 5.3.5) that there is a commutative diagram

$$(17.13.5.1) \quad \begin{array}{ccc} X & \xleftarrow{\quad} & T = Y \times_X Z \\ \Delta \uparrow & & \uparrow j \\ X \times_S X & \xleftarrow{\quad u \quad} & Y \times_S Z \end{array}$$

making T the product of the $(X \times_S X)$ -preschemes X and $Y \times_S Z$, where $u = f \times_S g$. As Δ is an S -immersion (I, 5.3.9), one is in the situation of the diagram (17.13.1.1); what corresponds to $\Omega_{X/S}^1$ in (17.13.1) is then $(\Omega_{Y/S}^1 \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y \times_S Z}) \oplus (\Omega_{Z/S}^1 \otimes_{\mathcal{O}_Z} \mathcal{O}_{Y \times_S Z})$ by (16.4.23). On the other hand, what corresponds to the $\mathcal{O}_Y$ -Module $\mathcal{N}_{Y/X}$ in (17.13.1) is here by definition $\Omega_{X/S}^1$ (16.3.1); there therefore corresponds to (17.13.1.3) an exact sequence

$$(17.13.5.2) \quad \Omega_{X/S}^1 \otimes_{\mathcal{O}_X} \mathcal{O}_T \xrightarrow{\rho} (\Omega_{Y/S}^1 \otimes_{\mathcal{O}_Y} \mathcal{O}_T) \oplus (\Omega_{Z/S}^1 \otimes_{\mathcal{O}_Z} \mathcal{O}_T) \xrightarrow{\sigma} \Omega_{T/S}^1 \rightarrow 0$$

where it remains to make the homomorphisms ρ and σ precise. Taking into account first (16.4.23) and (0, 20.5.2), one sees that if $p : T \rightarrow Y$, $q : T \rightarrow Z$ are the canonical projections, one has, with the notations of (16.4.19),

$$(17.13.5.3) \quad \sigma = p_{T/Y/S} + q_{T/Z/S}.$$

On the other hand, to evaluate ρ , one uses the commutativity of the left square in (17.13.1.2), which in the present case reduces first to explicating the canonical homomorphism $\rho' : \Omega_{X/S}^1 \rightarrow \Omega_{(X \times_S X)/S}^1 \otimes_{\mathcal{O}_{X \times_S X}} \mathcal{O}_X$ defined in (16.4.21) applied to the immersion Δ . One can reduce to the case where $S = \text{Spec}(A)$, $X = \text{Spec}(B)$ are affine, and then ρ' corresponds to the homomorphism δ of (0, 20.5.11.2); one sees that ρ' is the difference of the two homomorphisms $\pi^{(1)}$ and $\pi^{(2)}$ of $\Omega_{X/S}^1$ into $\Omega_{(X \times_S X)/S}^1 \otimes_{\mathcal{O}_{X \times_S X}} \mathcal{O}_X$, corresponding respectively to the first and second projections of $X \times_S X$ into X , by (16.4.3.3). To obtain ρ , one must then consider the homomorphism $\rho'' : \Omega_{(X \times_S X)/S}^1 \otimes_{\mathcal{O}_{X \times_S X}} \mathcal{O}_{(Y \times_S Z)} \rightarrow \Omega_{(Y \times_S Z)/S}^1$ corresponding by (16.4.3.3) to the morphism u of (17.13.5.1), then, after tensorisation by $\mathcal{O}_T$, form the composition $(\rho'' \otimes 1) \circ (\rho' \otimes 1)$; it follows from what precedes that one has, with the notations of (16.4.18),

$$(17.13.5.4) \quad \rho = (f_{Y(X/S)} \otimes 1_{\mathcal{O}_T}) \oplus (-g_{Z(X/S)} \otimes 1_{\mathcal{O}_T}).$$

Applying (17.13.2) to the situation of the diagram (17.13.5.1) (taking into account ((17.3.3), (iv))), which implies that $X \times_S X$ is smooth over S at x if X is so) gives the

Corollary (17.13.6). — *Let S be a prescheme, X, Y, Z three S -preschemes, $f : Y \rightarrow X$, $g : Z \rightarrow X$ two S -morphisms; set $T = Y \times_X Z$, and let $p : T \rightarrow Y$, $q : T \rightarrow Z$ be the canonical projections. Let t be a point of T , $y = p(t)$, $z = q(t)$, $x = f(y) = g(z)$. Suppose that X is smooth over S at the point x , of relative dimension m , Y (resp. Z) smooth over S at the point y (resp. z), of relative dimension $m + a$ (resp. $m + b$), a and b being positive or negative integers. Then the following conditions are equivalent:*

label=a) T is smooth over S at the point t , of relative dimension $m + a + b$.

lbbel=b) The homomorphism

$$\rho \otimes 1 : \Omega_{X/S}^1 \otimes_{\mathcal{O}_X} k(t) \longrightarrow (\Omega_{Y/S}^1 \otimes_{\mathcal{O}_Y} k(t)) \oplus (\Omega_{Z/S}^1 \otimes_{\mathcal{O}_Z} k(t))$$

where ρ is given by (17.13.5.4), is injective.

lcbel=c) The morphism $T = Y \times_S Z \rightarrow X \times_S X$ is transversal to the diagonal of $X \times_S X$ at the point t .
If t is rational over the residue field of its image s in S , these conditions are also equivalent to the following:

b') The homomorphism

$$(17.13.6.1) \quad T_y(f) - T_s(g) : T_{Y/S}(y) \oplus T_{Z/S}(z) \longrightarrow T_{X/S}(x)$$

(cf. (16.5.12.5)) is surjective.

Moreover, when conditions a) and b) are verified at t , they are so in a neighbourhood of t in T , and restricting T to such a neighbourhood, the sequence

$$(17.13.6.2) \quad 0 \longrightarrow \Omega_{X/S}^1 \otimes_{\mathcal{O}_X} \mathcal{O}_T \xrightarrow{\rho} (\Omega_{Y/S}^1 \otimes_{\mathcal{O}_Y} \mathcal{O}_T) \oplus (\Omega_{Z/S}^1 \otimes_{\mathcal{O}_Z} \mathcal{O}_T) \xrightarrow{\sigma} \Omega_{T/S}^1 \longrightarrow 0$$

(where σ is given by (17.13.5.3)) is exact.

The only point remaining to prove in (17.13.6) concerns *b')*, and results from the fact that, under the hypothesis that t is rational over $k(s)$, the homomorphism $\rho \otimes 1$ is the transpose of $T_y(f) - T_s(g)$, by (17.13.5.4).

When the equivalent conditions of (17.13.6) are satisfied, one says that f and g form a couple of transversal morphisms at the point t , relatively to S ; here again, one often suppresses the mention of S .

(17.13.7). Consider in particular the case where Y and Z are sub-preschemes of X , f and g being the canonical injections; T is then the sub-prescheme “intersection” $\inf(Y, Z)$ of X (I, 4.4.3), and one has $t = y = z = x$. Instead of saying that the couple (f, g) is transversal at the point x , one says then that Y and Z cut transversally at the point x (relatively to S). Noting X_s, Y_s, Z_s, T_s the fibres of X, Y, Z, T at the point s , image of x in S , and taking into account ((5.2.3)), ((5.1.9)) and (0, 16.5.12), one sees that for this (when X, Y, Z are smooth over S at the point x) it suffices that T be smooth over S at the point x , and that one has the relation

$$(17.13.7.1) \quad \text{codim}_x(T_s, X_s) = \text{codim}_x(Y_s, X_s) + \text{codim}_x(Z_s, X_s).$$

Proposition (17.13.8). — Let S be a prescheme, X an S -prescheme locally of finite presentation over S , Y, Z two sub-preschemes of X , $T = \inf(Y, Z)$ the sub-prescheme “intersection” of Y and Z , x a point of T . The following conditions are equivalent:

- a) The canonical injection $i : Y \rightarrow X$ is a transversal morphism to Z at the point x , relatively to S (17.13.3).
- a') The canonical injection $j : Z \rightarrow X$ is a transversal morphism to Y at the point x , relatively to S .
- a'') Y and Z cut transversally at the point x , relatively to S .
- b) The homomorphism

$$\rho \otimes 1 : \Omega_{X/S}^1 \otimes_{\mathcal{O}_X} k(x) \longrightarrow (\Omega_{Y/S}^1 \otimes_{\mathcal{O}_Y} k(x)) \oplus (\Omega_{Z/S}^1 \otimes_{\mathcal{O}_Z} k(x))$$

is injective.

When x is rational over the residue field $k(s)$ (s image of x in S), these conditions are also equivalent to the following:

b') The homomorphism

$$T_x(i) - T_x(j) : T_{Y/S}(x) \oplus T_{Z/S}(x) \longrightarrow T_{X/S}(x)$$

is surjective.

Moreover, when conditions a) and b) are verified at x , they are so in a neighbourhood of x in T , and restricting X to a neighbourhood of x , the sequence (17.13.6.2) is exact, and one has a canonical isomorphism

$$(17.13.8.1) \quad \mathcal{N}_{T/X} \xrightarrow{\sim} (\mathcal{N}_{Y/X} \otimes_{\mathcal{O}_Y} \mathcal{O}_T) \oplus (\mathcal{N}_{Z/X} \otimes_{\mathcal{O}_Z} \mathcal{O}_T).$$

Proof. The conditions a), a'), a'') all imply that X, Y, Z are smooth over S at the point x . Let moreover $m, m-a, m-b$ be the relative dimensions of X, Y, Z over S at the point x . It results then from (17.13.2) applied replacing X' by Z and Y' by $T = Y \times_X Z$, that the conditions a), a') are both equivalent to saying that T is smooth over S at the point x and of relative dimension $m-a-b$ at this point; but by (17.13.7), this means precisely that condition a'') is verified, whence the equivalence of a), a') and a''). The equivalence of a'') and b) (or b') when x is rational over $k(s)$ has been proved in (17.13.6), as well as the fact that if these conditions are satisfied at the point x , they are so in a neighbourhood of x , and the exactness of the sequence (17.13.6.2) in such

a neighbourhood. It remains to define the canonical isomorphism (17.13.8.1). Let us denote α and $-\beta$ the homomorphisms appearing in the second member of (17.13.5.4), γ and δ those appearing in the second member of (17.13.5.2). Saying that the sequence (17.13.6.2) is exact signifies that, in the category of $\mathcal{O}_T$ -Modules, $\Omega_{X/S}^1 \otimes_{\mathcal{O}_x} \mathcal{O}_T$ identifies canonically with the fibre product of $\Omega_{Y/S}^1 \otimes_{\mathcal{O}_y} \mathcal{O}_T$ and $\Omega_{Z/S}^1 \otimes_{\mathcal{O}_z} \mathcal{O}_T$ over $\Omega_{T/S}^1$, for the homomorphisms γ and δ . The same reasoning as in (0, 18.1.2 and 18.1.3), where one replaces rings and ideals respectively by Modules and sub-Modules, furnishes a commutative diagram where the 3rd and 4th rows and columns are exact by the smoothness hypotheses. The fact that the composed homomorphism $\varepsilon = \gamma \circ \alpha = \delta \circ \beta$ is the homomorphism corresponding to the canonical injection $T \rightarrow X$ results from (0, 20.5.2.7). As the diagonal of the preceding diagram is exact, $\text{Ker}(\varepsilon)$ identifies canonically with $\mathcal{N}_{T/X}$, taking into account that T is smooth over S (17.2.5); hence one has a canonical isomorphism (17.13.8.1) which is the reciprocal of ι . $\square$

(17.13.9). One can generalise the results of (17.13.5) and (17.13.6) to a finite number of S -morphisms $f_i : Y_i \rightarrow X$ ($i \in I$, I a finite set). For this, let X^I be the product of the S -preschemes Y_i (for the composed morphisms $Y_i \rightarrow X \rightarrow S$), T the product of the X -preschemes Y_i (for the morphisms f_i). Let then Y be the product of the X -preschemes Y_i (for the composed morphisms $Y_i \xrightarrow{f_i} X \rightarrow S$), T the product of the X -preschemes Y_i (for the morphisms f_i). One proves as in (I, 5.3.5) that one has a commutative diagram

$$(17.13.9.1) \quad \begin{array}{ccc} X & \xleftarrow{v} & T \\ \Delta \uparrow & & \uparrow j \\ X^I & \xleftarrow{u} & Y \end{array}$$

(where u is the product (over S) of the f_i), making T the product of the X^I -preschemes X and Y . By recurrence on the number of elements of I , it results from (16.4.23) that $\Omega_{X^I/S}^1$ identifies canonically to the direct sum of the $p_i^*(\Omega_{X/S}^1)$ (resp. of the $q_i^*(\Omega_{Y_i/S}^1)$), where $q_i : Y \rightarrow Y_i$ are the canonical projections. Suppose now that X is smooth over S at a point x (hence in a neighbourhood of this point ((17.3.3), (iv))), and restricting X to a neighbourhood of x , one has ((17.2.5)) an exact sequence of $\mathcal{O}_X$ -Modules locally free $0 \rightarrow \mathcal{J} \rightarrow (\Omega_{X/S}^1)^I \rightarrow \Omega_{X/S}^1 \rightarrow 0$. Setting $\mathcal{J} = \mathcal{J}_1(\Delta)$; one deduces from the preceding sequence an exact sequence of $\mathcal{O}_T$ -Modules locally free

$$0 \rightarrow \mathcal{J} \otimes_{\mathcal{O}_x} \mathcal{O}_T \rightarrow (\Omega_{X/S}^1)^I \otimes_{\mathcal{O}_x} \mathcal{O}_T \xrightarrow{\sigma} \Omega_{X/S}^1 \otimes_{\mathcal{O}_x} \mathcal{O}_T \rightarrow 0$$

where σ is none other than the canonical homomorphism which, to each family of sections $(t'_i)_{i \in I}$ of $\Omega_{X/S}^1$ above an open of T , sends their *sum*. On the other hand, what corresponds here to the second vertical arrow of the diagram (17.13.1.2) is the homomorphism

$$(17.13.9.2) \quad \tau : (\Omega_{X/S}^1)^I \otimes_{\mathcal{O}_x} \mathcal{O}_T \rightarrow \bigoplus_{i \in I} (\Omega_{Y_i/S}^1 \otimes_{\mathcal{O}_{y_i}} \mathcal{O}_T)$$

which, to every family $(t'_i \otimes 1)_{i \in I}$, associates the sum of the $((f_i)_{Y_i(X/S)}(t'_i)) \otimes 1$, with the notation of (16.4.18). The homomorphism ρ which corresponds to α of (17.13.1.3) is then the restriction to $\mathcal{J} \otimes_{\mathcal{O}_x} \mathcal{O}_T$ of the preceding homomorphism τ . These hypotheses and notations being specified, one can apply to the situation of (17.13.9.1) the proposition (17.13.2), which gives the

Corollary (17.13.10). — *Under the general hypotheses of (17.13.9), let t be a point of T , y_i ($i \in I$) its projections in the Y_i , x the point of X equal to each of the $f_i(y_i)$. Suppose that X is smooth over S at the point x and of relative dimension d , and that each of the Y_i is smooth over S at the point y_i ; let $d + c_i$ (c_i a positive or negative integer) be the relative dimension of Y_i at the point y_i . Then the following conditions are equivalent:*

label=a) *T is smooth over S at the point t , of relative dimension $d + \sum_{i \in I} c_i$.*

label=b) *The homomorphism*

$$\rho \otimes 1 : \mathcal{J} \otimes_{\mathcal{O}_x} k(t) \rightarrow \bigoplus_{i \in I} (\Omega_{Y_i/S}^1 \otimes_{\mathcal{O}_{y_i}} k(t))$$

is injective.

When t is rational over the residue field $k(s)$ (s the image of t in S), these conditions are also equivalent to the following: IV | 97

b') The homomorphism transpose of $\rho \otimes 1$

$$\bigoplus_{i \in I} T_{Y_i/X}(y_i) \longrightarrow (T_{X/S}(x))^I / \delta(T_{X/S}(x))$$

(where δ is the diagonal application) is surjective.

It suffices to note that X^I is smooth over S at the point x and of relative dimension $d \cdot \text{Card}(I)$, and Y smooth over S at the point t and of relative dimension $d \cdot \text{Card}(I) + \sum_{i \in I} c_i$.

When the equivalent conditions of (17.13.10) are verified, one says further that the f_i form a family of transversal morphisms at the point t , relatively to S . One sees further that the set of points $t \in T$ where this holds is open in T . The remark ((17.13.4), (ii)) shows moreover that if X and the Y_i are flat and locally of finite presentation over S , and if moreover T is proper over S , the set of $s \in S$ such that the f_i form a family of transversal morphisms at all the points of T_s , relatively to S , is open in S .

(17.13.11). Consider in particular the case, generalising (17.13.7), where the Y_i ($i \in I$) are sub-preschemes of X , the f_i being the canonical injections, so that $T = \inf_{i \in I} (Y_i)$ is also the sub-prescheme “intersection” of the Y_i , and $t = y_i = x$ for every i ; instead of saying that the f_i form a family of transversal morphisms at the point x , one says further that the Y_i cut transversally at the point x (relatively to S). The condition a) of (17.13.10) is expressed further by the relation that generalises (17.13.7.1):

$$(17.13.11.1) \quad \text{codim}_x(T_s, X_s) = \sum_{i \in I} \text{codim}_x((Y_i)_s, X_s).$$

Moreover, one has the following property which extends (17.13.8.1), and of which we give another demonstration when I has 2 elements:

Corollary (17.13.12). — When the Y_i cut transversally at the point x , one has a canonical isomorphism

$$(17.13.12.1) \quad \mathcal{N}_{T/X} \xrightarrow{\sim} \bigoplus_{i \in I} (\mathcal{N}_{Y_i/X} \otimes_{\mathcal{O}_{Y_i}} \mathcal{O}_T).$$

Proof. One can reduce to the case where the Y_i are closed sub-preschemes of X , defined by quasi-coherent Ideals $\mathcal{J}_i$, so that T is defined by the Ideal $\mathcal{J} = \sum_i \mathcal{J}_i$. By definition of the conormal sheaf of an immersion (16.1.3), the canonical homomorphism $\bigoplus_i \mathcal{J}_i \rightarrow \mathcal{J}$ gives, by passage to quotients, a surjective homomorphism

$$(17.13.12.2) \quad \bigoplus_{i \in I} (\mathcal{N}_{Y_i/X} \otimes_{\mathcal{O}_{Y_i}} \mathcal{O}_T) \longrightarrow \mathcal{N}_{T/X}.$$

But here the $\mathcal{O}_T$ -Modules on both sides of (17.13.12.2) are locally free and of the same rank (since c_i is the rank of $\mathcal{N}_{Y_i/X}$), by (17.2.5) and the condition a) of (17.13.10); one concludes then from Bourbaki, *Alg. comm.*, chap. II, §3, n° 2, cor. de la prop. 6, that (17.13.12.2) is bijective, and (17.13.12.1) is the reciprocal isomorphism. □

17.14. Local and infinitesimal characterisations of smooth morphisms, of unramified morphisms and of étale morphisms. IV | 98

Proposition (17.14.1). — Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, x a point of X , $y = f(x)$. For f to be smooth (resp. unramified, resp. étale) at the point x , it is necessary and sufficient that the following condition be satisfied:

For every local scheme $Y' = \text{Spec}(A')$ of closed point y' , every morphism $h : Y' \rightarrow Y$ such that $h(y') = y$, every closed sub-scheme $Y'_0 = \text{Spec}(A'/\mathfrak{J})$ of Y' , where $\mathfrak{J}$ is an Ideal of square zero, and every Y -morphism $g_0 : Y'_0 \rightarrow X$ such that $g_0(y') = x$, there exists at least one (resp. at most one, resp. a unique) Y -morphism $g : Y' \rightarrow X$ whose restriction to Y'_0 is g_0 .

Proof. Taking into account the definitions (17.1.1) and (17.3.1), it suffices to show that the condition is sufficient in the cases of smooth morphisms and unramified morphisms respectively.

(i) *Case of smooth morphisms.* One can restrict to the case where $Y = \text{Spec}(A)$ and $X = \text{Spec}(C)$ are affine, with $B = A[T_1, \dots, T_n]$ an A -algebra of polynomials and C a quotient $B/\mathfrak{R}$, $\mathfrak{R}$ being a finitely generated ideal of B . Using (17.7.2), (ii) and (17.7.9), one can further suppose S , X and Y

are Noetherian; finally by virtue of (0_{III}, 9.2.3), one is reduced to the Noetherian case. It suffices to establish that $\mathcal{O}_{X,x}$ is an $\mathcal{O}_{Y,y}$ -algebra formally smooth for the *pre-adic topologies*. But as $\mathfrak{m}'$ is nilpotent, one sees that the condition of the statement is satisfied when one replaces the condition $\mathfrak{m}'\mathfrak{J} = 0$ by $\mathfrak{J} \subset \mathfrak{m}'$; and considering the rings $A'/\mathfrak{m}'^i\mathfrak{J}$ and reasoning by recurrence on j , this identity results from the hypothesis applied, by recurrence on n , to $A' = C/\mathfrak{r}^n$ and $\mathfrak{J}' = (\mathfrak{r}^n + (\mathfrak{J}/\mathfrak{J}^2))/\mathfrak{r}^n$.

(ii) *Case of unramified morphisms.* We take up the notations of the proof of (17.14.1), (ii); to prove that $\mathfrak{J}/\mathfrak{J}^2$ of C is null, note that C is then a local Noetherian ring, being the local ring of an affine open sub-prescheme of $X \times_Y X$, which is locally of finite type on X , hence on Y . The hypothesis, applied to $A' = C$ and $\mathfrak{J}' = \mathfrak{J}/\mathfrak{J}^2$, shows that the two applications $B \xrightarrow{p_1} C \rightarrow C/\mathfrak{r}^n$ and $B \xrightarrow{p_2} C \rightarrow C/\mathfrak{r}^n$ are identical; but as $C/\mathfrak{r}^n$ is artinian, this identity results from the hypothesis applied to $A' = C/\mathfrak{r}^n$, hence $\Omega_{B/A}^1 = 0$ by (0, 20.4.7). $\square$

Proposition (17.14.2). — *Let Y be a prescheme locally Noetherian, $f : X \rightarrow Y$ a morphism locally of finite type, x a point of X , $y = f(x)$. For f to be smooth (resp. unramified, resp. étale) at the point x , it is necessary and sufficient that it satisfies the condition of (17.14.1) where one supposes that the local ring A' is artinian, that $\mathfrak{m}'\mathfrak{J} = 0$, where $\mathfrak{m}'$ is the maximal ideal of A' , and that the residue field $A'/\mathfrak{m}'$ of A' is equal to $k(x)$.*

Remark (17.14.3). — One notes that, under the hypotheses of (17.14.2), if moreover the residue field $k(x)$ is a finite extension of $k(y)$, then the A' -modules $\mathfrak{m}'^j/\mathfrak{m}'^{j+1}$ are $k(y)$ -vector spaces of finite rank, hence a fortiori A' is an $\mathcal{O}_{Y,y}$ -algebra finite.

17.15. The case of preschemes over a base field. We recall first (6.7.7), (6.7.8) and (6.8.1) the

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Proposition (17.15.1). — *Let k be a field, X a prescheme locally of finite type on k , x a point of X . For X to be smooth at a point x , it is necessary and sufficient that for every finite radiciel extension k' of k (or only for every finite radiciel extension k' of k), the local ring $\mathcal{O}_{X,x} \otimes_k k'$ be regular. If $k(x)$ is a separable extension of k (in particular if k is perfect), this amounts to saying that $\mathcal{O}_{X,x}$ is regular.*

Corollary (17.15.2). — *Under the conditions of (17.15.1), for X to be smooth on k , it is necessary and sufficient that X be geometrically regular on k (6.7.6), which implies that X is reduced. If k is perfect, X is smooth on k if and only if X is regular.*

Proposition (17.15.3). — *Let k be a field, X a prescheme locally of finite type on k , x a point of X , $n = \dim_x(X)$. Let $(g_i)_{1 \leq i \leq n}$ be a family of n sections of $\mathcal{O}_X$ above X , and let $f : X \rightarrow Y = \operatorname{Spec}(k[T_1, \dots, T_n])$ be the morphism corresponding to the k -homomorphism $k[T_1, \dots, T_n] \rightarrow \Gamma(X, \mathcal{O}_X)$ sending T_i to g_i . The following conditions are equivalent (and imply that X is smooth on k at the point x and hence regular at the point x):*

- a) f is étale at the point x .
- b) The images $d_{X/k}g_i$ form a base of the $\mathcal{O}_{X,x}$ -module $(\Omega_{X/k}^1)_x$.
- c) The images $d_{X/k}g_i$ engender the $\mathcal{O}_{X,x}$ -module $(\Omega_{X/k}^1)_x$.

If f is étale at the point x , X is smooth on k at the point x since $Y = \operatorname{Spec}(k[T_1, \dots, T_n])$ is smooth on k (17.3.8); the fact that a) implies b) is thus a particular case of (17.11.3). As b) trivially implies c), it remains to see that c) implies a).

Proof. Note first that hypothesis c) entails that the homomorphism $(f^*(\Omega_{Y/k}^1))_x \rightarrow (\Omega_{X/k}^1)_x$ is surjective, hence, replacing X by an open neighbourhood of x , one can suppose that the homomorphism $f^*(\Omega_{Y/k}^1) \rightarrow \Omega_{X/k}^1$ is surjective (Bourbaki, *Alg. comm.*, chap. II, § 5, n° 1, prop. 2); hence f is unramified ((17.2.2)). Let us first show that one can reduce to the case where x is rational over k . Indeed, if one sets $k' = k(x)$ and $X' = X \otimes_k k'$, $Y' = Y \otimes_k k' = \operatorname{Spec}(k'[T_1, \dots, T_n])$, there exists a point $x' \in X'$ above x such that $k(x') = k'$. To prove that f is étale at the point x , it suffices to show that $f' = f_{(k')} : X' \rightarrow Y'$ is étale at the point x' ((17.7.1), (ii)); moreover f' is unramified ((17.3.3), (iii)) and one has $\dim_{x'}(X') = n$ ((4.2.7)). In the same way, replacing k' by an algebraically closed extension of k' , one can suppose that k is algebraically closed. Set then $y = f(x)$, $A = \mathcal{O}_{Y,y}$, $B = \mathcal{O}_{X,x}$; as the residue field of B is equal to k , the same is true of that of A , hence x (resp. y) is a closed point of X (resp. Y) ((I, 6.4.2)) and one has therefore $\dim(A) = \dim(B) = n$. Since f is unramified, the homomorphism $\hat{A} \rightarrow \hat{B}$ is surjective ((17.4.4), f''); but as $\dim(\hat{A}) = \dim(\hat{B}) = n$, and $\hat{A}$, being a regular ring, is integral, the homomorphism $\hat{A} \rightarrow \hat{B}$ is also injective (0, 16.3.10).

This homomorphism is therefore bijective, which completes the proof that f is étale at the point x ((17.6.3), e''). $\square$

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Corollary (17.15.4). — *Under the general hypotheses of (17.15.3), suppose moreover that $k(x)$ is a finite and separable extension of k and that the germs $(g_i)_x$ belong to $\mathfrak{m}_x$. Then the conditions a), b), c) of (17.15.3) are also equivalent to each of the following:*

- d) *The n germs $(g_i)_x$ engender the maximal ideal $\mathfrak{m}_x$ of $\mathcal{O}_{X,x}$.*
- d') *The ring $\mathcal{O}_{X,x}$ is regular and the $(g_i)_x$ form a regular system of parameters for this ring (0, 17.1.6).*

Proof. Indeed, d') trivially implies d). On the other hand, as $k(x)$ is a finite extension and separable of k , one has $\Omega_{k(x)/k}^1 = 0$ (0, 20.6.20) and $k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$ is a k -algebra formally smooth for the discrete topologies (0, 19.6.1), hence the exact suite (0, 20.5.14.1) applies with $A = k$, $B = \mathcal{O}_{X,x}$, $\mathfrak{A} = \mathfrak{m}_x$, and furnishes an isomorphism canonical

$$\delta : \mathfrak{m}_x / \mathfrak{m}_x^2 \xrightarrow{\sim} (\Omega_{X/k}^1)_x \otimes_{\mathcal{O}_{X,x}} k(x).$$

From this one first deduces the equivalence of conditions c) and d), by virtue of the Nakayama lemma. On the other hand, if f is étale at the point x , the ring $\mathcal{O}_{X,x}$ is regular and of dimension n (I, 6.4.2), and n elements that engender this ideal maximal form necessarily a regular system of parameters for $\mathcal{O}_{X,x}$ (0, 17.1.6), which proves that a) implies d'). $\square$

Proposition (17.15.5). — *Let k be a field, X a prescheme locally of finite type on k , x a point of X , $n = \dim_x(X)$. The following conditions are equivalent:*

- a) *X is smooth on k at the point x .*
- b) *X is differentially smooth on k at the point x .*
- c) *$(\Omega_{X/k}^1)_x$ is an $\mathcal{O}_{X,x}$ -module libre of rank n .*
- d) *$(\Omega_{X/k}^1)_x$ is an $\mathcal{O}_{X,x}$ -module admitting a system of n generators.*

Proof. We have already proved that a) implies c) (17.10.2); c) implies d) trivially; c) also implies b) (17.12.4). The fact that d) implies a) results from (0, 20.4.7 and 17.15.3, c)). The fact that b) implies a) is shown in the proof of (17.15.3). $\square$

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Corollary (17.15.6). — *Let X be a prescheme of finite type on a field k . For X to be smooth on k , it is necessary and sufficient that the $\mathcal{O}_X$ -Module $\Omega_{X/k}^1$ be locally free, and that the local rings of X at maximal points be fields, separable extensions of k (the latter condition being automatically satisfied if k is a perfect field and X a reduced prescheme).*

Proof. The conditions are necessary, for if X is smooth on k , it results from (17.10.2) that $\Omega_{X/k}^1$ is locally free; on the other hand, X is regular, hence a fortiori reduced, hence at every maximal point x of X , $\mathcal{O}_{X,x}$ is a field, which must be a k -algebra formally smooth for the discrete topologies ((17.5.1)), hence a separable extension of k (0, 19.6.1).

The conditions are sufficient. One can indeed reduce to the case where X is connected, hence $\Omega_{X/k}^1$ is locally free of constant rank n . For every maximal point x of X , one then has $\mathrm{rg}_{k(x)} \Omega_{k(x)/k}^1 = n$ since $\mathcal{O}_{X,x} = k(x)$; as by hypothesis $k(x)$ is a separable extension of k , one has $\Omega_{k(x)/k}^1 = 0$ (0, 20.6.3), hence $\deg. \mathrm{tr}_k k(x) = n$ by Cartier's equality (0, 21.7.1). All the irreducible components of X therefore have the same dimension n ((5.2.1)), and one concludes that X is smooth on k at every point by virtue of the fact that c) implies a) in ((17.15.5)). $\square$

Corollary (17.15.7). — *If k is of characteristic 0, then, for X to be smooth on k at a point x , it is necessary and sufficient that $(\Omega_{X/k}^1)_x$ be an $\mathcal{O}_{X,x}$ -module libre.*

Proof. This results from (16.12.2) and the equivalence of b) and a) in (17.15.5). $\square$

Proposition (17.15.8). — *Let k be a field, X a prescheme locally of finite type on k , x a point of X ; set $n = \dim_x(X)$, $r = \dim(\mathcal{O}_{X,x})$, so that $n - r = \deg. \mathrm{tr}_k k(x)$ (5.2.3). Let $(g_i)_{1 \leq i \leq n}$ be a family of n sections of $\mathcal{O}_X$ above X , such that $(g_i)_x \in \mathfrak{m}_x$ for $1 \leq i \leq r$; let $f : X \rightarrow Y = \mathrm{Spec}(k[T_1, \dots, T_n])$ be the*

morphism corresponding to the k -homomorphism $k[T_1, \dots, T_n] \rightarrow \Gamma(X, \mathcal{O}_X)$ sending T_i to g_i . The following conditions are equivalent (and imply that X is smooth on k at the point x and that $k(x)$ is a separable extension of k):

- a) f is étale at the point x .
- b) The $(g_i)_x$ ($1 \leq i \leq r$) engender $\mathfrak{m}_x$ and the images in $\Omega_{k(x)/k}^1$ of the $(d_{X/k}g_i)_x$ for $r+1 \leq i \leq n$ engender $\Omega_{k(x)/k}^1$.

Corollary (17.15.9). — Let X be a prescheme locally of finite type on a field k , x a point of X ; set $n = \dim_x(X)$, $r = \dim(\mathcal{O}_{X,x})$. The following conditions are equivalent:

- a) The ring $\mathcal{O}_{X,x}$ is regular and $k(x)$ is a separable extension of k .
- b) X is smooth on k at the point x , and the canonical homomorphism

$$\mathfrak{m}_x / \mathfrak{m}_x^2 \longrightarrow (\Omega_{X/k}^1)_x \otimes_{\mathcal{O}_{X,x}} k(x) \longrightarrow \Omega_{k(x)/k}^1 \longrightarrow 0$$

is exact.

- c) There exist n sections g_i of $\mathcal{O}_X$ above an open neighbourhood U of x , such that $(g_i)_x \in \mathfrak{m}_x$ for $1 \leq i \leq r$ and the morphism $U \rightarrow \operatorname{Spec}(k[T_1, \dots, T_n])$ corresponding to the g_i (cf. (17.15.8)) is étale at the point x .
- d) There exist n sections g_i ($1 \leq i \leq n$) of $\mathcal{O}_X$ above an open neighbourhood U of x , such that $(g_i)_x$ ($1 \leq i \leq r$) engender $\mathfrak{m}_x$ and that the images in $\Omega_{k(x)/k}^1$ of the $(d_{U/k}g_i)_x$ for $r+1 \leq j \leq n$ engender $\Omega_{k(x)/k}^1$.

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Remarks (17.15.10). — label=(i) For a prescheme X of finite type over k to be smooth over k , it does not suffice that $\Omega_{X/k}^1$ be locally free, as shown by the example where $X = \operatorname{Spec}(K)$, K being a finite non separable extension of k .

lbbel=(ii) When k is not perfect, it may happen that X is smooth over k without $k(x)$ being separable over k . One has an example of this by taking $X = \operatorname{Spec}(k[T])$ and for x the point corresponding to the principal prime ideal $(T^p - \lambda)$ ($p > 0$ characteristic of k , $\lambda \in k - k^p$).

lcbel=(iii) However, if X is a prescheme smooth over k , the set of closed points of X such that $k(x)$ is separable (and finite) over k is dense in X . Indeed, let $f : X \rightarrow \operatorname{Spec}(k)$ be the structural morphism; for every $x_0 \in X$, there is an open neighbourhood U of x_0 in X and a factorisation of $f|U : U \xrightarrow{g} \operatorname{Spec}(k[T_1, \dots, T_n]) \xrightarrow{h} \operatorname{Spec}(k)$ where g is étale (17.11.4). Since one can restrict to the case where k is not perfect, hence infinite, the set of points of the open set $g(U)$ which are rational over k is non empty; if y is such a point and $x \in U$ a point above y , x is closed in X and $k(x)$ is separable over $k(y) = k$ (17.6.2).

Proposition (17.15.11). — Let X be a prescheme of finite type on a field k . The following conditions are equivalent:

- a) X is étale on k .
- b) X is unramified on k .
- c) X is isomorphic to $\operatorname{Spec}(A)$, where A is a finite and separable k -algebra.

Proof. This results from (17.6.2) and (17.4.2), taking into account that the condition b) implies here that X is finite on k . $\square$

Proposition (17.15.12). — Let k be a field, X a prescheme locally of finite type over k . For there to exist an everywhere dense open subset U of X which is smooth over k , it is necessary and sufficient that for every maximal point x of X , X be reduced at this point and that $k(x)$ be a separable extension of k . For there to exist an everywhere dense open subset U of X such that U_{red} be smooth over k , it is necessary and sufficient that for every maximal point x of X , $k(x)$ be a separable extension of k .

Proof. The second assertion results evidently from the first. To say that there exists an everywhere dense open U smooth over k signifies that X is smooth over k at each of its maximal points. It is necessary for this that $\mathcal{O}_{X,x}$ be regular (17.15.1) and a fortiori reduced, hence a field, equal to $k(x)$; moreover (17.15.1), $k(x) \otimes_k k'$ must be regular for every radiciel extension k' of k , hence $k(x)$ must be a separable extension of k (4.3.5); the reciprocal is immediate, by virtue of (17.15.1). $\square$

Corollary (17.15.13). — *Let X be an algebraic prescheme over a field k . Then there exist an everywhere dense open subset U in X and a finite radiciel extension k' of k such that $(U_{(k')})_{\text{red}}$ is smooth over k' .*

Proof. Indeed, there is such an extension k' such that $(X_{(k')})_{\text{red}}$ is geometrically reduced over k' (4.6.6), which is equivalent to saying (4.6.1) that for every maximal point x' of $X_{(k')}$, $k(x')$ is a separable extension of k' . One concludes therefore from (17.15.12) that there exists an everywhere dense open subset U' of $X_{(k')}$ such that U'_{red} is smooth over k' . But the morphism $X_{(k')} \rightarrow X$ is a homeomorphism (2.4.5), hence U' is of the form $U_{(k')}$, where U is an everywhere dense open subset in X . $\square$

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Proposition (17.15.14). — *Let X be an algebraic prescheme over a field k , of dimension ≤ 1 . Then there exists a finite radiciel extension k' of k such that the normalised $(X_{(k')})_{\text{red}}$ is smooth on k' .*

Proposition (17.15.15). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation. For f to be smooth at a point $x \in X$, it is necessary and sufficient that f be flat at the point x and that Ω_f^1 be a $\mathcal{O}_X$ -Module locally free in a neighbourhood of x , of rank at x equal to $\dim_x f$.*

Proof. The necessity of the conditions results from (17.5.1) and (17.10.2). Inversely, if these conditions are satisfied, and if $y = f(x)$, it suffices to show (17.5.1) that $f^{-1}(y)$ is smooth on $k(y)$ at the point x ; but this follows from the definition of $\dim_x f$ (17.10.1), from (16.4.5), and from (17.15.5). $\square$

17.16. Quasi-sections of flat or smooth morphisms. The statements of this number complement those of (14.5), by means of hypotheses of flatness. IV | 105

Proposition (17.16.1). — *Let $f : X \rightarrow S$ be a morphism flat locally of finite presentation. Let $s \in S$, x a closed point of X_s such that $\mathcal{O}_{X_s, x}$ is a Cohen-Macaulay ring; then there exists an open neighbourhood U of s in S and a sub-prescheme $X' \subset f^{-1}(U)$ of X such that $x \in X'$ and the morphism $X' \rightarrow U$, restriction of f , is flat, quasi-finite and of finite presentation.*

Corollary (17.16.2). — *Let $f : X \rightarrow S$ be a morphism faithfully flat and locally of finite presentation. Then there exists a morphism $g : S' \rightarrow S$, faithfully flat, locally of finite presentation and locally quasi-finite, such that there exists an S -morphism $S' \rightarrow X$ (in other words, such that there exists an S' -section of $X' = X \times_S S'$ (I, 3.3.14)). If S is quasi-compact (resp. quasi-compact and quasi-separated), one can suppose S' affine (resp. the morphism g of finite presentation and quasi-finite).*

Corollary (17.16.3). — (i) *Let $f : X \rightarrow S$ be a smooth morphism smooth. Let $s \in S$, x a closed point of X_s such that the residue field $k(x)$ is separable on $k(s)$; then, in the conclusion of (17.16.1), one can take the morphism $X' \rightarrow U$, restriction of f , to be étale.*

(ii) *Let $f : X \rightarrow S$ be a smooth morphism smooth surjective. Then, in the conclusions of (17.16.2), one can further suppose that $g : S' \rightarrow S$ be étale.*

Proposition (17.16.4). — *Let S be a prescheme quasi-compact and quasi-separated, $f : X \rightarrow S$ a morphism of finite presentation such that, for every $s \in S$, $X_s = f^{-1}(s)$ is a prescheme proper on $k(s)$. Then there exists a finite family $(S_i)_{i \in I}$ of sub-preschemes affines of S , of finite presentation on S , pairwise disjoint, of reunion S , and such that, for every $i \in I$, there exist two morphisms of finite presentation $S''_i \xrightarrow{g_i} S'_i \xrightarrow{h_i} S_i$, where h_i is étale and g_i flat and radiciel, and an S -morphism $S''_i \rightarrow X$ (in other words, such that there exists an S''_i -section of $X''_i = X \times_S S''_i \rightarrow S_i$).*

Corollary (17.16.5). — *Let $f : X \rightarrow S$ be a surjective morphism locally of finite presentation. Then there exists a morphism $g : S' \rightarrow S$ surjective, locally of finite presentation and locally quasi-finite, such that there exists an S -morphism $S' \rightarrow X$ (in other words, such that there exists an S' -section of $X' = X \times_S S'$). If S is quasi-compact (resp. quasi-compact and quasi-separated), one can suppose S' affine (resp. S' affine and the morphism g of finite presentation and quasi-finite).*

Proposition (17.16.6). — *Let S be a prescheme quasi-compact and quasi-separated, $f : X \rightarrow S$ a morphism of finite presentation such that, for every $s \in S$, $X_s = f^{-1}(s)$ is a prescheme proper on $k(s)$. Then there exists a finite family $(S_i)_{i \in I}$ of sub-preschemes affines of S , of finite presentation on S , pairwise disjoint, of reunion S , and such that, for every i , the morphism $X_i = X \times_S S_i \rightarrow S_i$ is proper and flat. If moreover, for every $s \in S$, X_s is finite on $k(s)$, one can take the S_i such that each of the morphisms $X_i \rightarrow S_i$ factors as*

$$X_i \xrightarrow{u_i} X'_i \xrightarrow{h_i} S_i$$

where u_i and h_i are finite and locally free, u_i is étale and surjective, and h_i is flat.

§18. SUPPLEMENT ON ÉTALE MORPHISMS. HENSELIAN LOCAL RINGS AND STRICTLY LOCAL RINGS

In this paragraph, we study various special properties of étale morphisms. Moreover, the notion of étale morphism allows us to develop in a natural way the theory of Nagata of Henselian rings, as well as that of strictly local rings. These rings play an important rôle in numerous recent developments, appearing each time one needs a procedure of “localisation” finer than that furnished by the Zariski topology (cf. for example [?], while awaiting the publication of the chapter of our Treatise devoted to the study of the “étale topology”).

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18.1. A remarkable equivalence of categories.

Proposition (18.1.1). — *Let S be a prescheme, S_0 a closed sub-prescheme of S , smooth (resp. étale) on S_0 , x_0 a point of X_0 . There exists then an open neighbourhood U_0 of x_0 in X_0 , an S -prescheme U smooth (resp. étale) on S and an S_0 -isomorphism $U \times_S S_0 \xrightarrow{\sim} U_0$.*

Proof. Note that if X_0 is étale on S_0 at the point x_0 , it is *a fortiori* unramified on S_0 at this point; if one constructs an S -prescheme U smooth on S such that $U \times_S S_0$ is isomorphic to X_0 in a neighbourhood of x_0 (17.4.1, *d*)), it will then follow that U is unramified on S at the point x_0 (17.4.1, *d*)), hence also étale in a neighbourhood of x_0 . It suffices therefore to prove the proposition when one only supposes X_0 smooth on S_0 .

The question being local on S and X_0 , one can suppose that $S = \text{Spec}(A)$ and $X_0 = \text{Spec}(C_0)$, where A_0 is a quotient ring of A , $C_0 = B_0/\mathfrak{J}_0$, where $B_0 = A_0[T_1, \dots, T_n]$ and $\mathfrak{J}_0$ is an ideal of finite type of B_0 ; finally, C_0 is an A_0 -algebra formally smooth for the discrete topologies. Let $\mathfrak{p}_0$ be the ideal corresponding to q_0 in B_0 . The Jacobian criterion (0, 22.6.4) applied to (0, 19.1.12) shows that there exist in $\mathfrak{J}_0$ a family of r polynomials u_i ($1 \leq i \leq r$) and r indices j_h ($1 \leq h \leq r$) such that the images of the u_i in $(\mathfrak{J}_0)_{q_0}/(\mathfrak{J}_0^2)_{q_0}$ engender this $(B_0)_{q_0}$ -module, and that one has

$$(18.1.1.1) \quad \det(\partial u_i / \partial T_{j_h}) \notin q_0.$$

As $(B_0)_{q_0}$ is a local ring, it results from the Nakayama lemma that one can suppose that the images of the u_i in $(\mathfrak{J}_0)_{q_0}$ engender this $(B_0)_{q_0}$ -module, then, replacing X_0 by an affine open neighbourhood of x_0 (0I, 5.2.2), that the u_i engender $\mathfrak{J}_0$. Set then $B = A[T_1, \dots, T_n]$; B_0 is thus a quotient ring of B . For every i , let $v_i \in B$ be an element whose u_i is the image in B_0 , and let $\mathfrak{J}$ be the ideal of B engendered by the v_i . The proposition will be established by taking U to be an open affine neighbourhood of the point of $\text{Spec}(B/\mathfrak{J})$ corresponding to the ideal prime $\mathfrak{p} = \mathfrak{q}/\mathfrak{J}$, provided that one proves that $B_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}$ is an A -algebra formally smooth for the discrete topologies. Or, by virtue of the Jacobian criterion (0, 22.6.1 and 20.5.12.1), it suffices to take $\det(\partial v_i / \partial T_{j_h}) \notin \mathfrak{q}$, which is possible since $\mathfrak{J}_0$ is the image of $\mathfrak{J}$ in B_0 . $\square$

Theorem (18.1.2). — *Let S be a prescheme, S_0 a closed sub-prescheme of S whose underlying space is identical to that of S . Then the functor*

$$X \longmapsto X \times_S S_0$$

from the category of S -preschemes étale and finite on S , to the category of S_0 -preschemes étale and finite on S_0 , is an equivalence of categories.

Proof. Let us show first that this functor is *pleinement fidèle*. Let X, Y be two S -preschemes étale on S , and set $X_0 = X \times_S S_0$, $Y_0 = Y \times_S S_0$. If $Z = X \times_S Y$, the set $\text{Hom}_S(X, Y)$ is in canonical bijection with the set of X -sections $\Gamma(Z/X)$, and likewise $\text{Hom}_{S_0}(X_0, Y_0)$ is in canonical bijection with $\Gamma(Z_0/X_0)$, where $Z_0 = Z \times_S S_0 = X_0 \times_{S_0} Y_0$. Or Z is étale on X , Z_0 étale on X_0 (resp. Z_0 is a closed sub-prescheme of X (resp. Z) with the same underlying space. The open parts of Z on which the restriction of the morphism $Z \rightarrow X$ is surjective and radiciel are thus the *same* as the parts of Z_0 having the corresponding properties and our assertion follows from (17.9.3).

To finish the proof of (18.1.2), it suffices to see that for every S_0 -prescheme X_0 étale and finite on S , there exists an $\mathcal{O}_X$ -Algebra $\mathcal{B}$ finite and étale and an S_0 -isomorphism $\mathcal{B}_0 \xrightarrow{\sim} \mathcal{B} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_0}$. Or, it results from prop. (18.1.1) that there is a projective system $(\mathcal{B}_n)$ such that B_n is an $\mathcal{O}_{X_n}$ -Algebra

finite and étale and the homomorphisms $B_{n+1} \otimes_{A_{n+1}} A_n \rightarrow B_n$ are bijectifs. It results from (18.3.2.1) and (18.3.1) that the A -algebra $B = \varprojlim B_n$ is a projective A -module of finite type and that B_0 is isomorphic to $B \otimes_A A_0$. To prove that B is an A -algebra étale, it suffices, by virtue of (18.2.5), to show that for every maximal ideal $\mathfrak{m}$ of A , $B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}}$ is a $k(y)$ -algebra separable. But since the morphism structural $f : X \rightarrow S$ is proper, $f(x)$ is a point belonging to S_0 , hence $\mathfrak{J}$ is contained in the radical of A (0_I, 7.1.10); the conclusion results from the fact that $X_0 = f^{-1}(S_0)$ and that $\mathcal{B}_0$ is an $\mathcal{O}_{X_0}$ -algebra finite and étale. $\square$

Corollary (18.1.3). — *Let S be a prescheme, X an S -prescheme étale on S , S' a sub-prescheme of S' having the same underlying space. Then the canonical application $X(S')_S \rightarrow X(S'_0)_S$ (I, 3.4.3) is bijective.*

Proof. One sets indeed $X' = X \times_S S'$, $X'_0 = X \times_S S'_0$, so that $X(S')_S = \Gamma(X'/S')$ and $X(S'_0)_S = \Gamma(X'_0/S'_0)$; the corollary results from the fact that the functor defined in (18.1.2) (with S and S_0 replaced by S' and S'_0 respectively) is pleinement fidèle (or directly from (17.9.3)). $\square$

18.2. Étale coverings.

(18.2.1). Given a ring A and a commutative A -algebra B which is finite and is an A -module libre, we recall (Bourbaki, Alg., chap. VIII, § 12, n° 2) that one defines on B an A -linear form $\text{Tr}_{B/A}$, the “trace form”; one deduces from this the definition of a symmetric A -bilinear form (also called “trace form”)

$$(18.2.1.1) \quad (x, y) \longmapsto \text{Tr}_{B/A}(xy)$$

whose data is equivalent to that of the A -linear application associated $\text{astr}_{B/A} : B \rightarrow \check{B}$, where $\check{B}$ is the dual of the A -module B , equal to its transposed. When A is a field, it is equivalent to say that this bilinear form is non degenerate or that B is a separable algebra (Bourbaki, Alg., chap. IX, § 2, prop. 5).

Let $f : A \rightarrow A'$ be a homomorphism of rings; set $B' = B \otimes_A A'$, and let $g : B \rightarrow B'$ be the canonical homomorphism; the image by g of a basis of the A -module B is then a basis of the A' -module B' , and it follows from the definitions that for every $x \in B$,

$$(18.2.1.2) \quad \text{Tr}_{B'/A'}(g(x)) = f(\text{Tr}_{B/A}(x)).$$

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(18.2.2). Consider now a ringed space $(X, \mathcal{O}_X)$ and let $\mathcal{B}$ be an $\mathcal{O}_X$ -Algebra which, as an $\mathcal{O}_X$ -Module, is locally free of finite rank; then, for every open subset $U \subset X$ such that $\mathcal{B}|_U$ is (as an $\mathcal{O}_U$ -Module) isomorphic to $\mathcal{O}_U^n$ (for some n depending on U), $\Gamma(U, \mathcal{B})$ is a $\Gamma(U, \mathcal{O}_X)$ -algebra which, as a $\Gamma(U, \mathcal{O}_X)$ -module, is free of finite rank and thus defines a $\Gamma(U, \mathcal{O}_X)$ -linear form $\text{Tr}_{\Gamma(U, \mathcal{B})/\Gamma(U, \mathcal{O}_X)}$, which we also denote $\text{Tr}_{\mathcal{B}/\mathcal{O}_X, U}$; one deduces from this an associated linear application

$$\text{astr}_{\mathcal{B}/\mathcal{O}_X, U} : \Gamma(U, \mathcal{B}) \longrightarrow \Gamma(U, \mathcal{B})^\vee = \Gamma(U, \check{\mathcal{B}}).$$

Furthermore, it follows from (18.2.1.2) that these linear applications are compatible with the restriction operations from U to a smaller open subset, and thus define, on the one hand, a homomorphism of $\mathcal{O}_X$ -Modules, also called trace homomorphism:

$$(18.2.2.1) \quad \text{Tr}_{\mathcal{B}/\mathcal{O}_X} : \mathcal{B} \longrightarrow \mathcal{O}_X$$

and on the other hand a homomorphism of $\mathcal{O}_X$ -Modules

$$(18.2.2.2) \quad \text{astr}_{\mathcal{B}/\mathcal{O}_X} : \mathcal{B} \longrightarrow \check{\mathcal{B}}$$

called associated to the trace, and equal to its transpose. It also follows from (18.2.1.2) that for every $x \in X$, we have

$$(18.2.2.3) \quad (\text{Tr}_{\mathcal{B}/\mathcal{O}_X})_x = \text{Tr}_{\mathcal{B}_x/\mathcal{O}_{X,x}}$$

$$(18.2.2.4) \quad (\text{astr}_{\mathcal{B}/\mathcal{O}_X})_x = \text{astr}_{\mathcal{B}_x/\mathcal{O}_{X,x}}.$$

Finally, under the conditions of (18.2.1), if we set $X = \text{Spec}(A)$ and if $\mathcal{B} = \tilde{B}$ is the $\mathcal{O}_X$ -Algebra corresponding to B , the form $\text{Tr}_{\mathcal{B}/\mathcal{O}_X}$ (resp. the homomorphism $\text{astr}_{\mathcal{B}/\mathcal{O}_X}$) corresponds to the form $\text{Tr}_{B/A}$ (resp. to the homomorphism of A -modules $\text{astr}_{B/A}$), as follows again from (18.2.1.2).

Proposition (18.2.3). — *Let $f : X \rightarrow Y$ be a finite morphism of preschemes and set $\mathcal{B} = f_*(\mathcal{O}_X)$. The following conditions are equivalent:*

- a) f is étale.
- a') f is a morphism flat of finite presentation, and for every $x \in X$, if one sets $y = f(x)$, $\mathcal{O}_{X,x}/\mathfrak{m}_y \mathcal{O}_{X,x}$ is a field, finite and separable extension of $k(y)$.
- b) $\mathcal{B}$ is an $\mathcal{O}_Y$ -Module locally free and for every $y \in Y$, $\mathcal{B}_y \otimes_{\mathcal{O}_{Y,y}} k(y)$ is a $k(y)$ -algebra finite separable (hence a direct composition of a finite number of fields, separable finite extensions of $k(y)$).
- c) $\mathcal{B}$ is an $\mathcal{O}_Y$ -Module locally free, and the homomorphism $\text{astr}_{\mathcal{B}/\mathcal{O}_Y} : \mathcal{B} \rightarrow \check{\mathcal{B}}$ (18.2.2) is bijective.

Proof. Since f is quasi-compact, the equivalence of a) and a') has already been demonstrated (17.6.2). To prove the rest of the proposition, we may restrict to the case where $Y = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affine, B being a finite A -algebra and $\mathcal{B} = \tilde{B}$. Since f is a morphism of finite presentation, saying that B is an A -module of finite presentation amounts to saying that B is a flat A -module ((1.4.7)). If moreover f is flat, then B is a flat A -module, and hence (Bourbaki, *Alg. comm.*, chap. II, § 5, n° 2, cor. 2 of th. 1) B is a projective A -module, hence $\mathcal{B}$ is an $\mathcal{O}_Y$ -Module locally free (*loc. cit.*, n° 2, th. 1), and the converse is immediate. On the other hand, $f^{-1}(y)$ is none other than the spectrum of the $k(y)$ -algebra $\mathcal{B}(y) = \mathcal{B}_y \otimes_{\mathcal{O}_{Y,y}} k(y)$, which completes the proof of the equivalence of a') and b). To see that b) is equivalent to c), note that the second assertion of b) amounts to the fact that the homomorphism $\text{astr}_{\mathcal{B}(y)/k(y)} : \mathcal{B}(y) \rightarrow \mathcal{B}(y)^\vee$ is bijective; since $\mathcal{B}(y) = \mathcal{B}_y/\mathfrak{m}_y \mathcal{B}_y$ and $\mathcal{B}_y$ and $\check{\mathcal{B}}_y$ are free $\mathcal{O}_{Y,y}$ -modules, it follows from (18.2.2.4) and Bourbaki, *Alg. comm.*, chap. II, § 3, n° 3, cor. of prop. 6, that the homomorphism $\text{astr}_{\mathcal{B}_y/\mathcal{O}_{Y,y}} : \mathcal{B}_y \rightarrow \check{\mathcal{B}}_y$ is also bijective; the converse being evident, this concludes the demonstration. $\square$

Corollary (18.2.4). — Let $f : X \rightarrow Y$ be a morphism finite and of finite presentation, and set $\mathcal{B} = f_*(\mathcal{O}_X)$. Let y be a point of Y . The following conditions are equivalent:

- a) There exists an open neighbourhood U of y in Y such that the restriction $f^{-1}(U) \rightarrow U$ of f is an étale morphism.
- b) $\mathcal{B}_y$ is an $\mathcal{O}_{X,y}$ -module libre of finite type and $\mathcal{B}_y \otimes_{\mathcal{O}_{Y,y}} k(y)$ is a $k(y)$ -algebra separable.

Proof. It is clear that a) implies b) by virtue of (18.2.3). On the other hand, $\mathcal{B}$ is an $\mathcal{O}_Y$ -Module of finite presentation ((1.6.3) and (1.4.7)), hence, if $\mathcal{B}_y$ is an $\mathcal{O}_{Y,y}$ -module libre, there exists an open neighbourhood U of y in Y such that $\mathcal{B}|_U$ is an $\mathcal{O}_U$ -Module locally libre (0_I, 5.2.7); moreover, by hypothesis, the homomorphism $\text{astr}_{\mathcal{B}_y/\mathcal{O}_{Y,y}} : \mathcal{B}_y \rightarrow \check{\mathcal{B}}_y$ is bijective, and it follows also from (0_I, 5.2.7) that one can suppose U chosen so that the homomorphism $\text{astr}_{\mathcal{B}|_U/\mathcal{O}_U}$ is bijective. The fact that b) implies a) then follows from (18.2.3). $\square$

Corollary (18.2.5). — Let Y be a prescheme quasi-compact or locally Noetherian, $f : X \rightarrow Y$ a morphism finite and of finite presentation; set $\mathcal{B} = f_*(\mathcal{O}_X)$. Suppose that for every closed point y of Y , $\mathcal{B}_y$ is an $\mathcal{O}_{Y,y}$ -module libre and $\mathcal{B}_y \otimes_{\mathcal{O}_{Y,y}} k(y)$ a $k(y)$ -algebra separable. Then f is étale.

Proof. Indeed, it follows from (18.2.4) that every closed point y of Y possesses an open neighbourhood U such that the restriction $f^{-1}(U) \rightarrow U$ of f is étale; the conclusion follows from the fact that in the two cases considered, every non-empty closed subset of Y contains a closed point ((5.1.11) and (0_I, 2.1.3)). $\square$

Corollary (18.2.6). — If $f : X \rightarrow Y$ is a morphism finite and étale, and if one sets $\mathcal{B} = f_*(\mathcal{O}_X)$, then the $\mathcal{O}_Y$ -homomorphism $\text{Tr}_{\mathcal{B}/\mathcal{O}_Y} : \mathcal{B} \rightarrow \mathcal{O}_Y$ (18.2.2) (also denoted Tr_f) is surjective.

Proof. The question being local, one can, by virtue of (18.2.3), suppose that $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$ with $\mathcal{B} = \tilde{B}$, B being a free A -module; since by virtue of (18.2.3) the bilinear form (18.2.1.1) is non-degenerate, this implies in particular that the linear form $\text{Tr}_{B/A}$ is surjective. $\square$

Remarks (18.2.7). — (i) When $f : X \rightarrow Y$ is a finite morphism such that $f_*(\mathcal{O}_X)$ is an $\mathcal{O}_Y$ -Module locally libre (resp. locally libre of rank n), one says also that f is a morphism *finite locally libre* (resp. *locally libre of rank n*). This condition, by virtue of (18.2.3), is satisfied if f is a finite étale morphism, but does not imply by itself that f be étale, as is shown by the example where $X = \text{Spec}(K)$ and $Y = \text{Spec}(k)$ are spectra of fields, K being a finite non separable extension of k . When $f : X \rightarrow Y$ is a finite étale morphism, one says also that X is an *étale covering* of Y . In particular $f(X)$ is an open and closed part of Y .

One says that a covering is *trivial* if X is the sum of a finite number of preschemes isomorphic to Y . One says that a covering $f : X \rightarrow Y$ is *locally trivial* if the morphism $f : X \rightarrow Y$ is such that for every point $y \in Y$ there is an open neighbourhood U such that $f^{-1}(U)$ is a trivial covering of U .

(ii) Let $f : X \rightarrow Y$ be a finite morphism, locally libre of rank n ; set $\mathcal{B} = f_*(\mathcal{O}_X)$ and let $u = \text{astr}_{\mathcal{B}/\mathcal{O}_Y} : \mathcal{B} \rightarrow \check{\mathcal{B}}$; one deduces from this an exterior n -th power homomorphism $\bigwedge^n u : \bigwedge^n \mathcal{B} \rightarrow (\bigwedge^n \mathcal{B})^\vee$ of $\mathcal{O}_Y$ -Modules *invertibles*, and hence (0_I, 5.4.2) an element

$$(18.2.7.1) \quad d_{X/Y} \in \Gamma(Y, (\bigwedge^n \check{\mathcal{B}}) \otimes_{\mathcal{O}_Y} (\bigwedge^n \check{\mathcal{B}}))$$

called the *discriminant* of X over Y . Since $(\bigwedge^n \mathcal{B}) \otimes_{\mathcal{O}_Y} (\bigwedge^n \mathcal{B})$ is the dual of $(\bigwedge^n \check{\mathcal{B}}) \otimes_{\mathcal{O}_Y} (\bigwedge^n \check{\mathcal{B}})$, $d_{X/Y}$ can also be identified with a homomorphism

$$(18.2.7.2) \quad (\bigwedge^n \mathcal{B}) \otimes_{\mathcal{O}_Y} (\bigwedge^n \mathcal{B}) \longrightarrow \mathcal{O}_Y$$

and one denotes $\mathcal{D}_{X/Y}$ the quasi-coherent Ideal of finite type of $\mathcal{O}_Y$, image of the homomorphism (18.2.7.2), also called the *Ideal discriminant* of X over Y .

Proposition (18.2.8). — *Let $f : X \rightarrow Y$ be a morphism étale, separated and of finite type, and for every $y \in Y$, let $n(y)$ be the geometric number of points of $f^{-1}(y)$. Then the function $y \mapsto n(y)$ is semi-continuous inferiorly in Y . For it to be continuous in a neighbourhood of y (and hence constant in a neighbourhood of y), it is necessary and sufficient that there exist an open neighbourhood U of y such that the restriction $f^{-1}(U) \rightarrow U$ of f is a finite (étale) morphism.*

Proof. As f is quasi-finite (17.6.1) and locally of finite presentation, it amounts to the same to say that f is finite or that f is proper (8.11.1): moreover each fibre $f^{-1}(y)$ is geometrically reduced on $k(y)$. The conclusions then result from (15.5.9), (i) and (ii) and from the fact that f is flat. $\square$

Corollary (18.2.9). — *Let Y be a connected prescheme, $f : X \rightarrow Y$ a morphism étale separated and of finite type. For f to be finite (in other words, for X to be an étale covering of Y), it is necessary and sufficient that all the fibres of f have the same geometric number of points.*

Proof. The condition is necessary since f , being finite and étale, is locally libre, and hence the function $y \mapsto n(y)$ is locally constant, thus constant on the connected space Y . Conversely, if $n(y)$ is constant, then it is continuous, hence by (18.2.8) there exists for every $y \in Y$ an open neighbourhood U such that $f^{-1}(U) \rightarrow U$ is finite. $\square$

Remarks (18.2.10). —

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- label=(i) The example of the “affine line with a doubled point” (I, 5.5.11) shows that a morphism étale of finite type of noetherian preschemes can be *non separated*; for this example, the first assertion of (18.2.8) is no longer exact.
- lbbel=(ii) For a morphism separated, of finite type and étale $f : X \rightarrow Y$ to make X a *locally trivial* covering, it is necessary and sufficient that for every $x \in X$ there exist an open neighbourhood U of $y = f(x)$ and a U -section g of $f^{-1}(U)$ such that $g(y) = x$. Indeed, the condition is evidently necessary; the fact that it is sufficient results from the fact that every fibre $f^{-1}(y)$ is finite ((17.6.1) and (I, 6.2.2)), from the characterisation of sections of a Y -scheme étale (17.9.3) and from the proposition (18.2.8).

18.3. Finite and étale algebras.

Proposition (18.3.1). — *Let A be a ring, B an A -algebra of finite presentation.*

label=(i) For B to be an unramified A -algebra, it is necessary and sufficient that B be an A -module of finite presentation and that B is a $(B \otimes_A B)$ -module projective.

lbbel=(ii) Suppose moreover that B is a finite A -algebra. For B to be an étale A -algebra, it is necessary and sufficient that B is an A -module projective and a $(B \otimes_A B)$ -module projective.

Proof. Of course, the $(B \otimes_A B)$ -module structure on B is that coming from the structure of $(B \otimes_A B)$ -algebra on B corresponding to the canonical A -homomorphism of rings $B \otimes_A B \rightarrow B$, which is surjective and has kernel $\mathfrak{J}_{B/A}$ (**0I**, 20.4.1).

(i) To say that the morphism $\text{Spec}(B) \rightarrow \text{Spec}(A)$ is locally of finite presentation is equivalent to saying that B is an A -algebra of finite presentation (1.4.6). To say that B is an unramified A -algebra means that $\mathfrak{J}_{B/A}$ induces an open and closed subscheme of $\text{Spec}(B \otimes_A B)$, and since one knows that $\text{Spec}((B \otimes_A B)/\mathfrak{J}_{B/A})$ is an open and closed part of $\text{Spec}(B \otimes_A B)$ (17.4.2) it is necessary and sufficient that $\mathfrak{J}_{B/A}$ be a direct factor ideal of $B \otimes_A B$ (Bourbaki, *Alg. comm.*, chap. II, § 4, n° 3, prop. 15); but this amounts to saying that the $(B \otimes_A B)$ -module quotient $(B \otimes_A B)/\mathfrak{J}_{B/A}$ is projective (Bourbaki, *Alg.*, chap. II, 3^e éd., § 2, n° 2, prop. 4).

(ii) If one recalls that a flat A -module of finite presentation is projective and reciprocally (Bourbaki, *Alg. comm.*, chap. II, § 5, n° 2, cor. 2 du th. 1), the assertion of (ii) results from that of (i) and from (17.6.2). $\square$

Proposition (18.3.2). — *Let A be a ring, $\mathfrak{J}$ an ideal of A such that, for the $\mathfrak{J}$ -pre-adic topology, A is separated and complete; set $A_0 = A/\mathfrak{J}$. Then the functor*

$$B \longmapsto B \otimes_A A_0$$

is an equivalence of the category of A -algebras finite and étale, to the category of A_0 -algebras finite and étale.

Proof. We shall first prove the following lemma:

Lemma (18.3.2.1). — *Let A be a ring, $\mathfrak{J}$ an ideal of A such that, for the $\mathfrak{J}$ -pre-adic topology, A is separated and complete.*

label=(i) Every projective A -module of finite type M is separated and complete for the $\mathfrak{J}$ -pre-adic topology, hence a projective limit of the projective $(A/\mathfrak{J}^{n+1})$ -modules $M/\mathfrak{J}^{n+1}M$.

lbbel=(ii) Conversely, set $A_n = A/\mathfrak{J}^{n+1}$, and let (M_n) be a projective system of A_n -modules, such that, for every n , the homomorphism $M_{n+1} \otimes_{A_{n+1}} A_n \rightarrow M_n$ deduced from the transition homomorphism $M_{n+1} \rightarrow M_n$ is bijective. Suppose moreover the M_n projective and of finite type. Then $M = \varprojlim M_n$ is a projective A -module of finite type such that $M \otimes_A A_0 \rightarrow M_0$ is bijective.

(i) There exists a free A -module of finite type L such that M is isomorphic to a direct factor of L ; since L is separated for the $\mathfrak{J}$ -pre-adic topology, the same is true of every submodule N of L since $\mathfrak{J}^{n+1}N \subset \mathfrak{J}^{n+1}L$; in particular M is separated for this topology, and since the surjective homomorphism $f : L \rightarrow M$ is continuous for the topology induced by that of L ; since L is complete and f a strict morphism, one concludes that M is complete (Bourbaki, *Top. gén.*, chap. IX, 2^e éd., § 3, n° 1, prop. 4).

(ii) It follows from Nakayama's lemma that if M_0 is generated by a finite family (x_{i0}) of r elements and if for every n , x_{in} is an element of M_n whose image in M_{n-1} is $x_{i,n-1}$, then (x_{in}) ($1 \leq i \leq r$) is a system of generators of M_n (Bourbaki, *Alg. comm.*, chap. II, § 3, cor. 2 de la prop. 4). This being so, for every n , set $L_n = A_n^r$; if $(\varepsilon_{in})_{1 \leq i \leq r}$ is the canonical basis of L_n , let $u_n : L_n \rightarrow M_n$ be the A -linear application such that $u_n(\varepsilon_{in}) = x_{in}$ for every i . By hypothesis one has a split exact sequence

$$0 \longrightarrow N_n \xrightarrow{v_n} L_n \xrightarrow{u_n} M_n \longrightarrow 0$$

and since $L_n = L_{n+1}/\mathfrak{J}^{n+1}L_{n+1}$ and $M_n = M_{n+1}/\mathfrak{J}^{n+1}M_{n+1}$, the vertical arrows in the commutative diagram IV | 116

$$\begin{array}{ccccccc} 0 & \longrightarrow & N_{n+1} & \xrightarrow{v_{n+1}} & L_{n+1} & \xrightarrow{u_{n+1}} & M_{n+1} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & N_n & \xrightarrow{v_n} & L_n & \xrightarrow{u_n} & M_n \longrightarrow 0 \end{array}$$

are all three surjective. Now we have $M = \varprojlim M_n$ and $L = A' = \varprojlim L_n$; if one sets $N = \varprojlim N_n$, $u = \varprojlim u_n$, $v = \varprojlim v_n$, one has, by virtue of (0_{III}, 13.2.2), the exact sequence

$$(18.3.2.2) \quad 0 \longrightarrow N \xrightarrow{v} L \xrightarrow{u} M \longrightarrow 0.$$

Furthermore, since for every n , v_n is left invertible and M_n is a projective A_n -module, it follows from (0, 19.1.8) that the exact sequence (18.3.2.2) is split, which proves the lemma.

This being so, let us first show that the functor of the statement of (18.3.2) is *fully faithful*. Set, as in the lemma, $A_n = A/\mathfrak{J}^{n+1}$; let B, C be two A -algebras finite and étale, and set, for every n , $B_n = B \otimes_A A_n$, $C_n = C \otimes_A A_n$; by virtue of (18.3.1) and (18.3.2.1), B and C are separated and complete for the $\mathfrak{J}$ -preadic topology, and one has $B = \varprojlim B_n$, $C = \varprojlim C_n$; moreover every homomorphism of A -algebras $u : B \rightarrow C$ is continuous for the $\mathfrak{J}$ -preadic topologies, and gives therefore a projective system of homomorphisms of A_n -algebras $u_n = u \otimes 1 : B_n \rightarrow C_n$, of which it is the projective limit; the reciprocal being evident, one has therefore a canonical bijection

$$\mathrm{Hom}_{A\text{-alg.}}(B, C) \xrightarrow{\sim} \varprojlim \mathrm{Hom}_{A_n\text{-alg.}}(B_n, C_n).$$

But since B and C are A -algebras étale, it also follows from (18.2.1.2) that the canonical application

$$\mathrm{Hom}_{A_{n+1}\text{-alg.}}(B_{n+1}, C_{n+1}) \longrightarrow \mathrm{Hom}_{A_n\text{-alg.}}(B_n, C_n)$$

is *bijective* for $n \geq 0$, which completes the proof that the canonical application $\mathrm{Hom}_{A\text{-alg.}}(B, C) \rightarrow \mathrm{Hom}_{A_0\text{-alg.}}(B_0, C_0)$ is bijective.

To complete the proof of (18.3.2), it suffices to show that for every A_0 -algebra B_0 finite and étale there exists a finite A -algebra B and an A_0 -isomorphism $B_0 \xrightarrow{\sim} B \otimes_A A_0$. Now, it follows from (18.2.1.2) that there is a projective system (B_n) such that B_n is an A_n -algebra finite and étale and the homomorphisms $B_{n+1} \otimes_{A_{n+1}} A_n \rightarrow B_n$ are bijective. It follows from (18.3.1) and (18.3.2.1) that the A -algebra $B = \varprojlim B_n$ is a projective A -module of finite type and that B_0 is isomorphic to $B \otimes_A A_0$. To prove that B is an A -algebra étale, it suffices, by virtue of (18.2.5), to show that for every maximal ideal $\mathfrak{m}$ of A , $B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}}$ is a $k(\mathfrak{m})$ -algebra separable. Now, since $\mathfrak{J}$ is contained in the radical of A (0_I, 7.1.10), the conclusion follows from the fact that B_0 is an A_0 -algebra finite and étale (18.2.5). $\square$

Example (18.3.3). — Prop. (18.3.2) applies in particular when A is a *local ring, separated and complete*, $\mathfrak{J}$ being the maximal ideal of A , so that A_0 is a field, and the category of A_0 -algebras finite and étale is identical to that of the A_0 -algebras finite and separable, hence isomorphic to direct compositions of fields, separable finite extensions of A_0 . In particular, if the field A_0 is *separably closed*, these étale coverings are all identical to A_0 , and hence every étale covering of $\mathrm{Spec}(A)$ is *trivial* (18.2.7) by virtue of (18.3.2).

Theorem (18.3.4). — Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A such that A is separated and complete for the $\mathfrak{J}$ -preadic topology, $A_0 = A/\mathfrak{J}$. Set $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(A_0)$. Let X be an S -scheme proper on S , and set $X_0 = X \times_S X_0$. Then the functor

$$Z \longmapsto Z \times_X X_0$$

from the category of X -schemes finite and étale on X , to the category of X_0 -schemes étale and finite on X_0 , is an equivalence of categories.

Proof. Let us first show that this functor is *fully faithful*. Let Z' and Z'' be two X -schemes finite and étale on X . Set $S_n = \mathrm{Spec}(A/\mathfrak{J}^{n+1})$, $X_n = X \times_S S_n$, $Z'_n = Z' \times_S S_n$, $Z''_n = Z'' \times_S S_n$ for every $n \geq 0$. It follows from (III, 5.4.1) that one has a canonical bijection

$$\mathrm{Hom}_X(Z', Z'') \xrightarrow{\sim} \varprojlim \mathrm{Hom}_{X_n}(Z'_n, Z''_n).$$

Now, by virtue of (18.1.2), the canonical application

$$\mathrm{Hom}_{X_{n+1}}(Z'_{n+1}, Z''_{n+1}) \longrightarrow \mathrm{Hom}_{X_n}(Z'_n, Z''_n)$$

is bijective, which proves our assertion.

It remains to prove that if $\mathcal{B}_0$ is an $\mathcal{O}_{X_0}$ -Algebra finite and étale $\mathcal{B}_0 \xrightarrow{\sim} \mathcal{B} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_0}$, there exists an $\mathcal{O}_X$ -Algebra $\mathcal{B}$ finite and étale and an isomorphism $\mathcal{B}_0 \xrightarrow{\sim} \mathcal{B} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_0}$. It follows from (18.1.2) that there is a projective system $(\mathcal{B}_n)$, where $\mathcal{B}_n$ is an $\mathcal{O}_{X_n}$ -Algebra finite and étale, and the second comparison theorem (III, 5.1.4) proves the existence of an $\mathcal{O}_X$ -Module coherent $\mathcal{B}$ and of a projective system of isomorphisms $\mathcal{B}_n \xrightarrow{\sim} \mathcal{B} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_n}$. The datum of an Algebra structure on a Module $\mathcal{F}$ being equivalent to that of a homomorphism $\mathcal{F} \otimes \mathcal{F} \rightarrow \mathcal{F}$ rendering commutative diagrams where only tensor powers of $\mathcal{F}$ intervene, it follows from (III, 5.1.3), (I, 10.11.4) and (I, 10.11.7) that $\mathcal{B}$ is naturally endowed with a structure of $\mathcal{O}_X$ -Algebra for which the isomorphism $\mathcal{B}_0 \xrightarrow{\sim} \mathcal{B} \otimes_{\mathcal{O}_X} \mathcal{O}_{X_0}$ is an isomorphism of Algebras. Moreover, $\mathcal{B}$ is an $\mathcal{O}_X$ -Module *locally free*; this follows still from (III, 5.1.3), (I, 10.11.4), (I, 10.11.7) and from the fact that, in the category of $\mathcal{O}_X$ -Modules coherent, the $\mathcal{O}_X$ -Modules locally free $\mathcal{F}$ can be defined by the fact that the $\mathcal{O}_{X_n}$ -Modules locally free are exactly those for which the functor $\mathcal{G} \mapsto \mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ is *exact*. Finally, to see that $\mathcal{B}$ is an $\mathcal{O}_X$ -Algebra étale, it suffices (18.2.5) to show that for every closed point $x \in X$, $\mathcal{B}_x \otimes_{\mathcal{O}_{X,x}} k(x)$ is a $k(x)$ -algebra separable. But since the morphism structural $f : X \rightarrow S$ is *proper*, $f(x)$ is a closed point of S , hence belongs to S_0 , since $\mathfrak{J}$ is contained in the radical of A (0_I, 7.1.10); the conclusion results from the fact that $\mathcal{B}_0$ is an $\mathcal{O}_{X_0}$ -Algebra finite and étale (18.2.5). $\square$

18.4. Local structure of unramified and étale morphisms.

Lemma (18.4.1). — *Let A be a ring, B an A -algebra finite monogene, u a generator of the A -algebra B , $F \in A[T]$ a polynomial such that $F(u) = 0$, F' the derived polynomial; set $u' = F'(u)$. Then the Ideal of B , annihilator of $\Omega_{B/A}^1$, contains $u'B$; it is equal to $u'B$ if the ideal $\mathfrak{J}$ of $A[T]$ consisting of the polynomials G such that $G(u) = 0$, is engendered by F , in other words if the surjective canonical homomorphism $\varphi : A[T]/F.A[T] \rightarrow B$ sending the image of T to u , is bijective.*

Proof. Set $C = A[T]$, so that $B = C/\mathfrak{J}$. We have the exact sequence (0, 20.5.12.1)

$$\mathfrak{J}/\mathfrak{J}^2 \longrightarrow \Omega_{C/A}^1 \otimes_C B \longrightarrow \Omega_{B/A}^1 \longrightarrow 0$$

and $\Omega_{B/A}^1$ therefore identifies with the quotient of B by the ideal $\mathfrak{J}'$ generated by the elements $G'(u)$, where G runs through a system of generators of $\mathfrak{J}$ (0, 20.5.13); the lemma follows immediately. $\square$

Proposition (18.4.2). — *With the notations of (18.4.1), let $\mathfrak{q}$ be a prime ideal of B . Then:*

- label=(i) If $\mathfrak{q}$ does not contain u' , $B_{\mathfrak{q}}$ is an $A_{\mathfrak{p}}$ -algebra formally unramified ($\mathfrak{p}$ being the inverse image of $\mathfrak{q}$ in A); in other words, $\mathrm{Spec}(B_u)$ is formally unramified on $\mathrm{Spec}(A)$.*
- lbbel=(ii) Suppose moreover that F is unitary and engenders $\mathfrak{J}$. Then, for B to be an A -algebra étale on $\mathrm{Spec}(A)$ at the point $\mathfrak{q}$, it is necessary and sufficient that $u' \notin \mathfrak{q}$.*

Proof. The hypothesis that $u' \notin \mathfrak{q}$ implies that $\Omega_{B_{\mathfrak{q}}/A_{\mathfrak{p}}}^1 = 0$ (0, 20.5.9), hence (i) follows from (17.2.1). Moreover, under the hypotheses of (ii), B is a free A -module by virtue of Euclidean division; as the annihilator $\mathfrak{J}'$ of $\Omega_{B/A}^1$ is then equal to $u'B$ by virtue of (18.4.1) and $\Omega_{B/A}^1$ is a B -module of finite presentation (16.4.22), the annihilator of $\Omega_{B_{\mathfrak{q}}/A_{\mathfrak{p}}}^1$ is equal to $u'B_{\mathfrak{q}}$ (Bourbaki, Alg. comm., chap. II, § 2, n° 4, formula (9)), and (ii) follows from (i) and from the implication $c) \Rightarrow a)$ in (17.6.1). $\square$

Corollary (18.4.3). — *With the notations of (18.4.2), suppose that F is unitary and engenders $\mathfrak{J}$. Then, for B to be an A -algebra étale, it is necessary and sufficient that u' be inversible in B (or, what amounts to the same, that the ideal of $A[T]$ engendered by F and F' be equal to $A[T]$).*

One says that a unitary polynomial $F \in A[T]$ such that the ideal of $A[T]$ engendered by F and F' is equal to $A[T]$ is separable; it is immediate that this definition coincides with the usual definition (Bourbaki, Alg., chap. V, § 7, n° 6) when A is a field.

Proof. Taking into account (18.4.2), (ii), to say that $\mathrm{Spec}(B)$ is étale over $\mathrm{Spec}(A)$ means that u' does not belong to any prime ideal of B , i.e. that it is inversible in B . $\square$

Lemma (18.4.4). — *Let A be a local ring, B an A -algebra finite monogene, u a generator of the A -algebra B . Let $\mathfrak{n}_i$ ($1 \leq i \leq r$) be the maximal ideals of B such that $\text{Spec}(B)$ is formally unramified on $\text{Spec}(A)$ at the points $\mathfrak{n}_i$. Then there exists a unitary polynomial $F \in A[T]$ such that $F(u) = 0$ and $F'(u) \notin \mathfrak{n}_i$ for every index i ($1 \leq i \leq r$). Moreover, if k is the residue field of A and $f \in k[T]$ the minimal polynomial of the image of u in $B \otimes_A k$ over k , there exists $F \in A[T]$ satisfying all the conditions, whose image canonical is f ; and F is necessarily of degree n .*

Proof. The maximal ideal $\mathfrak{m}$ of A is the inverse image of each of the $\mathfrak{n}_i$ ((II, 6.1.10)); set $L = B \otimes_A k$, which is a finite k -algebra. Let ζ be the image of u in L , and let n be the rank of L over k ; the minimal polynomial $f \in k[T]$ of ζ over k is therefore of degree n , and L is isomorphic to $k[T]/f.k[T]$. If $\mathfrak{n}'_i = \mathfrak{n}_i/\mathfrak{m}B$, the hypothesis implies that $\text{Spec}(L)$ is étale over k at the points $\mathfrak{n}'_i$ ($1 \leq i \leq r$) by virtue of (17.6.1, c)), hence $f'(\zeta) \notin \mathfrak{n}'_i$ by virtue of (18.4.2), (ii)). Note now that $\mathfrak{m}B$ is contained in the radical of B ; as $1, \zeta, \dots, \zeta^{n-1}$ form a base of L over k , it follows from Nakayama's lemma that $1, u, \dots, u^{n-1}$ generate the A -module B , and so there exists a unitary polynomial $F \in A[T]$ of degree n , such that $F(u) = 0$; moreover, as ζ is a root of the canonical image of F in $k[T]$, this image is necessarily equal to f . But then the image of $F'(u)$ in L is $f'(\zeta)$, and as $f'(\zeta) \notin \mathfrak{n}'_i$, we have $F'(u) \notin \mathfrak{n}_i$ for every index i . $\square$

Proposition (18.4.5). — *Let A be a local ring, k its residue field, B an A -algebra finite (resp. of finite presentation). Suppose moreover, either that the field k is infinite, or that B is a local ring. Let n be the rank of $L = B \otimes_A k$ over k . For B to be an A -algebra formally unramified (resp. étale), it is necessary and sufficient that there exist a unitary separable polynomial $F \in A[T]$ (18.4.3) such that B is isomorphic to a quotient of $A[T]/F.A[T]$ (resp. isomorphic to $A[T]/F.A[T]$). Moreover, one can suppose that F is of degree n (resp. F is necessarily of degree n).*

Proof. The conditions are sufficient by virtue of (18.4.2), without assuming k infinite or B local. To see that the conditions are necessary, note that if B is a formally unramified A -algebra, L is a finite separable algebra over k , hence a direct composition of a finite number of finite separable extensions k_j of k ($1 \leq j \leq r$), each k_j therefore being generated by an element ζ_j with minimal polynomial f_j ($1 \leq j \leq r$) (Bourbaki, Alg., chap. V, § 11, n° 4, prop. 4). Let us show that, under the hypotheses made on k or B , there exists an element ζ of L generating the k -algebra L . This is immediate if B is local, since then $r = 1$. Otherwise, k being assumed infinite, we may assume that the irreducible polynomials $f_j \in k[T]$ are all distinct (replacing if necessary each ζ_j by $\zeta_j + a_j$, for a suitable element $a_j \in k$); if we set $f = f_1 f_2 \cdots f_r$, it is clear that L is isomorphic to $k[T]/f.k[T]$ in both cases, hence is generated by an element ζ of minimal polynomial $f \in k[T]$ of degree n . If $u \in B$ is an element with image ζ in L , Nakayama's lemma shows that the elements $1, u, \dots, u^{n-1}$ generate the A -module B ; this already shows that there exists a unitary polynomial $F \in A[T]$ of degree n such that $F(u) = 0$, u generating the A -algebra B , which is therefore isomorphic to a quotient algebra of $A[T]/F.A[T]$; moreover B is a semi-local ring and at each of its maximal ideals $\mathfrak{n}_i$, we have $F'(u) \notin \mathfrak{n}_i$ by (18.4.4), which proves that $F'(u)$ is invertible in B , hence that F is a separable polynomial. Finally, if B is an étale A -algebra, B being a flat A -module of finite presentation (I, 1.4.7) is a free A -module, and $1, u, \dots, u^{n-1}$ form a base of the A -module B (Bourbaki, Alg. comm., chap. II, § 3, n° 2, prop. 5), that is to say the A -algebra B is isomorphic to $A[T]/F.A[T]$, and for every other unitary polynomial $G \in A[T]$ such that B is isomorphic to $A[T]/G.A[T]$, G is necessarily of degree n . $\square$

Theorem (18.4.6). — (Chevalley).

label=(i) Let $f : X \rightarrow Y$ be a morphism locally of finite type, x a point of X , $y = f(x)$, and set $A = \mathcal{O}_{Y,y}$. For $\mathcal{O}_{X,x}$ to be an A -algebra formally étale (resp. formally unramified), it is necessary and sufficient that there exist a unitary polynomial $F \in A[T]$ and a maximal ideal $\mathfrak{n}$ of $B = A[T]/F.A[T]$ (resp. an algebra quotient B of $A[T]/F.A[T]$) such that $\mathcal{O}_{X,x}$ is A -isomorphic to $B_{\mathfrak{n}}$ and that, if u is the image of T in B , one has $F'(u) \notin \mathfrak{n}$.

lbbel=(ii) Suppose moreover that f is locally of finite presentation. Then, for f to be étale at the point x , it is necessary and sufficient that one can take $B = A[T]/F.A[T]$.

Proof. The conditions are sufficient by virtue of (18.4.2). To see that they are necessary, we can evidently restrict to the case where X and Y are affine, and, taking into account the remark (17.1.2), to the case where f is formally unramified and quasi-finite. As f is affine, it follows from (8.12.8)

that there exist a finite A -algebra C and a maximal ideal $\mathfrak{r}$ of C (necessarily above the maximal ideal $\mathfrak{m}$ of A) such that $\mathcal{O}_{X,x}$ is A -isomorphic to $C_{\mathfrak{r}}$. Moreover (17.4.1.2) the residue field $C/\mathfrak{r} = C_{\mathfrak{r}}/\mathfrak{r}C_{\mathfrak{r}}$ is a finite separable extension of k . Let $\mathfrak{r}_i$ ($1 \leq i \leq h$) be the maximal ideals of the semi-local ring C other than $\mathfrak{r}$; let u be an element of C belonging to all the maximal ideals of C distinct from $\mathfrak{r}$, not belonging to $\mathfrak{r}$, and such that $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$ is a monogene algebra over $k = A/\mathfrak{m}$, generated by the image of u in $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$. Set $B = A[u]$, $\mathfrak{n} = \mathfrak{r} \cap B$. We shall show that the sub- A -algebra B of C and the ideal $\mathfrak{n}$ answer the question.

To treat the case where $\mathcal{O}_{X,x}$ is a formally unramified A -algebra, it will suffice to prove that $B_{\mathfrak{n}}$ is isomorphic to $C_{\mathfrak{r}}$; indeed, $B_{\mathfrak{n}}$ will then be formally unramified over A and the existence of the polynomial $F \in A[T]$ having the properties of the statement will follow from (18.4.4). Note now that, since f is formally unramified, $C/\mathfrak{r}$ is isomorphic to the A -algebra $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$ (17.4.1.2). We are thus reduced to proving the following lemma: $\square$

Lemma (18.4.6.1). — *Let A be a local ring, $\mathfrak{m}$ its maximal ideal, C a finite A -algebra, $\mathfrak{r}$ a maximal ideal of C . Let u be an element of C belonging to all the maximal ideals of C distinct from $\mathfrak{r}$, not belonging to $\mathfrak{r}$, and such that $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$ is a monogene algebra over $k = A/\mathfrak{m}$, generated by the image of u in $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$. Set $B = A[u]$, $\mathfrak{n} = \mathfrak{r} \cap B$. Then the canonical homomorphism $B_{\mathfrak{n}} \rightarrow C_{\mathfrak{r}}$ is an isomorphism.*

Proof. Set $R = B \setminus \mathfrak{n}$, $S = C \setminus \mathfrak{r}$, so that $B_{\mathfrak{n}} = R^{-1}B$ and $C_{\mathfrak{r}} = S^{-1}C$; the canonical homomorphism $B_{\mathfrak{n}} \rightarrow C_{\mathfrak{r}}$ can be written as the composite

$$R^{-1}B \xrightarrow{g} R^{-1}C \xrightarrow{h} S^{-1}C$$

and it suffices to show that each of these two homomorphisms is bijective.

Let us first show that $h : R^{-1}C \rightarrow S^{-1}C = C_{\mathfrak{r}}$ is bijective; it suffices to see that the images in $R^{-1}C$ of the elements of S are invertible, or equivalently that every maximal ideal $\mathfrak{p}$ of $R^{-1}C$ has an inverse image in C not meeting S , hence necessarily equal to $\mathfrak{r}$. Now, as $R^{-1}C$ is a finite $(R^{-1}B)$ -algebra, the inverse image of $\mathfrak{p}$ in $R^{-1}B = B_{\mathfrak{n}}$ is the unique maximal ideal $\mathfrak{n}B_{\mathfrak{n}}$ of this ring, hence the inverse image of $\mathfrak{p}$ in B is equal to $\mathfrak{n}$. But on the other hand, if q is the inverse image of $\mathfrak{p}$ in C , we have $q \cap B = \mathfrak{n}$, and as $\mathfrak{n}$ is maximal in B and C is a finite B -algebra, q is necessarily one of the maximal ideals of C ; moreover, we have $u \notin q$ since $u \notin \mathfrak{r}$ and $u \in B$, hence by hypothesis we necessarily have $q = \mathfrak{r}$.

On the other hand, as $B \subset C$, the homomorphism $g : R^{-1}B \rightarrow R^{-1}C$ is injective (0_I, 1.3.2); to see that it is surjective, note that $R^{-1}C$ is an $(R^{-1}B)$ -module of finite type, and on the other hand that $\mathfrak{m}R^{-1}B$ is contained in the maximal ideal of the local ring $R^{-1}B$; by virtue of Nakayama's lemma, it suffices to prove that the homomorphism $R^{-1}B/\mathfrak{m}R^{-1}B \rightarrow R^{-1}C/\mathfrak{m}R^{-1}C$ is surjective. But by virtue of the first part of the proof, $R^{-1}C/\mathfrak{m}R^{-1}C$ identifies with $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$; by hypothesis this k -algebra is generated by the image of u , and *a fortiori* it is equal to the image of $R^{-1}B/\mathfrak{m}R^{-1}B$. $\square$

Proof of (18.4.6), continued. Consider next the case where f is étale at the point x . By replacing X by a neighbourhood of x , we can suppose that X is a neighbourhood of $\text{Spec}(B)$ ((I, 1.7.2)). Set $B' = A[T]/F.A[T]$ and let $\mathfrak{n}'$ be the inverse image of $\mathfrak{n}$ in B' ; as the image of $F'(T)$ in B' does not belong to $\mathfrak{n}'$, the morphism $\text{Spec}(B') \rightarrow \text{Spec}(A)$ is étale at the point $\mathfrak{n}'$ by (18.4.2), (ii)). As by hypothesis $\text{Spec}(B) \rightarrow \text{Spec}(A)$ is étale at the point $\mathfrak{n}$, we conclude (17.3.4) that $\text{Spec}(B) \rightarrow \text{Spec}(B')$ is étale at the point $\mathfrak{n}$; but as this morphism is an immersion, it can only be étale at a point if it is a local isomorphism at that point (17.9.1), hence $B_{\mathfrak{n}}$ and $B'_{\mathfrak{n}'}$ are isomorphic.

Finally, suppose that $\mathcal{O}_{X,x}$ is a formally étale A -algebra; with the preceding notations, it follows from (17.1.5) that $B_{\mathfrak{n}}$ is a $B'_{\mathfrak{n}'}$ -algebra formally étale; but since the homomorphism $B'_{\mathfrak{n}'} \rightarrow B_{\mathfrak{n}}$ is surjective, this can only happen if this homomorphism is bijective (0, 19.10.3, (i)). This finishes the proof of (18.4.6). $\square$

Corollary (18.4.7). — *Let $f : X \rightarrow Y$ be a morphism locally of finite type, x a point of X . For f to be formally unramified at the point x , it is necessary and sufficient that there exist an open neighbourhood U of x such that $f|_U$ factors as $U \xrightarrow{j} X' \xrightarrow{h} Y$, where h is étale and j is a closed immersion.*

Proof. We can evidently restrict to the case where $Y = \text{Spec}(R)$ is affine and f of finite type. If $A = \mathcal{O}_{Y,y}$, with $y = f(x)$, the condition for f to be formally unramified at the point x is equivalent,

by virtue of (17.4.1.2), to saying that $\mathcal{O}_{X,x}$ is a formally unramified A -algebra. If this holds, we can apply (18.4.6), (i)); by replacing Y by an affine neighbourhood of y if necessary, we can suppose (with the notations of (18.4.6)) that the polynomial F is the image in $A[T]$ of a unitary polynomial $G \in R[T]$. Set $X' = \text{Spec}(R[T]/G.R[T])$; let x' be the image of the point n of $\text{Spec}(B)$ by the morphism corresponding to the composed homomorphism $R[T]/G.R[T] \rightarrow A[T]/F.A[T] \rightarrow B$. It follows from (18.4.2) that the morphism $h : X' \rightarrow Y$ corresponding to the canonical homomorphism $R \rightarrow R[T]/G.R[T]$ is étale at the point x' , hence, by restricting X' and Y to open neighbourhoods of x' and y respectively if necessary, we can suppose h étale. On the other hand, by virtue of (I, 6.5.1, (ii)) and (1.7.2), it follows from the fact that there is a local homomorphism $\varphi : \mathcal{O}_{X',x'} \rightarrow \mathcal{O}_{X,x}$ that this homomorphism corresponds to a morphism $j : U \rightarrow X'$ of an open neighbourhood U of x in X ; restricting U if necessary, we can, by applying (I, 6.5.1, (i)), suppose that $h \circ j = f|U$, hence j is of finite type (I, 6.3.4, (v)); finally, the homomorphism φ is surjective by (18.4.6), (i)), hence it follows from (I, 6.5.4, (i)) that we can, by restricting U and X' further, suppose that j is a closed immersion. This proves the necessity of the stated condition; the sufficiency is immediate (17.4.3), (i) and (ii)). $\square$

Corollary (18.4.8). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation. For f to be unramified (resp. étale), it is necessary and sufficient that there exist a family of flat morphisms $g_\alpha : Y'_\alpha \rightarrow Y$ and, for each α , an open set U_α in $X'_\alpha = X \times_Y Y'_\alpha$, such that, if $g'_\alpha : X'_\alpha \rightarrow X$ and $f'_\alpha : X'_\alpha \rightarrow Y'_\alpha$ are the canonical projections, the $g'_\alpha(U_\alpha)$ form a covering of X , and each of the composed morphisms $U_\alpha \rightarrow X'_\alpha \xrightarrow{f'_\alpha} Y'_\alpha$ is a closed immersion (resp. open immersion). One can moreover then take the g_α étale.*

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Proof. The necessity of the condition for étale morphisms is trivial, taking for each α a single Y'_α equal to X , the morphism g_α corresponding being equal to f , and the open set $U \subset X \times_Y X$ being the diagonal. When f is unramified, the necessity of the condition results from (18.4.7); one takes a covering open (V_α) of X such that for each α , $f|V_\alpha$ factors as $V_\alpha \xrightarrow{j_\alpha} Y'_\alpha \xrightarrow{g_\alpha} Y$, where j_α is a closed immersion and g_α an étale morphism. Then $j_\alpha : V_\alpha \rightarrow Y'_\alpha$, where s_α is a V_α -section of X'_α , and as the morphism $g'_\alpha : X'_\alpha \rightarrow X$ is étale, s_α is an open immersion (17.4.1), and it suffices to take $U_\alpha = s_\alpha(V_\alpha)$ to answer the question.

The sufficiency of the conditions results from (17.7.1); one concludes that f is unramified (resp. étale) at each point of $g'(U_\alpha)$, hence in all of X since the $g'(U_\alpha)$ cover X . $\square$

Proposition (18.4.9). — *Let S be a prescheme, $f : X \rightarrow S$ a morphism, $h : Y \rightarrow S$ a morphism locally of finite presentation, $g : X \rightarrow Y$ an S -morphism, x a point of X , $y = g(x)$. The following conditions are equivalent:*

- a) h is étale at the point y and g is flat at the point x .
- b) h is unramified at the point y and f is flat at the point x .

Proof. Since $f = h \circ g$, a) implies that f is flat at the point x (I, 2.1.6), and it is evident that h is unramified at the point y , hence a) implies b).

To prove that b) implies a), one can first suppose that h is unramified (by replacing Y by an open neighbourhood of y); then, by replacing S by an open neighbourhood of $s = h(y) = f(x)$, one can suppose that there exists an étale morphism $u : S' \rightarrow S$, a point y' of $Y' = Y_{(S')}$ above y and an open neighbourhood V of y' in Y' such that, if $h' = h_{(S')} : Y' \rightarrow S'$, the restriction of h' to V is a closed immersion (18.4.8). If one proves that h' is étale at the point y , it will result that h is étale at the point y (17.7.1, (ii)); moreover, $f' = f_{(S')} : X_{(S')} \rightarrow S'$ is flat at every point x' above x . As the projections $v : Y' \rightarrow Y$, $w : X' \rightarrow X$ are étale morphisms (hence flat), if one proves that $g' = g_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is flat at the point x' , it will result from (I, 2.2.11, (iv)) that g will be flat at the point x . One can hence restrict to the case where h is a closed immersion, f being supposed flat at the point x . Let $\mathcal{J}$ be the quasi-coherent Ideal of finite type of $\mathcal{O}_S$ which defines the closed sub-prescheme Y of S . The hypothesis that f is flat at the point x implies the homomorphism $\mathcal{O}_{S,s} \rightarrow \mathcal{O}_{X,x}$ is injective (0I, 6.5.1); a fortiori the homomorphism $\mathcal{O}_{S,s} \rightarrow \mathcal{O}_{Y,y}$ is injective, which means that $\mathcal{J}_s = 0$, since this is the kernel of the preceding homomorphism. Since $\mathcal{J}$ is of finite type, there is an open neighbourhood U of s in S such that $\mathcal{J}|U = 0$ (0I, 5.2.2). One can hence suppose that h is an open immersion, and then it is clear that g is flat at the point x . $\square$

Proposition (18.4.10). — Let us denote by $\mathbf{P}(f, x)$ a property verifying the following conditions:

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- 1° For every morphism $f : X \rightarrow Y$ and every local isomorphism $h : Y \rightarrow Z$, $\mathbf{P}(f, x)$ is equivalent to $\mathbf{P}(h \circ f, x)$ for $x \in X$.
- 2° For every morphism $f : X \rightarrow Y$, every étale morphism $g : Y' \rightarrow Y$, every point $x \in X$, if one sets $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$ and if $x' \in X'$ is above x , the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(f', x')$ are equivalent (“invariance by étale base change”).

Let S be a prescheme, $f : X \rightarrow S$ and $h : Y \rightarrow S$ two S -morphisms, $g : X \rightarrow Y$ an S -morphism, x a point of X , $y = g(x)$, and suppose h étale at the point y . Then the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(g, x)$ are equivalent.

Proof. With the same notations as in the proof of (18.4.9) and by replacing S (resp. Y) by an open neighbourhood of $s = h(y)$ (resp. of y), one can here, by virtue of (18.4.8), find an étale morphism $u : S' \rightarrow S$ such that h' is an open immersion; if $x' \in X'$ is above x , $\mathbf{P}(f', x')$ (resp. $\mathbf{P}(g', x')$) is then by hypothesis equivalent to $\mathbf{P}(f, x)$ (resp. $\mathbf{P}(g, x)$). As the projections $v : Y' \rightarrow Y$, $w : X' \rightarrow X$ are étale morphisms (hence flat), if one proves that $g' = g_{(S')} : X_{(S')} \rightarrow Y_{(S')}$ is flat at the point x' , one deduces the conclusion. The hypothesis implies that $\mathbf{P}(g, x)$ is then equivalent to $\mathbf{P}(h \circ g, x)$, from which one deduces the conclusion. $\square$

Examples (18.4.11). — One can take for property $\mathbf{P}(f, x)$ any one of the following, taking into account (17.7.4), (ii):

- label=(i) f is flat at the point x (I, 2.2.11), (iv));
- lbel=(ii) f is locally of finite presentation and of coprofondur $\leq n$ at the point x ((6.8.1) and (6.7.8));
- lbel=(iii) f is locally of finite presentation and Cohen–Macaulay at the point x ((6.8.1) and (6.7.8));
- ldbel=(iv) f is locally of finite presentation and possesses the property (S_n) at the point x ((6.8.1) and (6.7.8));
- label=(v) f is locally of finite presentation and possesses the property (R_n) at the point x ((6.8.1) and (6.7.8));
- lbel=(vi) f is locally of finite presentation and normal at the point x ((6.8.1) and (6.7.8));
- lgbel=(vii) f is locally of finite presentation and reduced at the point x ((6.8.1) and (6.7.8));
- lhbel=(viii) f is unramified at the point x (17.7.4);
- libel=(ix) f is smooth at the point x (17.7.4);
- ljbel=(x) f is étale at the point x (17.7.4).

Corollary (18.4.12). —

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label=(i) Let $f : X \rightarrow Y$ be a morphism locally of finite type, x a point of X . If f is flat and formally unramified at the point x , then, in every factorisation $f|_U : U \xrightarrow{j} X' \xrightarrow{h} Y$, where U is an open neighbourhood of x , j a closed immersion and h an étale morphism (18.4.7), the homomorphism $\mathcal{O}_{X', j(x)} \rightarrow \mathcal{O}_{X, x}$ corresponding to j is bijective (in particular $\mathcal{O}_{X, x}$ is an $\mathcal{O}_{Y, j(x)}$ -algebra essentially étale (18.6.1)).

lbel=(ii) For a morphism $f : X \rightarrow Y$ to be étale, it is necessary and sufficient that it be locally of finite type, formally unramified and flat.

Proof. (i) As the homomorphism $\mathcal{O}_{X', j(x)} \rightarrow \mathcal{O}_{X, x}$ is surjective, it suffices to prove that it makes $\mathcal{O}_{X, x}$ an $\mathcal{O}_{X', j(x)}$ -module faithfully flat, or only flat (0_I, 6.5.1 and 6.6.2); in other words, it suffices to show that j is a morphism flat at the point x ; but since $h \circ j = f$ is by hypothesis flat at the point x and h is étale, this results from (18.4.10) and (18.4.11), (i).

(ii) There is only to prove the suffisance of the conditions stated. For every $x \in X$, one has therefore in an open neighbourhood U of x a factorisation of $f|_U$ having the properties considered in (i). Since by hypothesis f is flat and formally unramified at all the points of U , the result of (i) applies not only at x but at all the points of U ; this means that if $\mathcal{J}$ is the quasi-coherent Ideal of $\mathcal{O}_{X'}$ corresponding to the closed sub-prescheme of X' associated to j , one has $\mathcal{J}_{j(z)} = 0$ for every $z \in U$, hence j is an open immersion, and f is therefore étale at every point of U , hence at every point of X . $\square$

Corollary (18.4.13). — Let $f : X \rightarrow Y$ be a morphism locally of finite type, x a point of X , $y = f(x)$. Suppose that y admits an open neighbourhood which is a reduced prescheme with only a finite number of

irreducible components. Then, for f to be étale at the point x , it is necessary and sufficient that f be flat and formally unramified at the point x .

Proof. There is only to show the suffisance of the condition. The question being local on X and Y , one can suppose (18.4.7) that f factors as $X \xrightarrow{j} X' \xrightarrow{h} Y$ where h is étale and j is a closed immersion. Then X' is reduced (17.5.7) and, by replacing X' by an open neighbourhood of $j(x)$, one can suppose that X' has only a finite number of irreducible components. Everything comes down to showing, with the notations of the proof of (18.4.12), that $\mathcal{J}_{x'} = 0$ for all the points x' of a neighbourhood of $j(x)$ in X' , knowing that $\mathcal{J}_{j(z)} = 0$ for every $z \in U$, one can replace X' by an affine neighbourhood of $j(x)$, one can suppose that all the irreducible components of X' contain $j(x)$; if $X' = \operatorname{Spec}(A')$, and if $\mathfrak{p}'$ is the prime ideal of A' corresponding to the point $j(x)$, the morphism $\operatorname{Spec}(A'_{\mathfrak{p}'}) \rightarrow \operatorname{Spec}(A')$ is dominant, hence $A' \rightarrow A'_{\mathfrak{p}'}$ is injective since A' is reduced (I, 1.2.7). If $\mathcal{J} = \tilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A' , $\mathfrak{J}$ identifies with a part of $\mathfrak{J}_{\mathfrak{p}'}$, and the hypothesis $\mathfrak{J}_{\mathfrak{p}'} = 0$ implies $\mathfrak{J} = 0$. $\square$

Corollary (18.4.14). — Let A be a local ring with maximal ideal $\mathfrak{m}$, residue field k , B a finite A -algebra.

label=(i) For B to be a formally unramified A -algebra, it is necessary and sufficient that $B \otimes_A k$ be an étale k -algebra.

lbbel=(ii) For B to be an étale A -algebra and that B be a flat A -module (which is equivalent to saying (0_I, 6.3.3) that for every maximal ideal $\mathfrak{n}$ of B (necessarily above $\mathfrak{m}$), $B_{\mathfrak{n}}$ is a flat A -module).

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Proof. One has evidently only to prove the suffisance of the stated conditions. It is clear that (ii) results from (i) and from (18.4.12), (ii)), taking into account (17.1.2), (i)). To prove (i) we remark that if $B \otimes_A k$ is an étale k -algebra, one has $\Omega_{(B \otimes_A k)/k}^1 = 0$ (17.2.1). But $\Omega_{(B \otimes_A k)/k}^1 = \Omega_{B/A}^1 \otimes_B k$ (0, 20.5.5) and since B is an A -algebra of finite type, $\Omega_{B/A}^1$ is a B -module of finite type (0, 20.4.7). But since B is a finite A -algebra, $\mathfrak{m}B$ is contained in the radical of B (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 1, prop. 1), hence the Nakayama lemma proves that $\Omega_{B/A}^1 = 0$, and hence B is an A -algebra formally unramified (17.2.1). $\square$

18.5. Henselian local rings.

(18.5.1). Let X be a prescheme, $\mathcal{E}$ an $\mathcal{O}_X$ -Module locally free of finite rank; the dual $\check{\mathcal{E}} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X)$ is therefore an $\mathcal{O}_X$ -Module locally free whose rank at every point of X is equal to that of $\mathcal{E}$ at that point, and the canonical homomorphism $\mathcal{E} \rightarrow \mathcal{H}om_{\mathcal{O}_X}(\check{\mathcal{E}}, \mathcal{O}_X) = (\check{\mathcal{E}})^\vee$ is bijective. For every morphism $X' \rightarrow X$, set $\mathcal{E}_{(X')} = \mathcal{E} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'}$, which is an $\mathcal{O}_{X'}$ -Module locally free, and consider the set $\Gamma(X', \mathcal{E}_{(X')})$ of sections of this $\mathcal{O}_{X'}$ -Module above X' . We will see that one thus defines a contravariant representable functor

$$(18.5.1.1) \quad {}^t\mathbf{V}_{\mathcal{E}} : X' \longmapsto \Gamma(X', \mathcal{E}_{(X')})$$

from the category of X -preschemes to that of sets (0_{III}, 8.1.8).

Indeed, we first see that the X -prescheme $\mathbf{V}(\check{\mathcal{E}})$ (II, 1.7.8) represents the functor ${}^t\mathbf{V}$. It is indeed immediate that $(\mathcal{E}_{(X')})^\vee = (\check{\mathcal{E}})_{(X')}$, hence $\mathbf{V}(\check{\mathcal{E}}) \times_X X' = \mathbf{V}((\mathcal{E}_{(X')})^\vee)$; taking into account (I, 3.3.14), one is led to define a bijection $\Gamma(\mathbf{V}(\check{\mathcal{E}})/X') \xrightarrow{\sim} \Gamma(X', \mathcal{E})$, and to verify that the bijection canonical of $\Gamma(\mathbf{V}(\check{\mathcal{E}})/X)$ to $\operatorname{Hom}_{\mathcal{O}_X}(\check{\mathcal{E}}, \mathcal{O}_X)$ (II, 1.7.8), and the transposition $u \mapsto {}^t u$ is a bijection of $\operatorname{Hom}_{\mathcal{O}_X}(\check{\mathcal{E}}, \mathcal{O}_X)$ to $\operatorname{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{E}) = \Gamma(X, \mathcal{E})$, by reason of the identification of $(\check{\mathcal{E}})^\vee$ and $\mathcal{E}$. The verification of the functorial property is immediate.

(18.5.2). Recall (II, 1.7.8) that one has by definition $\mathbf{V}(\check{\mathcal{E}}) = \operatorname{Spec}(\mathbf{S}_{\mathcal{O}_X}(\check{\mathcal{E}}))$. Let $\mathcal{J}$ be a quasi-coherent Ideal of $\mathbf{S}_{\mathcal{O}_X}(\check{\mathcal{E}})$, so that $\operatorname{Spec}(\mathbf{S}_{\mathcal{O}_X}(\check{\mathcal{E}})/\mathcal{J})$ is a closed sub-prescheme of $\mathbf{V}(\check{\mathcal{E}})$; we will interpret it as a representing functor from the category of X -preschemes to that of sets. Note that a section $u \in \Gamma(X, \mathcal{E})$ identifies canonically with an $\mathcal{O}_X$ -homomorphism $u : \mathcal{O}_X \rightarrow \mathcal{E}$, to which corresponds by transposition an $\mathcal{O}_X$ -homomorphism of $\mathcal{O}_X$ -Algebras $v : \mathbf{S}_{\mathcal{O}_X}(\check{\mathcal{E}}) \rightarrow \mathcal{O}_X$. Let $\operatorname{Al}(X, \mathcal{E}, \mathcal{J})$ denote the set of $u \in \Gamma(X, \mathcal{E})$ such that $\mathcal{J}$ is contained in the kernel of v ; it results immediately from these definitions that

$$X' \longmapsto \operatorname{Al}(X', \mathcal{E}_{(X')}, \mathcal{J} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'})$$

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is a functor represented by $\text{Spec}(\mathbf{S}_{\mathcal{O}_X}(\mathcal{E})/\mathcal{J})$. If $X' = \text{Spec}(A)$ is affine and such that $\mathcal{E}_{(X')}$ is isomorphic to $\mathcal{O}_{X'}^n$, $\Gamma(X', \mathcal{E}_{(X')})$ can be identified with A^n , and $\mathcal{J} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'}$ to an $\mathcal{O}_{X'}$ -Module of the form $\tilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of the polynomial ring $A[T_1, \dots, T_n]$; the set $\text{Al}(X', \mathcal{E}_{(X')}, \mathcal{J} \otimes_{\mathcal{O}_X} \mathcal{O}_{X'})$ identifies then with the part of A^n formed of the points $(t_1, \dots, t_n)$ such that $P(t_1, \dots, t_n) = 0$ for every $P \in \mathfrak{J}$; in imagery one can therefore say that the object $\text{Spec}(\mathbf{S}_{\mathcal{O}_X}(\mathcal{E})/\mathcal{J})$ represents “the algebraic subscheme of the affine twisted space X defined by $\mathcal{E}$, formed of the points annulling the ideal $\Gamma(X, \mathcal{J})$ ”. We note this X -prescheme also $\text{Al}(\mathcal{E}, \mathcal{J})$. One will note that if $\mathcal{J}$ is an Ideal of finite type of $\mathbf{S}_{\mathcal{O}_X}(\mathcal{E})$, $\text{Al}(\mathcal{E}, \mathcal{J})$ is an X -prescheme of finite presentation, since $\mathbf{S}_{\mathcal{O}_X}(\mathcal{E})$ is an $\mathcal{O}_X$ -Algebra of finite presentation.

Lemma (18.5.3). — *Let S be a prescheme, $f : X \rightarrow S$ a morphism finite and locally free (18.2.7). Consider the contravariant functor from the category of S -preschemes to the category of sets*

$$(18.5.3.1) \quad S' \longmapsto \text{Of}(X \times_S S')$$

where $\text{Of}(X \times_S S')$ is the set of parts at once open and closed of the underlying space of $X \times_S S'$. Then this functor is representable by an S -prescheme $\mathbf{Of}(X)$, which is affine, étale and of finite presentation over S .

Proof. By hypothesis $X = \text{Spec}(\mathcal{B})$, where $\mathcal{B} = f_*(\mathcal{O}_X)$ is a finite locally free $\mathcal{O}_S$ -Algebra. Set $X' = X \times_S S'$, $f' = f_{(S')}$, $\mathcal{B}' = \mathcal{B} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'} = f'_*(\mathcal{O}_{X'})$, so that $X' = \text{Spec}(\mathcal{B}')$; there is then a canonical bijection, functorial in S' , from the set $\text{Of}(X')$ to the set $\text{Id}(\mathcal{B}')$ of idempotents of the ring $\Gamma(S', \mathcal{B}') = \Gamma(X', \mathcal{O}_{X'})$. Indeed, by virtue of the equivalence of the category of S' -schemes affine over S' and of the opposite category of quasi-coherent $\mathcal{O}_{S'}$ -Algebras (II, 1.2.7 and 1.3.1), there is a canonical bijection between the decompositions of X' as a sum $X'_1 \amalg X'_2$ of two sub-preschemes induced on open sets of X' and the decompositions of $\mathcal{B}'$ as a direct composition of two Ideals $\mathcal{B}'_1, \mathcal{B}'_2$; these latter are in canonical bijective correspondence (and functorially in S') with $\text{Id}(\mathcal{B}')$. It suffices therefore to prove the lemma for the functor $S' \mapsto \text{Id}(\mathcal{B}')$.

To this end, we will show that there exists an Ideal of finite type $\mathcal{J}$ of $\mathbf{S}_{\mathcal{O}_S}(\mathcal{B})$ such that $\text{Id}(\mathcal{B}')$ is of the form $\text{Al}(S', \mathcal{B}_{(S')}, \mathcal{J} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'})$ (18.5.2). Note to this effect that since $\mathcal{B}$ is an $\mathcal{O}_S$ -Algebra, its image reciprocally $\mathcal{B}_{\mathbf{V}(\mathcal{B})}$ is an $\mathcal{O}_{\mathbf{V}(\mathcal{B})}$ -Algebra, and one can therefore form the square c^2 , in this Algebra, of the canonical section c (18.5.1); it corresponds canonically to a homomorphism $u^{(2)} : \mathbf{S}_{\mathcal{O}_S}(\mathcal{B}) \rightarrow \mathbf{S}_{\mathcal{O}_S}(\mathcal{B}) \otimes_{\mathcal{O}_S} \mathcal{B}$ of $\mathbf{S}_{\mathcal{O}_S}(\mathcal{B})$ -Modules, and one verifies immediately that for an affine open W of S , with the notations of (18.5.1), $u^{(2)}$ corresponds to the homomorphism of C -modules such that $u^{(2)}(1) = \sum_k (\sum_{i,j} c_{ijk} T_i T_j) \otimes e_k$, where (c_{ijk}) is the multiplication table of the algebra $\Gamma(W, \mathcal{B})$. Let us show that the Ideal $\mathcal{J}$ of $\mathbf{S}_{\mathcal{O}_S}(\mathcal{B})$ generated by the kernel of the homomorphism $u - u^{(2)}$ answers the question. It suffices indeed to note that the ideal $\Gamma(W, \mathcal{J})$ of C is generated by the polynomials $P_k(T_1, \dots, T_n) = T_k - \sum_{i,j} c_{ijk} T_i T_j$ and that the idempotents of $\Gamma(W, \mathcal{B})$ are precisely the elements $\sum t_i e_i$ of this algebra that annul all the polynomials P_k ($1 \leq i \leq n$). One deduces therefore from (18.5.2) that the S -prescheme affine of finite presentation $\mathbf{Of}(X) = \text{Al}(\mathcal{B}, \mathcal{J})$ represents well the functor (18.5.3.1). It remains to prove that $\mathbf{Of}(X)$ is étale over S , or, what amounts to the same, that it is formally étale over S . But if S' is an S -prescheme and S'_0 a closed sub-prescheme of S' defined by a locally nilpotent Ideal of $\mathcal{O}_{S'}$ (and having therefore the same underlying space as S'), it is clear that $X' = X \times_S S'$ and $X'_0 = X \times_S S'_0$ have the same underlying space, hence the canonical application $\text{Of}(X') \rightarrow \text{Of}(X'_0)$ is bijective, which concludes the proof (17.1.1). $\square$

Proposition (18.5.4). — *Let S be a prescheme, S_0 a closed sub-prescheme of S ; consider the following* IV | 127

a) *For every finite morphism $g : S' \rightarrow S$, the canonical application*

$$(18.5.4.1) \quad \text{Of}(S') \longrightarrow \text{Of}(S' \times_S S_0) \quad (\text{cf. (18.5.3)})$$

is bijective.

a') *For every morphism finite and locally free $g : S' \rightarrow S$, the application (18.5.4.1) is bijective.*

b) *For every étale and separated morphism $g : S' \rightarrow S$, the canonical application*

$$(18.5.4.2) \quad \Gamma(S'/S) \longrightarrow \Gamma(S' \times_S S_0/S_0)$$

is bijective.

The condition $b)$ implies $a')$; if moreover S is quasi-compact and quasi-separated, the condition $a)$ implies $b)$.

Proof. Let us show first that $b)$ implies $a')$. Suppose $b)$ verified, and let $g : S' \rightarrow S$ be a morphism finite and locally free; set $S'_0 = S' \times_S S_0$, so that $g_0 = g_{(S_0)} : S'_0 \rightarrow S_0$ is finite and locally free. Then it results from (18.5.3) that $P = \mathbf{Of}(S')$ is an S -prescheme étale and separated over S ; moreover, the definition of the functor $\mathbf{Of}$ shows that if one sets $P_0 = \mathbf{Of}(S'_0)$ (for the category of S_0 -preschemes), one has $P_0 = P \times_S S_0$. We have therefore by definition the commutative diagram

$$\begin{array}{ccc} \Gamma(P/S) & \longrightarrow & \Gamma(P_0/S_0) \\ \ell \downarrow & & \downarrow \ell \\ \mathbf{Of}(S') & \longrightarrow & \mathbf{Of}(S'_0) \end{array}$$

where the vertical arrows are the canonical bijections. As the hypothesis $b)$, applied to the morphism $P \rightarrow S$, implies that the first line is a bijection, the same is true of the second, which establishes our assertion.

Lemma (18.5.4.3). — If S and S_0 verify condition $a)$ of (18.5.4), then, for every finite morphism $g : S' \rightarrow S$, S' is the unique neighbourhood of $S'_0 = g^{-1}(S_0) = S' \times_S S_0$ in S' .

Indeed, it amounts to the same to say that if T' is a closed part of S' such that $T' \cap S'_0 = \emptyset$, then $T' = \emptyset$. If one also designates by T' a sub-prescheme of S' with underlying space T' , the composed morphism $h : T' \rightarrow S' \xrightarrow{g} S$ is finite and $h^{-1}(S_0)$ is empty; the condition $a)$ applied to the morphism h implies that T' is necessarily empty.

This lemma established, let us prove first that, under the hypothesis $a)$, the application (18.5.4.2) is *injective*. Indeed, if u', u'' are two S -sections of S' , the fact that the morphism $S' \rightarrow S$ is unramified implies the prescheme of coincidences of u' and u'' is induced on an open set U of S (17.4.6). If the restrictions of u' and u'' to S_0 are the same, the fact that u' and u'' are open immersions (17.4.1) implies that U contains S_0 , hence is equal to S by virtue of the lemma (18.5.4.3) applied to the case where $S' = S$.

It remains to show that under hypothesis $a)$, the application (18.5.4.2) is *surjective* (S being quasi-compact and quasi-separated). Let $u_0 : S_0 \rightarrow S'_0$ be an S_0 -section of S'_0 ; $u_0(S_0)$ being quasi-compact in S' , can be covered by a finite number of open affines V_i such that the restriction $g|_{V_i}$ is a morphism of finite type; if V is the reunion of the V_i , one concludes that $g|_V : V \rightarrow S$ is a morphism of *finite presentation*, being separated and locally of finite presentation by hypothesis ((1.6.1)). Replacing S' by V , one can hence suppose that g is of finite presentation. As S is quasi-compact and quasi-separated and g quasi-finite and separated (17.6.1), it results from the “Main theorem” (8.12.6) that g factors as $S' \xrightarrow{j} S'' \xrightarrow{f} S$, where j is an open immersion and f a finite morphism. Set $S''_0 = S'' \times_S S_0$, $j_0 = j_{(S_0)} : S'_0 \rightarrow S''_0$, which is an open immersion and $f_0 = f_{(S_0)} : S''_0 \rightarrow S_0$, which is a finite morphism. Then u_0 is also an S_0 -section of S''_0 . As $g_0 : S'_0 \rightarrow S_0$ is étale, u_0 is an open immersion of S_0 in S'_0 (17.4.1), hence *a fortiori* in S''_0 ; but on the other hand, as f_0 is a finite morphism, hence separated, u_0 is a closed immersion of S_0 in S''_0 (I, 5.4.6), hence $X_0 = u_0(S_0)$ is *at once open and closed* in S''_0 . By virtue of hypothesis $a)$, there exists an open and closed set X of S'' such that $X \cap S''_0 = X_0$. Let us show first that the morphism $f : S'' \rightarrow S$ is étale at the points of X ; indeed, the set U of points of X where $f|_X$ is étale is open and contains by hypothesis $X_0 \subset S'_0$. But the lemma (18.5.4.3) applied to the finite morphism $f|_X$ proves that $U = X$. On the other hand, $S' \cap X$ is open in X and contains X_0 by hypothesis, hence the same argument proves that $S' \cap X = X$, that is to say $X \subset S'$. It remains to show that, for every $s \in S$, the geometric number $n(s)$ of points of $X \cap f^{-1}(s)$ is equal to 1, for it will result that $f|_X$ is radiciel and surjective, and as $f|_X$ is étale, one will have shown (17.9.1) that $f|_X$ is an open isomorphism of $X \subset S'$ over S , whose inverse will be the S -section sought. But as $f|_X$ is étale and *finite*, $s \mapsto n(s)$ is continuous in S (18.2.8), and as $X \cap S'_0 = X_0$, one has $n(s) = 1$ in S_0 ; the set of points $s \in S$ such that $n(s) = 1$ being open in S and containing S_0 , it is equal to S by (18.5.4.3). Q.E.D. $\square$

Remark (18.5.4.4). — One can show that the statement (18.5.4) remains valid when, in condition $b)$, one supposes only the morphism g étale (but not necessarily separated) [?].

Definition (18.5.5). — One says that a prescheme S and a closed sub-prescheme S_0 of S form a *Henselian couple* if they verify condition *a*) of (18.5.4).

Taking into account (I, 5.1.8), it amounts to the same to say that (S, S_0) is a Henselian couple or that $(S_{\text{red}}, (S_0)_{\text{red}})$ is one.

Proposition (18.5.6). — *label=(i) If (S, S_0) is a Henselian couple, then, for every finite morphism $f : S' \rightarrow S$, if S'_0 is the sub-prescheme $f^{-1}(S_0)$ of S' , the couple (S', S'_0) is Henselian.*

lbbel=(ii) Let $S = \coprod S^{(\alpha)}$ be a sum of preschemes, S_0 a closed sub-prescheme of S , sum of the closed sub-preschemes $S_0^{(\alpha)}$ of the $S^{(\alpha)}$. For the couple (S, S_0) to be Henselian, it is necessary and sufficient that each of the couples $(S^{(\alpha)}, S_0^{(\alpha)})$ be Henselian.

Proof. The assertion (i) is an immediate consequence of the definition, since for every finite morphism $g : S'' \rightarrow S'$, $f \circ g : S'' \rightarrow S$ is a finite morphism. Similarly, under the conditions of (ii), for a morphism $g : S' \rightarrow S$ to be finite, it is necessary and sufficient that each of its restrictions $g^{(\alpha)} : S'^{(\alpha)} = g^{-1}(S^{(\alpha)}) \rightarrow S^{(\alpha)}$ be finite, and if one sets $S'_0 = g^{-1}(S_0)$, $S_0'^{(\alpha)} = (g^{(\alpha)})^{-1}(S_0^{(\alpha)})$, there is a bijective correspondence between the open and closed parts U (resp. U_0) of S' (resp. S'_0) and the families $(U^{(\alpha)})$ (resp. $U_0^{(\alpha)}$) where $U^{(\alpha)}$ (resp. $U_0^{(\alpha)}$) is an open and closed part of $S'^{(\alpha)}$ (resp. $S_0'^{(\alpha)}$), whence the assertion (ii). $\square$

Remark (18.5.7). — Let $S = \text{Spec}(A)$ be an affine scheme, S_0 a closed sub-prescheme of S defined by an ideal $\mathfrak{J}$ of A . Then, if the couple (S, S_0) is Henselian, the ideal $\mathfrak{J}$ is necessarily contained in the radical of A . Indeed, if $\mathfrak{m}$ is a maximal ideal of A , it must belong to $V(\mathfrak{J}) = S_0$, by virtue of (18.5.4.3), otherwise $\mathfrak{m}$ would not be a maximal ideal, which gives the conclusion. In particular, suppose that S_0 is reduced to a point, that is to say that the ideal $\mathfrak{J}$ is maximal; then $\mathfrak{J}$ must be the radical of A , in other words A must be a local ring and S_0 the unique closed point of $\text{Spec}(A)$.

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Definition (18.5.8). — One says that a ring A is *Henselian* if it is semi-local and if, denoting by τ the radical of A , the couple $(\text{Spec}(A), \text{Spec}(A/\tau))$ is Henselian. One calls a *Henselian scheme* a scheme isomorphic to the spectrum of a Henselian local ring.

Proposition (18.5.9). — *label=(i) For a semi-local ring A to be Henselian, it is necessary and sufficient that it be a direct composition of Henselian local rings.*

lbbel=(ii) For a local ring A to be Henselian, it is necessary and sufficient that every finite A -algebra B be isomorphic to a product of local rings.

Proof. (i) Indeed, the definition, applied to $S = \text{Spec}(A)$ and $S_0 = \text{Spec}(A/\tau)$, shows that, since S_0 is a finite discrete set of points and closed, S is the reunion of a finite number of open and closed parts S_i ($1 \leq i \leq n$), pairwise disjoint, each of which contains exactly one of the maximal ideals $\mathfrak{m}_i$ of A ; the conclusion results from (18.5.6), (ii) and from the remark (18.5.7).

(ii) Every finite morphism $S' \rightarrow \text{Spec}(A)$ is of the form $\text{Spec}(B) \rightarrow \text{Spec}(A)$, where B is a finite A -algebra. If k is the residue field of A , $\text{Spec}(B \otimes_A k)$ is a spectrum of an artinian ring, hence finite and discrete. To say that the couple $(\text{Spec}(A), \text{Spec}(k))$ is Henselian means therefore that B is a direct composition of local rings A_i such that $\text{Spec}(A_i \otimes_A k)$ is reduced to a point, that is to say that A_i (which is a finite A -algebra) must have only one maximal ideal (Bourbaki, *Alg. comm.*, chap. V, § 2, n° 1, prop. 1). $\square$

Proposition (18.5.10). — *If A is a Henselian ring, every finite A -algebra B is a Henselian ring (hence a direct composition of Henselian local rings (18.5.9)).*

Proof. Indeed, if τ' is the radical of B , the inverse image of τ' in A is the radical τ , and every ideal of B above a maximal ideal of A is a maximal ideal of B , hence $V(\tau')$ in $\text{Spec}(B)$ is the inverse image of $V(\tau)$ in $\text{Spec}(A)$. The proposition is then a consequence of (18.5.6), (i). $\square$

Theorem (18.5.11). — *Let A be a local ring, $\mathfrak{m}$ its maximal ideal. The following conditions are equivalent:*

- a) A is Henselian, in other words, every finite A -algebra B is isomorphic to a product of local rings.*
- a') The condition a) is satisfied for all A -algebras B of the form $A[T]/F.A[T]$, where $F \in A[T]$ is a unitary polynomial.*

- b) Let $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(k)$. For every étale morphism $g : S' \rightarrow S$, if one sets $S'_0 = S' \otimes_S S_0$, every S_0 -section u_0 of S'_0 is the restriction of an S -section u of S' .
- c) For every morphism $f : X \rightarrow S$, separated and locally of finite type, and every point $x \in X$ such that $f(x)$ is equal to the closed point s of S and that f is quasi-finite at the point x (**Err**_{III}, 20), X is the sum of two preschemes X', X'' such that $X' = \operatorname{Spec}(\mathcal{O}_{X,x})$ and that $f|_{X'} : X' \rightarrow S$ is a finite morphism.
- c') For every morphism locally of finite type $f : X \rightarrow S$, and every point $x \in X$ such that $f(x)$ is quasi-finite at the closed point s of S , $\mathcal{O}_{X,x}$ is a finite algebra over $\mathcal{O}_{S,s} = A$.
- c'') For every morphism locally of finite presentation $f : X \rightarrow S$, and every point $x \in X$ such that f is quasi-finite at the point x , and that $f(x)$ is equal to the closed point s of S , $\mathcal{O}_{X,x}$ is a finite algebra and of finite presentation over $\mathcal{O}_{S,s} = A$.

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Proof. Note first that the condition c') (resp. c'')) is equivalent to the same condition where one supposes moreover f separated, the question being local on X . In the same way, the condition b) is equivalent to the same condition where one supposes moreover g separated: indeed, it suffices to apply the latter to the restriction of g to an open affine neighbourhood of the point $u_0(S_0)$ in S' . We restrict henceforth to the case where, in b), c') and c''), the given morphisms are separated.

The fact that a) implies b) and that b) implies a') is a particular case of (18.5.4). Let us show moreover that a') implies a). It suffices to prove that if e_0 is an idempotent of $C = B \otimes_A k$, there exists an idempotent $e \in B$ whose image canonical is e_0 . If b is an element of B whose image in C is e_0 , the sub- A -algebra $A[b] = B'$ of B is finite, $B' \otimes_A k$ is a k -algebra finite, hence a direct composition of k -algebras local finites, and hence e_0 is the image in C' of an idempotent e'_0 of $B' \otimes_A k$. Or, $B' \otimes_A k$ is a k -algebra finite, hence a direct composition of fields, separable finite extensions of k ; and hence B is thus led to the case where B is monogene, and hence isomorphic to a quotient algebra of the form $A[T]/F.A[T]$, where F is a unitary polynomial. Or, by virtue of a'), $A[T]/F.A[T]$ is a direct composition of local rings, hence it is the same for all its quotient algebras, which achieves the proof of the existence of the idempotent e .

It is immediate that c) implies a), as one sees by reasoning by recurrence on the number of maximal ideals of the semi-local ring B . To show that a) implies c), one can, by virtue of (13.1.4), restrict to the case where f is an affine and quasi-finite morphism, se borner au cas où f est un morphisme affine et quasi-fini. Then, by application of the “Main theorem” (8.12.8), f can be written as a composed morphism $X \xrightarrow{j} Y \xrightarrow{g} S$, where g is a finite morphism and j an open immersion. As B is a finite A -algebra, it results from a) that B is a direct composition of local rings, and $\mathcal{O}_{X,x}$ identifies with one of these local rings; moreover, every open set of Y containing x contains necessarily $\operatorname{Spec}(\mathcal{O}_{X,x})$.

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It is trivial that c) implies c'), by virtue of the initial remark. Let us prove that c') implies c''). Suppose c') verified, and let us prove that under the conditions of c''), the set $Z = \operatorname{Spec}(\mathcal{O}_{X,x})$ identifies with an open and closed part of X , which will establish c'') (I, 2.4.2). First, the composed morphism $Z \xrightarrow{j} X \xrightarrow{f} S$, where j is the canonical morphism (I, 2.4.1), is finite and of finite presentation by hypothesis, and since f is separated and locally of finite presentation, j is also a morphism finite and of finite presentation ((II, 6.1.5) and (1.4.3)); j is therefore a closed morphism ((II, 6.1.10), which proves that Z is closed in X . But if $\mathcal{J}$ is the Ideal of $\mathcal{O}_X$ defining Z , on has by hypothesis also $\mathcal{J}_z = 0$ in every point z of an open neighbourhood V of x in X , since by hypothesis $\mathcal{J}$ is an $\mathcal{O}_X$ -Module quasi-coherent of finite type (((1.4.7) and (0_I, 5.2.2)). Now, such a neighbourhood contains Z , hence Z is open in X .

Finally, c'') implies a'): indeed, if $B = A[T]/F.A[T]$, the morphism $X = \operatorname{Spec}(B) \rightarrow \operatorname{Spec}(A) = Y$ is finite and of finite presentation, and the preceding demonstration shows that B is a direct composition of finite A -algebras $\mathcal{O}_{X,x_i}$, where the x_i are the points of the fibre of the closed point of Y . $\square$

Corollary (18.5.12). — Let A be a semi-local ring, $\mathfrak{r}$ its radical; set $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(A/\mathfrak{r})$. For A to be Henselian, it is necessary and sufficient that, for every finite morphism $f : X \rightarrow S$ and every étale and separated morphism $g : Y \rightarrow S$, if one sets $X_0 = X \times_S S_0$ and $Y_0 = Y \times_S S_0$, the canonical application

$$\operatorname{Hom}_S(X, Y) \longrightarrow \operatorname{Hom}_{S_0}(X_0, Y_0)$$

is bijective.

Proof. The suffisance of the condition results from the equivalence of $a)$ and $b)$ in (18.5.11), by applying this condition to the case $f = 1_S$. To see that the condition is necessary, note that if A is Henselian, the couple (X, X_0) is Henselian by (18.5.6), (i)); moreover $\text{Hom}_S(X, Y) = \Gamma(X \times_S Y/X)$ and $\text{Hom}_{S_0}(X_0, Y_0) = \Gamma(X_0 \times_{S_0} Y_0/X_0)$; as $X \times_S Y$ is étale and separated over X , the conclusion results from the fact that $a)$ implies $b)$ in (18.5.4). $\square$

Remarks (18.5.13). — The equivalent conditions $a), a'), b)$ of the theorem (18.5.11) are moreover equivalent to the following (“Hensel’s lemma”):

$a'')$ For every unitary polynomial $F \in A[T]$, with canonical image $F_0 \in k[T]$, and every decomposition $F_0 = G_0 H_0$ of F_0 into a product of two coprime unitary polynomials G_0, H_0 of $k[T]$, there exists a unique couple (G, H) of unitary polynomials of $A[T]$ possessing the following properties: G_0 and H_0 are the respective canonical images of G and H , one has $F = GH$, and the ideal of $A[T]$ generated by G and H is equal to $A[T]$.

Proposition (18.5.14). — Every semi-local ring A , separated and complete for the $\mathfrak{r}$ -preadic topology (where $\mathfrak{r}$ is the radical of A) is Henselian. IV | 133

Proof. Indeed, A is a direct composition of local rings separated and complete (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 13, cor. of the prop. 19), hence one is reduced to the case where A is a local ring. Let us verify the criterion $a')$ of (18.5.11). As B is a free A -module of finite type, B is evidently separated and complete for the $\mathfrak{r}$ -preadic topology (0_I, 7.3.6). Replacing B by A and $\mathfrak{J}B$ by $\mathfrak{J}$, everything comes down to seeing that the application which, to every idempotent of A , associates its class modulo $\mathfrak{J}$, is bijective. But one has $A = \varprojlim (A/\mathfrak{J}^n)$. Let us denote $\text{Idem}(A)$ the set of idempotents of A , and for every ring homomorphism $\varphi : A \rightarrow B$, let $\text{Idem}(\varphi)$ be the application of $\text{Idem}(A)$ into $\text{Idem}(B)$, the restriction of φ ; it results from the definition of the projective limit that $\text{Idem}(A) = \varprojlim \text{Idem}(A/\mathfrak{J}^n)$ for the applications $\psi_{nm} : \text{Idem}(A/\mathfrak{J}^m) \rightarrow \text{Idem}(A/\mathfrak{J}^n)$ restrictions of the canonical applications $A/\mathfrak{J}^m \rightarrow A/\mathfrak{J}^n$. But since $\text{Spec}(A/\mathfrak{J}^n) \rightarrow \text{Spec}(A/\mathfrak{J}^m)$ is a homeomorphism, the ψ_{nm} are bijections (as one saw in the proof of (18.5.3)); this proves our assertion. $\square$

Proposition (18.5.15). — Let A be a Henselian ring, $\mathfrak{r}$ its radical. Then the functor $B \mapsto B/\mathfrak{r}B$ is an equivalence of the category of finite and étale A -algebras with the category of $(A/\mathfrak{r})$ -algebras finite and étale. IV | 134

Proof. The fact that the stated functor is fully faithful is a particular case of (18.5.12). To show that this functor is an equivalence of categories, one can restrict to the case where A is local; it suffices then to apply (18.1.1) (for étale morphisms), to $S = \text{Spec}(A)$ and $S_0 = \text{Spec}(A/\mathfrak{r})$, reduced to a single point. $\square$

Remarks (18.5.16). — (i) We note that if the proposition (18.5.15) generalises to a Henselian couple (S, S_0) , even when S is affine and Noetherian.

(ii) Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A such that A is separated and complete for the $\mathfrak{J}$ -preadic topology. Then the couple $(\text{Spec}(A), \text{Spec}(A/\mathfrak{J}))$ is Henselian: indeed, for every finite A -algebra B , B is separated and complete for the $\mathfrak{J}$ -preadic topology (0_I, 7.3.6). Replacing B par A and $\mathfrak{J}B$ by $\mathfrak{J}$, everything comes down to seeing that the application which, to every idempotent of A , associates its class modulo $\mathfrak{J}$, is bijective. But one has $A = \varprojlim (A/\mathfrak{J}^n)$, and since $\text{Spec}(A/\mathfrak{J}^n) \rightarrow \text{Spec}(A/\mathfrak{J}^m)$ is a homeomorphism, the ψ_{nm} are bijections.

Theorem (18.5.17). — Let A be a Henselian local ring, $S = \text{Spec}(A)$, s the closed point of S . For every smooth morphism $f : X \rightarrow S$, the canonical application

$$\Gamma(X/S) \longrightarrow \Gamma(X_s/k(s))$$

(where $X_s = f^{-1}(s)$) is surjective.

Proof. The data of a $k(s)$ -section of X_s is equivalent to that of a rational point $x \in X$ above s over $k(s)$, and it suffices to prove that there exists an S -section $u : S \rightarrow X$ such that $u(s) = x$. Taking into account (17.16.3), (i)), one can suppose that f is étale. Then the conclusion results from the criterion (18.5.11), $b)$). (The reader will note that by virtue of this criterion, the validity of (18.5.17) for a given local ring A is a necessary and sufficient condition for A to be Henselian). $\square$

Remark (18.5.18). — The conditions of (18.5.11) are moreover equivalent to the following:

- d) For every morphism locally of finite type $f : X \rightarrow S$ and every point $x \in X$ such that $f(x)$ is the closed point s of S and that $\mathcal{O}_{X,x}/\mathfrak{m}_s\mathcal{O}_{X,x}$ is a field canonically isomorphic to $k(s) = k$, there exists an open neighbourhood U of x in X such that $f|_U$ is a closed immersion.

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Proposition (18.5.19). — Let A be a Noetherian and Henselian local ring, $S = \operatorname{Spec}(A)$, s the closed point of S , $f : X \rightarrow S$ a proper morphism; set $X_s = f^{-1}(s)$. Then the application $Y \mapsto Y \cap X_s$ is a bijection of the set of connected components of X to the set of connected components of X_s .

Proof. By considering a closed sub-prescheme of X having for underlying space a connected component of X , one is led to prove that if X is connected and non empty, then X_s is connected and non empty; the fact that X_s is non empty results from (II, 7.2.1); to prove that X_s is connected, reason by the absurd, by considering the Stein factorisation of f (III, 4.3.3); by hypothesis, the finite discrete set $g^{-1}(s)$ would contain at least two points. Since A is Henselian and g separated and of finite type, it would result from (18.5.11, c)) that S' would be a sum of two non empty preschemes S'_1, S'_2 (since the intersection of one of them with $g^{-1}(s)$ is reduced to a single point). Since f' is surjective (III, 4.3.1), one would conclude that X is a sum of two non empty preschemes, contradicting the hypothesis. $\square$

18.6. Henselisation.

(18.6.1). Given a local ring A , we will say that a local A -algebra B is *essentially étale* if there exists an étale A -algebra C and a prime ideal $\mathfrak{n}$ of C such that B is A -isomorphic to $C_{\mathfrak{n}}$ and that the composed homomorphism $A \rightarrow C \rightarrow C_{\mathfrak{n}}$ is *local* (in other words, that $\mathfrak{n}$ is above the maximal ideal $\mathfrak{m}$ of A); this implies (17.6.1) that $\mathfrak{n}$ is the *only* prime ideal of B above $\mathfrak{m}$ since B is a local ring; if B, B' are two local A -algebras essentially étale, every A -homomorphism $B \rightarrow B'$ is hence *local*.

If A' is a local A -algebra essentially étale and A'' an A' -algebra locally essentially étale, then A'' is a local A -algebra essentially étale. Indeed, one has by hypothesis $A' = B_{\mathfrak{n}}$ where B is an étale A -algebra and $\mathfrak{n}$ a prime ideal of B above the maximal ideal of A , and $A'' = B'_{\mathfrak{n}'}$, where B' is an A' -algebra étale and $\mathfrak{n}'$ a prime ideal of B' above the ideal $\mathfrak{n}B_{\mathfrak{n}}$. If one sets $S = B - \mathfrak{n}$, so that $A' = S^{-1}B$, B' is of the form $S^{-1}C$, where C is a B -algebra of finite presentation. Hence C is an A -algebra of finite presentation, and A'' is of the form $C_{\mathfrak{r}}$, where $\mathfrak{r}$ is a prime ideal of C above the maximal ideal of A . As A' is an A -algebra formally étale and A'' an A' -algebra formally étale (for the discrete topologies (17.1.3)); the morphism $\operatorname{Spec}(C) \rightarrow \operatorname{Spec}(A)$ is hence étale at the point $\mathfrak{r}$ (17.6.1), and hence there is an element $g \in C$ such that C_g is an A -algebra étale, which proves that A'' is a local A -algebra essentially étale.

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(18.6.1bis). Given a local ring A , there exists an *ensemble* $\mathfrak{E}$ of local A -algebras essentially étale such that every local A -algebra essentially étale is A -isomorphic to an algebra belonging to $\mathfrak{E}$. It suffices evidently for this to observe that there exists an ensemble $\mathfrak{F}$ of A -algebras of finite type such that every A -algebra of finite type is isomorphic to an algebra belonging to $\mathfrak{F}$; one can take $\mathfrak{F}$ equal to the set of quotients of polynomial algebras $A[T_1, \dots, T_n]$ ($n \in \mathbf{N}$).

In what follows, if A and B are two local rings, we will denote $\operatorname{Hom}.\operatorname{loc}(A, B)$ the set of *local* homomorphisms of A into B .

Lemma (18.6.2). — Let A, A' be two local rings, k, k' their respective residue fields, $\varphi : A \rightarrow A'$ a local homomorphism making A' a local A -algebra essentially étale (18.6.1) and such that the corresponding homomorphism $k \rightarrow k'$ is bijective. Then, for every Henselian local ring B , the canonical application

$$\operatorname{Hom}(\varphi, 1_B) : \operatorname{Hom}.\operatorname{loc}(A', B) \longrightarrow \operatorname{Hom}.\operatorname{loc}(A, B)$$

is bijective.

Proof. Set $S = \operatorname{Spec}(A)$, $Y = \operatorname{Spec}(B)$, and let s and y be the closed points of S and Y respectively. By hypothesis, A' is isomorphic to a local ring $\mathcal{O}_{X,x}$ of an S -scheme X étale on S , at a point $x \in X$ above s , and note that since $k(x) = k(s)$, it exists a unique point $x' \in X$ above x and y , and that

$k(x') = k(y)$. If one sets $Y_0 = \operatorname{Spec}(k(y))$, there exists hence a unique Y_0 -section f'_0 of X'_0 such that $f'_0(y) = x'$, and the conclusion results from this that B is supposed Henselian and from (18.5.11), b)). $\square$

We will say that a local A -algebra A' verifying the conditions of (18.6.2) is *strictly essentially étale*. Note that the criterion (18.5.11), b)) signifies that, for A to be Henselian, it is necessary and sufficient that every A -algebra strictly essentially étale be A -isomorphic to A .

Lemma (18.6.3). — *Let A be a local ring, A_1, A_2 two local A -algebras strictly essentially étale.*

label=(i) There exists at most one A -homomorphism (necessarily local) of A_1 in A_2 .

lbbel=(ii) There exists a local A -algebra strictly essentially étale A_3 and two A -homomorphisms $A_1 \rightarrow A_3$, $A_2 \rightarrow A_3$.

Proof. Set $S = \operatorname{Spec}(A)$; by hypothesis there are two S -schemes étale over S , X_1, X_2 , and two points $x_1 \in X_1, x_2 \in X_2$ above the closed point s of S and such that $A_1 = \mathcal{O}_{X_1, x_1}, A_2 = \mathcal{O}_{X_2, x_2}$. Set $X_3 = X_1 \times_S X_2$; the hypotheses $k(x_1) = k(x_2) = k(s)$ entail that there exists a unique point $x_3 \in X_3$ above x_1 and x_2 and that $k(x_3) = k(s)$ (I, 3.4.9). Moreover, X_3 is étale over S (17.3.3), hence $A_3 = \mathcal{O}_{X_3, x_3}$ verifies the conditions of (ii). Furthermore, we have seen that an A -homomorphism of A_1 in A_3 is necessarily local; it corresponds to an S -morphism f of $X'_2 = \operatorname{Spec}(A_2)$ in X_1 such that $f(x_2) = x_1$, or equivalently, setting $X'_3 = X_1 \times_S X'_2$, to an X'_2 -section f' of X'_3 such that $f'(y) = x'$. As $k(x_3) = k(x_2)$ and X'_2 is connected, the uniqueness of f' results from (17.4.9), whence (i). $\square$

(18.6.4). Let us designate by $\mathfrak{G}$ the sub-ensemble of the ensemble $\mathfrak{E}$ defined in (18.6.1) formed of A -algebras strictly essentially étale belonging to $\mathfrak{E}$. It results from (18.6.3) that the relation “there exists an A -homomorphism of A_1 in A_2 ” is a relation of *preorder* in $\mathfrak{G}$ and makes $\mathfrak{G}$ a *filtering directed* set. Indexing $\mathfrak{G}$ by itself via the identity application, it results from (18.6.3), (i) that if $\lambda \leq \mu$ in $\mathfrak{G}$, there is a unique A -homomorphism $\varphi_{\mu\lambda} : A_\lambda \rightarrow A_\mu$ and $(A_\lambda, \varphi_{\mu\lambda})$ is evidently an *inductive system* of local A -algebras, the $\varphi_{\mu\lambda}$ being local homomorphisms. Moreover, it results from (17.3.5) that for $\lambda \leq \mu$, A_μ is an A_λ -algebra strictly essentially étale. IV | 137

Definition (18.6.5). — One calls the *henselisation* of a local ring A and one denotes ${}^h A$ the A -algebra inductive limit of the inductive system $(A_\lambda, \varphi_{\mu\lambda})$ defined in (18.6.4).

This definition does not depend on the apparent choice of $\mathfrak{E}$; if $\mathfrak{E}'$ is another ensemble of local A -algebras essentially étale having the same property as $\mathfrak{E}$, $\mathfrak{G}'$ the sub-ensemble of $\mathfrak{E}'$ formed of strictly essentially étale A -algebras, it results from (18.6.3), (ii) that, for the preorder relations considered, $\mathfrak{G}$ and $\mathfrak{G}'$ are *cofinal* to $\mathfrak{G}'' = \mathfrak{G} \cup \mathfrak{G}'$, hence they give the same inductive limit up to a unique isomorphism.

One will remark that if A is a field, one has evidently ${}^h A = A$. In the general case, ${}^h A$ is also the henselisation of all the A -algebras strictly essentially étale A_λ ($\lambda \in \mathfrak{G}$), for if B is an A_λ -algebra strictly essentially étale, B is also an A -algebra strictly essentially étale (18.6.1), hence (up to A -isomorphism) one of the A_μ for $\mu \geq \lambda$ and such that A_μ is an A_λ -algebra strictly essentially étale, which forms a cofinal set in the preordered set of A_λ , by virtue of (18.6.1) and (18.6.3).

Theorem (18.6.6). — *Let A be a local ring, ${}^h A$ its henselisation and the structural homomorphism $A \rightarrow {}^h A$ is local.*

label=(i) ${}^h A$ is a Henselian local ring and the structural homomorphism $A \rightarrow {}^h A$ is local.

lbbel=(ii) For every Henselian local ring B , the canonical application

$$\operatorname{Hom}.\operatorname{loc}({}^h A, B) \longrightarrow \operatorname{Hom}.\operatorname{loc}(A, B)$$

is bijective.

lcbel=(iii) ${}^h A$ is an A -module faithfully flat, and if $\mathfrak{m}$ is the maximal ideal of A , $\mathfrak{m} \cdot {}^h A$ is the maximal ideal of ${}^h A$, and the homomorphism $A/\mathfrak{m}A \rightarrow {}^h A/\mathfrak{m} \cdot {}^h A$ of the residue fields is bijective.

ldbel=(iv) If $\hat{A}$ and $({}^h A)^\wedge$ are the separated completions of the local rings A and ${}^h A$, the homomorphism $\hat{A} \rightarrow ({}^h A)^\wedge$ deduced from the structural homomorphism $A \rightarrow {}^h A$ by completion is bijective.

lebel=(v) For ${}^h A$ to be Noetherian, it is necessary and sufficient that A be Noetherian.

lfbel=(vi) If A is Henselian, the canonical homomorphism $A \rightarrow {}^h A$ is bijective.

Proof. Let $\mathfrak{m}_\lambda$ be the maximal ideal of A_λ ; it results from the fact that, in (17.6.1), a), $\mathfrak{m}_\lambda = \mathfrak{m}_\lambda A_\mu$, the homomorphism $A_\lambda / \mathfrak{m}_\lambda \rightarrow A_\mu / \mathfrak{m}_\mu$ is bijective and A_μ is a flat A_λ -module. The fact that hA is local, that the homomorphism $A \rightarrow {}^hA$ is local, the assertion (iii) and the suffisance of (v) result therefore from (0_{III}, 10.3.1.3); the necessity of (v) results from the fact that hA is an A -module faithfully flat (0_I, 6.5.2).

To prove that hA is Henselian, apply the criterion (18.5.11), b). Set $S = \operatorname{Spec}({}^hA)$, $S_0 = \operatorname{Spec}({}^hA / \mathfrak{m} \cdot {}^hA)$, and let $g : S' \rightarrow S$ be a morphism étale; set $S'_0 = g^{-1}(S_0)$ and let $f_0 : S_0 \rightarrow S'_0$ be an S_0 -section of S'_0 . Reasoning as usual in (18.5.4), one can suppose that g is of finite presentation. It results from (8.8.2) and (17.7.5) that there exists an index $\lambda \in S$, a morphism étale $g_\lambda : S'^{(\lambda)} \rightarrow S^{(\lambda)} = \operatorname{Spec}(A_\lambda)$ and, if one sets $S_0^{(\lambda)} = \operatorname{Spec}(A_\lambda / \mathfrak{m} A_\lambda)$, an $S_0^{(\lambda)}$ -section $f_0^{(\lambda)} : S_0^{(\lambda)} \rightarrow S_0'^{(\lambda)}$ such that $S' = S'^{(\lambda)} \times_{g^{(\lambda)}} S$, $g = g^{(\lambda)} \times 1$ and $f_0 = f_0^{(\lambda)} \times 1$. Let s be the closed point of S , $x = f_0(s)$; as x_λ is above the closed point s_λ of $S^{(\lambda)}$, the local ring C_λ of $S'^{(\lambda)}$ at the point x_λ is an A_λ -algebra essentially étale; moreover, as $f_0^{(\lambda)}(s_\lambda) = x_\lambda$, one has $k(x_\lambda) = k(s_\lambda)$, in other words, C_λ is an A_λ -algebra strictly essentially étale, and hence by (18.6.1) an A_λ -isomorphism $C_\lambda \rightarrow {}^hA$, that is to say there is an A_λ -homomorphism $C_\lambda \rightarrow {}^hA$, and hence an $S^{(\lambda)}$ -morphism $h : S \rightarrow S'^{(\lambda)}$ such that $h(s) = x_\lambda$, and hence there is indeed an S -section f of S' such that $f(s) = x$, which achieves the proof of (i).

To prove (ii), it suffices to note that $\operatorname{Hom}.\operatorname{loc}({}^hA, B) \cong \varprojlim \operatorname{Hom}.\operatorname{loc}(A_\lambda, B)$ and that by (18.6.2) the canonical homomorphisms $\operatorname{Hom}.\operatorname{loc}(A_\lambda, B) \leftarrow \operatorname{Hom}.\operatorname{loc}(A, B)$ are bijective.

To demonstrate (iv), note that for every λ and every integer $n > 0$, $\mathfrak{m}_\lambda^n = \mathfrak{m}^n A_\lambda$, and $(\mathfrak{m} \cdot {}^hA)^n = \mathfrak{m}^n \cdot {}^hA = \varinjlim \mathfrak{m}_\lambda^n$. One concludes, by exactness of the functor $\varinjlim$ in the category of A -modules, that ${}^hA / (\mathfrak{m} \cdot {}^hA)^n = \varinjlim (A_\lambda / \mathfrak{m}_\lambda^n)$, and hence it suffices to show that for every n , and every index λ , the homomorphism $A / \mathfrak{m}^n \rightarrow A_\lambda / \mathfrak{m}_\lambda^n$ is bijective. This is true by hypothesis for $n = 1$; on the other hand, since A_λ is a flat A -module, one has

$$\mathfrak{m}_\lambda^n / \mathfrak{m}_\lambda^{n+1} = (\mathfrak{m}^n / \mathfrak{m}^{n+1}) \otimes_A A_\lambda = (\mathfrak{m}^n / \mathfrak{m}^{n+1}) \otimes_{A/\mathfrak{m}} (A_\lambda / \mathfrak{m} A_\lambda)$$

and as $A/\mathfrak{m} \rightarrow A_\lambda / \mathfrak{m}_\lambda = A_\lambda / \mathfrak{m} A_\lambda$ is bijective, the homomorphism $\mathfrak{m}^n / \mathfrak{m}^{n+1} \rightarrow \mathfrak{m}_\lambda^n / \mathfrak{m}_\lambda^{n+1}$ is also bijective; the conclusion results from Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 3 du th. 1.

Finally, if A is Henselian, it results from the remark following (18.6.3) that the homomorphisms $A \rightarrow A_\lambda$ are bijective, which proves (vi) by definition of hA . $\square$

(18.6.7). Let A be a *semi-local* ring, $\mathfrak{m}_i$ ($1 \leq i \leq r$) its maximal ideals. One calls the *henselisation* of A and denotes again hA the ring product $\prod_i {}^h(A_{\mathfrak{m}_i})$ of the henselisations of the local rings $A_{\mathfrak{m}_i}$. It is an A -module faithfully flat and a semi-local A -algebra, whose maximal ideals are the $\mathfrak{m}_i \cdot {}^hA$ by virtue of (18.6.6), (iii); moreover, if $\mathfrak{r} = \bigcap_i \mathfrak{m}_i$ is the radical of A , it results from the preceding that $\mathfrak{r} \cdot {}^hA$ is the radical of hA and that the application $A/\mathfrak{r} \rightarrow {}^hA/\mathfrak{r} \cdot {}^hA$ is bijective. As the separated completion $\hat{A}$ of A for the $\mathfrak{r}$ -preadic topology is the product of the separated completions $\hat{A}_{\mathfrak{m}_i}$ (Bourbaki, *Alg. comm.*, chap. III, § 2, n° 13, prop. 18) the canonical homomorphism $\hat{A} \rightarrow ({}^hA)^\wedge$ is bijective by (18.6.6), (iv)), and it is clear by (18.6.6), (v)) that for hA to be Noetherian, it is necessary and sufficient that A be Noetherian.

Proposition (18.6.8). — *Let A be a semi-local ring, B a semi-local and entire A -algebra. Then $B \otimes_A ({}^hA)$ is a B -algebra semi-local isomorphic to hB .* IV | 139

Proof. As $B \otimes_A ({}^hA)$ is entire over hA , each of its maximal ideals is above one of the maximal ideals of hA , hence above a maximal ideal of A , and as B is entire over A , the prime ideals of B above a maximal ideal of A are maximal ideals of B , so that finally a maximal ideal of $B \otimes_A ({}^hA)$ has for projections in $\operatorname{Spec}({}^hA)$ and $\operatorname{Spec}(B)$ maximal ideals; taking into account (18.6.6), (iii) and (I, 3.4.9), one concludes that the ring $B \otimes_A ({}^hA)$ is semi-local. Moreover, for every Henselian local ring C , $\operatorname{Hom}.\operatorname{sloc}(B \otimes_A ({}^hA), C)$ is in bijective correspondence with the set of couples (φ, ψ) of semi-local homomorphisms $\varphi : B \rightarrow C$, $\psi : {}^hA \rightarrow C$, such that the compositions $A \rightarrow B \xrightarrow{\varphi} C$ and $A \rightarrow {}^hA \xrightarrow{\psi} C$ are equal. But by virtue of the bijective correspondence between $\operatorname{Hom}.\operatorname{sloc}(A, C)$ and

$\text{Hom}.\text{sloc}({}^hA, C)$, one sees that for every $\varphi \in \text{Hom}.\text{sloc}(B, C)$ there exists a unique ψ having the preceding property, hence the application $\text{Hom}.\text{sloc}(B \otimes_A ({}^hA), C) \rightarrow \text{Hom}.\text{sloc}(B, C)$ is bijective, which proves the proposition by virtue of the uniqueness of the solution of a universal problem. $\square$

Theorem (18.6.9). — *Let A be a semi-local ring.*

- label=(i) For hA to be reduced (resp. normal), it is necessary and sufficient that A be reduced (resp. normal).*
lbbel=(ii) Suppose A Noetherian. Then, for every prime ideal $\mathfrak{p}$ of A , the ring $({}^hA)_{\mathfrak{p}}/\mathfrak{p}({}^hA)_{\mathfrak{p}}$ is a direct composition of a finite number of separable algebraic extensions of $k(\mathfrak{p})$ (which implies that the fibres of the canonical morphism $\text{Spec}({}^hA) \rightarrow \text{Spec}(A)$ are geometrically regular).

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Proof. (i) As hA is an A -module faithfully flat, it results from (I, 2.1.13) that if hA is reduced (resp. normal), so is A . To prove the converse, one can restrict to the case where A is local, so that with the notations of (18.6.4), ${}^hA = \varinjlim A_{\alpha}$. It results from the definition of the A_{α} and from (17.5.7) that if A is reduced, so are the A_{α} ; moreover, the homomorphisms $A_{\alpha} \rightarrow A_{\beta}$ for $\alpha \leq \beta$ are injective by virtue of (0_I, 6.5.1) since A_{β} is a flat A_{α} -module; hence the morphisms $\text{Spec}(A_{\beta}) \rightarrow \text{Spec}(A_{\alpha})$ are dominant, and one concludes from (5.13.2) (resp. (5.13.4)) that hA is reduced (resp. intègre and integrally closed).

(ii) One can restrict to the case where A is local. By virtue of the fact that the functor $\varinjlim$ permutes with tensor product, the fibre of the morphism $\text{Spec}({}^hA) \rightarrow \text{Spec}(A)$ at a point $\mathfrak{p}$ is the inductive limit of the fibres of the morphisms $\text{Spec}(A_{\alpha}) \rightarrow \text{Spec}(A)$ at that point. As A is Noetherian, one sees, taking into account (17.6.2), that one is reduced to proving the following lemma: $\square$

Corollary (18.6.10). — *For hA to possess one of the following properties:*

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- a) to be a Cohen–Macaulay ring (0, 16.5.3);*
- b) to verify the property (S_n) (5.7.3);*
- c) to be regular;*
- d) to verify the property (R_n) (5.8.2);*

it is necessary and sufficient that A possess this same property.

Proof. This follows immediately from the fact that the fibres of the morphism $\text{Spec}({}^hA) \rightarrow \text{Spec}(A)$ are geometrically regular (18.6.9) and from (6.4.1), (6.5.1) and (6.5.3). $\square$

Corollary (18.6.11). — *Let A be a semi-local Noetherian ring. For a prime ideal $\mathfrak{p}$ of hA to belong to $\text{Ass}({}^hA)$, it is necessary and sufficient that $\mathfrak{p} \cap A$ belong to $\text{Ass}(A)$.*

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Proof. Taking into account the description of the fibres of the morphism $\text{Spec}({}^hA) \rightarrow \text{Spec}(A)$, this follows immediately from (3.3.1). $\square$

Proposition (18.6.12). — *Let A be a local ring; the following properties are equivalent:*

- a) A is unibranch (0, 23.2.1) (resp. reduced and unibranch).*
- b) For every local A -algebra strictly essentially étale A' , $\text{Spec}(A')$ is irreducible (resp. intègre).*
- c) $\text{Spec}({}^hA)$ is irreducible (resp. intègre).*

Proof. Note first that ${}^h(A_{\text{red}}) = ({}^hA)_{\text{red}}$, since ${}^h(A_{\text{red}}) = ({}^hA) \otimes_A A_{\text{red}}$ (18.6.8), and as ${}^h(A_{\text{red}})$ is reduced (18.6.9), it is equal to $({}^hA)_{\text{red}}$. This allows us, in the demonstration, to consider only the case where A is reduced, hence also A' ((17.5.7)) and hA (18.6.9).

With the notations of (18.6.4), the condition *b)* signifies that all the A_{α} are intègre; one concludes that ${}^hA = \varinjlim A_{\alpha}$ is intègre ((5.13.3)), hence *b)* implies *c)*.

It is clear that if hA is intègre, the same holds for the $A_{\alpha} \subset {}^hA$, hence *c)* implies *b)*.

Let us show that *c)* implies *a)*. Note first that $A \subset {}^hA$ is intègre; let K and L be the fields of fractions of A and hA respectively. It suffices to see that for every finite A -algebra $B \subset K$, B is a local ring. Now B is a semi-local ring, whose henselisation is $B \otimes_A ({}^hA)$ (18.6.8). By flatness, $B \otimes_A ({}^hA)$ identifies with a subring of $K \otimes_A ({}^hA)$ and the latter with a subring of L , hence hB is intègre. But since hB is a semi-local henselian ring, it is a direct composition of local rings (18.5.9), hence can only be intègre if it is local, which indeed implies that B is local.

Let us prove finally that $a)$ implies $c)$. Let C be the integral closure of the integre ring A . Since C is a local ring by hypothesis, $C \otimes_A ({}^h A)$ is the henselisation of C (18.6.8), hence is integre and integrally closed (18.6.9). Since A is a subring of C , ${}^h A$ identifies by flatness with a subring of ${}^h C$, hence is itself integre, which concludes the demonstration. $\square$

Corollary (18.6.13). — *Let A be a Henselian local ring. Then, for A to be unibranch, it is necessary and sufficient that A_{red} be integre.*

Proof. This follows from (18.6.12) by noting that if A is Henselian, ${}^h A \cong A$ (18.6.6), (vi)), and hence $\text{Spec}({}^h A) = \text{Spec}(A)$. $\square$

Proposition (18.6.14). — *label=(i) Every filtering inductive limit of Henselian local rings (where the transition homomorphisms are local) is a Henselian local ring.*

lbbel=(ii) The functor $A \mapsto {}^h A$ in the category of local rings (where the morphisms are local homomorphisms) commutes with filtering inductive limits.

lcbel=(iii) Every Henselian local ring A is the inductive limit of an inductive system filtering of Henselian and Noetherian local rings A_λ (the transition homomorphisms $A_\lambda \rightarrow A_\mu$ being local).

Proof. (i) Let (A_λ) be a filtering inductive system of Henselian local rings, the transition homomorphisms being local. Let us show that $A' = \varinjlim A_\lambda$, which is local (0_{III}, 10.3.1.3). Let C be an A' -algebra étale, $\mathfrak{n}$ a prime ideal of C above the maximal ideal $\mathfrak{m}'$ of A' , such that $C_{\mathfrak{n}}$ is strictly essentially étale over A' (18.6.1), in other words that $k(\mathfrak{m}') \rightarrow k(\mathfrak{n})$ is an isomorphism. By virtue of (8.8.2), (ii) and of (17.7.8), there exists an index λ and an A_λ -algebra étale C_λ such that $C = C_\lambda \otimes_{A_\lambda} A'$ up to isomorphism; moreover, if $\mathfrak{n}_\lambda$ is the inverse image of $\mathfrak{n}$ in C_λ , and $\mathfrak{m}_\lambda$ the maximal ideal of A_λ , one can, by virtue of (8.8.2.4) and of the transitivity of the fibres (I, 3.6.4), suppose that $(C_\lambda)_{\mathfrak{n}_\lambda}$ is strictly essentially étale over A_λ ; since A_λ is Henselian, $(C_\lambda)_{\mathfrak{n}_\lambda}$ is necessarily isomorphic to A_λ (18.5.11, b)); there is hence an open neighbourhood of $\mathfrak{n}_\lambda$ in $\text{Spec}(C_\lambda)$ isomorphic to $\text{Spec}(A_\lambda)$ (I, 6.5.4); one concludes that there is an open neighbourhood of $\mathfrak{n}$ in $\text{Spec}(C)$ isomorphic to $\text{Spec}(A')$, which proves that A' is Henselian (18.6.2) and achieves the proof.

(ii) Suppose that $B = \varinjlim B_\lambda$, where the B_λ are local rings, the transition homomorphisms $B_\lambda \rightarrow B_\mu$ ($\lambda \leq \mu$) being local. Set $A_\lambda = {}^h B_\lambda$; the A_λ are Henselian local rings (18.6.6), (vi)), and by virtue of (18.6.6), (ii) the transition homomorphisms $B_\lambda \rightarrow B_\mu$ determine in a unique way local homomorphisms $A_\lambda \rightarrow A_\mu$, so that (A_λ) is an inductive system. Everything amounts to seeing that $A' = \varinjlim A_\lambda$ is canonically isomorphic to $A = {}^h B$. By virtue of (18.6.6), (ii) and of the definition of inductive limits, one has, for every Henselian local ring E ,

$$\text{Hom}.\text{loc}(A', E) = \varprojlim \text{Hom}.\text{loc}(A_\lambda, E) = \varprojlim \text{Hom}.\text{loc}(B_\lambda, E) = \text{Hom}.\text{loc}(B, E).$$

But as A' is Henselian by (i), this proves our assertion.

(iii) The ring A is the inductive limit of its sub-rings local Noetherian B_λ , the transition homomorphisms being local ((5.13.3), (iii)). As the ${}^h B_\lambda$ are Henselian local and Noetherian (18.6.6), (v)) and that $A = {}^h A$ (18.6.6), (vi)), it suffices to apply (ii) to the inductive system (B_λ) . $\square$

Corollary (18.6.15). — *Let A be a Henselian local ring, X an A -prescheme of finite presentation over A . There exists then a Noetherian local Henselian ring A_0 , a local homomorphism $A_0 \rightarrow A$, a prescheme X_0 of finite type over A_0 and an A -isomorphism $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.* IV | 142

Proof. This results from (18.6.14) and from (8.8.2), (ii)). $\square$

18.7. Henselisation and excellent rings.

(18.7.1). We will designate in this paragraph by $\mathbf{P}(Z, k)$ a property of the form considered in (7.3.1), where we suppose moreover that the property $\mathbf{Q}(A, k)$ satisfies the following condition:

For every separable algebraic extension k' of k , and every k' -algebra local Noetherian A , the property $\mathbf{Q}(A, k)$ is *equivalent* to $\mathbf{Q}(A, k')$.

It is immediate that all the properties $\mathbf{P}$ considered in (7.3.8) verify the preceding condition, taking into account the fact that if K is a finite extension of k , $k' \otimes_k K$ is a direct composition of separable algebraic extensions of K .

Proposition (18.7.2). — *Let A be a Noetherian local ring. For hA to be a $\mathbf{P}$ -ring (7.3.13), it is necessary and sufficient that A be one.* IV | 143

Proof. Indeed, by virtue of (18.6.6), (iv)), the completion $\hat{A}$ of A is also the completion of hA and one has $A \subset {}^hA \subset \hat{A}$. By (18.6.9), for every $x \in \text{Spec}(A)$, the fibre at x of the morphism $\text{Spec}({}^hA) \rightarrow \text{Spec}(A)$ is discrete and finite, and at each of the points y_i of this fibre, $k(y_i)$ is a separable algebraic extension of $k(x)$. The formal fibre of A at the point x is hence a prescheme *sum* of the formal fibres of hA at the points y_i . The conclusion results then from the hypothesis made on $\mathbf{P}$ in (18.7.1). $\square$

Corollary (18.7.3). — *Let A be a Noetherian local ring. For hA to be universally Japanese, it is necessary and sufficient that A be so.*

Proof. Indeed, for a Noetherian local ring B , it amounts to the same to say that B is universally Japanese, or that the formal fibres of B are geometrically reduced (7.6.4) and (7.7.1)); the conclusion results hence from (18.7.2). $\square$

Corollary (18.7.4). — *Let A be a Noetherian local ring. For the formal fibres of hA to be geometrically regular, it is necessary and sufficient that those of A be so.*

This is a particular case of (18.7.2).

Proposition (18.7.5). — *Let A be a Noetherian local ring. If A is universally catenaire (5.6.2) (resp. formally catenaire (7.1.9), resp. strictly formally catenaire (7.2.6)), so is hA .*

Proof. With the notations of (18.6.4), if A is universally catenaire, the same holds for the A_α , which are A -algebras essentially of finite type ((5.6.3)). To prove that in this case hA is universally catenaire, it will therefore suffice to establish the following lemma:

Lemma (18.7.5.1). — *Let $(B_\alpha, \varphi_{\beta\alpha})$ be an inductive system of Noetherian rings and suppose that, for $\alpha \leq \beta$, $\varphi_{\beta\alpha}$ makes B_β a faithfully flat B_α -module and that $B = \varinjlim B_\alpha$ is Noetherian. Then, if the B_α are catenaire (resp. universally catenaire), so is B .*

Proof of Lemma (18.7.5.1). To say that B is universally catenaire amounts to saying that $B[T_1, \dots, T_n]$ is catenaire for every n ((5.6.2)), and it is immediate that the inductive system $(B_\alpha[T_1, \dots, T_n])$ satisfies the same conditions as (B_α) ; it therefore suffices to prove that if the B_α are catenaire, so is B . If $\mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \dots \supset \mathfrak{p}_r$ is a saturated chain of prime ideals of B , the inverse images $\mathfrak{p}_{0\alpha} \supset \mathfrak{p}_{1\alpha} \supset \dots \supset \mathfrak{p}_{r\alpha}$ of these ideals in B_α form a chain of prime ideals, and it suffices to show that for α sufficiently large, this chain is *saturated*. Now, each $\mathfrak{p}_i$ is an ideal of finite type, hence there exists a λ such that, for $\alpha \geq \lambda$, one has $\mathfrak{p}_i = \mathfrak{p}_{i\alpha} \cdot B$. One therefore has $1 = \text{codim}(V(\mathfrak{p}_i), V(\mathfrak{p}_{i+1})) = \text{codim}(V(\mathfrak{p}_{i\alpha}), V(\mathfrak{p}_{i+1,\alpha}))$ since B is a flat B_α -module (((6.1.4) and (0_I, 6.2.3)), which finishes the proof of the lemma. $\square$

Suppose finally that A is strictly formally catenaire; then one has just seen that hA is formally catenaire, and it remains to prove that the fibres of the morphism $\text{Spec}(\hat{A}) \rightarrow \text{Spec}({}^hA)$ verify the property (S_1) ((7.2.5), *b*)). But this follows from the hypothesis on A and from (18.7.2) applied to the case where $\mathbf{P}(Z, k)$ is the property (S_1) for Z . $\square$

Corollary (18.7.6). — *If a Noetherian local ring A is excellent (7.8.2), so is hA .* IV | 144

Proof. This follows from (18.7.2), (18.7.3), (18.7.4) and (18.7.5). $\square$

Remark (18.7.7). — It can happen that hA is an excellent local ring without A being universally catenaire (nor *a fortiori* excellent).

18.8. Strictly local rings and strict henselisation.

Proposition (18.8.1). — *Let A be a local ring. The following conditions are equivalent:*

- a) *A is a Henselian ring and its residue field k is separably closed.*
- b) *A is a Henselian ring and every étale covering of $S = \operatorname{Spec}(A)$ is trivial.*
- c) *For every étale morphism $f : X \rightarrow S$ and every point $x \in X$ such that $f(x)$ is the closed point s of S , there exists an S -section $u : S \rightarrow X$ of X such that $u(s) = x$.*

Proof. The hypothesis c) implies that for every étale morphism $f : X \rightarrow S$, the residue fields at the points of $f^{-1}(s)$ are all isomorphic to k , and moreover that the condition b) of (18.5.11) is verified. Hence A is Henselian, and the fact that every étale covering of S is trivial results from (18.2.10), (ii); hence c) implies b). As the étale coverings of $\operatorname{Spec}(k)$ are trivially only if k is separably closed, b) implies a) by virtue of (18.5.11), b)). Finally, it results also from (18.5.11), b)) that a) implies c). $\square$

Definition (18.8.2). — One says that a local ring is a *strictly local ring* if it verifies the equivalent conditions of (18.8.1). One calls a *strictly local scheme* a scheme isomorphic to the spectrum of a strictly local ring.

Remark (18.8.3). — The conditions of (18.8.1) are moreover equivalent to the following:

- d) *For every morphism locally of finite type $f : X \rightarrow S$ and every point $x \in X$ such that $f(x)$ is the closed point s of S and that $\mathcal{O}_{X,x}/\mathfrak{m}_s\mathcal{O}_{X,x}$ is a field, a finite separable extension of $k(s) = k$, there exists an open neighbourhood U of x in X such that $f|_U$ is a closed immersion.*

(One will note that the hypothesis of d) is verified if f is formally unramified at the point x (17.4.1.2).) IV | 145

Lemma (18.8.4). — *Let A, A' be two local rings, $\varphi : A \rightarrow A'$ a local homomorphism making A' a local A -algebra essentially étale (18.6.1). Then, for every strictly local ring B , every local homomorphism $\psi : A \rightarrow B$ and every $k(A)$ -algebra homomorphism $\alpha : k(A') \rightarrow k(B)$, there exists a unique local homomorphism $\psi' : A' \rightarrow B$ such that $\psi = \psi' \circ \varphi$ and that α is equal to the homomorphism $\bar{\psi}'$ deduced from ψ' by passage to the quotients.*

Proof. With the notations of (18.6.2), the homomorphism α corresponds to a $k(s)$ -morphism $\operatorname{Spec}(k(y)) \rightarrow \operatorname{Spec}(k(x))$, hence to a well-determined point x' in X' above y . The existence of the Y -section f' of X' such that $f'(y) = x'$ then follows from (18.8.1), c) since f' is étale and B is strictly local, and its uniqueness follows from the fact that f' is separated and from (17.4.9). $\square$

Lemma (18.8.5). — *Let A be a local ring, A_1, A_2 two local A -algebras essentially étale, K a field, extension of $k(A)$.*

label=(i) *For every $k(A)$ -homomorphism $\gamma : k(A_1) \rightarrow k(A_2)$, there exists at most one A -homomorphism (necessarily local) $\psi : A_1 \rightarrow A_2$ such that γ is the homomorphism $\bar{\psi}$ deduced from ψ by passage to the quotients.*

lbel=(ii) *For every couple of local A -homomorphisms $\beta_1 : A_1 \rightarrow K, \beta_2 : A_2 \rightarrow K$ there exists a local A -algebra essentially étale A_3 , two A -homomorphisms $\varphi_1 : A_1 \rightarrow A_3, \varphi_2 : A_2 \rightarrow A_3$ and an A -homomorphism $\beta : A_3 \rightarrow K$ such that $\beta_1 = \beta \circ \varphi_1$ and $\beta_2 = \beta \circ \varphi_2$.*

Proof. With the notations of (18.6.3), the homomorphisms β_1, β_2 in (ii) correspond to two S -morphisms $\operatorname{Spec}(K) \rightarrow X_1, \operatorname{Spec}(K) \rightarrow X_2$, with respective images x_1, x_2 ; one deduces from this a well-determined S -morphism $\operatorname{Spec}(K) \rightarrow X_3$ (I, 3.2.1) with image x_3 above x_1 and x_2 ; the A -algebra $A_3 = \mathcal{O}_{X_3, x_3}$ and the homomorphism $A_3 \rightarrow K$ corresponding to these data satisfy the conditions of (ii). On the other hand, the data of γ in (i) corresponds to an S -morphism $\operatorname{Spec}(k(x_2)) \rightarrow \operatorname{Spec}(k(x_1))$ or equivalently to a point x'_2 of X'_2 above x_2 ; the uniqueness of ψ follows from the uniqueness of an X'_2 -section of X'_2 passing through x'_2 (17.4.9). $\square$

(18.8.6). Let A be a local ring, $\mathfrak{m}$ its maximal ideal, k its residue field, Ω a separable closure of k . Consider the ensemble $\mathfrak{E}$ of A -algebras essentially étale defined in (18.6.1), and designate by L the set of A -homomorphisms local $A' \rightarrow \Omega$, where A' ranges over $\mathfrak{E}$; one will note that for each such homomorphism, it factors as $\lambda : A' \rightarrow k(A') \xrightarrow{\omega_\lambda} \Omega$, where $\omega_\lambda : k(A') \rightarrow \Omega$ is a k -homomorphism (since $\mathfrak{m}A'$ is the maximal ideal of A'); inversely the data of such a homomorphism ω_λ determines uniquely a homomorphism $\lambda \in L$. For every $\lambda \in L$, we designate by A_λ the local A -algebra

essentially étale on which λ is defined. The set L is *preordered* by the relation “there exists an A -homomorphism $\varphi_{\mu\lambda} : A_\lambda \rightarrow A_\mu$ such that $\lambda = \mu \circ \varphi_{\mu\lambda}$ ”, and it results from (18.8.5), (i) that this homomorphism $\varphi_{\mu\lambda}$ is *unique*. Moreover, for the preceding preorder relation (which one denotes again $\lambda \leq \mu$), L is *filtering directed*, by virtue of (18.8.5), (ii). If $\bar{\varphi}_{\mu\lambda} : k(A_\lambda) \rightarrow k(A_\mu)$ is the k -homomorphism deduced from $\varphi_{\mu\lambda}$ by passage to quotients, $(k(A_\lambda), \bar{\varphi}_{\mu\lambda})$ is an *inductive system* of separable algebraic extensions of k , and the $\omega_\lambda : k(A_\lambda) \rightarrow \Omega$ form an inductive system of k -homomorphisms. Moreover, the inductive limit

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$$(18.8.6.1) \quad \omega : \varinjlim k(A_\lambda) \longrightarrow \Omega$$

is a k -isomorphism. It suffices indeed to prove that for every finite separable extension k' of k , there exists a local A -algebra essentially étale A' such that k' is the residue field of A' and that the homomorphism $k \rightarrow k'$ is deduced from $A \rightarrow A'$ by passage to quotients. But this results from (18.1.1) applied to étale morphisms, taking into account that $\text{Spec}(A)$ is the only open neighbourhood of its closed point.

Note now that if $\mathfrak{m}_\lambda$ is the maximal ideal of A_λ , it results from (17.6.1) that for $\lambda \leq \mu$, one has $\mathfrak{m}_\mu = \mathfrak{m}_\lambda A_\mu$ and that A_μ is a flat A_λ -module. Hence it results from (0III, 10.3.1.3) that the ring $\varinjlim A_\lambda$ is local, that the canonical homomorphism $w : A \rightarrow \varinjlim A_\lambda$ is *local*, and that one has a k -isomorphism canonical

$$\varinjlim k(A_\lambda) \xrightarrow{\sim} k(\varinjlim A_\lambda);$$

identifying these two fields, one deduces a k -isomorphism canonical

$$(18.8.6.2) \quad \omega^{-1} : \Omega \xrightarrow{\sim} k(\varinjlim A_\lambda).$$

Definition (18.8.7). — Let A be a local ring, $k = k(A)$ its residue field, $i : k \rightarrow \Omega$ a homomorphism of k into a separable closure Ω of k ; one calls the *strict henselisation* of A relative to i , and one denotes ${}^{hs}A_{(i)}$ (or ${}^{hs}A$ when this does not lead to confusion) the inductive limit of the inductive system $(A_\lambda, \varphi_{\mu\lambda})$ defined in (18.8.6).

As in (18.6.5) one sees that the definition given in (18.8.7) does not depend on the apparent choice of $\mathfrak{C}$; we will see below (18.8.8), (i) and (ii) that ${}^{hs}A_{(i)}$ is an object representing a functor entirely determined by the data of A and of i (0III, 8.1.8), and is hence determined up to a unique isomorphism.

Proposition (18.8.8). — Let A be a local ring, k its residue field, $i : k \rightarrow \Omega$ a homomorphism into a separable closure of k , ${}^{hs}A_{(i)}$ the strict henselisation of A relative to i .

- label=(i) ${}^{hs}A_{(i)}$ is a strictly local ring (18.8.2) and the structural homomorphism $w : A \rightarrow {}^{hs}A_{(i)}$ is local.
- lbbel=(ii) For every local homomorphism $u : A \rightarrow B$, where B is a strictly local ring, and every k -homomorphism $\psi : \Omega \rightarrow k(B)$, there exists a unique A -homomorphism $v : {}^{hs}A_{(i)} \rightarrow B$ such that ψ factors as $\Omega \xrightarrow{\omega^{-1}} k({}^{hs}A_{(i)}) \xrightarrow{\bar{v}} k(B)$, where $\bar{v}$ is deduced from v by passage to quotients.
- lcbel=(iii) ${}^{hs}A_{(i)}$ is an A -module faithfully flat, and if $\mathfrak{m}$ is the maximal ideal of A , $\mathfrak{m} \cdot {}^{hs}A_{(i)}$ is the maximal ideal of ${}^{hs}A_{(i)}$, and $k({}^{hs}A_{(i)})$ is a separable closure of k .
- ldbel=(iv) For ${}^{hs}A_{(i)}$ to be Noetherian, it is necessary and sufficient that A be so.
- lebel=(v) If A is strictly local (hence $\Omega = k$), one has ${}^{hs}A \cong A$.
- lfbel=(vi) If $i' : k \rightarrow \Omega'$ is a second homomorphism of k into a separable closure of k , for every k -isomorphism $\sigma : \Omega \xrightarrow{\sim} \Omega'$, the A -homomorphism corresponding (by (ii)) $v_\sigma : {}^{hs}A_{(i)} \rightarrow {}^{hs}A_{(i')}$ is an isomorphism.

Proof. We have already seen above that ${}^{hs}A_{(i)}$ is a local ring and that w is a local homomorphism; the fact that ${}^{hs}A_{(i)}$ is henselian (and hence strictly local) is demonstrated as in (18.6.6). The assertions of (iii) follow again from (0III, 10.3.1.3), as does the sufficiency of the condition (iv); the necessity of this condition follows from the fact that ${}^{hs}A$ is a faithfully flat A -module (0I, 6.5.2).

To prove (ii), remark that, with the notations of (18.8.6), for every $\lambda \in L$, there exists a unique local A -homomorphism $v_\lambda : A_\lambda \rightarrow B$, such that the composed $k(A_\lambda) \xrightarrow{\omega_\lambda} \Omega \xrightarrow{\psi} k(B)$ is deduced from v_λ by passage to quotients, by virtue of (18.8.4) and the hypothesis that B is strictly local. The uniqueness of v_λ implies moreover that the v_λ form an inductive system, whence the existence of

the homomorphism v satisfying the conditions of (ii); its uniqueness follows from (17.4.9). The uniqueness of v immediately implies assertion (vi), by considering the composites $v_\sigma \circ v_{\sigma^{-1}}$ and $v_{\sigma^{-1}} \circ v_\sigma$. Finally, if A is strictly local and $A' = B_n$ a local A -algebra essentially étale, where B is an étale A -algebra, it follows from (18.8.1) that there exists a $\text{Spec}(A)$ -section of $\text{Spec}(B)$ taking the value n at the closed point of $\text{Spec}(A)$, from which one deduces by (17.4.1) that the homomorphism $A \rightarrow A'$ is bijective, which proves (v). $\square$

(18.8.8.1). Let $\mathcal{C}$ be the category of strictly local rings, with local homomorphisms as morphisms; for every $B \in \mathcal{C}$, designate by $F(B)$ the set of couples (u, ψ) formed of a local homomorphism $u : A \rightarrow B$ and a k -isomorphism $\psi : \Omega \rightarrow k(B)$ such that the composed $k(A) = k \xrightarrow{i} \Omega \xrightarrow{\psi} k(B)$ is deduced from u by passage to quotients. One can say that the object ${}^{hs}A_{(i)}$ represents the covariant functor $F : \mathcal{C} \rightarrow \mathbf{Ens}$ (0III, 8.1.11).

One will remark that by virtue of (18.8.8), (vi)) the strict henselisations ${}^{hs}A_{(i)}$ are all A -isomorphic (for the diverse k -homomorphisms i of k into separable closures of k), but for given i and i' , there is in general an infinity of A -isomorphisms ${}^{hs}A_{(i)} \xrightarrow{\sim} {}^{hs}A_{(i')}$. To be precise, the group of A -automorphisms of ${}^{hs}A_{(i)}$ is isomorphic to the Galois group of Ω over k .

(18.8.9). Let A now be a *semi-local* ring, $\mathfrak{m}_j$ ($1 \leq j \leq r$) its maximal ideals, and for every index j , consider a homomorphism $i_j : k(A_{\mathfrak{m}_j}) \rightarrow \Omega_j$ of $k(A_{\mathfrak{m}_j})$ into a separable closure of this field. One calls the *strict henselisation* of A relative to the i_j and one denotes again ${}^{hs}A$ (when this does not lead to confusion) the ring product $\prod_{j=1}^r ({}^{hs}A_{\mathfrak{m}_j})$ of the strict henselisations of the local rings $A_{\mathfrak{m}_j}$ relative to the homomorphisms i_j . IV | 147

Proposition (18.8.10). — Let A be a semi-local ring, B a finite A -algebra. Let $\mathfrak{n}_j$ ($1 \leq j \leq r$) be the maximal ideals of B . Then there is, for each j , a maximal ideal $\mathfrak{q}_j$ of $B \otimes_A ({}^{hs}A)$ above $\mathfrak{n}_j$, such that ${}^{hs}B$ is isomorphic to the direct composition of the local rings $(B \otimes_A ({}^{hs}A))_{\mathfrak{q}_j}$ ($1 \leq j \leq r$). IV | 148

Proof. One can evidently restrict to the case where A is local, replacing A by $A_{\mathfrak{m}_j}$, where $\mathfrak{m}_j$ is the maximal ideal of A which is the inverse image of $\mathfrak{n}_j$. Set on the other hand $C = B_{\mathfrak{n}_j}$; by virtue of the definition of ${}^{hs}B$ (18.8.1), everything amounts to seeing that ${}^{hs}C$ is C -isomorphic to $(C \otimes_A ({}^{hs}A))_{\mathfrak{q}}$, where $\mathfrak{q}$ is one of the maximal ideals of $C \otimes_A ({}^{hs}A)$. Let k and K be the residue fields of A and C , Ω that of ${}^{hs}A$, which is by definition a separable closure of k ; since K is a finite extension of k , one can suppose K and Ω contained in the same algebraic closure of k ; it is then immediate that $K \otimes_k \Omega$ is a direct composition of a finite number of fields, all isomorphic to the separable closure $K\Omega$ of K . On the other hand, the ring ${}^{hs}A$ being henselian, $C \otimes_A ({}^{hs}A)$ is a direct composition of the local rings $(C \otimes_A ({}^{hs}A))_{\mathfrak{q}_i}$, where $\mathfrak{q}_i$ ($1 \leq i \leq s$) ranges over the maximal ideals of $C \otimes_A ({}^{hs}A)$, and these local rings are henselian (18.5.10) and have $K\Omega$ as residue field, hence are strictly local (18.8.1). Set for short $C' = C \otimes_A ({}^{hs}A)$ and $C'_i = C'_{\mathfrak{q}_i}$, which is thus a quotient ring of C' ; let $f : C \rightarrow C'$, $g : {}^{hs}A \rightarrow C'$, $\varphi_i : C' \rightarrow C'_i$ be the canonical maps.

Let $v : C \rightarrow {}^{hs}C$ be the canonical homomorphism; there exists a unique local homomorphism $u : {}^{hs}A \rightarrow {}^{hs}C$ which, by passage to quotients, gives the canonical injection $\Omega \rightarrow K\Omega$ and such that the composed $A \rightarrow C \xrightarrow{v} {}^{hs}C$ is equal to the composed $A \rightarrow {}^{hs}A \xrightarrow{u} {}^{hs}C$ (18.8.8); hence there exists a unique local homomorphism $w_0 : C' \rightarrow {}^{hs}C$ such that $w_0 \circ g = u$, $w_0 \circ f = v$, and since ${}^{hs}C$ is a local ring, this homomorphism factorises as $C' \xrightarrow{\varphi_i} C'_i \xrightarrow{w} {}^{hs}C$ for a well-determined index i , w being a local homomorphism. On the other hand, the ring C'_i being strictly local, there exists a local homomorphism $w' : {}^{hs}C \rightarrow C'_i$ which, by passage to quotients, gives the identity automorphism of $K\Omega$ and which is such that $w' \circ v = \varphi_i \circ f$ (18.8.8). One deduces first from this that $w \circ w'$ is an endomorphism of ${}^{hs}C$ which, by passage to quotients, gives the identity automorphism of $K\Omega$, hence is the identity (18.8.8). On the other hand, one has $w' \circ w \circ \varphi_i \circ f = w' \circ v = \varphi_i \circ f$, and $w' \circ w \circ \varphi_i \circ g = w' \circ u = \varphi_i \circ g$; whence $w' \circ w \circ \varphi_i = \varphi_i$, in other words $w' \circ w$ is the identity automorphism of C'_i . $\square$

Remark (18.8.11). — The proof of (18.8.10) uses the fact that B is *integral* over A , but only uses the fact that B is a *finite* A -algebra to establish that $K \otimes_k \Omega$ is a direct composition of a finite number of IV | 149

fields. Now, this latter property is still verified when one supposes that B is an *integral semi-local* A -algebra whose maximal ideals $\mathfrak{n}_j$ have residue fields of finite separable degrees $[k(\mathfrak{n}_j) : k]_s$.

Proposition (18.8.12). — *Let A be a semi-local ring.*

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- label=(i) For ${}^{\text{hs}}A$ to be reduced (resp. normal), it is necessary and sufficient that A be so.*
lbel=(ii) Suppose A Noetherian. Then, for every prime ideal $\mathfrak{p}$ of A , the ring $({}^{\text{hs}}A)_{\mathfrak{p}}/\mathfrak{p}({}^{\text{hs}}A)_{\mathfrak{p}}$ is a direct composition of a finite number of separable algebraic extensions of $k(\mathfrak{p})$ (which implies that the fibres of the canonical morphism $\text{Spec}({}^{\text{hs}}A) \rightarrow \text{Spec}(A)$ are geometrically regular and are discrete spaces).

Proof. The proofs are the same as those of (18.6.9) and (18.6.10) respectively. $\square$

Corollary (18.8.13). — *Let A be a Noetherian local ring. For ${}^{\text{hs}}A$ to possess one of the properties:*

- a) to be a Cohen–Macaulay ring (0, 16.5.3);*
- b) to verify the property (S_n) (5.7.3);*
- c) to be regular;*
- d) to verify the property (R_n) (5.8.2);*

it is necessary and sufficient that A possess this same property.

Proof. This follows from (18.8.12), (ii) and from (6.4.1), (6.5.1) and (6.5.3). $\square$

Corollary (18.8.14). — *Let A be a semi-local Noetherian ring. For a prime ideal $\mathfrak{p}$ of ${}^{\text{hs}}A$ to belong to $\text{Ass}({}^{\text{hs}}A)$, it is necessary and sufficient that $\mathfrak{p} \cap A$ belong to $\text{Ass}(A)$.*

Proof. This follows from the description of the fibres of $\text{Spec}({}^{\text{hs}}A) \rightarrow \text{Spec}(A)$ (18.8.12), (ii) and from (3.3.1). $\square$

Proposition (18.8.15). — *Let A be a local ring; the following properties are equivalent:*

- a) A is geometrically unibranch (resp. reduced and geometrically unibranch).*
- b) For every local A -algebra essentially étale A' , $\text{Spec}(A')$ is irreducible (resp. integre).*
- c) $\text{Spec}({}^{\text{hs}}A)$ is irreducible (resp. integre).*

Proof. One reduces as in (18.6.12) to the case where A is reduced, using the relation ${}^{\text{hs}}(A_{\text{red}}) = ({}^{\text{hs}}A) \otimes_A A_{\text{red}}$ (18.8.11), and the equivalence of b) and c) is proved as in (18.6.12); it is likewise for the fact that a) implies c), taking into account (18.8.11) and the definition of a geometrically unibranch integre local ring. Finally, to show that c) implies a), we keep the same notations as in (18.6.12); it remains to see that ${}^{\text{hs}}B$ is integre. But, by (18.8.10), ${}^{\text{hs}}B$ is a localised ring of the semi-local ring $B \otimes_A ({}^{\text{hs}}A)$, which identifies by flatness with a subring of the field L , hence is integre; it is therefore likewise for ${}^{\text{hs}}B$, which concludes the demonstration. $\square$

Corollary (18.8.16). — *Let A be a Henselian local ring (resp. strictly local). For A to be unibranch (resp. geometrically unibranch), it is necessary and sufficient that $\text{Spec}(A)$ be irreducible.*

Proof. This follows immediately from (18.8.15) and the definitions, since if A is Henselian (resp. strictly local), A is itself a local A -algebra essentially étale, and ${}^{\text{hs}}A$ is obtained from the henselisation hA (resp. from A itself). $\square$

Proposition (18.8.17). — *Let A be a Noetherian local ring. If A is universally catenaire, so is ${}^{\text{hs}}A$.*

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Proof. The proof is the same as that of (18.7.5). $\square$

Proposition (18.8.18). — *label=(i) Every filtering inductive limit of strictly local rings (where the transition homomorphisms are local) is a strictly local ring.*

lbel=(ii) The functor $A \mapsto {}^{\text{hs}}A$ in the category of local rings (where the morphisms are local homomorphisms) commutes with filtering inductive limits.

lbel=(iii) Every strictly local ring A is the inductive limit of an inductive system filtering of strictly local and Noetherian rings (the transition homomorphisms being local).

The proofs are modeled on those of (18.6.14).

18.9. Formal fibres of Noetherian Henselian rings.

Theorem (18.9.1). — *Let A be a Noetherian local Henselian ring, whose formal fibres are geometrically normal. Then the formal fibres of A are geometrically intègre (hence geometrically connected).*

Proof. The two assertions of the statement are in fact equivalent, a Noetherian connected and normal prescheme locally being intègre. As every finite A -algebra intègre is a local ring, hence the henselisation is hB (18.6.8), it results from (7.3.16.2) that one is reduced to proving that, if one supposes moreover A intègre, then the fibre of the morphism $\mathrm{Spec}(\hat{A}) \rightarrow \mathrm{Spec}(A)$ at the generic point is intègre, which results from the hypothesis. $\square$

Corollary (18.9.2). — *Let A be a Noetherian local Henselian and intègre ring, whose formal fibres are geometrically normal. Then $\hat{A}$ is intègre.*

Proof. This follows from (18.9.1): the formal fibres of A are geometrically intègre, and in particular the fibre at the generic point η of $\mathrm{Spec}(A)$ is geometrically intègre, hence $\hat{A} \otimes_A k(\eta)$ is intègre, which implies that $\hat{A}$ is intègre. $\square$

Corollary (18.9.3). — *Under the hypotheses of (18.9.2), the field of fractions L of $\hat{A}$ is a separable extension of the field of fractions K of A , and K is algebraically closed in L .*

Proof. The formal fibre at the generic point of $\mathrm{Spec}(A)$ is $\mathrm{Spec}(\hat{A} \otimes_A K)$; by (18.9.1), this is geometrically intègre, which means that $\hat{A} \otimes_A K \cong L$ is geometrically regular over K . It follows ((4.6.1) and (4.3.2)) that L is a separable extension of K and that K is algebraically closed in L . $\square$

Corollary (18.9.4). — *Let A be a Noetherian local ring, Henselian and whose formal fibres are geometrically normal (for example a Noetherian local Henselian and excellent ring), and set $A' = \hat{A}$. Let X be an A -prescheme, $X' = X \otimes_A A'$, $g : X' \rightarrow X$ the canonical projection. Then the application $U \mapsto g^{-1}(U)$ is a bijection of the set of open and closed parts of X to the set of open and closed parts of X' .*

Proof. Since g is a faithfully flat and quasi-compact morphism, one knows ((2.3.12)) that the topology of X is quotient of that of X' by the equivalence relation defined by f . It therefore amounts to seeing that every open and closed part U' of X' is *saturated* for this relation. But, since the morphism $\mathrm{Spec}(A') \rightarrow \mathrm{Spec}(A)$ has geometrically connected fibres (18.9.1), the fibres of g are connected, whence our assertion immediately.

In particular, if X is locally connected (which will be the case if X is locally Noetherian), then for every connected component U of X (which is open and closed in X), $g^{-1}(U)$ is connected (Bourbaki, *Top. gén.*, chap. I^{er}, 3^e éd., § 11, n^o 3, prop. 7). If one designates by $\pi_0(X)$ the set of connected components of X , the application $\pi_0(X') \rightarrow \pi_0(X)$ deduced canonically from g is therefore bijective. $\square$

Corollary (18.9.5). — *Under the hypotheses of (18.9.4), the functor*

$$(18.9.5.1) \quad Z \longmapsto Z \otimes_A A'$$

from the category of étale preschemes over X , to that of étale preschemes over X' , is fully faithful.

Proof. Let Z_1, Z_2 be two étale preschemes over X , and set $Z'_i = Z_i \otimes_A A'$ ($i = 1, 2$); it is a matter of proving that every X' -morphism $Z'_1 \rightarrow Z'_2$ comes, by base change, from a unique X -morphism $Z_1 \rightarrow Z_2$. Suppose first that Z_2 is *separated* over X ; then, since $\mathrm{Hom}_X(Z_1, Z_2)$ identifies with $\Gamma((Z_1 \times_X Z_2)/Z_1)$, and $Z_1 \times_X Z_2$ is étale and separated over Z_1 , $\mathrm{Hom}_X(Z_1, Z_2)$ identifies functorially, by virtue of (17.9.3), with the set of open and closed parts U of $Z_1 \times_X Z_2$ such that the restriction to U of the projection $p : Z_1 \times_X Z_2 \rightarrow Z_1$ is a surjective and radiciel morphism. The assertion therefore follows in this case from (18.9.4) and from the fact that $Z'_1 \times_{X'} Z'_2 = (Z_1 \times_X Z_2) \otimes_A A'$.

Let us pass to the general case: it will suffice to prove that the graph Γ' of an X' -morphism $Z'_1 \rightarrow Z'_2$, which is *open* (17.9.3) in $Z'_1 \times_{X'} Z'_2$, is of the form $\Gamma \otimes_A A'$, where Γ is induced on an open part of $Z = Z_1 \times_X Z_2$. Indeed, the restriction to Γ of the projection $p : Z \rightarrow Z_1$ is then an isomorphism, for by base change $A \rightarrow A'$ this restriction becomes the isomorphism restriction to Γ' of the projection $p' : Z' \rightarrow Z'_1$ (with $Z' = Z \otimes_A A'$), and it suffices to apply ((2.7.1), (viii)); the conclusion then results from the characterisation of graphs of morphisms (I, 5.3).

If $q : Z' \rightarrow Z$ is the canonical projection, to prove that Γ' is of the form $q^{-1}(\Gamma)$, where Γ is open in Z , it suffices, since q is a faithfully flat and quasi-compact morphism, to apply descent ((2.3.12)). Set $Z'' = Z' \times_Z Z' = Z \times_X X''$, where $X'' = X' \times_X X'$, and let $q_1 : Z'' \rightarrow Z'$, $q_2 : Z'' \rightarrow Z'$ be the canonical projections. Applying ((4.5.19.1)), it suffices to show that $q_1^{-1}(\Gamma') = q_2^{-1}(\Gamma')$. But this is a property that is true if and only if it is so after each base change $\text{Spec}(k(x)) \rightarrow X$, where x ranges over X . In other words, it suffices to prove the corollary when X is the spectrum of a field; but since every X -prescheme étale is then automatically separated over X ((17.6.2), c')), one is reduced to the case treated at the beginning of the demonstration. $\square$

Remarks (18.9.6). — (i) It is possible that, if the residue field of A is of characteristic 0, the functor (18.9.5.1) induces even an *equivalence* of the category of étale coverings of X and of the category of étale coverings of X' . One can in any case demonstrate this when A is *excellent*, by using the resolution of singularities of Hironaka (M. Artin).

(ii) One can also ask whether the formal fibres of A are geometrically connected (or even geometrically irreducible) when A is any Henselian local ring. It would suffice for this that, for every Noetherian local ring, Henselian and *intègre* A , $\text{Spec}(\widehat{A})$ be irreducible. This is not known in general.

(iii) One can also ask whether, when A is an excellent local ring, its strict henselisation ${}^{hs}A$ is excellent.

Theorem (18.9.7). — Let A, B be two Noetherian local rings, $\rho : A \rightarrow B$ a local homomorphism, $f : \text{Spec}(B) = X \rightarrow \text{Spec}(A) = Y$ the corresponding morphism. Suppose the following conditions verified: IV | 152

- label=(i) f is flat and all the fibres $X_y = f^{-1}(y)$ ($y \in Y$) are geometrically reduced (4.6.2).
 lbbel=(ii) Either the ring A is geometrically unibranch (0, 23.2.1), or else the ring A is unibranch and $k(B)$ is a primary extension ((4.3.1)) of $k(A)$.

Then, if η is the generic point of Y , X_η is connected, and for every closed part T of X such that $f(T)$ is rare in Y , $X - T$ is connected.

Proof. The two conclusions stated are in fact equivalent; indeed, the hypothesis that $f(T)$ is rare entails that $\overline{f(T)}$ does not contain η ; on the other hand, as $\{\eta\}$ is proconstructible (1.9.6) and f is flat, X_η is dense in X (2.3.10), and hence if X_η is connected, the same is true of $X - T$ which contains it. Conversely, when U ranges over the affine open neighbourhoods of η in Y , X_η is the projective limit of the subschemes induced on the open sets $f^{-1}(U)$ of X (8.1.2, a)); if one supposes that $X - T$ is connected for every closed subset T of X such that $f(T)$ is rare in Y , the $f^{-1}(U)$ are connected, and hence the same is true of X_η (8.4.1).

Let us first show that one can reduce to the case where A is intègre and $Y = \text{Spec}(A)$ geometrically unibranch (6.15.1) (which implies that A is geometrically unibranch, but is not equivalent to this condition (6.15.2)). It is clear that one can first suppose Y reduced, by considering the morphism $X \times_Y Y_{\text{red}} \rightarrow Y_{\text{red}}$, deduced from f by base change, which is flat and has the same fibres as f . One can therefore suppose A intègre and unibranch; if K is the field of fractions of A , there then exists a finite sub- A -algebra A'' of K such that, if A' is the integral closure of A , the morphism $\text{Spec}(A') \rightarrow \text{Spec}(A'')$ is *radiciel* (0, 23.2.5); one concludes (6.15.5) that $\text{Spec}(A'')$ is *geometrically unibranch*. As A is unibranch, A' is a local ring, and hence so is A'' ; let us show that the ring $B'' = B \otimes_A A''$ is also a local ring. Indeed, B'' is a finite B -algebra, hence a semi-local Noetherian ring; if k'' is the residue field of A'' , the maximal ideals of B'' are the points of $\text{Spec}(B'')$ above the closed point of $\text{Spec}(B)$, hence the points of $\text{Spec}(k(B) \otimes_{k(A)} k'')$ (I, 3.4.9), since the closed point of $\text{Spec}(A'')$ is the only point above the closed point of $\text{Spec}(A)$. Now, when A is geometrically unibranch, k'' is a radiciel extension of $k(A)$, so $\text{Spec}(k(B) \otimes_{k(A)} k'')$ is radiciel over $\text{Spec}(k(B))$ and therefore consists of a single point. When A is unibranch and $k(B)$ is a primary extension of $k(A)$, $\text{Spec}(k(B) \otimes_{k(A)} k'')$ is irreducible (4.3.2) and, being a finite discrete space (I, 6.4.4), is again reduced to a single point; this shows that in both cases considered, B'' is *local*. Setting $Y'' = \text{Spec}(A'')$, $X'' = \text{Spec}(B'') = X \times_Y Y''$, the morphism $f'' = f_{(Y'')} : X'' \rightarrow Y''$ is flat and has geometrically reduced fibres; moreover, if η'' is the generic point of Y'' , $k(\eta'') = k(\eta) = K$ by definition, hence the fibres X_η and $X''_{\eta''}$ are isomorphic. It therefore suffices to prove that $X''_{\eta''}$ is connected. IV | 153

(18.9.7.1) *Remark.*—When the ring A is *Japanese*, one can, in the above, take $A'' = A'$ by definition (0, 23.1.1); one sees therefore that in this case one is even reduced to proving the theorem when A is *integrally closed*.

(18.9.7.2) Suppose therefore henceforth that Y is *integre* and *geometrically unibranch*. Note that if T is a closed subset of X such that $f(T)$ is rare in Y , T contains no maximal point of X since f is flat (2.3.4), hence is rare in X ; as X is a local scheme, hence connected, to prove that $X - T$ is connected, it suffices, by virtue of (15.5.6.1), to show that for every $x \in X$ such that $f(x) \neq \eta$, $\text{Spec}(\mathcal{O}_{X,x}) - \{x\}$ is connected. Setting $y = f(x)$, $A_1 = \mathcal{O}_{Y,y}$, $B_1 = \mathcal{O}_{X,x}$, $Y_1 = \text{Spec}(A_1)$, $X_1 = \text{Spec}(B_1)$, and letting $f_1 : X_1 \rightarrow Y_1$ be the morphism corresponding to the local homomorphism $A_1 \rightarrow B_1$ deduced from ρ ; it is clear that f_1 is flat, and it follows from (4.6.1) that its fibres are geometrically reduced; moreover, A_1 is *integre* and *geometrically unibranch* (6.15.1) and $\dim(A_1) \geq 1$. We are thus reduced to proving the following lemma: $\square$

Lemma (18.9.7.3). — *Let A, B be two local Noetherian rings, $Y = \text{Spec}(A)$, $X = \text{Spec}(B)$, $\rho : A \rightarrow B$ a local homomorphism, $f : X \rightarrow Y$ the corresponding morphism. Suppose the following conditions verified:*

label=(i) f is flat and all the fibres X_y ($y \in Y$) are geometrically reduced.

*lbel=(ii) A is *integre*, *geometrically unibranch*, and $\dim(A) \geq 1$.*

Then, if b is the closed point of X , $X - \{b\}$ is connected.

Proof of Lemma (18.9.7.3). Note first that, by virtue of the connectedness theorem of Hartshorne (5.10.7), $X - \{b\}$ is connected if $\text{prof}(B) \geq 2$. Now by (6.3.1) we have

$$(18.9.7.4) \quad \text{prof}(B) = \text{prof}(A) + \text{prof}(B \otimes_A k)$$

where k denotes the residue field of A . On the other hand, since A is *integre* and $\dim(A) \geq 1$, we have $\text{prof}(A) \geq 1$, hence $\text{prof}(B) \geq 2$ *except* when $\text{prof}(B \otimes_A k) = 0$; but $B \otimes_A k$ is reduced by (i), so the relation $\text{prof}(B \otimes_A k) = 0$ signifies that $B \otimes_A k$ is a *field* (0, 16.4.7). We shall henceforth assume that this last condition is satisfied. We begin by treating a case where the proof is very simple.

Case A: A is integrally closed. If $\dim(A) \geq 2$, we have $\text{prof}(A) \geq 2$ (0, 16.5.1), hence $\text{prof}(B) \geq 2$. It remains only to consider the case where $\dim(A) = 1$ and $B \otimes_A k$ is a field; then A is a discrete valuation ring, and since f is flat, B is regular (0, 17.3.3); moreover, by (6.1.1.1), $\dim(B) = \dim(A) + \dim(B \otimes_A k) = \dim(A) = 1$, so B is also a discrete valuation ring. Hence $X - \{b\}$ is reduced to a single point, and the lemma (18.9.7.3) is proved in this case.

One will observe that this proves the statement of the theorem (18.9.7) when one supposes in addition that the ring A is *Japanese*, since by virtue of the remark (18.9.7.1), one is reduced to the case where A is *integrally closed*, and in the reduction preceding (18.9.7.3), the local ring $A_1 = \mathcal{O}_{Y,y}$ is itself *integrally closed*, so one can apply the result just proved. $\square$

Lemma (18.9.7.5). — *Let A be a semi-local Noetherian reduced ring, A' its integral closure in its total ring of fractions, $X = \text{Spec}(A)$, $X' = \text{Spec}(A')$; let S be the set of closed points of X and $U = X - S$. Suppose that for every point $x' \in X'$ above a point of S , we have $\dim(\mathcal{O}_{X',x'}) \geq 2$; then $A^{(1)} = \Gamma(U, \mathcal{O}_X)$ is integral over A and is isomorphic to a sub- A -algebra of A' .*

Proof of Lemma (18.9.7.5). Recall that if $\mathfrak{p}_i$ ($1 \leq i \leq m$) are the minimal ideals of A , A' is the direct product of the integral closures A'_i of the *integre* rings $A_i = A/\mathfrak{p}_i$, so that X' is the sum of the schemes $X'_i = \text{Spec}(A'_i)$. If X_0 is the sum of the schemes $X_i = \text{Spec}(A_i)$ and U_0 the inverse image of U in X_0 , the canonical homomorphism $\Gamma(U, \mathcal{O}_X) \rightarrow \Gamma(U_0, \mathcal{O}_{X_0})$ is injective since X is reduced and the morphism $X_0 \rightarrow X$ (and hence also $U_0 \rightarrow U$) is surjective. As $\Gamma(U_0, \mathcal{O}_{X_0})$ is the direct product of the $\Gamma(U_i, \mathcal{O}_{X_i})$ ($1 \leq i \leq m$), where U_i is the inverse image of U in X_i , it suffices to prove that for each i , $\Gamma(U_i, \mathcal{O}_{X_i})$ is isomorphic to a sub- A_i -algebra of A'_i . In other words, we are reduced to the case where A is a semi-local Noetherian *integre* ring. As X is reduced and the morphism $X' \rightarrow X$ surjective, the homomorphism $\Gamma(U, \mathcal{O}_X) \rightarrow \Gamma(U', \mathcal{O}_{X'})$ (where U' is the inverse image of U in X') is injective, and it suffices to show that the canonical homomorphism $A' = \Gamma(X', \mathcal{O}_{X'}) \rightarrow \Gamma(U', \mathcal{O}_{X'})$ is *bijective*. Now (I, 8.2.1.1), $\Gamma(U', \mathcal{O}_{X'})$ is the intersection of the local rings $A'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over U' , and by hypothesis, among these local rings feature all those for which $\mathfrak{p}'$ is of height 1. But the reasoning of (0, 23.2.7) applies also to a *semi-local* *integre* Noetherian ring, and A' is therefore the intersection of the local rings $A'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over the set of prime ideals of height 1 of A'

(Bourbaki, *Alg. comm.*, chap. VII, §1, n° 6, th. 4); *a fortiori*, A' is the intersection of the A'_p , for $p' \in U'$, which finishes the proof of the lemma. $\square$

Remark (18.9.7.6). — We know (0, 23.2.5) that there exists a finite sub- A -algebra A'' of the field of fractions K of A such that the morphism $\text{Spec}(A') \rightarrow \text{Spec}(A'')$ is *radiciel*; as this morphism is also integral and dominant, hence surjective (II, 6.1.10), it is a homeomorphism (2.4.5). By virtue of the geometric interpretation of $\dim(\mathcal{O}_{X',x'})$ (5.1.2), one has, for every point x'' of $X'' = \text{Spec}(A'')$ above a point $x \in X$, $\dim(\mathcal{O}_{X'',x''}) = \dim(\mathcal{O}_{X',x'})$ if x' is the unique point of X' above x . This being the case, it follows first from (0, 16.1.5) and (0, 16.1.4.1) that the relation $\dim(\mathcal{O}_{X',x'}) \geq 2$ implies $\dim(\mathcal{O}_{X'',x''}) \geq 2$. Conversely, suppose that $\dim(\mathcal{O}_{X,x}) \geq 2$ and that A is universally catenaire (5.6.2); then the same is true of $\mathcal{O}_{X,x}$ (5.6.3), hence $\dim(\mathcal{O}_{X'',x''}) = \dim(\mathcal{O}_{X,x})$ by (5.6.10) and we therefore have $\dim(\mathcal{O}_{X',x'}) \geq 2$. The same conclusion is valid if $\mathcal{O}_{X,x}$ is *geometrically unibranch*, since the fibre of the morphism $X' \rightarrow X$ at the point x is then reduced to a single point, and $\mathcal{O}_{X',x'}$ is an entire $\mathcal{O}_{X,x}$ -algebra; it suffices to apply (0, 16.1.5).

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Lemma (18.9.7.7). — Let A be a local Noetherian ring, geometrically unibranch, integre, and of dimension ≥ 2 ; set $Y = \text{Spec}(A)$, y the closed point of Y . Let X be a locally Noetherian connected prescheme, $f : X \rightarrow Y$ a flat morphism. Then, for every closed subset $T \subset f^{-1}(y)$, $X - T$ is connected.

Proof of Lemma (18.9.7.7). The same reasoning as at the beginning of (18.9.7.2) reduces to showing that for every point $x \in f^{-1}(y)$, $\text{Spec}(\mathcal{O}_{X,x}) - \{x\}$ is connected. One can therefore confine oneself to the case where $X = \text{Spec}(B)$, B being a local Noetherian ring and the homomorphism $A \rightarrow B$ corresponding to f being local; set $U = Y - \{y\}$, and note that $V = f^{-1}(U)$ is dense in X (2.3.10); it therefore suffices to prove that V is *connected*. Keeping the notations of (18.9.7.5), set $A^{(1)} = \Gamma(V, \mathcal{O}_X)$, $Y_1 = \text{Spec}(A^{(1)})$, $X_1 = X \times_Y Y_1 = \text{Spec}(B \otimes_A A^{(1)})$, so that one has the commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{\alpha_1} & X_1 \\ f \downarrow & & \downarrow f_1 \\ Y & \xleftarrow{g} & Y_1 \end{array}$$

By virtue of (2.3.1) and the definition of $A^{(1)}$, one has a canonical isomorphism $\Gamma(V, \mathcal{O}_X) \xrightarrow{\sim} B \otimes_A A^{(1)}$, and it therefore suffices to prove that this last ring is *local*, such a ring not being able to be the product of two non-zero rings (III, 7.8.6.1). Now by (18.9.7.5), $A^{(1)}$ is a sub- A -algebra of the integral closure A' of A . The hypothesis that A is geometrically unibranch entails that the morphism $\text{Spec}(A') \rightarrow \text{Spec}(A)$ is *radiciel* at the point y (6.15.3); *a fortiori* $g : Y_1 \rightarrow Y$ is *radiciel* at the point y ; as moreover g is an integral morphism, the closed points of X_1 lie above x , hence there exists only one closed point of X_1 , which finishes the proof of (18.9.7.7). $\square$

Proof of Lemma (18.9.7.3), continued. Case B: $\dim(A) \geq 2$. It is clear that (18.9.7.7) proves (18.9.7.3) when $\dim(A) \geq 2$ (even without supposing the fibres X_y geometrically reduced).

Case C: $\dim(A) = 1$. As we saw at the beginning of the proof of (18.9.7.3), we can suppose $\dim(B \otimes_A k) = 0$, and by flatness (6.1.1.1), we have

$$\dim(B) = \dim(A) + \dim(B \otimes_A k) = 1.$$

Since A is integre, Y consists of two points, the generic point η and the closed point a . Moreover, since f is flat, every irreducible component of X dominates Y (2.3.4), so $X - \{b\}$, which is the set of maximal points ξ_i of X , is equal to the underlying set of $f^{-1}(\eta)$, and the fibre $f^{-1}(\eta)$ is the sum of the $\text{Spec}(k(\xi_i))$. The hypothesis on the X_y and the fact that these fibres are of dimension 0 show that they are *geometrically regular* (6.7.6), and *a fortiori* geometrically normal; in other words, f is a *normal morphism*. Since A is integre and geometrically unibranch, it follows from (6.15.10) that B is integre and geometrically unibranch, hence $f^{-1}(\eta) = X - \{b\}$ is reduced to a single point, and *a fortiori* connected; this finishes the proof of (18.9.7.3) and of (18.9.7). $\square$

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Remark (18.9.7.8). — In Case C of the proof of (18.9.7.3), one can avoid making use of the delicate result (6.15.10) by reasoning as follows: since A is integre and of dimension 1, it follows from the theorem of Krull–Akizuki (Bourbaki, *Alg. comm.*, chap. VII, §2, n° 5, prop. 5) that its integral closure

A' is a Noetherian ring. The same reasoning as at the beginning of the proof of (18.9.7) then shows that $B' = B \otimes_A A'$ is a local ring; moreover, by flatness, B' is contained in the total ring of fractions R of the reduced ring B (3.3.5). But the reasoning which proves the theorem of Krull–Akizuki (Bourbaki, *loc. cit.*) also applies to a local Noetherian *reduced* ring of dimension 1, and shows that for such a ring B , every ring between B and its total ring of fractions is *Noetherian*. The ring B' , being a local Noetherian ring, and the morphism $\text{Spec}(B') \rightarrow \text{Spec}(A')$ being a normal morphism, B' is an *intègre* and *integrally closed* ring (6.5.4), and one concludes as at the beginning of the proof of (18.9.7).

Corollary (18.9.8). — *With the notations of (18.9.7), suppose verified the condition (i) of (18.9.7) and one of* IV | 156 *the two following conditions:*

(ii') *A is a strictly local ring (18.8.2).*

(ii'') *A is a Henselian local ring and $k(B)$ is a primary extension of $k(A)$.*

Then all the fibres X_y ($y \in Y$) are geometrically connected.

Proof. Let $\mathfrak{p}$ be the prime ideal of A corresponding to the point y , and set $A' = A/\mathfrak{p}$ and $B' = B \otimes_A A' = B/\mathfrak{p}B$, which are evidently Noetherian local rings; it is clear that the morphism $f' : \text{Spec}(B') \rightarrow \text{Spec}(A')$ verifies the condition (i) of (18.9.7); as y is the generic point of $\text{Spec}(A')$, $k(A') = k(A)$, $k(B') = k(B)$, one sees that it suffices to verify that, in case (ii'), A' is geometrically unibranch (resp. unibranch in case (ii'')). But A' is *intègre*, and in case (ii') (resp. (ii'')) it is strictly local (resp. Henselian) by virtue of (18.5.10). One concludes that A' is geometrically unibranch (resp. unibranch) by virtue of (18.8.15) (resp. (18.6.12)). $\square$

Corollary (18.9.9). — *Let A be a Noetherian local Henselian ring, universally Japanese (i.e. (7.6.4) and (7.7.2)), whose formal fibres (7.3.13) are geometrically reduced; then the formal fibres are also geometrically connected.*

Proof. It suffices to apply (18.9.8) to the case where $B = \widehat{A}$, since B is a flat A -module and $k(B) = k(A)$. $\square$

Remark (18.9.10). — The proofs of (18.9.4) and (18.9.5) do not use the fact that the formal fibres of A are geometrically connected. The conclusions of these two propositions are hence still valid when one supposes A a Noetherian local Henselian and universally Japanese ring.

Corollary (18.9.11). — *Let X, Y be two locally Noetherian preschemes, Y being supposed geometrically unibranch and X connected. Let $f : X \rightarrow Y$ be a reduced morphism (i.e. (6.8.1)) flat and with geometrically reduced fibres. Then, for every closed part T of X such that $f(T)$ is rare in Y , $X - T$ is connected.* IV | 157

Proof. As f is flat and $f(T)$, being rare, cannot contain a maximal point of Y , T cannot contain a maximal point of X (2.3.4), hence is rare in X . Using (15.5.6.1), it suffices to show that, for every $x \in T$, $\text{Spec}(\mathcal{O}_{X,x}) \setminus \{x\}$ is connected. Set $y = f(x)$, which is not a maximal point of Y . As the ring $\mathcal{O}_{Y,y}$ is geometrically unibranch, we can apply (18.9.7) to the morphism $f_1 : \text{Spec}(\mathcal{O}_{X,x}) \rightarrow \text{Spec}(\mathcal{O}_{Y,y})$, which verifies conditions (i) and (ii) of that theorem, and to the closed set $\{x\}$ of $\text{Spec}(\mathcal{O}_{X,x})$, since $f_1(\{x\}) = \{y\}$ is rare in $\text{Spec}(\mathcal{O}_{Y,y})$. $\square$

18.10. Étale preschemes over a geometrically unibranch or normal prescheme.

Theorem (18.10.1). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, x a point of X , $y = f(x)$. Suppose that Y is *intègre* and geometrically unibranch at the point y . Then, for f to be étale at the point x , it is necessary and sufficient that the two following conditions be verified:* IV | 157

1° *f is formally unramified at the point x ;*

2° *the homomorphism $\mathcal{O}_{Y,y} \rightarrow \mathcal{O}_{X,x}$ is injective.*

*Moreover, if this is so, X is *intègre* and geometrically unibranch at the point x .*

Finally, if $\mathcal{O}_{Y,y}$ is Noetherian, one can replace condition (ii) by the condition

(ii bis) $\dim \mathcal{O}_{Y,y} \leq \dim \mathcal{O}_{X,x}$.

Proof. The fact that, if f is étale, X is *intègre* and geometrically unibranch at the point x is a particular case of (17.5.7). It is clear on the other hand that conditions 1° and 2° are necessary for f to be

étale at the point x , since $\mathcal{O}_{X,x}$ is then a faithfully flat $\mathcal{O}_{Y,y}$ -module (17.6.1). Let us prove that these conditions are sufficient.

Set $A = \mathcal{O}_{Y,y}$, $A' = {}^h A$ (18.8.7), $Y' = \operatorname{Spec}(A')$, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$. Let y' be the closed point of Y' , x' a point of X' above x and y' ; since the morphism $g : Y' \rightarrow Y$ is flat (18.8.8), it suffices to show that f' is étale at the point x' ((17.7.1), (ii)). Hypothesis 2° implies that the morphism $\operatorname{Spec}(\mathcal{O}_{X,x}) \rightarrow \operatorname{Spec}(\mathcal{O}_{Y,y})$ is dominant (I, 1.2.7); as $\mathcal{O}_{Y,y}$ is intègre, $\operatorname{Spec}(\mathcal{O}_{Y,y})$ has a single generic point y_1 , and so there is a generisation x_1 of x in X above y_1 . The projection morphism $X' \rightarrow X$ being flat, there exists a generisation x'_1 of x' above x_1 (2.3.4); set $y'_1 = f'(x'_1)$; one therefore has $g(y'_1) = y_1$. Now, the hypothesis on y implies that A' is intègre (18.8.15); let η be the generic point of Y' ; η is also the generic point of $g^{-1}(y_1)$ (0I, 2.1.8), and one knows moreover that the fibres of g are discrete spaces (18.8.12), hence $g^{-1}(y_1) = \{\eta\}$, and thus $y'_1 = \eta$. The morphism $\operatorname{Spec}(\mathcal{O}_{X',x'}) \rightarrow \operatorname{Spec}(\mathcal{O}_{Y',y'})$ restriction of f' is therefore dominant, and since $\mathcal{O}_{Y',y'} = A'$ is intègre, the corresponding homomorphism $\mathcal{O}_{Y',y'} \rightarrow \mathcal{O}_{X',x'}$ is injective (I, 1.2.7). On the other hand, f' is unramified at the point x' , hence (by (18.8.3), d)), there is a neighbourhood U of x' in X' such that $f'|_U$ is a closed immersion into Y' ; *a fortiori* the homomorphism $\mathcal{O}_{Y',y'} \rightarrow \mathcal{O}_{X',x'}$ is surjective, and since it is injective, it is bijective. But then $\mathcal{O}_{X',x'}$ is a flat $\mathcal{O}_{Y',y'}$ -module, and f' is therefore étale at the point x' (17.6.1).

Suppose moreover that Y is locally intègre at the point y (I, 2.1.8). Then conditions 1° and 2° above (together with the fact that Y is geometrically unibranch at the point y and f locally of finite type) already imply that f is étale at the point x . Indeed, this results from what precedes and from (18.4.13).

Finally, when $\mathcal{O}_{Y,y}$ is Noetherian, one knows that if f is étale at the point x , one has $\dim \mathcal{O}_{Y,y} = \dim \mathcal{O}_{X,x}$ (17.6.4). Conversely, suppose that conditions (i) and (ii bis) are verified, and let us show that they imply conditions 1° and 2°. Let $\mathfrak{J}$ be the kernel of the canonical homomorphism $\mathcal{O}_{Y,y} \rightarrow \mathcal{O}_{X,x}$; restricting Y to a neighbourhood of y if necessary, we can suppose that $\mathfrak{J} = \mathcal{J}_y$, where $\mathcal{J}$ is an Ideal of $\mathcal{O}_Y$; if Y_1 is the closed sub-prescheme of Y defined by $\mathcal{J}$, we can, by further restricting X and Y to neighbourhoods of x and y respectively, suppose that f factors as $X \xrightarrow{f_1} Y_1 \xrightarrow{h} Y$ (I, 6.5.1). It is clear that f_1 is still unramified at the point x , hence (17.4.1) quasi-finite at this point; the proof of (5.4.1), (i) then shows that $\dim(\mathcal{O}_{X,x}) \leq \dim(\mathcal{O}_{Y,y}/\mathfrak{J})$. But if $\mathfrak{J} \neq 0$, one would conclude that $\dim(\mathcal{O}_{Y,y}/\mathfrak{J}) < \dim(\mathcal{O}_{Y,y})$ since Y is intègre (0, 16.1.2.2); one would then have $\dim \mathcal{O}_{X,x} < \dim \mathcal{O}_{Y,y}$ contrary to the hypothesis, which proves 2°. $\square$

Remarks (18.10.2). — (i) When $A = \mathcal{O}_{Y,y}$ is Noetherian, and one knows that its completion $\hat{A}$ is intègre (for example if A is regular), one can, in the preceding proof, replace A' by $\hat{A}$. IV | 158

(ii) When Y is locally Noetherian, one can give of (18.10.1) a proof more rapid, not using the strict henselisation, but making intervene the rather delicate results of §§ 14 and 15.

(iii) With the notations and hypotheses on Y being those of (18.10.1), suppose only that f is *locally of finite type* (and not necessarily locally of finite presentation). Then, for $\mathcal{O}_{X,x}$ to be an $\mathcal{O}_{Y,y}$ -algebra *essentially étale* (18.6.1), it is necessary and sufficient that f verify the two following conditions:

- 1° f is formally unramified at the point x ;
- 2° the homomorphism $\mathcal{O}_{Y,y} \rightarrow \mathcal{O}_{X,x}$ is injective.

Corollary (18.10.3). — *Let Y be an intègre prescheme and geometrically unibranch, η its generic point, X a connected prescheme, $f : X \rightarrow Y$ a morphism locally of finite type and such that $f^{-1}(\eta)$ is non empty. Then, if f is formally unramified, f is étale, and X is intègre and geometrically unibranch.* IV | 159

Proof. It follows from the hypothesis and from (18.4.13) that f is étale at all the points where it is flat, and in particular at the points of $f^{-1}(\eta)$, since $\mathcal{O}_{Y,\eta} = k(\eta)$ is a field. The open set U of points of X where f is étale is therefore non empty. Let us show on the other hand that for every $x \in \bar{U}$, the homomorphism $\mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$ is injective. Indeed, let s be a section of $\mathcal{O}_Y$ above an open neighbourhood of $f(x)$, which we can always suppose to be Y itself, and suppose that its image $t \in \Gamma(Y, f_*(\mathcal{O}_X)) = \Gamma(X, \mathcal{O}_X)$ has a zero germ at the point x , hence vanishes in an open neighbourhood V of x ; then the restriction of t to the open set $U \cap V \neq \emptyset$ is zero; but the restriction $f|(U \cap V)$ is étale, hence $W = f(U \cap V)$ is open in Y and therefore contains the generic point η ; the homomorphism $\mathcal{O}_{Y,\eta} \rightarrow \mathcal{O}_{X,z}$ being injective for every point $z \in f^{-1}(\eta)$, the hypothesis that

$t|(U \cap V) = 0$ implies that $s_\eta = 0$, and as Y is intègre, this implies $s = 0$, hence our assertion. Applying now (18.10.2), (iii)), we conclude that f is flat at the point x , i.e. $x \in U$, and U is therefore both open and closed in X ; as it is non empty and X is connected, we have $U = X$.

Note now that since f is étale, hence locally quasi-finite and Y irreducible, the maximal points of X belong to $f^{-1}(\eta)$ (2.3.4) and for every $x \in X$, there is a neighbourhood of x containing only a finite number of these maximal points; in other words the set of irreducible components of X is locally finite. But by (18.10.1) X is intègre and geometrically unibranch at every point, hence every point of X belongs to only one irreducible component, and since the irreducible components form a locally finite set, they are open (and closed) in X ; since X is connected, it is intègre. $\square$

Corollary (18.10.4). — *Let $g : Y \rightarrow S$, $h : X \rightarrow S$ be two morphisms locally of finite presentation, $f : X \rightarrow Y$ an S -morphism, x a point of X , $y = f(x)$, $s = g(y) = h(x)$. Suppose that h is flat at the point x and $g^{-1}(s)$ normal at the point y . Then, for f to be étale at the point x , it is necessary and sufficient that f be unramified at the point x and that $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$.* IV | 160

Proof. The conditions are necessary by virtue of the fact that if f is étale at the point x , the same is true of $f_s : h^{-1}(s) \rightarrow g^{-1}(s)$, and the conclusion then follows from the fact that f_s is flat and quasi-finite at this point (6.1.2). Conversely, if the conditions of the statement are verified, to see that f is étale at the point x , it suffices, by virtue of (17.8.3), to see that f_s is. One can therefore restrict to the case where S is the spectrum of a field. It then follows from the hypothesis $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$ and from the fact that f is unramified at the point x that one has (by (5.2.3) and (17.4.1)) $\dim \mathcal{O}_{X,x} = \dim \mathcal{O}_{Y,y}$; since f is quasi-finite at the point x and $\mathcal{O}_{Y,y}$ intègre, one deduces from (0_I, 7.4.4) and (0, 16.3.10) that the homomorphism $\mathcal{O}_{Y,y} \rightarrow \mathcal{O}_{X,x}$ is injective; it then suffices to apply (18.10.1) to conclude. $\square$

Corollary (18.10.5). — *With the notations of (18.10.4), suppose that h is flat and that $g^{-1}(s)$ is normal for every $s \in S$ (which will be the case for example if g is smooth). For f to be a monomorphism and an open immersion, it is necessary and sufficient that f be unramified and that, for every $x \in X$, $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$, where $y = f(x)$ and $s = g(y) = h(x)$.*

Proof. We have only to prove the sufficiency of the conditions; now, it follows from (17.1.3) that a monomorphism locally of finite presentation is unramified, hence the hypotheses imply that f is étale (18.10.4); one then concludes by (17.9.1). $\square$

Lemma (18.10.6). — *Let $f : X \rightarrow Y$ be a morphism flat and locally quasi-finite (Err_{III}, 20).*

- label=(i) The set $\text{Max}(X)$ of maximal points of X is in canonical bijection with the set $\coprod_{y \in \text{Max}(Y)} f^{-1}(y)$.*
- lbbel=(ii) If Y is irreducible with generic point η , the set of irreducible components of X is in canonical bijection with $f^{-1}(\eta)$, and in particular is finite if and only if $f^{-1}(\eta)$ is finite.*
- lcbel=(iii) If the set of irreducible components of Y is locally finite, the same is true of that of X , and in particular X is locally connected.*

Proof. Assertion (i) follows immediately from (2.3.4) and (0_I, 2.1.6) and from the fact that the fibres $f^{-1}(y)$ are discrete sets. It is clear that (ii) is a particular case of (i). Finally, to prove (iii), we can, by virtue of (i), restrict to the case where f is quasi-finite and where Y is irreducible (0_I, 2.1.6) and the conclusion then follows from (ii) and from the fact that the fibres of f are finite sets (II, 6.2.2). $\square$

Proposition (18.10.7). — *Let Y be a prescheme geometrically unibranch and irreducible (resp. intègre), $f : X \rightarrow Y$ an étale morphism. Then X is isomorphic to a sum of preschemes (resp. intègre) X_λ , where λ ranges over the fibre $f^{-1}(\eta)$ of the generic point η of Y . The X_λ are geometrically unibranch; if Y is normal, the X_λ are normal.*

Proof. Indeed, X is geometrically unibranch (17.5.7), hence every point of X belongs to only one irreducible component; moreover, by virtue of (18.10.6) and (17.6.1) the set of irreducible components of X is locally finite, hence (0_I, 2.1.6) they are the connected components X_λ , and X is the sum of the X_λ ; by virtue of (17.5.7), if Y is intègre (resp. normal), the X_λ are intègre (resp. normal). $\square$

Proposition (18.10.8). — *Let Y be a normal and intègre prescheme, with generic point η , $K = k(\eta)$ its field of rational functions, $f : X \rightarrow Y$ an étale morphism. Suppose that $f^{-1}(\eta)$ is finite and non empty, so that the K -scheme $f^{-1}(\eta)$ is the spectrum of a finite separable K -algebra L , direct composition of a finite number of fields K_i ($1 \leq i \leq n$), finite separable extensions of K (17.6.2). Let Y' be the integral closure* IV | 161

of Y in L , isomorphic to the sum of the integral closures Y_i of Y in the K_i (II, 6.3.6). Then the morphism f factors in a unique way as $f : X \xrightarrow{j} Y' \xrightarrow{g} Y$, such that the restriction $f'_\eta : f^{-1}(\eta) \rightarrow g^{-1}(\eta)$ is the canonical isomorphism. Moreover f' is an isomorphism local; for f' to be an open immersion, it is necessary and sufficient that f be separated.

Proof. By virtue of (18.10.7), we can restrict to the case where X is intègre and normal, L its field of rational functions, f being dominant. The existence and uniqueness of the factorisation of f then follow from (II, 6.3.9). As f' is birational and locally quasi-finite, the last assertions follow from (8.12.10). $\square$

Corollary (18.10.9). — Under the hypotheses of (18.10.8), for f to be a finite morphism (i.e. for X to be an étale covering of Y), it is necessary and sufficient that X be isomorphic to the integral closure Y' of Y in L .

Proof. If f is finite, the canonical injection $j : X \rightarrow Y'$ (which is an open immersion) is also a finite morphism (II, 6.1.5, (v)), hence a closed morphism (II, 6.1.10), and as $j(X)$ is dense in Y' by (18.10.8), we have $j(X) = Y'$. Conversely, as Y is normal and L a direct composition of finite separable extensions of K , Y' is finite over Y (Bourbaki, *Alg. comm.*, chap. V, § 1, n° 6, cor. 1 of prop. 18), hence the corollary. $\square$

(18.10.10). Let Y be a normal and intègre prescheme, $K = R(Y)$ its field of rational functions. We say that a K -algebra of finite rank L is *unramified over Y* if: 1° L is a finite separable K -algebra, hence a direct composition of finite separable extensions K_i of K ($1 \leq i \leq n$); 2° the integral closure Y' of Y in L (sum of the preschemes of integral closures of Y in the K_i) is *unramified over Y* (which, by (18.10.3), is equivalent to saying that Y' is étale over Y).

Remark (18.10.11). — Instead of saying that L is unramified over Y , some authors also say that L is “unramified over K ”; we will refrain from following this usage which lends itself to confusions.

One can then express (18.10.9) in the following form:

Corollary (18.10.12). — Let Y be a normal and intègre prescheme, $K = R(Y)$ its field of rational functions. Then the functor $X \mapsto R(X)$ is an equivalence of the category of étale coverings of Y , and of the category of K -algebras finite étale which are unramified over Y . One obtains a quasi-inverse functor by associating to every finite étale K -algebra L , unramified over Y , the integral closure Y' of Y in L .

Proof. This is a reformulation of (18.10.9) using the terminology of (18.10.10). $\square$

Proposition (18.10.13). — Let Y be a normal and intègre prescheme, $K = R(Y)$ its field of rational functions. IV | 162

- label=(i) The field K is a K -algebra unramified over Y .
- lbbel=(ii) Let L be a finite extension of K , unramified over Y , and let Z be the integral closure of Y in L . Then, if M is an L -algebra unramified over Z , M is also a K -algebra unramified over Y (“transitivity” of non-ramification).
- lcbel=(iii) Let Y' be a normal and intègre prescheme, K' its field of rational functions, $g : Y' \rightarrow Y$ a dominant morphism. If L is a K -algebra unramified over Y , then $L \otimes_K K'$ is a K' -algebra unramified over Y' (property of “translation”). If moreover $Y = \text{Spec}(A)$ and $Y' = \text{Spec}(A')$ are affine, and if C is the integral closure of A in L , then $C' = C \otimes_A A'$ is the integral closure of A' in $L \otimes_K K'$.

Proof. Assertion (i) is immediate, Y being by hypothesis its own normalisation in K . To prove (ii), let Z' be the integral closure of Z in M , which is by hypothesis an étale covering of Z . Since Z is étale and separated over Y , the same holds for Z' ((17.3.3)), which is evidently the integral closure of Y in M . Since M is a finite and separable K -algebra, it follows from (18.10.10) that M is unramified over Y . Finally, to prove (iii), note that if Z is the integral closure of Y in L , $Z \times_Y Y'$ is étale and separated over Y' ((17.3.3)), finite over Y' since Z is finite over Y , and has $L \otimes_K K'$ as its ring of rational functions (I, 3.4.9); since Y' is normal, the same holds for $Z \times_Y Y'$ ((17.5.7)), which is therefore the integral closure of Y' in $L \otimes_K K'$, which concludes the demonstration. $\square$

Corollary (18.10.14). — Let Y and Y' be two normal and intègre preschemes, K, K' their respective fields of rational functions, $g : Y' \rightarrow Y$ a dominant morphism, so that K' is an extension of K . Let L be a finite extension of K unramified over Y , L_1 a composed extension (Bourbaki, *Alg.*, chap. VIII, § 8, def. 1) of L and IV | 162

K' . Then L_1 is an extension of K' unramified over Y' . If moreover $Y = \operatorname{Spec}(A)$, $Y' = \operatorname{Spec}(A')$ are affine, and if C is the integral closure of A in L , then the sub-ring $C_1 = A[C, A']$ of L_1 is the integral closure of A' in L_1 .

Proof. Indeed, $L \otimes_K K'$ is a composed direct of finite separable extensions of K' , and L_1 is one of these extensions; hence the integral closure of Y in L is a sum of preschemes of which one is the integral closure of Y in L_1 . The corollary therefore results immediately from (18.10.13). $\square$

When for example A is the ring of integers of an algebraic number field K , K' and L algebraic extensions of K , there are classical examples showing that the relation $C_1 = A[C, A']$ is in default when L is not unramified over A .

(18.10.15). Suppose that $Y = \operatorname{Spec}(A)$ is affine, A being intègre and integrally closed; let K be its field of fractions, L a finite separable extension of K , and suppose that the integral closure C of A in L is an A -module projective of finite type (which will be the case for example if A is a Dedekind ring (Bourbaki, *Alg. comm.*, chap. VII, § 4, n° 10, prop. 22)). To say that L is unramified over A means that C is étale ((18.10.10)) and by virtue of ((18.2.7), (ii)), this is equivalent to saying that the discriminant $d_{C/A}$ of $\operatorname{Spec}(C)$ over $\operatorname{Spec}(A)$ is invertible in A . In the particular case where C is a free A -module and $(x_i)_{1 \leq i \leq n}$ a base of C over A , this means that $\det(\operatorname{Tr}_{C/A}(x_i x_j))$ is invertible in A .

Theorem (18.10.16). — *Let Y be an intègre prescheme, η its generic point, $f : X \rightarrow Y$ a separated morphism and locally of finite type. Suppose that every irreducible component of X dominates Y and that the generic fibre $X_\eta = f^{-1}(\eta)$ is a finite prescheme over $k(\eta)$, hence $(X_\eta)_{\text{red}}$ is equal to $\operatorname{Spec}(L)$, where $L = \prod L_i$ is a product of finite extensions L_i of k . Set* IV | 163

$$n = [L : K] = \sum [L_i : K], \quad n_s = \sum [L_i : K]_s \quad (\text{sum of separable degrees of the } L_i).$$

Let y be a geometrically unibranched point of Y , and designate by $n(y)$ the sum of separable degrees over $k(y)$ of the residue fields of the isolated points of $f^{-1}(y)$. Then:

- label=(i) One has $n(y) \leq n_s \leq n$.*
lbbel=(ii) Suppose X reduced and the point y normal in Y . For $n(y) = n$, it is necessary and sufficient that there exist an open neighbourhood U of y in Y such that the restriction $f^{-1}(U) \rightarrow U$ of f is an étale and finite morphism.

Proof. A) Reduction to the case where f is quasi-finite. One knows ((13.1.4)) that the set Z of isolated points $x \in X$ in $f^{-1}(f(x))$ is open in X , and by hypothesis Z contains $f^{-1}(\eta)$. Since $f^{-1}(\eta)$ is finite, Z is a filtered increasing union of quasi-compact opens V_λ containing $f^{-1}(\eta)$ (hence dense in X by hypothesis). If $n_\lambda(y)$ is the sum of separable degrees over $k(y)$ of the residue fields of the points of $V_\lambda \cap f^{-1}(y)$, one has $n(y) = \sup n_\lambda(y)$; to prove (i), it will thus suffice to show that $n_\lambda(y) \leq n_s$. On the other hand, if $n(y) = n$, there is an index λ such that $n_\lambda(y) = n$, and one shows that there exists an open neighbourhood U of y in Y such that the restriction $V_\lambda \cap f^{-1}(U) \rightarrow U$ of f is an étale and finite morphism. Since f is separated, the canonical injection $V_\lambda \cap f^{-1}(U) \rightarrow f^{-1}(U)$ is then also a finite morphism (II, 6.1.5, (v)), hence $V_\lambda \cap f^{-1}(U)$ is closed in $f^{-1}(U)$ (II, 6.1.10); but being everywhere dense in $f^{-1}(U)$, it is necessarily equal to it.

B) Reduction to the case where X is reduced, f finite and $Y = \operatorname{Spec}(\mathcal{O}_{Y,y})$. On the other hand, IV | 164
 by virtue of ((18.12.13))¹², one can (after having replaced Y by an affine open neighbourhood V of y and f by its restriction $f^{-1}(V) \rightarrow V$) factorise f as $X \xrightarrow{j} X' \xrightarrow{g} Y$, where g is finite and j an open immersion. Since X is reduced, one can replace X' by the reduced closed sub-prescheme of underlying space $\overline{j(X)}$ (I, 5.2.2), hence suppose X' reduced and such that every irreducible component of X' dominates Y . Moreover, $j(X)$ is a dense open in X' and has only a finite number of irreducible components by hypothesis; hence (0I, 1.2.7) the irreducible components of X' are the adherences of those of $j(X)$, and therefore the fibre $g^{-1}(\eta)$ identifies with $f^{-1}(\eta)$. If one supposes the theorem proved when in addition f is a finite morphism, one can apply it to g , and as $f^{-1}(y)$ identifies with a sub-prescheme of $g^{-1}(y)$, the relation (i) for g implies the same relation for f . Suppose on the other hand that $n(y) = n$; then, by virtue of what precedes, one can apply the

¹²The reader will verify that (18.10.16) is not used in the proof of (18.12.13). If f is supposed locally of finite presentation (resp. quasi-projective), one can replace (18.12.13) by (8.12.8) (resp. (8.12.6)).

conclusion of (ii) to the morphism g , and replacing Y by an open neighbourhood of y if needed, one can suppose g étale, and hence the same holds for f ; moreover, one then has $f^{-1}(y) = g^{-1}(y)$. As the morphism g is closed and $j(X)$ is open in X' , there exists an open neighbourhood U of y in Y such that $j(X) \cap g^{-1}(U) = j(f^{-1}(U))$, which proves that (ii) is also true for f .

One is thus reduced to the case where f is moreover a *finite* morphism. It is immediate that, to prove (i), one can restrict to the case where $Y = \operatorname{Spec}(\mathcal{O}_{Y,y})$. Let us show that the same holds for proving (ii). Indeed, suppose proved that the condition $n(y) = n$ implies that $X \times_Y \operatorname{Spec}(\mathcal{O}_{Y,y})$ is étale and finite over $\operatorname{Spec}(\mathcal{O}_{Y,y})$; as this implies that f is flat and formally unramified at the points of $f^{-1}(y)$, and that Y is intègre, one concludes from ((18.4.13)) that f is étale at the points of $f^{-1}(y)$, hence also in an open neighbourhood V of $f^{-1}(y)$ in X ; but since f is a finite morphism, hence closed, V contains an open of the form $f^{-1}(U)$, where U is an open neighbourhood of y in Y .

C) Reduction to the case where $\mathcal{O}_{Y,y}$ is strictly local. Set $A = \mathcal{O}_{Y,y}$, and let $A' = {}^{\text{hs}}A$ be the strict henselisation of A ((18.8.7)), which is a local intègre and geometrically unibranch ring ((18.8.16)). Let $Y' = \operatorname{Spec}(A')$, and let y' be the closed point and η' the generic point of Y' ; y' is above y and since the morphism $g : Y' \rightarrow Y$ is flat ((18.8.8)), η' is above η ((2.3.4)). Set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \rightarrow Y'$; f' is finite and since g is flat, every irreducible component of X' dominates Y' ((2.3.7)). If $n'(y')$, n' and n'_s are the numbers defined for f' in the same way as $n(y)$, n and n_s for f , one has $n'_s = n_s$ and $n'(y') = n(y)$ (I, 6.4.8), and it therefore amounts to the same to prove the assertion (i) of the statement for f or for f' . Moreover, if X is reduced and y a normal point of Y (hence A integrally closed), then A' is integrally closed ((18.8.12)). On the other hand it results from the definition ((18.8.7)), from the remark (8.1.2, a)) and from the theorem of the double inductive limit, that A' is an inductive limit of rings A_α such that the morphisms $\operatorname{Spec}(A_\alpha) \rightarrow \operatorname{Spec}(A) = Y$ are étale; hence X' is a projective limit of the $X_\alpha = X \otimes_A A_\alpha$, which are étale over X , hence reduced since X is ((17.5.7)); by passage to the limit, one deduces that X' is reduced (8.7.1). This being so, if $n(y) = n$, one has $n_s = n$ and hence all the residue fields of the points of X_s are necessarily separable over $k(\eta)$; the residue fields at the points of $X'_{\eta'}$ are therefore algebraic and separable over $k(\eta')$ (4.6.1), in other words $n'_s = n' = n$. The morphism f' has the same properties as f , and if one proves that f' is étale, it will follow that f is étale by ((17.7.1)).

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D) End of the proof. Suppose therefore A strictly local. Since A is henselian and the morphism f finite, it follows from ((18.5.11)) that if x_j ($1 \leq j \leq m$) are the distinct points of $f^{-1}(y)$, X is the *sum* of the open sub-preschemes $\operatorname{Spec}(\mathcal{O}_{X,x_j}) = X_j$, which are thus the connected components of X . By hypothesis, each of them dominates Y , hence meets $f^{-1}(\eta)$, and consequently the number of points of $f^{-1}(\eta)$ is $\geq m$; *a fortiori* we have $n_s \geq m$. But since $k(y)$ is separably closed and the $k(x_j)$ are algebraic over $k(y)$, they are necessarily *radiciel* extensions of $k(y)$, hence $m = n(y)$, and this proves (i). Suppose now that the hypotheses of (ii) are satisfied; the relations $n(y) = n_s = n$ entail first, since every irreducible component of X dominates Y , that each X_j is *irreducible*; moreover, since the X_j are *reduced*, they are *integral*; finally the relations $m = n_s = n$ entail that for the generic point ξ_j of X_j , $k(\xi_j)$ is a separable extension of $k(\eta)$ of degree 1, hence isomorphic to $k(\eta)$; in other words, the restriction $f|_{X_j} : X_j \rightarrow Y$ of f is a finite and *birational* morphism; moreover, X_j is integral and by hypothesis Y is *normal*; hence (8.12.10.1) $f|_{X_j}$ is an *isomorphism*, which proves that f is étale, and completes the proof of ((18.10.16)). $\square$

Remark (18.10.17). — The method of “étale localisation” used in the proof of (18.10.16) allows also to improve the results of (15.5.1) by eliminating the Noetherian hypothesis.

Lemma (18.10.18). — Let X, Y be two preschemes, $f : X \rightarrow Y$ a birational morphism¹³. For f to be an isomorphism local at the point x , it is necessary and sufficient that f be étale at the point x . For f to be an open immersion, it is necessary and sufficient that f be étale and separated.

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Proof. The stated conditions being trivially necessary, it suffices to see that they are sufficient. For the first assertion, the question is local on X and Y , so we can suppose f étale, X and Y affine, hence f separated, and we are reduced to proving the second assertion. By virtue of ((17.9.1)), it suffices to prove that f is *radiciel*. Let then $y \in Y$ and let us prove that $f^{-1}(y)$ is *radiciel* over $k(y)$. The base change $\operatorname{Spec}(\mathcal{O}_{Y,y}) \rightarrow Y$ not changing the fact that f is étale, separated and birational, we can

¹³We mean by this the notion defined in ((6.15.4)), but where we do not assume X and Y reduced.

restrict to the case where $Y = \operatorname{Spec}(A)$, with $A = \mathcal{O}_{Y,y}$. Let then $A' = {}^{\text{hs}}A$, which is a local ring and a faithfully flat A -module ((18.8.8)), the homomorphism $A \rightarrow A'$ being local. Set $Y' = \operatorname{Spec}(A')$, $X' = X_{(Y')}$, $f' = f_{(Y')} : X' \rightarrow Y'$, so that f' is étale and separated; moreover, since the morphism $Y' \rightarrow Y$ is flat, the reasoning of ((6.15.4.1)) shows that f' is also birational. If y' is the closed point of Y' , it will suffice, by virtue of ((2.6.1), (v)), to prove that $f'^{-1}(y')$ is radiciel over $k(y')$; since f' is étale, it will suffice to prove that $f'^{-1}(y')$ contains at most one point ((17.6.1), c')).

Suppose therefore $Y = \operatorname{Spec}(A)$, A being strictly local, f separated, étale and birational, and let us show that $f^{-1}(y)$ cannot contain more than one point. Indeed, if there existed two distinct points x_1, x_2 in $f^{-1}(y)$, there would exist two Y -sections u_1, u_2 of f such that $u_1(y) = x_1$ and $u_2(y) = x_2$ ((18.8.1)); but since f is étale and separated, u_1 and u_2 would be isomorphisms of Y onto two connected (open) components of X ((17.9.4)) and there would thus be at least two maximal points of X above every maximal point of Y , which contradicts the hypothesis that f is birational. $\square$

Proposition (18.10.19). — *Let Y be a reduced prescheme whose set of irreducible components is locally finite, $f : X \rightarrow Y$ a separated morphism, locally of finite type and formally unramified, g a Y -rational section of f (i.e. a Y -rational application of Y into X), U the domain of definition of g (I, 7.2.1), and let Z be the adherence of $g(U)$ in X . Then, for every point $y \in Y - U$ such that Y is geometrically unibranch at the point y , one has $Z_y = Z \cap f^{-1}(y) = \emptyset$. In particular, if Y is geometrically unibranch, $g(U)$ is an open and closed part of X . If, for every closed irreducible part T of X , $f(T)$ is closed in Y , g is defined at every geometrically unibranch point of Y .*

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Proof. Since Y is reduced and f separated, it follows from (I, 7.2.2) that there exists a U -section u of $f^{-1}(U)$ belonging to the class g ; moreover ((17.4.1.2)), since f is formally unramified, u is an isomorphism of U onto the prescheme induced on the open set $u(U)$ of X . If we also denote by Z the reduced sub-prescheme of X having Z for underlying space, $g(U) = u(U)$, being reduced, is also induced by Z , so the restriction $f' = f|_Z$ of f is a birational morphism of Z into Y , which is moreover formally unramified, since f is ((17.1.3)). The question being local on Y , since Y is reduced and geometrically unibranch at the point y , we can restrict ourselves (by replacing Y by an open neighbourhood of y) to the case where Y is intègre; then $u(U)$ is irreducible, hence also Z , and consequently Z is also intègre. Then (I, 8.2.7) since f' is dominant, the homomorphism $\mathcal{O}_{Y,y} \rightarrow \mathcal{O}_{Z,x}$ is injective for all $x \in Z_y$, and we conclude from ((18.10.2), (iii)) that f' is an étale morphism; being separated and birational, it is an isomorphism local at the point x by virtue of ((18.10.18)); but this would entail that the U -section u would be extendable to an open set distinct from U , contrary to the definition of the latter. We therefore necessarily have $Z_y = \emptyset$, which establishes the first assertion; the second is an obvious consequence. Moreover, under the hypotheses of the last assertion of the statement, since Z is closed and irreducible, $f(Z)$ is closed in Y , hence equal to Y , i.e. we have $Z_y \neq \emptyset$ for all $y \in Y$, which completes the proof. $\square$

Remark (18.10.20). — Let Y be a locally Noetherian prescheme. We say that a morphism $f : X \rightarrow Y$ is *essentially proper* if it is *locally of finite type* and if, for every Y -scheme of the form $Y' = \operatorname{Spec}(A)$ where A is a discrete valuation ring, the canonical map (II, 7.3.2.2) relative to the Y' -prescheme $X \times_Y Y'$, is bijective. The reasoning of (II, 7.3.8) proves that it amounts to the same to say that f (supposed locally of finite type) is *separated* and that for every base change $Y'' \rightarrow Y$, where Y'' is locally Noetherian, the image by $f_{(Y'')} : X_{(Y'')} \rightarrow Y''$ of every closed irreducible part of X'' is *closed*. To say that f is *proper* (for Y locally Noetherian) thus means that f is essentially proper and of finite type (II, 7.3.8); but one encounters important examples of essentially proper morphisms that are not of finite type (for example certain “Picard preschemes” or certain “Néron-Severi preschemes” (chap. VI)). It follows clearly from (18.10.19) that if Y is locally Noetherian, reduced and geometrically unibranch, and if the morphism $f : X \rightarrow Y$ is essentially proper and unramified, then every Y -rational section of f is defined everywhere.

The following proposition generalises the criterion (17.15.5) of smoothness of a morphism $X \rightarrow Y$ when Y is not the spectrum of a field:

Proposition (18.10.21). — *Let $f : X \rightarrow Y$ be a morphism locally of finite presentation, x a point of X , $y = f(x)$ and suppose the ring $\mathcal{O}_{Y,y}$ intègre and geometrically unibranch. Let r be the minimum number of generators of the $\mathcal{O}_{X,x}$ -module $(\Omega_{X/Y}^1)_x$ (equal to the rank of the $k(x)$ -vector space $(\Omega_{X/Y}^1)_x \otimes_{\mathcal{O}_{X,x}} k(x)$).*

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Then, for f to be smooth at the point x , it is necessary and sufficient that, if y' is the generic point of the unique irreducible component of Y containing y , there exists a generalisation x' of x such that $f(x') = y'$ and $\dim_{x'} f^{-1}(y') \geq r$.

Proof. If f is smooth at the point x , it is also smooth in an open neighbourhood U of x , hence at every generalisation x' of x , and $\dim_{x'}(f^{-1}(f(x')))) = r$ at each of these points, by virtue of ((17.10.2)). Since f is moreover flat in U , there exists a generalisation of x above every generalisation of y ((2.3.4)), which proves that the condition is necessary. To show that it is sufficient, note first that by ((17.5.1)), it suffices to prove that the morphism deduced from f by the base change $\mathrm{Spec}(\mathcal{O}_{Y,y}) \rightarrow Y$ is smooth at the point x ; we can therefore suppose $Y = \mathrm{Spec}(\mathcal{O}_{Y,y})$.

The question is local on X , so (((16.4.22)) and (0I, 5.2.2)) we can suppose that X is affine, and that there exist r sections s_i ($1 \leq i \leq r$) of $\mathcal{O}_X$ above Y such that the sections $d_{X/Y}(s_i)$ generate $\Omega_{X/Y}^1$. Let

$$g : X \longrightarrow Z = Y[T_1, \dots, T_r] = \mathbf{V}_Y^r$$

be the Y -morphism corresponding to the homomorphism $\mathcal{O}_Z \rightarrow f_*(\mathcal{O}_X)$ of $\mathcal{O}_Y$ -Modules defined by the s_i (considered as sections of $f_*(\mathcal{O}_X)$ above Y) (II, 1.2.7). We have the exact sequence ((16.4.19))

$$g^*(\Omega_{Z/Y}^1) \longrightarrow \Omega_{X/Y}^1 \longrightarrow \Omega_{X/Z}^1 \longrightarrow 0.$$

Since the dT_i ($1 \leq i \leq r$) generate $\Omega_{Z/Y}^1$, it follows from the definition of g that the homomorphism $g^*(\Omega_{Z/Y}^1) \rightarrow \Omega_{X/Y}^1$ is surjective, hence $\Omega_{X/Z}^1 = 0$, and g is consequently *unramified* ((17.4.1)). We shall see that g is étale at the point x , and taking into account ((17.3.8)), this will prove that f is smooth at the point x . Since $\mathcal{O}_{Y,y}$ is assumed intègre and geometrically unibranch, the same holds for $\mathcal{O}_{Z,z}$ ((17.5.7)) where $z = g(x)$, since the structural morphism $Z \rightarrow Y$ is smooth ((17.3.8)). By virtue of ((18.10.1)), it suffices to show that the homomorphism $\mathcal{O}_{Z,z} \rightarrow \mathcal{O}_{X,x}$ is injective; this amounts to seeing that the corresponding morphism $\mathrm{Spec}(\mathcal{O}_{X,x}) \rightarrow \mathrm{Spec}(\mathcal{O}_{Z,z})$ is dominant.

Let y' be the generic point of $Y = \mathrm{Spec}(\mathcal{O}_{Y,y})$ and let z' be the generic point of $Z = \mathrm{Spec}(\mathcal{O}_{Y,y}[T_1, \dots, T_r])$, which is intègre; if $h : Z \rightarrow Y$ is the structural morphism, we have $h(z') = y'$. It suffices to prove that there exists in $f^{-1}(y')$ a generalisation x' of x such that the image of x' by the morphism $g_{y'} : f^{-1}(y') \rightarrow h^{-1}(y')$ is equal to z' . Moreover, z' is also the generic point of $h^{-1}(y') = \mathrm{Spec}(k(y')[T_1, \dots, T_r])$; since the morphism $g_{y'}$ is unramified, hence quasi-finite ((17.4.2)), the same holds for its restriction to every irreducible component of $f^{-1}(y')$; since $f^{-1}(y')$ and $h^{-1}(y')$ are locally Noetherian and $\dim(h^{-1}(y')) = r$, it follows from ((5.4.1)) that the restriction of $g_{y'}$ to every irreducible component of *dimension* $\geq r$ of $f^{-1}(y')$ is a dominant morphism. Now, there exists by hypothesis such a component containing a generalisation of x ; *a fortiori* its generic point is a generalisation of x , whence the conclusion. $\square$

18.11. Application to complete local Noetherian algebras over a field.

Lemma (18.11.1). — *Let k be a field, A a local Noetherian complete ring which is a k -algebra, K the residue field of A .* IV | 169

- label=(i) If the K -vector space $\Omega_{K/k}^1$ is of finite rank, the A -module $\widehat{\Omega}_{A/k}^1$ is of finite type.*
lbbel=(ii) If moreover A is a k -algebra formally smooth (for the adic topology), $\widehat{\Omega}_{A/k}^1$ is a free A -module of rank equal to

$$\dim(A) + \mathrm{rg}_K(\Omega_{K/k}^1) - \mathrm{rg}_K(Y_{K/k}).$$

Moreover, for every sub-field k_0 of k such that Ω_{K/k_0}^1 is a K -vector space of finite rank, $\widehat{\Omega}_{A/k_0}^1$ is a free A -module of rank equal to

$$\dim(A) + \mathrm{rg}_K(\Omega_{K/k}^1) - \mathrm{rg}_K(Y_{K/k}) + \mathrm{rg}_K(\Omega_{k/k_0}^1).$$

Proof. Assertion (i) is only recalled for convenience, having been proved in (0, 20.7.15). To prove (ii), note that this assertion was in fact established by the proof of (0, 21.9.2), the statement of (0, 21.9.2) alone requiring the finiteness of the degree of transcendence of K over k (via Cartier's equality (0, 21.7.1)). $\square$

Lemma (18.11.2). — *Let k be a field of characteristic $p > 0$, C a local k -algebra, Noetherian complete, whose residue field is a finite extension of k , and which is intègre; let n be its dimension, L its field of fractions. Then $\Omega_{L/k}^1$ and $Y_{L/k}$ are L -vector spaces of finite rank, and one has*

$$(18.11.2.1) \quad \mathrm{rg}_L(\Omega_{L/k}^1) - \mathrm{rg}_L(Y_{L/k}) \geq n.$$

Suppose moreover that $[k : k^p] < +\infty$. Then one has the equality

$$(18.11.2.2) \quad \mathrm{rg}_L(\Omega_{L/k}^1) - \mathrm{rg}_L(Y_{L/k}) = n.$$

Moreover, $\Omega_{C/k}^1$ is then isomorphic to $\widehat{\Omega}_{C/k}^1$, hence (18.11.1) a C -module of finite type.

Proof. Indeed, one knows (0, 19.8.9) that there exists a sub- k -algebra C_0 of C isomorphic to a formal power series algebra $k[[T_1, \dots, T_n]]$ and such that C is a finite C_0 -algebra. Hence L is a finite extension of the field of fractions $L_0 = k((T_1, \dots, T_n))$ of C_0 .

Now, one has the exact sequence of L -vector spaces obtained by applying (0, 20.6.17.1) to the prime subfield of k and to the three fields $k \subset L_0 \subset L$ (and taking into account (0, 20.6.21, (i)))

$$0 \longrightarrow Y_{L_0/k} \otimes_{L_0} L \longrightarrow Y_{L/k} \longrightarrow Y_{L/L_0} \longrightarrow \Omega_{L_0/k}^1 \otimes_{L_0} L \longrightarrow \Omega_{L/k}^1 \longrightarrow \Omega_{L/L_0}^1 \longrightarrow 0.$$

Since L is a finite extension of L_0 , Ω_{L/L_0}^1 and Y_{L/L_0} are L -vector spaces of finite rank having the same rank, by virtue of Cartier's equality (0, 21.7.1). As L_0 is separable over k (0, 21.9.6.4), one has $Y_{L_0/k} = 0$ (0, 20.6.3); one therefore deduces from the preceding exact sequence that $Y_{L/k}$ is of finite rank and that one has in all cases (0_{III}, 11.10.2)

$$\mathrm{rg}_L(\Omega_{L/k}^1) - \mathrm{rg}_L(Y_{L/k}) = \mathrm{rg}_{L_0}(\Omega_{L_0/k}^1) - \mathrm{rg}_{L_0}(Y_{L_0/k}).$$

To prove (18.11.2.1) or (18.11.2.2), one can therefore reduce to proving these relations by replacing C and L by C_0 and L_0 . As L_0 is separable over k (0, 21.9.6.4), one has $Y_{L_0/k} = 0$ (0, 20.6.3); on the other hand $\Omega_{L_0/k}^1 = \Omega_{C_0/k}^1 \otimes_{C_0} L_0$ (0, 20.5.9). One knows (0, 21.9.4) that if $C_1 = k[[T_1^p, \dots, T_n^p]]$, $\widehat{\Omega}_{C_0/k}^1$ identifies with Ω_{C_0/C_1}^1 ; on the other hand, one has $\Omega_{C_0/k}^1 = \Omega_{C_0/k[C_1^p]}^1$ (0, 21.1.5), and as $C_1^p = k^p[[T_1^p, \dots, T_n^p]]$, one has $k[C_1^p] \subset C_1$, with equality when $[k : k^p] < +\infty$. Now, one has an exact sequence $\Omega_{C_0/k[C_1^p]}^1 \rightarrow \widehat{\Omega}_{C_0/k}^1 \otimes_{C_0} L_0 \rightarrow \Omega_{C_0/C_1}^1 \rightarrow 0$ (0, 20.5.7), whence a surjection $\Omega_{L_0/k}^1 \rightarrow \widehat{\Omega}_{C_0/k}^1 \otimes_{C_0} L_0 \rightarrow 0$. As $\widehat{\Omega}_{C_0/k}^1$ is a free C_0 -module of rank n (0, 21.9.3), one sees that one has in all cases $\mathrm{rg}_{L_0}(\Omega_{L_0/k}^1) \geq n$, with equality when $[k : k^p] < +\infty$; this already proves (18.11.2.1) and (18.11.2.2).

Finally, to see that $\Omega_{C/k}^1$ is isomorphic to $\widehat{\Omega}_{C/k}^1$ when $[k : k^p] < +\infty$, it suffices to prove that $\Omega_{C/k}^1$ is a C -module of finite type, since C is a complete local Noetherian ring (0_I, 7.3.3). Since $\Omega_{C/k}^1$ is isomorphic to $\Omega_{C/k[C_1^p]}^1$ and $k[C_1^p] \subset C_1$, it all reduces to proving that C is a finite C_0 -module and C_0 a $k[C_1^p]$ -module of finite type (0, 20.4.7); but this results from the fact that C is by hypothesis a finite C_0 -algebra and C_0 is a $k[C_1^p]$ -module of finite type by virtue of the hypothesis $[k : k^p] < +\infty$. $\square$

Lemma (18.11.3). — *Let k be a field, p its characteristic exponent, and suppose that $[k : k^p] < +\infty$. Let A be a local Noetherian complete k -algebra, whose residue field is a finite extension of k . Let $\mathfrak{p}$ be a prime ideal of A such that $A_{\mathfrak{p}}$ is geometrically regular over k (6.7.6), so that there exists only one minimal prime ideal $\mathfrak{q}$ of A contained in $\mathfrak{p}$. Then $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ is a free $A_{\mathfrak{p}}$ -module of rank equal to $\dim(A/\mathfrak{q})$.* IV | 170

Proof. Since A is Noetherian, and $\mathrm{Spec}(A)$ is regular (and a fortiori integral) at the point $\mathfrak{p}$, $\mathfrak{p}$ belongs to only one irreducible component of $\mathrm{Spec}(A)$, hence contains only one minimal prime ideal $\mathfrak{q}$ of A , and in addition one has $\mathfrak{q}_{\mathfrak{p}} = 0$. If one sets $B = A/\mathfrak{q}$, the suite (0, 20.7.20)

$$\mathfrak{q}/\mathfrak{q}^2 \longrightarrow \widehat{\Omega}_{A/k}^1 \otimes_A B \longrightarrow \widehat{\Omega}_{B/k}^1 \longrightarrow 0$$

is exact; indeed, $\widehat{\Omega}_{A/k}^1$ is an A -module of finite type (18.11.1), hence $\widehat{\Omega}_{A/k}^1 \otimes_A B = \widehat{\Omega}_{A/k}^1/\mathfrak{q} \cdot \widehat{\Omega}_{A/k}^1$, and since this B -module is of finite type, every sub- B -module of it is closed (0_I, 7.3.5), in view of (0, 20.4.5). Since the image of $\mathfrak{q}/\mathfrak{q}^2$ is dense in the kernel of the homomorphism $\widehat{\Omega}_{A/k}^1 \otimes_A B \rightarrow \widehat{\Omega}_{B/k}^1$ (0, 20.7.20), it is necessarily equal to this kernel, whence our assertion. Localising at $\mathfrak{p}$ the preceding exact suite, one sees furthermore that the canonical homomorphism $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}} \rightarrow (\widehat{\Omega}_{B/k}^1)_{\mathfrak{p}/\mathfrak{q}}$ is bijective.

One can thus restrict to the case where $\mathfrak{q} = 0$, in other words where A is *integral*. We distinguish two cases:

I) $p > 1$. One can apply (18.11.2) to the quotient algebra $C = A/\mathfrak{p}$, whose field of fractions K is none other than the residue field of $A_{\mathfrak{p}}$; one sees therefore that

$$(18.11.3.1) \quad \mathrm{rg}_K(\Omega_{K/k}^1) - \mathrm{rg}_K(Y_{K/k}) = \dim(A/\mathfrak{p}).$$

Note now (0, 19.6.6) that $A_{\mathfrak{p}}$ is a k -algebra *formally smooth* for its $\mathfrak{p}A_{\mathfrak{p}}$ -preadic topology; consequently $\Omega_{A_{\mathfrak{p}}/k}^1$ is formally projective (0, 20.4.9) for the $\mathfrak{p}$ -preadic topology (0, 20.4.5); on the other hand, $\Omega_{A_{\mathfrak{p}}/k}^1 = (\Omega_{A/k}^1)_{\mathfrak{p}}$ (0, 20.5.9) is an $A_{\mathfrak{p}}$ -module of finite type, by virtue of (18.11.2) applied to A . For every integer j , $\Omega_{A_{\mathfrak{p}}/k}^1/\mathfrak{p}^{j+1}\Omega_{A_{\mathfrak{p}}/k}^1$ is therefore a projective $(A_{\mathfrak{p}}/\mathfrak{p}^{j+1}A_{\mathfrak{p}})$ -module of rank $m = \mathrm{rg}_K(\Omega_{A_{\mathfrak{p}}/k}^1 \otimes_{A_{\mathfrak{p}}} K)$ (0, 19.2.4); one concludes from (0_{III}, 10.2.1 and 10.1.3) that $\Omega_{A_{\mathfrak{p}}/k}^1$ is a free $A_{\mathfrak{p}}$ -module of rank m . Let $A' = (A_{\mathfrak{p}})^{\wedge}$ be the completed algebra of $A_{\mathfrak{p}}$ for its $\mathfrak{p}A_{\mathfrak{p}}$ -preadic topology; A' is again a formally smooth k -algebra (0, 19.3.6) for its adic topology, and it follows from (0, 20.7.14) and (0, 20.4.5) that $\widehat{\Omega}_{A'/k}^1 = \widehat{\Omega}_{A/k}^1$; one concludes that $\widehat{\Omega}_{A'/k}^1$ is an A' -module *free of rank m* . But it follows then from (18.11.1) and from $\dim(A') = \dim(A_{\mathfrak{p}})$ (0, 16.2.4) that

$$(18.11.3.2) \quad m = \dim(A_{\mathfrak{p}}) + (\mathrm{rg}_K(\Omega_{K/k}^1) - \mathrm{rg}_K(Y_{K/k}))$$

whence, by virtue of (18.11.3.1),

$$m = \dim(A/\mathfrak{p}) + \dim(A_{\mathfrak{p}}).$$

But since A is a complete local Noetherian ring, it is a quotient of a regular ring (0, 19.8.8), hence (0, 16.5.12) $\dim(A) = \dim(A/\mathfrak{p}) + \dim(A_{\mathfrak{p}})$, which finishes the proof in this case.

II) $p = 1$. One has, as above, $\dim(A) = \dim(A/\mathfrak{p}) + \dim(A_{\mathfrak{p}})$; set $n = \dim(A)$, $r = \dim(A/\mathfrak{p})$, $s = \dim(A_{\mathfrak{p}})$; we will show that $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ is a free $A_{\mathfrak{p}}$ -module of rank n . We will first prove the following lemma:

Lemma (18.11.3.3). — *Let A be a local Noetherian ring, $\mathfrak{p}$ a prime ideal of A such that $A_{\mathfrak{p}}$ is a Cohen–Macaulay ring. For every system of parameters $(t_i)_{1 \leq i \leq s}$ of $A_{\mathfrak{p}}$, there exists a suite $(x_i)_{1 \leq i \leq s}$ of elements of $\mathfrak{p}$ such that the x_i form part of a system of parameters of A and that, for every i , $x_i/1$ is congruent to t_i modulo the ideal $t_1A_{\mathfrak{p}} + \dots + t_{i-1}A_{\mathfrak{p}}$. In particular, if $A_{\mathfrak{p}}$ is regular and the t_i form a regular system of parameters of $A_{\mathfrak{p}}$, the same is true of the $x_i/1$.*

This lemma will itself be a consequence of the following:

Lemma (18.11.3.4). — *Let A be a local Noetherian ring, $\mathfrak{p}$ a prime ideal of A , x an element of A ; then there exists an element $x' \in A$ such that $x'/1 = x/1$ in $A_{\mathfrak{p}}$ and such that x' does not belong to any of the minimal prime ideals of A not contained in $\mathfrak{p}$.*

Proof that ((18.11.3.4)) implies ((18.11.3.3)). Multiplying the t_i if needed by invertible elements of $A_{\mathfrak{p}}$, one can suppose they are of the form $x_i/1$, with $x_i \in \mathfrak{p}$ for $1 \leq i \leq s$. We reason by recurrence on s ; since t_1 forms part of a system of parameters of $A_{\mathfrak{p}}$, it does not belong to any of the minimal prime ideals of $A_{\mathfrak{p}}$ (0, 16.5.5), hence no $x'_1 \in A$ such that $x'_1/1 = t_1$ can belong to a minimal prime ideal of A contained in $\mathfrak{p}$. By virtue of (18.11.3.4), one can furthermore choose x'_1 such that $x'_1/1 = t_1$ (hence $x'_1 \in \mathfrak{p}$) and such that x'_1 does not belong to *any* minimal prime ideal of A , hence forms part of a system of parameters of A (0, 16.3.4 and 16.3.7). One then reasons by recurrence, considering the ring $A' = A/x'_1A$ and the prime ideal $\mathfrak{p}' = \mathfrak{p}/x'_1A$ of this ring; as $A'_{\mathfrak{p}'} = A_{\mathfrak{p}}/t_1A_{\mathfrak{p}}$, $A'_{\mathfrak{p}'}$ is also a Cohen–Macaulay ring (0, 16.5.5), and the images t'_i ($2 \leq i \leq s$) of the t_i in $A'_{\mathfrak{p}'}$ form a system of parameters of this ring; it suffices to apply to A' and the t'_i ($2 \leq i \leq s$) the recurrence hypothesis. The last assertion of (18.11.3.3) results from the fact that in $A_{\mathfrak{p}}$, a system of parameters that generates the maximal ideal is a regular system of parameters (0, 17.1.1). $\square$

Proof of ((18.11.3.4)). Let $(\mathfrak{p}'_k)_{1 \leq k \leq r}$ be the suite of minimal prime ideals of A not contained in $\mathfrak{p}$ and such that $x \in \mathfrak{p}'_k$, and let $(\mathfrak{p}_j)_{1 \leq j \leq n}$ be the suite of minimal prime ideals of A other than the $\mathfrak{p}'_k$; one can suppose $r \geq 1$. Since $\mathfrak{p}'_k$ does not contain $\bigcap_{1 \leq j \leq n} \mathfrak{p}_j$, there exists an element $y \in \bigcap_{1 \leq j \leq n} \mathfrak{p}_j$ which is not contained in any of the $\mathfrak{p}'_k$ (Bourbaki, *Alg. comm.*, chap. II, §1, no 1, prop. 2); moreover

$y/1$ belongs to all the minimal prime ideals of $A_{\mathfrak{p}}$, hence is nilpotent; if $(y/1)^h = 0$, the element $x' = x + y^h$ will satisfy the conditions of the statement, since on the one hand $y^h \notin \mathfrak{p}'_k$ for $1 \leq k \leq r$, and by definition of the $\mathfrak{p}'_k$ one indeed has $x' \notin \mathfrak{p}'_k$ for $1 \leq k \leq r$, and on the other hand, if $\mathfrak{p}_j$ is a minimal prime ideal of A not contained in $\mathfrak{p}$ but such that $x \notin \mathfrak{p}_j$, one also has $x' \notin \mathfrak{p}_j$ since $y \in \mathfrak{p}_j$. $\square$

Let us return to the proof of (18.11.3) when $p = 1$. By virtue of (18.11.3.3), there exists a regular system of parameters $(t_i)_{1 \leq i \leq s}$ of $A_{\mathfrak{p}}$ such that $t_i = x_i/1$, where the x_i ($1 \leq i \leq s$) belong to $\mathfrak{p}$ and form part of a system of parameters $(x_j)_{1 \leq j \leq n}$ of A . Set $A_0 = k[[T_1, \dots, T_n]]$; since A is a complete k -algebra and the x_j belong to the maximal ideal $\mathfrak{m}$ of A , there exists a local k -homomorphism $u : A_0 \rightarrow A$ such that $u(T_j) = x_j$ for $1 \leq j \leq n$ (Bourbaki, *Alg. comm.*, chap. III, §4, no 5, prop. 6); if $\mathfrak{n}$ is the ideal of A generated by the x_j ($1 \leq j \leq n$), it is by hypothesis an ideal of definition of A ; one deduces from (0_I, 7.4.4 and 7.4.3) that u makes A a finite A_0 -algebra.

Set $\mathfrak{p}_0 = \sum_{j=1}^s A_0 T_j$, $B_0 = (A_0)_{\mathfrak{p}_0}$ and $B = A \otimes_{A_0} B_0$, so that $u_{\mathfrak{p}_0} : B_0 \rightarrow B$ makes B a finite B_0 -algebra; furthermore, if $\mathfrak{p}'$ is the ideal of B generated by $\mathfrak{p}$, one has $B_{\mathfrak{p}'} = A_{\mathfrak{p}}$; since $\mathfrak{p}'$ contains $\mathfrak{p}_0 B$ by construction, it lies above the maximal ideal $\mathfrak{p}_0 B_0$ of B_0 . Let us show that the morphism $\text{Spec}(B) \rightarrow \text{Spec}(B_0)$ is unramified at the point $\mathfrak{p}'$: this results from the fact that $\kappa(\mathfrak{p}')$ is a finite extension of the field $\kappa(\mathfrak{p}_0)$ of characteristic 0, hence is necessarily separable, and that $B_{\mathfrak{p}'} / \mathfrak{p}_0 B_{\mathfrak{p}'} = \kappa(\mathfrak{p}')$ by virtue of the choice of the x_j for $1 \leq j \leq s$ (17.4.1).

Note now the following lemma:

Lemma (18.11.3.5). — *Let k be a field, R, S two local Noetherian complete k -algebras, such that, if K is the residue field of S , $\Omega_{K/k}^1$ is of finite rank over K ; let $u : R \rightarrow S$ be a k -homomorphism making S a finite R -algebra; then one has $\widehat{\Omega}_{S/R}^1 = \Omega_{S/R}^1$, and the suite*

$$(18.11.3.6) \quad \widehat{\Omega}_{R/k}^1 \otimes_R S \xrightarrow{v} \widehat{\Omega}_{S/k}^1 \xrightarrow{w} \Omega_{S/R}^1 \longrightarrow 0$$

(cf. (0, 20.7.17.3)) is exact.

Proof of (18.11.3.5). The first assertion results from the fact that $\Omega_{S/R}^1$ is an S -module of finite type (0, 20.4.7) and from (0_I, 7.3.6). On the other hand, one knows (0, 20.7.17.3) that the image of v is dense in $\text{Ker}(w)$ and that w is surjective; but it follows from (18.11.1) that $\widehat{\Omega}_{S/k}^1$ is an S -module of finite type; one concludes that every sub- S -module of $\widehat{\Omega}_{S/k}^1$ is closed (0_I, 7.3.5), whence the lemma. $\square$

Applying this lemma to the case where $R = A_0$, $S = A$, and noting that $\widehat{\Omega}_{A_0/k}^1$ is a free A_0 -module of rank n (0, 21.9.3); since $\text{Spec}(B) \rightarrow \text{Spec}(B_0)$ is unramified at the point $\mathfrak{p}'$, one has $(\Omega_{A/A_0}^1)_{\mathfrak{p}} = \Omega_{A_{\mathfrak{p}}/B_{\mathfrak{p}}}^1 = 0$ (16.4.15 and 17.4.1). If one localises the exact sequence (18.11.3.6) at $\mathfrak{p}$, one obtains hence a surjective homomorphism

$$v_{\mathfrak{p}} : (\widehat{\Omega}_{A_0/k}^1 \otimes_{A_0} A)_{\mathfrak{p}} \longrightarrow (\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$$

whence one concludes that the $A_{\mathfrak{p}}$ -module $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ admits a system of n generators. But the A -module $\widehat{\Omega}_{A/k}^1$ is of rank n , by virtue of (0, 21.9.5), which is applicable to the complete integral A since the field of fractions of A is of characteristic 0 by the theorem of Cohen (0, 19.8.8, (ii)). The $A_{\mathfrak{p}}$ -module $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ is hence also of rank n , and as its quotient by its torsion submodule admits a system of n generators, one deduces immediately that these n generators form a free system, whence the conclusion. $\square$

Lemma (18.11.4). — *Let k be a field, A a k -algebra local Noetherian complete, whose residue field is a finite extension of k . Let k' be any extension of k , and set $A' = A \widehat{\otimes}_k k'$. Then:*

label=(i) *A' is a semi-local Noetherian complete ring, direct composition of complete local rings A'_i ($1 \leq i \leq r$) which are faithfully flat A -modules, and whose residue fields are finite extensions of k' ; if $\mathfrak{m}$ is the maximal ideal of A , $\mathfrak{m}A'$ is an ideal of definition of A' ; one has $\dim(A'_i) = \dim(A)$ for every i .*

lbbel=(ii) *For every i , $\widehat{\Omega}_{A'/k'}^1$ is canonically isomorphic to $\widehat{\Omega}_{A/k}^1 \otimes_A A'_i$.*

Proof. (i) The first assertions result immediately from (7.5.5) and from (0_I, 6.6.2); the fact that $\mathfrak{m}A'$ is an ideal of definition of A' also results from (7.5.5), for since K is the residue field of A , $K \otimes_k k'$ is a

k' -algebra finite. Finally $A'_i/\mathfrak{m}A'_i$ is one of the artinian local rings of which $K \otimes_k k'$ is composed direct ((7.5.5)), hence is of dimension 0; since A'_i is a flat A -module, the equality of dimensions results from (6.1.2).

(ii) As $\widehat{\Omega}_{A/k}^1$ is, by virtue of the hypothesis on K , an A -module of finite type (18.11.1), $\widehat{\Omega}_{A/k}^1 \otimes_A A'$ is complete and identifies with the completed tensor product (0_I, 7.7.4), $\widehat{\Omega}_{A/k}^1 \otimes_A A' = \widehat{\Omega}_{A/k}^1 \widehat{\otimes}_A (A \widehat{\otimes}_k k')$ identifies with $\widehat{\Omega}_{A/k}^1 \widehat{\otimes}_k k'$. But $\Omega_{A/k}^1 \otimes_k k'$ identifies with $\Omega_{A''/k'}^1$ (0, 20.5.5), where $A'' = A \otimes_k k'$, and since A' is by definition the separated completion of $\Omega_{A''/k'}^1$, one has a canonical isomorphism $\widehat{\Omega}_{A'/k'}^1 \xrightarrow{\sim} \widehat{\Omega}_{A/k}^1 \otimes_A A'$. $\square$

Proposition (18.11.5). — *Let k be a field, A a local k -algebra Noetherian and complete, whose residue field is a finite extension of k , $\mathfrak{p}$ a prime ideal of A distinct from the maximal ideal $\mathfrak{m}$, such that there exists a unique minimal prime ideal $\mathfrak{q}$ of A contained in $\mathfrak{p}$ for which $\dim(A/\mathfrak{q}) = \dim(A)$ (which will take place in particular when A is equidimensional), m an integer ≥ 0 . The following conditions are equivalent:* IV | 174

- a) The $A_{\mathfrak{p}}$ -module $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ admits a system of m generators.
- b) There exists a k -homomorphism local $u : B \rightarrow A$, where $B = k[[T_1, \dots, T_m]]$, making A a B -algebra finite, and such that the corresponding morphism $\text{Spec}(A) \rightarrow \text{Spec}(B)$ is unramified at the point $\mathfrak{p}$.

Proof. To prove that b) implies a), note that by virtue of the lemma (18.11.3.5), one has then the exact sequence

$$\widehat{\Omega}_{B/k}^1 \otimes_B A \longrightarrow \widehat{\Omega}_{A/k}^1 \longrightarrow \Omega_{A/B}^1 \longrightarrow 0$$

since A is a finite B -algebra; localising at $\mathfrak{p}$ and noting that by hypothesis one has then $(\Omega_{A/B}^1)_{\mathfrak{p}} = 0$ (17.4.1), one obtains a surjective homomorphism $(\widehat{\Omega}_{B/k}^1 \otimes_B A)_{\mathfrak{p}} \rightarrow (\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$, and the conclusion follows from the fact that $\widehat{\Omega}_{B/k}^1$ is a free B -module of rank m (0, 21.9.3).

To prove that a) implies b), let us first prove the following lemmas.

Lemma (18.11.5.1). — *Let k be a field, A a k -algebra local Noetherian and complete, whose residue field is a finite extension of k , $\mathfrak{p}$ a prime ideal of A , $\mathfrak{q}$ a minimal ideal of A contained in $\mathfrak{p}$. Then the minimum number of generators of the $A_{\mathfrak{p}}$ -module $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ is at least equal to $\dim(A/\mathfrak{q})$.*

Set $n = \dim(A/\mathfrak{q})$, and let m be the minimum number of generators of the $A_{\mathfrak{p}}$ -module $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$, which is equal to $\text{rg}_{k(\mathfrak{p})}((\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p}))$ (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. 2 de la prop. 4). Note that $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{q}} = (\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{q}}$, hence the minimum number of generators of the $A_{\mathfrak{q}}$ -module $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{q}}$ is at most equal to m . It suffices hence to consider the case where $\mathfrak{p} = \mathfrak{q}$ is minimal. In second place, let us show that one can restrict to the case where k is algebraically closed. Indeed, let k' be an algebraic closure of k , and set $A' = A \otimes_k k'$, which is a direct composition of local k' -algebras A'_i ($1 \leq i \leq r$), whose residue fields are isomorphic to k' , and which are faithfully flat A -modules (18.11.4). Applying (2.3.4) and (6.1.1), one sees that in any given A'_i , there exists a minimal ideal $\mathfrak{q}'_i$ above $\mathfrak{q}$ such that $n' = \dim(A'/\mathfrak{q}'_i) \geq \dim(A/\mathfrak{q}) = n$. On the other hand (18.11.4), the minimum number m' of generators of the $(A'_i)_{\mathfrak{q}'_i}$ -module $(\widehat{\Omega}_{A'/k'}^1)_{\mathfrak{q}'_i}$ is at most equal to m ; whence our assertion.

Suppose therefore k algebraically closed, and let us show that one can moreover restrict to the case where $\mathfrak{q} = 0$. Indeed, one has $\Omega_{(A/\mathfrak{q})/A}^1 = 0$ (0, 20.4.12), hence the exact sequence (18.11.3.6) furnishes a surjective homomorphism $(\widehat{\Omega}_{A/k}^1) \otimes_A (A/\mathfrak{q}) \rightarrow \widehat{\Omega}_{(A/\mathfrak{q})/k}^1$, whence one deduces immediately that the minimum number of generators of the K -module $(\widehat{\Omega}_{(A/\mathfrak{q})/k}^1) \otimes_{A/\mathfrak{q}} K$, where K is the field of fractions of $A/\mathfrak{q}$, is at most equal to m .

But if A is integral and K is its field of fractions, K is separable over the algebraically closed field k , hence a geometrically regular k -algebra (6.7.6); one can apply (18.11.3) for $\mathfrak{p} = 0$ observing that $k = k^p$ since k is algebraically closed, and in this case one has $m = n$.

Lemma (18.11.5.2). — *Let k be a field, A a k -algebra local Noetherian complete, whose residue field is a finite extension of k , $\mathfrak{m}$ its maximal ideal, $\mathfrak{p} \subset \mathfrak{m}$ a prime ideal of A . Let m be the minimum number of generators of the $A_{\mathfrak{p}}$ -module $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$, and let $(x_i)_{1 \leq i \leq m}$ be a family of elements of $\mathfrak{m}$ not belonging to $\mathfrak{p}$.*

Then there exist m invertible elements u_i ($1 \leq i \leq m$) of A such that if one sets $y_i = u_i x_i$, the canonical images of the dy_i in $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ generate this $A_{\mathfrak{p}}$ -module.

For every $x \in A$, denote by $\delta(x)$ the canonical image of dx ($= d_{A/k}x$) in $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p})$; since this $k(\mathfrak{p})$ -vector space is of rank m , it suffices (by virtue of Nakayama's lemma) to prove that one can determine the u_i such that the $\delta(u_i x_i)$ form a free system. We argue by induction and suppose that, for some integer $r < m$, we have determined the u_i ($1 \leq i \leq r$) such that the $\delta(u_i x_i)$ for $1 \leq i \leq r$ are linearly independent over $k(\mathfrak{p})$; if $\delta(x_{r+1})$ is not a linear combination of the $\delta(u_i x_i)$ for $1 \leq i \leq r$, it suffices to take $u_{r+1} = 1$ to continue the induction. In the contrary case, note that for $u \in A$, $\delta(ux_{r+1})$ is the canonical image of $d(u)x_{r+1} + u(dx_{r+1})$; since $u(dx_{r+1})$ has a canonical image that is a linear combination of $\delta(u_i x_i)$ for $1 \leq i \leq r$, and since on the other hand the canonical image of x_{r+1} in $k(\mathfrak{p})$ is $\neq 0$, one sees that it suffices to prove that there exists an invertible element $u \in A$ such that $\delta(u)$ is not a linear combination of the $\delta(u_i x_i)$ for $1 \leq i \leq r$. Now, it follows from (0, 20.7.15) and (0_I, 7.2.9) that the $\delta(x)$ generate the $k(\mathfrak{p})$ -vector space $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p})$; since by hypothesis $r < m$, there exists therefore $z \in A$ such that $\delta(z)$ is not a linear combination of the $\delta(u_i x_i)$ for $1 \leq i \leq r$; if $z \notin \mathfrak{m}$, one takes $u = z$; otherwise, $1 + z$ is invertible and one has $\delta(1 + z) = \delta(z)$ since $d(1 + z) = d(z)$; one takes then $u = 1 + z$, which completes the proof of (18.11.5.2).

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Let us now return to the proof of the implication $a) \Rightarrow b)$ in (18.11.5). Note first that since $\mathfrak{p} \neq \mathfrak{m}$, there exists a system of parameters $(x_i)_{1 \leq i \leq n}$ of A (with $n = \dim(A)$) such that $x_i \notin \mathfrak{p}$ for $1 \leq i \leq n$: indeed, one cannot have $x_i \in \mathfrak{p}$ for every i , otherwise $A/\mathfrak{p}$ would be of finite length, and $\mathfrak{p}$ would be maximal, contrary to the hypothesis. But if for instance $x_1 \notin \mathfrak{p}$, it suffices, for each index i such that $x_i \in \mathfrak{p}$, to replace x_i by $x_1 + x_i$, to have a system of parameters none of whose elements belong to $\mathfrak{p}$. The hypothesis $a)$ and the relation $\dim(A/\mathfrak{q}) = n$ imply, by virtue of (18.11.5.1), that $m \geq n$; one can therefore consider a family $(x_i)_{1 \leq i \leq m}$ of elements $x_i \notin \mathfrak{p}$, whose n first form a system of parameters of A . Multiplying moreover the x_i by invertible elements u_i of A , one can, thanks to (18.11.5.2), suppose that the images of dx_i ($1 \leq i \leq m$) in $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ generate this $A_{\mathfrak{p}}$ -module, and the multiplication by the u_i has not altered the fact that the x_i for $1 \leq i \leq n$ form a system of parameters. Consider then the k -homomorphism local $u : B \rightarrow A$ such that $u(T_i) = x_i$ for $1 \leq i \leq m$ (Bourbaki, *Alg. comm.*, chap. III, § 4, n° 5, prop. 6); since the x_i generate an ideal of definition of A , it follows from (0_I, 7.4.4 and 7.4.3) that u makes A a finite B -algebra. One has therefore (18.11.3.5) the exact sequence

$$(\widehat{\Omega}_{B/k}^1) \otimes_B A \xrightarrow{v} \widehat{\Omega}_{A/k}^1 \xrightarrow{w} \Omega_{A/B}^1 \longrightarrow 0$$

But the $d_A x_i$ are the canonical images by v of the elements $d_B T_i \otimes 1$ (0, 20.5.2.6). If one localises the preceding exact sequence at $\mathfrak{p}$, one sees hence that $v_{\mathfrak{p}} : (\widehat{\Omega}_{B/k}^1) \otimes_B A_{\mathfrak{p}} \rightarrow (\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ is surjective, and hence one has $0 = (\Omega_{A/B}^1)_{\mathfrak{p}} = \Omega_{A_{\mathfrak{p}}/B_{\mathfrak{p}}}^1$ (where $\mathfrak{p}$ is the inverse image of $\mathfrak{p}$ in A , cf. (16.4.15)). By virtue of (17.4.1), this implies property $b)$ of (18.11.5). $\square$

Remark (18.11.6). — In the statement of (18.11.5), one cannot suppress the hypothesis $\mathfrak{p} \neq \mathfrak{m}$. Indeed, for every k -homomorphism local $u : B \rightarrow A$, where $B = k[[T_1, \dots, T_r]]$ (r any integer) making A a finite B -algebra, $\mathfrak{m}$ is the only point of $\text{Spec}(A)$ above the maximal ideal $\mathfrak{n}$ of B ; the morphism $\text{Spec}(A) \rightarrow \text{Spec}(B)$ can be unramified at $\mathfrak{m}$ only if $A/\mathfrak{n}A$ is a field, separable extension of k (17.4.1), which implies that the residue field K of A is a separable extension of k . If this condition is not satisfied, the conclusion of (18.11.5) can never be verified for $\mathfrak{p} = \mathfrak{m}$, whatever the integer m .

Corollary (18.11.7). — Let k be a field, A a k -algebra local Noetherian complete equidimensional, whose residue field is a finite extension of k , $\mathfrak{p}$ a prime ideal of A . Then the minimum number of generators of the $A_{\mathfrak{p}}$ -module $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ is at least equal to $\dim(A)$.

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This is a particular case of (18.11.5.1).

Corollary (18.11.8). — Let k be a field, A a k -algebra local Noetherian complete intègre, whose residue field is a finite extension of k . If K is the field of fractions of A , one has

$$\text{rg}_K((\widehat{\Omega}_{A/k}^1) \otimes_K K) \geq \dim(A).$$

It suffices to take $\mathfrak{p} = 0$ in ((18.11.7)).

Corollary (18.11.9). — *Let k be a field, A a k -algebra local Noetherian complete of dimension n , whose residue field is a finite extension of k . Let $\mathfrak{p}$ be a prime ideal of A distinct from the maximal ideal, and containing a unique minimal prime ideal $\mathfrak{q}$ such that $\dim(A/\mathfrak{q}) = n$. Suppose that the $A_{\mathfrak{p}}$ -module $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ admits n generators. Then:*

- label=(i) The $A_{\mathfrak{p}}$ -module $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ is free of rank n .*
- lbbel=(ii) There exists a k -homomorphism local $u : B \rightarrow A$, where $B = k[[T_1, \dots, T_n]]$, making A a B -algebra finite, and such that the corresponding morphism $\mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$ is étale at the point $\mathfrak{p}$.*
- lcbel=(iii) The k -algebra $A_{\mathfrak{p}}$ is geometrically regular.*

Proof. Let us first prove (ii); by virtue of ((18.11.5)), there exists a local homomorphism $u : B = k[[T_1, \dots, T_n]] \rightarrow A$ making A a finite B -algebra, and such that the corresponding morphism $\mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$ is unramified at the point $\mathfrak{p}$; set $\mathfrak{r} = u^{-1}(\mathfrak{p})$. The hypothesis on $\mathfrak{q} \subset \mathfrak{p}$, and the fact that A is a quotient of a regular ring (0, 19.8.8) imply that $\dim(A_{\mathfrak{p}}) = n - \dim(A/\mathfrak{p})$ (0, 16.5.12). Similarly $\dim(B_{\mathfrak{r}}) = n - \dim(B/\mathfrak{r})$. Finally, since the morphism $\mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$ is unramified at the point $\mathfrak{p}$, the fibre of this morphism at the point $\mathfrak{r}$ is of dimension 0, hence (0, 16.3.9) $\dim(A/\mathfrak{p}) \leq \dim(B/\mathfrak{r})$ and consequently $\dim(B_{\mathfrak{r}}) \leq \dim(A_{\mathfrak{p}})$. One therefore concludes from ((18.10.1)) that the morphism $\mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$ is étale at the point $\mathfrak{p}$.

Assertion (iii) results from the fact that $A_{\mathfrak{p}}$ is formally smooth over $B_{\mathfrak{r}}$ and that $B_{\mathfrak{r}}$ is formally smooth over k (for the preadic topologies) (0, 19.3.4 and 19.3.5), hence $A_{\mathfrak{p}}$ is formally smooth over k for its preadic topology, and consequently is geometrically regular over k (0, 22.5.8).

Let us finally prove (i). Let k' be an algebraically closed extension of k , and consider the k' -algebra semi-local $A' = A \otimes_k k'$. Viewing A as a B -module of finite type via the homomorphism u , one can write, up to canonical isomorphism, $A' = A \otimes_B (B \widehat{\otimes}_k k')$ ((7.5.7.1), where we recall that it is not necessary to suppose the residue field of B is a finite extension of k). Setting $B' = B \widehat{\otimes}_k k'$, which identifies canonically with the formal power series algebra $k'[[T_1, \dots, T_n]]$. Since the morphism $\mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$ is finite and étale at the point $\mathfrak{p}$, the morphism $\mathrm{Spec}(A') \rightarrow \mathrm{Spec}(B')$ is finite, and étale at every point $\mathfrak{p}'$ above $\mathfrak{p}$ ((17.3.3)); moreover, A' is a direct sum of local rings A'_i of dimension n ((18.11.4)) and $\mathfrak{p}'$ identifies with a prime ideal $\mathfrak{p}'_i$ of one of the A'_i . The same reasoning as above then proves that $(A'_i)_{\mathfrak{p}'_i}$ is regular. The same reasoning that proves (iii) shows that $(\widehat{\Omega}_{A'/k'}^1)_{\mathfrak{p}'_i}$ is a free $(A'_i)_{\mathfrak{p}'_i}$ -module of finite type. But since $(\widehat{\Omega}_{A'/k'}^1)_{\mathfrak{p}'}$ is isomorphic to $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} (A'_i)_{\mathfrak{p}'_i}$ ((18.11.4)) and $(A'_i)_{\mathfrak{p}'_i}$ is a faithfully flat $A_{\mathfrak{p}}$ -module, one sees that $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ is a free $A_{\mathfrak{p}}$ -module of finite type ((2.5.2)); moreover, by virtue of ((18.11.5.1)) and the hypothesis $\dim(A/\mathfrak{q}) = n$, one concludes that $(\widehat{\Omega}_{A/k}^1)_{\mathfrak{p}}$ is a free $A_{\mathfrak{p}}$ -module of rank n . $\square$

18.12. Applications of étale localisation to quasi-finite morphisms (generalisations of earlier results). The results of this section were communicated to us by P. Deligne.

Theorem (18.12.1). — *Let $f : X \rightarrow Y$ be a morphism locally of finite type, x a point of X , $y = f(x)$. Suppose that x is an isolated point of the space $f^{-1}(y)$. Then there exists an étale morphism $Y' \rightarrow Y$, a point x' of $X' = X \times_Y Y'$ above x , and an open neighbourhood V' of x' in X' such that, setting $f' = f|_{(V')} : X' \rightarrow Y'$, $f'|_{V'}$ is a finite morphism. If moreover f is separated, V' is both open and closed, and X' is thus the sum of two sub-preschemes induced on opens of X' , one of which is finite over Y and contains x' .*

Proof. The last assertion follows from the fact that, if f is separated, it suffices to note that f' is also separated; hence, if $j' : V' \rightarrow X'$ is the injection, the morphism $f' \circ j'$ is finite and therefore j' is also finite ((II, 6.1.5)), and consequently $V' = j'(V')$ is closed in X' ((II, 6.1.1)).

The question being local on X , we may suppose $Y = \mathrm{Spec}(B)$ and $X = \mathrm{Spec}(C)$ affine, B being an A -algebra of finite type, and $C = A[T_1, \dots, T_r]$ and $\mathfrak{J}$ is an ideal of type fini of C contained in $\mathfrak{J}$, so that $\mathfrak{J}$ is the union of the filtering family of the $\mathfrak{J}_{\lambda}$. If X and $X_{\lambda} = \mathrm{Spec}(C/\mathfrak{J}_{\lambda})$ are considered as closed sub-preschemes of $Z = \mathrm{Spec}(C)$, one has, for the underlying spaces, $X = \bigcap X_{\lambda}$; if $f_{\lambda} : X_{\lambda} \rightarrow Y$ is the structural morphism, one deduces that $f^{-1}(y) = \bigcap_{\lambda} f_{\lambda}^{-1}(y)$, and the sets $f_{\lambda}^{-1}(y)$ are closed in the Noetherian space $\mathrm{Spec}(k(y)[T_1, \dots, T_r])$. One can therefore suppose that $f^{-1}(y) = \{x\}$. One can moreover suppose that the set of points of X isolated in their fibre is open and that we can restrict to the case where f is quasi-finite. Let A'' be the henselisation of $\mathcal{O}_{Y,y}$, and set $Y'' = \mathrm{Spec}(A'')$. If

$X'' = X \times_Y Y''$, $f'' = f_{(Y'')} : X'' \rightarrow Y''$ is separated, quasi-finite, and of finite presentation; as A'' is henselian, there exists for every point $x'' \in X''$ above x an open neighbourhood, open and closed V'' of x'' in X'' which is finite over Y'' ((18.5.11, c)); as the immersion $V'' \rightarrow X''$ is open and closed, hence quasi-compact, of finite presentation ((1.6.2)), hence $f''|_{V''}$ is of finite presentation.

A'' being, by definition, the inductive limit of a filtering family of essentially étale strictly local (A_λ) of $\mathcal{O}_{Y,y}$ -algebras; each A_λ can be supposed to be an $\mathcal{O}_{Y,y}$ -algebra of finite presentation ((I, 8.1.2, a) and ((I, 8.1.2, a))). Finally, using the theorem of inductive limits ((IV, 8.10.5)), the fact that Y is the projective limit of a filtering family (Y'_α) of affine schemes over Y . If x'_α is the projection of x' onto $X'_\alpha = X \times_Y Y'_\alpha$, one can suppose the neighbourhood V'_α is of the form $V'_\alpha \times_{Y'_\alpha} Y''$ where V'_α is an open neighbourhood of x'_α in X'_α ((8.2.11)). Finally, V'' being of finite presentation over Y'' , we deduce that for α sufficiently large, V'_α is finite over Y'_α ((IV, 8.10.5, (x))) and that for any such α it is convenient. $\square$

Remark (18.12.2). — In the preceding demonstration, we see (taking into account (18.6.2)) that one has constructed a Y' answering the question, and such in addition that if $y' = f'(X')$, the homomorphism $k(y) \rightarrow k(y')$ is bijective.

Corollary (18.12.3). — *Let $f : X \rightarrow Y$ be a morphism locally of finite type and separated, y a point of Y such that the subspace $f^{-1}(y)$ is finite and discrete. Then there exists an étale morphism $Y' \rightarrow Y$, a point $y' \in Y'$ above y , such that $k(y') = k(y)$, and a decomposition of X' into a sum of two sub-preschemes X'_1, X'_2 induced on opens of X' , such that the restriction $f' = f_{(Y')} : X' \rightarrow Y'$ to X'_1 is a finite morphism and that $X'_2 \cap f'^{-1}(y') = \emptyset$.*

If n is the number of points of $f^{-1}(y)$, one argues by induction on n , the corollary being trivial for $n = 0$. Let x be a point of $f^{-1}(y)$; by virtue of (18.12.1) and (18.12.2) there is an étale morphism $Y_1 \rightarrow Y$, a point y_1 of Y_1 above y such that $k(y_1) = k(y)$, and if one sets $S_1 = X \times_Y Y_1$, $f_1 = f_{(S_1)} : S_1 \rightarrow Y_1$, there exists a point x_1 of $f_1^{-1}(y_1)$ such that S_1 is the sum of two opens V_1, X_1 , V_1 being finite over Y_1 and a neighbourhood of x_1 . By virtue of the relation $k(y_1) = k(y)$, the fibre $f_1^{-1}(y_1)$ in S_1 is isomorphic to $f^{-1}(y)$, hence $X_1 \cap f_1^{-1}(y_1)$ is finite, discrete and has $n - 1$ points. As $f_1|_{X_1}$ is locally of finite type and separated, one applies the induction hypothesis to this morphism: there is an étale morphism $Y' \rightarrow Y_1$, a point $y' \in Y'$ above y_1 such that $k(y') = k(y_1)$, and, if $X' = X \times_Y Y'$ and if $p : X' \rightarrow S_1$ is the canonical projection, one has a decomposition of $p^{-1}(X_1)$ into a sum of two sub-preschemes U' and X'_2 induced on opens of X' such that U' is finite over Y' . Setting $V' = p^{-1}(V_1)$ and X'_1 the sum of U' and V' , the answer to the question is obtained by taking X'_1 the sum of U' and V' . The following corollary improves ((IV, 8.11.1)):

Corollary (18.12.4). — *For a morphism $f : X \rightarrow Y$ to be finite, it suffices that it be separated, universally closed, locally of finite type, and that, for every $y \in Y$, $f^{-1}(y)$ be a finite discrete space; in particular, a morphism that is proper and quasi-finite is finite.*

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Indeed, one can apply the corollary (18.12.3) to any point y of Y . With the notations of this corollary, f' is a closed morphism, hence, since X'_2 is closed in X' , $f'(X'_2)$ is closed in Y' and does not contain y' ; it follows that there exists an open neighbourhood U' of y' in Y' such that $U' \rightarrow Y$ is of finite type and that $f^{-1}(U') = X \times_Y U'$ is finite on U' . Let U be the image of U' in Y , which is an open of Y by virtue of (11.3.1) since $Y' \rightarrow Y$ is étale, hence of finite presentation; it is clear that $X \times_Y U' = f^{-1}(U) \times_U U'$. Considering now the morphism $U' \rightarrow U$ which is faithfully flat and quasi-compact, one deduces from ((IV, 2.7.1, (xv))) that the morphism f , restriction of f , is finite, which proves the corollary.

Remark (18.12.5). — One notes that in the proof of (18.12.4) one has not used the hypothesis that f is universally closed in its general form, but only that $f_{(Y')}$ is a closed morphism for every étale morphism $Y' \rightarrow Y$.

Corollary (18.12.6). — *Let $f : X \rightarrow Y$ be a morphism of preschemes. The following conditions are equivalent:*

label=a) f is a closed immersion.

lbbel=b) f is a proper monomorphism.

lbel=c) f is proper and, for every $y \in Y$, the $k(y)$ -prescheme $X_y = f^{-1}(y)$ is radiciel and geometrically reduced over $\text{Spec}(k(y))$ (that is to say, empty or $k(y)$ -isomorphic to $\text{Spec}(k(y))$).

This follows from (18.12.4) together with ((IV, 8.11.5)) and ((IV, 8.11.1)).

Proposition (18.12.7). — Let $f : X \rightarrow Y$ be a morphism locally of finite type, y a point of Y . For there to exist an open neighbourhood U of y such that the restriction $f^{-1}(U) \rightarrow U$ of f is a closed immersion, it is necessary and sufficient that X_y be radiciel and geometrically reduced over $k(y)$ (that is to say, empty or $k(y)$ -isomorphic to $\text{Spec}(k(y))$) and that there exist an open neighbourhood V of y such that the restriction $f^{-1}(V) \rightarrow V$ of f is a universally closed morphism.

The conditions being evidently necessary, let us show they are sufficient. One can restrict to the case where $X_y \neq \emptyset$. By virtue of the second condition, one can already suppose f universally closed. We show furthermore that we can suppose f affine; this follows from the following lemma:

Lemma (18.12.7.1). — Let $f : X \rightarrow Y$ be a closed morphism, $y \in Y$ a point such that every affine open neighbourhood of X_y contains a single affine open neighbourhood of X_y (a condition that is always satisfied when X_y is empty or reduced to a single point). Then there exists an affine open neighbourhood U of y such that $f^{-1}(U) \rightarrow U$ of f is an affine morphism.

Indeed, let U_0 be an open affine neighbourhood of y in Y and V an affine open neighbourhood of X_y contained in $f^{-1}(U_0)$. Since f is closed, there exists an open affine neighbourhood $U \subset U_0$ of y such that $f^{-1}(U) \subset V$. As we can suppose Y and X affine, we can suppose f proper, and it suffices to show that there exists an open neighbourhood U of y such that $f^{-1}(U) \rightarrow U$ is a quasi-finite morphism ((18.12.4)). But there exists a neighbourhood V of the set X_y (reduced to a single point) such that $f|_V$ is quasi-finite ((13.1.4)), and since f is closed there is a neighbourhood U of y such that $f^{-1}(U) \subset V$, which finishes the proof of (18.12.6).

Proposition (18.12.8). — Let $f : X \rightarrow Y$ be a morphism of preschemes. For f to be entire, it is necessary and sufficient that f be affine and universally closed.

The conditions are necessary ((II, 6.1.10)); let us show they are sufficient. We may suppose $Y = \text{Spec}(A)$ affine, hence also $X = \text{Spec}(B)$, and it suffices to show that every element $b \in B$ is integral over A ((II, 6.1.1)). Let B' be the sub- A -algebra of B generated by b , which is an A -algebra of finite type; setting $X' = \text{Spec}(B')$, so that $f : X \rightarrow Y$ factorises as $X \xrightarrow{g} X' \xrightarrow{h} Y$, where h is of finite type, and g is dominant since the homomorphism $B' \rightarrow B$ is injective ((I, 1.2.7)). As h is separated and $h \circ g$ universally closed, g is universally closed ((II, 5.4.3) and (II, 5.4.9)); one concludes ((II, 5.4.3) and (II, 5.4.9)) that h is also universally closed, hence proper since it is of finite type and separated. But then, for every $y \in Y$, the morphism $h^{-1}(y) \rightarrow \text{Spec}(k(y))$ deduced of h is quasi-finite; it follows from (18.12.4) that h is quasi-finite; and therefore from (18.12.4) that h is finite, which proves that B' is an A -algebra finite, and hence that b is integral over A . Q.E.D.

Remark (18.12.9). — One could ask whether a morphism $f : X \rightarrow Y$ be entire when one supposes only that it is separated, universally closed, and such that for every $y \in Y$, the morphism $f^{-1}(y) \rightarrow \text{Spec}(k(y))$ deduced of f is entire. It would suffice for this to show that under these conditions f is affine, or else that every fibre X_y is contained in an affine open neighbourhood. IV | 183

Corollary (18.12.10). — A morphism $f : X \rightarrow Y$ that is injective and universally closed is entire.

It suffices, by (18.12.8), to show that f is affine, which follows from the lemma (18.12.7.1) and the hypothesis.

Corollary (18.12.11). — Let $f : X \rightarrow Y$ be a morphism of preschemes (resp. a morphism locally of finite type). For f to be a universal homeomorphism (I, 2.4.2), it is necessary and sufficient that f be entire (resp. finite), radiciel, and surjective.

We know that the sufficient conditions are sufficient; they are also necessary by virtue of (18.12.10) and (18.12.4). The following proposition improves ((IV, 8.11.2)):

Proposition (18.12.12). — Every morphism $f : X \rightarrow Y$ that is quasi-finite and separated is quasi-affine.

Set $\mathcal{A} = f_*(\mathcal{O}_X)$, which is a $\mathcal{O}_Y$ -Algebra quasi-coherent since f is quasi-compact and separated (I, 1.7.4); let $Z = \operatorname{Spec}(\mathcal{A})$ ((II, 1.3.1)) so that f factorises canonically as $X \xrightarrow{g} Z \xrightarrow{h} Y$, h being affine and g an *open immersion*, or, what amounts to the same ((IV, 17.9.1)), that g is étale and radiciel. It suffices evidently to show that, for every $y \in Y$, g is étale and radiciel at every point of $f^{-1}(y)$. For every $y \in Y$, one can apply f the result of (8.12.3), whose notation we retain; since the morphism $Y' \rightarrow Y$ is flat, and f is quasi-compact and separated, one has, up to canonical isomorphism, $f'_*(\mathcal{O}_{X'}) = \mathcal{A}' \simeq \mathcal{A}'$ ((II, 2.3.1)), hence $Z' = Z \times_Y Y'$ is identified with $\operatorname{Spec}(f'_*(\mathcal{O}_{X'}))$, and in the canonical factorisation $X' \xrightarrow{g'} Z' \xrightarrow{h'} Y'$ of f' ((II, 1.2.7)), one has $h' = h_{(Y')}$ and $g' = g_{(Y')}$. This being so, the decomposition of X' into two sub-preschemes X'_1, X'_2 induces the decomposition of $\mathcal{A}'$ into a direct product of two $\mathcal{O}_{Y'}$ -Algebras quasi-coherent $\mathcal{A}'_1, \mathcal{A}'_2$, direct images of $\mathcal{O}_{X'_1}$ and $\mathcal{O}_{X'_2}$, so that Z' is identified with the sum $Z'_1 \amalg Z'_2$, where $Z'_i = \operatorname{Spec}(\mathcal{A}'_i)$ and $g'(X'_i) \subset Z'_i$ for $i = 1, 2$. As X'_1 is finite over Y' , $g'_1 = g'|_{X'_1}$ is affine; as $X'_2 \cap f'^{-1}(y') = \emptyset$, one sees that $g'_1 = g'|_{X'_1} : X'_1 \rightarrow Z'_1$ is an isomorphism. The morphism $Y' \rightarrow Y$ being étale, flat, and locally of finite presentation, one deduces first from ((IV, 17.7.4)) that g is étale at all points of X which are projections of points of $f'^{-1}(y')$, that is to say at all points of $f^{-1}(y)$ ((II, 1.2.5)). On the other hand, the morphism $g'_y : f'^{-1}(y') \rightarrow h^{-1}(y')$ deduced from g' is also radiciel at every point of $f^{-1}(y)$, which proves the proposition. The following result improves (8.12.6):

Corollary (18.12.13). — (“Main Theorem” of Zariski.) *Let Y be a prescheme quasi-compact and quasi-separated, $f : X \rightarrow Y$ a quasi-finite and separated morphism. Then there exists a factorisation of f*

$$X \xrightarrow{g} Z \xrightarrow{u} Y$$

where g is an open immersion (not necessarily quasi-compact) and u a finite morphism.

Indeed, it follows from (18.12.12) that f is quasi-affine, hence quasi-projective. One can then apply (18.12.8), where the hypothesis on the existence of an ample $\mathcal{O}_Y$ -Module can in fact be replaced by the assumption that Y is quasi-compact and quasi-separated: indeed, it follows from ((8.12.3)) that the existence of the announced factorisation in (18.12.8) is a local property on Y , and it suffices therefore to prove it when Y is affine.

Remark (18.12.14). — One can give of (18.12.13) a proof analogous to that of (18.12.12) not using (18.12.8) (but using (18.12.3)), hence (18.5.11), which itself uses the “Main theorem” in its local form ((8.12.9)). Let us keep the notations of the proof of (18.12.12), and let $\mathcal{B}$ be the $\mathcal{O}_Y$ -Algebra quasi-coherent, integral closure of $\mathcal{O}_Y$ in $\mathcal{A}$ ((II, 6.3.2)). Setting $T = \operatorname{Spec}(\mathcal{B})$, it suffices, by virtue of ((8.12.3)), to show that the Y -morphism $g : X \rightarrow T$ corresponding to the canonical injection $\mathcal{B} \rightarrow \mathcal{A}$ is an open immersion, and for this it suffices to show ((IV, 17.9.1)) that g is étale and radiciel. With the notations of (18.12.12), one can suppose that $Y = \operatorname{Spec}(C)$ and $Y' = \operatorname{Spec}(C')$ are affine, C' being a C -algebra étale ((IV, 17.3.2)); hence $\mathcal{A} = \tilde{A}$, where A is a C -algebra, $\mathcal{A}' = \tilde{A}'$ with $A' = A \otimes_C C'$, and $\mathcal{B} = \tilde{B}$, where B is the integral closure of C in A . The algebra $\mathcal{A}'$ is isomorphic to the product $A'_1 \times A'_2$, where A'_1 is a finite C' -algebra. It suffices to show that $B' = B \otimes_C C'$, which contains A'_1 ; indeed, B' then decomposes into a product $A'_1 \times A'_2$ where A'_2 is a sub- C' -algebra of A'_2 , and if one sets $T'_2 = \operatorname{Spec}(A'_2)$, $T' = T \times_Y Y'$ will be a sum of $Z'_1 = \operatorname{Spec}(A'_1)$ and $T'_2 = \operatorname{Spec}(A'_2)$, and $g'|_{X'_1}$ will be an isomorphism of X'_1 onto Z'_1 , which will permit to conclude as in (18.12.12).

To show that B' contains A'_1 , it suffices evidently to show the following proposition, which partially extends ((6.14.4)) without the Noetherian hypotheses:

Proposition (18.12.15). — *Let C be a ring, C' a C -algebra étale, A a C -algebra, B the integral closure of C in A ; set $A' = A \otimes_C C'$, $B' = B \otimes_C C'$, B' identifying to a sub-algebra of A' ; then B' is the integral closure of C' in A' .*

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Proof. Considering the increasing filtering family of sub- C -algebras of finite type of A and reasoning as in ((IV, 6.14.4), (II,)), one can first suppose that A is a C -algebra of finite presentation, hence of the form $E/\mathfrak{J}$, where $E = C[T_1, \dots, T_r]$ and $\mathfrak{J}$ is an ideal of finite type of E ; one then has $A' = E'/\mathfrak{J}'E'$, where $E' = C'[T_1, \dots, T_r]$. Let $(\mathfrak{J}_\lambda)$ be the increasing filtering family of ideals of finite type of E contained in $\mathfrak{J}$, so that A is the inductive limit of the $E/\mathfrak{J}_\lambda$; if B_λ is the integral closure of C in $E/\mathfrak{J}_\lambda$, B is the inductive limit of the B_λ , as the reasoning of ((IV, 5.13.4)) shows. Similarly, the integral

closure of C' in $E'/\mathfrak{J}E'$ is the inductive limit of the integral closures of C' in $E'/\mathfrak{J}_\lambda E'$; if one proves that this last is equal to $B_\lambda \otimes_C C'$, it will follow that $B' = \varinjlim (B_\lambda \otimes_C C')$ is the integral closure of C' in A' . One is thus reduced to proving the proposition when A is a C -algebra of finite presentation.

Let us then show that one can reduce to the case where C is Noetherian. Indeed, C is the union of the filtering family of its sub- $\mathbb{Z}$ -algebras of finite type C_α , hence it follows from ((IV, 17.7.8)) that there exists an index α and a C_α -algebra étale C'_α such that $C' = C'_\alpha \otimes_{C_\alpha} C$; moreover C' is the inductive limit of the $C'_\beta = C'_\alpha \otimes_{C_\alpha} C_\beta$ for $\beta \geq \alpha$. One can furthermore suppose, since A is a C -algebra of finite presentation, that $A = A_\alpha \otimes_{C_\alpha} C$, where A_α is a C_α -algebra of finite type, and A is the inductive limit of the $A_\beta = A_\alpha \otimes_{C_\alpha} C_\beta$ for $\beta \geq \alpha$ ((8.9.1)). The reasoning of ((IV, 5.13.4)) shows that B is the inductive limit of the integral closures B_β of C_β in A_β . Similarly $A'_\beta = A_\beta \otimes_{C_\beta} C'_\beta$ for $\beta \geq \alpha$, hence by the same reasoning as above, it will suffice to prove that $B'_\alpha = B_\alpha \otimes_{C_\alpha} C'_\alpha$ is the integral closure of C'_α in A'_α .

Once reduced to the case where C is Noetherian, the proposition becomes a particular case of ((IV, 6.14.4)). One should however observe that, when C' is supposed étale over C Noetherian, the proof of ((IV, 6.14.4)) no longer requires the delicate theorem ((IV, 6.14.1)). Indeed, the successive reductions of the proof of ((IV, 6.14.4)) lead back, *without using* ((IV, 6.14.1)), to the case where C is integral, A the field of fractions of C , and B is then a normal ring and the morphism $\text{Spec}(B') \rightarrow \text{Spec}(B)$ is étale, hence B' is a normal ring by ((IV, 17.5.7)); the reasoning concludes by appealing only to the elementary lemma ((IV, 6.14.1.1)), and not to the difficult part of the proof of ((IV, 6.14.1)). $\square$

Proposition (18.12.16). — *Let $g : Y \rightarrow S$ be a morphism quasi-compact, $f : X \rightarrow Y$ a separated and quasi-finite morphism. For every $\mathcal{O}_Y$ -Module invertible $\mathcal{L}$, ample relatively to S ((II, 4.6.1)), the $\mathcal{O}_X$ -Module $f^*(\mathcal{L})$ is ample relatively to S .*

Indeed, by virtue of (18.12.12), the morphism f is quasi-affine, hence ((II, 5.1.6)) the $\mathcal{O}_X$ -Module $\mathcal{O}_X$ is ample for f . One deduces ((II, 4.6.13, (iii))) that $\mathcal{O}_X \otimes_{\mathcal{O}_Y} f^*(\mathcal{L}^{\otimes n}) = f^*(\mathcal{L}^{\otimes n}) = f^*(\mathcal{L})^{\otimes n}$ is ample relatively to S for n large enough. Since $f^*(\mathcal{L}^{\otimes n}) = (f^*(\mathcal{L}))^{\otimes n}$, this implies that $f^*(\mathcal{L})$ is ample relatively to S ((II, 4.6.9, (i))).

Corollary (18.12.17). — *Let Z be an affine scheme, $h : X \rightarrow Z$ a morphism of finite type, $\mathcal{L}$ an $\mathcal{O}_X$ -Module invertible. With the notations of ((II, 4.5.2)), the conditions of ((II, 4.5.2)) (which is equivalent to saying that $\mathcal{L}$ is ample) are also equivalent to each of the following:*

label=a") *h is separated, one has $G(\varepsilon) = X$ and the canonical morphism $u : G(\varepsilon) \rightarrow \text{Proj}(S)$ has finite and discrete fibres.*

label=b") *One has $G(\varepsilon) = X$ and the morphism canonical u is radiciel.*

If $Z = \text{Spec}(A)$, the ring S is canonically equipped with a structure of a graded A -algebra, and one has $h = g \circ u$ where $g : \text{Proj}(S) \rightarrow Z$ is a separated morphism ((II, 2.4.2)), and one has $h = g \circ u$. As every radiciel morphism is separated (I, 1.8.7.1, b'') implies that h is separated, hence b''). As the condition b) of ((II, 4.5.2)) obviously implies b''), it remains to show that b'') implies that $\mathcal{L}$ is ample. Since $h = g \circ u$ is of finite type by hypothesis and g is separated, u is also of finite type ((I, 6.3.4)), hence the hypothesis b'') implies that u is quasi-finite and separated ((I, 5.5.1)). To show that $\mathcal{L}$ is ample, we will apply (18.12.16). Setting $Y = \text{Proj}(S)$, since X is quasi-compact, there exists an integer $n > 0$ and a finite number of elements $f_j \in S_n$ such that the inverse images $u^{-1}(D_+(f_j))$ cover X ; one can therefore consider u as a morphism quasi-finite and separated from X to the open $Y' = \bigcup D_+(f_j)$ of Y . Consider then the $\mathcal{O}_X$ -Module invertible $\mathcal{L}' = \mathcal{O}_Y(n)|_{Y'}$ ((II, 2.5.8)); it is very ample relatively to Z ((II, 4.4.3)), and $u^*(\mathcal{L}')$ is none other by definition than $\mathcal{L}^{\otimes n}$. We leave to the reader the task of stating conditions equivalent to those of ((II, 4.6.3)) for $\mathcal{O}_X$ -Modules relatively ample.

§19. REGULAR IMMERSIONS AND NORMAL FLATNESS

The present paragraph is devoted, on the one hand, to a study of *regular immersions* ((16.9.2)) $Y \rightarrow X$ of preschemes, notably in the case where X and Y are flat over a same prescheme S ; on the other hand, it gives complements on M-regular suites and normal flatness, generalising notably results on flatness established by H. Hironaka in his theory of resolution of singularities [?].

19.1. Properties of regular immersions.

Proposition (19.1.1). — *Let X be a locally Noetherian prescheme, $j : Y \rightarrow X$ an immersion, y a regular point of Y . For j to be a regular immersion at the point y , it is necessary and sufficient that X be regular at the point $j(y)$.*

Taking into account ((16.9.10)), one is reduced to proving the

Corollary (19.1.2). — *Let A be a local Noetherian ring, $\mathfrak{J}$ an ideal of A ; suppose that the quotient ring $B = A/\mathfrak{J}$ is regular. For A to be regular, it is necessary and sufficient that the ideal $\mathfrak{J}$ be regular.*

Indeed, if A is regular, we know (0, (17.1.9)) that $\mathfrak{J}$ is generated by a part of a system of parameters of A , hence $\mathfrak{J}$ is regular. Inversely, if $\mathfrak{J}$ is regular, and if $(x_i)_{1 \leq i \leq r}$ is an A -regular suite generating $\mathfrak{J}$, (x_i) forms part of a system of parameters of A (0, (16.4.1)), hence one deduces from (0, (17.1.7)) that A is regular.

Definition (19.1.3). — Let X be a prescheme, Y a sub-prescheme of X , U an open of X containing Y and such that Y is defined by an Ideal $\mathcal{J}$ of $\mathcal{O}_U$; suppose that $\mathcal{J} / \mathcal{J}^2$ is an $\mathcal{O}_U / \mathcal{J}$ -Module locally free (which is the case if Y is quasi-regularly immersed in X). For every $y \in Y$, we call the *transverse codimension* of Y in X at the point y and write $\text{codim}_y^*(Y, X)$ the rank of the free $(\mathcal{O}_y / \mathfrak{J}_y)$ -module $\mathcal{J}_y / \mathcal{J}_y^2$. If $f : Z \rightarrow X$ is an immersion with image Y , we similarly call *transverse codimension* of f at the point $z \in Z$ the transverse codimension in X at the point $f(z)$ of the sub-prescheme Y .

Proposition (19.1.4). — *Let X be a locally Noetherian prescheme, Y a sub-prescheme of X regularly immersed; then, for every $y \in Y$, one has*

$$(6) \quad \text{codim}_y^*(Y, X) = \text{codim}_y(Y, X).$$

By virtue of ((5.1.3.2)), $\text{codim}_y(Y, X)$ is equal to the least value of the dimensions of the local rings $A = \mathcal{O}_{X,z}$, where z is a generic point of an irreducible component of Y containing y ; as such a point z is contained in every neighbourhood of y in X , $\mathcal{J}_z / \mathcal{J}_z^2$ is an A -module free of rank $n = \text{codim}_y^*(Y, X)$; it amounts to seeing that $\dim(A) = n$. But, since z is a maximal point of the sub-prescheme Y defined by $\mathcal{J}$, one has $\dim(\mathcal{O}_{X,z} / \mathcal{J}_z) = 0$, hence $\mathcal{J}_z$ is an *ideal of definition* of the local Noetherian ring A ; moreover, the hypothesis that $\mathcal{J}$ is quasi-regular implies that $\mathcal{J}_z^m / \mathcal{J}_z^{m+1}$ is an $(A / \mathcal{J}_z)$ -module free of rank $\binom{m+n-1}{n-1}$; if r is the length of the Artinian ring $A / \mathcal{J}_z$, the length of $\mathcal{J}_z^m / \mathcal{J}_z^{m+1}$ is thus $r \binom{m+n-1}{n-1}$; the Hilbert-Samuel polynomial of A for the $\mathcal{J}_z$ -adic filtration is thus of degree n , which proves the proposition (0, (16.2.3)).

By virtue of ((19.1.4)), we shall henceforth say “codimension” instead of “transverse codimension” when it concerns a sub-prescheme Y regularly immersed in X (even when X is not locally Noetherian).

Proposition (19.1.5). — (i) *For an immersion $j : Y \rightarrow X$ to be open, it is necessary and sufficient that it be regular and of codimension 0 at every point.* IV | 186

(ii) *Let $f : Y \rightarrow X$ be a regular immersion (resp. quasi-regular), $g : X' \rightarrow X$ a flat morphism, $Y' = Y \times_X X'$, $f' = f_{(X')} : Y' \rightarrow X'$. Then f' is a regular immersion (resp. quasi-regular), and for every $y' \in Y'$, if y is the projection of y' in Y , the codimension of f' at the point y' is equal to the codimension of f at the point y . Inversely, if f is an immersion and $g : X' \rightarrow X$ a faithfully flat and quasi-compact morphism, then, if f' is a quasi-regular immersion, so is f .*

(iii) *If $f : Y \rightarrow X$ and $g : Z \rightarrow Y$ are regular immersions, then $f \circ g : Z \rightarrow X$ is a regular immersion, and for every $z \in Z$, the codimension of $f \circ g$ at the point z is the sum of the codimension of g at the point z and of the codimension of f at the point $g(z)$. Moreover, the suite of canonical homomorphisms ((16.2.7.1))*

$$(7) \quad 0 \longrightarrow g^*(\mathcal{N}_{Y/X}) \longrightarrow \mathcal{N}_{Z/X} \longrightarrow \mathcal{N}_{Z/Y} \longrightarrow 0$$

is exact, and for every $z \in Z$, there exists a neighbourhood of z in Z in which the restriction homomorphisms of this suite form a split exact suite.

(iv) *Let X be a locally Noetherian prescheme, $f : Y \rightarrow X$ and $g : Z \rightarrow Y$ two immersions, z a point of Z . The following conditions are equivalent:*

label=a) g is a regular immersion at the point z and f a regular immersion at the point $g(z)$.

lbbel=b) g and $f \circ g$ are regular immersions at the point z .

lbel=c) $f \circ g$ is a regular immersion at the point z , f is a regular immersion at the point $g(z)$, and the suite of canonical homomorphisms ((19.1.5.1)), restricted to a sufficiently small open neighbourhood of z in Z , is exact and split.

(v) Let X be a locally Noetherian prescheme, $f : Y \rightarrow X$, $g : Z \rightarrow Y$ two immersions, z a point of Z ; suppose that $y = g(z)$ is a regular point of Y and $x = f(g(z))$ a regular point of X (which implies, by ((19.1.1)), that f is a regular immersion at the point $g(z)$); then, for $f \circ g$ to be a regular immersion at the point z , it is necessary and sufficient that g be a regular immersion at the point z .

Proof. Assertion (i) is another way of expressing ((16.1.10)). To prove the first assertion of (ii), one can restrict to the case where Y is a closed sub-prescheme of X defined by a regular (resp. quasi-regular) Ideal $\mathcal{J}$; then Y' is the closed sub-prescheme of X' defined by the Ideal $g^*(\mathcal{J})\mathcal{O}_{X'}$ ((I, 4.4.5)) which here identifies with $g^*(\mathcal{J})$ since g is flat; by replacing X by a sufficiently small affine open, one can suppose that $\mathcal{J}$ admits a regular (resp. quasi-regular) suite of generators f_i (sections of $\mathcal{O}_X$ above X). If the suite (f_i) is regular, so is the suite $(f_i \otimes 1)$ (which generates $g^*(\mathcal{J})\mathcal{O}_X$) by virtue of (0, 15.2.5); the analogous result for quasi-regular suites follows at once from the flatness of g and the criterion ((16.9.4)). Similarly, if g is faithfully flat and quasi-compact, and if $g^*(\mathcal{J})$ is quasi-regular, $\mathcal{J}$ is of finite type by virtue of ((2.5.2)); by flatness, one has $g^*(\mathcal{J})/g^*(\mathcal{J})^2 = g^*(\mathcal{J}/\mathcal{J}^2)$, so, since $g^*(\mathcal{J}/\mathcal{J}^2)$ is locally free by hypothesis ((16.9.4)), the same holds for $\mathcal{J}/\mathcal{J}^2$ ((2.5.2)). Finally, condition (iii) of ((16.9.4)) relative to $g^*(\mathcal{J})$ entails the same condition for $\mathcal{J}$ by virtue of the faithful flatness of g ((2.2.7)). One concludes therefore from ((16.9.4)) that $\mathcal{J}$ is quasi-regular.

To prove (iii), one can similarly suppose that Y and Z are closed sub-preschemes of X , defined respectively by Ideals $\mathcal{J}, \mathcal{K}$ of $\mathcal{O}_X$ such that $\mathcal{J} \subset \mathcal{K}$; moreover, one can suppose that there exist sections f_i ($1 \leq i \leq m$) of $\mathcal{J}$ above X generating $\mathcal{J}$ and such that for $1 \leq i \leq m$, the endomorphism of $\mathcal{O}_X/(\sum_{j=1}^{i-1} f_j \mathcal{O}_X)$ defined by f_i is injective, and sections $\bar{f}_i$ ($m+1 \leq i \leq n$) of $\mathcal{K}/\mathcal{J}$ above X generating $\mathcal{K}/\mathcal{J}$ and such that for $m+1 \leq i \leq n$, the endomorphism of $(\mathcal{O}_X/\mathcal{J})/(\sum_{j=m+1}^{i-1} \bar{f}_j (\mathcal{O}_X/\mathcal{J}))$ defined by $\bar{f}_i$ is injective. For every $z \in Z$, let U be an open neighbourhood of z in X such that there exists a section f_i of $\mathcal{K}$ above U , whose class $\bar{f}_i|_U$ in $\Gamma(U, \mathcal{K}/\mathcal{J})$ for $m+1 \leq i \leq n$; then $(\mathcal{O}_X/\mathcal{J}|_U)/(\sum_{j=m+1}^{i-1} f_j \mathcal{O}_U)$, restricted to U , is isomorphic to $\mathcal{O}_U/((\mathcal{J}|_U) + \sum_{j=m+1}^{i-1} f_j \mathcal{O}_U) = \mathcal{O}_U/(\sum_{j=1}^{i-1} f_j \mathcal{O}_U)$, and one sees therefore that the suite $(f_i)_{1 \leq i \leq n}$ is regular in $\mathcal{O}_U$; since it generates $\mathcal{K}|_U$, this proves the first assertion of (iii).

To prove the last assertion of (iii), one can restrict (with the same notations) to the case where the images t_i of the f_i in $\mathcal{K}/\mathcal{K}^2$ (for $1 \leq i \leq n$) form a base of this $\mathcal{O}_X/\mathcal{K}$ -Module ((16.9.5)); for the same reason, one can suppose that the t_i such that $1 \leq i \leq m$ are the canonical images of elements of a base of the $\mathcal{O}_X/\mathcal{K}$ -Module $g^*(\mathcal{J}/\mathcal{J}^2)$, and that the canonical images $\bar{t}_i$ of the t_i in $(\mathcal{K}/\mathcal{J})/(\mathcal{K}/\mathcal{J})^2$ for $m+1 \leq i \leq n$ form a base of this $\mathcal{O}_X/\mathcal{K}$ -Module. This completes the proof of (iii).

Let us pass to the proof of (iv). The fact that a) implies c) results from (iii). Let us prove that c) implies a). Set $A = \mathcal{O}_{X,f(g(z))}$, so that $\mathcal{O}_{Y,g(z)} = A/\mathfrak{J}$, $\mathcal{O}_{Z,z} = A/\mathfrak{K}$ with $\mathfrak{J} \subset \mathfrak{K}$. By hypothesis, $\mathfrak{J}$ and $\mathfrak{K}$ are regular ideals of A and one has a split exact sequence

$$0 \longrightarrow \mathfrak{J}/\mathfrak{J}\mathfrak{K} \longrightarrow \mathfrak{K}/\mathfrak{K}^2 \longrightarrow (\mathfrak{K}/\mathfrak{J})/(\mathfrak{K}/\mathfrak{J})^2 \longrightarrow 0$$

(cf. ((16.2.7))). This entails that there exist elements f_j ($1 \leq j \leq r+s$) of $\mathfrak{K}$ whose images f'_j in $\mathfrak{K}/\mathfrak{K}^2$ form a base of this $(A/\mathfrak{K})$ -module, and such moreover that the f_j of index $j \leq r$ belong to $\mathfrak{J}$ and are such that their images in $\mathfrak{J}/\mathfrak{J}\mathfrak{K}$ form a base of this $(A/\mathfrak{K})$ -module. The f_j for $1 \leq j \leq r+s$ generate hence $\mathfrak{K}$, and the f_j for $1 \leq j \leq r$ generate $\mathfrak{J}$ since A is Noetherian (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, prop. 5); since $\mathfrak{K}$ is a regular ideal, it follows from ((16.9.5)) that the suite $(f_j)_{1 \leq j \leq r+s}$ is regular. By definition, the images of $f_{r+1}, \dots, f_{r+s}$ in $\mathfrak{K}/\mathfrak{J}$ form hence a regular suite in this $(A/\mathfrak{J})$ -module, which completes the proof that c) entails a) in (iv). $\square$

Corollary (19.1.6). — Let A be a local Noetherian ring, $\mathfrak{J}, \mathfrak{K}$ two ideals of A contained in the maximal ideal $\mathfrak{m}$ and such that $\mathfrak{J} \subset \mathfrak{K}$, $A' = A/\mathfrak{J}$, $\mathfrak{K}' = \mathfrak{K}/\mathfrak{J}$. Then:

label=(i) If $\mathfrak{J}$ and $\mathfrak{K}'$ are regular ideals, so is $\mathfrak{K}$.

lbel=(ii) If $\mathfrak{K}$ and $\mathfrak{K}'$ are regular ideals, so is $\mathfrak{J}$.

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lcbel=(iii) Suppose the rings A and $A' = A/\mathfrak{J}$ are regular. For the ideal $\mathfrak{K}$ to be regular, it is necessary and sufficient that $\mathfrak{K}'$ be so.

Assertion (i) is a particular case of ((19.1.5, iii)). To prove (ii), consider the surjective canonical homomorphism $u : \mathfrak{K}/\mathfrak{K}^2 \rightarrow \mathfrak{K}'/\mathfrak{K}'^2 = \mathfrak{K}/(\mathfrak{K}^2 + \mathfrak{J})$. Since by hypothesis $\mathfrak{K}/\mathfrak{K}^2$ and $\mathfrak{K}'/\mathfrak{K}'^2$ are $(A/\mathfrak{K})$ -modules free, whose ranks we denote p and q respectively, the kernel of u is an $(A/\mathfrak{K})$ -module projective (Bourbaki, *Alg. comm.*, chap. II, § 2, n° 2, prop. 4), hence free of rank p since $A/\mathfrak{K}$ is a local Noetherian ring (**0III**, (10.1.3)); in other words, there exists a base of $\mathfrak{K}/\mathfrak{K}^2$ whose first p elements form a base of $(\mathfrak{K}^2 + \mathfrak{J})/\mathfrak{K}^2$. This means (by virtue of Nakayama's lemma) that there exists a system of generators $(f_i)_{1 \leq i \leq p+q}$ of $\mathfrak{K}$, forming a regular suite in A , such that the suite $(f_i)_{1 \leq i \leq p}$ is formed of elements of $\mathfrak{J}$; it remains to show that this last suite generates $\mathfrak{J}$. For this, denoting by $\mathfrak{J}' \subset \mathfrak{J}$ the ideal it generates, one sees that one is reduced to proving the following lemma:

Lemma (19.1.6.1). — Let B be a local Noetherian ring with maximal ideal $\mathfrak{n}$, $(g_j)_{1 \leq j \leq q}$ a regular suite in B of elements of $\mathfrak{n}$, $\mathfrak{b}$ the ideal that it generates, $\mathfrak{c} \subset \mathfrak{b}$ an ideal such that the canonical images of the g_j in $B/\mathfrak{c}$ form a regular suite in this ring. Then $\mathfrak{c} = 0$.

The lemma being trivial for $q = 0$, one argues by induction on q . The induction hypothesis, applied to the ring B/g_1B and to the canonical images of the g_j ($2 \leq j \leq q$), shows that $\mathfrak{c} \subset g_1\mathfrak{b}$. Let $\mathfrak{b}$ be the ideal of B of elements $x \in B$ such that $g_1x \in \mathfrak{c}$; one therefore has $\mathfrak{c} = g_1\mathfrak{b}$, but since by hypothesis g_1 is regular in $B/\mathfrak{c}$, one has necessarily $\mathfrak{b} = \mathfrak{c}$, hence $\mathfrak{c} = g_1\mathfrak{c}$. Since $\mathfrak{c}$ is of finite type and g_1 is contained in the radical of B , Nakayama's lemma shows that $\mathfrak{c} = 0$.

Finally, taking into account ((16.9.10)), it is immediate that (v), as well as the equivalence of a) and b) in (iv), result from Corollary ((19.1.6)).

Remark (19.1.7). — (i) We ignore whether the composition of two quasi-regular immersions is quasi-regular.

(ii) In an annelated space in local rings X , for every Ideal $\mathcal{J}$ of $\mathcal{O}_X$, $\mathcal{O}_X/\mathcal{J}$ is of finite type and thus has closed support in X (**0I**, (5.2.2)). One can thus define as in ((I, 4.1.3) and (I, 4.2.1)) the notions of closed subspace (locally closed subspace) of X defined by an Ideal $\mathcal{J}$ of a sheaf $\mathcal{O}_U$, where U is an open of X , and the notion of immersion. The definitions of quasi-regular and regular immersions, of transverse codimension, and the results of ((19.1.4, i) to [iv]) remain valid without modification, assuming in the second assertion of (i) and in (iv) that the sheaf $\mathcal{O}_X$ is coherent and the local rings $\mathcal{O}_{X,x}$ are Noetherian; as for (ii), one must define Y' as the closed subspace defined by the $\mathcal{O}_X$ -Ideal $g^*(\mathcal{J})$; the proofs are unchanged.

(19.1.8). We have already pointed out (notably in ((16.9.6, ii))) that in non-Noetherian rings, the notion of “regular suite” of elements of a ring does not possess the properties that would render it usable. Recent researches seem to show that the notion which should in this case be substituted for that of regular suite is that of a suite $\mathbf{f} = (f_1, \dots, f_n)$ having the property that the Koszul complex $K_\bullet(\mathbf{f}, M)$ ((III, 1.1.3)) has zero homology in degrees > 0 (cf. ((III, 1.1.4))); see in particular A. Grothendieck, Séminaire de géométrie algébrique, 1966, to appear shortly.

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19.2. Transversally regular immersions.

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Definition (19.2.1). — Let $u : X \rightarrow S$ be a morphism of preschemes, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent. One says that a suite $(f_i)_{1 \leq i \leq n}$ of sections of $\mathcal{F}$ above X is *transversally $\mathcal{F}$ -regular relatively to S* (or relatively to u) if the suite (f_i) is $\mathcal{F}$ -regular and such that the $\mathcal{O}_X$ -Modules $\mathcal{F}/(\sum_{i < j} f_i \mathcal{F})$ are S -flat for $0 \leq i \leq n$. One says that an Ideal quasi-coherent $\mathcal{J}$ of $\mathcal{O}_X$ is *transversally regular relatively to S* (or relatively to u) at a point $x \in \text{Supp}(\mathcal{O}_X/\mathcal{J})$ if there exists an open neighbourhood U of x and a finite suite (f_i) of sections of $\mathcal{J}$ which is $\mathcal{O}_U$ -regular relatively to S and which generates $\mathcal{J}|_U$.

In an A -algebra B , one says that an ideal $\mathfrak{J}$ is *transversally regular relatively to A* if $\mathcal{J} = \tilde{\mathfrak{J}}$ is transversally regular relatively to $S = \text{Spec}(A)$ at every point of $V(\mathfrak{J})$ in $\text{Spec}(B)$.

Definition (19.2.2). — Let S be a prescheme, X, Y two S -preschemes, $u : Y \rightarrow X$ an S -morphism which is an immersion. One says that u is an immersion *transversally regular relatively to S* at a point $y \in Y$ if there exists a neighbourhood V of $u(y)$ in X such that the sub-prescheme of X associated to u is defined by an Ideal of $\mathcal{O}_U$ transversally regular relatively to S at the point $u(y)$.

It is clear that the set of points of Y where an immersion is transversally regular relatively to S is open in Y ; if it is equal to Y , one says simply that u is an immersion transversally regular relatively to S .

If $S \rightarrow T$ is a flat morphism, then, if $\mathcal{J}$ is an Ideal of $\mathcal{O}_X$ transversally regular relatively to S at $x \in X$, it is so also relatively to T ($\mathbf{0_I}$, (6.2.1)). A Y -immersion $Y \rightarrow X$ transversally regular relatively to S at a point $y \in Y$ relatively to S is therefore also a T -immersion transversally regular at this point.

Proposition (19.2.3). — *Let S be a prescheme, X, Y two S -preschemes, $u : Y \rightarrow X$ an S -morphism which is an immersion transversally regular relatively to S at a point $y \in Y$. Then, for every change of base $S' \rightarrow S$, if one sets $X' = X \times_S S'$, $Y' = Y \times_S S'$, the immersion $u' = u_{(S')} : Y' \rightarrow X'$ is transversally regular relatively to S' at every point $y' \in Y'$ above y .*

This follows immediately from the definition and ($\mathbf{0}$, (15.1.15)).

Proposition (19.2.4). — *Let S be a prescheme, X, Y two S -preschemes, $u : Y \rightarrow X$ an S -morphism which is an immersion. Suppose, either that S and X are locally Noetherian, or that the structural morphisms $g : X \rightarrow S$, $h : Y \rightarrow S$ are locally of finite presentation. Let y be a point of Y , $s = g(x) = h(y)$. The following conditions are equivalent:*

- label=a) *u is transversally regular relatively to S at the point y .*
- lbbel=b) *The morphisms g and h are flat at the neighbourhoods of the points $u(y)$ and y respectively, and if one sets $X_s = g^{-1}(s)$, $Y_s = h^{-1}(s)$, the immersion $u_s : Y_s \rightarrow X_s$ is regular at the point y .*
- b') *The morphisms g and h are flat at the neighbourhoods of the points $u(y)$ and y respectively, and, for every change of base $S' \rightarrow S$, if one sets $X' = X \times_S S'$, $Y' = Y \times_S S'$, the immersion $u' = u_{(S')} : Y' \rightarrow X'$ is regular at every point of Y' above y .*
- lcbel=c) *The morphism h is flat at the neighbourhood of the point y , and the immersion u is regular at the neighbourhood of the point y .*
- c') *The morphism h is flat at the neighbourhood of the point y , and the immersion u is quasi-regular at the neighbourhood of the point y .*

Proof. One has already proved in ((19.2.3)) that a) implies $b')$ without any finiteness hypothesis, and it is trivial that $b')$ implies $b)$. Let us show that $b)$ implies $c)$. For this one can (the question being local on X and on S) suppose that Y is a closed sub-prescheme of X defined by an Ideal quasi-coherent $\mathcal{J}$ of $\mathcal{O}_X$, and u the canonical injection. The hypothesis that $\mathcal{O}_X / \mathcal{J}$ is g -flat entails that the suite

$$0 \longrightarrow \mathcal{J} \otimes_{\mathcal{O}_g} k(s) \longrightarrow \mathcal{O}_X \otimes_{\mathcal{O}_g} k(s) \longrightarrow (\mathcal{O}_X / \mathcal{J}) \otimes_{\mathcal{O}_g} k(s) \longrightarrow 0$$

is exact ($\mathbf{0_I}$, 6.1.2), hence Y_s is the closed sub-prescheme of X_s defined by the Ideal $\mathcal{J}_s$ of $\mathcal{O}_{X_s} = \mathcal{O}_X \otimes_{\mathcal{O}_g} k(s)$ which identifies with $\mathcal{J} / \mathfrak{m}_s \mathcal{J}$. Consider a finite suite (f_i) of sections of $\mathcal{J}$ above a neighbourhood of y in X , whose images in $\mathcal{J}_y / (\mathfrak{m}_s \mathcal{O}_{X,y} + \mathcal{J}_y^2)$ form a base of this $(\mathcal{O}_{X,y} / (\mathfrak{m}_s \mathcal{O}_{X,y} + \mathcal{J}_y))$ -module (which is free by virtue of the hypothesis $b)$). Since X_s is locally Noetherian, it follows from ((16.9.11)), and from the hypothesis $b)$ that the images $(\tilde{f}_i)_s = f_i \otimes_{\mathcal{O}_g} k(s)$ above a neighbourhood of y in X_s , form a regular suite generating the restriction of $\mathcal{J}_s$ to this neighbourhood. Since $\mathfrak{m}_s \mathcal{O}_{X,y}$ is contained in the radical of $\mathcal{O}_{X,y}$, and since in both cases under consideration $\mathcal{J}_y$ is an ideal of finite type of $\mathcal{O}_{X,y}$, it follows from Nakayama's lemma that the $(f_i)_y$ also generate $\mathcal{J}_y$; since $\mathcal{J}$ is an Ideal of finite type of $\mathcal{O}_X$ in both cases under consideration, there is a neighbourhood U of y in X such that the $f_i|_U$ generate $\mathcal{J}|_U$. The fact that $b)$ implies $c)$ is then a consequence of ($\mathbf{0}$, 15.1.16) when S and X are locally Noetherian, and of ((11.3.8)) when g and h are locally of finite presentation; in this latter case, ((11.3.8)) also proves the equivalence of $c)$ and $c')$ (which is trivial when X and S are locally Noetherian ($\mathbf{0}$, 15.1.11)). Finally, if $c)$ is satisfied, to prove $a)$, one can again restrict to the case where Y is a closed sub-prescheme of X defined by $\mathcal{J}$, and by hypothesis, there is a regular suite (f_i) of sections of $\mathcal{J}$ above a neighbourhood of y in X such that the f_i generate $\mathcal{J}|_U$ and that h is flat in U ; to see that $c)$ implies $a)$, it suffices again to apply ($\mathbf{0}$, 15.1.6) when S and X are locally Noetherian, and ((11.3.8)) when g and h are locally of finite presentation. $\square$

Remark (19.2.5). — *(i) When in condition $b)$ one suppresses the hypothesis that h is flat at the neighbourhood of y , the condition thus modified no longer implies $a)$. It suffices to take for S the spectrum of a local ring which is not a field, for s its closed point, and $Y = X_s$.*

(ii) Suppose that Y is a closed sub-prescheme of X defined by an Ideal quasi-coherent $\mathcal{J}$ of $\mathcal{O}_X$. One has proved in the course of the proof of ((19.2.4)) that, when the equivalent conditions of this proposition are satisfied, then, for every system (f_i) of sections of $\mathcal{J}$ from a neighbourhood of y in X , such that the images of f_i in $\mathcal{J}_y / (\mathfrak{m}_s \mathcal{O}_{X,y} + \mathcal{J}_y^2)$ form a base of this $(\mathcal{O}_{X,y} / (\mathfrak{m}_s \mathcal{O}_{X,y} + \mathcal{J}_y))$ -module (which is free by virtue of the hypothesis b)), there exists an open neighbourhood U of y in X such that the $f_i|_U$ satisfy the conditions of the definition ((19.2.1)).

Corollary (19.2.6). — Suppose, either that S and X are locally Noetherian, or that g and h are locally of finite presentation. Suppose furthermore that the fibres X_s and Y_s are regular preschemes at the points $u(y)$ and y respectively, and that the morphisms g and h are flat at the neighbourhoods of the points $u(y)$ and y respectively. Then the immersion $Y \rightarrow X$ is transversally regular at the point x . IV | 192

Indeed, it follows from ((19.1.1)) that the immersion $Y_s \rightarrow X_s$ is regular at the point y , and it suffices to apply ((19.2.4, b)).

Proposition (19.2.7). — (i) Every open S -immersion is transversally regular relatively to S .

(ii) Let $f : Y \rightarrow X$ be an S -immersion, $g : S' \rightarrow S$ a morphism, and set $X' = X_{(S')}$, $f' = f_{(S')} : Y' \rightarrow X'$. If f is an immersion transversally regular relatively to S , f' is transversally regular relatively to S' . The converse is true if furthermore g is faithfully flat, and if moreover X and Y are locally of finite presentation over S .

(iii) If $f : Y \rightarrow X$ and $g : Z \rightarrow Y$ are immersions transversally regular relatively to S , so is $f \circ g : Z \rightarrow X$.

(iv) Let X be an S -prescheme, $f : Y \rightarrow X$, $g : Z \rightarrow Y$ two S -immersions. Suppose, either that S and X are locally Noetherian, or that X , Y and Z are locally of finite presentation over S . Then, if g and $f \circ g$ are transversally regular relatively to S at the point $z \in Z$, f is transversally regular relatively to S at the point $g(z)$.

(v) Under the general conditions of (iv), suppose that X and Y are S -flat, Y_s regular at the point $g(z)$ and X_s regular at the point $f(g(z))$; then, if $f \circ g$ is transversally regular at the point z , so is g .

Proof. Assertion (i) is immediate, and the first assertion of (ii) follows immediately from (19.2.3). To prove the second assertion of (ii), we apply (19.2.4): X' and Y' being locally of finite presentation over S' , it follows first from this criterion that X' and Y' are flat over S' , hence X and Y are flat over S (2.5.1); on the other hand, for every $s \in S$, if $s' \in S'$ is a point above s , $Y'_{s'} \rightarrow X'_{s'}$ is by hypothesis a regular immersion, and one deduces from (19.1.5, (ii)) that the same is true of $Y_s \rightarrow X_s$, since these preschemes are locally Noetherian and it then amounts to the same to say that the immersion $Y_s \rightarrow X_s$ is regular or quasi-regular. To prove (iii), one reduces to the case where Z is a closed subprescheme of Y and Y a closed subprescheme of X , and one forms as in (19.1.5, (iii)) a system of generators of the Ideal of $\mathcal{O}_X$ defining Z . To prove (iv), one remarks that the hypothesis together with (19.2.4) shows first that X , Y , and Z are S -flat in a neighbourhood of z , and it suffices moreover, if s is the image of z in S , to prove that the immersion $f_s : Y_s \rightarrow X_s$ is regular at the point z , knowing that the same is true of $g_s : Z_s \rightarrow Y_s$ and of $f_s \circ g_s$; but this results from (19.1.5, (iv)). Finally, for (v), one reasons in the same manner as for (iv); the hypothesis on $f \circ g$ already implies by (19.2.4, b)) that Z is S -flat in a neighbourhood of z , and the same is true of X and Y by hypothesis; one is then reduced to proving that g_s is regular at the point z , knowing that the same is true of $f_s \circ g_s$, which follows from (19.1.5, (v)), given the hypotheses on X_s and Y_s . □

Proposition (19.2.8). — Let X be an S -prescheme, Y a closed sub-prescheme of X defined by an Ideal quasi-coherent $\mathcal{J}$ of $\mathcal{O}_X$, $u : Y \rightarrow X$ the canonical injection. If u is transversally regular relatively to S , the $\mathcal{O}_X$ -Modules $\mathcal{J}^n$ and $\mathcal{J}^n / \mathcal{J}^m$ ($0 \leq n < m$) are S -flat at all points of Y . For every change of base $S' \rightarrow S$, if $X' = X_{(S')}$, $Y' = Y_{(S')}$ and $u' = u_{(S')}$, one has $\mathcal{G}_n(u') = \mathcal{G}_n(u) \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$ as isomorphism for every $n > 0$. IV | 193

Indeed, $\mathcal{O}_X$ and $\mathcal{O}_X / \mathcal{J}$ are by hypothesis S -flat at points of Y , and $\mathcal{J}^n / \mathcal{J}^{n+1}$ is thus S -flat at points of Y . The first assertion follows then from (0_I, (6.1.2)) and the second from ((11.2.9.2)).

The following proposition generalises and makes precise ((17.16.1)):

Proposition (19.2.9). — Let $f : X \rightarrow S$ be a morphism, x a point of X , $s = f(x)$. Suppose f is of finite presentation at the point x , and is a Cohen-Macaulay morphism at the point x ((6.8.1)). Then there exists a sub-prescheme Y of X containing x and having the following properties:

label=(i) The canonical injection $j : Y \rightarrow X$ is transversally regular relatively to S .

lbbel=(ii) The morphism $g = f \circ j : Y \rightarrow S$ is de Cohen-Macaulay (which is equivalent to saying, taking into account (i), that all the fibres $g^{-1}(g(y))$ of g are Cohen-Macaulay preschemes).

lcbel=(iii) The point x is a maximal point of $Y_s = g^{-1}(s)$.

Set $X_s = f^{-1}(s)$; the ring $\mathcal{O}_{X_s, x}$ is Cohen-Macaulay; let $(t_i)_{1 \leq i \leq n}$ be a system of parameters of this ring, which is therefore a regular suite (0, (16.5.7)). There is a neighbourhood open U of x in X and sections h_i ($1 \leq i \leq n$) of $\mathcal{O}_X$ above U whose canonical images $(h_i)_x$ in the quotient ring $\mathcal{O}_{X_s, x}$ coincide with t_i ; and $\mathcal{O}_{Y_s, x}$ is flat over $\mathcal{O}_{S, s}$; by virtue of ((11.3.1)) f and g are flat in a neighbourhood of x and it follows from ((19.2.4)) that j is an immersion transversally regular relatively to S (replacing U by a suitable open neighbourhood of x in X). Finally, the ring $\mathcal{O}_{Y_s, x}$, quotient of $\mathcal{O}_{X_s, x}$ by the ideal generated by the t_i , is Artinian by construction (0, (16.3.6)), so x is a maximal point of Y_s . Finally, the ring $\mathcal{O}_{Y_s, x}$, being Artinian, is a Cohen-Macaulay ring (0, (16.5.1)), and, replacing Y by its intersection with a suitable open neighbourhood V of x in X , one can suppose that g is a Cohen-Macaulay morphism, by virtue of ((12.1.7, iii)).

In particular, one recovers the existence of “quasi-sections” flat of a morphism $f : X \rightarrow S$ at a point $x \in X_s$ which is closed in X_s and such that f is flat at the point x and that $\mathcal{O}_{X_s, x}$ is a Cohen-Macaulay ring ((17.16.1)); one sees moreover that the immersion $Y \rightarrow X$ is transversally regular relatively to S . This last complement is also valid for the quasi-section étale defined in ((17.16.3, i)).

19.3. Relative complete intersections (flat case).

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Definition (19.3.1). — Let A be a local Noetherian ring. One says that A is a ring of complete intersection (or equivalently, a ring of absolute complete intersection, if there is risk of confusion) if the completion $\hat{A}$ is isomorphic to the quotient of a local Noetherian regular complete ring B by a regular ideal (i.e. ((16.9.7)) an ideal generated by a regular suite of elements of B).

Every regular local ring is a ring of complete intersection.

One says that a locally Noetherian prescheme X is a complete intersection (absolute) at a point $x \in X$ if the local ring $\mathcal{O}_{X, x}$ is a ring of complete intersection.

Proposition (19.3.2). — Let B be a local Noetherian regular ring, $\mathfrak{J}$ an ideal of B . For $A = B/\mathfrak{J}$ to be a ring of complete intersection, it is necessary and sufficient that the ideal $\mathfrak{J}$ be regular (i.e. ((16.9.7)) generated by a regular suite of elements of B).

One can write $\hat{A} = \hat{B}/\mathfrak{J}\hat{B}$ and since $\hat{B}$ is a B -module faithfully flat and B Noetherian, $\mathfrak{J}\hat{B}$ is regular if and only if $\mathfrak{J}$ is ((19.1.5, ii)). This shows (by virtue of (0, (17.1.5, ii))) that the statement is sufficient, and on the other hand that to prove it is necessary, one can restrict to the case where A and B are complete. By hypothesis, there exists then a local Noetherian complete regular ring B' such that A is isomorphic to a quotient $B'/\mathfrak{J}'$, where $\mathfrak{J}'$ is a regular ideal of B' . Consider then the fibred product $B' = B \times_A B''$ for the surjective homomorphisms $B \rightarrow A, B' \rightarrow A$ (0, (18.1.2)); we shall first prove the following lemma:

Lemma (19.3.2.1). — Let A, E, F be three rings, $f : E \rightarrow A, g : F \rightarrow A$ two homomorphisms, and set $G = E \times_A F$ (0, (18.1.2)).

label=(i) If f is surjective and if E and F are Noetherian rings, so is G .

lbbel=(ii) If A, E and F are local rings and if f and g are local homomorphisms, G is local, and the canonical homomorphisms $G \rightarrow E, G \rightarrow F$ are local.

lcbel=(iii) If the conditions of (i) and (ii) are satisfied, if g is surjective, and if E and F are complete local Noetherian rings, so is G .

This lemma being established, it follows from the Cohen theorem (0, (19.8.8)) that B'' is isomorphic to a quotient of a local Noetherian complete regular ring C . It follows then from ((19.1.2)) that the immersion $\text{Spec}(B') \rightarrow \text{Spec}(C)$ is regular; since by hypothesis, we conclude from ((19.1.5, iii)) that the immersion $\text{Spec}(A) \rightarrow \text{Spec}(C)$ also writes as the composed $\text{Spec}(A) \rightarrow \text{Spec}(B) \rightarrow \text{Spec}(C)$. But since B is regular by hypothesis, we conclude from ((19.1.5, iii)) that this immersion is also composed as $\text{Spec}(A) \rightarrow \text{Spec}(B) \rightarrow \text{Spec}(C)$; hence it is also the same of $\text{Spec}(A) \rightarrow \text{Spec}(B)$ by ((19.1.2)); hence it is also of the same of $\text{Spec}(A) \rightarrow \text{Spec}(B)$ by ((19.1.5, iv)).

Corollary (19.3.3). — Let X be a locally Noetherian prescheme locally immersible in a regular prescheme. Then the set of $x \in X$ where X is a complete intersection is open in X and contains the set of regular points of X .

One can indeed restrict to the case where $X = \operatorname{Spec}(B/\mathfrak{J})$, where B is a regular ring and $\mathfrak{J}$ an ideal of B . The set of $\mathfrak{p} \supset \mathfrak{J}$ in $\operatorname{Spec}(B)$ such that $\mathfrak{J}_{\mathfrak{p}}$ is regular in $B_{\mathfrak{p}}$ is open ((16.9.5)), and this is the set of points of $\operatorname{Spec}(B)$ where X is a complete intersection by virtue of ((19.3.2)).

Corollary (19.3.4). — *Let k be a field, k' an extension of k , $X' = X \otimes_k k'$. Let x' be a point of X' , x its projection in X . For X to be a complete intersection at x , it is necessary and sufficient that X' be a complete intersection at x' .*

The question being local on X , one can suppose that $X = \operatorname{Spec}(A)$, where A is a k -algebra of finite type, hence a quotient of a polynomial algebra $k[T_1, \dots, T_n]$, so that X is a closed sub-prescheme of the regular scheme $Y = \operatorname{Spec}(k[T_1, \dots, T_n])$ (0, (17.3.7)). If one sets $Y' = Y \otimes_k k' = \operatorname{Spec}(k'[T_1, \dots, T_n])$, Y' is also a regular scheme. Since $\operatorname{Spec}(k')$ is faithfully flat over $\operatorname{Spec}(k)$, it follows from ((19.1.5, ii)) and the fact that these are locally Noetherian preschemes that the immersion $Y \rightarrow X$ is regular at a point x if and only if the immersion $Y' \rightarrow X'$ is regular at every point x' above x . The conclusion follows from the criterion ((19.3.2)).

Corollary (19.3.5). — *Let A, C be two local Noetherian rings, $\rho : A \rightarrow C$ a local homomorphism, $B = C/\mathfrak{J}$ a quotient of C , k the residue field of A . Suppose that ρ makes C a flat A -module, and that the ring $C \otimes_A k$ is regular. Then the following conditions are equivalent:*

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label=a) *The ideal $\mathfrak{J}$ is transversally regular relatively to A .*

lbbel=b) *B is a flat A -module, and $B \otimes_A k$ is a ring of complete intersection.*

By virtue of ((19.2.4)), the condition a) is equivalent to saying that B is a flat A -module and that $\mathfrak{J} \otimes_A k$ is a regular ideal of $C \otimes_A k$ (by virtue of the flatness of B over A (0_I, (6.1.2))); as the ring $C \otimes_A k$ is regular, it follows at once, by virtue of ((19.3.2)), that $\mathfrak{J} \otimes_A k$ is a regular ideal of $C \otimes_A k$, or that $B \otimes_A k = (C \otimes_A k) / (\mathfrak{J} \otimes_A k)$ is a ring of complete intersection; whence the corollary.

Definition (19.3.6). — *Let $f : X \rightarrow S$ be a flat morphism locally of finite presentation. One says that X is a complete intersection relatively to S at a point x , if the fibre $f^{-1}(f(x))$ is a complete intersection (absolute) at the point x . One says that X is a complete intersection relatively to S , or that the morphism f is a morphism of complete intersection, if X is a complete intersection relatively to S at each of its points.*

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Proposition (19.3.7). — *Let $g : X \rightarrow S, h : Y \rightarrow S$ be two flat morphisms locally of finite presentation, $f : Y \rightarrow X$ an S -immersion, y a point of $Y, x = f(y), s = g(x) = h(y)$. Suppose that the fibre $X_s = g^{-1}(s)$ is regular at the point x . Then, for f to be an immersion transversally regular at the point x , it is necessary and sufficient that Y be a complete intersection relatively to S at the point y .*

To say that f is transversally regular at the point y amounts, by virtue of ((19.2.4)), to saying that $f_s : Y_s \rightarrow X_s$ is a regular immersion at the point y . But since X_s is regular at the point $x = f(y)$, this is equivalent, by virtue of ((19.3.2)), to saying that Y_s is a complete intersection at the point y , whence the proposition.

Corollary (19.3.8). — *Let $f : X \rightarrow S$ be a flat morphism locally of finite presentation. The set U of $x \in X$ at which X is a complete intersection relatively to S is open in X . If moreover f is proper, the set E of $s \in S$ such that $X_s = f^{-1}(s)$ is a complete intersection is an open subset of S .*

Since $E = Y - f(X - U)$, the second assertion follows from the first and the fact that f is a closed map. The first assertion is local on X , hence one can suppose that X is a closed sub-prescheme of a prescheme of the form $Z = S[T_1, \dots, T_n]$ (T_i indeterminates). The latter being smooth over S ((17.3.8)), its fibres Z_s are regular, and it amounts, by virtue of ((19.3.7)), to saying that $x \in U$ if and only if the immersion $X \rightarrow Z$ is transversally regular at the point x ; one concludes thus from the corollary by ((19.2.2)).

Proposition (19.3.9). — (i) *Every open immersion is a morphism of complete intersection.*

(ii) *Let $f : X \rightarrow S$ be a flat morphism locally of finite presentation which is a morphism of complete intersection. For every change of base $g : S' \rightarrow S, f' = f_{(S')} : X_{(S')} \rightarrow S'$ is a morphism of complete intersection. The converse is true if g is faithfully flat and quasi-compact.*

(iii) *If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are two flat morphisms locally of finite presentation which are morphisms of complete intersection, so is $g \circ f : X \rightarrow Z$.*

Assertion (i) follows immediately from the definition ((19.3.1)). To prove (ii), we first note that if f is flat and locally of finite presentation, so is f' ((2.1.4)), the converse being true if g is faithfully flat ((2.5.1) and ((2.7.1))); moreover for every $s' \in S'$, if $s = g(s')$, it is equivalent to say that $f'^{-1}(s')$ is a complete intersection or that $f'^{-1}(s')$ is a complete intersection; whence (ii), taking into account of ((19.3.4)). Finally, under the hypotheses of (iii), $g \circ f$ is flat and locally of finite presentation. By definition ((19.3.6)), one is thus reduced to the case where Z is the spectrum of a field, and to prove that the local rings $\mathcal{O}_{X,x}$ are rings of complete intersection, which is contained in the following more general result:

Corollary (19.3.10). — *Let $f : X \rightarrow S$ be a flat morphism locally of finite presentation, x a point of X , $s = f(x)$. If $\mathcal{O}_{S,s}$ is a local Noetherian ring of complete intersection, so is $\mathcal{O}_{X,x}$.* IV | 197

Proof. One can evidently restrict to the case where $S = \text{Spec}(A)$ with $A = \mathcal{O}_{S,s}$. We show moreover that one can restrict to the case where the local ring A is complete. Indeed, set $A' = \hat{A}$, $X' = X \otimes_A A'$, and let s' be the unique closed point of $S' = \text{Spec}(A')$, which is the unique point of S' above s ; if $f' = f_{(S')} : X' \rightarrow S'$, the fibre $f'^{-1}(s')$ is canonically isomorphic to $f^{-1}(s)$ ((I, 3.6.4)); there is thus a single point $x' \in X'$ above x and s' , and one has by suite $\mathcal{O}_{X',x'} = \mathcal{O}_{X,x} \otimes_A A'$. One deduces that the local rings $\mathcal{O}_{X,x}$ and $\mathcal{O}_{X',x'}$ have isomorphic completions, for, if E is a local ring, the separated completion $(E \otimes_A \hat{A})^\wedge$ of the ring $E \otimes_A \hat{A}$ equipped with the topology induced by the $\mathfrak{m}$ -preadic topology of G , is isomorphic to $\hat{E} \otimes_A \hat{A} \cong \hat{E}$, hence to the separated completion $\hat{E}$ of E (0_I, (7.7.1)). By virtue of the definition of complete intersection ((19.3.1)), it follows that $\mathcal{O}_{X,x}$ is a ring of complete intersection at the same time as $\mathcal{O}_{X',x'}$.

Suppose therefore A complete; one can moreover suppose that $X = \text{Spec}(B)$, B being a quotient of a polynomial algebra $C = A[T_1, \dots, T_r]$. Since polynomial rings over fields are regular rings (0, (17.3.7)), it follows from ((19.3.7)) that the immersion $X \rightarrow Z = \text{Spec}(C)$ can be supposed transversally regular relatively to S , so that one can suppose that $B = C/\mathfrak{J}$, where $\mathfrak{J}$ is generated by a regular suite $(g_i)_{1 \leq i \leq n}$ of elements of C . On the other hand, since A is complete, one can by hypothesis write $A = A'/\mathfrak{K}$, where A' is a local regular ring and $\mathfrak{K}$ an ideal of A' (0, (19.8.8)), and the hypothesis that A is a ring of complete intersection implies that $\mathfrak{K}$ is generated by a regular suite $(f_j)_{1 \leq j \leq m}$ ((19.3.2)). In the polynomial ring $C' = A'[T_1, \dots, T_r]$, the elements f_j ($1 \leq j \leq m$) form a regular suite $\mathfrak{K}' = \mathfrak{K}[T_1, \dots, T_r]$ (0, (15.1.4)); and the suite formed of the f_j ($1 \leq j \leq m$) and the g_i ($1 \leq i \leq n$) is regular in C' ; if $\mathfrak{J}'$ is the ideal of C' that it generates, one has $B = C'/\mathfrak{J}'$, whence the conclusion by virtue of ((19.3.2)), since C' is a regular ring (0, (17.3.7)). □

19.4. Application: criteria of regularity and smoothness for blow-ups.

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(19.4.1). Let X be a prescheme, Y a closed sub-prescheme of X , defined by an Ideal quasi-coherent $\mathcal{J}$ of $\mathcal{O}_X$. Recall ((II, 8.1.3)) that the X -scheme X' obtained by blowing up Y is the prescheme

$$X' = \text{Proj}(\mathcal{S}), \quad \text{where} \quad \mathcal{S} = \bigoplus_{n \geq 0} \mathcal{J}^n.$$

If $\mathcal{J}$ is of finite type, the structural morphism $f : X' \rightarrow X$ is projective. Without further hypothesis on $\mathcal{J}$, the closed sub-prescheme $Y' = f^{-1}(Y)$ of X' is defined by the Ideal quasi-coherent $\mathcal{J} \mathcal{O}_{X'}$ of $\mathcal{O}_{X'}$, canonically isomorphic to $\mathcal{O}_{X'}(1)$; more precisely, one has an exact suite

$$(8) \quad 0 \longrightarrow \mathcal{O}_{X'}(1) \xrightarrow{\tilde{u}_0} \mathcal{O}_{X'} \longrightarrow \mathcal{O}_{Y'} \longrightarrow 0$$

where $\tilde{u}_0$ is the image of $\tilde{u}_0$ ((II, 8.1.7) and (II, 8.1.8)). Since, in a neighbourhood of a point of X' , the $\mathcal{O}_{X'}$ -Module $\mathcal{O}_{X'}(1)$ is free of rank 1, $\mathcal{J} \mathcal{O}_{X'}$ is generated in such a neighbourhood by an invertible section of $\mathcal{O}_{X'}$ and $\mathcal{J} \mathcal{O}_{X'} / \mathcal{J}^2 \mathcal{O}_{X'}$ is therefore an $\mathcal{O}_{X'} / \mathcal{J} \mathcal{O}_{X'}$ -Module free of rank 1. This proves the first assertion of the following lemma:

Lemma (19.4.2). — *The canonical immersion $Y' \rightarrow X'$ is regular and of codimension 1, and Y' is canonically Y -isomorphic to the Y -scheme $\text{Proj}(\text{gr}^*_{\mathcal{J}}(\mathcal{O}_X))$.*

The second assertion follows from ((II, 3.5.3)) applied to the canonical injection $Y \rightarrow X$, taking into account that by definition, $\mathcal{O}_Y$ is isomorphic to $\mathcal{O}_X / \mathcal{J}$ and $\mathcal{J}^n / \mathcal{J}^{n+1}$.

Corollary (19.4.3). — *If the canonical immersion $Y \rightarrow X$ is quasi-regular, the restriction $g : Y' \rightarrow Y$ of the morphism f is smooth.*

Indeed, $\mathcal{N}_{Y/X} = \mathcal{I} / \mathcal{I}^2$ is then a $\mathcal{O}_Y$ -Module locally free of finite rank, hence Y' is Y -isomorphic to $\mathbf{P}(\mathcal{N}_{Y/X})$ ((16.9.8)), hence Y' is smooth over Y ((17.3.9)).

We shall see later (chap. V) that g is also smooth when the immersion $Y \rightarrow X$ “presents only ordinary singularities”.

Proposition (19.4.4). — *With the notations of ((19.4.1)), suppose that X is locally Noetherian and that the morphism $g : Y' \rightarrow Y$, restriction of f , is smooth. Let x' be a point of Y' , x its image in Y . For Y to be regular at the point x , it is necessary and sufficient that Y' be regular at the point x' ; when this is the case, X' is also regular at the point x' .*

The first assertion follows from ((17.5.8)) and the second from ((19.1.1)) and ((19.4.2)).

Corollary (19.4.5). — *Under the general hypotheses of ((19.4.4)), suppose that Y is regular at the point x . Then, for every generalisation x_1 of x in X not belonging to Y , X is regular at the point x_1 . If $\text{Reg}(X)$ is open ((6.12)), for example if X is excellent ((7.8.6, iii)) and if Y is regular, then there exists an open neighbourhood U of Y in X such that X is regular at the points of $U - Y$.*

To prove the first assertion, one can reduce to the case where X is a local scheme with closed point x ; the hypothesis then implies that Y is regular (0, 17.3.2), hence the same holds for Y' by ((19.4.4)). Note now that the morphism $f : X' \rightarrow X$ is projective, hence proper, and consequently (II, 7.2.1), if one considers the unique point x'_1 of X' above x_1 (II, 8.1.3), there exists a specialisation of x'_1 belonging to Y' ; applying ((19.4.4)) to it and using (0, 17.3.2), one concludes that X' is regular at the point x'_1 , hence X is regular at the point x_1 since $X' - Y'$ is isomorphic to $X - Y$ (II, 8.1.3). If Y is regular, every point of $X - Y$ that is a generalisation of a point of Y therefore belongs to $\text{Reg}(X)$; if $\text{Reg}(X)$ is open, $U = Y \cup \text{Reg}(X)$ is then locally constructible and stable under generalisation, hence open (0_{III}, 9.2.5), and answers the question.

Proposition (19.4.6). — *Let $g : X \rightarrow S$ be a morphism, Y a closed sub-prescheme of X defined by an Ideal quasi-coherent $\mathcal{I}$, X' the X -scheme obtained by blowing up Y , Y' the inverse image of Y in X' . The following conditions are equivalent:*

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label=a) $\text{gr}^*_{\mathcal{I}}(\mathcal{O}_X)$ is a g -flat $\mathcal{O}_X$ -Algebra.

lbel=b) For every $n \geq 0$, the $\mathcal{O}_X$ -Module $\mathcal{O}_X / \mathcal{I}^{n+1}$ is g -flat (in other words, the n -th infinitesimal neighbourhood $Y^{(n)}$ of Y in X ((16.1.2)) is S -flat).

lbel=c) For every change of base $S_1 \rightarrow S$ and every integer $n \geq 1$, if one sets $X_1 = X \times_S S_1$, the canonical homomorphism $\mathcal{I}^n \otimes_{\mathcal{O}_S} \mathcal{O}_{S_1} \rightarrow \mathcal{I}^n \mathcal{O}_{X_1}$ is bijective.

When this is the case, Y' is S -flat, and for every change of base $S_1 \rightarrow S$, if Y_1 is the inverse image of Y in X_1 , the prescheme X'_1 obtained by blowing up Y_1 in X_1 is canonically isomorphic to $X' \times_S S_1$.

Proof. (i). The equivalence of a) and b) follows immediately from (0_I, 6.1.2); to show the equivalence of b) and c), one can reduce to the case where $S = \text{Spec}(A)$ and $X = \text{Spec}(B)$ are affine, with $\mathcal{I} = \tilde{J}$ where J is an ideal of B ; the exact Tor sequence then shows that condition c) implies $\text{Tor}_1^A(B/J^{n+1}, A') = 0$ for all integers $n \geq 0$ and all A -algebras A' ; taking for A' the direct sum algebra $A \oplus M$, where M is any A -module and the multiplication is defined by $(a, m)(a', m') = (aa', am' + a'm)$, one sees that the preceding condition is equivalent to $\text{Tor}_1^A(B/J^{n+1}, M) = 0$, hence to the fact that B/J^{n+1} is a flat A -module. Setting $\mathcal{I}_1 = \mathcal{I} \otimes_{\mathcal{O}_S} \mathcal{O}_{S_1}$, one then has $\bigoplus \mathcal{I}_1^n = (\bigoplus \mathcal{I}^n) \otimes_{\mathcal{O}_S} \mathcal{O}_{S_1}$, and the assertion relative to the blown-up prescheme X'_1 follows immediately. Finally, to prove that Y' is S -flat, one can again reduce to the case where S and X are affine; setting $C = \text{gr}^*_J(B)$, every point of Y' then admits, by virtue of ((19.4.2)) and ((II, 2.3.6)), an affine open neighbourhood whose ring is of the form $C_{(t)}$, where t is a homogeneous element of degree 1 of C ; since by hypothesis C is a flat A -module, the same holds for its ring of fractions C_t and for the degree 0 component $C_{(t)}$ of this graded A -module, whence our assertion.

(ii). One can always reduce to the case where S and X are affine, B being an A -algebra of finite presentation, J an ideal of finite type of B . By virtue of ((8.9.1)), ((8.6.3)), and ((11.2.9)), there exists a Noetherian subring A_0 of A , an A_0 -algebra of finite type B_0 , an ideal J_0 of B_0 such that $B = B_0 \otimes_{A_0} A$,

$J = J_0 B$, $\mathrm{gr}_{J_0}^*(B_0)$ is a flat A_0 -module, and $\mathrm{gr}_J^*(B) = \mathrm{gr}_{J_0}^*(B_0) \otimes_{A_0} A$. By virtue of (i), one then has $\bigoplus J^n = (\bigoplus J_0^n) \otimes_{A_0} A = (\bigoplus J_0^n) \otimes_{B_0} B$; since $\bigoplus J_0^n$ is a B_0 -algebra generated by the ideal of finite type J_0 of B_0 , hence a B_0 -algebra of finite type, and consequently of finite presentation since B_0 is Noetherian, one deduces that $\bigoplus J^n$ is a B -algebra of finite presentation. Similarly, the morphism $\mathrm{Proj}(\bigoplus J_0^n) \rightarrow \mathrm{Spec}(B_0)$ is of finite type ((II, 2.7.1)), hence of finite presentation; thus the morphism $X' \rightarrow X$, which is obtained by base change, is of finite presentation. Since we have already seen ((19.4.2)) that the canonical immersion $Y' \rightarrow X'$ is regular, and Y is S -flat as seen in (i), we deduce from ((19.2.4)) that the immersion $Y' \rightarrow X'$ is transversally regular relative to S and that X' is S -flat at the points of Y' , hence ((11.3.1)) in an open neighbourhood of Y' . On the other hand, the fact that $\mathrm{gr}_{\mathcal{J}}^*(\mathcal{O}_X)$ is an S -flat $\mathcal{O}_X$ -Algebra implies, by ((11.3.4)), that X is S -flat at the points of Y , hence ((11.3.1)) in a neighbourhood of Y . $\square$

Corollary (19.4.7). — *Let $g : X \rightarrow S$ be a morphism locally of finite presentation, Y a closed sub-prescheme of X such that the composed morphism $Y \rightarrow X \rightarrow S$ is locally of finite presentation; suppose moreover that Y is S -flat and that $\mathcal{O}_X$ is normally flat along Y ((11.3.4)). Then (with the notations of ((19.4.1))), the morphism $X' \rightarrow X$ is of finite presentation, the canonical immersion $Y' \rightarrow X'$ is transversally regular of codimension 1 relatively to S , and the set of points of X (resp. X') where X is S -flat (resp. Y') is a neighbourhood of Y (resp. Y').*

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Indeed, by hypothesis $\mathrm{gr}_{\mathcal{J}}^*(\mathcal{O}_X)$ is a flat $(\mathcal{O}_X/\mathcal{J})$ -Module, hence is S -flat since Y is supposed to be S -flat. One can therefore apply the results of ((19.4.6), (i) and (ii)). As Y' is Y -isomorphic to $\mathrm{Proj}(\mathrm{gr}_{\mathcal{J}}^*(\mathcal{O}_X))$ ((19.4.2)), the same reasoning as in ((19.4.6)) shows that Y' is Y -flat.

Proposition (19.4.8). — *Suppose verified the hypotheses of ((19.4.7)) and suppose in addition that the morphism $Y' \rightarrow Y$ is smooth. Let x' be a point of Y' , x its image in Y . For Y to be smooth over S at the point x , it is necessary and sufficient that Y' be smooth over S at the point x' ; when this is the case, X' is also smooth over S at the point x' .*

The preceding conclusions hold in particular when the morphism $X \rightarrow S$ is locally of finite presentation and the immersion $Y \rightarrow X$ is transversally regular relatively to S .

Taking into account the compatibility of the formation of a blown-up prescheme with base changes in the case under consideration ((19.4.6), (i) and ((17.5.1) and ((17.7.1))), it suffices to consider the case where S is the spectrum of an algebraically closed field. But then it is equivalent to say that an S -prescheme locally of finite type is smooth over S at a point or that it is regular at that point ((17.15.1)), hence the conclusions follow from ((19.4.4)).

When $X \rightarrow S$ is locally of finite presentation and the canonical immersion $Y \rightarrow X$ is quasi-regular, this immersion is by definition of finite presentation since the Ideal defining Y is of finite type, hence the composed morphism $Y \rightarrow X \rightarrow S$ is locally of finite presentation. We have already seen ((19.4.3)) that under these conditions the morphism $Y' \rightarrow Y$ is smooth; moreover, one then knows that $\mathcal{N}_{Y/X}$ is a locally free $\mathcal{O}_Y$ -Module and that $\mathrm{gr}_{\mathcal{J}}^*(\mathcal{O}_X)$ is isomorphic to $\mathbf{S}_{\mathcal{O}_Y}^*(\mathcal{N}_{Y/X})$, hence is a flat $\mathcal{O}_Y$ -Module. The hypotheses of ((19.4.7)) are therefore all verified and one can apply the conclusions of the statement of ((19.4.8)).

Corollary (19.4.9). — *Under the general hypotheses of ((19.4.8)), suppose in addition that Y is smooth over S at the point x . Then, for every generalisation x_1 of x in X not belonging to Y , X is smooth over S at the point x_1 . If Y is smooth over S , there exists an open neighbourhood U of Y in X such that X is smooth over S at the points of $U - Y$.*

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For the first assertion, one can reduce to the case where X is a local scheme with closed point x , and the hypothesis then implies that Y is smooth over S , since every neighbourhood of the closed point of Y is necessarily Y itself; one concludes therefore by the same reasoning as in ((19.4.5)), using the fact that the set of points where X' is smooth over S is open in X' . To prove the second assertion, one can reduce to the case where S is Noetherian and X is of finite type over S , thanks to ((8.9.1)), ((8.6.3)), ((11.2.9)), and ((17.7.6)); one concludes then as in ((19.4.5)), using the fact that the set of points where X is smooth over S is open in X .

Remark (19.4.10). — In ((19.4.6, ii)), replacing the hypothesis that X and Y are locally of finite presentation over S by that of S and X (hence also Y) being *locally Noetherian*. Then it is clear

that $\bigoplus_{n \geq 0} \mathcal{I}^n$ is an $\mathcal{O}_X$ -Algebra of finite type, that the morphism $X' \rightarrow X$ is of finite type, and it follows still from ((19.2.4)) applied to locally Noetherian preschemes, that the immersion $Y' \rightarrow X'$ is transversally regular relatively to S and that X' is S -flat at points of Y' .

Proposition (19.4.11). — *Let X be a prescheme, $\mathcal{E}$ an $\mathcal{O}_X$ -Module quasi-coherent, $u : \mathcal{E} \rightarrow \mathcal{O}_X$ a homomorphism of $\mathcal{O}_X$ -Modules, $\mathcal{I} = u(\mathcal{E})$ the Ideal quasi-coherent of $\mathcal{O}_X$ image of u , Y the closed sub-prescheme of X defined by $\mathcal{I}$. Let X' be the X -scheme obtained by blowing up Y . On the other hand, let $P = \mathbf{P}(\mathcal{E})$ be the projective fibre defined by $\mathcal{E}$ ((II, 4.1.1)), $p : P \rightarrow X$ the structural morphism, $\alpha_1^\# : p^*(\mathcal{E}) \rightarrow \mathcal{O}_P(1)$ the canonical homomorphism ((II, 4.1.5.1)) and $\mathcal{H}$ its kernel, so that one has the exact suite*

$$0 \longrightarrow \mathcal{H} \longrightarrow p^*(\mathcal{E}) \xrightarrow{\alpha_1^\#} \mathcal{O}_P(1) \longrightarrow 0.$$

Finally, let $\mathcal{K}$ the Ideal quasi-coherent of $\mathcal{O}_P$ image of the restriction $v : \mathcal{H} \rightarrow \mathcal{O}_P$ of $p^*(u)$, and let Z be the closed sub-prescheme of P defined by $\mathcal{K}$. Then:

- label=(i) The image of X' by the closed immersion $j : X' \rightarrow \mathbf{P}(\mathcal{E})$ corresponding to the surjective graduated homomorphism $\mathbf{S}_{\mathcal{O}_X}(\mathcal{E}) \rightarrow \mathbf{S}_{\mathcal{O}_X}(\mathcal{I}) \rightarrow \bigoplus_{n \geq 0} \mathcal{I}^n$ ((II, 3.6.2)) is a closed sub-prescheme of Z .
- lbbel=(ii) If $\mathcal{E}$ is an $\mathcal{O}_X$ -Module locally free of rank m , $\mathcal{H}$ is an $\mathcal{O}_P$ -Module locally free of rank $m - 1$.
- lcbel=(iii) Suppose that X is locally Noetherian, that $\mathcal{E}$ is locally free of finite type, that the sub-prescheme Y is regular, and that at every point $x \in Y$, the canonical immersion $i : Y \rightarrow X$ is regular and of codimension equal to $\text{rg}_{\mathcal{O}_x}(\mathcal{E}_x)$ (which we express by saying that the homomorphism u is regular at the point x). Then the image of X' by the closed immersion $j : X' \rightarrow P$ is the closed sub-prescheme Z of P ; at all points of $X' - Y'$ (Y' the inverse image of Y in X') the immersion j is transversally regular relatively to X ; moreover at all points x' of Y' , X' is regular, the immersion j is regular and of codimension $\text{rg}_{\mathcal{O}_{P,z}}(\mathcal{K}_z)$ where $z = j(x')$ (in other words the homomorphism v is also regular at the point z).

Proof. All questions being local on X , one can reduce to the case where $X = \text{Spec}(A)$ is affine and $\mathcal{E} = \tilde{E}$, where E is an A -module. To verify (i), one can furthermore restrict to examining what happens in an affine open of P of the form $D_+(t)$, where $t \in E = \mathbf{S}_A^1(E)$; setting $S = \mathbf{S}_A(E)$, one has $\Gamma(D_+(t), \mathcal{O}_P) = S_{(t)}$ ((II, 2.3.14)). Referring to the definition of $\alpha_1^\#$ ((II, 2.6.2.2)), one sees that for any element $a = \sum_i x^{(i)} \otimes ((x_1^{(i)} \cdots x_n^{(i)})/t^n)$ of $E \otimes_A S_{(t)}$ (where the $x^{(i)}$ and $x_j^{(i)}$ are in E), $\alpha_1^\#$ maps it to $\sum_i (x^{(i)} x_1^{(i)} \cdots x_n^{(i)})/t^{n+1}$ in $(S(1))_{(t)}$. To prove (i), it suffices to show that if this last element is zero, then the image $a' = v(a) = \sum_i u(x^{(i)})(x_1^{(i)} \cdots x_n^{(i)})/t^n$ is also zero under the canonical homomorphism $S_{(t)} \rightarrow (u(E))^n_{(t)}$ ((II, 3.6.2)). But this image is none other than the product of the element $u(t)$ and the canonical image of $\sum_i (x^{(i)} x_1^{(i)} \cdots x_n^{(i)})/t^{n+1}$ by the algebra homomorphism, canonical extension to $S_{(t)}$ of the homomorphism $u : E \rightarrow A$ of A -modules. This proves (i).

To establish (ii), one can suppose that E is a free A -module of rank m ; since $\alpha_1^\# : E \otimes_A S_{(t)} \rightarrow (S(1))_{(t)}$ is surjective and $(S(1))_{(t)}$ is a free $S_{(t)}$ -module of rank 1, the exact suite

$$0 \longrightarrow \mathcal{H}_{(t)} \longrightarrow E \otimes_A S_{(t)} \xrightarrow{\alpha_1^\#} (S(1))_{(t)} \longrightarrow 0$$

is *split*, and $\mathcal{H}_{(t)}$ is therefore a projective $S_{(t)}$ -module of rank $m - 1$, hence (ii).

To prove (iii), let us first examine what happens above a point $x \in X - Y$. Then $u : \mathcal{E} \rightarrow \mathcal{O}_X$, restricted to a neighbourhood of x , is *surjective*, so one can reduce to the case where $\mathcal{I} = \mathcal{O}_X$, in which case X' identifies canonically with X . To see that the closed sub-prescheme of P image of X' identifies with Z , one can reduce as above to the case where $X = \text{Spec}(A)$ is affine and moreover $E = A^m$, so that $S = A[T_0, \dots, T_{m-1}]$. The elements of $\mathcal{H}$ are then the elements of the form $\sum_\alpha L_\alpha(T) \otimes T^\alpha$ of $E \otimes_A A[T_1, \dots, T_{m-1}]$ where the $L_\alpha(T) = \lambda_{\alpha 0} + \lambda_{\alpha 1} T_1 + \cdots + \lambda_{\alpha, m-1} T_{m-1}$ must be such that $\sum_\alpha L_\alpha(T) T^\alpha = 0$ in $A[T_1, \dots, T_{m-1}]$. One verifies easily that when the condition $\lambda_{\alpha 0} = 0$ is satisfied, the values of the $\lambda_{\alpha i}$ can be taken arbitrarily for $|\alpha| \geq 1$, and one can always determine the $\lambda_{\alpha i}$ for $i \geq 1$ so as to verify the condition $\sum_\alpha L_\alpha(T) T^\alpha = 0$. The elements of the corresponding ideal $\mathcal{K}$ are then the polynomials $\sum_\alpha \lambda_{\alpha 0} T^\alpha$ *without constant term*, and as $(\mathbf{S}_A(A))_{(t)}$ identifies with A , these polynomials are exactly the elements of the kernel of the homomorphism $(\mathbf{S}_A(E))_{(t)} \rightarrow (\mathbf{S}_A(A))_{(t)}$.

One has thus proved ((19.2.1)) that the immersion $j : X' \rightarrow P$ is *transversally regular* relatively to X at all points of $X' - Y'$ and that $j(X') = Z$ at these points.

It remains to examine what happens above a point $x \in Y$. The hypothesis of “regularity” of u at this point made in (iii) is equivalent (by (19.1.1) and (0, 17.1.7)) to saying that X is regular at the point x and that the homomorphism $u_x \otimes 1 : \mathcal{E}_x \otimes_{\mathcal{O}_{X,x}} \kappa(x) \rightarrow \mathfrak{m}_x / \mathfrak{m}_x^2$ is *injective*. As $p : P \rightarrow X$ is smooth (17.3.9), for every $z \in P$ above x , P is regular at the point z (17.5.8); moreover (0, 17.3.3) the canonical homomorphism

$$(19.4.11.1) \quad (\mathfrak{m}_x / \mathfrak{m}_x^2) \otimes_{\kappa(x)} \kappa(z) \longrightarrow \mathfrak{m}_z / \mathfrak{m}_z^2$$

is injective. One deduces that the homomorphism $(\mathcal{E}_z \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{P,z}) \otimes_{\mathcal{O}_{P,z}} \kappa(z) \rightarrow \mathfrak{m}_z / \mathfrak{m}_z^2$, deduced from $p^*(u)$, is also injective, for it factors as the composite

$$(\mathcal{E}_z \otimes_{\mathcal{O}_{X,x}} \kappa(x)) \otimes_{\kappa(x)} \kappa(z) \longrightarrow (\mathfrak{m}_x / \mathfrak{m}_x^2) \otimes_{\kappa(x)} \kappa(z) \longrightarrow \mathfrak{m}_z / \mathfrak{m}_z^2$$

where the first arrow is injective by flatness and the second is the injective homomorphism (19.4.11.1). As $\mathcal{H}$ is (in a neighbourhood of z in P) a direct factor of $p^*(\mathcal{E})$, the homomorphism $v_z \otimes 1 : \mathcal{H}_z \otimes_{\mathcal{O}_{P,z}} \kappa(z) \rightarrow \mathfrak{m}_z / \mathfrak{m}_z^2$ is also injective, establishing the “regularity” of the homomorphism $v : \mathcal{H} \rightarrow \mathcal{O}_P$ at the point z .

It remains to show that the image of X' by j is the closed sub-prescheme Z of P . It will suffice to show on one hand that the image by j of the points of $q^{-1}(x)$ (where $q : X' \rightarrow X$ is the structural morphism) are exactly those of $Z \cap p^{-1}(x)$, and on the other hand that for each of these points $z = j(x')$, the surjective homomorphism $\mathcal{O}_{Z,z} \rightarrow \mathcal{O}_{X',x'}$ is bijective. But, taking into account the hypothesis and ((19.4.3)), the restriction $Y' \rightarrow Y$ of q is smooth, and since $\mathcal{O}_{Y,z}$ is supposed regular, so is $\mathcal{O}_{Y',x'}$ (17.5.8); moreover, the immersion $Y' \rightarrow X'$ is regular ((19.4.2)), hence $\mathcal{O}_{X',x'}$ is also a regular ring ((19.1.1)). To prove that $\mathcal{O}_{Z,z} \rightarrow \mathcal{O}_{X',x'}$ is bijective, it suffices to show that these two regular local rings have the same dimension (0, 17.1.9). But by (0, 17.3.3), this is equivalent to the equality of the dimensions of the two rings $\mathcal{O}_{Z,z} \otimes_{\mathcal{O}_{X,x}} \kappa(x)$ and $\mathcal{O}_{X',x'} \otimes_{\mathcal{O}_{X,x}} \kappa(x)$, so that one can replace X' and Z by the fibres $q^{-1}(x)$ and $Z \cap p^{-1}(x)$ respectively, and reduce to proving that the morphism $q^{-1}(x) \rightarrow Z \cap p^{-1}(x)$ deduced from j is an *isomorphism*. One can here restrict to the case where $X = \text{Spec}(B)$ with $B = \mathcal{O}_{X,x}$ and $E = B^m$, whence $S = B[T_0, \dots, T_m]$; moreover, the hypothesis means that there exists a regular system of parameters $(t_i)_{1 \leq i \leq n}$ of B such that $u(T_i) = t_i$ for $1 \leq i \leq m$ (0, 17.1.7); let $\mathfrak{J}$ be the image of u ; if $\mathfrak{m}$ is the maximal ideal of B , one has $\mathfrak{J}^n \otimes_B (B/\mathfrak{m}) = \mathfrak{J}^n / \mathfrak{m} \mathfrak{J}^n = (\mathfrak{J}^n / \mathfrak{J}^{n+1}) \otimes_B (B/\mathfrak{m})$ and by (16.9.4.1), $(\bigoplus_{n \geq 0} \mathfrak{J}^n) \otimes_B \kappa(x)$ identifies with $\kappa(x)[T_1, \dots, T_m]$, whence it follows that the immersion $q^{-1}(x) \rightarrow p^{-1}(x)$ is an isomorphism; this implies *a fortiori* that the closed sub-prescheme $Z \cap p^{-1}(x)$ is identical to the fibre $q^{-1}(x)$, and finishes the proof. $\square$

19.5. Criteria of M-regularity. We revisit here, in complementing them, the criteria for a suite of elements of a ring A to be M-regular or M-quasi-regular (M being an A -module), already studied in (0, (15.1)). IV | 204

Theorem (19.5.1). — *Let A be a ring, $\mathbf{f} = (f_i)_{1 \leq i \leq n}$ a suite of elements of A , M an A -module. Consider the following properties:*

label=a) The suite $\mathbf{f}$ is M-regular.

lbbel=b) $H_i(\mathbf{f}, M) = 0$ for every $i > 0$ ((III, 1.1.3)).

b') $H_1(\mathbf{f}, M) = 0$.

lcbel=c) The suite $\mathbf{f}$ is M-quasi-regular.

Then one has the implications

$$a) \Rightarrow b) \Rightarrow b') \quad \text{and} \quad a) \Rightarrow c).$$

Set $\mathfrak{J} = \sum_{i=1}^n f_i A$. If the modules $M_i = M / (\sum_{j=1}^i f_j M)$ are separated for the $\mathfrak{J}$ -preadic topology (resp. if every quotient module of a sub-module of M is separated for the $\mathfrak{J}$ -preadic topology), then one also has the implication $c) \Rightarrow a)$ (resp. $b') \Rightarrow a)$).

Proof. The implications $a) \Rightarrow c)$, and $c) \Rightarrow a)$ when the M_i are separated for the $\mathfrak{J}$ -preadic topology, have already been proved (0, 15.1.9), and are given only as a reminder. It has also been shown ((III, 1.1.4) and (III, 1.1.3.3)) that $a)$ implies $b)$, and $b)$ trivially implies $b')$. It remains therefore to

see that when every quotient module of a sub-module of M is separated for the $\mathfrak{J}$ -preadic topology, $b')$ implies $a)$. We reason by induction on n . For $n = 1$, the assertion follows trivially from the definitions, since then $H_1(\mathbf{f}, M)$ is none other than the kernel of the homothety $z \mapsto f_1 z$ in M ((III, 1.1.1) and (III, 1.1.2)). Suppose therefore $n \geq 2$, and set $\mathbf{f}' = (f_i)_{1 \leq i \leq n-1}$; with the notations of ((III, 1.1.2)), one has $K_*(\mathbf{f}, M) = K_*(f_n) \otimes_A K_*(\mathbf{f}', M)$. One therefore has ((III, 1.1.4.1)) the exact sequence

$$(9) \quad 0 \longrightarrow H_0(f_n, H_1(\mathbf{f}', M)) \longrightarrow H_1(\mathbf{f}, M) \longrightarrow H_1(f_n, H_0(\mathbf{f}', M)) \longrightarrow 0.$$

The relation $H_1(\mathbf{f}, M) = 0$ therefore implies $H_0(f_n, H_1(\mathbf{f}', M)) = 0$ and $H_1(f_n, H_0(\mathbf{f}', M)) = 0$. The first of these two relations signifies that $f_n H_1(\mathbf{f}', M) = H_1(\mathbf{f}', M)$ ((III, 1.1.3.5)). Now, by definition ((III, 1.1.1)), $H_1(\mathbf{f}', M)$ is isomorphic to a quotient of a sub-module of M^{n-1} ; taking on M^{n-1} the filtration of the M^j for $j \leq n-1$, on N the induced filtration, and on $H_1(\mathbf{f}', M)$ the quotient filtration of that of N , one obtains on $H_1(\mathbf{f}', M)$ a finite filtration whose quotients are isomorphic to quotients of sub-modules of M , hence separated for the $\mathfrak{J}$ -preadic topology by hypothesis. Since the relation $f_n H_1(\mathbf{f}', M) = H_1(\mathbf{f}', M)$ implies *a fortiori* $\mathfrak{J}H_1(\mathbf{f}', M) = H_1(\mathbf{f}', M)$, one deduces from the hypothesis and the preceding remarks that $H_1(\mathbf{f}', M) = 0$. The induction hypothesis then proves that the suite $\mathbf{f}'$ is M -regular. On the other hand, the relation $H_1(f_n, H_0(\mathbf{f}', M)) = 0$ signifies that f_n is $(M/(\sum_{i=1}^{n-1} f_i M))$ -regular, hence the suite $\mathbf{f}$ is M -regular. $\square$

Corollary (19.5.2). — Suppose that A is a Noetherian ring, that M is an A -module of finite type, and that the f_i ($1 \leq i \leq n$) belong to the radical of A . Then the four conditions a), b), b'), c) of ((19.5.1)) are equivalent.

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Indeed, one knows in this case (0_I , (7.3.5)) that every A -module of finite type is separated for the $\mathfrak{J}$ -preadic topology.

Proposition (19.5.3). — Let

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be an exact suite of A -modules, $\mathbf{f} = (f_i)_{1 \leq i \leq n}$ a suite M -regular, $\mathfrak{J} = \sum_{i=1}^n f_i A$. Consider the following properties:

label=a) The suite $\mathbf{f}$ is M'' -regular.

lbel=b) The $\mathfrak{J}$ -preadic filtration of M' is induced by the $\mathfrak{J}$ -preadic filtration of M (in other words, the canonical homomorphism $\mathrm{gr}_{\mathfrak{J}}^*(M') \rightarrow \mathrm{gr}_{\mathfrak{J}}^*(M)$ is injective).

lbel=c) The canonical homomorphism $M' / (\sum_{i=1}^n f_i M') \rightarrow M / (\sum_{i=1}^n f_i M)$ is injective.

Then one has the implications $a) \Rightarrow b) \Rightarrow c)$, and $a)$ implies furthermore that the suite $\mathbf{f}$ is M' -regular. If every quotient of a sub-module of M'' is separated for the $\mathfrak{J}$ -preadic topology, the conditions a), b) and c) are equivalent.

Proof. It is clear that c) is a consequence of b). Let us prove that under the hypotheses of the last assertion, c) implies a): since $\mathbf{f}$ is M -regular, one has, by ((19.5.1)), $H_1(\mathbf{f}, M) = 0$, whence an exact sequence, a portion of the exact homology sequence

$$0 \longrightarrow H_1(\mathbf{f}, M'') \longrightarrow H_0(\mathbf{f}, M') \longrightarrow H_0(\mathbf{f}, M).$$

Now, condition c) expresses that the homomorphism $H_0(\mathbf{f}, M') \rightarrow H_0(\mathbf{f}, M)$ is injective ((III, 1.1.3.5)); it is therefore equivalent to $H_1(\mathbf{f}, M'') = 0$, whence, by virtue of the separation hypothesis and ((19.5.1)), the fact that $\mathbf{f}$ is M'' -regular.

Let us show next that a) implies c) and the fact that $\mathbf{f}$ is M' -regular, by proceeding by induction on n . For $n = 1$, f_1 being M -regular is also M' -regular and property c) is none other than the lemma ((3.4.1.4)). For $n > 1$, the induction hypothesis shows that if one sets $\mathbf{f}' = (f_1, \dots, f_{n-1})$, the suite $\mathbf{f}'$ is M' -regular and one has, by virtue of c) applied to $\mathbf{f}'$, an exact sequence

$$0 \longrightarrow M' / \left(\sum_{i=1}^{n-1} f_i M' \right) \longrightarrow M / \left(\sum_{i=1}^{n-1} f_i M \right) \longrightarrow M'' / \left(\sum_{i=1}^{n-1} f_i M'' \right) \longrightarrow 0.$$

By hypothesis, f_n is $(M/(\sum_{i=1}^{n-1} f_i M))$ -regular and $(M''/(\sum_{i=1}^{n-1} f_i M''))$ -regular, hence the same reasoning shows on the one hand that f_n is $(M' / (\sum_{i=1}^{n-1} f_i M'))$ -regular, and on the other hand, by

virtue of ((3.4.1.4)), that the sequence

$$0 \longrightarrow M' / \left(\sum_{i=1}^n f_i M' \right) \longrightarrow M / \left(\sum_{i=1}^n f_i M \right) \longrightarrow M'' / \left(\sum_{i=1}^n f_i M'' \right) \longrightarrow 0$$

is exact, whence c).

Let us show finally that a) implies b). Since the suite $\mathbf{f}$ is then M -regular and M' -regular, hence M -quasi-regular and M' -quasi-regular, one has canonical isomorphisms

$$\mathrm{gr}_{\mathfrak{J}}^*(M') \xrightarrow{\sim} \mathrm{gr}_{\mathfrak{J}}^0(M')[T_1, \dots, T_n], \quad \mathrm{gr}_{\mathfrak{J}}^*(M) \xrightarrow{\sim} \mathrm{gr}_{\mathfrak{J}}^0(M)[T_1, \dots, T_n]$$

(0, 15.1.7), and since we have seen above that $\mathrm{gr}_{\mathfrak{J}}^0(M') \rightarrow \mathrm{gr}_{\mathfrak{J}}^0(M)$ is injective, the same holds for $\mathrm{gr}_{\mathfrak{J}}^*(M') \rightarrow \mathrm{gr}_{\mathfrak{J}}^*(M)$.

One notes moreover that the equivalence of conditions a), b), c) of ((19.5.3)) holds in particular when A is Noetherian, M'' an A -module of finite type, and the f_i belong to the radical of A . $\square$

(19.5.4). In the remainder of this section, we keep the notations $A, \mathbf{f}, M, \mathfrak{J}$ and suppose moreover **IV** | 206
 M equipped with a decreasing filtration (M_k) formed of sub- A -modules, with $M_0 = M$. Recall (0, (15.1.5)) that one defines then on M a second decreasing filtration formed of the sub-modules

$$M'_k = M_k + \mathfrak{J}M_{k-1} + \dots + \mathfrak{J}^{k-1}M_1 + \mathfrak{J}^kM_0$$

and that, if $\mathrm{gr}_{\cdot}(M)$ and $\mathrm{gr}'_{\cdot}(M)$ are the graded A -modules associated to the filtrations (M_k) and (M'_k) respectively, one defines a surjective graded homomorphism of degree 0

$$\psi_M : (\mathrm{gr}_{\cdot}(M) \otimes_A (A/\mathfrak{J}))[T_1, \dots, T_n] \longrightarrow \mathrm{gr}'_{\cdot}(M)$$

(0, (15.1.5.2)). Recall also that one has the surjective canonical homomorphism (0, (15.1.1.1))

$$\varphi_{\mathrm{gr}_{\cdot}(M)} : (\mathrm{gr}_{\cdot}(M) \otimes_A (A/\mathfrak{J}))[T_1, \dots, T_n] \longrightarrow \mathrm{gr}_{\mathfrak{J}}^*(\mathrm{gr}_{\cdot}(M)).$$

Theorem (19.5.5). — With the notations of ((19.5.4)), consider the following properties:

label=a) The suite $\mathbf{f}$ is $\mathrm{gr}_{\cdot}(M)$ -regular.

lbbel=b) The canonical homomorphism ψ_M is bijective.

lcbel=c) The canonical homomorphism $\varphi_{\mathrm{gr}_{\cdot}(M)}$ is bijective (in other words (0, (15.1.7)), the suite $\mathbf{f}$ is $\mathrm{gr}_{\cdot}(M)$ -quasi-regular).

One has the implications

$$a) \Rightarrow b) \Rightarrow c).$$

Furthermore, if, for every integer $k \geq 0$, the quotient modules of sub-modules of $\mathrm{gr}_k(M)$ are separated for the $\mathfrak{J}$ -preadic topology (which is the case when A is Noetherian, the f_i are in the radical of A and the $\mathrm{gr}_k(M)$ are A -modules of finite type), then the conditions a), b) and c) are equivalent.

Proof. The implication $a) \Rightarrow b)$ was shown in (0, (15.1.8)); the implication $c) \Rightarrow a)$ under the separation hypothesis is a special case of (0, (15.1.9)). It remains to prove that $b)$ implies $c)$, which will be done in several steps.¹⁴

We will denote by $\mathfrak{J}$ the ideal $\sum_{i=1}^n f_i A$ generated by the f_i ; for every suite $\mathbf{q} = (q_1, \dots, q_n)$ of n integers ≥ 0 , we set $|\mathbf{q}| = \sum_{i=1}^n q_i$ and $\mathbf{f}^{\mathbf{q}} = f_1^{q_1} f_2^{q_2} \dots f_n^{q_n}$.

(19.5.5.1) For ψ_M to be injective (and hence bijective), it is necessary and sufficient that for $p \geq 0$ and $q \geq 0$ the following condition be satisfied:

(I_{q,p}) For every family $(x_{\mathbf{q}}^p)_{|\mathbf{q}|=q}$ of elements of M_p , the relation

$$\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x_{\mathbf{q}}^p \in \sum_{i=1}^q \mathfrak{J}^{q-i} M_{p+i} + M'_{p+q+1}$$

implies $x_{\mathbf{q}}^p \in M_{p+1} + \mathfrak{J}M_p$ for every $\mathbf{q}$ such that $|\mathbf{q}| = q$.

¹⁴The implication $b) \Rightarrow c)$, without hypothesis of separation on the $\mathrm{gr}_k(M)$, is due to P. Deligne, whose proof we reproduce below.

One proceeds as in (0, (15.1.6)) by considering the sub- A -module Q_k (resp. Q'_k) of the degree- k terms of the first (resp. second) member of (0, (15.1.5.2)). One equips Q_k with the filtration

$$(Q_k)_i = \sum_{j \leq k-i} \left(\sum_{|\mathbf{l}|=j} (\mathrm{gr}_{k-j}(M) \otimes_A (A/\mathfrak{J})) \mathbf{T}^{\mathbf{l}} \right)$$

and Q'_k with the filtration formed by the images $\psi_M((Q_k)_i) = (Q'_k)_i$; it suffices to prove that the homomorphisms $(\psi_M)_{ki} : \mathrm{gr}_i(Q_k) \rightarrow \mathrm{gr}_i(Q'_k)$ are injective. Now, we have

$$\mathrm{gr}_i(Q_k) = \sum_{|\mathbf{l}|=k-i} ((M_i/M_{i+1}) \otimes_A (A/\mathfrak{J})) \mathbf{T}^{\mathbf{l}} = \sum_{|\mathbf{l}|=k-i} (M_i/(\mathfrak{J}M_i + M_{i+1})) \mathbf{T}^{\mathbf{l}};$$

on the other hand, $(Q'_k)_{i+1}$ is the image of $\sum_{h=0}^{k-i-1} \mathfrak{J}^{k-i-1-h} M_{i+h+1}$ in M_k/M'_{k+1} . Writing that $(\psi_M)_{ki}$ is injective thus amounts to writing the condition $(I_{k-i,i})$; whence our assertion.

(19.5.5.2) For the suite $\mathbf{f}$ to be $\mathrm{gr}_\bullet(M)$ -quasi-regular, it suffices that, for $p \geq 0$ and $q \geq 0$, the following condition be satisfied:

$(S_{q,p})$ For every family $(x_{\mathbf{q}}^p)_{|\mathbf{q}|=q}$ of elements of M_p , the relation $\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x_{\mathbf{q}}^p \in M_{p+1}$ implies $x_{\mathbf{q}}^p \in M_{p+1} + \mathfrak{J}M_p$ for every $\mathbf{q}$ such that $|\mathbf{q}| = q$.

By definition, saying that $\mathbf{f}$ is $\mathrm{gr}_\bullet(M)$ -quasi-regular means that, for every $p \geq 0$ and every $q \geq 0$, a relation

$$\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x_{\mathbf{q}}^p \in M_{p+1} + \mathfrak{J}^{q+1}M_p$$

for a family $(x_{\mathbf{q}}^p)_{|\mathbf{q}|=q}$ of elements of M_p implies $x_{\mathbf{q}}^p \in M_{p+1} + \mathfrak{J}M_p$ for every $\mathbf{q}$ such that $|\mathbf{q}| = q$. Now the hypothesis on the suite $(x_{\mathbf{q}}^p)$ means that there exists a family $(y_{\mathbf{q}}^p)_{|\mathbf{q}|=q}$ of elements of $\mathfrak{J}M_p$ such that $\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} (x_{\mathbf{q}}^p - y_{\mathbf{q}}^p) \in M_{p+1}$. The condition $(S_{q,p})$ then implies $x_{\mathbf{q}}^p - y_{\mathbf{q}}^p \in M_{p+1} + \mathfrak{J}M_p$ for every $\mathbf{q}$ such that $|\mathbf{q}| = q$, whence $x_{\mathbf{q}}^p \in M_{p+1} + \mathfrak{J}M_p$; we see therefore that the conditions $(S_{q,p})$ imply that the suite $\mathbf{f}$ is $\mathrm{gr}_\bullet(M)$ -quasi-regular.

It suffices to prove the following proposition:

(19.5.5.3) For every $r > 0$, if the conditions $(I_{q,p})$ are satisfied for $p + q < r$, then the conditions $(S_{q,p})$ are satisfied for $p + q < r$.

Introduce the conditions:

$(A_{q,p})$ For every family $(x_{\mathbf{q}}^p)_{|\mathbf{q}|=q}$ of elements of M_p , the relation $\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x_{\mathbf{q}}^p \in M_{p+1}$ implies $\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x_{\mathbf{q}}^p \in \sum_{i=1}^q \mathfrak{J}^{q-i} M_{p+i}$.

It is clear that the conjunction of $(A_{q,p})$ and $(I_{q,p})$ implies $(S_{q,p})$. On the other hand, the conditions $(A_{1,p})$ and $(S_{0,p})$ are trivial. Consequently, **(19.5.5.3)** will follow from the proposition:

(19.5.5.4) For all $p \geq 0$ and $q \geq 1$, the condition $(A_{q+1,p})$ is entailed by the conjunction of conditions $(A_{q,p})$, $(I_{q-1,p+1})$, $(I_{q-2,p+2})$, $\dots$, $(I_{0,p+q})$.

Indeed, once this proposition is proved, the conditions $(I_{q,p})$, assumed satisfied for $p + q < r$, will entail, for each $p \leq r$, the condition $(A_{q,p})$ for $1 \leq q < r - p$, by recurrence on q .

Let us prove **(19.5.5.4)**. Note that the condition $(A_{q,p})$ is also equivalent to

$$(10) \quad \mathfrak{J}^q M_p \cap M_{p+1} \subset \sum_{i=1}^q \mathfrak{J}^{q-i} M_{p+i}.$$

Let $m \in \mathfrak{J}^{q+1} M_p \cap M_{p+1} \subset \mathfrak{J}^q M_p \cap M_{p+1}$. Applying the condition $(A_{q,p})$, we see that there exists, for each i such that $1 \leq i \leq q$, a family $(y_{\mathbf{q}}^{p+i})_{|\mathbf{q}|=q-i}$ of elements of M_{p+i} such that

$$m = \sum_{i=1}^q \left(\sum_{|\mathbf{q}|=q-i} \mathbf{f}^{\mathbf{q}} y_{\mathbf{q}}^{p+i} \right).$$

Suppose that, for $1 \leq i < j$ ($j \geq 1$), we have proved that

$$y_{\mathbf{q}}^{p+i} \in \mathfrak{J}M_{p+i} + M_{p+i+1} \quad (\text{for } |\mathbf{q}| = q - i).$$

We deduce, by the definition ((19.5.4)), that

$$m - \sum_{i < j} \left(\sum_{|\mathbf{q}|=q-i} \mathbf{f}^{\mathbf{q}} y_{\mathbf{q}}^{p+i} \right) \in M'_{p+q+1},$$

since $m \in \mathfrak{J}^{q+1} M_p \subset M'_{p+q+1}$ and, for $i < j$, each term $\mathbf{f}^{\mathbf{q}} y_{\mathbf{q}}^{p+i}$ (with $|\mathbf{q}| = q - i$) belongs to $\mathfrak{J}^{q-i+1} M_{p+i} + \mathfrak{J}^{q-i} M_{p+i+1} \subset M'_{p+q+1}$. Now the left-hand side equals $\sum_{i \geq j} \left(\sum_{|\mathbf{q}|=q-i} \mathbf{f}^{\mathbf{q}} y_{\mathbf{q}}^{p+i} \right)$, whence

$$\sum_{|\mathbf{q}|=q-j} \mathbf{f}^{\mathbf{q}} y_{\mathbf{q}}^{p+j} \in \sum_{l=1}^{q-j} \mathfrak{J}^{q-j-l} M_{p+j+l} + M'_{p+q+1}.$$

Since by hypothesis $(I_{q-j, p+j})$ is satisfied, we deduce

$$y_{\mathbf{q}}^{p+j} \in \mathfrak{J} M_{p+j} + M_{p+j+1}$$

and, by recurrence on j , we see that this condition is satisfied for every j such that $1 \leq j \leq q$. We have thus

$$m \in \sum_{i=1}^q \mathfrak{J}^{q-i} (\mathfrak{J} M_{p+i} + M_{p+i+1}) = \sum_{i=1}^{q+1} \mathfrak{J}^{q-i+1} M_{p+i}$$

and we have thus proved $(A_{q+1, p})$ and finished the proof. $\square$

Remark (19.5.6). — (i) The results of this section remain valid when, instead of an A -module M and elements $f_i \in A$, one takes an object M of an arbitrary abelian category, and n endomorphisms f_i of M which commute two by two; the proofs adapt without difficulty to this more general case. The hypothesis of separation for an A -module N relatively to the $\mathfrak{J}$ -preadic topology should here be replaced by the hypothesis that every sub-object of N contained in all the $\sum f_i^r(N)$ (r an arbitrary integer) should be zero. The same remark applies to the results of (19.7).

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(ii) Suppose that A is a graded ring with degrees bounded below ≥ 0 . If M (resp. each $\text{gr}_p(M)$) is a graded module with degrees bounded below and if the f_i contain no homogeneous component of degree 0, the hypothesis of separation of $(0, (15.1.9))$ is *ipso facto* verified for M (resp. each $\text{gr}_p(M)$), and one sees thus that in this case one has the implication $c) \Rightarrow a)$ in ((19.5.1)) (resp. ((19.5.5))).

(iii) Recall that, without hypothesis of separation, the implication $c) \Rightarrow b)$ is no longer valid even for $n = 1$ (0, (15.1.12, iii)).

19.6. Regular suites relatively to a filtered quotient module.

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(19.6.1). Let A be a ring, $\mathbf{f} = (f_1, \dots, f_n)$ a finite suite of elements of A , $\mathfrak{J} = f_1 A + \dots + f_n A$ the ideal that it generates, and consider an exact suite of A -modules

$$0 \longrightarrow R \xrightarrow{j} N \xrightarrow{p} M \longrightarrow 0$$

where we suppose N equipped with a decreasing filtration (N_k) formed of sub- A -modules, with $N_0 = N$. One sets $R_k = R \cap N_k$ (filtration induced by (N_k)), $M_k = p(N_k)$ (quotient filtration of (N_k)) and one designates by $\text{gr}(N)$, $\text{gr}(R)$ and $\text{gr}(M)$ the graded A -modules associated to these three filtrations. One sets on the other hand for every k ,

$$N'_k = N_k + \mathfrak{J} N_{k-1} + \dots + \mathfrak{J}^{k-1} N_1 + \mathfrak{J}^k N_0$$

so that $M'_k = p(N'_k) = M_k + \mathfrak{J} M_{k-1} + \dots + \mathfrak{J}^{k-1} M_1 + \mathfrak{J}^k M_0$; one sets on the other hand $R'_k = R \cap N'_k$ (filtration induced by (N'_k)), and one notes $\text{gr}'(N)$, $\text{gr}'(M)$ and $\text{gr}'(R)$ the graded A -modules associated to these three filtrations; one has thus a commutative diagram of exact suites

$$(11) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \text{gr}(R) & \xrightarrow{\text{gr}(j)} & \text{gr}(N) & \xrightarrow{\text{gr}(p)} & \text{gr}(M) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \text{gr}'(R) & \xrightarrow{\text{gr}'(j)} & \text{gr}'(N) & \xrightarrow{\text{gr}'(p)} & \text{gr}'(M) \longrightarrow 0 \end{array}$$

where the vertical arrows are the canonical homomorphisms of a graded module associated to a filtration to the graded module associated to a finer filtration.

(19.6.2). One has defined in (0, (15.1.5.2)) surjective graded homomorphisms of degree 0

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$$\begin{aligned}\psi_N &: (\operatorname{gr}_\bullet(N) \otimes_A (A/\mathfrak{J}))[\mathbf{T}_1, \dots, \mathbf{T}_n] \longrightarrow \operatorname{gr}'_\bullet(N) \\ \psi_M &: (\operatorname{gr}_\bullet(M) \otimes_A (A/\mathfrak{J}))[\mathbf{T}_1, \dots, \mathbf{T}_n] \longrightarrow \operatorname{gr}'_\bullet(M) \\ \psi_R &: (\operatorname{gr}_\bullet(R) \otimes_A (A/\mathfrak{J}))[\mathbf{T}_1, \dots, \mathbf{T}_n] \longrightarrow \operatorname{gr}''_\bullet(R)\end{aligned}$$

where the first two are deduced from the two last vertical arrows of ((19.6.1.1)). We denote by ψ'_R the homomorphism composed $\gamma \circ \psi_R$.

Proposition (19.6.3). — *With the preceding notations, suppose that the suite $\mathbf{f}$ is $\operatorname{gr}_\bullet(N)$ -regular. Consider the following properties:*

label=a) $\mathbf{f}$ is $\operatorname{gr}_\bullet(M)$ -regular.

lbbel=b) ψ_M is bijective.

lcbel=c) ψ'_R is surjective (in other words $\operatorname{gr}'_\bullet(R)$ is generated as $\operatorname{gr}^*_{\mathfrak{J}}(A)$ -module by $\operatorname{Im}(\operatorname{gr}_\bullet(R) \rightarrow \operatorname{gr}'_\bullet(R))$).

ldbel=d) ψ'_R is injective.

d') The homomorphism $(\psi'_R)_0 : \operatorname{gr}_\bullet(R) \otimes_A (A/\mathfrak{J}) \rightarrow \operatorname{gr}'_\bullet(R)$ obtained by restricting ψ'_R to polynomials of degree 0, is injective.

lebel=e) ψ'_R is bijective.

f) j' is injective.

f') The homomorphism $\operatorname{gr}_\bullet(R) \otimes_A (A/\mathfrak{J}) \rightarrow \operatorname{gr}_\bullet(N) \otimes_A (A/\mathfrak{J})$ obtained by restricting j' to polynomials of degree 0, is injective.

lfbel=f) The $\mathfrak{J}$ -preadic filtration of $\operatorname{gr}_\bullet(R)$ is induced by that of $\operatorname{gr}_\bullet(N)$.

lgbel=g) The filtrations (R'_k) and (R''_k) on R are identical.

One has the implications:

$$a) \Rightarrow e) \Rightarrow b) \Leftrightarrow c) \Leftrightarrow h)$$

$$g) \Rightarrow d) \Leftrightarrow d') \Leftrightarrow f) \Leftrightarrow f')$$

Moreover, when every module quotient of a sub-module of $\operatorname{gr}_k(M)$ is separated for the $\mathfrak{J}$ -preadic topology (which will be the case when A is Noetherian, the f_i belong to the radical of A and the $\operatorname{gr}_k(M)$ are A -modules of finite type), then all the conditions a) to h) are equivalent.

Proof. Note that by (19.5.5), the hypothesis that $\mathbf{f}$ is $\operatorname{gr}_\bullet(N)$ -regular implies that ψ_N is bijective; since $\operatorname{gr}'_\bullet(j)$ is injective, the equivalence of d) and f) follows from the diagram (19.6.2.1), as does that of d') and f'); moreover f) and f') are trivially equivalent, which shows the equivalence of d), d'), f) and f').

Since one already knows that ψ_M is surjective and ψ_N bijective, the equivalence of b) and c) also follows from the diagram (19.6.2.1) by a particular case of the five lemma.

The relation $\psi'_R = \gamma \circ \psi_R$ (19.6.2) and the fact that ψ_R is surjective show that condition c) is equivalent to saying that γ is surjective, which is equivalent to condition h); one has thus proved the equivalence of b), c) and h).

It is trivial that e) is equivalent to the conjunction of c) and d), hence also of b) and d).

Note now that a) implies b) and that b) implies a) under the hypothesis of separation ((19.5.5)). On the other hand, the implications a) $\Rightarrow$ g) $\Rightarrow$ f'), and the implication f') $\Rightarrow$ a) under the hypothesis of separation, result from ((19.5.3)). One has thus proved that a) implies e) (equivalent to the conjunction of b) and f')), and also that all conditions are equivalent under the hypothesis of separation. $\square$

Corollary (19.6.4). — *Let $\mathfrak{K}$ be an ideal of A , $\mathfrak{L} = \mathfrak{J} + \mathfrak{K}$. Suppose that the filtration (N_k) (hence also (M_k) and (N'_k)) is the $\mathfrak{K}$ -preadic filtration. Consider $\operatorname{gr}_\bullet(R)$ (resp. $\operatorname{gr}'_\bullet(R)$) as a graded module over $\operatorname{gr}^*_{\mathfrak{K}}(A)$ (resp. $\operatorname{gr}^*_{\mathfrak{L}}(A)$).*

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label=(i) Let S be a part of $\operatorname{gr}_\bullet(R)$ which generates $\operatorname{gr}_\bullet(R)$ as $\operatorname{gr}^*_{\mathfrak{K}}(A)$ -module. Then the condition c) of ((19.6.3)) is equivalent to the following condition:

label=c') The image of S in $\operatorname{gr}'_\bullet(R)$ by ψ'_R generates $\operatorname{gr}'_\bullet(R)$ as $\operatorname{gr}^*_{\mathfrak{L}}(A)$ -module.

lbbel=(ii) Suppose verified the condition e) of ((19.6.3)), and suppose moreover that the quotients of the $\operatorname{gr}_k(R)$ are separated for the $\mathfrak{J}$ -preadic topology (which will be the case when A is Noetherian, the f_i are in the radical of A and the $\operatorname{gr}_k(M)$ are A -modules of finite type). Then, for a part S of

homogeneous elements to generate $\mathrm{gr}_\bullet(R)$ as $\mathrm{gr}_\mathfrak{K}^*(A)$ -module, it is necessary and sufficient that its image by ψ'_R generate $\mathrm{gr}'_\bullet(R)$ as $\mathrm{gr}_\mathfrak{L}^*(A)$ -module.

Proof. (i) Consider the degree-0 homomorphism

$$\psi_A : (\mathrm{gr}_\mathfrak{K}^*(A) \otimes_A (A/\mathfrak{J}))[\mathrm{T}_1, \dots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(A) = \mathrm{gr}_\mathfrak{L}^*(A)$$

(0, (15.1.5.2)). The fact that this homomorphism is surjective implies that the sub- $\mathrm{gr}_\mathfrak{L}^*(A)$ -algebra of $\mathrm{gr}'_\bullet(R)$ generated by $S' = \psi'_R(S)$ is none other than $\mathrm{Im}(\psi'_R)$, whence the equivalence of c) and c').

(ii) Since e) is verified, the same is true of c), and by virtue of (i), it remains to prove the sufficiency of the condition of the statement. As ψ'_R is bijective, to say that S generates $\mathrm{gr}'_\bullet(R)$ considered as $\mathrm{gr}_\mathfrak{L}^*(A)$ -module is equivalent to saying that S generates $(\mathrm{gr}_\bullet(R) \otimes_A (A/\mathfrak{J}))[\mathrm{T}_1, \dots, \mathrm{T}_n]$ as $\mathrm{gr}_\mathfrak{L}^*(A)$ -module, and by the surjectivity of ψ_A , one can in this statement replace the ring $\mathrm{gr}_\mathfrak{L}^*(A)$ by $(\mathrm{gr}_\mathfrak{K}^*(A) \otimes_A (A/\mathfrak{J}))[\mathrm{T}_1, \dots, \mathrm{T}_n]$. This clearly amounts to saying that S generates $\mathrm{gr}_\bullet(R) \otimes_A (A/\mathfrak{J})$ as module over the ring $\mathrm{gr}_\mathfrak{K}^*(A) \otimes_A (A/\mathfrak{J})$, or equivalently as $\mathrm{gr}_\mathfrak{K}^*(A)$ -module. Now, if T is the sub- $\mathrm{gr}_\mathfrak{K}^*(A)$ -module of $\mathrm{gr}_\bullet(R)$ generated by S and if one sets $T' = \mathrm{gr}_\bullet(R)/T$, one has by hypothesis $T'/\mathfrak{J}T' = 0$, i.e. $\mathfrak{J}T' = T'$; but T' is a graded module, and each T'_p is a quotient of $\mathrm{gr}_p(R)$, hence separated for the $\mathfrak{J}$ -preadic topology; the hypothesis $\mathfrak{J}T'_p = T'_p$ therefore implies $T'_p = 0$ for every integer p , hence $T' = 0$, which completes the proof. $\square$

19.7. Hironaka's criterion of normal flatness.

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Theorem (19.7.1). — (Hironaka.) — Let A be a ring, $\mathfrak{K}$ an ideal of A , $\mathbf{f} = (f_1, \dots, f_n)$ a suite of elements of A which is $(A/\mathfrak{K})$ -regular, $\mathfrak{J} = f_1A + \dots + f_nA$ the ideal generated by $\mathbf{f}$, $\mathfrak{L} = \mathfrak{J} + \mathfrak{K}$, M an A -module. One has in this case $(\mathrm{gr}_\mathfrak{K}^*(M)) \otimes_A (A/\mathfrak{J}) = (\mathrm{gr}_\mathfrak{K}^*(M)) \otimes_A (A/\mathfrak{L}) = (\mathrm{gr}_\mathfrak{K}^*(M)) \otimes_{A/\mathfrak{K}} (A/\mathfrak{L})$, since $(\mathfrak{K}^n M / \mathfrak{K}^{n+1} M) \otimes_A (A/\mathfrak{J}) = \mathfrak{K}^n M / (\mathfrak{K}^{n+1} + \mathfrak{J}\mathfrak{K}^n) M = \mathfrak{K}^n M / (\mathfrak{K}^{n+1} + \mathfrak{L}\mathfrak{K}^n) M$. Consider the following conditions:

label=a) $\mathrm{gr}_\mathfrak{K}^*(M)$ is a flat $(A/\mathfrak{K})$ -module.

lbbel=b) $\mathrm{gr}_\mathfrak{L}^*(M)$ is a flat $(A/\mathfrak{L})$ -module and the canonical homomorphism (0, (15.1.5.2))

$$\psi_M : ((\mathrm{gr}_\mathfrak{K}^*(M)) \otimes_A (A/\mathfrak{L}))[\mathrm{T}_1, \dots, \mathrm{T}_n] \longrightarrow \mathrm{gr}_\mathfrak{L}^*(M)$$

is bijective.

lcbel=c) $(\mathrm{gr}_\mathfrak{K}^*(M)) \otimes_{A/\mathfrak{K}} (A/\mathfrak{L})$ is a flat $(A/\mathfrak{L})$ -module and the suite $\mathbf{f}$ is $(\mathrm{gr}_\mathfrak{K}^*(M))$ -regular.

ldbel=d) $\mathrm{gr}_\mathfrak{L}^*(M)$ is a flat $(A/\mathfrak{L})$ -module, and for every $\mathfrak{p} \in \mathrm{Ass}_{A/\mathfrak{K}}(A/\mathfrak{L})$, $(\mathrm{gr}_\mathfrak{K}^*(M))_{\mathfrak{p}}$ is a flat $(A/\mathfrak{K})_{\mathfrak{p}}$ -module.

One then has the implications

$$a) \Rightarrow c) \Rightarrow b)$$

and $a) \Rightarrow d)$.

When every quotient of a sub-module of $\mathrm{gr}_\mathfrak{K}^p(M)$ ($p \geq 0$) is ideally separated for $\mathfrak{J}$ (Bourbaki, Alg. comm., chap. III, § 5, n° 1), the conditions a), b) and c) are equivalent.

Finally, when A is Noetherian, the f_i belong to the radical of A , the suite $\mathbf{f}$ is $\mathrm{gr}_\mathfrak{K}^*(A)$ -regular and M is an A -module of finite type, the conditions a), b), c), d) are equivalent.

Remark (19.7.2). — (i) When one supposes that $\mathrm{gr}_\mathfrak{K}^*(A)$ is a flat $(A/\mathfrak{K})$ -module, the hypothesis that the suite $\mathbf{f}$ is $(A/\mathfrak{K})$ -regular implies that it is also $\mathrm{gr}_\mathfrak{K}^*(A)$ -regular (0, (15.1.14)). One notes in particular that this will be the case when A and $A/\mathfrak{K}$ are regular rings (0, (17.3.6)): indeed, $\mathfrak{K}$ is then a regular ideal of A ((19.1.2)), hence quasi-regular, and localising at the maximal ideals of A containing $\mathfrak{K}$ and using (0, (15.1.7)), one sees that $\mathrm{gr}_\mathfrak{K}^*(A)$ is a flat $(A/\mathfrak{K})$ -module projective.

(ii) It would be interesting to be able to prove the last conclusion of ((19.7.1)) by only supposing that there exists a B -algebra B which is a Noetherian ring, for which $\mathfrak{J}B$ is contained in the radical of B , and M is a B -module of finite type.

Corollary (19.7.3). — Let A be a local Noetherian ring, $\mathfrak{m}$ its maximal ideal, $\mathfrak{K}$ an ideal of A such that the ring $A/\mathfrak{K}$ is regular and of dimension n , M an A -module of finite type. The following conditions are equivalent:

label=a) $\mathrm{gr}_\mathfrak{K}^*(M)$ is a flat $(A/\mathfrak{K})$ -module.

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label=b) If $P(T)$ and $Q(T)$ are the Poincaré series of the $(A/\mathfrak{m})$ -graded modules $\mathrm{gr}_{\mathfrak{m}}^*(M)$ and $\mathrm{gr}_{\mathfrak{R}}^*(M) \otimes_{A/\mathfrak{R}} (A/\mathfrak{m})$, one has

$$(12) \quad Q(T) = P(T)(1 - T)^{-n}.$$

Proof. Let $\mathbf{f} = (f_1, \dots, f_n)$ be a sequence of elements of A such that the images of the f_i in $A/\mathfrak{R}$ form a regular system of parameters of $A/\mathfrak{R}$ (0, 17.1.6), so that $\mathbf{f}$ is an $(A/\mathfrak{R})$ -regular sequence; moreover, if $\mathfrak{J}$ is the ideal generated by $\mathbf{f}$, then $\mathfrak{J} + \mathfrak{R} = \mathfrak{m}$, and since $A/\mathfrak{m}$ is a field, $\mathrm{gr}_{\mathfrak{m}}^*(M)$ is a flat $(A/\mathfrak{m})$ -module. Since A is Noetherian and M an A -module of finite type, one can apply the equivalence of *a*) and *b*) in ((19.7.1)). Moreover, since $A/\mathfrak{m}$ is a field, the fact that ψ_M is surjective amounts to saying that for all $h \geq 0$, $\mathrm{gr}_{\mathfrak{m}}^h(M)$ and the sub-module of elements of degree h of $(\mathrm{gr}_{\mathfrak{R}}^*(M) \otimes_{A/\mathfrak{R}} (A/\mathfrak{m}))[T_1, \dots, T_n]$ have the same rank over $A/\mathfrak{m}$; but for the second of these modules, the rank in question is evidently equal to

$$\sum_{r=0}^h \binom{r+n-1}{n-1} \mathrm{rg}(\mathrm{gr}_{\mathfrak{R}}^{h-r}(M) \otimes_{A/\mathfrak{R}} (A/\mathfrak{m})).$$

Since $(1 - T)^{-n} = \sum_{r=0}^{\infty} \binom{r+n-1}{n-1} T^r$, this proves that the relation (19.7.3.1) is necessary and sufficient for ψ_M to be bijective. $\square$

Corollary (19.7.4). — *Let X be a locally Noetherian prescheme, Y a closed sub-prescheme of X , regular and connected, $\mathcal{F}$ an $\mathcal{O}_X$ -Module coherent normally flat along Y ((6.10.1)). Then there exists a formal power series $R \in \mathbb{Q}[[T]]$ (independent of $x \in Y$) such that, for every $x \in Y$, the Poincaré series of $\mathrm{gr}_{\mathfrak{m}_x}^*(\mathcal{F}_x)$ is given by the formula*

$$(13) \quad P_x(\mathcal{F})(T) = R(T)(1 - T)^{-n} \quad \text{where } n = \dim(\mathcal{O}_{X,x}).$$

Proof. Let $\mathcal{J}$ be the coherent Ideal of $\mathcal{O}_X$ defining Y ; the hypothesis on $\mathcal{F}$ means that $\mathrm{gr}_{\mathcal{J}}^*(\mathcal{F})$ is a flat $\mathcal{O}_Y$ -Module. One can therefore, for every $x \in Y$, apply ((19.7.3)) by taking $A = \mathcal{O}_{X,x}$, $\mathfrak{R} = \mathcal{J}_x$, $M = \mathcal{F}_x$; hypothesis *a*) being satisfied by hypothesis, the Poincaré series $P_x(\mathcal{F})(T)$ is therefore equal to $R_x(T)(1 - T)^{-n}$, where $R_x(T)$ is the Poincaré series of $\mathrm{gr}_{\mathcal{J}_x}^*(\mathcal{F}_x) \otimes_{\mathcal{O}_{Y,x}} k(x)$. Now, each of the $\mathcal{O}_Y$ -Modules $\mathcal{J}^h \mathcal{F} / \mathcal{J}^{h+1} \mathcal{F}$ is flat and coherent by hypothesis, and since Y is supposed connected, each of the $\mathcal{J}^h \mathcal{F} / \mathcal{J}^{h+1} \mathcal{F}$ is of constant rank in Y , locally free ((2.1.12)), and this proves that the Poincaré series $R_x(T)$ is independent of $x \in Y$. $\square$

Corollary (19.7.5). — *Let X be a locally Noetherian prescheme, Y a closed sub-prescheme of X , Z a closed sub-prescheme of Y ; suppose Y and Z regular. Let $\mathcal{F}$ be an $\mathcal{O}_X$ -Module coherent.*

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label=i) If $\mathcal{F}$ is normally flat along Y at the points of Z (cf. ((11.3.4)) for the definition of normal flatness at a point), then $\mathcal{F}$ is normally flat along the points of Z .

label=ii) Suppose moreover that Z is connected and that $\mathcal{O}_X$ is normally flat along Y at the points of Z (which will take place in particular if X is regular at the points of Z). If $\mathcal{F}$ is normally flat along Z , and if there exists a point $z \in Z$ such that $\mathcal{F}$ is normally flat along Y at the point z , then $\mathcal{F}$ is normally flat along Y at all points of Z .

Proof. Let $\mathcal{K}$ and $\mathcal{L} \supset \mathcal{K}$ be the coherent Ideals of $\mathcal{O}_X$ defining respectively Y and Z . For every point $z \in Z$, the canonical immersion $Z \rightarrow Y$ is regular at the point z , since Y and Z are regular at this point ((19.1.1)); in other words, there exists a sequence $(f_i)_{1 \leq i \leq n}$ of elements of $\mathcal{O}_{X,z}$ which is $\mathcal{K}_z$ -regular and is such that $\mathcal{L}_z = \mathcal{K}_z + \sum f_i \mathcal{O}_{X,z}$.

(i). The hypothesis means that $\mathrm{gr}_{\mathcal{K}_z}^*(\mathcal{F}_z)$ is a flat $\mathcal{O}_{Y,z}$ -module; it then entails that $\mathrm{gr}_{\mathcal{L}_z}^*(\mathcal{F}_z)$ is a flat $\mathcal{O}_{Z,z}$ -module, by virtue of the fact that in ((19.7.1)), *a*) entails *b*).

(ii). The supplementary hypothesis on X entails that the sequence (f_i) is $\mathrm{gr}_{\mathcal{K}_z}^*(\mathcal{F}_z)$ -regular at every point $z \in Z$ ((19.7.2)). On the other hand, Z being regular and connected, is integral, and if $\mathrm{gr}_{\mathcal{K}_z}^*(\mathcal{F}_z)$ is $\mathcal{O}_{Y,z}$ -flat at a point $z \in Z$, then, as the point is a generisation of the generic point ζ of Z , one concludes that $\mathrm{gr}_{\mathcal{K}_\zeta}^*(\mathcal{F}_\zeta)$ is $\mathcal{O}_{Y,\zeta}$ -flat (0I, 6.3.1). One deduces that if at every point z' of Z , $\mathrm{gr}_{\mathcal{L}_{z'}}^*(\mathcal{F}_{z'})$ is a flat $\mathcal{O}_{Z,z'}$ -module, then $\mathrm{gr}_{\mathcal{K}_{z'}}^*(\mathcal{F}_{z'})$ is a flat $\mathcal{O}_{Y,z'}$ -module, because one can apply the equivalence of *a*) and *d*) in ((19.7.1)). $\square$

19.8. Properties of passage to the projective limit. In this section, the notations and conventions on projective limits are those of ((8.5.1)) and ((8.1.1)). IV | 219

Proposition (19.8.1). — *Suppose that the transition morphisms $S_\mu \rightarrow S_\lambda$ ($\lambda \leq \mu$) are flat, and in addition that one of the two following hypotheses is verified:*

- 1° *The preschemes S_λ are locally Noetherian.*
- 2° *The transition morphisms $S_\mu \rightarrow S_\lambda$ are surjective (hence faithfully flat).*

In these conditions:

- label=(i) *Suppose S_α quasi-compact. Let $\mathcal{F}_\alpha$ be an $\mathcal{O}_{S_\alpha}$ -Module quasi-coherent, which we suppose moreover of finite type when hypothesis 1° is satisfied. Let $(f_{i\alpha})_{1 \leq i \leq n}$ be a finite suite of sections of $\mathcal{F}_\alpha$ above S_α . For the suite of sections f_i of the $\mathcal{O}_S$ -Module $\mathcal{F}$ above S , corresponding to the $f_{i\alpha}$, to be $\mathcal{F}$ -regular, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that the suite of sections $f_{i\lambda}$ of $\mathcal{F}_\lambda$ above S_λ be $\mathcal{F}_\lambda$ -regular.*
- lbbel=(ii) *Suppose X_α quasi-compact, and let $j_\alpha : Y_\alpha \rightarrow X_\alpha$ a locally finite immersion, which we suppose moreover locally of finite presentation when hypothesis 2° is satisfied. Then, for the corresponding immersion $j : X \rightarrow S$ to be regular, it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that $j_\lambda : X_\lambda \rightarrow S_\lambda$ be regular.*

Proof. (i). If $p_\lambda : S \rightarrow S_\lambda$ is the canonical projection, one has $\mathcal{F} / (\sum_{j=1}^{i-1} f_j \mathcal{F}) = p_\lambda^*(\mathcal{F}_\lambda / (\sum_{j=1}^{i-1} f_{j\lambda} \mathcal{F}_\lambda))$, hence one is reduced to the case $n = 1$, in which case one suppresses the index i . The sufficiency of the condition results from the fact that p_λ is flat ((8.3.8)) and from (0, 15.1.5). In the case 2°, p_λ is faithfully flat ((8.3.8)) and the necessity of the condition results again from (0, 15.2.5), with $\lambda = \alpha$. In the case 1°, let $\mathcal{N}_\lambda$ (resp. $\mathcal{N}$) be the kernel of the homomorphism $\mathcal{F}_\lambda \rightarrow \mathcal{F}_\lambda$ (resp. $\mathcal{F} \rightarrow \mathcal{F}$), multiplication by f_λ (resp. by f); since $\mathcal{F}_\lambda$ is coherent by hypothesis, so is $\mathcal{N}_\lambda$, hence the hypothesis $\mathcal{N} = 0$ entails $\mathcal{N}_\lambda = 0$ for some $\lambda \geq \alpha$ by virtue of ((8.5.8), (ii)).

(ii). Since p_λ is flat, the sufficiency of the condition results from ((19.1.5), (ii)). To prove its necessity, note that, since $j_\alpha(X_\alpha)$ is quasi-compact, it is contained in a quasi-compact open subset of S_α , and one can therefore limit to the case where S_α is also quasi-compact and j_α a closed immersion, so that the image of X_α is defined by a quasi-coherent Ideal $\mathcal{J}_\alpha$ of $\mathcal{O}_{S_\alpha}$, which one can moreover suppose of finite type in cases 1° and 2° (j_α being locally of finite presentation in case 2°); the image of X_λ (resp. X) in S_λ (resp. S) is then defined by $\mathcal{J}_\lambda = \mathcal{J}_\alpha \mathcal{O}_{S_\lambda}$ (resp. $\mathcal{J} = \mathcal{J}_\alpha \mathcal{O}_S$) which is again of finite type. One can moreover, by virtue of ((8.2.11)), suppose that $\mathcal{J}$ is generated by a regular suite $(f_i)_{1 \leq i \leq n}$ of sections above S , which define a surjective homomorphism $u : \mathcal{O}_S^n \rightarrow \mathcal{J}$. By virtue of ((8.5.2), (i)) and ((8.5.7)), there exists $\lambda \geq \alpha$ and a surjective homomorphism $u_\lambda : \mathcal{O}_{S_\lambda}^n \rightarrow \mathcal{J}_\lambda$ such that $u = p_\lambda^*(u_\lambda)$, hence the f_i are the canonical images of sections $f_{i\lambda}$ of $\mathcal{J}_\lambda$ above S_λ generating this Ideal. By virtue of (i), there thus exists $\mu \geq \lambda$ such that the suite $(f_{i\mu})$ is $\mathcal{O}_{S_\mu}$ -regular, hence the immersion j_μ is regular. □

Proposition (19.8.2). — *Let X_α be an S_α -prescheme quasi-compact and locally of finite presentation.*

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- label=(i) *Let $\mathcal{F}_\alpha$ be an $\mathcal{O}_{X_\alpha}$ -Module quasi-coherent of finite presentation, $(f_{i\alpha})_{1 \leq i \leq n}$ a suite of sections of $\mathcal{O}_{X_\alpha}$ above X_α . For the suite of sections f_i of $\mathcal{O}_X$ above X , corresponding to the $f_{i\alpha}$, to be $\mathcal{F}_X$ -transversally regular relatively to S ((19.2.1)), it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that the suite of sections $f_{i\lambda}$ of $\mathcal{O}_{X_\lambda}$ above X_λ be $\mathcal{F}_\lambda$ -transversally regular relatively to S_λ .*
- lbbel=(ii) *Let Y_α be an S_α -prescheme quasi-compact, $j_\alpha : Y_\alpha \rightarrow X_\alpha$ an S_α -immersion locally of finite presentation. For the S -immersion corresponding $j : Y \rightarrow X$ to be transversally regular relatively to S , it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that $j_\lambda : Y_\lambda \rightarrow X_\lambda$ be transversally regular relatively to S_λ .*

Proof. (i). The sufficiency of the condition results from (0, 15.1.15). To prove its necessity, it will suffice to show that every point $x \in X$ possesses an open neighbourhood $V(x)$ for which there exist an index $\lambda = \lambda(x) \geq \alpha$ and an open neighbourhood V_λ of the image x_λ of x in X_λ such that $V(x)$ is the inverse image of V_λ and that the suite of restrictions to V_λ of the $f_{i\lambda}$ is $(\mathcal{F}_\lambda|_{V_\lambda})$ -transversally regular relatively to S_λ : indeed, it will then suffice to apply ((8.3.4)) to the open subset U_λ of X_λ

defined by the condition of being the largest open such that the restrictions to U_λ of the $f_{i\lambda}$ form a suite $(\mathcal{F}_\lambda|_{U_\lambda})$ -transversally regular relatively to S_λ . Let s be the image of x in S , and for all $\lambda \geq \alpha$, let s_λ be the image of s in S_λ . Since X_s is the projective limit of the fibres $(X_\lambda)_{s_\lambda}$ and is locally Noetherian since the morphism $X \rightarrow S$ is locally of finite presentation; moreover $(X_\alpha)_{s_\alpha}$ is quasi-compact and the transition morphisms $(X_\mu)_{s_\mu} \rightarrow (X_\lambda)_{s_\lambda}$ are flat (since this is the case for $\text{Spec}(k(s_\mu)) \rightarrow \text{Spec}(k(s_\lambda))$), one can apply to the sections $f_i \otimes 1$ of $\mathcal{F}_s = \mathcal{F} \otimes_{\mathcal{O}_S} k(s)$ above X_s the result of ((19.8.1), (i)); whence the existence of $\lambda \geq \alpha$ such that the sections $f_{i\lambda} \otimes 1$ of $(\mathcal{F}_\lambda)_{s_\lambda} = \mathcal{F}_\lambda \otimes_{\mathcal{O}_{S_\lambda}} k(s_\lambda)$ above $(X_\lambda)_{s_\lambda}$ are $(\mathcal{F}_\lambda)_{s_\lambda}$ -regular. Moreover ((11.2.6)), one can suppose that $\mathcal{F}_\lambda$ is S_λ -flat at the point x_λ . It then results from ((11.3.8)) that there exists an open neighbourhood V_λ of x_λ in X_λ satisfying the required conditions.

(ii). The sufficiency of the condition results from ((19.2.7), (ii)). To prove its necessity, it again suffices to show that every point $x \in j(Y)$ admits an open neighbourhood $V(x)$ for which there exist an index $\lambda = \lambda(x) \geq \alpha$ and an open neighbourhood V_λ of the image x_λ of x in X_λ such that $V(x)$ is the inverse image of V_λ and that the restriction $j_\lambda^{-1}(V_\lambda) \rightarrow V_\lambda$ of j_λ is transversally regular relatively to S_λ . Let s be the image of x in S , and for all $\lambda \geq \alpha$, let s_λ be the image of s in S_λ . Since X_s (resp. Y_s) is the projective limit of the fibres $(X_\lambda)_{s_\lambda}$ (resp. $(Y_\lambda)_{s_\lambda}$), one can as in (i) use ((19.8.1), (i), (ii)), and since by hypothesis the immersion $Y_s \rightarrow X_s$ is regular, there exists an index $\lambda(s)$ such that for $\lambda \geq \lambda(s)$ the immersion $(Y_\lambda)_{s_\lambda} \rightarrow (X_\lambda)_{s_\lambda}$ is regular. Moreover, by virtue of ((19.2.4)) there exists a quasi-compact open neighbourhood $W(x)$ of x in X such that the structural morphisms $W(x) \rightarrow S$ and $Y \cap W(x) \rightarrow S$ are flat; applying ((11.2.6)) and ((8.2.11)), one can suppose that there exists $\lambda \geq \lambda(s)$ and a quasi-compact open neighbourhood $W(x_\lambda)$ of x_λ in X_λ , such that $W(x)$ is the inverse image of $W(x_\lambda)$ and that the restrictions to $W(x_\lambda)$ and to $Y_\lambda \cap W(x_\lambda)$ respectively of the structural morphisms $X_\lambda \rightarrow S_\lambda$ and $Y_\lambda \rightarrow S_\lambda$ are flat. One therefore concludes from ((19.2.4)) that there exists an open neighbourhood V_λ of x_λ answering the question. $\square$

Remark (19.8.3). — The preceding proofs show that the statements of ((19.8.1)) and ((19.8.2)) subsist when one replaces the conditions concerning regularity or transverse regularity in X (resp. X_λ) by these conditions in a neighbourhood of a point $x \in X$ (resp. in a neighbourhood of x_λ , projection of x).

19.9. $\mathcal{F}$ -regular suites and depth.

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(19.9.1). Recall that if X is a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, T a part of X , we set (0_I, 5.10.1)

$$(19.9.1.1) \quad \text{prof}_T(\mathcal{F}) = \inf_{x \in T} \text{prof}(\mathcal{F}_x).$$

On the other hand, for every point $t \in T$, we set

$$(19.9.1.2) \quad \text{prof}_{T,t}(\mathcal{F}) = \inf_{x \in T \cap \text{Spec}(\mathcal{O}_{X,t})} \text{prof}(\mathcal{F}_x),$$

and we say that $\text{prof}_{T,t}(\mathcal{F})$ is the T -depth of $\mathcal{F}$ at the point t ; it is clear that one has

$$(19.9.1.3) \quad \text{prof}_T(\mathcal{F}) = \inf_{t \in T} \text{prof}_{T,t}(\mathcal{F}).$$

Lemma (19.9.2). — Let A be a local Noetherian ring, M an A -module of finite type, $\mathfrak{J}$ an ideal of A . In order that, for every prime ideal $\mathfrak{p} \supset \mathfrak{J}$, we have $\text{prof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \geq r$, it is necessary and sufficient that there exist an M -regular suite of r elements belonging to $\mathfrak{J}$.

Proof. The sufficiency of the condition follows immediately from (0, 16.4.5); let us prove its necessity. We reason by induction on r (there is nothing to prove if $r = 0$): since $\text{prof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) \geq r - 1$ for every $\mathfrak{p} \supset \mathfrak{J}$, there exists by hypothesis an M -regular suite $(g_i)_{1 \leq i \leq r-1}$ formed of elements of $\mathfrak{J}$. Set $N = M/(g_1 M + \cdots + g_{r-1} M)$; for $\mathfrak{p} \supset \mathfrak{J}$, the images of the g_i in $A_{\mathfrak{p}}$ belong to the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$, and form an $M_{\mathfrak{p}}$ -regular suite (0, 15.1.14); we thus conclude from (0, 16.4.6) that $\text{prof}_{A_{\mathfrak{p}}}(N_{\mathfrak{p}}) = \text{prof}_{A_{\mathfrak{p}}}(M_{\mathfrak{p}}) - (r - 1) \geq 1$, and we see that it all comes down to proving the lemma for $r = 1$. But the hypothesis then means that, for every $\mathfrak{p} \supset \mathfrak{J}$, $\mathfrak{p}$ is not associated to M , hence $\mathfrak{p}$ is not contained in any of the prime ideals of $\text{Ass}(M)$. In other words, $\mathfrak{J}$ is not contained in their union (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2), which proves the lemma. $\square$

Proposition (19.9.3). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a coherent $\mathcal{O}_X$ -Module, Y a closed part of X , y a point of Y , n an integer > 0 . The following conditions are equivalent:*

- label=a) *There exists an open neighbourhood U of y in X such that $\text{prof}_{Y \cap U}(\mathcal{F}|_U) \geq n$.*
- lbbel=b) *There exists an open neighbourhood U of y in X and an $(\mathcal{F}|_U)$ -regular suite of n sections f_i of $\mathcal{O}_U$ over U , such that $f_i(x) = 0$ (**0I**, 5.5.1) for $1 \leq i \leq n$ at every point $x \in U \cap Y$.*
- lcbel=c) *We have $\text{prof}_{Y,y}(\mathcal{F}) \geq n$.*

Moreover, the set of $y \in Y$ satisfying these conditions is open in Y .

Proof. The last assertion follows trivially from a). The equivalence of a) and b) follows from (19.9.2) applied to an open affine neighbourhood of y in X and from (0, 15.1.14). It is clear that a) implies c); let us show conversely that c) implies b). It follows from c) and from the definition (19.9.1) that one can apply (19.9.2) to the ring $\mathcal{O}_{X,y}$, to the $\mathcal{O}_{X,y}$ -module $\mathcal{F}_y$ and to the ideal $\mathfrak{J}_y$ of $\mathcal{O}_{X,y}$. There thus exists an $\mathcal{F}_y$ -regular suite $(t_i)_{1 \leq i \leq n}$ of n elements of $\mathfrak{J}_y$. There thus exists an open neighbourhood U of y in X and a suite $(f_i)_{1 \leq i \leq n}$ of sections of $\mathfrak{J}$ over U such that the t_i are the germs of the f_i at the point y ; using (0, 15.2.4), we deduce that there exists an open neighbourhood $U' \subset U$ of y in X such that the $f_i|_{U'}$ form an $(\mathcal{F}|_{U'})$ -regular suite. $\square$

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Corollary (19.9.4). — *With the notations of (19.9.3), the function $y \mapsto \text{prof}_{Y,y}(\mathcal{F})$ is lower semi-continuous in Y .*

Proposition (19.9.5). — *With the notations of (19.9.3), let X' be a locally Noetherian prescheme, $g : X' \rightarrow X$ a flat morphism, $Y' = g^{-1}(Y)$, $\mathcal{F}' = g^*(\mathcal{F})$. Then, for every point $y' \in Y'$ above y , we have $\text{prof}_{Y',y'}(\mathcal{F}') = \text{prof}_{Y,y}(\mathcal{F})$.*

Proof. Indeed, for every $z' \in \text{Spec}(\mathcal{O}_{X',y'})$, it follows from (6.3.1) that $\text{prof}(\mathcal{F}'_{z'}) \geq \text{prof}(\mathcal{F}_{g(z')})$, and if z'' is a maximal point of the fibre $g^{-1}(g(z'))$, a generisation of z' (hence also belonging to $\text{Spec}(\mathcal{O}_{X',y'})$), we have (loc. cit.) $\text{prof}(\mathcal{F}'_{z''}) = \text{prof}(\mathcal{F}_{g(z'')})$. The proposition then follows from the fact that $g(\text{Spec}(\mathcal{O}_{X',y'})) = \text{Spec}(\mathcal{O}_{X,g(y)})$ since g is flat (2.3.4). $\square$

Proposition (19.9.6). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, Y a closed part of X , $\mathcal{F}$ an $\mathcal{O}_X$ -Module of finite presentation and f -flat, n an integer > 0 . For every point $y \in Y$, the following conditions are equivalent (denoting by $X_{f(y)}$ the fibre of f at the point $f(y)$, by $\mathcal{F}_{f(y)}$ the inverse image of $\mathcal{F}$ in $X_{f(y)}$, and setting $Y_{f(y)} = X_{f(y)} \cap Y$):*

- label=a) *We have $\text{prof}_{Y_{f(y)},y}(\mathcal{F}_{f(y)}) \geq n$.*
- lbbel=b) *There exists an open neighbourhood U of y in X and a suite $(g_i)_{1 \leq i \leq n}$ of n sections of $\mathcal{O}_X$ over U , which is transversally $\mathcal{F}$ -regular relative to S (cf. (19.2.1) for the terminology) and such that $g_i(x) = 0$ for $1 \leq i \leq n$ and for every $x \in U \cap Y$.*

The set V_n of $y \in Y$ satisfying these conditions is open in Y ; if moreover Y is locally constructible in X (**0III**, 9.1.1), V_n is locally constructible.

Proof. By virtue of (19.9.3), condition a) amounts to the existence of an open neighbourhood U of y in X and of a $(\mathcal{F}_{f(y)}|(U \cap X_{f(y)}))$ -regular suite $(h_i)_{1 \leq i \leq n}$ of sections of $\mathcal{O}_{X_{f(y)}}$ over $U \cap X_{f(y)}$ such that $h_i(z) = 0$ for every $z \in U \cap Y_{f(y)}$. If we denote also by Y a closed sub-prescheme of X having Y as underlying space, by $\mathfrak{J}$ the quasi-coherent Ideal of $\mathcal{O}_X$ defining Y , we can further say that $h_i \in \mathfrak{J}_{f(y)}$ for $1 \leq i \leq n$. As we can replace U by a smaller affine neighbourhood of y , we can suppose that the h_i are images of a suite $(g_i)_{1 \leq i \leq n}$ of sections of $\mathfrak{J}$ over U , so that the germs $(g_i)_y$ belong to the maximal ideal $\mathfrak{m}_y$ of $\mathcal{O}_{X,y}$. The equivalence of a) and b) then follows from (11.3.8), which also proves that the set V_n is open in Y ; since it concerns local properties on X , we can suppose $S = \text{Spec}(A)$ affine and X of finite presentation over S ; there then exists a Noetherian sub-ring A_0 of A , a prescheme X_0 of finite type over $S_0 = \text{Spec}(A_0)$ and a coherent $\mathcal{O}_{X_0}$ -Module $\mathcal{F}_0$ such that $X = X_0 \times_{S_0} S$ and $\mathcal{F} = \mathcal{F}_0 \otimes_{\mathcal{O}_{X_0}} \mathcal{O}_X$ that $Y = p^{-1}(Y_0)$, where $p : X \rightarrow X_0$ is the canonical projection. Then, for every $y \in Y$, it follows from (19.9.5) and from the transitivity of fibres that

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$$\text{prof}_{Y_{f(y)},y}(\mathcal{F}_{f(y)}) = \text{prof}_{(Y_0)_{f_0(y_0)},y_0}((\mathcal{F}_0)_{f_0(y_0)}),$$

where we set $y_0 = p(y)$; if V_n^0 is the set of $y_0 \in Y_0$ for which the second member of this relation is $\geq n$, we thus have $V_n = p^{-1}(V_n^0)$. As X_0 is Noetherian, V_n^0 is a retrocompact open in X_0 , and thus (cf. demonstration of (1.8.2)) V_n is retrocompact in X . $\square$

Corollary (19.9.7). — *Under the general hypotheses of (19.9.6), the function $y \mapsto \text{prof}_{Y_{f(y)}, Y}(\mathcal{F}_{f(y)})$ is lower semi-continuous in Y . If moreover Y is locally constructible, this function is locally constructible.*

Proposition (19.9.8). — *Let $f : X \rightarrow S$ be a morphism locally of finite presentation, Z a closed part of X , $\mathcal{F}$ an $\mathcal{O}_X$ -Module of finite presentation and f -flat, $\mathcal{G}$ an $\mathcal{O}_S$ -Module quasi-coherent. Suppose that, for every $x \in Z$, we have $\text{prof}((\mathcal{F}_{f(x)})_x) \geq 1$ (resp. $\text{prof}((\mathcal{F}_{f(x)})_x) \geq 2$). Then, if $j : X - Z \rightarrow X$ is the canonical injection, and if we set $\mathcal{H} = \mathcal{F} \otimes_{\mathcal{O}_S} f^*(\mathcal{G})$, the canonical homomorphism $\rho_{\mathcal{H}} : \mathcal{H} \rightarrow j_*(j^*(\mathcal{H}))$ relative to j is injective (resp. bijective).*

Proof. The question is evidently local on X and S , and it suffices to prove the proposition in a neighbourhood of a point $x \in Z$. In other words, we can restrict to the case where X and S are affine. Thanks to (19.9.6), we can suppose that there exists a transversally $\mathcal{F}$ -regular section g_1 of $\mathcal{O}_X$ over X relative to S (resp. a transversally $\mathcal{F}$ -regular suite (g_1, g_2) of two sections of $\mathcal{O}_X$ over X relative to S) such that if Z' is the set of $x' \in X$ for which $g_1(x') = 0$ (resp. $g_1(x') = g_2(x') = 0$), we have $Z \subset Z'$.

This being so, let us return to the proof of (19.9.8); by virtue of (19.9.6) we still have $\text{prof}((\mathcal{F}_{f(x)})_x) \geq 1$ (resp. $\text{prof}((\mathcal{F}_{f(x)})_x) \geq 2$) for every $x \in Z'$. If the proposition is proved by replacing the pair (X, Z) by (X, Z') and $(X - Z, Z' \cap (X - Z))$, it is immediate that it will also hold for X and Z' . In other words, we can restrict to proving (19.9.8) when $S = \text{Spec}(A)$, $X = \text{Spec}(B)$, where B is an A -algebra of finite presentation, $Z = V(\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of finite type of B , and a B_0 -module N_0 of finite type M_0 which is an A_0 -module flat and an A_0 -module N_0 such that $B = B_0 \otimes_{A_0} A$, $\mathfrak{J} = \mathfrak{J}_0 B$, $M = M_0 \otimes_{A_0} A$ and $N = N_0 \otimes_{A_0} A$ (8.9.1), (8.5.11) and (11.2.7). Moreover, let (A_λ) be the family of sub- A_0 -algebras of finite type of A , so that $A = \varinjlim A_\lambda$; set $B_\lambda = B_0 \otimes_{A_0} A_\lambda$, $\mathfrak{J}_\lambda = \mathfrak{J}_0 B_\lambda$, $M_\lambda = M_0 \otimes_{A_0} A_\lambda$, $N_\lambda = N_0 \otimes_{A_0} A_\lambda$ and $S_\lambda = \text{Spec}(A_\lambda)$, $X_\lambda = \text{Spec}(B_\lambda)$, $Z_\lambda = V(\mathfrak{J}_\lambda)$; if $p_\lambda : X \rightarrow X_\lambda$ is the canonical projection, we thus have $Z = p_\lambda^{-1}(Z_\lambda)$, and $X - Z = p_\lambda^{-1}(X_\lambda - Z_\lambda)$. As $X_\lambda - Z_\lambda$ is quasi-compact and quasi-separated, we conclude (8.5.2) that if we set $\mathcal{F}_\lambda = \tilde{M}_\lambda$ and $\mathcal{H}_\lambda = (M_\lambda \otimes_{A_\lambda} N_\lambda)^\sim$, the homomorphisms $\varinjlim \Gamma(X_\lambda, \mathcal{H}_\lambda) \rightarrow \Gamma(X, \mathcal{H})$ and $\varinjlim \Gamma(X_\lambda - Z_\lambda, \mathcal{H}_\lambda) \rightarrow \Gamma(X - Z, \mathcal{H})$ are bijective; by virtue of the exactness of the functor $\varinjlim$, it therefore suffices to prove that for λ sufficiently large, the canonical homomorphism $\Gamma(X_\lambda, \mathcal{H}_\lambda) \rightarrow \Gamma(X_\lambda - Z_\lambda, \mathcal{H}_\lambda)$ is injective (resp. bijective). But for every $x \in Z$, if x_λ is the projection of x in Z_λ , we have $\text{prof}((\mathcal{F}_{f(x)})_x) = \text{prof}((\mathcal{F}_\lambda)_{f_\lambda(x_\lambda)})_{x_\lambda}$ by virtue of (4.2.7) and (6.7.1); to reduce to proving (19.9.8) in the case where A is Noetherian, we must show that, for λ sufficiently large, we have $\text{prof}((\mathcal{F}_\lambda)_{f_\lambda(x_\lambda)})_{x_\lambda} \geq 1$ (resp. ≥ 2) for every $x_\lambda \in Z_\lambda$. Now, denoting by Z'_λ the set of points $x_\lambda \in Z_\lambda$ such that, for every generisation z_λ of x_λ in $Z_\lambda \cap (X_\lambda)_{f_\lambda(x_\lambda)}$, we have $\text{prof}((\mathcal{F}_\lambda)_{f_\lambda(z_\lambda)})_{z_\lambda} \geq 1$ (resp. ≥ 2) for every $x_\lambda \in Z_\lambda$. On the other hand, it follows from (19.9.6) that Z'_λ is open in Z_λ ; on the other hand, it follows from the hypothesis and from (19.9.5) that for λ sufficiently large we have $\text{prof}((\mathcal{F}_\lambda)_{f_\lambda(x_\lambda)})_{x_\lambda} \geq 1$ (resp. ≥ 2) for every point $x_\lambda \in Z_\lambda$.

To finish the proof when A is Noetherian, note that by virtue of the flatness hypothesis and of (6.3.1), we then have $\text{prof}_Z(\mathcal{H}) \geq 1$ (resp. $\text{prof}_Z(\mathcal{H}) \geq 2$); but the conclusion of (19.9.8) follows in this case from (5.10.2) (resp. (5.10.4)). $\square$

Remark (19.9.9). — The same proof, combined with the results of chap. III, 3rd part on depth and local cohomology, allows one to establish the following generalisation of (19.9.8):

Under the general conditions of (19.9.8), suppose that, for every $x \in Z$, we have $\text{prof}((\mathcal{F}_{f(x)})_x) \geq k$; then the canonical homomorphism

$$H^i(X, \mathcal{H}) \longrightarrow H^i(X - Z, \mathcal{H}|_{(X - Z)})$$

is bijective for $i \leq k - 2$ and injective for $i = k - 1$ (which, expressed using the general cohomological notion of depth introduced in chap. III, is also written $\text{prof}_Z(\mathcal{H}) \geq k$).

§20. MEROMORPHIC FUNCTIONS AND PSEUDO-MORPHISMS

20.0. Introduction Most of the concepts and results of §§20 and 21 directly relate to Chapter I, and hardly depend on Chapters I and IV, except for the occasional usage of the notion of depth and of a local regular ring (in (20.6), (21.11), (21.13), and (21.15)), of Zariski’s “Main theorem” in (20.4) and (21.12), and the properties of transversely regular immersions in (20.6) and (21.15).

In §20, we introduce several variants of the concept of a rational map, already studied in (I, 7) from a point of view still fairly close to the classical point of view, and for this reason quite ill-suited to the case of not necessarily reduced preschemes. The notions and results of §20 are used in §21 (n^{os} (21.1) and (21.7)) to develop the general notion of a divisor and its most elementary properties. This notion is especially convenient when the local rings of the preschemes considered are Noetherian and integrally closed, and especially when they are also *factorial* ((21.6) and (21.7)), because of their identification in the latter case with the notion of a 1-codimensional cycle (a linear combination of irreducible subpreschemes of codimension 1). In (21.9), we determine the divisors on a Noetherian prescheme of dimension 1 but not necessarily normal, which is useful for various applications. The (21.11) and (21.12) give two important theorems, due respectively to Auslander–Buchsbaum and Van der Waerden, and relate the notion of a factorial ring (the n^{os} (21.9), (21.11), and (21.12) are independent of each other). In the n^{os} (21.13) and (21.14), also independent of the previous three, we study a useful variant of the notion of a local factorial ring, that of a local *parafactorial* ring, which is introduced in particular [?] in the development of comparison theorem of the Picard group of a projective prescheme X over a field k and a “hyperplane section”. We will see in (21.14.1) (Ramanujam–Samuel theorem) that the local parafactorial rings are much more numerous than one would have expected *a priori*.

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In (20.5), (20.6), and (21.15), we review the previous notions but from a point of view “relative” to a fixed base prescheme. For the moment these notions are used only relatively rarely; in particular, the concept of a relative divisor is hardly used except when it comes to positive divisors, and in this case it is explained advantageously without the help of the notion of a relative meromorphic functions, using the notion of a transversely regular immersion of codimension 1. It will therefore be advantageous to omit these sections on a first reading.

20.1. Meromorphic functions

(20.1.1). Let $(X, \mathcal{O}_X)$ be a ringed space, and let $\mathcal{S}$ be a subsheaf of *sets* of $\mathcal{O}_X$. For every open U of X , consider the *ring of fractions* $\Gamma(U, \mathcal{O}_X)[\Gamma(U, \mathcal{S})^{-1}]$ (Bourbaki, *Alg. comm.*, chap. II, §2, $n^{\circ}1$). It is immediate that the map $U \mapsto \Gamma(U, \mathcal{O}_X)[\Gamma(U, \mathcal{S})^{-1}]$ is a *presheaf of rings* ((0, 1.5.1) and (0, 1.5.7)). We denote by $\mathcal{O}_X[\mathcal{S}^{-1}]$ the *sheaf of rings* associated to this presheaf and we say that this is the *sheaf of rings of fractions of $\mathcal{O}_X$ with denominators in $\mathcal{S}$* ; this is a *flat* $\mathcal{O}_X$ -module. It is immediate that for every $x \in X$, we have a canonical isomorphism

$$(20.1.1.1) \quad (\mathcal{O}_X[\mathcal{S}^{-1}])_x \simeq \mathcal{O}_x[\mathcal{S}_x^{-1}],$$

since the reasoning of (0, 1.4.5) generalizes immediately in the case where we have an inductive system $(A_\alpha, \phi_{\beta\alpha})$ of rings, and for each index α a subset S_α of A_α such that $\phi_{\beta\alpha}(S_\alpha) \subset S_\beta$ for $\alpha \leq \beta$; we then take for S the inductive limit in $A = \varinjlim A_\alpha$ of the inductive limit of the subsets (S_α) .

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(20.1.2). Now let $\mathcal{F}$ be an $\mathcal{O}_X$ -module. We then set

$$(20.1.2.1) \quad \mathcal{F}[\mathcal{S}^{-1}] = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X[\mathcal{S}^{-1}],$$

and we say that this is the *sheaf of modules of fractions of $\mathcal{F}$ with denominators in $\mathcal{S}$* ; it is immediate that it is associated to the presheaf of modules $U \mapsto \Gamma(U, \mathcal{F})[\Gamma(U, \mathcal{S})^{-1}]$, and that for every $x \in X$, we have a canonical isomorphism

$$(20.1.2.2) \quad (\mathcal{F}[\mathcal{S}^{-1}])_x \simeq \mathcal{F}_x[\mathcal{S}_x^{-1}].$$

(20.1.3). We will focus here on the case where $\mathcal{S}$ is the subsheaf $\mathcal{S}(\mathcal{O}_X)$ of $\mathcal{O}_X$ such that for every open U , $\Gamma(U, \mathcal{S})$ is the *set of regular elements* of the ring $\Gamma(U, \mathcal{O}_X)$; it is immediate that it is a sheaf (and not only a presheaf), the regularity of a section of $\mathcal{O}_X$ over U being verified “fibre by fibre” (i.e.

meaning that the germ of the section in x is regular in $\mathcal{O}_X$ for all $x \in U$, in other words $\mathcal{S}(\mathcal{O}_X)_x$ is none other than the set of regular elements of $\mathcal{O}_{X,x}$. The corresponding sheaf of rings

$$\mathcal{M}_X = \mathcal{O}_X[\mathcal{S}^{-1}]$$

is called the *sheaf of germs of meromorphic functions on X* , and the sections of $\mathcal{M}_X$ over X are called the *meromorphic functions on X* ; they form a ring which we denote by $M(X)$. For every $\mathcal{O}_X$ -module $\mathcal{F}$,

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{M}_X = \mathcal{F}[\mathcal{S}^{-1}]$$

is also denoted $\mathcal{M}_X(\mathcal{F})$ and called the *sheaf of germs of meromorphic sections of $\mathcal{F}$* ; its sections over X form an $M(X)$ -module denoted by $M(X, \mathcal{F})$, whose elements are called *meromorphic sections of $\mathcal{F}$ over X* . These definitions imply that for every open U of X , we have a canonical isomorphism $\mathcal{M}_X(\mathcal{F})|_U \simeq \mathcal{M}_U(\mathcal{F}|_U)$, in particular $\mathcal{M}_X|_U \simeq \mathcal{M}_U$.

(20.1.3.1). If X is a *reduced prescheme*, then we note that if an element $s \in \Gamma(U, \mathcal{O}_X)$ is such that $s_\xi \neq 0$ for every maximal point ξ of U , then s is *regular*. Indeed, if $st = 0$ for a $t \in \Gamma(U, \mathcal{O}_X)$, then we have $s_\xi t_\xi = 0$, so $t_\xi = 0$ since $\mathcal{O}_{X,\xi}$ is a field, and say that $t_\xi = 0$ for every maximal point ξ of X means that $t = 0$: we are immediately reduced to the case where U is affine, and an element of a reduced ring which belongs to all the minimal prime ideals is zero by definition. The converse is true if the set of irreducible components of X is *locally finite*. We immediately reduce to the case where $X = \text{Spec}(A)$ is affine; if $\mathfrak{p}_i$ ($1 \leq i \leq n$) are the minimal prime ideals of A and if $s \in \mathfrak{p}_i$ for an index i , then there exists $t \in A$ such that $t \in \mathfrak{p}_j$ for $j \neq i$ and $t \notin \mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. II, §1, n°1, Prop. 1); so we have $st \in \mathfrak{p}_i$ for all i , and as a result $st = 0$ since A is reduced; s is therefore nonregular.

(20.1.4). For every open U of X , the homomorphism $t \mapsto t/1$ from $\Gamma(U, \mathcal{O}_X)$ to $\Gamma(U, \mathcal{O}_X)[\Gamma(U, \mathcal{S})^{-1}]$ (which is none other than the *total ring of fractions* of $\Gamma(U, \mathcal{O}_X)$) is injective; these homomorphisms thus define a *canonical injective homomorphism*

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$$(20.1.4.1) \quad i : \mathcal{O}_X \longrightarrow \mathcal{M}_X$$

which allows us to identify $\mathcal{O}_X$ with a subsheaf of $\mathcal{M}_X$. Given a meromorphic function $\phi \in M(X)$, we say that ϕ is *defined* on an open U of X if $\phi|_U$ is a *section of $\mathcal{O}_U$ over U* ; the axioms of sheaves show that there is, for a given section ϕ , a *largest* open on which ϕ is defined; we call this the *domain of definition* of ϕ and denote it by $\text{dom}(\phi)$.

(20.1.5). For every $\mathcal{O}_X$ -module $\mathcal{F}$, we obtain from (20.1.4.1) a di-homomorphism consisting of i and the homomorphism of sheaves of additive groups

$$(20.1.5.1) \quad 1_{\mathcal{F}} \otimes i : \mathcal{F} \longrightarrow \mathcal{M}_X(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{M}_X.$$

We note that the latter is not injective in general; when it is injective, we say that $\mathcal{F}$ is *strictly torsion-free*: this means that for every open U of X and every section $s \in \Gamma(U, \mathcal{O}_X)$ which is a regular element in this ring the homothety $z \mapsto sz$ of $\Gamma(U, \mathcal{F})$ is injective; this condition is evidently satisfied if $\mathcal{F}$ is *locally free*.

Proposition (20.1.6). — *Let X be a locally Noetherian prescheme, $\mathcal{F}$ a quasi-coherent $\mathcal{O}_X$ -module. For $\mathcal{F}$ to be strictly torsion-free, it is necessary and sufficient that $\text{Ass}(\mathcal{F}) \subset \text{Ass}(\mathcal{O}_X)$.*

Proof. We immediately reduce to the case where $X = \text{Spec}(A)$ is affine, $\mathcal{F} = \tilde{M}$, and we know that the elements s of A belonging to an ideal of $\text{Ass}(M)$ are exactly those for which the homothety $z \mapsto sz$ is not injective (Bourbaki, *Alg. comm.*, chap. IV, §1, n°1, cor. 2 of prop. 2). $\square$

(20.1.7). If u is a section of $\mathcal{M}_X(\mathcal{F})$ over X , then we say that u is *defined* at a point $x \in X$ if there exists an open neighbourhood V of x in X such that $u|_V$ is the image of a section of $\mathcal{F}$ over V under the di-homomorphism (20.1.5.1). We say that u is *defined* on an open U of X if it is defined at every point in U ; there is still a larger open in which u is defined, called the *domain of definition* of u and denoted $\text{dom}(u)$. When $\mathcal{F}$ is strictly torsion-free, such that $\mathcal{F}$ is identified by (20.1.5.1) with a subsheaf of $\mathcal{M}_X(\mathcal{F})$, then saying u is defined on U means that $u|_U$ is a *section of $\mathcal{F}$ over U* .

(20.1.8). In accordance with the general notation of (0_I, 5.4.7), we denote by $\mathcal{M}_X^*$ the sheaf of multiplicative groups such that $\Gamma(U, \mathcal{M}_X^*)$ is (for every open U of X) the group of *invertible elements* of $\Gamma(U, \mathcal{M}_X)$. This sheaf is none other than the sheaf $\mathcal{S}(\mathcal{M}_X)$ defined in (20.1.3): indeed, if

$s \in \Gamma(U, \mathcal{S}(\mathcal{M}_X))$, then for every $x \in U$, there exists an open neighbourhood $V \subset U$ of x such that $s|_V$ is a regular element in the *total ring of fractions* of $\Gamma(V, \mathcal{O}_X)$, and we know that such an element is necessarily invertible in this ring of fractions. We say that the sections of $\mathcal{M}_X^*$ over X are the *regular meromorphic functions* (note that we are deviating here from the terminology followed by certain authors, who call “regular” meromorphic functions those which are *sections* of $\mathcal{O}_X$, identified with a subsheaf of $\mathcal{M}_X$).

Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module (0_I, 5.4.1); then it is clear that $\mathcal{M}_X(\mathcal{L}) = \mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{M}_X$ is an invertible $\mathcal{M}_X$ -module. Let U be an open such that $\mathcal{L}|_U$ is isomorphic to $\mathcal{O}_U$; as every automorphism of $\mathcal{M}_U$ is multiplication by an invertible element of $\Gamma(U, \mathcal{M}_X)$ (0_I, 5.4.7), it is equivalent to say that a section $s \in \Gamma(U, \mathcal{M}_X(\mathcal{L}))$ has an invertible image in $\Gamma(U, \mathcal{M}_X)$ under an isomorphism or by any isomorphism on $\Gamma(U, \mathcal{M}_X)$; we say in this case that s is a *regular meromorphic section* of $\mathcal{L}$ over U ; a section s of $\mathcal{L}$ over X is called a *regular meromorphic section* of $\mathcal{L}$ if, for every open U such that $\mathcal{L}|_U$ is isomorphic to $\mathcal{O}_U$, $s|_U$ is a regular meromorphic section of $\mathcal{L}$ over U . We denote by $(\mathcal{M}_X(\mathcal{L}))^*$ the subsheaf of $\mathcal{M}_X(\mathcal{L})$ such that for every open U , $\Gamma(U, (\mathcal{M}_X(\mathcal{L}))^*)$ is the set of regular meromorphic sections of $\mathcal{L}$ over U . Let s be a meromorphic section of $\mathcal{L}$ over X (i.e. a section of $\mathcal{M}_X(\mathcal{L})$); it defines a homomorphism $h_s : \mathcal{M}_X \rightarrow \mathcal{M}_X(\mathcal{L})$ which sends every section t of $\mathcal{M}_X$ over an open U to $(s|_U)t$. It follows immediately from the above that for s to be *regular*, it is necessary and sufficient for h_s to be *injective*, and in fact h_s is then a *bijective* homomorphism from $\mathcal{M}_X$ to $\mathcal{M}_X(\mathcal{L})$, and its restriction to $\mathcal{M}_X^*$ is a bijection to $(\mathcal{M}_X(\mathcal{L}))^*$. We conclude that the homothety $t \mapsto ts$ is an isomorphism from $M(X)$ to $M(X, \mathcal{L})$.

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(20.1.9). Let s be a regular meromorphic section of an invertible $\mathcal{O}_X$ -module $\mathcal{L}$ over X ; then for every $\mathcal{O}_X$ -module $\mathcal{F}$, s similarly defines a homomorphism $h_s \otimes 1_{\mathcal{F}} : \mathcal{M}_X(\mathcal{F}) \rightarrow \mathcal{M}_X(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L})$, which is again *bijective*.

(20.1.10). Let s be a meromorphic section of an invertible $\mathcal{O}_X$ -module $\mathcal{L}$ over X ; for s to be regular, it is necessary and sufficient for there to exist a meromorphic section s' of $\mathcal{L}^{-1}$ over X such that the canonical image of $s \otimes s'$ in $\mathcal{M}_X$ (0_I, 5.4.3) is the unit section, and this section s' is then unique: indeed, the necessity of the local existence of such a section is evident, and its local uniqueness implies its global (and unique) existence; moreover, the existence of s' is trivially sufficient for s to be regular. We will take $s' = s^{-1}$.

Finally, if $\mathcal{L}'$ is a second invertible $\mathcal{O}_X$ -module, s (resp. s') a regular meromorphic section of $\mathcal{L}$ (resp. $\mathcal{L}'$) over X , then $s \otimes s'$ is evidently a regular meromorphic section of $\mathcal{L} \otimes \mathcal{L}'$ over X .

(20.1.11). If $f : X' \rightarrow X$ is a morphism of ringed spaces, then there is in general no natural map sending a meromorphic function on X to a meromorphic function on X' . For example, if X is the spectrum of a local integral domain A , X' its residue field k , then there is no natural homomorphism from the field of fractions K of A to k , and we can only send an element of K to an element of k if it is already in A .

In general, if $f = (\psi, \theta)$, then for every open U of X , denote by $\mathcal{S}_f(U)$ the set of *regular* sections $s \in \Gamma(U, \mathcal{O}_X)$ such that the image of s under

$$\Gamma(\theta^\#) : \Gamma(U, \mathcal{O}_X) \longrightarrow \Gamma(f^{-1}(U), \mathcal{O}_{X'})$$

is a *regular* section. It is immediate that $U \mapsto \mathcal{S}_f(U)$ is a *subsheaf* of the sheaf of sets $\mathcal{S}(\mathcal{O}_X)$, which we denote by $\mathcal{S}_f$. We set $\mathcal{M}_f = \mathcal{O}_X[\mathcal{S}_f^{-1}]$; this is a subsheaf of rings of $\mathcal{M}_X$, and we canonically obtain from $\theta^\# : \psi^*(\mathcal{O}_X) \rightarrow \mathcal{O}_{X'}$ a homomorphism of sheaves of rings $\theta'^\# : \psi^*(\mathcal{M}_f) \rightarrow \mathcal{M}_{X'}$ extending $\theta^\#$ (Bourbaki, *Alg. comm.*, chap. II, §2, n°1, prop. 2); hence, recalling that $f^*(\mathcal{M}_f) = \psi^*(\mathcal{M}_f) \otimes_{\psi^*(\mathcal{O}_X)} \mathcal{O}_{X'}$, we get a canonical homomorphism of $\mathcal{O}_{X'}$ -algebras

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$$(20.1.11.1) \quad f^*(\mathcal{M}_f) \longrightarrow \mathcal{M}_{X'}.$$

For every meromorphic function ϕ on X that is a section of $\mathcal{M}_f$, $\Gamma(\theta'^\#)(\phi)$ is a meromorphic function on X' , called the *inverse image* of ϕ under f , and denoted by $\phi \circ f$ if there is no cause for confusion.

Similarly, if $\mathcal{F}$ is an $\mathcal{O}_X$ -module, then we set $\mathcal{M}_f(\mathcal{F}) = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{M}_f$, and we immediately obtain from $\theta'^\#$ a canonical homomorphism (which is also written as $u \mapsto u \circ f$)

$$\Gamma(X, \mathcal{M}_f(\mathcal{F})) \longrightarrow \Gamma(X', \mathcal{M}_{X'}(f^*(\mathcal{F}))).$$

In addition, if $u \in \Gamma(X, \mathcal{M}_f(\mathcal{F}))$ is defined (20.1.7) at a point x , then u coincides, on a neighbourhood U of x , with a section of the form $\sum_i h_i \otimes (t_i/s_i)$, where the h_i belong to $\Gamma(U, \mathcal{F})$, the t_i to $\Gamma(U, \mathcal{O}_X)$, and the s_i to $\Gamma(U, \mathcal{S}_f)$. As by hypothesis the images of the s_i in $\Gamma(f^{-1}(U), \mathcal{O}_{X'})$ are regular, we see that $u \circ f$ is defined at every point of $f^{-1}(U)$; in other words, we have

$$(20.1.11.2) \quad f^{-1}(\text{dom}(u)) \subset \text{dom}(u \circ f).$$

We will see later (20.6.5, (i)) examples (with $\mathcal{F} = \mathcal{O}_X$) where the two sides of (20.1.11.2) can be different.

Consider in particular the case where $\mathcal{M}_f = \mathcal{M}_X$; then, if $\mathcal{L}$ is an invertible $\mathcal{O}_X$ -module, the image in $\mathcal{M}_{X'}(f^*(\mathcal{L}))$, under $\Gamma(\theta^\#)$, of a regular meromorphic section of $\mathcal{L}$ over X (20.1.8) is a regular meromorphic section of $f^*(\mathcal{L})$ over X' , as it follows immediately from the definition of its sections, and from the fact that a homomorphism of rings sends an invertible element to an invertible element.

Let $f' : X'' \rightarrow X'$ be a second morphism of ringed spaces, and suppose that $\mathcal{M}_f = \mathcal{M}_X$ and $\mathcal{M}_{f'} = \mathcal{M}_{X'}$; then, if we set $f'' = f \circ f'$, we also have $\mathcal{M}_{f''} = \mathcal{M}_X$, and we immediately see that for every meromorphic section u of $\mathcal{F}$ over X , we have $u \circ f'' = (u \circ f) \circ f'$.

Proposition (20.1.12). — *If the morphism $f : X' \rightarrow X$ is flat (0I, 6.7.1), then we have $\mathcal{M}_f = \mathcal{M}_X$, and the homomorphism $\phi \mapsto \phi \circ f$ is defined on all of $M(X)$. In addition, if f is a (flat) morphism of locally ringed spaces, then we have $\text{dom}(\phi \circ f) = f^{-1}(\text{dom}(\phi))$; if in addition f is surjective (thus faithfully flat), then the homomorphism $\phi \mapsto \phi \circ f$ is injective.*

Proof. The first assertion follows from the fact that, if B is an A -algebra which is a flat A -module, then every element of A not a divisor of 0 in A is not a divisor of 0 in B (0I, 6.3.4). To prove the other assertions, note that, for every $x' \in X'$, if $x = f(x')$, then $\mathcal{O}_{X',x'}$ is a flat $\mathcal{O}_{X,x}$ -module, and as the homomorphism $\mathcal{O}_{X,x} \rightarrow \mathcal{O}_{X',x'}$ is local by hypothesis, it is injective ((0I, 6.5.1) and (0I, 6.6.2)); if we set $A = \mathcal{O}_{X,x}$, $B = \mathcal{O}_{X',x'}$, such that A identifies with a subring of B , then $(f^*(\mathcal{M}_X))_{x'}$ is equal to $S^{-1}A \otimes_A B = S^{-1}B$, where S is the set of regular elements of A , $(\mathcal{M}_{X'})_{x'}$ is equal to $T^{-1}B$, where T is the set of regular elements of B , and as we have seen that $S \subset T$, the homomorphism $S^{-1}B \rightarrow T^{-1}B$ is injective; in other words, this proves that the homomorphism (20.1.11.1) $f^*(\mathcal{M}_X) \rightarrow \mathcal{M}_{X'}$ is injective (hence the last assertion of the statement). The quotient $f^*(\mathcal{M}_X)/\mathcal{O}_{X'}$ identifies with an $\mathcal{O}_{X'}$ -submodule of $\mathcal{M}_{X'}/\mathcal{O}_{X'}$, and $(f^*(\mathcal{M}_X)/\mathcal{O}_{X'})_{x'}$ identifies with $(\mathcal{M}_X/\mathcal{O}_X)_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{X',x'}$. Then suppose that $x \notin \text{dom}(\phi)$; the image of ϕ_x in $(\mathcal{M}_X/\mathcal{O}_X)_x$ is therefore $\neq 0$; by faithful flatness, we deduce that it is the same for the image of $(\phi \circ f)_{x'}$, so $x' \notin \text{dom}(\phi \circ f)$, which finishes the proof. $\square$

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Remark (20.1.13). — Let X be a reduced complex analytic space; then the notion of a meromorphic function on X defined above coincides with the usual notion. Consider on the other hand a prescheme Y , locally of finite type over the field $\mathbf{C}$; we then know that we can associate to Y an analytic space Y^{an} having the same underlying topological space, and the canonical morphism $f : Y^{\text{an}} \rightarrow Y$ is flat [?]; by virtue of (20.1.12), the canonical homomorphism $u \mapsto u \circ f$ from $M(Y)$ to $M(Y^{\text{an}})$ is therefore always defined and is injective; but it is not surjective in general. For example, when $Y = \mathbf{V}_0^r$ (ErrIII, 14) is the affine space of dimension r over $\mathbf{C}$, $M(Y)$ canonically identifies with the field $R(Y)$ of rational functions on Y (20.2.13, (i)), while $M(Y^{\text{an}})$ is the field of usual meromorphic functions on $\mathbf{C}^r$. Because of this fact, it is often preferable, in algebraic geometry, to abstain from the terminology introduced in this section, and to use the equivalent terminology of “pseudo-function” which will be defined below.

20.2. Pseudo-morphisms and pseudo-functions The only ringed spaces discussed in this section are *preschemes*.

(20.2.1). Recall (11.10.2) that in a prescheme X , we say that an open U is *schematically dense* if, for every open V of X , the canonical homomorphism $\Gamma(V, \mathcal{O}_X) \rightarrow \Gamma(V \cap U, \mathcal{O}_X)$ is injective. IV-4 | 231

Consider two preschemes X, Y , two *schematically dense* opens U, U' of X ; we say that two morphisms $u : U \rightarrow Y, u' : U' \rightarrow Y$ are *equivalent* if there exists an open $U'' \subset U \cap U'$, *schematically dense* in X , such that $u|_{U''} = u'|_{U''}$. As it results immediately from the definition of schematically dense opens that the intersection of two such opens is again one, it is immediate that the preceding relation is indeed an equivalence relation. An equivalence class for this relation is called a *pseudo-morphism* of X in Y , or a *strict rational application* of X in Y .

If S is a prescheme and X, Y two S -preschemes, we call *pseudo- S -morphism* the equivalence class (for the preceding relation) of an S -morphism of a schematically dense open of X in Y . A pseudo-morphism is therefore nothing other than a pseudo- S -morphism for $S = \text{Spec}(\mathbf{Z})$.

We denote by $\text{Ps. hom}(X, Y)$ (resp. $\text{Ps. hom}_S(X, Y)$) the set of pseudo-morphisms (resp. pseudo- S -morphisms) of X in Y .

(20.2.2). It follows from the definition recalled above that if U is a schematically dense open in X , then, for every open V of X , $U \cap V$ is schematically dense in V . If two morphisms $u : U \rightarrow Y, u' : U' \rightarrow Y$ of schematically dense opens of X in Y are equivalent, it follows that their restrictions $u|(V \cap U)$ and $u'|(V \cap U')$ are also equivalent morphisms (for the prescheme induced on V); their class is called the *restriction to V of the pseudo-morphism ω* , class of u , and we denote this pseudo-morphism $\omega|U$. If $W \subset V$ is an open of X , it is clear that $(\omega|V)|W = \omega|W$. This shows that the restriction maps define a *presheaf* of sets $U \mapsto \text{Ps. hom}(U, Y)$; we note that this presheaf is *not* in general a sheaf (cf. (20.2.16)); the associated sheaf is denoted $\mathcal{P}\text{s. hom}(X, Y)$. For pseudo- S -morphisms, we see that $U \mapsto \text{Ps. hom}_S(U, Y)$ is a presheaf of sets, whose associated sheaf we denote $\mathcal{P}\text{s. hom}_S(X, Y)$. IV-4 | 232

If V is a *schematically dense* open in X , then for every open U of X , $U \cap V$ is also schematically dense in X , so the map $\omega \mapsto \omega|V$ is a *bijection* of $\text{Ps. hom}(X, Y)$ (resp. $\text{Ps. hom}_S(X, Y)$) onto $\text{Ps. hom}(V, Y)$ (resp. $\text{Ps. hom}_S(V, Y)$).

(20.2.3). Given a pseudo- S -morphism ω of X in Y , we say that it is *defined* at a point $x \in X$ if there exists an open neighbourhood V of x in X , an open $U \subset V$ containing x and *schematically dense* in V , and finally an S -morphism $u : U \rightarrow Y$ whose class in $\text{Ps. hom}_S(V, Y)$ is equal to $\omega|V$ (20.2.2); we call the *domain of definition* of ω and denote $\text{dom}_S(\omega)$ (or simply $\text{dom}(\omega)$ if $S = \text{Spec}(\mathbf{Z})$) the set of $x \in X$ where ω is defined; this is evidently an *open* of X . Moreover, for every open W of X , we have

$$(20.2.3.1) \quad \text{dom}_S(\omega|W) = (\text{dom}_S(\omega)) \cap W$$

by virtue of the property of schematically dense opens recalled in (20.2.2).

Proposition (20.2.4). — Suppose X, Y are S -preschemes such that the structural morphism $Y \rightarrow S$ is separated; then, if ω is a pseudo- S -morphism of X in Y , $\text{dom}_S(\omega)$ is the largest of the schematically dense opens U of X such that there exists an S -morphism $u : U \rightarrow Y$ belonging to the class ω .

Proof. It suffices to prove that if U, U' are two schematically dense opens in X such that two S -morphisms $u : U \rightarrow Y$ and $u' : U' \rightarrow Y$ are equivalent, then the restrictions of u and u' to $U \cap U'$ are equal. But there exists by hypothesis an open $U'' \subset U \cap U'$, schematically dense in X , in which u and u' coincide; as U'' is also schematically dense in $U \cap U'$, our assertion results from (11.10.1, d)). □

Corollary (20.2.5). — Let S be an S_0 -scheme, X, Y two S -preschemes such that the composed morphism $Y \rightarrow S \rightarrow S_0$ is separated (which implies ((I, 5.5.1)) that $Y \rightarrow S$ is also). For every pseudo- S -morphism ω of X in Y , one then has $\text{dom}_S(\omega) = \text{dom}_{S_0}(\omega)$. In particular, if Y is a scheme, one has $\text{dom}_S(\omega) = \text{dom}(\omega)$.

Proof. Indeed, it is clear that $\text{dom}_S(\omega) \subset \text{dom}_{S_0}(\omega)$ without the hypothesis of separation; by virtue of (20.2.4), it suffices to prove that if $u_0 : U_0 \rightarrow Y$ is an S_0 -morphism of a schematically dense open U_0 of X in Y such that the composed $v_0 : U_0 \xrightarrow{u_0} Y \rightarrow S$ coincides with the restriction of the structural morphism $w_0 : U_0 \rightarrow S$ in an open $U \subset U_0$ schematically dense in X , then one has $v_0 = w_0$. But by virtue of the hypothesis that the morphism $S \rightarrow S_0$ is separated, this again results from (11.10.1, d)). IV-4 | 233

Corollary (20.2.6). — *Under the hypotheses of (20.2.4), the presheaf $U \mapsto \text{Ps. hom}_S(U, Y)$ is a sheaf.*

Proof. Indeed, let U be an open of X , (U_α) a covering of U by opens of X , and suppose given for each α a pseudo- S -morphism ω_α of U_α in Y , such that for every pair of indices α, β , the restrictions (20.2.2) of ω_α and ω_β to $U_\alpha \cap U_\beta$ are equal; by virtue of (20.2.3.1), this implies that $\text{dom}_S(\omega_\alpha) \cap U_\beta = \text{dom}_S(\omega_\beta) \cap U_\alpha$. Let W be the union of the opens $\text{dom}_S(\omega_\alpha)$, and, for each α , let u_α be the unique S -morphism $\text{dom}_S(\omega_\alpha) \rightarrow Y$ belonging to the class ω_α (20.2.4); by virtue of (20.2.4), the restrictions of u_α and u_β to $\text{dom}_S(\omega_\alpha) \cap \text{dom}_S(\omega_\beta)$ are equal, so there exists a morphism $u : W \rightarrow Y$ whose restriction to each $\text{dom}_S(\omega_\alpha)$ is equal to u_α ; it is clear that W is schematically dense in U , so it defines a pseudo- S -morphism ω of U in Y whose restrictions to the U_α are the ω_α . $\square$

Remark (20.2.7). — We know (11.10.4) that when X is *reduced*, it amounts to the same to say that an open of X is dense in X or that it is schematically dense in X ; the notion of pseudo-morphism (resp. pseudo- S -morphism) of X in Y then coincides with that of *rational application* (resp. *S -rational application*) of X in Y (I, 7.1.2). In the general case, the notion of pseudo-morphism seems more useful than that of rational application.

(20.2.8). We call *pseudo-function* on X a pseudo-morphism of X in $\text{Spec}(\mathbf{Z}[T])$ (T indeterminate), or, what amounts to the same, an X -pseudo-morphism of X in $X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$; this amounts also to the same (I, 3.3.15) as saying that a pseudo-function on X is an *equivalence class of sections of $\mathcal{O}_X$ over schematically dense opens of X* , two sections $g \in \Gamma(U, \mathcal{O}_X)$, $g' \in \Gamma(U', \mathcal{O}_X)$ over such opens being equivalent if there exists an open $U'' \subset U \cap U'$, schematically dense in X , where g and g' coincide. We can apply (20.2.4) with $S = X$ and $Y = X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$; then, for every pseudo-function φ on X , there is a *largest* schematically dense open $\text{dom}(\varphi)$ in X such that there is a section of $\mathcal{O}_X$ over $\text{dom}(\varphi)$ belonging to the class φ . It is clear that the sheaf $\mathcal{P}\text{s. hom}_X(X, X \otimes_{\mathbf{Z}} \mathbf{Z}[T])$ is here a *sheaf of rings*, and even an $\mathcal{O}_X$ -Algebra, which we denote $\mathcal{M}'_X$; it results from (20.2.6) that, for every open U of X , $\Gamma(U, \mathcal{M}'_X)$ is equal to the set of pseudo-morphisms of U in $\text{Spec}(\mathbf{Z}[T])$; we set $M'(X) = \Gamma(X, \mathcal{M}'_X)$. To say that a section φ of $\mathcal{M}'_X$ over U is *invertible* in the ring $\Gamma(U, \mathcal{M}'_X)$ means that there exists an open U' schematically dense in $\text{dom}(\varphi)$, thus in U , such that, if g is the unique section of $\mathcal{O}_X$ over $\text{dom}(\varphi)$ belonging to the class φ , $g|_{U'}$ is *invertible* in $\Gamma(U', \mathcal{O}_X)$. It results from (I, 3.3.15) that, in the canonical correspondence between $\Gamma(V, \mathcal{O}_X)$ and $\text{Hom}(V, \mathbf{Z}[T])$ (V open of X), the invertible elements of $\Gamma(V, \mathcal{O}_X)$ correspond to the morphisms which *factor through* $V \rightarrow \text{Spec}(\mathbf{Z}[T, T^{-1}]) \rightarrow \text{Spec}(\mathbf{Z}[T])$. We conclude that the sheaf $\mathcal{M}^{*'}_X$ of invertible germs of sections of $\mathcal{M}'_X$ is canonically identified with the sheaf $\mathcal{P}\text{s. hom}_X(X, X \otimes_{\mathbf{Z}} \mathbf{Z}[T, T^{-1}])$. IV-4 | 234

Lemma (20.2.9). — *Let U be an open of X , s a regular element of the ring $\Gamma(U, \mathcal{O}_X)$; then the open U_s of the $x \in U$ such that $s(x) \neq 0$ is schematically dense in U .*

Proof. Indeed, let V be an open of U , t a section of $\mathcal{O}_X$ over V such that $t|(V \cap U_s) = 0$. For every affine open $W \subset V$, there exists an integer n such that $s^n t|_W = 0$ (I, 1.4.1); but since s is a *regular* element of $\Gamma(U, \mathcal{O}_X)$, this implies $t|_W = 0$, hence $t = 0$. $\square$

(20.2.10). Consider a meromorphic function $f \in M(X)$ (20.1.4); then $\text{dom}(f)$ is *schematically dense* in X : indeed, every point of X admits an open neighbourhood U such that there exists a section $s \in \Gamma(U, \mathcal{O}_X)$ which is a regular element of this ring, and for which $s(f|_U) \in \Gamma(U, \mathcal{O}_X)$; as $s|_{U_s}$ is invertible, we conclude that $f|_{U_s} \in \Gamma(U_s, \mathcal{O}_X)$, hence $U_s \subset \text{dom}(f)$ by definition (20.1.4), and our assertion results from the lemma (20.2.9). We can thus associate to f the pseudo-function equal to the class of the section of $\mathcal{O}_X$ over $\text{dom}(f)$, restriction of f ; operating in the same way replacing X by an arbitrary open of X , we thus obtain a canonical *homomorphism of $\mathcal{O}_X$ -Algebras*

$$(20.2.10.1) \quad \mathcal{M}_X \longrightarrow \mathcal{M}'_X$$

which, by restriction, gives evidently a homomorphism of sheaves of abelian groups

$$(20.2.10.2) \quad \mathcal{M}^{*'}_X \longrightarrow \mathcal{M}'^{*'}_X$$

for the sheaves of invertible germs of sections of $\mathcal{M}_X$ and $\mathcal{M}'_X$.

Proposition (20.2.11). — *label=(i) The canonical homomorphism (20.2.10.1) (and hence also (20.2.10.2)) is injective.*

lbbl=(ii) Suppose either that X is locally Noetherian, or that X is reduced and the set of its irreducible components is locally finite. Then the canonical homomorphism (20.2.10.1) (and hence also (20.2.10.2)) is bijective.

Proof. The questions being local on X , we can restrict to the case where $X = \operatorname{Spec}(A)$ is affine, and show that the canonical homomorphism $M(X) \rightarrow M'(X)$ is injective, and that it is bijective in the cases considered in (ii). Taking into account the definition of the sheaf $\mathcal{M}_X$ (20.1.3), we can in fact remark that (20.2.10.1) comes from a homomorphism of presheaves

$$\Gamma(U, \mathcal{O}_X)[\Gamma(U, \mathcal{S})^{-1}] \longrightarrow \Gamma(U, \mathcal{M}'_X)$$

and it suffices to show that the latter is injective (resp. bijective under the hypotheses of (ii)). Denoting by S the set of regular elements of A (so that $S^{-1}A$ is the total ring of fractions of A), we must therefore consider the canonical homomorphism

$$(20.2.11.1) \quad S^{-1}A \longrightarrow M'(X).$$

We can write $S^{-1}A = \varinjlim A_t$, where t runs through the set of regular elements of A , ordered by the relation “ t is a divisor of t' ” (0_I, 1.4.5); we can also write $A_t = \Gamma(D(t), \mathcal{O}_X)$. On the other hand, by definition $M'(X) = \varinjlim \Gamma(U, \mathcal{O}_X)$, where U runs through the set of schematically dense opens of X (ordered by the relation $\supset$), and the map (20.2.11.1) is none other than the canonical map coming from the fact that the $D(t)$ constitute a part of the set of schematically dense opens in X (20.2.9). Note that by definition of a schematically dense open, the homomorphism $\Gamma(U, \mathcal{O}_X) \rightarrow \Gamma(U', \mathcal{O}_X)$ for any two such opens $U' \subset U$ is always injective, hence the homomorphisms $\Gamma(U, \mathcal{O}_X) \rightarrow M'(X)$ are also injective, and we conclude that (20.2.11.1) is injective. To prove that (20.2.11.1) is bijective, it suffices to show that the $D(t)$ are cofinal in the set of schematically dense opens, i.e., that for such an open U , there exists t regular in A such that $D(t) \subset U$. Now, when X is reduced and the irreducible components of X form a locally finite set, this set is finite since X has been supposed affine, so A has only a finite number of minimal primes $\mathfrak{p}_i$; as A is reduced, the intersection of the $\mathfrak{p}_i$ is zero, and to say that t is regular is equivalent to saying that t does not belong to any of the $\mathfrak{p}_i$; we conclude by the reasoning of (I, 7.1.9, .1). When A is Noetherian, to say that $U = X - Y$ (where $Y = V(\mathfrak{J})$ is closed in X) is schematically dense means (5.10.2) that Y does not meet $\operatorname{Ass}(\mathcal{O}_X)$, and by virtue of Bourbaki, *Alg. comm.*, chap. IV, §1, n° 4, prop. 8, this implies the existence of a $t \in \mathfrak{J}$ such that t is A -regular, hence $U \supset D(t)$.

We have moreover proved in the course of this demonstration the □

Corollary (20.2.12). — If $X = \operatorname{Spec}(A)$, where A is Noetherian, or reduced and having only a finite number of minimal prime ideals, the schematically dense opens in X are those which contain an open of the form $D(t)$, where t is regular in A , and $M(X) = M'(X)$ is the total ring of fractions $S^{-1}A$, where S is the set of regular elements of X .

Remarks (20.2.13). — label=(i) Let φ be an element of $M(X)$, φ' its image in $M'(X)$; by definition ((20.1.4) and (20.2.8)) $\operatorname{dom}(\varphi) \subset \operatorname{dom}(\varphi')$; but in fact one has the equality $\operatorname{dom}(\varphi) = \operatorname{dom}(\varphi')$, because there exists a section of $\mathcal{O}_X$ over $\operatorname{dom}(\varphi')$, belonging to the class φ' , and the corresponding meromorphic function equals φ by (20.2.11, (i)), hence $\operatorname{dom}(\varphi') \subset \operatorname{dom}(\varphi)$.

lbbl=(ii) We have already noted that when X is reduced, one has $\mathcal{M}'_X = \mathcal{R}(X)$ (sheaf of rational functions on X (I, 7.3.2)); if moreover the irreducible components of X form a locally finite set, one has $\mathcal{M}_X = \mathcal{M}'_X = \mathcal{R}(X)$. In general, since every schematically dense open is dense, one has a canonical homomorphism $\mathcal{M}'_X \rightarrow \mathcal{R}(X)$, but even when X is locally Noetherian, this homomorphism is not necessarily injective. For example, if $X = \operatorname{Spec}(A)$, where A is a Noetherian ring such that $\operatorname{Ass}(A)$ contains embedded prime ideals (which implies that A is not reduced), where S is the set of regular elements of A , complementary of the union of the ideals $\mathfrak{p} \in \operatorname{Ass}(A)$; on the other hand, $\mathcal{R}(X)$ is the complementary of the union of the minimal ideals of A (I, 7.1.9), and the canonical homomorphism $A \rightarrow Q^{-1}A$ (and a fortiori $S^{-1}A \rightarrow Q^{-1}A$) is therefore not injective, since there exists in $A - Q$ elements $\neq 0$ of A annihilated by elements of Q (Bourbaki, *Alg. comm.*, chap. IV, §1, n° 1, cor. 2 of prop. 1).

lcbel=(iii) We note also that when X is locally Noetherian, the $\mathcal{O}_X$ -Module $\mathcal{M}'_X$ is *not necessarily quasi-coherent*. Consider for example a local Noetherian ring A of dimension ≥ 2 , with maximal ideal $\mathfrak{m} \in \text{Ass}(A)$, and if we set $X = \text{Spec}(A)$, the scheme induced on the open U of X , complementary of $\{\mathfrak{m}\}$, is *integral*. The only regular elements of A are the invertible elements, so $\Gamma(X, \mathcal{M}_X) = M(X) = A$; but as U is an integral scheme, $\mathcal{M}_X|_U = R(U)$ is a simple sheaf, so $\mathcal{O}_U$ is not a simple sheaf since $\dim(A) \geq 2$.

It remains to give an example of a ring A having the above properties. It suffices to consider an integral local ring B of dimension ≥ 2 and residue field k , and to take $A = B \otimes k$ with the law of multiplication $(b, x)(b', x') = (bb', bx' + b'x)$.

ldbel=(iv) If X is locally Noetherian, it results from (5.10.2) that the schematically dense opens in X are those which contain $\text{Ass}(\mathcal{O}_X)$.

(20.2.14). Let X be a prescheme, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent and *strictly torsion-free* (20.1.5), so that $\mathcal{F}$ identifies via (20.1.5.1) with a sub- $\mathcal{O}_X$ -Module of $\mathcal{M}_X(\mathcal{F})$. For every meromorphic section u of $\mathcal{F}$ over X , we call the *ideal of denominators* of u the Ideal $\mathcal{J}$ of the section $\bar{u}$, image of u in $\mathcal{M}_X(\mathcal{F})/\mathcal{F}$. The Ideal $\mathcal{J}$ is *quasi-coherent*: indeed, the question being local on X , we can restrict to the case where X is affine, and there exists a section $s \in \Gamma(X, \mathcal{O}_X)$ such that $v = su \in \Gamma(X, \mathcal{F})$; to say that a section $f \in \Gamma(U, \mathcal{O}_X)$ belongs to $\Gamma(U, \mathcal{J})$ means that $f(u|_U) \in \Gamma(U, \mathcal{F})$, and since $s|_U$ is a regular element of $\Gamma(U, \mathcal{O}_X)$ and $\mathcal{F}$ is strictly torsion-free, the preceding relation is equivalent to $f((su)|_U) \in \Gamma(U, s\mathcal{F})$; so if $\bar{v}$ is the section of $\mathcal{F}/s\mathcal{F}$, canonical image of v , we see that $\mathcal{J}$ is the kernel of the homomorphism $\mathcal{O}_X \rightarrow \mathcal{F}/s\mathcal{F}$ obtained by multiplication by the section $\bar{v}$. As $\mathcal{F}/s\mathcal{F}$ is quasi-coherent, so is $\mathcal{J}$.

It results immediately from the preceding definition that $\text{dom}(u)$ is the open complement of the closed sub-prescheme of X defined by the Ideal of denominators of u .

Proposition (20.2.15). — *Let $f : X' \rightarrow X$ be a morphism, $\mathcal{F}$ an $\mathcal{O}_X$ -Module quasi-coherent, φ a section of $\mathcal{M}_f(\mathcal{F})$ over X (20.1.11). Then $f^{-1}(\text{dom}(\varphi))$ is schematically dense in X' .*

Proof. The question being evidently local on X and X' , we can suppose $X = \text{Spec}(A)$, $X' = \text{Spec}(A')$ affines, $\mathcal{F} = \tilde{M}$, and $\varphi = h \otimes (1/s)$, where $h \in M$ and s is a regular element of A whose image s' in A' is a regular element. We know then (20.2.9) that $D(s')$ is a schematically dense open in X' , and it is the inverse image by f of $D(s)$, which is contained in $\text{dom}(\varphi)$. $\square$

Remark (20.2.16). — When Y is not separated, the presheaf $U \mapsto \text{Ps. hom}_S(U, Y)$ on X is not necessarily a sheaf, even when X is Noetherian, as the following example shows. We take for X the spectrum of a semi-local Noetherian ring A of dimension ≥ 2 , having exactly two maximal ideals $\mathfrak{m}'$, $\mathfrak{m}''$ (so that X has exactly two closed points x' , x''), such that $\mathfrak{m}'$ and $\mathfrak{m}''$ belong to $\text{Ass}(A)$ and that the open $U = X - \{x', x''\}$ is *integral*. Note that the only schematically dense opens in X are those containing x' and x'' (20.2.13, (iv)), and that $U = U' \cap U''$. Let $U' = X - \{x'\}$, $U'' = X - \{x''\}$, so that $U = U' \cap U''$. To have a counterexample, it suffices to define two S -morphisms $u' : U' \rightarrow Y$, $u'' : U'' \rightarrow Y$ (with $S = X$) whose restrictions to U belong to the same pseudo-morphism, but which for every neighbourhood V' of x' in U' and every neighbourhood V'' of x'' in U'' , the restrictions of u' and u'' to $V' \cap V''$ are distinct. For this, let us consider a closed irreducible part Z of X , distinct from X ; let Y be the X -prescheme obtained by gluing two copies Y' , Y'' of X along the open $X - Z$ (I, 2.3.1). We take for u' and u'' the restrictions to U' and U'' respectively of the structural isomorphisms $Y' \xrightarrow{\sim} X$, $Y'' \xrightarrow{\sim} X$. As V' and V'' contain the generic point of Z , and $V' \cap V''$ are distinct, but the restrictions of u' and u'' to $U - (U \cap Z)$ are equal, and this is a dense open in U , hence schematically dense since U is reduced.

It remains therefore only to define A and Z so as to have the preceding properties. Let $X_0 = \text{Spec}(B)$ be an integral affine scheme (for example the affine plane over a field k), Y an irreducible closed subvariety of X_0 (for example a line) containing two distinct closed points x' , x'' , corresponding to maximal ideals $\mathfrak{n}'$, $\mathfrak{n}''$ of B . Consider the ring $C = B \oplus (B/\mathfrak{n}' \oplus B/\mathfrak{n}'')$ with the multiplication $(b, z)(b', z') = (bb', bz' + b'z)$. If $X_1 = \text{Spec}(C)$, we have $X_0 = (X_1)_{\text{red}}$ and X_1 is reduced except at the points x' , x'' . It then suffices to take $A = R^{-1}C$, where R is the complement of the union of the maximal ideals of C at the points x' , x'' , and for Z the trace of Y on $X = \text{Spec}(A)$.

20.3. Composition of pseudo-morphisms

(20.3.1). Let X, Y, Z be three preschemes, ω a pseudo-morphism of X in Y , $f : Y \rightarrow Z$ a morphism. It is clear that if U', U'' are two schematically dense opens in X , $u' : U' \rightarrow Y$, $u'' : U'' \rightarrow Y$ two morphisms of the class ω , the morphisms fu' and fu'' are equivalent (for the relation defined in (20.2.1)), so, for all morphisms u of the class ω , the morphisms fu belong to a single equivalence class, which we call the pseudo-morphism of X in Z *composed* of f and ω . One has $\text{dom}(f\omega) = \text{dom}(\omega)$. If $g : Z \rightarrow T$ is a morphism, it is clear that $g \circ (f\omega) = (g \circ f)\omega$.

(20.3.2). Let us now suppose given a pseudo- S -morphism ω of X in Y , where Y is *separated over S* , so that there exists an S -morphism $u : \text{dom}_S(\omega) \rightarrow Y$ of the class ω (20.2.4). Let $f : X' \rightarrow X$ be an S -morphism such that the open $V' = f^{-1}(\text{dom}_S(\omega))$ is schematically dense in X' ; then we say that the class (for the equivalence relation of (20.2.1)) of the S -morphism $u \circ (f|U')$ (class which is defined by virtue of the preceding hypothesis) is the pseudo- S -morphism *composed* of ω and f , and we denote it $\omega \circ f$; it is clear that one has

$$(20.3.2.1) \quad f^{-1}(\text{dom}_S(\omega)) \subset \text{dom}_S(\omega \circ f).$$

For $f : X' \rightarrow X$ given, we denote by $\text{Ps. hom}_S(X, Y)^f$ the set of pseudo- S -morphisms ω of X in Y which verify the preceding condition. If ω is such a pseudo- S -morphism, it is clear that for every open V of X ,

$$f^{-1}(\text{dom}_S(\omega|V)) = f^{-1}(V \cap \text{dom}_S(\omega)) = f^{-1}(V) \cap f^{-1}(\text{dom}_S(\omega))$$

is schematically dense in $f^{-1}(V)$, so, if $f^V : f^{-1}(V) \rightarrow V$ is the restriction of f , the composed $(\omega|V) \circ f^V$ is defined and equal to $(\omega \circ f)|f^{-1}(V)$. One thus defines a canonical restriction map $\text{Ps. hom}_S(X, Y)^f \rightarrow \text{Ps. hom}_S(V, Y)^{f'}$, which allows to define a *presheaf* of sets $V \mapsto \text{Ps. hom}_S(V, Y)^{f'}$ on X , sub-presheaf of $V \mapsto \text{Ps. hom}_S(V, Y)$, which gives a sheaf of sets that we denote $\mathcal{P}\text{s. hom}_S(X, Y)^f$. Moreover, for every open V of X , we have a map $\omega \mapsto \omega \circ f^V$ in $\text{Ps. hom}_S(f^{-1}(V), Y)$, which defines an f -morphism of sheaves of sets

$$\mathcal{P}\text{s. hom}_S(X, Y)^f \longrightarrow \mathcal{P}\text{s. hom}_S(X', Y).$$

(20.3.3). Let now $f' : X'' \rightarrow X'$ be an S -morphism such that the open $f'^{-1}(f^{-1}(\text{dom}_S(\omega)))$ is schematically dense in X'' ; then $\omega \circ (f \circ f')$ is defined and $u \circ f \circ f'$ belongs to this pseudo- S -morphism; moreover, by virtue of (20.3.2.1), $f'^{-1}(\text{dom}_S(\omega \circ f))$ is *a fortiori* schematically dense in X'' , so $(\omega \circ f) \circ f'$ is also defined and $u \circ f \circ f'$ belongs to this pseudo- S -morphism, hence $(\omega \circ f) \circ f' = \omega \circ (f \circ f')$. On the other hand, for every S -morphism $g : Y \rightarrow Z$, one has $\text{dom}_S(g\omega) = \text{dom}_S(\omega)$ (20.3.1), so $(g\omega) \circ f$ is defined, and $g \circ u \circ f$ belongs to this pseudo- S -morphism, which shows that $(g\omega) \circ f = g \circ (\omega \circ f)$.

(20.3.4). The most important case in the definition (20.3.2) is that where one has $\mathcal{P}\text{s. hom}_S(X, Y)^f = \mathcal{P}\text{s. hom}_S(X, Y)$: it suffices for this that for every open U of X and every open V schematically dense in U , $f^{-1}(V)$ is schematically dense in $f^{-1}(U)$; when this holds, then, for *every* open U of X and *every* pseudo- S -morphism $\omega : U \rightarrow Y$, one can define the composed $\omega \circ f^U$, even if Y is not separated over S . Indeed, if $u' : U' \rightarrow Y$ and $u'' : U'' \rightarrow Y$ are two morphisms of the class ω , they coincide in an open U_0 schematically dense in U , so the composed morphisms $f^{-1}(U') \xrightarrow{u'} Y$ and $f^{-1}(U'') \xrightarrow{u''} Y$ coincide in $f^{-1}(U_0)$, which is by hypothesis schematically dense in $f^{-1}(U)$; one can therefore define $\omega \circ f^U$ as the class of any of the morphisms $f^{-1}(U') \xrightarrow{u'} Y$, where u' belongs to ω .

Proposition (20.3.5). — *Let X, X' be two preschemes, $f : X' \rightarrow X$ a morphism. Then, in each of the three following cases, for every open U of X and every open V schematically dense in U , $f^{-1}(V)$ is schematically dense in $f^{-1}(U)$:*

- label=(i) f is flat and locally of finite presentation.
- lbbel=(ii) f is flat and the underlying space of X is locally Noetherian.
- lcbel=(iii) X' is reduced, the set of irreducible components of X' is locally finite, and every irreducible component of X' dominates an irreducible component of X .

Proof. Assertion (i) results from (11.10.5, (ii), b)); assertion (ii) results from (11.10.5, (ii), a)), since then every open V in U is retrocompact, i.e. the canonical injection $j : V \rightarrow U$ is a quasi-compact

morphism. Finally, to prove (iii), note that since X' is reduced, it suffices to show that for every open U of X and every open V dense in U , $f^{-1}(V)$ is dense in $f^{-1}(U)$. Now, for $f^{-1}(V)$ to be dense in $f^{-1}(U)$, it suffices that $f^{-1}(V)$ contain all the maximal points of X' belonging to $f^{-1}(U)$; the conclusion therefore results from the hypothesis that the image by f of every maximal point of X' belonging to $f^{-1}(V)$ is a maximal point of X belonging to U , hence to V since the set of irreducible components of X is locally finite. $\square$

(20.3.6). Let X, Y be two S -preschemes; suppose that X satisfies one of the two following hypotheses:

label=a) X is locally Noetherian;

lbbel=b) X is reduced and the set of its irreducible components is locally finite.

Then, for every $x \in X$, the canonical S -morphism $j : \text{Spec}(\mathcal{O}_{X,x}) \rightarrow X$ is flat and satisfies condition (ii) of (20.3.5) in case a), condition (iii) of (20.3.5) in case b). For every pseudo- S -morphism ω of X in Y , the composed is therefore defined and is a pseudo- S -morphism of $\text{Spec}(\mathcal{O}_{X,x})$ in Y , called the *restriction* of ω to $\text{Spec}(\mathcal{O}_{X,x})$. Note now that if X satisfies condition a) (resp. b)) of (20.3.6), the same holds for every prescheme induced on an open U of X containing x . By passage to the inductive limit, we deduce from the canonical maps $\text{Ps. hom}_S(U, Y) \rightarrow \text{Ps. hom}_S(\text{Spec}(\mathcal{O}_{X,x}), Y)$ thus obtained, a canonical map

$$(20.3.6.1) \quad (\mathcal{P}\text{s. hom}_S(X, Y))_x \longrightarrow \text{Ps. hom}_S(\text{Spec}(\mathcal{O}_{X,x}), Y)$$

where the left-hand side is the fibre at the point x of the sheaf $\mathcal{P}\text{s. hom}_S(X, Y)$, the set of germs at x of pseudo- S -morphisms of open neighbourhoods of x in Y .

Proposition (20.3.7). — *Under the hypotheses of (20.3.6), the canonical map (20.3.6.1) is injective. If moreover Y is locally of finite presentation over S , the map (20.3.6.1) is bijective.*

Proof. By application of the method of (8.1.2, a)), this proposition will be a particular case of the following: $\square$

Proposition (20.3.8). — *The notations being those of (8.8.1), suppose X_α quasi-compact (resp. quasi-compact and quasi-separated) and Y_α locally of finite type (resp. locally of finite presentation) over S_α . Suppose moreover that one of the following conditions is satisfied:*

label=(i) *The transition morphisms $S_\mu \rightarrow S_\lambda$ (for $\lambda \leq \mu$) are flat, and the X_λ and X are Noetherian.*

lbbel=(ii) *The X_λ are reduced, the set of irreducible components of X is locally finite, and, for $\lambda \leq \mu$, every irreducible component of X_μ dominates an irreducible component of X_λ .*

Then, the canonical map

$$(20.3.8.1) \quad \varinjlim \text{Ps. hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \text{Ps. hom}_S(X, Y)$$

is injective (resp. bijective).

Proof. Note first that, in case (i), the morphisms $X_\mu \rightarrow X_\lambda$ (for $\lambda \leq \mu$) and $X \rightarrow X_\lambda$ are flat, so by IV-4 | 240 (20.3.4) and (20.3.5) the canonical maps

$$\text{Ps. hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \text{Ps. hom}_{S_\mu}(X_\mu, Y_\mu)$$

are defined (without hypothesis of separation on the Y_λ or Y); the same is true in case (ii) by the same reasoning. The proposition will result from the following lemmas.

Lemma (20.3.8.2). — *The notations being those of (8.8.1), suppose X_α quasi-compact, and let U_α be an open in X_α ; let U_λ and U be its inverse images in X_λ and X . Suppose that one of the conditions (i), (ii) of (20.3.8) is satisfied. Then, for U to be schematically dense in X , it is necessary and sufficient that there exist $\lambda \geq \alpha$ such that U_λ is schematically dense in X_λ , and in this case U_μ is schematically dense in X_μ for $\mu \geq \lambda$.*

We first verify condition (i); denote by $j_\lambda : U_\lambda \rightarrow X_\lambda$ and $j : U \rightarrow X$ the canonical injections, by $\mathcal{J}_\lambda$ the kernel of the canonical homomorphism $\mathcal{O}_{X_\lambda} \rightarrow (j_\lambda)_*(\mathcal{O}_{U_\lambda})$, and $\mathcal{J}$ is coherent, of finite type, hence it results from (2.3.1) that one can identify $(j_\lambda)_*(\mathcal{O}_{U_\lambda})$ to $p_\lambda^*((j_\lambda)_*(\mathcal{O}_{U_\lambda}))$ (resp. $j_*(\mathcal{O}_U)$ to $(j)_*(\mathcal{O}_U)$). The assertion results then from the definition of a schematically dense open, and of (8.5.8, (ii)).

To establish (20.3.8.2) when condition (ii) is verified, and noting that X is then *reduced* by virtue of (8.7.1); it amounts to the same to say that U_λ (resp. U) is schematically dense in X_λ (resp. X) or that it is dense in X_λ (resp. X), and the conclusion results from (20.3.8.4) applied to X_λ and X .

Lemma (20.3.8.3). — *Under the hypotheses of (8.2.2), let M_α (resp. M) be the set of maximal points of S_α (resp. S). Suppose that for every λ , the set M_λ is finite, and that the M_λ form a projective system of sets. Then M is the projective limit of the M_λ .*

Let us first show that a point $s \in \varprojlim M_\lambda$ is maximal in S : indeed, if s' is a generalisation of s , the image s'_λ of s' in S_λ is a generalisation of the image s_λ of s , hence equal to s_λ for all λ , which implies $s' = s$, since the set of points underlying S is the projective limit of the sets underlying the S_λ (8.2.9). Inversely, let s be a maximal point of S and let us prove that it belongs to $\varprojlim M_\lambda$. Let s_λ be the image of s in S_λ , M'_λ the set of maximal points of S_λ that are generalisations of s_λ ; the M'_λ are non-empty finite sets forming a projective system, hence $M' = \varprojlim M'_\lambda$ is non-empty and the map $M' \rightarrow M'_\lambda$ is surjective (Bourbaki, *Ens.*, chap. III, 2^e éd., § 7, n^o 4, Example I). On the other hand, $\text{Spec}(\mathcal{O}_{S,s}) = \varprojlim \text{Spec}(\mathcal{O}_{S_\lambda, s_\lambda})$ by virtue of (8.2.12) and (8.2.9), hence the points of $\varprojlim M'_\lambda$ are also maximal points of $\text{Spec}(\mathcal{O}_{S,s})$ by the first part of the reasoning. Thus $M' = \varprojlim M'_\lambda$ necessarily reduces to the point s ; one concludes that M'_λ reduces to the point s_λ and consequently the s_λ are maximal in the S_λ , which completes the proof of the lemma.

Lemma (20.3.8.4). — *The hypotheses being those of (20.3.8.3), suppose moreover S_α quasi-compact, and let V_α be an open of S_α , and V its inverse images in S_λ and S . If U_α is dense in S_α , then U_λ is dense in S_λ for $\lambda \geq \alpha$ and U is dense in S . Inversely, if U is dense in S and if moreover the set M of maximal points of S is finite, there exists $\lambda \geq \alpha$ such that U_λ is dense in S_λ .*

Indeed, since M_α is finite, the hypothesis that U_α is dense in S_α implies that $M_\alpha \subset U_\alpha$, hence $M_\lambda \subset U_\lambda$ for $\lambda \geq \alpha$ and $M \subset U$ by virtue of (20.3.8.3), which proves the first assertion. Inversely, suppose M finite and U dense in S , hence $M \subset U$; note that the U_λ are open, hence ind-constructible, and the M_λ finite, hence pro-constructible (1.9.6). The second assertion then follows from (8.3.2).

Proof of (20.3.8.5) (end of proof of (20.3.8)). Suppose condition (ii) is verified, and note that X is then reduced by virtue of (8.7.1); it amounts to the same to say that U_λ (resp. U) is schematically dense in X_λ (resp. X) or that it is dense, and the conclusion results from (20.3.8.4) applied to X_λ and X .

Proof of (20.3.8.6) (injectivity of (20.3.8.1)). To show that the map (20.3.8.1) is injective, consider two morphisms u'_λ, u''_λ of a schematically dense open $U_\lambda \subset X_\lambda$ in Y_λ , and suppose that their images u', u'' , morphisms of U of X in Y , coincide in a schematically dense open V of U . Under either hypothesis (i) or (ii), one can suppose V quasi-compact; otherwise, since X has only a finite number of maximal points and is reduced, it suffices to replace V by a union of a finite number of affine open neighbourhoods of these maximal points (contained in V by hypothesis). Then there exists $\lambda \geq \alpha$ tel such that V is the inverse image of a quasi-compact open V_λ of X_λ (8.2.11), and by (20.3.8.2) we can suppose that V_λ is schematically dense in X_λ . It results from (8.8.2, (i)) that taking λ sufficiently large, u'_λ and u''_λ coincide in V_λ , hence belong to the same pseudo-homomorphism.

Let us prove finally that the map (20.3.8.1) is surjective. Consider now a morphism u of a schematically dense open U of X in Y ; as above, we can suppose U quasi-compact and quasi-separated (U being replaceable, in case (ii), by a union of a finite number of disjoint affine opens). We can furthermore suppose there exists $\lambda \geq \alpha$ such that U is the inverse image of a quasi-compact open U_λ of X_λ (8.2.11) which is automatically quasi-separated, and by (20.3.8.2) we can suppose that U_λ is schematically dense in X_λ . Since the Y_λ are supposed locally of finite presentation, it results from (8.8.2, (i)) that there exists λ such that u is the image of a morphism $u_\lambda : U_\lambda \rightarrow Y_\lambda$; whence the conclusion.

Remarks (20.3.8.7). — label=(i) To show that the map (20.3.8.1) is injective, it is not necessary, in the hypothesis (i) of (20.3.8), to suppose X Noetherian. Indeed, the lemma (20.3.8.2) does not use this hypothesis. With the notations of (20.3.8.6), let Z_λ be the sub-prescheme of coincidences of u'_λ and u''_λ , and let Z be the sub-prescheme of coincidences of u' and u'' ; it results from the definition (17.4.5) and from (I, 3.3.10.1) that Z is the projective limit of the Z_μ for $\mu \geq \lambda$. Now, by hypothesis, Z majorizes a schematically dense open of X ; it follows that Z is itself induced on an open of X by virtue of the following lemma:

Lemma (20.3.8.8). — *Let T be a prescheme. Then every sub-prescheme W of T which majorizes a schematically dense open V of T is induced on an open (schematically dense) of T .*

Indeed, the subspace of T underlying W is locally closed, hence open in its closure, hence open in T ; the conclusion results from (11.10.1, , c)).

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This lemma being established, we conclude that for $\mu \geq \lambda$ sufficiently large, Z_μ is induced on an open of X_μ by virtue of (8.6.3), and Z_λ en tant que sub-prescheme of a Noetherian prescheme, is of finite presentation (1.6.2), and so it is the same for Z_μ for $\mu \geq \lambda$ on X_μ and of Z on X . One can now apply (20.3.8.2) which shows that for $\mu \geq \lambda$ sufficiently large, Z is schematically dense in X , whence the conclusion.

lbel=(ii) If, in the hypothesis (i) of (20.3.8), we drop the condition that X is Noetherian, the reasoning of (20.3.8.6) shows yet that the image of (20.3.8.1) is formed of the pseudo- S -morphisms having a representative which is an S -morphism $U \rightarrow Y$, where U is schematically dense in X and *quasi-compact and quasi-separated*.

□

Corollary (20.3.9). — Suppose verified either hypothesis a), b) of (20.3.6) on X , and let Y locally of finite presentation over S . Then, for a pseudo- S -morphism ω of X in Y to be defined at a point x (20.2.3), it is necessary and sufficient that its restriction to $\text{Spec}(\mathcal{O}_{X,x})$ be everywhere defined (in other words, that it be an S -morphism of $\text{Spec}(\mathcal{O}_{X,x})$ in Y).

Lemma (20.3.10). — Let $f : X' \rightarrow X$ be an S -morphism faithfully flat and quasi-compact, $X'' = X' \times_X X'$, p_1 and p_2 the canonical projections of X'' in X' , Y an S -prescheme separated on S . Let U be an open of X , $U' = f^{-1}(U)$, $U'' = p_1^{-1}(U') = p_2^{-1}(U')$, and suppose that U'' is schematically dense in X'' . Let $g : U \rightarrow Y$ be an S -morphism; then, if $g \circ (f|_U)$ extends to an S -morphism $X' \rightarrow Y$, g extends to an S -morphism $X \rightarrow Y$.

Proof. To prove (20.3.10), denote by g' an S -morphism $X' \rightarrow Y$ which extends $g \circ (f|_U)$, and set $g''_i = g' \circ p_i : X'' \rightarrow Y$ ($i = 1, 2$). If one sets $f'' = f \circ p_1 = f \circ p_2 : X'' \rightarrow X$, it is clear that g''_1 and g''_2 coincide in U'' with $g \circ (f''|_{U''})$. But as Y is separated on S and U'' schematically dense in X'' , we have $g''_1 = g''_2$ (11.10.1, , d)). As f is faithfully flat and quasi-compact, it results from descent theory (chap. VI) that there exists a unique S -morphism $h : X \rightarrow Y$ such that $h \circ f = g'$; as the restriction $U' \rightarrow U$ of f is a faithfully flat and quasi-compact morphism and $U'' = U' \times_U U'$, the result of uniqueness applied to U instead of X , shows that $h|_U = g$, which proves the lemma. □

Proposition (20.3.11). — Let Y be an S -prescheme separated on S , ω a pseudo- S -morphism of X in Y , $f : X' \rightarrow X$ an S -morphism. Suppose that f is flat, and that one of the following conditions is satisfied:

label=(i) f is an open morphism, or surjective and quasi-compact, and $\text{dom}_S(\omega)$ contains an open U schematically dense in X and retrocompact in X .

lbel=(ii) f is locally of finite presentation.

lbel=(iii) Y is locally of finite presentation over S , and X satisfies one of the conditions a), b) of (20.3.6).

Then, $f^{-1}(\text{dom}_S(\omega))$ is schematically dense in X' , so that $\omega \circ f$ is defined (20.3.2) and one has

$$(20.3.11.1) \quad \text{dom}_S(\omega \circ f) = f^{-1}(\text{dom}_S(\omega)).$$

Proof. Let us first prove that $f^{-1}(\text{dom}_S(\omega))$ is schematically dense in X' . The question being local on X' , we can suppose X and X' affines and since f is flat, it suffices to verify, by (11.10.5, (ii), a)), that $\text{dom}_S(\omega)$ contains a retrocompact open U schematically dense in X . This is the hypothesis in case (i), and of (20.2.12) in case (iii), since in an affine scheme every open of the form $D(t)$ is retrocompact; in case (ii), we can directly apply (20.3.5, (i)).

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Let us now prove (20.3.11.1), i.e., that, for every point $x' \in \text{dom}_S(\omega \circ f)$, one has $x = f(x') \in \text{dom}_S(\omega)$. Note first that we can restrict to the case where f is faithfully flat and quasi-compact. This is indeed the second case of (i) and in case (ii) f is an open morphism (11.3.1), so replacing X by the open $f(X')$, we may suppose f surjective, hence faithfully flat. In case (iii), by (20.3.9), it suffices to prove that the restriction of ω to $\text{Spec}(\mathcal{O}_{X,x})$ is everywhere defined, and we can therefore replace X by $\text{Spec}(\mathcal{O}_{X,x})$, X' by $X' \times_X \text{Spec}(\mathcal{O}_{X,x})$ and f by its restriction to this last prescheme, which is a surjective morphism (2.3.4), hence faithfully flat.

Suppose therefore f faithfully flat and quasi-compact; with the notations of the lemma (20.3.10), it suffices to see that U'' is schematically dense in X'' , by taking for U an open schematically dense in X , contained in $\text{dom}_S(\omega)$; this will be the case, by virtue of (11.10.5, (ii), a)), if U is taken retrocompact in

X (since the morphism $f'' : X'' \rightarrow X$ is flat). The existence of such an open U is part of the hypothesis in case (i); in case (iii) it results from (20.2.12) and the fact that in an affine scheme $\text{Spec}(A)$, every open of the form $D(t)$ is retrocompact. Finally, in case (ii), we take $U = \text{dom}_S(\omega)$ and show directly that U'' is schematically dense in X'' : it suffices for this to remark that $f'' : X'' \rightarrow X$ is flat and locally of finite presentation and to apply (20.3.5, (i)). $\square$

Corollary (20.3.12). — *Let φ be a pseudo-function on a prescheme X . Then, for every morphism flat and locally of finite presentation $f : X' \rightarrow X$, the pseudo-function $\varphi \circ f$ is defined and one has $\text{dom}(\varphi \circ f) = f^{-1}(\text{dom}(\varphi))$.*

Remark (20.3.13). — When X satisfies one of the conditions a), b) of (20.3.6), then we have seen (20.2.11) that the pseudo-functions on X are identified with the meromorphic functions on X . By virtue of (20.1.12) and of (20.2.13, (i)), if one only supposes that the morphism $f : X' \rightarrow X$ is flat, then, for every pseudo-function φ on X , $\varphi \circ f$ is defined and one has $\text{dom}(\varphi \circ f) = f^{-1}(\text{dom}(\varphi))$.

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20.4. Properties of domains of definition of rational functions

(20.4.1). Let X, Y be two S -preschemes, ω a pseudo- S -morphism of X in Y . Let u be an S -morphism $U \rightarrow Y$ belonging to the class ω , where U is schematically dense in X , and consider the graph Γ_u , sub-prescheme of $U \times_S Y$ (I, 5.3.11), and therefore sub-prescheme of $X \times_S Y$. Suppose that this sub-prescheme admits an *adherence* (I, 9.5.11) in $X \times_S Y$; if $j : \Gamma_u \rightarrow X \times_S Y$ is the canonical injection, this will be the case when the $\mathcal{O}_{X \times_S Y}$ -Module $j_*(\mathcal{O}_{\Gamma_u})$ is quasi-coherent; it results from the definition of the class of equivalence (20.2.1) that $j_*(\mathcal{O}_{\Gamma_u})$ does not depend on the representative u considered, hence the adherence of this sub-prescheme is well determined by ω ; we say that this closed sub-prescheme is the *graph* of the pseudo- S -morphism ω and we denote it Γ_ω . We note that Γ_ω is defined if X is locally Noetherian; it will also be defined when X is *reduced*, and it is then none other than the *reduced* closed sub-prescheme of $X \times_S Y$ having for underlying space the adherence in $X \times_S Y$ (I, 5.2.1) and (5.2.2). Note finally that if Y is *separated* on S , then Γ_u is *closed* in $U \times_S Y$ (I, 5.4.3), hence Γ_ω (whenever it exists) is the open $\Gamma_\omega \cap (U \times_S Y)$ (I, 9.5.10); by contrast, the prescheme induced by Γ_ω is in general different from Γ_u when Y is not separated. In particular, if $v : X \rightarrow Y$ is an S -morphism, the graph Γ_ω of the class ω of v may be distinct from the graph Γ_v . In particular also in what follows, when the question is about the graph of a pseudo- S -morphism, we will restrict to the case where Y is *separated over S* .

(20.4.2). Suppose therefore Γ_ω defined and Y separated on S ; denote by p and q the restrictions to Γ_ω of the canonical projections.

$$\begin{array}{ccc} & \Gamma_\omega & \\ p \swarrow & & \searrow q \\ X & & Y \end{array}$$

Then, if $U \subset \text{dom}_S(\omega)$, the restriction $p^{-1}(U) \rightarrow U$ of p is an isomorphism (I, 5.3.11); reciprocally, if U is an open of X having this property, and if u is the isomorphism reciprocal of the restriction $p^{-1}(U) \rightarrow U$ of p , $q \circ u$ is an S -morphism of U in Y which coincides with a morphism of the class ω in $U \cap \text{dom}_S(\omega)$. We conclude that $\text{dom}_S(\omega)$ is the *largest open* U of X such that the restriction $p^{-1}(U) \rightarrow U$ of p is an isomorphism; from the class of v (whose rational application associated (20.2.13, (ii)) is then *birational*); as $p^{-1}(\text{dom}_S(\omega))$ is the graph of the S -morphism $u = q \circ v : \text{dom}_S(\omega) \rightarrow Y$, it is schematically dense in Γ_ω (11.10.3, (iv)), so ω can be considered as the *composed* $q \circ p^{-1}$ (20.3.2). For every part M of the underlying space of Γ_ω set $\omega(M) = q(p^{-1}(M))$, which amounts to considering ω as a “multi-valued function”. We note that when $x \in \text{dom}_S(\omega)$, $\omega(\{x\})$ is the singleton $\{u(x)\}$; in general, for any $x \in X$, if one denotes by s the image of x in S , via the structural morphism, $\omega(\{x\})$ is a part of the prescheme Y_s , the fibre at the point s of the structural morphism.

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(20.4.3). In all the rest of this n^o, we will restrict to the case where X is *reduced*, so that there is then an identity between pseudo- S -morphisms and S -rational applications (20.2.7). Moreover, the *graph* of the S -rational application ω of X in Y exists and is none other than the *reduced* closed sub-prescheme of $X \times_S Y$ having for underlying space the adherence of the graph of any representative

$u : \text{dom}_S(\omega) \rightarrow Y$ in the space underlying $X \times_S Y$ (I, 5.2.1) and (5.2.2). It is then said to be *defined* at x (20.4.1).

Proposition (20.4.4). — *Let X be an S -prescheme locally integral, Y an S -prescheme locally of finite type and separated on S , ω an S -rational application of X in Y , $p : \Gamma_\omega \rightarrow X$ the canonical projection. Then, if $x \in X$ is a normal point such that $p^{-1}(x)$ contains an isolated point, ω is defined at the point x .*

Proof. Indeed, the first projection $p_1 : X \times_S Y \rightarrow X$ is then a separated and locally of finite type morphism, and so is its restriction $p : \Gamma_\omega \rightarrow X$, which is moreover *birational*, and as Γ_ω is reduced and X integral, Γ_ω is *integral*; it results then from (ErrIV, 30) that the hypothesis on x implies the existence of an open neighbourhood U of x such that the restriction $p^{-1}(U) \rightarrow U$ of p is an isomorphism; whence the conclusion by (20.4.2).

We note that (20.4.4) is the original formulation given by Zariski of his “Main theorem” (up to the fact that it was originally restricted to algebraic schemes over a base field). $\square$

Proposition (20.4.5). — *The hypotheses and notations being those of (20.4.4), suppose moreover X normal, and let X' be a reduced prescheme, $f : X' \rightarrow X$ a morphism locally of finite type and universally open. Then $\omega \circ f$ is defined, and one has $\text{dom}_S(\omega \circ f) = f^{-1}(\text{dom}(\omega))$ (in other words, for $x' \in X'$ and $x = f(x')$, so that ω is defined at x , it suffices that $\omega \circ f$ is at x').*

Proof. As X' is reduced and $f^{-1}(\text{dom}_S(\omega))$ is everywhere dense in X' by virtue of (2.4.11), the composite $\Gamma_{\omega \circ f}$ is a reduced prescheme (20.4.1), and as by hypothesis the composed morphism $\Gamma_{\omega \circ f} \rightarrow \Gamma_\omega \times_X X'$ is an isomorphism, $p_{(X')}$ is surjective; as f is surjective, it suffices to apply (20.4.4) that ω is defined at x . $\square$

Proposition (20.4.6). — *Let S be a prescheme, X, Y two S -preschemes; suppose X locally Noetherian, Y locally of finite type on S . Let U be an open schematically dense in X , $h : U \rightarrow Y$ an S -morphism, $x \in X - U$ a normal point of X , $h'_x : \text{Spec}(k(x)) \rightarrow Y$ an S -morphism. In order that h can be extended to an S -morphism $h' : U' \rightarrow Y$, where U' is an open of X containing U and x , and where the composed morphism $\text{Spec}(k(x)) \rightarrow U' \xrightarrow{h'} Y$ is the S -morphism given by h'_x , it is necessary and sufficient that the following condition be satisfied:*

(P) For every spectrum S_1 of a discrete valuation ring, of closed point a and generic point b , and for every morphism $u : S_1 \rightarrow X$ such that $u(a) = x$, $u(b) \in U$, the composed morphism $h_1 = h \circ (u|_{\{b\}}) : \{b\} = u^{-1}(U) \rightarrow Y$ extends to a morphism $h'_1 : S_1 \rightarrow Y$ such that the diagram

$$\begin{array}{ccc} \text{Spec}(k(a)) & \longrightarrow & S_1 \\ & & \uparrow h'_1 \\ \text{Spec}(k(x)) & \xrightarrow{h'_x} & Y \end{array}$$

is commutative.

Moreover, if this condition is satisfied, and if $h'' : U'' \rightarrow Y$ is a morphism satisfying the same conditions as h' , then there exists an open set containing $U \cup \{x\}$, in which h' and h'' coincide.

Proof. Let us first prove the last assertion; we can suppose h' and h'' defined in the same open $U_0 \supset U \cup \{x\}$. The sub-prescheme Z of coincidences of h' and h'' (17.4.5) contains U and x , so there is an open neighbourhood V of x in U_0 such that the sub-prescheme induced by Z on the open $Z \cap V$ is a *closed* sub-prescheme of the prescheme induced by V on V ; as this sub-prescheme $Z \cap V$ majorizes the prescheme induced by V on the open $U \cap V$, and the latter is schematically dense in V , $Z \cap V$ is necessarily equal to V (20.3.8.8), which proves that h' and h'' coincide in $U \cup V$.

The necessity of the condition is evident, let us prove that it is sufficient. By the bijective correspondence between S -morphisms of X in Y and X -sections of $X \times_S Y$ (I, 3.3.15), we can restrict to the case where $S = X$ and h is therefore a U -section of Y ; we note that then Y is locally Noetherian, and we can evidently (since X is locally intègre and locally Noetherian) suppose $X = S$ is irreducible.

Moreover, by virtue of (20.3.7), we can suppose that $X = \operatorname{Spec}(A)$, where A is a local Noetherian ring, integral and integrally closed.

Note that for every $x' \in X \cap U$, $h(x')$ is a specialisation of $h'_x(x) = y$. As Γ_ω is formed of the adherences of $h(U)$ and $h(U)$ formed of points adherent to $h(\xi)$, where ξ is the generic point of X , Γ_ω is formed of adherences of $h(\xi)$; indeed there exists a spectrum S_1 of a discrete valuation ring and a morphism $u : S_1 \rightarrow X$ such that $u(a) = x$, $u(b) = \xi$ (II, 7.1.9), and we set $v = p \circ u$, so that $v(a) = z$, $v(b) = h(\xi)$. Applying the condition of the statement to u , we see that one obtains a morphism $h'_1 : S_1 \rightarrow Y$ such that $h'_1(a) = y$ and $h'_1(b) = h(\xi)$; but this means $z = y$ by virtue of (II, 7.2.3), since Y is separated on X and v and h'_1 must coincide, since they are equal at b . C.Q.F.D. $\square$

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Corollary (20.4.7). — *The hypotheses on S , X , Y , U and h being those of (20.4.6), let E be a part of $X - U$ such that X is normal at every point of E , and for each $x \in E$, let $h'_x : \operatorname{Spec}(k(x)) \rightarrow Y$ be an S -morphism, such that the condition (P) of (20.4.6) is satisfied. Suppose moreover that the union F of the graphs of the h'_x (identified with parts of $X \times_S Y$) for $x \in E$, is contained in a union of a finite number of opens V_i of $X \times_S Y$ which are separated on X (a condition automatically satisfied if Y is separated on S , or if X is quasi-compact and Y of finite type on S). Then there exists an S -morphism $h' : U' \rightarrow Y$, where U' is an open of X containing $U \cup E$, such that, for every $x \in E$, the composed*

$$\operatorname{Spec}(k(x)) \longrightarrow U' \xrightarrow{h'} Y$$

is equal to h'_x .

Proof. Note first that, in (20.4.6), if Y is supposed separated, there is a largest open $U_0 \supset U$, equal to the domain of the S -rational application corresponding to h , in which h can be extended, and this extension is unique (I, 7.2.2); whence the conclusion in this case, by virtue of (20.4.6). In the general case, let E_i be the set of $x \in E$ such that $(x, h'_x(x)) \in V_i$. By virtue of (20.4.6), for every $x \in E$, there is an extension $h^{(x)}$ of h to $U \cup W^{(x)}$, where $W^{(x)}$ is an open neighbourhood of x in X such that the graph of $h^{(x)}|_{W^{(x)}}$ is contained in all the V_i such that $x \in E_i$. Since V_i is separated on X and reduced, for two points x', x'' of E_i , $h^{(x')}$ and $h^{(x'')}$ coincide in $W^{(x')} \cap W^{(x'')}$ since the preceding morphisms coincide in an open everywhere dense $U \cap W^{(x')} \cap W^{(x'')}$ of $U_i \cap U_j$, they are equal. The morphism h' , equal to h in U , to h_i in each U_i , answers the question. $\square$

Remark (20.4.8). — When $E = X - U$, it is clear that if, for every open affine T of X , there exists an S -morphism $h'_T : T \rightarrow Y$ extending $h|_{(U \cap T)}$ and such that for every $x \in T - (U \cap T)$, the h'_T are the restrictions of an S -morphism $h' : X \rightarrow Y$ extending h (everywhere defined) by the assertion of uniqueness in (20.4.6). To prove the existence of h' , one is therefore reduced to the case where X is affine, and it then suffices that the union F of the graphs of the h'_x , is quasi-compact in $X \times_S Y$ so that one can apply (20.4.7). The latter will hold in particular when the h'_x are of the form $\operatorname{Spec}(k(x)) \rightarrow Z \xrightarrow{h_0} Y$, where Z is a closed sub-prescheme of X having $X - U$ for underlying space, and h_0 an S -morphism.

Corollary (20.4.9). — *Let S be a prescheme locally Noetherian, $f : X \rightarrow S$ a morphism flat, $g : Y \rightarrow S$ a morphism locally of finite type. Let U be an open schematically dense in S , $h : f^{-1}(U) \rightarrow Y$ an S -morphism, $T = S - U$, Z the closed sub-prescheme reduced of X having $f^{-1}(T)$ as underlying space, $h_0 : Z \rightarrow Y$ an S -morphism. Suppose X normal at all points of Z . In order that there exist a (necessarily unique) S -morphism $h' : X \rightarrow Y$ extending h and h_0 , it is necessary and sufficient that the following condition be satisfied:*

For every spectrum S_1 of a discrete valuation ring, of closed point a and generic point b , and for every morphism $u : S_1 \rightarrow S$ such that $u(a) \in T$ and $u(b) \in U$, there exists an S_1 -morphism $h'_1 : X_{(S_1)} \rightarrow Y_{(S_1)}$ which extends $h_{(S_1)} : f_{(S_1)}^{-1}(b) \rightarrow Y_{(S_1)}$ and is such that the diagram

$$\begin{array}{ccc} \operatorname{Spec}(k(z_1)) & \longrightarrow & Z_{(S_1)} \\ & & \uparrow \\ X_{(S_1)} & \xrightarrow{h'_1} & Y_{(S_1)} \end{array}$$

is commutative for every $z_1 \in Z_{(S_1)}$.

The hypothesis that f is flat implies indeed that $f^{-1}(U)$ is dense in X (2.3.10), and it then suffices to apply (20.4.8).

Corollary (20.4.10). — Under the hypotheses of (20.4.6), suppose moreover Y separated and locally quasi-finite on S . Let U be an open schematically dense in X , $h : U \rightarrow Y$ an S -morphism, ω the S -rational application corresponding to it, $x \in X - U$ a normal point of X . In order that ω be defined at the point x , it is necessary and sufficient that the following condition be satisfied: there exists a spectrum S_1 of a discrete valuation ring, of closed point a and generic point b , and a morphism $u : S_1 \rightarrow X$ such that $u(a) = x$, $u(b) \in U$, and the S -rational application $\omega \circ u$ is everywhere defined.

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Indeed, by hypothesis all the fibres of the projection morphism $X \times_S Y \rightarrow X$ are sets of isolated points, and to apply (20.4.4) it suffices to know that the fibre $p^{-1}(x)$ is non-empty, whence the conclusion.

Proposition (20.4.11). — Let X be an S -prescheme locally Noetherian, Y an S -prescheme affine on S , $Z = X - U$ an open of X . Suppose that $\text{prof}_Z(\mathcal{O}_X) \geq 2$ (5.10.1); then every S -morphism $f : U \rightarrow Y$ extends uniquely to an S -morphism of X in Y .

Proof. We can restrict to the case where S and X are affines, and (by virtue of (I, 3.3.14)) to the case $S = X$; so one has $X = \text{Spec}(A)$, $Y = \text{Spec}(B)$, B being a finite type A -algebra. As B is quotient of a polynomial algebra $A[(T_\lambda)]_{\lambda \in L}$, Y is a closed sub-prescheme of $Y' = X[(T_\lambda)]_{\lambda \in L}$. On the other hand, the hypothesis on Z implies that U is schematically dense in X by virtue of (20.2.13, (iv)) and (5.10.2). If we prove that every X -morphism $u : U \rightarrow Y$ extends uniquely to an X -morphism $v' : X \rightarrow Y'$, it will result that v' will factorise as $X \xrightarrow{v} Y \rightarrow Y'$: indeed, the sub-prescheme $v'^{-1}(Y)$ is closed and majorizes U (20.3.8.8). In these conditions, v will be the unique S -morphism of X in Y extending u . But then there is a bijective correspondence between the X -morphisms of an open $U \subset X$ in Y' and the families $(s_\lambda)_{\lambda \in L}$ of sections of $\mathcal{O}_X$ over U (II, 1.7.9); the conclusion results therefore from (5.10.5). $\square$

Corollary (20.4.12). — Let X be an S -prescheme locally Noetherian reduced and satisfying the condition (S_2) (5.7.2) (for example a locally Noetherian and normal prescheme (5.8.6)), Y an S -prescheme affine on S , f an S -rational application of X in Y ; then every irreducible component of $X - \text{dom}(f)$ is of codimension 1 in X .

Proof. This amounts to the same as saying that if $Z^{(2)}$ is the set of $x \in X$ such that $\dim(\mathcal{O}_{X,x}) \geq 2$, then, for every closed part $Z \subset Z^{(2)}$ of X , every S -morphism of $X - Z$ in Y extends to an S -morphism of X in Y ; but this results from the hypothesis on X (5.7.2) and from (20.4.11). $\square$

20.5. Relative pseudo-morphisms

(20.5.1). Let X, Y be two S -preschemes. It results from the definitions (11.10.8) that the intersection of two opens U, U' of X , universally schematically dense with respect to S , possesses this same property. We therefore define an equivalence relation between S -morphisms $u : U \rightarrow Y$ by replacing in (20.2.1) “schematically dense” by “universally schematically dense with respect to S ”. An equivalence class for this relation is called a *pseudo-morphism of X in Y relatively to S* , and the set of these classes is denoted $\text{Ps. hom}_{X/S}(X, Y)$.

(20.5.2). As every open of X universally schematically dense with respect to S is in particular schematically dense, the elements of a pseudo-morphism of X in Y relatively to S are equivalent in the sense of (20.2.1), whence a canonical map

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$$(20.5.2.1) \quad \text{Ps. hom}_{X/S}(X, Y) \longrightarrow \text{Ps. hom}_S(X, Y).$$

Proposition (20.5.3). — Suppose Y separated on S . Then:

label=(i) The map (20.5.2.1) is injective and identifies $\text{Ps. hom}_{X/S}(X, Y)$ with the subset of $\text{Ps. hom}_S(X, Y)$ formed of the pseudo- S -morphisms ω such that $\text{dom}_S(\omega)$ is universally schematically dense relatively to S .

lbbel=(ii) The presheaf $U \mapsto \text{Ps. hom}_{U/S}(U, Y)$ on X is a sheaf.

Proof. Assertion (i) is immediate, for two S -morphisms $u : U \rightarrow Y$, $u' : U' \rightarrow Y$ are equivalent in the sense of (20.2.1) means that they are the restrictions of a single morphism $\text{dom}_S(\omega) \rightarrow Y$ (20.2.4), and if U and U' are universally schematically dense with respect to S , the same holds

a fortiori for $\text{dom}_S(\omega)$. To prove (ii), note that $U \mapsto \text{Ps. hom}_S(U, Y)$ is then a sheaf (20.2.6); on the other hand, given a covering open (U_α) of U , and a pseudo- S -morphism ω of U in Y , for $\text{dom}_S(\omega)$ to be universally schematically dense in U relatively to S , it results immediately from the definitions (cf. (20.2.1)) that it suffices that each of the sets $\text{dom}_S(\omega) \cap U_\alpha = \text{dom}_S(\omega|_{U_\alpha})$ be universally schematically dense in U_α relatively to S ; whence (ii). $\square$

When Y is separated, we denote by $\mathcal{P}\text{s. hom}_{X/S}(X, Y)$ the sub-sheaf

$$U \longmapsto \text{Ps. hom}_{U/S}(U, Y)$$

of $\mathcal{P}\text{s. hom}_S(X, Y)$.

When X is flat and locally of finite presentation over S and Y separated on S , the pseudo-morphisms of X in Y relatively to S are identified with the pseudo- S -morphisms ω such that, for every fibre X_s of the morphism $X \rightarrow S$, $\text{dom}_S(\omega) \cap X_s$ is schematically dense in X_s (11.10.10).

(20.5.4). A particularly important case is that where Y is separated on S and $Y = S[T] = S \otimes_{\mathbf{Z}} \text{Spec}(\mathbf{Z}[T])$ (T indeterminate). There is then a bijective correspondence between the pseudo- S -morphisms of X in Y and the pseudo-functions on X (20.2.8) by the definition of a product of $\mathbf{Z}$ -preschemes. The pseudo-morphisms of X in Y relatively to S are identified then, by virtue of (20.5.3), with the pseudo-functions φ on X such that $\text{dom}(\varphi)$ is universally schematically dense relatively to S . The sheaf $\mathcal{P}\text{s. hom}_{X/S}(X, Y)$ is a sub-sheaf of rings of $\mathcal{M}'_X$, that we denote $\mathcal{M}'_{X/S}$.

We then set $\text{Ps. hom}_{X/S}(X, Y) = M'(X/S)$ and we say that its elements are the pseudo-functions on X relative to S .

(20.5.5). Let X, Y, Z be three S -preschemes, $f : Y \rightarrow Z$ an S -morphism. We can, in the reasoning of (20.3.1), replace everywhere “schematically dense” by “universally schematically dense relatively to S ”; for every pseudo-morphism $\omega \in \text{Ps. hom}_{X/S}(X, Y)$, the morphisms $f\omega$, where ω runs through ω , belong therefore to a single equivalence class (for the relation defined in (20.5.1)), which we call the pseudo-morphism of X in Z , relatively to S , composed of f and ω and denoted $f\omega$. If $g : Z \rightarrow T$ is a morphism, it is immediate that $g \circ (f\omega) = (g \circ f)\omega$.

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(20.5.6). Suppose Y separated on S , and let ω be a pseudo-morphism of X in Y relatively to S . If $f : X' \rightarrow X$ is an S -morphism such that $f^{-1}(\text{dom}_S(\omega))$ is universally schematically dense in X' relatively to S , it results from (20.3.2) that the pseudo- S -morphism $\omega \circ f$ has a domain universally schematically dense relatively to S , hence (20.5.3) may be considered as a pseudo-morphism relatively to S . When X' is flat and of locally of finite presentation over S , the condition that $f^{-1}(\text{dom}_S(\omega))$ be universally schematically dense in X' relatively to S is equivalent to saying that for every $s \in S$, $(f^{-1}(\text{dom}_S(\omega)))_s$ (notation of (11.10.10)) is schematically dense in X'_s . This latter condition will in particular be satisfied if, for every $s \in S$, X'_s and f_s satisfy one of the three conditions (i), (ii), (iii) of (20.3.5).

(20.5.7). Suppose now that X and X' are both S -preschemes flat and locally of finite presentation over S , and that $f : X' \rightarrow X$ is an S -morphism flat (or, what amounts to the same (11.3.10), that for every $s \in S$, $f_s : X'_s \rightarrow X_s$ is a flat morphism). Then, for every open U of X and every open $V \subset U$ universally schematically dense in U relatively to S , it results from (11.10.5) and (11.10.9) that $f^{-1}(V)$ is universally schematically dense in $f^{-1}(U)$ relatively to S . For every pseudo- S -morphism ω of X in Y relatively to S , it results from (20.3.4) that the pseudo- S -morphism $\omega \circ f$ is defined and is a pseudo-morphism of X' in Y relatively to S , even when Y is not supposed separated on S . We deduce from this that for every S -morphism $g : Y \rightarrow Z$, $(g\omega) \circ f$ is still defined and equal to $g \circ (\omega \circ f)$ (with the definitions of (20.5.1)), and that it is therefore a pseudo-morphism relatively to S .

(20.5.8). Let X be an S -prescheme, $S' \rightarrow S$ a morphism, $g : X' \rightarrow X$ the canonical projection, $X' = X \times_S S'$. Then, for every open U of X and every open $V \subset U$ universally schematically dense in U relatively to S , $V' = g^{-1}(V)$ is universally schematically dense in $U' = g^{-1}(U)$ relatively to S' by definition (11.10.8). Let ω be a pseudo- S -morphism of X in an S -prescheme Y relatively to S ; if u_1, u_2 are S -morphisms of the class ω , defined respectively on opens U_1, U_2 of X , universally schematically dense in X relatively to S , it results from what precedes that $u'_1 = u_1 \circ (g|_{g^{-1}(U_1)})$ and $u'_2 = u_2 \circ (g|_{g^{-1}(U_2)})$ coincide in an open U'_3 universally schematically dense relatively to S' , hence the S' -morphisms v_1 and v_2 correspond to the same element of $\text{Ps. hom}_{X'/S'}(X', Y')$, which

we call the *image reciprocal* of ω by the base change $S' \rightarrow S$ and denote $\omega_{(S')}$. It is clear that if $S'' \rightarrow S'$ is a second morphism, $(\omega_{(S')})_{(S'')} = \omega_{(S'')} (for the composed morphism $S'' \rightarrow S' \rightarrow S$).$

20.6. Relative meromorphic functions

(20.6.1). Let S be a prescheme, X an S -prescheme which is *flat and locally of finite presentation* over S ; for every $s \in S$, we denote by X_s the fibre at the point s of the structural morphism $X \rightarrow S$. In general, if φ is a meromorphic function on X , it is not possible, for each $s \in S$, to associate to it in a “natural” way a meromorphic function “induced” on X_s (20.1.11). For every open U of X , denote by $T_{X/S}(U)$ the set of sections $t \in \Gamma(U, \mathcal{O}_X)$ such that, for every $s \in S$, the image t_s of t by the canonical homomorphism $\Gamma(U, \mathcal{O}_X) \rightarrow \Gamma(U \cap X_s, \mathcal{O}_{X_s})$ is a *regular* section; this also implies, by the equivalence of a) and b) in (11.3.7), that t is itself a *regular* section. It is clear that $U \mapsto T_{X/S}(U)$ is a *sub-sheaf* of the sheaf of sets $\mathcal{S} = \mathcal{S}(\mathcal{O}_X)$ (notation of (20.1.3)), that we denote $\mathcal{T}_{X/S}$; we set

$$(20.6.1.1) \quad \mathcal{M}_{X/S} = \mathcal{O}_X[\mathcal{T}_{X/S}^{-1}]$$

(notation of (20.1.1)) and we say that this sheaf of rings is the *sheaf of germs of meromorphic functions on X relative to S* ; its sections over X are called *meromorphic functions on X relative to S* and their set is denoted $M(X/S)$. It is clear that $\mathcal{M}_{X/S}$ is a *sub-sheaf* of $\mathcal{M}_X = \mathcal{O}_X[\mathcal{S}^{-1}]$; for every meromorphic function $\varphi \in M(X/S)$ and every $s \in S$, the inverse image of φ by the canonical injection morphism $j_s : X_s \rightarrow X$ is then defined (20.1.11), and denoted φ_s .

(20.6.2). Let now $\mathcal{F}$ be an $\mathcal{O}_X$ -Module quasi-coherent; we set

$$(20.6.2.1) \quad \mathcal{M}_{X/S}(\mathcal{F}) = \mathcal{F}[\mathcal{T}_{X/S}^{-1}] = \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{M}_{X/S};$$

the sections of $\mathcal{M}_{X/S}(\mathcal{F})$ are called *meromorphic sections of $\mathcal{F}$ over X , relative to S* and their set is denoted $M(X/S, \mathcal{F})$. The canonical homomorphism $\mathcal{F} \rightarrow \mathcal{M}_{X/S}(\mathcal{F})$ is not necessarily injective; when it is, we say that $\mathcal{F}$ is *without torsion relatively to S* : this means that for every open U of X and every section $t \in T_{X/S}(U)$, t is $(\mathcal{F}|_U)$ -regular; this condition is satisfied *a fortiori* when $\mathcal{F}$ is strictly torsion-free (20.1.5). In this last case, it results immediately from the definitions (20.1.2) that the canonical homomorphism of $\mathcal{O}_X$ -Modules

$$(20.6.2.2) \quad \mathcal{M}_{X/S}(\mathcal{F}) \longrightarrow \mathcal{M}_X(\mathcal{F})$$

is *injective*, so that the meromorphic sections of $\mathcal{F}$ relative to S are meromorphic sections of $\mathcal{F}$ in the sense of (20.1.3).

Proposition (20.6.3). — *The image by the injective homomorphism (20.2.10.1) of the sub-sheaf $\mathcal{M}_{X/S}$ of $\mathcal{M}_X$ is the sub-sheaf $\mathcal{M}'_{X/S}$ of pseudo-functions on X relative to S (i.e. of the pseudo-functions whose domain of definition is universally schematically dense relatively to S (20.5.4)).*

Proof. We can evidently restrict to proving that the image of $M(X/S)$ by the canonical homomorphism $M(X) \rightarrow M'(X)$ is equal to $M'(X/S)$; the proposition is then a consequence of the following more general proposition: $\square$

Proposition (20.6.4). — *Let $\mathcal{F}$ be an $\mathcal{O}_X$ -Module quasi-coherent of finite presentation and strictly torsion-free. Then, for a meromorphic section φ of $\mathcal{F}$ over X to be a meromorphic section relative to S , it is necessary and sufficient that $\text{dom}(\varphi)$ be universally schematically dense relatively to S .*

Proof. The necessity of the condition results from (20.2.15) applied to each canonical injection $X_s \rightarrow X$ ($s \in S$), taking into account (11.10.9). To see that the condition is sufficient, it suffices to prove that for every $x \in X$, there exists a neighbourhood U of x and a section of $\mathcal{M}_{X/S}(\mathcal{F})$ over U whose restriction to $U \cap \text{dom}(\varphi)$ coincides with φ in an open schematically dense of $U \cap \text{dom}(\varphi)$. Consider the Ideal of denominators $\mathcal{J}$ of φ (20.2.14), which is quasi-coherent, and which defines the sub-prescheme whose underlying space is $X - \text{dom}(\varphi)$. By hypothesis, if s is the image of x in S , $\text{dom}(\varphi) \cap X_s$ is schematically dense in the prescheme locally Noetherian X_s , hence (20.2.13, (iv)) contains $\text{Ass}(\mathcal{O}_{X_s})$; this implies that the Ideal $\mathcal{J}_x$ has an image in $\mathcal{O}_{X_s, x} = \mathcal{O}_{X, x} \otimes_{\mathcal{O}_{S, s}} k(s)$ which is not contained in any of the prime ideals $\mathfrak{p}_i \in \text{Ass}(\mathcal{O}_{X_s, x})$ (finitely many); hence (Bourbaki, Alg. comm., chap. II, §1, n° 1, prop. 2) there exists an element $t_x \in \mathcal{J}_x$ whose image in $\mathcal{O}_{X_s, x}$ does not belong to any of the $\mathfrak{p}_i$, and is therefore *regular* in this Noetherian ring. Let t be a section of $\mathcal{J}$ over an

affine neighbourhood U of x , whose germ at x is t_x ; since X is flat and locally of finite presentation over S , we can suppose (11.3.8) that t is a regular section of $\mathcal{O}_X$ over U and that, for every $s' \in S$, the image of t in $\Gamma(U \cap X_{s'}, \mathcal{O}_{X_{s'}})$ is also regular; in other words, $t \in T_{X/S}(U)$. But then, since $\mathcal{F}$ is strictly torsion-free, $t(\varphi|(U \cap \text{dom}(\varphi)))$ is a section u of $\mathcal{F}$ over $U \cap \text{dom}(\varphi)$; on the other hand, $U \cap \text{dom}(\varphi)$ contains the open U_t of $x' \in U$ where $t(x') \neq 0$, and this last is schematically dense in U (20.2.9). We see that in U_t , φ coincides with the restriction to U_t of the section u/t of $\mathcal{M}_{X/S}(\mathcal{F})$ over U . C.Q.F.D. $\square$

Remarks (20.6.5). — label=(i) Let φ be a meromorphic function on X relative to S , so that for every $s \in S$, φ_s is a meromorphic function on X_s (20.6.1); by virtue of (20.1.11.1), one has

$$(20.6.5.1) \quad \text{dom}(\varphi) \cap X_s \subset \text{dom}(\varphi_s).$$

But it is worth noting that even when S is the spectrum of a discrete valuation ring A and $X = S[T]$ (T indeterminate), the two sides of (20.6.5.1) are not necessarily equal: for example, if π is a uniformizer of A , it is immediate that $\varphi = \pi/T$ is a meromorphic function on X relative to S , since if a and b are the closed and generic points of S , T is regular in $\Gamma(X_a, \mathcal{O}_{X_a}) = k[T]$ and in $\Gamma(X_b, \mathcal{O}_{X_b}) = K[T]$, k being the residue field and K the field of fractions of A . One has $\text{dom}(\varphi) = D(T)$ in X , but $\text{dom}(\varphi_a) = X_a$ since $\varphi_a = 0$.

lbel=(ii) For a meromorphic function φ relative to S to be *invertible in the ring $M(X/S)$* , it is necessary and sufficient that for every $s \in S$, φ_s be invertible in $M(X_s)$ (in other words, that φ_s be a regular meromorphic function on X_s (20.1.8)). The condition is trivially necessary. Conversely, if it is satisfied, and if x is any point of X , s its image in S , there exists by hypothesis a neighbourhood U of x and two sections u, t of $\mathcal{O}_X$ over U such that $t \in T_{X/S}(U)$ and $\varphi|U = u/t$; the hypothesis implies that u_s is regular at x in $\Gamma(U \cap X_s, \mathcal{O}_{X_s})$, so u_s is regular at point x . Restricting U , we can therefore suppose, by virtue of (11.3.8), that $u \in T_{X/S}(U)$, whence the conclusion.

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When φ is invertible in $M(X/S)$, we also say that φ is a *regular meromorphic function relative to S* . We note that $\varphi \in M(X/S)$ can be invertible in $M(X)$ (in other words, be *regular* in the sense of (20.1.8)) without being invertible in $M(X/S)$, as the above example of (i) shows.

lbel=(iii) Let $\mathcal{L}$ be an $\mathcal{O}_X$ -Module *invertible*, and let s be a meromorphic *regular* section of $\mathcal{L}$ over X (20.1.8); we say that φ is *regular relatively to S* if, for every open U of X such that $\mathcal{L}|U$ is isomorphic to $\mathcal{O}_U$, $\varphi|U$ corresponds to an element of $\Gamma(U, \mathcal{M}_{X/S})$ which is *regular relatively to S* (20.1.8); by virtue of (ii), it is immediate that it suffices that for every $s \in S$, φ_s be a regular meromorphic section (20.1.8) of the $\mathcal{O}_{X_s}$ -Module invertible $\mathcal{L} \otimes_{\mathcal{O}_S} k(s)$. If φ' is the inverse of φ in $\mathcal{L}^{-1}$ (20.1.10), φ' is then also regular relatively to S , and if $\mathcal{L}_1$ is a second $\mathcal{O}_X$ -Module invertible, φ_1 a meromorphic section of $\mathcal{L}_1$ over X , regular relatively to S , then $\varphi \otimes \varphi_1$ is a meromorphic section of $\mathcal{L} \otimes \mathcal{L}_1$ over X , regular relatively to S .

Proposition (20.6.6). — *Let X be an S -prescheme flat and locally of finite presentation, S an $\mathcal{O}_X$ -Module locally free of finite type; for every $s \in S$, denote by X_s the fibre at the point s of the structural morphism $f : X \rightarrow S$. Let φ be a meromorphic section of $\mathcal{F}$ over X , relatively to S , and suppose that φ is defined at all points $x \in X$ such that $\text{prof}(\mathcal{O}_{X_{f(x)},x}) = 1$. Then φ is everywhere defined.*

Proof. By hypothesis, $\text{dom}(\varphi) \cap X_s$ is schematically dense in X_s for every $s \in S$, hence contains the points x of X_s such that $\text{prof}(\mathcal{O}_{X_s,x}) = 0$ (5.10.2); the hypothesis therefore signifies that if one sets $Z = X - \text{dom}(\varphi)$, one has $\text{prof}(\mathcal{O}_{X_{f(x)},x}) \geq 2$ for all points x of Z . It suffices then to apply (19.9.8). $\square$

(20.6.7). Let X, X' be two S -preschemes flat and locally of finite presentation, $f = (\psi, \theta) : X' \rightarrow X$ an S -morphism. For every open U of X , denote by $T_f(U)$ the set of sections $t \in T_{X/S}(U)$ whose image in $\Gamma(f^{-1}(U), \mathcal{O}_{X'})$ belongs to $T_{X'/S}(f^{-1}(U))$; it is immediate that $U \mapsto T_f(U)$ is a sub-sheaf of the sheaf of sets $\mathcal{T}_{X/S}$, that we denote $\mathcal{T}_f$. We set $\mathcal{M}_{X/S,f} = \mathcal{O}_X[\mathcal{T}_f^{-1}]$; this is a sub-sheaf of rings of $\mathcal{M}_{X/S}$, and we canonically obtain from $\theta^\# : \psi^*(\mathcal{O}_X) \rightarrow \mathcal{O}_{X'}$ a homomorphism of sheaves of rings $\theta'^\# : \psi^*(\mathcal{M}_{X/S,f}) \rightarrow \mathcal{M}_{X'/S}$ extending $\theta^\#$. A meromorphic function φ on X , relative to S , that is

a section of $\mathcal{M}_{X/S,f}$, then has an inverse image $\Gamma(\theta^\#)(\varphi)$ which is a meromorphic function on X' relative to S' , called the *inverse image of φ by f* , and denoted $\varphi \circ f$ if this causes no confusion. We extend immediately in the same way the definitions of (20.1.11) relative to $\mathcal{O}_X$ -Modules.

Proposition (20.6.8). — *With the notations of (20.6.7), if the S -morphism $f : X' \rightarrow X$ is flat, one has $\mathcal{M}_{X/S,f} = \mathcal{M}_{X/S}$, and the homomorphism $\varphi \mapsto \varphi \circ f$ is defined on all of $M(X/S)$.*

Proof. Indeed, the hypothesis implies, by virtue of (11.3.10), that for every $s \in S$, $f_s : X'_s \rightarrow X_s$ is flat; hence, for every section $t \in T_{X/S}(U)$, if t' is its inverse image, t'_s is a regular section of $\mathcal{O}_{X'_s}$ over $f^{-1}(U) \cap X'_s$; by virtue of (20.1.12), we conclude that by definition $t' \in T_{X'/S}(f^{-1}(U))$, whence the proposition. $\square$

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(20.6.9). Consider finally a morphism $S' \rightarrow S$, and set $X' = X \times_S S'$, which is flat and locally of finite presentation over S' ; let $p : X' \rightarrow X$ be the canonical projection. Let U be an open of X , t a section belonging to $T_{X/S}(U)$, t' its image in $\Gamma(p^{-1}(U), \mathcal{O}_{X'})$; for every $s' \in S'$, if s is the image of s' , we have $X'_{s'} = X_s \otimes_{k(s)} k(s')$, hence the morphism $X'_{s'} \rightarrow X_s$ is flat, and by virtue of (20.1.12) the inverse image $t'_{s'}$ of t_s is regular; this proves that $t' \in T_{X'/S'}(p^{-1}(U))$. We deduce from this, as in (20.1.12), a canonical homomorphism of $\mathcal{O}_{X'}$ -Algebras

$$(20.6.9.1) \quad p^*(\mathcal{M}_{X/S}) \longrightarrow \mathcal{M}_{X'/S'}.$$

Using the identification permitted by (20.6.3), this notion of base change for relative meromorphic functions is a particular case of the analogous notion for relative pseudo-morphisms (20.5.8).

§21. DIVISORS

For the content of the present paragraph, see the comments in the introduction of §20. For the *global* properties of divisors, the reader will refer to the paragraph dedicated to them in chap. V.

21.1. Divisors on a ringed space

(21.1.1). Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{M}_X$ the sheaf of germs of meromorphic functions on X (20.1.3), $\mathcal{M}_X^*$ the sheaf (of multiplicative groups) of germs of regular meromorphic functions on X (20.1.8). It is clear that the sheaf (of multiplicative groups) $\mathcal{O}_X^*$ of germs of *invertible* sections of $\mathcal{O}_X$ identifies canonically with a sub-sheaf (of multiplicative groups) of $\mathcal{M}_X^*$.

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Definition (21.1.2). — We call *sheaf of divisors* on X and denote $\mathcal{D}iv_X$ the quotient sheaf (of commutative groups) $\mathcal{M}_X^* / \mathcal{O}_X^*$; the sections of this sheaf over X are called the *divisors* on X ; they form a commutative group denoted $\text{Div}(X)$. For every section f of $\mathcal{M}_X^*$ over X (in other words, every *regular meromorphic function* on X (20.1.8)) we call the *divisor of f* and denote $\text{div}(f)$ (or $\text{div}_X(f)$) the divisor on X which is the image of f by the canonical homomorphism $\Gamma(X, \mathcal{M}_X^*) \rightarrow \Gamma(X, \mathcal{D}iv_X)$.

The *support* of a divisor D is the closed set of $x \in X$ such that $D_x \neq 0$. We denote this by $\text{Supp}(D)$.

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For every open U of X , one has evidently $\mathcal{M}_X^*|_U = \mathcal{M}_U^*$, $\mathcal{O}_X^*|_U = \mathcal{O}_U^*$, hence $\mathcal{D}iv_X|_U = \mathcal{D}iv_U$, and consequently the sheaf $\mathcal{D}iv_X$ is equal to the presheaf $U \mapsto \text{Div}(U)$. When $X = \text{Spec}(A)$ is affine, we write $\text{Div}(A)$ instead of $\text{Div}(\text{Spec}(A))$.

(21.1.3). We will always write *additively* the group $\text{Div}(X)$ of divisors on X . For two regular meromorphic functions f, g on X , one thus has

$$(21.1.3.1) \quad \text{div}(fg) = \text{div}(f) + \text{div}(g),$$

$$(21.1.3.2) \quad \text{div}(f^{-1}) = -\text{div}(f).$$

By definition, for every regular meromorphic function f on X , one has the equivalence

$$(21.1.3.3) \quad \text{div}(f) = 0 \iff f \in \Gamma(X, \mathcal{O}_X^*),$$

whence, for two regular meromorphic functions f, g on X ,

$$(21.1.3.4) \quad \text{div}(f) = \text{div}(g) \iff fg^{-1} \in \Gamma(X, \mathcal{O}_X^*).$$

(21.1.4). Let now $\mathcal{L}$ be an $\mathcal{O}_X$ -Module *invertible*, and let s be a *regular* meromorphic section of $\mathcal{L}$ over X (20.1.8). Every $x \in X$ possesses a neighbourhood U such that $\mathcal{L}|_U$ is isomorphic to $\mathcal{O}_U$, hence $\mathcal{M}_X(\mathcal{L})|_U$ isomorphic to $\mathcal{M}_X|_U$; by one of these isomorphisms, $s|_U$ corresponds to a section $f \in \Gamma(U, \mathcal{M}_X^*)$, and since two such isomorphisms differ only by multiplication by an element of $\Gamma(U, \mathcal{O}_X^*)$ (0_I, 5.4.3), the element $\text{div}_U(f)$ of $\Gamma(U, \mathcal{D}iv_X)$ is *independent* of the isomorphism chosen; it is clear that these elements (for U variable) are the restrictions of a section of $\mathcal{D}iv_X$ over X , that we call the *divisor* of s and that we denote $\text{div}(s)$ (such a divisor is not necessarily of the form $\text{div}(g)$ for a regular meromorphic function g on X ; see (21.2.9)). For $\mathcal{L} = \mathcal{O}_X$, the definition of $\text{div}(s)$ coincides with that of (21.1.2). If $\mathcal{L}, \mathcal{L}'$ are two $\mathcal{O}_X$ -Modules invertibles, s (resp. s') a regular meromorphic section of $\mathcal{L}$ (resp. $\mathcal{L}'$) over X , it is immediate that one has

$$(21.1.4.1) \quad \text{div}(s \otimes s') = \text{div}(s) + \text{div}(s')$$

$$(21.1.4.2) \quad \text{div}(s^{\otimes n}) = n \cdot \text{div}(s) \quad \text{for all } n \in \mathbb{Z}$$

(s^{-1} being the regular meromorphic section of $\mathcal{L}^{-1}$ over X defined in (20.1.10)) and, for two regular meromorphic sections s, s' of $\mathcal{L}$ over X , one has the relation

$$(21.1.4.3) \quad \text{div}(s) = \text{div}(s') \iff s' = ts \quad \text{with } t \in \Gamma(X, \mathcal{O}_X^*).$$

(21.1.5). The sheaf $\mathcal{S}(\mathcal{O}_X)$ (20.1.3) whose sections over an open U of X are the regular elements of $\Gamma(U, \mathcal{O}_X)$ is a *sub-sheaf of monoids* of $\mathcal{M}_X^*$; one can write

$$(21.1.5.1) \quad \mathcal{S}(\mathcal{O}_X) = \mathcal{O}_X \cap \mathcal{M}_X^*.$$

If we denote $\mathcal{S}(\mathcal{O}_X)^{-1}$ the sheaf whose sections over U are the *inverses* in $\Gamma(U, \mathcal{M}_X^*)$ of the elements of $\Gamma(U, \mathcal{S}(\mathcal{O}_X))$, it is clear that one has $\Gamma(U, \mathcal{S}(\mathcal{O}_X)) \cap \Gamma(U, \mathcal{S}(\mathcal{O}_X)^{-1}) = \Gamma(U, \mathcal{O}_X^*)$, hence

$$(21.1.5.2) \quad \mathcal{S}(\mathcal{O}_X) \cap \mathcal{S}(\mathcal{O}_X)^{-1} = \mathcal{O}_X^*.$$

Definition (21.1.6). — The sub-sheaf of sets of $\mathcal{D}iv_X$, canonical image of the sub-sheaf $\mathcal{S}(\mathcal{O}_X)$ of $\mathcal{M}_X^*$, is denoted $\mathcal{D}iv_X^+$; its sections over X are called *positive divisors* on X and their set is denoted $\text{Div}^+(X)$.

Since $\mathcal{S}(\mathcal{O}_X)$ is a sheaf of monoids (multiplicative), one has

$$(21.1.6.1) \quad \text{Div}^+(X) + \text{Div}^+(X) \subset \text{Div}^+(X)$$

and on the other hand, by virtue of (21.1.5.2) and (21.1.3.3),

$$(21.1.6.2) \quad \text{Div}^+(X) \cap (-\text{Div}^+(X)) = \{0\}.$$

These two relations show that $\text{Div}^+(X)$ is the set of positive elements for a structure of *order* on the group $\text{Div}(X)$, compatible with this group structure; we denote this relation of order $D \leq D'$, in other words, one has

$$(21.1.6.3) \quad D \geq 0 \iff D \in \text{Div}^+(X).$$

We will always assume in what follows that $\text{Div}(X)$ is equipped with this order structure. The fibre $(\mathcal{D}iv_X^+)_x$ at a point x of the sheaf $\mathcal{D}iv_X^+$ is a sub-monoid of the group $(\mathcal{D}iv_X)_x$, the set of elements ≥ 0 for an order structure compatible with the group structure; for a divisor D on X , it amounts to the same to say that $D \geq 0$ or that $D_x \geq 0$ for every $x \in X$.

By definition, for every regular meromorphic function f on X , one has the relation

$$(21.1.6.4) \quad \text{div}(f) \geq 0 \iff f \in \Gamma(X, \mathcal{S}(\mathcal{O}_X))$$

in other words, $\text{div}(f) \geq 0$ means that f is a *regular* section of $\mathcal{O}_X$, or equivalently a section of $\mathcal{O}_X$ invertible in $M(X)$.

More generally, given a divisor D on X , the relation $\text{div}(f) \geq D$ is equivalent to the following: for every open U of X such that $D|_U = \text{div}(g)$, where $g \in \Gamma(U, \mathcal{M}_X^*)$, there exists a regular element t of $\Gamma(U, \mathcal{O}_X)$ such that $f|_U = tg$.

(21.1.7). Let $\mathcal{L}$ be an $\mathcal{O}_X$ -Module invertible, s a regular meromorphic section of $\mathcal{L}$ over X ; one has the relation

$$(21.1.7.1) \quad \operatorname{div}(s) \geq 0 \iff s \in \Gamma(X, \mathcal{L}) \cap \Gamma(X, (\mathcal{M}_X(\mathcal{L}))^*)$$

as it results immediately from the definitions (21.1.4) and (21.1.6).

Proposition (21.1.8). — *Let X be a locally Noetherian prescheme, D a divisor on X . Suppose that for every $x \in X$ such that $\operatorname{prof}(\mathcal{O}_{X,x}) = 1$ one has $D_x \geq 0$ (resp. $D_x = 0$). Then one has $D \geq 0$ (resp. $D = 0$).*

Proof. The question being local on X , we can suppose that $D = \operatorname{div}(f)$, f being a regular meromorphic function on X ; the relation $D_x \geq 0$ is equivalent to $x \in \operatorname{dom}(f)$, hence the hypothesis signifies that if $T = X - \operatorname{dom}(f)$, one has $\operatorname{prof}(\mathcal{O}_{X,x}) \geq 2$ for every $x \in T$ (since $\operatorname{dom}(f)$ contains the maximal points of X). By (5.10.5) the restriction homomorphism $\Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(X - T, \mathcal{O}_X)$ is bijective, which shows that there exists a section s of $\mathcal{O}_X$ over X such that $f = s|(X - T)$. But by definition of T , this implies $T = \emptyset$, hence $f = s$ and $D \geq 0$. The assertion relative to $D_x = 0$ follows immediately by applying the preceding to $-D$. $\square$

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Corollary (21.1.9). — *Let X be a locally Noetherian prescheme, D a divisor on X . Let S be the support of D . Then, for every maximal point x of S , $\operatorname{prof}(\mathcal{O}_{X,x}) = 1$.*

Proof. Set $X_1 = \operatorname{Spec}(\mathcal{O}_{X,x})$; taking into account (20.3.6), the sheaf $\mathcal{M}_{X_1}^*$ is induced on X_1 by $\mathcal{M}_X^*$, and we can restrict to the case where $X = X_1$, to which case, as $x \in S$ and x is a maximal point of S , S is necessarily $S = \{x\}$. If one had then $\operatorname{prof}(\mathcal{O}_{X,x}) \neq 1$, we would conclude, by virtue of (21.1.8), that $D = 0$, which contradicts the definition of S . $\square$

Proposition (21.1.10). — *Let A be a local Noetherian ring; for $\operatorname{Div}(A) = 0$, it is necessary and sufficient that $\operatorname{prof}(A) = 0$ (in other words, that the maximal ideal $\mathfrak{m}$ of A be associated to A (0, 16.4.6)).*

Proof. Indeed, to say that $\operatorname{Div}(A) = 0$ means that every regular element is invertible, or equivalently that all elements of $\mathfrak{m}$ are zero divisors, which means that $\mathfrak{m} \in \operatorname{Ass}(A)$ (Bourbaki, Alg. comm., chap. IV, §1, n° 1, cor. 3 of prop. 2). $\square$

21.2. Divisors and invertible fractional ideals

(21.2.1). Let $(X, \mathcal{O}_X)$ be a ringed space. We call *fractional Ideal* on X a sub- $\mathcal{O}_X$ -Module of the $\mathcal{O}_X$ -Module $\mathcal{M}_X$ of germs of meromorphic functions on X . A fractional Ideal $\mathcal{J}$ on X which is an $\mathcal{O}_X$ -Module invertible is called a *fractional Ideal invertible*.

Proposition (21.2.2). — *For a fractional Ideal $\mathcal{J}$ on X to be invertible, it is necessary and sufficient that for every $x \in X$, there exists a neighbourhood U of x and a section $f \in \Gamma(U, \mathcal{M}_X^*)$ such that $\mathcal{J}|U = \mathcal{O}_U f$.*

Proof. The condition is evidently sufficient, the map $s \mapsto s(f|V)$ from $\Gamma(V, \mathcal{O}_X)$ being evidently bijective for every open V contained in U . To see that it is necessary, note that there exists by hypothesis a neighbourhood U of x and an isomorphism of $\mathcal{O}_X$ -Modules $\mathcal{O}_U \xrightarrow{\sim} \mathcal{J}|U$. If f is the image of the section $1 \in \Gamma(U, \mathcal{O}_X)$ by this isomorphism, we can suppose, restricting U , that $f = u/s$, where $u \in \Gamma(U, \mathcal{O}_X)$ and $s \in \Gamma(U, \mathcal{S}(\mathcal{O}_X))$, and the isomorphism considered corresponds, to every section $v \in \Gamma(V, \mathcal{O}_X)$ (where V is an open contained in U) the homomorphism $t(f|V) \mapsto st$ from $\Gamma(V, \mathcal{J})$ to $\Gamma(V, \mathcal{O}_X)$; to say that this map is bijective means that $u|V$ is a regular element of $\Gamma(V, \mathcal{O}_X)$, hence $f \in \Gamma(U, \mathcal{M}_X^*)$.

We note that for every open U of X such that $\mathcal{J}|U = \mathcal{O}_U f$ with $f \in \Gamma(U, \mathcal{M}_X^*)$, the section f is determined up to multiplication by an element of $\Gamma(U, \mathcal{O}_X^*)$, since the multiplication by these elements provides all automorphisms of the $\mathcal{O}_U$ -Module $\mathcal{O}_U$ (0_I, 5.4.3). $\square$

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Corollary (21.2.3). — *label=(i) Let $\mathcal{J}$ be an invertible fractional Ideal; then the $\mathcal{O}_X$ -Module invertible $\mathcal{J}^{-1}$ identifies canonically with the fractional Ideal $\mathcal{J}'$ (transporter of $\mathcal{J}$ in $\mathcal{O}_X$) defined as follows: for every open U of X such that $\mathcal{J}|U = \mathcal{O}_U f$, where $f \in \Gamma(U, \mathcal{M}_X^*)$, one has $\mathcal{J}'|U = \mathcal{O}_U f^{-1}$.*

label=(ii) If $\mathcal{J}_1$ and $\mathcal{J}_2$ are two invertible fractional Ideals, the canonical map $\mathcal{J}_1 \otimes \mathcal{J}_2 \rightarrow \mathcal{J}_1 \mathcal{J}_2$ is an isomorphism of $\mathcal{O}_X$ -Modules.

Proof. Assertion (ii) results immediately from (21.2.2). On the other hand, the remark made at the end of (21.2.2) proves that there exists a well-defined fractional Ideal $\mathcal{J}'$ satisfying the condition of the statement; the isomorphism canonique of $\mathcal{J}'$ on $\mathcal{J}^{-1} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{J}, \mathcal{O}_X)$ is obtained by making correspond, to every section $s(f^{-1}|V)$ of $\mathcal{J}^{-1}|V$ (where V is an open contained in U and $s \in \Gamma(V, \mathcal{O}_X)$) the homomorphism $t(f|V) \mapsto st$ from $\Gamma(V, \mathcal{J})$ to $\Gamma(V, \mathcal{O}_X)$.

By virtue of (21.2.3, (i)), we will henceforth identify generally the $\mathcal{O}_X$ -Modules $\mathcal{J}$ and $\mathcal{J}^{-1}$, considering $\mathcal{J}^{-1}$ as a sub- $\mathcal{O}_X$ -Module of $\mathcal{M}_X$. $\square$

(21.2.4). It results from (21.2.3) that the set $\text{Id. inv}(X)$ of invertible fractional Ideals on X is equipped with a structure of *commutative group* for the law of composition $(\mathcal{J}_1, \mathcal{J}_2) \mapsto \mathcal{J}_1 \mathcal{J}_2$, the neutral element of this group being $\mathcal{O}_X$. It is clear that for every open U of X , one has $(\mathcal{J}_1 \mathcal{J}_2)|U = (\mathcal{J}_1|U)(\mathcal{J}_2|U)$, hence the map of restriction $\mathcal{J} \mapsto \mathcal{J}|U$ is a homomorphism of groups $\text{Id. inv}(X) \rightarrow \text{Id. inv}(U)$; we thus define a *presheaf of commutative groups* $U \mapsto \text{Id. inv}(U)$; it is immediate that in fact this presheaf is a *sheaf of commutative groups*, that we denote $\mathcal{J}d.\text{inv}_X$.

(21.2.5). For every regular meromorphic function $f \in \Gamma(X, \mathcal{M}_X^*)$, it results from (21.2.2) that $\mathcal{J}(f) = \mathcal{O}_X f$ is an invertible fractional Ideal, and one has evidently $\mathcal{J}(f_1 f_2) = \mathcal{J}(f_1) \mathcal{J}(f_2)$, so the map $f \mapsto \mathcal{J}(f)$ is a homomorphism of the commutative group $\Gamma(X, \mathcal{M}_X^*)$ in the commutative group $\text{Id. inv}(X)$. Replacing X by an arbitrary open U of X and noting that the homomorphisms obtained are compatible with the restriction operations, one obtains a canonical homomorphism of sheaves of commutative groups:

$$(21.2.5.1) \quad I_0 : \mathcal{M}_X^* \longrightarrow \mathcal{J}d.\text{inv}_X.$$

Note that if $f \in \Gamma(X, \mathcal{O}_X^*)$, one has $\mathcal{J}(f) = \mathcal{O}_X$; we deduce immediately that the homomorphism I_0 factorises as

$$(21.2.5.2) \quad \mathcal{M}_X^* \longrightarrow \mathcal{M}_X^* / \mathcal{O}_X^* = \mathcal{D}iv_X \xrightarrow{I} \mathcal{J}d.\text{inv}_X$$

where I is a homomorphism of the (additive) sheaf of groups $\mathcal{D}iv_X$ in the (multiplicative) sheaf $\mathcal{J}d.\text{inv}_X$; one has therefore for every open U a homomorphism $\mathcal{J}_U : \text{Div}(U) \rightarrow \text{Id. inv}(U)$ of commutative groups, such that, for every section $f \in \Gamma(U, \mathcal{M}_X^*)$, one has

$$(21.2.5.3) \quad \mathcal{J}_U(\text{div}_U(f)) = \mathcal{O}_U f.$$

We conclude that $\mathcal{J}_X(D)$, for every divisor $D \in \text{Div}(X)$, is the invertible fractional Ideal defined in the following way: for every open U of X such that $D|U = \text{div}_U(f)$, where $f \in \Gamma(U, \mathcal{M}_X^*)$, $\mathcal{J}_X(D)|U$ is the invertible fractional Ideal $\mathcal{O}_U \cdot f$. We thus have, by virtue of (21.1.6), for every regular meromorphic function $f \in \Gamma(X, \mathcal{M}_X^*)$, the relation

$$(21.2.5.4) \quad f \in \Gamma(X, \mathcal{J}_X(D)) \iff \text{div}(f) \geq D.$$

Proposition (21.2.6). — *The homomorphism $I : \mathcal{D}iv_X \rightarrow \mathcal{J}d.\text{inv}_X$ is bijective.*

Proof. We define indeed a homomorphism I'_X of $\text{Id. inv}(X)$ in $\text{Div}(X)$ by making correspond to every invertible fractional Ideal $\mathcal{J}$ on X the divisor $I'_X(\mathcal{J})$ defined in the following way: for every open U of X such that $\mathcal{J}|U = \mathcal{O}_U f$, where $f \in \Gamma(U, \mathcal{M}_X^*)$ (21.2.2), we take $I'_X(\mathcal{J})|U = \text{div}_U(f)$; by virtue of the remark following (21.2.2), this definition is independent of the generator f chosen in $\mathcal{J}|U$, and determines well a divisor on X . Furthermore, this definition shows immediately that the homomorphisms $\mathcal{J}_X$ and I'_X are reciprocal of one another. Replacing X by an arbitrary open U , we deduce the definition of the isomorphism $I' : \mathcal{J}d.\text{inv}_X \rightarrow \mathcal{D}iv_X$, reciprocal of I , whence the proposition. We set $I'_X(\mathcal{J}) = \text{div}(\mathcal{J})$, so that one has, for every regular meromorphic function f on X

$$(21.2.6.1) \quad \text{div}(\mathcal{O}_X \cdot f) = \text{div}(f).$$

$\square$

(21.2.7). We will often identify the sheaves $\mathcal{D}iv_X$ and $\mathcal{J}d.\text{inv}_X$ (resp. the groups $\text{Div}(X)$ and $\text{Id. inv}(X)$) by means of the isomorphisms I and I' (resp. $\mathcal{J}_X$ and I'_X) above. We note that one has the relation

$$(21.2.7.1) \quad D \geq 0 \iff \mathcal{J}_X(D) \subset \mathcal{O}_X \quad \text{for } D \in \text{Div}(X)$$

as it results immediately from the definitions (21.1.6) and from (21.1.5.1); in other terms, the image $\mathcal{I}_X(\text{Div}^+(X))$ is the set of Ideals of $\mathcal{O}_X$ (also sometimes called *integral* Ideals) which are $\mathcal{O}_X$ -Modules invertibles: such an Ideal $\mathcal{I}$ is further characterised by the fact that for every $x \in X$, there exists an open neighbourhood U of x in X such that $\mathcal{I}|_U = \mathcal{O}_U f$, where f is a regular element of $\Gamma(U, \mathcal{O}_X)$. The set $\mathcal{I}_X(\text{Div}^+(X))$ of these Ideals is thus a sub-monoid of $\text{Id. inv}(X)$, equal to the set of positive elements for an order relation compatible with the group structure of $\text{Id. inv}(X)$, and it is immediate that this relation is none other than the relation *opposite* to inclusion; in other words, one has

$$(21.2.7.2) \quad \mathcal{I}_1 \subset \mathcal{I}_2 \iff \text{div}(\mathcal{I}_1) \geq \text{div}(\mathcal{I}_2)$$

for $\mathcal{I}_1, \mathcal{I}_2$ in $\text{Id. inv}(X)$.

(21.2.8). For every divisor D on X , we set

$$(21.2.8.1) \quad \mathcal{O}_X(D) = (\mathcal{I}_X(D))^{-1};$$

$\mathcal{O}_X(D)$ is thus an invertible fractional Ideal, defined in the following way: for every open U of X such that $D|_U = \text{div}_U(f)$, where $f \in \Gamma(U, \mathcal{M}_X^*)$, $\mathcal{O}_X(D)|_U$ is the invertible fractional Ideal $\mathcal{O}_U \cdot f^{-1}$; by virtue of (21.1.6), for every regular meromorphic function f , one has the relation

$$(21.2.8.2) \quad f \in \Gamma(X, \mathcal{O}_X(D)) \iff \text{div}(f) \geq -D.$$

Furthermore, it is clear that one has canonical isomorphisms (21.2.3)

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$$(21.2.8.3) \quad \begin{cases} \mathcal{O}_X(0) = \mathcal{O}_X, & \mathcal{O}_X(D + D') = \mathcal{O}_X(D) \mathcal{O}_X(D') \simeq \mathcal{O}_X(D) \otimes \mathcal{O}_X(D') \\ \mathcal{O}_X(nD) = (\mathcal{O}_X(D))^n \simeq (\mathcal{O}_X(D))^{\otimes n} \end{cases}$$

for every integer $n \in \mathbb{Z}$, and for any two divisors D, D' on X .

(21.2.9). Let $\mathcal{I}$ be an invertible fractional Ideal on X . The canonical injection $\mathcal{I} \rightarrow \mathcal{M}_X$ defines by tensorisation a homomorphism of $\mathcal{O}_X$ -Modules

$$(21.2.9.1) \quad \mathcal{M}_X(\mathcal{I}) = \mathcal{I} \otimes_{\mathcal{O}_X} \mathcal{M}_X \longrightarrow \mathcal{M}_X \otimes_{\mathcal{O}_X} \mathcal{M}_X = \mathcal{M}_X$$

which is an *isomorphism*: indeed, if U is an open of X such that $\mathcal{I}|_U = \mathcal{O}_U \cdot f$, where $f \in \Gamma(U, \mathcal{M}_X^*)$, the homomorphism (21.2.9.1) restricted to U is none other than the isomorphism which to every section t of $\mathcal{M}_X(\mathcal{I})|_U = \mathcal{M}_X|_U$ over $V \subset U$, makes correspond the section $t(f|_V)^{-1}$ of the same sheaf. In the isomorphism (21.2.9.1), to the regular meromorphic sections of $\mathcal{I}$ over X correspond the regular meromorphic functions on X .

Consider in particular the case where $\mathcal{I} = \mathcal{O}_X(D)$, where D is a divisor on X ; one then has a canonical isomorphism

$$(21.2.9.2) \quad \mathcal{M}_X(\mathcal{O}_X(D)) \simeq \mathcal{M}_X$$

and one denotes by s_D the regular meromorphic section of $\mathcal{O}_X(D)$ over X which corresponds by this isomorphism to the section 1 of $\mathcal{M}_X$. If U is an open of X such that $\mathcal{O}_X(D)|_U = \mathcal{O}_U \cdot f^{-1}$, where $f \in \Gamma(U, \mathcal{M}_X^*)$, one has $s_D|_U = 1$ in $\Gamma(U, \mathcal{M}_X)$; since then $D|_U = \text{div}_U(f)$, we deduce (21.1.4) that one has

$$(21.2.9.3) \quad \text{div}(s_D) = D.$$

On the other hand, one deduces immediately from the canonical isomorphisms (21.2.8.3) the formulas

$$(21.2.9.4) \quad s_0 = 1, \quad s_{D+D'} = s_D \otimes s_{D'}, \quad s_{nD} = s_D^{\otimes n} \quad (n \in \mathbb{Z}).$$

(21.2.10). Consider, between two couples $(\mathcal{L}, s), (\mathcal{L}', s')$, where $\mathcal{L}$ and $\mathcal{L}'$ are two $\mathcal{O}_X$ -Modules invertibles, s (resp. s') a regular meromorphic section of $\mathcal{L}$ (resp. $\mathcal{L}'$) over X , the relation: “there exists an isomorphism $u : \mathcal{L} \xrightarrow{\sim} \mathcal{L}'$ such that $\bar{u}(s) = s'$ ”, where $\bar{u} : \Gamma(X, \mathcal{M}_X(\mathcal{L})) \xrightarrow{\sim} \Gamma(X, \mathcal{M}_X(\mathcal{L}'))$ is the isomorphism deduced from u (we note that the isomorphism u satisfying this condition is then determined in a unique way). It is clear that this is an equivalence relation, and since there exists a set of $\mathcal{O}_X$ -Modules invertibles such that every $\mathcal{O}_X$ -Module invertible is isomorphic to an element of this set (0_I, 5.4.7), one can speak of the set $D(X)$ of *equivalence classes* of couples $(\mathcal{L}, s)$ for the preceding relation. For every $\mathcal{O}_X$ -Module invertible $\mathcal{L}$ and every regular meromorphic section s of $\mathcal{L}$ over X , one denotes $\text{cl}(\mathcal{L}, s)$ the element of $D(X)$ corresponding to the couple $(\mathcal{L}, s)$. It

results from (0_I, 5.4.3) that if s, s' are two regular meromorphic sections of $\mathcal{L}$ over X , the relation $\text{cl}(\mathcal{L}, s) = \text{cl}(\mathcal{L}, s')$ is equivalent to the existence of a section $t \in \Gamma(X, \mathcal{O}_X^*)$ such that $s' = ts$.

It is immediate that if $(\mathcal{L}, s)$ is equivalent to $(\mathcal{L}_1, s_1)$ and $(\mathcal{L}', s')$ to $(\mathcal{L}'_1, s'_1)$, the couples $(\mathcal{L} \otimes \mathcal{L}', s \otimes s')$ and $(\mathcal{L}_1 \otimes \mathcal{L}'_1, s_1 \otimes s'_1)$ are equivalent; one defines thus in $\text{D}(X)$ a law of composition by setting

$$(21.2.10.1) \quad \text{cl}(\mathcal{L}, s) \text{cl}(\mathcal{L}', s') = \text{cl}(\mathcal{L} \otimes \mathcal{L}', s \otimes s');$$

it is immediate that this is a law of *commutative group*, whose neutral element is $\text{cl}(\mathcal{O}_X, 1)$ and where the inverse of $\text{cl}(\mathcal{L}, s)$ is $\text{cl}(\mathcal{L}^{-1}, s^{\otimes(-1)})$.

Proposition (21.2.11). — *The applications*

$$(21.2.11.1) \quad D \longmapsto \text{cl}(\mathcal{O}_X(D), s_D), \quad \text{cl}(\mathcal{L}, s) \longmapsto \text{div}(s)$$

are reciprocal isomorphisms of $\text{Div}(X)$ on $\text{D}(X)$ and of $\text{D}(X)$ on $\text{Div}(X)$ respectively.

Proof. Taking into account (21.2.8.3), (21.2.9.4) and (21.2.10.1), it suffices to see that the composites of these two applications are the identities in $\text{Div}(X)$ and $\text{D}(X)$ respectively. The first assertion is none other than (21.2.9.3). On the other hand, let $D = \text{div}(s)$, where s is a regular meromorphic section of $\mathcal{L}$ over X , and let U be an open of X such that there exists an isomorphism of $\mathcal{L}|_U$ on $\mathcal{O}_U$, transforming $s|_U$ in $f \in \Gamma(U, \mathcal{M}_X^*)$, so that $D|_U = \text{div}_U(f)$, $\mathcal{O}_X(D)|_U = \mathcal{O}_U \cdot f^{-1}$ and $s_D|_U$ the unit element of $\Gamma(U, \mathcal{M}_X)$. There is thus an isomorphism $v_U : \mathcal{L}|_U \rightarrow \mathcal{O}_U \cdot f^{-1} = \mathcal{O}_X(D)|_U$ such that $\bar{v}_U$ transforms $s|_U$ in $f \cdot f^{-1} = 1$; one sees immediately that these isomorphisms are compatible with the restriction operations, hence define an isomorphism $v : \mathcal{L} \rightarrow \mathcal{O}_X(D)$ such that $\bar{v}(s) = s_D$. Q.E.D.

One can transport by the first of the isomorphisms (21.2.11.1) the structure of ordered group of $\text{Div}(X)$ to $\text{D}(X)$; the elements ≥ 0 of $\text{D}(X)$ are thus the classes $\text{cl}(\mathcal{L}, s)$ such that $\text{div}(s) \geq 0$, that is to say (21.1.7.1) such that

$$s \in \Gamma(X, \mathcal{L}) \cap \Gamma(X, (\mathcal{M}_X(\mathcal{L}))^*).$$

□

(21.2.12). Let D be a positive divisor on a prescheme X ; the Ideal $\mathcal{J}_X(D)$ is an Ideal of $\mathcal{O}_X$, which is an $\mathcal{O}_X$ -Module invertible; let $Y(D)$ be the closed subscheme of X that it defines. For every $x \in Y(D)$, there exists by hypothesis an open neighbourhood U of x in X and a regular section $t \in \Gamma(U, \mathcal{O}_X)$ such that $\mathcal{J}_X(D)|_U = \mathcal{O}_U \cdot t$ (21.2.7); in other terms, the canonical immersion $Y(D) \rightarrow X$ is regular and of codimension 1 (19.1.4) at every point of $Y(D)$. Conversely, if Y is a closed subscheme of X , regularly immersed in X and of codimension 1 at every point of Y , there exists a positive divisor and only one D such that $Y(D) = Y$, since for every $x \in Y$, there exists a neighbourhood U of x in X such that $Y \cap U$ is defined by an Ideal of $\mathcal{O}_U$ of the form $\mathcal{O}_U \cdot t$, where t is regular in $\Gamma(U, \mathcal{O}_X)$.

We note that one then has $\text{Supp}(D) = Y(D)$, since to say that $D_x \neq 0$ means that t_x is not invertible, that is to say that $x \in Y(D)$.

21.3. Linear equivalence of divisors

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(21.3.1). We say that a divisor D on X is *principal* if it is of the form $\text{div}(f)$, where f is a regular meromorphic function on X ; the regular meromorphic functions f' such that $\text{div}(f') = D$ are then all those of the form tf , where $t \in \Gamma(X, \mathcal{O}_X^*)$ (21.1.3.4). The set of principal divisors is a subgroup of $\text{Div}(X)$, denoted $\text{Div.princ}(X)$, isomorphic to $\Gamma(X, \mathcal{M}_X^*)/\Gamma(X, \mathcal{O}_X^*)$. Two divisors D, D' are said to be *linearly equivalent* if $D - D'$ is a principal divisor; the principal divisors are thus the divisors linearly equivalent to 0.

(21.3.2). Recall (0_I, 5.4.7) that one can speak of equivalence classes of $\mathcal{O}_X$ -Modules invertibles for the isomorphism relation; one denotes this set by $\text{Pic}(X)$, and for every $\mathcal{O}_X$ -Module invertible $\mathcal{L}$, one denotes $\text{cl}(\mathcal{L})$ the equivalence class of the $\mathcal{O}_X$ -Modules isomorphic to $\mathcal{L}$; furthermore, $\text{Pic}(X)$ is a commutative group for the multiplication defined by $\text{cl}(\mathcal{L}) \text{cl}(\mathcal{L}') = \text{cl}(\mathcal{L} \otimes \mathcal{L}')$. It is clear that the application

$$(21.3.2.1) \quad r : \text{cl}(\mathcal{L}, s) \longmapsto \text{cl}(\mathcal{L})$$

is a homomorphism of the group $D(X)$ (21.2.10) in the group $\text{Pic}(X)$. By composition, one deduces thus a homomorphism

$$(21.3.2.2) \quad l : \text{Div}(X) \xrightarrow{\sim} D(X) \longrightarrow \text{Pic}(X)$$

(also denoted l_X) such that, for every divisor D , one has

$$(21.3.2.3) \quad l(D) = \text{cl}(\mathcal{O}_X(D)).$$

Note finally that, if $u : X' \rightarrow X$ is a morphism of ringed spaces, $\mathcal{L}_1, \mathcal{L}_2$ two $\mathcal{O}_X$ -Modules invertibles isomorphes, the $\mathcal{O}_{X'}$ -Modules invertibles $u^*(\mathcal{L}_1)$ and $u^*(\mathcal{L}_2)$ (0_I, 5.4.5) are isomorphic; since on the other hand, for any two $\mathcal{O}_X$ -Modules invertibles $\mathcal{L}_1, \mathcal{L}_2$ one has $u^*(\mathcal{L}_1 \otimes \mathcal{L}_2) = u^*(\mathcal{L}_1) \otimes u^*(\mathcal{L}_2)$ up to canonical isomorphism (0_I, 4.3.3), one sees that the morphism u defines canonically a homomorphism of commutative groups

$$(21.3.2.4) \quad \text{Pic}(u) : \text{Pic}(X) \longrightarrow \text{Pic}(X').$$

Proposition (21.3.3). — *label=(i) The kernel of the canonical homomorphism $l : \text{Div}(X) \rightarrow \text{Pic}(X)$ is the subgroup $\text{Div. princ}(X)$, in other words, for $\mathcal{O}_X(D)$ and $\mathcal{O}_X(D')$ to be isomorphic, it is necessary and sufficient that D and D' be linearly equivalent. One thus has a canonical injective homomorphism*

$$(21.3.3.1) \quad \text{Div}(X) / \text{Div. princ}(X) \longrightarrow \text{Pic}(X)$$

deduced from l .

lbel=(ii) For an $\mathcal{O}_X$ -Module invertible $\mathcal{L}$ to be such that $\text{cl}(\mathcal{L})$ is of the form $l(D)$, or equivalently for $\mathcal{L}$ to be isomorphic to an $\mathcal{O}_X$ -Module of the form $\mathcal{O}_X(D)$, it is necessary and sufficient that there exists a regular meromorphic section of $\mathcal{L}$.

Proof. The proposition results immediately from the definitions and from (21.2.10). $\square$

Proposition (21.3.4). — *Let X be a prescheme satisfying one of the two following hypotheses:*

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label=a) X is locally Noetherian and $\text{Ass}(\mathcal{O}_X)$ is contained in an affine open of X .

lbel=b) X is reduced and the set of its irreducible components is locally finite.

Then the canonical homomorphism $l : \text{Div}(X) \rightarrow \text{Pic}(X)$ is surjective, and gives by passage to the quotient an isomorphism

$$\text{Div}(X) / \text{Div. princ}(X) \xrightarrow{\sim} \text{Pic}(X).$$

Proof. It suffices to show that every $\mathcal{O}_X$ -Module invertible $\mathcal{L}$ admits a regular meromorphic section over X (21.3.3, (ii)). In both cases, it will suffice, thanks to (20.2.11, (ii)), to define a section s of $(\mathcal{M}_X(\mathcal{L}))^*$ over a schematically dense open U of X , or else in case a), of an open U containing $\text{Ass}(\mathcal{O}_X)$ (20.2.13, (iv)). Indeed, let then V be an arbitrary open of X such that $\mathcal{L}|_V$ is isomorphic to $\mathcal{O}_V$, so that there exists an isomorphism of $\mathcal{M}_X(\mathcal{L})|_V$ on $\mathcal{M}_X|_V$, transforming $s|(U \cap V)$ in a section f_V of $\mathcal{M}_X^*$ over $U \cap V$. Since $U \cap V$ is schematically dense in V , it results from (20.2.11) that there exists a regular meromorphic function and only one g_V in V such that $g_V|(U \cap V) = f_V$, and this section thus corresponds by the considered isomorphism to a regular meromorphic section u_V of $\mathcal{L}$ over V such that $u_V|(U \cap V) = s|(U \cap V)$. Furthermore, if V' is a second open of X such that $\mathcal{L}|_{V'}$ is isomorphic to $\mathcal{O}_{V'}$, the restrictions of u_V and $u_{V'}$ to $V \cap V'$ are equal, since they correspond by isomorphism to two meromorphic functions which coincide in a schematically dense open $U \cap V \cap V'$, and we conclude again by (20.2.11); the u_V are thus indeed the restrictions of a section of $(\mathcal{M}_X(\mathcal{L}))^*$ over X .

This being so, in case b), we take for each of the maximal points x_λ of X an open U_λ containing x_λ , not meeting any irreducible component of X distinct from $\overline{\{x_\lambda\}}$ and such that $\mathcal{L}|_{U_\lambda}$ is isomorphic to $\mathcal{O}_{U_\lambda}$; we take for s the section of $\mathcal{L}|_{U_\lambda}$ which corresponds by the preceding isomorphism to the unit section of $\mathcal{O}_{U_\lambda}$.

In case a), one can take for U by hypothesis an affine open (hence Noetherian), in other words, suppose that $X = \text{Spec}(A)$, where A is Noetherian, and $\mathcal{L} = \tilde{P}$, where P is a projective A -module of rank 1. If S is the set of regular elements of A , one has $\Gamma(X, \mathcal{M}_X) = S^{-1}A$ (20.2.12) and $\Gamma(X, \mathcal{M}_X(\mathcal{L})) = S^{-1}P$; now S is the set of elements not belonging to any of the ideals associated to A , hence $S^{-1}A$ is a semi-local ring whose maximal ideals come from the maximal elements of $\text{Ass}(A)$, and $S^{-1}P$ is a projective $S^{-1}A$ -module of rank 1, hence here a free module of rank 1

(Bourbaki, *Alg. comm.*, chap. II, §5, n° 3, prop. 5); an element forming a base of this $S^{-1}A$ -module is thus by (20.1.8) a regular meromorphic section of $\mathcal{L}$ over X . $\square$

Corollary (21.3.5). — *If X is a Noetherian prescheme such that there exists an ample $\mathcal{O}_X$ -Module invertible (II, 4.5.3) (for example a quasi-projective prescheme over a Noetherian ring spectrum (II, 5.3.1 and 4.6.6)), the canonical homomorphism $l : \text{Div}(X) \rightarrow \text{Pic}(X)$ is surjective.*

Proof. Indeed (II, 4.5.4), there then exists an affine open neighbourhood of the finite set $\text{Ass}(\mathcal{O}_X)$. $\square$ **IV-4 | 265**

Remark (21.3.6). — Recall (0_I, 5.4.7) that one has a canonical isomorphism $\pi : H^1(X, \mathcal{O}_X^*) \xrightarrow{\sim} \text{Pic}(X)$ defined in the following way. One starts from a 1-cocycle $(c_{\alpha\beta})$ with values in $\mathcal{O}_X^*$ corresponding to an open covering (U_α) of X , $c_{\alpha\beta}$ being an element of $\Gamma(U_\alpha \cap U_\beta, \mathcal{O}_X^*)$, and one associates to it the class of the $\mathcal{O}_X$ -Module invertible obtained by gluing the $\mathcal{O}_{U_\alpha}$ along the isomorphisms $\mathcal{O}_{U_\alpha}|_{(U_\alpha \cap U_\beta)} \simeq \mathcal{O}_{U_\beta}|_{(U_\alpha \cap U_\beta)}$ defined by multiplication by $c_{\alpha\beta}$. On the other hand, one deduces from the short exact sequence of sheaves of commutative groups

$$1 \longrightarrow \mathcal{O}_X^* \longrightarrow \mathcal{M}_X^* \longrightarrow \mathcal{D}iv_X \longrightarrow 0$$

the boundary homomorphism of the exact cohomology sequence

$$(21.3.6.1) \quad \partial : \text{Div}(X) \longrightarrow H^1(X, \mathcal{O}_X^*).$$

Let us show that the composed homomorphism

$$\text{Div}(X) \xrightarrow{\partial} H^1(X, \mathcal{O}_X^*) \xrightarrow{\pi} \text{Pic}(X)$$

is none other than the homomorphism l defined in (21.3.2.2). Indeed one must start from a divisor D and an open covering (U_α) of X , where $D|_{U_\alpha} = \text{div}_{U_\alpha}(g_\alpha)$, where g_α is a regular meromorphic function over U_α ; $\partial(D)$ is the cohomology class of the cocycle $(c_{\alpha\beta})$, where $c_{\alpha\beta} = g_{\alpha|\beta} g_{\beta|\alpha}^{-1}$, denoting the restriction of g_α to $U_\alpha \cap U_\beta$. It is clear that the image by π of this cohomology class is the class of the invertible fractional Ideal $\mathcal{J}$ such that for every α , $\mathcal{J}|_{U_\alpha} = \mathcal{O}_{U_\alpha} \cdot g_\alpha^{-1}$, which is none other by definition than $\mathcal{O}_X(D)$ (21.2.8).

21.4. Reciprocal images of divisors

(21.4.1). Let $f : X' \rightarrow X$ be a morphism of ringed spaces; let us give conditions allowing us to associate to a divisor D on X a divisor D' on X' , *reciprocal image* of D by f . Note first for this that for every section $t \in \Gamma(X, \mathcal{O}_X^*)$, the image of t by the canonical homomorphism $\Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(X', \mathcal{O}_{X'})$ is still invertible, in other words belongs to $\Gamma(X', \mathcal{O}_{X'}^*)$. Consider then D as given by the equivalence class of a couple $(\mathcal{L}, s)$, where $\mathcal{L}$ is an $\mathcal{O}_X$ -Module invertible and s a regular meromorphic section of $\mathcal{L}$ (20.1.11). Form the $\mathcal{O}_{X'}$ -Module invertible $f^*(\mathcal{L}) = \mathcal{L}'$; to say that the reciprocal image $s_0 f$ of s by f exists (20.1.11) and is a regular meromorphic section of $\mathcal{L}'$ over X' amounts to saying that the reciprocal images by f of s and of $s^{\otimes(-1)}$ exist, in other words that $s \in \Gamma(X, \mathcal{M}_f(\mathcal{L}))$ and $s^{\otimes(-1)} \in \Gamma(X, \mathcal{M}_f(\mathcal{L}^{-1}))$; the remark made above shows then that if $(\mathcal{L}_1, s_1)$ is a couple equivalent to $(\mathcal{L}, s)$ in the sense of (21.2.10), the reciprocal image $s_1 \circ f$ exists and the couples $(\mathcal{L}'_1, s_1 \circ f)$ and $(\mathcal{L}', s_0 \circ f)$ are equivalent. We can thus give the following definition:

Definition (21.4.2). — Given a morphism $f : X' \rightarrow X$ of ringed spaces, we say that the *reciprocal image* by f of a divisor D on X exists if one has $s_D \in \Gamma(X, \mathcal{M}_f(\mathcal{O}_X(D)))$ and $s_{-D} \in \Gamma(X, \mathcal{M}_f(\mathcal{O}_X(-D)))$ (cf. (20.1.11)). We then call *reciprocal image* of D by f and denote $f^*(D)$ the divisor on X' which corresponds canonically (21.2.11) to the class of the couple $(f^*(\mathcal{O}_X(D)), s_D \circ f)$.

It results immediately from this definition that if D and D' have reciprocal images by f , the same holds for $-D$ and $D + D'$ (taking into account (21.2.9.4)) and that one has $f^*(D + D') = f^*(D) + f^*(D')$.

In other words, the set $\text{Div}^f(X)$ of divisors on X whose reciprocal image by f exists is a subgroup of $\text{Div}(X)$, and the application $D \mapsto f^*(D)$ is an *increasing homomorphism* of the ordered subgroup **IV-4 | 266**

$\text{Div}^f(X)$ in the ordered group $\text{Div}(X')$, making commutative the diagram

$$(21.4.2.1) \quad \begin{array}{ccc} \text{Div}^f(X) & \xrightarrow{l_X} & \text{Pic}(X) \\ f^* \downarrow & & \downarrow \text{Pic}(f) \\ \text{Div}(X') & \xrightarrow{l_{X'}} & \text{Pic}(X') \end{array}$$

(21.4.3). The definition (21.4.2) shows immediately that, for $f^*(D)$ to exist, it is necessary and sufficient that for every open U of X , the reciprocal image by $f^U : f^{-1}(U) \rightarrow U$ (restriction of f) of $D|_U$ exists. Now, if $D = \text{div}(g)$, where g is a regular meromorphic function on X , to say that $f^*(\mathcal{O}_X(D))$ is defined means (21.2.9) that the reciprocal image of g by f exists and is a regular meromorphic function on X' . We deduce immediately a second description of $\text{Div}^f(X)$ and of $f^*(D)$: one considers the sub-sheaf of groups of $\mathcal{M}_X^*$, denoted $\mathcal{M}_f^*$, formed of the germs of regular meromorphic functions on an open of X whose reciprocal image by f exists and is regular on the reciprocal image open (20.1.11). Then if $f = (\psi, \theta)$, the canonical homomorphism (20.1.11) $\psi^*(\mathcal{M}_f) \rightarrow \mathcal{M}_{X'}$ gives by restriction a homomorphism of sheaves of groups $\psi^*(\mathcal{M}_f^*) \rightarrow \mathcal{M}_{X'}^*$. Setting $\mathcal{D}iv_X^f = \mathcal{M}_f^* / \mathcal{O}_X^*$, one has $\text{Div}^f(X) = \Gamma(X, \mathcal{D}iv_X^f)$, and the application $D \mapsto f^*(D)$ corresponds to the homomorphism of sheaves of groups $\psi^*(\mathcal{M}_f^*) / \psi^*(\mathcal{O}_X^*) \rightarrow \mathcal{M}_{X'}^* / \mathcal{O}_{X'}^* = \mathcal{D}iv_{X'}$ deduced from the preceding by passage to quotients.

(21.4.4). It results immediately from the preceding definitions that if $f' : X'' \rightarrow X'$ is a second morphism of ringed spaces, D a divisor on X such that the reciprocal images $f^*(D)$ and $f'^*(f^*(D))$ exist, then $(f' \circ f)^*(D)$ exists and is equal to $f'^*(f^*(D))$.

Proposition (21.4.5). — *Let $f : X' \rightarrow X$ be a morphism of ringed spaces. In each of the three following cases, the reciprocal image by f of every divisor on X is defined:*

- label=(i) f is flat.
- lbbel=(ii) X and X' are locally Noetherian preschemes and one has $f(\text{Ass}(\mathcal{O}_{X'})) \subset \text{Ass}(\mathcal{O}_X)$.
- lcbel=(iii) X and X' are preschemes, the set of irreducible components of X is locally finite, X' is reduced and every irreducible component of X' dominates an irreducible component of X .

Proof. It suffices indeed to show that in the three cases one has $\mathcal{M}_f = \mathcal{M}_X$; in case (i), this results from (20.1.12). In case (iii), one can restrict to the case where $X = \text{Spec}(A)$ and $X' = \text{Spec}(A')$ are affine; if $s \in A$ is regular, it does not belong to any minimal prime ideal of A (20.3.1), hence the hypothesis implies that its image in A' does not belong to any minimal prime ideal of A' , and is thus a regular element of A' (20.1.3.1). In case (ii) the meromorphic functions on X' identify with the pseudo-functions on X' (20.2.11), and the hypothesis, together with (20.2.13, (iv)) ensures that the reciprocal image by f of every schematically dense open in X is a schematically dense open in X' ; we thus conclude by (20.3.12). $\square$

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Corollary (21.4.6). — *Let X be a prescheme having one of the following properties:*

- label=(i) X is locally Noetherian.
- lbbel=(ii) X is reduced and the set of its irreducible components is locally finite.

Then, for every $x \in X$, one has a canonical isomorphism

$$(21.4.6.1) \quad (\mathcal{D}iv_X)_x \xrightarrow{\sim} \text{Div}(\mathcal{O}_{X,x}).$$

Proof. This results indeed from (20.2.11), (20.3.7) and from the fact that $(\mathcal{O}_X^*)_x$ identifies with the group of invertible elements of the ring $\mathcal{O}_{X,x}$. $\square$

(21.4.7). Let X, X' be two preschemes, $f : X' \rightarrow X$ a morphism. If D is a *positive* divisor on X such that the reciprocal image $f^*(D)$ is defined (21.4.2), then the closed subprescheme $Y(f^*(D))$ of X' is none other than the reciprocal image $f^{-1}(Y(D))$; this results immediately from the definitions (21.4.2) and (21.2.12).

Proposition (21.4.8). — *Let X, Y be two preschemes, $f : X \rightarrow Y$ a faithfully flat morphism. Then, if D is a divisor on Y such that $f^*(D) \geq 0$ (the existence of $f^*(D)$ resulting from (21.4.5)), one has $D \geq 0$. In particular, the application $D \mapsto f^*(D)$ of $\text{Div}(Y)$ in $\text{Div}(X)$ is injective.*

Proof. The question being local on Y , one can restrict to the case where $D = \operatorname{div}(w)$, with $w = uv^{-1}$, u and v being two regular sections of $\mathcal{O}_Y$ over Y . By hypothesis one has $u_y \mathcal{O}_{X,x} \subset v_y \mathcal{O}_{X,x}$; hence, for every $x \in X$, if one sets $y = f(x)$, one has $u_y \mathcal{O}_{X,x} \subset v_y \mathcal{O}_{X,x}$; one thus concludes that $\mathcal{O}_{X,x}$ is a faithfully flat $\mathcal{O}_{Y,y}$ -module by virtue of the hypothesis and of Bourbaki, *Alg. comm.*, chap. I, §3, n° 5, prop. 10, (ii); whence $u \mathcal{O}_Y \subset v \mathcal{O}_Y$ and since f is surjective and therefore $D \geq 0$. $\square$

21.5. Direct images of divisors

(21.5.1). Let X, X' be two preschemes, $f : X' \rightarrow X$ a morphism. We will, in this section, give sufficient conditions for one to be able to associate to every divisor D' on X' a divisor D on X , *direct image* of D' by f . We will restrict ourselves to the case where f is a *finite* morphism (for more general conditions, see the chapter of this Treatise dedicated to intersection theory).

Lemma (21.5.2). — *Let A be a ring, E an A -module free of finite rank. For an endomorphism u of E to be injective, it is necessary and sufficient that $\det(u)$ be a regular element of A .*

Proof. This is proved in Bourbaki, *Alg.*, chap. III, 3rd ed., §8, n° 2, prop. 3. $\square$

(21.5.3). Suppose now that the morphism $f : X' \rightarrow X$ is finite, and moreover that f satisfies one of the two following properties:

- I) f is a finite locally free morphism, in other words (18.2.7) the $\mathcal{O}_X$ -Module quasi-coherent of finite type $f_*(\mathcal{O}_{X'})$ is locally free.
- II) X is a locally Noetherian reduced prescheme, the $\mathcal{R}(X)$ -Module $f_*(\mathcal{O}_{X'}) \otimes_{\mathcal{O}_X} \mathcal{R}(X)$ is locally free, and for every section $s' \in \Gamma(f^{-1}(U), \mathcal{O}_{X'})$ (U open

in X), $N_{f_*(\mathcal{O}_{X'})/\mathcal{O}_X}(s')$ is a section of $\mathcal{O}_X$ over U (cf. (II, 6.5.1)). (Recall that condition II) is satisfied for every finite morphism f when X is a locally Noetherian *normal* prescheme (*loc. cit.*)).

One knows then (II, 6.5.5) that for every $\mathcal{O}_{X'}$ -Module invertible $\mathcal{L}'$ one defines the *norm* $\mathcal{L} = N_{X'/X}(\mathcal{L}')$ (which we will also write $N(\mathcal{L}')$), which is an $\mathcal{O}_X$ -Module invertible. Furthermore, for every open U of X and every regular section $s' \in \Gamma(f^{-1}(U), \mathcal{O}_{X'})$, the norm $N_{X'/X}(s') = N_{f_*(\mathcal{O}_{X'})/\mathcal{O}_X}(s')$ (which we will also write $N(s')$) is a regular element of $\Gamma(U, \mathcal{O}_X)$; one is indeed reduced immediately to the case where $U = X$ is affine, and then the conclusion results from (21.5.2) in hypothesis I); on the other hand, in hypothesis II), the fact that $\mathcal{R}(X)$ is a flat $\mathcal{O}_X$ -Module implies that the section $s' \otimes 1$ of $\Gamma(U, f_*(\mathcal{O}_{X'}) \otimes_{\mathcal{O}_X} \mathcal{R}(X))$ is also regular (01, 6.1.2), and the conclusion results again from (21.5.2) applied to the ring $\Gamma(U, \mathcal{R}(X))$, taking into account the definition of the norm of a section (II, 6.5.3). This being so, let u' be a meromorphic section of $\mathcal{L}'$ over X' ; the morphism f being affine, every point $x \in X$ admits an open neighbourhood U such that $u'|f^{-1}(U)$ is of the form t'/s' , where $t' \in \Gamma(f^{-1}(U), \mathcal{L}')$ and s' is a regular section in $\Gamma(f^{-1}(U), \mathcal{O}_{X'})$; the element $N(t')/N(s')$ is then a meromorphic section of $\mathcal{L} = N(\mathcal{L}')$ over U by virtue of what precedes, and it is immediately verified that it is independent of the representation t'/s' chosen for $u'|f^{-1}(U)$; the meromorphic sections of $\mathcal{L}$ thus defined, for variable U , are the restrictions of a single meromorphic section of $\mathcal{L}$ over X , that we call the *norm* of u' and that we denote $N_{X'/X}(u')$ (or simply $N(u')$). If u' is a *regular* meromorphic section of $\mathcal{L}'$ over X' , it results immediately from what precedes that $N(u')$ is a regular meromorphic section of $N(\mathcal{L}')$ over X . The application thus defined

$$(21.5.3.1) \quad N_{X'/X} : \Gamma(X', \mathcal{M}_{X'}(\mathcal{L}')) \longrightarrow \Gamma(X, \mathcal{M}_X(N_{X'/X}(\mathcal{L}')))$$

prolongs the norm defined in (II, 6.5.3); if u' is a regular meromorphic section of $\mathcal{L}'$ over X' , it results immediately from what precedes that $N(u')$ is a regular meromorphic section of $N(\mathcal{L}')$ over X , since (with the same notations) $N(t')$ is regular if t' is. Finally, if $\mathcal{L}'_1, \mathcal{L}'_2$ are two $\mathcal{O}_{X'}$ -Modules invertibles, s'_1 (resp. s'_2) a regular meromorphic section of $\mathcal{L}'_1$ (resp. $\mathcal{L}'_2$) over X' , one has, by virtue of what precedes and of (II, 6.5.3.1)

$$(21.5.3.2) \quad N(s'_1 \otimes s'_2) = N(s'_1) \otimes N(s'_2).$$

(21.5.4). Suppose always that f satisfies one of the hypotheses I), II) of (21.5.3). If $(\mathcal{L}'_1, s'_1), (\mathcal{L}'_2, s'_2)$ are two couples each formed of an $\mathcal{O}_{X'}$ -Module invertible and of a regular meromorphic section of $\mathcal{L}'$ over X' , which are moreover *equivalent* in the sense of (21.2.10), then the couples $(N(\mathcal{L}'_1), N(s'_1))$ and $(N(\mathcal{L}'_2), N(s'_2))$ are also equivalent, since an isomorphism of $\mathcal{O}_{X'}$ -Modules invertibles has for

norm an isomorphism of $\mathcal{O}_X$ -Modules invertibles, and one has seen above that $N_{X'/X}$ transforms the sections of $\Gamma(X', \mathcal{O}_{X'}^*)$ in sections of $\Gamma(X, \mathcal{O}_X^*)$. One can thus give the following definition:

Definition (21.5.5). — Given a finite morphism $f : X' \rightarrow X$ of preschemes, satisfying one of the conditions I), II) of (21.5.3), we call *direct image* (or *norm*) of a divisor D' on X' by f , and denote $f_*(D')$ (or $N_{X'/X}(D')$), the divisor on X which corresponds canonically (21.2.11) to the class of the couple $(N_{X'/X}(\mathcal{O}_{X'}(D')), N_{X'/X}(s_{D'}))$.

It results from this definition, taking into account (21.2.9.4) and (21.5.3.2), that

$$(21.5.5.1) \quad f_*(D'_1 + D'_2) = f_*(D'_1) + f_*(D'_2)$$

and $D' \geq 0$ implies $f_*(D') \geq 0$, in other words $D' \mapsto f_*(D')$ is an *increasing homomorphism* of the ordered group $\text{Div}(X')$ in the ordered group $\text{Div}(X)$. The definition (21.5.5) shows moreover that for every open U of X , one has $f_*^U(D'|f^{-1}(U)) = f_*(D')|U$ (f^U being the restriction $f^{-1}(U) \rightarrow U$ of f), and the homomorphisms f_*^U , for variable U , define an *homomorphism of sheaves of ordered groups*

$$(21.5.5.2) \quad N_{X'/X} : f_*(\mathcal{D}iv_{X'}) \longrightarrow \mathcal{D}iv_X.$$

Furthermore, for every open U of X , every $\mathcal{O}_{X'}$ -Module invertible $\mathcal{L}'$ and every regular meromorphic section s' of $\mathcal{L}'$ over $f^{-1}(U)$, one has, by the preceding definitions and (21.1.4)

$$(21.5.5.3) \quad \text{div}_U(N(s')) = f_*^U(\text{div}_{f^{-1}(U)}(s')).$$

Proposition (21.5.6). — Let $f : X' \rightarrow X$ be a finite locally free morphism and suppose that $f_*(\mathcal{O}_{X'})$ is of constant rank n . Then, for every divisor D on X , $f_*(D)$ is defined and one has

$$(21.5.6.1) \quad f_*(f^*(D)) = nD.$$

Proof. The first assertion results from the fact that f is flat (21.4.5), and the second is an immediate consequence of the definitions and of (II, 6.5.3.2). $\square$

Proposition (21.5.7). — Let $f : X' \rightarrow X$ be a finite morphism satisfying one of the hypotheses I), II) of (21.5.3), $f' : X'' \rightarrow X'$ a finite locally free morphism of constant rank n . Then $f'' = f \circ f' : X'' \rightarrow X$ satisfies the same hypothesis as f , and for every divisor D'' on X'' , one has

$$(21.5.7.1) \quad f''_*(D'') = f_*(f'_*(D'')).$$

Taking into account the definition (21.5.5), it suffices to prove the following result:

Lemma (21.5.7.2). — Under the hypotheses of (21.5.7), one has a functorial isomorphism

$$(21.5.7.3) \quad N_{X''/X}(\mathcal{L}'') \xrightarrow{\sim} N_{X'/X}(N_{X''/X'}(\mathcal{L}''))$$

in the category of $\mathcal{O}_X$ -Modules invertibles.

The question is evidently local on X , and one can thus restrict to the case where $X = \text{Spec}(A)$ is affine; one then has $X' = \text{Spec}(A')$ and $X'' = \text{Spec}(A'')$, and one can suppose that A'' is a projective A' -module of rank n . When f is locally free, one can suppose that A' is a free A -module of rank m , and then A'' is a free A -module of rank mn ; the formula (21.5.7.3) results then from the transitivity of the norm (Bourbaki, *Alg.*, chap. VIII, §12, n° 2, prop. 7). When f satisfies hypothesis II), A is Noetherian reduced, and if R is its total ring of fractions, $A' \otimes_A R$ is a free R -module of rank m , and then $A'' \otimes_A R$ is a free R -module of rank mn ; and since R is also a semi-local ring, this R -module is free; the proposition results then also from the transitivity of the norms.

Proposition (21.5.8). — Let $f : X' \rightarrow X$ be a finite morphism, $g : Y \rightarrow X$ a morphism; set $Y' = X' \times_X Y$, $f' = f_{(Y)} : Y' \rightarrow Y$, $g' = g_{(X)} : Y' \rightarrow X'$. Suppose satisfied one of the following conditions:

- label=(i) f is locally free and g is flat.
- lbbel=(ii) f is locally free, X and Y are locally Noetherian, $g(\text{Ass}(\mathcal{O}_Y)) \subset \text{Ass}(\mathcal{O}_X)$ and $g'(\text{Ass}(\mathcal{O}_{Y'})) \subset \text{Ass}(\mathcal{O}_{X'})$.
- lcbel=(iii) f satisfies hypothesis II) of (21.5.3), Y is locally Noetherian, Y and Y' are reduced and every irreducible component of Y (resp. Y') dominates an irreducible component of X (resp. X').

Then, for every divisor D' on X' , $g'^*(D')$ is defined, $g^*(f_*(D'))$ is defined and one has

$$(21.5.8.1) \quad g^*(f_*(D')) = f'_*(g'^*(D')).$$

Proof. Indeed, in all the cases, it results from (II, 6.5.8) that one has a functorial isomorphism

$$g^*(N_{X'/X}(\mathcal{L}')) \xrightarrow{\sim} N_{Y'/Y}(g'^*(\mathcal{L}'))$$

in the category of $\mathcal{O}_X$ -Modules invertibles; furthermore (II, 6.5.4), if s' is a section of $\mathcal{O}_{X'}$ over $f^{-1}(U)$, s'' the corresponding section of $\mathcal{O}_{Y'}$ over $g'^{-1}(f^{-1}(U))$ (U open of X), $N_{Y'/Y}(s'')$ is the section of $\mathcal{O}_Y$ over $g^{-1}(U)$ which corresponds to the section $N_{X'/X}(s')$ of $\mathcal{O}_X$ over U . The formula (21.5.8.1) results then from the definitions if $g'^*(D')$ and $g^*(D)$ are defined, whatever the divisors D' on X' and D on X . As for what concerns D , in case (i) g' is flat, hence in all cases $g'^*(D')$ is defined by virtue of (21.4.5). As for what concerns D' , in case (i) g is flat, hence in all cases $g^*(f_*(D'))$ is defined by virtue of (21.4.5). $\square$

21.6. The 1-codimensional cycle associated to a divisor

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(21.6.1). Let X be a locally Noetherian prescheme, and let $\mathfrak{I}(X)$ be the set of *irreducible closed subsets* of X (which is in bijective correspondence with X by the application $x \mapsto \overline{\{x\}}$). In the product group $\mathbf{Z}^X$, consider the subgroup $\mathfrak{Z}(X)$ of the elements $(n_x)_{x \in X}$ such that the set of $\overline{\{x\}} \in \mathfrak{I}(X)$ such that $n_x \neq 0$ is *locally finite*. It is clear that $\mathfrak{Z}(X)$ is a *subgroup* of $\mathbf{Z}^X$, which contains the direct sum group $\mathbf{Z}^{(X)}$ (free group having $\mathfrak{I}(X)$ for basis), and is equal to it when X is Noetherian. The elements of $\mathfrak{Z}(X)$ are called *cycles* on X and those of $\mathfrak{I}(X)$ *prime cycles* (they do not in general form a basis of $\mathfrak{Z}(X)$ when X is not Noetherian). One always considers $\mathfrak{Z}(X)$ as *ordered* by the order induced on this subgroup by the product order of $\mathbf{Z}^X$, and one denotes $\mathfrak{Z}^+(X)$ the set of cycles ≥ 0 .

For every cycle $Z \in \mathfrak{Z}(X)$, equal to $(n_x)_{x \in X}$, one will write by abuse of language,

$$Z = \sum_{x \in X} n_x \cdot \overline{\{x\}};$$

one calls *multiplicity* of Z at the point x and one denotes $\text{mult}_x(Z)$ the integer n_x (positive or negative). To say that $Z \geq 0$ means that $\text{mult}_x(Z) \geq 0$ for every $x \in X$. One calls *support* of Z and one denotes $\text{Supp}(Z)$ the union of the $\overline{\{x\}}$ such that $\text{mult}_x(Z) \neq 0$; by definition of $\mathfrak{Z}(X)$, this is a closed set of X , being the union of a locally finite family of closed subsets. One calls *dimension* (resp. *codimension*) of Z and one denotes $\dim(Z)$ (resp. $\text{codim}(Z)$) the dimension (resp. codimension in X) of $\text{Supp}(Z)$.

(21.6.2). One says that a closed subset Y of X is *purely of codimension p* (in X) if *each* of its irreducible components is of codimension p in X . One says that a cycle is *purely of codimension p* , or (by abuse of language) is a *p -codimensional cycle* if its support is purely of codimension p . One denotes $X^{(p)}$ the set of $x \in X$ such that $\text{codim}(\overline{\{x\}}, X) = p$, or, which amounts to the same (5.1.2.1), $\dim(\mathcal{O}_{X,x}) = p$: the purely p -codimensional cycles form a subgroup $\mathfrak{Z}^p(X)$ of $\mathfrak{Z}(X)$, isomorphic to the subgroup of $\mathbf{Z}^{X^{(p)}}$ formed of the $(n_x)_{x \in X^{(p)}}$ such that the set of $\overline{\{x\}}$ (or the set of x) for which $n_x \neq 0$ is locally finite; this subgroup contains the free group $\mathbf{Z}^{(X^{(p)})}$ and is equal to it when X is Noetherian. One denotes $\mathfrak{Z}^{p+}(X)$ the set of elements ≥ 0 of $\mathfrak{Z}^p(X)$. It is clear that the ordered group $\mathfrak{Z}(X)$ contains the direct sum of the ordered subgroups $\mathfrak{Z}^p(X)$, and is identical to this direct sum when X is Noetherian; in this last case, one considers $\mathfrak{Z}(X)$ as *graded* by the $\mathbf{Z}^p(X)$ for $p \in \mathbf{N}$.

(21.6.3). Let $Z = \sum_{x \in X} n_x \cdot \overline{\{x\}}$ be a cycle on X , U an open of X ; one calls *restriction* of Z to U and one denotes $Z|U$ the cycle $\sum_{x \in U} n_x \cdot (U \cap \overline{\{x\}})$ on U ; one has $\text{Supp}(Z|U) = \text{Supp}(Z) \cap U$, hence $Z \mapsto Z|U$ is an homomorphism of ordered groups $\mathfrak{Z}(X) \rightarrow \mathfrak{Z}(U)$ (resp. $\mathfrak{Z}^{p+}(U) \rightarrow \mathfrak{Z}^{p+}(U)$). The sheaf $\mathcal{Z}_X$ (resp. $\mathcal{Z}_X^p$) is evidently the *direct sum* of the sheaves $(i_x)_*(\mathbf{Z}_{\overline{\{x\}}})$, where x ranges over X (resp. $X^{(p)}$) and i_x is the canonical injection $\overline{\{x\}} \rightarrow X$ and $\mathbf{Z}_{\overline{\{x\}}}$ is the simple constant sheaf $\mathbf{Z}$ over the space $\overline{\{x\}}$: this results immediately from the description of the sections of a direct sum of sheaves of commutative groups (G, II, 2.7). One denotes this sheaf $\mathcal{Z}_X$ (resp. $\mathcal{Z}_X^p$). One denotes $\mathcal{Z}_X^+$ (resp. $\mathcal{Z}_X^{p+}$) the sub-sheaf of monoids (resp. $U \mapsto \mathfrak{Z}^{p+}(U)$) of $\mathcal{Z}_X$ (resp. $\mathcal{Z}_X^p$).

Note finally that the sheaves $\mathcal{Z}_X^p$ (and hence also $\mathcal{Z}_X$) are *flasque*. Suppose in effect given a section Z of $\mathcal{Z}_X^p$ over an open U , so that the set S of $x \in U$ such that $\text{mult}_x(Z) \neq 0$ is locally finite in U ; this set is also locally finite in X , since for every $z \in X$ and every Noetherian open neighbourhood V of z , $U \cap V$ is Noetherian, hence contains only a finite number of points of S . One defines thus a

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section Z' of $\mathcal{Z}_X^p$ over X by setting $Z' = \sum_{x \in U} \text{mult}_x(Z) \cdot \overline{\{x\}}$ and since the adherence of x in U is $\overline{\{x\}} \cap U$, one has $Z'|_U = Z$, whence our assertion.

(21.6.4). We now propose to define a canonical homomorphism

$$(21.6.4.1) \quad c : \mathcal{D}iv_X \longrightarrow \mathcal{Z}_X^1$$

of sheaves of commutative groups. It suffices evidently to define a homomorphism of $\mathcal{M}_X^*$ in $\mathcal{Z}_X^1$, which is zero on $\mathcal{O}_X^*$ (21.1.2); or, by definition, $\mathcal{M}_X^*$ is the *symmetrised* of the sheaf of monoids $\mathcal{S}(\mathcal{O}_X) = \mathcal{O}_X \cap \mathcal{M}_X^*$, and a homomorphism $\mathcal{M}_X^* \rightarrow \mathcal{Z}_X^1$ is determined in a unique way by its restriction $\mathcal{S}(\mathcal{O}_X) \rightarrow \mathcal{Z}_X^1$, which is a homomorphism of sheaves of *monoids* subject to the sole condition of being zero on $\mathcal{O}_X^*$; it amounts to the same to say that to define c , it suffices to define a homomorphism of *sheaves of monoids*

$$(21.6.4.2) \quad c : \mathcal{D}iv_X^+ \longrightarrow \mathcal{Z}_X^{1+}.$$

(21.6.5). Now, one has seen that every positive divisor D on X determines a closed subscheme $Y(D)$ of X , defined by the Ideal $\mathcal{I}_X(D) \subset \mathcal{O}_X$, and which is regularly immersed and of codimension 1. In each of the maximal points x of $Y(D)$, one has ((19.1.4) and (5.1.2)) $\text{codim}(\overline{\{x\}}, X) = \dim(\mathcal{O}_{X,x}) = 1$, in other words $x \in X^{(1)}$; moreover the set of these maximal points is locally finite in X (3.1.6), and $\mathcal{O}_{Y(D),x}$ is an *Artinian* ring. At every point $x \in X^{(1)}$ which is not a maximal point of $Y(D)$, one necessarily has $x \notin Y(D)$, hence $\mathcal{O}_{Y(D),x} = 0$. Set

$$(21.6.5.1) \quad \text{cyc}(D) = \sum_{x \in X^{(1)}} \text{long}(\mathcal{O}_{Y(D),x}) \cdot \overline{\{x\}}$$

which is thus an element of $\mathfrak{Z}^{1+}(X)$.

Proposition (21.6.6). — *The application $D \mapsto \text{cyc}(D)$ is a homomorphism of the monoid $\text{Div}^+(X)$ in $\mathfrak{Z}^{1+}(X)$.*

Proof. It amounts to seeing that for two positive divisors D, D' , one has

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$$\text{cyc}(D + D') = \text{cyc}(D) + \text{cyc}(D').$$

Now, one has (21.2.5) $\mathcal{I}_X(D + D') = \mathcal{I}_X(D) \mathcal{I}_X(D')$; if one sets $A = \mathcal{O}_{X,x}$, and if t and t' are two regular elements of A , one has $\text{long}(A/tt'A) = \text{long}(A/tA) + \text{long}(A/t'A)$; but since t is regular, $tA/tt'A$ is isomorphic to $A/t'A$, whence the proposition immediately. $\square$

It results immediately from the definitions that for every open $U \subset X$, one has $Y(D)|_U = Y(D) \cap U$, hence $\text{cyc}(D|_U) = \text{cyc}(D)|_U$, and the homomorphisms $\text{cyc} : \text{Div}^+(U) \rightarrow \mathfrak{Z}^{1+}(U)$ define a homomorphism of sheaves of monoids (21.6.4.2), whence the sought homomorphism (21.6.4.1) of sheaves of groups.

One has $\text{Supp}(\text{cyc}(D)) \subset \text{Supp}(D)$ for every divisor D and

$$(21.6.6.2) \quad \text{Supp}(\text{cyc}(D)) = \text{Supp}(D) \quad \text{when } D \geq 0;$$

one has already seen the second relation (21.2.12); when D is any divisor, the relation $D_x = 0$ implies the existence of an open neighbourhood U of x such that $D|_U = 0$, whence $\text{cyc}(D)|_U = 0$, which proves our assertion.

(21.6.7). The homomorphism (21.6.4.1) gives, by passage to the groups of sections over X , an *increasing homomorphism of ordered groups* $\text{Div}(X) \rightarrow \mathfrak{Z}^1(X)$, that we denote again $D \mapsto \text{cyc}(D)$; the number $\text{mult}_x(\text{cyc}(D))$ for $x \in X^{(1)}$ is also denoted $\text{mult}_x(D)$ or $\text{mult}_{\overline{\{x\}}}(D)$ and is called the *multiplicity of the divisor D at the point x* , or *multiplicity of the prime cycle $\overline{\{x\}}$ in D* ; this is a positive or negative integer, equal as we have seen to $\text{long}(\mathcal{O}_{Y(D),x})$ when D is a *positive* divisor; one has therefore by definition

$$(21.6.7.1) \quad \text{cyc}(D) = \sum_{x \in X^{(1)}} \text{mult}_x(D) \cdot \overline{\{x\}},$$

and by virtue of the fact that $D \mapsto \text{cyc}(D)$ is a homomorphism, one has

$$(21.6.7.2) \quad \text{mult}_x(-D) = -\text{mult}_x(D), \quad \text{mult}_x(D + D') = \text{mult}_x(D) + \text{mult}_x(D')$$

for any two divisors D, D' .

For every regular meromorphic function f on X , one sets in particular, for $x \in X^{(1)}$

$$(21.6.7.3) \quad o_x(f) = \text{mult}_x(\text{div}(f))$$

and one says that $o_x(f)$ (a positive or negative integer) is the *order of f at the point $x \in X^{(1)}$* . If $f_x \in \mathcal{O}_{X,x}$ (regular element of this local ring by hypothesis), one thus has

$$(21.6.7.4) \quad o_x(f) = \text{long}(\mathcal{O}_{X,x}/f_x\mathcal{O}_{X,x});$$

for two regular meromorphic functions f, f' on X , one has

$$(21.6.7.5) \quad o_x(ff') = o_x(f) + o_x(f'), \quad o_x(f^{-1}) = -o_x(f)$$

for every $x \in X^{(1)}$. The 1-codimensional cycles

$$Z^+(f) = \sum_{x \in X^{(1)}} (o_x(f))^+ \cdot \overline{\{x\}}, \quad Z^-(f) = \sum_{x \in X^{(1)}} (o_x(f))^- \cdot \overline{\{x\}}$$

are called respectively the *cycle of zeros* (or *polar cycle*) of f , and one has evidently $\text{cyc}(\text{div}(f)) = Z^+(f) - Z^-(f)$.

The 1-codimensional cycles of the form $\text{cyc}(\text{div}(f))$ are called *principal* (or *linearly equivalent to 0*) IV-4 | 274 and form a subgroup $\mathfrak{Z}_{\text{princ}}^1(X)$ of $\mathfrak{Z}^1(X)$. The sections over X of the image $c(\mathcal{D}iv_X) \subset \mathcal{Z}_X^1$ are

called *locally principal cycles*; such a cycle Z is thus characterised by the fact that, for every $y \in X$, there exists an open neighbourhood U of y in X and a regular meromorphic function f on U such that $Z|_U = \sum_{x \in U \cap X^{(1)}} o_x(f) \cdot (\overline{\{x\}} \cap U)$. If $Z = \sum_{x \in X^{(1)}} n_x \cdot \overline{\{x\}}$, it amounts to the same to say that, for every $y \in X$, if one sets $T_y = \text{Spec}(\mathcal{O}_{X,x})$ and $Z_y = \sum_{x \in X^{(1)} \cap T_y} n_x (\overline{\{x\}} \cap T_y)$, Z_y is *principal*, in other words there exists a regular meromorphic function g on T_y such that $Z_y = \text{cyc}(\text{div}(g))$. This results immediately from the preceding definition and from (20.3.7) which establishes a bijective correspondence between regular meromorphic functions on T_y and the germs of regular meromorphic functions on the open neighbourhoods of y in X when X is locally Noetherian. One expresses also the fact that Z_y is principal by saying that Z is *principal at the point y* . The set of points where Z is principal is evidently *open*, by virtue of what precedes.

One deduces from the canonical homomorphism $\text{cyc} : \text{Div}(X) \rightarrow \mathfrak{Z}^1(X)$ a homomorphism $\text{Div}(X)/\text{Div.princ}(X) \rightarrow \mathfrak{Z}^1(X)/\mathfrak{Z}_{\text{princ}}^1(X)$, by virtue of the definition of $\mathfrak{Z}_{\text{princ}}^1(X)$. One says that the group $\mathfrak{Z}^1(X)/\mathfrak{Z}_{\text{princ}}^1(X)$ is the *group of classes of 1-codimensional cycles* of X and one denotes it $\mathfrak{Cl}(X)$. Two elements of $\mathfrak{Z}^1(X)$ having the same image in $\mathfrak{Cl}(X)$ are called *linearly equivalent*.

(21.6.8). Consider in particular the case where $X = \text{Spec}(A)$, where A is an *integrally closed* Noetherian local ring. Then $X^{(1)}$ is the set of *prime ideals of height 1* of A , and $\mathfrak{Z}^1(X)$ identifies with the *group of divisors* (in the sense of N. Bourbaki) of the Krull ring A (Bourbaki, *Alg. comm.*, chap. VII, §1, n° 3, cor. of th. 2 and n° 6, th. 3).

On the other hand, the regular meromorphic functions on X identify with the elements $\neq 0$ of the field of fractions K of A , the application $f \mapsto \text{cyc}(\text{div}(f))$ of $M(X)$ in $\mathfrak{Z}^1(X)$ identifies with the application denoted $f \mapsto \text{div}(f)$ in Bourbaki (*loc. cit.*, §1, n° 1); $\mathfrak{Z}_{\text{princ}}^1(X)$ identifies thus with the *group of principal divisors* of A in the sense of Bourbaki, and $\mathfrak{Cl}(X)$ with the *group of classes of divisors* of A in the sense of Bourbaki (*loc. cit.*, §1, n° 2 and n° 10).

Theorem (21.6.9). — *Let X be a locally Noetherian normal prescheme.*

label=(i) *The canonical homomorphism $\text{cyc} : \text{Div}(X) \rightarrow \mathfrak{Z}^1(X)$ is injective and its image is formed of locally principal cycles.*

lbbel=(ii) *The following conditions are equivalent:*

label=a) *The homomorphism $\text{cyc} : \text{Div}(X) \rightarrow \mathfrak{Z}^1(X)$ is bijective.*

lbbel=b) *Every 1-codimensional cycle is locally principal.*

lcbel=c) *For every $x \in X$, the local ring $\mathcal{O}_{X,x}$ is factorial.*

(One says then that X is a locally factorial prescheme.)

Proof. (i) It suffices to show that

$$(21.6.9.1) \quad \text{cyc}^{-1}(\mathfrak{Z}^{1+}(X)) = \text{Div}^+(X),$$

since one has $\text{Div}^+(X) \cap (-\text{Div}^+(X)) = 0$ and $\mathfrak{Z}^{1+}(X) \cap (-\mathfrak{Z}^{1+}(X)) = 0$. One is thus reduced to proving that if D is a divisor on X such that $\text{mult}_x(D) \geq 0$ for every $x \in X^{(1)}$, one has $D \geq 0$. Now, for every $x \notin X^{(1)}$, the local ring $\mathcal{O}_{X,x}$ is integrally closed and of dimension 0 or ≥ 2 , hence one has $\text{prof}(\mathcal{O}_{X,x}) = 0$ or $\text{prof}(\mathcal{O}_{X,x}) \geq 2$ (0, 16.5.1). On the other hand, at the points $x \in X^{(1)}$, the ring $\mathcal{O}_{X,x}$ is a discrete valuation ring (II, 7.1.6), hence $\text{prof}(\mathcal{O}_{X,x}) = 1$ (0, 16.5.1); the only points of X such that $\text{prof}(\mathcal{O}_{X,x}) = 1$ are thus the points of $X^{(1)}$, and the conclusion results by (21.1.8).

(ii) The equivalence of a) and b) is clear from (i). By the characterisation of locally principal cycles given in (21.6.7), and the relation between 1-codimensional cycles on $\text{Spec}(A)$ and divisors (in the sense of Bourbaki) of a Noetherian integrally closed ring A (21.6.8), condition b) is equivalent to saying that for every $x \in X$, every divisor of the ring $\mathcal{O}_{X,x}$ is principal, in other words that the ring $\mathcal{O}_{X,x}$ is factorial (Bourbaki, *Alg. comm.*, chap. VII, §3, n° 1), whence the equivalence of b) and c).

(21.6.9.2). When A is a Noetherian factorial ring, it is clear that $X = \text{Spec}(A)$ is locally factorial, since every localisation of A is also locally factorial (Bourbaki, *Alg. comm.*, chap. VII, §3, n° 4, prop. 3). If, in this case, one writes an element $f \neq 0$ of the field of fractions K of A in the form r/s , where r and s are two coprime elements of A , then the divisors of r and s are well determined (*loc. cit.*, §3, n° 3), and these two divisors identify respectively with the divisor of zeros and the divisor of poles of f (21.6.7).

□

Corollary (21.6.10). — *Let X be a locally Noetherian normal prescheme.*

label=(i) There exists a canonical injective homomorphism

$$(21.6.10.1) \quad \text{Pic}(X) \longrightarrow \mathfrak{Cl}(X).$$

lbel=(ii) If X is locally factorial, the homomorphism (21.6.10.1) is bijective, and reciprocally.

Proof. One has seen (21.6.7) that the image of $\text{Div. princ}(X)$ by the homomorphism cyc is $\mathfrak{Z}_{\text{princ}}^1(X)$; one deduces thus from (21.6.9) that the homomorphism $\text{Div}(X) / \text{Div. princ}(X) \rightarrow \mathfrak{Z}^1(X) / \mathfrak{Z}_{\text{princ}}^1(X) = \mathfrak{Cl}(X)$ deduced from cyc is injective, and that it is bijective if and only if X is locally factorial. The conclusion results from this and from the fact that, when X is locally Noetherian and reduced, the canonical homomorphism $\text{Div}(X) / \text{Div. princ}(X) \rightarrow \text{Pic}(X)$ (21.3.3.1) is bijective (21.3.4, (ii)). □

Corollary (21.6.11). — *Let X be a locally Noetherian and locally factorial prescheme. Then the sheaf $\mathcal{D}iv_X$ is flasque, and for every open U of X , the canonical homomorphism $\text{Pic}(X) \rightarrow \text{Pic}(U)$ is surjective.*

Proof. Taking into account (21.6.9, (ii)), the first assertion is equivalent to saying that the sheaf $\mathcal{Z}_X^1$ is flasque, which was seen above (21.6.3). For every open U of X , the homomorphism $\mathfrak{Z}^1(X) \rightarrow \mathfrak{Z}^1(U)$ is thus surjective; since by virtue of (21.6.10), the homomorphism $\text{Pic}(X) \rightarrow \text{Pic}(U)$ identifies canonically with $\mathfrak{Z}^1(X) / \mathfrak{Z}_{\text{princ}}^1(X) \rightarrow \mathfrak{Z}^1(U) / \mathfrak{Z}_{\text{princ}}^1(U)$, it is also surjective. □

Proposition (21.6.12). — *Let X be a Noetherian reduced prescheme. Let $(U_\lambda)_{\lambda \in L}$ be a decreasing filtered family of opens of X satisfying the following conditions:*

- 1° *If $Y_\lambda = X - U_\lambda$, one has $\text{codim}(Y_\lambda, X) \geq 2$ for every $\lambda \in L$.*
- 2° *For every $x \in \bigcap_{\lambda \in L} U_\lambda$, the ring $\mathcal{O}_{X,x}$ is factorial.*

Then there exist canonical isomorphisms

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$$(21.6.12.1) \quad \varinjlim \text{Div}(U_\lambda) \xrightarrow{\sim} \mathfrak{Z}^1(X), \quad \varinjlim \text{Pic}(U_\lambda) \xrightarrow{\sim} \mathfrak{Cl}(X)$$

such that, for every open V of X , the diagrams

$$(21.6.12.2) \quad \begin{array}{ccc} \varinjlim \text{Div}(U_\lambda) & \xrightarrow{\sim} & \mathfrak{Z}^1(X) \\ \downarrow & & \downarrow \\ \varinjlim \text{Div}(U_\lambda \cap V) & \xrightarrow{\sim} & \mathfrak{Z}^1(V) \end{array} \quad \begin{array}{ccc} \varinjlim \text{Pic}(U_\lambda) & \xrightarrow{\sim} & \mathfrak{Cl}(X) \\ \downarrow & & \downarrow \\ \varinjlim \text{Pic}(U_\lambda \cap V) & \xrightarrow{\sim} & \mathfrak{Cl}(V) \end{array}$$

are commutative.

Proof. Hypothesis 1° on U_λ implies that for every $\lambda \in L$, one has $X^{(1)} \subset U_\lambda$ (5.1.3.1) hence the restriction homomorphism $\mathfrak{Z}^1(X) \rightarrow \mathfrak{Z}^1(U_\lambda)$ is bijective and therefore one has a canonical isomorphism $\varinjlim \mathfrak{Z}^1(U_\lambda) \xrightarrow{\sim} \mathfrak{Z}^1(X)$. The canonical homomorphisms $\text{cyc} : \text{Div}(U_\lambda) \rightarrow \mathfrak{Z}^1(U_\lambda)$ define thus, by passage to the inductive limit, the first of the canonical homomorphisms (21.6.12.1), and the second is deduced from it by passage to quotients. Furthermore, it results from hypothesis 1° that the U_λ are dense in X , hence schematically dense since X is reduced (11.10.4), and therefore every meromorphic function on X is entirely determined by its restriction to each U_λ ; one deduces immediately from this that in the isomorphism $\mathfrak{Z}^1(X) \xrightarrow{\sim} \mathfrak{Z}^1(U_\lambda)$, the image of $\mathfrak{Z}_{\text{princ}}^1(X)$ is $\mathfrak{Z}_{\text{princ}}^1(U_\lambda)$, hence one also has a canonical isomorphism $\varinjlim \mathfrak{C}l(U_\lambda) \xrightarrow{\sim} \mathfrak{C}l(X)$. The second of the canonical homomorphisms (21.6.12.1) is thus also an isomorphism if the first is, and it remains therefore to prove the commutativity of the diagrams (21.6.12.2), which is trivial.

Let us show first that the homomorphism $\varinjlim \text{Div}(U_\lambda) \rightarrow \mathfrak{Z}^1(X)$ is *injective*. Set $T = \bigcap_\lambda U_\lambda$, and note that the U_λ form a *fundamental system of neighbourhoods* of T . Indeed, for every open $V \supset T$, $X - V$ is a closed subset of X , hence a Noetherian space whose every closed irreducible subset admits a generic point; since the sets $(X - V) \cap (X - U_\lambda)$ are closed in $X - V$, form an increasing filtered family and have for union $X - V$, our assertion results from (0III, 9.2.4). This being so, it suffices to see that if $D \in \text{Div}(U_\lambda)$ is such that $\text{cyc}(D) = 0$ in $\mathfrak{Z}^1(U_\lambda)$, then there exists $\mu \geq \lambda$ such that $D|_{U_\mu} = 0$. By virtue of what precedes, it will suffice to see that for every $x \in T$, $D_x = 0$, and we are reduced to the case where $X = \text{Spec}(\mathcal{O}_{X,x})$ with $x \in T$; but by hypothesis $\mathcal{O}_{X,x}$ is factorial, hence $\text{Spec}(\mathcal{O}_{X,x})$ is normal; hence the conclusion results from (21.6.9, (i)).

To prove that the homomorphism $\varinjlim \text{Div}(U_\lambda) \rightarrow \mathfrak{Z}^1(X)$ is bijective, it will suffice to prove that for every $Z \in \mathfrak{Z}^1(X)$ and every $x \in T$, there exists a neighbourhood $W(x)$ of x in X and a divisor $D^{(x)}$ on $W(x)$ such that $\text{cyc}(D^{(x)}) = Z|_{W(x)}$. Indeed, one can then cover the quasi-compact set T by a finite number of the $W(x_i)$, and by virtue of the first part of the proof, since the couples (i, j) are finite in number, there will exist an index λ such that in each of the $W(x_i) \cap W(x_j) \cap U_\lambda$, the restrictions of $D^{(x_i)}$ and $D^{(x_j)}$ coincide; since one can suppose moreover that U_λ is contained in the union of the $W(x_i)$, one sees that the restrictions $D^{(x_i)}|_{(W(x_i) \cap U_\lambda)}$ are the restrictions of a single $D \in \text{Div}(U_\lambda)$ which will be such that $\text{cyc}(D) = Z|_{U_\lambda}$. We are thus reduced again to the case where $X = \text{Spec}(\mathcal{O}_{X,x})$ with $x \in T$; but by hypothesis $\mathcal{O}_{X,x}$ is factorial, hence the conclusion results from (21.6.9, (ii)). $\square$

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Corollary (21.6.13). — *Let A be an integral integrally closed Noetherian local ring of dimension ≥ 2 . Set $X = \text{Spec}(A)$, and let a be the closed point of X , $U = X - \{a\}$. For A to be factorial, it is necessary and sufficient that U be locally factorial and that $\text{Pic}(U) = 0$.*

Proof. Indeed, to say that A is factorial is equivalent to saying that $\mathfrak{C}l(X) = 0$ (Bourbaki, *Alg. comm.*, chap. VII, §1, n° 4, cor. of th. 2 and §3, n° 2, th. 1); it thus suffices to use the existence of the second isomorphism (21.6.12.1), taking the family (U_λ) restricted to the single open U . $\square$

Corollary (21.6.14). — *Let A be a Noetherian local ring of dimension ≥ 2 , $X = \text{Spec}(A)$, a the closed point of X , $U = X - \{a\}$. The following conditions are equivalent:*

label=a) *A is factorial.*

label=b) *$\text{Pic}(U) = 0$ and $\text{prof}(A) \geq 2$ (conditions that we will express later (21.13) by saying that the ring A is parafactorial), and U is locally factorial.*

Proof. It is clear that if A is factorial, it is normal, hence $\dim(A) \geq 2$ since $\text{prof}(A) \geq 2$ (0, 16.5.1) and (21.6.13) shows that a) implies b). Conversely, if b) is satisfied, it suffices to see that A is integrally closed to conclude by (21.6.13) that b) implies a). By virtue of Serre's criterion (5.8.6), it suffices to verify for X the conditions (R_1) and (S_2) . Now, U being locally factorial satisfies these conditions, and the hypothesis $\text{prof}(A) \geq 2$ implies that X also satisfies them. $\square$

21.7. Interpretation of positive 1-codimensional cycles in terms of subschemes

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(21.7.1). Let X be a locally Noetherian prescheme, $C = \sum_{x \in X^{(1)}} n_x \cdot \overline{\{x\}}$ a positive 1-codimensional cycle (so that one has $n_x \geq 0$ for every $x \in X^{(1)}$, and $n_x = 0$ except on a locally finite set of points). Denote by $Y(C)$ the closed subscheme of X , closed image (I, 9.5.3 and 9.5.1) of the sum prescheme $Y'(C) = \coprod_{x \in X^{(1)}} \text{Spec}(\mathcal{O}_{X,x}/\mathfrak{m}_x^{n_x})$ by the canonical morphism, and by $\mathcal{J}_X(C)$ (or $\mathcal{J}(C)$) the Ideal of $\mathcal{O}_X$ defining $Y(C)$.

Proposition (21.7.2). — *Let X be a locally Noetherian prescheme satisfying the condition (R_1) (5.8.2). For a closed subscheme Y of X to be of the form $Y(C)$, where C is a positive 1-codimensional cycle, it is necessary and sufficient that it satisfy the two following conditions:*

label=(i) Y is purely of codimension 1.

lbel=(ii) Y satisfies the condition (S_1) , in other words (5.7.5) has no associated immersed prime cycle.

One then has

$$(21.7.2.1) \quad \text{mult}_x(C) = \text{long } \mathcal{O}_{X,x}.$$

The application $C \mapsto Y(C)$ is a bijection of $\mathfrak{Z}^{+}(X)$ on the set of closed subschemes of X satisfying conditions (i) and (ii).

Proof. The conditions are necessary (without assuming that X satisfies (R_1)). Indeed, the question being local on X , one can suppose that $X = \text{Spec}(A)$, where A is a Noetherian ring; one then has $Y(C) = \text{Spec}(A/\mathfrak{J})$, where $\mathfrak{J}$ is, by definition (I, 9.5.1) the kernel of the canonical homomorphism $A \rightarrow \bigoplus A_{\mathfrak{p}_i}/\mathfrak{p}_i^{n_i} A_{\mathfrak{p}_i}$, where the $\mathfrak{p}_i$ are the prime ideals corresponding to $x \in X^{(1)}$ for which $n_x \geq 0$, and where one has set $n_i = n_x$. The reciprocal image $\mathfrak{q}_i$ of $\mathfrak{p}_i^{n_i} A_{\mathfrak{p}_i}$ in A is a $\mathfrak{p}_i$ -primary ideal and one has $\mathfrak{J} = \bigcap_i \mathfrak{q}_i$. Moreover, since the $x \in X^{(1)}$ are such that $\overline{\{x\}}$ is of codimension 1, no point of $X^{(1)}$ can be adherent to another point of $X^{(1)}$. Hence the $\mathfrak{p}_i$ are the minimal prime ideals of $A/\mathfrak{J}$ and the set of the $\mathfrak{p}_i$ is equal to $\text{Ass}(A/\mathfrak{J})$, which proves conditions (i) and (ii).

The conditions are sufficient; denoting by $\mathcal{J}$ the Ideal of $\mathcal{O}_X$ defining Y , hypothesis (ii) implies that $\text{Ass}(\mathcal{O}_X/\mathcal{J})$ is identical to the set F of maximal points of Y , and hypothesis (i) implies that F is contained in $X^{(1)}$; hence (3.2.6) $\mathcal{O}_X/\mathcal{J}$ identifies with a sub- $\mathcal{O}_X$ -Module of $\bigoplus_{x \in F} \mathcal{G}(x)$, where for each $x \in F$, $\mathcal{G}(x)$ is an irreducible $\mathcal{O}_X$ -Module such that $\text{Ass}(\mathcal{G}(x)) = \{x\}$. Now, since X satisfies (R_1) , for each $x \in F$, $\mathcal{O}_{X,x}$ is a discrete valuation ring and therefore the primary ideals $\neq 0$ of this ring are the powers $\mathfrak{m}_x^{n_x}$ of the maximal ideal; supposing further that $X = \text{Spec}(A)$, one concludes that $\mathcal{J} = \mathfrak{J}$, where $\mathfrak{J} = \bigcap_i \mathfrak{q}_i$, the $\mathfrak{q}_i$ being the reciprocal images of $\mathfrak{p}_i^{n_i} A_{\mathfrak{p}_i}$ at the corresponding points of F . One sees well that Y is of the form $Y(C)$. $\square$

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(21.7.3.1). When X is normal, $\text{Div}^+(X)$ identifies canonically with the set of closed subschemes Y of X satisfying conditions (i) and (ii) of (21.7.2) and regularly immersed in X .

Corollary (21.7.3). — *Let X be a locally Noetherian prescheme. For every positive divisor D on X such that X satisfies (R_1) at the maximal points of the closed subscheme $Y(D)$ (21.2.12), $Y(D)$ majorises the closed subscheme $Y(\text{cyc}(D))$ (21.7.1) and has the same underlying space; for these two subschemes to be equal, it is necessary and sufficient that $Y(D)$ satisfy the condition (S_1) , or equivalently that, for every $z \in Y(D)$ distinct from a maximal point of $Y(D)$, one has $\text{prof}(\mathcal{O}_{X,z}) \geq 2$ (a condition always satisfied when X is normal).*

Proof. Indeed, the question being local, one can always suppose that $\mathcal{J}_X(D) = t\mathcal{O}_X$, t being a regular section of $\mathcal{O}_X$ over X ; at every maximal point x of $Y(D)$, belonging necessarily to $X^{(1)}$, $\mathcal{O}_{X,x}$ is by hypothesis a discrete valuation ring, hence $t_x \mathcal{O}_{X,x} = \mathfrak{m}_{X,x}^{n_x}$, where $n_x = \text{mult}_x(D)$. One can suppose $X = \text{Spec}(A)$, and then, setting $\mathcal{J}_X(\text{cyc}(D)) = \tilde{\mathfrak{J}}$ and $\mathcal{J}_X(D) = \tilde{\mathfrak{J}}'$, it follows from the above and from (21.7.1) that $\tilde{\mathfrak{J}}$ and $\tilde{\mathfrak{J}}'$ have the same non-embedded primary ideals, hence $\tilde{\mathfrak{J}}' \subset \tilde{\mathfrak{J}}$, since $\tilde{\mathfrak{J}}$ is the intersection of these primary ideals (21.7.2). This proves that $Y(D)$ majorises $Y(\text{cyc}(D))$ and that these two subschemes are equal if and only if $Y(D)$ has no associated embedded prime cycle (in other words satisfies (S_1)). Since for every $x \in Y(D)$, there exists by hypothesis an open neighbourhood U of x in X and a regular element $t \in \Gamma(U, \mathcal{O}_X)$ such that $\mathcal{O}_{Y(D)}|_{(U \cap Y(D))}$ is

the restriction to $Y(D) \cap U$ of $\mathcal{O}_U/t\mathcal{O}_U$, to say that $Y(D)$ satisfies (S_1) also means that at every non-maximal point $z \in Y(D)$, one has $\text{prof}(\mathcal{O}_{X,z}/t_z\mathcal{O}_{X,z}) \geq 1$, that is to say (0, 16.4.6) $\text{prof}(\mathcal{O}_{X,z}) \geq 2$. The assertion concerning the case where X is normal is then immediate since X satisfies (S_2) and at non-maximal points z of $Y(D)$ one has $\dim(\mathcal{O}_{X,z}) \geq 2$ (0, 16.3.4). $\square$

Proposition (21.7.4). — *Let X be a locally Noetherian prescheme, reduced at each of its isolated points. The following conditions are equivalent:*

label=a) *The canonical homomorphism $c : \mathcal{D}iv_X \rightarrow \mathcal{Z}_X^1$ (21.6.4) is an isomorphism of sheaves of ordered groups, and on $\text{Div}(X) = \text{Div. princ}(X)$.*

label=a') *The canonical homomorphism $c : \mathcal{D}iv_X \rightarrow \mathcal{Z}_X^1$ is injective, and every positive 1-codimensional cycle is the image by cyc of a positive divisor principal, in other words is of the form $\text{cyc}(\text{div}(f))$, where f is a regular section of $\mathcal{O}_X$ over X .*

label=a'') *For every closed integral subprescheme Y of X , of codimension 1, there exists a regular section f of $\mathcal{O}_X$ over X such that Y is defined by the Ideal $f\mathcal{O}_X$.*

b) *X is normal and every 1-codimensional cycle on X is principal.*

c) *X is locally factorial and $\text{Pic}(X) = 0$.*

d) *X is normal, and for every open U of X , one has $\text{Pic}(U) = 0$.*

Proof. The equivalence of a), b) and c) results immediately from (21.7.4) and from (21.6.11). Furthermore, it results immediately from c) that every non-empty open U of X satisfies also the same conditions, so that c) implies d). It remains to see that d) implies b). Now, denote by U_λ the opens that are complementary of finite unions of closed sets of the form $\overline{\{x\}}$, where $\dim(\mathcal{O}_{X,x}) \geq 2$; it is clear that the U_λ form a decreasing filtered family and that for every $x \in \bigcap_\lambda U_\lambda$, one has $\dim(\mathcal{O}_{X,x}) \leq 1$, hence $\mathcal{O}_{X,x}$ is a principal ring under the hypothesis. One can thus apply to the family (U_λ) the proposition (21.6.12), which shows that $\mathcal{C}l(X)$ is isomorphic to $\varinjlim \text{Pic}(U_\lambda)$, hence $\mathcal{C}l(X) = 0$ by virtue of the hypothesis, which proves b) by definition. $\square$

Remark (21.7.7). — When $X = \text{Spec}(A)$, where A is an integral Noetherian ring, the conditions equivalent of (21.7.6) are equivalent to saying that A is a *factorial* ring.

21.8. Divisors and normalisation

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Lemma (21.8.1). — *Let $f : X' \rightarrow X$ be an integral morphism of preschemes. For every $\mathcal{O}_{X'}$ -Module locally free $\mathcal{E}'$ of constant rank n and every $x \in X$, there exists an open neighbourhood U of x in X such that, if one sets $U' = f^{-1}(U)$, $\mathcal{E}'|_{U'}$ is isomorphic to $\mathcal{O}_{U'}^n$.*

Proof. The question being local on X , one can restrict to the case where $X = \text{Spec}(A)$ is affine, in which case $X' = \text{Spec}(A')$, where A' is an integral A -algebra. Then A' is the inductive limit of its finite sub- A -algebras A'_λ . Setting $X'_\lambda = \text{Spec}(A'_\lambda)$ and denoting by $\rho_\lambda : X' \rightarrow X'_\lambda$ the structural morphism, it results from (8.5.2) and (8.5.5) that there exists an index λ and an $\mathcal{O}_{X'_\lambda}$ -Module locally free $\mathcal{E}'_\lambda$ of rank n such that $\mathcal{E}' = \rho_\lambda^*(\mathcal{E}'_\lambda)$, and it will suffice to prove the lemma for X'_λ and $\mathcal{E}'_\lambda$; in other words, one can restrict to the case where the morphism f is *finite*. Setting $B = \mathcal{O}_{X,x}$ and let B' be the ring of the affine scheme $X'_0 = X' \times_X \text{Spec}(\mathcal{O}_{X,x})$; since B is a local ring and B' is a finite B -algebra, B' is a semi-local ring (Bourbaki, *Alg. comm.*, chap. V, §2, n° 1, prop. 3); one concludes that the $\mathcal{O}_{X'_0}$ -Module locally free $\mathcal{E}'_0 = \mathcal{E}' \otimes_{\mathcal{O}_{X'}} \mathcal{O}_{X,x}$ is isomorphic to $\mathcal{O}_{X'_0}^n$ (Bourbaki, *Alg. comm.*, chap. II, §5, n° 3, prop. 5). Considering X'_0 as the projective limit of preschemes induced by X' on the open affine neighbourhoods of x in X , following the method of (8.1.2, a)), and applying (8.5.2.5), one obtains the sought conclusion. $\square$

Corollary (21.8.2). — *Let $f : X' \rightarrow X$ be an integral morphism of preschemes. Then:*

label=(i) $R^1 f_*(\mathcal{O}_{X'}^*) = 0$ (0III, 12.2.1).

label=(ii) *The group $\text{Pic}(X')$ is canonically isomorphic to $H^1(X, f_*(\mathcal{O}_{X'}^*))$.*

Proof. (i) $R^1 f_*(\mathcal{O}_{X'}^*)$ is the sheaf associated to the presheaf of commutative groups $U \mapsto H^1(f^{-1}(U), \mathcal{O}_{X'}^*)$ (*loc. cit.*); the fibre of this sheaf at a point x can therefore (0I, 5.4.7) be identified with the commutative group $\varinjlim \text{Pic}(f^{-1}(U))$, where U ranges over the open neighbourhoods of x , the transition homomorphisms $\text{Pic}(f^{-1}(U')) \rightarrow \text{Pic}(f^{-1}(U))$ for two open $U \subset U'$ being defined by (21.3.2.4).

Now, for every $x \in X$, every open neighbourhood U' of x in X , and every $\zeta \in \text{Pic}(f^{-1}(U'))$, it results from (21.8.1) that there exists an open $U \subset U'$ of x such that the image of ζ in $\text{Pic}(f^{-1}(U))$ is zero. Hence the fibre of $R^1 f_*(\mathcal{O}_{X'}^*)$ at the point x is zero.

(ii) The Leray spectral sequence for the composed functor $\mathcal{F}' \mapsto \Gamma(X, f_*(\mathcal{F}'))$ of sheaves of commutative groups on X' (T, 2.4) gives the exact sequence of low degree terms

$$0 \longrightarrow H^1(X, f_*(\mathcal{O}_{X'}^*)) \longrightarrow H^1(X', \mathcal{O}_{X'}^*) \longrightarrow H^0(X, R^1 f_*(\mathcal{O}_{X'}^*))$$

and the conclusion results from (i) and from the isomorphism of $\text{Pic}(X')$ and $H^1(X', \mathcal{O}_{X'}^*)$ (0I, 5.4.7). $\square$

Proposition (21.8.3). — *Let $f = (\psi, \theta) : X' \rightarrow X$ be an integral morphism of preschemes; suppose that, for every open U of X , the homomorphism $\Gamma(\theta) : \Gamma(U, \mathcal{O}_X) \rightarrow \Gamma(U, f_*(\mathcal{O}_{X'}^*))$ transforms regular elements in regular elements, which allows to canonically prolong the homomorphism $\theta : \mathcal{O}_X \rightarrow f_*(\mathcal{O}_{X'}^*)$ in a homomorphism of sheaves of rings $\theta' : \mathcal{M}_X \rightarrow f_*(\mathcal{M}_{X'})$, whence homomorphisms of sheaves of multiplicative groups $\theta^* : \mathcal{O}_X^* \rightarrow f_*(\mathcal{O}_{X'}^*)$ and $\theta'^* : \mathcal{M}_X^* \rightarrow f_*(\mathcal{M}_{X'}^*)$, giving by passage to quotients a homomorphism $\theta''' : \mathcal{D}iv_X \rightarrow f_*(\mathcal{D}iv_{X'})$. One then has a commutative diagram*

$$(21.8.3.1) \quad \begin{array}{ccccccc} 1 & \longrightarrow & \mathcal{O}_X^* & \longrightarrow & \mathcal{M}_X^* & \longrightarrow & \mathcal{D}iv_X \longrightarrow 0 \\ & & \downarrow \theta^* & & \downarrow \theta'^* & & \downarrow \theta''' \\ 1 & \longrightarrow & f_*(\mathcal{O}_{X'}^*) & \longrightarrow & f_*(\mathcal{M}_{X'}^*) & \longrightarrow & f_*(\mathcal{D}iv_{X'}) \longrightarrow 0 \end{array}$$

in which the two rows are exact.

Proof. The only point to verify is the exactness of the second row of the diagram, which results from the application of the exact cohomology sequence for the functor f_* to the short exact sequence of sheaves of commutative groups on X'

$$1 \longrightarrow \mathcal{O}_{X'}^* \longrightarrow \mathcal{M}_{X'}^* \longrightarrow \mathcal{D}iv_{X'} \longrightarrow 0$$

and from (21.8.2). $\square$

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Corollary (21.8.4). — *If, in addition to the hypotheses of (21.8.3), the homomorphism θ' is an isomorphism of sheaves of rings, then $\theta^* : \mathcal{O}_X^* \rightarrow f_*(\mathcal{O}_{X'}^*)$ is injective, $\theta''' : \mathcal{D}iv_X \rightarrow f_*(\mathcal{D}iv_{X'})$ is surjective and $\text{Ker}(\theta''')$ is isomorphic to $\text{Coker}(\theta^*)$.*

Proof. This is an immediate consequence of the snake diagram (Bourbaki, Alg. comm., chap. I, §2, n° 4, prop. 2) applied to diagram (21.8.3.1). $\square$

Proposition (21.8.5). — *Let X, X' be two locally Noetherian preschemes, $f : X' \rightarrow X$ an integral morphism; suppose that there exists a schematically dense open U in X such that $f^{-1}(U)$ is schematically dense in X' (cf. (20.3.5)), and that the morphism $f^{-1}(U) \rightarrow U$, restriction of f , is an isomorphism. Then:*

label=(i) *The homomorphism $\theta' : \mathcal{M}_X \rightarrow f_*(\mathcal{M}_{X'})$ is defined and bijective, the homomorphism*

$$\theta^* : \mathcal{O}_X^* \longrightarrow f_*(\mathcal{O}_{X'}^*)$$

is injective, the homomorphism $\theta''' : \mathcal{D}iv_X \rightarrow f_(\mathcal{D}iv_{X'})$ is surjective, $\text{Ker}(\theta''')$ is isomorphic to $\mathcal{N} = \text{Coker}(\theta^*)$ and the support of the sheaf of multiplicative groups $\mathcal{N}$ is contained in $S = X - U$.*

lbbel=(ii) *Suppose moreover the set S is discrete and set for brevity $\mathcal{O}_X^l = f_*(\mathcal{O}_{X'})$. Then one has a commutative diagram*

$$(21.8.5.1) \quad \begin{array}{ccccccc} 1 & \longrightarrow & \prod_{s \in S} \mathcal{O}_{X,s}^* / \mathcal{O}_{X,s}^* & \xrightarrow{j} & \text{Div}(X) & \longrightarrow & \text{Div}(X') \longrightarrow 0 \\ & & \downarrow & & \downarrow l_X & & \downarrow l_{X'} \\ 1 & \longrightarrow & \left(\prod_{s \in S} \mathcal{O}_{X,s}^* / \mathcal{O}_{X,s}^* \right) / \text{Im } j' & \xrightarrow{\bar{j}} & \text{Pic}(X) & \longrightarrow & \text{Pic}(X') \longrightarrow 0 \end{array}$$

where the two rows are exact and the left vertical arrow is the canonical homomorphism.

Proof. (i) The hypothesis implies that one has a canonical isomorphism $\mathcal{M}_X^* \xrightarrow{\sim} f_*(\mathcal{M}_{X'}^*)$ for the sheaves of germs of pseudo-functions (20.2.2), whence the assertion relative to θ' , the assertion relative to θ' following from (20.2.11). The rest of assertion (i) follows from (21.8.4), except for what concerns the support of $\mathcal{N}$, which results directly from the hypothesis on U .

(ii) Applying the exact cohomology sequence to the two short exact sequences of sheaves of commutative groups

$$(21.8.5.2) \quad \begin{aligned} 1 &\longrightarrow \mathcal{N} \longrightarrow \mathcal{D}iv_X \longrightarrow f_*(\mathcal{D}iv_{X'}) \longrightarrow 0 \\ 1 &\longrightarrow \mathcal{O}_X^* \longrightarrow f_*(\mathcal{O}_{X'}^*) \longrightarrow \mathcal{N} \longrightarrow 1 \end{aligned}$$

one obtains respectively the exact sequences

$$(21.8.5.3) \quad \begin{aligned} 1 &\longrightarrow \Gamma(X, \mathcal{N}) \xrightarrow{j} \text{Div}(X) \longrightarrow \text{Div}(X') \longrightarrow H^1(X, \mathcal{N}) \\ \Gamma(X', \mathcal{O}_{X'}^*) &\longrightarrow \Gamma(X, \mathcal{N}) \xrightarrow{\beta} \text{Pic}(X) \longrightarrow \text{Pic}(X') \longrightarrow H^1(X, \mathcal{N}) \end{aligned}$$

taking into account the canonical isomorphism $\text{Pic}(X') \xrightarrow{\sim} H^1(X, f_*(\mathcal{O}_{X'}^*))$ (21.8.2). Now, since the support of $\mathcal{N}$ is contained in the discrete set S , closed in X , one has $H^1(X, \mathcal{N}) = 0$, and the conclusion results from (i) and from the isomorphism of $\text{Pic}(X')$ and $H^1(X, f_*(\mathcal{O}_{X'}^*))$ (21.8.2). $\square$

21.9. Divisors on preschemes of dimension ≤ 1

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(21.9.1). Let X be a topological space, x a point of X , $i_x : \{x\} \rightarrow X$ the canonical injection. If $A(x)$ is a commutative group, one can consider it as a sheaf of commutative groups on the space $\{x\}$ reduced to a single point, and take its direct image $(i_x)_*(A(x))$ which is a sheaf of commutative groups on X (0_I, 3.4.1); it results immediately from the definitions that for every open U of X , $\Gamma(U, (i_x)_*(A(x))) = A(x)$ if $x \in U$, and is reduced to 0 in the contrary case; hence, for $y \in \overline{\{x\}}$, $((i_x)_*(A(x)))_y = A(x)$, and for $y \notin \overline{\{x\}}$, $((i_x)_*(A(x)))_y = 0$.

If now $\mathcal{F}$ is a sheaf of commutative groups on X and Y a subset of X , for every open U of X , one has a canonical homomorphism

$$(21.9.1.1) \quad \Gamma(U, \mathcal{F}) \longrightarrow \prod_{x \in U \cap Y} \mathcal{F}_x = \prod_{x \in Y} \Gamma(U, (i_x)_*(\mathcal{F}_x))$$

and since these homomorphisms commute with restriction operators, they define a canonical homomorphism of sheaves

$$(21.9.1.2) \quad j_Y : \mathcal{F} \longrightarrow \prod_{x \in Y} (i_x)_*(\mathcal{F}_x).$$

Lemma (21.9.2). — Let X be a locally Noetherian topological space, X_0 the set of its closed points, $\mathcal{F}$ a sheaf of commutative groups on X . The following conditions are equivalent:

label=a) The canonical homomorphism $j_{X_0} : \mathcal{F} \rightarrow \prod_{x \in X_0} (i_x)_*(\mathcal{F}_x)$ is injective and has for image $\bigoplus_{x \in X_0} (i_x)_*(\mathcal{F}_x)$.

label=a') There exists a family of commutative groups $(A(x))_{x \in X_0}$ such that $\mathcal{F}$ is isomorphic to $\bigoplus_{x \in X_0} (i_x)_*(A(x))$.

b) Every section of $\mathcal{F}$ over an open U of X has discrete support contained in X_0 .

When b) is thus satisfied, for every $x \in X_0$, the group $A(x)$ is determined up to unique isomorphism and is isomorphic to $\mathcal{F}_x$. Furthermore, the sheaf $\mathcal{F}$ is then flasque.

Proof. Since the points of X_0 are closed in X , one has $((i_x)_*(A(x)))_y = 0$ for every $y \neq x \in X_0$; this remark proves the uniqueness assertion relative to the groups $A(x)$, and it is clear moreover that a) implies a'). To see that a') implies b), one can restrict to the case where U is Noetherian; then (G, II, 3.10) one has

$$\Gamma(U, \bigoplus_{x \in X_0} (i_x)_*(A(x))) = \bigoplus_{x \in X_0} \Gamma(U, (i_x)_*(A(x))),$$

hence every section s of $\mathcal{F}$ over U is a direct sum of a finite number of sections $s_k \in \Gamma(U, (i_{x_k})_*(A(x_k)))$ ($1 \leq k \leq n$) with $x_k \in X_0$. Since x_k is a closed point of X , the support of s_k is reduced to x_k , hence the support of s is contained in the finite set of the x_k , which is evidently discrete since the points x_k are closed in X . Let us show finally that b) implies a) and that $\mathcal{F}$ is flasque.

Let us show that $\mathcal{F}$ is flasque. Consider a section s of $\mathcal{F}$ over an open U of X ; since the support of s is a discrete closed subset of X , we prolong s to a section s' of $\mathcal{F}$ over X by setting $s'_x = 0$ in $X - U$, whence our assertion. $\square$

Proposition (21.9.4). — *Let X be a locally Noetherian prescheme, $X^{(1)}$ the set of points $x \in X$ such that $\dim(\mathcal{O}_{X,x}) = 1$. One then has a canonical isomorphism*

$$(21.9.4.1) \quad \mathcal{D}iv_X \xrightarrow{\sim} \bigoplus_{x \in X^{(1)}} (i_x)_*(\text{Div}(\mathcal{O}_{X,x}))$$

and $\mathcal{D}iv_X$ is flasque.

Proof. Taking into account the isomorphism (21.4.6.1), the homomorphism (21.9.4.1) is defined by (21.9.1.1); let us prove that c is a bijection. Since $\dim(X) \leq 1$, the points of $X^{(1)}$ are the *non-isolated closed points* of X . One is reduced to proving that: 1° $\mathcal{D}iv_X$ satisfies condition b) of (21.9.2); 2° for every isolated point $x \in X$, one has $(\mathcal{D}iv_X)_x = 0$. The second point results from the fact that $\mathcal{O}_{X,x}$ is an Artinian ring and that every regular element of $\mathcal{O}_{X,x}$ is invertible. For the 1°, it suffices to note that for every open U of X , every divisor $D \in \text{Div}(U)$, the maximal points of the support S of D are such that $\text{prof}(\mathcal{O}_{X,x}) = 1$ (21.1.9), hence *a fortiori* belong to $X^{(1)} \cap U$; since $\dim(X) \leq 1$, the set of these points is equal to S , and S is thus discrete, the set of irreducible components of S being locally finite. $\square$

Corollary (21.9.5). — *Let X be a locally Noetherian prescheme of dimension ≤ 1 . For every discrete subset $E \subset X^{(1)}$ and every family $(D(x))_{x \in E}$ with $D(x) \in \text{Div}(\mathcal{O}_{X,x})$, there exists a divisor D on X and only one such that the support of D is contained in E and that $D_x = D(x)$ for every $x \in E$.*

Proof. This results from (21.9.4) and from (21.9.3, (ii)). $\square$

Corollary (21.9.6). — *Let X be a locally Noetherian prescheme of dimension ≤ 1 ; every divisor D on X is of the form $D' - D''$, where D', D'' are two divisors ≥ 0 of disjoint supports.*

Proof. By virtue of (21.9.5), one is immediately reduced to the case where $X = \text{Spec}(A)$, where A is a Noetherian local ring; it then suffices to observe that $M(X)$ is the total ring of fractions of A , and that a section of $\mathcal{M}_X^*$ over X can be written as a quotient b/a of two regular elements of A . $\square$

Corollary (21.9.7). — *Let X be a locally Noetherian prescheme of dimension ≤ 1 , without isolated point, and U an open dense in X . There then exists a divisor $D \geq 0$ on X , of support contained in U and meeting each of the irreducible components of X .*

Proof. One can suppose that U is the union of disjoint non-empty opens U_α , each of which is contained in a single irreducible component of X ; it suffices then to take in each U_α a closed point x_α (there exists one since the unique generic point of U_α cannot be isolated by hypothesis), and to apply (21.9.5) to the discrete set of the x_α . $\square$

(21.9.8). For locally Noetherian preschemes X of dimension ≤ 1 , the proposition (21.9.4) reduces the determination of $\mathcal{D}iv_X$ to the case where $X = \text{Spec}(A)$, A being a Noetherian local ring of dimension 1.

When A is a regular local ring of dimension 1 (in other words, a discrete valuation ring), the group $\text{Div}(A)$ identifies canonically with $\mathbf{Z}$ (21.6.8); hence, by virtue of (21.9.2):

Proposition (21.9.9). — *Let X be a locally Noetherian regular prescheme of dimension ≤ 1 . Then the sheaf $\mathcal{D}iv_X$ is canonically isomorphic to $\bigoplus_{x \in X^{(1)}} (i_x)_*(\mathbf{Z}(x))$, where $\mathbf{Z}(x)$ is the additive group $\mathbf{Z}$ considered as a sheaf of groups on the space $\{x\}$.*

Proof. By (21.9.4), the determination of $\mathcal{D}iv_X$ reduces to the case where $X = \text{Spec}(A)$, A being a local Noetherian ring of dimension 1. When A is a regular local ring of dimension 1 (i.e. a discrete valuation ring), $\text{Div}(A)$ identifies canonically with $\mathbf{Z}$ (21.6.8); the conclusion then follows from (21.9.2). $\square$

Proposition (21.9.11). — *Let X be a Noetherian prescheme, X_0 a closed subprescheme of X having the following properties:*

1° $\dim(X_0) \leq 1$.

2° For every locally closed subset Y of X such that $Y \cap X_0$ is discrete, there exists a part Y' of Y , closed in X and open in Y , containing $Y \cap X_0$.

Under these conditions:

- label=(i) Let Z_0 be the union of the sets $\overline{\{x\}} \cap X_0$ when x ranges over the part of $\text{Ass}(\mathcal{O}_X)$ formed of points such that $\{x\} \cap X_0$ is finite. Then, for every divisor D_0 on X_0 , whose support does not meet Z_0 , there exists a divisor D on X whose reciprocal image by the canonical injection $X_0 \rightarrow X$ exists (21.4.5) and is equal to D_0 ; if moreover $D_0 \geq 0$, one can suppose $D \geq 0$.
- lbbel=(ii) Suppose in addition that there exists in X_0 an affine open U_0 containing $(\text{Ass}(\mathcal{O}_{X_0})) \cup Z_0$ (condition automatically satisfied when there exists an $\mathcal{O}_{X_0}$ -Module ample (II, 4.5.4)). Then the canonical homomorphism (21.3.2.4) $\text{Pic}(X) \rightarrow \text{Pic}(X_0)$ is surjective.

Proof. (i) In view of (21.9.6), one can restrict to proving the assertion relative to divisors $D_0 \geq 0$; by virtue of (21.9.4), the support T of D_0 is a finite discrete closed subset of X_0 . One can suppose $D_0 \neq 0$, i.e. $T \neq \emptyset$, since otherwise there is nothing to prove. For every $x \in T$, there exists an element $s_x \in \mathcal{O}_{X_0, x}$ which is a non-zero-divisor, belonging to its maximal ideal, and whose image in $(\mathcal{D}iv_{X_0})_x$ equals $(D_0)_x$. There exists an affine open neighbourhood $U^{(x)}$ of x in X and a section $g^{(x)}$ of $\mathcal{O}_X$ over $U^{(x)}$ such that s_x is the image of the germ $(g^{(x)})_x \in \mathcal{O}_{X, x}$; by taking $U^{(x)}$ sufficiently small, one can arrange that $g^{(x)}$ is not a zero divisor of $\Gamma(U^{(x)}, \mathcal{O}_X)$, in other words ((3.1.9)), denoting by $V^{(x)}$ the closed part of $U^{(x)}$ formed of the y such that $g^{(x)}(y) = 0$, that $V^{(x)} \cap \text{Ass}(\mathcal{O}_X) = \emptyset$. Indeed, if $z \in \text{Ass}(\mathcal{O}_X)$, the hypothesis $x \notin Z_0$ implies either $x \notin \overline{\{z\}}$, or x is not isolated in $\overline{\{z\}} \cap X_0$. In the first case, by replacing $U^{(x)}$ by a smaller neighbourhood, one can suppose $V^{(x)} \cap \text{Ass}(\mathcal{O}_X) = \emptyset$.

By the choice of $U^{(x)}$, one can define a divisor $D'^{(x)}$ on $U^{(x)}$ by $D'^{(x)} = \text{div}(g^{(x)})$; moreover, as one saw above, x is necessarily isolated in $V^{(x)} \cap X_0$, so by replacing $U^{(x)}$ by a smaller open, one can suppose $V^{(x)} \cap X_0 = \{x\}$. But then by condition 2°, there exists $W^{(x)}$, a part of $V^{(x)}$, closed in X and open in $V^{(x)}$, such that $W^{(x)} \cap X_0 = \{x\}$. Replacing $U^{(x)}$ by a smaller open, one can suppose $W^{(x)} = V^{(x)}$, i.e. $V^{(x)}$ is closed in X . One can then define a divisor $D^{(x)}$ on X by $D^{(x)}|_{U^{(x)}} = D'^{(x)}$, and $D^{(x)}|(X - V^{(x)}) = 0$, which makes sense since the restriction of $g^{(x)}$ is an invertible section on $U^{(x)} \cap (X - V^{(x)})$, so the restrictions of the two divisors to this open are equal. It then suffices to take $D = \sum_{x \in T} D^{(x)}$, which answers the question since T is finite.

(ii) In view of the commutative diagram (21.4.2.1), the conclusion will follow from (i) if one proves that every invertible $\mathcal{O}_{X_0}$ -Module $\mathcal{L}_0$ is isomorphic to an $\mathcal{O}_{X_0}$ -Module $\mathcal{O}_{X_0}(D_0)$ (21.2.8), where D_0 is a divisor on X_0 whose support does not meet Z_0 . Now, since U_0 is schematically dense in X_0 (20.2.13, (iv)), it suffices to prove that there exists $t \in \Gamma(U_0, \mathcal{L}_0)$ such that $t(z_0) \neq 0$ at the points of $(\text{Ass}(\mathcal{O}_{X_0})) \cup Z_0$. One can therefore restrict to the case where $X_0 = \text{Spec}(A_0)$ is affine; but then $\mathcal{L}_0$ is ample (II, 5.1.4) and since $\text{Ass}(\mathcal{O}_{X_0}) \cup Z_0$ is finite, the conclusion follows from (II, 4.5.4). $\square$

Corollary (21.9.12). — Let A be a henselian local ring (18.5.8), $S = \text{Spec}(A)$, s_0 the closed point of S , $f : X \rightarrow S$ a separated morphism of finite presentation, and suppose that the set $X_0 = f^{-1}(s_0)$ is of dimension ≤ 1 . Then, for every closed subscheme X'_0 of X , having X_0 as underlying set and of finite presentation over S , the canonical homomorphism (21.3.2.4) $\text{Pic}(X) \rightarrow \text{Pic}(X'_0)$ is surjective. IV-4 | 288

Proof. Let us first show that it suffices to prove the corollary when the henselian local ring A is moreover Noetherian. Indeed ((18.6.15)), one can write $A = \varinjlim A_\lambda$, where the A_λ are Noetherian henselian local rings, the homomorphisms $A_\lambda \rightarrow A$ being local; there moreover exist an index α and a separated morphism of finite presentation $f_\alpha : X_\alpha \rightarrow S_\alpha = \text{Spec}(A_\alpha)$ such that X and f are deduced from X_α and f_α by the base change $S \rightarrow S_\alpha$ ((8.10.5), (v)); with the usual notation ((8.8.1)), the morphisms $f_\lambda : X_\lambda \rightarrow S_\lambda$ will be separated for $\lambda \geq \alpha$ and one will have $X = X_\lambda \times_{S_\lambda} S$. Furthermore, one can suppose that, if $s_{0\alpha}$ is the unique closed point of S_α , there is a closed subscheme $X'_{0\alpha}$ of X_α , having the same underlying set as $f_\alpha^{-1}(s_{0\alpha})$, such that $X'_0 = X'_{0\alpha} \times_{S_\alpha} S$ ((8.6.3)); one has, for $\lambda \geq \alpha$, $\dim(X'_{0\lambda}) = \dim(X'_0) \leq 1$ by transitivity of fibres and (4.1.4). It suffices to see that if $\text{Pic}(X_\lambda) \rightarrow \text{Pic}(X'_{0\lambda})$ is surjective for all $\lambda \geq \alpha$, the same is true of $\text{Pic}(X) \rightarrow \text{Pic}(X'_0)$; this follows

from the commutative diagram

$$\begin{array}{ccc} \mathrm{Pic}(X_\lambda) & \longrightarrow & \mathrm{Pic}(X) \\ \downarrow & & \downarrow \\ \mathrm{Pic}(X'_{0\lambda}) & \longrightarrow & \mathrm{Pic}(X'_0) \end{array}$$

Indeed, for every invertible $\mathcal{O}_{X'_0}$ -Module $\mathcal{L}'_0$, there exist $\lambda \geq \alpha$ and an invertible $\mathcal{O}_{X'_{0\lambda}}$ -Module $\mathcal{L}'_{0\lambda}$ such that $\mathcal{L}'_0$ is deduced from it by base change $S \rightarrow S_\lambda$ ((8.5.2) and (8.5.5)); it suffices to consider an invertible $\mathcal{O}_{X_\lambda}$ -Module $\mathcal{L}_\lambda$ whose class maps to $\mathrm{cl}(\mathcal{L}'_{0\lambda})$, then to take the $\mathcal{O}_X$ -Module $\mathcal{L}$ deduced from $\mathcal{L}_\lambda$ by base change.

Suppose therefore A Noetherian, hence also X , and let us verify that X and X_0 satisfy the conditions of (21.9.11), (ii). One has by hypothesis $\dim(X'_0) \leq 1$; on the other hand, to verify condition 2° of (21.9.11), consider a locally closed subscheme Y of X such that $Y \cap X_0$ is discrete, i.e. the morphism $g : Y \rightarrow S$, restriction of f , is quasi-finite at each of the points x_i of $Y \cap X_0$ ($1 \leq i \leq n$); one can apply ((18.5.11), c)) and one sees that Y is the sum of open subschemes $Y_i = \mathrm{Spec}(\mathcal{O}_{Y, x_i})$ ($1 \leq i \leq n$) and of a prescheme Y'' such that $Y'' \cap X_0 = \emptyset$, and moreover the canonical injections $j_i : Y_i \rightarrow X$ are such that $f \circ j_i$ is a finite morphism, hence Y_i is closed in X ; one answers the question by taking $Y' = \bigcup_{1 \leq i \leq n} Y_i$.

It remains to verify the supplementary hypothesis of (21.9.11), (ii). Now, X'_0 being a separated curve over the field $k(s_0)$, is a $k(s_0)$ -scheme quasi-projective (chap. V)¹⁵, hence there exists an ample $\mathcal{O}_{X'_0}$ -Module (II, 5.3.1), and this finishes the proof. $\square$

21.10. Reciprocal and direct images of 1-codimensional cycles In a later chapter dedicated to intersection theory, the notions of inverse and direct image of cycles will be developed systematically. In the present section, we will content ourselves with defining these notions in certain particular cases useful, and for 1-codimensional cycles, these definitions being chosen in such a way that they are compatible with the corresponding definitions for divisors (21.4) and (21.5), taking into account the homomorphism cyc defined in (21.6).

(21.10.1). Let X, X' be two locally Noetherian preschemes, $f : X' \rightarrow X$ a morphism, T a part of X ; suppose that the image by f of every maximal point of X' is either a maximal point of X , and that, for every $x' \in X'^{(1)}$ (i.e. such that $\dim \mathcal{O}_{X', x'} = 1$), setting $x = f(x')$, one has one of the three following conditions:

- 1° $x \notin T$, one takes $n_{x'} = 0$;
- 2° $x \in X^{(1)}$ and $\mathcal{O}_{X', x'}$ is a flat $\mathcal{O}_{X, x}$ -module, one knows (6.1.1) that $\mathcal{O}_{X', x'} / \mathfrak{m}_x \mathcal{O}_{X', x'}$ is an $\mathcal{O}_{X', x'}$ -module of finite length $\lambda_{x'}$, and one takes $n_{x'} = \lambda_{x'} n_x$;
- 3° $\mathcal{O}_{X, x}$ is factorial and $\mathfrak{m}_x \notin \mathrm{Ass}(\mathcal{O}_{X', x'})$, one knows (21.6.9) that the canonical homomorphism $\mathrm{cyc} : \mathrm{Div}(\mathcal{O}_{X, x}) \rightarrow \mathfrak{Z}^1(\mathrm{Spec}(\mathcal{O}_{X, x}))$ is bijective, and moreover since $\dim(\mathcal{O}_{X', x'}) = 1$ and $\mathrm{Ass}(\mathcal{O}_{X', x'})$ is composed uniquely of the maximal points of $\mathrm{Spec}(\mathcal{O}_{X', x'})$, x' being the unique closed point of $\mathrm{Spec}(\mathcal{O}_{X', x'})$, $\mathfrak{Z}^1(\mathrm{Spec}(\mathcal{O}_{X', x'}))$ is canonically isomorphic to $\mathbf{Z}$.

Under these conditions, we propose, for every 1-codimensional cycle Z on X of support contained in T , to define a 1-codimensional cycle $Z' = f^*(Z)$, so that $Z \mapsto f^*(Z)$ is an homomorphism of ordered groups of the subgroup of $\mathfrak{Z}^1(X)$ of support contained in T , in the ordered group $\mathfrak{Z}^1(X')$. For this, let

$$(21.10.1.1) \quad Z = \sum_{x \in T \cap X^{(1)}} n_x \cdot \overline{\{x\}}$$

where the family of the $x \in T \cap X^{(1)}$ such that $n_x \neq 0$ is locally finite. For every $x' \in X'^{(1)}$, define an integer $n_{x'}$ in the following way, setting $x = f(x')$: in case 1° take $n_{x'} = 0$; in case 2° take $n_{x'} = \lambda_{x'} n_x$; in case 3°, since $\mathcal{O}_{X, x}$ is factorial and $\mathfrak{m}_x \notin \mathrm{Ass}(\mathcal{O}_{X', x'})$, the homomorphism $f^* : \mathrm{Div}(\mathcal{O}_{X, x}) \rightarrow \mathrm{Div}(\mathcal{O}_{X', x'})$ is defined, and it results from (21.4.5, (ii)) that the homomorphism $f^* : \mathrm{Div}(\mathcal{O}_{X, x}) \rightarrow \mathrm{Div}(\mathcal{O}_{X', x'})$ is defined; finally, x' being the unique closed point of $\mathrm{Spec}(\mathcal{O}_{X', x'})$, $\mathfrak{Z}^1(\mathrm{Spec}(\mathcal{O}_{X', x'}))$ is canonically isomorphic to $\mathbf{Z}$.

¹⁵The reader will verify that (21.9.12) has not been used to prove this property in chap. V.

One can now set

$$(21.10.3) \quad f^*(Z) = \sum_{x' \in X^{(1)}} n_{x'} \cdot \overline{\{x'\}},$$

and one says that the 1-codimensional cycle $f^*(Z)$ is the *reciprocal image* of Z by f .

Proposition (21.10.4). — Suppose the conditions of (21.10.1) satisfied. Then, for every divisor D on X such that $\text{Supp}(D) \subset T$ and such that $f^*(D)$ is defined (21.4.2), one has

$$(21.10.4.1) \quad \text{cyc}(f^*(D)) = f^*(\text{cyc}(D)).$$

Proof. The question being local on X , one can restrict to the case where $X = \text{Spec}(A)$ is affine, where $D = \text{div}(t)$, where t is a regular non-invertible element of A , and where the subscheme $Y(D)$ (21.2.12) has only one maximal point y , so that $\text{cyc}(D) = n_y \cdot \overline{\{y\}}$, where n_y is the length of $\mathcal{O}_{X,y}/t_y \mathcal{O}_{X,y}$ (21.6.5, .1). One has seen in the proof of (21.10.2, , B)) that the points $x' \in X'$ such that $\text{mult}_{x'}(f^*(\text{cyc}(D))) \neq 0$ are maximal points of $f^{-1}(\overline{\{y\}})$. If the point x' is in case 3° of (21.10.1), the equality of the multiplicities in x' of the two members of (21.10.4.1) results from the definition of $n_{x'}$ by means of the homomorphism (21.10.1.1). Suppose on the contrary that x' is in case 2° of (21.10.1), and let $x = f(x')$; since $x \in X^{(1)} \cap \overline{\{y\}}$, one has necessarily $x = y$. Note now that $f^*(D) = \text{div}(t')$, where t' is the image of t in $\Gamma(X', \mathcal{O}_{X'})$, and $Y(f^*(D)) = f^{-1}(Y(D))$; the multiplicity of $f^*(D)$ at the point x' is therefore the length $n_{x'}$ of the $\mathcal{O}_{X',x'}$ -module $\mathcal{O}_{X',x'}/t'_{x'} \mathcal{O}_{X',x'}$; as $\mathcal{O}_{X',x'}/t'_{x'} \mathcal{O}_{X',x'} = (\mathcal{O}_{X,y}/t_y \mathcal{O}_{X,y}) \otimes_{\mathcal{O}_{X,y}} \mathcal{O}_{X',x'}$, it results from (4.7.1) that one has $n_{x'} = \lambda_{x'} n_y$, hence the multiplicities at the point x' of the two members of (21.10.4.1) are still equal, which completes the proof. $\square$

(21.10.5). Suppose now that $f : X' \rightarrow X$ is a morphism of locally Noetherian preschemes, transforming every maximal point of X' into a maximal point of X , and suppose moreover that for every rare closed part T of X , f satisfies the conditions of (21.10.1); this means also that for every $x' \in X'^{(1)}$, either $x = f(x')$ is a maximal point of X , or x' satisfies one of the conditions (ii), (iii) of (21.10.1). Bearing in mind that every 1-codimensional cycle has rare support in X , one sees that $f^*(Z)$ is defined for every 1-codimensional cycle Z on X ; by virtue of (21.10.3, .1), one has thus defined a homomorphism of sheaves of ordered commutative groups

$$\psi^*(\mathcal{Z}_X^1) \longrightarrow \mathcal{Z}_{X'}^1.$$

If moreover, for every divisor D on X , $f^*(D)$ is defined (21.4.5), the fact that the support of D is rare in X (21.6.6) implies that (21.10.4) is applicable, and one has therefore the formula (21.10.4.1) for every divisor D on X . In particular:

Proposition (21.10.6). — Let X, X' be two locally Noetherian preschemes, $f : X' \rightarrow X$ a flat morphism. Then $f^*(Z)$ is defined for every 1-codimensional cycle Z on X , $f^*(D)$ is defined for every divisor D on X and one has the relation (21.10.4.1).

Proof. Indeed, if $x' \in X'^{(1)}$ is such that $x = f(x')$ is not maximal, it results from (6.1.1) that one has necessarily $x \in X^{(1)}$, hence one is in case (ii) of (21.10.1). One can therefore apply (21.10.5), hence one is in case (ii). One can therefore apply (21.4.5) and (2.3.4). $\square$

Remark (21.10.7). — The existence of $f^*(Z)$ for every 1-codimensional cycle Z on X already results from the hypothesis that f is flat at the points x' of X' of codimension ≤ 1 in X' (i.e. such that $\dim \mathcal{O}_{X',x'} \leq 1$); indeed, for every maximal point $z' \in X'$, it results from (6.1.1) that $z = f(z')$ is a maximal point of X since $\dim \mathcal{O}_{X,z} \leq \dim \mathcal{O}_{X',z'} = 0$. Similarly, if $x' \in X'^{(1)}$, $x = f(x')$ is maximal or belongs to $X^{(1)}$ by (6.1.1); one can therefore still apply (21.10.5).

Proposition (21.10.8). — Let X, X', X'' be three locally Noetherian preschemes, $f : X' \rightarrow X, g : X'' \rightarrow X'$ two morphisms; suppose that f (resp. g) is flat at every point of codimension ≤ 1 in X' (resp. X''). Then $f \circ g$ is flat at every point of codimension ≤ 1 in X'' and for every 1-codimensional cycle Z on X , one has $(f \circ g)^*(Z) = g^*(f^*(Z))$.

Proof. The first assertion results from (6.1.1) and (2.1.6). The second results from the fact that f (resp. g , resp. $f \circ g$) transforms the maximal points of X' (resp. X'' , resp. X''') into maximal points of X (resp. X' , resp. X) and from the fact that, if $x'' \in X'''^{(1)}$ is such that $x' = g(x'') \in X'^{(1)}$ and $x = f(x') \in X^{(1)}$, one has

$$\text{long}(\mathcal{O}_{X'',x''}/\mathfrak{m}_x \mathcal{O}_{X'',x''}) = \text{long}(\mathcal{O}_{X'',x''}/\mathfrak{m}_{x'} \mathcal{O}_{X'',x''}) \cdot \text{long}(\mathcal{O}_{X',x'}/\mathfrak{m}_x \mathcal{O}_{X',x'}).$$

Indeed, if one sets $A = \mathcal{O}_{X,x}$, $A' = \mathcal{O}_{X',x'}$, $A'' = \mathcal{O}_{X'',x''}$, $\mathfrak{m} = \mathfrak{m}_x$, $\mathfrak{m}' = \mathfrak{m}_{x'}$, the formula

$$\text{long}_{A''}(A''/\mathfrak{m}A'') = \text{long}_{A''}(A''/\mathfrak{m}'A'') \cdot \text{long}_{A'}(A'/\mathfrak{m}A')$$

results from (4.7.1) and the hypothesis of flatness of A'' over A' . $\square$

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(21.10.9). Let X be a locally Noetherian prescheme, Λ a commutative group, written additively, and therefore considered as a $\mathbf{Z}$ -module; one will further denote by $\mathbf{Z}$ and Λ the simple sheaves on X associated to the constant presheaves equal respectively to $\mathbf{Z}$ and Λ ($\mathbf{0}_I$, 3.6). One calls the *sheaf of germs of 1-codimensional cycles with coefficients in Λ* the sheaf of commutative groups $\mathcal{Z}_X^1 \otimes_{\mathbf{Z}} \Lambda$. If $\Lambda = \mathbf{Q}$, one says that the sections of $\mathcal{Z}_X^1 \otimes_{\mathbf{Z}} \mathbf{Q}$ over X are the *1-codimensional cycles with rational coefficients*. As the fibres of $\mathcal{Z}_X^1$ are torsion-free $\mathbf{Z}$ -modules (21.6.3), the canonical homomorphism $\mathcal{Z}_X^1 \rightarrow \mathcal{Z}_X^1 \otimes_{\mathbf{Z}} \mathbf{Q}$ is injective, so that the 1-codimensional cycles are identified with 1-codimensional cycles with rational coefficients.

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(21.10.10). We shall see that under certain conditions, one can enlarge the definition of $f^*(Z)$ given in (21.10.3) for a 1-codimensional cycle Z on X , but on condition of taking for $f^*(Z)$ a 1-codimensional cycle *with rational coefficients* on X' . The more general case where we place ourselves is that where f transforms every maximal point of X' into a maximal point of X , and where, at every point $x' \in X'^{(1)}$, one has one of the conditions (i), (ii), (iii) of (21.10.1) or a fourth condition:

- (iv) $x \in X^{(1)}$, $\mathfrak{m}_x \notin \text{Ass}(\mathcal{O}_{X,x})$ and moreover, if one sets $A = \widehat{\mathcal{O}}_{X,x}$, $A' = \widehat{\mathcal{O}}_{X',x'}$, and if K denotes the total ring of fractions of A , then A' is a finite A -algebra and $K' = A' \otimes_A K$ is a free K -module.

Let us then denote by $r_{x'}$ the rank of the free K -module K' , $q_{x'}$ the degree of $k(x') = k(A')$ over $k(x) = k(A)$, and set $\mu_{x'} = r_{x'}/q_{x'}$. For a 1-codimensional cycle with support contained in T , given by (21.10.1.1) and every $x' \in X'^{(1)}$, one defines $c_{x'} \in \mathbf{Q}$ as equal to the number $n_{x'}$ when one is in one of the cases 1°, 2°, 3° of (21.10.1); but there remains here a fourth possibility:

- 4° if x' satisfies condition (iv) above, one takes $c_{x'} = \mu_{x'} n_{x'} \in \mathbf{Q}$.

(21.10.11). One must still prove that when condition 4° of (21.10.10) is satisfied *at the same time* as one of the conditions 1°, 2°, 3° of (21.10.1), one has $n_{x'} = c_{x'}$. This is evident if $x \notin T$ since then $n_x = 0$. To study the two other cases, let us note that $\mathfrak{m}_x \mathcal{O}_{X',x'}$ is closed for the $\mathfrak{m}_{x'}$ -preadic topology, hence the completion of $\mathcal{O}_{X',x'}/\mathfrak{m}_x \mathcal{O}_{X',x'}$ for this latter topology is $A'/\mathfrak{m}_x A'$; if one is at the same time in case 2° and case 4°, $\mathcal{O}_{X,x}$ is a discrete valuation ring, hence regular, and $\mathcal{O}_{X',x'}$ is also discrete for the $\mathfrak{m}_{x'}$ -preadic topology, hence isomorphic to $A'/\mathfrak{m}_x A'$. As A' is a finite A -algebra and *flat* ($\mathbf{0}_{III}$, 10.2.3), it is a free A -module ($\mathbf{0}_{III}$, 10.1.3), and the rank of K' over K is equal to the rank of A' over A , hence also to the length $\lambda_{x'}$ of $\mathcal{O}_{X',x'}/\mathfrak{m}_x \mathcal{O}_{X',x'}$ (as an $\mathcal{O}_{X',x'}$ -module, or $k(x')$ -module) times the degree $[k(x') : k(x)] = q_{x'}$, which proves the relation $\mu_{x'} = r_{x'}/q_{x'} = \lambda_{x'}$ in this case.

Suppose finally that one is at the same time in case 3° and case 4°. Then, since $\dim(\mathcal{O}_{X,x}) = 1$, $\mathcal{O}_{X,x}$ is a discrete valuation ring, hence regular. On the other hand, $\mathcal{O}_{X',x'}$ is of dimension 1, and since $\mathfrak{m}_{x'} \notin \text{Ass}(\mathcal{O}_{X',x'})$, $\mathcal{O}_{X',x'}$ is a Cohen–Macaulay ring ($\mathbf{0}$, 16.4.6); finally, since A' is an A -module of finite type, $A'/\mathfrak{m}_x A'$ is a finite $k(x)$ -vector space; since $\mathcal{O}_{X',x'}/\mathfrak{m}_x \mathcal{O}_{X',x'}$ is contained in $A'/\mathfrak{m}_x A'$, it is also a $k(x)$ -vector space of finite rank, hence an Artinian ring. The application of (6.1.5) shows then that $\mathcal{O}_{X',x'}$ is a flat $\mathcal{O}_{X,x}$ -module, hence one is also in case 2°, and one concludes by what has been seen above.

In the case considered, one will set

$$f^*(Z) = \sum_{x' \in X'^{(1)}} c_{x'} \cdot \overline{\{x'\}}$$

which is therefore a 1-codimensional cycle on X' with *rational* coefficients. One has again defined thus a homomorphism f^* of ordered groups, satisfying (21.10.3.1), and hence a homomorphism of sheaves of commutative groups

$$\psi^*(F_T(\mathcal{Z}_X^1)) \longrightarrow \mathcal{Z}_{X'}^1 \otimes_{\mathbf{Z}} \mathbf{Q}.$$

When f satisfies the preceding conditions for every rare closed part T in X , i.e. for every $x' \in X'^{(1)}$, **IV-4 | 294** either $x = f(x')$ is a maximal point of X , or x' satisfies one of the conditions (ii), (iii) or (iv), $f^*(Z)$ is then defined for every 1-codimensional cycle Z on X , and one has defined a homomorphism of sheaves of ordered commutative groups

$$\psi^*(\mathcal{Z}_X^1) \longrightarrow \mathcal{Z}_{X'}^1 \otimes_{\mathbf{Z}} \mathbf{Q}$$

whence, by tensorisation, a homomorphism of sheaves of ordered $\mathbf{Q}$ -vector spaces

$$(21.10.11.1) \quad \psi^*(\mathcal{Z}_X^1 \otimes_{\mathbf{Z}} \mathbf{Q}) \longrightarrow \mathcal{Z}_{X'}^1 \otimes_{\mathbf{Z}} \mathbf{Q}.$$

Remark (21.10.12). — When one is in the situation of (21.10.10), one can effectively, for 1-codimensional cycles Z on X , have for $f^*(Z)$ cycles with *non-integer* coefficients; in other words the numbers $\mu_{x'}$ can be non-integer. One has an example by taking the integral complete ring A of (6.15.11, (ii)) and its integral closure A' : the closed point x' of $\text{Spec}(A')$ satisfies condition (iv) of (21.10.10) and one has $\mu_{x'} = \frac{1}{2}$.

Lemma (21.10.13). — Let A be a local Noetherian ring of dimension 1, t a regular element of A belonging to the maximal ideal $\mathfrak{m}$ (which implies that $\mathfrak{m} \notin \text{Ass}(A)$).

- label=(i) For every A -module M of finite type, the kernel $N_t(M)$ and the cokernel $P_t(M)$ of the homothety $t_M : M \rightarrow M$ of ratio t are of finite length. Set $d_t(M) = \text{long } P_t(M) - \text{long } N_t(M)$.
 lbbel=(ii) If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is an exact sequence of A -modules of finite type, one has $d_t(M) = d_t(M') + d_t(M'')$.
 lcbel=(iii) One has $d_t(M) \geq 0$ for every A -module M of finite type; for $d_t(M) = 0$, it is necessary and sufficient that M be of finite length.
 ldbel=(iv) If K is the total ring of fractions of A and if $M \otimes_A K$ is a free K -module of rank n , one has $d_t(M) = n \cdot d_t(A) = n \cdot \text{long}(A/tA)$.
 lebel=(v) If M satisfies the hypotheses of (iv) and if moreover t is M -regular, one has $\text{long}(M/tM) = n \cdot \text{long}(A/tA)$.

Proof. (i) $\text{Spec}(A)$ is formed of the point $\mathfrak{m}$ and the minimal prime ideals $\mathfrak{p}_i$; since by hypothesis $t \notin \mathfrak{p}_i$ for all i (Bourbaki, *Alg. comm.*, chap. IV, §1, n° 1, cor. 3 of prop. 2), the image of t in each $A_{\mathfrak{p}_i}$ is invertible, and the supports of the A -modules $N_t(M)$ and $P_t(M)$ of finite type are thus empty or reduced to $\mathfrak{m}$; one concludes (0, 16.1.10) that these modules are of finite length.

(ii) Since t is regular, one has an exact sequence

$$0 \longrightarrow A \xrightarrow{t_A} A \longrightarrow A/tA \longrightarrow 0$$

and since $\text{Tor}_i^A(M, A) = 0$ for $i \geq 1$, the Tor exact sequence gives on the one hand the exact sequence

$$0 \longrightarrow \text{Tor}_1^A(M, A/tA) \longrightarrow M \xrightarrow{t_M} M \longrightarrow M/tM \longrightarrow 0$$

and on the other hand, for $i \geq 2$,

$$0 = \text{Tor}_i^A(M, A) \longrightarrow \text{Tor}_i^A(M, A/tA) \longrightarrow \text{Tor}_{i-1}^A(M, A) = 0$$

hence $N_t(M) = \text{Tor}_1^A(M, A/tA)$ and $\text{Tor}_i^A(M, A/tA) = 0$ for $i \geq 2$; the Tor exact sequence applied to $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ gives the exact sequence

$$\begin{aligned} 0 \longrightarrow \text{Tor}_1^A(M', A/tA) \longrightarrow \text{Tor}_1^A(M, A/tA) \longrightarrow \text{Tor}_1^A(M'', A/tA) \longrightarrow \\ M' \otimes_A (A/tA) \longrightarrow M \otimes_A (A/tA) \longrightarrow M'' \otimes_A (A/tA) \longrightarrow 0 \end{aligned}$$

and this sequence can be written, by what precedes

$$0 \longrightarrow N_t(M') \longrightarrow N_t(M) \longrightarrow N_t(M'') \longrightarrow P_t(M') \longrightarrow P_t(M) \longrightarrow P_t(M'') \longrightarrow 0$$

which proves (ii).

(iii) To prove (iii), note that there is a composition series (M_h) of M whose quotients M_h/M_{h+1} are isomorphic to $A/\mathfrak{m}$ or to one of the $A/\mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. IV, §1, n° 4, th. 1), and for M to be of finite length, it suffices that all these quotients be isomorphic to $A/\mathfrak{m}$. By (ii), it reduces to proving that $d_t(A/\mathfrak{m}) = 0$ and $d_t(A/\mathfrak{p}_i) > 0$. Now, the image of t in $A/\mathfrak{m}$ being 0, one has $N_t(A/\mathfrak{m}) = A/\mathfrak{m}$ and $P_t(A/\mathfrak{m}) = A/\mathfrak{m}$, hence the first assertion; on the other hand, the image of t in $A/\mathfrak{p}_i$ is regular, hence $N_t(A/\mathfrak{p}_i) = 0$ and $P_t(A/\mathfrak{p}_i) = A/(tA + \mathfrak{p}_i)$ which is not reduced to 0, hence the second assertion.

(iv) There is a base of $M \otimes_A K$ of the form $(x_j/s)_{1 \leq j \leq n}$, where s is a regular element of A and $x_j \in M$. Consider the homomorphism $u : A^n \rightarrow M$ that transforms e_j (the canonical base element of A^n) to x_j and show that $\text{Ker}(u)$ and $\text{Coker}(u)$ are of finite length; indeed, for each i , the image of s in $A_{\mathfrak{p}_i}$ is invertible, and since the images x_j/s in $M_{\mathfrak{p}_i}$ form a base of this $A_{\mathfrak{p}_i}$ -module, $u_{\mathfrak{p}_i} : A_{\mathfrak{p}_i}^n \rightarrow M_{\mathfrak{p}_i}$ is bijective. Since (i), $\text{Ker}(u)$ and $\text{Coker}(u)$ have supports contained in $\{\mathfrak{m}\}$, hence are of finite length. It then results from (ii) and (iii) that $d_t(M) = d_t(A^n) = n \cdot d_t(A) = n \cdot \text{long}(A/tA)$ since t is regular.

(v) It is clear that (v) follows immediately from (iv), since then $N_t(M) = 0$. $\square$

This lemma allows to generalise (21.10.4):

Proposition (21.10.13). — Suppose that f satisfies the conditions of (21.10.10). Then, for every divisor D on X such that $\text{Supp}(D) \subset T$ and such that $f^*(D)$ is defined, one has

$$(21.10.13.1) \quad \text{cyc}(f^*(D)) = f^*(\text{cyc}(D)).$$

Proof. Reasoning as in (21.10.4), everything reduces to seeing (with the same notations) that if x' is in case 4° of (21.10.10) and if n_y is the length of $\mathcal{O}_{X,y}/t_y \mathcal{O}_{X,y}$, then the length $n_{x'}$ of $\mathcal{O}_{X',x'}/t'_{x'} \mathcal{O}_{X',x'}$ is equal to $\mu_{x'} n_y$, $\mu_{x'}$ being the rational number defined in (21.10.10). As $\mathcal{O}_{X',x'}/t'_{x'} \mathcal{O}_{X',x'}$ is of finite length, it has the same length as its $\mathfrak{m}_{x'}$ -preadic completion $A'/t'_{x'} A'$, which one can also write $A'/t_y A'$; moreover, as $t'_{x'}$ is regular by hypothesis in $\mathcal{O}_{X',x'}$, it is also regular in A' by flatness (0I, 6.3.4), and when A' is considered as an A -module, one can also say that t_y is A' -regular. As A is of dimension 1 and A' is an A -module of finite type such that $A' \otimes_A K$ is a free K -module of rank $r_{x'}$, one can apply (21.10.13, (v)) to $M = A'$ and to t_y , the length of $A'/t_y A'$ as an A -module is therefore $r_{x'} n_y$. As $k(x')$ is a $k(x)$ -vector space of rank $q_{x'}$, the length of $A'/t_y A'$ as an A' -module is therefore $r_{x'} n_y / q_{x'} = \mu_{x'} n_y$, which completes the proof. $\square$

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(21.10.14). Let now X, X' be two locally Noetherian preschemes, $f : X' \rightarrow X$ a morphism having the two following properties:

label=a) f is finite;

lbbel=b) the image by f of every maximal point of X' is a maximal point of X .

For every $x \in X^{(1)}$, the points $x' \in f^{-1}(x)$ then all belong to $X'^{(1)}$, since, as it results from hypothesis b) and from the inequality (0, 16.3.9.1), the fibre $f^{-1}(x)$ is discrete. Let then

$$Z' = \sum_{x' \in X'^{(1)}} n_{x'} \cdot \overline{\{x'\}}$$

be a 1-codimensional cycle on X' . For every $x \in X^{(1)}$, let us define an integer n_x by the formula

$$n_x = \sum_{x' \in f^{-1}(x)} n_{x'} \cdot [k(x') : k(x)]$$

which makes sense, the points of $f^{-1}(x)$ being finite in number and $k(x')$ being a field of finite degree over $k(x)$ ((I, 6.4.4)). Moreover, the set of $x \in X^{(1)}$ such that $n_x \neq 0$ is *locally finite* in X , for it is contained in the image by f of the set of $x' \in X'^{(1)}$ such that $n_{x'} \neq 0$, and the conclusion results from the fact that the morphism f is quasi-compact. One can therefore define a 1-codimensional cycle on X by setting

$$(21.10.14.1) \quad f_*(Z') = \sum_{x \in X^{(1)}} n_x \cdot \overline{\{x\}}$$

and one says that $f_*(Z')$ is the *direct image* of Z' by f . It is clear that the application $f_* : \mathfrak{Z}^1(X') \rightarrow \mathfrak{Z}^1(X)$ is thus defined as a homomorphism of ordered groups. Moreover, if U is an open of X , $f_U : f^{-1}(U) \rightarrow U$ the restriction of f , it results immediately from the definitions that one has

$$(21.10.14.2) \quad (f_U)_*(Z'|f^{-1}(U)) = f_*(Z')|U$$

hence, denoting by ψ the continuous map underlying the morphism f , the maps $\Gamma(f^{-1}(U), \mathcal{Z}_{X'}^1) \rightarrow \Gamma(U, \mathcal{Z}_X^1)$ that one has just defined define a homomorphism of sheaves of ordered commutative groups on X

$$\psi_*(\mathcal{Z}_{X'}^1) \longrightarrow \mathcal{Z}_X^1.$$

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Proposition (21.10.15). — *Let X, X', X'' be three locally Noetherian preschemes, $f : X' \rightarrow X, g : X'' \rightarrow X'$ two morphisms satisfying conditions a) and b) of (21.10.14). Then $f \circ g$ satisfies the same conditions, and for every 1-codimensional cycle Z'' on X'' , one has*

$$(f \circ g)_*(Z'') = f_*(g_*(Z'')).$$

Proof. This results immediately from the definitions. $\square$

Proposition (21.10.16). — *Let X, X', X_1 be three locally Noetherian preschemes, $f : X' \rightarrow X$ a morphism satisfying conditions a) and b) of (21.10.14), $g : X_1 \rightarrow X$ a flat morphism. Set $X'_1 = X' \times_X X_1$ (so that X'_1 is locally Noetherian) and denote by $f_1 : X'_1 \rightarrow X_1$ and $g_1 : X'_1 \rightarrow X'$ the canonical projections. Then f_1 satisfies conditions a) and b) of (21.10.14), and for every 1-codimensional cycle Z' on X' , one has*

$$(21.10.16.1) \quad g^*(f_*(Z')) = (f_1)_*(g_1^*(Z')).$$

Proof. It is clear that f_1 is finite, and it satisfies condition b) of (21.10.14) by virtue of (2.3.7). To prove (21.10.16.1), one is immediately reduced to the case where X, X' and X_1 are spectra of local Noetherian rings of dimension 1, of closed points x, x' and x_1 respectively, A, A', A_1 being the local rings at these points, and A_1 a flat A -module. Denoting by x, x' and x_1 the closed points of X, X' and X_1 respectively, it is a matter of seeing that one has

$$(21.10.16.2) \quad \sum_{x'_1} \lambda_{x'_1} [k(x'_1) : k(x_1)] = \lambda_{x_1} [k(x') : k(x)]$$

where x'_1 ranges over the set of closed points of X'_1 (i.e. the set of points above both x' and x_1) and where $\lambda_{x'_1} = \text{long}(\mathcal{O}_{X'_1, x'_1} / \mathfrak{m}_{x'} \mathcal{O}_{X'_1, x'_1})$ and $\lambda_{x_1} = \text{long}(A_1 / \mathfrak{m}_x A_1)$. As one has

$$[k(x'_1) : k(x')] \cdot [k(x') : k(x)] = [k(x'_1) : k(x)] = [k(x'_1) : k(x_1)] \cdot [k(x_1) : k(x)]$$

the first member of (21.10.16.2) can also be written

$$\frac{[k(x') : k(x)]}{[k(x_1) : k(x)]} \text{long}_{A_1}(A'_1 / \mathfrak{m}_{x'} A'_1)$$

where one has set $A'_1 = A' \otimes_A A_1$. Since $A'_1 / \mathfrak{m}_{x'} A'_1 = (A' / \mathfrak{m}_{x'} A') \otimes_A A_1$, and as A_1 is a flat A -module, one has by (4.7.1)

$$\text{long}_{A_1}(A'_1 / \mathfrak{m}_{x'} A'_1) = \text{long}_A(A' / \mathfrak{m}_{x'} A') \cdot \text{long}_{A_1}(A_1 / \mathfrak{m}_x A_1) = [k(x') : k(x)] \cdot \lambda_{x_1}$$

which gives the result since $[k(x') : k(x)] / [k(x_1) : k(x)] \cdot [k(x') : k(x)] \cdot \lambda_{x_1} = \lambda_{x_1} \cdot [k(x') : k(x)]$. $\square$

Proposition (21.10.17). — *Let X, X' be two locally Noetherian preschemes, $f : X' \rightarrow X$ a finite locally free morphism. Then f satisfies condition b) of (21.10.14), and for every divisor D' on X' , one has*

$$(21.10.17.1) \quad f_*(\text{cyc}(D')) = \text{cyc}(f_*(D')).$$

Proof. As f is flat and finite, condition b) of (21.10.14) and the relation $f(X'^{(1)}) \subset X^{(1)}$ result from (6.1.1). The definition (21.10.14.1) shows that one can reduce to the case where $X = \text{Spec}(A) = \text{Spec}(\mathcal{O}_{X, x})$ for an $x \in X^{(1)}$; then $X' = \text{Spec}(A')$, where A' is an A -algebra that is a free A -module of finite rank; moreover, one can suppose that $D' = \text{div}(t')$, where t' is a regular element of A' ; one has then $f_*(D') = \text{div}(t)$, where $t = N_{A'/A}(t')$ is a regular element of A (21.5.2). One can restrict to the case where t' is not invertible in A' , the ring A/tA is then composed of local Artinian rings A'_i ($1 \leq i \leq r$), the residue field of A'_i being $k(x'_i)$, where x'_i ($1 \leq i \leq r$) are the points of X'

above x . If t'_i ($1 \leq i \leq r$) is the image of t' in A'_i , $A'/t'A'$ is a direct sum of the $A'_i/t'_i A'_i$; as the product $\text{long}_{A'_i}(A'_i/t'_i A'_i) \cdot [k(x'_i) : k(x)]$ is equal to $\text{long}_A(A'_i/t'_i A'_i)$, one sees that the formula to prove reduces to

$$(21.10.17.2) \quad \text{long}_A(A'/t'A') = \text{long}_A(A/tA).$$

This relation follows from the following more general lemma: □ IV-4 | 298

Lemma (21.10.17.3). — *Let A be a local Noetherian ring of dimension 1, M a free A -module of finite rank, u an injective endomorphism of M . Then one has*

$$(21.10.17.4) \quad \text{long}_A(\text{Coker } u) = \text{long}_A(A/(\det u)A).$$

Proof. Let us distinguish several cases.

I) A is a discrete valuation ring. Let π be a uniformiser of A , and note that $\text{long}_A(A/\pi^k A) = k$; if n is the rank of M and if π^{m_i} ($1 \leq i \leq n$) are the invariant factors of u , $\text{Coker}(u)$ is a direct sum of the A -modules $A/\pi^{m_i} A$, hence has length $m = \sum_{i=1}^n m_i$, and $\det(u)$ is a product of an invertible element and of π^m , whence the conclusion in this case (Bourbaki, *Alg.*, chap. VII, §4, n° 5, cor. 1 of prop. 4).

II) A is a complete integral ring (of dimension 1). One knows then (0, 19.8.8, (ii)) that there is a subring B of A , which is a discrete valuation ring, such that $B \rightarrow A$ is a local homomorphism making A a B -module of finite type; as this B -module is evidently torsion-free, it is free (Bourbaki, *Alg. comm.*, chap. VI, §3, n° 6, lemme 1). Denote by M' the set M endowed with its structure of (free) B -module, and by u' the endomorphism of M' (the endomorphism u considered as a B -endomorphism). It results from I) that one has

$$(21.10.17.5) \quad \text{long}_B(\text{Coker } u') = \text{long}_B(B/(\det u')B).$$

But one has (Bourbaki, *Alg.*, chap. VIII, §12, n° 2, prop. 7)

$$\det u' = N_{A/B}(\det u),$$

hence, applying (21.10.17.4) to the homothety $x \mapsto (\det u)x$ of the B -module free A , one obtains

$$\text{long}_B(B/(\det u')B) = \text{long}_B(A/(\det u)A),$$

whence, substituting in (21.10.17.5) and dividing by $[k(A) : k(B)]$, the length of the B -module $k(A)$, one obtains (21.10.17.4) in the case considered.

III) A is a complete ring. Let us note that one can moreover suppose that $\mathfrak{m} \notin \text{Ass}(A)$; indeed, since $\det(u)$ is a regular element of A , if one had $\mathfrak{m} \in \text{Ass}(A)$, one would deduce $\det(u) \notin \mathfrak{m}$, hence $\det(u)$ would be invertible, u an automorphism of M , and the formula (21.10.17.4) would become trivial, both members being zero. In what follows, for an endomorphism v of a module N over a ring R , such that $\text{Ker } v$ and $\text{Coker } v$ are of finite length, one sets

$$\chi(N, v) = \text{long}_R(\text{Ker}(v)) - \text{long}_R(\text{Coker}(v)).$$

One will note that the hypothesis on v amounts to saying that the complex

$$K^0 : 0 \longrightarrow N \xrightarrow{v} N \longrightarrow 0$$

has its cohomology modules of finite length and that $\chi(N, v) = \chi(H^0(K^0))$ (0III, 11.10). One deduces that if N' , N , N'' are three R -modules, if one has a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & N' & \xrightarrow{r} & N & \xrightarrow{s} & N'' \longrightarrow 0 \\ & & \downarrow v' & & \downarrow v & & \downarrow v'' \\ 0 & \longrightarrow & N' & \xrightarrow{r} & N & \xrightarrow{s} & N'' \longrightarrow 0 \end{array}$$

whose rows are exact, and if two of the three numbers $\chi(N, v)$, $\chi(N', v')$ and $\chi(N'', v'')$ are defined, then so is the third and one has

$$(21.10.17.6) \quad \chi(N, v) = \chi(N', v') + \chi(N'', v'').$$

This results indeed immediately from the exact cohomology sequence. Finally, if N is an R -module of finite length, one has $\chi(N, v) = 0$.

With these notations, one has the following lemma: $\square$

Lemma (21.10.17.7). — *Let A be a local Noetherian ring of dimension 1 whose maximal ideal $\mathfrak{m}$ is such that $\mathfrak{m} \notin \text{Ass}(A)$; let $\mathfrak{p}_i$ ($1 \leq i \leq n$) be the minimal prime ideals of A . Let M be a free A -module of finite type, u an endomorphism of M such that $\chi(M, u)$ is defined; for each i , set $M_i = M \otimes_A (A/\mathfrak{p}_i)$ and let u_i be the endomorphism $u \otimes 1_{A/\mathfrak{p}_i}$ of M_i ; then if $\chi(M_i, u_i)$ is defined for each i , one has*

$$(21.10.17.8) \quad \chi(M, u) = \sum_{i=1}^n \text{long}(A_{\mathfrak{p}_i}) \cdot \chi(M_i, u_i).$$

Proof. Since $\mathfrak{m} \notin \text{Ass}(A)$, one has a unique reduced primary decomposition $(0) = \bigcap_{1 \leq i \leq n} \mathfrak{q}_i$, where the ideal $\mathfrak{q}_i$ is $\mathfrak{p}_i$ -primary for $1 \leq i \leq n$. If one sets $M'_i = M \otimes_A (A/\mathfrak{q}_i)$, one has therefore an exact sequence

$$0 \longrightarrow M \longrightarrow \bigoplus_i M'_i \longrightarrow M'' \longrightarrow 0$$

of A -modules, where M'' is of finite length: indeed, localising at each of the $\mathfrak{p}_i$, one obtains $M''_{\mathfrak{p}_i} = 0$ since $(\mathfrak{q}_i)_{\mathfrak{p}_i} = 0$ and $(\mathfrak{q}_j)_{\mathfrak{p}_i} = A_{\mathfrak{p}_i}$ for $j \neq i$ (Bourbaki, *Alg. comm.*, chap. IV, §2, n° 4, prop. 6), hence $(M'_i)_{\mathfrak{p}_i} = M_{\mathfrak{p}_i}$ and $(M'_j)_{\mathfrak{p}_i} = 0$; the support of M'' is therefore reduced to $\mathfrak{m}$, so M'' is of finite length (0, 16.1.10). Setting $u'_i = u \otimes 1_{A/\mathfrak{q}_i}$, and if u'' is the endomorphism of M'' deduced from u by restriction to quotients, one deduces from (21.10.17.6) that $\chi(M, u) + \chi(M'', u'') = \sum_i \chi(M'_i, u'_i)$, and since M'' is of finite length, $\chi(M'', u'') = 0$. To prove (21.10.17.8), it therefore suffices to show that, for $n = 1$, i.e. A integral, if one designates by $\mathfrak{p}$ the unique minimal prime ideal of A , and if $A_0 = A/\mathfrak{p}$, $M_0 = M \otimes_A A_0$, $u_0 = u \otimes 1_{A_0}$, one has, assuming $\chi(M_0, u_0)$ is defined,

$$(21.10.17.9) \quad \chi(M, u) = \text{long}(A_{\mathfrak{p}}) \cdot \chi(M_0, u_0).$$

Let $\mathfrak{n}_j$ ($0 \leq j \leq r$) be the “symbolic j -th power” of $\mathfrak{p}$, the preimage in A of the ideal $(\mathfrak{p}A_{\mathfrak{p}})^j$ of $A_{\mathfrak{p}}$ (IV-4 | 300 ($1 \leq j \leq r$), with $\mathfrak{n}_0 = A$ and $\mathfrak{n}_r = 0$; set

$$M_j = \mathfrak{n}_j M / \mathfrak{n}_{j+1} M \quad (0 \leq j \leq r-1),$$

and denote by v_j the endomorphism of M_j deduced from u by restriction to $\mathfrak{n}_j M$ and passage to quotients. We shall first show that each of the numbers $\chi(M_j, v_j)$ is defined and that one has

$$(21.10.17.10) \quad \chi(M, u) = \sum_j \chi(M_j, v_j).$$

The first assertion will imply the second, by applying (21.10.17.6) to each of the exact sequences

$$0 \longrightarrow \mathfrak{n}_j M / \mathfrak{n}_{j+1} M \longrightarrow M / \mathfrak{n}_{j+1} M \longrightarrow M / \mathfrak{n}_j M \longrightarrow 0.$$

To prove the first assertion, note that if m is the rank of the free A -module M , M_j is isomorphic to $(\mathfrak{n}_j / \mathfrak{n}_{j+1})^m$, or further (since $\mathfrak{p}$ annihilates each of the quotients $\mathfrak{n}_j / \mathfrak{n}_{j+1}$), M_j is an A_0 -module isomorphic to $M_0 \otimes_{A_0} (\mathfrak{n}_j / \mathfrak{n}_{j+1})^m$. Denote by l_j the rank of the A_0 -module $\mathfrak{n}_j / \mathfrak{n}_{j+1}$; as the field of fractions K_0 of A_0 is the residue field of $A_{\mathfrak{p}}$, there is a system of generators of the A_0 -module M_j which contains a base of $M_j \otimes_{A_0} K_0$; hence there is an A_0 -homomorphism

$$w_j : M_0^{l_j} \longrightarrow M_j$$

whose localisation at the ideal (0) of A_0 is an isomorphism, so that the supports of $\text{Ker}(w_j)$ and $\text{Coker}(w_j)$ are reduced to the maximal ideal $\mathfrak{m}/\mathfrak{p}$ of A_0 ; $\text{Ker}(w_j)$ and $\text{Coker}(w_j)$ are therefore A_0 -modules of finite length (0, 16.1.10). As by hypothesis $\chi(M_0, u_0)$ is defined, so is $\chi(M_0^{l_j}, u_0^{l_j}) = l_j \chi(M_0, u_0)$ and by virtue of (21.10.17.6) and the fact that $\text{Ker}(w_j)$ and $\text{Coker}(w_j)$ are of finite length, one sees that $\chi(M_j, v_j)$ is defined and equal to $l_j \chi(M_0, u_0)$; the relation (21.10.17.10) then gives

$$\chi(M, u) = \left(\sum_j l_j \right) \chi(M_0, u_0),$$

and by virtue of a preceding remark, $\sum_j l_j$ is none other than the length of $A_{\mathfrak{p}}$, which completes the proof of the lemma. $\square$

To apply this lemma when A is a complete ring and $\mathfrak{m} \notin \text{Ass}(A)$, observe that if u is injective, it is the same for each of the u_i (with the notations of the lemma): indeed, since $\det(u)$ is a regular element of A , it does not belong to any of the $\mathfrak{p}_i$, and its image $\det u_i$ in $A/\mathfrak{p}_i$ is by consequence a non-zero element of this integral ring, which proves that u_i is injective. As $\text{Coker}(u_i)$ is an image of $\text{Coker}(u)$, it is also of finite length and $\chi(M_i, u_i)$ is therefore defined for each i ; one has therefore by the formula (21.10.17.8). On the other hand, since $\det(u)$ is a regular element of A , it is not contained in any $\mathfrak{p}_i$; the ideal $(\det u)A$ is therefore $\mathfrak{m}$ -primary and the quotient $A/(\det u)A$ is of finite length. Applying the same reasoning as above to the injective homothety $t : \xi \rightarrow (\det u)\xi$ of A and to its images t_i in the $A/\mathfrak{p}_i$, one obtains

$$\chi(A, t) = \sum_i \text{long}(A/\mathfrak{p}_i) \cdot \chi(A/\mathfrak{p}_i, t_i).$$

But as the $A/\mathfrak{p}_i$ are integral rings and complete, and as each $A/\mathfrak{p}_i$ has also $\text{long}(A/\mathfrak{p}_i)$, one obtains by applying the result of each $A/\mathfrak{p}_i$ the relation (21.10.17.4) for M and u in this case.

IV) *General case.* Set $A' = \hat{A}$, $M' = M \otimes_A A'$, $u' = u \otimes 1_{A'}$; one has $\det(u') = \det(u)$, and by flatness, $\text{Coker}(u') = (\text{Coker}(u)) \otimes_A A'$ and $A' / (\det u)A' = (A / (\det u)A) \otimes_A A'$; as the formula (21.10.17.4) is true for A' and u' by virtue of III), it is also true for A and u by virtue of (4.7.1).

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This completes the proof of (21.10.17).

Proposition (21.10.18). — *Let X, X' be two locally Noetherian preschemes, $f : X' \rightarrow X$ a finite morphism, transforming every maximal point of X' into a maximal point of X , and satisfying for every $x' \in X'$ one of the conditions (ii), (iii), (iv) of (21.10.10). Suppose moreover that there exists an integer n such that, for every maximal point x of X , $(f_*(\mathcal{O}_{X'}))_x$ is a free $\mathcal{O}_{X,x}$ -module of rank n . Then, for every 1-codimensional cycle Z on X , one has*

$$(21.10.18.1) \quad f_*(f^*(Z)) = n \cdot Z$$

(“projection formula”).

Proof. By virtue of the definitions, one is immediately reduced to the case where X is the spectrum of a local Noetherian ring A of dimension 1, of closed point x , and where $Z = \overline{\{x\}}$; set $X' = \text{Spec}(A')$ where A' is a finite A -algebra. Let us make the base change $h : Y \rightarrow X$, where $Y = \text{Spec}(B)$, $B = \hat{A}$, and set $Y' = X' \times_X Y = \text{Spec}(B \otimes_A A')$ and $g = f_{(Y)} : Y' \rightarrow Y$; the morphism g satisfies the same conditions as f ; if one proves the formula (21.10.18.1) for g and $\overline{\{y\}}$, the same formula will be valid for f and $\overline{\{x\}}$, by virtue of (21.10.16.1). One can therefore suppose that A is complete; then so is A' which is therefore a direct product of complete local rings; one can thus restrict to the case where A' is a local ring, and it remains to verify (21.10.18.1) in each of the cases (ii), (iii), (iv) for the closed point x' of X' .

In case (ii), A' being a flat A -module of finite type, is a free A -module (0III, 10.1.3), of rank n by virtue of the hypothesis. Now, by definition (21.10.1) and (21.10.3), $f^*(Z) = \lambda_{x'} \cdot \overline{\{x'\}}$, where $\lambda_{x'}$ is the length of the A' -module $A'/\mathfrak{m}_x A'$, then $f_*(f^*(Z)) = \lambda_{x'} \cdot [k(x') : k(x)] \cdot \overline{\{x\}}$; but $\lambda_{x'} \cdot [k(x') : k(x)]$ is the length of $A'/\mathfrak{m}_x A'$ as an A -module, or equivalently its rank as a $k(x)$ -vector space, hence is equal to n .

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In case (iii), A is a discrete valuation ring, hence regular, and the hypothesis $\mathfrak{m}_{x'} \notin \text{Ass}(\mathcal{O}_{X',x'})$ implies that A' is a Cohen–Macaulay ring (0, 16.4.6); as $\dim(A') = 1$ and $A'/\mathfrak{m}_x A'$ is an Artinian ring, it results from (6.1.5) that A' is a flat A -module, and one is reduced to case (ii).

In case (iv), if K is the total ring of fractions of A , $K' = A' \otimes_A K$ is by hypothesis a free K -module of rank n and by definition (21.10.10), one has $f^*(Z) = (n/[k(x') : k(x)]) \cdot \overline{\{x'\}}$, whence $f_*(f^*(Z)) = n \cdot \overline{\{x\}}$. $\square$

One will note that the formula (21.10.18.1) is applicable in particular when the morphism f is finite and flat and such that for every maximal point x of X , $(f_*(\mathcal{O}_{X'}))_x$ is a free $\mathcal{O}_{X,x}$ -module of rank n .

Corollary (21.10.19). — *Under the hypotheses of (21.10.18), let D be a divisor on X such that $f^*(D)$ is defined (21.4.5); then one has*

$$(21.10.19.1) \quad f_*(\text{cyc}(f^*(D))) = n \cdot \text{cyc}(D).$$

Proof. This results from (21.10.18) and (21.10.13.prop). $\square$

21.11. Factoriality of regular local rings

Theorem (21.11.1). — (Auslander–Buchsbaum) *A regular local Noetherian ring is factorial.*

Proof. The proof that follows is due to I. Kaplansky.

Let A be a regular local Noetherian ring of dimension n ; we shall reason by induction on n . For $n = 0$, A is a field and for $n = 1$, a discrete valuation ring, hence a principal ring (and *a fortiori* factorial). Suppose therefore $n \geq 2$ and the theorem proved for regular rings of dimension $< n$. Set $X = \text{Spec}(A)$, designate by a the closed point of X and set $U = X - \{a\}$. For every $y \in U$, one has $\dim(\mathcal{O}_{X,y}) \leq n - 1$, and since A is regular, the rings $\mathcal{O}_{X,y}$ are also regular (0, 17.3.2), hence the induction hypothesis implies that they are factorial. Moreover one has $\text{prof}(A) = \dim(A) \geq 2$ since A is regular, hence Cohen–Macaulay (0, 17.1.3).

Using (21.6.14), one is reduced to proving that $\text{Pic}(U) = 0$. Consider therefore an $\mathcal{O}_U$ -Module $\mathcal{L}$, and prove that it is isomorphic to $\mathcal{O}_U$. It results from ((I, 9.4.5)) that there exists an $\mathcal{O}_X$ -Module coherent $\mathcal{F}$ such that $\mathcal{F}|_U = \mathcal{L}$. As A is regular, hence of finite cohomological dimension (0, 17.3.1), there exists a finite left resolution of $\mathcal{F}$:

$$0 \leftarrow \mathcal{F} \leftarrow \mathcal{O}_X^{n_1} \leftarrow \mathcal{O}_X^{n_2} \leftarrow \cdots \leftarrow \mathcal{O}_X^{n_h} \leftarrow 0$$

(0, 17.2.8 and 0, 17.2.2, (iii)). By restriction to U , one has therefore a finite resolution

$$(21.11.1.1) \quad 0 \leftarrow \mathcal{L} \leftarrow \mathcal{O}_U^{n_1} \leftarrow \mathcal{O}_U^{n_2} \leftarrow \cdots \leftarrow \mathcal{O}_U^{n_h} \leftarrow 0.$$

The theorem will then result from the following general considerations. On a ringed space X , let $\mathcal{E}$ be an $\mathcal{O}_X$ -Module locally free of finite rank; one will designate by $\Lambda^{\max} \mathcal{E}$ the $\mathcal{O}_X$ -Module invertible which, in a neighbourhood of each point of X , is equal to $\Lambda^p \mathcal{E}$, designating by p the rank of $\mathcal{E}$ in this neighbourhood (which may vary with the connected component of X). With this notation, one has the $\square$

Lemma (21.11.1.2). — *Let X be a ringed space in local rings, and*

$$0 \leftarrow \mathcal{E}_0 \xleftarrow{u_0} \mathcal{E}_1 \leftarrow \cdots \leftarrow \mathcal{E}_h \leftarrow 0$$

an exact sequence of $\mathcal{O}_X$ -Modules locally free of finite rank; then the $\mathcal{O}_X$ -Module invertible $\bigotimes_{0 \leq i \leq h} (\Lambda^{\max} \mathcal{E}_i)^{\otimes (-1)^i}$ is isomorphic to $\mathcal{O}_X$.

Let us show how this lemma allows to conclude the proof of (21.11.1). It suffices to note for this that, for every integer n , $\Lambda^{\max}(\mathcal{O}_U^n)$ is isomorphic to $\mathcal{O}_U$, so that, applying the lemma to the exact sequence (21.11.1.1), one obtains that $\mathcal{L}$ is isomorphic to $\mathcal{O}_U$.

Proof of Lemma (21.11.1.2). We proceed by induction on h , the lemma being trivial for $h = 1$. For $h > 1$, $\mathcal{N} = \text{Ker}(u_0)$ is an $\mathcal{O}_X$ -Module locally free of finite rank (0I, 5.5.5), and one has the two exact sequences

$$\begin{aligned} 0 \leftarrow \mathcal{N} \leftarrow \mathcal{E}_1 \leftarrow \mathcal{E}_2 \leftarrow \cdots \leftarrow \mathcal{E}_h \leftarrow 0 \\ 0 \leftarrow \mathcal{E}_0 \leftarrow \mathcal{E}_1 / \mathcal{N} \leftarrow 0. \end{aligned}$$

By virtue of the induction hypothesis, $(\Lambda^{\max} \mathcal{N}) \otimes \left(\bigotimes_{2 \leq i \leq h} (\Lambda^{\max} \mathcal{E}_i)^{\otimes (-1)^{i-1}} \right)$ is isomorphic to $\mathcal{O}_X$; it will therefore suffice to define a canonical isomorphism $(\Lambda^{\max} \mathcal{N}) \otimes (\Lambda^{\max} \mathcal{E}_0) \xrightarrow{\sim} \Lambda^{\max} \mathcal{E}_1$ to complete the proof. Now, there exists an open covering (U_α) of X such that in every U_α , $\mathcal{E}_1|_{U_\alpha}$ is a direct sum of $\mathcal{N}|_{U_\alpha}$ and of an $\mathcal{O}_{U_\alpha}$ -Module locally free $\mathcal{M}_\alpha$ (0I, 5.5.5), whence a canonical isomorphism $v_\alpha : \mathcal{M}_\alpha \xrightarrow{\sim} \mathcal{E}_0|_{U_\alpha}$. As one has

$$r_\alpha : (\Lambda^{\max} \mathcal{N}|_{U_\alpha}) \otimes (\Lambda^{\max} \mathcal{M}_\alpha) \xrightarrow{\sim} (\Lambda^{\max} \mathcal{E}_1)|_{U_\alpha}$$

one deduces by means of v_α an isomorphism

$$u_\alpha : (\Lambda^{\max} \mathcal{N}|_{U_\alpha}) \otimes (\Lambda^{\max} \mathcal{E}_0|_{U_\alpha}) \xrightarrow{\sim} (\Lambda^{\max} \mathcal{E}_1)|_{U_\alpha}$$

and it is a matter of showing that u_α and u_β coincide in $U_\alpha \cap U_\beta$ for any two indices α, β . Now, if v'_α and v'_β are the restrictions to $U_\alpha \cap U_\beta$ of v_α and v_β respectively, one has $v'_\alpha = v'_\beta \circ w_{\beta\alpha}$, where $w_{\beta\alpha} : \mathcal{M}_\alpha|(U_\alpha \cap U_\beta) \xrightarrow{\sim} \mathcal{M}_\beta|(U_\alpha \cap U_\beta)$ is the “parallel projection” along $\mathcal{N}$ such that for every

section $s \in \Gamma(U_\alpha \cap U_\beta, \mathcal{M}_\alpha)$, $w_{\beta\alpha}(s) = s + t$ with $t \in \Gamma(U_\alpha \cap U_\beta, \mathcal{N})$; the identity of u_α and u_β results immediately from this fact and from the definition of the canonical isomorphism r_α (Bourbaki, *Alg.*, chap. III, 3^e éd.). $\square$

21.12. The Van der Waerden purity theorem for the ramification locus of a birational morphism IV-4 | 304

(21.12.1). Let X and U be two preschemes, $f : U \rightarrow X$ a quasi-compact and quasi-separated morphism, so that $f_*(\mathcal{O}_U)$ is a quasi-coherent $\mathcal{O}_X$ -Algebra (1.7.4). One calls the *affine envelope* of the X -prescheme U the X -prescheme affine over X

$$U^0 = \text{Aff}(U/X) = \text{Spec}(f_*(\mathcal{O}_U)) = \text{Spec}(\mathcal{A}(U)) \quad ((\text{II}, 1.1.1)).$$

If $f^0 : U^0 \rightarrow X$ is the structural morphism, one has therefore by definition

$$\mathcal{A}(U^0) = f_*^0(\mathcal{O}_{U^0}) = f_*(\mathcal{O}_U) = \mathcal{A}(U),$$

and to the identity isomorphism of $f_*(\mathcal{O}_U)$ there corresponds by ((II, 1.2.7)) a unique canonical X -morphism

$$(21.12.1.1) \quad i_U : U \rightarrow U^0.$$

For i_U to be an *isomorphism*, it is necessary and sufficient that the morphism $f : U \rightarrow X$ be *affine*. For every X -prescheme V affine over X , the map $u \mapsto u \circ i_U$ is a *bijection*

$$\text{Hom}_X(U^0, V) \xrightarrow{\sim} \text{Hom}_X(U, V)$$

functorial in V : this results from the existence of the canonical bijections $\text{Hom}_X(U, V) \xrightarrow{\sim} \text{Hom}_{\mathcal{O}_X}(\mathcal{A}(V), \mathcal{A}(U))$ and $\text{Hom}_X(U^0, V) \xrightarrow{\sim} \text{Hom}_{\mathcal{O}_X}(\mathcal{A}(V), \mathcal{A}(U^0))$ ((II, 1.2.7)). One can therefore say that U^0 *represents* the covariant functor $V \mapsto \text{Hom}_X(U, V)$ in the category of X -preschemes affine over X (0_{III}, 8.1.11). One deduces (0_{III}, 8.1.7) that for X fixed, $U \mapsto \text{Aff}(U/X)$ is a covariant functor from the category of quasi-compact and quasi-separated X -preschemes over X , to the category of X -preschemes affine over X ; more precisely, if U_1, U_2 are two quasi-compact and quasi-separated X -preschemes, to every X -morphism $g : U_1 \rightarrow U_2$ corresponds the unique X -morphism $g^0 : U_1^0 \rightarrow U_2^0$ making commutative the diagram

$$\begin{array}{ccc} U_1 & \xrightarrow{g} & U_2 \\ i_{U_1} \downarrow & & \downarrow i_{U_2} \\ U_1^0 & \xrightarrow{g^0} & U_2^0 \end{array}$$

More generally, consider a commutative diagram

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$$\begin{array}{ccc} U & \xrightarrow{v} & U' \\ f \downarrow & & \downarrow f' \\ X & \xrightarrow{u} & X' \end{array}$$

where the morphisms f, f' are quasi-compact and quasi-separated and the morphism u is *affine*. If one sets $h = u \circ f$, one has $h_*(\mathcal{O}_U) = u_*(f_*(\mathcal{O}_U)) = u_*(f_*^0(\mathcal{O}_{U^0}))$ and $u \circ f^0$ is an affine morphism; one has therefore $U^0 = \text{Aff}(U/X')$ (relative to the morphism h), whence a unique X' -morphism $v^0 : U^0 \rightarrow U'^0$ making commutative the diagram

$$\begin{array}{ccc} U & \xrightarrow{v} & U' \\ i_U \downarrow & & \downarrow i_{U'} \\ U^0 & \xrightarrow{v^0} & U'^0 \end{array}$$

Proposition (21.12.2). — *Let $f : U \rightarrow X$ be a quasi-compact and quasi-separated morphism, $h : X' \rightarrow X$ a flat morphism; set $U' = U \times_X X'$, $f' = f_{(X')} : U' \rightarrow X'$. Then one has a canonical X' -isomorphism*

$$(21.12.2.1) \quad \text{Aff}(U'/X') \xrightarrow{\sim} \text{Aff}(U/X) \times_X X'.$$

Proof. Indeed, $\text{Aff}(U'/X') = \text{Spec}(f'_*(\mathcal{O}_{U'}))$, and $\text{Aff}(U/X) \times_X X' = \text{Spec}(h^*(f_*(\mathcal{O}_U)))$ ((II, 1.5.1)); the isomorphism of the statement comes from the canonical isomorphism $h^*(f_*(\mathcal{O}_U)) \xrightarrow{\sim} f'_*(\mathcal{O}_{U'})$ (2.3.1). $\square$

Corollary (21.12.3). — *For every quasi-compact and quasi-separated morphism $f : U \rightarrow X$ and every $x \in X$, one has a canonical isomorphism near*

$$U^0 \times_X \text{Spec}(\mathcal{O}_{X,x}) = (U \times_X \text{Spec}(\mathcal{O}_{X,x}))^0.$$

It results also from (21.12.2) that one has, for every open V of X , a canonical isomorphism near

$$(21.12.4) \quad (f^{-1}(V))^0 = (f^0)^{-1}(V).$$

(21.12.5). We shall consider in particular the case where $f : U \rightarrow X$ is an *open immersion*, U thus being identified with an open subprescheme induced on an open of X . As the morphism $f^{-1}(U) \rightarrow U$ restriction of f^0 is an isomorphism, $i_U : U \rightarrow U^0$ is therefore an *open immersion* allowing to identify U with an open subprescheme induced on an open of U^0 .

More generally, for an open V of X , the restriction $(f^0)^{-1}(V) \rightarrow V$ of f^0 is an isomorphism of $(f^0)^{-1}(V)$ on the subprescheme induced on the open $V \cap U$ if and only if the open immersion $U \cap V \rightarrow V$ is an *affine* morphism. It is clear ((II, 1.2.1)) that the reunion of these opens V is the *largest* among them, U_1 , which contains U ; by virtue of what precedes, U_1 is also the largest open not meeting the set

$$\text{Daf}(U/X) = f^0(U^0) - U$$

(“affinity defect” of the open U relative to X , which is empty if and only if U is affine over X); in other words, the closed set $Z = X - U_1$ is the *adherence* of the set $\text{Daf}(U/X)$.

One will note that for every *flat* morphism $h : X' \rightarrow X$, if one sets $U' = h^{-1}(U)$, one has

$$(21.12.5.1) \quad \text{Daf}(U'/X') = h^{-1}(\text{Daf}(U/X))$$

as results immediately from (21.12.2.1) and ((I, 3.4.8)). In particular, for every open V of X , one has

$$(21.12.5.2) \quad \text{Daf}((U \cap V)/V) = \text{Daf}(U/X) \cap V$$

and for every $x \in X$,

$$(21.12.5.3) \quad \text{Daf}((U \cap \text{Spec}(\mathcal{O}_{X,x}))/\text{Spec}(\mathcal{O}_{X,x})) = \text{Daf}(U/X) \cap \text{Spec}(\mathcal{O}_{X,x}).$$

We shall now, when X is locally Noetherian, give precisions on the nature of $\text{Daf}(U/X)$, which will show for example that when U is everywhere dense in X , $\text{Daf}(U/X)$ is not a rare closed set, whatever the case:

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Theorem (21.12.6). — *Let X be a locally Noetherian prescheme, U an open non-void subset of X , $f : U \rightarrow X$ the canonical injection. Then:*

label=(i) *The adherence $Z = X - U_1$ of $\text{Daf}(U/X)$ is of codimension ≥ 2 in X .*

lbbel=(ii) *If $T = X - U \supset Z$ is of codimension ≥ 2 , the morphism $f^0 : U^0 \rightarrow X$ is surjective.*

Proof. (i) Let us first show that for every $x \in \text{Daf}(U/X)$ one has necessarily $\dim(\mathcal{O}_{X,x}) > 1$. Indeed, x cannot evidently be a maximal point of X , being contained in $\bar{U} - U$; it is a matter therefore of seeing that one cannot have $\dim(\mathcal{O}_{X,x}) = 1$. Now, this relation would imply, by (21.12.5.3), $x \in \text{Daf}(U^{(x)}/\text{Spec}(\mathcal{O}_{X,x}))$, where $U^{(x)} = U \cap \text{Spec}(\mathcal{O}_{X,x})$. But the only opens of $\text{Spec}(\mathcal{O}_{X,x})$ are $\text{Spec}(\mathcal{O}_{X,x})$ itself and the parts of the set (finite) of maximal points of $\text{Spec}(\mathcal{O}_{X,x})$. Now, the open set reduced to a maximal point of $\text{Spec}(\mathcal{O}_{X,x})$ is affine, hence by definition of preschemes; one concludes that *all* the open sets in $\text{Spec}(\mathcal{O}_{X,x})$ are affine, hence $\text{Daf}(U^{(x)}/\text{Spec}(\mathcal{O}_{X,x})) = \emptyset$ contrary to the hypothesis.

To prove (i) it must be shown further, to know that if $x \in X$ is such that $\dim(\mathcal{O}_{X,x}) = 1$, there exists an open neighbourhood W of x in X such that W does not meet $\text{Daf}(U/X)$, i.e. such that the canonical injection $f_W : U \cap W \rightarrow W$ is affine. But, with the same notations as above, one has just seen that the injection $f^{(x)} : U^{(x)} \rightarrow \text{Spec}(\mathcal{O}_{X,x})$ is affine. One can evidently restrict to the case where X is Noetherian; as the morphism f is of finite presentation, the conclusion results from (8.10.5, (viii)) applied following the method of (8.1.2, a)).

(ii) It is a matter of proving that for every point $x \in T$, one has $x \in f^0(U^0)$; we shall first show that one can reduce to the case where $X = \text{Spec}(A)$, where A is a complete local Noetherian ring, and x the closed point of X . For this, it suffices to make the base change $h : X' = \text{Spec}(A) \rightarrow X$, where $A = \widehat{\mathcal{O}_{X,x}}$; if one sets $U' = h^{-1}(U)$, $f' = f_{(X')}$ is the canonical injection $U' \rightarrow X'$, and since the morphism h is *flat*, it results from (21.12.2) that if one proves that x belongs to $f'^0(U'^0)$, one will deduce that $x \in f^0(U^0)$. By virtue of (6.1.1), one has operated the sought reduction.

Let then X_1 be a closed irreducible subscheme of X having for underlying space an irreducible component of X , of *maximal* dimension, among those that contain a maximal component of T , and set $U_1 = U \cap X_1$, $T_1 = T \cap X_1 = X_1 - U_1$; one has therefore $\text{codim}(T_1, X_1) \geq 2$ (0, 14.2.1), and the couple (U_1, X_1) satisfies the same hypotheses as the couple (U, X) (X_1 being the spectrum of a quotient ring of A , hence local Noetherian and complete). One has moreover a commutative diagram

$$\begin{array}{ccc} U_1 & \xrightarrow{j} & U \\ h_1 \downarrow & & \downarrow f \\ X_1 & \xrightarrow{i} & X \end{array}$$

where i and j are the canonical injections, i being therefore an *affine* morphism; one deduces from (21.12.1) the existence of a morphism $j^0 : U_1^0 \rightarrow U^0$ making commutative the diagram

$$\begin{array}{ccc} U_1^0 & \xrightarrow{j^0} & U^0 \\ f_1^0 \downarrow & & \downarrow f^0 \\ X_1 & \xrightarrow{i} & X \end{array}$$

and hence, to prove that $x \in f^0(U^0)$, it suffices to prove that $x \in f_1^0(U_1^0)$. One can therefore replace X by X_1 , and one can consequently suppose that the ring A is moreover *integral*. But A is a quotient of a regular local Noetherian ring by virtue of the theorem of Cohen (0, 19.8.8, (i)) and since it is integral, one can apply (5.11.1) with the family (x_α) reduced to the single maximal point of X ; since $\text{codim}(T, X) \geq 2$ by hypothesis, one can suppose that U contains none of the irreducible components of Y whose adherence contains y . By replacing X by $X' = X - \bigcup_i T_i$, one sees that the image by g of $X' - X'_y$ does not contain the images $g(t_i)$, hence does not contain any of the irreducible components of Y' whose adherence contains y . As X and Y are affine, the morphism $g : X \rightarrow Y$ is affine, hence so is $f_*(\mathcal{O}_U)$ is a *coherent* $\mathcal{O}_X$ -Module, hence the morphism $f^0 : U^0 \rightarrow X$ is *finite*; as this morphism is dominant (U being everywhere dense), it is surjective ((II, 6.1.10)), and hence $x \in f^0(U^0)$. $\square$

Corollary (21.12.7). — *Let X be a locally Noetherian prescheme, U a sub-prescheme induced on an open of X , $j : U \rightarrow X$ the canonical immersion; suppose that j is an affine morphism. Then every irreducible component of $T = X - U$ is of codimension ≤ 1 (and in particular is of codimension 1 if U is everywhere dense).*

Proof. Suppose indeed that one of the irreducible components T_1 of T is of codimension ≥ 2 in X . Replacing X if needed by a neighbourhood of the generic point of T_1 , one can suppose T irreducible and of codimension ≥ 2 . But then the hypothesis of (21.12.6, (ii)) implies that j^0 must be surjective. But then $j : U \rightarrow X$ is affine implies that U^0 is identified with U and j^0 with j , in other words $j^0(U^0) = U$; or, this contradicts the conclusion of (21.12.6) that, under the hypothesis of dimension made, j^0 must be surjective. $\square$

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(21.12.8). Let A be a local Noetherian ring, $Y = \text{Spec}(A)$, y the unique closed point of Y , $Y' = Y - \{y\}$. Consider the following condition:

(W) *For every open U of Y contained in Y' , not containing any irreducible component of Y' and such that the canonical immersion $U \rightarrow Y'$ is affine, U is itself an affine open.*

This condition is implied by the following:

(W') For every closed part T of Y whose every irreducible component is of codimension 1 in Y , and such that for every irreducible component Y_i of Y , one has $Y_i \cap T \neq \{y\}$, the open $U = Y - T$ is affine.

Indeed, if (W') is satisfied and if U is an open of Y satisfying the hypothesis of (W), it results from (21.12.7) that no irreducible component of $T = Y - U$ can be of codimension ≥ 2 ; as moreover U contains none of the irreducible components of the open $Y' - \{y\}$ of Y' , the condition (W') shows that U is affine.

One will note that condition (W') when Y is irreducible and is then equivalent to the following:

(W'') For every irreducible closed part T of Y , of codimension 1 in Y , $Y - T$ is an affine open.

Indeed, it is clear that (W') implies (W'') when Y is irreducible, and the reciprocal comes from considering the irreducible components of T and noting that the intersection of a finite number of affine opens is an affine open ((I, 5.5.6)).

(21.12.9). Examples (21.12.9). — If A is a local Noetherian factorial ring, it satisfies (W') (and a fortiori (W)), since every prime ideal of height 1 is principal ((I, 1.3.6)). But there exist local Noetherian rings that are not factorial satisfying (W'), for example those of dimension ≤ 1 : indeed, one has observed in the proof of (21.12.6, (i)) that all the opens of Y are then affine. One can on the other hand prove using the theory of local duality (chap. III, 3^e part) that every local Noetherian ring of dimension 2 satisfies (W').

Proposition (21.12.10). — Let X, Y be two locally Noetherian preschemes, $g : X \rightarrow Y$ a morphism, y a closed point of Y , $Y' = Y - \{y\}$, $X' = g^{-1}(Y')$; suppose that the morphism $g' : X' \rightarrow Y'$ restriction of g is an open immersion, and that the local ring $\mathcal{O}_{Y,y}$ satisfies condition (W) (21.12.8). Then, for every irreducible component Z of $X_y = g^{-1}(y)$, either Z is of codimension ≤ 1 in X , or its generic point is isolated in Z . If Z is locally of finite type over $k(y)$, the second alternative implies that Z is reduced to a single point.

Proof. The last assertion of the statement results from this that Z is then a prescheme of Jacobson (10.4.7), and as the set of closed points of Z is then dense in Z , the generic point of Z cannot be isolated in Z unless Z is reduced to a single point.

Suppose that there exists in X_y an irreducible component Z whose generic point z is not isolated in Z and such that $\text{codim}(Z, X) \geq 2$. The question being local on X and Y , one can, by replacing X by a neighbourhood of z in X , suppose X and Y affine, $X_y = Z$ irreducible, not reduced to a point and such that $\text{codim}(Z, X) \geq 2$. The image $g(X - X_y) = U$ is an open of Y' isomorphic to X' by hypothesis; let us show by replacing if needed X by a neighbourhood of y , that U does not contain any of the irreducible components of Y' whose adherence contains y . Indeed, one can first suppose that all the maximal points of X are generisations of z , the set of these points being finite; hence the set of maximal points of U is the set of images y_i by g of the maximal points x_i of X (no maximal point of X can be contained in X_y , since $\dim(\mathcal{O}_{X,x}) \geq 2$). Setting $X_i = \overline{\{x_i\}}$, since by hypothesis, one has a $z \in X_i$ and since z is not isolated in X_y , there exists in X_y a point $x \neq z$. As $X_i \neq Z$, there exists in X_i a point t_i such that if one sets $T_i = \overline{\{t_i\}}$, one has $\dim(\mathcal{O}_{T_i,z}) = 1$ (10.5.9); this implies by hypothesis $t_i \notin Z$, hence t_i is not a generisation of z . Replacing X by $X' = X - \bigcup_i T_i$, one sees that the image by g of $X' - X'_y$ does not contain the images $g(t_i)$, hence does not contain any of the irreducible components of Y' whose adherence contains y . Since X and Y are affine, the morphism $g : X \rightarrow Y$ is affine, hence so is the restriction $g' : X' \rightarrow Y'$ which is by hypothesis an immersion, hence the canonical immersion $U \rightarrow Y'$ is affine. In view of condition (W), $U = U \cap Y'$ is then an affine open. Setting $Y_1 = \text{Spec}(\mathcal{O}_{Y,y})$, $Y'_1 = Y_1 - \{y\}$, $U_1 = U \cap Y_1$, one has therefore $\text{codim}(T_1, X_1) \geq 2$, and the couple (U_1, X_1) satisfies the same hypotheses. We have a commutative diagram, and by (21.12.1) the existence of a morphism $f^0 : U_1^0 \rightarrow U^0$. But A is a quotient of a regular local ring by the theorem of Cohen, and since it is integral, one can apply (5.11.1); since $\text{codim}(T, X) \geq 2$, $f_*(\mathcal{O}_U)$ is a coherent $\mathcal{O}_X$ -Module, hence the morphism $f^0 : U^0 \rightarrow X$ is finite; as this morphism is dominant, it is surjective ((II, 6.1.10)), and hence $x \in f^0(U^0)$; Q.E.D. $\square$

Corollary (21.12.11). — Let X be a locally Noetherian prescheme, U a sub-prescheme induced on an open of X , $j : U \rightarrow X$ the canonical immersion; suppose that j is an affine morphism. Then every irreducible component of $T = X - U$ is of codimension ≤ 1 (and in particular is of codimension 1 if U is everywhere dense).

Theorem (21.12.12). — (van der Waerden) *Let X, Y be two locally Noetherian integral preschemes, $g : X \rightarrow Y$ a birational morphism and locally of finite type. Suppose moreover that Y is normal and that for every $y \in Y$, the local ring $\mathcal{O}_{Y,y}$ satisfies condition (W) of (21.12.8) (conditions satisfied in particular when Y is locally factorial (21.12.9)). If V is the largest open of X such that the restriction $g|_V : V \rightarrow Y$ is a local isomorphism, then all the irreducible components of $T = X - V$ are of codimension 1 in X .*

Proof. One will note that since g is birational, the open V is not empty; it therefore suffices to prove that no maximal point z of T can be isolated in X_y , where $y = g(z)$. But, by restricting to a neighbourhood of z , one can suppose $T = X_y$; as moreover $z \in X_y$, one can restrict to the case where $T = X_y$, and one can suppose $g : X \rightarrow Y$ is an affine morphism. On the other hand, since A is integral, A is a discrete valuation ring, hence regular, and the hypothesis $\mathfrak{m}_{x'} \notin \text{Ass}(\mathcal{O}_{X',x'})$ implies A' is Cohen–Macaulay (0, 16.4.6); as $\dim(A') = 1$, it results from (6.1.5) that A' is flat. Then all the fibres $g^{-1}(y')$ ($y' \in Y$) would be empty or reduced to a point and it would result from the “Main theorem” (8.12.10) that g would be a local isomorphism, contrary to the hypothesis $z \in T$. $\square$

Corollary (21.12.13). — *Suppose satisfied the hypotheses of (21.12.12) and moreover, suppose that g is quasi-finite in each of the points of $X^{(1)}$ (one recalls that these are the points x where $\dim(\mathcal{O}_{X,x}) = 1$); then g is a local isomorphism, and if moreover g is separated, g is an open immersion.*

Proof. It suffices to prove the first assertion, the second being a consequence of the first and of ((I, 8.2.8)). Everything reduces to proving, with the notations of (21.12.12), that $T = \emptyset$; in the contrary case, a generic point z of an irreducible component of T would belong to $X^{(1)}$ by virtue of (21.12.12), and by hypothesis it would be isolated in X_y with $y = g(z)$; but one has seen in the proof of (21.12.12) that this is not possible. $\square$

Remark (21.12.14). — (i) The conclusion of (21.12.12) applies when Y is regular and integral and X integral, by virtue of the theorem of Auslander–Buchsbaum (21.11.1), Y being then locally factorial. On the other hand, one knows examples where X and Y are normal algebraic schemes of dimension 3, over an algebraically closed field of characteristic 3, and where the conclusion of (21.12.12) is no longer valid.

(ii) The set T of (21.12.12) is also the set of points where g is ramified. Indeed, if g is unramified at a point $x \in X$, it is also unramified in a neighbourhood affine open U of x (17.3.7), hence $g|_U$ is a separated, quasi-finite and birational morphism (17.4.3); since Y is normal, one concludes from the “Main theorem” ((III, 4.4.9)) that g is a local isomorphism at the point x , hence $x \in X - T$. Reciprocally, g is evidently unramified at every point where it is a local isomorphism. This justifies the title given to this section.

(iii) Without supposing that Y is locally factorial, but supposing on the other hand that X is a complete intersection relative to Y (chap. V), we will see in chap. V that one has a more precise result than (21.12.12), expressing T as a 1-codimensional cycle of a section of an invertible $\mathcal{O}_X$ -Module, canonically associated to g . This applies notably when X and Y are both regular, or when g is an S -morphism, X and Y being smooth preschemes over S .

(iv) One can ask whether (21.12.10) admits a converse; in other words, for a given local Noetherian ring A , if one sets $Y = \text{Spec}(A)$, y denoting the closed point of Y , and if, for every locally Noetherian prescheme X and every morphism $g : X \rightarrow Y$ such that $g' : X' \rightarrow Y'$ is an open immersion, every irreducible component of X_y is of codimension ≤ 1 in X or is reduced to a point, then is it true that A satisfies condition (W) of (21.12.8)?

(v) Let Y be a regular integral prescheme, X a locally Noetherian normal prescheme, $g : X \rightarrow Y$ a morphism locally of finite type; suppose moreover that, if η is the generic point of Y , the fibre X_η is étale over $k(\eta)$, and let T' be the complement of the largest open V' of X such that $g|_{V'} : V' \rightarrow Y$ is étale; is it true then that all the irreducible components of T' are of codimension 1? This is indeed the case when one supposes moreover that g is locally quasi-finite (“purity theorem” of Zariski–Nagata). One can show that the preceding conjecture would be a consequence of the following (and would even be equivalent to it if the conjecture of (iv) were verified): for a regular local ring A contained in an integral and integrally closed local ring B , such that B is a finite A -module, the open U of the

points of $\mathrm{Spec}(B)$ at which $\mathrm{Spec}(B)$ is unramified over $\mathrm{Spec}(A)$ (or, what amounts to the same by (18.10.1), étale over $\mathrm{Spec}(A)$) is affine.

Proposition (21.12.15). — *Let S be a prescheme, $g : X \rightarrow S$ a flat morphism and locally of finite presentation, whose fibres $X_s = g^{-1}(s)$ are geometrically irreducible (4.5.2), $h : Y \rightarrow S$ a smooth morphism; we denote by $Y_s = h^{-1}(s)$ the fibres of this morphism. Let $f : X \rightarrow Y$ be a proper S -morphism such that, for every maximal point η of S , the morphism $f_\eta : X_\eta \rightarrow Y_\eta$ is an isomorphism. Then f is an isomorphism.*

Proof. Since h is flat, every maximal point of Y is above a maximal point of S (2.3.4), hence the union of the Y_η , as η runs through the maximal points of S , is dense in Y ; as the f_η are isomorphisms, f is dominant and hence surjective since it is proper. By (17.9.5), it suffices to prove that for every $s \in S$, $f_s : X_s \rightarrow Y_s$ is an open immersion. As s is a generisation of a maximal point η_s , one can, by making the base change $\mathrm{Spec}(\mathcal{O}_{S,s}) \rightarrow S$, reduce to the case where S is the spectrum of a local integral Noetherian ring A_0 of $A = \mathcal{O}_{S,s}$ such that $A_0 \rightarrow A$ is a local homomorphism, two morphisms of finite type $g_0 : X_0 \rightarrow S_0 = \mathrm{Spec}(A_0)$, $h_0 : Y_0 \rightarrow S_0$ and an S_0 -morphism $f_0 : X_0 \rightarrow Y_0$ such that g, h, f are deduced from g_0, h_0, f_0 by the base change $S \rightarrow S_0$ ((8.9.1) and (5.13.3)); one can moreover suppose g_0 flat (11.2.7), h_0 smooth (17.7.9) and f_0 proper (8.10.5, (xii)). On the other hand, the projection S_0 of the generic point η of S is the closed point of S_0 , and it results from (2.7.1), (viii)) that the morphism $(f_0)_{\eta_0} : (X_0)_{\eta_0} \rightarrow (Y_0)_{\eta_0}$ is an isomorphism; finally, the closed point s_0 of S_0 is above the generic point η of S , hence one can reduce to the case where moreover S is Noetherian.

On the other hand, the projection of the generic point η of S is the closed point of S_0 ; we can therefore reduce to the case where S is the spectrum of a discrete valuation ring. One can then finish by reducing to the case where Y is integral. Since $g : X \rightarrow S$ is flat, the points x of X are contained in X_η (2.3.4) and since f_η is an isomorphism, X is irreducible and the local ring at its generic point is a field; moreover, by flatness (3.3.2), X has no embedded cycles, hence X is reduced (3.2.1) and hence f is a birational morphism and separated. To prove that f is an open immersion, one can apply the criterion (21.12.13); to see that f is quasi-finite at the points of $X^{(1)}$, one remarks that the assertion is evident for those of these points that belong to X_η ; the only point of $X^{(1)}$ belonging to X_s is the generic point x of X_s (6.1.1). Now, it results from (2.4.6) and of (14.2.2) that the morphisms g and h are equidimensional; as X_η and Y_η are isomorphic, the irreducible components of X_s and of Y_s have all the same dimension. But since by hypothesis X_s is irreducible and as one has seen that f is surjective, hence it is also the case for $f_s : X_s \rightarrow Y_s$, which entails that Y_s is also irreducible. Hence $f(x) = y$, and the relation $\dim(X_s) = \dim(Y_s)$ entails then, by virtue of (5.6.6), that $\dim(f_s^{-1}(y)) = 0$, which says that f_s (and hence also f) is quasi-finite at the point x , which finishes the proof. $\square$

Corollary (21.12.16). — *Let $g : X \rightarrow S$ be a proper morphism, flat and of finite presentation, $h : Y \rightarrow S$ a smooth morphism, proper and of finite presentation, $f : X \rightarrow Y$ an S -morphism. Suppose that the fibres $X_s = g^{-1}(s)$ of g are geometrically irreducible. Let U be the set of $s \in S$ such that $f_s : X_s \rightarrow Y_s$ is an isomorphism. Then U is open and closed in S , and the morphism $g^{-1}(U) \rightarrow h^{-1}(U)$, restriction of f , is an isomorphism.*

Proof. The last assertion results from the first and from (17.9.5). We already know (9.6.1), (xi)) that U is locally constructible in S . By virtue of (1.10.1), it suffices to prove that U is stable by specialisation and by generisation. To prove these assertions, one reasons as in the proof of (21.12.15) (noting, with the same notations, that the closed point of S is the projection of the closed point of S_0) to reduce to the case where moreover S is Noetherian; then the conclusion results from (III, 4.6.7), (ii)). $\square$

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Remark (21.12.17). — (i) The conclusion of the proposition (21.12.15) is no longer valid if one eliminates the hypothesis that the fibres X_s are irreducible. Take for example $S = \mathrm{Spec}(A)$, where A is a discrete valuation ring, $Y = \mathbf{P}_k^1$, which is proper and smooth over S (17.3.9). Let s and η be the closed point and the generic point of S ; let z be a closed point of Y_s , say a rational point over $k = k(s)$, and let X be the Y -prescheme obtained by blowing up z in the construction of (15.1.1.6) that, if $f : X \rightarrow Y$ is the structural morphism, $f^{-1}(z)$ is isomorphic to $\mathbf{P}_k^1$, and on the other hand, the complement of $f^{-1}(z)$ in X_s , i.e. the two irreducible components of X_s . Since X is integral ((II, 8.1.4)) and for every affine open U of X , the homomorphism $A \rightarrow \Gamma(U, \mathcal{O}_X)$ is injective (1.2.7),

hence $\Gamma(U, \mathcal{O}_X)$ is an A -module without torsion, and hence flat since A is a discrete valuation ring (01, 6.3.4). It is clear that $f_\eta : X_\eta \rightarrow Y_\eta$ is an isomorphism, though $f_s : X_s \rightarrow Y_s$ is not.

(ii) The conclusion of (21.12.15) also fails if, under the hypotheses, one drops the hypothesis that f is *proper*. It suffices to take to see this to define S and Y as in (i), f by the restriction $f' : X' \rightarrow Y$ of f ; the morphism $g' : X' \rightarrow S$, restriction of g , is still flat, and this time X'_s is geometrically irreducible; X'_s is moreover isomorphic to the complement of a closed point in $\mathbb{A}_k^1$; hence X'_s is an affine scheme isomorphic to $\mathbb{A}_k^1$; as its image in Y_s is reduced to the point z , f' is not an isomorphism, even though f' is one.

(iii) It is possible that the statement of the proposition (21.12.15) remains valid when one replaces the word “isomorphism” by “étale morphism” (cf. (21.12.14), (v)).

21.13. Parafactorial pairs. Parafactorial local rings.

Definition (21.13.1). — Let X be a ringed space, Y a closed subset of X , $U = X - Y$. We say that the couple (X, Y) is *parafactorial* if, for every open V of X , the restriction functor $\mathcal{L} \mapsto \mathcal{L}|_{(U \cap V)}$, from the category of invertible $\mathcal{O}_V$ -Modules to that of invertible $\mathcal{O}_{U \cap V}$ -Modules, is an equivalence of categories.

To say that the couple (X, Y) is parafactorial thus means that, for every open V of X , the following conditions are satisfied:

- 1° the functor $\mathcal{L} \mapsto \mathcal{L}|_{(U \cap V)}$ is *fully faithful*, in other words, for two invertible $\mathcal{O}_V$ -Modules $\mathcal{L}, \mathcal{L}'$, the restriction map

$$\mathrm{Hom}_{\mathcal{O}_V}(\mathcal{L}, \mathcal{L}') \longrightarrow \mathrm{Hom}_{\mathcal{O}_{U \cap V}}(\mathcal{L}|_{(U \cap V)}, \mathcal{L}'|_{(U \cap V)})$$

is bijective;

- 2° the functor $\mathcal{L} \mapsto \mathcal{L}|_{(U \cap V)}$ is *essentially surjective*, in other words, for every invertible $\mathcal{O}_{U \cap V}$ -Module $\mathcal{L}_0$, there exists an invertible $\mathcal{O}_V$ -Module $\mathcal{L}$ such that $\mathcal{L}_0$ is isomorphic to $\mathcal{L}|_{(U \cap V)}$; this is also expressed by saying that the canonical homomorphism (21.3.2.4)

$$\mathrm{Pic}(V) \longrightarrow \mathrm{Pic}(U \cap V)$$

is *surjective*.

Lemma (21.13.2). — Let $f : X' \rightarrow X$ be a morphism of ringed spaces; for every open V of X , we denote by $f_V : f^{-1}(V) \rightarrow V$ the restriction of f . The following conditions are equivalent:

- a) For every open V of X , the functor $\mathcal{E} \mapsto f_V^*(\mathcal{E})$ from the category of $\mathcal{O}_V$ -Modules locally free of finite rank to the category of $\mathcal{O}_{f^{-1}(V)}$ -Modules locally free of finite rank, is faithful (resp. fully faithful).
- a') For every open V of X , the functor $\mathcal{L} \mapsto f_V^*(\mathcal{L})$ from the category of $\mathcal{O}_V$ -Modules locally free of rank 1 to the category of $\mathcal{O}_{f^{-1}(V)}$ -Modules locally free of rank 1 is faithful (resp. fully faithful).
- b) For every open V of X , the canonical homomorphism (01, 4.4.3.2) $\mathcal{E} \rightarrow (f_V)_*(f_V^*(\mathcal{E}))$ is a monomorphism (resp. an isomorphism) for every $\mathcal{O}_V$ -Module locally free of finite rank $\mathcal{E}$.
- b') The canonical homomorphism $\mathcal{O}_X \rightarrow f_*(\mathcal{O}_{X'})$ is a monomorphism (resp. an isomorphism).

Suppose that the canonical homomorphism $\mathcal{O}_X \rightarrow f_*(\mathcal{O}_{X'})$ is bijective; then, for an $\mathcal{O}_{X'}$ -Module locally free of finite rank $\mathcal{E}'$ to be isomorphic to an $\mathcal{O}_{X'}$ -Module of the form $f^*(\mathcal{E})$, where $\mathcal{E}$ is an $\mathcal{O}_X$ -Module locally free of finite rank, it is necessary and sufficient that the two following conditions be satisfied:

- (i) $f_*(\mathcal{E}')$ is an $\mathcal{O}_X$ -Module locally free;
- (ii) the canonical homomorphism (01, 4.4.3.3) $f^*(f_*(\mathcal{E}')) \rightarrow \mathcal{E}'$ is an isomorphism.

When these two conditions are satisfied, $\mathcal{E}$ is isomorphic to $f_*(\mathcal{E}')$.

To say that the functor $\mathcal{E} \mapsto f_V^*(\mathcal{E})$ is faithful (resp. fully faithful) means that for two $\mathcal{O}_V$ -Modules locally free of finite rank $\mathcal{E}_1, \mathcal{E}_2$, the map $\mathrm{Hom}_{\mathcal{O}_V}(\mathcal{E}_1, \mathcal{E}_2) \rightarrow \mathrm{Hom}_{\mathcal{O}_{f^{-1}(V)}}(f_V^*(\mathcal{E}_1), f_V^*(\mathcal{E}_2))$ is injective (resp. bijective); as this must hold on replacing V by an open $W \subset V$ and $\mathcal{E}_1, \mathcal{E}_2$ by $\mathcal{E}_1|_W, \mathcal{E}_2|_W$, and as $\mathrm{Hom}_{\mathcal{O}_W}(\mathcal{E}_1|_W, \mathcal{E}_2|_W) = \Gamma(W, \mathcal{H}om_{\mathcal{O}_V}(\mathcal{E}_1, \mathcal{E}_2))$, the condition means also that the canonical sheaf homomorphism

$$(21.13.2.1) \quad \mathcal{H}om_{\mathcal{O}_V}(\mathcal{E}_1, \mathcal{E}_2) \longrightarrow (f_V)_*(f_V^*(\mathcal{H}om_{\mathcal{O}_V}(\mathcal{E}_1, \mathcal{E}_2)))$$

is injective (resp. bijective). But since $\mathcal{E}_1$ and $\mathcal{E}_2$ are locally free of finite rank, $\mathcal{H}om_{\mathcal{O}_V}(\mathcal{E}_1, \mathcal{E}_2)$, isomorphic to $\mathcal{E}_1^\vee \otimes_{\mathcal{O}_V} \mathcal{E}_2$ (01, 5.4.2) is also locally of finite rank; this already shows that a) implies b), and

conversely b') is a particular case of a), taking into account the isomorphism $\mathcal{E}^\vee \cong \mathcal{H}om_{\mathcal{O}_V}(\mathcal{E}, \mathcal{O}_V)$. It is clear that $a')$ is a particular case of a) and $b')$ a particular case of $a')$. Conversely, as $b)$, the question being local, one can, to verify it, reduce to the case where $\mathcal{E} = \mathcal{O}_V^n$, and this shows that $b')$.

If the canonical homomorphism $\mathcal{O}_X \rightarrow f_*(\mathcal{O}_{X'})$ is bijective, and if conditions (i) and (ii) are satisfied, it is clear that $\mathcal{E}' = f^*(\mathcal{E})$ with $\mathcal{E} = f_*(\mathcal{E}')$, up to isomorphism. Conversely, suppose that $\mathcal{E}' = f^*(\mathcal{E})$, with $\mathcal{E}$ locally free; the question being local on X , one can suppose that $\mathcal{E} = \mathcal{O}_X^n$, whence $\mathcal{E}' = \mathcal{O}_{X'}^n$, and conditions (i) and (ii) result from the hypothesis that $\mathcal{O}_X \rightarrow f_*(\mathcal{O}_{X'})$ is an isomorphism.

In this number, we will apply the preceding lemma to a canonical injection $j : U \rightarrow X$, where U is the ringed space induced by X on an open of X . With these notations, (21.13.2) translates as:

Corollary (21.13.3). — *Let X be a ringed space, Y a closed subset of X , $U = X - Y$, $j : U \rightarrow X$ the canonical injection. The following conditions are equivalent:*

- a) For every open V of X , the restriction functor $\mathcal{E} \mapsto \mathcal{E}|(U \cap V)$ from the category of $\mathcal{O}_V$ -Modules locally free of finite rank to the category of $\mathcal{O}_{U \cap V}$ -Modules locally free of finite rank is faithful (resp. fully faithful).*
- a') For every open V of X , the restriction functor $\mathcal{L} \mapsto \mathcal{L}|(U \cap V)$ from the category of $\mathcal{O}_V$ -Modules locally free of rank 1 to the category of $\mathcal{O}_{U \cap V}$ -Modules locally free of rank 1 is faithful (resp. fully faithful).*
- b) For every open V of X , the canonical homomorphism $\mathcal{E} \rightarrow (j_V)_*(\mathcal{E}|(U \cap V))$ is injective (resp. bijective).*
- b') The canonical homomorphism $\mathcal{O}_X \rightarrow j_*(\mathcal{O}_U)$ is injective (resp. bijective).*

Suppose that the canonical homomorphism $\mathcal{O}_X \rightarrow j_*(\mathcal{O}_U)$ is bijective. Then, for an $\mathcal{O}_U$ -Module locally free of finite rank $\mathcal{F}$ to be of the form $\mathcal{E}|U$, where $\mathcal{E}$ is an $\mathcal{O}_X$ -Module locally free of finite rank, it is necessary and sufficient that $j_*(\mathcal{F})$ be an $\mathcal{O}_X$ -Module locally free, and in this case, one can take $\mathcal{E} = j_*(\mathcal{F})$.

Lemma (21.13.4). — *Let X be a locally Noetherian prescheme, Y a closed subset of X , $U = X - Y$, $j : U \rightarrow X$ the canonical injection. For the canonical homomorphism $\mathcal{O}_X \rightarrow j_*(\mathcal{O}_U)$ to be injective (resp. bijective), it is necessary and sufficient that $\text{prof}_Y(\mathcal{O}_X) \geq 1$ (resp. $\text{prof}_Y(\mathcal{O}_X) \geq 2$) (in other words, that $\text{prof}(\mathcal{O}_{X,x}) \geq 1$ (resp. $\text{prof}(\mathcal{O}_{X,x}) \geq 2$) for every $x \in Y$).*

Proof. These assertions are indeed particular cases of (5.10.2) and (5.10.5). $\square$

Proposition (21.13.5). — *Let X be a ringed space, Y a closed subset, $U = X - Y$, $j : U \rightarrow X$ the canonical injection. For the couple (X, Y) to be parafactorial, it is necessary and sufficient that for every open V of X and every invertible $\mathcal{O}_{U \cap V}$ -Module $\mathcal{L}$, $(j_V)_*(\mathcal{L})$ be an invertible $\mathcal{O}_V$ -Module (notation of (21.13.3)).*

Proof. This is an immediate consequence of the definition (21.13.1) and of (21.13.3). $\square$

Corollary (21.13.6). — *Let X be a ringed space, Y a closed subset of X .*

- (i) If the couple (X, Y) is parafactorial, so is $(W, Y \cap W)$ for every open W of X . Conversely, if (W_α) is an open covering of X such that each of the couples $(W_\alpha, Y \cap W_\alpha)$ is parafactorial, then the couple (X, Y) is parafactorial.*
- (ii) If the couple (X, Y) is parafactorial, so is (X, Y') for every closed subset Y' of Y .*
- (iii) Suppose that X is a prescheme, and let X' be a prescheme, $f : X' \rightarrow X$ a faithfully flat and quasi-compact morphism; set $Y' = f^{-1}(Y)$. Suppose that the couple (X', Y') is parafactorial and that the open $U = X - Y$ is retrocompact in X (0_{III}, 9.1.1). Then the couple (X, Y) is parafactorial.*

Proof. (i) As the fact that the canonical homomorphism $\mathcal{O}_X \rightarrow j_*(\mathcal{O}_U)$ is bijective is a local property on X , everything reduces to seeing (by virtue of (21.13.5)), denoting by j_α the canonical injection $U \cap W_\alpha \rightarrow W_\alpha$, that one has the following property: if, for every α and every invertible $\mathcal{O}_U$ -Module $\mathcal{L}$, $(j_\alpha)_*(\mathcal{L}|(U \cap W_\alpha))$ is an invertible $\mathcal{O}_{W_\alpha}$ -Module, then $j_*(\mathcal{L})$ is an invertible $\mathcal{O}_X$ -Module. But the property of being an invertible $\mathcal{O}_X$ -Module is local on X ; as one has $j_*(\mathcal{L})|W_\alpha = (j_\alpha)_*(\mathcal{L}|(U \cap W_\alpha))$, this proves our assertion.

(ii) Set $U' = X - Y'$ and let $j' : U' \rightarrow X$ be the canonical injection. To say that the canonical homomorphism $\mathcal{O}_X \rightarrow j'_*(\mathcal{O}_{U'})$ is bijective amounts to saying that for every open V of X , the homomorphism $\Gamma(V, \mathcal{O}_X) \rightarrow \Gamma(V \cap U', \mathcal{O}_X)$ is bijective; but the composed homomorphism

$$\Gamma(V, \mathcal{O}_X) \longrightarrow \Gamma(V \cap U', \mathcal{O}_X) \longrightarrow \Gamma(V \cap U, \mathcal{O}_X)$$

is bijective by hypothesis (21.13.5) and on replacing V by $V \cap U'$, one sees that $\Gamma(V \cap U', \mathcal{O}_X) \rightarrow \Gamma(V \cap U, \mathcal{O}_X)$ is bijective, hence $\Gamma(V, \mathcal{O}_X) \rightarrow \Gamma(V \cap U', \mathcal{O}_X)$ is bijective. Note next that if $j'' : U \rightarrow U'$ is the canonical injection and if $\mathcal{L}'$ is an invertible $\mathcal{O}_{U'}$ -Module, then $j_*(\mathcal{L}'|_U) = j'_*(j''_*(\mathcal{L}'|_U))$ is by hypothesis an invertible $\mathcal{O}_X$ -Module. As the couple $(U', U' \cap Y)$ is parafactorial by virtue of (i), this proves our assertion.

(iii) Set $U' = X' - Y'$ and let $j' : U' \rightarrow X'$ be the canonical injection. The hypothesis that $\mathcal{O}_{X'} \rightarrow j'_*(\mathcal{O}_{U'})$ is bijective means that for every open V of X , the homomorphism $\Gamma(V, \mathcal{O}_X) \rightarrow \Gamma(V \cap U, \mathcal{O}_X)$ is bijective. By making the base change $\text{Spec}(\mathcal{O}_{S,s}) \rightarrow S$, one can reduce to the case where S is a local scheme, and the conclusion follows from (21.13.5) and from (2.5.2). $\square$

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Definition (21.13.7). — We say that a local ring A is *parafactorial* if, setting $X = \text{Spec}(A)$ and denoting by a the closed point of X , the couple $(X, \{a\})$ is parafactorial.

Proposition (21.13.8). — *The notations being those of (21.13.7), set $U = X - \{a\}$. For A to be parafactorial, it is necessary and sufficient that it satisfy the following conditions:*

- (i) *The canonical homomorphism $A = \Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(U, \mathcal{O}_X)$ is bijective.*
- (ii) *$\text{Pic}(U) = 0$.*

If moreover A is Noetherian, condition (i) is equivalent to

- (i bis) $\text{prof}(A) \geq 2$.*

Proof. Indeed, the only open of X containing a is X itself, and hence every invertible $\mathcal{O}_X$ -Module is isomorphic to $\mathcal{O}_X$, in other words $\text{Pic}(X) = 0$; the first assertion results therefore from the definition (21.13.1) and from (21.13.3). The second assertion is nothing other than a particular case of (21.13.4). $\square$

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Example (21.13.9). — (i) A Noetherian parafactorial local ring is necessarily of dimension ≥ 2 by virtue of (21.13.8); in other words a Noetherian local ring of dimension ≤ 1 is not parafactorial.

(ii) A Noetherian *factorial* local ring A of dimension ≥ 2 is parafactorial, as results from (21.13.8) and from (21.6.14).

(iii) If A is a Noetherian local ring of dimension ≥ 3 and parafactorial, it is not necessarily normal nor even reduced. Consider a regular local ring B of dimension ≥ 3 , and set $A = D_B(B)$ (0, 18.2.3), isomorphic to $B[T]/(T^2)$; one sees immediately that one has $\text{prof}(A) = \text{prof}(B) = \dim(B) \geq 3$. To show that A is parafactorial, it suffices, with the notations of (21.13.8), to prove that $\text{Pic}(U) = 0$. Let $\mathfrak{J}$ be the kernel of the augmentation $A \rightarrow B$, which is such that $\mathfrak{J}^2 = 0$ and which, as a B -module, is isomorphic to B . As $B = A/\mathfrak{J}$, $X = \text{Spec}(A)$ and $X_0 = \text{Spec}(B)$ have the same underlying space; if $\mathcal{J} = \widetilde{\mathfrak{J}}$, X_0 is the sub-prescheme defined by $\mathcal{J}$; we denote by U_0 the sub-prescheme induced by X_0 on the open U . For every $z \in \mathfrak{J}$, set $\varphi(z) = 1 + z$; since $(1 + z)(1 - z) = 1$, $1 + z$ is invertible in A and $\varphi(\mathfrak{J})$ is the kernel of the surjective canonical homomorphism of multiplicative groups $A^* \rightarrow B^*$; in other terms, one has an exact sequence of groups

$$0 \longrightarrow \mathfrak{J} \xrightarrow{\varphi} A^* \longrightarrow B^* \longrightarrow 1$$

(the last three groups being multiplicative, the first two additive). For the same reason, for every $t \in A$, one has an exact sequence

$$0 \longrightarrow \mathfrak{J}_t \xrightarrow{\varphi_t} (A_t)^* \longrightarrow (B_t)^* \longrightarrow 1$$

denoting by t_0 the image of t in B , since $B_t = A_t/\mathfrak{J}_t$; in other words, one has an exact sequence of sheaves of groups on the topological space X

$$0 \longrightarrow \mathcal{J} \xrightarrow{u} \mathcal{O}_X^* \longrightarrow \mathcal{O}_{X_0}^* \longrightarrow 1.$$

By the exact cohomology sequence, one deduces an exact sequence

$$(21.13.9.1) \quad H^1(U, \mathcal{J}) \longrightarrow H^1(U, \mathcal{O}_U^*) \xrightarrow{u_*} H^1(U, \mathcal{O}_{U_0}^*).$$

But since $\mathfrak{J}$ is a B -module isomorphic to B , one has $H^1(U, \mathcal{J}) = H^1(U_0, \mathcal{O}_{U_0})$, and it results from chap. III, 3^e partie (see also [?, III, Exemple III-1]) that one has $H^1(U_0, \mathcal{O}_{U_0}) = 0$ by reason of the relation $\text{prof}(B) \geq 3$. Furthermore, as B is factorial, one has $H^1(U, \mathcal{O}_{U_0}^*) = \text{Pic}(U_0) = 0$; one therefore deduces from the preceding exact sequence that $\text{Pic}(U) = H^1(U, \mathcal{O}_U^*) = 0$.

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(iv) There also exist Noetherian local rings of dimension 3 that are integral, integrally closed and parafactorial, but not factorial. Let indeed B be a Noetherian local ring, complete, integral and integrally closed, of dimension ≥ 2 and not factorial (for example the completion of the localised ring of the integral algebra $k[U, V, W]/(W^2 - UV)$ at the image of the ideal $(U) + (V) + (W)$). Then it results from what one will see further below (21.14.2) that the local ring $A = B[[T]]$ of formal power series in T over B is parafactorial, but it is not factorial, for otherwise B would be so by virtue of (21.13.12) below.

(v) One can show that a local ring of absolute complete intersection (19.3.1) of dimension ≥ 4 is parafactorial (cf. [?, XI, 3.13 (i)]).

(vi) We have seen (Remark (ii)) that every factorial Noetherian local ring A of dimension 2 is parafactorial. But there exist Noetherian local rings of dimension 2 that are parafactorial and not factorial. One can show that, for a Noetherian local ring A of dimension 2 to be parafactorial and not factorial, it suffices that it satisfy the following conditions:

- 1° A is a Cohen–Macaulay ring (in other words $\text{prof}(A) = 2$).
- 2° A is integral and if A' is its integral closure, A' is factorial and is a finite A -algebra.
- 3° Let $\mathfrak{J}$ be the conductor of A in A' (annihilator of the A -module A'/A , or equivalently the largest ideal of A' contained in A); set $B = A/\mathfrak{J}$, $B' = A'/\mathfrak{J}$; then $\dim(B) = 1$ (which implies $A' \neq A$, in other words A is not integrally closed), and the canonical map $\text{Div}(B) \rightarrow \text{Div}(B')$ (21.4.5) is *surjective*.

One can show moreover that these conditions entail the following property:

- 4° The ring B' (and *a fortiori* $B \subset B'$) is reduced, and the morphism $\text{Spec}(B') \rightarrow \text{Spec}(B)$ is bijective.

If one sets $X = \text{Spec}(A)$, $X' = \text{Spec}(A')$, $Y = \text{Spec}(B)$, $Y' = \text{Spec}(B')$, so that Y (resp. Y') is defined by the ideal $\mathfrak{J}$ of A (resp. A'), the structural morphism $f : X' \rightarrow X$ is an isomorphism of $X' - Y'$ onto $X - Y$ (Bourbaki, *Alg. comm.*, chap. V, § 1, n° 5, cor. 5 of prop. 16). One sees then by virtue of 4° that f is a *bijective* morphism; in other words, X is a unibranch prescheme (6.15.1) (and in particular A' is a *local* Noetherian ring); in general X is not geometrically unibranch. The space Y , of dimension 1, is constituted by the closed point x of X and the maximal points y_i ($1 \leq i \leq r$) of Y , and the unique point y'_i of Y' above y_i is also a maximal point of Y' ; as B and B' are reduced, one deduces that one has $\mathfrak{m}_{Y', y'} = \mathfrak{m}_{y'}$ for $z = y_i$ and $z' = y'_i$; this relation is also evidently satisfied for $z \neq Y$ and z' the unique point of $f^{-1}(z)$, hence for every point $z \in U = X - \{x\}$ and z' the unique point of $f^{-1}(z)$. In particular, if the ring A is of characteristic 0 (0, 11.1.1) one sees that U' is *unramified* over U (hence *étale* in general).

We will limit ourselves here to proving that conditions 1°, 2°, 3° above are *sufficient* for A to be parafactorial. By virtue of condition 1° and of (21.13.8), it suffices to prove that $\text{Pic}(U) = 0$. Since A' is factorial by virtue of 2°, one has $\text{Pic}(U') = 0$, and one deduces from (21.8.5), (ii) a surjection exact sequence

$$1 \longrightarrow \left(\prod_{s \in S} \mathcal{O}_{U, s}^* / \mathcal{O}_{U', s}^* / \text{Im } \Gamma(U', \mathcal{O}_{U'}^*) \right) \longrightarrow \text{Pic}(U) \longrightarrow \text{Pic}(U') \longrightarrow 0$$

and it thus suffices to show that the second term of this sequence is reduced to the neutral element, by taking S the set of y_i ($1 \leq i \leq r$) and noting that $\mathcal{O}'_X = f_*(\mathcal{O}_{X'})$ is the $\mathcal{O}_X$ -Algebra $\widetilde{A'}$.

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Proposition (21.13.10). — *Let X be a locally Noetherian prescheme, Y a closed subset of X . For the couple (X, Y) to be parafactorial, it is necessary and sufficient that, for every $y \in Y$, the local ring $\mathcal{O}_{X, y}$ be parafactorial.*

Proof. Set $U = X - Y$ and let $j : U \rightarrow X$ be the canonical injection. By virtue of (21.13.4), to say that the canonical homomorphism $\mathcal{O}_X \rightarrow j_*(\mathcal{O}_U)$ is injective is equivalent to saying that each of the local rings $\mathcal{O}_{X, y}$ for $y \in Y$ satisfies condition (i bis) of (21.13.8).

Let us show first that if the couple (X, Y) is parafactorial, each of the local rings $\mathcal{O}_{X,y}$ ($y \in Y$) is parafactorial. Taking into account the remark from the beginning, everything reduces to seeing that, if one sets $T_y = \operatorname{Spec}(\mathcal{O}_{X,y})$ and $U_y = T_y - \{y\}$, the hypothesis that $\mathcal{O}_{X,y}$ is parafactorial entails $\operatorname{Pic}(U_y) = 0$; but as T_y is a projective limit of the preschemes induced by X on the affine open neighbourhoods $V - (V \cap \overline{\{y\}})$ of y in X , which are quasi-compact since X is locally Noetherian, that the prescheme U_y is a projective limit of preschemes $U_V = V - (V \cap \overline{\{y\}})$ which are separated, it results from (8.5.2, (ii)) and (8.5.5) that there exists a affine open neighbourhood V of y in X and an $\mathcal{O}_{U_V}$ -Module invertible $\mathcal{L}'_V$ on U_V inducing $\mathcal{L}'$ on U_y ; applying this (8.5.2) and (8.5.2, b)) that $j_*(\mathcal{L})|_{V_1}$ is isomorphic to $\mathcal{L}_1$, hence invertible. But this contradicts the definition of V , and one thus concludes the proof of (21.13.10).

Conversely, let us prove that if all the $\mathcal{O}_{X,y}$ ($y \in Y$) are parafactorial the couple (X, Y) is parafactorial. One is evidently reduced, taking into account the remark from the beginning, to showing that for every invertible $\mathcal{O}_U$ -Module $\mathcal{L}$, $j_*(\mathcal{L})$ is an invertible $\mathcal{O}_X$ -Module (21.13.5). The question being local on X , one can suppose X Noetherian. The set of $x \in X$ such that the restriction of $j_*(\mathcal{L})$ to a neighbourhood of x in X is invertible is evidently an open in X ; it suffices to show that $V = X$. Suppose the contrary and set $Z = X - V \neq \emptyset$. Let y be a maximal point of Z ; since $Z \subset Y$ by definition, $\mathcal{O}_{X,y}$ is by hypothesis a parafactorial ring. Replacing X as needed by a neighbourhood of y in X , one can suppose that the restriction of $j_*(\mathcal{L})$ to $V - (V \cap \overline{\{y\}})$ is invertible; hence, with the notations introduced above, the $\mathcal{O}_{U_y}$ -Module induced by $j_*(\mathcal{L})$ on U_y is invertible. Since $\mathcal{O}_{X,y}$ is parafactorial, $\mathcal{L}_0$ is induced by an $\mathcal{O}_{T_y}$ -Module invertible $\mathcal{L}'$; but as T_y is projective limit of affine open neighbourhoods W of y in X , one can again apply (8.5.2, (ii)) and (8.5.5); replacing W by a smaller neighbourhood of y , one can suppose that $\mathcal{L}_V$ is isomorphic to $\mathcal{O}_V$, hence $\mathcal{L}_0$ is isomorphic to $\mathcal{O}_{U_y}$. One concludes then, from (21.13.2, b')), that $j_*(\mathcal{L})$ induces on W an invertible sheaf $\mathcal{L}_1$. But this contradicts the definition of V , which concludes the proof. $\square$

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Corollary (21.13.11). — *Let X be a locally Noetherian prescheme, Y a closed subset of X , $U = X - Y$. Suppose that the couple (X, Y) is parafactorial and that U is locally factorial; in other terms (21.13.10), for every $x \in X$, the ring $\mathcal{O}_{X,x}$ is: 1° parafactorial if $x \in Y$; 2° factorial if $x \notin Y$. Then X is locally factorial (in other words, $\mathcal{O}_{X,x}$ is in fact factorial for every $x \in X$).*

Proof. Suppose on the contrary, and let y be a point of Y such that $\mathcal{O}_{X,y}$ is not factorial and having the least dimension *minima* amongst the points of Y with this property. Then, if one sets $T_y = \operatorname{Spec}(\mathcal{O}_{X,y})$, $U_y = T_y - \{y\}$, the hypothesis that $\mathcal{O}_{X,y}$ is parafactorial entails $\operatorname{Pic}(U_y) = 0$ and $\operatorname{prof}(\mathcal{O}_{X,y}) \geq 2$ (21.13.8); moreover the choice of y entails that U_y is locally factorial. But then (21.6.14) proves that $\mathcal{O}_{X,y}$ is factorial, contrary to the hypothesis. $\square$

Proposition (21.13.12). — *Let A, B be two local Noetherian rings, $\rho : A \rightarrow B$ a local homomorphism making B a flat A -module. Then, if B is factorial, so is A .*

Proof. One already knows that A is integral and integrally closed (6.5.4), hence one can reduce to the case where $\dim(A) \geq 2$, and reason by induction on $n = \dim(A)$. Let $X = \operatorname{Spec}(A)$, a the closed point of X , $X' = \operatorname{Spec}(B)$, $f : X' \rightarrow X$ the structural morphism, $U = X - \{a\}$, $U' = f^{-1}(U)$.

The local rings $\mathcal{O}_{X',x'}$ at the points $x' \in f^{-1}(a)$ are factorial and of dimension ≥ 2 (6.1.2), hence parafactorial (21.13.9, (ii)); one thus concludes (21.13.10) that the couple $(X', f^{-1}(a))$ is parafactorial. By virtue of (21.13.6), (iii)), this shows that the ring A is parafactorial, hence $\operatorname{prof}(A) \geq 2$ and $\operatorname{Pic}(U) = 0$ (21.13.8). Finally, the local rings of U being of dimension $< n$, the induction hypothesis on the recursion proves that U is locally factorial; the conclusion results then from (21.6.14). $\square$

(21.13.12.1). One will note that the statement of (21.13.12) where one replaces “factorial” by “parafactorial” is no longer exact, as shown by the example where $A = k$ is a field and B the local ring $k[[T_1, T_2]]$. However, it results from (21.13.12) that, under the conditions of (21.13.6), (iii)), if $\mathfrak{m}$ denotes the maximal ideal of A and if $\mathfrak{m}B$ is an ideal of definition of B (in other words, if $\dim(B/\mathfrak{m}B) = 0$), then, when B is parafactorial, so is A . In particular, if the completion $\hat{A}$ of a Noetherian local ring is parafactorial, so is A .

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Remark (21.13.13). — A part of the preceding definitions and results relates to more general considerations on the cohomology of sheaves of commutative groups on a topological space, developed in chap. III, 3^e partie, and we have only given here an independent account for the convenience of the reader. Indeed, if X is a ringed space, one has a canonical *equivalence* between the category of invertible $\mathcal{O}_X$ -Modules and that of *principal homogeneous sheaves under the sheaf of groups* $\mathcal{O}_X^*$ (16.5.15). This equivalence is defined by making correspond to every invertible $\mathcal{O}_X$ -Module $\mathcal{L}$ the sheaf $\mathcal{L}^* = \mathcal{I} \text{som}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{L})$ of germs of isomorphisms of $\mathcal{O}_X$ onto $\mathcal{L}$; it is immediate that $\mathcal{L}^*$ is canonically endowed with a structure of principal homogeneous sheaf under $\mathcal{O}_X^* \cong \mathcal{I} \text{som}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{O}_X)$. The verification that the functor $\mathcal{L} \mapsto \mathcal{L}^*$ is an equivalence of categories is immediate. This being so, let us consider in a general way a topological space X ; let Y be a closed subset of X , set $U = X - Y$, and let i be an integer such that $0 \leq i \leq 2$; we will say that the couple (X, Y) is *i -pure* for $\mathcal{G}$ if, for every open V of X , the restriction functor $\mathcal{P} \mapsto \mathcal{P}|_{(U \cap V)}$, from the category of principal homogeneous sheaves under the sheaf of groups $\mathcal{G}|_V$, to the analogous category under the sheaf of groups $\mathcal{G}|_{(U \cap V)}$, is faithful ($i = 0$), resp. fully faithful ($i = 1$), resp. an equivalence of categories ($i = 2$). In the case where X is a ringed space and $\mathcal{G} = \mathcal{O}_X^*$, to say that (X, Y) is *i -pure* thus means for $i = 0$, that the homomorphism $\mathcal{O}_X \rightarrow j_*(\mathcal{O}_U)$ is injective; for $i = 1$, that this homomorphism is bijective; and finally, for $i = 2$, that the couple (X, Y) is *parafactorial* (21.13.3).

Returning to the general case, recall that the morphisms of the category of principal sheaves under $\mathcal{G}$ are by definition isomorphisms. To say that the couple (X, Y) is *0-pure* thus means that for every open V of X , the canonical homomorphism $H^0(V, \mathcal{G}) \rightarrow H^0(U \cap V, \mathcal{G})$ is injective; to say that the couple (X, Y) is *1-pure* means that the canonical homomorphism $H^0(V, \mathcal{G}) \rightarrow H^0(U \cap V, \mathcal{G})$ is bijective (and then the canonical homomorphism $H^1(V, \mathcal{G}) \rightarrow H^1(U \cap V, \mathcal{G})$ is injective); finally, one can show that for the couple (X, Y) to be *2-pure*, it is necessary and sufficient that the homomorphisms $H^i(V, \mathcal{G}) \rightarrow H^i(U \cap V, \mathcal{G})$ be bijective for $i = 0$ and $i = 1$. When $\mathcal{G}$ is a sheaf of commutative groups, by introducing the cohomology sheaves $\mathcal{H}_Y^i(\mathcal{G})$ defined in chap. III, 3^e partie, to say that (X, Y) is *i -pure* for $i \leq 2$ means that $\mathcal{H}_Y^j(\mathcal{G}) = 0$ for $j \leq i$; under this form, the notion generalises immediately for every integer i .

Proposition (21.13.14). — *Let X be a locally Noetherian prescheme, $(U_\lambda)_{\lambda \in L}$ a decreasing filtering family of open subsets of X . The following conditions are equivalent:*

- a) *Every 1-codimensional cycle Z on X such that there exists an index λ for which $Z|_{U_\lambda}$ is locally principal (21.6.7), is locally principal.*
- b) *For every $x \in X$ such that $\dim(\mathcal{O}_{X,x}) \geq 2$ and such that x does not belong to $\bigcap_\lambda U_\lambda$, the ring $\mathcal{O}_{X,x}$ is parafactorial.*

Property a) can also be expressed as follows: if the closed set N of points $x \in X$ where Z is not locally principal is contained in one of the $X - U_\lambda$, then Z is locally principal. As X is normal, the condition $\dim(\mathcal{O}_{X,x}) \leq 1$ entails that $\mathcal{O}_{X,x}$ is a field or a discrete valuation ring, hence Z_x is principal; hence necessarily $\text{codim}(N, X) \geq 2$ (5.1.3). Property a) is therefore equivalent to the following:

- a') *If there exists a closed set Y contained in one of the $X - U_\lambda$, such that $\text{codim}(Y, X) \geq 2$ and such that $Z|_{X - Y}$ is locally principal, then Z is locally principal.*

Let us note on the other hand that b) implies b') by virtue of (21.13.10); conversely, if b') is satisfied and if $x \notin U_\lambda$, then $Y = \{x\}$ is contained in $X - U_\lambda$ and one has $\text{codim}(Y, X) \geq 2$, which shows that $\mathcal{O}_{X,x}$ is parafactorial, hence the equivalence of a') and of b'). Let us note that since X is normal, on has, for every closed subset Y of X such that $\text{codim}(Y, X) \geq 2$, $\text{prof}_Y(\mathcal{O}_X) \geq 2$ (5.8.6); if one sets $U = X - Y$, the homomorphism $\mathcal{O}_X \rightarrow j_*(\mathcal{O}_U)$ is therefore bijective (21.13.4); as conditions a') and b') are local, on has, for every closed subset Y of X such that $\text{codim}(Y, X) \geq 2$ and $Y \subset X - U_\lambda$, the couple (X, Y) is parafactorial, that is to say that the equivalence of conditions a') and b').

Let us first prove that a'') entails c). Inversely, let us prove that b'') entails a''). Let Z be a 1-codimensional cycle on X such that $Z|_U$ is locally principal; the image of $Z|_U$ in $\mathcal{C}l(U)$ (21.6.11) is the class of a 1-codimensional cycle Z_0 on U , which is locally principal. Since Z_0 is locally principal, it is also the case of Z on U , by virtue of a''); it results from (21.6.11) that the image of Z in $\mathcal{C}l(X)$ is the image of a unique element $\zeta \in \text{Pic}(X)$, and it is clear then that ζ_0 is the image of ζ .

Let us prove first that a'') entails c). Let Z be a 1-codimensional cycle on X which is locally principal; by hypothesis $Z|_{U_\lambda}$ is also locally principal. Conversely c) entails a''): if $Z|_U$ is locally

principal, the image of $Z|U$ in $\mathcal{C}l(U)$ is therefore the class of a 1-codimensional cycle Z' on X such that $Z|U$ and $Z'|U$ are linearly equivalent. But since U is schematically dense in X by the isomorphism $\mathfrak{z}^1(X) \rightarrow \mathfrak{z}^1(U)$, Z and Z' are linearly equivalent, and since Z' is locally principal by virtue of a''), hence so is Z .

Corollary (21.13.15). — *Let X be a locally Noetherian prescheme and normal, S a subset of X . The following conditions are equivalent:*

- a) Every 1-codimensional cycle on X which is principal at the points of S is locally principal.
- b) For every $x \in X$ which is not a generisation of a point of S (in other words, such that $\overline{\{x\}} \cap S = \emptyset$) and which is such that $\dim(\mathcal{O}_{X,x}) \geq 2$, the ring $\mathcal{O}_{X,x}$ is parafactorial.

It is clear that the set S' of the points of X which are generisations of points of S is the intersection of the open neighbourhoods of S . One can reduce to the case where X is Noetherian (the properties a) and b) being local on X); as every 1-codimensional cycle on X which is principal at the points of S is also such in a neighbourhood open of S , one sees therefore that, in the statement of (21.13.15), if $S = \{z\}$, the conditions a) and b) are also equivalent to the following:

- c) Every 1-codimensional cycle on X whose support does not contain z is locally principal.

If moreover $\text{Pic}(X) = 0$, one concludes that this condition entails that every 1-codimensional cycle whose support does not contain z is *principal*.

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Remark (21.13.16). — Under the general conditions of (21.13.14), suppose moreover that one has $\varinjlim \text{Pic}(U_\lambda) = 0$. Then condition a) of (21.13.14) is also equivalent to the following:

- c) Every 1-codimensional cycle on X whose support is contained in one of the sets $X - U_\lambda$ is locally principal.

One can reduce to the case where X is irreducible and all the U_λ are non-empty. It is clear that a) entails c), for if the 1-codimensional cycle Z on X has its support in $X - U_\lambda$, $Z|U_\lambda = 0$, hence $Z|U_\lambda$ is locally principal. Conversely c) entails a): let Z be a 1-codimensional cycle on X such that $Z|U_\lambda$ is locally principal; by hypothesis there exists $U_\mu \subset U_\lambda$ such that $D_\mu = \text{div}(f_\mu)$, where f_μ is a rational function regular on U_μ , and f_μ can be viewed as a rational function regular on X . If $Z' = \text{cyc}(\text{div}(f_\mu))$, then $Z'' = Z - Z'$ has its support in $X - U_\mu$; by virtue of the hypothesis, Z'' is locally principal, hence so is the conclusion.

The condition $\varinjlim \text{Pic}(U_\lambda) = 0$ is fulfilled in particular if (U_λ) is the family of open neighbourhoods of a point $z \in X$, for then for every $\mathcal{O}_{U_\lambda}$ -Module invertible $\mathcal{L}_\lambda$, there exists $U_\mu \subset U_\lambda$ such that $\mathcal{L}_\lambda|U_\mu$ is isomorphic to $\mathcal{O}_{U_\mu}$. One sees therefore that, in the statement of (21.13.15), if $S = \{z\}$, the conditions a) and b) are also equivalent to the following:

- c) Every 1-codimensional cycle on X whose support does not contain z is locally principal.

If moreover $\text{Pic}(X) = 0$, one concludes that this condition entails that every 1-codimensional cycle whose support does not contain z is *principal*.

21.14. The Ramanujam–Samuel theorem.

Theorem (21.14.1). — (Ramanujam–Samuel) *Let A be a Noetherian local ring of maximal ideal $\mathfrak{m}$, such that its completion $\hat{A}$ is integral and integrally closed. Let B be a Noetherian ring such that $\dim(B) > \dim(A)$, $\rho : A \rightarrow B$ a local homomorphism making B a A -algebra formally smooth (for the preadic topologies (0, 19.3.1)) and such that the residue field of B is finite over that of A . Then every 1-codimensional cycle on $\text{Spec}(B)$ which is principal at the point $\mathfrak{p} = \mathfrak{m}B$, is a 1-codimensional principal cycle.*

Proof. If $k = A/\mathfrak{m}$ is the residue field of A , $B/\mathfrak{m}B$ is a k -algebra formally smooth (for its preadic topology) (0, 19.3.5), hence regular, and in particular integral, in other words $\mathfrak{p} = \mathfrak{m}B$ is indeed a prime ideal of B , which justifies the statement. Everything evidently reduces to proving that every prime ideal $\mathfrak{q}$ of B not contained in $\mathfrak{p}$ is *principal*.

Let $\hat{A}, \hat{B}$ be the completions of A and B respectively, so that the maximal ideal of $\hat{A}$ is $\mathfrak{m}\hat{A}$; one knows (0, 19.3.6) that $\hat{B}$ is an $\hat{A}$ -algebra formally smooth for the adic topologies. Let k' be the residue field of $\hat{B}$, finite extension of k ; there exists a homomorphism of local rings $\hat{A} \rightarrow C$, where C is a local Noetherian ring which is a *finite* and *flat* $\hat{A}$ -module and is such that $C/\mathfrak{m}C$ is isomorphic to

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k' (0_{III}, 10.3.1); one deduces then from (7.5.1), (7.5.3) and (6.5.4), (ii)) that C is *integral and integrally closed*. Moreover, $D = \widehat{B} \otimes_{\widehat{A}} C$ is a complete semi-local ring, composed direct of complete local rings of which one, D_0 , has for residue field k' (since $k' \otimes_k k'$ is composed direct of local rings of which one is isomorphic to k'). As D is formally smooth over C , so is $D_0/\mathfrak{m}D_0$, which is a k' -algebra formally smooth, hence is k' -isomorphic to an algebra of formal power series $k'[[T_1, \dots, T_n]]$ (0, 19.6.4); one concludes, by (0, 19.7.1.5), that D_0 is C -isomorphic to $C[[T_1, \dots, T_n]]$, and hence *integral and integrally closed* (Bourbaki, *Alg. comm.*, chap. V, § 1, n° 4, prop. 14). As the morphisms $\text{Spec}(D_0) \rightarrow \text{Spec}(\widehat{B})$ and $\text{Spec}(D_0) \rightarrow \text{Spec}(\widehat{C})$ are faithfully flat, one deduces that B and $\widehat{B}$ are also *integral and integrally closed* (2.1.13). This proves that $\mathfrak{q}D_0$ is divisorial (Bourbaki, *Alg. comm.*, chap. VII, § 1, n° 10, prop. 15) and not contained in $\mathfrak{m}D_0$, hence in particular that no prime ideal $\mathfrak{r}_h$ of height 1 in D_0 which contains $\mathfrak{q}D_0$ can be contained in $\mathfrak{m}D_0$; since $\mathfrak{q}D_0$ is principal by faithful flatness (0_I, 6.5.1). One concludes that the divisors (in the sense of Bourbaki) of $\mathfrak{q}D_0$ and of a product of powers of the $\mathfrak{r}_h$ are equal (Bourbaki, *Alg. comm.*, chap. VII, § 1, n° 4, prop. 5). As $\mathfrak{q}D_0 \cap B = \mathfrak{q}$, one deduces by faithful flatness that $\mathfrak{q}$ is principal (2.5.2), using the fact that B is a local ring.

One is thus reduced to proving the theorem in the particular case where A is *complete* and $B = A[[T_1, \dots, T_n]]$. Let us show first that one can reduce to the case where $n = 1$. Indeed, we proceed by induction on n , and let $f \in \mathfrak{q}$ be an element not belonging to $\mathfrak{p} = \mathfrak{m}B$; it will suffice to show that there exists an A -automorphism σ of B such that $\sigma(f)$ does not belong to the ideal $\mathfrak{p}' = \mathfrak{p} + BT_1 + \dots + BT_{n-1}$: indeed, if $B' = A[[T_1, \dots, T_{n-1}]]$, B' is complete and has for maximal ideal $\mathfrak{m}B' + B'T_1 + \dots + B'T_{n-1} = \mathfrak{m}'$. One has $B = B'[[T_n]]$ and $\mathfrak{p}' = \mathfrak{m}'B$, and it will result from the induction hypothesis (taking into account that B' is integral and integrally closed) that the ideal $\sigma(\mathfrak{q})$ is principal in B , hence also $\mathfrak{q}$. Now, if $\bar{f} \in k[[T_1, \dots, T_n]]$ is the image of f (“reduced series” of f), one knows (Bourbaki, *Alg. comm.*, chap. VII, § 3, n° 7, lemme 3) that there is an A -automorphism σ of B such that $(\sigma(f))(0, \dots, 0, T_n) \neq 0$, which implies evidently $\sigma(f) \notin \mathfrak{p}'$.

Suppose then $n = 1$ and write T in place of T_1 , so that $B = A[[T]]$. It suffices to show that in the polynomial ring $A[T]$, the ideal $\mathfrak{q} \cap A[T]$ is principal; indeed, let f_0 be an element of $\mathfrak{q}$ not belonging to $\mathfrak{m}B$; this is a power series whose reduced series $\bar{f}_0$ is non-zero; hence, by virtue of the preparation theorem (Bourbaki, *Alg. comm.*, chap. VII, § 3, n° 8, prop. 5), for every $f \in \mathfrak{q}$, there exist $g \in B$ and a polynomial $r \in A[T]$ such that $f = gf_0 + r$ and one has thus $r \in \mathfrak{q} \cap A[T]$; on the other hand (*loc. cit.*, prop. 6) there exist a distinguished non-constant polynomial $F_0 \in A[T]$ and an invertible $u \in B$ such that $f_0 = uF_0$, hence one also has $F_0 \in \mathfrak{q} \cap A[T]$, which proves that $\mathfrak{q}$ is generated by $\mathfrak{q} \cap A[T] = \mathfrak{q}_1$. Since B is flat over $A[T]$ (0_I, 7.3.3), it results from Bourbaki, *Alg. comm.*, chap. VII, § 1, n° 10, prop. 15, that $\mathfrak{q}_1$ is a prime ideal of height 1 in $A[T]$. Moreover, one has necessarily $\mathfrak{q}_1 \cap A = 0$; otherwise, $\mathfrak{q}_1 \cap A$ would necessarily be of height ≥ 1 , and it would result from (5.5.3) that one would have $\mathfrak{q}_1 = (\mathfrak{q}_1 \cap A)A[T]$. But then, since $\mathfrak{q}_1 \cap A \subset \mathfrak{m}$, one would have $\mathfrak{q}_1 \subset \mathfrak{m}A[T]$, contrary to the hypothesis on $\mathfrak{q}$. If K is the field of fractions of A , $\mathfrak{q}_1K[T]$ is therefore a prime ideal distinct from 0 and of $K[T]$, hence of the form $h \cdot K[T]$, where $h(T) = T^m + a_1T^{m-1} + \dots + a_m$ with $m \geq 1$ and the $a_i \in K$, h being irreducible in $K[T]$. But one has seen above that there exists in $\mathfrak{q}_1$ a distinguished non-constant polynomial $F_0 \in A[T]$. If t is the class of T in $K[T]/\mathfrak{q}_1K[T]$, t is therefore a root of the polynomial F_0 in an extension of K and hence h divides F_0 in $K[T]$; but since h and F_0 are unitary, this entails that the coefficients a_i of h are integers over A (Bourbaki, *Alg. comm.*, chap. V, § 1, n° 3, prop. 11), hence belong to A since A is integrally closed. In other words, one has $h \in A[T] \cap \mathfrak{q}_1K[T] = \mathfrak{q}_1$ (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 5, prop. 11); as every polynomial $g \in \mathfrak{q}_1$ is divisible by h in $K[T]$ and h is unitary, the coefficients of g/h belong to A , hence $\mathfrak{q}_1 = h \cdot A[T]$. Q.E.D. $\square$

The statement (21.14.1) is *equivalent* to the following:

Corollary (21.14.2). — *Under the hypotheses of (21.14.1) on A , B and ρ , for every prime ideal $\mathfrak{q}$ of B not contained in $\mathfrak{p} = \mathfrak{m}B$ and such that $\dim(B_{\mathfrak{q}}) \geq 2$, the ring $B_{\mathfrak{q}}$ is *parafactorial*. In particular, if $\dim(B \otimes_A k) \geq 0$ (i.e. $\dim B > \dim A$ ((0, 19.7.1) and (6.1.1))), the ring B is *parafactorial*.*

Proof. Indeed, one has seen in the proof of (21.14.1) that the conditions of (21.14.1) entail that B is integral and integrally closed; the equivalence of the statements (21.14.1) and (21.14.2) results then from (21.13.15) applied to $X = \text{Spec}(B)$ and $S = \{\mathfrak{p}\}$. $\square$

Proposition (21.14.3). — *Let S be a normal prescheme, $f : X \rightarrow S$ a smooth morphism.*

- (i) *If S is locally Noetherian (hence also X), then, for every $x \in X$ such that $\dim(\mathcal{O}_{X,x}) \geq 2$ and such that x is not the maximal point of its fibre $f^{-1}(f(x))$, the ring $\mathcal{O}_{X,x}$ is local factorial. Every 1-codimensional cycle Z on X such that $\text{Supp}(Z)$ does not contain any irreducible component of a fibre $f^{-1}(s) = X_s$ is locally principal.*
- (ii) *Let Y be a closed subset of X not containing any irreducible component of a fibre X_s , and such that for every maximal point η of S , $Y_\eta = Y \cap X_\eta$ is of codimension ≥ 2 in X_η . Then the couple (X, Y) is local factorial.*

One will note that the conditions of (ii) are satisfied if Y is a closed subset of X such that, for every $s \in S$, $Y_s = Y \cap X_s$ is of codimension ≥ 2 in X_s .

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Proof. To prove (21.14.3), let us note first that under the hypotheses of (i), if one sets $Y = \overline{\{x\}}$, there exists a neighbourhood V of x in X such that $Y \cap V$ and V verify the conditions of (ii): indeed, it results from the hypothesis and from (9.5.3) that one can take V such that $V \cap Y$ does not contain any irreducible component of an X_s ; on the other hand, for every maximal point η of S such that $Y \cap X_\eta \neq \emptyset$, the points of $Y \cap X_\eta$ are specialisations of x , hence there is at most a single point z and one has $\dim(\mathcal{O}_{X,z}) \geq 2$, and as $\mathcal{O}_{X,z} = \mathcal{O}_{X_\eta,z}$ (the reduction being at point η by virtue of (2.1.13)), $Y \cap X_\eta$ is indeed of codimension ≥ 2 . As the couple $(V, Y \cap V)$ is then local factorial by virtue of (ii), one concludes from (21.13.10) that $\mathcal{O}_{X,x}$ is local factorial, i.e. the first assertion of (i). To prove the second, one can reduce to the case where $Z = \{\bar{z}\}$, where $z \in X^{(1)}$; as $\mathcal{O}_{X,z}$ is a discrete valuation ring, Z is therefore principal at the point z ; for every other point x of $\text{Supp}(Z)$, one has $\dim(\mathcal{O}_{X,x}) \geq 2$ and x is not the maximal point of its fibre by hypothesis, hence $\mathcal{O}_{X,x}$ is local factorial. We then apply (21.13.15) taking for S the set formed by z and the maximal points of the fibres of f ; it is clear that Z is principal at the points of S ; as on the other hand condition b) of (21.13.15) is evidently satisfied, one concludes that Z is locally principal.

We are therefore reduced to proving (ii). Set $U = X - Y$ and let $j : U \rightarrow X$ be the canonical injection. As the hypotheses and the conclusion are local on S and on X by virtue of (21.13.6), (i), one can reduce to the case where S and X are affine, f being then of finite presentation. As the fibres of f are regular, the hypothesis on Y entails that, for every point $x \in Y$, $\text{prof}(\mathcal{O}_{X_{f(x)},x}) = \dim(\mathcal{O}_{X_{f(x)},x}) \geq 1$; it results from (1998) that the canonical homomorphism $\mathcal{O}_X \rightarrow j_*(\mathcal{O}_U)$ is injective. Moreover, replacing Y by a larger closed subset following the method of the proof of (1998), and using (21.13.6), (ii), one can suppose that Y is defined by an ideal of finite type of the ring of X , hence the open $U = X - Y$ is quasi-compact and quasi-separated and the closed set Y is constructible.

To prove (ii), it suffices, for every invertible $\mathcal{O}_U$ -Module $\mathcal{L}$ and every point $x \in Y$, to establish the existence of a neighbourhood open V of x in X such that: 1° the canonical homomorphism $\mathcal{O}_V \rightarrow (j_V)_*(\mathcal{O}_{U \cap V})$ is surjective; 2° there exists an invertible $\mathcal{O}_V$ -Module $\mathcal{L}'_V$ such that $\mathcal{L}'_V|_{(U \cap V)} \cong \mathcal{L}|_{(U \cap V)}$ ((21.13.5) and (21.13.3)). Set $s = f(x)$; we will show that one can reduce to the case where $S = S_0 = \text{Spec}(\mathcal{O}_{S,s})$. Let (S_ν) be a fundamental system of affine open neighbourhoods of s in S , and set $X_\nu = f^{-1}(S_\nu)$, $Y_\nu = Y \cap X_\nu$, $U_\nu = X_\nu - Y_\nu$, $j_\nu : U_\nu \rightarrow X_\nu$ being the canonical injection; $X_0 = X \times_S S_0$ is then a projective limit of the X_ν (8.1.2, a)); suppose the proposition true for the morphism $f_0 : X_0 \rightarrow S_0$ and set $Y_0 = Y \cap X_0$, and set $U_0 = X_0 - Y_0$, $j_0 : U_0 \rightarrow X_0$ being the canonical injection. The canonical projection $p_\nu : X_0 \rightarrow X_\nu$ being an affine morphism, one has $(j_0)_*(\mathcal{O}_{U_0}) = p_\nu^*((j_\nu)_*(\mathcal{O}_{U_\nu}))$ ((II, 1.5.2)) and the canonical homomorphism $\rho_0 : \mathcal{O}_{X_0} \rightarrow (j_0)_*(\mathcal{O}_{U_0})$ is nothing other than $p_\nu^*(\rho_\nu)$, where $\rho_\nu : \mathcal{O}_{X_\nu} \rightarrow (j_\nu)_*(\mathcal{O}_{U_\nu})$ is the canonical homomorphism; as by hypothesis ρ_0 is surjective, so is ρ_ν for ν large enough (8.5.2, (ii)); moreover (8.5.5) one can suppose that $\mathcal{L}'_\nu$ is invertible. Finally, as the U_ν are quasi-compact and quasi-separated, and as $\mathcal{L}'_0|_{U_0} = \mathcal{L}_0$, there exists ν large enough such that $\mathcal{L}'_\nu|_{U_\nu}$ and $\mathcal{L}|_{U_\nu}$ are isomorphic (8.5.2.5).

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One is thus reduced to the case where $S = \text{Spec}(A)$, $X = \text{Spec}(B)$, where A is a local ring. Since X is normal and f smooth, S is normal (17.5.7), hence A is integral and integrally closed. Consider then A as an inductive limit of its sub- $\mathbf{Z}$ -algebras of finite type; as the integral closure of such a sub- $\mathbf{Z}$ -algebra is a sub-ring of A and is also a $\mathbf{Z}$ -algebra of finite type (7.8.3, (ii), (iii) and (vi)), A is an inductive filtering limit of sub- $\mathbf{Z}$ -algebras of type A_λ of A which are rings integrally closed. By virtue of (1.8.4.2), there exists an index λ and an A_λ -algebra of finite type B_λ such that $B = B_\lambda \otimes_{A_\lambda} A$ up to

an A -isomorphism. Set $S_\lambda = \operatorname{Spec}(A_\lambda)$, $X_\lambda = \operatorname{Spec}(B_\lambda)$; if $p_\lambda : X \rightarrow X_\lambda$ is the canonical projection, one can moreover suppose (since Y is *constructible*) that $Y = p_\lambda^{-1}(Y_\lambda)$, where Y_λ is a closed subset of X_λ (8.3.11). Let f_λ be the morphism $X_\lambda \rightarrow S_\lambda$; since f is smooth, one can also suppose that f_λ is smooth (17.7.8); finally, the image of the generic point η of S in S_λ is the generic point η_λ of S_λ , and one has $\operatorname{codim}((Y_\lambda)_{\eta_\lambda}, (X_\lambda)_{\eta_\lambda}) = \operatorname{codim}(Y_\eta, X_\eta) \geq 2$ by virtue of (6.1.4). One sees therefore that S_λ , X_λ and Y_λ verify all the hypotheses of (ii), and on setting $X_\mu = X_\lambda \times_{S_\lambda} S_\mu$ for $\lambda \leq \mu$, it is the same of S_μ , X_μ and Y_μ . Let us show that if the couple (X_μ, Y_μ) is parafactorial for every $\mu \geq \lambda$, then it is the same of (X, Y) . Indeed, let $U = X - Y$, $U_\lambda = X_\lambda - Y_\lambda$, $j : U \rightarrow X$, $j_\lambda : U_\lambda \rightarrow X_\lambda$ the canonical injections; the projection $p_\lambda : X \rightarrow X_\lambda$ being an affine morphism, one has $j_*(\mathcal{O}_U) = p_\lambda^*((j_\lambda)_*(\mathcal{O}_{U_\lambda}))$ ((II, 1.5.2)), and hence the canonical homomorphism $\rho : \mathcal{O}_X \rightarrow j_*(\mathcal{O}_U)$ is nothing other than $p_\lambda^*(\rho_\lambda)$; this latter being by hypothesis bijective, so is ρ . On the other hand, for every invertible $\mathcal{O}_U$ -Module $\mathcal{L}$, there exists a $\mu \geq \lambda$ and an invertible $\mathcal{O}_{U_\mu}$ -Module $\mathcal{L}_\mu$ such that $\mathcal{L} = p_\mu^*(\mathcal{L}_\mu)$ (denoting by $p'_\mu : U \rightarrow U_\mu$ the restriction of p_μ) (8.5.2 and 8.5.5), U_λ being quasi-compact since X_λ is Noetherian. By hypothesis, there exists an $\mathcal{O}_{X_\mu}$ -Module invertible $\mathcal{L}'_\mu$ such that $\mathcal{L}'_\mu|_{U_\mu}$ is isomorphic to $\mathcal{L}_\mu$; $\mathcal{L}' = p_\mu^*(\mathcal{L}'_\mu)$ is then an $\mathcal{O}_X$ -Module invertible such that $\mathcal{L}'|_U$ is isomorphic to $\mathcal{L}$, which proves our assertion.

One is thus reduced to proving (ii) when A is a $\mathbf{Z}$ -algebra of finite type integrally closed; as the local rings $\mathcal{O}_{S,s}$ of S are then rings which are excellent integrally closed (7.8.3, (ii), (iii) and (vii)), their completions are also integrally closed.

For every $x \in Y$, the ring $\mathcal{O}_{X,y}$ is an $\mathcal{O}_{S,s}$ -algebra formally smooth for the preadic topologies (17.5.3); since Y_s does not contain any irreducible component of X_s , one has $\dim(\mathcal{O}_{X,x}) > \dim(\mathcal{O}_{S,s})$ (17.5.8). If $s \neq \eta$, one has $\dim(\mathcal{O}_{S,s}) \geq 1$; if on the contrary $s = \eta$, one has by hypothesis $\dim(\mathcal{O}_{X,x}) \geq 2$; hence in all cases $\dim(\mathcal{O}_{X,x}) \geq 2$. Moreover, the prime ideal of $\mathcal{O}_{X,y}$ corresponding to the point x is not contained in $\mathfrak{m}_s \mathcal{O}_{X,y}$ since Y does not contain any irreducible component of X_s . Finally, since y is a closed point of X_s , $k(y)$ is a finite extension of $k(s)$ ((I, 6.4.2)). In all cases, one can apply to $A = \mathcal{O}_{X,y}$ and $B_q = \mathcal{O}_{X,x}$ the result of (21.14.2), which finishes the proof. $\square$

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Remark (21.14.4). — (i) It is possible that the statements (21.14.1) and (21.14.2) remain valid *without hypothesis* on the residue field of B . Stated with this generality, the result would be equivalent, by virtue of (21.13.15) and (21.13.12.1), to the following: let A be a Noetherian local ring, complete, integral and integrally closed, B a Noetherian local ring which is an A -algebra formally smooth, such that $\dim(B) > \dim(A)$; then B is parafactorial.

(ii) We will see in chap. VI, using a technique of “finite descent” applied to the morphism $Y' \rightarrow Y = \operatorname{Spec}(A)$, where Y' is the normalised of Y , that the conclusion of (21.14.1) (or of (21.14.2)) remains valid on replacing the hypothesis that $\hat{A}$ is integrally closed by the hypothesis that $\hat{A}$ is *reduced*, provided that $\dim B \geq \dim A + 2$. If one replaces this last condition by $\dim B \geq \dim A + 3$, one can even suppress the hypothesis that $\hat{A}$ is reduced. Similarly, the conclusion of (21.14.3) remains valid on replacing the hypothesis that X is normal by the hypothesis that X is reduced, provided that $\dim(\mathcal{O}_{X,x}) \geq \dim(\mathcal{O}_{S,f(x)}) + 2$.

(iii) In chap. III, 3^e partie, one proves that if $f : X \rightarrow S$ is a smooth morphism, Y a locally constructible closed subset of X such that, for every $s \in S$, one has (with the conditions of (21.14.3)) $\operatorname{codim}(Y_s, X_s) \geq 3$, then the couple (X, Y) is parafactorial ([?, XII, 4.8]). This conclusion is no longer valid when one supposes only that $\operatorname{codim}(Y_s, X_s) \geq 2$ for every $s \in S$ and when one does not suppose X reduced. Take for example k a field, $A = k[T]/(T^2)$, algebra of dual numbers over k , $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(A[T_1, T_2])$ (T_1, T_2 indeterminate), so that $X = \mathbf{V}_S^2$, Y being the “zero section” of this fibre; if s is the unique closed point of S , one has $X_s = \mathbf{V}_k^2$ and Y_s is the “zero section” of X_s . To see that the couple (X, Y) is not parafactorial, it suffices to show that the ring $B = A[T_1, T_2]_{\mathfrak{m}}$, where $\mathfrak{m}$ is the ideal $(T_1) + (T_2)$ which defines Y , is not parafactorial (21.13.10). Let $B_0 = B_{\text{red}}$, and denote by U and U_0 the complements of the closed point in $\operatorname{Spec}(B)$ and $\operatorname{Spec}(B_0)$; by reasoning as in (21.13.9), one has the exact sequence, extension of (21.13.9.1)

$$\Gamma(U, \mathcal{O}_U)^* \longrightarrow \Gamma(U_0, \mathcal{O}_{U_0})^* \longrightarrow H^1(U_0, \mathcal{O}_{U_0}) \longrightarrow \operatorname{Pic}(U) \longrightarrow \operatorname{Pic}(U_0).$$

Now, one has $\Gamma(U, \mathcal{O}_U) = B$, $\Gamma(U_0, \mathcal{O}_{U_0}) = B_0$ since $\text{prof}(B) = \text{prof}(B_0) = 2$ (5.10.5). Furthermore $\text{Pic}(U_0) = 0$ since B_0 is factorial, and $H^1(U_0, \mathcal{O}_{U_0}) \neq 0$ [?, 3, Exemple III-1]; as the homomorphism $B^* \rightarrow B_0^*$ is surjective, one concludes that $\text{Pic}(U) \neq 0$, hence B is not parafactorial.

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